CHAPTER

32

LANDING GEAR

CHAPTER 32
LANDING GEAR

Subject/Page	Date	COC
32-EFFECTIVE PAGES		
1 thru 20	Sep 15/2026	
32-CONTENTS		
1	May 15/2025	
2	Sep 15/2025	
3	Sep 15/2025	
4	Sep 15/2024	
5	May 15/2024	
6	May 15/2024	
7	May 15/2025	
8	May 15/2024	
9	May 15/2024	
10	Jan 15/2026	
O 11	Sep 15/2026	
12	Jan 15/2026	
13	Jan 15/2026	
O 14	Sep 15/2026	
15	Jan 15/2026	
16	Jan 15/2026	
17	Jan 15/2026	
18	May 15/2026	
19	Jan 15/2026	
20	Jan 15/2026	
21	Jan 15/2026	
22	Jan 15/2026	
O 23	Sep 15/2026	
24	May 15/2026	
25	Jan 15/2026	
26	Jan 15/2026	
O 27	Sep 15/2026	
28	Jan 15/2026	
29	May 15/2026	
30	Jan 15/2026	
31	Jan 15/2026	
32	Jan 15/2026	
33	Jan 15/2026	
34	Jan 15/2026	

Subject/Page	Date	COC
32-CONTENTS (cont)		
35	Jan 15/2026	
36	Jan 15/2026	
O 37	Sep 15/2026	
38	Jan 15/2026	
39	Jan 15/2026	
40	Jan 15/2026	
32-00-00		
901	Sep 15/2025	
902	May 15/2024	
903	May 15/2024	
904	May 15/2024	
905	May 15/2024	
906	May 15/2025	
907	May 15/2025	
908	May 15/2025	
909	May 15/2024	
910	May 15/2024	
911	Sep 15/2025	
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914	Sep 15/2025	
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916	Sep 15/2025	
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919	Sep 15/2025	
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921	Sep 15/2025	
922	Sep 15/2025	
923	Sep 15/2025	
924	Sep 15/2025	
925	May 15/2026	
926	May 15/2026	
927	Sep 15/2025	
928	Sep 15/2025	
929	Sep 15/2025	

Subject/Page	Date	COC
32-00-00 (cont)		
930	Sep 15/2025	
931	Sep 15/2025	
932	Sep 15/2025	
933	Sep 15/2025	
934	Sep 15/2025	
935	Sep 15/2025	
936	Sep 15/2025	
937	Sep 15/2025	
938	Sep 15/2025	
32-00-01		
201	Jan 15/2025	
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206	May 15/2024	
207	May 15/2024	
208	Jan 15/2025	
209	Jan 15/2025	
210	BLANK	
32-00-10		
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602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
32-00-10		
701	May 15/2024	
702	May 15/2024	
703	May 15/2024	
704	May 15/2024	
32-00-10		
801	May 15/2024	
802	May 15/2024	
32-00-11		
701	May 15/2024	
702	May 15/2024	
703	May 15/2024	

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32-00-11 (cont)		
704	May 15/2024	
32-00-37		
201	May 15/2024	
202	BLANK	
32-05-03		
601	Jan 15/2026	
602	May 15/2024	
603	May 15/2024	
604	Jan 15/2026	
605	May 15/2024	
606	May 15/2024	
607	Jan 15/2026	
608	Jan 15/2026	
609	May 15/2024	
610	May 15/2024	
32-09-00		
201	Sep 15/2024	
202	Jan 15/2025	
203	Jan 15/2025	
204	Jan 15/2025	
205	Jan 15/2025	
206	Jan 15/2025	
207	May 15/2026	
R 208	Sep 15/2026	
209	Sep 15/2024	
210	Sep 15/2024	
211	Sep 15/2024	
212	Sep 15/2024	
213	Jan 15/2025	
214	Jan 15/2025	
215	Sep 15/2024	
216	BLANK	
32-09-01		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
32-09-01 (cont)		
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
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32-09-01		
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502	May 15/2024	
503	May 15/2024	
504	May 15/2024	
R 505	Sep 15/2026	
R 506	Sep 15/2026	
32-09-02		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
32-09-02		
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503	May 15/2024	
504	May 15/2024	
505	May 15/2024	
506	May 15/2024	
32-09-10		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
R 404	Sep 15/2026	
405	May 15/2024	
406	May 15/2024	
32-09-10		
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511	Jan 15/2025	
512	Jan 15/2025	
513	Jan 15/2025	
514	Jan 15/2025	
515	May 15/2024	
516	May 15/2024	
32-09-12		
501	Sep 15/2024	
502	May 15/2026	
503	Jan 15/2026	
504	May 15/2024	
505	May 15/2024	
506	Sep 15/2024	
507	May 15/2024	
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515	May 15/2024	
516	May 15/2024	
517	May 15/2024	
518	May 15/2024	
519	May 15/2024	
520	BLANK	
32-11-00		
201	Sep 15/2025	
202	Sep 15/2025	
203	May 15/2024	

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206	May 15/2024	
207	May 15/2024	
208	May 15/2024	
209	May 15/2024	
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211	May 15/2024	
212	BLANK	
32-11-00		
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420	May 15/2024	
421	May 15/2024	
422	May 15/2024	
423	May 15/2024	
424	May 15/2025	
R 425	Sep 15/2026	
426	Jan 15/2025	

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32-11-00 (cont)		
427	Jan 15/2025	
428	May 15/2024	
429	May 15/2025	
430	May 15/2025	
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433	Jan 15/2025	
434	May 15/2025	
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438	May 15/2025	
439	May 15/2026	
440	May 15/2025	
441	May 15/2025	
442	May 15/2025	
32-11-00		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
32-11-21		
801	May 15/2024	
802	Jan 15/2025	
803	May 15/2024	
804	May 15/2024	
805	Jan 15/2025	
806	May 15/2024	
807	May 15/2024	
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809	May 15/2024	
810	May 15/2025	
811	May 15/2025	
812	May 15/2025	
813	May 15/2024	
814	May 15/2024	
815	May 15/2024	

Subject/Page	Date	COC
32-11-21 (cont)		
816	May 15/2024	
817	May 15/2024	
818	May 15/2025	
819	May 15/2024	
820	May 15/2024	
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822	May 15/2024	
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824	May 15/2024	
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829	May 15/2024	
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833	May 15/2025	
834	May 15/2025	
835	Sep 15/2024	
836	Sep 15/2024	
837	Sep 15/2024	
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843	May 15/2025	
844	Jan 15/2026	
R 845	Sep 15/2026	
846	May 15/2025	
847	May 15/2025	
848	May 15/2025	
849	May 15/2025	
850	May 15/2025	
851	May 15/2025	

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852	Sep 15/2025	
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R 854	Sep 15/2026	
855	Sep 15/2025	
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32-11-37		
R 201	Sep 15/2026	
R 202	Sep 15/2026	
R 203	Sep 15/2026	
204	May 15/2024	
205	May 15/2024	
206	May 15/2024	
32-11-51		
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32-11-51		
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32-11-61		
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407	May 15/2024	
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417	May 15/2024	
418	Jan 15/2026	
419	Sep 15/2025	
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427	Sep 15/2025	
428	Jan 15/2026	
429	Sep 15/2025	
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431	Jan 15/2025	
432	BLANK	
32-11-61		
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603	May 15/2024	
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32-11-62		
401	Jan 15/2025	
402	May 15/2024	
R 403	Sep 15/2026	
R 404	Sep 15/2026	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
411	May 15/2024	
R 412	Sep 15/2026	
O 413	Sep 15/2026	
414	May 15/2024	
R 415	Sep 15/2026	
R 416	Sep 15/2026	
R 417	Sep 15/2026	
418	May 15/2025	
32-11-71		
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407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2025	
411	May 15/2024	
412	May 15/2024	
413	May 15/2024	

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416	May 15/2024	
417	May 15/2024	
418	May 15/2024	
419	May 15/2024	
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32-11-81		
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207	May 15/2024	
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210	May 15/2024	
211	Sep 15/2025	
212	Sep 15/2025	
213	Sep 15/2025	
214	BLANK	
32-11-81		
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404	May 15/2024	
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406	May 15/2024	
407	May 15/2024	
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409	May 15/2024	
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32-11-81		
601	Jan 15/2026	
602	Jan 15/2026	
603	Jan 15/2026	

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32-11-81 (cont)		
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605	Jan 15/2026	
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32-11-83		
401	May 15/2026	
402	May 15/2024	
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404	May 15/2024	
32-11-85		
401	May 15/2024	
402	Sep 15/2024	
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405	May 15/2024	
406	May 15/2024	
R 407	Sep 15/2026	
408	Sep 15/2025	
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410	Sep 15/2025	
411	Sep 15/2025	
412	Sep 15/2025	
413	Sep 15/2025	
414	Sep 15/2025	
32-11-85		
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32-11-89		
401	Jan 15/2025	
402	Jan 15/2025	
403	May 15/2024	
404	May 15/2024	

Subject/Page	Date	COC
32-11-89 (cont)		
405	May 15/2024	
406	May 15/2024	
407	Sep 15/2025	
408	Sep 15/2025	
409	Sep 15/2025	
410	Sep 15/2025	
32-12-12		
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403	May 15/2026	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
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32-12-12		
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603	May 15/2024	
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606	BLANK	
32-13-11		
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402	May 15/2026	
403	May 15/2026	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
R 411	Sep 15/2026	
O 412	Sep 15/2026	
O 413	Sep 15/2026	
O 414	Sep 15/2026	

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O 416	Sep 15/2026	
R 417	Sep 15/2026	
O 418	Sep 15/2026	
O 419	Sep 15/2026	
O 420	Sep 15/2026	
O 421	Sep 15/2026	
R 422	Sep 15/2026	
O 423	Sep 15/2026	
O 424	Sep 15/2026	
O 425	Sep 15/2026	
O 426	Sep 15/2026	
32-13-11		
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32-13-21		
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402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	

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32-13-21 (cont)		
408	May 15/2026	
409	May 15/2024	
410	May 15/2024	
32-13-21		
501	Jan 15/2026	
502	Jan 15/2026	
503	Jan 15/2026	
504	Jan 15/2026	
505	Jan 15/2026	
506	May 15/2024	
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32-21-00		
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32-21-00		
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405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
411	May 15/2024	

Subject/Page	Date	COC
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413	May 15/2024	
414	May 15/2024	
415	May 15/2024	
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427	Jan 15/2025	
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429	Jan 15/2025	
430	BLANK	
32-21-00		
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603	May 15/2024	
604	May 15/2024	
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606	Jan 15/2026	
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611	May 15/2024	
612	May 15/2024	
613	Jan 15/2026	
614	Jan 15/2026	
615	May 15/2024	
616	May 15/2024	

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32-21-00		
801	May 15/2024	
802	May 15/2024	
803	May 15/2024	
804	BLANK	
32-21-11		
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802	Sep 15/2025	
803	May 15/2024	
804	May 15/2024	
805	Jan 15/2025	
806	May 15/2024	
807	Sep 15/2025	
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822	May 15/2024	
823	May 15/2024	
824	Sep 15/2024	
825	Jan 15/2025	
826	Jan 15/2025	
827	Sep 15/2024	
828	Sep 15/2024	
829	Jan 15/2025	

Subject/Page	Date	COC
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831	Sep 15/2024	
832	Sep 15/2024	
833	Sep 15/2024	
834	Sep 15/2024	
835	Sep 15/2024	
836	Sep 15/2024	
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850	May 15/2025	
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854	May 15/2025	
32-21-14		
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403	May 15/2024	
404	May 15/2024	
32-21-21		
401	Jan 15/2025	
402	Jan 15/2025	
403	Jan 15/2025	
404	Jan 15/2025	
405	May 15/2024	
406	Jan 15/2025	

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409	May 15/2024	
410	May 15/2024	
411	May 15/2024	
412	Jan 15/2025	
413	Jan 15/2025	
R 414	Sep 15/2026	
415	Jan 15/2025	
416	Jan 15/2025	
32-21-21		
R 601	Sep 15/2026	
602	May 15/2024	
32-21-24		
401	Jan 15/2025	
402	Jan 15/2025	
403	Jan 15/2025	
404	May 15/2026	
405	May 15/2026	
406	May 15/2024	
407	May 15/2024	
408	May 15/2026	
409	May 15/2026	
410	May 15/2026	
411	May 15/2026	
412	May 15/2026	
413	May 15/2026	
414	May 15/2026	
415	May 15/2026	
416	May 15/2026	
417	May 15/2026	
418	May 15/2026	
419	May 15/2026	
420	May 15/2026	
421	May 15/2026	
422	May 15/2026	
423	May 15/2026	

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Subject/Page	Date	COC
32-21-24 (cont)		
424	May 15/2026	
425	May 15/2026	
426	May 15/2026	
32-21-31		
R 401	Sep 15/2026	
R 402	Sep 15/2026	
O 403	Sep 15/2026	
O 404	Sep 15/2026	
O 405	Sep 15/2026	
O 406	Sep 15/2026	
R 407	Sep 15/2026	
R 408	Sep 15/2026	
R 409	Sep 15/2026	
R 410	Sep 15/2026	
R 411	Sep 15/2026	
R 412	Sep 15/2026	
R 413	Sep 15/2026	
R 414	Sep 15/2026	
R 415	Sep 15/2026	
R 416	Sep 15/2026	
A 417	Sep 15/2026	
A 418	Sep 15/2026	
32-21-31		
801	Sep 15/2025	
802	Sep 15/2025	
803	May 15/2024	
804	May 15/2024	
32-21-37		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
204	May 15/2024	
205	May 15/2024	
206	May 15/2024	
207	May 15/2024	
208	BLANK	
32-21-71		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
32-22-11		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2026	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
32-22-11		
501	May 15/2024	
502	Sep 15/2025	
503	May 15/2024	
504	May 15/2024	
505	Sep 15/2025	
506	May 15/2024	
507	Sep 15/2025	
508	Sep 15/2025	
32-31-00		
501	May 15/2026	
R 502	Sep 15/2026	
503	May 15/2024	
504	May 15/2024	
505	May 15/2024	
506	May 15/2026	
507	May 15/2024	
508	May 15/2024	
R 509	Sep 15/2026	
510	May 15/2024	
511	May 15/2024	
512	BLANK	
32-31-11		
R 401	Sep 15/2026	
R 402	Sep 15/2026	
403	May 15/2024	
404	May 15/2024	
R 405	Sep 15/2026	
R 406	Sep 15/2026	
32-31-11		
501	May 15/2026	
502	Jan 15/2026	
503	Jan 15/2026	
504	Jan 15/2026	
32-31-51		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2026	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
411	May 15/2024	
412	May 15/2024	
32-31-51		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
605	May 15/2024	
606	May 15/2024	
32-31-71		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2026	

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CHAPTER 32
LANDING GEAR

Subject/Page	Date	COC
32-31-71 (cont)		
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
411	May 15/2024	
412	May 15/2024	
413	May 15/2024	
414	Jan 15/2025	
415	Jan 15/2025	
416	BLANK	
32-31-71		
501	May 15/2024	
502	May 15/2024	
503	May 15/2024	
504	BLANK	
32-32-00		
501	May 15/2024	
502	May 15/2024	
503	May 15/2024	
504	May 15/2024	
505	May 15/2024	
506	May 15/2024	
507	May 15/2024	
508	May 15/2024	
509	May 15/2024	
510	May 15/2024	
511	May 15/2024	
512	May 15/2024	
513	May 15/2024	
514	BLANK	
32-32-11		
401	May 15/2026	
402	May 15/2026	
403	May 15/2026	
404	May 15/2024	
405	May 15/2024	

Subject/Page	Date	COC
32-32-11 (cont)		
406	May 15/2026	
407	May 15/2024	
408	May 15/2024	
409	Jan 15/2025	
410	May 15/2024	
411	May 15/2024	
412	May 15/2024	
413	May 15/2024	
414	May 15/2024	
415	May 15/2024	
416	BLANK	
32-32-21		
401	Jan 15/2025	
402	Jan 15/2025	
403	Jan 15/2025	
404	May 15/2025	
405	Jan 15/2025	
406	BLANK	
32-32-31		
R 401	Sep 15/2026	
R 402	Sep 15/2026	
403	May 15/2024	
R 404	Sep 15/2026	
405	Jan 15/2025	
R 406	Sep 15/2026	
407	Jan 15/2025	
408	Jan 15/2025	
R 409	Sep 15/2026	
410	Jan 15/2025	
R 411	Sep 15/2026	
R 412	Sep 15/2026	
R 413	Sep 15/2026	
R 414	Sep 15/2026	
R 415	Sep 15/2026	
R 416	Sep 15/2026	
D 417	Sep 15/2026	
D 418	Sep 15/2026	

Subject/Page	Date	COC
32-32-31 (cont)		
D 419	Sep 15/2026	
D 420	BLANK	
32-32-41		
401	May 15/2026	
402	May 15/2026	
403	May 15/2024	
404	May 15/2026	
405	May 15/2026	
406	May 15/2026	
407	May 15/2026	
408	May 15/2026	
409	May 15/2026	
410	BLANK	
32-32-51		
401	Jan 15/2026	
402	Jan 15/2026	
403	Jan 15/2026	
404	Jan 15/2026	
405	May 15/2024	
406	Jan 15/2026	
407	Jan 15/2026	
R 408	Sep 15/2026	
409	Jan 15/2026	
410	Jan 15/2026	
32-32-52		
401	May 15/2026	
402	Jan 15/2025	
403	May 15/2024	
404	May 15/2024	
405	May 15/2026	
406	Sep 15/2025	
407	May 15/2026	
408	May 15/2026	
409	May 15/2024	
410	BLANK	

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CHAPTER 32
LANDING GEAR

Subject/Page	Date	COC
32-32-71		
401	May 15/2026	
402	May 15/2026	
403	May 15/2026	
404	May 15/2026	
405	May 15/2026	
406	May 15/2026	
407	May 15/2026	
408	BLANK	
32-32-72		
401	May 15/2026	
402	May 15/2026	
403	May 15/2026	
404	May 15/2024	
405	May 15/2024	
406	BLANK	
32-32-91		
401	May 15/2026	
402	May 15/2026	
403	May 15/2026	
404	May 15/2026	
405	May 15/2026	
406	May 15/2026	
407	May 15/2026	
408	May 15/2026	
32-32-93		
401	Jan 15/2026	
402	Jan 15/2026	
403	May 15/2024	
404	May 15/2024	
405	May 15/2026	
406	May 15/2026	
407	Jan 15/2026	
408	BLANK	
32-33-00		
501	May 15/2024	
502	May 15/2024	
503	May 15/2024	
32-33-00 (cont)		
504	May 15/2024	
505	May 15/2024	
506	May 15/2024	
507	May 15/2024	
508	May 15/2024	
32-33-11		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	Sep 15/2024	
408	Sep 15/2024	
409	Sep 15/2024	
410	BLANK	
32-33-21		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2025	
406	May 15/2024	
407	May 15/2024	
408	BLANK	
32-33-31		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
32-33-41		
401	May 15/2024	
402	May 15/2024	
32-33-41 (cont)		
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
32-33-51		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
R 409	Sep 15/2026	
410	May 15/2026	
411	May 15/2026	
412	May 15/2024	
32-33-52		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
32-33-61		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	

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CHAPTER 32
LANDING GEAR

Subject/Page	Date	COC
32-33-61		
501	May 15/2024	
502	May 15/2024	
503	May 15/2024	
504	May 15/2024	
505	May 15/2024	
506	May 15/2024	
507	May 15/2024	
508	May 15/2024	
32-33-61		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
605	May 15/2024	
606	May 15/2024	
607	May 15/2024	
608	May 15/2024	
32-33-71		
401	May 15/2024	
402	May 15/2025	
403	May 15/2025	
404	May 15/2025	
405	May 15/2025	
406	May 15/2025	
407	May 15/2024	
408	May 15/2025	
409	May 15/2025	
410	May 15/2025	
32-33-72		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	BLANK	

Subject/Page	Date	COC
32-34-00		
R 501	Sep 15/2026	
R 502	Sep 15/2026	
R 503	Sep 15/2026	
504	May 15/2024	
505	May 15/2024	
506	May 15/2024	
507	Jan 15/2026	
508	May 15/2024	
509	May 15/2024	
510	May 15/2025	
R 511	Sep 15/2026	
R 512	Sep 15/2026	
R 513	Sep 15/2026	
514	May 15/2024	
R 515	Sep 15/2026	
516	May 15/2024	
R 517	Sep 15/2026	
R 518	Sep 15/2026	
R 519	Sep 15/2026	
R 520	Sep 15/2026	
R 521	Sep 15/2026	
O 522	Sep 15/2026	
A 523	Sep 15/2026	
A 524	BLANK	
32-34-11		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
411	May 15/2024	

Subject/Page	Date	COC
32-34-11 (cont)		
412	May 15/2024	
413	May 15/2024	
414	May 15/2024	
415	May 15/2024	
416	May 15/2024	
417	May 15/2024	
418	BLANK	
32-34-21		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
32-34-31		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
32-35-00		
501	May 15/2024	
502	May 15/2024	
503	May 15/2024	
504	May 15/2025	
505	May 15/2026	
R 506	Sep 15/2026	
507	May 15/2026	

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CHAPTER 32
LANDING GEAR

Subject/Page	Date	COC
32-35-00 (cont)		
508	May 15/2026	
R 509	Sep 15/2026	
R 510	Sep 15/2026	
R 511	Sep 15/2026	
O 512	Sep 15/2026	
513	May 15/2026	
514	May 15/2026	
32-35-11		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	BLANK	
32-35-20		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	BLANK	
32-35-21		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	Sep 15/2025	
410	Sep 15/2025	
411	Sep 15/2025	
412	BLANK	
32-41-00		
201	May 15/2024	
R 202	Sep 15/2026	
32-41-00 (cont)		
R 203	Sep 15/2026	
O 204	Sep 15/2026	
R 205	Sep 15/2026	
R 206	Sep 15/2026	
R 207	Sep 15/2026	
R 208	Sep 15/2026	
R 209	Sep 15/2026	
R 210	Sep 15/2026	
R 211	Sep 15/2026	
R 212	Sep 15/2026	
O 213	Sep 15/2026	
R 214	Sep 15/2026	
R 215	Sep 15/2026	
R 216	Sep 15/2026	
O 217	Sep 15/2026	
O 218	BLANK	
32-41-00		
501	Sep 15/2025	
R 502	Sep 15/2026	
R 503	Sep 15/2026	
R 504	Sep 15/2026	
R 505	Sep 15/2026	
R 506	Sep 15/2026	
O 507	Sep 15/2026	
O 508	Sep 15/2026	
509	May 15/2024	
510	May 15/2024	
511	May 15/2024	
512	May 15/2024	
513	May 15/2024	
514	May 15/2024	
515	Sep 15/2025	
516	Jan 15/2025	
517	Jan 15/2025	
518	May 15/2024	
519	May 15/2024	
32-41-00 (cont)		
520	Sep 15/2025	
521	Sep 15/2025	
522	BLANK	
32-41-11		
401	May 15/2024	
402	Sep 15/2025	
403	May 15/2025	
404	May 15/2024	
405	Sep 15/2025	
406	Sep 15/2025	
407	Sep 15/2025	
408	BLANK	
32-41-21		
401	May 15/2026	
402	May 15/2026	
403	May 15/2026	
404	May 15/2026	
405	May 15/2026	
406	May 15/2026	
407	May 15/2026	
408	May 15/2026	
409	May 15/2026	
410	May 15/2026	
411	May 15/2026	
412	May 15/2026	
413	May 15/2026	
414	May 15/2026	
32-41-22		
401	Jan 15/2025	
402	Jan 15/2025	
403	Jan 15/2025	
404	Jan 15/2025	
405	Jan 15/2025	
406	Jan 15/2025	
407	Jan 15/2025	
408	Jan 15/2025	

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CHAPTER 32
LANDING GEAR

Subject/Page	Date	COC
32-41-31		
401	Jan 15/2026	
402	Jan 15/2026	
403	Jan 15/2026	
404	Jan 15/2026	
405	Jan 15/2026	
406	Jan 15/2026	
407	Jan 15/2026	
408	May 15/2024	
409	Jan 15/2026	
R 410	Sep 15/2026	
R 411	Sep 15/2026	
O 412	Sep 15/2026	
O 413	Sep 15/2026	
414	BLANK	
32-41-31		
601	Jan 15/2026	
602	Jan 15/2026	
32-41-41		
R 401	Sep 15/2026	
402	Sep 15/2025	
403	Sep 15/2025	
404	Sep 15/2025	
405	Sep 15/2025	
406	Sep 15/2025	
407	Sep 15/2025	
408	Sep 15/2025	
409	Sep 15/2025	
410	Sep 15/2025	
R 411	Sep 15/2026	
412	Sep 15/2025	
413	Sep 15/2025	
414	Jan 15/2026	
415	May 15/2026	
416	May 15/2026	
32-41-41		
601	Sep 15/2024	
602	Sep 15/2024	
32-41-41 (cont)		
603	Sep 15/2024	
604	May 15/2024	
R 605	Sep 15/2026	
R 606	Sep 15/2026	
607	Jan 15/2026	
608	Jan 15/2026	
609	Jan 15/2026	
610	Jan 15/2026	
611	Sep 15/2024	
612	Sep 15/2024	
613	Sep 15/2024	
614	Jan 15/2026	
615	Jan 15/2026	
616	Sep 15/2024	
32-41-42		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	Sep 15/2024	
405	Sep 15/2024	
406	BLANK	
32-41-51		
401	May 15/2026	
402	May 15/2026	
403	May 15/2024	
404	May 15/2026	
405	May 15/2024	
406	May 15/2024	
407	May 15/2025	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
411	May 15/2025	
412	May 15/2024	
413	May 15/2024	
414	BLANK	
32-41-53		
401	May 15/2026	
402	May 15/2026	
403	May 15/2026	
404	May 15/2026	
405	May 15/2026	
406	BLANK	
32-41-54		
401	May 15/2026	
402	May 15/2024	
403	May 15/2026	
404	May 15/2026	
32-41-71		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	BLANK	
32-41-72		
401	Jan 15/2026	
402	May 15/2024	
403	May 15/2024	
R 404	Sep 15/2026	
405	Sep 15/2025	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	Sep 15/2025	
411	May 15/2024	
412	BLANK	
32-41-81		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	

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Subject/Page	Date	COC
32-41-81 (cont)		
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407	May 15/2024	
408	May 15/2024	
409	Sep 15/2025	
410	May 15/2024	
411	May 15/2024	
412	BLANK	
32-41-81		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
605	May 15/2024	
606	May 15/2024	
32-41-91		
401	May 15/2024	
402	May 15/2026	
403	May 15/2024	
404	May 15/2026	
R 405	Sep 15/2026	
R 406	Sep 15/2026	
O 407	Sep 15/2026	
408	BLANK	
32-41-93		
R 401	Sep 15/2026	
R 402	Sep 15/2026	
R 403	Sep 15/2026	
R 404	Sep 15/2026	
405	May 15/2025	
406	May 15/2025	
R 407	Sep 15/2026	
408	May 15/2025	
409	Jan 15/2026	
410	BLANK	
32-41-93		
501	May 15/2024	
502	Jan 15/2026	

Subject/Page	Date	COC
32-41-93 (cont)		
503	Jan 15/2026	
504	Jan 15/2026	
505	Jan 15/2026	
506	Jan 15/2026	
507	Jan 15/2026	
508	BLANK	
32-41-95		
401	May 15/2026	
R 402	Sep 15/2026	
R 403	Sep 15/2026	
R 404	Sep 15/2026	
R 405	Sep 15/2026	
O 406	Sep 15/2026	
A 407	Sep 15/2026	
A 408	BLANK	
32-41-96		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	Jan 15/2025	
405	Jan 15/2025	
406	Jan 15/2025	
32-42-00		
501	May 15/2025	
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503	Sep 15/2025	
504	May 15/2024	
505	May 15/2024	
506	Jan 15/2025	
507	May 15/2024	
508	May 15/2024	
509	May 15/2024	
510	May 15/2024	
511	May 15/2024	
512	May 15/2024	
513	Jan 15/2025	
514	Jan 15/2025	

Subject/Page	Date	COC
32-42-00 (cont)		
515	Sep 15/2025	
516	May 15/2025	
517	May 15/2024	
518	Jan 15/2025	
519	Sep 15/2025	
520	Sep 15/2025	
521	Sep 15/2025	
522	Sep 15/2025	
523	Sep 15/2025	
524	Sep 15/2025	
525	Sep 15/2025	
526	Sep 15/2025	
527	Sep 15/2025	
528	Sep 15/2025	
529	Sep 15/2025	
530	Sep 15/2025	
531	Sep 15/2025	
532	Sep 15/2025	
533	Sep 15/2025	
534	May 15/2024	
535	May 15/2024	
536	BLANK	
32-42-11		
401	May 15/2025	
402	May 15/2025	
403	May 15/2025	
404	May 15/2025	
405	May 15/2025	
406	May 15/2026	
32-42-21		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	

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CHAPTER 32
LANDING GEAR

Subject/Page	Date	COC
32-42-31		
R 401	Sep 15/2026	
R 402	Sep 15/2026	
R 403	Sep 15/2026	
404	May 15/2024	
R 405	Sep 15/2026	
R 406	Sep 15/2026	
R 407	Sep 15/2026	
R 408	Sep 15/2026	
R 409	Sep 15/2026	
410	May 15/2024	
411	May 15/2026	
R 412	Sep 15/2026	
R 413	Sep 15/2026	
R 414	Sep 15/2026	
R 415	Sep 15/2026	
R 416	Sep 15/2026	
A 417	Sep 15/2026	
A 418	BLANK	
32-42-61		
401	May 15/2025	
402	May 15/2025	
403	May 15/2024	
404	May 15/2025	
405	May 15/2025	
406	May 15/2025	
407	May 15/2025	
408	May 15/2025	
409	May 15/2025	
410	BLANK	
32-42-71		
R 401	Sep 15/2026	
R 402	Sep 15/2026	
R 403	Sep 15/2026	
404	May 15/2024	
405	May 15/2025	
406	May 15/2025	
407	Jan 15/2026	
32-42-71 (cont)		
408	May 15/2025	
409	May 15/2025	
410	BLANK	
32-42-81		
401	May 15/2026	
402	May 15/2026	
403	May 15/2026	
404	May 15/2026	
405	May 15/2026	
406	May 15/2026	
407	May 15/2026	
408	May 15/2026	
409	May 15/2026	
410	May 15/2026	
411	May 15/2026	
412	May 15/2026	
413	May 15/2026	
414	May 15/2026	
415	May 15/2026	
416	May 15/2026	
32-44-00		
501	Sep 15/2025	
502	Sep 15/2025	
503	Sep 15/2025	
504	May 15/2024	
505	May 15/2024	
506	May 15/2024	
507	May 15/2024	
508	May 15/2024	
509	May 15/2024	
510	May 15/2024	
511	May 15/2024	
512	May 15/2024	
513	May 15/2024	
514	May 15/2024	
515	May 15/2024	
516	May 15/2025	
32-44-00 (cont)		
517	May 15/2025	
518	May 15/2024	
519	May 15/2024	
520	BLANK	
32-44-11		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
411	Jan 15/2026	
412	Jan 15/2026	
413	May 15/2024	
414	May 15/2024	
415	Jan 15/2026	
416	May 15/2024	
417	May 15/2024	
418	May 15/2024	
419	May 15/2024	
420	May 15/2024	
421	May 15/2024	
422	BLANK	
32-44-21		
401	May 15/2025	
402	May 15/2025	
403	May 15/2024	
404	May 15/2024	
405	May 15/2026	
R 406	Sep 15/2026	
R 407	Sep 15/2026	
R 408	Sep 15/2026	
O 409	Sep 15/2026	

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CHAPTER 32
LANDING GEAR

Subject/Page	Date	COC
32-44-21 (cont)		
410	BLANK	
32-45-00		
201	May 15/2024	
202	Sep 15/2025	
203	May 15/2024	
204	May 15/2024	
32-45-00		
601	May 15/2024	
602	Sep 15/2025	
603	Sep 15/2025	
604	Sep 15/2025	
605	May 15/2024	
606	May 15/2024	
607	May 15/2024	
608	May 15/2024	
609	May 15/2024	
610	May 15/2024	
611	May 15/2024	
612	May 15/2024	
613	May 15/2024	
614	May 15/2024	
615	May 15/2024	
616	May 15/2024	
617	May 15/2024	
618	May 15/2024	
32-45-08		
401	May 15/2024	
402	May 15/2025	
403	May 15/2025	
404	Jan 15/2026	
405	Jan 15/2026	
406	Jan 15/2026	
32-45-11		
401	Sep 15/2025	
R 402	Sep 15/2026	
403	Sep 15/2025	
404	Sep 15/2025	

Subject/Page	Date	COC
32-45-11 (cont)		
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
R 410	Sep 15/2026	
O 411	Sep 15/2026	
O 412	Sep 15/2026	
O 413	Sep 15/2026	
O 414	Sep 15/2026	
32-45-21		
401	Sep 15/2025	
R 402	Sep 15/2026	
403	Sep 15/2025	
404	Sep 15/2025	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
R 409	Sep 15/2026	
O 410	Sep 15/2026	
O 411	Sep 15/2026	
O 412	Sep 15/2026	
O 413	Sep 15/2026	
414	BLANK	
32-45-31		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
204	May 15/2024	
205	May 15/2024	
206	BLANK	
32-46-00		
R 501	Sep 15/2026	
O 502	Sep 15/2026	

Subject/Page	Date	COC
32-46-21		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	BLANK	
32-46-31		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	BLANK	
32-49-00		
R 501	Sep 15/2026	
502	May 15/2024	
503	May 15/2024	
504	May 15/2024	
32-49-03		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
411	May 15/2024	
412	May 15/2024	
413	May 15/2024	
414	May 15/2024	
415	May 15/2024	
416	May 15/2024	
417	May 15/2024	
418	May 15/2024	

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LANDING GEAR

Subject/Page	Date	COC
32-49-03 (cont)		
419	Sep 15/2025	
420	Sep 15/2025	
421	Sep 15/2025	
422	Sep 15/2025	
423	Sep 15/2025	
424	Sep 15/2025	
425	Sep 15/2025	
426	Sep 15/2025	
32-51-00		
501	May 15/2024	
502	May 15/2024	
503	May 15/2024	
504	May 15/2024	
505	May 15/2024	
506	May 15/2024	
507	May 15/2024	
508	May 15/2024	
509	May 15/2024	
510	May 15/2024	
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512	May 15/2024	
513	May 15/2024	
514	May 15/2024	
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516	May 15/2024	
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518	May 15/2024	
519	May 15/2024	
520	May 15/2024	
521	May 15/2024	
522	May 15/2024	
523	May 15/2024	
524	May 15/2024	
525	Sep 15/2025	
526	May 15/2025	
527	May 15/2025	
32-51-00 (cont)		
528	May 15/2025	
529	May 15/2025	
530	May 15/2025	
531	May 15/2025	
532	May 15/2025	
533	May 15/2025	
534	May 15/2025	
535	May 15/2025	
R 536	Sep 15/2026	
O 537	Sep 15/2026	
O 538	Sep 15/2026	
539	May 15/2025	
540	May 15/2025	
R 541	Sep 15/2026	
542	May 15/2025	
543	May 15/2025	
544	May 15/2025	
545	May 15/2025	
546	May 15/2025	
547	May 15/2025	
548	Jan 15/2026	
549	Jan 15/2026	
550	May 15/2025	
551	May 15/2025	
552	May 15/2025	
553	May 15/2025	
554	May 15/2025	
32-51-00		
601	Sep 15/2025	
602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
605	May 15/2024	
606	May 15/2024	
607	May 15/2024	
608	May 15/2024	
32-51-11		
201	Sep 15/2025	
202	Sep 15/2025	
203	Sep 15/2025	
204	BLANK	
32-51-11		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	Jan 15/2026	
408	Jan 15/2026	
409	Jan 15/2026	
410	Jan 15/2026	
32-51-21		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
411	May 15/2024	
412	May 15/2024	
413	May 15/2024	
414	May 15/2024	
32-51-31		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	

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Subject/Page	Date	COC
32-51-31 (cont)		
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
R 410	Sep 15/2026	
411	May 15/2024	
R 412	Sep 15/2026	
413	May 15/2024	
414	Sep 15/2024	
415	May 15/2024	
416	May 15/2024	
417	May 15/2024	
418	May 15/2024	
419	May 15/2024	
420	May 15/2024	
421	May 15/2024	
422	May 15/2024	
423	May 15/2024	
424	May 15/2024	
425	May 15/2026	
426	BLANK	
32-51-31		
601	May 15/2024	
602	May 15/2024	
32-51-41		
401	May 15/2024	
402	Sep 15/2024	
403	Sep 15/2024	
404	Sep 15/2024	
405	Sep 15/2024	
406	Sep 15/2024	
407	Sep 15/2024	
408	Sep 15/2024	
409	May 15/2026	
410	Sep 15/2024	
411	Sep 15/2024	
412	BLANK	
32-51-51		
401	Jan 15/2025	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	Jan 15/2026	
410	Jan 15/2026	
411	Jan 15/2026	
412	Jan 15/2026	
413	May 15/2024	
414	May 15/2024	
415	May 15/2024	
416	BLANK	
32-51-51		
601	May 15/2024	
602	May 15/2024	
32-51-52		
401	May 15/2024	
402	May 15/2024	
403	May 15/2025	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2026	
408	May 15/2024	
409	May 15/2025	
410	May 15/2024	
411	May 15/2025	
412	May 15/2024	
413	May 15/2025	
414	May 15/2024	
415	May 15/2024	
416	May 15/2024	
417	May 15/2025	
32-51-52 (cont)		
418	May 15/2024	
419	May 15/2025	
420	May 15/2025	
32-51-61		
401	Jan 15/2025	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
411	May 15/2024	
412	May 15/2024	
413	May 15/2024	
414	May 15/2024	
415	Jan 15/2025	
416	May 15/2024	
417	May 15/2024	
418	May 15/2024	
419	May 15/2024	
420	Jan 15/2025	
421	May 15/2024	
422	May 15/2024	
423	May 15/2024	
424	May 15/2024	
425	May 15/2024	
426	May 15/2024	
427	May 15/2024	
428	May 15/2024	
429	May 15/2024	
430	May 15/2024	
431	May 15/2024	
432	May 15/2024	

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LANDING GEAR

Subject/Page	Date	COC
32-51-61 (cont)		
433	May 15/2024	
434	Jan 15/2025	
435	May 15/2024	
436	May 15/2024	
437	May 15/2024	
438	May 15/2024	
439	May 15/2024	
440	May 15/2024	
441	May 15/2024	
442	BLANK	
32-51-81		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2026	
407	May 15/2026	
408	May 15/2024	
409	May 15/2024	
410	BLANK	
32-61-00		
501	May 15/2024	
502	May 15/2024	
503	May 15/2024	
504	May 15/2024	
505	May 15/2024	
506	May 15/2024	
507	May 15/2024	
508	BLANK	
32-61-21		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
R 404	Sep 15/2026	
R 405	Sep 15/2026	
R 406	Sep 15/2026	
32-61-21 (cont)		
O 407	Sep 15/2026	
408	BLANK	
32-61-21		
R 501	Sep 15/2026	
R 502	Sep 15/2026	
R 503	Sep 15/2026	
O 504	Sep 15/2026	
O 505	Sep 15/2026	
O 506	Sep 15/2026	
R 507	Sep 15/2026	
R 508	Sep 15/2026	
R 509	Sep 15/2026	
R 510	Sep 15/2026	
R 511	Sep 15/2026	
R 512	Sep 15/2026	
A 513	Sep 15/2026	
A 514	Sep 15/2026	
32-61-31		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
R 406	Sep 15/2026	
407	May 15/2024	
408	BLANK	
32-61-31		
501	May 15/2024	
502	May 15/2024	
503	May 15/2024	
504	May 15/2024	
R 505	Sep 15/2026	
R 506	Sep 15/2026	
507	May 15/2024	
508	May 15/2024	
32-61-41		
401	Sep 15/2024	
402	Sep 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	Sep 15/2024	
408	Jan 15/2026	
409	Sep 15/2024	
410	May 15/2024	
32-61-41		
501	May 15/2024	
502	May 15/2024	
503	May 15/2024	
504	May 15/2024	
505	May 15/2024	
506	Jan 15/2026	
507	May 15/2024	
508	May 15/2024	
32-61-51		
401	Sep 15/2024	
402	Sep 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	Sep 15/2024	
408	Jan 15/2026	
409	Sep 15/2024	
410	May 15/2024	
32-61-51		
501	May 15/2024	
502	May 15/2024	
503	May 15/2024	
504	May 15/2024	
505	Jan 15/2026	
506	May 15/2024	

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Subject/Page	Date	COC
32-61-51 (cont)		
507	May 15/2024	
508	BLANK	
32-71-00		
R 201	Sep 15/2026	
O 202	Sep 15/2026	
203	May 15/2024	
204	May 15/2024	
205	May 15/2024	
206	May 15/2024	
207	May 15/2026	
208	BLANK	
32-71-00		
401	May 15/2024	
402	May 15/2024	
403	May 15/2026	
404	May 15/2024	
405	May 15/2026	
406	May 15/2026	
407	May 15/2026	
408	May 15/2024	
409	May 15/2026	
410	May 15/2026	
411	May 15/2026	
412	May 15/2024	
413	May 15/2024	
414	BLANK	
32-71-00		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	
604	May 15/2026	
32-71-21		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
32-71-21 (cont)		
406	May 15/2024	

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LANDING GEAR

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
LANDING GEAR - DDG MAINTENANCE PROCEDURES	32-00-00		901	PTF ALL
DDG 32-30-01 Landing Gear Actuation System Inoperative - Preparation TASK 32-00-00-040-801			902	PTF ALL
DDG 32-30-01 Landing Gear Actuation System Inoperative - Restoration TASK 32-00-00-440-801			903	PTF ALL
DDG 32-31-01 Landing Gear Selector Bypass Valve Inoperative - Preparation TASK 32-00-00-040-802			904	PTF ALL
DDG 32-31-01 Landing Gear Selector Bypass Valve Inoperative - Restoration TASK 32-00-00-440-802			905	PTF ALL
DDG 32-31-02B Both Landing Gear Control Systems Inoperative - Preparation TASK 32-00-00-040-803			906	PTF ALL
DDG 32-31-02B Both Landing Gear Control Systems Inoperative - Restoration TASK 32-00-00-440-803			907	PTF ALL
DDG 32-32-01 Main Landing Gear Uplock Springs Inoperative - Preparation TASK 32-00-00-040-804			908	PTF ALL
DDG 32-32-01 Main Landing Gear Uplock Springs Inoperative - Restoration TASK 32-00-00-440-804			909	PTF ALL
DDG 32-32-02 Main Landing Gear Frangible Fitting Inoperative - Preparation TASK 32-00-00-040-805			910	PTF ALL
DDG 32-32-02 Main Landing Gear Frangible Fitting Inoperative - Restoration TASK 32-00-00-440-821			911	PTF ALL
DDG 32-35-01 Landing Gear Manual Extension System Inoperative - Preparation TASK 32-00-00-040-806			912	PTF ALL
DDG 32-35-01 Landing Gear Manual Extension System Inoperative - Restoration TASK 32-00-00-440-806			913	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
DDG 32-41-01-02 Flight Deck Brake Pressure Indicating System Inoperative - Preparation TASK 32-00-00-040-807			914	PTF ALL
DDG 32-41-01-02 Flight Deck Brake Pressure Indicating System Inoperative - Restoration TASK 32-00-00-440-807			916	PTF ALL
DDG 32-42-01-01 Antiskid System Inoperative - Preparation TASK 32-00-00-040-808			916	PTF ALL
DDG 32-42-01-01 Antiskid System Inoperative - Restoration TASK 32-00-00-440-808			918	PTF ALL
DDG 32-42-02-01 Alternate Antiskid Valves Inoperative - Preparation TASK 32-00-00-040-809			918	PTF ALL
DDG 32-42-02-01 Alternate Antiskid Valves Inoperative - Restoration TASK 32-00-00-440-809			920	PTF ALL
DDG 32-42-03 Automatic Brake System Inoperative - Preparation TASK 32-00-00-040-810			921	PTF ALL
DDG 32-42-03 Automatic Brake System Inoperative - Restoration TASK 32-00-00-440-810			927	PTF ALL
DDG 32-44-01-01 Parking Brake Valve Inoperative - Preparation TASK 32-00-00-040-811			930	PTF ALL
DDG 32-44-01-01 Parking Brake Valve Inoperative - Restoration TASK 32-00-00-440-811			931	PTF ALL
DDG 32-44-02-01 Flight Deck Parking Brake Light Inoperative - Preparation TASK 32-00-00-040-812			933	PTF ALL
DDG 32-44-02-01 Flight Deck Parking Brake Light Inoperative - Restoration TASK 32-00-00-440-812			934	PTF ALL

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LANDING GEAR

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
DDG 32-49-01A Tire Pressure Indication System - Preparation TASK 32-00-00-040-834			934	PTF ALL
DDG 32-51-01 Rudder Pedal Nose Wheel Steering System Inoperative - Preparation TASK 32-00-00-040-813			935	PTF ALL
DDG 32-51-01 Rudder Pedal Nose Wheel Steering System Inoperative - Restoration TASK 32-00-00-440-813			936	PTF ALL
DDG 32-51-02 Nose Wheel Steering Switch Inoperative - Preparation TASK 32-00-00-040-814			937	PTF ALL
DDG 32-51-02 Nose Wheel Steering Switch Inoperative - Restoration TASK 32-00-00-440-814			938	PTF ALL
LANDING GEAR DOWNLOCK PINS - MAINTENANCE PRACTICES	32-00-01		201	PTF ALL
Landing Gear Downlock Pins Installation TASK 32-00-01-480-801			201	PTF ALL
Landing Gear Downlock Pins Removal TASK 32-00-01-080-801			208	PTF ALL
LANDING GEAR INNER CYLINDER CHROME - INSPECTION	32-00-10		601	PTF ALL
Landing Gear Inner Cylinder Chrome Inspection TASK 32-00-10-211-801			601	PTF ALL
LANDING GEAR INNER CYLINDER CHROME - CLEANING/PAINTING	32-00-10		701	PTF ALL
Landing Gear Inner Cylinder Chrome Cleaning - with Strut Extended TASK 32-00-10-100-801			701	PTF ALL
Landing Gear Inner Cylinder Chrome Cleaning TASK 32-00-10-100-802			703	PTF ALL
LANDING GEAR INNER CYLINDER CHROME - REPAIR	32-00-10		801	PTF ALL
Landing Gear Inner Cylinder Chrome Repair TASK 32-00-10-300-801			801	PTF ALL

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Main Landing Gear Retract Actuators Chrome Surfaces Cleaning TASK 32-00-11-100-801			701	PTF ALL
Nose Landing Gear Steering Actuators Chrome Surfaces Cleaning TASK 32-00-11-100-802			702	PTF ALL
Nose Landing Gear Retract Actuator Chrome Surfaces Cleaning TASK 32-00-11-100-803			704	PTF ALL
LANDING GEAR - CORROSION PREVENTION	32-00-37		201	PTF ALL
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INTERNAL - GENERAL VISUAL: LEFT MAIN LANDING GEAR ASSEMBLY TASK 32-05-03-210-801			601	PTF ALL
INTERNAL - GENERAL VISUAL: RIGHT MAIN LANDING GEAR ASSEMBLY TASK 32-05-03-210-802			604	PTF ALL
INTERNAL - GENERAL VISUAL: NOSE LANDING GEAR ASSEMBLY TASK 32-05-03-210-803			607	PTF ALL
AIR/GROUND SYSTEM - MAINTENANCE PRACTICES	32-09-00		201	PTF ALL
Prepare to Put the Airplane in the Air Mode TASK 32-09-00-840-801			201	PTF ALL
Put the Airplane in the Air Mode TASK 32-09-00-860-801			206	PTF ALL
Return the Airplane to the Ground Mode TASK 32-09-00-860-802			211	PTF ALL
Return the Airplane Systems Back to Their Normal on Ground Condition TASK 32-09-00-840-802			212	PTF ALL

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MAIN LANDING GEAR AIR/GROUND SENSOR - REMOVAL/INSTALLATION	32-09-01		401	PTF ALL
Main Landing Gear Air/Ground Sensor Removal TASK 32-09-01-020-801			401	PTF ALL
Main Landing Gear Air/Ground Sensor Installation TASK 32-09-01-400-801			405	PTF ALL
MAIN LANDING GEAR AIR/GROUND SENSOR - ADJUSTMENT/TEST	32-09-01		501	PTF ALL
Main Landing Gear Air/Ground Sensor Clearance Measurement TASK 32-09-01-400-802			501	PTF ALL
Main Landing Gear Air/Ground Sensor Clearance Adjustment TASK 32-09-01-820-801			503	PTF ALL
Main Landing Gear Air/Ground Sensor Operational Test TASK 32-09-01-710-801			505	PTF ALL
NOSE LANDING GEAR AIR/GROUND SENSOR - REMOVAL/INSTALLATION	32-09-02		401	PTF ALL
Nose Landing Gear Air/Ground Sensor Removal TASK 32-09-02-000-801			401	PTF ALL
Nose Landing Gear Air/Ground Sensor Installation TASK 32-09-02-400-801			405	PTF ALL
NOSE LANDING GEAR AIR/GROUND SENSOR - ADJUSTMENT/TEST	32-09-02		501	PTF ALL
Nose Landing Gear Air/Ground Sensor Clearance Measurement TASK 32-09-02-220-801			501	PTF ALL
Nose Landing Gear Air/Ground Sensor Clearance Adjustment TASK 32-09-02-400-802			504	PTF ALL
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Proximity Switch Electronics Unit Removal TASK 32-09-10-000-801			401	PTF ALL
Proximity Switch Electronics Unit Installation TASK 32-09-10-400-801			404	PTF ALL
PROXIMITY SWITCH ELECTRONICS UNIT (PSEU) - ADJUSTMENT/TEST	32-09-10		501	PTF ALL
Proximity Switch Electronics Unit - LRU Replacement Test TASK 32-09-10-710-801			501	PTF ALL
Proximity Switch Electronics Unit - Operational Test TASK 32-09-10-740-801			509	PTF ALL
Proximity Switch Electronics Unit Special Functions TASK 32-09-10-750-801			511	PTF ALL
Proximity Switch Electronics Unit Test Using UN-ATE Tool by Technologies Harness Scanner TASK 32-09-10-740-802			513	PTF ALL
PROXIMITY SENSOR - ADJUSTMENT/TEST	32-09-12		501	PTF ALL
PSEU Proximity Sensor - Functional Test with a LCR Meter TASK 32-09-12-700-801			501	PTF ALL
PSEU Proximity Sensor - Functional Test with a Breakout Box and a LCR Meter TASK 32-09-12-700-802			510	PTF ALL
MAIN LANDING GEAR - MAINTENANCE PRACTICES	32-11-00		201	PTF ALL
Main Landing Gear Gas Charging Valve Removal TASK 32-11-00-000-801			201	PTF ALL
Main Landing Gear Gas Charging Valve Installation TASK 32-11-00-400-801			204	PTF ALL
Main Landing Gear Oil Charging Valve Removal TASK 32-11-00-000-806			205	PTF ALL

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MAIN LANDING GEAR - REMOVAL/INSTALLATION	32-11-00		401	PTF ALL
Main Landing Gear Removal TASK 32-11-00-000-802			401	PTF ALL
Main Landing Gear Installation TASK 32-11-00-400-802			424	PTF ALL
MAIN LANDING GEAR - INSPECTION/CHECK	32-11-00		601	PTF ALL
Main Landing Gear Inspection TASK 32-11-00-200-801			601	PTF ALL
MAIN LANDING GEAR SHOCK STRUT SEALS - REPAIRS	32-11-21		801	PTF ALL
Active Seals with the Spare Seals Replacement TASK 32-11-21-960-801			801	PTF ALL
Active and the Spare Seals Replacement TASK 32-11-21-960-802			829	PTF ALL
Scraper Ring Replacement TASK 32-11-21-960-803			845	PTF ALL
Main Landing Gear Shock Strut Seal Leakage Check TASK 32-11-21-200-801			852	PTF ALL
MAIN LANDING GEAR CORROSION PREVENTION - MAINTENANCE PRACTICES	32-11-37		201	PTF ALL
Main Landing Gear Corrosion Prevention TASK 32-11-37-600-801			201	PTF ALL
MAIN LANDING GEAR TORSION LINKS - REMOVAL/INSTALLATION	32-11-51		401	PTF ALL
Main Landing Gear Upper Torsion Link Removal TASK 32-11-51-000-801			401	PTF ALL
Main Landing Gear Lower Torsion Link Removal TASK 32-11-51-000-802			408	PTF ALL
Main Landing Gear Torsion Link Disconnection TASK 32-11-51-020-801			409	PTF ALL
Main Landing Gear Upper Torsion Link Installation TASK 32-11-51-400-801			410	PTF ALL

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Main Landing Gear Lower Torsion Link Installation TASK 32-11-51-400-802			414	PTF ALL
Main Landing Gear Torsion Link Connection TASK 32-11-51-400-803			417	PTF ALL
MAIN LANDING GEAR TORSION LINKS - INSPECTION/CHECK	32-11-51		601	PTF ALL
Main Landing Gear Torsion Links Inspection TASK 32-11-51-200-801			601	PTF ALL
MAIN LANDING GEAR SIDE STRUT ASSEMBLY - REMOVAL/INSTALLATION	32-11-61		401	PTF ALL
Main Landing Gear Upper Side Strut Removal TASK 32-11-61-000-801			401	PTF ALL
Main Landing Gear Upper Side Strut Installation TASK 32-11-61-400-801			406	PTF ALL
Main Landing Gear Lower Side Strut Removal TASK 32-11-61-000-802			410	PTF ALL
Main Landing Gear Lower Side Strut Installation TASK 32-11-61-400-802			415	PTF ALL
Main Landing Gear Side Strut Removal TASK 32-11-61-000-803			419	PTF ALL
Main Landing Gear Side Strut Installation TASK 32-11-61-400-803			426	PTF ALL
MAIN LANDING GEAR SIDE STRUT ASSEMBLY - INSPECTION/CHECK	32-11-61		601	PTF ALL
Main Landing Gear Side Strut Assembly Inspection TASK 32-11-61-200-801			601	PTF ALL
MAIN LANDING GEAR SIDE BRACE ASSEMBLY - REMOVAL/INSTALLATION	32-11-62		401	PTF ALL
Main Landing Gear Side Brace Assembly Removal TASK 32-11-62-000-801			401	PTF ALL
Main Landing Gear Side Brace Assembly Installation TASK 32-11-62-400-801			412	PTF ALL

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MAIN LANDING GEAR REACTION LINK ASSEMBLY - REMOVAL/INSTALLATION	32-11-71		401	PTF ALL
Main Landing Gear Reaction Link Assembly Removal TASK 32-11-71-000-801			401	PTF ALL
Main Landing Gear Reaction Link Assembly Installation TASK 32-11-71-420-801			412	PTF ALL
MAIN LANDING GEAR HYDRAULIC SHIMMY DAMPER COMPONENTS - REMOVAL/INSTALLATION	32-11-81		201	PTF ALL
Main Landing Gear Shimmy Damper Compensator Removal TASK 32-11-81-000-801			201	PTF ALL
Main Landing Gear Shimmy Damper Compensator Installation TASK 32-11-81-400-801			206	PTF ALL
Main Landing Gear Shimmy Damper Manifold Assembly Removal TASK 32-11-81-000-802			207	PTF ALL
Main Landing Gear Shimmy Damper Manifold Assembly Installation TASK 32-11-81-400-802			208	PTF ALL
Main Landing Gear Shimmy Damper End Gland Assembly Removal TASK 32-11-81-000-803			210	PTF ALL
Main Landing Gear Shimmy Damper End Gland Assembly Installation TASK 32-11-81-400-803			211	PTF ALL
MAIN LANDING GEAR HYDRAULIC SHIMMY DAMPER - REMOVAL/INSTALLATION	32-11-81		401	PTF ALL
Main Landing Gear Shimmy Damper Removal TASK 32-11-81-000-804			401	PTF ALL
Main Landing Gear Shimmy Damper Installation TASK 32-11-81-400-804			405	PTF ALL
Hydraulic Shimmy Damper Bleeding TASK 32-11-81-870-801			407	PTF ALL

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MAIN LANDING GEAR SHIMMY DAMPER AND TORSION LINK APEX ASSEMBLY – INSPECTION/CHECK	32-11-81		601	PTF ALL
Main Landing Gear Hydraulic Shimmy Damper and Torsion Link Assembly Inspection TASK 32-11-81-200-801			601	PTF ALL
MAIN LANDING GEAR FORWARD TRUNNION BEARING ASSEMBLY - REMOVAL/INSTALLATION	32-11-83		401	PTF ALL
Main Landing Gear Forward Trunnion Bearing Assembly Removal TASK 32-11-83-000-801			401	PTF ALL
Main Landing Gear Forward Trunnion Bearing Assembly Installation TASK 32-11-83-400-801			403	PTF ALL
MAIN LANDING GEAR AXLE - REMOVAL/INSTALLATION	32-11-85		401	PTF ALL
Main Landing Gear Axle Removal TASK 32-11-85-000-801			401	PTF ALL
Main Landing Gear Axle Installation TASK 32-11-85-420-801			407	PTF ALL
Main Landing Gear Axle Sleeve Removal TASK 32-11-85-000-802			411	PTF ALL
Main Landing Gear Axle Sleeve Installation TASK 32-11-85-420-802			412	PTF ALL
MAIN LANDING GEAR AXLE - REPAIRS	32-11-85		801	PTF ALL
Main Landing Gear Axle Repair TASK 32-11-85-300-801			801	PTF ALL
Repair of the Axle Sleeve of the Main Landing Gear TASK 32-11-85-300-802			804	PTF ALL
Repair of the Brake Sleeve of the Main Landing Gear TASK 32-11-85-300-803			804	PTF ALL

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MAIN LANDING GEAR DOWNLOCK STRUT - REMOVAL/INSTALLATION	32-11-89		401	PTF ALL
Main Landing Gear Downlock Strut Removal TASK 32-11-89-000-801			401	PTF ALL
Main Landing Gear Downlock Strut Installation TASK 32-11-89-420-801			407	PTF ALL
MAIN LANDING GEAR WHEEL WELL BLADE SEAL - REMOVAL/INSTALLATION	32-12-12		401	PTF ALL
Main Landing Gear Wheel Well Blade Seal - Removal TASK 32-12-12-000-801			401	PTF ALL
Main Landing Gear Wheel Well Blade Seal Installation TASK 32-12-12-400-801			406	PTF ALL
MAIN LANDING GEAR WHEEL WELL BLADE SEAL - INSPECTION/CHECK	32-12-12		601	PTF ALL
Main Landing Gear Wheel Well Blade Seal Inspection TASK 32-12-12-200-801			601	PTF ALL
MAIN LANDING GEAR SHOCK STRUT DOORS - REMOVAL/INSTALLATION	32-13-11		401	PTF ALL
Main Landing Gear Center Door Removal TASK 32-13-11-000-801			401	PTF ALL
Main Landing Gear Center Door Installation TASK 32-13-11-420-801			411	PTF ALL
Main Landing Gear Inner Door Removal TASK 32-13-11-000-802			415	PTF ALL
Main Landing Gear Inner Door Installation TASK 32-13-11-420-802			417	PTF ALL
Main Landing Gear Door Assembly Removal TASK 32-13-11-000-803			420	PTF ALL
Main Landing Gear Door Assembly Installation TASK 32-13-11-420-803			422	PTF ALL

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MAIN LANDING GEAR SHOCK STRUT DOOR - ADJUSTMENT/TEST	32-13-11		501	PTF ALL
Main Landing Gear Shock Strut Door Adjustment TASK 32-13-11-820-801			501	PTF ALL
MAIN LANDING GEAR WING DOOR - REMOVAL/INSTALLATION	32-13-21		401	PTF ALL
Main Landing Gear Wing Door Removal TASK 32-13-21-000-801			401	PTF ALL
Main Landing Gear Wing Door Installation TASK 32-13-21-420-801			406	PTF ALL
MAIN LANDING GEAR WING DOOR - ADJUSTMENT/TEST	32-13-21		501	PTF ALL
Main Landing Gear Wing Door Adjustment TASK 32-13-21-000-802			501	PTF ALL
NOSE LANDING GEAR - MAINTENANCE PRACTICES	32-21-00		201	PTF ALL
Nose Landing Gear Gas Charging Valve Removal TASK 32-21-00-000-801			201	PTF ALL
Nose Landing Gear Gas Charging Valve Installation TASK 32-21-00-400-801			204	PTF ALL
Nose Landing Gear Oil Charging Valve Removal TASK 32-21-00-000-803			205	PTF ALL
Nose Landing Gear Oil Charging Valve Installation TASK 32-21-00-400-803			209	PTF ALL
NOSE LANDING GEAR - REMOVAL/INSTALLATION	32-21-00		401	PTF ALL
Nose Landing Gear Removal TASK 32-21-00-000-802			401	PTF ALL
Nose Landing Gear Installation TASK 32-21-00-400-802			419	PTF ALL
NOSE LANDING GEAR - INSPECTION/CHECK	32-21-00		601	PTF ALL
Nose Landing Gear - Inspection TASK 32-21-00-200-801			601	PTF ALL

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Nose Landing Gear Torsional Freeplay Inspection TASK 32-21-00-700-801			606	PTF ALL
Nose Landing Gear Steering Broken Cable Compensator Operational Check TASK 32-21-00-710-801			613	PTF ALL
NOSE LANDING GEAR - REPAIRS	32-21-00		801	PTF ALL
Flexible Conduit on the Nose Landing Gear Repair TASK 32-21-00-300-801			801	PTF ALL
NOSE LANDING GEAR SHOCK STRUT SEALS - REPAIRS	32-21-11		801	PTF ALL
Active Seals with the Spare Seals Replacement TASK 32-21-11-960-801			801	PTF ALL
Seals and Spare Seals Replacement TASK 32-21-11-960-802			824	PTF ALL
Scraper Ring Replacement TASK 32-21-11-960-803			844	PTF ALL
Nose Landing Gear Shock Strut Seal Leakage Check TASK 32-21-11-200-801			848	PTF ALL
Nose Landing Gear Shock Strut Metering Pin Seal Leakage Check TASK 32-21-11-200-802			851	PTF ALL
NOSE LANDING GEAR TOW FITTING - REMOVAL/INSTALLATION	32-21-14		401	PTF ALL
Nose Landing Gear Tow Fitting Removal TASK 32-21-14-000-801			401	PTF ALL
Nose Landing Gear Tow Fitting Installation TASK 32-21-14-400-801			403	PTF ALL
NOSE LANDING GEAR DRAG STRUT - REMOVAL/INSTALLATION	32-21-21		401	PTF ALL
Nose Landing Gear Drag Strut Removal TASK 32-21-21-000-801			401	PTF ALL
Nose Landing Gear Drag Strut Installation TASK 32-21-21-400-801			411	PTF ALL

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Nose Landing Gear Drag Strut Inspection TASK 32-21-21-200-801			601	PTF ALL
NLG SHOCK STRUT INNER CYLINDER - REMOVAL/INSTALLATION	32-21-24		401	PTF ALL
NLG Inner Cylinder Removal TASK 32-21-24-000-801			401	PTF ALL
NLG Inner Cylinder Installation TASK 32-21-24-400-801			420	PTF ALL
NOSE LANDING GEAR TORSION LINKS - REMOVAL/INSTALLATION	32-21-31		401	PTF ALL
Nose Landing Gear Upper Torsion Link Removal TASK 32-21-31-000-801			401	PTF ALL
Nose Landing Gear Upper Torsion Link Installation TASK 32-21-31-400-801			407	PTF ALL
Nose Landing Gear Lower Torsion Link Removal TASK 32-21-31-000-802			411	PTF ALL
Nose Landing Gear Lower Torsion Link Installation TASK 32-21-31-400-802			412	PTF ALL
Nose Landing Gear Torsion Link Disconnection TASK 32-21-31-000-803			415	PTF ALL
Nose Landing Gear Torsion Link Connection TASK 32-21-31-400-803			416	PTF ALL
NOSE LANDING GEAR TORSION LINK - REPAIRS	32-21-31		801	PTF ALL
Nose Landing Gear Lower Torsion Link Repair TASK 32-21-31-350-801			801	PTF ALL
NOSE LANDING GEAR CORROSION PREVENTION - MAINTENANCE PRACTICES	32-21-37		201	PTF ALL
Nose Landing Gear Corrosion Prevention TASK 32-21-37-600-801			201	PTF ALL

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Nose Gear Axle Detailed Inspection for Corrosion TASK 32-21-71-200-801			601	PTF ALL
NOSE LANDING GEAR WHEEL WELL DOORS - REMOVAL/INSTALLATION	32-22-11		401	PTF ALL
Nose Landing Gear Wheel Well Doors Removal TASK 32-22-11-000-801			401	PTF ALL
Nose Landing Gear Wheel Well Doors Installation TASK 32-22-11-400-801			405	PTF ALL
NOSE LANDING GEAR WHEEL WELL DOORS - ADJUSTMENT/TEST	32-22-11		501	PTF ALL
Nose Landing Gear Wheel Well Doors Adjustment TASK 32-22-11-000-802			501	PTF ALL
LANDING GEAR CONTROL SYSTEM - ADJUSTMENT/TEST	32-31-00		501	PTF ALL
System Test - Landing Gear Control System TASK 32-31-00-730-801			501	PTF ALL
Landing Gear Control Electrical Circuits - Operational Test TASK 32-31-00-710-801			506	PTF ALL
LANDING GEAR CONTROL LEVER MODULE - REMOVAL/INSTALLATION	32-31-11		401	PTF ALL
Landing Gear Control Lever Module Removal TASK 32-31-11-020-801			401	PTF ALL
Landing Gear Control Lever Module Installation TASK 32-31-11-400-801			405	PTF ALL
LANDING GEAR CONTROL LEVER MODULE - ADJUSTMENT/TEST	32-31-11		501	PTF ALL
Landing Gear Control Lever Module - Operational Test TASK 32-31-11-710-801			501	PTF ALL
LANDING GEAR SELECTOR VALVE - REMOVAL/INSTALLATION	32-31-51		401	PTF ALL
Landing Gear Selector Valve Removal TASK 32-31-51-020-801			401	PTF ALL

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Landing Gear Selector Valve Installation TASK 32-31-51-400-801			405	PTF ALL
Selector/Bypass Valve Operational Test TASK 32-31-51-700-801			407	PTF ALL
LANDING GEAR SELECTOR VALVE - INSPECTION/CHECK	32-31-51		601	PTF ALL
Landing Gear Selector Valve - Inspection TASK 32-31-51-200-801			601	PTF ALL
Landing Gear Selector Valve Secondary End Cap - Inspection TASK 32-31-51-200-802			604	PTF ALL
LANDING GEAR TRANSFER VALVE - REMOVAL/INSTALLATION	32-31-71		401	PTF ALL
Landing Gear Transfer Valve Removal TASK 32-31-71-020-801			401	PTF ALL
Landing Gear Transfer Valve Installation TASK 32-31-71-400-801			405	PTF ALL
Landing Gear Transfer Valve System Test TASK 32-31-71-730-801			408	PTF ALL
Landing Gear Transfer Valve Operational Test TASK 32-31-71-700-801			414	PTF ALL
LANDING GEAR TRANSFER VALVE - ADJUSTMENT/TEST	32-31-71		501	PTF ALL
Alternate Landing Gear Transfer Valve Operational Test TASK 32-31-71-700-802			501	PTF ALL
MAIN GEAR EXTENSION AND RETRACTION - ADJUSTMENT/TEST	32-32-00		501	PTF ALL
Main Landing Gear Test - Component Replacement TASK 32-32-00-710-801			501	PTF ALL
Main Landing Gear - Operational Test TASK 32-32-00-710-802			505	PTF ALL
Transfer Valve - Operational Test TASK 32-32-00-710-803			510	PTF ALL

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Main Gear Actuator Assembly Removal TASK 32-32-11-000-801			401	PTF ALL
Main Gear Actuator Assembly Installation TASK 32-32-11-400-801			409	PTF ALL
MAIN GEAR RETRACT PRESSURE FUSES - REMOVAL/INSTALLATION	32-32-21		401	PTF ALL
Main Gear Retract Pressure Fuses Removal TASK 32-32-21-000-801			401	PTF ALL
Main Gear Retract Pressure Fuses - Installation TASK 32-32-21-400-801			404	PTF ALL
MAIN GEAR UPLOCK ASSEMBLY - REMOVAL/INSTALLATION	32-32-31		401	PTF ALL
Main Gear Uplock Assembly Removal TASK 32-32-31-000-801			401	PTF ALL
Main Gear Uplock Assembly Installation TASK 32-32-31-400-801			411	PTF ALL
MAIN GEAR UPLOCK ACTUATOR - REMOVAL/INSTALLATION	32-32-41		401	PTF ALL
Main Gear Uplock Actuator Removal TASK 32-32-41-000-801			401	PTF ALL
Main Gear Uplock Actuator Installation TASK 32-32-41-400-801			405	PTF ALL
MAIN GEAR DOWNLOCK ACTUATOR - REMOVAL/INSTALLATION	32-32-51		401	PTF ALL
Main Gear Downlock Actuator Removal TASK 32-32-51-000-801			401	PTF ALL
Main Gear Downlock Actuator Installation TASK 32-32-51-400-801			406	PTF ALL
MAIN GEAR FRANGIBLE FITTING ASSEMBLY - REMOVAL/INSTALLATION	32-32-52		401	PTF ALL
Main Gear Frangible Fitting Assembly Removal TASK 32-32-52-000-801			401	PTF ALL

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Main Gear Frangible Fitting Removal TASK 32-32-52-000-802			406	PTF ALL
Main Gear Frangible Fitting Installation TASK 32-32-52-400-802			408	PTF ALL
MAIN GEAR TRANSFER CYLINDER - REMOVAL/INSTALLATION	32-32-71		401	PTF ALL
Main Gear Transfer Cylinder Removal TASK 32-32-71-000-801			401	PTF ALL
Main Gear Transfer Cylinder Installation TASK 32-32-71-400-801			404	PTF ALL
MAIN LANDING GEAR HYDRAULIC FLOW REGULATOR - REMOVAL/INSTALLATION	32-32-72		401	PTF ALL
Main Gear Hydraulic Flow Regulator Removal TASK 32-32-72-000-801			401	PTF ALL
Main Gear Hydraulic Flow Regulator Installation TASK 32-32-72-400-801			404	PTF ALL
MAIN GEAR DOWNLOCK SPRINGS - REMOVAL/INSTALLATION	32-32-91		401	PTF ALL
Main Gear Downlock Springs Removal TASK 32-32-91-000-801			401	PTF ALL
Main Gear Downlock Spring Installation TASK 32-32-91-400-801			406	PTF ALL
MAIN GEAR BEAM HANGER PIN - REMOVAL/INSTALLATION	32-32-93		401	PTF ALL
Main Gear Beam Hanger Pin Removal TASK 32-32-93-000-801			401	PTF ALL
Main Gear Beam Hanger Pin Installation TASK 32-32-93-400-801			405	PTF ALL
NOSE GEAR EXTENSION AND RETRACTION - ADJUSTMENT/TEST	32-33-00		501	PTF ALL
Nose Landing Gear Test - Component Replacement TASK 32-33-00-710-801			501	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Nose Landing Gear Operational Test TASK 32-33-00-710-802			504	PTF ALL
NOSE GEAR RETRACT ACTUATOR - REMOVAL/INSTALLATION	32-33-11		401	PTF ALL
Nose Gear Retract Actuator Removal TASK 32-33-11-000-801			401	PTF ALL
Nose Gear Retract Actuator Installation TASK 32-33-11-400-801			406	PTF ALL
NOSE GEAR LOCK ACTUATOR - REMOVAL/INSTALLATION	32-33-21		401	PTF ALL
Nose Landing Gear Lock Actuator Removal TASK 32-33-21-000-801			401	PTF ALL
Nose Landing Gear Lock Actuator Installation TASK 32-33-21-400-801			405	PTF ALL
NOSE GEAR LOCK VALVE MANIFOLD - REMOVAL/INSTALLATION	32-33-31		401	PTF ALL
Nose Gear Lock Valve Manifold Removal TASK 32-33-31-000-801			401	PTF ALL
Nose Gear Lock Valve Manifold Installation TASK 32-33-31-400-801			405	PTF ALL
NOSE GEAR TRANSFER CYLINDER - REMOVAL/INSTALLATION	32-33-41		401	PTF ALL
Nose Gear Transfer Cylinder Removal TASK 32-33-41-000-801			401	PTF ALL
Nose Gear Transfer Cylinder Installation TASK 32-33-41-400-801			404	PTF ALL
NOSE GEAR LOCK MECHANISM - REMOVAL/INSTALLATION	32-33-51		401	PTF ALL
Nose Gear Lock Mechanism Removal TASK 32-33-51-000-801			401	PTF ALL
Nose Gear Lock Mechanism Installation TASK 32-33-51-400-801			408	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
NOSE GEAR LOCK SPRING - REMOVAL/INSTALLATION	32-33-52		401	PTF ALL
Nose Gear Lock Spring Removal TASK 32-33-52-000-801			401	PTF ALL
Nose Gear Lock Spring Installation TASK 32-33-52-400-801			404	PTF ALL
NOSE GEAR WHEEL WELL DOOR LINKAGE - REMOVAL/INSTALLATION	32-33-61		401	PTF ALL
Nose Gear Wheel Well Door Linkage - Removal TASK 32-33-61-000-801			401	PTF ALL
Nose Gear Wheel Well Door Linkage - Installation TASK 32-33-61-400-801			407	PTF ALL
NOSE GEAR WHEEL WELL DOOR LINKAGE - ADJUSTMENT/TEST	32-33-61		501	PTF ALL
Nose Gear Wheel Well Door Linkage Adjustment and Check TASK 32-33-61-820-801			501	PTF ALL
NOSE GEAR WHEEL WELL DOOR LINKAGE - INSPECTION/CHECK	32-33-61		601	PTF ALL
Nose Gear Wheel Well Door Linkage Inspection TASK 32-33-61-200-801			601	PTF ALL
NOSE GEAR PRESSURE FUSE - REMOVAL/INSTALLATION	32-33-71		401	PTF ALL
Nose Gear Retract Pressure Fuse Removal TASK 32-33-71-000-801			401	PTF ALL
Nose Gear Retract Pressure Fuse Installation TASK 32-33-71-400-801			404	PTF ALL
Nose Gear Extend Pressure Fuse Removal TASK 32-33-71-000-802			405	PTF ALL
Nose Gear Extend Pressure Fuse Installation TASK 32-33-71-400-802			409	PTF ALL
NOSE GEAR HYDRAULIC FLOW REGULATOR - REMOVAL/INSTALLATION	32-33-72		401	PTF ALL
Nose Gear Hydraulic Flow Regulator Removal TASK 32-33-72-000-801			401	PTF ALL

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Nose Gear Hydraulic Flow Regulator Installation TASK 32-33-72-400-801			404	PTF ALL
MAIN GEAR MANUAL EXTENSION SYSTEM - ADJUSTMENT/TEST	32-34-00		501	PTF ALL
Main Landing Gear Manual Extension System Adjustment and Test TASK 32-34-00-820-801			501	PTF ALL
Main Gear Manual Extension System Test (Airplane on Jacks) TASK 32-34-00-730-801			508	PTF ALL
Main Gear Manual Extension System Test (Airplane not on Jacks) TASK 32-34-00-730-802			517	PTF ALL
Manual Extension System Access Door Switch Adjustment TASK 32-34-00-820-802			519	PTF ALL
MAIN GEAR MANUAL EXTENSION CONTROL MECHANISM - REMOVAL/INSTALLATION	32-34-11		401	PTF ALL
Main Gear Manual Extension Control Mechanism Removal TASK 32-34-11-000-801			401	PTF ALL
Main Gear Manual Extension Control Mechanism Installation TASK 32-34-11-400-801			410	PTF ALL
Manual Extension System Access Door Switch Removal TASK 32-34-11-820-801			412	PTF ALL
Manual Extension System Access Door Switch Installation TASK 32-34-11-820-802			416	PTF ALL
MAIN GEAR MANUAL EXTENSION MECHANISM - REMOVAL/INSTALLATION	32-34-21		401	PTF ALL
Main Gear Manual Extension Mechanism Removal TASK 32-34-21-000-801			401	PTF ALL
Main Gear Manual Extension Mechanism Installation TASK 32-34-21-400-801			408	PTF ALL

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MAIN GEAR MANUAL EXTENSION SYSTEM CABLES - REMOVAL/INSTALLATION	32-34-31		401	PTF ALL
Main Gear Manual Extension System Cables Removal TASK 32-34-31-000-801			401	PTF ALL
Main Gear Manual Extension System Cables Installation TASK 32-34-31-400-801			409	PTF ALL
NOSE GEAR MANUAL EXTENSION SYSTEM - ADJUSTMENT/TEST	32-35-00		501	PTF ALL
Nose Landing Gear Manual Extension System Adjustment TASK 32-35-00-820-801			501	PTF ALL
Nose Gear Manual Extension System Test - Airplane on Jacks TASK 32-35-00-730-801			505	PTF ALL
Nose Gear Manual Extension System Test - Airplane Not on Jacks TASK 32-35-00-730-802			512	PTF ALL
NOSE GEAR MANUAL EXTENSION RELEASE MECHANISM - REMOVAL/INSTALLATION	32-35-11		401	PTF ALL
Nose Gear Manual Extension Release Mechanism Removal TASK 32-35-11-000-801			401	PTF ALL
Nose Gear Manual Extension Release Mechanism Installation TASK 32-35-11-400-801			404	PTF ALL
NOSE GEAR MANUAL EXTENSION SPRING - REMOVAL/INSTALLATION	32-35-20		401	PTF ALL
Nose Gear Manual Extension Spring Removal TASK 32-35-20-000-801			401	PTF ALL
Nose Gear Manual Extension Spring Installation TASK 32-35-20-400-801			404	PTF ALL

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NOSE GEAR MANUAL EXTENSION SYSTEM CABLES - REMOVAL/INSTALLATION	32-35-21		401	PTF ALL
Nose Gear Manual Extension System Cables Removal TASK 32-35-21-000-801			401	PTF ALL
Nose Gear Manual Extension System Cables Installation TASK 32-35-21-400-801			409	PTF ALL
HYDRAULIC BRAKE SYSTEM - MAINTENANCE PRACTICES	32-41-00		201	PTF ALL
Hydraulic Brake System Upstream of the Antiskid Valves Bleeding TASK 32-41-00-870-801			201	PTF ALL
Normal (System B) Hydraulic Brake System Bleeding TASK 32-41-00-870-802			203	PTF ALL
Alternate (System A) Hydraulic Brake System Bleeding TASK 32-41-00-870-803			210	PTF ALL
Normal (System B) Hydraulic System Brake Fuse Test TASK 32-41-00-870-804			212	PTF ALL
Alternate (System A) Hydraulic System Brake Fuse Test TASK 32-41-00-870-805			215	PTF ALL
HYDRAULIC BRAKE SYSTEM - ADJUSTMENT/TEST	32-41-00		501	PTF ALL
Hydraulic Brake System Adjustment TASK 32-41-00-820-801			501	PTF ALL
Normal and Alternate Brake Application Functional Test TASK 32-41-00-710-801			515	PTF ALL
Accumulator Operational Test TASK 32-41-00-720-801			518	PTF ALL
Alternate Brake System Operational Test TASK 32-41-00-710-803			520	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
BRAKE DISCONNECT - REMOVAL/INSTALLATION	32-41-11		401	PTF ALL
Brake Disconnect Removal TASK 32-41-11-000-801			401	PTF ALL
Brake Disconnect Installation TASK 32-41-11-400-801			405	PTF ALL
BRAKE METERING VALVE CONTROL QUADRANT - REMOVAL/INSTALLATION	32-41-21		401	PTF ALL
Brake Metering Valve Control Quadrant Removal TASK 32-41-21-000-801			401	PTF ALL
Brake Metering Valve Control Quadrant Installation TASK 32-41-21-400-801			411	PTF ALL
BRAKE PEDAL BUS CRANK ASSEMBLY - REMOVAL/INSTALLATION	32-41-22		401	PTF ALL
Brake Pedal Bus Crank Assembly Removal TASK 32-41-22-000-801			401	PTF ALL
Brake Pedal Bus Crank Assembly Installation TASK 32-41-22-400-801			406	PTF ALL
BRAKE METERING VALVE ASSEMBLY - REMOVAL/INSTALLATION	32-41-31		401	PTF ALL
Brake Metering Valve Removal TASK 32-41-31-000-801			401	PTF ALL
Brake Metering Valve Installation TASK 32-41-31-400-801			409	PTF ALL
BRAKE METERING VALVE ASSEMBLY - INSPECTION/CHECK	32-41-31		601	PTF ALL
Brake Metering Valve Detailed Inspection TASK 32-41-31-200-801			601	PTF ALL
MAIN LANDING GEAR BRAKE - REMOVAL/INSTALLATION	32-41-41		401	PTF ALL
Main Landing Gear Brake Removal TASK 32-41-41-000-801			401	PTF ALL
Main Landing Gear Brake Installation TASK 32-41-41-400-801			411	PTF ALL

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MAIN LANDING GEAR BRAKE - INSPECTION/CHECK	32-41-41		601	PTF ALL
Examine the Main Landing Gear Brakes for Wear TASK 32-41-41-700-801			601	PTF ALL
Main Landing Gear Brake Fast Check (Wheel Installed on the Airplane) TASK 32-41-41-700-802			604	PTF ALL
Main Landing Gear Brake Check (Wheel Removed from the Airplane) TASK 32-41-41-700-803			606	PTF ALL
MAIN LANDING GEAR BRAKE SLEEVE - REMOVAL/INSTALLATION	32-41-42		401	PTF ALL
Main Landing Gear Brake Sleeve Removal TASK 32-41-42-000-801			401	PTF ALL
Main Landing Gear Brake Sleeve Installation TASK 32-41-42-420-801			404	PTF ALL
HYDRAULIC BRAKE ACCUMULATOR - REMOVAL/INSTALLATION	32-41-51		401	PTF ALL
Hydraulic Brake Accumulator Removal TASK 32-41-51-000-801			401	PTF ALL
Hydraulic Brake Accumulator Installation TASK 32-41-51-420-801			407	PTF ALL
Brake Accumulator Pressure Gage Removal TASK 32-41-51-000-802			409	PTF ALL
Brake Accumulator Pressure Gage Installation TASK 32-41-51-420-802			411	PTF ALL
BRAKE ACCUMULATOR PRESSURE TRANSMITTER - REMOVAL/INSTALLATION	32-41-53		401	PTF ALL
Brake Accumulator Pressure Transmitter Removal TASK 32-41-53-000-801			401	PTF ALL
Brake Accumulator Pressure Transmitter Installation TASK 32-41-53-400-801			404	PTF ALL

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BRAKE ACCUMULATOR PRESSURE GAGE (FLIGHT COMPARTMENT) - REMOVAL/INSTALLATION	32-41-54		401	PTF ALL
Brake Accumulator Pressure Gage (Flight Compartment) Removal TASK 32-41-54-000-801			401	PTF ALL
Brake Accumulator Pressure Gage (Flight Compartment) Installation TASK 32-41-54-400-801			403	PTF ALL
BRAKE ACCUMULATOR PRESSURE RELIEF VALVE - REMOVAL/INSTALLATION	32-41-71		401	PTF ALL
Brake Accumulator Pressure Relief Valve Removal TASK 32-41-71-000-801			401	PTF ALL
Brake Accumulator Pressure Relief Valve Installation TASK 32-41-71-420-801			404	PTF ALL
BRAKE HYDRAULIC FUSES - REMOVAL/INSTALLATION	32-41-72		401	PTF ALL
Brake Hydraulic Fuse (Alternate Brake System) Removal TASK 32-41-72-020-801			401	PTF ALL
Brake Hydraulic Fuse (Alternate Brake System) Installation TASK 32-41-72-420-801			405	PTF ALL
Brake Hydraulic Fuse (Normal Brake System) Removal TASK 32-41-72-000-801			406	PTF ALL
Brake Hydraulic Fuse (Normal Brake System) Installation TASK 32-41-72-400-801			410	PTF ALL
HYDRAULIC BRAKE CONTROL CABLE - REMOVAL/INSTALLATION	32-41-81		401	PTF ALL
Hydraulic Brake Control Cable Removal TASK 32-41-81-000-801			401	PTF ALL
Hydraulic Brake Control Cable Installation TASK 32-41-81-400-801			409	PTF ALL

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HYDRAULIC BRAKE CONTROL CABLE - INSPECTION/CHECK	32-41-81		601	PTF ALL
Brake Control System Detailed Inspection TASK 32-41-81-211-801			601	PTF ALL
BRAKE ACCUMULATOR ISOLATION VALVE - REMOVAL/INSTALLATION	32-41-91		401	PTF ALL
Accumulator Isolation Valve Removal TASK 32-41-91-000-801			401	PTF ALL
Accumulator Isolation Valve Installation TASK 32-41-91-420-801			405	PTF ALL
ALTERNATE BRAKE SELECTOR VALVE - REMOVAL/INSTALLATION	32-41-93		401	PTF ALL
Alternate Brake Selector Valve Removal TASK 32-41-93-000-801			401	PTF ALL
Alternate Brake Selector Valve Installation TASK 32-41-93-420-801			405	PTF ALL
ALTERNATE BRAKE SELECTOR VALVE - ADJUSTMENT/TEST	32-41-93		501	PTF ALL
Alternate Brake Selector Valve Operational Test TASK 32-41-93-710-801			501	PTF ALL
Gear Retract Braking Operational Test TASK 32-41-93-710-802			502	PTF ALL
BRAKE SHUTTLE VALVES - REMOVAL/INSTALLATION	32-41-95		401	PTF ALL
Brake Shuttle Valve - Removal TASK 32-41-95-000-801			401	PTF ALL
Brake Shuttle Valve Installation TASK 32-41-95-400-801			406	PTF ALL
BRAKE FLOW LIMITERS - REMOVAL/INSTALLATION	32-41-96		401	PTF ALL
Brake Flow Limiter Removal TASK 32-41-96-000-801			401	PTF ALL
Brake Flow Limiter Installation TASK 32-41-96-400-801			405	PTF ALL

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ANTISKID/AUTOBRAKE SYSTEM - ADJUSTMENT/TEST	32-42-00		501	PTF ALL
Antiskid/Autobrake Control Unit Operational Test TASK 32-42-00-720-801			501	PTF ALL
Antiskid Valve Functional Test TASK 32-42-00-720-802			506	PTF ALL
Transducer Functional Test TASK 32-42-00-710-801			513	PTF ALL
Antiskid Valve Operational Test TASK 32-42-00-720-803			518	PTF ALL
Autobrake Pressure Control Module Functional Test TASK 32-42-00-400-801			520	PTF ALL
Autobrake Shuttle Valve Operational Test TASK 32-42-00-710-802			525	PTF ALL
Transducer Operational Test TASK 32-42-00-710-803			527	PTF ALL
Antiskid/Autobrake Control Unit BITE Procedure TASK 32-42-00-740-801			530	PTF ALL
ANTISKID WHEEL SPEED TRANSDUCER - REMOVAL/INSTALLATION	32-42-11		401	PTF ALL
Antiskid Wheel Speed Transducer Removal TASK 32-42-11-000-801			401	PTF ALL
Antiskid Wheel Speed Transducer Installation TASK 32-42-11-400-801			405	PTF ALL
ANTISKID/AUTOBRAKE CONTROL UNIT - REMOVAL/INSTALLATION	32-42-21		401	PTF ALL
Antiskid/Autobrake Control Unit Removal TASK 32-42-21-000-801			401	PTF ALL
Antiskid/Autobrake Control Unit Installation TASK 32-42-21-400-801			405	PTF ALL
ANTISKID VALVE - REMOVAL/INSTALLATION	32-42-31		401	PTF ALL
Antiskid Valve - Removal TASK 32-42-31-000-802			401	PTF ALL

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Antiskid Valve - Installation TASK 32-42-31-400-802			411	PTF ALL
AUTOBRAKE SELECTOR SWITCH - REMOVAL/INSTALLATION	32-42-61		401	PTF ALL
Autobrake Selector Switch Removal TASK 32-42-61-000-801			401	PTF ALL
Autobrake Selector Switch Installation TASK 32-42-61-400-801			406	PTF ALL
AUTOBRAKE SHUTTLE VALVE - REMOVAL/INSTALLATION	32-42-71		401	PTF ALL
Autobrake Shuttle Valve Removal TASK 32-42-71-000-801			401	PTF ALL
Autobrake Shuttle Valve Installation TASK 32-42-71-400-801			406	PTF ALL
AUTOBRAKE PRESSURE CONTROL MODULE - REMOVAL/INSTALLATION	32-42-81		401	PTF ALL
Autobrake Pressure Control Module Valve Removal TASK 32-42-81-000-801			401	PTF ALL
Autobrake Pressure Control Module Valve Installation TASK 32-42-81-400-801			407	PTF ALL
Autobrake Pressure Control Module Line Replaceable Units Removal TASK 32-42-81-000-802			410	PTF ALL
Autobrake Pressure Control Module Line Replaceable Units Installation TASK 32-42-81-400-802			412	PTF ALL
PARKING BRAKE SYSTEM - ADJUSTMENT/TEST	32-44-00		501	PTF ALL
Parking Brake System Adjustment TASK 32-44-00-820-801			501	PTF ALL
Parking Brake System Operational Test TASK 32-44-00-710-801			513	PTF ALL
Parking Brake Hold Check TASK 32-44-00-790-801			516	PTF ALL

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PARKING BRAKE LINKAGE - REMOVAL/INSTALLATION	32-44-11		401	PTF ALL
Parking Brake Lever Removal TASK 32-44-11-000-801			401	PTF ALL
Parking Brake Lever Installation TASK 32-44-11-400-801			405	PTF ALL
Parking Brake Jackshaft Removal TASK 32-44-11-000-802			406	PTF ALL
Parking Brake Jackshaft Installation TASK 32-44-11-400-802			410	PTF ALL
Parking Brake Latch Removal TASK 32-44-11-000-803			411	PTF ALL
Parking Brake Latch Installation TASK 32-44-11-400-803			415	PTF ALL
Parking Brake Latch Switch Removal TASK 32-44-11-000-804			416	PTF ALL
Parking Brake Latch Switch Installation TASK 32-44-11-400-804			420	PTF ALL
Parking Brake Mechanism Visual Inspection TASK 32-44-11-000-805			421	PTF ALL
PARKING BRAKE SHUTOFF VALVE - REMOVAL/INSTALLATION	32-44-21		401	PTF ALL
Parking Brake Shutoff Valve Removal TASK 32-44-21-000-801			401	PTF ALL
Parking Brake Shutoff Valve Installation TASK 32-44-21-400-801			405	PTF ALL
TIRES AND WHEELS - MAINTENANCE PRACTICES	32-45-00		201	PTF ALL
Landing Gear Tire - Deflation TASK 32-45-00-910-801			201	PTF ALL
Landing Gear Tire - Inflation TASK 32-45-00-910-802			204	PTF ALL
TIRES AND WHEELS - INSPECTION/CHECK	32-45-00		601	PTF ALL
Wheels Fast Check (Wheel Installed on the Airplane) TASK 32-45-00-700-801			601	PTF ALL

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Wheels Inspection (Wheel Removed from the Airplane) TASK 32-45-00-700-802			602	PTF ALL
Landing Gear Tire Inspection TASK 32-45-00-700-803			603	PTF ALL
Wheel Bearings Inspection TASK 32-45-00-211-801			614	PTF ALL
TIRE INFLATION VALVE - REMOVAL/INSTALLATION	32-45-08		401	PTF ALL
Tire Inflation Valve Removal TASK 32-45-08-000-801			401	PTF ALL
Tire Inflation Valve Installation TASK 32-45-08-400-801			404	PTF ALL
MAIN LANDING GEAR WHEEL AND TIRE ASSEMBLY - REMOVAL/INSTALLATION	32-45-11		401	PTF ALL
Main Landing Gear Wheel and Tire Assembly Removal TASK 32-45-11-000-801			401	PTF ALL
Main Landing Gear Wheel and Tire Assembly - Installation TASK 32-45-11-400-801			410	PTF ALL
NOSE LANDING GEAR WHEEL AND TIRE ASSEMBLY - REMOVAL/INSTALLATION	32-45-21		401	PTF ALL
Nose Landing Gear Wheel and Tire Assembly Removal TASK 32-45-21-000-801			401	PTF ALL
Nose Landing Gear Wheel and Tire Assembly Installation TASK 32-45-21-400-801			409	PTF ALL
NOSE WHEEL SPIN BRAKE LINING - MAINTENANCE PRACTICES	32-45-31		201	PTF ALL
Nose Wheel Spin Brake Lining Removal TASK 32-45-31-000-801			201	PTF ALL
Nose Wheel Spin Brake Lining Installation TASK 32-45-31-400-801			204	PTF ALL
Nose Wheel Spin Brake Lining Inspection TASK 32-45-31-700-801			204	PTF ALL

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TIRE AND BRAKE MONITORING SYSTEM - ADJUSTMENT/TEST	32-46-00		501	PTF ALL
Tire and Brake Monitoring System - System Test TASK 32-46-00-730-801			501	PTF ALL
BRAKE TEMPERATURE SENSOR - REMOVAL/ INSTALLATION	32-46-21		401	PTF ALL
Brake Temperature Sensor Removal TASK 32-46-21-000-801			401	PTF ALL
Brake Temperature Sensor Installation TASK 32-46-21-400-801			404	PTF ALL
TIRE AND BRAKE MONITORING UNIT - REMOVAL/ INSTALLATION	32-46-31		401	PTF ALL
Tire and Brake Monitoring Unit Removal TASK 32-46-31-000-801			401	PTF ALL
Tire and Brake Monitoring Unit Installation TASK 32-46-31-400-801			404	PTF ALL
Tire Pressure Indication System - Adjustment/Test	32-49-00		501	PTF ALL
Tire Pressure Sensor Functional Test TASK 32-49-00-720-801			501	PTF ALL
TIRE PRESSURE INDICATION SYSTEM - REMOVAL/INSTALLATION	32-49-03		401	PTF ALL
Nose Landing Gear Hubcap Assembly Removal TASK 32-49-03-000-801			401	PTF ALL
Nose Landing Gear Hubcap Assembly Installation TASK 32-49-03-400-801			405	PTF ALL
Main Landing Gear Hubcap Assembly Removal TASK 32-49-03-000-802			406	PTF ALL
Main Landing Gear Hubcap Assembly Installation TASK 32-49-03-400-802			410	PTF ALL
Nose Landing Gear In-Axle Adapter Removal TASK 32-49-03-000-803			411	PTF ALL
Nose Landing Gear In-Axle Adapter Installation TASK 32-49-03-400-803			414	PTF ALL
Main Landing Gear In-Axle Adapter Removal TASK 32-49-03-000-804			415	PTF ALL

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Main Landing Gear In-Axle Adapter Installation TASK 32-49-03-400-804			418	PTF ALL
Tire Pressure Sensor/Fill Valve Removal TASK 32-49-03-000-805			419	PTF ALL
Tire Pressure Sensor/Fill Valve Installation TASK 32-49-03-400-805			424	PTF ALL
NOSE WHEEL STEERING SYSTEM - ADJUSTMENT/TEST	32-51-00		501	PTF ALL
General Check of the Nose Wheel Steering System Adjustment TASK 32-51-00-820-801			501	PTF ALL
Rudder Pedal Interconnect Mechanism Adjustment Check TASK 32-51-00-820-802			513	PTF ALL
Rudder Pedal Interconnect Mechanism Adjustment TASK 32-51-00-820-803			515	PTF ALL
Nose Wheel Steering System Adjustment Check TASK 32-51-00-820-804			519	PTF ALL
Nose Wheel Steering System Adjustment TASK 32-51-00-820-805			524	PTF ALL
Nose Wheel Steering System Adjustment (Adjusting Steering Metering Valve Input Rod Method) TASK 32-51-00-820-806			532	PTF ALL
Nose Wheel Steering System Test (Airplane on Ground) TASK 32-51-00-700-801			536	PTF ALL
Nose Wheel Steering System Test (Airplane on Jacks) TASK 32-51-00-700-802			544	PTF ALL
NOSE WHEEL STEERING SYSTEM - INSPECTION/CHECK	32-51-00		601	PTF ALL
Nose Wheel Steering System Mechanism Inspection TASK 32-51-00-200-801			601	PTF ALL

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CHAPTER 32
LANDING GEAR

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
NOSE WHEEL STEERING - MAINTENANCE PRACTICES	32-51-11		201	PTF ALL
Clean the Exterior Surfaces of the Towing Lever Assembly TASK 32-51-11-100-801			201	PTF ALL
STEERING METERING VALVE - REMOVAL/INSTALLATION	32-51-11		401	PTF ALL
Steering Metering Valve Removal TASK 32-51-11-000-801			401	PTF ALL
Steering Metering Valve Installation TASK 32-51-11-400-801			407	PTF ALL
RUDDER PEDAL STEERING MECHANISM - REMOVAL/INSTALLATION	32-51-21		401	PTF ALL
Interconnect Spring Removal TASK 32-51-21-000-801			401	PTF ALL
Interconnect Spring Installation TASK 32-51-21-400-801			402	PTF ALL
Rudder Pedal Steering Mechanism Removal TASK 32-51-21-000-802			403	PTF ALL
Rudder Pedal Steering Mechanism Installation TASK 32-51-21-400-802			409	PTF ALL
Rudder Pedal Steering Mechanism Visual Inspection TASK 32-51-21-000-803			410	PTF ALL
STEERING SYSTEM CABLES - REMOVAL/INSTALLATION	32-51-31		401	PTF ALL
Nose Wheel Steering Position Cables Removal TASK 32-51-31-000-801			401	PTF ALL
Nose Gear Steering Position Cables Installation TASK 32-51-31-400-801			415	PTF ALL
Nose Gear Steering System Cables Removal TASK 32-51-31-000-802			416	PTF ALL
Nose Gear Steering System Cables Installation TASK 32-51-31-400-802			420	PTF ALL

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CHAPTER 32
LANDING GEAR

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
STEERING SYSTEM CABLES - INSPECTION/CHECK	32-51-31		601	PTF ALL
Nose Gear Steering System Cables Inspection TASK 32-51-31-220-801			601	PTF ALL
NOSE LANDING GEAR CONTROL WHEEL AND TILLER ASSEMBLY - REMOVAL/INSTALLATION	32-51-41		401	PTF ALL
Nose Landing Gear Steering Control Wheel Removal TASK 32-51-41-000-801			401	PTF ALL
Nose Landing Gear Steering Control Wheel Installation TASK 32-51-41-400-801			404	PTF ALL
Nose Landing Gear Tiller Assembly Removal TASK 32-51-41-000-802			405	PTF ALL
Nose Landing Gear Tiller Assembly Installation TASK 32-51-41-400-802			410	PTF ALL
NOSE GEAR STEERING ACTUATOR - REMOVAL/INSTALLATION	32-51-51		401	PTF ALL
Nose Gear Steering Actuator Removal TASK 32-51-51-000-801			401	PTF ALL
Nose Gear Steering Actuator Installation TASK 32-51-51-400-801			409	PTF ALL
Nose Gear Hydraulic Swivel Removal TASK 32-51-51-000-802			412	PTF ALL
Nose Gear Hydraulic Swivel Installation TASK 32-51-51-400-802			414	PTF ALL
NOSE GEAR STEERING ACTUATOR - INSPECTION/CHECK	32-51-51		601	PTF ALL
Nose Gear Steering Actuator Rod End Detailed Inspection TASK 32-51-51-200-801			601	PTF ALL
NOSE LANDING GEAR STEERING SUMMING MECHANISM - REMOVAL/INSTALLATION	32-51-52		401	PTF ALL
Summing Mechanism Removal (Without Using Cable Clamp Assembly) TASK 32-51-52-020-801			401	PTF ALL

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CHAPTER 32
LANDING GEAR

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Summing Mechanism Installation (Without Using Cable Clamp Assembly) TASK 32-51-52-420-801			410	PTF ALL
Summing Mechanism Removal (Using Cable Clamp Assembly) TASK 32-51-52-020-802			412	PTF ALL
Summing Mechanism Installation (Using Cable Clamp Assembly) TASK 32-51-52-420-802			419	PTF ALL
STEERING COLLAR - REMOVAL/INSTALLATION	32-51-61		401	PTF ALL
Steering Collar Removal (Without Using Cable Clamp Assembly) TASK 32-51-61-000-801			401	PTF ALL
Steering Collar Installation (Without Using Cable Clamp Assembly) TASK 32-51-61-400-801			415	PTF ALL
Steering Collar Removal (Using Cable Clamp Assembly) TASK 32-51-61-000-802			419	PTF ALL
Steering Collar Installation (Using Cable Clamp Assembly) TASK 32-51-61-400-802			434	PTF ALL
Upper Steering Sleeve Removal TASK 32-51-61-000-803			438	PTF ALL
Upper Steering Sleeve Installation TASK 32-51-61-400-803			439	PTF ALL
RUDDER PEDAL STEERING ROTARY ACTUATOR - REMOVAL/INSTALLATION	32-51-81		401	PTF ALL
Rudder Pedal Steering Rotary Actuator Removal TASK 32-51-81-000-801			401	PTF ALL
Rudder Pedal Steering Rotary Actuator Installation TASK 32-51-81-400-801			406	PTF ALL

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CHAPTER 32
LANDING GEAR

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
LANDING GEAR POSITION INDICATING AND WARNING SYSTEM - ADJUSTMENT/TEST	32-61-00		501	PTF ALL
Landing Gear Position Indicating and Warning System Test TASK 32-61-00-710-801			501	PTF ALL
MAIN LANDING GEAR UPLOCK SENSOR - REMOVAL/INSTALLATION	32-61-21		401	PTF ALL
Main Landing Gear Uplock Sensor Removal TASK 32-61-21-020-801			401	PTF ALL
Main Landing Gear Uplock Sensor Installation TASK 32-61-21-400-801			405	PTF ALL
MAIN LANDING GEAR UPLOCK SENSOR - ADJUSTMENT/TEST	32-61-21		501	PTF ALL
Main Landing Gear Uplock Sensor Clearance Measurement (Airplane on Jacks) TASK 32-61-21-400-802			501	PTF ALL
Main Landing Gear Uplock Sensor Clearance Measurement (Airplane not on Jacks) TASK 32-61-21-400-803			507	PTF ALL
Main Landing Gear Uplock Sensor Clearance Adjustment TASK 32-61-21-820-801			511	PTF ALL
Main Landing Gear Up-and-Locked Sensor Operational Test TASK 32-61-21-710-801			512	PTF ALL
MAIN LANDING GEAR DOWN-AND-LOCKED SENSOR - REMOVAL/INSTALLATION	32-61-31		401	PTF ALL
Main Landing Gear Down-and-Locked Sensor Removal TASK 32-61-31-020-801			401	PTF ALL
Main Landing Gear Down-and-Locked Sensor Installation TASK 32-61-31-400-801			405	PTF ALL

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CHAPTER 32
LANDING GEAR

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
MAIN LANDING GEAR DOWN-AND-LOCKED SENSOR - ADJUSTMENT/TEST	32-61-31		501	PTF ALL
Main Landing Gear Down-and-Locked Sensor Clearance Measurement TASK 32-61-31-220-801			501	PTF ALL
Main Landing Gear Down-and-Locked Sensor Clearance Adjustment TASK 32-61-31-400-802			505	PTF ALL
Main Landing Gear Down-and-Locked Sensor Operational Test TASK 32-61-31-710-801			507	PTF ALL
NOSE LANDING GEAR LOCK SENSOR - REMOVAL/INSTALLATION	32-61-41		401	PTF ALL
Nose Landing Gear Lock Sensor Removal TASK 32-61-41-020-801			401	PTF ALL
Nose Landing Gear Lock Sensor Installation TASK 32-61-41-400-801			407	PTF ALL
NOSE LANDING GEAR LOCK SENSOR - ADJUSTMENT/TEST	32-61-41		501	PTF ALL
Nose Landing Gear Lock Sensor Clearance Measurement TASK 32-61-41-220-801			501	PTF ALL
Nose Landing Gear Lock Sensor Clearance Adjustment TASK 32-61-41-400-802			505	PTF ALL
Nose Landing Gear Lock Sensor Operational Test TASK 32-61-41-710-801			507	PTF ALL
NOSE LANDING GEAR DOWN POSITION SENSOR - REMOVAL/INSTALLATION	32-61-51		401	PTF ALL
Nose Landing Gear Down Position Sensor Removal TASK 32-61-51-020-801			401	PTF ALL
Nose Landing Gear Down Position Sensor Installation TASK 32-61-51-400-801			407	PTF ALL

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CHAPTER 32
LANDING GEAR

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
NOSE LANDING GEAR DOWN POSITION SENSOR - ADJUSTMENT/TEST	32-61-51		501	PTF ALL
Nose Landing Gear Down Position Sensor Clearance Measurement TASK 32-61-51-220-801			501	PTF ALL
Nose Landing Gear Down Position Sensor Clearance Adjustment TASK 32-61-51-400-802			504	PTF ALL
Nose Landing Gear Down Position Sensor Operational Test TASK 32-61-51-710-801			506	PTF ALL
TAILSKID ASSEMBLY - MAINTENANCE PRACTICES	32-71-00		201	PTF ALL
Tailskid Energy Absorber Cartridge Replacement TASK 32-71-00-960-801			201	PTF ALL
Tailskid Shoe Replacement TASK 32-71-00-960-802			204	PTF ALL
Tailskid Warning Decal Replacement TASK 32-71-00-960-803			206	PTF ALL
TAILSKID ASSEMBLY - REMOVAL/INSTALLATION	32-71-00		401	PTF ALL
Tailskid Skirt Removal TASK 32-71-00-000-801			401	PTF ALL
Tailskid Skirt Installation TASK 32-71-00-400-801			404	PTF ALL
Tailskid Drag Lever Removal TASK 32-71-00-000-802			405	PTF ALL
Tailskid Drag Lever Installation TASK 32-71-00-400-802			408	PTF ALL
Tailskid Cartridge Assembly Removal TASK 32-71-00-000-803			409	PTF ALL
Tailskid Cartridge Assembly Installation TASK 32-71-00-400-803			412	PTF ALL
TAILSKID - INSPECTION/CHECK	32-71-00		601	PTF ALL
Tailskid Cartridge Assembly Inspection/Check TASK 32-71-00-210-801			601	PTF ALL

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CHAPTER 32
LANDING GEAR

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Tailskid Shoe Inspection TASK 32-71-00-211-801			602	PTF ALL
TAIL SKID PIVOT LINKAGE - REMOVAL/INSTALLATION	32-71-21		401	PTF ALL; AIRPLANES WITH FIXED POSITION TAIL SKID
Tailskid Pivot Linkage Removal TASK 32-71-21-000-801			401	PTF ALL; AIRPLANES WITH FIXED POSITION TAIL SKID
Tailskid Pivot Linkage Installation TASK 32-71-21-400-801			405	PTF ALL; AIRPLANES WITH FIXED POSITION TAIL SKID

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LANDING GEAR - DDG MAINTENANCE PROCEDURES

1. General

A. This procedure has the maintenance tasks for the Master Minimum Equipment List (MMEL) maintenance requirements as shown in the Dispatch Deviations Guide (DDG). These tasks prepare the airplane for flight with systems/components that are inoperative.

B. This procedure also has the tasks that put the airplane back to its usual condition.

C. These are the tasks for the components in the landing gear system:

(1) DDG 32-30-01 Landing Gear Actuation System Inoperative - Preparation
(2) DDG 32-30-01 Landing Gear Actuation System Inoperative - Restoration
(3) DDG 32-31-01 Landing Gear Selector Bypass Valve Inoperative - Preparation
(4) DDG 32-31-01 Landing Gear Selector Bypass Valve Inoperative - Restoration
(5) DDG 32-31-02B Both Landing Gear Control Systems Inoperative - Preparation
(6) DDG 32-31-02B Both Landing Gear Control Systems Inoperative - Restoration
(7) DDG 32-32-01 Main Landing Gear Uplock Springs Inoperative - Preparation
(8) DDG 32-32-01 Main Landing Gear Uplock Springs Inoperative - Restoration
(9) DDG 32-32-02 Main Landing Gear Frangible Fitting Inoperative - Preparation
(10) DDG 32-32-02 Main Landing Gear Frangible Fitting Inoperative - Restoration
(11) DDG 32-35-01 Landing Gear Manual Extension System Inoperative - Preparation
(12) DDG 32-35-01 Landing Gear Manual Extension System Inoperative - Restoration
(13) DDG 32-41-01-02 Flight Deck Brake Pressure Indicating System Inoperative - Preparation
(14) DDG 32-41-01-02 Flight Deck Brake Pressure Indication System Inoperative - Restoration
(15) DDG 32-42-01-01 Antiskid System Inoperative - Preparation
(16) DDG 32-42-01-01 Antiskid System Inoperative - Restoration
(17) DDG 32-42-02-01 Alternate Antiskid Valves Inoperative - Preparation
(18) DDG 32-42-02-01 Alternate Antiskid Valves Inoperative - Restoration
(19) DDG 32-42-03 Automatic Brake System Inoperative - Preparation
(20) DDG 32-42-03 Automatic Brake System Inoperative - Restoration
(21) DDG 32-44-01-01 Parking Brake Valve Inoperative - Preparation
(22) DDG 32-44-01-01 Parking Brake Valve Inoperative - Restoration
(23) DDG 32-44-02-01 Flight Deck Parking Brake Light Inoperative - Preparation
(24) DDG 32-44-02-01 Flight Deck Parking Brake Light Inoperative - Restoration
(25) DDG 32-49-01A Tire Pressure Indication System - Preparation
(26) DDG 32-51-01 Rudder Pedal Nose Wheel Steering System Inoperative - Preparation
(27) DDG 32-51-01 Rudder Pedal Nose Wheel Steering System Inoperative - Restoration
(28) DDG 32-51-02 Nose Wheel Steering Switch Inoperative - Preparation
(29) DDG 32-51-02 Nose Wheel Steering Switch Inoperative - Restoration

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TASK 32-00-00-040-801

2. DDG 32-30-01 Landing Gear Actuation System Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the landing gear actuation system inoperative.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Consumable Materials

Reference	Description	Specification
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G51674	Kit - Safety Cable, 321 CRES - 0.032 Inch (0.81 mm) Diameter, (Contains both Cable and Ferrule), 12 Inches Long	BACC13AT3K12

D. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
211	Flight Compartment - Left
212	Flight Compartment - Right
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Landing Gear Actuation System Preparation

SUBTASK 32-00-00-480-046

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-00-00-480-047

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-00-00-860-005

(3) Set the ALTERNATE VMO/MMO switch in the EE bay service panel, Maintenance Zone 118, to the ALTN position, do these steps:

(a) Cut the safety cable or lockwire in the switch guard.

(b) Lift the switch guard and move the switch to the ALTN position.

(c) Move the guard down and lockwire the switch guard using safety cable kit, G51674 (preferred), or MS20995NC32 lockwire, G01912 (alternate), and double twist method.

(d) Make sure that the ADR label 270 bit 23 is set to “1”.

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SUBTASK 32-00-00-860-125

(4) If the ARINC 429 Data Bus analyzer is not available, check the ADR input monitoring page on the FMC CDU page.

(a) Navigate to FMC CDU MAINT BITE Index page, then select ADIRS.

(b) Push LSK to select ADIRS L for the left ADIRU or ADIRS R for the right ADIRU.

(c) Push LSK 5L to select INPUT MONITORING.

(d) Push LSK 2L to select ADR PROGRAM PIN STATUS.

(e) Navigate to page 3/6.

NOTE: There are 6 pages of ADR program pin status.

1) Make sure that the VMO/MMO 2 pin F3 discrete is displayed as GD (ground).

NOTE: In normal operating condition, the VMO/MMO 1 through 4 are open.

(f) Check both left and right ADR program pins.

SUBTASK 32-00-00-480-005

(5) Attach a GEAR LOCKED DOWN placard to the landing gear lever.

——— END OF TASK ———

TASK 32-00-00-440-801

3. DDG 32-30-01 Landing Gear Actuation System Inoperative - Restoration

A. General

(1) This task restores the airplane back to its usual condition after operation with the landing gear actuation system inoperative.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Consumable Materials

Reference	Description	Specification
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G51674	Kit - Safety Cable, 321 CRES - 0.032 Inch (0.81 mm) Diameter, (Contains both Cable and Ferrule), 12 Inches Long	BACC13AT3K12

D. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
211	Flight Compartment - Left
212	Flight Compartment - Right
734	Left Main Landing Gear
744	Right Main Landing Gear

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E. Landing Gear Actuation System Restoration

SUBTASK 32-00-00-480-052

> **WARNING** MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-00-00-480-053

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-00-00-860-006

(3) Set the ALTERNATE VMO/MMO switch in the EE bay service panel, Maintenance Zone 118, to the NORM position, do these steps:

(a) Cut the safety cable or lockwire in the switch guard.

(b) Lift the switch guard and move the switch to the NORM position.

(c) Move the guard down and lockwire the switch guard using safety cable kit, G51674 (preferred), or MS20995NC32 lockwire, G01912 (alternate), and double twist method.

(d) Make sure that the ADR label 270 bit 23 is set to “0”.

SUBTASK 32-00-00-080-005

(4) Remove the GEAR LOCKED DOWN placard from the landing gear lever.

——— END OF TASK ———

TASK 32-00-00-040-802

4. DDG 32-31-01 Landing Gear Selector Bypass Valve Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the landing gear selector bypass valve inoperative.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
734	Left Main Landing Gear
744	Right Main Landing Gear

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Landing Gear Selector Bypass Valve Preparation

SUBTASK 32-00-00-480-088

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-00-00-480-007

(2) Make sure that the chocks are installed at the wheels (CHOCK INSTALLATION, PAGEBLOCK 10-11-05/201).

SUBTASK 32-00-00-860-007

(3) Open this circuit breaker and install safety lock:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	16	C01432	LANDING GEAR ALTN EXTEND SOL

SUBTASK 32-00-00-710-001

(4) Make sure that the bypass valve is in the NORMAL position:

(a) Push and hold the LOCK OVRD button.

(b) Put the landing gear lever temporarily in the UP position.

(c) Make sure that the main landing gear downlock actuator tries to extend.

(d) Put the landing gear lever in the DN position.

(e) Release the LOCK OVRD button.

END OF TASK

TASK 32-00-00-440-802

5. DDG 32-31-01 Landing Gear Selector Bypass Valve Inoperative - Restoration

A. General

(1) This task restores the airplane back to its usual condition after operation with the landing gear selector bypass valve inoperative.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
734	Left Main Landing Gear
744	Right Main Landing Gear

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Landing Gear Selector Bypass Valve Restoration

SUBTASK 32-00-00-480-089

> **WARNING** MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-00-00-480-090

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-00-00-860-008

(3) Remove the safety lock and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	16	C01432	LANDING GEAR ALTN EXTEND SOL

——— END OF TASK ———

TASK 32-00-00-040-803

6. DDG 32-31-02B Both Landing Gear Control Systems Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with both landing gear control systems inoperative.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Consumable Materials

Reference	Description	Specification
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G51674	Kit - Safety Cable, 321 CRES - 0.032 Inch (0.81 mm) Diameter, (Contains both Cable and Ferrule), 12 Inches Long	BACC13AT3K12

D. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
211	Flight Compartment - Left
212	Flight Compartment - Right
734	Left Main Landing Gear
744	Right Main Landing Gear

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Landing Gear Control Systems Preparation

SUBTASK 32-00-00-480-048

> **WARNING** MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-00-00-480-049

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-00-00-860-009

(3) Set the ALTERNATE VMO/MMO switch in the Electronic Equipment (EE) bay service panel, Maintenance Zone 118, to the ALTN position, do these steps:

(a) Cut the safety cable or lockwire in the switch guard.

(b) Lift the switch guard and move the switch to the ALTN position.

(c) Move the guard down and lockwire the switch guard using safety cable kit, G51674 (preferred), or MS20995NC32 lockwire, G01912 (alternate), and double twist method.

(d) Make sure that the Air Data Reference (ADR) label 270 bit 23 is set to “1”.

SUBTASK 32-00-00-480-009

(4) Attach a GEAR LOCKED DOWN placard to the landing gear lever.

SUBTASK 32-00-00-810-009

(5) If it is necessary, extinguish the MAINT light on the P5-1 panel, do this task: Fault Interrogation for a Maintenance Message, TASK 31-65-00-750-801.

——— END OF TASK ———

TASK 32-00-00-440-803

7. DDG 32-31-02B Both Landing Gear Control Systems Inoperative - Restoration

A. General

(1) This task restores the airplane back to its usual condition after operation with both landing gear control systems inoperative.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Consumable Materials

Reference	Description	Specification
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G51674	Kit - Safety Cable, 321 CRES - 0.032 Inch (0.81 mm) Diameter, (Contains both Cable and Ferrule), 12 Inches Long	BACC13AT3K12

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
211	Flight Compartment - Left
212	Flight Compartment - Right
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Landing Gear Control Systems Restoration

SUBTASK 32-00-00-480-050

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-00-00-480-051

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-00-00-860-010

(3) Set the ALTERNATE VMO/MMO switch in the Electronic Equipment (EE) bay service panel, Maintenance Zone 118, to the NORM position, do these steps:

(a) Cut the safety cable or lockwire in the switch guard.

(b) Lift the switch guard and move the switch to the NORM position.

(c) Move the guard down and lockwire the switch guard using safety cable kit, G51674 (preferred), or MS20995NC32 lockwire, G01912 (alternate), and double twist method.

(d) Make sure that the Air Data Reference (ADR) label 270 bit 23 is set to “0”.

SUBTASK 32-00-00-080-008

(4) Remove the GEAR LOCKED DOWN placard from the landing gear lever.

F. MAINT Light Status Message

SUBTASK 32-00-00-750-002

(1) Uncheck the applicable MAINT light Status Message(s) on the MAINT LIGHT menu (TASK 31-65-00-750-801).

——— END OF TASK ———

TASK 32-00-00-040-804

8. DDG 32-32-01 Main Landing Gear Uplock Springs Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the main landing gear uplock springs inoperative.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-32-31-400-801	Main Gear Uplock Assembly Installation (P/B 401)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Main Landing Gear Uplock Springs Preparation

SUBTASK 32-00-00-480-091

WARNING: MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-00-00-480-092

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-00-00-200-001

(3) Do the applicable steps to remove the applicable main landing gear uplock spring (TASK 32-32-31-400-801).

——— END OF TASK ———

TASK 32-00-00-440-804

9. DDG 32-32-01 Main Landing Gear Uplock Springs Inoperative - Restoration

A. General

(1) This task restores the airplane back to its usual condition after operation with the main landing gear uplock springs inoperative.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-32-31-400-801	Main Gear Uplock Assembly Installation (P/B 401)

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Main Landing Gear Uplock Springs Restoration

SUBTASK 32-00-00-480-054

WARNING: MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-00-00-480-055

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-00-00-420-017

(3) Do the applicable steps to install the applicable main landing uplock spring (TASK 32-32-31-400-801).

——— END OF TASK ———

TASK 32-00-00-040-805

10. DDG 32-32-02 Main Landing Gear Frangible Fitting Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the main landing gear frangible fitting inoperative.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Main Landing Gear Frangible Fitting Preparation

SUBTASK 32-00-00-480-058

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-00-00-480-059

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-00-00-860-011

WARNING MAKE SURE TO REMOVE ALL PRESSURE FROM THE HYDRAULIC SYSTEM. DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

(3) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-00-00-020-011

(4) Disconnect the nut from the frangible fitting assembly.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-00-00-020-012

(5) Remove the frangible fitting from the frangible fitting assembly.

SUBTASK 32-00-00-420-002

(6) Install a standard BACC14AD06JL or equivalent cap on the frangible fitting assembly.

SUBTASK 32-00-00-860-107

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(7) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

——— END OF TASK ———

TASK 32-00-00-440-821

11. DDG 32-32-02 Main Landing Gear Frangible Fitting Inoperative - Restoration

A. General

This task restores the airplane back to its usual condition after operation with the main landing gear frangible fitting inoperative.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-32-52-000-802	Main Gear Frangible Fitting Removal (P/B 401)
32-32-52-400-802	Main Gear Frangible Fitting Installation (P/B 401)

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Main Landing Gear Frangible Fitting Restoration

SUBTASK 32-00-00-480-065

WARNING

MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-00-00-480-066

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-00-00-860-110

WARNING MAKE SURE TO REMOVE ALL PRESSURE FROM THE HYDRAULIC SYSTEM. DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

(3) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-00-00-020-013

(4) Remove the cap from the frangible fitting assembly.

SUBTASK 32-00-00-420-015

(5) Replace the frangible fitting:

(a) Do this task: Main Gear Frangible Fitting Removal, TASK 32-32-52-000-802.

(b) Do this task: Main Gear Frangible Fitting Installation, TASK 32-32-52-400-802.

SUBTASK 32-00-00-860-119

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(6) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

——— END OF TASK ———

TASK 32-00-00-040-806

12. DDG 32-35-01 Landing Gear Manual Extension System Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the landing gear manual extension system inoperative.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Consumable Materials

Reference	Description	Specification
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G51674	Kit - Safety Cable, 321 CRES - 0.032 Inch (0.81 mm) Diameter, (Contains both Cable and Ferrule), 12 Inches Long	BACC13AT3K12

D. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Landing Gear Manual Extension System Preparation

SUBTASK 32-00-00-480-062

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-00-00-480-063

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-00-00-860-013

(3) Set the ALTERNATE VMO/MMO switch in the EE bay service panel, Maintenance Zone 118, to the ALTN position, do these steps:

(a) Cut the safety cable or lockwire in the switch guard.

(b) Lift the switch guard and move the switch to the ALTN position.

(c) Move the guard down and lockwire the switch guard using safety cable kit, G51674 (preferred), or MS20995NC32 lockwire, G01912 (alternate), and double twist method.

(d) Make sure that the ADR label 270 bit 23 is set to “1”.

SUBTASK 32-00-00-480-014

(4) Attach a GEAR LOCKED DOWN placard to the landing gear lever.

——— END OF TASK ———

TASK 32-00-00-440-806

13. DDG 32-35-01 Landing Gear Manual Extension System Inoperative - Restoration

A. General

(1) This task restores the airplane back to its usual condition after operation with the landing gear manual extension system inoperative.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Consumable Materials

Reference	Description	Specification
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description	Specification
G51674	Kit - Safety Cable, 321 CRES - 0.032 Inch (0.81 mm) Diameter, (Contains both Cable and Ferrule), 12 Inches Long	BACC13AT3K12

D. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
211	Flight Compartment - Left
212	Flight Compartment - Right
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Landing Gear Manual Extension System Restoration

SUBTASK 32-00-00-480-060

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-00-00-480-061

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-00-00-860-014

(3) Set the ALTERNATE VMO/MMO switch in the EE bay service panel, Maintenance Zone 118, to the NORM position, do these steps:

(a) Cut the safety cable or lockwire in the switch guard.

(b) Lift the switch guard and move the switch to the NORM position.

(c) Move the guard down and lockwire the switch guard using safety cable kit, G51674 (preferred), or MS20995NC32 lockwire, G01912 (alternate), and double twist method.

(d) Make sure that the ADR label 270 bit 23 is set to “0”.

SUBTASK 32-00-00-080-011

(4) Remove the GEAR LOCKED DOWN placard from the landing gear lever.

——— END OF TASK ———

TASK 32-00-00-040-807

14. DDG 32-41-01-02 Flight Deck Brake Pressure Indicating System Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the flight deck brake pressure indicating system inoperative.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
12-15-11-420-801	Hydraulic Brake Accumulator Servicing (P/B 301)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
211	Flight Compartment - Left
212	Flight Compartment - Right
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Flight Deck Brake Pressure Indicating System Preparation

SUBTASK 32-00-00-480-093

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-00-00-480-064

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-00-00-860-015

(3) Release the parking brakes.

SUBTASK 32-00-00-860-108

WARNING MAKE SURE TO REMOVE ALL PRESSURE FROM THE HYDRAULIC SYSTEM. DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

(4) For hydraulic systems A and B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-00-00-200-005

(5) Apply and release the brakes 10 times with approximately 3 seconds between each application.

SUBTASK 32-00-00-200-006

(6) Make sure that the brake accumulator pressure gauge is within the limits (TASK 12-15-11-420-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-00-00-860-109

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(7) For hydraulic systems A and B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-00-00-200-007

(8) Make sure that the brake accumulator pressure gauge indication is approximately 3000 psi (20,684 kPa).

SUBTASK 32-00-00-860-017

(9) Set the parking brakes.

SUBTASK 32-00-00-860-019

(10) Attach an INOP placard to the brake accumulator pressure indicator.

——— END OF TASK ———

TASK 32-00-00-440-807

15. DDG 32-41-01-02 Flight Deck Brake Pressure Indicating System Inoperative - Restoration

A. General

(1) This task restores the airplane back to its usual condition after operation with the flight deck brake pressure indicating system inoperative.

B. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Put the Airplane Back to Its Usual Condition

SUBTASK 32-00-00-080-013

(1) Remove the INOP placard from the applicable brake pressure indicator.

——— END OF TASK ———

TASK 32-00-00-040-808

16. DDG 32-42-01-01 Antiskid System Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the antiskid system inoperative.

NOTE: The applicable Antiskid Channel Deactivation procedure uses the BITE test to find which antiskid channel is inoperative.

B. References

Reference	Title
32-42-00-740-801	Antiskid/Autobrake Control Unit BITE Procedure (P/B 501)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Antiskid System Preparation

SUBTASK 32-00-00-740-006

(1) Find which antiskid channel is inoperative:

(a) To get access to the antiskid/autobrake control unit (AACU) on the E1-3 shelf, do this step:

Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(b) Do this task: Antiskid/Autobrake Control Unit BITE Procedure, TASK 32-42-00-740-801.

SUBTASK 32-00-00-040-001

(2) Deactivate the applicable antiskid channel:

NOTE: An operative antiskid channel must not be deactivated. Antiskid channels that are faulted must be deactivated.

(a) If message XDCR 1, XDCR 4, BOX 1, BOX 4, VLV 1, or VLV 4 shows on the AACU display, then do this step:

1) Open this circuit breaker and install safety lock:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	18	C00195	LANDING GEAR ANTISKID OUTBD

(b) If message XDCR 2, XDCR 3, BOX 2, BOX 3, VLV 2, or VLV 3 shows on the AACU display, then do this step:

1) Open this circuit breaker and install safety lock:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	16	C00196	LANDING GEAR ANTISKID INBD

(c) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 32-00-00-480-016

(3) Attach an AUTOBRAKE INOP - LEAVE OFF placard to the autobrake selector switch on the P9-82 control panel.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-00-00-440-808

17. DDG 32-42-01-01 Antiskid System Inoperative - Restoration

A. General

(1) This task restores the airplane back to its usual condition after operation with the antiskid system inoperative.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

C. Antiskid System Restoration

SUBTASK 32-00-00-860-020

(1) Remove the safety locks and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	16	C00196	LANDING GEAR ANTISKID INBD
E	18	C00195	LANDING GEAR ANTISKID OUTBD

SUBTASK 32-00-00-080-042

(2) Remove the AUTOBRAKE INOP - LEAVE OFF placard from the autobrake selector switch on the P9-82 control panel.

——— END OF TASK ———

TASK 32-00-00-040-809

18. DDG 32-42-02-01 Alternate Antiskid Valves Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the alternate antiskid valves inoperative.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
734	Left Main Landing Gear
744	Right Main Landing Gear

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Manual Braking Operational Test

SUBTASK 32-00-00-480-069

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-00-00-480-070

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-00-00-860-022

(3) Release the parking brake.

SUBTASK 32-00-00-860-024

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(4) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-00-00-860-120

WARNING MAKE SURE TO REMOVE ALL PRESSURE FROM THE HYDRAULIC SYSTEM. DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

(5) For hydraulic system B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-00-00-860-025

(6) Make sure that you can operate the brakes manually:

(a) Push and release the brake pedals several times.

(b) Make sure that all brakes operate correctly.

NOTE: To make sure that the brake operates correctly, look for the movement of the brake wear indicator pins.

(c) If all of the brakes operate correctly, then attach an ALT ANTISKID VALVE INOP placard to the antiskid inop light.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-00-00-860-027

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(7) Supply pressure to hydraulic system B. Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

——— END OF TASK ———

TASK 32-00-00-440-809

19. DDG 32-42-02-01 Alternate Antiskid Valves Inoperative - Restoration

A. General

(1) This task restores the airplane back to its usual condition after operation with the alternate antiskid valves inoperative.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
733	Left Main Landing Gear - Inboard Door
734	Left Main Landing Gear

D. Alternate Antiskid Valves Restoration

SUBTASK 32-00-00-480-067

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-00-00-480-068

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-00-00-080-015

(3) Remove the ALT ANTISKID VALVE INOP placard from the ANTISKID INOP light.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-00-00-040-810

20. DDG 32-42-03 Automatic Brake System Inoperative - Preparation

(Figure 901, Figure 902)

A. General

(1) This task gives the maintenance steps that prepare the airplane for flight with the Automatic Brake System inoperative.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-42-81-000-801	Autobrake Pressure Control Module Valve Removal (P/B 401)

C. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
130	Subzone - Body Station 540.00 to Body Station 727.00
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

E. Automatic Brake System Inoperative Preparation

SUBTASK 32-00-00-480-071

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-00-00-480-072

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-00-00-860-123

(3) Set the AUTO BRAKE selector switch, on the P9 forward electronic panel, to the OFF position (Figure 901).

SUBTASK 32-00-00-860-124

(4) If the AUTO BRAKE DISARM light, on the P9 forward electronic panel, goes off, then no more deactivation procedures are necessary (Figure 901).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Autobrake Pressure Control Module Deactivation - Method 1

SUBTASK 32-00-00-860-028

WARNING MAKE SURE TO REMOVE ALL PRESSURE FROM THE HYDRAULIC SYSTEM. DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

(1) Remove the pressure from the hydraulic system B (TASK 29-11-00-860-805).

SUBTASK 32-00-00-860-029

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1

SUBTASK 32-00-00-020-003

(3) Disconnect, cap, and stow the electrical connectors from the autobrake pressure control module in the main gear wheel well (Figure 902).

SUBTASK 32-00-00-480-019

(4) Remove the autobrake pressure control module (TASK 32-42-81-000-801).

(a) Install a cap/plug, STD-14686, on the hydraulic supply pressure line.

(b) Connect the brake pressure line and the return line with a 3/8 inch section of tubing or a flexible hose.

NOTE: This prevents hydraulically locking the brake shuttle valves in the brakes off position.

SUBTASK 32-00-00-860-030

(5) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1

SUBTASK 32-00-00-860-031

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(6) Pressurize the hydraulic system B (TASK 29-11-00-860-801).

SUBTASK 32-00-00-210-002

(7) Make sure that there is no leakage at the autobrake pressure control module.

SUBTASK 32-00-00-480-020

(8) Attach an INOP placard to the AUTO BRAKE selector switch on the P9 forward electronic panel.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Autobrake Pressure Control Module Deactivation - Method 2

SUBTASK 32-00-00-860-034

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1

SUBTASK 32-00-00-020-004

(2) Disconnect, cap and stow the electrical connectors from the autobrake pressure control module (Figure 902).

SUBTASK 32-00-00-860-126

(3) Remove pressure from the normal (system B) and the alternate (system A) hydraulic systems, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 32-00-00-860-128

(4) Release the PARKING BRAKE.

SUBTASK 32-00-00-840-001

(5) Operate the brake pedals until the gage at the brake accumulator shows no change in the pressure.

NOTE: You will have to operate the brake pedals approximately 12 times to fully release the pressure from the accumulator.

SUBTASK 32-00-00-020-005

(6) Disconnect the hydraulic supply pressure line.

(a) Install a cap/plug, STD-14686, on the hydraulic supply pressure line.

(b) Install a cap/plug, STD-14686, on the hydraulic supply pressure port on the module.

SUBTASK 32-00-00-860-035

(7) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1

SUBTASK 32-00-00-860-036

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(8) Pressurize the hydraulic system A and B (TASK 29-11-00-860-801).

SUBTASK 32-00-00-210-003

(9) Make sure that there is no leakage at the autobrake pressure control module.

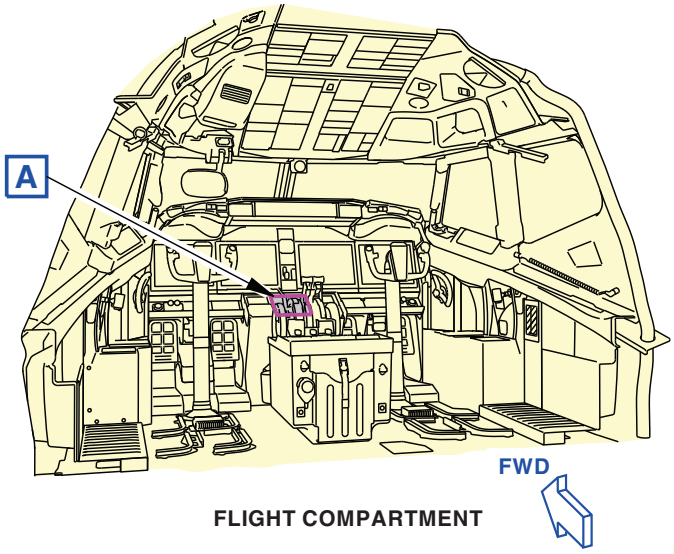
ECCN 9E991 BOEING PROPRIETARY - See title page for details

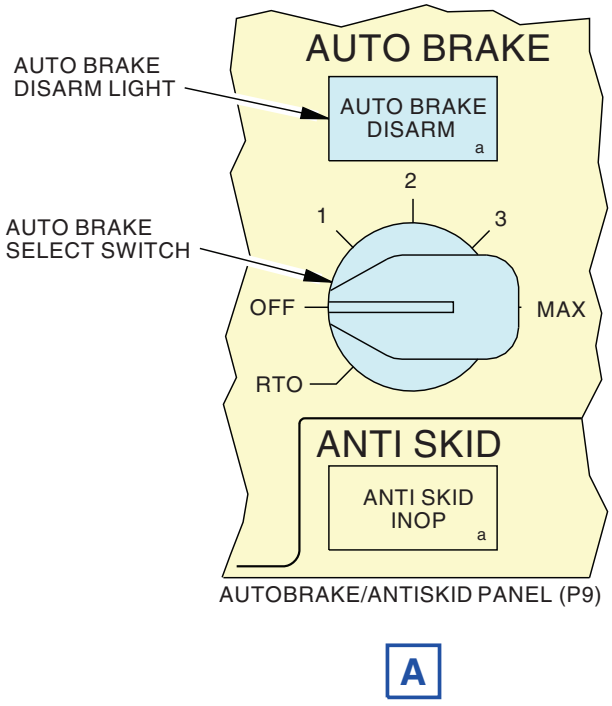
SUBTASK 32-00-00-480-021

(10) Attach an INOP placard to the AUTO BRAKE selector switch on the P9 forward electronic panel.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

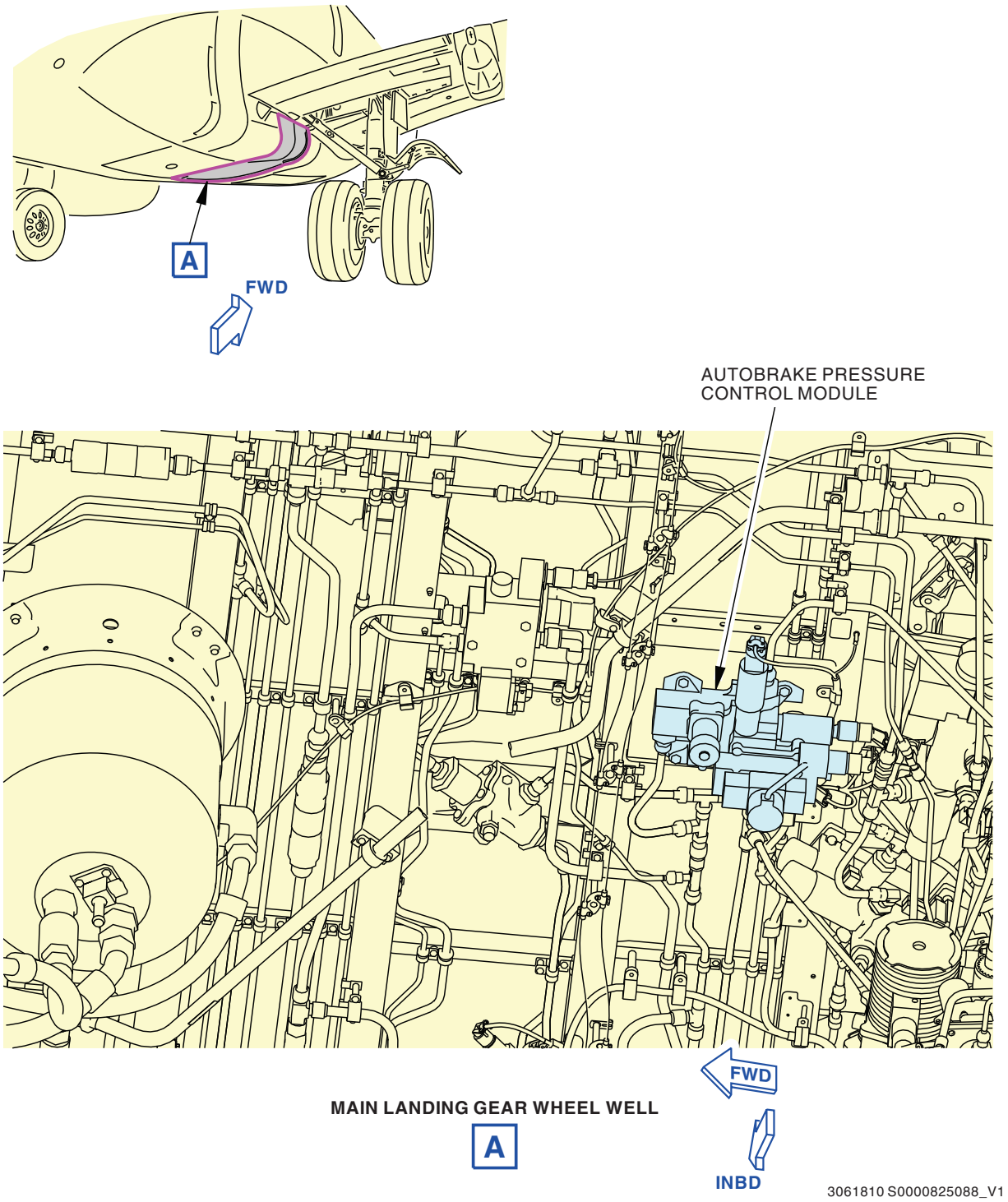




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Autobrake Selector Switch and Autobrake Disarm Light
Figure 901/32-00-00-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Autobrake Pressure Control Module
Figure 902/32-00-00-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-00-00-440-810

21. DDG 32-42-03 Automatic Brake System Inoperative - Restoration

(Figure 901, Figure 902)

A. General

(1) This task restores the airplane back to its usual condition after operation with the Automatic Brake System inoperative.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-42-81-400-801	Autobrake Pressure Control Module Valve Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
130	Subzone - Body Station 540.00 to Body Station 727.00
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

E. Autobrake Pressure Control Module Restoration - Method 1

SUBTASK 32-00-00-480-075

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-00-00-480-076

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-00-00-860-111

WARNING MAKE SURE TO REMOVE ALL PRESSURE FROM THE HYDRAULIC SYSTEM. DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

(3) Remove the pressure from the hydraulic system B (TASK 29-11-00-860-805).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-00-00-860-112

(4) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1

SUBTASK 32-00-00-080-016

(5) Remove the 3/8 inch section of tubing or the flexible hose from the brake pressure line and the return line.

SUBTASK 32-00-00-080-046

(6) Remove the cap/plug, STD-14686, from the hydraulic supply pressure line.

SUBTASK 32-00-00-040-007

(7) Install the autobrake pressure control module (TASK 32-42-81-400-801).

SUBTASK 32-00-00-420-004

(8) Connect the electrical connectors to the autobrake pressure control module.

SUBTASK 32-00-00-860-113

(9) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1

SUBTASK 32-00-00-860-114

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(10) Pressurize the hydraulic system B (TASK 29-11-00-860-801).

SUBTASK 32-00-00-080-019

(11) Remove the INOP placard from the AUTO BRAKE selector switch on the P9 forward electronic panel.

F. Autobrake Pressure Control Module Restoration - Method 2

SUBTASK 32-00-00-480-081

WARNING

MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-00-00-480-082

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-00-00-860-116

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1

SUBTASK 32-00-00-860-127

(4) Remove pressure from the normal (system B) and the alternate (system A) hydraulic systems, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 32-00-00-860-129

(5) Release the PARKING BRAKE.

SUBTASK 32-00-00-840-002

(6) Operate the brake pedals until the gage at the brake accumulator shows no change in the pressure.

NOTE: You will have to operate the brake pedals approximately 12 times to fully release the pressure from the accumulator.

SUBTASK 32-00-00-080-017

(7) Remove the cap/plug, STD-14686, from the autobrake pressure control module and the cap/plug, STD-14686, from the hydraulic supply pressure line.

(a) Connect the hydraulic pressure line to the autobrake pressure control module.

SUBTASK 32-00-00-420-016

(8) Connect the electrical connectors to the autobrake pressure control module.

SUBTASK 32-00-00-860-117

(9) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1

SUBTASK 32-00-00-860-118



WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(10) Pressurize the hydraulic system A and B (TASK 29-11-00-860-801).

SUBTASK 32-00-00-020-014

(11) Remove the INOP placard from the AUTO BRAKE selector switch on the P9 forward electronic panel.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-00-00-040-811

22. DDG 32-44-01-01 Parking Brake Valve Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the parking brake valve inoperative.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

D. Parking Brake Valve Deactivation in the Closed Position

SUBTASK 32-00-00-480-079

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-00-00-480-080

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-00-00-860-040

(3) Release the parking brake.

SUBTASK 32-00-00-860-041

(4) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	16	C01346	LANDING GEAR PARKING BRAKE

SUBTASK 32-00-00-020-006

(5) Disconnect, cap, and stow the electrical connector from the parking brake valve in the main wheel well.

SUBTASK 32-00-00-860-042

(6) Manually position the override lever on the parking brake valve to the POS 2 (closed) position.

SUBTASK 32-00-00-860-043

(7) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	16	C01346	LANDING GEAR PARKING BRAKE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-00-00-080-045

WARNING MAKE SURE THE LANDING GEAR DOWNLOCK PINS ARE REMOVED, BEFORE THE LANDING GEAR CONTROL LEVER CAN BE MOVED TO THE UP POSITION. DAMAGE TO EQUIPMENT AND INJURIES TO PERSONNEL CAN OCCUR.

(8) Make sure that the downlock pins are removed on the nose and main landing gear (Landing Gear Downlock Pins Removal, TASK 32-00-01-080-801).

SUBTASK 32-00-00-860-122

(9) Open these circuit breakers and install safety locks:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	16	C00196	LANDING GEAR ANTISKID INBD
E	18	C00195	LANDING GEAR ANTISKID OUTBD

SUBTASK 32-00-00-860-044

(10) Make sure that the ANTISKID INOP light is on.

SUBTASK 32-00-00-860-045

(11) Set the AUTO BRAKE selector switch to OFF.

SUBTASK 32-00-00-480-024

(12) Attach a LIGHT INOP placard to the parking brake light on the center aisle stand.

SUBTASK 32-00-00-480-025

(13) Attach a PARK BRK VLV CLOSED placard near the parking brake lever.

SUBTASK 32-00-00-480-026

(14) Attach an AUTOBRAKE INOP - LEAVE OFF placard to the autobrake selector switch.

SUBTASK 32-00-00-480-027

(15) Attach an AUTO SPOILERS INOP placard near the auto speed brake control lever.

END OF TASK

TASK 32-00-00-440-811

23. DDG 32-44-01-01 Parking Brake Valve Inoperative - Restoration

A. General

(1) This task restores the airplane back to its usual condition after operation with the parking brake valve inoperative.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Parking Brake Valve Restoration

SUBTASK 32-00-00-480-077

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-00-00-480-078

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-00-00-860-046

(3) Release the parking brake.

SUBTASK 32-00-00-860-047

(4) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	16	C01346	LANDING GEAR PARKING BRAKE

SUBTASK 32-00-00-420-005

(5) Connect the electrical connector to the parking brake valve in the main wheel well.

SUBTASK 32-00-00-860-048

(6) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	16	C01346	LANDING GEAR PARKING BRAKE

SUBTASK 32-00-00-080-044

WARNING MAKE SURE THE LANDING GEAR DOWNLOCK PINS ARE REMOVED, BEFORE THE LANDING GEAR CONTROL LEVER CAN BE MOVED TO THE UP POSITION. DAMAGE TO EQUIPMENT AND INJURIES TO PERSONNEL CAN OCCUR.

(7) Make sure that the downlock pins are removed on the nose and main landing gear (Landing Gear Downlock Pins Removal, TASK 32-00-01-080-801).

SUBTASK 32-00-00-860-121

(8) Remove the safety locks and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	16	C00196	LANDING GEAR ANTISKID INBD
E	18	C00195	LANDING GEAR ANTISKID OUTBD

SUBTASK 32-00-00-080-022

(9) Remove the LIGHT INOP placard from the parking brake light on the center aisle stand.

SUBTASK 32-00-00-080-023

(10) Remove the PARK BRK VLV CLOSED placard near the parking brake lever.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-00-00-080-024

(11) Remove the AUTOBRAKE INOP - LEAVE OFF placard from the autobrake selector switch.

SUBTASK 32-00-00-080-025

(12) Remove the AUTO SPOILERS INOP placard near the auto speed brake control lever.

END OF TASK

TASK 32-00-00-040-812

24. DDG 32-44-02-01 Flight Deck Parking Brake Light Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the flight deck parking brake light inoperative.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

D. Flight Deck Parking Brake Light Preparation

SUBTASK 32-00-00-480-096

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-00-00-480-095

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-00-00-860-050

(3) Make sure that the parking brake shutoff valve operates normally:

(a) Set the parking brake.

(b) Make sure that the parking brake shutoff valve moves to the POS 2 (closed) position.

NOTE: The parking brake shutoff valve is installed on the wheel well aft bulkhead.

(c) Release the parking brake.

(d) Make sure that the parking brake shutoff valve moves to POS 1 (open) position.

(e) Make sure that the ANTISKID INOP light turns off when the parking brake is released.

NOTE: The ANTISKID INOP light can come on when the parking brake is set. If ANTISKID INOP light does not go off when the parking brake is released, refer to DDG 32-42-01-01 Antiskid System Inoperative - Preparation, TASK 32-00-00-040-808.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-00-00-080-047

WARNING DO NOT REMOVE THE DOWNLOCK PINS UNLESS ALL MAINTENANCE AND SERVICING OF THE AIRPLANE IS COMPLETE. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(4) If all maintenance and servicing of the airplane is complete, do this task: Landing Gear Downlock Pins Removal, TASK 32-00-01-080-801.

SUBTASK 32-00-00-480-032

(5) Attach a LIGHT INOP placard to the parking brake light on the center aisle stand.

——— END OF TASK ———

TASK 32-00-00-440-812

25. DDG 32-44-02-01 Flight Deck Parking Brake Light Inoperative - Restoration

A. General

(1) This task restores the airplane back to its usual condition after operation with the flight deck parking brake light inoperative.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

C. Put the Airplane Back to Its Usual Condition

SUBTASK 32-00-00-960-005

(1) Replace the parking brake light.

SUBTASK 32-00-00-080-027

(2) Remove the LIGHT INOP placard from the parking brake light on the center aisle stand.

——— END OF TASK ———

TASK 32-00-00-040-834

26. DDG 32-49-01A Tire Pressure Indication System - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the tire pressure indication system inoperative.

B. References

Reference	Title
12-15-51-780-801	Landing Gear Tire Pressure Check and Tire Servicing (P/B 301)

C. Location Zones

Zone	Area
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Tire Pressure Indication System Preparation

SUBTASK 32-00-00-700-001

(1) Check the landing gear tire pressures (Landing Gear Tire Pressure Check and Tire Servicing, TASK 12-15-51-780-801).

NOTE: The landing gear tire pressure check must be performed daily. The time between checks must not be more than 48 clock hours.

——— END OF TASK ———

TASK 32-00-00-040-813

27. DDG 32-51-01 Rudder Pedal Nose Wheel Steering System Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the rudder pedal nose wheel steering system inoperative.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-09-00 P/B 201	AIR/GROUND SYSTEM - MAINTENANCE PRACTICES
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
324	Vertical Fin - Rear Spar To Trailing Edge
713	Nose Landing Gear

D. Rudder Pedal Nose Wheel Steering System Preparation

SUBTASK 32-00-00-730-001

(1) Put the airplane in the air mode (PAGEBLOCK 32-09-00/201).

(a) Wait a minimum of 5 seconds.

SUBTASK 32-00-00-860-051

(2) Open this circuit breaker and install safety lock:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	17	C01027	LANDING GEAR NOSE GEAR STEER

SUBTASK 32-00-00-860-052

(3) Return the airplane to the ground mode (TASK 32-09-00-860-802).

SUBTASK 32-00-00-860-053

WARNING

KEEP PERSONNEL, AND EQUIPMENT AWAY FROM THE RUDDER. KEEP YOUR HANDS WHERE THE RUDDER CANNOT CATCH THEM. THE RUDDER WILL MOVE AUTOMATICALLY DURING THIS TEST. THE RUDDER CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(4) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

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SUBTASK 32-00-00-860-104

WARNING KEEP PERSONNEL, AND EQUIPMENT AWAY FROM THE RUDDER. KEEP YOUR HANDS WHERE THE RUDDER CANNOT CATCH THEM. THE RUDDER WILL MOVE AUTOMATICALLY DURING THIS TEST. THE RUDDER CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(5) Operate the rudder pedals.

(a) Make sure that the nose gear does not turn.

(b) Make sure that the rudder operates correctly

SUBTASK 32-00-00-860-055

WARNING KEEP ALL PERSONNEL AND EQUIPMENT CLEAR OF THE AREA AROUND THE NOSE GEAR. THE MOVEMENT OF THE NOSE GEAR CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(6) Operate the tiller.

(a) Make sure that the nose gear steering operates correctly.

SUBTASK 32-00-00-860-106

WARNING MAKE SURE TO REMOVE ALL PRESSURE FROM THE HYDRAULIC SYSTEM. DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

(7) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-00-00-860-056

(8) Attach a RUDDER PEDAL NWS INOP placard adjacent to each pilots airspeed indicator.

END OF TASK

TASK 32-00-00-440-813

28. DDG 32-51-01 Rudder Pedal Nose Wheel Steering System Inoperative - Restoration

A. General

(1) This task restores the airplane back to its usual condition after operation with the rudder pedal nose wheel steering system inoperative.

B. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

C. Rudder Pedal Nose Wheel Steering System Restoration

SUBTASK 32-00-00-860-057

(1) Remove the safety lock and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	17	C01027	LANDING GEAR NOSE GEAR STEER

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Put the Airplane Back to Its Usual Condition

SUBTASK 32-00-00-080-041

(1) Remove the RUDDER PEDAL NWS INOP placard from each pilots airspeed indicator.

——— END OF TASK ———

TASK 32-00-00-040-814

29. DDG 32-51-02 Nose Wheel Steering Switch Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the nose wheel steering switch inoperative.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-09-10-740-801	Proximity Switch Electronics Unit - Operational Test (P/B 501)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
211	Flight Compartment - Left
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Nose Wheel Steering Switch Preparation

SUBTASK 32-00-00-860-058

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Pressurize the hydraulic system A. Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-00-00-710-002

(2) Do the PSEU-LGTV Ground Test in this task: Proximity Switch Electronics Unit - Operational Test, TASK 32-09-10-740-801.

(a) Make sure that the landing gear transfer valve operational test shows TEST PASS.

NOTE: The airplane can be dispatched if the landing gear transfer valve operational test shows TEST PASS.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-00-00-860-059

WARNING MAKE SURE TO REMOVE ALL PRESSURE FROM THE HYDRAULIC SYSTEM. DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

(3) Remove the pressure from the hydraulic system A. Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-00-00-860-060

(4) Set the nose wheel steering switch to the NORM position.

SUBTASK 32-00-00-480-033

(5) Attach an INOP placard to the nose wheel steering switch.

——— END OF TASK ———

TASK 32-00-00-440-814

30. DDG 32-51-02 Nose Wheel Steering Switch Inoperative - Restoration

A. General

(1) This task restores the airplane back to its usual condition after operation with the nose wheel steering switch inoperative.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
211	Flight Compartment - Left

C. Put the Airplane Back to Its Usual Condition

SUBTASK 32-00-00-080-029

(1) Remove the INOP placard from the nose wheel steering switch.

——— END OF TASK ———

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LANDING GEAR DOWNLOCK PINS - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) An installation of the downlock pins for the main and the nose landing gear

(2) A removal of the downlock pins for the main and the nose landing gear.

B. The downlock pins are items of safety equipment. You install the downlock pins to make sure that the landing gear does not retract when you do maintenance on the airplane.

C. You also install the downlock pins (with the safety hardware) when the airplane must fly with the landing gear locked in the extended position.

TASK 32-00-01-480-801

2. Landing Gear Downlock Pins Installation

(Figure 201 or Figure 202)

A. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205
SPL-1880	Downlock Equipment - NLG and MLG Part #: C32026-15 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

B. Location Zones

Zone	Area
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

C. Prepare for the Installation

SUBTASK 32-00-01-860-001

(1) Make sure that the nose and main landing gears are down and locked.

SUBTASK 32-00-01-860-002

(2) Make sure that the control lever for the landing gear on the center instrument panel is in the DN position.

WARNING

MAKE SURE THAT THE NOSE AND MAIN LANDING GEAR ARE ALL IN THE DOWN AND LOCKED POSITION. THEN TRY TO INSTALL THE GROUND LOCK ASSEMBLY IN THE NOSE AND MAIN LANDING GEAR. IF THE NOSE AND MAIN LANDING GEAR ARE NOT IN THE DOWN AND LOCKED POSITION, THE LANDING GEAR COULD RETRACT. THIS CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Attach a DO NOT OPERATE tag, STD-858, on the control lever for the landing gear.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Landing Gear Downlock Pins Installation

SUBTASK 32-00-01-480-001

(1) Do the steps below for each of the main landing gear to install the lock pin [4] (downlock equipment, SPL-1880):

(a) If the airplane must fly with the main landing gear locked in the extended position, do the step below:

1) Remove the split rings and REMOVE BEFORE FLIGHT streamer from each of the lock pin [4] (downlock equipment, SPL-1880).

(b) Put the lock pin [4] (downlock equipment, SPL-1880) through the upper lock link [1] and lower lock link [3] for the main landing gear.

NOTE: If the lock pin will not go into the lock links, it is possible an adjustment of the lock links is necessary.

(c) If you installed the lock pin [4] (downlock equipment, SPL-1880) for maintenance, make sure that the REMOVE BEFORE FLIGHT streamers are extended and can be easily seen.

SUBTASK 32-00-01-480-002

(2) Do the steps below to install the lock pin [21] (downlock equipment, SPL-1880) for the nose landing gear:

(a) If the airplane must fly with the landing gear locked in the extended position, do the step below:

1) Remove the split rings and REMOVE BEFORE FLIGHT streamer from the lock pin [21] (downlock equipment, SPL-1880).

(b) Put the lock pin [21] (downlock equipment, SPL-1880) through the forward lock link [24] and aft lock link [22] for the nose landing gear.

NOTE: If the lock pin will not go into the lock links, it is possible an adjustment of the lock links is necessary.

(c) If you installed the lock pin [21] (downlock equipment, SPL-1880) for maintenance, make sure that the REMOVE BEFORE FLIGHT streamer is extended and can be easily seen.

SUBTASK 32-00-01-480-003

WARNING ONLY USE THE CORRECT PIN FOR THE AIRPLANE MODEL. IF YOU USE AN INCORRECT PIN, THE HYDRAULIC STEERING CAN OPERATE. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(3) To install the lock pin [25] (NLG towing lever pin, SPL-1499) for towing the airplane, do the steps below:

NOTE: The towing lever lets you depressurize the nose wheel steering system. Installation of the NLG towing lever pin, SPL-1499, in the towing lever lets you tow the airplane without depressurizing the hydraulic system A.

(a) Move the towing lever [27] forward to the tow position.

(b) Put the lock pin [25] (NLG towing lever pin, SPL-1499) through the pin hole in the towing lever [27] to hold the lever in the towing position.

(c) If you installed the lock pin [25] (NLG towing lever pin, SPL-1499) for maintenance, make sure that the REMOVE BEFORE FLIGHT streamer is extended and can be easily seen.

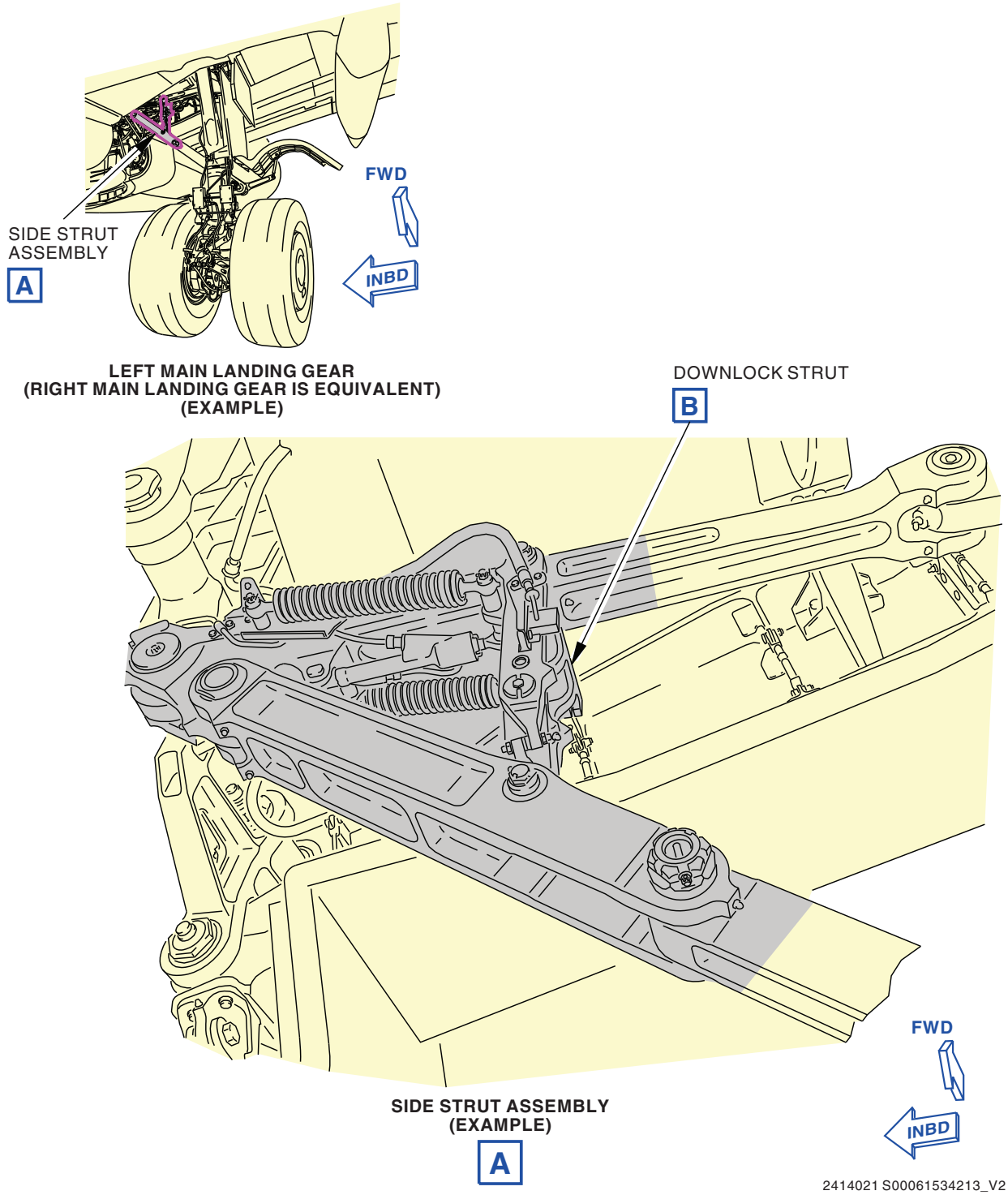
(d) If the airplane must fly with the landing gear locked in the extended position, do the steps below:

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1) Remove the lock pin [25] (NLG towing lever pin, SPL-1499) from the towing lever [27] for the nose wheel steering mechanism.

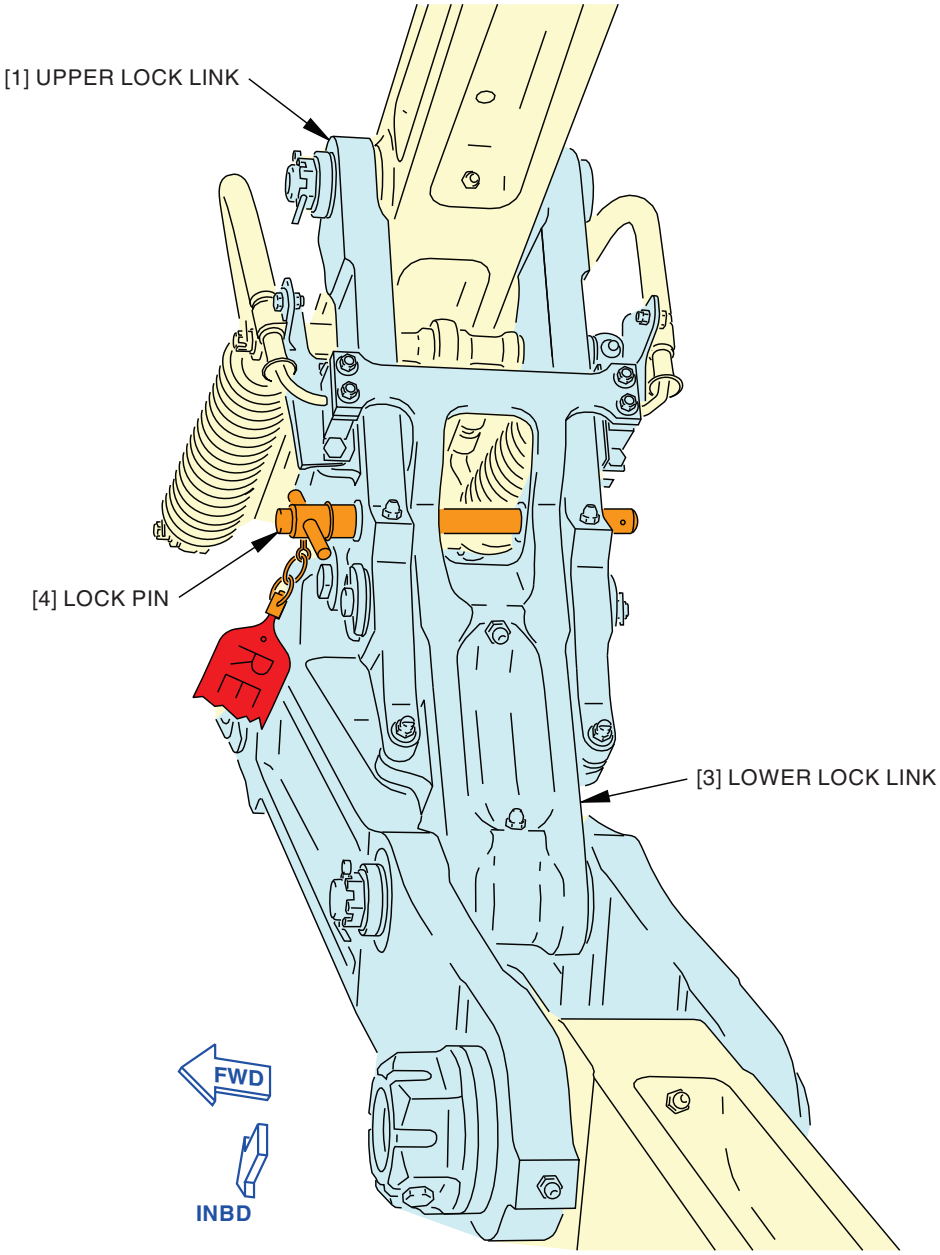
——— END OF TASK ———

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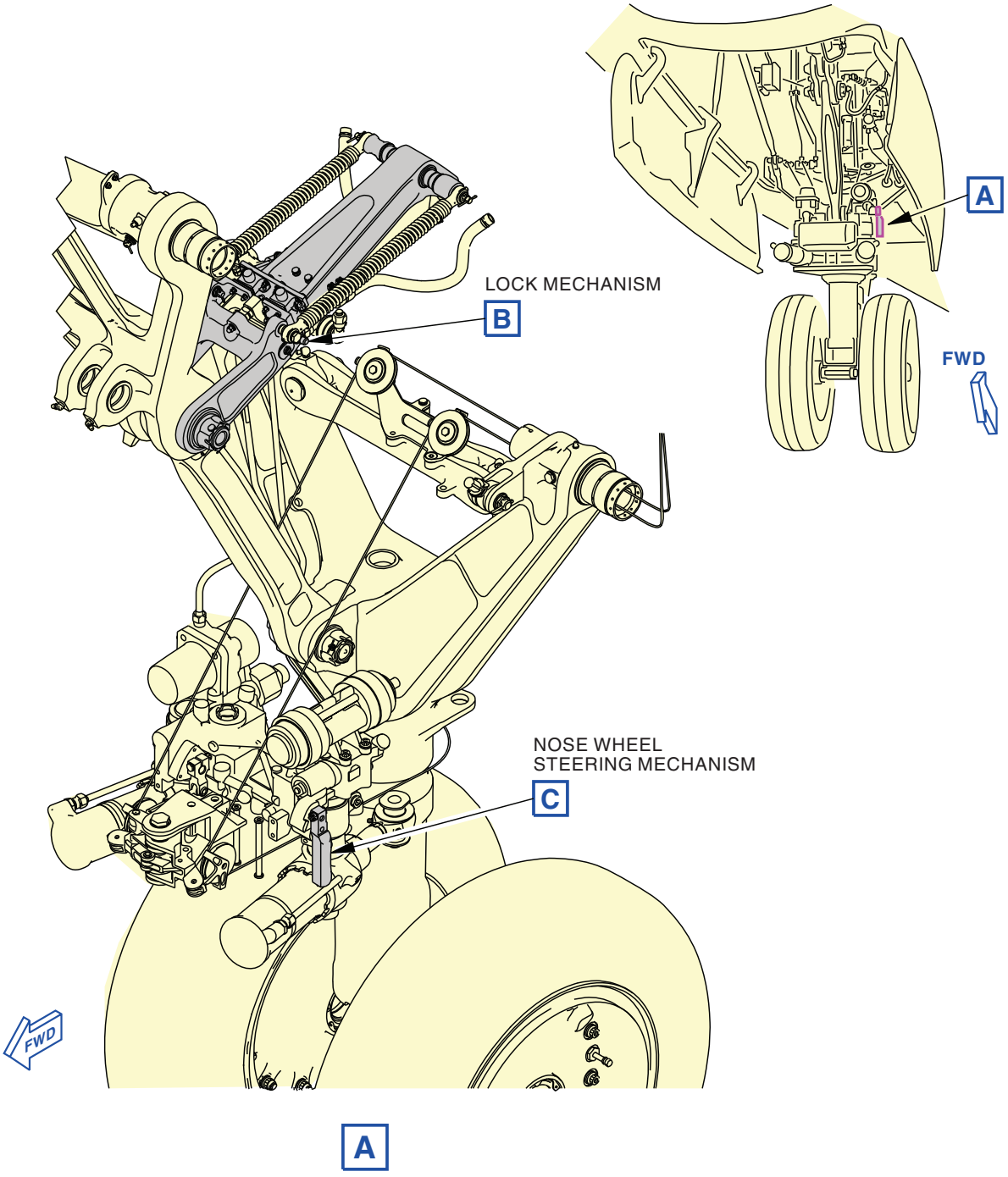
Main Landing Gear Downlock Pin Installation
Figure 201/32-00-01-990-801 (Sheet 1 of 2)

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Main Landing Gear Downlock Pin Installation
Figure 201/32-00-01-990-801 (Sheet 2 of 2)

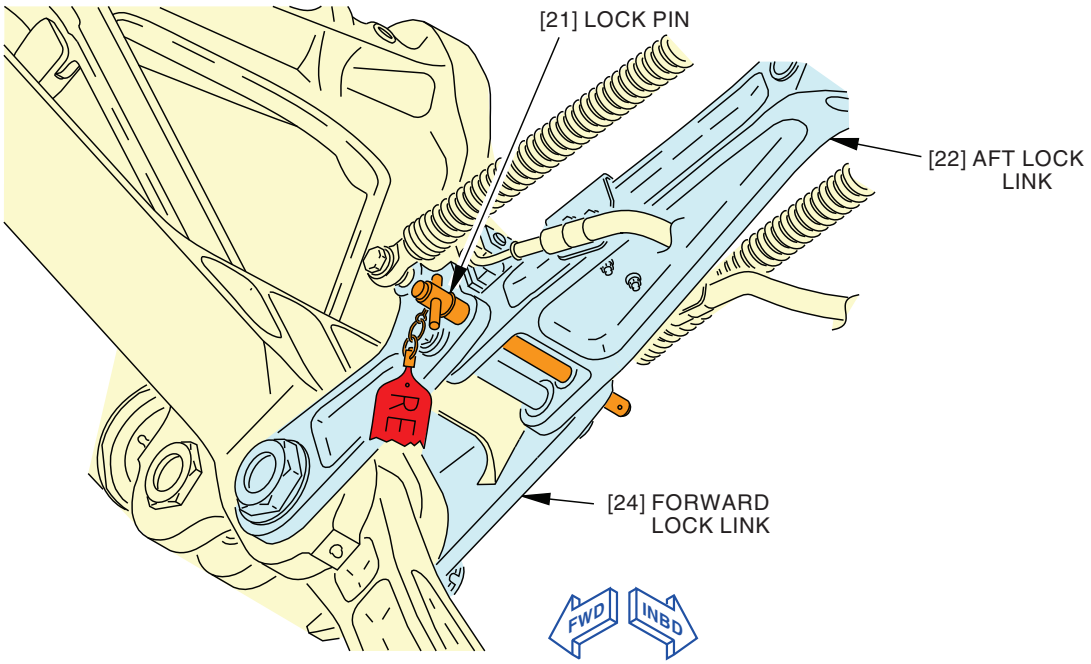
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Nose Landing Gear Downlock Pins Installation
Figure 202/32-00-01-990-802 (Sheet 1 of 2)

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Nose Landing Gear Downlock Pins Installation
Figure 202/32-00-01-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-00-01-080-801

3. Landing Gear Downlock Pins Removal

(Figure 201 or Figure 202)

A. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205
SPL-1880	Downlock Equipment - NLG and MLG Part #: C32026-15 Supplier: 81205
STD-858	Tag - DO NOT OPERATE
STD-3731	Streamer - REMOVE BEFORE FLIGHT

B. Location Zones

Zone	Area
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

C. Prepare for the Removal

SUBTASK 32-00-01-860-003

(1) Make sure that the control lever for the landing gear on the center instrument panel is in the DN position.

WARNING

MAKE SURE THAT THE NOSE AND MAIN LANDING GEAR ARE ALL IN THE DOWN AND LOCKED POSITION. THEN TRY TO INSTALL THE GROUND LOCK ASSEMBLY IN THE NOSE AND MAIN LANDING GEAR. IF THE NOSE AND MAIN LANDING GEAR ARE NOT IN THE DOWN AND LOCKED POSITION, THE LANDING GEAR COULD RETRACT. THIS CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Attach a DO NOT OPERATE tag, STD-858, on the control lever for the landing gear.

D. Landing Gear Downlock Pins Removal

SUBTASK 32-00-01-080-001

(1) Do the steps below for each main landing gear to remove the lock pin [4] (downlock equipment, SPL-1880):

(a) Remove the lock pin [4] (downlock equipment, SPL-1880), from the upper lock link [1] and lower lock link [3].

(b) If it is necessary, attach the split rings and REMOVE BEFORE FLIGHT streamer, STD-3731, to the lock pin [4] (downlock equipment, SPL-1880).

SUBTASK 32-00-01-080-002

(2) Do the steps below to remove the lock pin [21] (downlock equipment, SPL-1880), from the nose landing gear:

(a) Remove the lock pin [21] (downlock equipment, SPL-1880), from the forward lock link [24] and aft lock link [22] for the nose landing gear.

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(b) If it is necessary, attach the split rings and REMOVE BEFORE FLIGHT streamer, STD-3731, to the lock pin [21] (downlock equipment, SPL-1880).

SUBTASK 32-00-01-080-003

(3) Do the steps below to remove the lock pin [25] (NLG towing lever pin, SPL-1499), from the towing lever [27]:

(a) Remove the lock pin [25] (NLG towing lever pin, SPL-1499), from the towing lever [27] for the nose wheel steering mechanism.

(b) If it is necessary, attach the split rings and REMOVE BEFORE FLIGHT streamer, STD-3731, to the lock pin [25] (NLG towing lever pin, SPL-1499).

SUBTASK 32-00-01-010-001

(4) Remove the DO NOT OPERATE tag, STD-858, on the control lever for the landing gear.

——— END OF TASK ———

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LANDING GEAR INNER CYLINDER CHROME - INSPECTION

1. General

A. This procedure provides instructions to examine the chrome surface of the landing gear inner cylinder.

B. This procedure is applicable to the nose landing gear and the main landing gear.

TASK 32-00-10-211-801

2. Landing Gear Inner Cylinder Chrome Inspection

A. References

Reference	Title
12-15-31-610-802	Main Landing Gear Shock Strut Servicing, Airplane on the Ground (P/B 301)
12-15-41-610-802	Nose Landing Gear Shock Strut Servicing, Airplane on the Ground (P/B 301)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-00-10-100-801	Landing Gear Inner Cylinder Chrome Cleaning - with Strut Extended (P/B 701)
32-00-10-300-801	Landing Gear Inner Cylinder Chrome Repair (P/B 801)
32-11-21-200-801	Main Landing Gear Shock Strut Seal Leakage Check (P/B 801)
32-11-21-960-802	Active and the Spare Seals Replacement (P/B 801)
32-11-21-960-803	Scraper Ring Replacement (P/B 801)
32-21-11-200-801	Nose Landing Gear Shock Strut Seal Leakage Check (P/B 801)
32-21-11-960-802	Seals and Spare Seals Replacement (P/B 801)
32-21-11-960-803	Scraper Ring Replacement (P/B 801)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-15067	Laser Gauge Sensor Part #: TS800-F08 Supplier: 1L0G0
COM-15068	Portable Coating Thickness Analyzer Part #: 1007330 Supplier: 61325 Part #: 1007331 Supplier: 61325 Opt Part #: 605-023 Supplier: 61325

C. Location Zones

Zone	Area
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Prepare for the procedure

SUBTASK 32-00-10-420-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED IN ALL OF THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801

SUBTASK 32-00-10-212-001

(2) Inspect the scraper ring and shock strut seals for damage.

(a) Visually examine the chrome surface of the inner cylinder before you clean it.

NOTE: If you find vertical streaks of fluid or accumulated debris on the chrome surface, it is an indication that part of the scraper ring may be defective.

(b) Make sure the scraper ring is in the installed position. Normal operation can force the scraper ring down the inner cylinder.

NOTE: A mirror and flashlight can help you to examine the scraper ring. Put the mirror below the outer cylinder and against the inner cylinder to see the condition and position of the scraper ring.

(c) If one of the two inspections finds a fault with the scraper ring, do the steps that follow:

1) Examine the shock strut for seal leakage (TASK 32-11-21-200-801 or TASK 32-21-11-200-801).

2) Deflate the strut and drain the oil (TASK 12-15-31-610-802TASK 12-15-41-610-802).

3) Disconnect the gland nut from the outer cylinder (TASK 32-11-21-960-802 or TASK 32-21-11-960-802), then lower the gland nut to examine the scraper ring.

4) If the scraper ring is damaged then replace the scraper ring with a new production scraper ring (TASK 32-11-21-960-802 or TASK 32-21-11-960-802.

NOTE: This task requires the removal of the inner cylinder from the outer cylinder.

a) If replacement of the scraper ring is required and removal of the inner cylinder is not possible, the installation of a split scraper ring can be used as a temporary repair (TASK 32-11-21-960-803 or TASK 32-21-11-960-803).

NOTE: The installation of a split scraper ring should be replaced with a new scraper ring when the landing gear is disassembled for other maintenance or overhaul.

(d) Examine the shock strut for seal leakage (TASK 32-11-21-200-801 or TASK 32-21-11-200-801).

SUBTASK 32-00-10-615-001

(3) Do this task: Landing Gear Inner Cylinder Chrome Cleaning - with Strut Extended, TASK 32-00-10-100-801.

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E. Procedure

SUBTASK 32-00-10-212-002

(1) Examine the chrome surface on the outer diameter of the shock strut inner cylinder:

NOTE: Chrome plating on large diameter parts will show very small and narrow lines that look like cracks. These lines (chicken-wire cracks) naturally occur when the chrome plate was initially applied on the base metal. When you look at these parts in dim light or with a light source angled to the surface for the inspection, you will see these lines.

(a) Shine light at an angle to the chrome surface of the inner cylinder.

NOTE: You can use a flashlight for this check.

(b) Look at the surface from different directions to make sure you see all of the cracks or scratches.

NOTE: Do not use a magnifying glass to do this inspection. A scratch on the chrome will have sharp edges along it's length and can be seen without the aide of a magnifying glass or plastic scraper. A wide crack on the chrome surface has a gradual contour to it's edges which makes it difficult to see and requires the use of the plastic scraper to identify.

(c) Move the sharp edge of a new hard plastic scraper perpendicular over all the exposed chrome plated surface with a smooth movement and light hand pressure.

NOTE: You will feel a "grab" or a "catch" when you go over a wide crack or scratch with this method. Make sure the plastic scraper is new, as used scrapers will not satisfactorily find the wide cracks.

(d) If you see any possible area of corrosion, chipping, lifting of the chrome plate, or if you felt a wide crack with the plastic scraper, reinspect the area with a metal dental explorer. Move the sharp point of the explorer with medium hand pressure to go over the area of suspected damage. Go over the area with the dental explorer from multiple directions. If the chrome is lifting and/or loose try to pick it off with the metal explorer.

NOTE: If you feel a sudden high resistance when you go over the area, but do not see signs of possible corrosion, then this is a site with a wide crack. It is an area that must have more protection and be closely monitored. Repair this area with the crack repair procedure. If you are able to remove or lift small pieces of chrome plate from the base metal, if there are bubbles in the chrome, or if the crack extends to the base metal the inner cylinder is not serviceable.

SUBTASK 32-00-10-211-001

(2) If you find scratches or wide cracks, do the steps that follow:

(a) Look for one or more of the conditions that follow, if any of the conditions that follow are present then the inner cylinder is not serviceable:

1) Corrosion exists in the damaged area of the chrome.

NOTE: You must use a 4x (minimum) magnifier to examine the area of damage.

2) Poor chrome bonds with the base metal.

3) Bubbles in the chrome plating.

4) An area of chrome plate that moves above the smooth surface of the adjacent chrome plate when you move the dental explorer across it.

5) Chrome plate that comes off in flakes when you move the dental explorer over the chrome plate.

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(b) Make sure the scratches or wide cracks have not penetrated to the base metal of the strut.

1) Repair the scratches on the chrome surface (TASK 32-00-10-300-801), if identified.

(c) If the base metal is exposed you must measure the depth of the damage.

(d) Replace or overhaul the inner cylinder, if any of these conditions are present:

1) The measured depth of damage to the base metal exceeds 0.0005 in. (0.0127 mm), excluding chrome thickness.

a) Measure the coating thickness using portable coating thickness analyzer, COM-15068.

b) Measure the scratch depth using laser gauge sensor, COM-15067.

c) Subtract the coating thickness from the scratch depth measurement. Ensure the result is positive 0.001 in. (0.025 mm) or larger.

NOTE: The 0.001 in. (0.025 mm) or larger difference above accounts for instrument measuring errors to ensure a 0.0005 in. (0.0127 mm) maximum scratch depth into base material.

2) Damage to the base metal area exceeds 1 in^2 (645 mm^2).

(e) If the base metal is exposed by a scratch then the scratch will then be considered a gouge and be repaired as a crack.

(f) Measure the size of the area that has the cracks, and make a sketch of the location and shape of each area, and mark the dimensions on the sketch. Make sure you keep this data so it can be used for the next inspection.

(g) Repair the cracks on the chrome surface (TASK 32-00-10-300-801), if identified.

(h) If you repaired the chrome surface for cracks then you must do this inspection task again in four months or 400 cycles (whichever comes first), and then again until the shock strut is removed and replaced. Once chrome is damaged, there is no permanent repair, and the damage can continue to spread. The only permanent repair is to strip and replate the chrome. If a repair for minor scratches was preformed, it is not necessary to perform the repeat inspection provided that there are no shock strut fluid leaks and that the minor repair did not affect the base material.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LANDING GEAR INNER CYLINDER CHROME - CLEANING/PAINTING

1. General

A. This procedure provides instruction to clean the dirt and grit from the chrome surfaces of the landing gear inner cylinders. Cleaning these chrome surfaces will make it easier to determine if any seals leak and it will also prolong the life of the seals.

B. This procedure is applicable to the nose landing gear and the main landing gear.

TASK 32-00-10-100-801

2. Landing Gear Inner Cylinder Chrome Cleaning - with Strut Extended

A. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
12-15-31 P/B 301	MAIN LANDING GEAR SHOCK STRUT - SERVICING
12-15-31-610-802	Main Landing Gear Shock Strut Servicing, Airplane on the Ground (P/B 301)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Consumable Materials

Reference	Description	Specification
D00070	Fluid - Hydraulic, Petroleum Base	MIL-PRF-5606 (Replaces MIL-H-5606)
D00106	Fluid - Hydraulic, Petroleum Base, For Preservation And Operation	MIL-PRF-6083 (NATO C-635)
D00467	Fluid - Landing Gear Shock Strut	BMS3-32 Type II
D50022	Fluid - Landing Gear Shock Strut (Specifically For Preservation)	BMS3-32 Type I

C. Location Zones

Zone	Area
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

D. General

SUBTASK 32-00-10-940-001

(1) This task is used with the inspection and repair tasks for the inner cylinder chrome.

E. Prepare for the procedure

SUBTASK 32-00-10-420-002



WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED IN ALL OF THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-00-10-582-001

(2) To clean as much of the chrome surface possible, do one of these steps:

(a) Clean the chrome before you fill the airplane with fuel.

(b) Over-inflate the shock strut with dry-air or nitrogen (TASK 12-15-31-610-802).

CAUTION

MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION

MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(c) Use airplane jacks to lift the airplane until the tires do not touch the ground (TASK 07-10-01-580-801).

F. Procedure

SUBTASK 32-00-10-100-001

CAUTION

DO NOT RUB THE SMEAR MARKS TO TRY TO REMOVE THEM. THIS COULD CAUSE DAMAGE TO THE WEAR SURFACE.

(1) On the shock strut inner cylinder chrome, you may find dark smear marks that have a small amount of texture and are difficult to remove. These smear marks are created by oxidation of the self lubricating material that is deposited on the chrome by the shock strut bearings. These smear marks are considered acceptable.

SUBTASK 32-00-10-100-002

(2) Clean the dirt, oil and other unwanted materials from the exposed chrome surface of the inner cylinders with a clean cloth that is soaked in one of the these:

(a) fluid, D50022

(b) fluid, D00467

(c) fluid, D00070

(d) fluid, D00106.

G. Put the aircraft back to its usual condition.

SUBTASK 32-00-10-610-001

(1) Service the shock strut to the proper inflation (PAGEBLOCK 12-15-31/301).

SUBTASK 32-00-10-580-001

CAUTION

MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(2) Lower the airplane off the jacks (TASK 07-10-01-580-802).

——— END OF TASK ———

TASK 32-00-10-100-802

3. Landing Gear Inner Cylinder Chrome Cleaning

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Consumable Materials

Reference	Description	Specification
D00070	Fluid - Hydraulic, Petroleum Base	MIL-PRF-5606 (Replaces MIL-H-5606)
D00106	Fluid - Hydraulic, Petroleum Base, For Preservation And Operation	MIL-PRF-6083 (NATO C-635)
D00467	Fluid - Landing Gear Shock Strut	BMS3-32 Type II
D50022	Fluid - Landing Gear Shock Strut (Specifically For Preservation)	BMS3-32 Type I

C. Location Zones

Zone	Area
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Prepare for the Cleaning

SUBTASK 32-00-10-420-003

WARNING

MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Procedure

SUBTASK 32-00-10-160-001



CAUTION

DO NOT RUB THE SMEAR MARKS TO TRY TO REMOVE THEM. THIS COULD CAUSE DAMAGE TO THE WEAR SURFACE.

(1) On the shock strut inner cylinder chrome, you may find dark smear marks that have a small amount of texture and are difficult to remove. These smear marks are created by oxidation of the self lubricating material that is deposited on the chrome by the shock strut bearings. These smear marks are considered acceptable.

SUBTASK 32-00-10-610-002

(2) Clean the dirt, oil and other unwanted materials from the chrome surface of the inner cylinder. Use a clean cloth that is soaked in one of these:

NOTE: To clean the maximum exposed chrome, do this task before you fill the airplane with fuel.

(a) fluid, D50022

(b) fluid, D00467

(c) fluid, D00070

(d) fluid, D00106

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LANDING GEAR INNER CYLINDER CHROME - REPAIR

1. General

A. This procedure provides instructions to repair cracks and minor scratches to the chrome surface of the landing gear inner cylinder.

B. This procedure is applicable to the nose landing gear and the main landing gear.

TASK 32-00-10-300-801

2. Landing Gear Inner Cylinder Chrome Repair

A. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-00-10-100-801	Landing Gear Inner Cylinder Chrome Cleaning - with Strut Extended (P/B 701)
32-00-10-211-801	Landing Gear Inner Cylinder Chrome Inspection (P/B 601)

B. Consumable Materials

Reference	Description	Specification
C00259	Coating - Chemical And Solvent Resistant Finish, Corrosion Inhibiting Primer	BMS10-11 Type I
D00467	Fluid - Landing Gear Shock Strut	BMS3-32 Type II
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G01601	Abrasive - Aluminum Oxide Coated Cloth, 400 to 600 grit	
G50220	Pad - Abrasive, Scotch- Brite 7447	
G50338	Abrasive Paper - Aluminum Oxide, 800 Grit	

C. Location Zones

Zone	Area
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Prepare for the procedure

SUBTASK 32-00-10-420-004

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED IN ALL OF THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801

SUBTASK 32-00-10-615-002

(2) Make sure these tasks are accomplished before you do the repair:

(a) Landing Gear Inner Cylinder Chrome Cleaning - with Strut Extended, TASK 32-00-10-100-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Landing Gear Inner Cylinder Chrome Inspection, TASK 32-00-10-211-801.

E. Procedure

SUBTASK 32-00-10-350-001

CAUTION

DO NOT USE POWER TOOLS TO DO THIS TASK. YOU MUST USE HAND TOOLS. IF YOU USE POWER TOOLS, YOU CAN CAUSE MORE DAMAGE TO COMPONENTS.

(1) To repair cracks to the chrome surface, do these steps:

(a) Use 400 to 600 grit abrasive cloth, G01601, that is soaked in fluid, D00467, to hand blend the exposed base metal and remove any corrosion and sharp edges to the chrome.

NOTE: Ensure reworked edges can not be caught with the sharp point of the dental explorer.

(b) Use a cotton wiper, G00034 to apply a thick layer of primer, C00259 to the exposed base metal with adequate overlap on the chrome surface.

(c) Before the primer dries, remove the excess primer with a clean, dry cloth.

NOTE: Make sure no primer gets on the seals or scraper, as they can be damaged by the primer and may start to leak.

(d) Let the primer cure for at least 4 hours., then apply a second layer of primer and allow to dry an additional 4 hours.

(e) Use Scotch-Brite 7447 pad, G50220, that is soaked in fluid, D00467, to blend and smooth out the primer surface to achieve a similar contour as the chrome surface.

(f) Use abrasive paper, G50338, that is soaked in fluid, D00467, to polish the chrome surface.

SUBTASK 32-00-10-350-002

CAUTION

DO NOT USE POWER TOOLS TO DO THIS TASK. YOU MUST USE HAND TOOLS. IF YOU USE POWER TOOLS, YOU CAN CAUSE MORE DAMAGE TO COMPONENTS.

(2) To repair minor scratches to the chrome surface, do these steps:

(a) Use 400 to 600 grit abrasive cloth, G01601, that is soaked in fluid, D00467, to smooth out the chrome surface until you can not see evidence of a scratch.

(b) Use abrasive paper, G50338, that is soaked in fluid, D00467, to polish the chrome surface.

SUBTASK 32-00-10-615-003

(3) Do this task: Landing Gear Inner Cylinder Chrome Cleaning - with Strut Extended, TASK 32-00-10-100-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LANDING GEAR RETRACT/STEERING ACTUATOR CHROME - CLEANING

1. General

A. This procedure has these task:

(1) Main Landing Gear Retract Actuator Chrome Cleaning.

(2) Nose Landing Gear Steering Actuator Chrome Surfaces Cleaning.

(3) Nose Landing Gear Retract Actuator Chrome Cleaning.

TASK 32-00-11-100-801

2. Main Landing Gear Retract Actuators Chrome Surfaces Cleaning

A. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735

C. Location Zones

Zone	Area
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Access Panels

Number	Name/Location
551BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel
651BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel

E. Prepare to Clean the Main Landing Gear Retract Actuators Chrome Surfaces

SUBTASK 32-00-11-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-00-11-010-001

(2) To gain access to the left or right wing, do this step:

Open these access panels:

Number	Name/Location
551BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel
651BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Clean the Main Landing Gear Retract Actuators Chrome Surfaces

SUBTASK 32-00-11-100-001

(1) Clean the dirt, oil and other unwanted materials from the chrome cylinder surface of the main landing gear retract actuators with a clean cloth that is soaked in isopropyl alcohol, B00130.

NOTE: It is not necessary to retract the landing gear to expose more of the chrome surface.

SUBTASK 32-00-11-840-001

(2) Clean the dirt, oil and other unwanted materials from the main landing gear retract actuator exterior surfaces where the chrome piston rod passes through the end gland with a clean cloth that is soaked in isopropyl alcohol, B00130.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-00-11-410-001

(1) Close the applicable access panel, do this step:

Close these access panels:

Number	Name/Location
551BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel
651BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel

——— END OF TASK ———

TASK 32-00-11-100-802

3. Nose Landing Gear Steering Actuators Chrome Surfaces Cleaning

A. General

(1) There are two methods to clean the chrome surfaces of the steering actuators. Each method is optional to the other. The method that is used is determined by the ground equipment that is available.

(a) Method 1:

1) This method disconnects the apex joint of the torsion link.

(b) Method 2:

1) This method uses hydraulic system A to turn the nose wheel tires.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-21-31-000-803	Nose Landing Gear Torsion Link Disconnection (P/B 401)
32-21-31-400-803	Nose Landing Gear Torsion Link Connection (P/B 401)

C. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735

D. Location Zones

Zone	Area
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare to Clean the Nose Landing Gear Steering Actuator Chrome Surfaces

SUBTASK 32-00-11-480-002

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-00-11-020-001

(2) For Method 1, do this task: Nose Landing Gear Torsion Link Disconnection, TASK 32-21-31-000-803.

SUBTASK 32-00-11-860-001

(3) For Method 1, push or pull on the upper torsion link to turn the steering collar until the chrome surface of the steering actuator to be cleaned is exposed.

SUBTASK 32-00-11-840-002

(4) For Method 2, do the following if you will use greased plates to turn the nose landing gear.

NOTE: The use of greased plates is optional for method 2.

(a) Put greased plates on the ground, such that the greased plates are close to the nose landing gear.

(b) Move the airplane until the nose landing gear tires are on the greased plates.

SUBTASK 32-00-11-860-002

(5) For Method 2, pressurize hydraulic system A: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801..

SUBTASK 32-00-11-820-001

(6) For Method 2, use the steering tiller to operate the steering system of the nose wheel full left and full right or until the chrome surface of the actuator to be cleaned is fully exposed.

NOTE: The steering system has a 1 - 2 degree dead band in the center position.

F. Clean the Nose Landing Gear Steering Actuators Chrome Surfaces

SUBTASK 32-00-11-100-002

(1) Clean the dirt, oil and other unwanted materials from the chrome cylinder surface of the steering actuator with a clean cloth that is soaked in isopropyl alcohol, B00130.

NOTE: Make sure that the chrome cylinder surface of the steering actuators are exposed when you do this step.

SUBTASK 32-00-11-100-003

(2) Clean the dirt, oil and other unwanted materials from the steering actuator exterior surfaces where the chrome piston rod passes through the end gland with a clean cloth that is soaked in isopropyl alcohol, B00130.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-00-11-420-001

(1) If you used Method 1, do this task: Nose Landing Gear Torsion Link Connection, TASK 32-21-31-400-803.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-00-11-840-003

(2) If you used Method 2, move the airplane away from the greased plates if they were used.

END OF TASK

TASK 32-00-11-100-803

4. Nose Landing Gear Retract Actuator Chrome Surfaces Cleaning

A. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735

C. Location Zones

Zone	Area
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Prepare to Clean the Nose Landing Gear Retract Actuator Chrome Surfaces

SUBTASK 32-00-11-480-003

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

E. Clean the Nose Landing Gear Retract Actuator Chrome Surfaces

SUBTASK 32-00-11-100-004

(1) Clean the dirt, oil and other unwanted materials from the chrome cylinder surface of the retract actuator of the nose landing gear with a clean cloth that is soaked in isopropyl alcohol, B00130.

NOTE: It is not necessary to retract the landing gear to expose more of the chrome surface.

(2) Clean the dirt, oil and other unwanted materials from the nose landing gear retract actuator exterior surfaces where the chrome piston rod passes through the end gland with a clean cloth that is soaked in isopropyl alcohol, B00130.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LANDING GEAR - CORROSION PREVENTION

1. General

Table 201/32-00-37-993-801 specific Corrosion Problems - Landing Gear

AREA	PROBLEM	INDEX
Main Landing Gear	Corrosion because of environmental exposure.	32-11-37
	Damage to the side strut finish caused by interference with center door.	
	Corrosion on MLG torsion pins.	
	Stress corrosion cracks on H11 brake attachment bolts.	
Nose Landing Gear	Corrosion on torsion link pin and shaft.	32-21-37
	Corrosion found in steering collar area.	
	Corrosion on torsion link pin and shaft.	

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-05-03-210-801

1. INTERNAL - GENERAL VISUAL: LEFT MAIN LANDING GEAR ASSEMBLY

Figure 601

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
51-05-01-210-802	737-7/8/8200/9 Basic Task Description (P/B 201)

B. Location Zones

Zone	Area
734	Left Main Landing Gear

C. Access Panels

Number	Name/Location
551BB	Lower Inboard Fixed Trailing Edge, Gear Adjustment Door
551BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel
551DT	Upper Inboard Fixed Trailing Edge, MLG Beam Access Panel
551EB	Lower Inboard Fixed Trailing Edge, MLG Attach Fitting Access Panel
551ET	Upper Inboard Fixed Trailing Edge, MLG Beam Access Panel
551FB	Lower Inboard Fixed Trailing Edge, Landing Gear Access Panel

D. Inspection

SUBTASK 32-05-03-010-001

(1) Open these access panels:

Number	Name/Location
551BB	Lower Inboard Fixed Trailing Edge, Gear Adjustment Door
551BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel
551DT	Upper Inboard Fixed Trailing Edge, MLG Beam Access Panel
551EB	Lower Inboard Fixed Trailing Edge, MLG Attach Fitting Access Panel
551ET	Upper Inboard Fixed Trailing Edge, MLG Beam Access Panel
551FB	Lower Inboard Fixed Trailing Edge, Landing Gear Access Panel

SUBTASK 32-05-03-210-001

(2) Do a General Visual inspection of the left main landing gear assembly, including outer cylinder, inner cylinder, axle, torsion links, side strut, lock links and retraction linkage. Landing gear removal is required. Disassemble as required to accomplish CPCP basic task on all fittings, lugs, lug bores, bolts and pins. Normal overhaul procedures, applied with the landing gear removed, at intervals not exceeding 10 years are adequate to maintain corrosion at safe levels on main landing gear components.

SUBTASK 32-05-03-910-001

(3) Do this Task: 737-7/8/8200/9 Basic Task Description, TASK 51-05-01-210-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

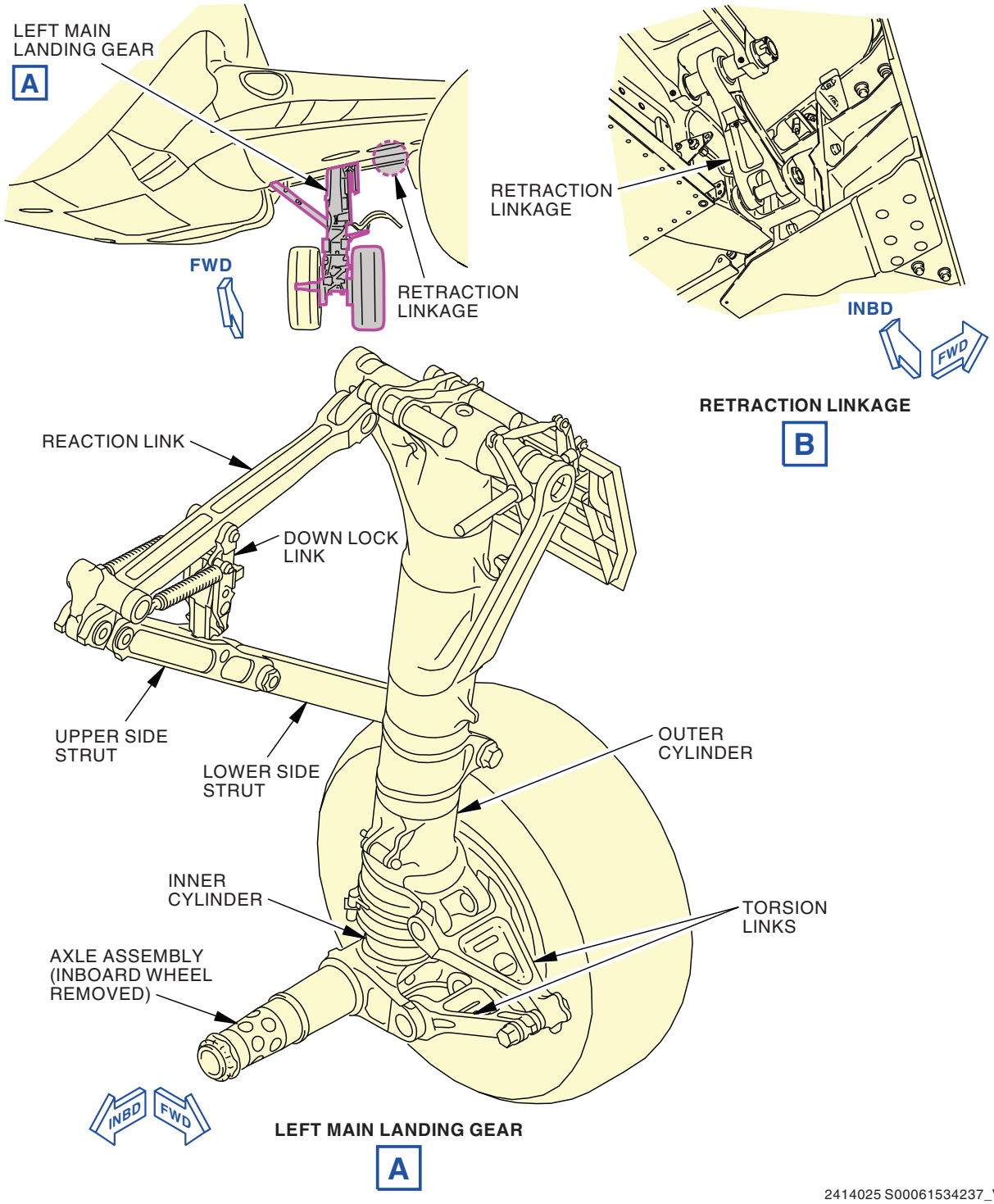
SUBTASK 32-05-03-410-001

(4) Close these access panels:

Number	Name/Location
551BB	Lower Inboard Fixed Trailing Edge, Gear Adjustment Door
551BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel
551DT	Upper Inboard Fixed Trailing Edge, MLG Beam Access Panel
551EB	Lower Inboard Fixed Trailing Edge, MLG Attach Fitting Access Panel
551ET	Upper Inboard Fixed Trailing Edge, MLG Beam Access Panel
551FB	Lower Inboard Fixed Trailing Edge, Landing Gear Access Panel

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Left Main Landing Gear Assembly
Figure 601/32-05-03-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-05-03-210-802

2. INTERNAL - GENERAL VISUAL: RIGHT MAIN LANDING GEAR ASSEMBLY

Figure 602

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
51-05-01-210-802	737-7/8/8200/9 Basic Task Description (P/B 201)

B. Location Zones

Zone	Area
744	Right Main Landing Gear

C. Access Panels

Number	Name/Location
651BB	Lower Inboard Fixed Trailing Edge, Gear Adjustment Door
651BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel
651DT	Upper Inboard Fixed Trailing Edge, MLG Beam Access Panel
651EB	Lower Inboard Fixed Trailing Edge, MLG Attach Fitting Access Panel
651ET	Upper Inboard Fixed Trailing Edge, MLG Beam Access Panel
651FB	Lower Inboard Fixed Trailing Edge, Landing Gear Access Panel

D. Inspection

SUBTASK 32-05-03-010-002

(1) Open these access panels:

Number	Name/Location
651BB	Lower Inboard Fixed Trailing Edge, Gear Adjustment Door
651BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel
651DT	Upper Inboard Fixed Trailing Edge, MLG Beam Access Panel
651EB	Lower Inboard Fixed Trailing Edge, MLG Attach Fitting Access Panel
651ET	Upper Inboard Fixed Trailing Edge, MLG Beam Access Panel
651FB	Lower Inboard Fixed Trailing Edge, Landing Gear Access Panel

SUBTASK 32-05-03-210-002

(2) Do a General Visual inspection of the right main landing gear assembly, including outer cylinder, inner cylinder, axle, torsion links, side strut, lock links and retraction linkage. Landing gear removal is required. Disassemble as required to accomplish CPCP basic task on all fittings, lugs, lug bores, bolts and pins. Normal overhaul procedures, applied with the landing gear removed, at intervals not exceeding 10 years, are adequate to maintain corrosion at safe levels on main landing gear components.

SUBTASK 32-05-03-910-002

(3) Do this Task: 737-7/8/8200/9 Basic Task Description, TASK 51-05-01-210-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

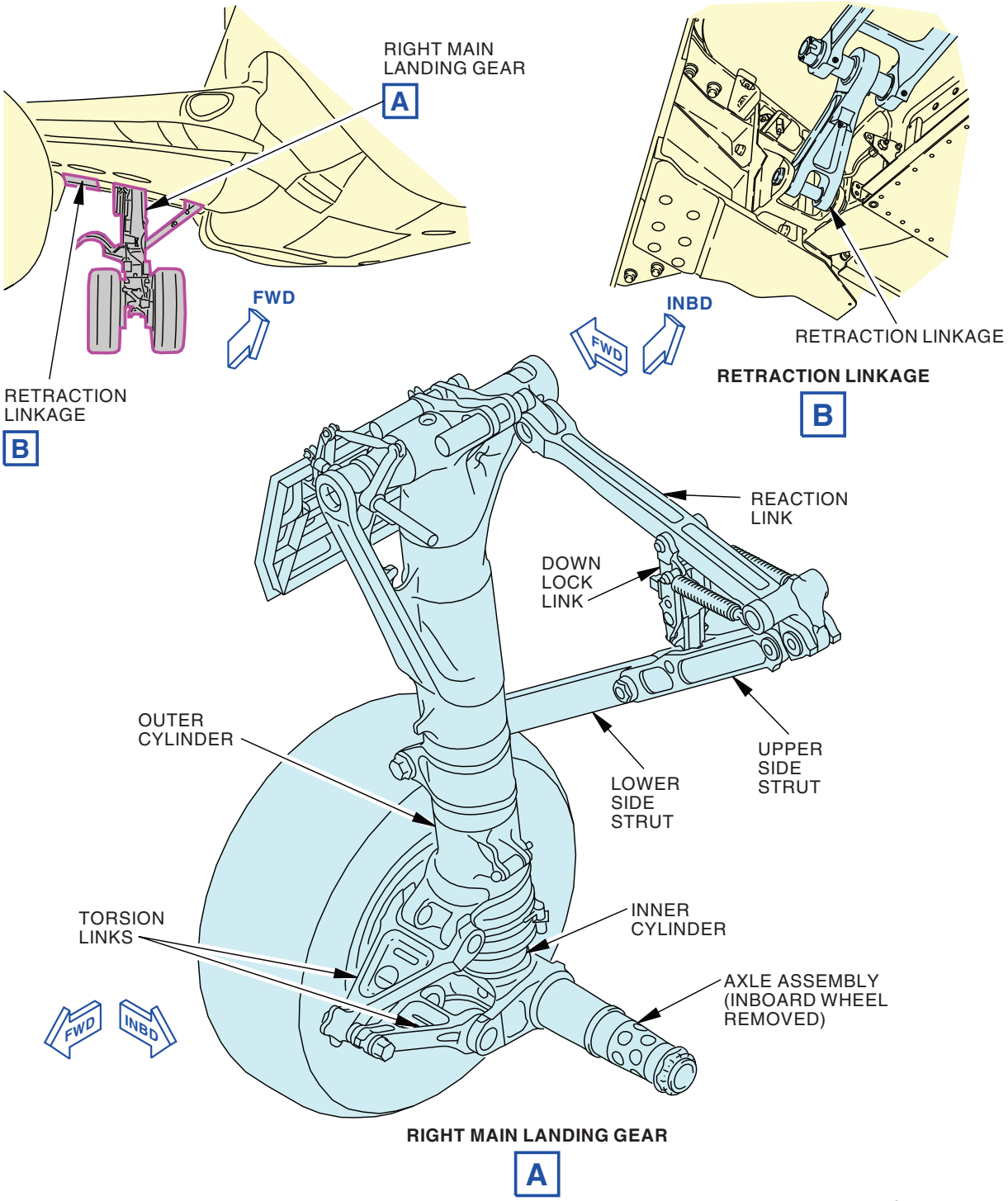
SUBTASK 32-05-03-410-002

(4) Close these access panels:

Number	Name/Location
651BB	Lower Inboard Fixed Trailing Edge, Gear Adjustment Door
651BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel
651DT	Upper Inboard Fixed Trailing Edge, MLG Beam Access Panel
651EB	Lower Inboard Fixed Trailing Edge, MLG Attach Fitting Access Panel
651ET	Upper Inboard Fixed Trailing Edge, MLG Beam Access Panel
651FB	Lower Inboard Fixed Trailing Edge, Landing Gear Access Panel

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Right Main Landing Gear Assembly
Figure 602/32-05-03-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-05-03-210-803

3. INTERNAL - GENERAL VISUAL: NOSE LANDING GEAR ASSEMBLY

Figure 603

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
05-51-91-790-801	Cabin Pressure Leak Test (P/B 201)
51-05-01-210-802	737-7/8/8200/9 Basic Task Description (P/B 201)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

B. Location Zones

Zone	Area
713	Nose Landing Gear

C. Access Panels

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel
114AC	Fwd Nose Wheel Well Upper Access Panel
114AW	Forward Nose Wheel Well Panel
114BW	Forward Nose Wheel Well Panel

D. Inspection

SUBTASK 32-05-03-010-004

(1) Open these access panels:

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel
114AC	Fwd Nose Wheel Well Upper Access Panel
114AW	Forward Nose Wheel Well Panel
114BW	Forward Nose Wheel Well Panel

(a) Do this task: Nose Wheel Well Access Panels Removal, TASK 53-14-01-020-801.

SUBTASK 32-05-03-210-003

(2) Do a General Visual inspection of the nose landing gear assembly, including outer cylinder, inner cylinder, drag strut, lock links, torsion links, and steering mechanism (plates and collar). Landing gear removal is required. Disassemble as required to accomplish CPCP basic task on all fittings, lugs, lug bores, bolts and pins. Normal overhaul procedures, applied with the landing gear removed, at intervals not exceeding 10 years, are adequate to maintain corrosion at safe levels on nose landing gear components.

SUBTASK 32-05-03-910-003

(3) Do this Task: 737-7/8/8200/9 Basic Task Description, TASK 51-05-01-210-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-05-03-410-004

(4) Close these access panels:

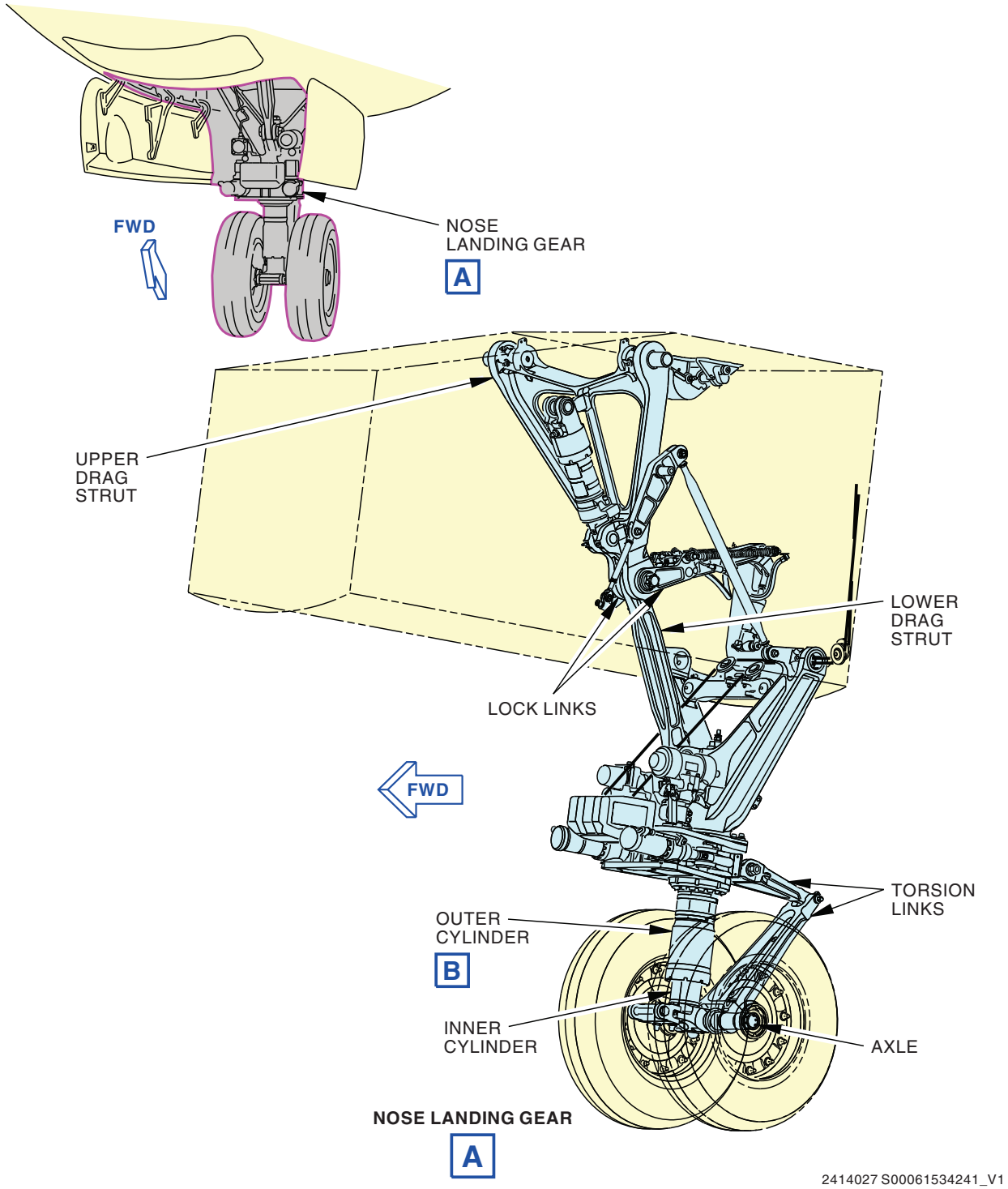
Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel
114AC	Fwd Nose Wheel Well Upper Access Panel
114AW	Forward Nose Wheel Well Panel
114BW	Forward Nose Wheel Well Panel

(a) Do this task: Nose Wheel Well Access Panels Installation, TASK 53-14-01-420-801.

(b) Do this task: Cabin Pressure Leak Test, TASK 05-51-91-790-801.

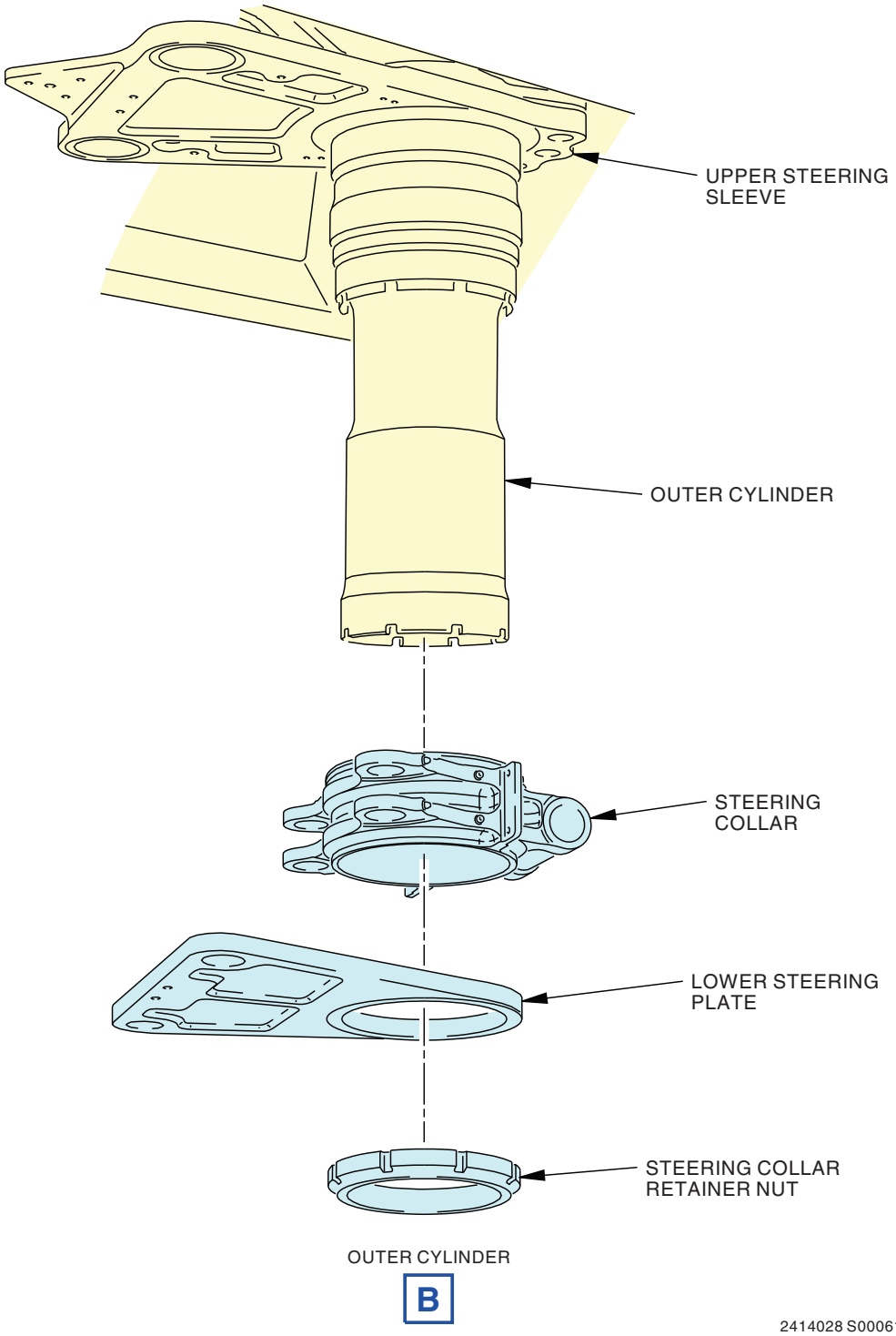
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Landing Gear Assembly
Figure 603/32-05-03-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Landing Gear Assembly
Figure 603/32-05-03-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1. General

A. This procedure has these tasks:

(1) Prepare to put the airplane in the air mode

(2) Put the airplane in the air mode

(3) Return the airplane to the ground mode

(4) Put the airplane systems back to their normal on ground condition.

TASK 32-09-00-840-801

2. Prepare to Put the Airplane in the Air Mode

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare to Put the Airplane in the Air Mode

SUBTASK 32-09-00-420-001

CAUTION

DO NOT START ENGINES WITH THE AIRPLANE IN THE AIR MODE. IF YOU START THE ENGINES WITH THE AIRPLANE IN THE AIR MODE, DAMAGE TO THE ENGINE COMPRESSORS CAN OCCUR.

(1) Do the following steps to make sure that the engines cannot be started:

(a) Do these steps to make sure that the Left and Right ENGINE START switches are not operated:

1) On the P5 overhead panel, make sure that both the Left and Right ENGINE START switches are in the OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Put the DO NOT OPERATE tag, STD-858, on both the Left and Right ENGINE START switches.

(b) Do these steps to make sure that the engine fuel control valve and spar valve stay in the closed position:

1) Make sure that the Left and Right START LEVERS are in the CUTOFF position.

2) Install the DO NOT OPERATE tag, STD-858, on the Left and Right START LEVERS.

SUBTASK 32-09-00-480-003

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(2) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-09-00-480-005

WARNING MAKE SURE LANDING GEAR WHEELS ARE CHOCKED, BEFORE PARKING BRAKE LEVER IS RELEASED. INJURIES TO PERSONNEL AND DAMAGE TO THE AIRCRAFT CAN OCCUR.

(3) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-09-00-040-012

(4) Make sure that the right stick shaker will not operate.

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	6	C01393	STICK SHAKER RIGHT

SUBTASK 32-09-00-040-013

(5) Make sure that the weather radar will not operate.

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

SUBTASK 32-09-00-040-014

(6) Make sure that the Aircraft Communications Addressing And Reporting System (ACARS) will not operate.

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC

ECCN 9E991 BOEING PROPRIETARY - See title page for details

This circuit breaker is inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	9	C01500	CMU/ACARS DC (INOP)

SUBTASK 32-09-00-040-022

WARNING

AVOID THE AUTOMATIC ACTIVATION OF THE PTU WHEN AIRPLANE IS IN AIR MODE, HYDRAULIC SYSTEM B EDP LOW PRESSURE, AND TRAILING EDGE FLAP NOT UP OR BETWEEN UP AND 15. AN UNINTENDED PRESSURIZATION OF HYDRAULIC SYSTEM B BY THE PTU CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(7) Make sure that the hydraulic power transfer unit will not operate.

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	15	C01081	HYDRAULIC SYSTEM PTU VALVE CONT 1
A	16	C01085	HYDRAULIC SYSTEM PTU VALVE CONT 2

SUBTASK 32-09-00-040-023

(8) Make sure that the take off warning horn will not operate.

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN

SUBTASK 32-09-00-040-024

(9) Make sure that the left stick shaker will not operate.

(a) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	4	C01392	STICK SHAKER LEFT

SUBTASK 32-09-00-860-012

(10) Make sure that the pitot probe, temperature probe, and Angle of Attack (AOA) vane heaters will not operate.

(a) Set the PROBE HEAT switches to the AUTO position.

NOTE: If the PROBE HEAT switches are set to the AUTO position, the air data heat system will only come on if an engine is started.

NOTE: If an engine is started or the PROBE HEAT switches are set to ON, it is necessary to open the circuit breakers.

1) Install a DO NOT OPERATE tag, STD-858, on the switches.

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(b) Open these circuit breakers and install safety tags if an engine is started or the PROBE HEAT switches are set to ON.

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	1	C00523	HEATERS CAPT PITOT
C	2	C00238	HEATERS TEMP PROBE
C	3	C01072	HEATERS ALPHA VANE LEFT
C	4	C00236	HEATERS ELEV PITOT LEFT
D	3	C01071	HEATERS ALPHA VANE RIGHT
D	4	C00237	HEATERS ELEV PITOT RIGHT
D	5	C00525	HEATERS F/O PITOT
D	6	C00524	HEATERS AUX PITOT

SUBTASK 32-09-00-040-025

(11) Make sure that the leading edge autoslat system will not operate.

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	14	C01070	FLIGHT CONTROL AUTOSLAT DC 2
C	14	C01068	FLIGHT CONTROL AUTOSLAT DC 1

SUBTASK 32-09-00-040-015

(12) Make sure that the drain masts do not get too hot:

NOTE: If you will put the airplane in the air mode for less than a minute, you do not need to de-activate the drain mast.

(a) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	4	C00700	HEATERS DRAIN MAST AIR

SUBTASK 32-09-00-040-004

(13) Make sure that the door area heaters will not operate.

(a) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	11	C01274	AIR CONDITIONING DOOR AREA HEAT CONT

SUBTASK 32-09-00-010-001

(14) To get access to the P91 and P92 panels, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-09-00-040-021

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(15) Make sure that the door area heaters will not operate.

(a) Open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	7	C01279	DOOR AREA HTR-AFT
D	9	C01280	DOOR AREA HTR-FWD

SUBTASK 32-09-00-040-006

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(16) Make sure that the standby hydraulic pump will not operate.

(a) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
F	2	C01449	STANDBY HYDRAULIC PUMP

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SUBTASK 32-09-00-840-001

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(17) Make sure that persons and equipment are clear of flight control surfaces.

SUBTASK 32-09-00-840-005

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE RAM INLET DOOR. THIS DEFLECTOR DOOR MOVES WHEN THE AIRPLANE IS SWITCHED TO AIR MODE. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(18) Make sure that persons and equipment are clear of the ram inlet door.

SUBTASK 32-09-00-040-008

WARNING DO THE DEACTIVATION PROCEDURE TO PREVENT THE OPERATION OF THE THRUST REVERSER. THE ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(19) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

——— END OF TASK ———

TASK 32-09-00-860-801

3. Put the Airplane in the Air Mode

(Figure 201)

A. General

(1) This task has three options to put the airplane in the air mode.

NOTE: Using the MAX Display System to simulate air mode is the recommended method. Using deactuators or jacking the airplane are the alternate methods. The use of deactuators or jacking the airplane will cause nuisance faults.

(a) You can put the airplane in the air mode with MAX Display System (MDS) (Recommended Method).

(b) You can also put deactuators on the nose and main landing gear to put the airplane in air mode (Alternate Method).

NOTE: A total of six (6) deactuators are needed.

(c) You can also lift the airplane on jacks to put the airplane in air mode (Alternate Method).

B. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

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Reference	Description
SPL-1690	Actuators/Deactuators Set - Proximity Sensor Part #: 8-758-01 Supplier: 08748 Part #: A27092-139 Supplier: 81205 Part #: A27092-146 Supplier: 81205 Opt Part #: A27092-106 Supplier: 81205

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left
212	Flight Compartment - Right
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Prepare to Put the Airplane in the Air Mode

SUBTASK 32-09-00-860-001

WARNING OBEY THE PROCEDURE THAT PUTS THE AIRPLANE IN THE AIR MODE. IF YOU DO THE PROCEDURE INCORRECTLY, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do this task: Prepare to Put the Airplane in the Air Mode, TASK 32-09-00-840-801.

SUBTASK 32-09-00-860-013

(2) Access the MDS to enable AIR MODE.

(a) Set the GND TESTS switch, on the maintenance control panel, to ENABLE position.

(b) Select SYS on the Multi-Function Display Panel (MFP).

(c) Select MAINT -> ONBD MAINT -> OTHER FUNCTIONS -> SPECIAL FUNCTIONS -> ATA System 32: Proximity Sensor Electronics Unit -> CONTINUE -> SELECT.

(d) Select PSEU -> AIR/GND OVERRIDE and then select CONTINUE.

(e) Select CONTINUE.

(f) Select START FUNCTION.

F. Put the Airplane in the Air Mode With the MAX Display System (Recommended Method)

SUBTASK 32-09-00-860-002

(1) Access the MDS to enable AIR MODE.

(a) Set SYS 1 and SYS 2 IN AIR and then select CONTINUE.

(b) Make sure that the Multi-Function Display indicates AIR/GND OVERRIDE is ACTIVE and both SYS 1 and SYS 2 are in AIR mode.

(c) Select STOP FUNCTION.

NOTE: The airplane should be now in air mode.

G. Put the Airplane in the Air Mode With Deactuators (Alternate Method)

SUBTASK 32-09-00-860-003

NOTE: This is not the recommended method to simulate air mode. This method will induce nuisance faults.

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(1) Attach deactuators to the face of the air/ground sensors for the nose and main landing gear.

NOTE: Use the deactuator plate that is provided with the proximity sensor test set, SPL-1690, to put the airplane in the air mode.

NOTE: As an option, deactuator plate for rectangular sensor can be made from 5052, 6061, 7075 or 2024 aluminum with a dimension of 1.5 in. (38 mm) x 0.75 in. (19 mm) x 0.05 in. (1.27 mm). Deactuator plate for round sensor can be made from copper sheet with a dimension of 1.0 in. (25 mm) x 1.0 in. (25 mm) x 0.025 in. (0.64 mm).

SUBTASK 32-09-00-860-004

(2) Make sure that the airplane is in the air mode.

(a) Use the MDS to access the Proximity Sensor Electronics Unit (PSEU) Maintenance Page 1.

(b) Make sure that AIR is shown for PSEU SYS 1 and SYS 2.

NOTE: When the AIR/GND override is used, “SIM” for simulated will appear next to the AIR/GND state.

(c) Select STOP FUNCTION.

H. Put the Airplane in the Air Mode by Lifting the Airplane on Jacks (Alternate Method)

SUBTASK 32-09-00-860-005

CAUTION

MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION

MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

NOTE: This is not the recommended method to simulate air mode. This method will induce nuisance faults.

(1) Lift the airplane on jacks (TASK 07-10-01-580-801).

SUBTASK 32-09-00-860-006

(2) Make sure that the airplane is in the air mode.

(a) Use the MDS to access the PSEU Maintenance Page 1.

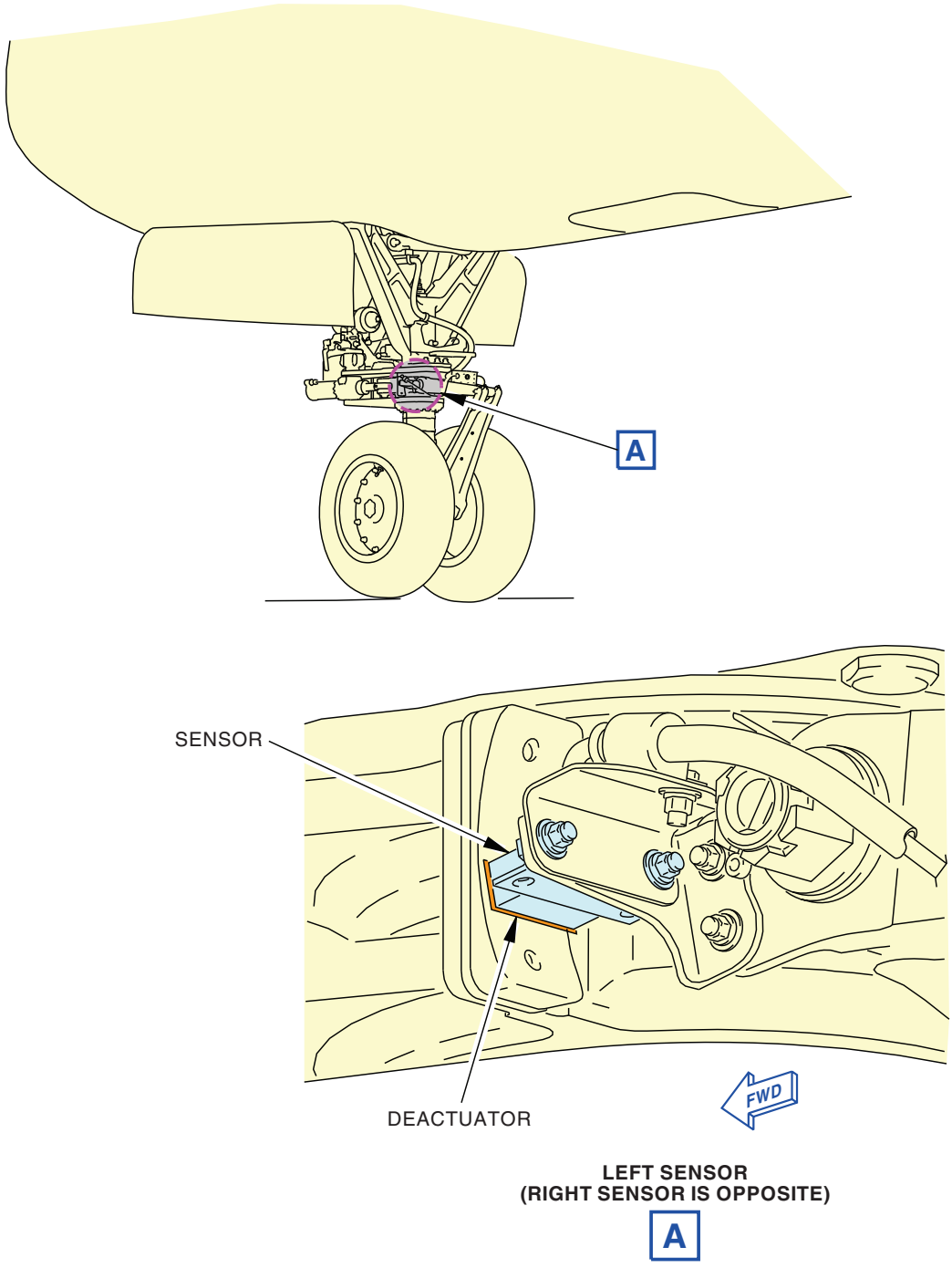
(b) Make sure that AIR is shown for PSEU SYS 1 and SYS 2.

NOTE: When the AIR/GND override is used, “SIM” for simulated will appear next to the AIR/GND state.

(c) Select STOP FUNCTION.

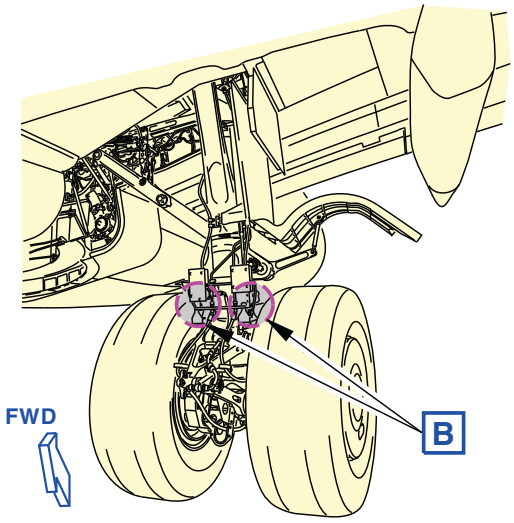
END OF TASK

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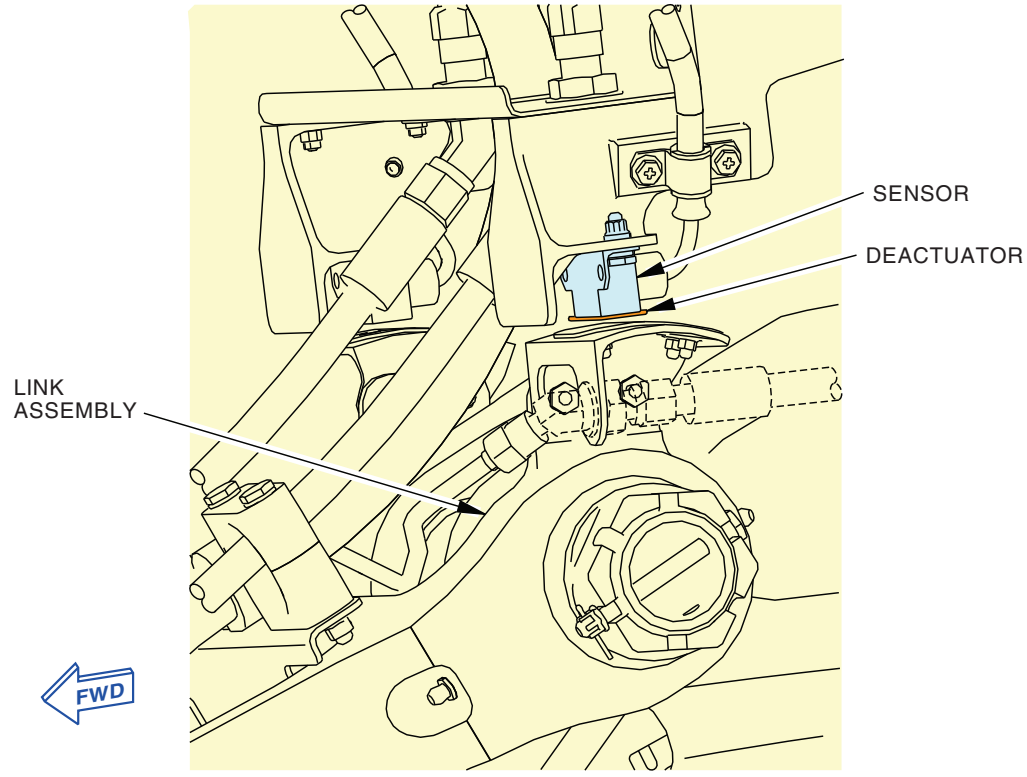


Air Mode Simulation with Deactuators
Figure 201/32-09-00-990-802 (Sheet 1 of 2)

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LEFT MAIN LANDING GEAR
(RIGHT MAIN LANDING GEAR IS OPPOSITE)



(EXAMPLE)

B

2414031 S00061534250_V3

Air Mode Simulation with Deactuators
Figure 201/32-09-00-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-09-00-860-802

4. Return the Airplane to the Ground Mode

(Figure 201)

A. References

Reference	Title
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)

B. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left
212	Flight Compartment - Right
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

C. Put the Airplane in Ground Mode

SUBTASK 32-09-00-860-014

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE RAM INLET DOOR. THIS DEFLECTOR DOOR MOVES WHEN THE AIRPLANE IS SWITCHED TO AIR MODE. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Make sure that persons and equipment are clear of the RAM inlet door.

SUBTASK 32-09-00-860-016

(2) Make sure that the Ground Test Enable switch on the Maintenance BITE Panel is in the ENABLE position.

SUBTASK 32-09-00-860-007

(3) If the airplane was put in the air mode with the MAX Display System (MDS), do these steps:

(a) Return to the special function page on the MDS:

1) Select MAINT -> ONBD MAINT -> OTHER FUNCTIONS -> SPECIAL FUNCTIONS -> ATA System 32: Proximity Sensor Electronics Unit -> CONTINUE -> SELECT.

2) Select PSEU -> AIR/GND OVERRIDE and then select CONTINUE.

3) Select CONTINUE.

(b) Select START FUNCTION.

(c) Select RESET OVERRIDE and then select CONTINUE.

(d) Make sure that the Multi-Function Display indicates AIR/GND OVERRIDE is NOT ACTIVE and both SYS 1 and SYS 2 are in GND mode.

(e) Select STOP FUNCTION.

SUBTASK 32-09-00-860-008

(4) If the airplane was put in the air mode with deactuators, remove the deactuators from the nose and main landing gear.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-09-00-860-009

CAUTION: MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION: MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(5) If the airplane was put in the air mode by jacking the airplane, do this task: Lower the Airplane off Jacks, TASK 07-10-01-580-802.

SUBTASK 32-09-00-860-015

(6) Move the Ground Test Enable switch to the NORM position.

SUBTASK 32-09-00-840-002

(7) Do this task: Return the Airplane Systems Back to Their Normal on Ground Condition, TASK 32-09-00-840-802.

——— END OF TASK ———

TASK 32-09-00-840-802

5. Return the Airplane Systems Back to Their Normal on Ground Condition

A. References

Reference	Title
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

B. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Put the Airplane Systems Back to Their Normal on Ground Condition

SUBTASK 32-09-00-440-007

(1) Return the right stick shaker to its original condition.

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	6	C01393	STICK SHAKER RIGHT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-09-00-440-008

(2) Return the weather radar to its original condition.

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

SUBTASK 32-09-00-440-009

(3) Return Aircraft Communications Addressing And Reporting System (ACARS) to its original condition.

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC

This circuit breaker is inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	9	C01500	CMU/ACARS DC (INOP)

SUBTASK 32-09-00-440-010

(4) Return the hydraulic power transfer unit to it's original condition.

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	15	C01081	HYDRAULIC SYSTEM PTU VALVE CONT 1
A	16	C01085	HYDRAULIC SYSTEM PTU VALVE CONT 2

SUBTASK 32-09-00-440-001

(5) Return the take off warning system to its original condition.

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN

SUBTASK 32-09-00-440-002

(6) Return the left stick shaker to its original condition.

(a) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	4	C01392	STICK SHAKER LEFT

SUBTASK 32-09-00-040-020

(7) Return the pitot probe, temperature probe, and Angle of Attack (AOA) vane heaters to their original condition.

(a) If necessary, remove the DO NOT OPERATE tag from the PROBE HEAT switches.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) If necessary, remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	1	C00523	HEATERS CAPT PITOT
C	2	C00238	HEATERS TEMP PROBE
C	3	C01072	HEATERS ALPHA VANE LEFT
C	4	C00236	HEATERS ELEV PITOT LEFT
D	3	C01071	HEATERS ALPHA VANE RIGHT
D	4	C00237	HEATERS ELEV PITOT RIGHT
D	5	C00525	HEATERS F/O PITOT
D	6	C00524	HEATERS AUX PITOT

SUBTASK 32-09-00-440-015

(8) Return the leading edge autoslat system to its original condition.

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	14	C01070	FLIGHT CONTROL AUTOSLAT DC 2
C	14	C01068	FLIGHT CONTROL AUTOSLAT DC 1

SUBTASK 32-09-00-440-011

(9) Return the drain mast heat to the normal condition.

(a) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	4	C00700	HEATERS DRAIN MAST AIR

SUBTASK 32-09-00-440-013

(10) Return the door area heaters to their original condition.

(a) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	11	C01274	AIR CONDITIONING DOOR AREA HEAT CONT

SUBTASK 32-09-00-440-014

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(11) Return the door area heaters to their original condition.

(a) Remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	7	C01279	DOOR AREA HTR-AFT
D	9	C01280	DOOR AREA HTR-FWD

SUBTASK 32-09-00-840-003

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(12) Return the standby hydraulic pump to the normal condition.

(a) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
F	2	C01449	STANDBY HYDRAULIC PUMP

SUBTASK 32-09-00-410-001

(13) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 32-09-00-440-006

(14) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 32-09-00-020-002

(15) Remove the DO NOT OPERATE tag from the Left and Right ENGINE START switches.

SUBTASK 32-09-00-020-003

(16) Remove the DO NOT OPERATE tag from the Left and Right START LEVERS.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN LANDING GEAR AIR/GROUND SENSOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the main landing gear air/ground sensor

(2) Installation of the main landing gear air/ground sensor.

B. The procedures for the left and right installations are similar.

TASK 32-09-01-020-801

2. Main Landing Gear Air/Ground Sensor Removal

(Figure 401)

A. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-840-801	Prepare to Put the Airplane in the Air Mode (P/B 201)

B. Tools/Equipment

Reference	Description
STD-764	Scraper - Non-metallic

C. Location Zones

Zone	Area
731	Left Main Landing Gear - Outboard Door
741	Right Main Landing Gear - Outboard Door

D. Prepare for the Removal

SUBTASK 32-09-01-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-09-01-840-001

WARNING OBEY THE PROCEDURE THAT PUTS THE AIRPLANE IN THE AIR MODE. IF YOU DO THE PROCEDURE INCORRECTLY, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do this task: Prepare to Put the Airplane in the Air Mode, TASK 32-09-00-840-801.

SUBTASK 32-09-01-860-001

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

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E. Main Landing Gear Air/Ground Sensor Removal

SUBTASK 32-09-01-210-001

(1) Inspect the sensor [4] and target assembly for damage or other evidence that the sensor clearance may be out of tolerance.

(a) Write a note to check the sensor clearance if there is evidence that the sensor clearance may be out of tolerance.

SUBTASK 32-09-01-020-001

(2) Disconnect the sensor wires in the junction box.

(a) Remove and discard the lockwire [22] from the screws [7].

(b) Remove the screws [7] and washers [5] securing the junction box cover [6].

(c) Remove the junction box cover [6].

(d) Remove the sensor wire contacts from the terminal block in the junction box.

(e) Attach a string to the sensor wires.

NOTE: This string will be used to pull the wires of the new sensor through the conduit. The string can be tied to the landing gear structure to ensure that the string is not pulled through the conduit.

SUBTASK 32-09-01-020-002

(3) Remove the sensor [4].

(a) Remove the nuts [1], washers [2], washers [8], and bolts [3] which secure the sensor [4] to the mounting bracket.

(b) Remove the sensor [4] from the bracket.

1) Use a non-metallic scraper, STD-764, to “lift” the sensor off the mounting bracket.

NOTE: Any damage to the paint must be repaired prior to installing the new sensor.

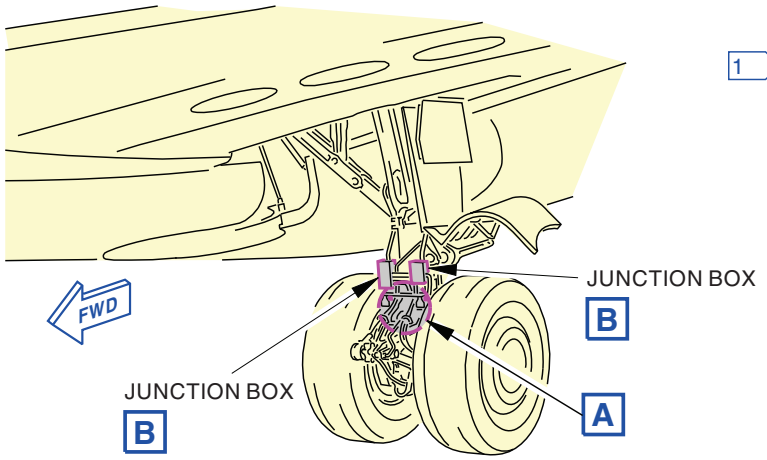
(c) Pull the sensor wires out of the conduit.

1) Make sure that you do not pull the string through the conduit.

(d) Remove the string from the sensor wires.

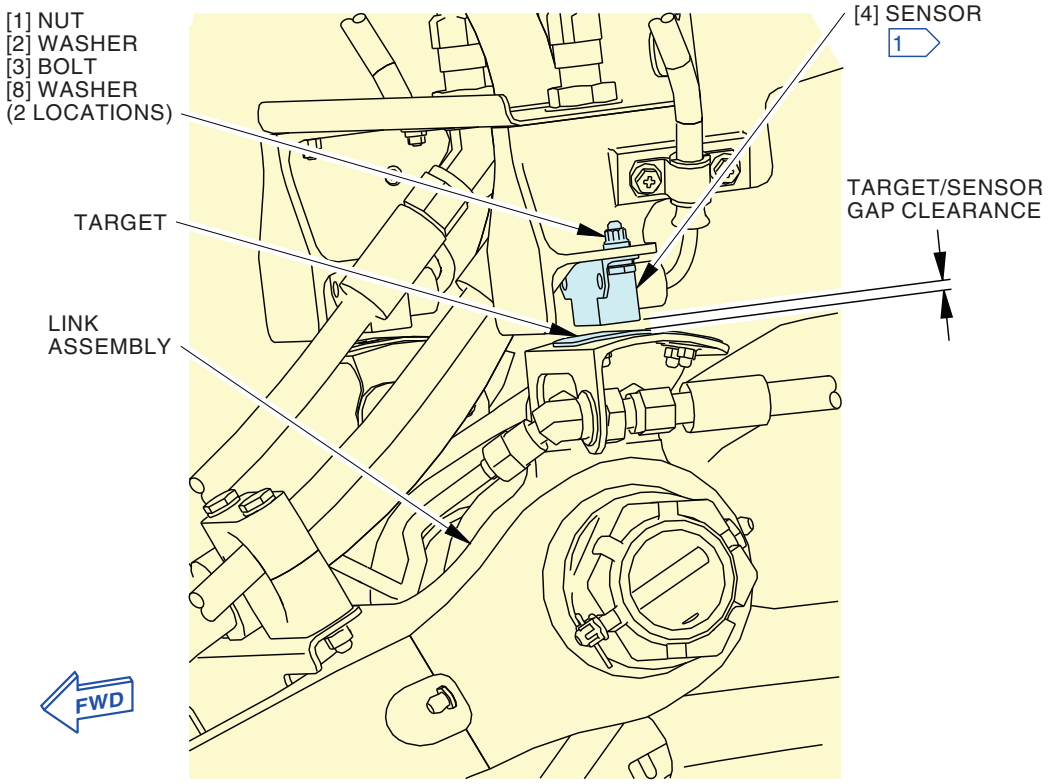
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ITEM	SENSOR NO.	MLG	LOCATION
[4]	S1010	R	OUTBD
[4]	S1011	R	INBD
[4]	S1012	L	OUTBD
[4]	S1013	L	INBD

LEFT MAIN LANDING GEAR
(RIGHT MAIN LANDING GEAR IS OPPOSITE)



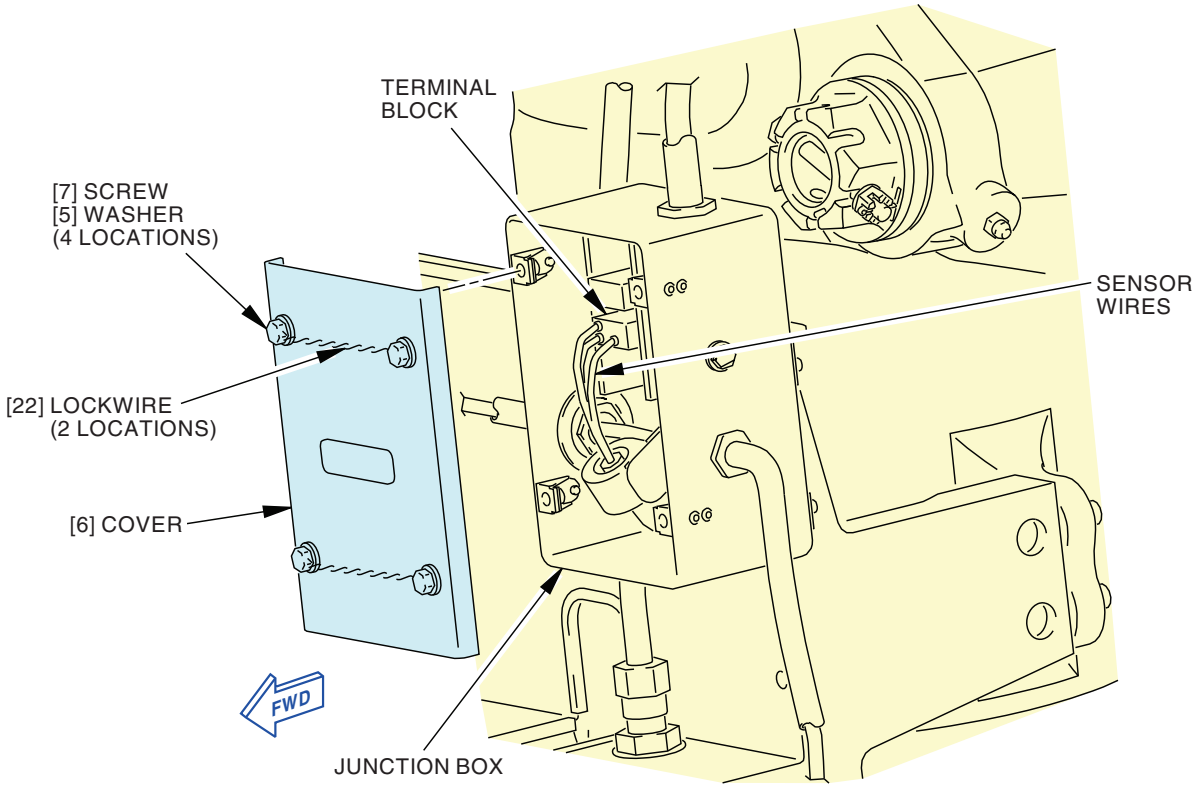
(EXAMPLE)

A

2414033 S00061534257_V3

Main Landing Gear Air/Ground Sensor Installation
Figure 401/32-09-01-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



OUTBOARD JUNCTION BOX
(INBOARD JUNCTION BOX IS EQUIVALENT)
(EXAMPLE)

B

2414034 S00061534258_V2

Main Landing Gear Air/Ground Sensor Installation
Figure 401/32-09-01-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-09-01-400-801

3. Main Landing Gear Air/Ground Sensor Installation

(Figure 401)

A. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-840-801	Prepare to Put the Airplane in the Air Mode (P/B 201)
32-09-00-840-802	Return the Airplane Systems Back to Their Normal on Ground Condition (P/B 201)
32-09-01-400-802	Main Landing Gear Air/Ground Sensor Clearance Measurement (P/B 501)
32-09-01-710-801	Main Landing Gear Air/Ground Sensor Operational Test (P/B 501)
51-21-99-300-802	Decorative Exterior Paint System Repair (P/B 701)
51-22-00-390-808	Fastener Seal Application (P/B 201)

B. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Sensor	32-09-01-01-015	PTF ALL
		32-09-01-01-215	PTF ALL

D. Location Zones

Zone	Area
731	Left Main Landing Gear - Outboard Door
741	Right Main Landing Gear - Outboard Door

E. Prepare for the Installation

SUBTASK 32-09-01-480-002

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-09-01-840-002

WARNING OBEY THE PROCEDURE THAT PUTS THE AIRPLANE IN THE AIR MODE. IF YOU DO THE PROCEDURE INCORRECTLY, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) If the airplane systems are not prepared for air mode simulation, do this task: Prepare to Put the Airplane in the Air Mode, TASK 32-09-00-840-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Main Landing Gear Air/Ground Sensor Installation

SUBTASK 32-09-01-370-001

(1) Make sure that the Sensor bracket paint is not scratched.

(a) Repair all damage to the finish (TASK 51-21-99-300-802).

SUBTASK 32-09-01-350-001

(2) Cut the sensor wires to the correct length:

(a) You can use the sensor that was removed to find the wire length on the replacement sensor.

SUBTASK 32-09-01-420-001

(3) Attach the contacts to the sensor wires.

SUBTASK 32-09-01-410-001

(4) Fay seal the bottom of the sensor [4] with sealant, A00247.

SUBTASK 32-09-01-420-002

(5) Attach the sensor [4] to the mounting bracket.

(a) Put the sensor [4] in position on the bracket.

(b) Wet install the bolts [3], nuts [1], washers [2], and washers [8] with sealant, A00247 (TASK 51-22-00-390-808, Procedure 3 for wet install process).

(c) Install the bolts [3], washers [2], washers [8], and nuts [1].

NOTE: Use CRES washers below bolt heads with titanium sensors. Use cad-plated washers below nuts. CRES washers are optional below bolt heads with stainless steel sensors.

SUBTASK 32-09-01-410-002

(6) Attach the sensor wires to the airplane.

(a) Attach the string to the sensor wires.

(b) Pull the wires through the conduit to the junction box.

(c) Remove the string from the sensor wires

(d) Install the sensor contacts into the terminal block.

1) Red wire goes into position A.

2) Yellow wire goes into position E.

3) Blue wire goes into position C.

SUBTASK 32-09-01-410-003

(7) Instal the junction box cover [6].

(a) Make sure that the junction box cover [6] is in satisfactory condition.

1) If necessary, replace the junction box cover [6].

(b) Install the washers [5] and screws [7] to attach the junction box cover [6].

(c) Install the lockwire [22] (MS20995C32 lockwire, G01048) between the screws [7].

G. Main Landing Gear Air/Ground Sensor Test

SUBTASK 32-09-01-200-001

(1) Do a post installation check of the sensor [4]:

(a) Do this task: Main Landing Gear Air/Ground Sensor Clearance Measurement, TASK 32-09-01-400-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

(c) Do this task: Return the Airplane Systems Back to Their Normal on Ground Condition, TASK 32-09-00-840-802.

(d) Do this task: Main Landing Gear Air/Ground Sensor Operational Test, TASK 32-09-01-710-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN LANDING GEAR AIR/GROUND SENSOR - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) Measurement of the main landing gear air/ground sensor clearance

(2) Adjustment of the main landing gear air/ground sensor clearance

(3) Operational test of the main landing gear air/ground sensor

B. The procedures for the left and right adjustment/tests are similar.

TASK 32-09-01-400-802

2. Main Landing Gear Air/Ground Sensor Clearance Measurement

(Figure 501)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
731	Left Main Landing Gear - Outboard Door
741	Right Main Landing Gear - Outboard Door

C. Prepare for the Measurement

SUBTASK 32-09-01-480-003

(1) Make sure that chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-09-01-480-004

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) If the downlock pins are not installed on the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

D. Main Landing Gear Air/Ground Sensor Clearance Measurement

SUBTASK 32-09-01-220-001

(1) Measure the clearance between the sensor [1] and target [6] with a feeler gage.

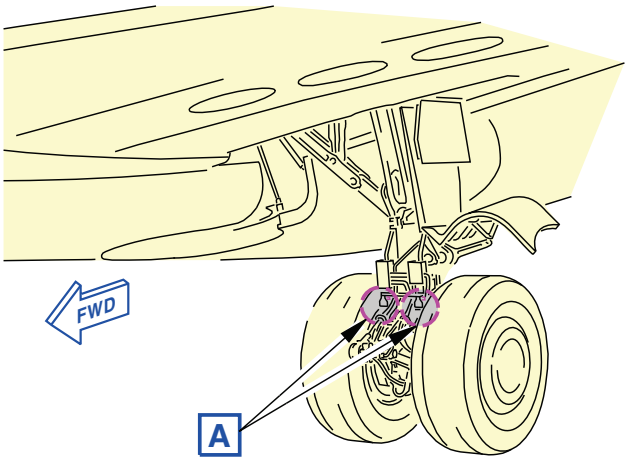
(a) Make a record of the clearance dimension.

SUBTASK 32-09-01-820-001

(2) If the sensor clearance is outside the tolerance shown in Figure 501, do this task: Main Landing Gear Air/Ground Sensor Clearance Adjustment, TASK 32-09-01-820-801.

END OF TASK

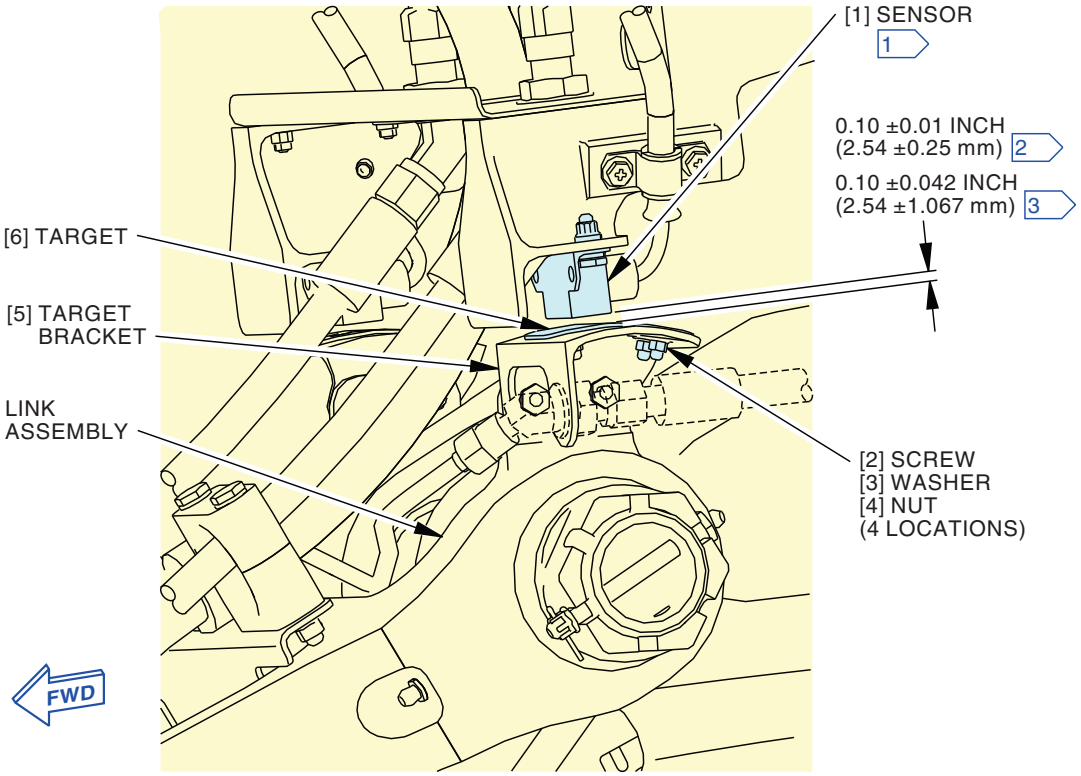
ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 SYSTEM NO.	SENSOR NO.	MLG	LOCATION
1	S1010	R	OUTBD
2	S1011	R	INBD
1	S1012	L	OUTBD
2	S1013	L	INBD

2 PHYSICAL GAP TOLERANCE
3 PSEU TOLERANCE

LEFT MAIN LANDING GEAR
(RIGHT MAIN LANDING GEAR IS OPPOSITE)



(EXAMPLE)

A

2414035 S00061534262_V6

Main Landing Gear Air/Ground Sensor Test
Figure 501/32-09-01-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-09-01-820-801

3. Main Landing Gear Air/Ground Sensor Clearance Adjustment

(Figure 501)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

C. Location Zones

Zone	Area
731	Left Main Landing Gear - Outboard Door
741	Right Main Landing Gear - Outboard Door

D. Prepare for the Adjustment

SUBTASK 32-09-01-480-005

(1) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-09-01-480-006

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Make sure the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

E. Main Landing Gear Air/Ground Sensor Clearance Adjustment

SUBTASK 32-09-01-020-003

(1) Use the target bracket [5] to adjust the sensor clearance.

(a) Loosen the attaching hardware on the target bracket [5].

(b) Adjust the target bracket [5] to get the correct clearance shown in Figure 501.

(c) If you can adjust the target bracket [5] to get the correct clearance, do the steps to re-tighten the bracket fasteners.

(d) If you cannot adjust the target bracket [5] to get the correct clearance, continue the adjustment procedure.

(e) Do these steps to re-tighten the bracket fasteners:

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WARNING USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

1) Apply a thin coat of corrosion inhibiting material, G50136, to the thread reliefs and threads of the nuts and the faces of the washers that secure the target bracket [5].
   a) Wipe off the excess of corrosion inhibiting material, G50136.
2) Tighten the target bracket [5] fasteners.

SUBTASK 32-09-01-020-004

(2) Do these steps to remove the main landing gear air/ground sensor [1]:
   (a) Remove the attaching hardware from the sensor [1].
   (b) Move the sensor [1] out of the way.

SUBTASK 32-09-01-020-005

(3) Do these steps to remove the target [6]:
   (a) Remove the nuts [4], washers [3], and screws [2].
   (b) Remove the target [6].

SUBTASK 32-09-01-110-001

(4) Clean the mounting surfaces.

SUBTASK 32-09-01-420-003

(5) Do these steps to secure the target [6]:

WARNING DO NOT GET PRIMER AND SEALANT IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM PRIMER AND SEALANT. PUT ON GOGGLES, AND GLOVES WHEN YOU USE PRIMER AND SEALANT. KEEP PRIMER AND SEALANT AWAY FROM SPARKS, FLAME, AND HEAT. PRIMER AND SEALANT IS POISONOUS AND FLAMMABLE, AND CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

   (a) Apply a light coat of sealant, A00247, to the fay surface of the target [6].
   (b) Position the target [6] on the target bracket [5].

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

   (c) Apply a thin coat of corrosion inhibiting material, G50136, to the thread reliefs and threads of the nuts [4] and faces of the washers [3].
      1) Wipe off the excess of corrosion inhibiting material, G50136.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Install the screws [2], washers [3], and nuts [4].

SUBTASK 32-09-01-020-006

(6) Do these steps to re-install the main landing gear air/ground sensor [1]:

(a) Position the sensor [1] sensor on the link assembly.

WARNING

DO NOT GET PRIMER AND SEALANT IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM PRIMER AND SEALANT. PUT ON GOGGLES, AND GLOVES WHEN YOU USE PRIMER AND SEALANT. KEEP PRIMER AND SEALANT AWAY FROM SPARKS, FLAME, AND HEAT. PRIMER AND SEALANT IS POISONOUS AND FLAMMABLE, AND CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(b) Apply a thin coat of sealant, A00247, to the fay surfaces of the sensor [1] and to the thread reliefs, threads, and washer faces of the attaching hardware.

1) Wipe off the excess sealant, A00247.

(c) Secure the sensor [1] with the attaching hardware.

SUBTASK 32-09-01-220-002

(7) Measure the sensor clearance (TASK 32-09-01-400-802).

(a) If the sensor clearance is not correct, do the steps to adjust the sensor clearance again.

(b) If the sensor clearance is correct, check the operational functionality and values displayed on the maintenance data pages (TASK 32-09-01-710-801).

——— END OF TASK ———

TASK 32-09-01-710-801

4. Main Landing Gear Air/Ground Sensor Operational Test

(Figure 501)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) On Proximity Sensor Electronics Unit (PSEU) Maintenance Page 1, the STATE column shows the current condition (NEAR, FAR or MARG) of each sensor.

(a) Target-Near (or NEAR) refers to the condition where the sensor is considered actuated.

(b) Target-Far (or FAR) refers to the unactuated condition.

(c) Marginal (or MARG) refers to a target-near condition that is not within the acceptable range specified in Table 1, and re-rigging of the sensor/target installation should be performed at the next maintenance opportunity.

(2) On PSEU Maintenance Page 1, the current clearance value of each sensor is shown in the MILS column, if the target is near or marginal.

(a) If the target is far, the value will be shown as “---”.

(3) On PSEU Maintenance Page 1, the serviceable sensor clearance rigging range is shown in the RANGE column.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. **Location Zones**

Zone	Area
731	Left Main Landing Gear - Outboard Door
741	Right Main Landing Gear - Outboard Door

D. **Prepare for the Operational Test**

SUBTASK 32-09-01-480-007

(1) Make sure that chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-09-01-480-008

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) If the downlock pins are not installed on the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

E. **Main Landing Gear Air/Ground Sensor Operational Test**

SUBTASK 32-09-01-740-001

(1) Use the PSEU Maintenance Page to examine the adjustment of the sensor.

(a) Set the GND TESTS switch, on the P61-4 maintenance bite panel, to ENABLE position.

(b) Select SYS on the Multi-Function Display panel.

(c) Select MAINT -> MAINT DATA PGS -> ATA System 32: Proximity Sensor Electronics Unit -> DISPLAY.

(d) Make sure that the sensor state is NEAR.

SUBTASK 32-09-01-710-001

(2) Do the PSEU SELF-TEST.

(a) Set the GND TESTS, on the P61-4 maintenance bite panel, to ENABLE position.

(b) Select SYS on the Multi-Function Display panel.

(c) Select MAINT-> ONBD MAINT-> LINE MAINT -> GROUND TESTS -> ATA SYSTEM 32: Proximity Sensor Electronics Unit -> OPERATIONAL TEST.

(d) Select PSEU - SELF TEST and then select CONTINUE.

(e) Follow the instructions shown on the Multi-Function Display and select CONTINUE.

(f) Select START TEST.

1) Make sure that PASS shows on the MAX Display System (MDS).

(g) Set the GND TESTS, on the P61-4 maintenance bite panel, to NORM position.

——— **END OF TASK** ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE LANDING GEAR AIR/GROUND SENSOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Nose landing gear air/ground sensor removal

(2) Nose landing gear air/ground sensor installation.

B. The procedures for the left and right sensors are same.

TASK 32-09-02-000-801

2. Nose Landing Gear Air/Ground Sensor Removal

(Figure 401)

A. General

(1) This task is for the removal of the nose landing gear air/ground sensor.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-840-801	Prepare to Put the Airplane in the Air Mode (P/B 201)

C. Location Zones

Zone	Area
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

D. Prepare for the Removal

SUBTASK 32-09-02-480-011

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-09-02-840-001

WARNING OBEY THE PROCEDURE THAT PREPARES TO PUT THE AIRPLANE IN THE AIR MODE. IN THE AIR MODE, MANY OF THE AIRPLANE SYSTEMS CAN OPERATE. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do this task: Prepare to Put the Airplane in the Air Mode, TASK 32-09-00-840-801.

SUBTASK 32-09-02-860-001

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Nose Landing Gear Air/Ground Sensor Removal

SUBTASK 32-09-02-210-001

(1) Examine the sensor [1] and the target assembly as follows:

(a) Look for damage or an indication that the sensor clearance is possibly out of tolerance.

NOTE: Write a note to examine the sensor clearance if there is an indication that the clearance is not in limits.

SUBTASK 32-09-02-020-001

(2) Disconnect the connector from the airplane.

(a) Unpin the sensor wire contacts from the connector.

(b) Attach a string to the sensor wires.

NOTE: This string will be used to pull the wires of the new sensor through the conduit and junction box. You can attach the string to the landing gear structure. This will make sure that you do not pull the string through the conduit and the junction box.

SUBTASK 32-09-02-020-002

(3) Remove the sensor [1] as follows:

(a) Remove the nuts [2], washers [3], washers [8], and bolts [4] which secure the sensor [1] to the mounting bracket.

(b) Remove the sensor [1] from the bracket.

NOTE: Use a non-metallic scrapper to remove the sensor from the bracket. Make sure that you do not cause damage to the paint on the sensor bracket.

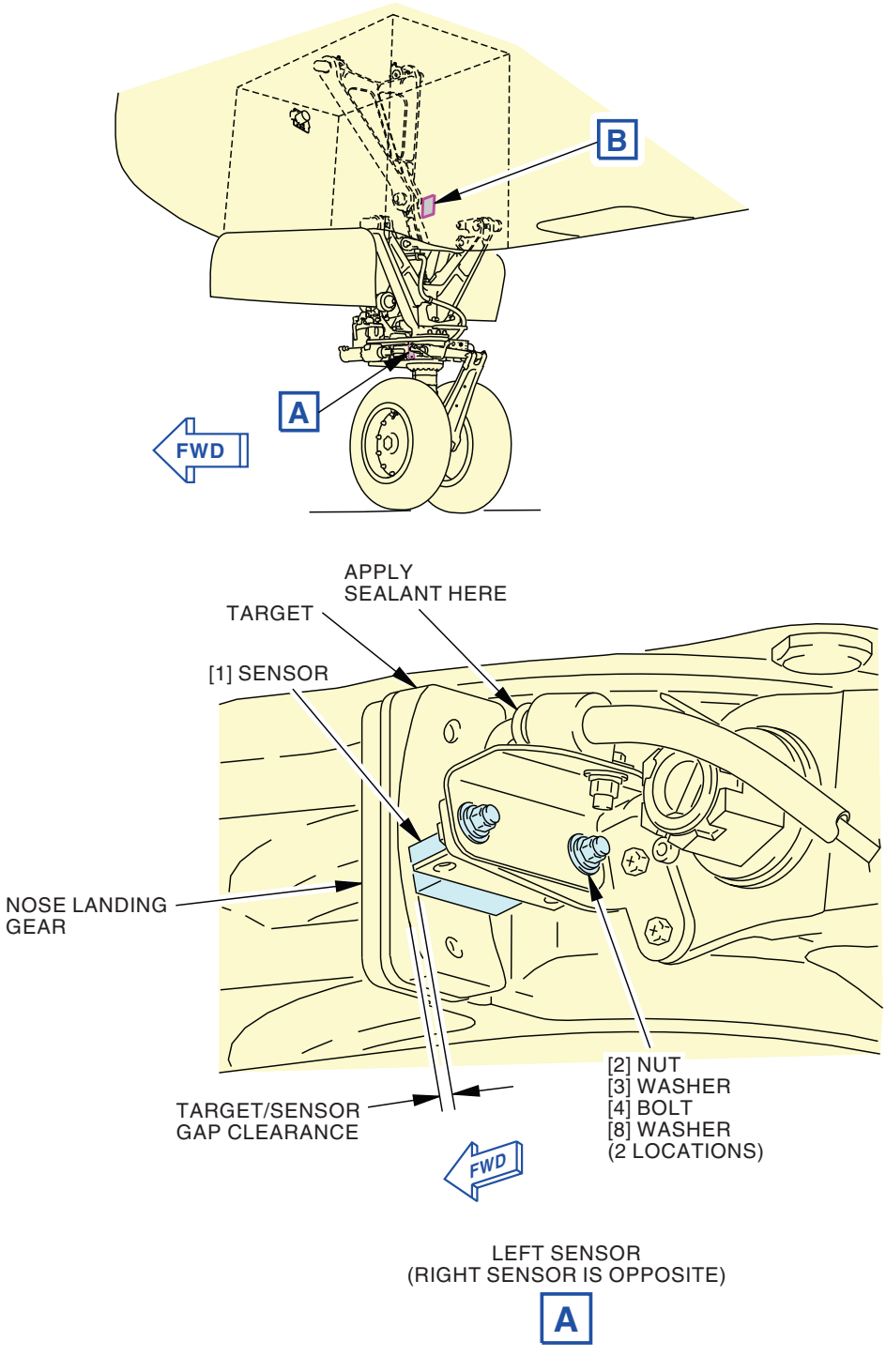
(c) Pull the sensor wires out of the conduit and junction box.

NOTE: Make sure that you do not pull the string through the conduit.

(d) Remove the string from the sensor wires.

——— END OF TASK ———

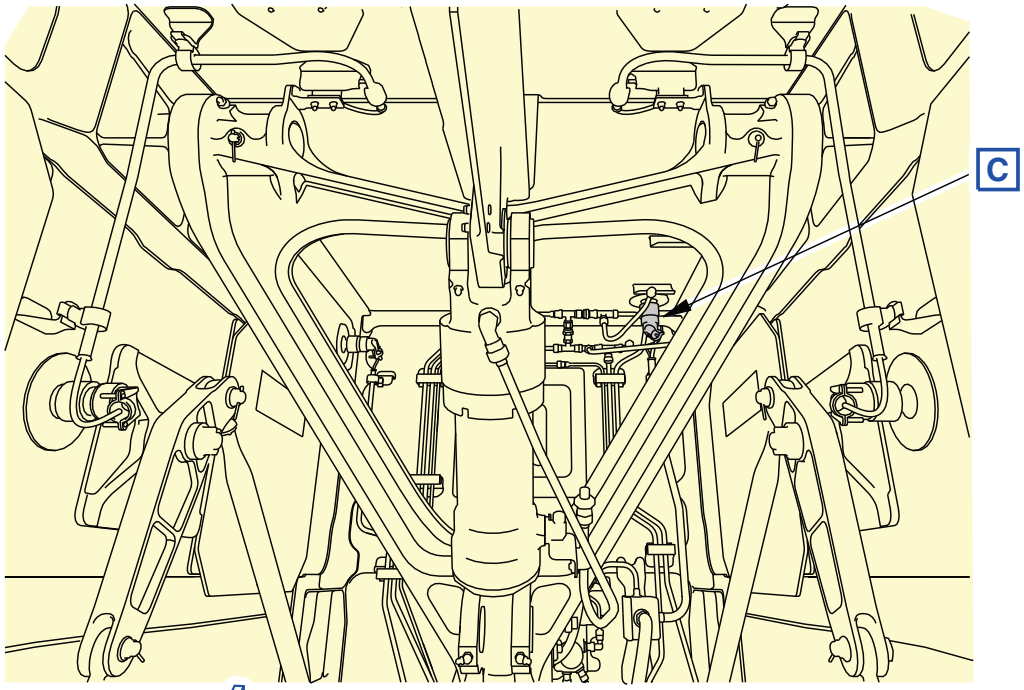
ECCN 9E991 BOEING PROPRIETARY - See title page for details

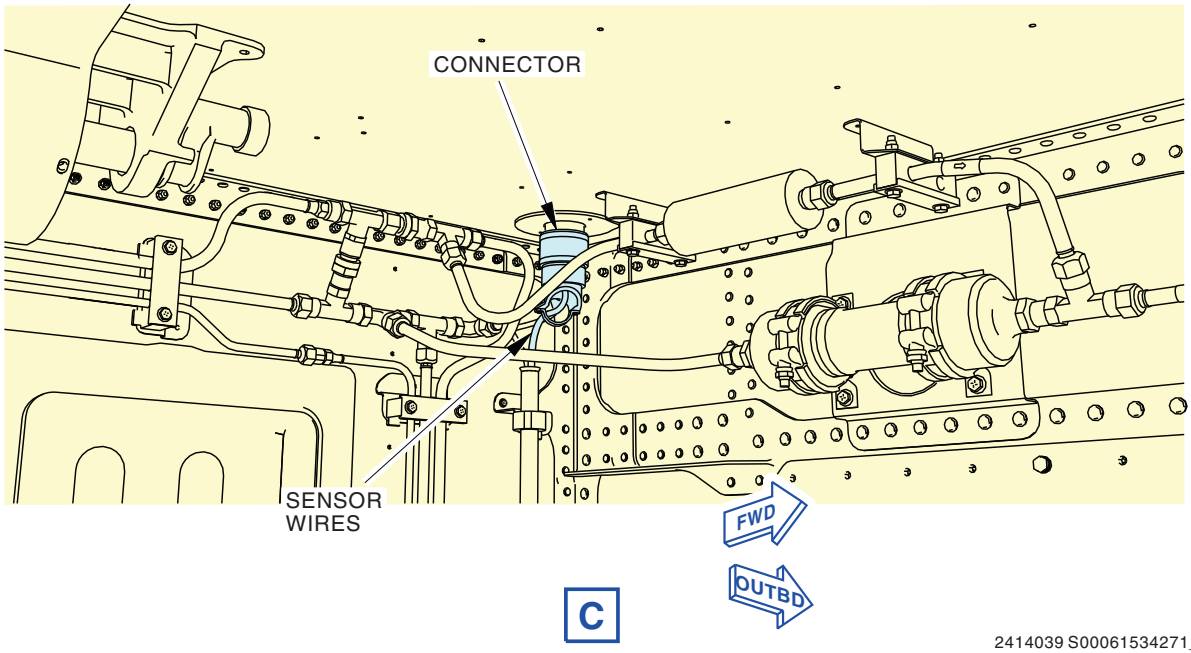


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Nose Landing Gear Air/Ground Sensor Installation
Figure 401/32-09-02-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





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Nose Landing Gear Air/Ground Sensor Installation
Figure 401/32-09-02-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-09-02-400-801

3. Nose Landing Gear Air/Ground Sensor Installation

(Figure 401)

A. General

(1) This task is for the installation of the nose landing gear air/ground sensor.

B. References

Reference	Title
32-09-00-840-802	Return the Airplane Systems Back to Their Normal on Ground Condition (P/B 201)
32-09-02-220-801	Nose Landing Gear Air/Ground Sensor Clearance Measurement (P/B 501)
32-09-02-710-801	Nose Landing Gear Air/Ground Sensor Operational Test (P/B 501)
51-22-00-390-808	Fastener Seal Application (P/B 201)

C. Consumable Materials

Reference	Description	Specification
A00027	Adhesive - Silicone Rubber, 1 Part, RTV	BAC5010 Type 60
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Sensor	32-09-02-02-065	PTF ALL
		32-61-51-01-005	PTF ALL

E. Location Zones

Zone	Area
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

F. Nose Landing Gear Air/Ground Sensor Installation

SUBTASK 32-09-02-410-001

(1) Fay seal the bottom of the sensor [1] with sealant, A00247.

SUBTASK 32-09-02-420-001

(2) Attach the sensor [1] to the mounting bracket.

(a) Position the sensor [1] on the bracket.

(b) Apply wet sealant, A00247 to the two bolts [4], two washers [3], two washers [8], and the two nuts [2] (TASK 51-22-00-390-808).

(c) Install the sensor [1] with the bolts [4], washers [3], washers [8] and nuts [2].

NOTE: Use CRES washers below bolt heads with titanium sensors. Use cad-plated washers below nuts. CRES washers are optional below bolt heads with stainless steel sensors.

SUBTASK 32-09-02-970-001

(3) Do this task: Nose Landing Gear Air/Ground Sensor Clearance Measurement, TASK 32-09-02-220-801.

SUBTASK 32-09-02-350-001

(4) Do these steps to pull the sensor wires through the conduit:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Attach the string to the sensor wires.

(b) Pull the wires through the conduit and junction box.

(c) Remove the string from the sensor wires.

SUBTASK 32-09-02-420-002

(5) Re-pin the sensor wires.

SUBTASK 32-09-02-410-002

(6) Connect the connector to the airplane.

SUBTASK 32-09-02-390-001

(7) Seal the end of the conduit near the sensor with 1 in. (25.4 mm) of adhesive, A00027.

NOTE: Make sure that the sensor wiring does not touch the flared end of the conduit.

G. Put the Airplane Back to its Usual Condition

SUBTASK 32-09-02-860-003

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

SUBTASK 32-09-02-860-004

(2) Do this task: Return the Airplane Systems Back to Their Normal on Ground Condition, TASK 32-09-00-840-802.

H. Nose Landing Gear Air/Ground Sensor Installation Test

SUBTASK 32-09-02-710-002

(1) Do this task: Nose Landing Gear Air/Ground Sensor Operational Test, TASK 32-09-02-710-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE LANDING GEAR AIR/GROUND SENSOR - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) Nose landing gear air/ground sensor clearance measurement

(2) Nose landing gear air/ground sensor clearance adjustment

(3) Nose landing gear air/ground sensor operational test.

B. The adjustment/tests procedures for the left and right sensors are similar.

TASK 32-09-02-220-801

2. Nose Landing Gear Air/Ground Sensor Clearance Measurement

(Figure 501)

A. General

(1) This task is for the measurement of the clearance between the sensor and the target.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

Reference	Description
STD-371	Gauge - Feeler, 0.0 to 0.5 in (0.0 to 12.7 mm), Readable to 1/1000 in (.025 mm)

D. Location Zones

Zone	Area
713	Nose Landing Gear

E. Prepare for the Measurement

SUBTASK 32-09-02-480-009

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-09-02-480-010

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

F. Measure the Nose Landing Gear Air/Ground Sensor Clearance

SUBTASK 32-09-02-220-001

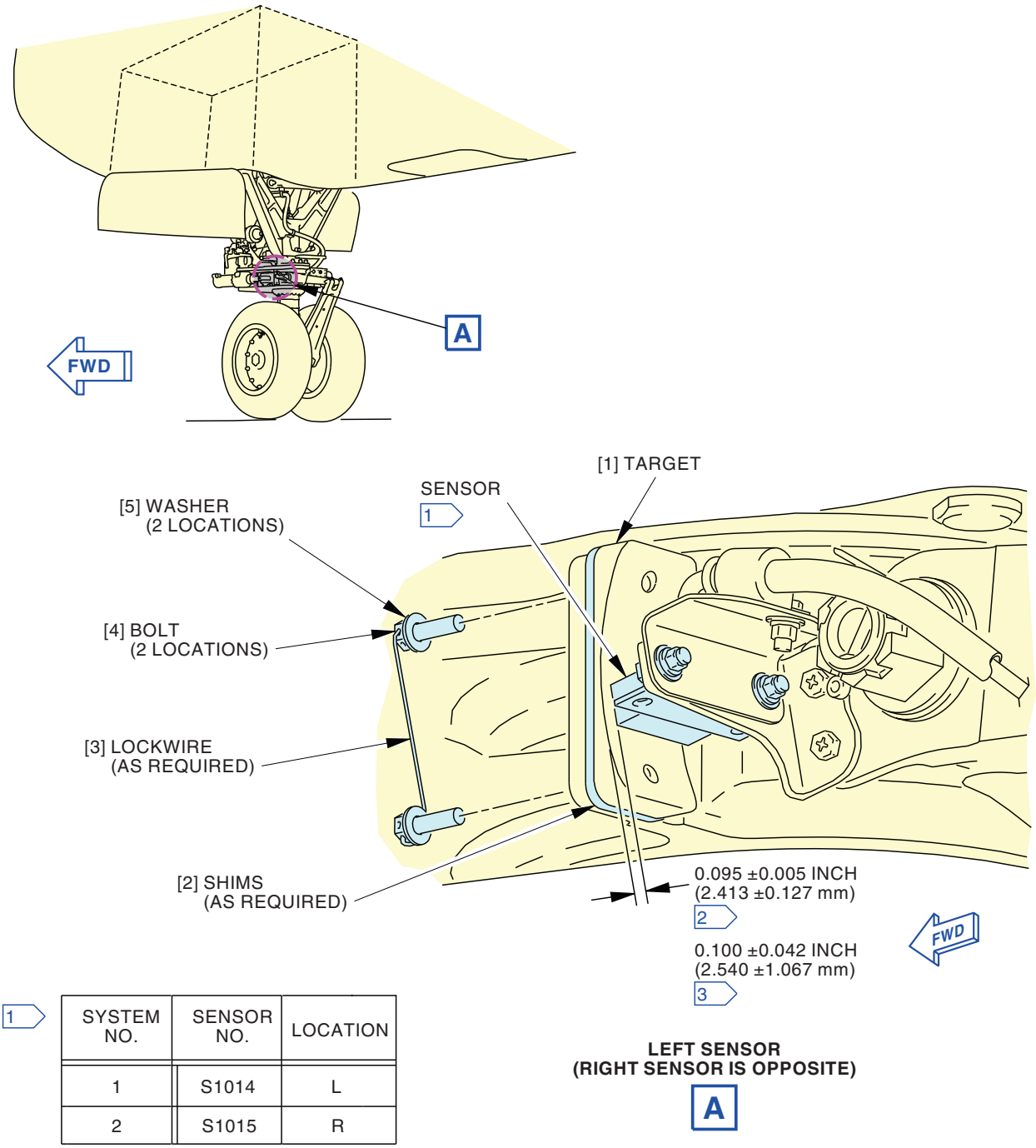
(1) Measure the clearance between the sensor and target with a 0.0 to 0.5 in (0.0 to 12.7 mm) feeler gauge, STD-371.

SUBTASK 32-09-02-820-001

(2) If the sensor clearance is outside the tolerance shown in Figure 501, do this task: Nose Landing Gear Air/Ground Sensor Clearance Adjustment, TASK 32-09-02-400-802.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1

SYSTEM NO.	SENSOR NO.	LOCATION
1	S1014	L
2	S1015	R

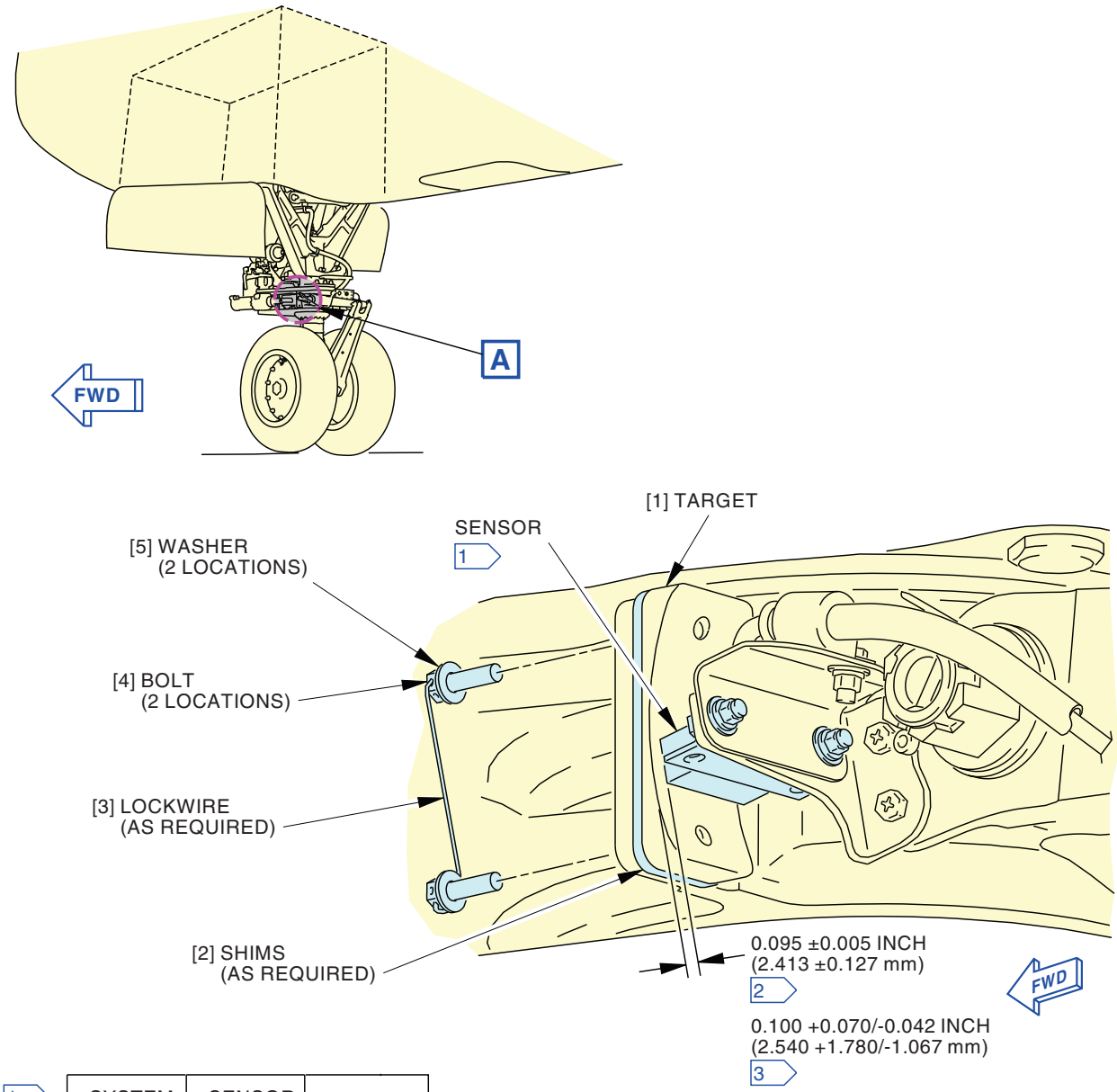
2 PHYSICAL GAP TOLERANCE

3 PSEU TOLERANCE

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Nose Landing Gear Air/Ground Sensor Clearance Measurement
Figure 501/32-09-02-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1

SYSTEM NO.	SENSOR NO.	LOCATION
1	S1014	L
2	S1015	R

2 PHYSICAL GAP TOLERANCE

3 PSEU TOLERANCE

2971986 S0000750780_V2

Nose Landing Gear Air/Ground Sensor Clearance Measurement
Figure 501/32-09-02-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-09-02-400-802

3. Nose Landing Gear Air/Ground Sensor Clearance Adjustment

(Figure 501)

A. General

(1) This task is for the adjustment of the gap clearance between the Nose Landing Gear (NLG) air/ground sensor and target.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

D. Location Zones

Zone	Area
713	Nose Landing Gear

E. Prepare for the Adjustment

SUBTASK 32-09-02-480-007

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-09-02-480-008

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

F. Nose Landing Gear Air/Ground Sensor Clearance Adjustment

SUBTASK 32-09-02-020-003

(1) Remove the lockwire [3], bolts [4], and washers [5].

SUBTASK 32-09-02-020-004

(2) Remove the target [1] and shim [2] or shims [2].

SUBTASK 32-09-02-110-001

(3) Clean the shim [2] or shims [2] and mounting surfaces.

SUBTASK 32-09-02-820-002

(4) Add a new shim [2] or remove laminations from the original shim [2] to get the necessary clearance (Figure 501).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-09-02-620-001

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(5) Apply a thin coat of corrosion inhibiting material, G50136, to the thread reliefs and threads of the bolts [4] and faces of the washers [5].

NOTE: Wipe off the excess material.

SUBTASK 32-09-02-420-003

(6) Install the target [1] as follows:

(a) Position the shim [2] or shims [2] and target [1] on the mounting bracket.

(b) Install the bolts [4] and washers [5].

SUBTASK 32-09-02-220-002

(7) Measure the sensor clearance (TASK 32-09-02-220-801).

(a) If the sensor clearance is not correct, do the steps to adjust the sensor clearance again.

(b) If the sensor clearance is correct, check the operational functionality and values displayed on the maintenance data pages (TASK 32-09-02-710-801).

SUBTASK 32-09-02-390-002

(8) Seal the shim stack as follows:

(a) Clean the shim [2] or shims [2].

(b) Apply fay surface sealant, A00247, to the shim [2] or shims [2] surfaces.

SUBTASK 32-09-02-620-002

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(9) Apply a thin coat of corrosion inhibiting material, G50136, to the lockwire [3].

SUBTASK 32-09-02-420-004

(10) Install the lockwire [3] on the bolts [4].

END OF TASK

TASK 32-09-02-710-801

4. Nose Landing Gear Air/Ground Sensor Operational Test

(Figure 501)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task is for the operational test of the Nose Landing Gear (NLG) air/ground sensor.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Nose Landing Gear Air/Ground Sensor Operational Test

SUBTASK 32-09-02-740-001

(1) Use the Proximity Sensing Electronics Unit (PSEU) Maintenance Page to examine the adjustment of the sensor as follows:

(a) Set the GND TESTS switch, on the P61-4 maintenance bite panel, to ENABLE position.

(b) Select SYS on the Multi-Function Display panel.

(c) Select MAINT -> MAINT DATA PGS -> ATA System 32: Proximity Sensor Electronics Unit -> DISPLAY.

(d) Make sure that the sensor state is NEAR.

SUBTASK 32-09-02-070-001

(2) On PSEU Maintenance Page 1, the STATE column shows the current condition (NEAR, FAR, or MARG) of each sensor.

(a) Target-Near (or NEAR) refers to the condition where the sensor is considered actuated, while Target-Far (or FAR) refers to the unactuated condition.

(b) Marginal (or MARG) refers to a target-near condition that is not within the acceptable range required by the PSEU, and re-rigging of the sensor/target installation should be performed at the next maintenance opportunity.

SUBTASK 32-09-02-070-002

(3) On PSEU Maintenance Page 1, the current clearance value of each sensor is shown in the MILS column, if the target is near or marginal.

(a) If the target is far, the value will be shown as “---”.

SUBTASK 32-09-02-070-003

(4) On PSEU Maintenance Page 1, the serviceable sensor clearance rigging range is shown in the RANGE column.

SUBTASK 32-09-02-710-001

(5) Do the PSEU SELF-TEST as follows:

(a) Set the GND TESTS switch, on the P61-4 maintenance bite panel, to ENABLE position.

(b) Select SYS on the Multi-Function Display panel.

(c) Select MAINT-> ONBD MAINT-> LINE MAINT -> GROUND TESTS -> ATA SYSTEM 32: Proximity Sensor Electronics Unit -> OPERATIONAL TEST.

(d) Select PSEU - SELF TEST and then select CONTINUE.

(e) Follow the instructions shown on the Multi-Function Display and select CONTINUE.

(f) Select START TEST.

1) Make sure that PASS shows on the MAX Display System (MDS).

D. Put the Airplane Back to Its Usual Condition

SUBTASK 32-09-02-860-005

(1) Set the GND TEST switch, on the P61-4 maintenance bite panel, to the NORM position.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PROXIMITY SWITCH ELECTRONICS UNIT (PSEU) - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Proximity Switch Electronics Unit removal

(2) Proximity Switch Electronics Unit installation.

TASK 32-09-10-000-801

2. Proximity Switch Electronics Unit Removal

(Figure 401)

A. General

(1) This task gives the instructions on how to remove the Proximity Switch Electronics Unit.

(2) The Proximity Switch Electronics Unit is referred to as the PSEU.

B. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-840-801	Prepare to Put the Airplane in the Air Mode (P/B 201)

C. Tools/Equipment

Reference	Description
STD-7394	Cap - Dust, Electrical Connector

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well

E. Access Panels

Number	Name/Location
112A	Forward Access Door

F. Prepare for the Removal

SUBTASK 32-09-10-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Make sure the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-09-10-840-001

(2) Prepare for air mode simulation (TASK 32-09-00-840-801).

SUBTASK 32-09-10-010-001

(3) To get access to the PSEU [1], do this step:

(a) Open this access panel:

Number	Name/Location
112A	Forward Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-09-10-860-001

(4) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	17	C00129	LANDING GEAR LATCH & PRESS WARN
C	15	C01355	LANDING GEAR AIR/GND SYS 2
C	16	C01356	LANDING GEAR AIR/GND SYS 1
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

G. Proximity Switch Electronics Unit Removal

SUBTASK 32-09-10-480-004

CAUTION

DO NOT TOUCH THE PSEU BEFORE YOU DO THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE PSEU.

(1) Do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

SUBTASK 32-09-10-020-001

(2) Remove the electrical connectors [7] from the PSEU [1].

SUBTASK 32-09-10-860-002

(3) Install dust caps, STD-7394, on the electrical connectors [7].

SUBTASK 32-09-10-020-002

(4) Remove the bolt [4] and washers [5] to disconnect the jumper [6] from the PSEU [1].

SUBTASK 32-09-10-020-003

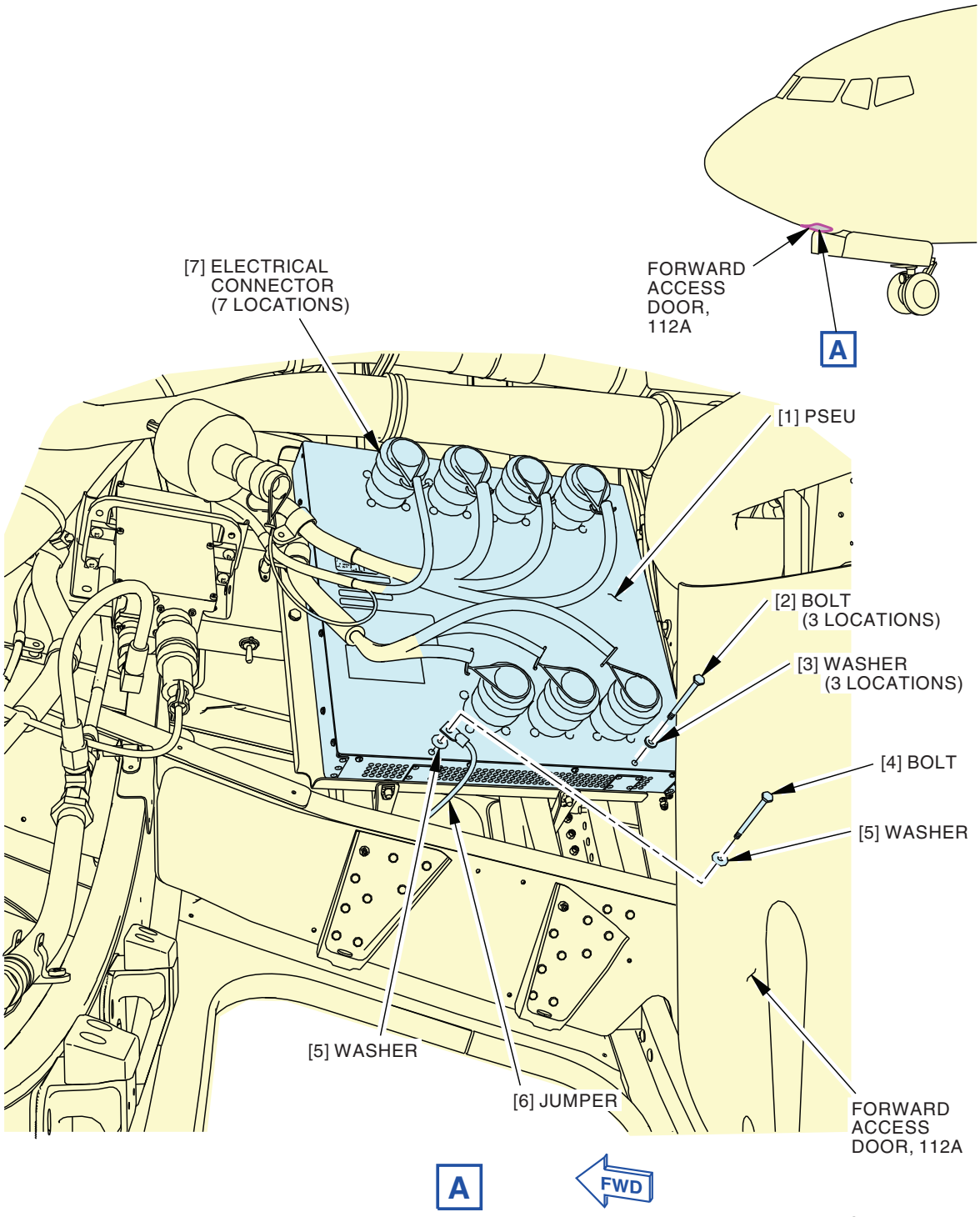
(5) Remove the bolts [2] and washers [3].

SUBTASK 32-09-10-020-004

(6) Remove the PSEU [1] from the mounting surface.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Proximity Switch Electronics Unit (PSEU) Installation
Figure 401/32-09-10-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-09-10-400-801

3. Proximity Switch Electronics Unit Installation

(Figure 401)

A. General

(1) This task gives the instructions on how to install the Proximity Switch Electronics Unit.

(2) The Proximity Switch Electronics Unit is referred to as the PSEU.

B. References

Reference	Title
20-10-34-110-802	Clean Bare, Clad, or Plated Metal with Solvent (P/B 701)
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
32-09-00-840-802	Return the Airplane Systems Back to Their Normal on Ground Condition (P/B 201)
32-09-10-710-801	Proximity Switch Electronics Unit - LRU Replacement Test (P/B 501)
52-11-00 P/B 201	FORWARD ENTRY DOOR - MAINTENANCE PRACTICES
52-13-00 P/B 201	AFT ENTRY DOOR - MAINTENANCE PRACTICES
52-22-00 P/B 201	EMERGENCY EXIT DOOR - MAINTENANCE PRACTICES
52-41-00 P/B 201	GALLEY SERVICE DOORS - MAINTENANCE PRACTICES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
STD-7394	Cap - Dust, Electrical Connector

D. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
B00130	Alcohol - Isopropyl	TT-I-735
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
G50398	Pad - Adbrasive, Scotch-Brite Type S, Abrasive Pad	
G50492	Pad - 3M Scotch Brite 7448 Ultra Fine Pad	MIL-A-9962A Type III Grade AAA

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well

F. Proximity Switch Electronics Unit Installation

SUBTASK 32-09-10-480-003

CAUTION DO NOT TOUCH THE PSEU BEFORE YOU DO THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE PSEU.

(1) Do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

SUBTASK 32-09-10-480-002

(2) Do these steps to prepare the PSEU [1] for the installation:

(a) To remove the sealant, do this task: Clean Bare, Clad, or Plated Metal with Solvent, TASK 20-10-34-110-802.

(b) Clean the mating surfaces on the PSEU [1], washers [5], bolt [4], and bonding jumper [6] (SWPM 20-20-00, Section 2):

1) Use Cleaning Procedure 1 to clean the mating surface on the PSEU [1].

a) Use a Scotch-Brite Type S pad, G50398, or Scotch-Brite 7448 pad, G50492.

2) Use Cleaning Procedure 5 to clean the remaining PSEU [1] surfaces and the washers [5], bolt [4], and bonding jumper [6] with solvent, B00184, or alcohol, B00130.

SUBTASK 32-09-10-410-001

(3) Put the PSEU [1] in position on the mounting brackets.

SUBTASK 32-09-10-410-002

(4) Install the washers [3] and bolts [2].

SUBTASK 32-09-10-420-001

(5) Connect the bonding jumper [6] to the PSEU [1] with the washers [5] and bolt [4].

(a) Tighten the bolt [4] to 35 ±3 in-lb (3.95 ±0.34 N·m).

SUBTASK 32-09-10-760-001

(6) Do a check of the bonding resistance between the PSEU [1] and the airplane structure with an intrinsically safe approved bonding meter, COM-1550 (SWPM 20-20-00).

(a) Make sure that the bonding resistance is less than 0.001 ohm (1 milliohm).

SUBTASK 32-09-10-390-001

(7) Seal the bonding jumper [6] connector.

(a) Clean the bonding jumper [6] connector with solvent, B00184, or alcohol, B00130.

(b) Seal the bonding jumper [6] connector with sealant, A02315.

SUBTASK 32-09-10-020-005

(8) Remove the dust caps, STD-7394, from the electrical connectors [7].

SUBTASK 32-09-10-410-003

(9) Install the electrical connectors [7] to the PSEU [1].

SUBTASK 32-09-10-840-002

(10) Put the airplane systems back to normal on ground condition (TASK 32-09-00-840-802).

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G. Proximity Switch Electronics Unit Installation Test

SUBTASK 32-09-10-840-003

(1) Make sure these doors are closed:

(a) Forward Entry Door (PAGEBLOCK 52-11-00/201).

(b) Aft Entry Door (PAGEBLOCK 52-13-00/201).

(c) Left and Right Overwing Exit Doors (PAGEBLOCK 52-22-00/201).

(d) Galley Service Doors (PAGEBLOCK 52-41-00/201).

SUBTASK 32-09-10-860-003

(2) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	17	C00129	LANDING GEAR LATCH & PRESS WARN
C	15	C01355	LANDING GEAR AIR/GND SYS 2
C	16	C01356	LANDING GEAR AIR/GND SYS 1
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

SUBTASK 32-09-10-200-001

(3) Do this task: Proximity Switch Electronics Unit - LRU Replacement Test, TASK 32-09-10-710-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PROXIMITY SWITCH ELECTRONICS UNIT (PSEU) - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) Proximity Sensor Electronics Unit (PSEU) operational test

(a) This task is used for this Line Replaceable Unit (LRU) replacement test on the MAX Display System (MDS):

- PSEU - replacement test.

(2) PSEU ground test

(a) This task is used for these operational tests on the MDS:

- PSEU - self test
- PSEU - LGTV test.

(3) PSEU special functions test.

TASK 32-09-10-710-801

2. Proximity Switch Electronics Unit - LRU Replacement Test

(Figure 501)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task provides the instructions on how to do the replacement test for the Proximity Sensor Electronics Unit (PSEU).

(2) This task is used for the Line Replaceable Unit (LRU) replacement test, PSEU - replacement test on the MAX Display System (MDS).

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
73-21-00-800-801-G00	EEC Maintenance Power Selection (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
821	Forward Cargo Door
822	Aft Cargo Door
831	Forward Entry Door
832	Emergency Exit Door (AOE)
833	Emergency Exit Door (AOE)
834	Aft Entry Door

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(Continued)

Number	Name/Location
841	Forward Galley Service Door
842	Emergency Exit Door (AOE)
843	Emergency Exit Door (AOE)
844	Aft Galley Service Door

E. Prepare for the Test

SUBTASK 32-09-10-840-023

(1) Make sure that these doors are closed:

Number	Name/Location
117A	Electronic Equipment Access Door
821	Forward Cargo Door
822	Aft Cargo Door
831	Forward Entry Door
832	Emergency Exit Door (AOE)
833	Emergency Exit Door (AOE)
834	Aft Entry Door
841	Forward Galley Service Door
842	Emergency Exit Door (AOE)
843	Emergency Exit Door (AOE)
844	Aft Galley Service Door

SUBTASK 32-09-10-840-004

(2) Make sure that there is no hydraulic pressure on the airplane.

(a) If there is hydraulic pressure, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-09-10-840-005

(3) Make sure that the parking brake is set and chocks are installed on the nose and main landing gear (CHOCK INSTALLATION, PAGEBLOCK 10-11-05/201).

SUBTASK 32-09-10-840-006

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(4) Make sure that the downlock pins are installed on the nose and main landing gear.

(a) If the downlock pins are not installed, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-09-10-840-007

(5) Make sure that the flaps are in the FULL UP position (TASK 27-51-00-860-804).

SUBTASK 32-09-10-840-017

(6) Make sure that the stab trim is in the green band.

NOTE: The stabilizer indicator is on the control stand.

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SUBTASK 32-09-10-840-019

(7) If installed, make sure that the airstairs are in the stowed position.

NOTE: This will prevent warning messages concerning the airstairs from displaying when the airplane is placed in the air mode.

F. Proximity Switch Electronics Unit Operational Test

SUBTASK 32-09-10-710-001

(1) Set the GND TEST switch, on the P61-4 maintenance bite panel, to the ENABLE position.

SUBTASK 32-09-10-710-002

(2) Do the PSEU replacement test as follows:

(a) Set the EEC MAINT ENG 1 POWER switch and EEC MAINT ENG 2 POWER switch to the TEST position.

1) Do this task: EEC Maintenance Power Selection, TASK 73-21-00-800-801-G00.

(b) Select SYS on the Multi Function Display (MFD) at the P-9 multi-function panel.

1) Do these steps to simulate the left engine running:

a) Select MAINT

b) Select ONBD MAINT

c) Select OTHER FUNCTIONS

d) Select SPECIAL FUNCTIONS

e) Select ATA System 71-80: ENGINE-1

f) Select Engine Running Simulation

g) Select CONTINUE

h) Follow the instructions that show on the MFD to enable the engine running simulation.

2) Do these steps to simulate the right engine running:

a) Select MAINT

b) Select ONBD MAINT

c) Select OTHER FUNCTIONS

d) Select SPECIAL FUNCTIONS

e) Select ATA System 71-80: ENGINE-2

f) Select Engine Running Simulation

g) Select CONTINUE

h) Follow the instructions that show on the MFD to enable the engine running simulation.

3) Do these steps to start the PSEU replacement test:

a) Select MAINT

b) Select ONBD MAINT

c) Select LINE MAINT

d) Select GROUND TESTS

e) Select ATA System 32: Proximity Switch Electronic Unit

f) Select LRU REPLACEMENT TEST

ECCN 9E991 BOEING PROPRIETARY - See title page for details

g) Select PSEU - LRU REPLACEMENT TEST

h) Select CONTINUE

i) Do the instructions that show on the MFD PRECONDITIONS screen.

<1> Make sure that the inhibited conditions instructions are correct and followed closely.

NOTE: If one of the inhibited conditions is not met, the test will be failed.

j) Make these selections on the MFD:

<1> CONTINUE

<2> START TEST.

k) Do the instructions that show on the MFD to change these PSEU inputs:

NOTE: The LRU REPLACEMENT TEST uses signals from the following systems to test the PSEU. The PSEU looks for a change in these inputs. These inputs can be changed in any order.

<1> Move the landing gear lever from the DN position to the UP position and back to the DN position.

<2> Release and reset the parking brake.

<3> Put the NOSE WHEEL STEER switch to the ALT position and back to the NORM position.

<4> Momentarily push the HORN CUTOUT SWITCH.

NOTE: If the switch is pushed for more than 5 seconds, then the test will fail.

<5> Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

<6> Push the left thrust lever fully forward and back to idle.

<7> Push the right thrust lever fully forward and back to idle.

<8> Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

l) When steps are finished, select CONFIRM.

m) When the test is completed, make sure that the screen displays TEST CONDITION: PASS.

n) If FAILED shows, refer to the IFIM and do the applicable procedure(s) for the maintenance message.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-09-10-860-006

(1) Set the EEC MAINT ENG 1 POWER switch and EEC MAINT ENG 2 POWER switch to the NORM position.

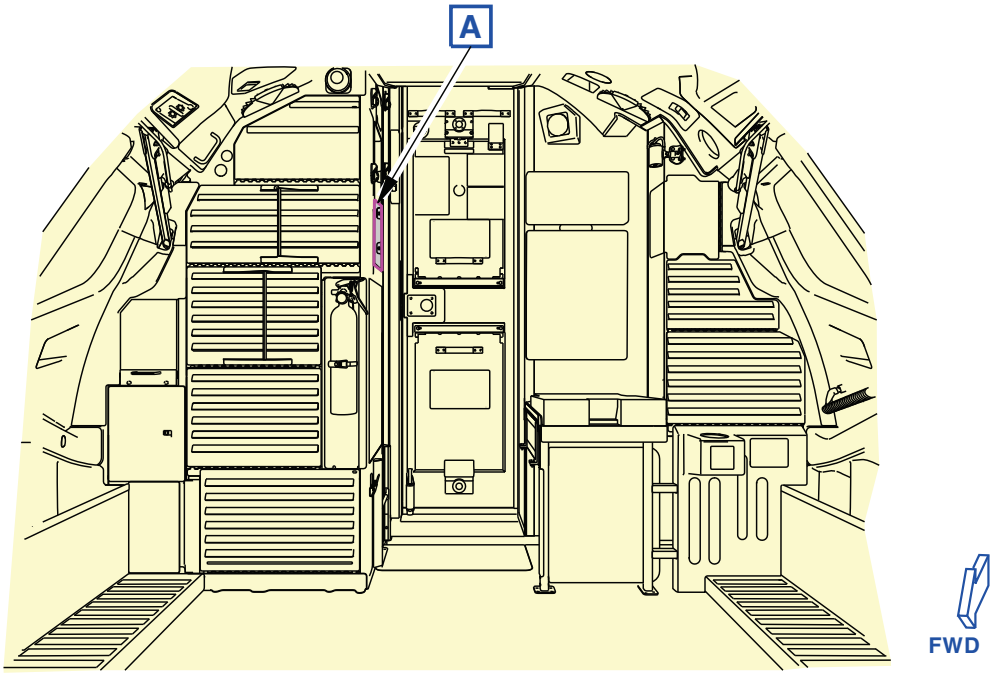
(a) Do this task: EEC Maintenance Power Selection, TASK 73-21-00-800-801-G00.

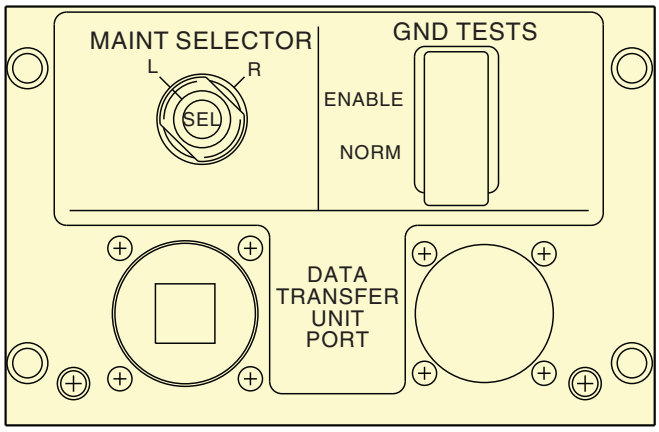
ECCN 9E991 BOEING PROPRIETARY - See title page for details

(2) Set the GND TEST switch, on the P61-4 maintenance bite panel, to the NORM position.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





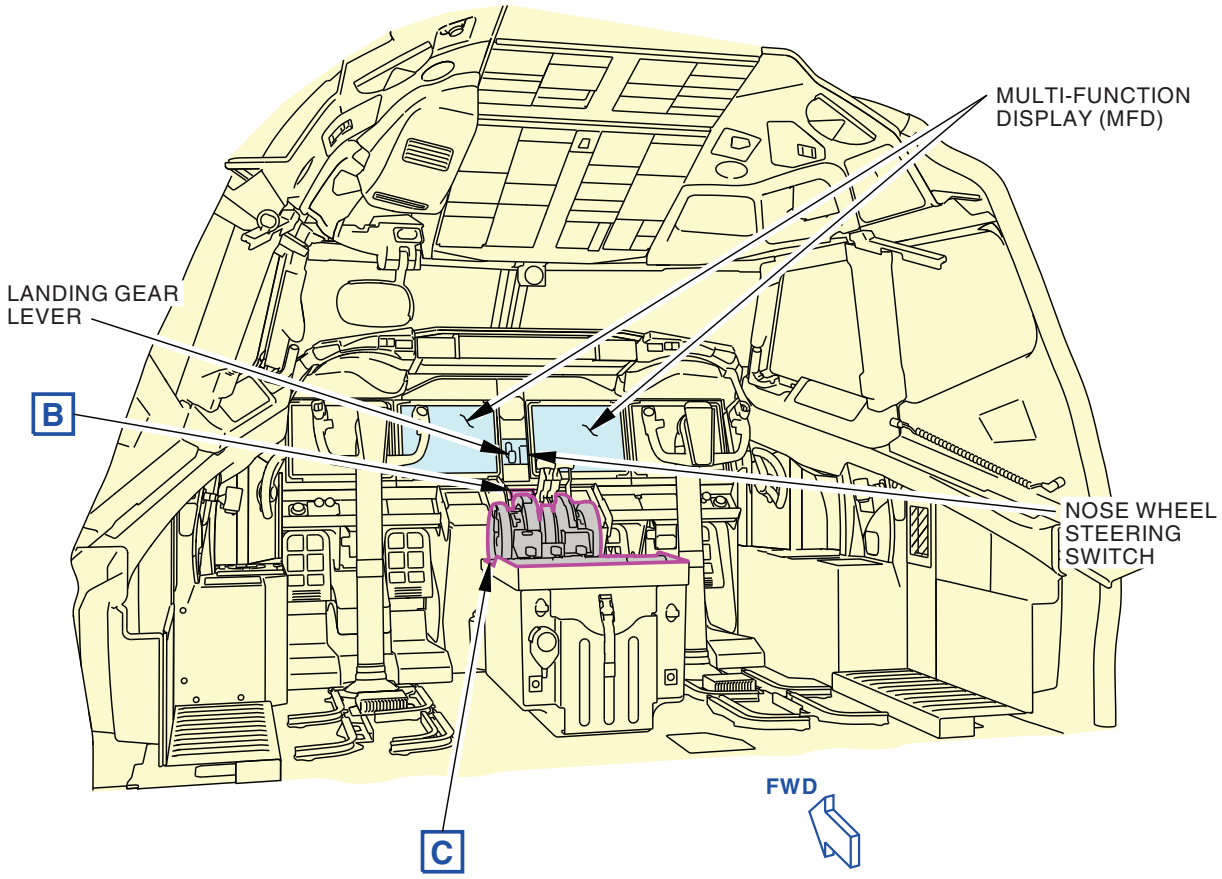
P61-4 MAINTENANCE BITE PANEL

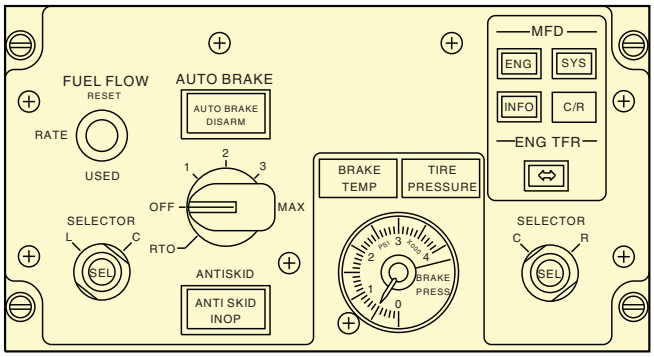
A

2414044 S00061534287_V2

Proximity Switch Electronics Unit (PSEU) Test
Figure 501/32-09-10-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





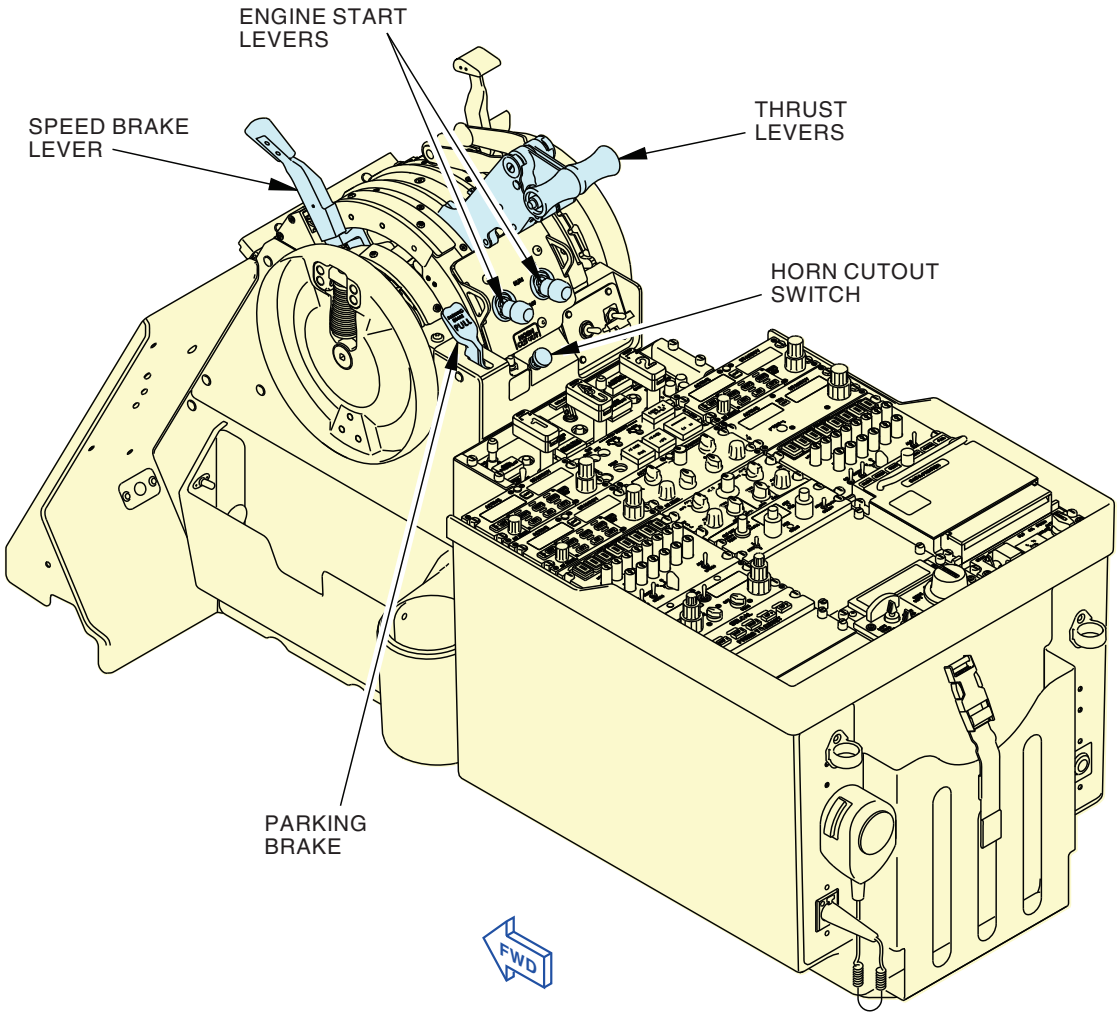
P9-82-MULTI-FUNCTIONAL PANEL (MFP)

B

2414045 S00061534288_V3

Proximity Switch Electronics Unit (PSEU) Test
Figure 501/32-09-10-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CONTROL STAND, P10

C

2476847 S0000581060_V1

Proximity Switch Electronics Unit (PSEU) Test
Figure 501/32-09-10-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-09-10-740-801

3. Proximity Switch Electronics Unit - Operational Test

(Figure 501)

A. General

(1) This task provides the instructions on how to do the ground test of the Proximity Switch Electronic Unit (PSEU).

(2) This task is used for the Operational Tests on the MAX Display System. The PSEU has the following ground tests:

(a) SELF-TEST - This performs a self test sequence of the PSEU.

(b) Landing Gear Transfer Valve (LGTV) Test - This performs a landing gear transfer valve (LGTV) test.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Tests

SUBTASK 32-09-10-860-010

(1) Select the left or right MAX Display System (MDS) Multi Function Display (MFD).

(a) Press SEL button to activate the screen on the MFD.

SUBTASK 32-09-10-860-011

(2) Set the GND TEST switch, on the P61-4 Maintenance Bite Panel, to the ENABLE position.

E. SELF-TEST

SUBTASK 32-09-10-740-001

(1) Do the SELF-TEST on the PSEU as follows:

(a) Press SYS button on the P9 Multi-Function Panel.

(b) Make these selections on the MFD to start the PSEU SELF-TEST:

1) MAINT
2) ONBD MAINT
3) LINE MAINT
4) GROUND TESTS
5) 32 PROXIMITY SWITCH ELECTRONIC UNIT
6) OPERATIONAL TEST
7) SELF-TEST
8) CONTINUE.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Do the instructions that show on the MFD PRECONDITIONS and then make these selections:

NOTE: Make sure that you verify and follow the inhibited conditions instructions closely. If one of the inhibited conditions is not met, the test will be failed.

1) CONTINUE
2) START TEST.

(d) When the test is completed, make sure that the screen displays TEST CONDITION: PASS.

(e) If FAILED shows, refer to the applicable Maintenance Message Reference in the FAULT GUIDANCE or select the maintenance message to see the recommended maintenance actions.

F. LGTV TEST

SUBTASK 32-09-10-840-008

(1) For Hydraulic System A and System B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-09-10-740-002

(2) Do the LGTV TEST on the PSEU as follows:

(a) Press SYS button on the P9 Multi-Function Panel.

(b) Make these selections on the MFD to start the PSEU LGVT TEST:

1) MAINT
2) ONBD MAINT
3) LINE MAINT
4) GROUND TESTS
5) 32 PROXIMITY SWITCH ELECTRONIC UNIT
6) OPERATIONAL TEST
7) LGTV TEST
8) CONTINUE.

(c) Do the instructions that show on the MFD PRECONDITIONS and then make these selections:

NOTE: Make sure that you verify and follow the inhibited conditions instructions closely. If one of the inhibited conditions is not met, the test will be failed.

1) CONTINUE
2) START TEST.

(d) When the test is completed, make sure that the screen displays TEST CONDITION: PASS.

(e) If FAILED shows, refer to the applicable Maintenance Message Reference in the FAULT GUIDANCE or select the maintenance message to see the recommended maintenance actions.

SUBTASK 32-09-10-840-009

(3) For Hydraulic System A and System B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-09-10-860-012

(1) Set the GND TEST switch, on the P61-4 Maintenance Bite Panel, to the NORM position.

——— END OF TASK ———

TASK 32-09-10-750-801

4. Proximity Switch Electronics Unit Special Functions

(Figure 501)

A. General

(1) This task provides the instructions on how to do the special functions of the Proximity Switch Electronics Unit.

(2) The Proximity Switch Electronics Unit is referred to as the PSEU.

(3) The PSEU has the following special functions:

(a) Air/Gnd Override - This special function allows the operator change the output of the air/ground systems.

(b) Tailskid Extend - This special function allows the operator to extend and retract the two-position tailskid for test, repair or inspection purposes.

(c) Reset Latches - This special function allows the operator to reset the PSEU faults that are latched.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Functions

SUBTASK 32-09-10-860-015

(1) Select the left or right Multi-Function Display (MFD).

(a) Press SEL button to active the screen MFD.

SUBTASK 32-09-10-860-016

(2) Set the GND TESTS switch, on the P61 maintenance bite panel, to the ENABLE position.

D. Air/Gnd Override Special Function

SUBTASK 32-09-10-750-001

(1) Do the Air/Gnd Override special function on the PSEU as follows:

(a) Make these selections on the MFD main function menu:

1) ONBOARD MAINTENANCE
2) OTHER FUCNTIONS
3) SPECIAL FUNCTIONS
4) 32 PROXIMITY SWITCH ELECTRONIC UNIT
5) PSEU - AIR/GND OVERRIDE
6) CONTINUE.

(b) Do the instructions that show on the MFD PRECONDITIONS and then make these selections:

NOTE: Make sure that you verify and follow the inhibited conditions instructions closely.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) CONTINUE
2) START FUNCTION.

(c) Do the instructions that show on the MFD to change the output of the applicable air/ground systems.

1) When you are finished, select CONFIRM.

E. Tailskid Extend Special Function

SUBTASK 32-09-10-750-002

(1) Do the Tailskid Extend special function on the PSEU as follows:

(a) Make these selections on the MFD main function menu:

1) ONBOARD MAINTENANCE
2) OTHER FUNCTIONS
3) SPECIAL FUNCTIONS
4) 32 PROXIMITY SWITCH ELECTRONIC UNIT
5) PSEU - TAILSKID EXTEND
6) CONTINUE.

(b) Do the instructions that show on the MFD PRECONDITIONS and then make these selections:

NOTE: Make sure that you verify and follow the inhibited conditions instructions closely.

1) CONTINUE
2) START FUNCTION.

(c) Do the instructions that show on the MFD to extend or retract the tailskid.

1) When you are finished, select CONFIRM.

F. Reset Latches Special Function

SUBTASK 32-09-10-750-003

(1) Do the Reset Latches special function on the PSEU as follows:

(a) Make these selections on the MFD main function menu:

1) ONBOARD MAINTENANCE
2) OTHER FUNCTIONS
3) SPECIAL FUNCTIONS
4) 32 PROXIMITY SWITCH ELECTRONIC UNIT
5) PSEU - RESET LATCHES
6) CONTINUE.

(b) Do the instructions that show on the MFD PRECONDITIONS and then make these selections:

NOTE: Make sure that you verify and follow the inhibited conditions instructions closely.

1) CONTINUE
2) START FUNCTION.

(c) Do the instructions that show on the MFD to reset latches.

1) When you are finished, select CONFIRM.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-09-10-860-017

(1) Set the GND TESTS switch, on the P61 maintenance bite panel, to the NORM position.

——— END OF TASK ———

TASK 32-09-10-740-802

5. Proximity Switch Electronics Unit Test Using UN-ATE Tool by Technologies Harness Scanner

A. General

(1) UN-ATE (Universal Automated Tooling, Electrical) is a device that can be used to examine and show the health of wire bundles and proximity sensors. UN-ATE provides a detailed evaluation of the wire bundles and the proximity sensors that are faulted and degrading.

(a) It has these functions:

1) Gives a condition of system wire bundles and proximity sensors
2) Identifies opened links
3) Identifies shorted links
4) Identifies out of tolerance links.

B. References

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)
24-22-00-860-802	Remove Electrical Power (P/B 201)
32-09-10-000-801	Proximity Switch Electronics Unit Removal (P/B 401)
32-09-10-400-801	Proximity Switch Electronics Unit Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-17989	Wiring Tester - Automated Wireless Part #: UN-ATE BASIC Supplier: 87Z60
SPL-18162	Interface Cable - UN-ATE Proximity Sensor Electronics Unit Part #: F2300+F2301+C30 Supplier: 87Z60
SPL-18164	Software Module - UN-ATE PSEU Part #: SS-B737MX-PSEU Supplier: 87Z60
STD-15075	Cord - Bungee

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well

E. Prepare for the Test

SUBTASK 32-09-10-020-006

(1) Do this task: Remove Electrical Power, TASK 24-22-00-860-802.

NOTE: Electrical power must be OFF to prevent incorrect fault indications or fault messages from occurring.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-09-10-860-022

(2) Make sure that the battery switch is in the OFF position on the P-5 overhead panel.

SUBTASK 32-09-10-760-002

(3) Make sure that all ground power is disconnected.

SUBTASK 32-09-10-020-007

(4) Remove the Proximity Switch Electronic Unit (PSEU) (TASK 32-09-10-000-801).

SUBTASK 32-09-10-480-005

(5) Connect the PSEU connectors on the airplane to the interface cable, SPL-18162 with the optional electronic dust cap to verify if there is any power on the connector.

Table 501/32-09-10-993-801 PSEU Connector to Interface Cable Connections

PSEU Connector	Interface Cable Number	Branch
D10982	2300	A
D10984	2300	B
D10986	2301	A
D10988	2301	B
D11142	C30P61A	N/A

(a) If the electronic dust cap has power, remove the power.

SUBTASK 32-09-10-020-008

(6) If no power is confirmed, remove the electronic dust cap and connect the wiring tester, COM-17989 to the interface cable, SPL-18162.

NOTE: Use a bungee cord, STD-15075 to secure the wiring tester, COM-17989 in place if required.

SUBTASK 32-09-10-480-007

(7) Connect the ground from the airplane to the wiring tester, COM-17989.

SUBTASK 32-09-10-480-008

(8) Log into to the tablet to provide testing and traceability.

(a) Start up the software module, SPL-18164 and login with your user ID.

(b) Identify the location of the aircraft being maintained.

(c) Select the aircraft model

(d) Select the aircraft subsystem, example: PSEU A1 AND A2.

(e) Identify the reason for the test (training/diagnostic/preventative).

(f) The software will now guide you into the set-up with pictograph.

SUBTASK 32-09-10-800-002

(9) Make sure that the handheld tablet has a wireless connection with the wiring tester, COM-17989.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Procedure for PSEU Test with UN-ATE

SUBTASK 32-09-10-700-001

(1) Press “Scan” on the handheld tablet's software module, SPL-18164.

NOTE: The UN-ATE will automatically make measurements and present results as per wiring diagram.

NOTE: UN-ATE will highlight in color code links that are open (Blue), shorted (Red), and out of tolerance (Resistance - Orange, Impedance - Purple)

SUBTASK 32-09-10-700-002

(2) Save A1 test results.

SUBTASK 32-09-10-700-003

(3) Load A2 test and traceability for the A2 test.

SUBTASK 32-09-10-700-004

(4) Press “Scan” on the handheld tablet's software module, SPL-18164.

NOTE: The UN-ATE will automatically make measurements and present results as per wiring diagram.

NOTE: UN-ATE will highlight in color code links that are open (Blue), shorted (Red), and out of tolerance (Resistance - Orange, Impedance - Purple)

SUBTASK 32-09-10-700-005

(5) Save A2 test results.

G. Put the Airplane Back to its Usual Condition

SUBTASK 32-09-10-080-001

(1) Disconnect the wiring tester, COM-17989 from the interface cable, SPL-18162.

NOTE: Remove the bungee cord, STD-15075 if applicable.

SUBTASK 32-09-10-080-002

(2) Remove the interface cable, SPL-18162 from the PSEU connectors.

SUBTASK 32-09-10-420-002

(3) Install the PSEU (TASK 32-09-10-400-801).

SUBTASK 32-09-10-840-020

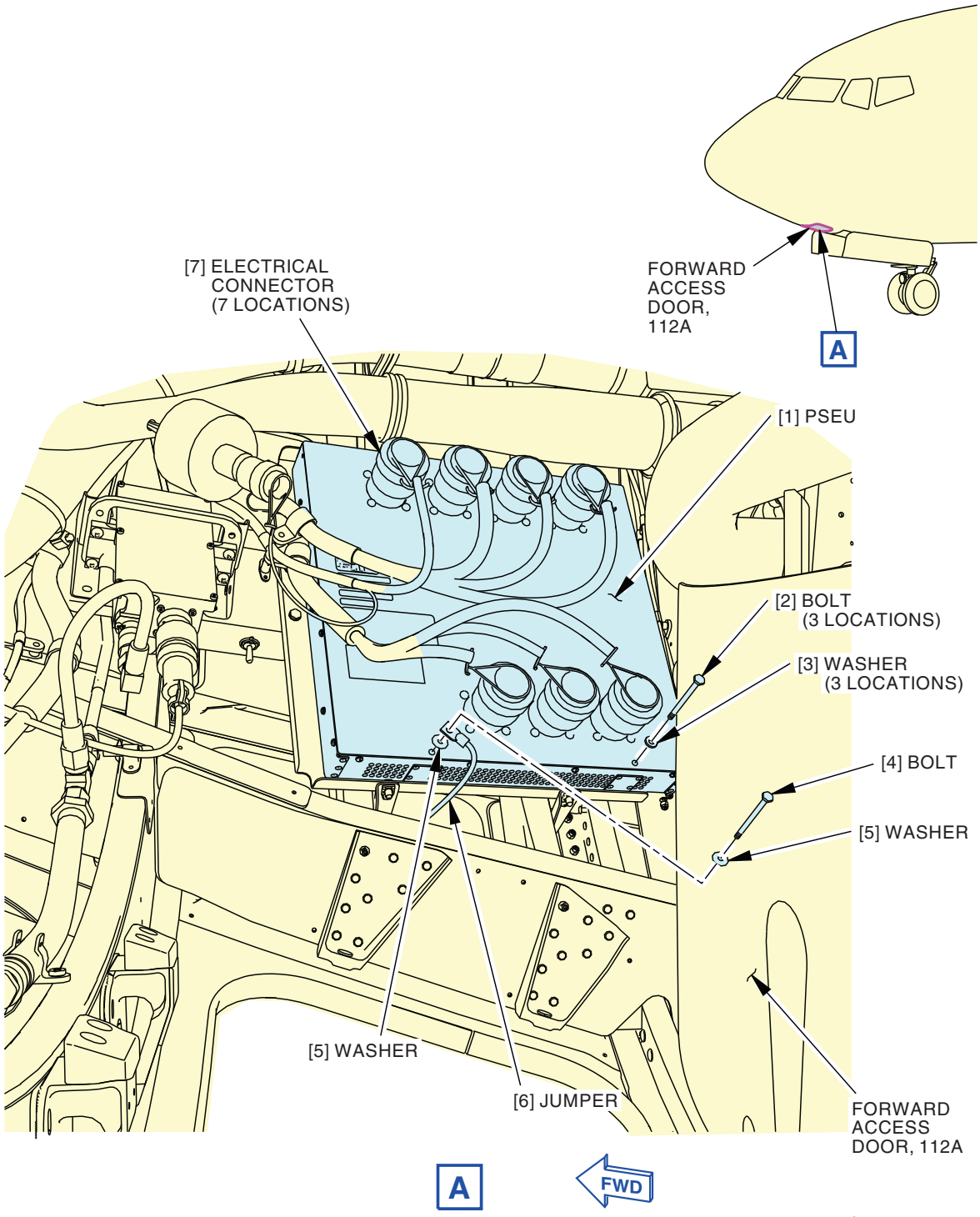
(4) Make sure that the battery switch is in the ON position on the P-5 overhead panel.

SUBTASK 32-09-10-840-021

(5) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Proximity Switch Electronics Unit (PSEU) Location
Figure 502/32-09-10-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PROXIMITY SENSOR - ADJUSTMENT/TEST

1. General

A. This procedure contains these tasks:

(1) The PSEU proximity sensor functional test with a LCR Meter

(2) The PSEU proximity sensor functional test with a Breakout Box and a LCR Meter.

B. These tasks are functional tests of the proximity switch electronic unit's sensors.

NOTE: If available, use the B737 PSEU Breakout Box (MPN 38-812-01) to measure the inductance and resistance.

C. These tests help to find a proximity sensor's health to prevent the replacement of the PSEU.

D. The proximity sensors supply position data of these systems:

(1) Entry/Service Doors

(2) The Nose and MLG

(3) The Airstairs Door (optional equipment)

E. The proximity sensors stay installed on the airplane during the test.

TASK 32-09-12-700-801

2. PSEU Proximity Sensor - Functional Test with a LCR Meter

(Figure 501, Figure 502)

A. General

(1) This test will measure the resistance and inductance of the 3-wire proximity sensor and wiring and compare the values to a known range. This test also evaluates the target to sensor relationship.

(2) Observe all applicable warnings, cautions, and maintenance practice instructions as required to gain access to the Proximity Sensor under test.

(3) Use an LCR meter (Inductance, Capacitance, Resistance), COM-1741 that has a Universal Serial Bus (USB) port to record the data with a PC automatically. For example 878B or 879B have a USB port. Use the instructions included with the LCR meter (Inductance, Capacitance, Resistance), COM-1741.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-840-801	Prepare to Put the Airplane in the Air Mode (P/B 201)
32-09-00-840-802	Return the Airplane Systems Back to Their Normal on Ground Condition (P/B 201)
32-09-10-710-801	Proximity Switch Electronics Unit - LRU Replacement Test (P/B 501)
46-13-02-700-803	Existing Flight Deck Effects Check (P/B 201)
52-11-00 P/B 201	FORWARD ENTRY DOOR - MAINTENANCE PRACTICES
52-13-00 P/B 201	AFT ENTRY DOOR - MAINTENANCE PRACTICES
52-22-00 P/B 201	EMERGENCY EXIT DOOR - MAINTENANCE PRACTICES
52-41-00 P/B 201	GALLEY SERVICE DOORS - MAINTENANCE PRACTICES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
WDM 31-53-11	Wiring Diagram Manual
WDM 32-31-11	Wiring Diagram Manual
WDM 32-31-12	Wiring Diagram Manual
WDM 32-64-11	Wiring Diagram Manual
WDM 32-64-12	Wiring Diagram Manual
WDM 52-71-11	Wiring Diagram Manual

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1741	Meter - LCR (Inductance, Capacitance, Resistance; Use COM-20443 for OH bleed loop testing) Part #: 878B Supplier: 08098 Part #: 879B Supplier: 08098 Part #: 880 Supplier: 08098 Part #: U1733P Supplier: 1LQK8 Opt Part #: 875B Supplier: 08098 Opt Part #: 878A Supplier: 08098 Opt Part #: 879 Supplier: 08098
SPL-1690	Actuators/Deactuators Set - Proximity Sensor Part #: 8-758-01 Supplier: 08748 Part #: A27092-139 Supplier: 81205 Part #: A27092-146 Supplier: 81205 Opt Part #: A27092-106 Supplier: 81205
STD-6481	Voltmeter - Digital
STD-7394	Cap - Dust, Electrical Connector

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
112A	Forward Access Door

F. Prepare for the Functional Test

SUBTASK 32-09-12-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-09-12-480-011

(2) Make sure that chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-09-12-700-003

(3) Check existing flight deck affects (TASK 46-13-02-700-803).

(a) Make a note of Proximity Switch Electronic Unit (PSEU), Nose Landing Gear (NLG), Main Landing Gear (MLG) or Door faults.

SUBTASK 32-09-12-740-001

(4) Use the PSEU Maintenance Page to make a note of each sensor's state (Figure 501):

(a) Set the GND TESTS switch to ENABLE on the P61-4 Panel.

(b) Select SYS on the Multi-Function Display panel.

(c) Select MAINT -> MAINT DATA PGS -> ATA System 32: PSEU -> DISPLAY.

(d) Make a note of each sensor's state.

1) Select NEXT PAGE if necessary to see all sensors.

(e) Set the GND TESTS switch to NORM on the P61-4 Panel.

SUBTASK 32-09-12-970-001

(5) On PSEU Maintenance Page 1, the STATE column shows the current condition (NEAR, FAR or MARG) of each sensor.

(a) Target-Near (NEAR) refers to the condition where the sensor is actuated. Target-Far (FAR) refers to the unactuated condition.

(b) Marginal (MARG) refers to a target-near condition which is not within the acceptable range. And rigging of the sensor/target installation is necessary.

SUBTASK 32-09-12-970-003

(6) On PSEU Maintenance Page 1, the current clearance value of each sensor is shown in the MILS column, if the target is near or marginal. If the target is far, the value will be shown as "---".

SUBTASK 32-09-12-970-004

(7) On PSEU Maintenance Page 1, the serviceable sensor clearance rigging range is shown in the RANGE column.

SUBTASK 32-09-12-840-001

(8) Prepare for air mode simulation (TASK 32-09-00-840-801).

SUBTASK 32-09-12-010-001

(9) To get access to the PSEU, do this step (Figure 502):

(a) Open this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 32-09-12-860-001

(10) Remove electrical power from the system for the test.

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	17	C00129	LANDING GEAR LATCH & PRESS WARN
C	15	C01355	LANDING GEAR AIR/GND SYS 2
C	16	C01356	LANDING GEAR AIR/GND SYS 1
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

SUBTASK 32-09-12-480-002

CAUTION

DO NOT TOUCH THE PSEU BEFORE YOU DO THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE PSEU.

(11) Do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

SUBTASK 32-09-12-020-001

(12) Remove the electrical connector [7] J2 through J6 from the PSEU [1].

SUBTASK 32-09-12-480-003

(13) Install dust caps, STD-7394, on the electrical connector [7].

SUBTASK 32-09-12-210-001

(14) Examine the Aeronautical Radio Incorporated (ARINC) connector pins for damage or corrosion.

(a) Repair if necessary.

G. Measure the Proximity Sensor and Wiring Installation Resistance

SUBTASK 32-09-12-760-001

(1) Refer to the applicable Wiring Diagram Manual (WDM) (WDM 31-53-11, WDM 32-31-11, WDM 32-31-12,WDM 32-64-11, WDM 32-64-12 and WDM 52-71-11) for the 3-wire sensor to PSEU [1] ARINC connector pin callout.

SUBTASK 32-09-12-760-006

(2) Connect the LCR meter (Inductance, Capacitance, Resistance), COM-1741 (a digital voltmeter, STD-6481, may be used to measure the resistance).

(a) First do the Blue to Yellow measurement.

(b) Then the Red to Yellow measurement.

(c) Alternate connections as needed.

SUBTASK 32-09-12-860-015

(3) Supply power to the LCR meter (Inductance, Capacitance, Resistance), COM-1741.

SUBTASK 32-09-12-760-007

(4) Set the LCR meter (Inductance, Capacitance, Resistance), COM-1741, to measure series resistance (Rs) at 120 Hz.

SUBTASK 32-09-12-760-008

(5) Measure the resistance at the ARINC connector pins on the applicable J-connector for each sensor under test and record in the log sheet (Figure 501).

(a) The “DATA HOLD” button is used to "capture" your current measured result.

(b) To make a new measurement, the, "DATA HOLD" must be released, by pressing the button once, then the new measurement can be made.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Record the resistance of sensors in the appropriate space on log sheet. Values to be recorded in hundredths (two numbers right of the decimal point), for example XX.XX or XXX.XX.

SUBTASK 32-09-12-760-009

(6) Compare the measured resistance value to Table 502.

Table 501/32-09-12-993-801 SENSOR TABLE

BOEING SENSOR PART NUMBER	CRANE SENSOR PART NUMBER	TARGET NEAR INDUCTANCE (WITH GO-GAUGE)	TARGET FAR INDUCTANCE (WITH GO-GAUGE)	COIL RESISTANCE RED-YELLOW (TARGET FAR)	RESISTANCE BLUE-YELLOW (TARGET FAR)
10-61226-29	1-899-29	38.0 mH minimum	36.0-37.5 mH	27.0-39.0 Ω	344.5-351.5 Ω
10-61226-30	1-899-30	38.0 mH minimum	36.0-37.5 mH	27.0-33.0 Ω	344.5-351.5 Ω
10-61226-45	80-207-01	38.0 mH minimum	36.0-37.5 mH	27.0-33.0 Ω	344.5-351.5 Ω
10-61226-47	80-207-02	38.0 mH minimum	36.0-37.5 mH	27.0-33.0 Ω	344.5-351.5 Ω

SUBTASK 32-09-12-760-010

(7) If the measured resistance value is less than 5.00 ohms, this indicates a short circuit condition.

(a) If there is a short circuit, inspect the associated wiring and connectors for contamination or damage.

SUBTASK 32-09-12-760-011

(8) If the resistance value as measured between the Red and Yellow leads is greater than 100.00 ohms, this indicates a loose contact or wire damage or a defective sensor.

(a) If there is loose contact or wire damage, inspect the associated wiring and connectors for contamination or damage.

SUBTASK 32-09-12-760-012

(9) If the resistance value as measured between the Blue and Yellow leads is greater than 500.00 ohms, this indicates a loose contact or wire damage or a defective sensor.

(a) If there is loose contact or wire damage, inspect the associated wiring and connectors for contamination or damage.

H. Measure the Inductance at the ARINC Connector Pins on the applicable J-connector for each Sensor Under Test - Red-Blue

SUBTASK 32-09-12-760-002

(1) Refer to the applicable WDM (WDM 31-53-11, WDM 32-31-11, WDM 32-31-12,WDM 32-64-11, WDM 32-64-12 and WDM 52-71-11) for the 3-wire sensor to PSEU [1] ARINC connector pin callout.

SUBTASK 32-09-12-760-013

(2) Connect the LCR meter (Inductance, Capacitance, Resistance), COM-1741, for Red to Blue measurement.

SUBTASK 32-09-12-860-016

(3) Supply power to the LCR meter (Inductance, Capacitance, Resistance), COM-1741.

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SUBTASK 32-09-12-760-014

(4) Set the LCR meter (Inductance, Capacitance, Resistance), COM-1741, to read series inductance (Ls) at 1 kHz.

SUBTASK 32-09-12-760-015

(5) Measure the inductance and record in the log sheet (Figure 501) for each sensor.

(a) The “DATA HOLD” button is used to "capture" your current measured result.

(b) To make a new measurement, the, "DATA HOLD" must be released, by pressing the button once, then the new measurement can be made.

(c) Record the inductance of sensors in the appropriate space on log sheet. Values to be recorded in hundredths (two numbers right of the decimal point), for example XX.XX or XXX.XX.

(d) If you suspect a faulty sensor, use the applicable proximity sensor test set, SPL-1690, 8-758-01, Proximity Sensor Test Kit – Go-Gauge to simulate a configuration change of the sensors current position to a target FAR or a target NEAR position.

(e) If GO-GAUGES are used compare the measured inductance value to Table 501.

SUBTASK 32-09-12-760-016

(6) Compare the measured inductance value to Table 502.

Table 502/32-09-12-993-802 NOMINAL VALUES

VALUE	SENSOR CONDITION
Three-wire Sensors	
Less than 40 mH	Target FAR
40 mH - 50 mH	Correctly rigged NEAR sensor
Greater than 50 mH	Target may be rigged too near to the sensor
Resistance on aircraft (Red-Yellow resistance plus Blue-Yellow resistance): ≤ 390 Ω	
Typical Sensor Inductance Values Using Go Gauges	
Three-wire sensor - Go side 38.20 mH, No-Go side - 36.20 mH	

NOTE: Sensor inductance values will vary between sensor positions because of differences in operation and design requirements. In some condition, the inductance value cannot be achieved. Inductance values outside of the nominal range are permitted if the minimum value of 38.00 mH (Target NEAR) is obtained and no system anomaly is observed.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 32-09-12-080-001

(1) Remove the dust caps, STD-7394, from the electrical connectors [7].

SUBTASK 32-09-12-420-001

(2) Install the J2 through J6 electrical connector [7] to the PSEU [1].

SUBTASK 32-09-12-840-002

(3) Put the airplane systems back to normal on ground condition (TASK 32-09-00-840-802).

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SUBTASK 32-09-12-410-001

(4) Close this access door:

Number	Name/Location
112A	Forward Access Door

SUBTASK 32-09-12-840-003

(5) Make sure that these doors are closed:

(a) Forward Entry Door (PAGEBLOCK 52-11-00/201).

(b) Aft Entry Door (PAGEBLOCK 52-13-00/201).

(c) Left and Right Overwing Exit Doors (PAGEBLOCK 52-22-00/201).

(d) Galley Service Doors (PAGEBLOCK 52-41-00/201).

SUBTASK 32-09-12-860-003

(6) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	17	C00129	LANDING GEAR LATCH & PRESS WARN
C	15	C01355	LANDING GEAR AIR/GND SYS 2
C	16	C01356	LANDING GEAR AIR/GND SYS 1
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

SUBTASK 32-09-12-700-001

(7) Do this task: Proximity Switch Electronics Unit - LRU Replacement Test, TASK 32-09-10-710-801.

——— END OF TASK ———

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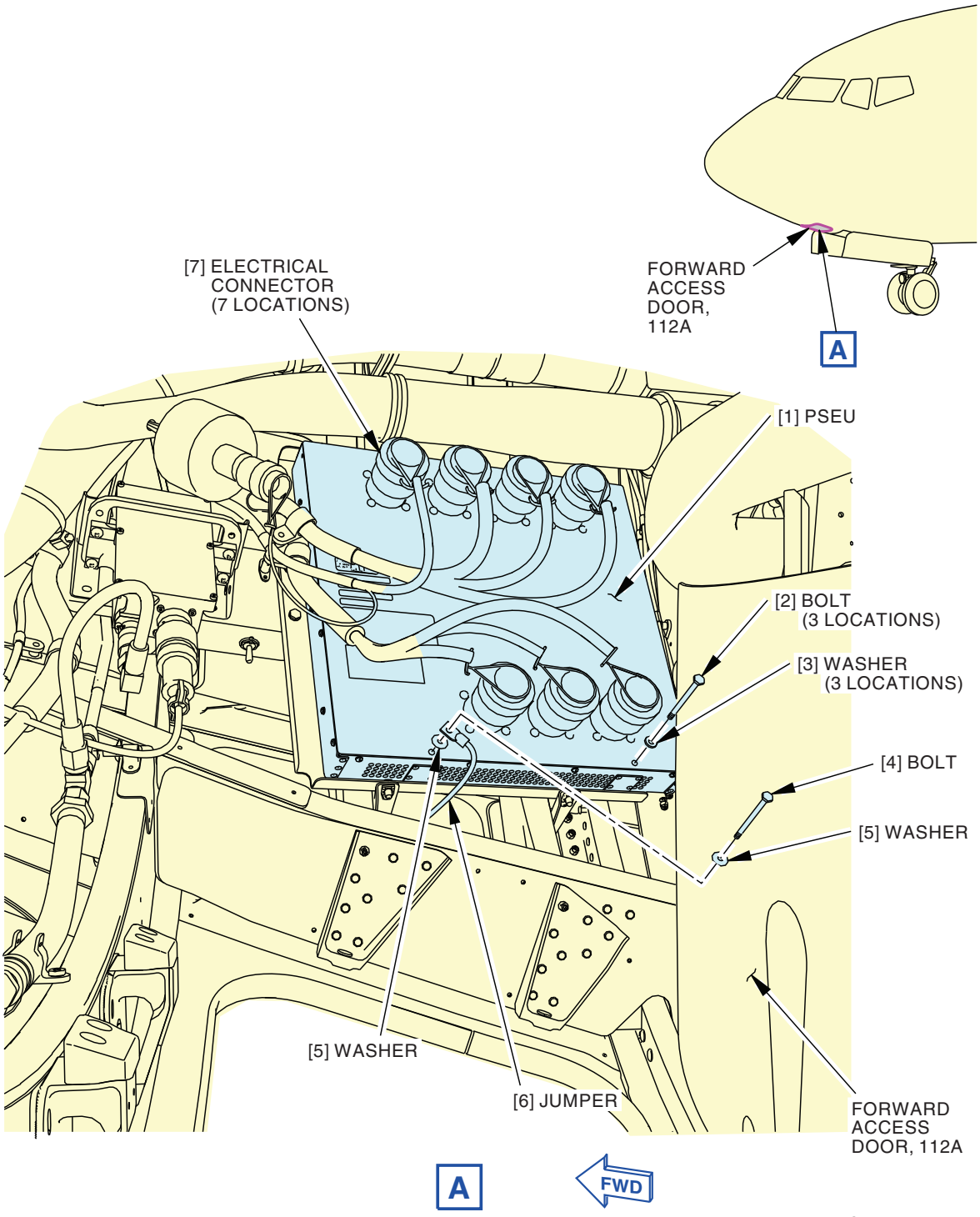
DATE:	LOC:		GROUND TEMP:		
A/C:			FUSELAGE TEMP:		
			RESISTANCE OHMS (MEASURE AT 120HZ)		INDUCTANCE MH (MEAS. AT 1KHZ)
PSEU CABLE	SENSOR EQUIP. NO.	SYSTEM 1 SENSOR INSTALLATION	BLUE -YELLOW (TC RESISTOR)	RED -YELLOW (COILS)	RED-BLUE
J2	S1050 1	GSBV CLOSED			
J2	S1010	RIGHT ON GND A			
J2	S1014	NOSE ON GND A			
J2	S1012	LEFT ON GND A			
J3	S846	NOSE LKD A			
J3	S73	RIGHT DOWN LKD A			
J3	S845	NOSE DWN A			
J3	S71	LEFT DWN LKD A			
J3	S74	RIGHT UP LKD A			
J3	S72	LEFT UP LKD A			
J4	S200	AFT ENT LKD			
J4	S195	AFT SER LKD			
			RESISTANCE OHMS (MEASURE AT 120HZ)		INDUCTANCE MH (MEAS. AT 1KHZ)
PSEU CABLE	SENSOR EQUIP. NO.	SYSTEM 2 SENSOR INSTALLATION	BLUE -YELLOW (TC RESISTOR)	RED -YELLOW (COILS)	
J5	S854	NOSE LKD B			
J5	S1016	LEFT UP LKD B			
J5	S301	RIGHT DWN LKD B			
J5	S853	NOSE DWN B			
J5	S302	LEFT DWN LKD B			
J5	S1017	RIGHT UP LKD B			
J6	S1011	RIGHT ON GND B			
J6	S1015	NOSE ON GND B			
J6	S199	FWD ENT LKD			
J6	S1013	LEFT ON GND B			
J6	S194	FWD SER LKD			
J6	S282 2	AIRSTRS LKD			

1 NOT USED ON MAX
2 ONLY IF OPTION SELECTED

2910517 S0000697733_V1

Proximity Sensor Performance Values - Log Sheet
Figure 501/32-09-12-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Proximity Switch Electronics Unit (PSEU)
Figure 502/32-09-12-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-09-12-700-802

3. PSEU Proximity Sensor - Functional Test with a Breakout Box and a LCR Meter

(Figure 503, Figure 504)

A. General

CAUTION

THE PROXIMITY SENSOR BREAKOUT BOX IS USED ONLY TO EXAMINE BASIC SENSOR HEALTH AND INSTALLATION. DO NOT USE AS A RIGGING TOOL. RIGGING OF SENSORS AND OTHER MAINTENANCE ACTIONS SHOULD BE DONE ONLY IN ACCORDANCE WITH AIRLINE OPERATING AND MAINTENANCE PROCEDURES, APPROVED MAINTENANCE MANUALS, OR DIRECTIVES OF THE APPROPRIATE CERTIFICATION AGENCY.

(1) The test will measure the inductance and resistance of the 3-wire sensors, sensor wires, and connectors. And compare the values to a known range. This test also examines the target to sensor conditions.

(2) This test uses the Crane Sensor Test System made up of the following parts:

(a) Proximity Switch Electronic Unit (PSEU) - proximity sensor breakout box, SPL-18742, and LCR meter (Inductance, Capacitance, Resistance), COM-1741.

(b) Sensors Test Gauge Set.

1) The controls on the face of the proximity sensor breakout box, SPL-18742, are used to select the desired sensor for testing.

a) The proximity sensor breakout box, SPL-18742, does not test two-wire sensors. Two-wire sensors, do not need to be tested.

2) The Sensor Test Gauge Kit consists of six go-gauges, sensor de-actuator tool and an instruction booklet. Use the go-gauges in place of a sensor, if you suspect the sensor is faulty.

(3) Obey all applicable warnings, cautions, and maintenance practice instructions when necessary to measure the Proximity Sensor in the test.

(4) Use an LCR meter (Inductance, Capacitance, Resistance), COM-1741, that has a USB port to record the data with a PC automatically. For example 878B or 879B have a USB port. Use the specified instructions included with the LCR meter (Inductance, Capacitance, Resistance), COM-1741. Full instructions are included in the Breakout Box (BOB)'s Sensor Test System.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-840-801	Prepare to Put the Airplane in the Air Mode (P/B 201)
32-09-00-840-802	Return the Airplane Systems Back to Their Normal on Ground Condition (P/B 201)
32-09-10-710-801	Proximity Switch Electronics Unit - LRU Replacement Test (P/B 501)
46-13-02-700-803	Existing Flight Deck Effects Check (P/B 201)
52-11-00 P/B 201	FORWARD ENTRY DOOR - MAINTENANCE PRACTICES
52-13-00 P/B 201	AFT ENTRY DOOR - MAINTENANCE PRACTICES
52-22-00 P/B 201	EMERGENCY EXIT DOOR - MAINTENANCE PRACTICES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
52-41-00 P/B 201	GALLEY SERVICE DOORS - MAINTENANCE PRACTICES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1741	Meter - LCR (Inductance, Capacitance, Resistance; Use COM-20443 for OH bleed loop testing) Part #: 878B Supplier: 08098 Part #: 879B Supplier: 08098 Part #: 880 Supplier: 08098 Part #: U1733P Supplier: 1LQK8 Opt Part #: 875B Supplier: 08098 Opt Part #: 878A Supplier: 08098 Opt Part #: 879 Supplier: 08098
SPL-1690	Actuators/Deactuators Set - Proximity Sensor Part #: 8-758-01 Supplier: 08748 Part #: A27092-139 Supplier: 81205 Part #: A27092-146 Supplier: 81205 Opt Part #: A27092-106 Supplier: 81205
SPL-18742	Breakout Box - Proximity Sensor Part #: 38-812-01 Supplier: 08748
STD-6481	Voltmeter - Digital
STD-7394	Cap - Dust, Electrical Connector

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
112A	Forward Access Door

F. Prepare for the Functional Test

SUBTASK 32-09-12-480-004

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the Nose Landing Gear (NLG) and Main Landing Gear (MLG) (TASK 32-00-01-480-801).

SUBTASK 32-09-12-480-012

(2) Make sure that chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-09-12-700-004

(3) Check existing flight deck affects (TASK 46-13-02-700-803).

(a) Make a note of the PSEU, NLG, MLG or Door faults.

SUBTASK 32-09-12-740-002

(4) Use the PSEU Maintenance Page to make a note of each sensor's state (Figure 504):

(a) Set the GND TESTS switch to ENABLE on the P61-4 Panel.

(b) Select SYS on the Multi-Function Display panel.

(c) Select MAINT -> MAINT DATA PGS -> ATA System 32: PSEU -> DISPLAY.

(d) Make a note of each sensor's state.

1) Select NEXT PAGE if necessary to see all sensors.

(e) Set the GND TESTS switch to NORM on the P61-4 Panel.

SUBTASK 32-09-12-970-002

(5) On the PSEU Maintenance Page 1, the STATE column shows the current condition (NEAR, FAR or MARG) of each sensor.

(a) Target-Near (NEAR) refers to the condition where the sensor is actuated.

(b) Target-Far (FAR) refers to the unactuated condition.

(c) Marginal (MARG) refers to a target-near condition which is not within the acceptable range and rigging of the sensor/target installation is necessary.

SUBTASK 32-09-12-970-005

(6) On the PSEU Maintenance Page 1, the current clearance value of each sensor is shown in the MILS column, if the target is near or marginal.

(a) If the target is far, the value will be shown as “---”.

SUBTASK 32-09-12-970-006

(7) On the PSEU Maintenance Page 1, the serviceable sensor clearance rigging range is shown in the RANGE column.

SUBTASK 32-09-12-840-004

(8) Prepare for the air mode simulation (TASK 32-09-00-840-801).

SUBTASK 32-09-12-010-002

(9) To get access to the PSEU, do this step (Figure 502):

(a) Open this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 32-09-12-860-004

(10) Remove the electrical power from the system for the test.

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

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F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	17	C00129	LANDING GEAR LATCH & PRESS WARN
C	15	C01355	LANDING GEAR AIR/GND SYS 2
C	16	C01356	LANDING GEAR AIR/GND SYS 1
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

SUBTASK 32-09-12-480-005

CAUTION

DO NOT TOUCH THE PSEU BEFORE YOU DO THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE PSEU.

(11) Do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

SUBTASK 32-09-12-020-002

(12) Remove the electrical connector [7] J2 through J6 from the PSEU [1].

SUBTASK 32-09-12-480-006

(13) Install the dust caps, STD-7394, on the electrical connectors [7].

SUBTASK 32-09-12-210-002

(14) Examine the Aeronautical Radio Incorporated (ARINC) connector pins for damage or corrosion.

(a) Repair if necessary.

SUBTASK 32-09-12-480-007

(15) Install the proximity sensor breakout box, SPL-18742, cables to the related PSEU [1] cables (for example J2 to J2).

(a) If a fault is suspected in the PSEU [1], test each pin of the ARINC connector for continuity.

1) Set the sensor selector switch to each sensor in its turn.

2) Measure for continuity between the sensor inputs of the ARINC connector and the front panel connectors.

NOTE: No calibration or test is necessary for the proximity sensor BOB.

G. Measure the Proximity Sensor and Wiring Installation Resistance

SUBTASK 32-09-12-480-008

(1) Connect the LCR meter (Inductance, Capacitance, Resistance), COM-1741, leads to the BLUE and YELLOW jacks on the front of the proximity sensor breakout box, SPL-18742.

(a) A digital voltmeter, STD-6481, may be used to measure the resistance.

SUBTASK 32-09-12-860-005

(2) Set the LCR meter (Inductance, Capacitance, Resistance), COM-1741, to measure series resistance (Rs) at 120 Hz.

SUBTASK 32-09-12-860-006

(3) Push the L/C/R button to change to resistance mode.

SUBTASK 32-09-12-860-007

(4) Push and hold the LCR meter (Inductance, Capacitance, Resistance), COM-1741, button for more than one second to change to series mode.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) SER is shown in the middle-right side of the LCR meter (Inductance, Capacitance, Resistance), COM-1741, display.

SUBTASK 32-09-12-760-003

(5) Measure the resistance (BLUE to YELLOW) and record in the log sheet (Figure 504) for each sensor.

(a) The “DATA HOLD” button is used to "capture" your current measured result.

(b) To make a new measurement, the "DATA HOLD" must be released, by pressing the button once, then the new measurement can be made.

(c) Record the resistance of sensors in the appropriate space on log sheet.

1) Values to be recorded in hundredths (two numbers right of the decimal point), for example XX.XX or XXX.XX.

SUBTASK 32-09-12-860-008

(6) Use the proximity sensor breakout box, SPL-18742, controls to set the next sensor.

(a) Measure the resistance (BLUE to YELLOW) and record for each of the remaining sensors.

SUBTASK 32-09-12-480-009

(7) Connect the LCR meter (Inductance, Capacitance, Resistance), COM-1741, leads to the RED and YELLOW jacks on the front of the proximity sensor breakout box, SPL-18742.

(a) A digital voltmeter, STD-6481, may be used to measure the resistance.

SUBTASK 32-09-12-760-004

(8) Measure the resistance (RED to YELLOW) and record in the log sheet (Figure 504) for each sensor.

(a) The “DATA HOLD” button is used to "capture" your current measured result.

(b) To make a new measurement, the "DATA HOLD" must be released, by pressing the button once, then the new measurement can be made.

(c) Record the resistance of sensors in the appropriate space on log sheet.

1) Values should be recorded in hundredths (two numbers right of the decimal point), for example XX.XX or XXX.XX.

SUBTASK 32-09-12-200-002

(9) Compare the combined measured resistance value to Table 505.

SUBTASK 32-09-12-860-009

(10) Use the proximity sensor breakout box, SPL-18742, controls to set the next sensor.

(a) Measure the resistance (RED to YELLOW) and record for each of the remaining sensors.

H. Measure the Inductance (Red-Blue) for each Sensor Under Test

SUBTASK 32-09-12-480-010

(1) Connect the LCR meter (Inductance, Capacitance, Resistance), COM-1741, leads to the RED and BLUE jacks on the front of the proximity sensor breakout box, SPL-18742.

SUBTASK 32-09-12-860-010

(2) Supply power to the LCR meter (Inductance, Capacitance, Resistance), COM-1741.

SUBTASK 32-09-12-860-011

(3) Use the Table 503 to set the BOB controls to the appropriate sensor for test.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 503/32-09-12-993-803 PSEU PROXIMITY SENSOR BREAKOUT BOX (BOB)

SENSOR EQUIP. NO.	BOB SETTING	PSEU AND BOB CABLE	P/N 38-812-01
	SYSTEM 1 - SW1 POSITION	CONNECTOR	FUNCTION
S1050*[1]	1	J2	GSBV CLOSED
S1010	2	J2	RIGHT ON GND A
S1014	3	J2	NOSE ON GND A
S1012	4	J2	LEFT ON GND A
S846	5	J3	NOSE LKD A
S73	6	J3	RIGHT DOWN LKD A
S845	7	J3	NOSE DWN A
S71	8	J3	LEFT DWN LKD A
S74	9	J3	RIGHT UP LKD A
S72	10	J3	LEFT UP LKD A
S200	11	J4	AFT ENT LKD
S195	12	J4	AFT SER LKD
	SYSTEM 2 - SW3 POSITION	CONNECTOR	FUNCTION
S854	1	J5	NOSE LKD B
S1016	2	J5	LEFT UP LKD B
S301	3	J5	RIGHT DWN LKD B
S853	4	J5	NOSE DWN B
S302	5	J5	LEFT DWN LKD B
S1017	6	J5	RIGHT UP LKD B
S1011	7	J6	RIGHT ON GND B
S1015	8	J6	NOSE ON GND B
S199	9	J6	FWD ENT LKD
S1013	10	J6	LEFT ON GND B
S194	11	J6	FWD SER LKD
S282*[2]	12	J6	AIRSTRS LKD

*[1] Not used on MAX.

*[2] Only if airstairs option selected.

SUBTASK 32-09-12-860-012

(4) Set the LCR meter (Inductance, Capacitance, Resistance), COM-1741, to read series inductance (Ls) at 1 kHz.

SUBTASK 32-09-12-760-005

(5) Measure the inductance and record in the log sheet (Figure 504) for each sensor.

(a) The “DATA HOLD” button is used to "capture" your current measured result.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) To make a new measurement, the "DATA HOLD" must be released, by pressing the button once, then the new measurement can be made.

(c) Record the inductance of sensors in the appropriate space on log sheet.

1) Values to be recorded in hundredths (two numbers right of the decimal point), for example XX.XX or XXX.XX.

(d) If you suspect a faulty sensor, use the applicable proximity sensor test set, SPL-1690, 8-758-01, Proximity Sensor Test Kit – GO-GAUGE to simulate a configuration change of the sensors current position to a target FAR or a target NEAR position.

SUBTASK 32-09-12-760-017

(6) If the GO-GAUGES are used, compare the measured inductance value to Table 504.

Table 504/32-09-12-993-804 SENSOR TABLE

BOEING SENSOR PART NUMBER	CRANE SENSOR PART NUMBER	TARGET NEAR INDUCTANCE (WITH GO-GAUGE)	TARGET FAR INDUCTANCE (WITH GO-GAUGE)	COIL RESISTANCE RED-YELLOW (TARGET FAR)	RESISTANCE BLUE-YELLOW (TARGET FAR)
10-61226-29	1-899-29	38.0 mH minimum	36.0-37.5 mH	27.0-39.0 Ω	344.5-351.5 Ω
10-61226-30	1-899-30	38.0 mH minimum	36.0-37.5 mH	27.0-33.0 Ω	344.5-351.5 Ω
10-61226-45	80-207-01	38.0 mH minimum	36.0-37.5 mH	27.0-33.0 Ω	344.5-351.5 Ω
10-61226-47	80-207-02	38.0 mH minimum	36.0-37.5 mH	27.0-33.0 Ω	344.5-351.5 Ω

SUBTASK 32-09-12-200-003

(7) Compare the measured inductance value to Table 505.

Table 505/32-09-12-993-805 SENSOR NOMINAL VALUES

VALUE	SENSOR CONDITION
Three-wire Sensors	
35 mH	Target FAR
40 mH - 50 mH	Correctly rigged NEAR sensor
Greater than 50 mH	Target may be rigged too near to the sensor
Resistance on aircraft (Red-Yellow resistance plus Blue-Yellow resistance): ≤ 390 Ω	
Typical Sensor Inductance Values Using Go Gauges	
Three-wire sensor - Go side 38.20 mH, No-Go side - 36.20 mH	

NOTE: Sensor inductance values will vary between sensor positions because of differences in operation and design requirements. In some condition, the inductance value cannot be achieved. Inductance values outside of the nominal range are permitted if the minimum value of 38.00 mH (Target NEAR) is obtained and no system anomaly is observed.

SUBTASK 32-09-12-860-013

(8) Use the proximity sensor breakout box, SPL-18742, controls to set the next sensor.

(a) Measure the inductance (RED to BLUE) and record for each of the remaining sensors.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Put the Airplane Back to Its Usual Condition

SUBTASK 32-09-12-080-002

(1) Remove the proximity sensor breakout box, SPL-18742, cables from the J2 through J6 cables of the PSEU [1].

SUBTASK 32-09-12-080-003

(2) Remove the dust caps, STD-7394, from the electrical connectors [7].

SUBTASK 32-09-12-420-002

(3) Install the J2 through J6 electrical connector [7] to the PSEU [1].

SUBTASK 32-09-12-840-005

(4) Put the airplane systems back to normal on ground condition (TASK 32-09-00-840-802).

SUBTASK 32-09-12-410-002

(5) Close this access door:

Number	Name/Location
112A	Forward Access Door

SUBTASK 32-09-12-840-007

(6) Make sure that these doors are closed:

(a) Forward Entry Door (PAGEBLOCK 52-11-00/201).

(b) Aft Entry Door (PAGEBLOCK 52-13-00/201).

(c) Left and Right Overwing Exit Doors (PAGEBLOCK 52-22-00/201).

(d) Galley Service Doors (PAGEBLOCK 52-41-00/201).

SUBTASK 32-09-12-860-014

(7) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

F/O Electrical System Panel, P6-3

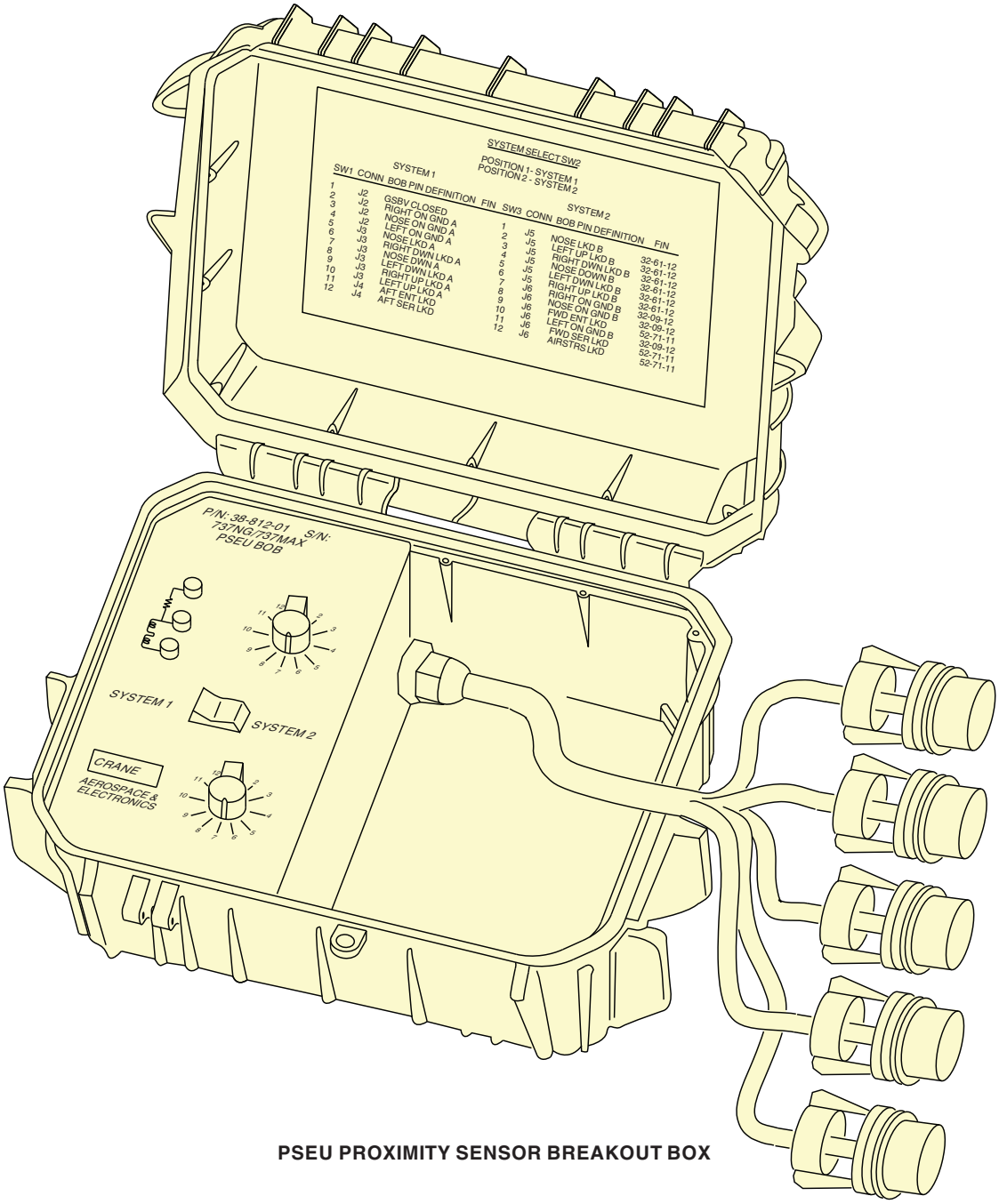
Row	Col	Number	Name
B	17	C00129	LANDING GEAR LATCH & PRESS WARN
C	15	C01355	LANDING GEAR AIR/GND SYS 2
C	16	C01356	LANDING GEAR AIR/GND SYS 1
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

SUBTASK 32-09-12-700-002

(8) Do this task: Proximity Switch Electronics Unit - LRU Replacement Test, TASK 32-09-10-710-801.

END OF TASK

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PSEU PROXIMITY SENSOR BREAKOUT BOX

2911888 S0000697772_V1

PSEU Proximity Sensor Breakout Box (BOB)
Figure 503/32-09-12-990-803

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DATE:		LOC:		GROUND TEMP:		
A/C:				FUSELAGE TEMP:		
				RESISTANCE OHMS (MEASURE AT 120Hz)		INDUCTANCE mH (MEAS. AT 1KHz)
SENSOR EQUIP. NO.	BREAKOUT BOX SWITCH (SW1) POSITION	PSEU/BOB CABLE CONNECTOR	SYSTEM 1 SENSOR INSTALLATION	BLUE -YELLOW (TC RESISTOR)	RED -YELLOW (COILS)	RED-BLUE
S1050 1	1	J2	GSBV CLOSED			
S1010	2	J2	RIGHT ON GND A			
S1014	3	J2	NOSE ON GND A			
S1012	4	J2	LEFT ON GND A			
S846	5	J3	NOSE LKD A			
S73	6	J3	RIGHT DOWN LKD A			
S845	7	J3	NOSE DWN A			
S71	8	J3	LEFT DWN LKD A			
S74	9	J3	RIGHT UP LKD A			
S72	10	J3	LEFT UP LKD A			
S200	11	J4	AFT ENT LKD			
S195	12	J4	AFT SER LKD			
				RESISTANCE OHMS (MEASURE AT 120Hz)		INDUCTANCE mH (MEAS. AT 1KHz)
SENSOR EQUIP. NO.	BREAKOUT BOX SWITCH (SW3) POSITION	PSEU/BOB CABLE CONNECTOR	SYSTEM 2 SENSOR INSTALLATION	BLUE -YELLOW (TC RESISTOR)	RED -YELLOW (COILS)	RED-BLUE
S854	1	J5	NOSE LKD B			
S1016	2	J5	LEFT UP LKD B			
S301	3	J5	RIGHT DWN LKD B			
S853	4	J5	NOSE DWN B			
S302	5	J5	LEFT DWN LKD B			
S1017	6	J5	RIGHT UP LKD B			
S1011	7	J6	RIGHT ON GND B			
S1015	8	J6	NOSE ON GND B			
S199	9	J6	FWD ENT LKD			
S1013	10	J6	LEFT ON GND B			
S194	11	J6	FWD SER LKD			
S282 2	12	J6	AIRSTRS LKD			

1 NOT USED ON MAX
2 ONLY IF OPTION SELECTED

2910522 S0000697767_V2

Proximity Sensor Performance Values - Log Sheet
Figure 504/32-09-12-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN LANDING GEAR - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) A removal of the main landing gear gas charging valve

(2) An installation of the main landing gear gas charging valve

(3) A removal of the main landing gear oil charging valve

(4) An installation of the main landing gear oil charging valve.

B. The gas charging valve is installed on the inboard side of the shock strut. The gas charging valve is between the trunnion and the lower side strut.

TASK 32-11-00-000-801

2. Main Landing Gear Gas Charging Valve Removal

(Figure 201)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

C. Prepare for the Removal

SUBTASK 32-11-00-480-024

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-11-00-860-002

(2) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-11-00-860-003

(3) Make sure that there is no pressure in the hydraulic systems A and B (TASK 29-11-00-860-805).

D. Main Landing Gear Gas Charging Valve Removal

SUBTASK 32-11-00-630-001

WARNING DO NOT REMOVE THE VALVE BODY UNTIL YOU DEFLATE THE SHOCK STRUT FULLY. THE AIR PRESSURE CAN BLOW THE VALVE BODY OUT AND CAUSE INJURIES TO PERSONNEL.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING CLEAR THE AREA BELOW THE WING BEFORE YOU DEFLATE THE SHOCK STRUT. IF YOU DEFLATE ONE SHOCK STRUT, THE WING-TIP CAN MOVE DOWN AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(1) Deflate the shock strut [1] for the applicable main landing gear fully:

(a) Remove the lockwire [6] from the gas charging valve [4] and shock strut [1].

1) Discard the lockwire [6].

(b) Remove the dust cap [2] for the gas charging valve [4].

(c) Loosen the swivel nut [3] of the gas charging valve [4] for a maximum of two turns.

NOTE: Fluid in the shock strut will have bubbles when you release the pressure.

1) Deflate the shock strut [1] slowly to prevent the leakage of the fluid through the gas charging valve [4].

(d) Loosen the swivel nut [3] of the gas charging valve [4] fully when all of the pressure in the shock strut [1] is released.

NOTE: The shock strut is fully deflated when the dimension from the lower surface of the outer cylinder to the upper surface of the axle clevis on the inner cylinder is 0.81 in. (20.57 mm) - 1.11 in. (28.19 mm).

SUBTASK 32-11-00-020-001

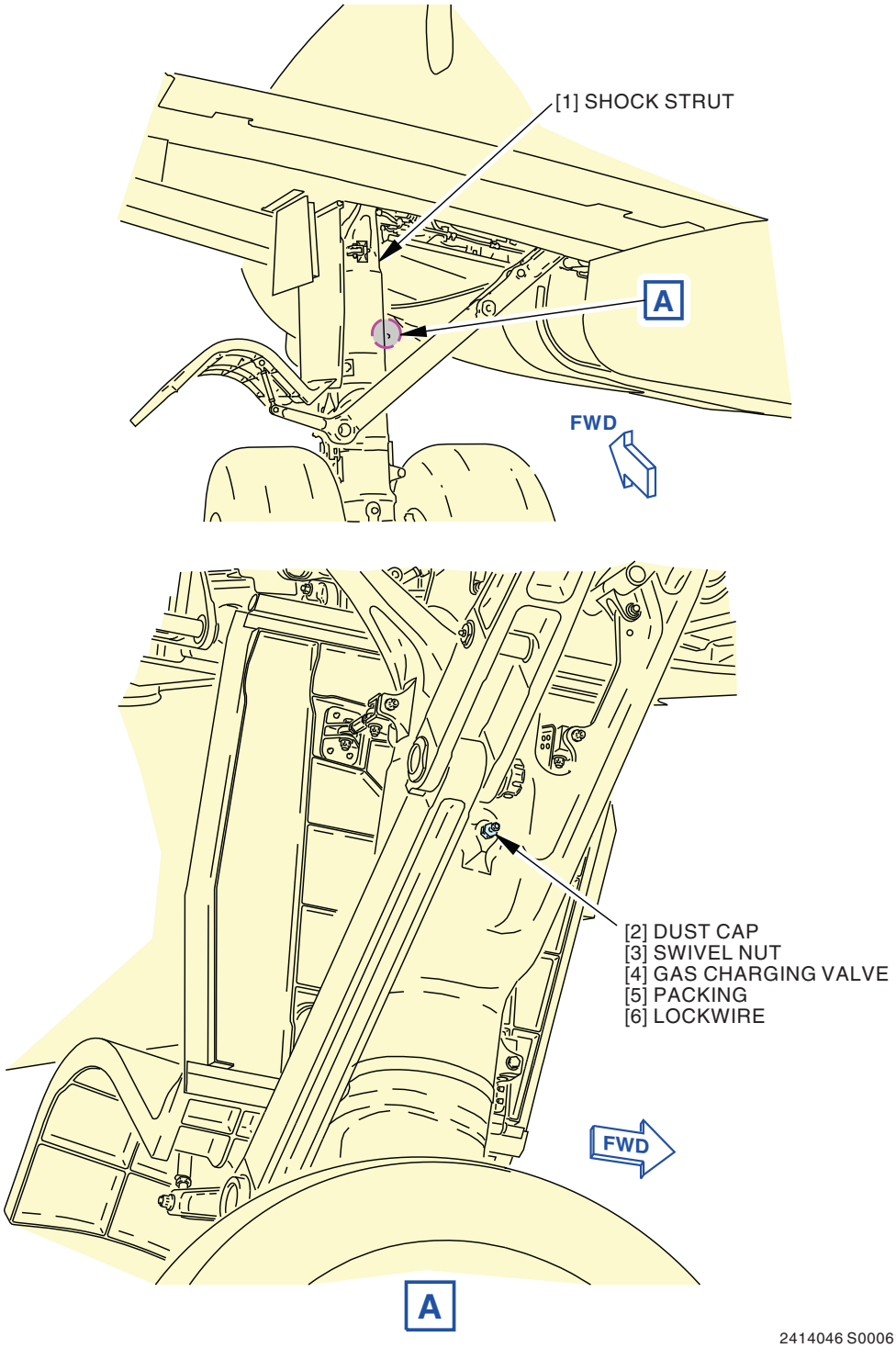
(2) Remove the gas charging valve [4].

(a) Remove the packing [5] from the gas charging valve [4].

(b) Discard the packing [5].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Gas Charging Valve
Figure 201/32-11-00-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-11-00-400-801

3. Main Landing Gear Gas Charging Valve Installation

(Figure 201)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
12-15-31-610-802	Main Landing Gear Shock Strut Servicing, Airplane on the Ground (P/B 301)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Consumable Materials

Reference	Description	Specification
A00226	Compound - Tamper-Proof Putty	BMS8-45
D00467	Fluid - Landing Gear Shock Strut	BMS3-32 Type II
D50022	Fluid - Landing Gear Shock Strut (Specifically For Preservation)	BMS3-32 Type I
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Gas charging valve	32-11-21-01-020	PTF ALL
5	Packing	32-11-21-01-023	PTF ALL

D. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Prepare for the Installation

SUBTASK 32-11-00-480-025

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-11-00-860-080

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

F. Main Landing Gear Gas Charging Valve Installation

SUBTASK 32-11-00-600-001

(1) To install the new packing [5] on the gas charging valve [4], do these steps:

(a) Lubricate the new packing [5] with fluid, D00467, or fluid, D50022.

(b) Lubricate the surface of the gas charging valve [4] that the packing slides against with fluid, D00467, or fluid, D50022.

(c) Install the packing [5] on the gas charging valve [4].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-11-00-600-002

(2) Lubricate the threads of the gas charging valve [4] with the fluid for the shock strut [1] (TASK 12-15-31-610-802).

SUBTASK 32-11-00-420-001

(3) Install the gas charging valve [4] on the shock strut [1].

NOTE: The dust cap and swivel nut of the gas charging valve are installed when the shock strut is serviced.

(a) Tighten the body of the gas charging valve [4] to 12 ±1 ft-lb (16 ±2 N·m).

(b) Install MS20995C32 lockwire, G01048, on the gas charging valve [4] and shock strut [1] with the double twist method.

(c) Apply compound, A00226, on the gas charging valve [4] and shock strut [1].

NOTE: Apply it to make sure that the seal made by compound, A00226, will break if the body of the gas charging valve is loosened.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-11-00-610-001

(1) Service the shock strut [1] (TASK 12-15-31-610-802).

——— END OF TASK ———

TASK 32-11-00-000-806

4. Main Landing Gear Oil Charging Valve Removal

(Figure 202)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1829	Oil Drain Valve - Shock Strut, LG Part #: J32108-16 Supplier: 81205 Opt Part #: A32066-1 Supplier: 81205
STD-1110	Container - Hydraulic Fluid Resistant, 5 Gallon (19 Liter)
STD-1140	Hose - Flexible, 1/2 Inch ID, BMS 3-11 Resistant, 10 Foot

C. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Prepare for the Removal

SUBTASK 32-11-00-800-001

CAUTION DO NOT DEFLATE OR SERVICE THE SHOCK STRUTS WHILE THE TAIL STAND IS INSTALLED. IF YOU DEFLATE OR SERVICE THE SHOCK STRUTS, THE LOAD ON THE TAIL STAND CAN BE TOO LARGE. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Do not deflate or service the shock struts while the tail stand is installed.

SUBTASK 32-11-00-480-060

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(2) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-11-00-860-071

(3) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-11-00-860-058

(4) Make sure that there is no pressure in the hydraulic systems A and B (TASK 29-11-00-860-805).

E. Main Landing Gear Oil Charging Valve Removal

SUBTASK 32-11-00-630-004

WARNING DO NOT REMOVE THE VALVE BODY UNTIL YOU DEFLATE THE SHOCK STRUT FULLY. THE AIR PRESSURE CAN BLOW THE VALVE BODY OUT AND CAUSE INJURIES TO PERSONNEL.

WARNING CLEAR THE AREA BELOW THE WING BEFORE YOU DEFLATE THE SHOCK STRUT. IF YOU DEFLATE ONE SHOCK STRUT, THE WING-TIP CAN MOVE DOWN AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(1) Deflate the shock strut [1] for the Main Landing Gear (MLG) fully:

(a) Remove the dust cap [2] for the gas charging valve [4].

(b) Loosen the swivel nut [3] of the gas charging valve [4] for a maximum of two turns.

NOTE: Fluid in the shock strut [1] will have bubbles when you release the pressure. Deflate the shock strut [1] slowly to prevent the leakage of the fluid through the gas charging valve [4].

(c) Loosen the swivel nut [3] of the gas charging valve [4] fully when all of the pressure in the shock strut [1] is released.

NOTE: The shock strut [1] is fully deflated when the dimension from the lower surface of the outer cylinder to the upper surface of the axle clevis on the inner cylinder is 0.81 in. (20.57 mm) to 1.11 in. (28.19 mm).

SUBTASK 32-11-00-680-001

(2) Drain the oil from the shock strut [1]:

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(a) Remove the cap [29] from the oil charging valve [30].

(b) Put a 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110 in a position to catch the oil when the oil charging valve [30] is opened.

(c) Install the drain equipment on the oil charging valve [30]:

NOTE: The tool, landing gear shock strut drain valve, SPL-1829, could also be used, but it will take much longer to drain the shock strut.

1) Cut a length of plastic ten foot hose (1/2 inch ID), STD-1140, long enough to reach the container on the floor.

2) Insert a small allen wrench in the end of the cut length of tubing so that the long end of the allen wrench is flush with the end of the tube and the short end goes through the wall of the tube.

3) Install the cut length of tubing on the oil charging valve [30] so that the allen wrench goes into the check valve and holds it open to drain the oil.

(d) Remove the drain equipment when you have removed all of the oil.

SUBTASK 32-11-00-020-066

(3) Remove the oil charging valve [30] as follows:

(a) Remove the lockwire [28] from the oil charging valve [30].

1) Discard the lockwire [28].

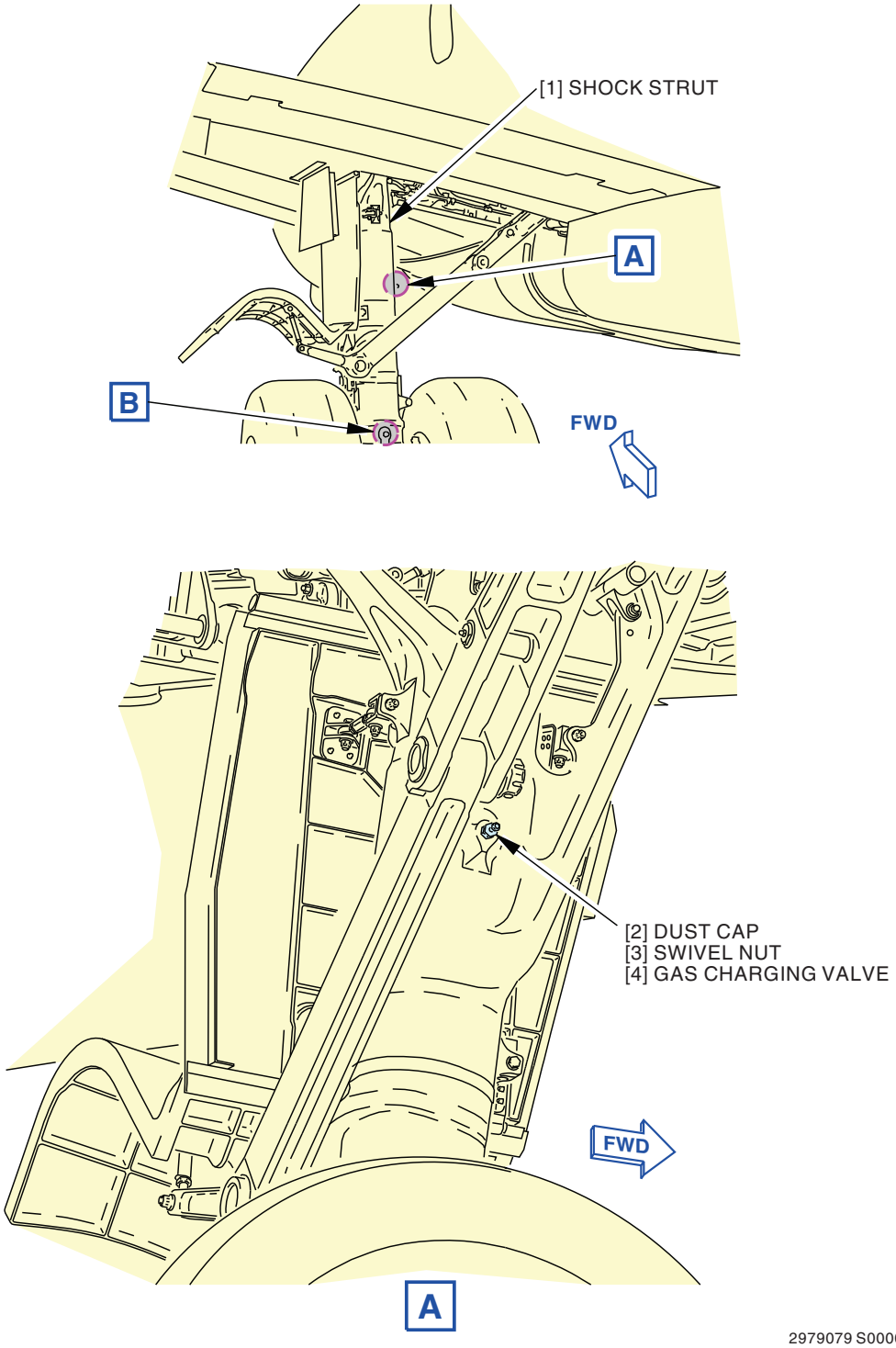
(b) Remove the oil charging valve [30].

(c) Remove the packing [31] from the oil charging valve [30].

1) Discard the packing [31].

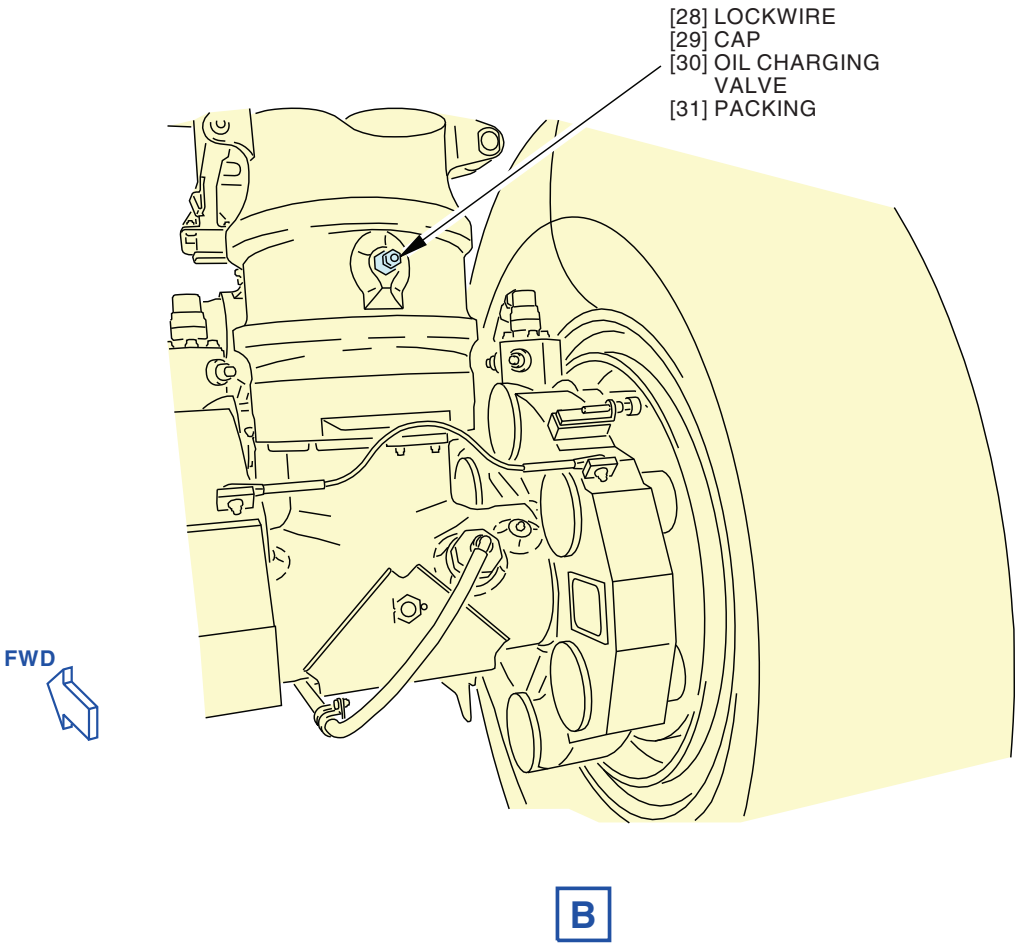
——— END OF TASK ———

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Main Landing Gear Oil Charging Valve Installation
Figure 202/32-11-00-990-818 (Sheet 1 of 2)

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Main Landing Gear Oil Charging Valve Installation
Figure 202/32-11-00-990-818 (Sheet 2 of 2)

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TASK 32-11-00-400-806

5. Main Landing Gear Oil Charging Valve Installation

(Figure 202)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
12-15-31-610-802	Main Landing Gear Shock Strut Servicing, Airplane on the Ground (P/B 301)
24-22-00-860-802	Remove Electrical Power (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Consumable Materials

Reference	Description	Specification
A00226	Compound - Tamper-Proof Putty	BMS8-45
D00467	Fluid - Landing Gear Shock Strut	BMS3-32 Type II
D50022	Fluid - Landing Gear Shock Strut (Specifically For Preservation)	BMS3-32 Type I
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
30	Oil charging valve	32-11-21-01-025	PTF ALL
31	Packing	32-11-21-01-040	PTF ALL

D. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Prepare for the Installation

SUBTASK 32-11-00-480-061

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-11-00-860-064

(2) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

F. Main Landing Gear Oil Charging Valve Installation

SUBTASK 32-11-00-420-056

(1) To install the new packing [31] on the oil charging valve [30], do these steps:

(a) Lubricate the new packing [31] with fluid, D00467 or fluid, D50022.

(b) Lubricate the surface of the oil charging valve [30] which the packing slides against with fluid, D00467 or fluid, D50022.

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(c) Install the new packing [31] on the oil charging valve [30].

SUBTASK 32-11-00-640-009

(2) Lubricate the threads of the oil charging valve [30] with the fluid for the shock strut [1].

(a) Refer to Main Landing Gear Shock Strut Servicing, Airplane on the Ground, TASK 12-15-31-610-802 for the fluid.

SUBTASK 32-11-00-420-057

(3) Install the oil charging valve [30] on the shock strut [1].

NOTE: The cap [29] is installed when the shock strut [1] is serviced.

(a) Torque the body of the oil charging valve [30] to 22 ft-lb (30 N·m) - 25 ft-lb (34 N·m).

(b) Install the lockwire [28] (MS20995C32 lockwire, G01048) on the oil charging valve [30] and the shock strut [1] using the double twist method.

(c) Apply compound, A00226 on the oil charging valve [30] and the shock strut [1].

NOTE: Apply it to make sure that the seal made by the compound, A00226, will break if the body of the oil charging valve [30] is loosened.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-11-00-600-004

(1) Service the shock strut [1], do this task: Main Landing Gear Shock Strut Servicing, Airplane on the Ground, TASK 12-15-31-610-802.

SUBTASK 32-11-00-860-061

(2) If electrical power is not necessary, do this task: Remove Electrical Power, TASK 24-22-00-860-802.

SUBTASK 32-11-00-860-063

(3) Remove the downlock pins, do this task: Landing Gear Downlock Pins Removal, TASK 32-00-01-080-801.

——— END OF TASK ———

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MAIN LANDING GEAR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the main landing gear

(2) An installation of the main landing gear.

TASK 32-11-00-000-802

2. Main Landing Gear Removal

(Figure 401, Figure 402, Figure 403, and Figure 405)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
10-11-05 P/B 201	CHOCK INSTALLATION
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-62-00-800-801	Apply Pressure to the Speed Brake Hydraulic Systems A and B (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-11-61-000-802	Main Landing Gear Lower Side Strut Removal (P/B 401)
32-11-62-000-801	Main Landing Gear Side Brace Assembly Removal (P/B 401)
32-11-71-000-801	Main Landing Gear Reaction Link Assembly Removal (P/B 401)
32-13-11-000-801	Main Landing Gear Center Door Removal (P/B 401)
32-13-11-000-802	Main Landing Gear Inner Door Removal (P/B 401)
32-13-21-000-801	Main Landing Gear Wing Door Removal (P/B 401)
32-41-41-000-801	Main Landing Gear Brake Removal (P/B 401)
32-45-11-000-801	Main Landing Gear Wheel and Tire Assembly Removal (P/B 401)
57-16-01-000-801	MLG Forward Trunnion Housing Assembly Removal (P/B 401)
57-16-01-400-801	MLG Forward Trunnion Housing Assembly Installation (P/B 401)
57-16-02-000-801	MLG Aft Trunnion Bearing Assembly Removal (P/B 401)
57-16-02-400-801	MLG Aft Trunnion Bearing Assembly Installation (P/B 401)
57-16-06-200-802	Main Landing Gear Beam Hanger Link Free Play Check (P/B 601)
57-50-02-000-801	Inboard Fixed Trailing Edge Lower Panel Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1362	Wrench Adapter, C32029-6 or C32029-96 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205

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(Continued)

Reference	Description
SPL-1363	Wrench Adapter, C32029-7 or C32029-98 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1365	Thread Protector, C32029-9 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1366	Thread Protector, C32029-10 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1371	Thread Protector, C32029-15 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1372	Thread Protector, C32029-16 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1744	Lock Set - Inboard Ground Spoiler Actuator (Contains 4 Lock Assemblies) Part #: C27046-13 Supplier: 81205 Opt Part #: C27046-1 Supplier: 81205
SPL-1862	Removal/Installation Equipment - Pin, Aft Trunnion MLG Part #: C32031-22 Supplier: 81205 Opt Part #: C32031-1 Supplier: 81205
SPL-1869	Removal/Installation Equipment - Main Landing Gear Part #: C32034-343 Supplier: 81205 Opt Part #: C32034-227 Supplier: 81205 Opt Part #: C32034-273 Supplier: 81205 Opt Part #: C32034-339 Supplier: 81205
SPL-1871	Retention Straps - Shock Strut, NLG/MLG Part #: C32030 -31 Supplier: 81205 Opt Part #: C32030-10 Supplier: 81205
SPL-14021	Wrench Adapter, C32029-48 or C32029-90 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-14022	Wrench Adapter, C32029-49 or C32029-88 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-19644	Dolly - Side Strut, MLG Part #: C32078-1 Supplier: 81205

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
734	Left Main Landing Gear
744	Right Main Landing Gear

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D. Access Panels

Number	Name/Location
551BB	Lower Inboard Fixed Trailing Edge, Gear Adjustment Door
551BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel
551CB	Lower Inboard Fixed Trailing Edge, Gear Access Panel
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel
651BB	Lower Inboard Fixed Trailing Edge, Gear Adjustment Door
651BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel
651CB	Lower Inboard Fixed Trailing Edge, Gear Access Panel
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

E. Prepare for the Removal

SUBTASK 32-11-00-480-022

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-11-00-480-056

(2) Make sure that the chocks installed around the tires (PAGEBLOCK 10-11-05/201).

SUBTASK 32-11-00-860-009

(3) Extend the trailing edge flaps fully (TASK 27-51-00-860-803).

NOTE: This step will give access to the aft trunnion pin.

SUBTASK 32-11-00-860-010

(4) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 32-11-00-860-052

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(5) Do this task: Apply Pressure to the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-801.

SUBTASK 32-11-00-860-012

(6) Put the speed brake lever in the UP position to lift the ground spoilers.

SUBTASK 32-11-00-480-002

(7) Install the lock set, SPL-1744, for the applicable ground spoiler.

SUBTASK 32-11-00-860-011

(8) Put the SPOILER A and B switches, on the P5-3 flight control panel, to the OFF position to remove the hydraulic power from the flight spoilers.

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SUBTASK 32-11-00-860-072

(9) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	4	C00700	HEATERS DRAIN MAST AIR

SUBTASK 32-11-00-860-013

(10) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1
B	16	C01346	LANDING GEAR PARKING BRAKE
B	17	C00129	LANDING GEAR LATCH & PRESS WARN
C	15	C01355	LANDING GEAR AIR/GND SYS 2
C	16	C01356	LANDING GEAR AIR/GND SYS 1
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN
D	15	C01401	LANDING GEAR AIR/GND RELAY
D	16	C01432	LANDING GEAR ALTN EXTEND SOL
E	16	C00196	LANDING GEAR ANTISKID INBD
E	18	C00195	LANDING GEAR ANTISKID OUTBD

SUBTASK 32-11-00-860-014

(11) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-11-00-860-015

(12) Push the brake pedals a minimum of twelve times to remove the pressure from the brake lines.

SUBTASK 32-11-00-020-002

WARNING DO NOT REMOVE THE VALVE BODY UNTIL YOU DEFLATE THE SHOCK STRUT FULLY. THE AIR PRESSURE CAN BLOW THE VALVE BODY OUT AND CAUSE INJURIES TO PERSONNEL.

(13) Deflate the shock strut [1] for the two main landing gear fully.

NOTE: This task is necessary keep the airplane level when you do the subsequent steps.

(a) Remove the dust cap [4] for the gas charging valve [2].

(b) Loosen the swivel nut [3] a maximum of two turns.

(c) Deflate the shock strut [1] slowly to prevent the leakage of the fluid through the gas charging valve [2].

NOTE: Fluid in the shock strut will have bubbles when you release the pressure.

(d) Loosen the swivel nut [3] fully when all of the pressure in the shock strut [1] is released.

NOTE: The shock strut is fully deflated when the dimension from the lower surface of the outer cylinder to the upper surface of the axle clevis on the inner cylinder is 0.9 in. (2.3 cm).

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SUBTASK 32-11-00-480-003

WARNING DO AN INSPECTION OF THE STRAP FOR CORRECT INSTALLATION AND DAMAGE BEFORE YOU USE IT. INCORRECT INSTALLATION OR DAMAGE CAN CAUSE THE STRAP TO BREAK. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(14) Install the retention strap, SPL-1871, with the marker facing up, to hold the shock strut [1] in its compressed position.

NOTE: Two straps are necessary to hold the inner cylinder secure.

SUBTASK 32-11-00-480-004

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(15) Lift the airplane with the jacks until the tires are off of the ground (TASK 07-10-01-580-801).

SUBTASK 32-11-00-200-002

(16) If necessary, do this task: Main Landing Gear Beam Hanger Link Free Play Check, TASK 57-16-06-200-802.

NOTE: Free play check is necessary, when the landing gear is replaced according to the scheduled restoration task (MPD Item Number 32-040-01 or 32-040-02). If the landing gear is not being replaced as a scheduled restoration task (MPD Item Number 32-040-01 or 32-040-02), then the free play check is optional.

SUBTASK 32-11-00-010-001

(17) Remove the applicable access panel:

(a) For the left landing gear, remove these access panels:

(TASK 57-50-02-000-801)

Number	Name/Location
551BB	Lower Inboard Fixed Trailing Edge, Gear Adjustment Door
551CB	Lower Inboard Fixed Trailing Edge, Gear Access Panel

(b) For the right landing gear, remove these access panels:

(TASK 57-50-02-000-801)

Number	Name/Location
651BB	Lower Inboard Fixed Trailing Edge, Gear Adjustment Door
651CB	Lower Inboard Fixed Trailing Edge, Gear Access Panel

(c) Open these access panels:

Number	Name/Location
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

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SUBTASK 32-11-00-020-003

(18) Do this task: Main Landing Gear Wheel and Tire Assembly Removal, TASK 32-45-11-000-801.

SUBTASK 32-11-00-020-004

(19) Do this task: Main Landing Gear Brake Removal, TASK 32-41-41-000-801.

SUBTASK 32-11-00-020-005

(20) Do this task: Main Landing Gear Inner Door Removal, TASK 32-13-11-000-802.

SUBTASK 32-11-00-020-006

(21) Do this task: Main Landing Gear Center Door Removal, TASK 32-13-11-000-801.

SUBTASK 32-11-00-020-007

(22) Do this task: Main Landing Gear Wing Door Removal, TASK 32-13-21-000-801.

SUBTASK 32-11-00-020-064

(23) Do one of these steps to remove the MLG side strut and the MLG reaction link assembly:

(a) If the side strut dolly, SPL-19644 is used, do this task: Main Landing Gear Side Brace Assembly Removal, TASK 32-11-62-000-801.

(b) Remove the MLG side strut and the MLG reaction link assembly as follows:

1) Do the applicable steps in this task to disconnect the lower side strut from the outer cylinder: Main Landing Gear Lower Side Strut Removal, TASK 32-11-61-000-802.

2) Do the applicable steps in this task to disconnect the reaction link assembly [30] from the outer cylinder trunnion [33]: Main Landing Gear Reaction Link Assembly Removal, TASK 32-11-71-000-801.

SUBTASK 32-11-00-020-010

(24) Remove the nuts [37], washers [35], washers [36], and bolts [34] to disconnect the turnbuckle [38].

SUBTASK 32-11-00-020-062

(25) Remove the nuts [83], washers [82], and bolts [81] that attach the bracket [80] to the MLG beam.

SUBTASK 32-11-00-020-063

(26) Remove the bolts [85] that attach the cable clamps [86] and clamp block [84] to the bracket.

SUBTASK 32-11-00-010-009

(27) Remove the clamps for the electrical conduit on the inboard side of the nut [45] and move the conduit to get clearance to remove the nut [45].

SUBTASK 32-11-00-020-011

(28) Disconnect the retract actuator [43] from the main gear trunnion.

(a) Remove the cotter pin [58], nut [47], and washer [46] from the bolt [44].

(b) Remove the bolt [44] from the actuator attach pin [50].

(c) Use the wrench adapter, C32029-7 or C32029-98, SPL-1363, to hold the head of the actuator attach pin [50].

(d) Remove the nut [45] and washer [48] from the actuator attach pin [50].

(e) Remove the wrench adapter, C32029-7 or C32029-98, SPL-1363, from the head of the actuator attach pin [50].

(f) Install the thread protector, C32029-16, SPL-1372, on the actuator attach pin [50].

(g) Remove the actuator attach pin [50] to disconnect the retract actuator [43] from the main gear trunnion.

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(h) Remove the thread protector, C32029-16, SPL-1372, from the actuator attach pin [50].

(i) Remove the spacers [49].

SUBTASK 32-11-00-410-010

(29) Put the electrical conduit back in its position and install the clamps.

SUBTASK 32-11-00-020-012

(30) Disconnect the walking beam [51] from the main gear trunnion.

(a) Remove the cotter pin [59], nut [47], and washer [46] from the bolt [54].

(b) Remove the bolt [54] from the walking beam attach pin [52].

(c) Use the wrench adapter, C32029-6 or C32029-96, SPL-1362, to hold the head of the walking beam attach pin [52].

(d) Remove the nut [55] and washer [53].

(e) Remove the wrench adapter, C32029-6 or C32029-96, SPL-1362, from the head of the walking beam attach pin [52].

(f) Install the thread protector, C32029-15, SPL-1371, on the walking beam attach pin [52].

(g) Remove the walking beam attach pin [52] to disconnect the walking beam [51] from the main gear trunnion.

(h) Remove the thread protector, C32029-15, SPL-1371, from the walking beam attach pin [52].

SUBTASK 32-11-00-480-005

(31) Use rope to hold the disconnected ends of the walking beam [51], and retract actuator [43], such that the ends will be out of the way when you remove the main landing gear [93].

SUBTASK 32-11-00-020-013

(32) Remove the hose guide [18].

SUBTASK 32-11-00-010-002

(33) Remove the applicable access panel:

(a) For the left wing, open this access panel:

Number	Name/Location
551BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel

(b) For the right wing, open this access panel:

Number	Name/Location
651BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel

F. Main Landing Gear Removal

SUBTASK 32-11-00-020-015

(1) Disconnect the brake hose assembly [6] and hydraulic return shimmy damper line [114].

(a) Install tags to identify the brake hose assembly [6] and hydraulic return shimmy damper line [114].

NOTE: This will prevent the crossing of the hoses during the installation.

(b) Disconnect the brake hose assembly [6] and hydraulic return shimmy damper line [114] from the elbow [5].

(c) Put plugs in the brake hose assembly [6] and hydraulic return shimmy damper line [114].

(d) Put caps on the elbow [5].

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(e) Move the brake hose assembly [6] and hydraulic return shimmy damper line [114] away from the work area.

SUBTASK 32-11-00-020-016

(2) Disconnect the electrical connector [11].

(a) Remove the lockwire from the electrical connector [11].

(b) Remove the screws [7], washers [8], clamps [9], and nuts [10] that hold the electrical connector [11] to the structure.

(c) Disconnect the electrical connector [11] from the electrical connector.

(d) Put cap on the electrical connector [11].

(e) Pull the electrical connector [11] out of the hose guide [12].

(f) Wind the electrical connector [11] into a loop.

SUBTASK 32-11-00-020-017

(3) Disconnect the electrical connector [13].

(a) Remove the lockwire from the electrical connector [13].

(b) Remove the screws [14], washers [15], clamps [16], and nut [17] that hold the electrical connector [13] to the hose guide [18].

(c) Disconnect the electrical connector [13] from the electrical connector.

(d) Put cap on the electrical connector [13].

(e) Wind the electrical connector [13] into a loop.

SUBTASK 32-11-00-020-019

(4) Remove the aft trunnion cross bolt [23].

(a) Remove the pin [20], nut [21], and washer [22] from the aft trunnion cross bolt [23].

1) If it is necessary, do these steps to get access to the nut [21]:

a) Remove the nuts [104], washers [103], washers [102], and bolts [101] to disconnect the turnbuckle [100].

b) Remove the nuts [90], washers [89], and bolts [88] that attach the bracket [87] to the MLG beam.

2) Use the wrench adapter, C32029-48 or C32029-90, SPL-14021, to hold the head of the aft trunnion cross bolt [23].

(b) Install the thread protector, C32029-10, SPL-1366, on the aft trunnion cross bolt [23].

(c) Remove the aft trunnion cross bolt [23] to disconnect the aft trunnion pin [19] from the aft trunnion.

(d) Remove the thread protector, C32029-10, SPL-1366, from the aft trunnion cross bolt [23].

SUBTASK 32-11-00-020-020

(5) Remove the forward trunnion cross bolt [28].

(a) Remove the pin [24], nut [25], and washer [26] from the forward trunnion cross bolt [28].

1) Use the wrench adapter, C32029-49 or C32029-88, SPL-14022, to hold the head of the forward trunnion cross bolt [28].

(b) Install the thread protector, C32029-9, SPL-1365, on the forward trunnion cross bolt [28].

(c) Remove the forward trunnion cross bolt [28] to disconnect the forward trunnion from the forward trunnion bearing pin [29].

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(d) Remove the thread protector, C32029-9, SPL-1365, from the forward trunnion cross bolt [28].

SUBTASK 32-11-00-480-006

CAUTION

MAKE SURE THAT THE FORWARD TRUNNION BEARING PIN DOES NOT MOVE ON THE PROTECTOR ASSEMBLY DURING THIS PROCEDURE. IF THE PIN MOVES DURING THE REMOVAL OR INSTALLATION OF THE MAIN LANDING GEAR, DAMAGE TO THE PIN CAN OCCUR.

(6) Install the protector assembly [92] (C32034-228/-235).

NOTE: The protector assembly (C32034–228/-235) is part of this tool set R/I equipment, SPL-1869.

NOTE: The protector assembly (C32034–228/-235) will prevent damage to the forward trunnion bearing pin during removal or installation of the main landing gear.

(a) Put the protector assembly [92] (C32034-228/-235) over the forward trunnion bearing pin [29].

(b) Secure in place with tape.

SUBTASK 32-11-00-480-007

(7) Attach the support assembly [96] (C32034-212) to the main landing gear [93].

NOTE: A minimum of two persons are necessary to do this task.

(a) Make sure that the side strut attach assembly [94] (C32034-14) is not attached to the support assembly [96] (C32034-212), and the jackball pin is removed from the hole in the jackball fitting.

NOTE: The jackball fitting is located on the support assembly (C32034-212).

(b) Put the side strut attach assembly [94] (C32034-14) into the lower side strut attach point on the main landing gear [93].

1) Make sure that the stenciled plate of the side strut attach assembly [94] (C32034-14) is in the vertical and up position.

2) Install the washer and nut, finger tight, on the threaded end of the side strut attach assembly [94] (C32034-14).

(c) Put the support assembly [96] (C32034-212) in its position on the aft side of the main landing gear [93].

1) Make sure that the jackball fitting on the support assembly is fully engaged on the jacking ball of the main landing gear [93].

(d) If it is necessary, align the holes in the support assembly [96] (C32034-212) with the jacking ball hole for the installation of the jackball pin.

(e) Put the jackball pin through the jackball fitting and the jacking ball of the main landing gear [93].

(f) Install the lynch pin in the jackball pin.

(g) Loosen the nut that holds the side strut attach assembly [94] (C32034-14) such that the stenciled plate can be attached to the studs on the side strut support assembly [97] (C32034-12).

1) If it is necessary, adjust the adjustable square support tube of the side strut support assembly [97] (C32034-12).

(h) Tighten the two hand knobs to hold the stenciled plate to the studs on the side strut support assembly [97] (C32034-12).

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(i) Tighten the nut that holds the side strut attach assembly [94] (C32034-14) to the lower side strut attach point.

SUBTASK 32-11-00-840-001

CAUTION

MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION

MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(8) Lift the airplane with the jacks until the lower surface of the housing assembly, for the forward trunnion bearing pin [29], is 97 ±2 in. (246 ±5 cm) above the ground (TASK 07-10-01-580-801) (View B-B, Figure 401).

(a) Make sure that to keep the airplane level.

SUBTASK 32-11-00-480-008

(9) Attach the dolly assembly [98] (C32034-213) to the support assembly [96] (C32034-212).

NOTE: A minimum of two persons are necessary to do this task.

(a) If it is installed, remove the transportation support assembly and the aft tire support assembly from the dolly assembly [98] (C32034-213).

(b) From the aft side, carefully move the dolly assembly [98] (C32034-213) in the forward direction under the wing.

(c) Push the dolly assembly [98] (C32034-213) aft such that the boom arm [91] (C32034-214) points forward of the main landing gear [93] and the legs are parallel to the fuselage and in the center with the main landing gear [93].

(d) Lift or lower the boom arm [91] (C32034-214) until you can connect the lift links [95] (C32034-34) to the support assembly [96] (C32034-212).

(e) Install the pins to hold the lift links [95] (C32034-34) to the support assembly [96] (C32034-212).

(f) Connect the winch assembly [99] (C32034-207) to the support assembly [96] (C32034-212) with the snapper pin.

SUBTASK 32-11-00-020-022

WARNING

MAKE SURE THAT THE HYDRAULIC PRESSURE IS BELOW 5000 PSIG (34,474 KPA). IF YOU CONTINUE WHEN THE HYDRAULIC PRESSURE IS 5000 (34,474 KPA) OR HIGHER, A BINDING CONDITION WILL OCCUR. A BINDING CONDITION CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(10) Prepare the dolly assembly [98] (C32034-213) to disconnect the aft trunnion.

(a) Operate the hydraulic pump on the dolly assembly [98] (C32034-213) until approximately 1500 psi (10,342 kPa) is shown on the gage.

1) Make sure that the dolly assembly [98] (C32034-213) stays in center with the main landing gear [93].

(b) Make sure that the swivel locks for the single caster wheels are released.

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(c) Use the pry lever bar to remove the load off of the caster wheels one at a time.

(d) Use the lever bar assemblies to turn the single caster wheels parallel to the legs of the dolly assembly [98] (C32034-213).

1) Lock the swivel locks for the single wheel casters.

(e) Operate the hydraulic pump on the dolly assembly [98] (C32034-213) to decrease the weight of the main landing gear from the aft trunnion pin [19].

NOTE: The amount of pressure required to perform this step will depend on how many components are installed on the gear at the time of removal, and it will vary approximately between 2500 psi (17,237 kPa) to 4500 psi (31,026 kPa).

(f) Put chocks on all the caster wheels.

SUBTASK 32-11-00-480-009

(11) Use the pin R/I equipment, SPL-1862, to push the aft trunnion pin [19] into the aft trunnion.

(a) Attach the push adapter [75] to the shaft [71].

(b) Put the push adapter [75] through the pin access hole in the main landing gear beam.

(c) Push the slide hammer [72] until the aft trunnion pin [19] is disconnected from the aft trunnion bearing and it is inside the aft trunnion.

SUBTASK 32-11-00-480-010

(12) Use tie wraps to hold the aft trunnion pin [19] inside the aft trunnion.

NOTE: This will prevent the aft trunnion pin from falling out when you do the subsequent steps.

SUBTASK 32-11-00-950-001

(13) Put protective tape on the outer edge of the aft trunnion to protect its surface finish from scratching.

SUBTASK 32-11-00-420-002

WARNING

MAKE SURE THAT THE HYDRAULIC PRESSURE IS BELOW 5000 PSIG (34,474 KPA). IF YOU CONTINUE WHEN THE HYDRAULIC PRESSURE IS 5000 (34,474 KPA) OR HIGHER, A BINDING CONDITION WILL OCCUR. A BINDING CONDITION CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(14) Lower the aft trunnion.

NOTE: The initial main landing gear removal path is done with the forward trunnion bearing pin attached to the forward trunnion. This removal path follows a combination of movements from lowering and moving the aft trunnion inboard while the main landing gear pivots about the forward trunnion bearing pin.

(a) Make sure to lock the swivel locks for the caster wheels when the main landing gear [93] pivots about the forward trunnion bearing pin [29].

1) Use the pry lever bar to remove the load off of the caster wheels one at a time.

2) Lock the single wheel casters perpendicular to the forward trunnion pivot axis approximately 38 degrees, and 90 degrees for the double wheel caster.

(b) Put the fork end of the lever bar assemblies on the rectangular tubing above the single wheel casters.

(c) Hold the lever bar assemblies to slowly push the aft end of the dolly assembly [98] (C32034-213) inboard.

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1) Make sure that the swivel locks for all the caster wheels are released.

(d) To lower the aft trunnion inboard while at the same time the aft end of the main landing gear [93] moves inboard, do these steps:

NOTE: When lifting or lowering the aft trunnion, put the single caster wheels parallel to the dolly assembly legs and lock the swivel locks.

1) Use the winch assembly [99] (C32034-207) to pull the bottom of the main landing gear [93] forward.

2) Use the manual valve on the hydraulic pump to lower the boom arm [91] (C32034-214).

3) Continue to lower and to move the aft trunnion inboard until it clears these components:

a) The lower flange of the main landing gear beam

b) The seals for the lower trailing edge panel

c) The seal for the support beam

d) And the aft trunnion is just below the lower trailing edge panel.

(e) If the forward tire support assembly touches the tires at this time, remove it.

SUBTASK 32-11-00-020-023

WARNING MAKE SURE THAT THE HYDRAULIC PRESSURE IS BELOW 5000 PSIG (34,474 KPA). IF YOU CONTINUE WHEN THE HYDRAULIC PRESSURE IS 5000 (34,474 KPA) OR HIGHER, A BINDING CONDITION WILL OCCUR. A BINDING CONDITION CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(15) Disconnect the forward trunnion.

(a) Make sure that the swivel locks for the caster wheels are released.

(b) Use the lever bar assemblies to push the dolly assembly [98] (C32034-213) outboard such that the axis of the forward trunnion bearing pin [29] is perpendicular to the housing assembly and the rear spar.

(c) Use the pry lever bar to remove the load off of the caster wheels one at a time.

(d) If it is necessary, use the lever bar assemblies and put the single caster wheels parallel to the legs of the dolly assembly [98] (C32034-213).

1) Lock the swivel locks for the single caster wheels.

(e) Attach the outer cylinder strap assembly to the hoist ring on the end of the boom arm [91] (C32034-214) and around the outer cylinder at the aft trunnion.

(f) Remove the tension from the strap.

(g) Put the chocks approximately 2 in. (5 cm) aft of the single wheel casters, and in contact with the double wheel caster to limit the movement of the dolly assembly [98] (C32034-213).

(h) Make sure that the chock is in contact with the double wheel caster to prevent the movement of the dolly assembly [98] (C32034-213).

(i) Put the fork end of the lever bar assemblies on the rectangular tubing above the single wheel casters.

(j) Hold the lever bar assemblies to slowly push on the dolly assembly [98] (C32034-213), such that the forward trunnion will come off of the forward trunnion bearing pin [29].

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1) Make sure that the swivel locks for all the caster wheels are released.

(k) Continue to push the dolly assembly [98] (C32034-213) while lowering the aft end of the main landing gear [93] with the use of the winch assembly [99] (C32034-207) and the boom arm [91] (C32034-214).

(l) Continue this movement until the aft trunnion will clear the lower trailing edge panel seals, the forward trunnion is disconnected from the forward trunnion bearing pin [29].

NOTE: The retaining ring and the seal are still on the airplane until the forward trunnion is disconnected.

(m) Remove the outer cylinder strap assembly from the hoist ring and the outer cylinder.

(n) Use the boom arm [91] (C32034-214) and winch assembly [99] (C32034-207) to lower the main landing gear [93], such that the dolly assembly [98] (C32034-213), with the main landing gear [93] can be removed below the wing.

SUBTASK 32-11-00-580-001

(16) Lower the main landing gear [93] on the tire support assemblies such that the support assembly [96] (C32034-212) rests in the cradle of the transportation support assembly.

SUBTASK 32-11-00-480-011

(17) Use the retention strap, SPL-1871, to hold the tires and the main landing gear [93].

SUBTASK 32-11-00-480-012

(18) Tighten the adjustable square support tube of the side strut support assembly [97] (C32034-12) with the two adjustable handles.

SUBTASK 32-11-00-020-024

CAUTION

DO NOT TOW THE MAIN LANDING-GEAR SUPPORT EQUIPMENT MORE THAN 5 MPH (8 KM/H). THE MAIN LANDING-GEAR SUPPORT EQUIPMENT IS FOR SHOP OPERATION ONLY. IF YOU DO NOT OBEY, DAMAGE TO THE EQUIPMENT CAN OCCUR.

CAUTION

INSTALL THE RETENTION STRAP OR EQUIVALENT ON THE SHOCK STRUT WHEN YOU REMOVE THE MAIN LANDING-GEAR FIXTURE. THE SHOCK STRUT CAN EXTEND AND DAMAGE TO THE EQUIPMENT CAN OCCUR.

(19) Remove the dolly assembly [98] (C32034-213), with the main landing gear [93], from the work area.

SUBTASK 32-11-00-210-013

(20) Do a general visual inspection of the Main Landing Gear (MLG) at the forward trunnion bearing pin [29] and aft trunnion pin [19] locations and the visible surfaces of the forward trunnion bearing pin [29] and aft trunnion pin [19] for signs of corrosion and chrome plating damage.

(a) If any sign of corrosion or chrome plating damage is found for the forward trunnion bearing pin [29], do these tasks:

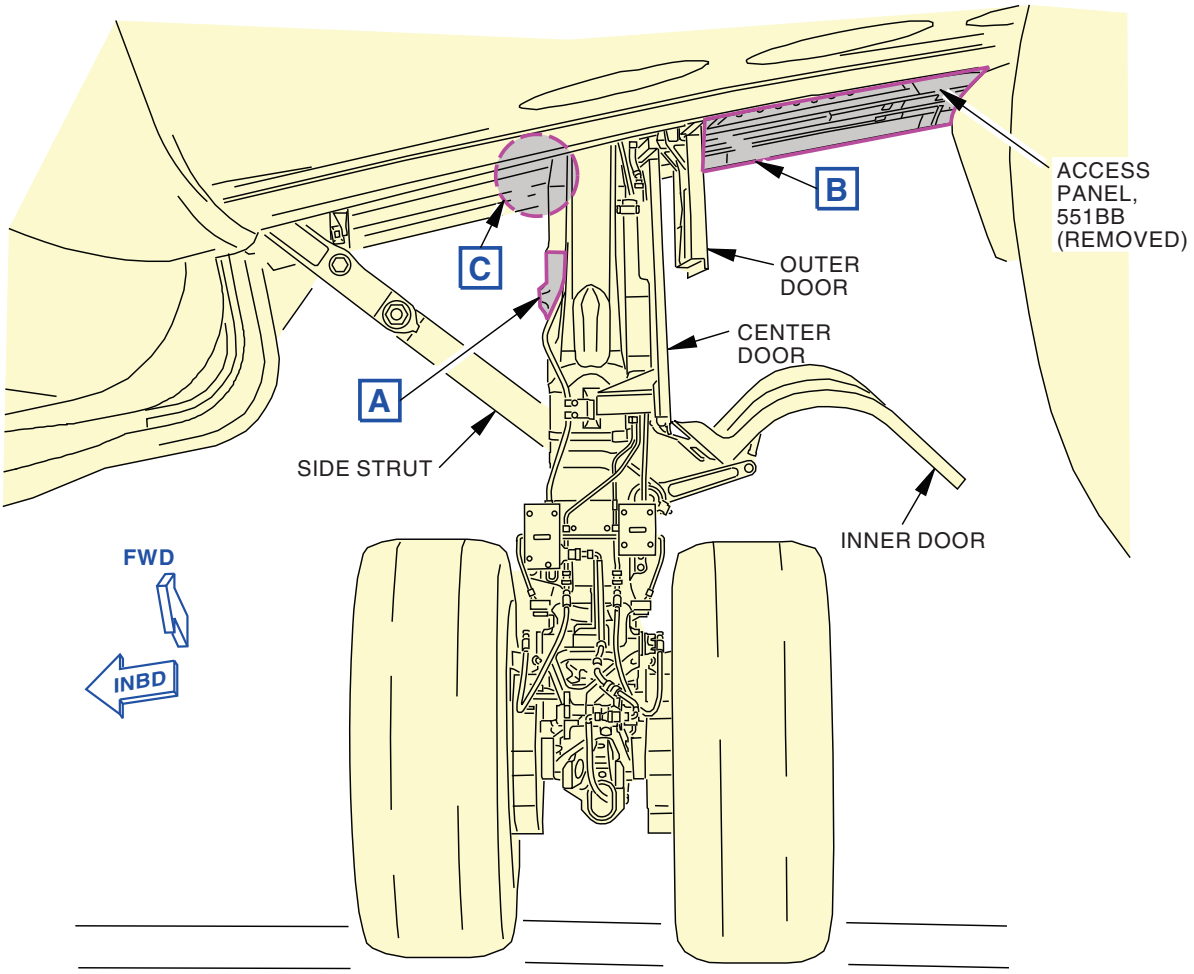
- MLG Forward Trunnion Housing Assembly Removal, TASK 57-16-01-000-801
- MLG Forward Trunnion Housing Assembly Installation, TASK 57-16-01-400-801.

(b) If any sign of corrosion or chrome plating damage is found for the aft trunnion pin [19], do these tasks:

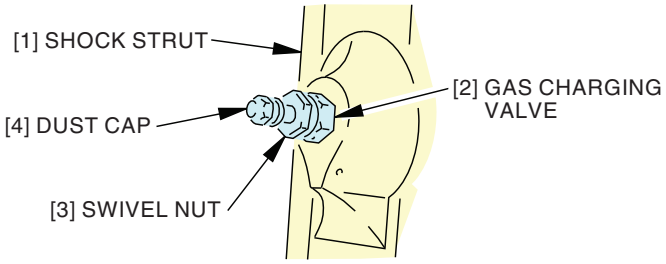
- MLG Aft Trunnion Bearing Assembly Removal, TASK 57-16-02-000-801
- MLG Aft Trunnion Bearing Assembly Installation, TASK 57-16-02-400-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT MAIN LANDING GEAR
(RIGHT MAIN LANDING GEAR IS EQUIVALENT)



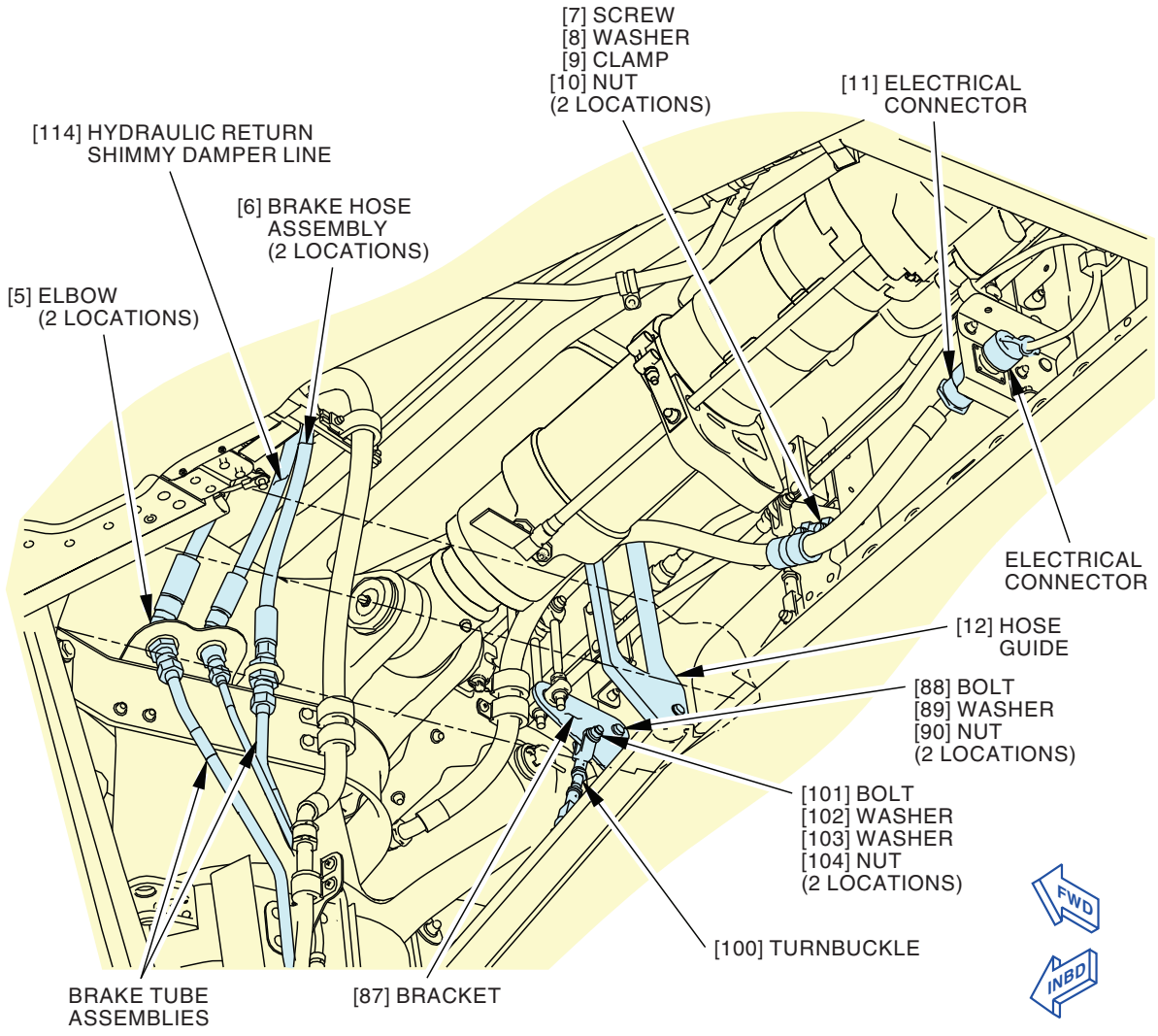
SHOCK STRUT INFLATION VALVE

A

2414047 S00061534299_V2

Main Landing Gear Installation
Figure 401/32-11-00-990-802 (Sheet 1 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



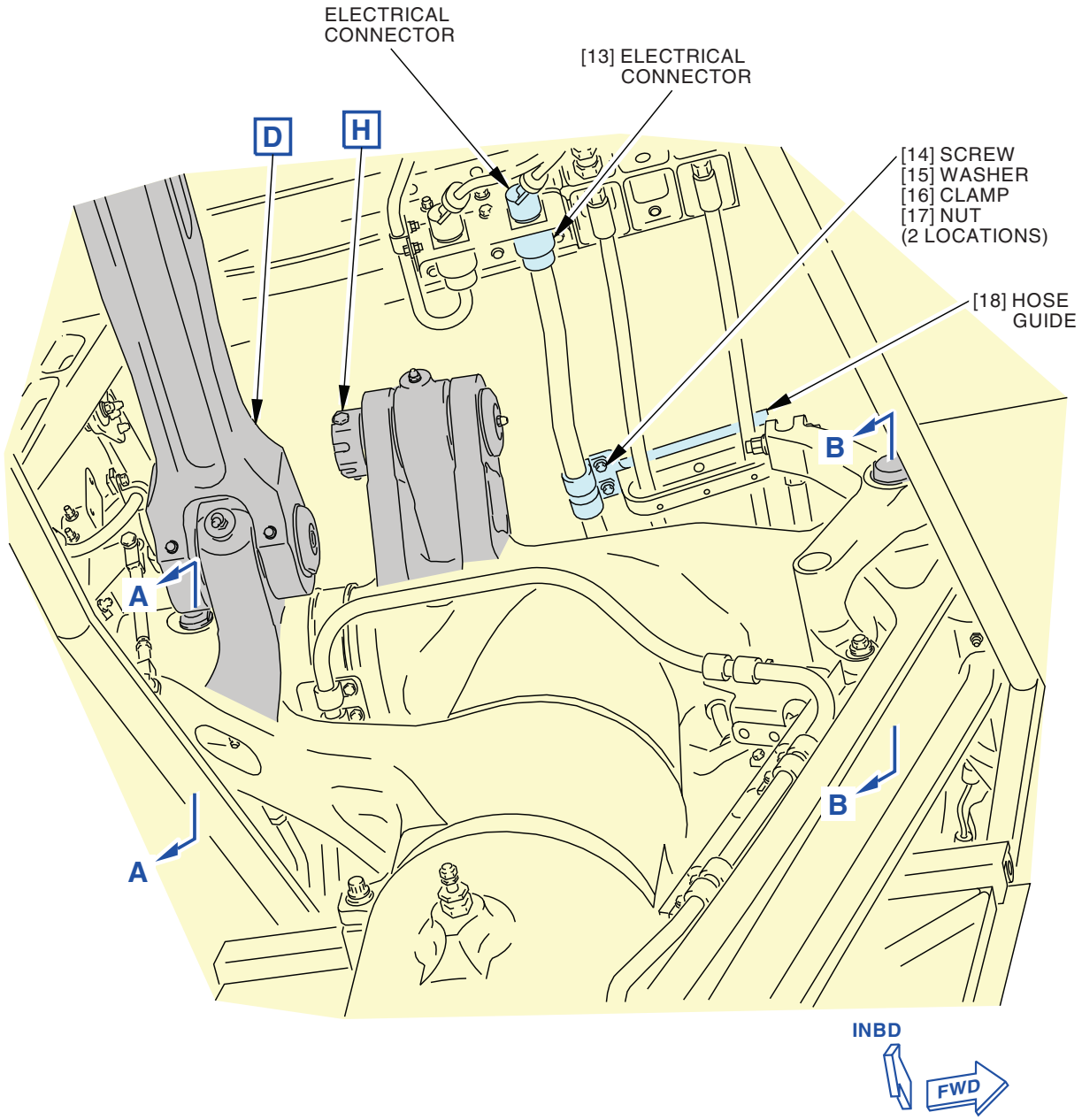
HYDRAULIC LINES AND ELECTRICAL CONNECTORS

B

2414048 S00061534300_V3

Main Landing Gear Installation
Figure 401/32-11-00-990-802 (Sheet 2 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

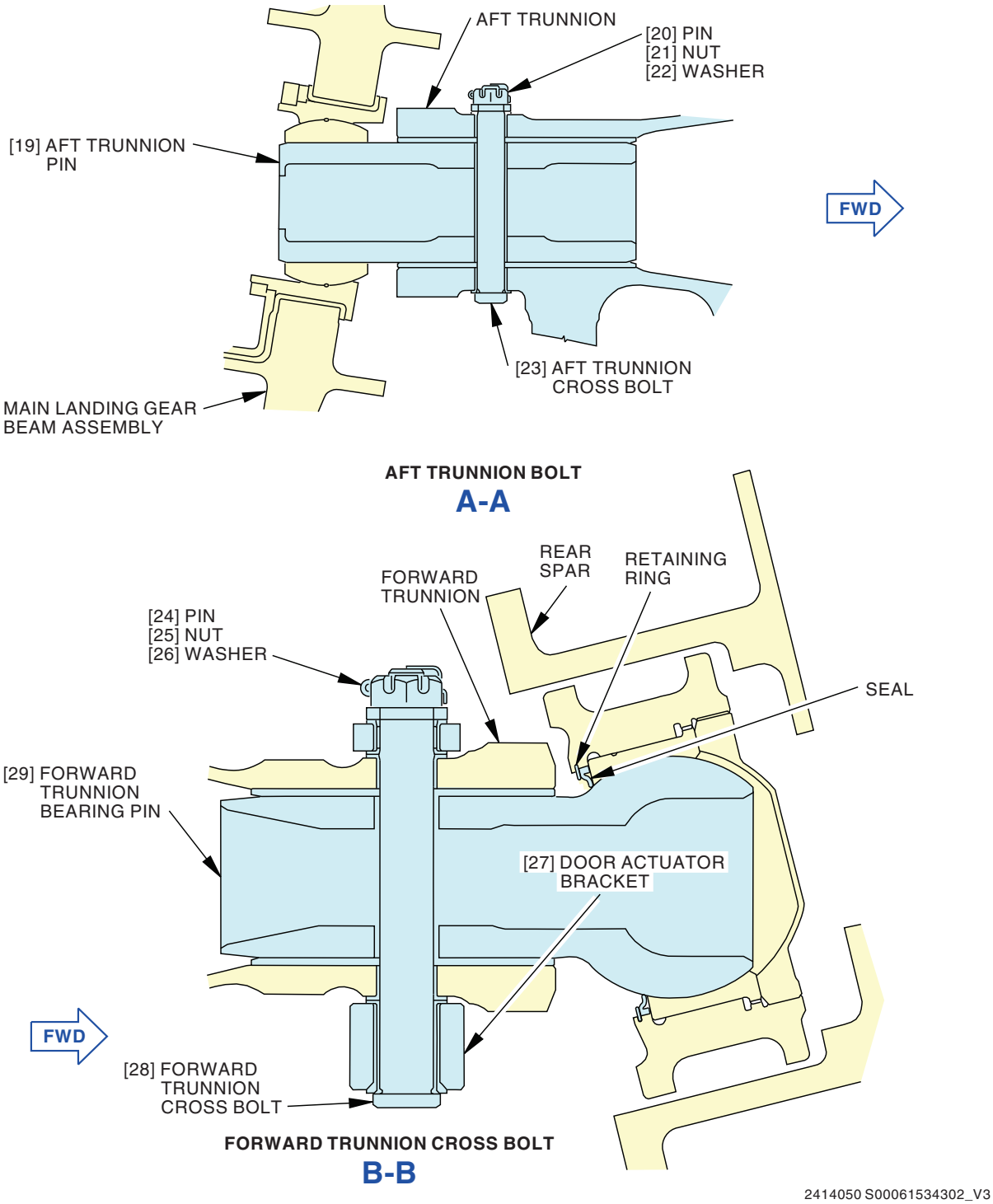


ELECTRICAL CONNECTOR AND TRUNNION SUPPORTS

2414049 S00061534301_V3

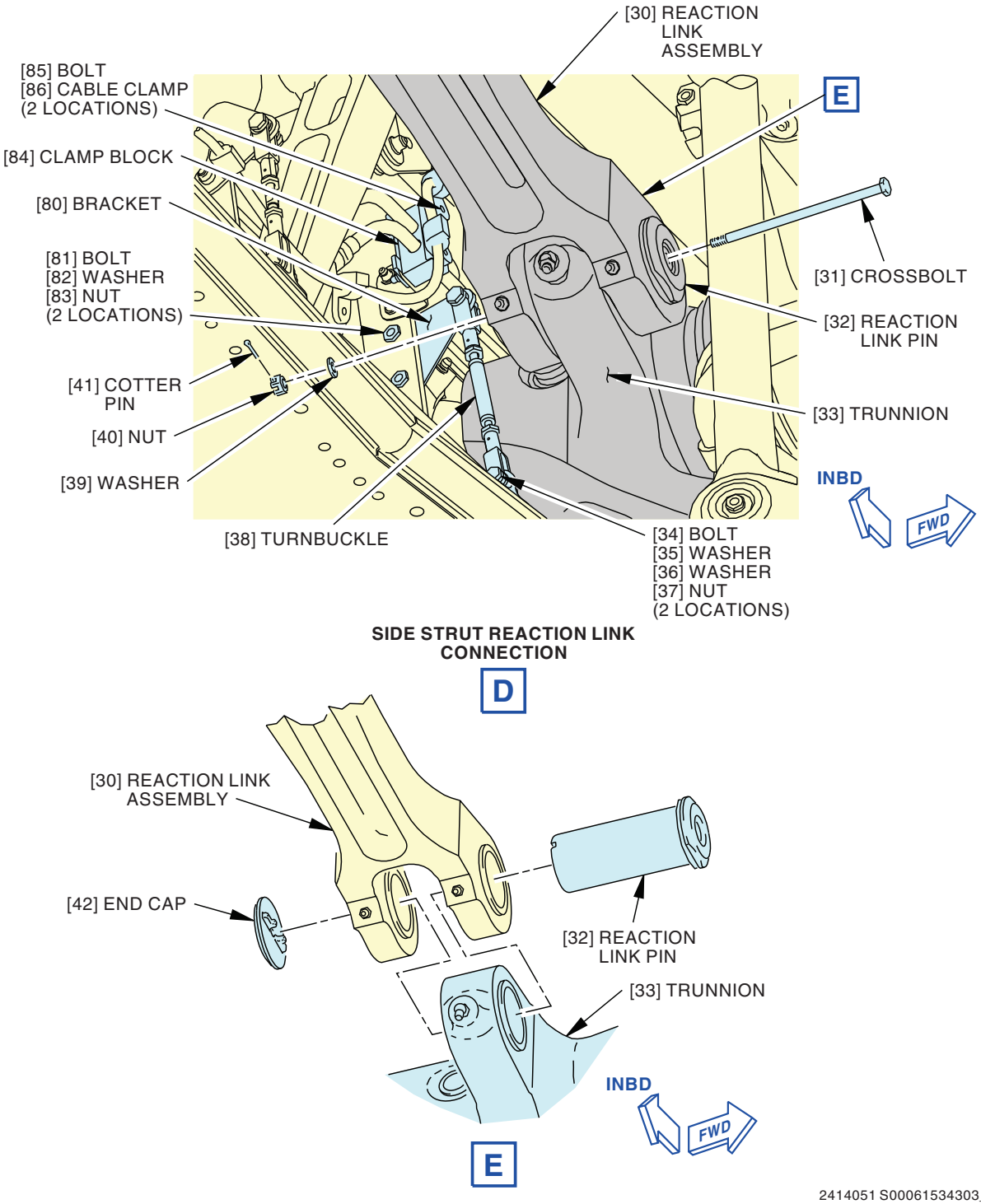
Main Landing Gear Installation
Figure 401/32-11-00-990-802 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



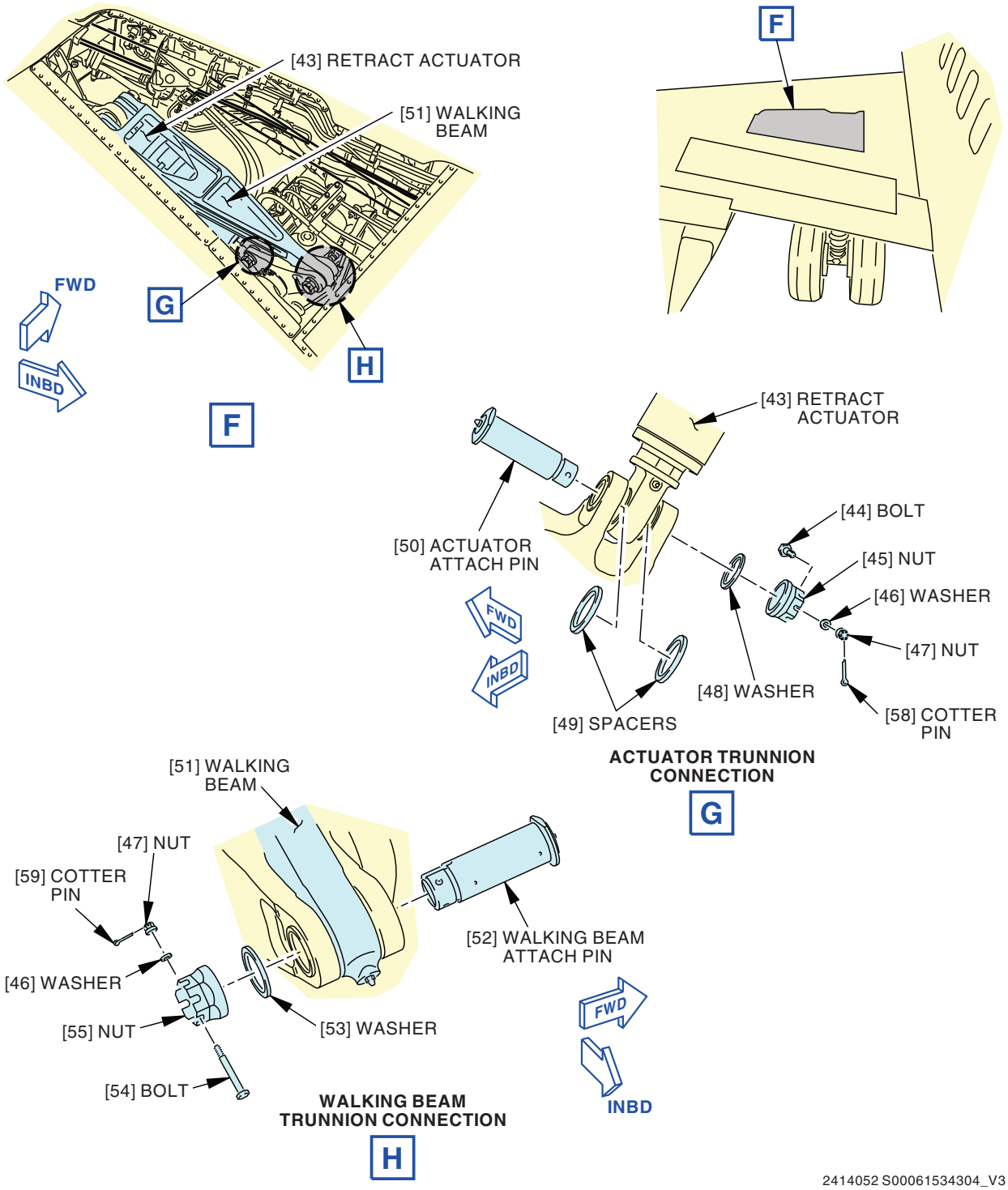
Main Landing Gear Installation
Figure 401/32-11-00-990-802 (Sheet 4 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Installation
Figure 401/32-11-00-990-802 (Sheet 5 of 6)

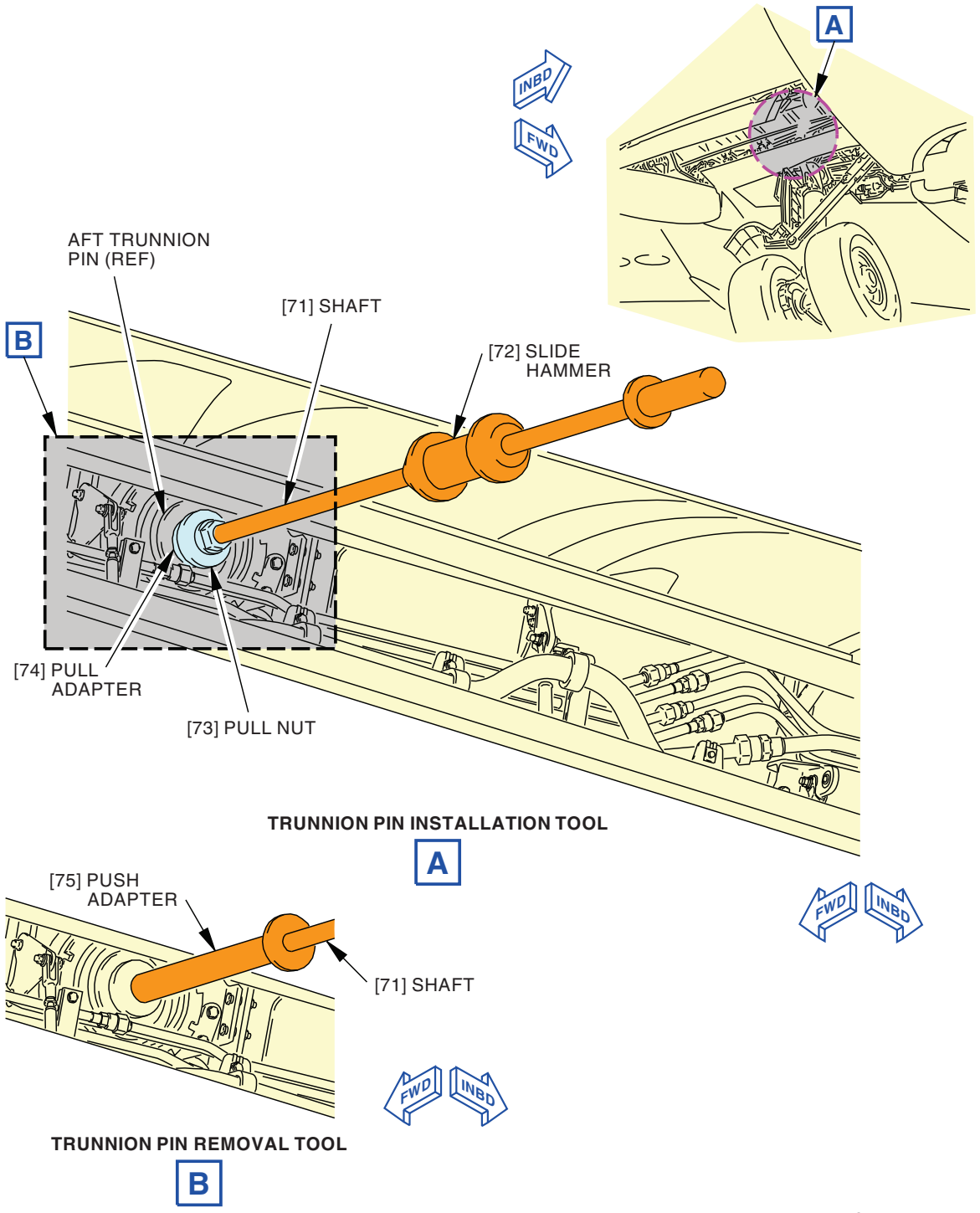
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414052 S00061534304_V3

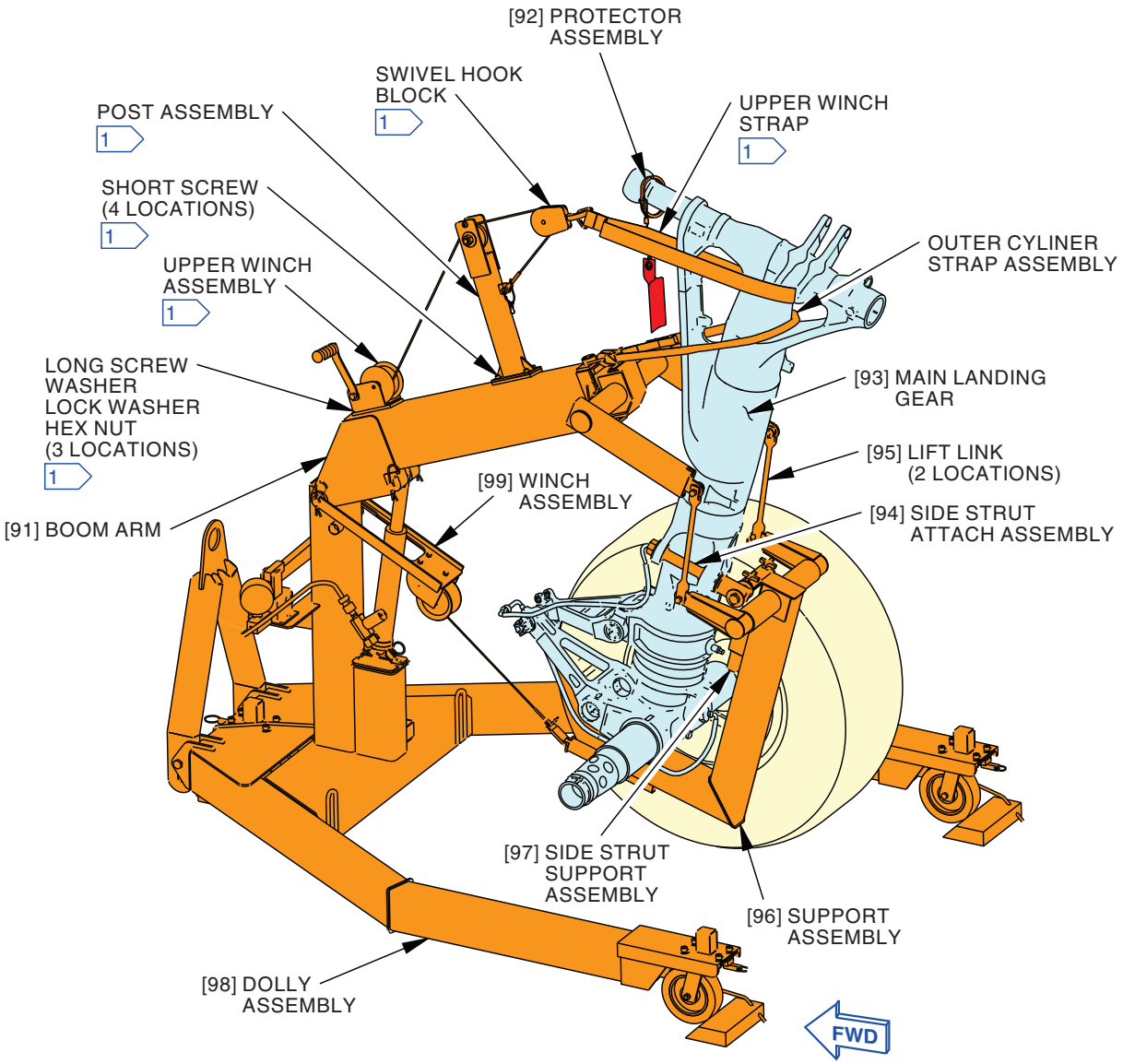
Main Landing Gear Installation
Figure 401/32-11-00-990-802 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Trunnion Pin Puller Equipment
Figure 402/32-11-00-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

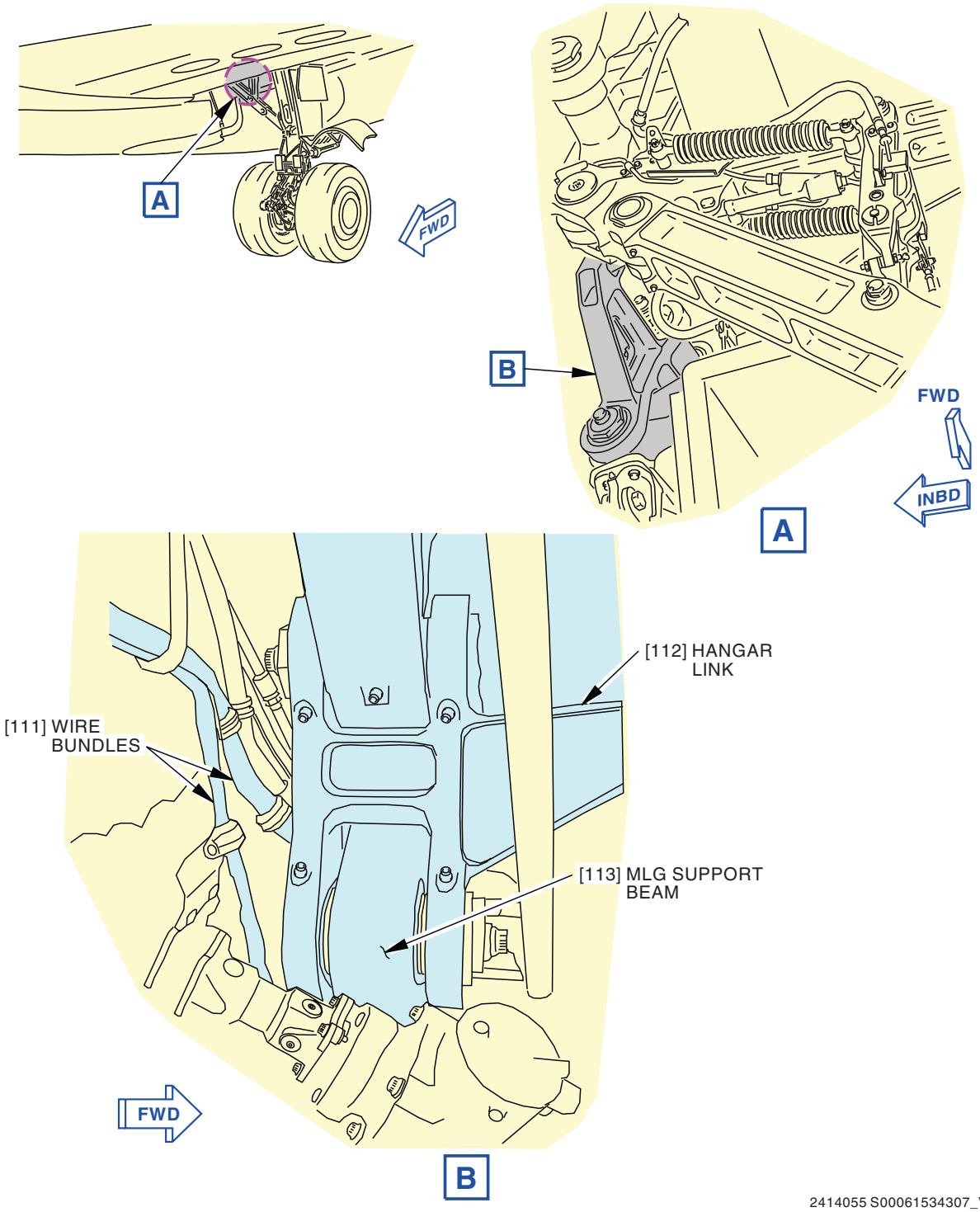


1 REMOVABLE PARTS TO SUPPORT MLG INSTALLATION. NOT NEEDED DURING MLG REMOVAL.

2414054 S00061534306_V3

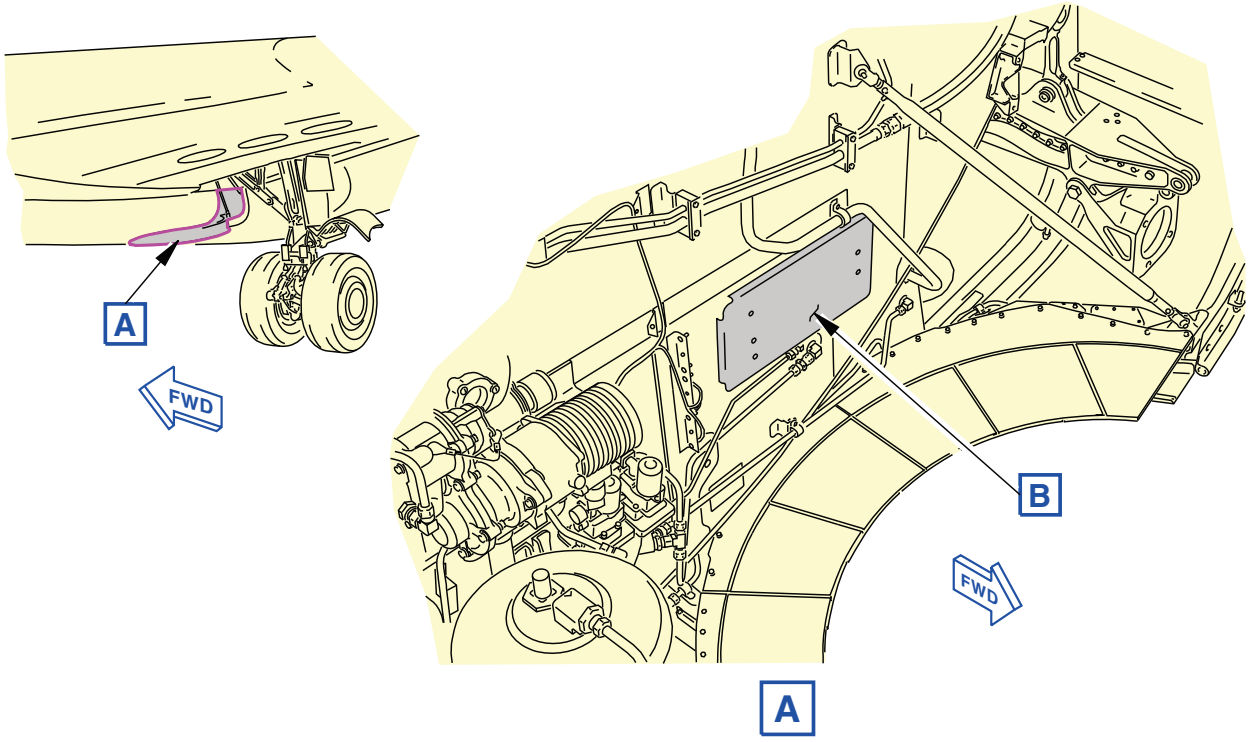
Main Landing Gear Support Equipment
Figure 403/32-11-00-990-804

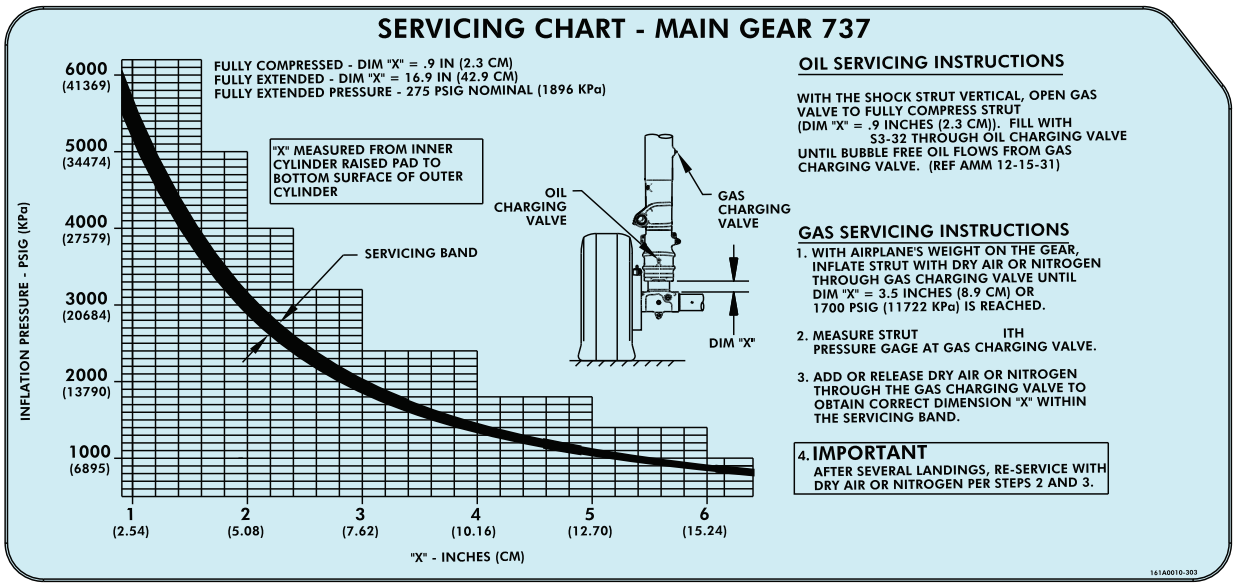
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Clearance Requirements
Figure 404/32-11-00-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details





2410803 S00061526401_V2

Main Landing Gear Shock Strut Servicing Chart
Figure 405/32-11-00-990-816

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-11-00-400-802

3. Main Landing Gear Installation

(Figure 401, Figure 402, Figure 403, Figure 404, and Figure 405)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
12-15-31-610-802	Main Landing Gear Shock Strut Servicing, Airplane on the Ground (P/B 301)
12-15-31-610-803	Main Landing Gear Strut Servicing, Airplane on Jacks (P/B 301)
12-21-11-640-801	Main Landing Gear Upper End Components Servicing (P/B 301)
12-21-11-640-802	Main Landing Gear Lower End Components Servicing (P/B 301)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-62-00-800-801	Apply Pressure to the Speed Brake Hydraulic Systems A and B (P/B 201)
27-62-00-840-801	Shut Down the Speed Brake Hydraulic Systems A and B (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-11-61-400-802	Main Landing Gear Lower Side Strut Installation (P/B 401)
32-11-62-400-801	Main Landing Gear Side Brace Assembly Installation (P/B 401)
32-11-71-420-801	Main Landing Gear Reaction Link Assembly Installation (P/B 401)
32-11-81-870-801	Hydraulic Shimmy Damper Bleeding (P/B 401)
32-11-83-000-801	Main Landing Gear Forward Trunnion Bearing Assembly Removal (P/B 401)
32-11-83-400-801	Main Landing Gear Forward Trunnion Bearing Assembly Installation (P/B 401)
32-13-11-420-801	Main Landing Gear Center Door Installation (P/B 401)
32-13-11-420-802	Main Landing Gear Inner Door Installation (P/B 401)
32-13-11-820-801	Main Landing Gear Shock Strut Door Adjustment (P/B 501)
32-13-21-000-802	Main Landing Gear Wing Door Adjustment (P/B 501)
32-13-21-420-801	Main Landing Gear Wing Door Installation (P/B 401)
32-32-00-710-802	Main Landing Gear - Operational Test (P/B 501)
32-34-00-730-801	Main Gear Manual Extension System Test (Airplane on Jacks) (P/B 501)
32-41-00-870-802	Normal (System B) Hydraulic Brake System Bleeding (P/B 201)
32-41-41-400-801	Main Landing Gear Brake Installation (P/B 401)
32-42-00-710-803	Transducer Operational Test (P/B 501)
32-42-00-720-803	Antiskid Valve Operational Test (P/B 501)

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(Continued)

Reference	Title
32-45-11-400-801	Main Landing Gear Wheel and Tire Assembly - Installation (P/B 401)
32-46-00-730-801	Tire and Brake Monitoring System - System Test (P/B 501)
57-50-02-400-801	Inboard Fixed Trailing Edge Lower Panel Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1362	Wrench Adapter, C32029-6 or C32029-96 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1363	Wrench Adapter, C32029-7 or C32029-98 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1365	Thread Protector, C32029-9 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1366	Thread Protector, C32029-10 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1371	Thread Protector, C32029-15 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1372	Thread Protector, C32029-16 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1521	Strut Inflation Tool - Landing Gear Part #: F70200-35 Supplier: 81205 Opt Part #: F70200-1 Supplier: 81205 Opt Part #: F70200-14 Supplier: 81205 Opt Part #: F70200-17 Supplier: 81205 Opt Part #: F70200-18 Supplier: 81205
SPL-1744	Lock Set - Inboard Ground Spoiler Actuator (Contains 4 Lock Assemblies) Part #: C27046-13 Supplier: 81205 Opt Part #: C27046-1 Supplier: 81205
SPL-1862	Removal/Installation Equipment - Pin, Aft Trunnion MLG Part #: C32031-22 Supplier: 81205 Opt Part #: C32031-1 Supplier: 81205
SPL-1869	Removal/Installation Equipment - Main Landing Gear Part #: C32034-343 Supplier: 81205 Opt Part #: C32034-227 Supplier: 81205 Opt Part #: C32034-273 Supplier: 81205 Opt Part #: C32034-339 Supplier: 81205

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(Continued)

Reference	Description
SPL-1871	Retention Straps - Shock Strut, NLG/MLG Part #: C32030 -31 Supplier: 81205 Opt Part #: C32030-10 Supplier: 81205
SPL-14021	Wrench Adapter, C32029-48 or C32029-90 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-14022	Wrench Adapter, C32029-49 or C32029-88 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-14023	Wrench Adapter, C32029-50 or C32029-89 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-14024	Thread Protector , C32029-51 (included in C32029 Kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-19644	Dolly - Side Strut, MLG Part #: C32078-1 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33
G00018	Nitrogen - Gaseous, Pressurizing, 99.5 Percent Pure	A-A-59503 Type I Grade B, MIL-PRF-27401 Type I Grade A
G02166	Lockwire - MS20995NC20, Monel - 0.020 Inch (0.508 mm) Diameter	NASM20995
G02314	Air - Compressed, Breathing	BB-A-1034 Source I Grade A
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
20	Pin	32-11-00-05-010	PTF ALL
24	Pin	32-11-00-05-010	PTF ALL
58	Cotter pin	32-32-11-01-010	PTF ALL
59	Cotter pin	32-32-93-01-030	PTF ALL

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
734	Left Main Landing Gear
744	Right Main Landing Gear

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F. Access Panels

Number	Name/Location
551BB	Lower Inboard Fixed Trailing Edge, Gear Adjustment Door
551BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel
551CB	Lower Inboard Fixed Trailing Edge, Gear Access Panel
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel
651BB	Lower Inboard Fixed Trailing Edge, Gear Adjustment Door
651BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel
651CB	Lower Inboard Fixed Trailing Edge, Gear Access Panel
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

G. Prepare for the Installation

SUBTASK 32-11-00-640-001

(1) Apply the lubricant grease, D00633, to the following parts and areas:

(a) Apply to the aft trunnion pin [19] and forward trunnion bearing pin [29] bushing or ball inner diameter to completely fill all lubrication grooves.

(b) Apply to the AFT and FORWARD trunnion bushing lubrication fittings until grease appears at the bushing inner diameter through holes.

(c) If not already done, remove the aft trunnion pin [19] from the aft trunnion and lubricate the chrome plated surfaces.

1) Make sure that the aft trunnion pin [19] moves easily.

2) If it is necessary, lubricate the aft trunnion pin [19] again.

3) Put the aft trunnion pin [19] into the aft trunnion.

(d) The chrome plated surfaces of the forward trunnion bearing pin [29].

SUBTASK 32-11-00-160-013

(2) Remove all unwanted grease, D00633.

SUBTASK 32-11-00-480-013

CAUTION MAKE SURE THAT THE FORWARD TRUNNION BEARING PIN DOES NOT MOVE ON THE PROTECTOR ASSEMBLY DURING THIS PROCEDURE. IF THE PIN MOVES DURING THE REMOVAL OR INSTALLATION OF THE MAIN LANDING GEAR, DAMAGE TO THE PIN CAN OCCUR.

(3) If the protector assembly [92] (C32034-228/-235) from the R/I equipment, SPL-1869, is not already installed, do the steps that follow:

NOTE: The protector assembly will prevent damage to the forward trunnion bearing pin during the installation of the main landing gear.

NOTE: For more protection of the transition radius of the forward trunnion bearing pin, apply tape around the forward trunnion bearing pin before the main landing gear installation. Remove the tape after installation.

NOTE: The retaining ring and the seal should be attached on the forward trunnion bearing pin before the forward trunnion is connected.

(a) Put the upper end of the protector assembly [92] (C32034-228/-235) in the large clearance between the forward trunnion bearing pin [29] and the housing assembly.

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(b) Attach the lower end of the protector assembly [92] (C32034-228/-235) over the bottom of the housing assembly, such that you hear a click when you attach it.

SUBTASK 32-11-00-480-014

(4) Install the retention strap, SPL-1871, with the marker facing up, to hold the shock strut [1] in its compressed position.

NOTE: Two straps are necessary to hold the inner cylinder secure.

SUBTASK 32-11-00-480-015

(5) Use tie wraps to hold the aft trunnion pin [19] inside the aft trunnion.

NOTE: This will prevent the aft trunnion pin from falling out of the outer cylinder trunnion when you do the subsequent steps.

SUBTASK 32-11-00-950-002

(6) Put protective tape on the outer edge of the aft trunnion to protect its surface finish from scratching.

SUBTASK 32-11-00-860-016

(7) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

H. Main Landing Gear Installation

SUBTASK 32-11-00-480-016

(1) Attach the support assembly [96] and the side strut attach assembly [94] to the main landing gear [93].

NOTE: A minimum of two persons are necessary to do this task.

(a) Make sure that the side strut attach assembly [94] is not attached to the support assembly [96].

(b) Make sure that the jackball pin is removed from the hole in the jackball fitting of the support assembly [96].

(c) Put the side strut attach assembly [94] into the lower side strut attach point on the main landing gear [93], and such that the stenciled plate is in the vertical position.

1) Install the washer on the threaded end of the side strut attach assembly [94].

2) Make sure that the stenciled plate is still in a vertical position and install the nut finger tight.

(d) Put the support assembly [96] in its position on the aft side of the main landing gear [93].

1) Make sure that the jackball fitting on the support assembly [96] is fully engaged on the jacking ball of the main landing gear [93].

(e) If it is necessary, align the holes for the installation of the jackball pin.

(f) Put the jackball pin through the jackball fitting of the support assembly [96] and the jacking ball of the main landing gear [93].

(g) Install the lynch pin in the jackball pin.

(h) Loosen the nut that holds the side strut attach assembly [94], such that the stenciled plate can be attached to the studs on the side strut support assembly [97].

1) If it is necessary, adjust the adjustable square support tube of the side strut support assembly [97].

(i) Tighten the two hand knobs to hold the stenciled plate to the studs on the side strut support assembly [97].

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(j) Tighten the nut that holds the side strut attach assembly [94] to the lower side strut attach point.

(k) Tighten the two adjustable handles to hold the adjustable square support tube of the side strut support assembly [97].

SUBTASK 32-11-00-480-017

(2) Attach the dolly assembly [98] to the support assembly [96].

NOTE: A minimum of two persons are necessary to do this task.

(a) Put the dolly assembly [98] in its position such that the boom arm [91] is on the forward side of the main landing gear [93] and the legs are in center with the main landing gear [93].

(b) Use the manual valve on the hydraulic pump to lift or lower the boom arm [91] until you can connect the lift link [95] to the support assembly [96].

(c) Connect the lift links [95] to the support assembly [96].

(d) Install the pins to hold the lift links [95] to the support assemblies [96].

(e) Connect the winch assembly [99] to the support assembly [96] with the snapper pin.

SUBTASK 32-11-00-480-018

CAUTION

MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION

MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(3) Make sure that the airplane is lifted with the jacks and the lower surface of the housing, for the forward trunnion bearing pin [29], measures 97 ±2 in. (246 ±5 cm) above the ground (TASK 07-10-01-580-801) (View A, Figure 403).

(a) Make sure that to keep the airplane level.

SUBTASK 32-11-00-580-002

(4) Use the boom arm [91] to lift the main landing gear [93], such that the transportation support assembly and the aft tire support assembly can be removed from the dolly assembly [98].

SUBTASK 32-11-00-980-001

(5) From the aft side, carefully move the dolly assembly [98] with the main landing gear [93] forward below the wing and put it in its position below the wheel well area.

(a) Move the dolly assembly [98] such that the axis of the forward trunnion bearing pin [29] will be perpendicular to the housing assembly and the rear spar.

SUBTASK 32-11-00-210-001

(6) Visually examine the protective coating on the transition radius of the forward trunnion pin for chipping, cracks, or corrosion.

(a) If the transition radius is damaged, replace the trunnion pin, do these tasks:

- Main Landing Gear Forward Trunnion Bearing Assembly Removal, TASK 32-11-83-000-801
- Main Landing Gear Forward Trunnion Bearing Assembly Installation, TASK 32-11-83-400-801.

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SUBTASK 32-11-00-020-025

WARNING

MAKE SURE THAT THE HYDRAULIC PRESSURE IS BELOW 5000 PSIG (34,474 KPA). IF YOU CONTINUE WHEN THE HYDRAULIC PRESSURE IS 5000 (34,474 KPA) OR HIGHER, A BINDING CONDITION WILL OCCUR. A BINDING CONDITION CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(7) To connect the forward trunnion, do these steps:

(a) Lift the main landing gear [93] with the boom arm [91] and the winch assembly [99], such that the forward trunnion and the forward trunnion bearing pin [29] are aligned.

1) Remove the transportation support assembly and the aft tire support assembly from the dolly assembly [98].

(b) Attach the outer cylinder strap assembly to the hoist ring on the end of the boom arm [91] and around the outer cylinder at the aft trunnion.

(c) Remove the tension from the strap.

(d) Use the pry lever bar to remove the load off of the caster wheels one at a time.

(e) Use the lever bar assemblies to turn all the caster wheels parallel to the dolly assembly [98] legs, such that the forward trunnion can be inserted straight onto the forward trunnion bearing pin [29].

1) Make sure that the swivel locks on the caster wheels are not locked.

(f) Make sure that the forward trunnion stays aligned with the forward trunnion bearing pin [29] and lock the swivel locks.

(g) Put chocks on all caster wheels but put the forward chock approximately 2 in. (5 cm) forward of the double wheel caster.

(h) Put the fork end of the lever bar assemblies on the rectangular tubing above the single wheel casters.

(i) Use the lever bar assemblies to slowly push the dolly assembly [98] forward.

1) Make sure that the aft trunnion will clear the lower trailing edge panel seals when you do this task.

2) Make sure that the chocks are in contact with the single wheel casters to prevent the movement of the dolly assembly [98].

(j) Slowly lift the aft end of the main landing gear [93] with the winch assembly [99] and lifting the boom arm [91] at the same time while you continue to push the dolly assembly [98] forward.

1) Use the manual valve on the hydraulic pump to lift the boom arm [91].

2) Tighten the outer cylinder strap if necessary to remove the tension, such that the forward trunnion will not come off of the forward trunnion bearing pin [29].

(k) Continue the movement of the dolly assembly [98] until the forward trunnion is fully inserted onto the forward trunnion bearing pin [29].

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SUBTASK 32-11-00-620-001

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(8) Apply a thin layer of corrosion inhibiting material, G50136, to these items:

(a) The shank, thread, and thread relief of the forward trunnion cross bolt [28]

(b) The thread of the nut [25]

(c) The spline, and face of the washer [26]

(d) The pin [24]

(e) Around the bushing holes on the forward trunnion bearing pin [29].

SUBTASK 32-11-00-160-001

(9) Remove all unwanted corrosion inhibiting material, G50136.

SUBTASK 32-11-00-420-003

(10) Do these steps to install the forward trunnion cross bolt [28]:

(a) Install the thread protector, C32029-9, SPL-1365, or thread protector, C32029-51, SPL-14024, on the forward trunnion cross bolt [28].

(b) Put the forward trunnion cross bolt [28] through the door actuator bracket [27], forward trunnion, and the forward trunnion bearing pin [29].

(c) Remove the thread protector, C32029-9, SPL-1365, or thread protector, C32029-51, SPL-14024, from the forward trunnion cross bolt [28].

(d) Install the washer [26] and nut [25] on the forward trunnion cross bolt [28].

(e) Tighten the nut [25] to 21 ft-lb (28 N·m) - 25 ft-lb (34 N·m) plus the run-on torque.

1) Use the wrench adapter, C32029-49 or C32029-88, SPL-14022, or wrench adapter, C32029-50 or C32029-89, SPL-14023, to hold the head of the forward trunnion cross bolt [28].

(f) Install the new pin [24] in the forward trunnion cross bolt [28].

1) If it is necessary, loosen the nut [25] to the nearest castellation to align the holes for the pin [24].

SUBTASK 32-11-00-080-001

(11) Remove the outer cylinder strap assembly from the hoist ring and the outer cylinder.

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SUBTASK 32-11-00-420-004

WARNING MAKE SURE THAT THE HYDRAULIC PRESSURE IS BELOW 5000 PSIG (34,474 KPA). IF YOU CONTINUE WHEN THE HYDRAULIC PRESSURE IS 5000 (34,474 KPA) OR HIGHER, A BINDING CONDITION WILL OCCUR. A BINDING CONDITION CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(12) Operate the dolly assembly [98] to put the aft trunnion in its position.

NOTE: The main landing gear [93] installation path is done with the forward trunnion bearing pin [29] attached to the forward trunnion. This installation path is a combination of moving the aft trunnion inboard, lifting the aft trunnion, and moving the aft trunnion outboard as the main landing gear pivots about the forward trunnion bearing pin [29].

(a) Make sure that to lock the caster wheels when the main landing gear [93] pivots about the forward trunnion bearing pin [29].

1) Use the pry lever bar to remove the load off of the caster wheels one at a time.

2) Lock the single wheel casters perpendicular to the forward trunnion pivot axis approximately 38 degrees, and 90 degrees for the double wheel caster.

(b) Put the fork end of the lever bar assemblies on the rectangular tubing above the single wheel casters.

(c) Do a check to make sure that the locks for the caster wheels are released.

(d) Hold the lever bar assemblies to slowly push the aft end of the dolly assembly [98] inboard, such that the aft trunnion will clear the main landing gear beam lower flange, lower trailing edge panel seals, and the seal support beam when the aft trunnion is lifted.

(e) If it is necessary, use the lever bar assemblies and the pry lever bar to put the caster wheels parallel to the legs of the dolly assembly [98].

(f) Lock the caster wheels.

(g) Lift and move the aft trunnion outboard at the same time while you clear the main landing gear [93] from the main landing gear beam.

1) Release the tension on the winch assembly [99].

2) Use the manual valve on the hydraulic pump to lift the boom arm [91].

(h) Put the aft trunnion in its position directly below the aft trunnion bearing.

1) If the forward tire support assembly is on the way of the tires, remove it.

(i) Put chocks on all the casters.

(j) Continue to slowly lift the aft trunnion of the main landing gear [93] until the aft trunnion and the aft trunnion bearing are aligned.

1) Make sure that the chock is in contact with the double wheel caster to prevent the forward movement of the dolly assembly [98].

SUBTASK 32-11-00-080-002

(13) Remove the two tie wraps that hold the aft trunnion pin [19].

SUBTASK 32-11-00-080-003

(14) Remove the protective tape from the outer edge of the aft trunnion.

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SUBTASK 32-11-00-620-002

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(15) Apply a thin layer of corrosion inhibiting material, G50136, to these items:

(a) The shank, thread, and thread relief of the aft trunnion cross bolt [23]

(b) The thread of the nut [21]

(c) The spline, and face of the washer [22]

(d) The pin [20]

(e) Around the bushing holes on the aft trunnion cross bolt [23].

SUBTASK 32-11-00-160-003

(16) Remove all unwanted corrosion inhibiting material, G50136.

SUBTASK 32-11-00-420-005

WARNING

MAKE SURE THAT THE HYDRAULIC PRESSURE IS BELOW 5000 PSIG (34,474 KPA). IF YOU CONTINUE WHEN THE HYDRAULIC PRESSURE IS 5000 (34,474 KPA) OR HIGHER, A BINDING CONDITION WILL OCCUR. A BINDING CONDITION CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(17) Connect the aft trunnion.

(a) Put a flashlight, or other bright lights, at the top of the main landing gear [93] and shine them back past the aft trunnion.

(b) Look through the aft trunnion pin access hole so you can see the aft trunnion pin [19] through the aft trunnion bearing.

(c) Move the dolly assembly [98] from side to side or lift and lower it until you can see equal quantities of light all the way around the aft trunnion pin [19] through the aft trunnion bearing.

NOTE: When you have the aft trunnion pin and the aft trunnion bearing correctly aligned, the aft trunnion pin will install very easily.

(d) Use the aft trunnion pin R/I equipment, SPL-1862, to pull the aft trunnion pin [19] from the aft trunnion:

1) Put the pull adapter [74] through the pin access hole in the main landing gear beam.

2) Attach the pull adapter [74] to the aft trunnion pin [19] with the pull nut [73].

3) Pull the slide hammer [72] until the aft trunnion pin [19] is out of the aft trunnion bearing such that the aft trunnion will connect to the main landing gear beam.

(e) Move the aft trunnion bearing until it is in line with the aft trunnion pin [19].

(f) If it is necessary, use the aft trunnion pin R/I equipment, SPL-1862, to turn the aft trunnion pin [19] in the clockwise direction until the holes for the aft trunnion cross bolt [23] are aligned.

(g) Install the thread protector, C32029-10, SPL-1366, on the aft trunnion cross bolt [23].

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(h) Put the aft trunnion cross bolt [23] through the aft trunnion and the aft trunnion pin [19].

(i) Remove the thread protector, C32029-10, SPL-1366, from the aft trunnion cross bolt [23].

(j) Install the washer [22] and nut [21] on the aft trunnion cross bolt [23].

(k) Tighten the nut [21] to 160 in-lb (18 N·m) - 190 in-lb (21 N·m) plus the run-on torque.

1) Use long wrench with narrow body, ratchet wrench or crowfoot with 4 in. (102 mm) extension to tighten the nut [21].

2) Use the wrench adapter, C32029-48 or C32029-90, SPL-14021, to hold the head of the aft trunnion cross bolt [23].

(l) Install the new pin [20] in the aft trunnion cross bolt [23].

1) If it is necessary, loosen the nut [21] to the nearest castellation to align the holes for the pin [20].

(m) If removed to get access to the nut [21], do these steps:

1) Install the bolts [88], washers [89], and nuts [90] that attach the bracket [87] to the MLG beam.

2) Install the bolts [101], washers [102], washers [103], and nuts [104] to connect the turnbuckle [100].

SUBTASK 32-11-00-080-004

(18) Remove the main landing gear R/I equipment, SPL-1869, from the main landing gear [93].

SUBTASK 32-11-00-080-005

(19) Remove the protector assembly [92].

SUBTASK 32-11-00-210-002

(20) Visually examine the protective coating on the transition radius of the forward trunnion pin for chipping, cracks, or corrosion.

(a) If the transition radius is damaged, replace the trunnion pin, do these tasks:

- Main Landing Gear Forward Trunnion Bearing Assembly Removal, TASK 32-11-83-000-801
- Main Landing Gear Forward Trunnion Bearing Assembly Installation, TASK 32-11-83-400-801.

SUBTASK 32-11-00-212-001

(21) Before you connect the electrical connectors, examine the electrical connectors for corrosion.

WARNING

DO THE STEPS BELOW IF THE AIRPLANE OPERATES WHERE DEICING FLUID THAT CONTAINS POTASSIUM FORMATE OR POTASSIUM ACETATE IS USED. ALSO, DO THE STEPS FOR ALL AIRPLANES THAT YOU FOUND CORROSION IN THE ELECTRICAL CONNECTORS IN THE MAIN WHEEL WELL. THE ELECTRICAL CONNECTORS ARE IN A SYSTEM THAT IS NECESSARY FOR SAFE FLIGHT.

(a) If there is corrosion, correct the problem (SWPM 20-60-03).

(b) Apply D5026NS or ZC-026 compound, G50171, to the electrical connector (SWPM 20-60-03).

SUBTASK 32-11-00-420-007

(22) To connect the electrical connector [13], do these steps:

(a) Attach the clamps [16] to the electrical connector [13].

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(b) Install the screws [14], washers [15], and nuts [17] to the clamps [16] that hold the electrical connector [13] to the hose guide [18].

(c) Remove the cap from the electrical connector [13].

(d) Connect the electrical connector [13] to the electrical connector.

(e) Install MS20995NC20 lockwire, G02166, on the electrical connector [13] (TASK 20-10-44-400-801).

SUBTASK 32-11-00-420-008

(23) To connect the electrical connector [11], do these steps:

(a) Put the electrical connector [11] through the hose guide [12].

(b) Attach the clamps [9] to the electrical connector [11].

(c) Install the screws [7], washers [8], and nuts [10] to the clamps [9] that hold the electrical connector [11] to the structure.

(d) Remove the cap from the electrical connector [11].

(e) Connect the electrical connector [11] to the electrical connector.

(f) Install MS20995NC20 lockwire, G02166, on the electrical connector [11] (TASK 20-10-44-400-801).

SUBTASK 32-11-00-420-009

(24) To connect the brake hose assembly [6] and hydraulic return shimmy damper line [114], do these steps:

(a) Remove the caps from the brake hose assembly [6] and hydraulic return shimmy damper line [114].

(b) Remove the caps from the elbow [5].

(c) Connect the brake hose assembly [6] and hydraulic return shimmy damper line [114] to the elbow [5].

(d) Remove the tags from the brake hose assembly [6] and hydraulic return shimmy damper line [114].

SUBTASK 32-11-00-100-001

(25) Clean the surfaces of these items that will attach the retract actuator [43] to the main gear trunnion:

(a) Attachment fitting on the main gear trunnion

(b) actuator attach pin [50]

(c) spacers [49]

(d) washer [48]

(e) nut [45]

(f) cotter pin [58], nut [47], washer [46], and bolt [44].

SUBTASK 32-11-00-100-002

(26) Clean the surfaces of these items that will attach the walking beam [51] and the retract actuator [43] to the main gear trunnion:

(a) Attachment fitting on the main gear trunnion

(b) walking beam attach pin [52]

(c) washer [53]

(d) nut [55]

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(e) cotter pin [59], nut [47], washer [46], and bolt [54].

SUBTASK 32-11-00-600-003

(27) Apply grease, D00633, to the chrome-plated surfaces of the reaction link pin [32], actuator attach pin [50], and walking beam attach pin [52].

SUBTASK 32-11-00-620-003

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(28) Apply a thin layer of corrosion inhibiting material, G50136, to these items:

NOTE: Do not apply corrosion preventive compound to the chrome plated surfaces.

(a) The threads and thread reliefs on the actuator attach pin [50], and walking beam attach pin [52]

(b) The threads and thread reliefs of the bolt [44] and bolt [54]

(c) The threads and thread reliefs of the nut [45], nut [47], and nut [55]

(d) The faces of the washer [46], washer [48], and washer [53]

(e) The cotter pin [58] and cotter pin [59].

SUBTASK 32-11-00-160-004

(29) Remove all unwanted corrosion inhibiting material, G50136.

SUBTASK 32-11-00-080-007

(30) Remove the rope that holds the disconnected ends of the walking beam [51] and retract actuator [43].

SUBTASK 32-11-00-010-003

(31) Remove the clamps for the electrical conduit on the inboard side of the nut [45] and move the conduit to get clearance to tighten the nut [45].

SUBTASK 32-11-00-420-011

(32) To connect the retract actuator [43] to the main gear trunnion, do these steps:

(a) Put the retract actuator [43] in its position for connection to the main gear trunnion.

(b) Install the thread protector, C32029-16, SPL-1372, on the actuator attach pin [50].

(c) Put the actuator attach pin [50] through the retract actuator [43], spacers [49] and main gear trunnion.

(d) Remove the thread protector, C32029-16, SPL-1372, from the actuator attach pin [50].

(e) Install the washer [48] and nut [45] on the actuator attach pin [50].

(f) Use the wrench adapter, C32029-7 or C32029-98, SPL-1363, to hold the head of the actuator attach pin [50].

(g) Tighten the nut [45] to 50 ft-lb (67.79 N·m) - 75 ft-lb (101.69 N·m).

(h) If it is necessary, loosen the nut [45] to the nearest castellation to align the holes for the bolt [44].

(i) Remove the wrench adapter, C32029-7 or C32029-98, SPL-1363, from the actuator attach pin [50].

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(j) Put the bolt [44] through the actuator attach pin [50].

(k) Install the washer [46] and nut [47] on the actuator attach pin [50].

(l) Tighten the nut [47] to 45 in-lb (5.08 N·m) - 55 in-lb (6.21 N·m).

(m) If it is necessary, loosen the nut [47] to the nearest castellation to align the holes for the cotter pin [58].

(n) Install the new cotter pin [58] in the bolt [44].

SUBTASK 32-11-00-410-001

(33) Put the electrical conduit back in its position and install the clamps.

SUBTASK 32-11-00-420-012

(34) To connect the walking beam [51] to the main gear trunnion, do these steps:

(a) Put the walking beam [51] in its position for the connection to the main gear trunnion.

(b) Install the thread protector, C32029-15, SPL-1371, on the walking beam attach pin [52].

(c) Put the walking beam attach pin [52] through the walking beam [51] and main gear trunnion.

(d) Remove the thread protector, C32029-15, SPL-1371, from the walking beam attach pin [52].

(e) Install the washer [53] and nut [55] on the walking beam attach pin [52].

(f) Install the wrench adapter, C32029-6 or C32029-96, SPL-1362, to hold the head of the walking beam attach pin [52].

(g) Tighten the nut [55] to 50 ft-lb (67.79 N·m) - 75 ft-lb (101.69 N·m).

(h) If it is necessary, loosen the nut [55] to the nearest castellation to align the holes for the bolt [54].

(i) Remove the wrench adapter, C32029-6 or C32029-96, SPL-1362, from the walking beam attach pin [52].

(j) Put the bolt [54] through the walking beam attach pin [52].

(k) Install the washer [46] and nut [47] on the walking beam attach pin [52].

(l) Tighten the nut [47] to 45 in-lb (5.08 N·m) - 55 in-lb (6.21 N·m).

(m) If it is necessary, loosen the nut [47] to the nearest castellation to align the holes for the cotter pin [59].

(n) Install the new cotter pin [59] in the bolt [54].

SUBTASK 32-11-00-420-013

(35) Install the bolts [81], washers [82], and nuts [83] that attach the bracket [80] to the MLG beam.

SUBTASK 32-11-00-420-052

(36) Install the nuts [37], washers [35], washers [36], and bolts [34] to connect the turnbuckle [38].

SUBTASK 32-11-00-420-053

(37) Install the bolts [85] that attach the cable clamps [86], clamp block [84], and hydraulic tubes to the bracket.

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SUBTASK 32-11-00-620-004

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(38) Apply a thin layer of the corrosion inhibiting material, G50136, to these items:

NOTE: Do not apply corrosion preventive compound to the chrome plated surfaces.

(a) The threads and thread reliefs of the crossbolt [31]

(b) The threads of the nut [40]

(c) The faces of the washer [39].

SUBTASK 32-11-00-640-003

(39) Apply grease, D00633, to the chrome plated surfaces of the reaction link pin [32] and end cap [42].

SUBTASK 32-11-00-160-005

(40) Remove all unwanted corrosion inhibiting material, G50136, and grease, D00633.

SUBTASK 32-11-00-420-054

(41) Do one of these steps to install the MLG side strut and the MLG reaction link assembly:

(a) If the side strut dolly, SPL-19644 is used, do this task: Main Landing Gear Side Brace Assembly Installation, TASK 32-11-62-400-801.

(b) Install the MLG side strut and the MLG reaction link assembly as follows:

1) Do the applicable steps in this task to connect the lower side strut to the outer cylinder: Main Landing Gear Lower Side Strut Installation, TASK 32-11-61-400-802.

2) Do the applicable steps in this task to connect the reaction link assembly [30] to the outer cylinder trunnion [33]: Main Landing Gear Reaction Link Assembly Installation, TASK 32-11-71-420-801.

SUBTASK 32-11-00-200-001

WARNING

MAKE SURE THAT THE SYSTEM-TO-SYSTEM SEPARATIONS ARE CORRECT. SEPARATIONS THAT ARE NOT CORRECT COULD CAUSE UNWANTED CONDITIONS, WHICH COULD INCLUDE CHAFING, FIRE OR ELECTROMAGNETIC INTERFERENCE. THIS CAN CAUSE INJURIES TO PERSONNEL, OR CAN MAKE FLIGHT DANGEROUS, OR CAN CAUSE DAMAGE TO THE SYSTEMS.

(42) Make sure that there is a minimum distance of 0.13 in. (3.30 mm) between the wire bundles [111] and hanger link [112] (Figure 404).

SUBTASK 32-11-00-420-016

(43) Install the hose guide [18].

SUBTASK 32-11-00-410-002

(44) Install the applicable panel:

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(a) For the left wing, close this access panel:

Number	Name/Location
551BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel

(b) For the right wing, close this access panel:

Number	Name/Location
651BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel

SUBTASK 32-11-00-420-018

(45) Do this task: Main Landing Gear Wing Door Installation, TASK 32-13-21-420-801.

SUBTASK 32-11-00-420-019

(46) Do this task: Main Landing Gear Center Door Installation, TASK 32-13-11-420-801.

SUBTASK 32-11-00-420-020

(47) Do this task: Main Landing Gear Inner Door Installation, TASK 32-13-11-420-802.

SUBTASK 32-11-00-420-021

(48) Do this task: Main Landing Gear Brake Installation, TASK 32-41-41-400-801.

SUBTASK 32-11-00-420-022

(49) Do this task: Main Landing Gear Wheel and Tire Assembly - Installation, TASK 32-45-11-400-801.

SUBTASK 32-11-00-080-006

(50) Remove retention strap, SPL-1871, from the main landing gear [93].

SUBTASK 32-11-00-410-003

(51) Install the applicable panel:

(a) For the left landing gear, install these access panels:

(TASK 57-50-02-400-801)

Number	Name/Location
551BB	Lower Inboard Fixed Trailing Edge, Gear Adjustment Door
551CB	Lower Inboard Fixed Trailing Edge, Gear Access Panel

(b) For the right landing gear, install these access panels:

(TASK 57-50-02-400-801)

Number	Name/Location
651BB	Lower Inboard Fixed Trailing Edge, Gear Adjustment Door
651CB	Lower Inboard Fixed Trailing Edge, Gear Access Panel

(c) Close these access panels:

Number	Name/Location
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

I. Put the Airplane Back to Its Usual Condition

SUBTASK 32-11-00-080-008

(1) Remove the ground spoiler actuator lock lock set, SPL-1744.

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SUBTASK 32-11-00-860-017

(2) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1
B	16	C01346	LANDING GEAR PARKING BRAKE
B	17	C00129	LANDING GEAR LATCH & PRESS WARN
C	15	C01355	LANDING GEAR AIR/GND SYS 2
C	16	C01356	LANDING GEAR AIR/GND SYS 1
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN
D	15	C01401	LANDING GEAR AIR/GND RELAY
D	16	C01432	LANDING GEAR ALTN EXTEND SOL
E	16	C00196	LANDING GEAR ANTISKID INBD
E	18	C00195	LANDING GEAR ANTISKID OUTBD

SUBTASK 32-11-00-860-076

(3) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	4	C00700	HEATERS DRAIN MAST AIR

SUBTASK 32-11-00-860-018

(4) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

SUBTASK 32-11-00-860-019

(5) Do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

SUBTASK 32-11-00-860-021

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(6) Do this task: Apply Pressure to the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-800-801.

SUBTASK 32-11-00-860-022

(7) Move the speed brake lever to the DOWN position to lower the spoilers.

SUBTASK 32-11-00-860-055

(8) Do this task: Shut Down the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-840-801.

SUBTASK 32-11-00-820-001

(9) If you changed the length of one or more of the door rods, do the steps that follow:

(a) Adjust the shock strut doors of the main landing gear (TASK 32-13-11-820-801).

(b) Adjust the outer door of the main landing gear (TASK 32-13-21-000-802).

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SUBTASK 32-11-00-610-002

(10) Lubricate the main landing gear (TASK 12-21-11-640-801 and TASK 12-21-11-640-802).

SUBTASK 32-11-00-619-001

(11) To fully service the shock strut [1] with the airplane on jacks, do these steps to service the shock strut [1] with fluid:

(a) Hold the shock strut [1] in a vertical position.

(b) Make sure that the shock strut is fully deflated, refer to service placard for the fully deflated dimension (Figure 405).

(c) Do the applicable steps to fill the shock strut [1] with fluid in this task: Main Landing Gear Strut Servicing, Airplane on Jacks, TASK 12-15-31-610-803.

(d) Close the gas charging valves [2], and install their swivel nuts [3] and dust caps [4].

SUBTASK 32-11-00-619-002

(12) To inflate the shock strut [1] with the airplane on jacks, do these steps:

NOTE: To do these steps, you must have done the steps to add fluid to the shock strut, at the start of this task.

(a) Remove the cap on the gas valve.

(b) Install the inflation tool, SPL-1521, on the gas valve.

(c) Inflate the shock strut [1] with nitrogen, G00018, until you reach the fully extended pressure, refer to the servicing placard for the correct pressure (Figure 405).

NOTE: If nitrogen is not available, you can use air, G02314, as an alternative, to inflate the shock strut.

(d) Remove the inflation tool, SPL-1521, and install the cap on the gas valve.

SUBTASK 32-11-00-619-003

(13) If you did not do the steps to service the shock strut [1] with fluid, at the start of this task, do this task: Main Landing Gear Shock Strut Servicing, Airplane on the Ground, TASK 12-15-31-610-802.

SUBTASK 32-11-00-080-009

(14) If the downlock pins are installed on all the landing gear, do this task: Landing Gear Downlock Pins Removal, TASK 32-00-01-080-801.

SUBTASK 32-11-00-710-001

(15) Do this task: Main Landing Gear - Operational Test, TASK 32-32-00-710-802.

SUBTASK 32-11-00-710-002

(16) Do this task: Main Gear Manual Extension System Test (Airplane on Jacks), TASK 32-34-00-730-801.

SUBTASK 32-11-00-480-023

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(17) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

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SUBTASK 32-11-00-580-003

CAUTION

MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION

MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(18) Do this task: Lower the Airplane off Jacks, TASK 07-10-01-580-802.

SUBTASK 32-11-00-860-023

(19) For hydraulic systems A and B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-11-00-870-001

(20) Bleed the brake hydraulic system (TASK 32-41-00-870-802).

SUBTASK 32-11-00-790-001

(21) Examine the connections of the hydraulic brake line for leakage.

SUBTASK 32-11-00-870-002

(22) Bleed the shimmy damper (TASK 32-11-81-870-801).

SUBTASK 32-11-00-790-002

(23) Examine the connections of the shimmy damper for leakage.

SUBTASK 32-11-00-710-009

(24) Do this task: Antiskid Valve Operational Test, TASK 32-42-00-720-803.

SUBTASK 32-11-00-710-003

(25) Do a test of the operation of the antiskid transducer (TASK 32-42-00-710-803).

SUBTASK 32-11-00-730-001

(26) Do this task: Tire and Brake Monitoring System - System Test, TASK 32-46-00-730-801.

SUBTASK 32-11-00-610-003

(27) Do a check of the pressure and extension of the shock strut [1].

(a) Measure the pressure and extension of the shock strut [1].

(b) Make sure that the pressure you measure is still correct for the extension of the shock strut [1] (Figure 405).

(c) If the pressure is not correct for the extension of the shock strut [1], then the shock strut [1] does not have the correct quantity of fluid and you must fully service the shock strut [1] (TASK 12-15-31-610-802).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN LANDING GEAR - INSPECTION/CHECK

1. General

A. This procedure has this task:

(1) Main Landing Gear Inspection.

TASK 32-11-00-200-801

2. Main Landing Gear Inspection

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task provides the regular inspection of the main landing gear.

NOTE: This inspection can be done with the airplane on the ground or on jacks.

(2) Refer to TIME LIMITS/MAINTENANCE CHECKS, CHAPTER 05 for special inspection procedure on the following conditions:

(a) After a hard or overweight landing

(b) After a brake seizure when you tow the airplane, or the airplane is moving at high speed on the ground

(c) After the gear has collided with an obstacle

(d) After a high energy stop.

B. References

Reference	Title
05	TIME LIMITS/MAINTENANCE CHECKS
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-00-10-211-801	Landing Gear Inner Cylinder Chrome Inspection (P/B 601)
32-11-21-200-801	Main Landing Gear Shock Strut Seal Leakage Check (P/B 801)
32-45-00-700-801	Wheels Fast Check (Wheel Installed on the Airplane) (P/B 601)
32-45-00-700-803	Landing Gear Tire Inspection (P/B 601)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Access Panels

Number	Name/Location
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

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E. Prepare for the Inspection

SUBTASK 32-11-00-480-020

WARNING OBEY THE PROCEDURE FOR THE INSTALLATION OF THE DOWNLOCK PINS. IF YOU MOVE THE CONTROL LEVER FOR THE LANDING GEAR TO THE UP POSITION, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-11-00-480-021

(2) Put chocks around the tires of all the landing gear (PAGEBLOCK 10-11-05/201).

SUBTASK 32-11-00-010-004

(3) Remove the applicable access panel:

(a) For the left landing gear, do this step:

Open this access panel:

Number	Name/Location
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

(b) For the right landing gear, do this step:

Open this access panel:

Number	Name/Location
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

F. Main Landing Gear Inspection

SUBTASK 32-11-00-210-003

(1) Examine the upper components of the main landing gear:

(a) Examine the retract actuator assembly as follows:

1) Damage, looseness and security of actuator components and attachment bolts.

2) Dirt, scratches and debris on exposed piston chrome and actuator body rod end.

3) Leaks and loose hydraulic connections.

(b) Examine the walking beam and the attachment bolts for damage, looseness, and security.

(c) Examine the attachments of the beam hanger for security and damage.

(d) Examine the uplock mechanism for damage and security of attachments.

(e) Examine the hydraulic connections of the uplock and the downlock actuator assemblies for leaks and security of attachments.

(f) Do these steps to examine the shock strut:

1) Examine the shock strut for leaks and specified extension, do this task: Main Landing Gear Shock Strut Seal Leakage Check, TASK 32-11-21-200-801.

2) Examine the exposed surface of the inner cylinder for dirt, scratches or galling, do this task: Landing Gear Inner Cylinder Chrome Inspection, TASK 32-00-10-211-801.

(g) Examine the attachments of the side strut assembly for corrosion, damage or missing pins and cross bolts.

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1) Examine the components of the side strut assembly for damage or missing pins and cross bolt.

(h) Make sure that the wing door is correctly attached to the push rod and the stabilizer link assembly.

1) Examine the attaching hardware for condition and security.

(i) Make sure that the shock strut doors are correctly attached to the shock strut.

1) Examine the attaching hardware for condition and security.

SUBTASK 32-11-00-210-004

(2) Examine the lower end components of the main landing gear:

(a) Do this task: Landing Gear Tire Inspection, TASK 32-45-00-700-803.

(b) Do this task: Wheels Fast Check (Wheel Installed on the Airplane), TASK 32-45-00-700-801.

(c) Examine the brake hydraulic hoses for chafing and leaks.

(d) Examine the safety sensors and all the electrical wiring for security and chafing.

(e) Make sure that the torsion links are correctly attached to the shock strut.

1) Examine the attachment lugs of the torsion links for cracks.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-11-00-410-004

(1) Close the applicable access panel:

(a) For the left landing gear, do this step:

Close this access panel:

Number	Name/Location
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

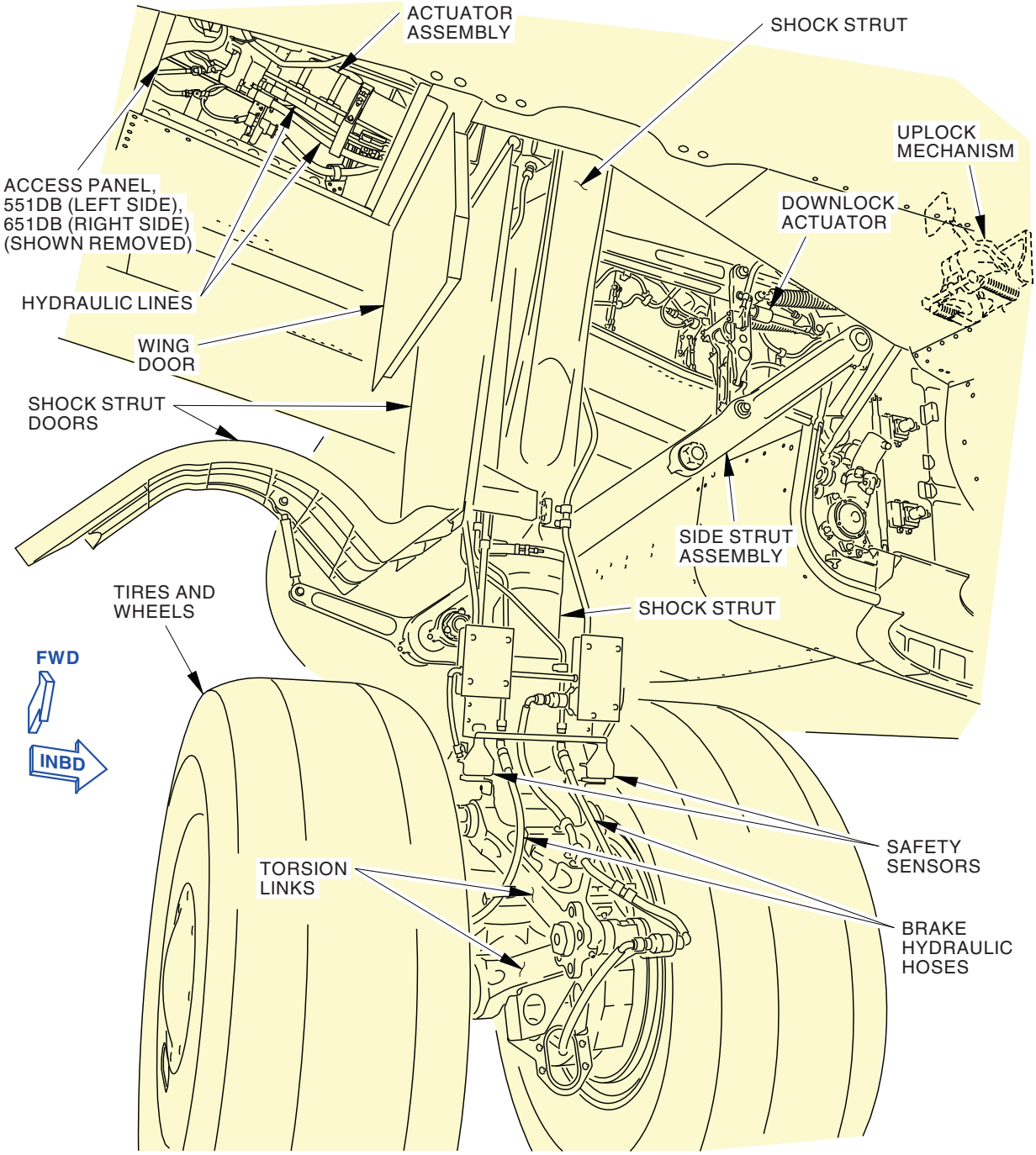
(b) For the right landing gear, do this step:

Close this access panel:

Number	Name/Location
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

——— END OF TASK ———

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RIGHT MAIN LANDING GEAR
(LEFT MAIN LANDING GEAR IS EQUIVALENT)

2414056 S00061534311_V2

Main Landing Gear Inspection
Figure 601/32-11-00-990-806

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MAIN LANDING GEAR SHOCK STRUT SEALS - REPAIRS

1. General

A. There are three tasks in this procedure that supply instructions to replace the seals on the shock strut of the main landing gear:

(1) The first task is to replace the active seals with the spare seals on the lower bearing carrier.

(2) The second task is to install new dynamic seals, new static seals, and new spare seals.

(3) The third task is to replace the scraper ring with a split scraper ring. This task would be used when the scraper ring is damaged and removing the inner cylinder is not possible.

B. The fourth task does a check of hydraulic leaks on the shock strut of the main landing gear.

TASK 32-11-21-960-801

2. Active Seals with the Spare Seals Replacement

(Figure 801, Figure 802, Figure 803)

A. General

(1) This task replaces the active static seal [15] and the active dynamic seal [14] with the spare static seal [18] and spare dynamic seal [17] on the lower bearing carrier [11] of the main landing gear.

B. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
07-10-03-580-801	Lift the Airplane on Axle Jacks-Handling (P/B 201)
12-15-31-610-802	Main Landing Gear Shock Strut Servicing, Airplane on the Ground (P/B 301)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-11-51-020-801	Main Landing Gear Torsion Link Disconnection (P/B 401)
32-11-51-400-803	Main Landing Gear Torsion Link Connection (P/B 401)
32-41-41-000-801	Main Landing Gear Brake Removal (P/B 401)
32-41-41-400-801	Main Landing Gear Brake Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1340	Spacer Assy, C32017-4 (included in C32017 Eqpt) Part #: C32017-67 Supplier: 81205 Opt Part #: C32017-52 Supplier: 81205 Opt Part #: C32017-53 Supplier: 81205
SPL-1341	Carrier Puller Assy, C32017-5 (included in C32017 Eqpt) Part #: C32017-67 Supplier: 81205 Opt Part #: C32017-52 Supplier: 81205 Opt Part #: C32017-53 Supplier: 81205

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(Continued)

Reference	Description
SPL-1342	Dynamic Seal Clamp Assy, C32017-7 (included in C32017 Eqpt) Part #: C32017-67 Supplier: 81205 Opt Part #: C32017-52 Supplier: 81205 Opt Part #: C32017-53 Supplier: 81205
SPL-1343	Thread Protector Retainer Assy, C32017-8 (included in C32017 Eqpt) Part #: C32017-67 Supplier: 81205 Opt Part #: C32017-52 Supplier: 81205 Opt Part #: C32017-53 Supplier: 81205
SPL-1344	Thread Protector, C32017-9 (included in C32017 Eqpt) Part #: C32017-67 Supplier: 81205 Opt Part #: C32017-52 Supplier: 81205 Opt Part #: C32017-53 Supplier: 81205
SPL-1347	Dynamic Seal Plunger, C32017-23 (included in C32017 Eqpt) Part #: C32017-67 Supplier: 81205 Opt Part #: C32017-52 Supplier: 81205 Opt Part #: C32017-53 Supplier: 81205
SPL-1829	Oil Drain Valve - Shock Strut, LG Part #: J32108-16 Supplier: 81205 Opt Part #: A32066-1 Supplier: 81205
SPL-3927	Strap Assy, C32017-60 (included in C32017 Eqpt) Part #: C32017-67 Supplier: 81205 Opt Part #: C32017-52 Supplier: 81205 Opt Part #: C32017-53 Supplier: 81205
SPL-10997	Wrench Adapter - Gland Nut, Shock Strut MLG Part #: F80033-8 Supplier: 81205
STD-1110	Container - Hydraulic Fluid Resistant, 5 Gallon (19 Liter)
STD-1140	Hose - Flexible, 1/2 Inch ID, BMS 3-11 Resistant, 10 Foot

D. Consumable Materials

Reference	Description	Specification
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
17	Spare dynamic seal	32-11-21-01-090	PTF ALL

F. Location Zones

Zone	Area
731	Left Main Landing Gear - Outboard Door
741	Right Main Landing Gear - Outboard Door

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G. Prepare for the Replacement

SUBTASK 32-11-21-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-11-21-860-001

(2) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-11-21-860-002

(3) Operate the brakes eight times to remove pressure from the brake accumulator.

SUBTASK 32-11-21-480-002

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(4) Set the wing jacks correctly to keep the airplane level prior to deflating the Main Landing Gear (MLG) shock strut [1] (TASK 07-10-01-580-801).

NOTE: The airplane is to be lifted further in a subsequent step.

SUBTASK 32-11-21-860-003

(5) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1
B	16	C01346	LANDING GEAR PARKING BRAKE
B	17	C00129	LANDING GEAR LATCH & PRESS WARN
C	15	C01355	LANDING GEAR AIR/GND SYS 2
C	16	C01356	LANDING GEAR AIR/GND SYS 1
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN
D	15	C01401	LANDING GEAR AIR/GND RELAY
D	16	C01432	LANDING GEAR ALTN EXTEND SOL
E	16	C00196	LANDING GEAR ANTISKID INBD
E	18	C00195	LANDING GEAR ANTISKID OUTBD

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SUBTASK 32-11-21-020-001

WARNING DO NOT REMOVE THE VALVE BODY UNTIL YOU DEFLATE THE SHOCK STRUT FULLY. THE AIR PRESSURE CAN BLOW THE VALVE BODY OUT AND CAUSE INJURIES TO PERSONNEL.

(6) Deflate the shock strut [1] for the main landing gear.

(a) Remove the dust cap [2] for the gas charging valve [3].

(b) Loosen the swivel nut [4] a maximum of two turns.

1) Deflate the shock strut [1] slowly to prevent the leakage of the fluid through the gas charging valve [3].

NOTE: Fluid in the shock strut [1] will have bubbles when you release the pressure.

(c) Loosen the swivel nut [4] fully when all of the pressure in the shock strut [1] is released.

SUBTASK 32-11-21-680-001

(7) Drain the oil from the shock strut [1] for the main landing gear.

(a) Remove the dust cap [5] from the oil charging valve [6].

(b) Put a 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110, in a position to catch the shock strut fluid when the oil charging valve [6] is opened.

(c) Install the drain equipment on the oil charging valve [6].

1) Install the landing gear shock strut drain valve, SPL-1829, on the oil charging valve [6] to drain the shock strut [1].

2) If you wish to drain the shock strut [1] with a quicker method, do the steps that follow:

a) Cut a length of plastic ten foot hose (1/2 inch ID), STD-1140, long enough to reach the container on the floor.

b) Insert a small Allen wrench in the end of the length of tubing, such that the end of the long part of the Allen wrench is flush with the end of the tube and the short part penetrates the side wall of the tube.

c) Install the ten foot hose (1/2 inch ID), STD-1140, on the oil charging valve [6] such that the Allen wrench goes into the oil charging valve [6] and holds it open to drain the hydraulic fluid.

(d) Remove the drain equipment when you have removed all of the shock strut oil.

SUBTASK 32-11-21-020-002

(8) Do this task: Main Landing Gear Torsion Link Disconnection, TASK 32-11-51-020-801.

SUBTASK 32-11-21-480-003

(9) Use rope to hold the disconnected end of the upper torsion link and attach it to the outer cylinder [8].

NOTE: This step will hold the upper torsion link in its position when you do the subsequent steps.

SUBTASK 32-11-21-480-004

(10) Lift the airplane until there is more than enough room to install the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]), on the inner cylinder [12] after the gland nut [19] has been removed (TASK 07-10-01-580-801).

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SUBTASK 32-11-21-020-003

(11) Do this task: Main Landing Gear Brake Removal, TASK 32-41-41-000-801.

NOTE: This procedure will also remove the wheel and the tire assemblies. Also, this will give better access to the inner cylinder [12] and outer cylinder [8].

SUBTASK 32-11-21-580-001

(12) Use an axle jack to hold the inner cylinder [12] in its position in the outer cylinder [8] (TASK 07-10-03-580-801).

SUBTASK 32-11-21-020-004

(13) Remove the gland nut [19].

(a) Remove the nuts [24], washers [25], and bolts [26] for the lockplate [23] to release the gland nut [19].

(b) Wind a cloth around the chrome surface of the inner cylinder [12].

NOTE: This will prevent damage to the inner cylinder [12] when the gland nut [19] moves down.

(c) Use the gland nut wrench adapter, SPL-10997, to remove the gland nut [19] from the outer cylinder [8].

1) Loosen the gland nut [19] until it is disconnected from the outer cylinder [8].

a) Do not move the gland nut [19] down the inner cylinder [12].

SUBTASK 32-11-21-580-002

(14) Lower the axle jack until the gland nut [19] is approximately 3 in. (7.6 cm) below the outer cylinder [8].

NOTE: This will let you install the spacer assy, C32017-4, SPL-1340 (spacer assembly, SPL-1340 [51]), such that the gland nut [19] can slide down the inner cylinder [12].

SUBTASK 32-11-21-480-005

(15) Install the spacer assy, C32017-4, SPL-1340 (spacer assembly, SPL-1340 [51]).

(a) Put the spacer assy, C32017-4, SPL-1340 (spacer assembly, SPL-1340 [51]), around the gland nut [19].

(b) Tighten the retainer screws to hold the spacer assy, C32017-4, SPL-1340 (spacer assembly, SPL-1340 [51]), in its position.

(c) Move the spacer assy, C32017-4, SPL-1340 (spacer assembly, SPL-1340 [51]), with the gland nut [19] down the inner cylinder [12].

SUBTASK 32-11-21-020-005

(16) Remove the lower bearing carrier [11] from the outer cylinder [8].

(a) Loosely install the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]), around the inner cylinder [12]

(b) Push the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]), up, such that it is installed correctly at the bottom of the outer cylinder [8].

(c) Align the holes in the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]), with the holes in the lower bearing carrier [11] for the carrier puller screws [52].

(d) Install the carrier puller screws [52] to hold the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]), to the lower bearing carrier [11].

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(e) Tighten the retainer screws on each side of the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]).

(f) Install the strap assy, C32017-60, SPL-3927 (strap assembly, SPL-3927 [47]), around the axle and connect the ends to the shoulder bolts on each side of the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]).

(g) Lift the airplane until there is enough room to install the retainer assy, C32017-8, SPL-1343 (retainer assembly, SPL-1343 [49]), at the bottom of the outer cylinder [8] (TASK 07-10-01-580-801).

SUBTASK 32-11-21-480-006

WARNING MAKE SURE THAT YOU INSTALL THE RETAINER ASSEMBLY CORRECTLY. IF YOU DO NOT INSTALL THE RETAINER ASSEMBLY CORRECTLY, THE SPACER CAN MOVE DOWN THE INNER CYLINDER QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(17) If the spacer [10] has moved down the inner cylinder [12] with the lower bearing carrier [11], push it back into the outer cylinder [8], then install the retainer assy, C32017-8, SPL-1343 (retainer assembly, SPL-1343 [49]), as follows:

NOTE: The retainer assy, C32017-8, SPL-1343 (retainer assembly, SPL-1343 [49]), will hold the spacer [10] inside the outer cylinder [8] and prevent it from moving down the inner cylinder [12].

(a) Loosely install the retainer assy, C32017-8, SPL-1343 (retainer assembly, SPL-1343 [49]), around the inner cylinder [12].

(b) Push the retainer assy, C32017-8, SPL-1343 (retainer assembly, SPL-1343 [49]), up, such that it is installed correctly at the bottom of the outer cylinder [8].

(c) Tighten the retainer screws on each side of the retainer assy, C32017-8, SPL-1343 (retainer assembly, SPL-1343 [49]).

(d) Move the two retainer plates on each side of the retainer assy, C32017-8, SPL-1343 (retainer assembly, SPL-1343 [49]), such that they touch the chrome surface of the inner cylinder [12].

(e) Tighten the four shoulder bolts to hold the two retainer plates in their position.

(f) Put the snapper pin to prevent movement of the retainers.

(g) Install the strap assy, C32017-60, SPL-3927 (strap assembly, SPL-3927 [47]), around the outer cylinder [8] and connect the ends to the shoulder bolts on each side of the retainer assy, C32017-8, SPL-1343 (retainer assembly, SPL-1343 [49]).

SUBTASK 32-11-21-580-003

(18) Continue to lift the airplane until the lower bearing carrier [11] is completely out of the outer cylinder [8] (TASK 07-10-01-580-801).

(a) Lift the airplane a minimum of 77 in. (1956 mm) from the floor to the lowest point on the MLG inner door.

NOTE: This will have the airplane high enough from the floor.

SUBTASK 32-11-21-020-006

(19) Push the lower bearing carrier [11] down the inner cylinder [12] until it rests on the spacer assy, C32017-4, SPL-1340 (spacer assembly, SPL-1340 [51]), with the gland nut [19].

NOTE: The lower bearing carrier [11] is coated with a dry lube film. The film may be worn and appear uneven. This is not a cause for bearing carrier replacement.

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H. Active Static and the Active Dynamic Seal with the Spare Seals Replacement

SUBTASK 32-11-21-900-001

(1) Replace the active static seal [15] with a spare static seal [16].

(a) Make sure that your hands are clean when you handle the elastomer seals.

NOTE: Some greases can cause the elastomer seals to swell too much. This can cause the seals to leak.

(b) Remove the brown rings from around the active static seal [15].

(c) Make sure that the brown rings are serviceable.

(d) Install new brown rings if there are nicks, scratches or other damage to the old brown rings.

NOTE: The recommended installation is to replace all the old brown rings with new brown rings. Use the old brown rings only after a full inspection, and new rings are not available.

(e) Use a tool for cutting plastic that has a back-up strip to cut the active static seal [15] from its outer groove in the lower bearing carrier [11].

(f) Move the spare static seal [16] out of the spare seal groove of the lower bearing carrier [11].

CAUTION

DO NOT APPLY VASELINE OR PETROLATUM TO THE SEALS. THIS CAN CAUSE THE SEAL MATERIAL TO BECOME LARGER. DAMAGE TO THE SEALS CAN OCCUR.

(g) Wipe the spare static seal [16] and the brown rings to remove any unwanted assembly lubricant or hydraulic fluid.

(h) Install the spare static seal [16] on the active static seal groove of the lower bearing carrier [11].

(i) Install the brown rings on the spare static seal [16], with the rounded edge of the brown rings against the seal.

SUBTASK 32-11-21-020-007

(2) Remove the seal retainer [22].

(a) Remove the O-ring [13] for the seal retainer [22].

(b) Remove the retainer pins [21].

(c) Move the seal retainer [22] away from the lower bearing carrier [11] to get access to the location for the active dynamic seal [14].

(d) Use tape to hold the seal retainer [22] around the inner cylinder [12], and prevent it from moving down on the lower bearing carrier [11].

SUBTASK 32-11-21-900-002

(3) Replace the active dynamic seal [14] with a spare dynamic seal [17].

(a) Make sure that your hands are clean when you handle the elastomer seals.

NOTE: Some greases can cause the elastomer seals to swell too much. This can cause the seals to leak.

(b) Remove the active dynamic seal [14]:

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1) Move the lower bearing carrier [11] up and down the inner cylinder [12] to move the active dynamic seal [14], backup ring, and the cap from the inner groove of the lower bearing carrier [11].

2) If the active dynamic seal [14], backup ring, and the cap does not come out of the inner groove of the lower bearing carrier [11], remove them with a tool which is not sharp.

NOTE: Hardened steel cotter pin removal tools work well for removing the active dynamic seal [14].

3) Use a tool for cutting plastic that has a back-up strip to cut the old active dynamic seal [14], backup ring, and the cap from the inner groove of the lower bearing carrier [11].

(c) Move one of the spare dynamic seal [17] from the spare seals groove of the lower bearing carrier [11] and move it into the inner cylinder [12].

NOTE: The round sealing surface of the seal must be against the chrome on the inner cylinder.

(d) Remove the spare backup rings from the spare seals groove of the lower bearing carrier [11].

CAUTION

DO NOT APPLY VASELINE OR PETROLATUM TO THE SEALS. THIS CAN CAUSE THE SEAL MATERIAL TO BECOME LARGER. DAMAGE TO THE SEALS CAN OCCUR.

(e) Wipe the spare dynamic seal [17] and the spare backup rings to remove any unwanted assembly lubricant or hydraulic fluid.

CAUTION

MAKE SURE THAT THE BACKUP RINGS AND THE SEAL STAY TOGETHER WHEN YOU INSTALL THE SEAL. IF THE BACKUP RINGS AND THE SEAL MOVE APART, THEY WILL NOT INSTALL CORRECTLY. THIS CAN CAUSE DAMAGE TO THE SEAL AND THE BACKUP RINGS.

(f) Install the spare dynamic seal [17].

1) Install one of the brown backup rings [29] in the inner groove for the dynamic seal.

2) Install one of the white backup rings [28] in the inner groove for the dynamic seal, with the split in an opposite position on the diameter of the inner cylinder [12] as the brown backup ring [29], and with the rounded edge facing up.

3) Make sure that the backup rings are tight against the inner cylinder [12].

4) Use the dynamic seal clamp assy, C32017-7, SPL-1342 (dynamic seal clamp assembly, SPL-1342 [54]), to install the spare dynamic seal [17] in the inner groove for the dynamic seal, with the rounded edge against the inner cylinder [12].

5) Use the dynamic seal plunger, C32017-23, SPL-1347 (dynamic seal plunger, SPL-1347 [53]), to push the spare dynamic seal [17] in the inner groove of the lower bearing carrier [11].

6) Install the other white backup ring [28] in the inner groove for the dynamic seal, with the rounded edge facing the spare dynamic seal [17].

7) Install the other brown backup ring [29] in the inner groove for the dynamic seal, with the split in an opposite position on the diameter of the inner cylinder [12] as the white backup ring [28].

8) Make sure that the dynamic seal assembly is together, without any twists or bulges.

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SUBTASK 32-11-21-420-001

(4) Install the seal retainer [22].

(a) Remove the tape that holds the seal retainer [22] from around the inner cylinder [12].

(b) Move the seal retainer [22] down on the lower bearing carrier [11].

(c) Install the retainer pins [21] to hold the seal retainer [22] on the lower bearing carrier [11].

(d) Install the O-ring [13] to hold the retainer pins [21] in their positions.

SUBTASK 32-11-21-020-008

WARNING MAKE SURE THAT YOU INSTALL THE RETAINER ASSEMBLY CORRECTLY. IF YOU DO NOT INSTALL THE RETAINER ASSEMBLY CORRECTLY, THE SPACER CAN MOVE DOWN THE INNER CYLINDER QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(5) Move the retainer plates for the retainer assy, C32017-8, SPL-1343 (retainer assembly, SPL-1343 [49]).

(a) Remove the two snapper pins.

(b) Loosen the four shoulder bolts that hold the two retainer plates in their positions.

(c) Move the two retainer plates away from the chrome surface of the inner cylinder [12].

SUBTASK 32-11-21-480-007

(6) If the spacer [10] has moved down the inner cylinder [12] with the lower bearing carrier [11], push it back into the outer cylinder [8].

SUBTASK 32-11-21-420-002

(7) Carefully put the thread protector, C32017-9, SPL-1344 (thread protector, SPL-1344 [48]), between the outer cylinder [8] and the inner cylinder [12].

NOTE: The thread protector, C32017-9, SPL-1344 (thread protector, SPL-1344 [48]), will expand and push against the outer cylinder [8]. This will protect the static seals from the gland nut [19] threads on the outer cylinder [8].

SUBTASK 32-11-21-080-001

(8) Remove the strap assy, C32017-60, SPL-3927 (strap assembly, SPL-3927 [47]), around the axle and from the shoulder bolts on each side of the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]).

SUBTASK 32-11-21-580-020

(9) Slowly lower the airplane so that the top of the lower bearing carrier [11] is approximately 1 in. (3 cm) to 3 in. (8 cm) from the bottom of the outer cylinder [8]. If the seals were replaced on one side only, then depressurize the opposite shock strut [1] prior to lowering the airplane.

SUBTASK 32-11-21-580-004

(10) To lift the inner cylinder [12] slowly with the axle jack until the bottom of the lower bearing carrier [11] is about 1.5 in. (3.8 cm) from the bottom of the outer cylinder [8], do this task: Lift the Airplane on Axle Jacks-Handling, TASK 07-10-03-580-801.

(a) Make sure that the static seals stay in their positions as the lower bearing carrier [11] is installed.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 32-11-21-080-002

(1) Remove the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]).

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(a) Remove the carrier puller screws [52] to disconnect the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]), from the lower bearing carrier [11].

(b) Loosen the two retainer screws and pull the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]), out of the outer cylinder [8].

SUBTASK 32-11-21-080-003

(2) Remove the retainer assy, C32017-8, SPL-1343 (retainer assembly, SPL-1343 [49]).

(a) Remove the strap assy, C32017-60, SPL-3927 (strap assembly, SPL-3927 [47]), that holds the retainer assy, C32017-8, SPL-1343 (retainer assembly, SPL-1343 [49]), to the outer cylinder [8].

(b) Loosen the two retainer screws and remove the retainer assy, C32017-8, SPL-1343 (retainer assembly, SPL-1343 [49]).

SUBTASK 32-11-21-020-009

(3) Remove the thread protector, C32017-9, SPL-1344 (thread protector, SPL-1344 [48]), inside the outer cylinder [8].

SUBTASK 32-11-21-640-001

(4) Remove the spacer assy, C32017-4, SPL-1340 (spacer assembly, SPL-1340 [51]), from around the gland nut [19].

SUBTASK 32-11-21-620-001

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(5) Apply a thin layer of corrosion inhibiting material, G50136, to these items.

(a) The threads and thread reliefs of the bolt [26].

(b) The threads of the nut [24].

(c) The splines and faces of the washers [25].

SUBTASK 32-11-21-620-007

WARNING

DO NOT GET THE CORROSION-INHIBITING COMPOUND ON YOUR SKIN, IN YOUR EYES, OR ON YOUR CLOTHES. THE COMPOUND IS POISONOUS AND FLAMMABLE. USE APPROVED GLOVES, AND EYE PROTECTION. MAKE SURE THAT THERE IS GOOD AIRFLOW IN THE WORK AREA. KEEP THE COMPOUND AWAY FROM SPARKS AND FLAMES. THE COMPOUND CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(6) Apply a thick layer of corrosion preventive Cor-Ban 27L Compound, G50237 to these components:

(a) The gland nut [19] thread major and minor diameters.

(b) The mating outer cylinder [8] thread major and minor diameters.

(c) The outer cylinder [8] thread relief.

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SUBTASK 32-11-21-160-001

WARNING

DO NOT GET THE CORROSION-INHIBITING COMPOUND ON YOUR SKIN, IN YOUR EYES, OR ON YOUR CLOTHES. THE COMPOUND IS POISONOUS AND FLAMMABLE. USE APPROVED GLOVES, AND EYE PROTECTION. MAKE SURE THAT THERE IS GOOD AIRFLOW IN THE WORK AREA. KEEP THE COMPOUND AWAY FROM SPARKS AND FLAMES. THE COMPOUND CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(7) Remove all unwanted corrosion inhibiting material, G50136, or corrosion preventive Cor-Ban 27L Compound, G50237.

SUBTASK 32-11-21-420-003

(8) Install the gland nut [19].

(a) Move the gland nut [19] up to the outer cylinder [8] and engage the gland nut [19] threads.

(b) Use the gland nut wrench adapter, SPL-10997, to torque the gland nut [19] to 125 ft-lb (169 N·m) - 150 ft-lb (203 N·m).

SUBTASK 32-11-21-420-004

(9) Install the lockplate [23] for the gland nut [19].

(a) Install the lockplate [23] in the holes which will put the tab of the lockplate [23] as close to a slot in the gland nut [19] as possible.

1) If it is necessary, loosen the gland nut [19] a small amount to help with the alignment.

(b) Install the nuts [24], washers [25] and bolts [26].

SUBTASK 32-11-21-420-057

(10) Remove the rope that holds the disconnected end of the upper torsion link to the outer cylinder [8].

SUBTASK 32-11-21-420-058

(11) Do this task: Main Landing Gear Torsion Link Connection, TASK 32-11-51-400-803.

SUBTASK 32-11-21-020-010

(12) Do this task: Main Landing Gear Brake Installation, TASK 32-41-41-400-801.

NOTE: This procedure will also install the wheel and tire assemblies.

SUBTASK 32-11-21-080-004

(13) Remove the axle jack (TASK 07-10-03-580-801).

SUBTASK 32-11-21-080-005

CAUTION

MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION

MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(14) Lower the airplane and remove the jacks (TASK 07-10-01-580-802).

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SUBTASK 32-11-21-420-007

(15) Install the dust cap [5] loosely.

SUBTASK 32-11-21-600-001

(16) Do this task: Main Landing Gear Shock Strut Servicing, Airplane on the Ground, TASK 12-15-31-610-802.

SUBTASK 32-11-21-860-004

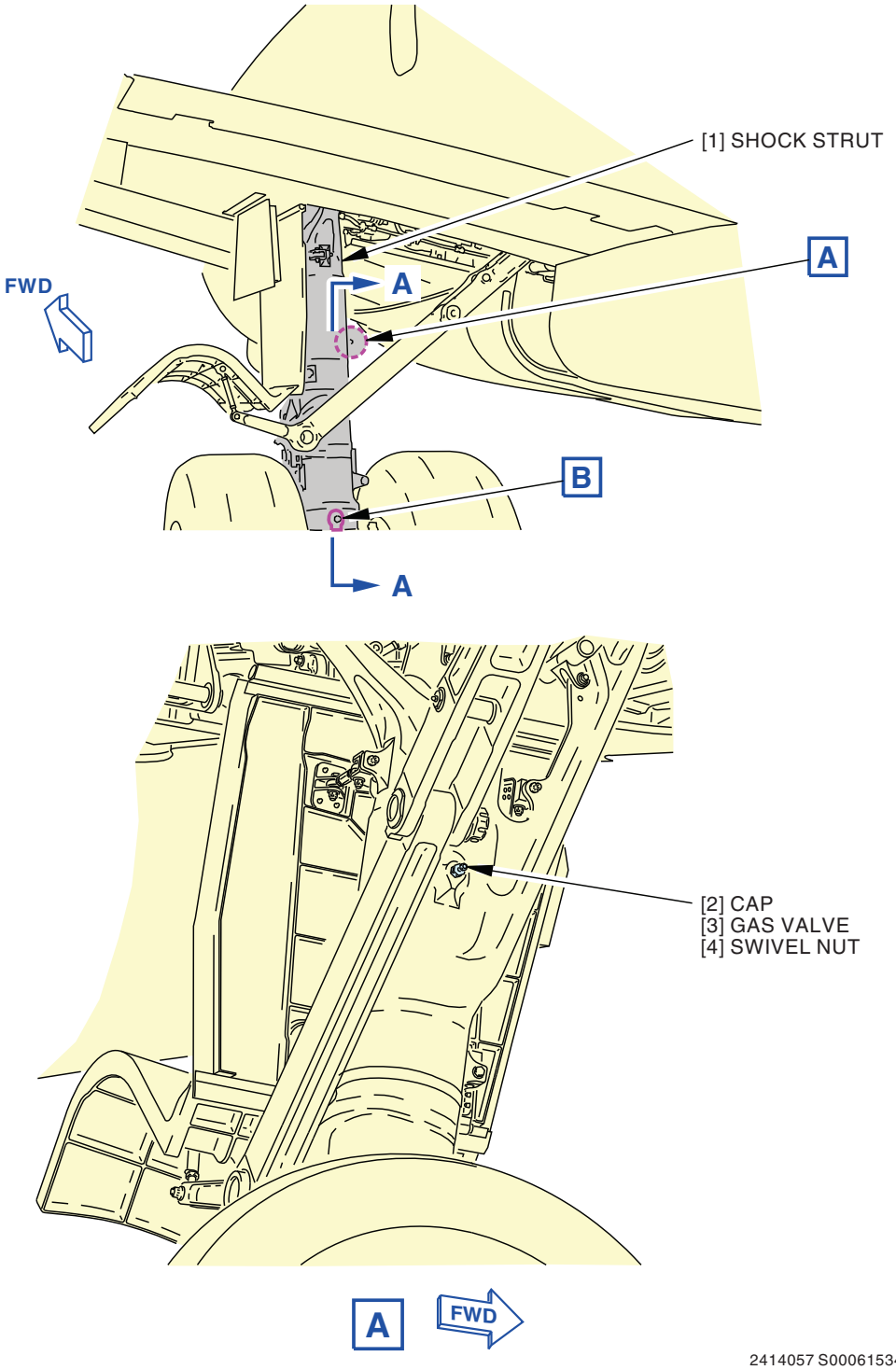
(17) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1
B	16	C01346	LANDING GEAR PARKING BRAKE
B	17	C00129	LANDING GEAR LATCH & PRESS WARN
C	15	C01355	LANDING GEAR AIR/GND SYS 2
C	16	C01356	LANDING GEAR AIR/GND SYS 1
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN
D	15	C01401	LANDING GEAR AIR/GND RELAY
D	16	C01432	LANDING GEAR ALTN EXTEND SOL
E	16	C00196	LANDING GEAR ANTISKID INBD
E	18	C00195	LANDING GEAR ANTISKID OUTBD

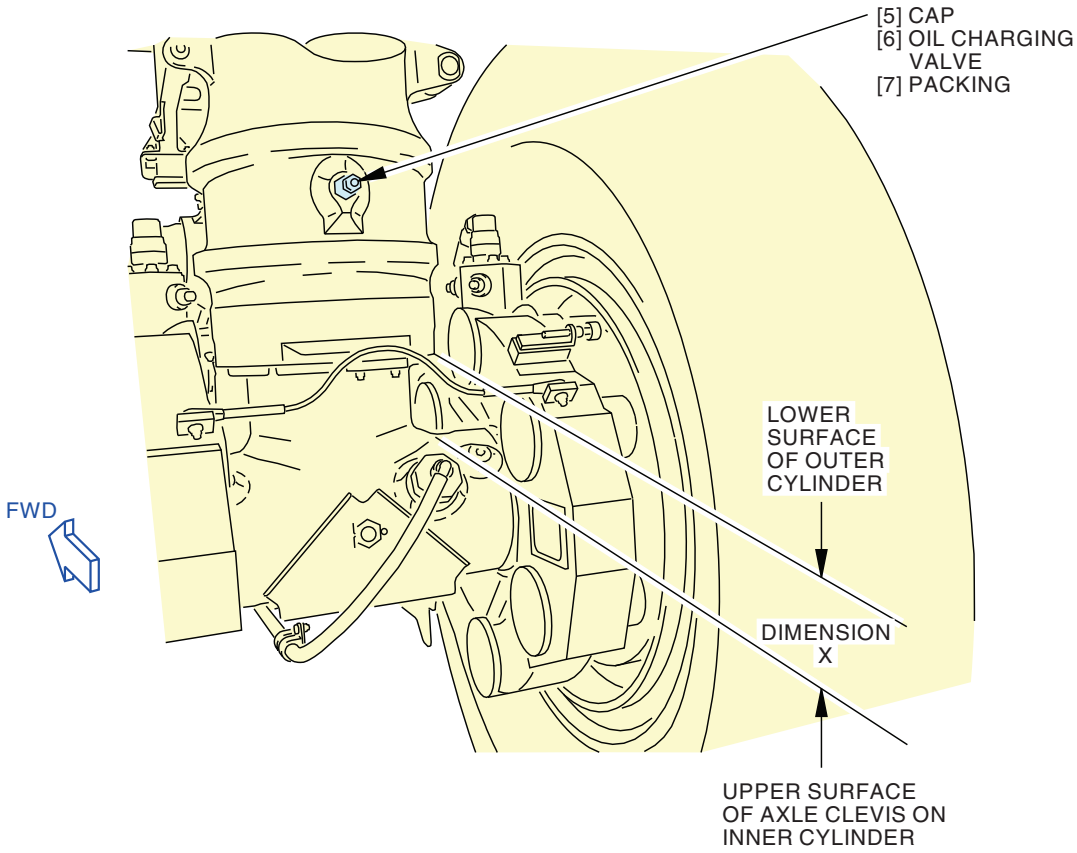
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Shock Strut Seal Replacement
Figure 801/32-11-21-990-801 (Sheet 1 of 5)

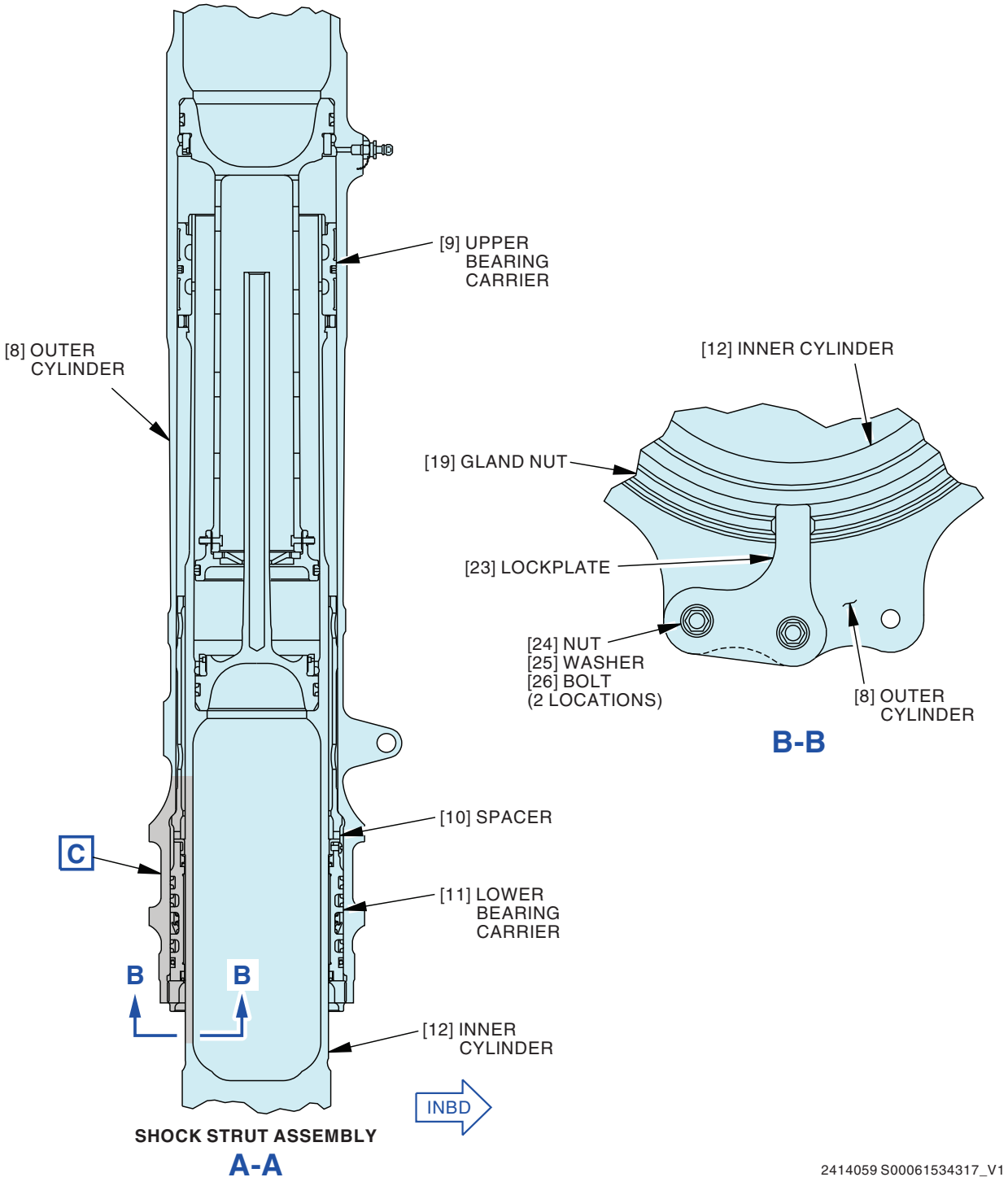
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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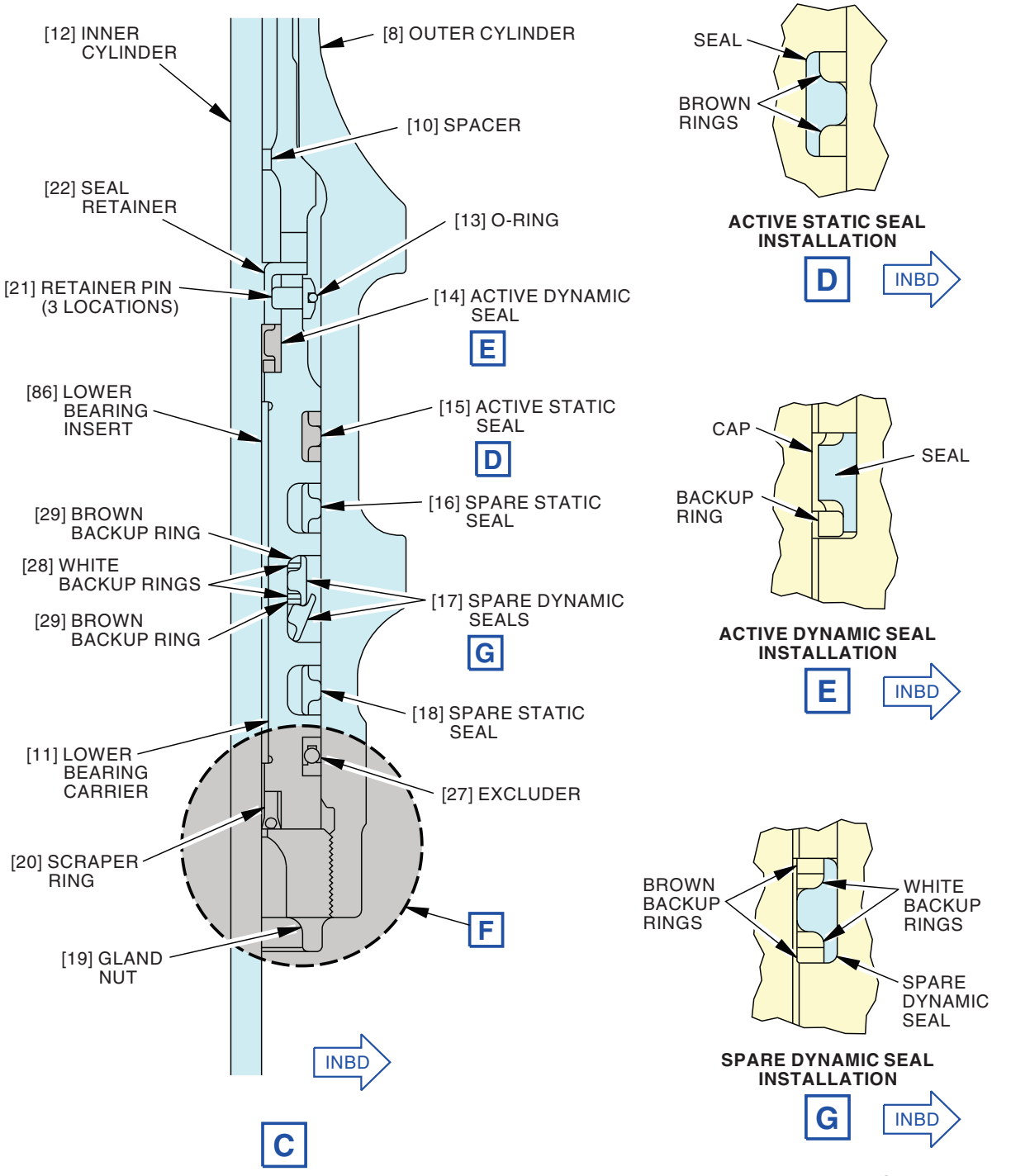
Main Landing Gear Shock Strut Seal Replacement
Figure 801/32-11-21-990-801 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



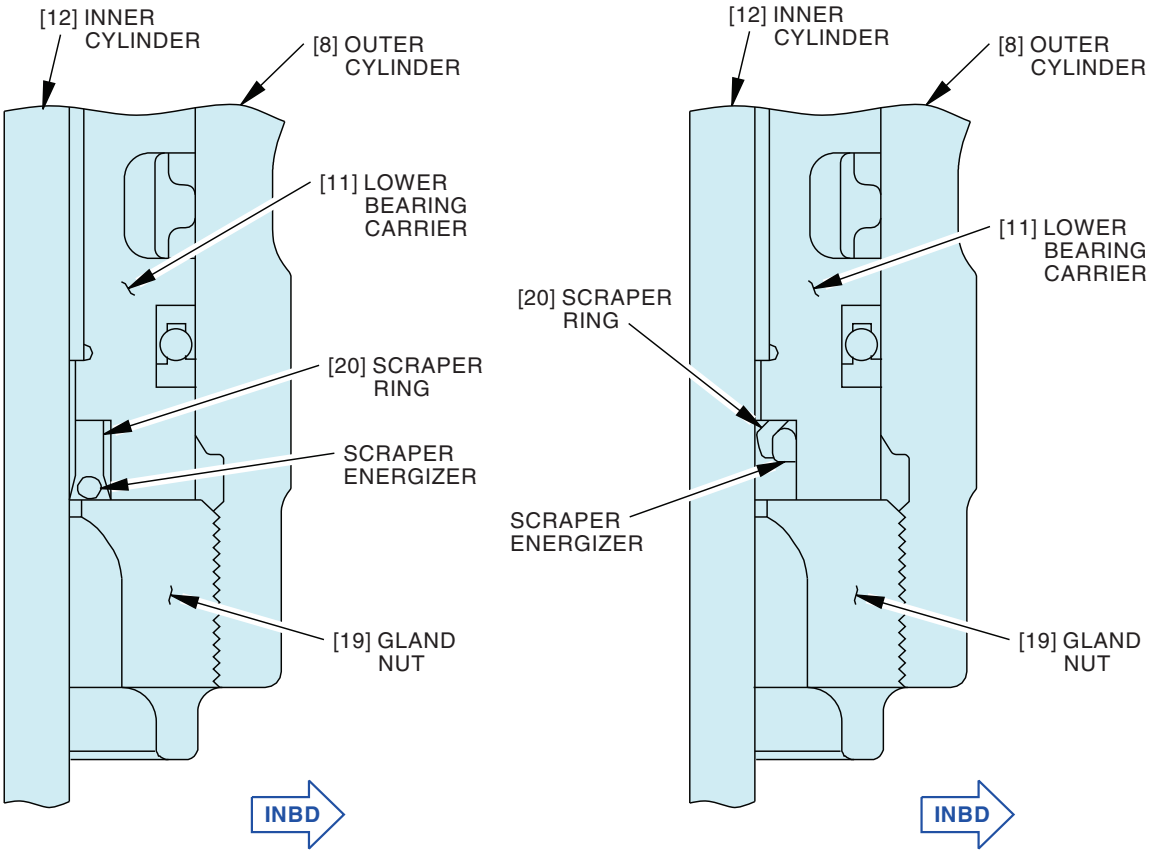
Main Landing Gear Shock Strut Seal Replacement
Figure 801/32-11-21-990-801 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Shock Strut Seal Replacement
Figure 801/32-11-21-990-801 (Sheet 4 of 5)

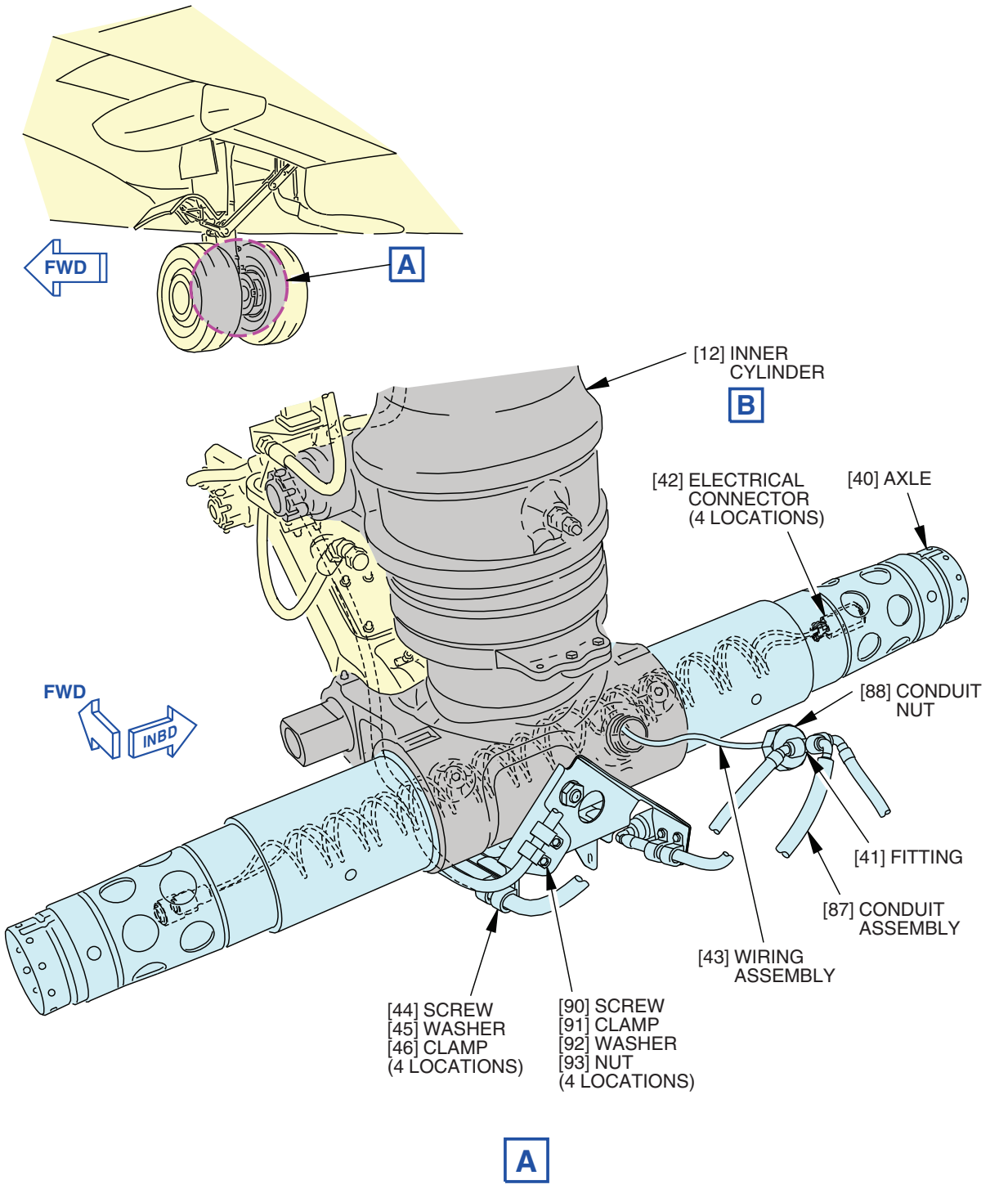
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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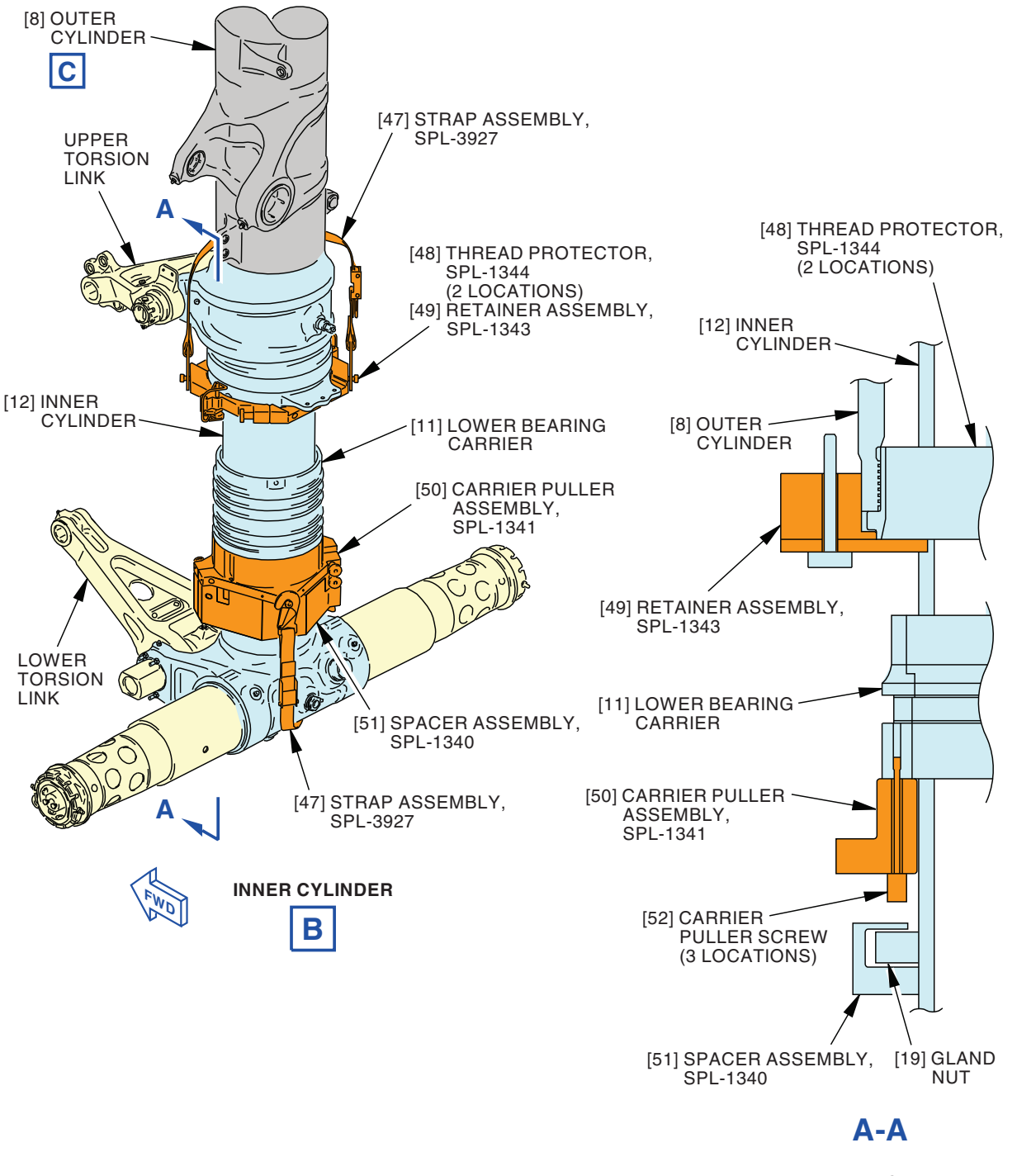
Main Landing Gear Shock Strut Seal Replacement
Figure 801/32-11-21-990-801 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



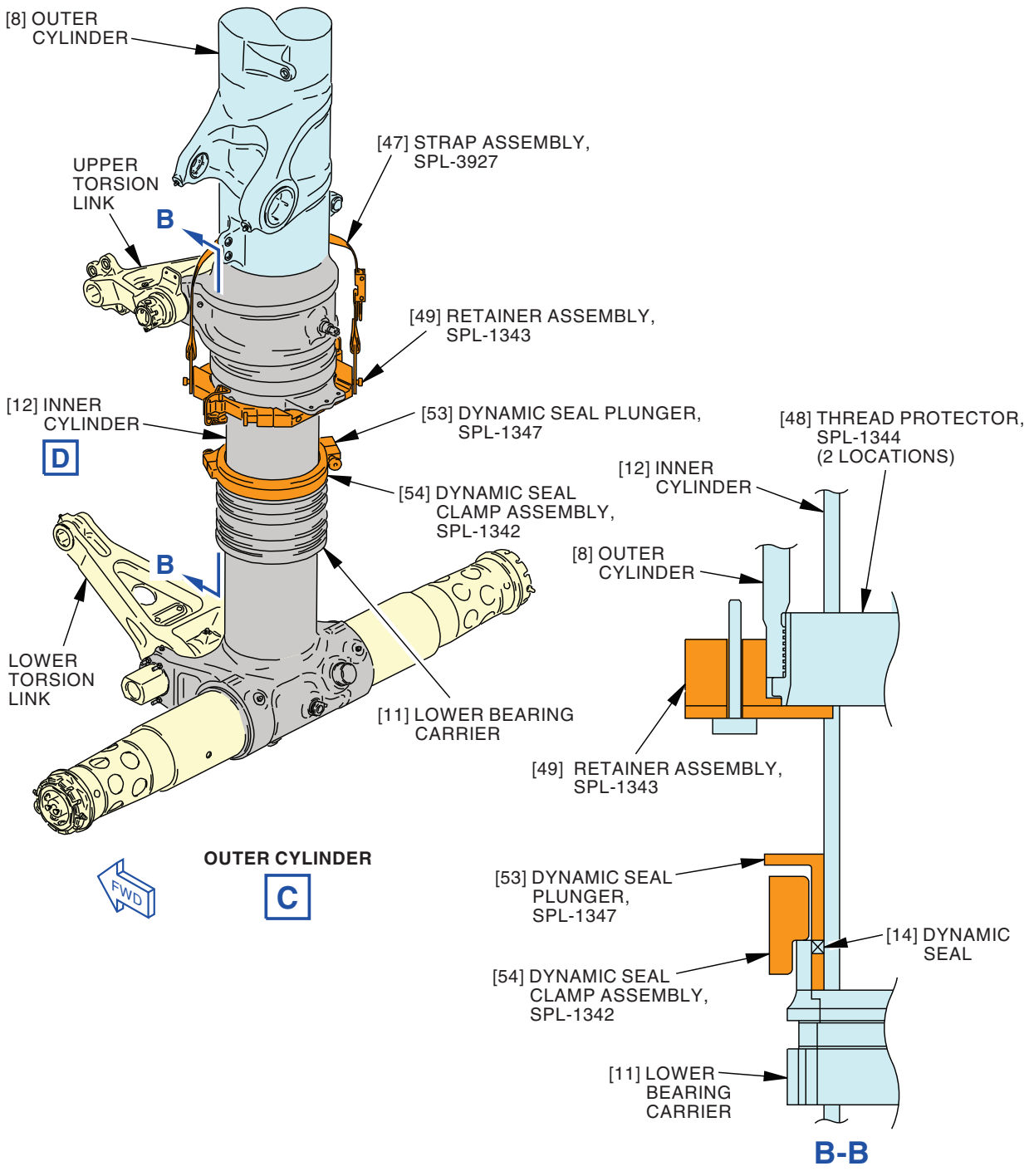
Main Landing Gear Lower Bearing Seal Replacement
Figure 802/32-11-21-990-802 (Sheet 1 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



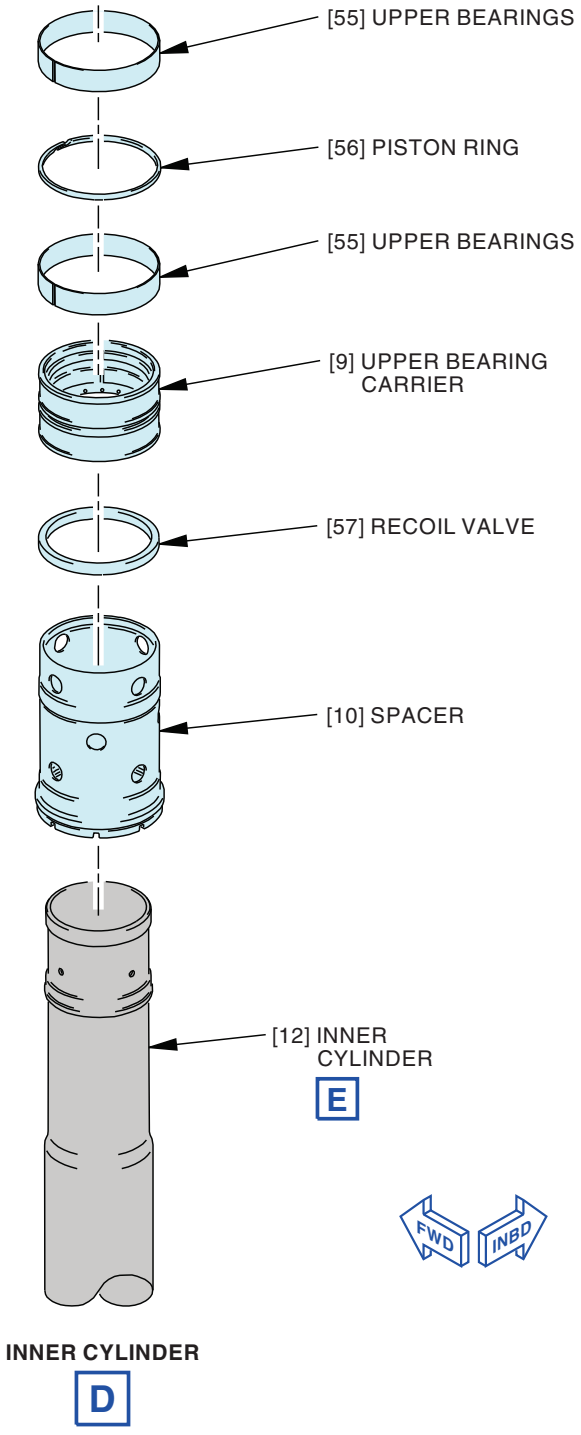
Main Landing Gear Lower Bearing Seal Replacement
Figure 802/32-11-21-990-802 (Sheet 2 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Lower Bearing Seal Replacement
Figure 802/32-11-21-990-802 (Sheet 3 of 10)

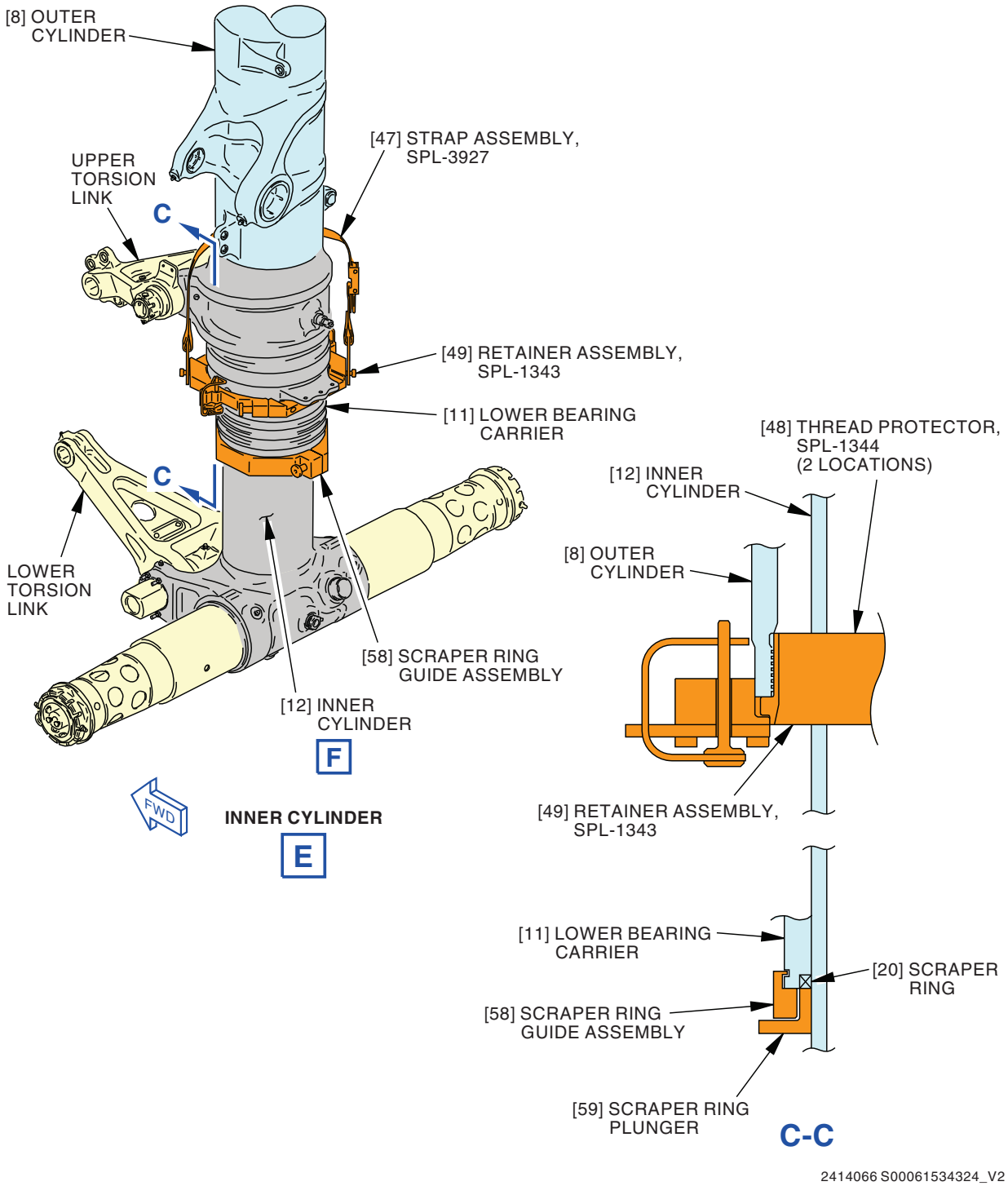
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414065 S00061534323_V2

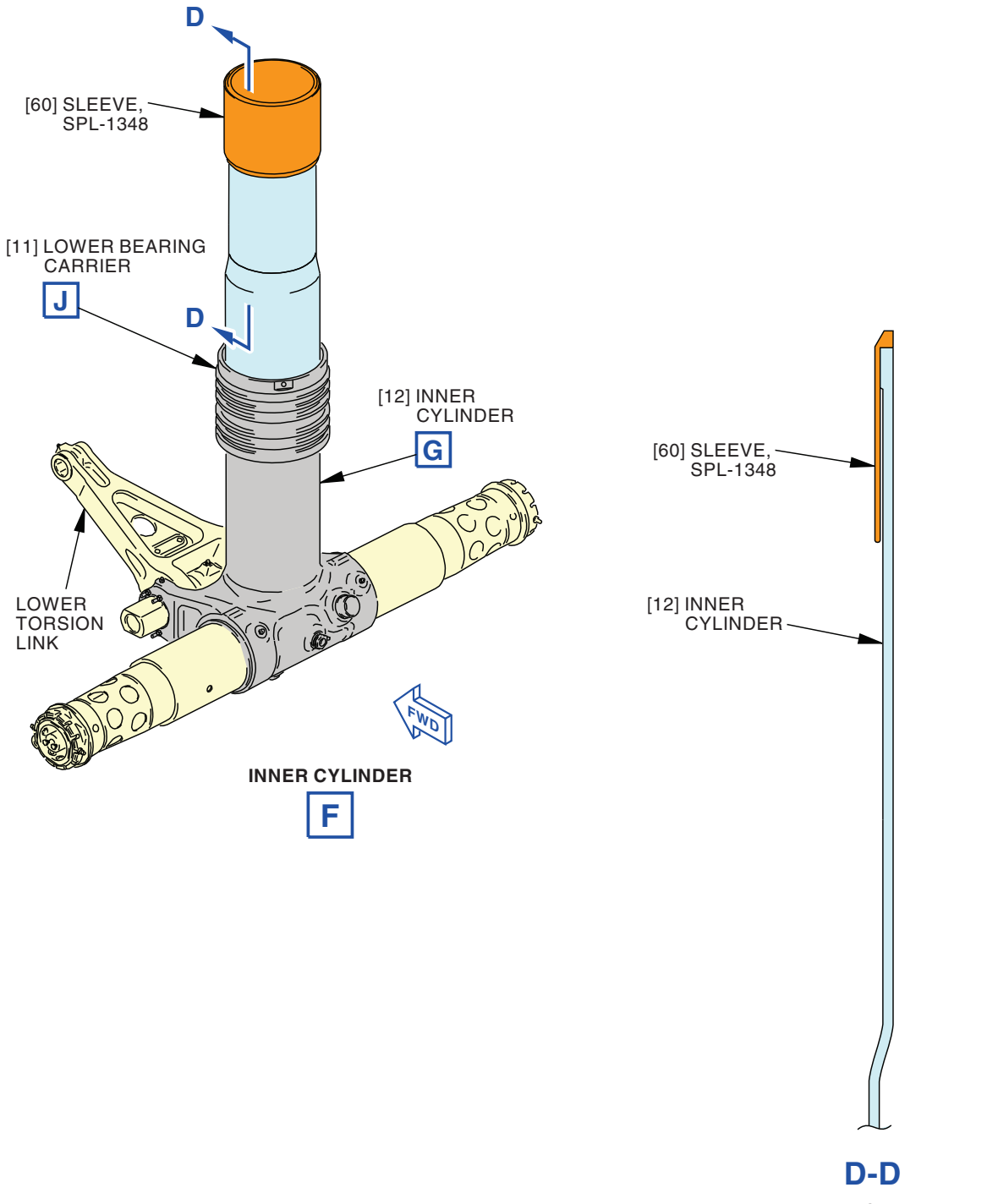
Main Landing Gear Lower Bearing Seal Replacement
Figure 802/32-11-21-990-802 (Sheet 4 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



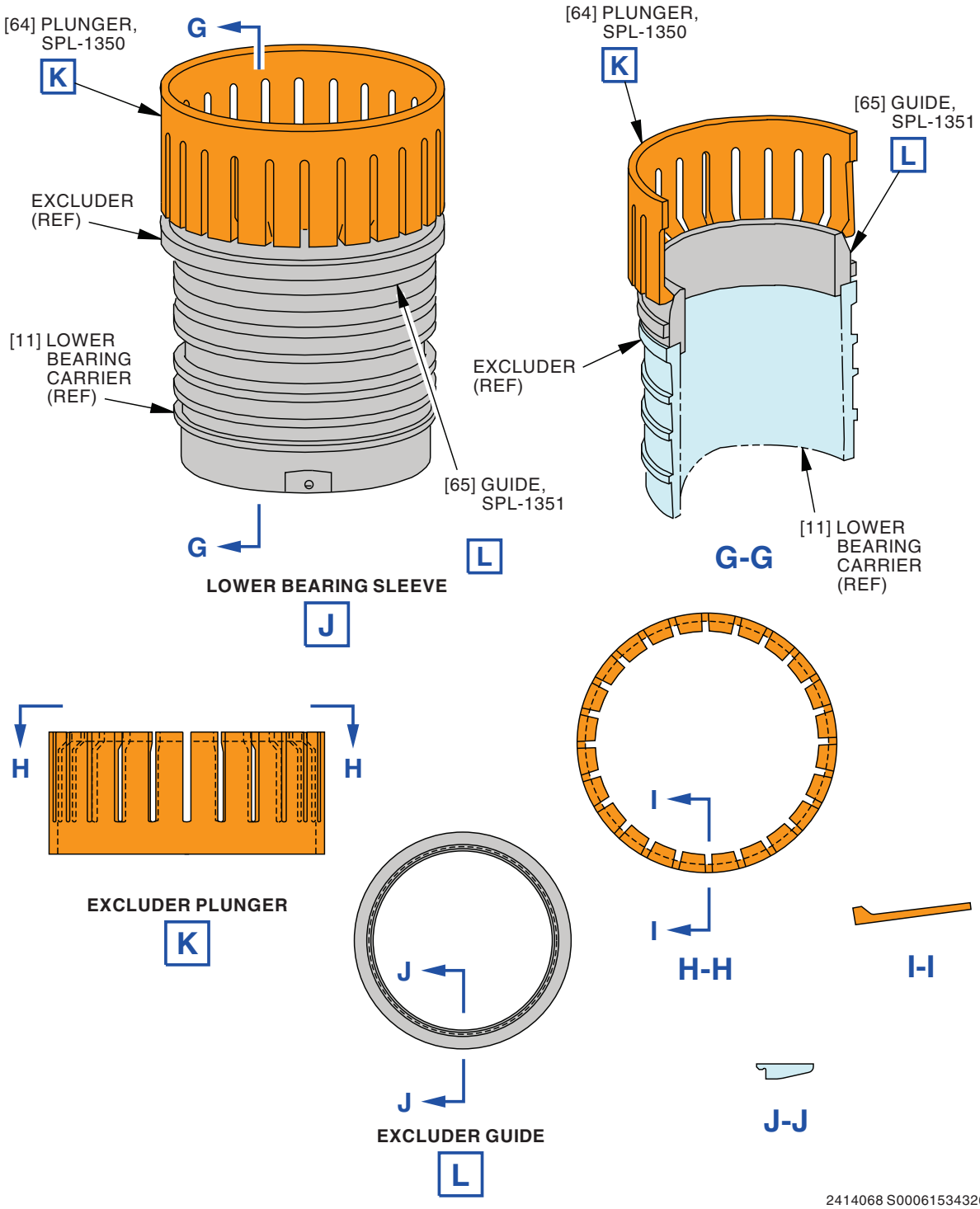
Main Landing Gear Lower Bearing Seal Replacement
Figure 802/32-11-21-990-802 (Sheet 5 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



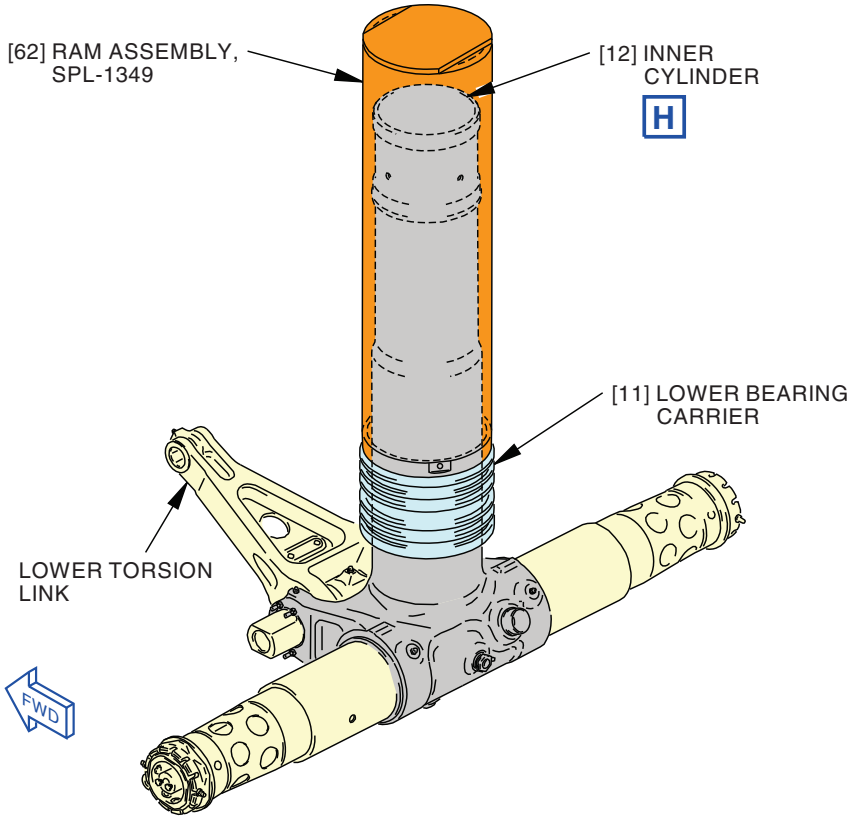
Main Landing Gear Lower Bearing Seal Replacement
Figure 802/32-11-21-990-802 (Sheet 6 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Lower Bearing Seal Replacement
Figure 802/32-11-21-990-802 (Sheet 7 of 10)

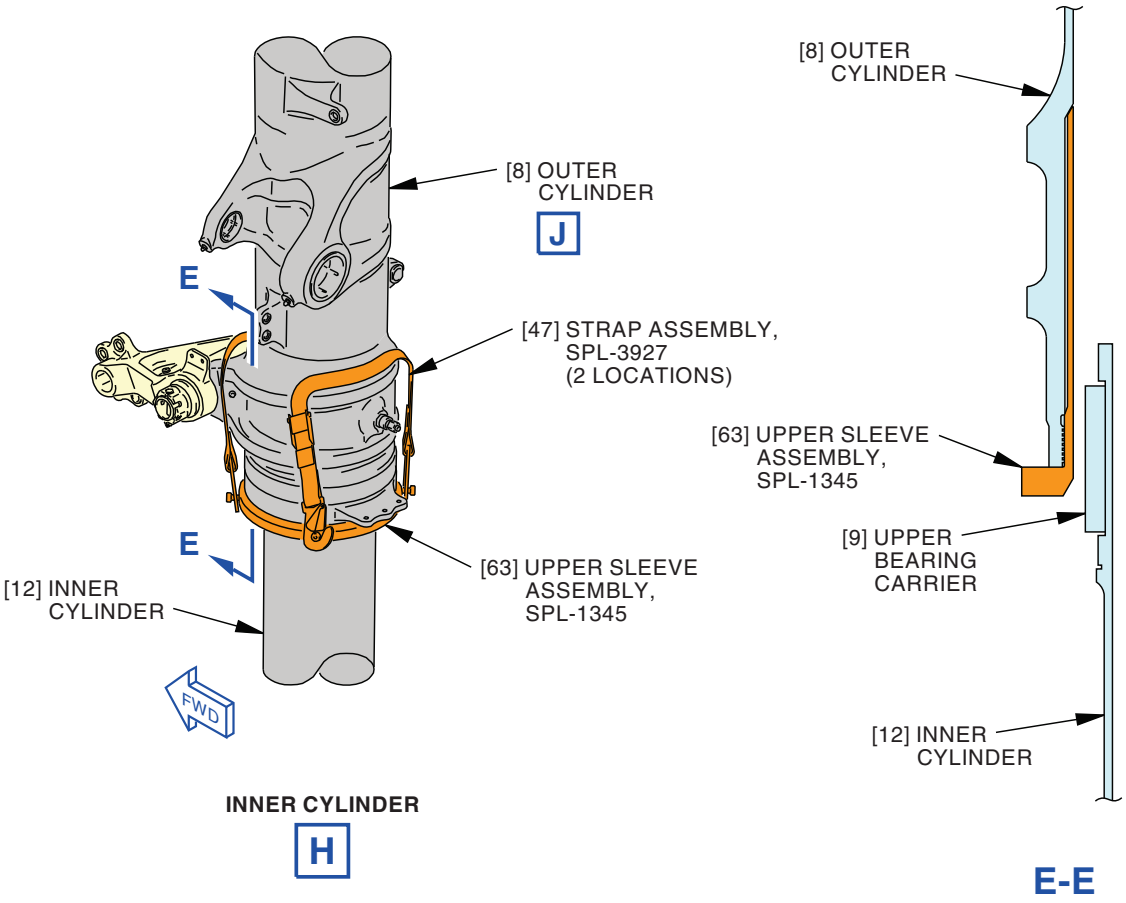
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Main Landing Gear Lower Bearing Seal Replacement
Figure 802/32-11-21-990-802 (Sheet 8 of 10)

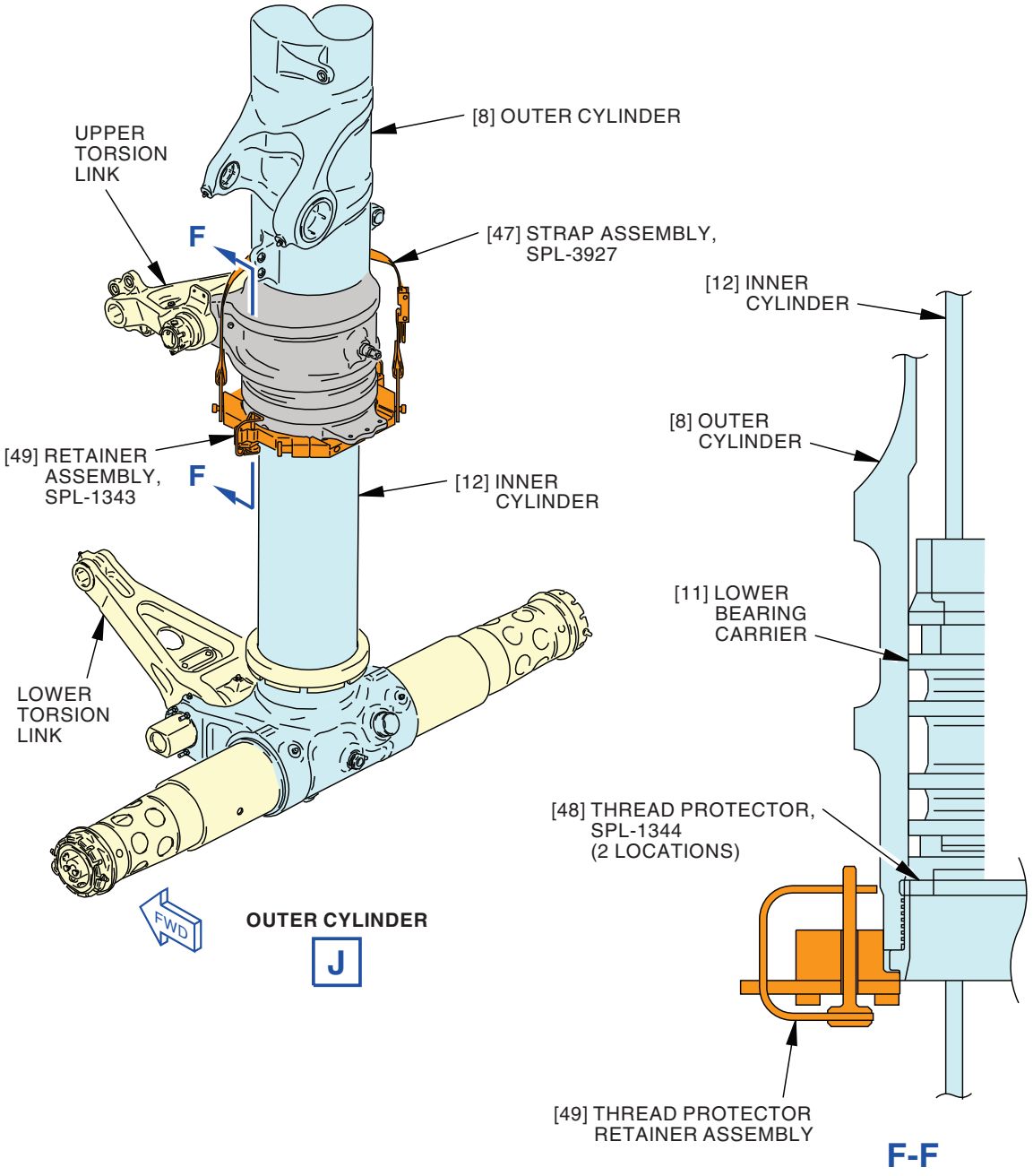
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414070 S00061534328_V2

Main Landing Gear Lower Bearing Seal Replacement
Figure 802/32-11-21-990-802 (Sheet 9 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

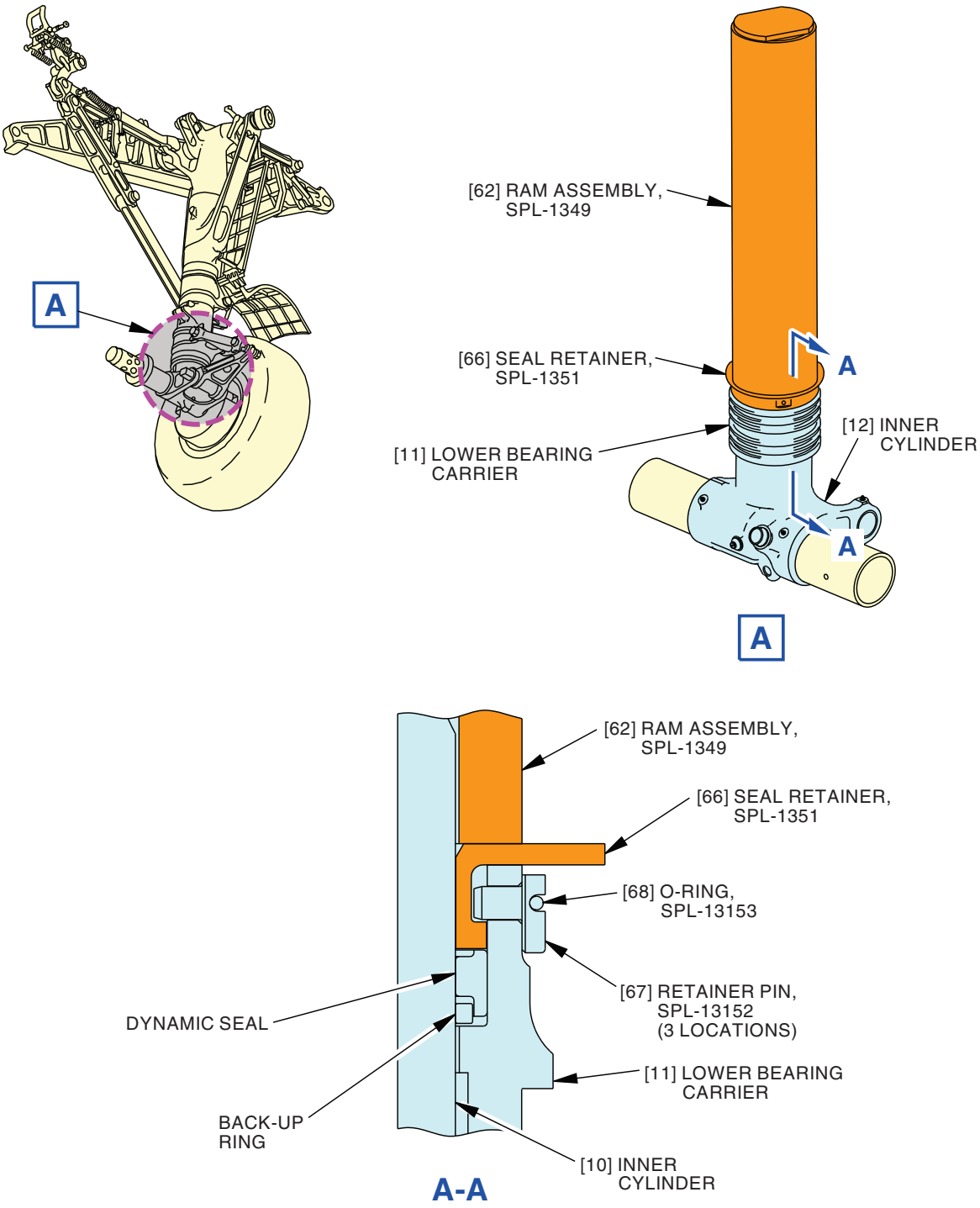


1 USE THREAD PROTECTOR FOR LOWER BEARING CARRIER INSTALLATION ONLY.

2414071 S00061534329_V2

Main Landing Gear Lower Bearing Seal Replacement
Figure 802/32-11-21-990-802 (Sheet 10 of 10)

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Lower Bearing Carrier Installation Using Optional Tooling
Figure 803/32-11-21-990-803

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TASK 32-11-21-960-802

3. Active and the Spare Seals Replacement

(Figure 801, Figure 802)

A. General

(1) This procedure supplies instructions to install new active dynamic seals [14], new active static seals [15], new spare static seal [16], spare static seals [18], and new spare dynamic seals [17].

NOTE: To do this procedure it is necessary to disconnect the inner cylinder [12] from the outer cylinder [8].

(2) Use this procedure when there are no spare seals available on the lower bearing carrier [11] to replace the active seals, and you will install a new 3-piece active dynamic seal.

B. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
07-10-03-580-801	Lift the Airplane on Axle Jacks-Handling (P/B 201)
12-15-31-610-802	Main Landing Gear Shock Strut Servicing, Airplane on the Ground (P/B 301)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-11-51-020-801	Main Landing Gear Torsion Link Disconnection (P/B 401)
32-11-51-400-803	Main Landing Gear Torsion Link Connection (P/B 401)
32-41-41-000-801	Main Landing Gear Brake Removal (P/B 401)
32-41-41-400-801	Main Landing Gear Brake Installation (P/B 401)
32-42-00-710-803	Transducer Operational Test (P/B 501)
32-42-11-000-801	Antiskid Wheel Speed Transducer Removal (P/B 401)
32-42-11-400-801	Antiskid Wheel Speed Transducer Installation (P/B 401)
32-46-00-730-801	Tire and Brake Monitoring System - System Test (P/B 501)
SWPM 20-60-03	Special Protection of Electrical Connectors

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1340	Spacer Assy, C32017-4 (included in C32017 Eqpt) Part #: C32017-67 Supplier: 81205 Opt Part #: C32017-52 Supplier: 81205 Opt Part #: C32017-53 Supplier: 81205
SPL-1341	Carrier Puller Assy, C32017-5 (included in C32017 Eqpt) Part #: C32017-67 Supplier: 81205 Opt Part #: C32017-52 Supplier: 81205 Opt Part #: C32017-53 Supplier: 81205

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(Continued)

Reference	Description
SPL-1343	Thread Protector Retainer Assy, C32017-8 (included in C32017 Eqpt) Part #: C32017-67 Supplier: 81205 Opt Part #: C32017-52 Supplier: 81205 Opt Part #: C32017-53 Supplier: 81205
SPL-1344	Thread Protector, C32017-9 (included in C32017 Eqpt) Part #: C32017-67 Supplier: 81205 Opt Part #: C32017-52 Supplier: 81205 Opt Part #: C32017-53 Supplier: 81205
SPL-1345	Upper Sleeve Assy, C32017-10 (included in C32017 Eqpt) Part #: C32017-67 Supplier: 81205 Opt Part #: C32017-52 Supplier: 81205 Opt Part #: C32017-53 Supplier: 81205
SPL-1348	Dynamic Seal Sleeve, C32017-54 (included in C32017 Eqpt) Part #: C32017-67 Supplier: 81205 Opt Part #: C32017-52 Supplier: 81205 Opt Part #: C32017-53 Supplier: 81205
SPL-1349	Ram Assy, C32017-30 (included in C32017 Eqpt) Part #: C32017-67 Supplier: 81205 Opt Part #: C32017-52 Supplier: 81205 Opt Part #: C32017-53 Supplier: 81205
SPL-1350	Excluder Plunger, C32017-33 (included in C32017 Eqpt) Part #: C32017-67 Supplier: 81205 Opt Part #: C32017-52 Supplier: 81205 Opt Part #: C32017-53 Supplier: 81205
SPL-1351	Excluder Guide, C32017-34 (included in C32017 Eqpt) Part #: C32017-67 Supplier: 81205 Opt Part #: C32017-52 Supplier: 81205 Opt Part #: C32017-53 Supplier: 81205
SPL-1829	Oil Drain Valve - Shock Strut, LG Part #: J32108-16 Supplier: 81205 Opt Part #: A32066-1 Supplier: 81205
SPL-3927	Strap Assy, C32017-60 (included in C32017 Eqpt) Part #: C32017-67 Supplier: 81205 Opt Part #: C32017-52 Supplier: 81205 Opt Part #: C32017-53 Supplier: 81205
SPL-8264	Transportation Equipment - Tower Jack or Holding Fixture, MLG Part #: J32062-26 Supplier: 81205 Opt Part #: J32062-1 Supplier: 81205
SPL-10997	Wrench Adapter - Gland Nut, Shock Strut MLG Part #: F80033-8 Supplier: 81205
SPL-13151	Seal Retainer, C32017-45 (included in C32017 Eqpt) Part #: C32017-67 Supplier: 81205 Opt Part #: C32017-53 Supplier: 81205
SPL-13152	Retainer Pin, C32017-46 (included in C32017 Eqpt) Part #: C32017-67 Supplier: 81205 Opt Part #: C32017-53 Supplier: 81205

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(Continued)

Reference	Description
SPL-13153	O-Ring Assy, C32017-47 (included in C32017 Eqpt) Part #: C32017-67 Supplier: 81205 Opt Part #: C32017-53 Supplier: 81205
STD-1110	Container - Hydraulic Fluid Resistant, 5 Gallon (19 Liter)
STD-1140	Hose - Flexible, 1/2 Inch ID, BMS 3-11 Resistant, 10 Foot
STD-1200	Sandbag
STD-3906	Mallet - Rubber

D. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735
D00467	Fluid - Landing Gear Shock Strut	BMS3-32 Type II
D00589	Lubricant - Hydraulic Assembly - AFS-682	
D50004	Compound - Antiseize	BMS3-28
G02166	Lockwire - MS20995NC20, Monel - 0.020 Inch (0.508 mm) Diameter	NASM20995
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38
G51677	Cloth - 100% Synthetic or Blended Synthetic, Cotton or Cellulose Material	AMS3819 Class 1, 2, or 4, Grade A or B, Form 1

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
14	Active dynamic seal	32-11-21-01-120	PTF ALL
15	Active static seal	32-11-21-01-080	PTF ALL
16	Spare static seal	32-11-21-01-085	PTF ALL
17	Spare dynamic seal	32-11-21-01-090	PTF ALL
18	Spare static seal	32-11-21-01-085	PTF ALL
20	Scraper ring	32-11-21-01-100	PTF ALL
		32-11-21-01-105	PTF ALL
27	Excluder	32-11-21-01-095	PTF ALL

F. Location Zones

Zone	Area
731	Left Main Landing Gear - Outboard Door
741	Right Main Landing Gear - Outboard Door

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G. Prepare for the Replacement

SUBTASK 32-11-21-480-008

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-11-21-860-005

(2) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-11-21-860-006

(3) Operate the brakes eight times to remove pressure from the brake accumulator.

SUBTASK 32-11-21-480-009

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(4) Set the wing jacks correctly to keep the airplane level prior to deflating the Main Landing Gear (MLG) shock strut [1] (TASK 07-10-01-580-801).

NOTE: Lift the airplane to a sufficient height that will let you install the spacer assy, C32017-4, SPL-1340.

NOTE: The airplane is to be lifted further in a subsequent step.

SUBTASK 32-11-21-860-007

(5) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1
B	16	C01346	LANDING GEAR PARKING BRAKE
B	17	C00129	LANDING GEAR LATCH & PRESS WARN
C	15	C01355	LANDING GEAR AIR/GND SYS 2
C	16	C01356	LANDING GEAR AIR/GND SYS 1
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN
D	15	C01401	LANDING GEAR AIR/GND RELAY
D	16	C01432	LANDING GEAR ALTN EXTEND SOL
E	16	C00196	LANDING GEAR ANTISKID INBD
E	18	C00195	LANDING GEAR ANTISKID OUTBD

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SUBTASK 32-11-21-020-011

WARNING DO NOT REMOVE THE VALVE BODY UNTIL YOU DEFLATE THE SHOCK STRUT FULLY. THE AIR PRESSURE CAN BLOW THE VALVE BODY OUT AND CAUSE INJURIES TO PERSONNEL.

(6) Deflate the shock strut [1] for the main landing gear.

(a) Remove the dust cap [2] for the gas charging valve [3].

(b) Loosen the swivel nut [4] a maximum of two turns.

1) Deflate the shock strut [1] slowly to prevent the leakage of the fluid through the gas charging valve [3].

NOTE: Fluid in the shock strut [1] will have bubbles when you release the pressure.

(c) Loosen the swivel nut [4] fully when all of the pressure in the shock strut [1] is released.

SUBTASK 32-11-21-680-002

(7) Drain the oil from the shock strut [1] for the main landing gear.

(a) Remove the dust cap [5] from the oil charging valve [6].

(b) Put a 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110, in a position to catch the shock strut fluid when the oil charging valve [6] is opened.

(c) Install the drain equipment on the oil charging valve [6].

1) Install the tool, landing gear shock strut drain valve, SPL-1829, on the oil charging valve [6] to drain the shock strut [1].

2) If you wish to drain the shock strut [1] with a quicker method, do these steps as follows:

a) Cut a length of plastic ten foot hose (1/2 inch ID), STD-1140, long enough to reach the container on the floor.

b) Insert a small Allen wrench in the end of the length of tubing, such that the end of the long part of the Allen wrench is flush with the end of the tube and the short part penetrates the side wall of the tube.

c) Install the ten foot hose (1/2 inch ID), STD-1140, on the oil charging valve [6] such that the Allen wrench goes into the oil charging valve [6] and holds it open to drain the hydraulic fluid.

(d) Remove the drain equipment when you have removed all of the shock strut oil.

H. Prepare to Disconnect the Inner Cylinder from the Outer Cylinder

SUBTASK 32-11-21-020-012

(1) Do this task: Main Landing Gear Brake Removal, TASK 32-41-41-000-801.

NOTE: This procedure will also remove the wheel and tire assemblies.

SUBTASK 32-11-21-020-013

(2) Remove the in-axle assemblies (TASK 32-42-11-000-801).

(a) Make sure that you attach a label to the wires to indicate the left and right connectors.

SUBTASK 32-11-21-020-014

(3) Disconnect the conduit assembly [87] and the wiring assembly [43]:

(a) Remove the screws [44] and washers [45] to disconnect the conduit assembly [87] from the clamps [46].

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(b) Remove the screws [90], nuts [93], and washers [92] to disconnect the conduit assembly [87] from the clamps [91].

(c) Remove the lockwires from the fitting [41] and conduit nut [88].

(d) Loosen the conduit nut [88] to disconnect the conduit assembly [87] from the inner cylinder [12].

(e) Loosen the fitting [41].

(f) Pull the wiring assembly [43] with the electrical connectors [42] completely out of the axle [40] and inner cylinder [12].

1) Make sure that you attach a label to the wires to indicate the left and right connectors.

SUBTASK 32-11-21-020-015

(4) Do this task: Main Landing Gear Torsion Link Disconnection, TASK 32-11-51-020-801.

SUBTASK 32-11-21-480-010

(5) Use rope to hold the disconnected end of the lower torsion link, electrical connector [42], and wiring assembly [43], such that they will be out of the way when you do the subsequent steps.

SUBTASK 32-11-21-580-005

(6) Use the axle jack to hold the inner cylinder [12] when you do the subsequent steps (TASK 07-10-03-580-801).

SUBTASK 32-11-21-020-016

(7) Remove the gland nut [19].

(a) Remove the nuts [24], washers [25], and bolts [26] for the lockplate [23] to release the gland nut [19].

(b) Wind a cloth around the chrome surface of the inner cylinder [12].

NOTE: This will prevent damage to the inner cylinder [12] when the gland nut [19] moves down.

(c) Use the gland nut wrench adapter, SPL-10997, to remove the gland nut [19] from the outer cylinder [8].

1) Loosen the gland nut [19] until it is disconnected from the outer cylinder [8].

a) Do not move the gland nut [19] down the inner cylinder [12].

SUBTASK 32-11-21-480-011

(8) Install the spacer assy, C32017-4, SPL-1340 (spacer assembly, SPL-1340 [51]):

(a) Put the spacer assy, C32017-4, SPL-1340 (spacer assembly, SPL-1340 [51]), around the gland nut [19].

(b) Tighten the retainer screws to hold the spacer assy, C32017-4, SPL-1340 (spacer assembly, SPL-1340 [51]), in its position.

(c) Move the spacer assy, C32017-4, SPL-1340 (spacer assembly, SPL-1340 [51]), with the gland nut [19] down the inner cylinder [12].

SUBTASK 32-11-21-580-006

(9) Lift the airplane with the jacks until the measurement from the bottom of the outer cylinder [8] to the ground is 60 in. (1524 mm) (TASK 07-10-01-580-801).

SUBTASK 32-11-21-480-012

(10) Prepare sandbags, STD-1200, to put the inner cylinder [12] down after its disconnected from the outer cylinder [8].

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SUBTASK 32-11-21-020-017

WARNING

USE A MINIMUM OF FOUR PERSONS TO HOLD THE INNER CYLINDER WHEN YOU DO THE SUBSEQUENT STEPS. THE INNER CYLINDER WEIGHS APPROXIMATELY 320 LB (145 KG). INJURIES TO PERSONNEL AND DAMAGE TO THE EQUIPMENT CAN OCCUR IF YOU DO NOT HOLD THE INNER CYLINDER CORRECTLY.

(11) Remove the inner cylinder [12].

(a) Lower the axle jack slowly until the inner cylinder [12] is completely disconnected from the outer cylinder [8] (TASK 07-10-03-580-801).

(b) Hold the inner cylinder [12] in a vertical position.

SUBTASK 32-11-21-020-018

(12) Remove the upper bearing carrier [9].

(a) Remove the upper bearings [55] from the upper bearing carrier [9].

1) Pull the upper bearing [55] apart slightly and slide it up and off the upper bearing carrier [9].

2) Inspect the upper bearings [55] for damage and excessive wear.

a) Make sure that the upper bearing [55] thickness is equal to or more than 0.0865 in. (2.2 mm).

b) If it is necessary, replace the upper bearing [55].

(b) Remove the piston ring [56] from the upper bearing carrier [9].

(c) Remove the upper bearing carrier [9] halves from the inner cylinder [12].

SUBTASK 32-11-21-020-019

(13) Remove the recoil valve [57] from the inner cylinder [12].

SUBTASK 32-11-21-020-020

(14) Remove the spacer [10] from the inner cylinder [12].

SUBTASK 32-11-21-020-021

(15) Remove the seal retainer [22].

(a) Remove the O-ring [13] for the seal retainer [22].

(b) Remove the retainer pins [21] to disconnect the seal retainer [22] from the lower bearing carrier [11].

(c) Remove the seal retainer [22] from the lower bearing carrier [11].

SUBTASK 32-11-21-020-022

(16) Remove the lower bearing carrier [11] from the inner cylinder [12].

(a) Loosely install the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]), around the inner cylinder [12].

(b) Align the holes in the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]), with the holes in the lower bearing carrier [11] for the carrier puller screws [52].

(c) Put the carrier puller screws [52] through the holes in the bottom of the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]), and in the lower bearing carrier [11].

(d) Tighten the carrier puller screws [52] to attach the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]), to the lower bearing carrier [11].

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(e) Tighten the retainer screws on each side of the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]).

(f) Use a rubber mallet, STD-3906, and lightly hit the bottom of the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]), until the lower bearing carrier [11] can be pulled out off the top of the inner cylinder [12].

NOTE: The lower bearing carrier [11] is coated with a dry lube film. The film may be worn and appear uneven, this is not cause for lower bearing carrier [11] replacement.

(g) Remove the carrier puller screws [52] to disconnect the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]), from the lower bearing carrier [11].

(h) Remove the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]), from the lower bearing carrier [11].

SUBTASK 32-11-21-840-001

WARNING

USE A MINIMUM OF FOUR PERSONS TO HOLD THE INNER CYLINDER WHEN YOU DO THE SUBSEQUENT STEPS. THE INNER CYLINDER WEIGHS APPROXIMATELY 320 LB (145 KG). INJURIES TO PERSONNEL AND DAMAGE TO THE EQUIPMENT CAN OCCUR IF YOU DO NOT HOLD THE INNER CYLINDER CORRECTLY.

(17) Slowly put the inner cylinder [12] down on sandbags, STD-1200.

NOTE: If necessary, attach a strap to the inner cylinder [12] to assist when lowering onto the sandbags.

SUBTASK 32-11-21-640-002

CAUTION

DO NOT APPLY VASELINE OR PETROLATUM TO THE SEALS. THIS CAN CAUSE THE SEAL MATERIAL TO BECOME LARGER. DAMAGE TO THE SEALS CAN OCCUR.

(18) Lubricate the replacement seals with fluid, D00467, or AFS-682 lubricant, D00589.

(a) These are the seals that can be necessary:

1) 1 active dynamic seal [14]
2) 1 active static seal [15]
3) 1 spare static seal [16]
4) 2 spare dynamic seals [17]
5) 1 spare static seal [18]
6) 1 excluder [27]
7) 1 scraper ring [20].

I. New Active Static Seal, Spare Static Seals, Spare Dynamic Seals, Active Dynamic Seal, Excluder, and Scraper Ring Installation

SUBTASK 32-11-21-840-009

(1) Remove the lower bearing insert [86] from the lower bearing carrier [11].

(a) Inspect the lower bearing insert [86] for damage and excessive wear.

1) The lower bearing insert [86] thickness must be greater than or equal to 0.0865 in. (2.1971 mm).
2) If it is necessary, replace the lower bearing insert [86].

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SUBTASK 32-11-21-420-056

(2) Install the lower bearing insert [86] on the lower bearing carrier [11].

SUBTASK 32-11-21-900-003

(3) Install the new scraper ring [20] on the lower bearing carrier [11].

NOTE: The scraper ring [20] prevents dirt and debris from entering the shock strut [1] when the inner cylinder [12] and outer cylinder [8] compress.

(a) Remove the old scraper ring [20] from its groove on the lower bearing carrier [11].

(b) Install the new scraper ring [20] in its groove on the lower bearing carrier [11].

NOTE: There are two possible scraper configurations used for installation (Figure 801).

(c) Make sure that to use the correct scraper configuration that will be installed in its groove on the lower bearing carrier [11].

SUBTASK 32-11-21-900-004

(4) Install the new excluder [27] on the lower bearing carrier [11].

(a) Remove the old excluder [27] from its groove on the lower bearing carrier [11].

(b) Use the guide, C32017-34, SPL-1351 (guide, SPL-1351 [65]), and plunger, C32017-33, SPL-1350 (plunger, SPL-1350 [64]), to install the new excluder [27] in its groove on the lower bearing carrier [11].

SUBTASK 32-11-21-840-003

(5) Put the lower bearing carrier [11] in the up position.

SUBTASK 32-11-21-900-005

(6) Install the new spare static seal [18] on the lower bearing carrier [11].

(a) Remove the old spare static seal [18] from its groove on the lower bearing carrier [11].

(b) Install the new spare static seal [18] in its groove on the lower bearing carrier [11].

SUBTASK 32-11-21-420-008

(7) Install the two spare dynamic seals [17], two brown backup rings [29], and two white backup rings [28].

(a) Install the first spare dynamic seal [17] at the top of its groove on the lower bearing carrier [11].

(b) Apply a thin layer of fluid, D00467, or AFS-682 lubricant, D00589, to the brown backup ring [29], and white backup ring [28].

NOTE: The AFS-682 lubricant, D00589, will help keep the brown backup ring [29] and white backup ring [28] in their positions.

(c) Install the brown backup ring [29] and white backup ring [28] between the first spare dynamic seal [17] installed, with the rounded edge of the white backup ring [28] against the seal.

(d) Install the second spare dynamic seal [17] from its groove in the lower bearing carrier [11], with the edge overlaps to the first spare dynamic seal [17] installed.

SUBTASK 32-11-21-900-006

(8) Install the new spare static seal [16] on the lower bearing carrier [11].

(a) Remove the old spare static seal [16] from its groove on the lower bearing carrier [11].

(b) Install the new spare static seal [16] in its groove on the lower bearing carrier [11].

SUBTASK 32-11-21-900-007

(9) Install the new active static seal [15] on the lower bearing carrier [11].

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(a) Remove the two brown rings from around the active static seal [15].

(b) Make sure that the brown rings are serviceable.

NOTE: The recommended installation is to replace all the old brown rings with new brown rings. Use the old brown rings only after a full inspection, and new rings are not available.

1) Install new brown rings if there are nicks, scratches, or other damage.

(c) Remove the old active static seal [15] from its groove on the lower bearing carrier [11].

(d) Install the new active static seal [15] in its groove on the lower bearing carrier [11].

(e) Apply a thin layer of fluid, D00467, or AFS-682 lubricant, D00589, to the brown rings.

NOTE: The AFS-682 lubricant, D00589, will help keep the brown rings in their positions.

(f) Install the brown rings on the new active static seal [15] with the rounded edge of the brown rings against the active static seal [15].

SUBTASK 32-11-21-900-008

(10) Install the new active dynamic seal [14] in the lower bearing carrier [11].

(a) Remove the cap, active dynamic seal [14], and backup ring from the inner groove for the dynamic seal.

CAUTION

DO NOT APPLY VASELINE OR PETROLATUM TO THE SEALS. THIS CAN CAUSE THE SEAL MATERIAL TO BECOME LARGER. DAMAGE TO THE SEALS CAN OCCUR.

(b) Apply AFS-682 lubricant, D00589, to the active dynamic seal [14] to help keep the seal parts together.

(c) Install the backup ring part of the new active dynamic seal [14] in the inner groove for the active dynamic seal [14].

1) Put the backup ring in position toward the bottom end of the shock strut [1] with the curved corners of the backup ring against the active dynamic seal [14] (Figure 801).

(d) Install the seal part of the new active dynamic seal [14] in inner groove for the active dynamic seal [14].

(e) Install the cap part of the new active dynamic seal [14].

SUBTASK 32-11-21-420-009

(11) Install the seal retainer [22] or optional seal retainer, C32017-45, SPL-13151 (seal retainer, SPL-13151 [66]).

NOTE: You may use optional seal retainer, C32017-45, SPL-13151 (seal retainer, SPL-13151 [66]), to help correctly install the dynamic seal

(a) Install the production seal retainer [22].

1) Put the seal retainer [22] in its position on the lower bearing carrier [11].

2) Install the retainer pins [21] to connect the seal retainer [22] to the lower bearing carrier [11].

3) Install the O-ring [13] to hold the retainer pins [21] in their position.

(b) Install the optional seal retainer, C32017-45, SPL-13151 (seal retainer, SPL-13151 [66]).

1) Put the seal retainer, C32017-45, SPL-13151 (seal retainer, SPL-13151 [66]), in its position on the lower bearing carrier [11].

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2) Install the retainer pin, C32017-46, SPL-13152 (retainer pin, SPL-13152 [67]), to connect the seal retainer, C32017-45, SPL-13151 (seal retainer, SPL-13151 [66]), to the lower bearing carrier [11].

3) Install the o-ring assy, C32017-47, SPL-13153 (O-ring, SPL-13153 [68]), to hold the retainer pin, C32017-46, SPL-13152 (retainer pin, SPL-13152 [67]), in position.

SUBTASK 32-11-21-840-004

WARNING

USE A MINIMUM OF FOUR PERSONS TO HOLD THE INNER CYLINDER WHEN YOU DO THE SUBSEQUENT STEPS. THE INNER CYLINDER WEIGHS APPROXIMATELY 320 LB (145 KG). INJURIES TO PERSONNEL AND DAMAGE TO THE EQUIPMENT CAN OCCUR IF YOU DO NOT HOLD THE INNER CYLINDER CORRECTLY.

(12) Hold the inner cylinder [12] in a vertical position.

NOTE: If necessary, attach a strap to the inner cylinder [12] to assist with holding in a vertical position.

SUBTASK 32-11-21-480-013

(13) Install the sleeve, C32017-54, SPL-1348 (sleeve, SPL-1348 [60]), at the top of the inner cylinder [12].

SUBTASK 32-11-21-020-023

(14) Install the lower bearing carrier [11].

NOTE: You can install the lower bearing carrier [11] onto the sleeve, C32017-54, SPL-1348 (sleeve, SPL-1348 [60]), at the workbench and then install the sleeve, C32017-54, SPL-1348 (sleeve, SPL-1348 [60]), with the lower bearing carrier [11] onto the top of the cylinder.

CAUTION

MAKE SURE THAT THE BACKUP RINGS AND THE SEAL STAY TOGETHER WHEN YOU INSTALL THE SEAL. IF THE BACKUP RINGS AND THE SEAL MOVE APART, THEY WILL NOT INSTALL CORRECTLY. THIS CAN CAUSE DAMAGE TO THE SEAL AND THE BACKUP RINGS.

(a) Put the lower bearing carrier [11] in its position for installation on the inner cylinder [12].

(b) Use a rubber mallet, STD-3906, and lightly hit the top of the lower bearing carrier [11] until it passes the sleeve, C32017-54, SPL-1348 (sleeve, SPL-1348 [60]).

(c) Use the ram assy, C32017-30, SPL-1349 (ram assembly, SPL-1349 [62]), to push the lower bearing carrier [11] down to the inner cylinder [12].

NOTE: A 1 in. (25 mm) webbing may be used to help move the ram assy, C32017-30, SPL-1349 (ram assembly, SPL-1349 [62]).

NOTE: Leave enough room for the installation of the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]).

SUBTASK 32-11-21-020-024

(15) Remove the ram assy, C32017-30, SPL-1349 (ram assembly, SPL-1349 [62]), and sleeve, C32017-54, SPL-1348 (sleeve, SPL-1348 [60]), from the inner cylinder [12].

(a) If you used optional seal retainer, C32017-45, SPL-13151 (seal retainer, SPL-13151 [66]), for the installation, then remove the o-ring assy, C32017-47, SPL-13153 (O-ring, SPL-13153 [68]), retainer pin, C32017-46, SPL-13152 (retainer pin, SPL-13152 [67]), and seal retainer, C32017-45, SPL-13151 (seal retainer, SPL-13151 [66]).

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(b) After you remove the optional seal retainer, C32017-45, SPL-13151 (seal retainer, SPL-13151 [66]), look down into the lower bearing carrier [11] and examine the dynamic seal.

1) Make sure that the dynamic seal and cap should appear fully seated.

(c) Install the production seal retainer [22], retainer pins [21], and O-ring [13] onto the lower bearing carrier [11].

SUBTASK 32-11-21-420-010

(16) Put the spacer [10] down the inner cylinder [12] and on top of the lower bearing carrier [11].

SUBTASK 32-11-21-420-011

(17) Install the recoil valve [57] on the inner cylinder [12].

SUBTASK 32-11-21-420-012

(18) Install the upper bearing carrier [9].

(a) Install the upper bearing carrier [9] halves on the inner cylinder [12].

(b) Install the piston ring [56] to hold the upper bearing carrier [9] halves around the inner cylinder [12].

(c) Install one of the upper bearings [55] on the lower groove of the upper bearing carrier [9] halves, with the split about 90 degrees from the split of the upper bearing carrier [9] halves.

1) Make an overlap of 0.25 in. (0.64 cm) to the upper bearing [55] to keep their round shape.

(d) Install the other upper bearing [55] on the top groove of the upper bearing carrier [9] halves, with the split opposite the split of the first upper bearing [55] installed.

SUBTASK 32-11-21-420-013

(19) Install the upper sleeve assy, C32017-10, SPL-1345 (upper sleeve assembly, SPL-1345 [63]).

(a) Install the upper sleeve assy, C32017-10, SPL-1345 (upper sleeve assembly, SPL-1345 [63]), such that it fits at the bottom of the outer cylinder [8].

(b) Install the strap assy, C32017-60, SPL-3927 (strap assembly, SPL-3927 [47]), around the outer cylinder [8] and connect the ends to the shoulder bolts on the side of the upper sleeve assy, C32017-10, SPL-1345 (upper sleeve assembly, SPL-1345 [63]).

1) Tighten the strap assy, C32017-60, SPL-3927 (strap assembly, SPL-3927 [47]).

SUBTASK 32-11-21-420-014

CAUTION

MAKE SURE THAT YOU PUT THE INNER AND OUTER CYLINDER TOGETHER CAREFULLY. THE NEW SEALS CAN CATCH ON THE OUTER CYLINDER, OR THE INNER AND OUTER CYLINDERS CATCH WHEN YOU ASSEMBLE THEM. THIS CAN CAUSE DAMAGE TO THE EQUIPMENT.

(20) Slowly lift the inner cylinder [12] with the axle jack such that the inner cylinder [12] goes into the outer cylinder [8] (TASK 07-10-03-580-801).

(a) Leave enough chrome visible on the inner cylinder [12] such that you can install the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]).

SUBTASK 32-11-21-080-006

(21) Remove the upper sleeve assy, C32017-10, SPL-1345 (upper sleeve assembly, SPL-1345 [63]), from the outer cylinder [8] after it goes on the upper bearing carrier [9].

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SUBTASK 32-11-21-420-015

(22) Carefully put the thread protector, C32017-9, SPL-1344 (thread protector, SPL-1344 [48]), between the outer cylinder [8] and inner cylinder [12].

NOTE: The thread protector, C32017-9, SPL-1344 (thread protector, SPL-1344 [48]), will expand and push against the outer cylinder [8]. This will protect the static seals from the gland nut [19] threads on the outer cylinder [8].

SUBTASK 32-11-21-480-014

(23) Install the retainer assy, C32017-8, SPL-1343 (retainer assembly, SPL-1343 [49]).

(a) Install the retainer assy, C32017-8, SPL-1343 (retainer assembly, SPL-1343 [49]), around the inner cylinder [12], such that it fits at the bottom of the outer cylinder [8].

(b) Tighten the retainer screws on each side of the retainer assy, C32017-8, SPL-1343 (retainer assembly, SPL-1343 [49]).

(c) Move the two retainer plates for the retainer assy, C32017-8, SPL-1343 (retainer assembly, SPL-1343 [49]), such that it does not touch the chrome surface of the inner cylinder [12].

(d) Install the strap assy, C32017-60, SPL-3927 (strap assembly, SPL-3927 [47]), around the outer cylinder [8] and connect the ends to the shoulder bolts on each side of the retainer assy, C32017-8, SPL-1343 (retainer assembly, SPL-1343 [49]).

SUBTASK 32-11-21-480-015

(24) Install the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]).

(a) Loosely install the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]), around the inner cylinder [12], such that its below the lower bearing carrier [11].

(b) Align the holes in the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]), with the holes in the lower bearing carrier [11] for the carrier puller screws [52].

(c) Put the carrier puller screws [52] through the holes in the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]), and in the lower bearing carrier [11].

(d) Tighten the carrier puller screws [52] to attach the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]), to the lower bearing carrier [11].

(e) Tighten the retainer screws on each side of the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]).

SUBTASK 32-11-21-580-021

(25) Slowly lower the airplane so that the top of the lower bearing carrier [11] is approximately 1 in. (3 cm) to 3 in. (8 cm) from the bottom of the outer cylinder [8].

(a) If the seals were replaced on one side only, then depressurize the opposite shock strut [1] before the airplane is lowered.

SUBTASK 32-11-21-580-007

(26) Lift the inner cylinder [12] slowly with the axle jack or transportation equipment, SPL-8264, until the bottom of the lower bearing carrier [11] is approximately 1.5 in. (3.8 cm) from the bottom of the outer cylinder [8] (TASK 07-10-03-580-801).

(a) Make sure that the static seals stay in their positions as the lower bearing carrier [11] is installed.

SUBTASK 32-11-21-080-007

(27) Remove the retainer assy, C32017-8, SPL-1343 (retainer assembly, SPL-1343 [49]).

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(a) Remove the strap assy, C32017-60, SPL-3927 (strap assembly, SPL-3927 [47]), that holds the retainer assy, C32017-8, SPL-1343 (retainer assembly, SPL-1343 [49]), to the outer cylinder [8].

(b) Loosen the retainer screws and remove the retainer assy, C32017-8, SPL-1343 (retainer assembly, SPL-1343 [49]).

SUBTASK 32-11-21-080-008

(28) Remove the thread protector, C32017-9, SPL-1344 (thread protector, SPL-1344 [48]), from the outer cylinder [8].

SUBTASK 32-11-21-480-016

(29) Remove the spacer assy, C32017-4, SPL-1340 (spacer assembly, SPL-1340 [51]), from around the gland nut [19].

SUBTASK 32-11-21-080-009

(30) Remove the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]).

(a) Remove the carrier puller screws [52] to disconnect the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]), from the lower bearing carrier [11].

(b) Loosen the retainer screws and pull the carrier puller assy, C32017-5, SPL-1341 (carrier puller assembly, SPL-1341 [50]), out of the outer cylinder [8].

SUBTASK 32-11-21-620-002

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(31) Apply a thin layer of corrosion inhibiting material, G50136, to these items:

(a) The threads and thread reliefs of the bolts [26].

(b) The threads of the nuts [24].

(c) The splines and faces of the washers [25].

SUBTASK 32-11-21-620-006

WARNING

DO NOT GET THE CORROSION-INHIBITING COMPOUND ON YOUR SKIN, IN YOUR EYES, OR ON YOUR CLOTHES. THE COMPOUND IS POISONOUS AND FLAMMABLE. USE APPROVED GLOVES, AND EYE PROTECTION. MAKE SURE THAT THERE IS GOOD AIRFLOW IN THE WORK AREA. KEEP THE COMPOUND AWAY FROM SPARKS AND FLAMES. THE COMPOUND CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(32) Apply a thick layer of corrosion preventive Cor-Ban 27L Compound, G50237 to these components:

(a) The gland nut [19] thread major and minor diameters.

(b) The mating outer cylinder [8] thread major and minor diameters.

(c) The outer cylinder [8] thread relief.

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SUBTASK 32-11-21-160-002

WARNING

DO NOT GET THE CORROSION-INHIBITING COMPOUND ON YOUR SKIN, IN YOUR EYES, OR ON YOUR CLOTHES. THE COMPOUND IS POISONOUS AND FLAMMABLE. USE APPROVED GLOVES, AND EYE PROTECTION. MAKE SURE THAT THERE IS GOOD AIRFLOW IN THE WORK AREA. KEEP THE COMPOUND AWAY FROM SPARKS AND FLAMES. THE COMPOUND CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(33) Remove all unwanted corrosion inhibiting material, G50136, or corrosion preventive Cor-Ban 27L Compound, G50237.

SUBTASK 32-11-21-420-016

(34) Install the gland nut [19].

(a) Move the gland nut [19] up to the outer cylinder [8] and engage the gland nut [19] threads.

(b) Use the gland nut wrench adapter, SPL-10997, to torque the gland nut [19] to 125 ft-lb (169 N·m) - 150 ft-lb (203 N·m).

(c) Install the lockplate [23] in the holes that will put the tab of the lockplate [23] as close to a slot in the gland nut [19] as possible.

1) If it is necessary, loosen the gland nut [19] a small quantity to help with the alignment.

(d) Install the bolts [26], washers [25], and nuts [24].

(e) Remove excess corrosion inhibiting material, G50136, from gland nut [19] and lock plate.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 32-11-21-420-017

(1) Before you connect the electrical connector [42] and wiring assembly [43], examine the electrical connector [42] for corrosion.

WARNING

DO THE STEPS BELOW IF THE AIRPLANE OPERATES WHERE DEICING FLUID THAT CONTAINS POTASSIUM FORMATE OR POTASSIUM ACETATE IS USED. ALSO, DO THE STEPS FOR ALL AIRPLANES THAT YOU FOUND CORROSION IN THE ELECTRICAL CONNECTORS IN THE MAIN WHEEL WELL. THE ELECTRICAL CONNECTORS ARE IN A SYSTEM THAT IS NECESSARY FOR SAFE FLIGHT.

(a) If there was corrosion, correct the problem (SWPM 20-60-03).

(b) Apply D5026NS or ZC-026 compound, G50171, to the electrical connector [42] (SWPM 20-60-03).

SUBTASK 32-11-21-420-061

(2) Connect the wiring assembly [43] and conduit assembly [87].

(a) Put the wiring assembly [43] with the electrical connectors [42] into the inner cylinder [12] and axle [40].

(b) Clean the mating surfaces of the fitting [41] and conduit nut [88] with alcohol, B00130, and cloth, G51677.

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WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(c) Apply compound, D50004 (preferred), or Cor-Ban 27L Compound, G50237 (alternative), to the threads of the fitting [41] and conduit nut [88].

(d) Tighten the fitting [41] to 50 ±5 in-lb (5.6 ±0.6 N·m).

(e) Tighten the conduit nut [88] to 70 in-lb (7.9 N·m) - 90 in-lb (10.2 N·m) to connect the conduit assembly [87] to the inner cylinder [12].

(f) Install MS20995NC20 lockwire, G02166 on the fitting [41] and the conduit nut [88].

(g) Put the conduit assembly [87] in its position for attachment to the clamps [46] on the bracket.

(h) Install the washers [45] and screws [44] to hold the electrical harness to the clamps [46].

(i) Put the conduit assembly [87] in its position for attachment to the clamps [91] on the bracket.

(j) Install the washers [92], nuts [93], and screws [90] to hold the conduit assembly [87] to the clamps [91].

SUBTASK 32-11-21-420-053

(3) Lower the inner cylinder [12] with the MLG axle jack to install the brake assembly and wheel-tire assembly.

SUBTASK 32-11-21-420-059

(4) Remove the rope that holds the disconnected end of the upper torsion link to the outer cylinder [8].

SUBTASK 32-11-21-420-060

(5) Do this task: Main Landing Gear Torsion Link Connection, TASK 32-11-51-400-803.

SUBTASK 32-11-21-420-054

(6) Do this task: Main Landing Gear Brake Installation, TASK 32-41-41-400-801.

NOTE: This procedure will also install the wheel and tire assemblies.

(a) Make sure that you connect the correct wires to the correct in-axle assembly.

SUBTASK 32-11-21-420-019

(7) Do this task: Antiskid Wheel Speed Transducer Installation, TASK 32-42-11-400-801.

SUBTASK 32-11-21-730-003

(8) Do this task: Tire and Brake Monitoring System - System Test, TASK 32-46-00-730-801.

SUBTASK 32-11-21-860-019

(9) If seals were replaced on one side only, then depressurize the opposite shock strut before the airplane is lowered.

NOTE: This is to ensure symmetrical loading at the jackpoints as the airplane lowers to the ground.

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SUBTASK 32-11-21-860-020

(10) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1
B	16	C01346	LANDING GEAR PARKING BRAKE
B	17	C00129	LANDING GEAR LATCH & PRESS WARN
C	15	C01355	LANDING GEAR AIR/GND SYS 2
C	16	C01356	LANDING GEAR AIR/GND SYS 1
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN
D	15	C01401	LANDING GEAR AIR/GND RELAY
D	16	C01432	LANDING GEAR ALTN EXTEND SOL
E	16	C00196	LANDING GEAR ANTISKID INBD
E	18	C00195	LANDING GEAR ANTISKID OUTBD

SUBTASK 32-11-21-420-020

CAUTION

MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION

MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(11) Do this task: Lower the Airplane off Jacks, TASK 07-10-01-580-802.

SUBTASK 32-11-21-420-023

(12) Install the dust cap [5] loosely.

SUBTASK 32-11-21-600-002

(13) Do this task: Main Landing Gear Shock Strut Servicing, Airplane on the Ground, TASK 12-15-31-610-802.

SUBTASK 32-11-21-700-001

(14) Do this task: Transducer Operational Test, TASK 32-42-00-710-803.

——— END OF TASK ———

TASK 32-11-21-960-803

4. Scraper Ring Replacement

A. General

(1) Use the task to replace a damaged scraper ring [20] with a split scraper ring [20]. This task is done when replacement of the scraper ring [20] is necessary and removal of the inner cylinder [12] is not possible.

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(2) It is acceptable to operate the aircraft with a missing section of the energizer ring or other damage to the scraper ring. The scraper ring should be replaced with a new solid scraper ring or split scraper ring at the earliest maintenance opportunity to avoid excessive damage to the dynamic seal. If you do not have a split scraper ring available, you can cut a solid scraper ring or the damaged scraper ring to re-install it.

NOTE: The installation of a split scraper ring is also a temporary repair and should also be replaced at the earliest maintenance opportunity.

B. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
07-10-03-580-801	Lift the Airplane on Axle Jacks-Handling (P/B 201)
12-15-31-610-802	Main Landing Gear Shock Strut Servicing, Airplane on the Ground (P/B 301)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-11-51-020-801	Main Landing Gear Torsion Link Disconnection (P/B 401)
32-11-51-400-803	Main Landing Gear Torsion Link Connection (P/B 401)
32-41-41-000-801	Main Landing Gear Brake Removal (P/B 401)
32-41-41-400-801	Main Landing Gear Brake Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1829	Oil Drain Valve - Shock Strut, LG Part #: J32108-16 Supplier: 81205 Opt Part #: A32066-1 Supplier: 81205
SPL-9507	Replacement Equipment - Lower Bearing Seals, MLG Part #: C32017-67 Supplier: 81205 Opt Part #: C32017-52 Supplier: 81205 Opt Part #: C32017-53 Supplier: 81205
SPL-10997	Wrench Adapter - Gland Nut, Shock Strut MLG Part #: F80033-8 Supplier: 81205
STD-1110	Container - Hydraulic Fluid Resistant, 5 Gallon (19 Liter)
STD-1140	Hose - Flexible, 1/2 Inch ID, BMS 3-11 Resistant, 10 Foot

D. Consumable Materials

Reference	Description	Specification
D00467	Fluid - Landing Gear Shock Strut	BMS3-32 Type II
D00589	Lubricant - Hydraulic Assembly - AFS-682	
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

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E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
20	Scraper ring	32-11-21-01-100	PTF ALL
		32-11-21-01-105	PTF ALL

F. Location Zones

Zone	Area
731	Left Main Landing Gear - Outboard Door
741	Right Main Landing Gear - Outboard Door

G. Prepare for the Replacement

SUBTASK 32-11-21-910-001

WARNING MAKE SURE THAT THE GROUND LOCK PINS ARE INSTALLED IN ALL OF THE LANDING GEAR. WITHOUT THE GROUND LOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-11-21-730-001

(2) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-11-21-730-002

(3) Operate the brakes eight times to remove pressure from the brake accumulator.

SUBTASK 32-11-21-580-008

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(4) Set the wing jacks correctly to keep the airplane level prior to deflating the Main Landing Gear (MLG) shock strut [1] (TASK 07-10-01-580-801).

NOTE: The airplane is to be lifted further in a subsequent step.

SUBTASK 32-11-21-760-001

(5) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1
B	16	C01346	LANDING GEAR PARKING BRAKE
B	17	C00129	LANDING GEAR LATCH & PRESS WARN
C	15	C01355	LANDING GEAR AIR/GND SYS 2
C	16	C01356	LANDING GEAR AIR/GND SYS 1
D	2	C01400	PSEU ALTN

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(Continued)

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	15	C01401	LANDING GEAR AIR/GND RELAY
D	16	C01432	LANDING GEAR ALTN EXTEND SOL
E	16	C00196	LANDING GEAR ANTISKID INBD
E	18	C00195	LANDING GEAR ANTISKID OUTBD

H. Scraper Ring Replacement

SUBTASK 32-11-21-610-001

WARNING DO NOT REMOVE THE VALVE BODY UNTIL YOU DEFLATE THE SHOCK STRUT FULLY. THE AIR PRESSURE CAN BLOW THE VALVE BODY OUT AND CAUSE INJURIES TO PERSONNEL.

(1) Deflate the shock strut [1] for the main landing gear (Figure 801).

(a) Remove the dust cap [2] for the gas charging valve [3].

(b) Loosen the swivel nut [4] a maximum of two turns.

1) Deflate the shock strut [1] slowly to prevent the leakage of the fluid through the gas charging valve [3].

NOTE: Fluid in the shock strut [1] will have bubbles when you release the pressure.

(c) Loosen the swivel nut [4] fully when all of the pressure in the shock strut [1] is released.

SUBTASK 32-11-21-610-002

(2) Drain the oil from the shock strut [1] for the main landing gear.

(a) Remove the dust cap [5] from the oil charging valve [6].

(b) Put a 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110, in a position to catch the shock strut fluid when the oil charging valve [6] is opened.

(c) Install the drain equipment on the oil charging valve [6].

1) Install the landing gear shock strut drain valve, SPL-1829, on the oil charging valve [6] to drain the shock strut [1].

2) If you wish to drain the shock strut [1] with a quicker method, do these steps as follows:

a) Cut a length of plastic ten foot hose (1/2 inch ID), STD-1140, long enough to reach the container on the floor.

b) Insert a small Allen wrench in the end of the length of tubing, such that the end of the long part of the Allen wrench is flush with the end of the tube and the short part penetrates the side wall of the tube.

c) Install the ten foot hose (1/2 inch ID), STD-1140, on the oil charging valve [6] such that the Allen wrench goes into the oil charging valve [6] and holds it open to drain the hydraulic fluid.

(d) Remove the drain equipment when you have removed all of the shock strut oil.

SUBTASK 32-11-21-580-009

(3) Lift the airplane until there is approximately 5 in. (127 mm) of chrome exposed on the inner cylinder [12] after the gland nut [19] has been removed (TASK 07-10-01-580-801).

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SUBTASK 32-11-21-020-027

(4) Do this task: Main Landing Gear Brake Removal, TASK 32-41-41-000-801.

NOTE: This procedure will also remove the wheel and the tire assemblies. Also, this will give better access to the inner cylinder [12] and outer cylinder [8].

SUBTASK 32-11-21-020-025

(5) Do this task: Main Landing Gear Torsion Link Disconnection, TASK 32-11-51-020-801.

SUBTASK 32-11-21-020-026

(6) Use rope to hold the disconnected end of the upper torsion link and attach it to the outer cylinder [8].

NOTE: This step will hold the upper torsion link in its position when you do the subsequent steps.

SUBTASK 32-11-21-580-010

(7) Use axle jack to hold the inner cylinder [12] in its position in the outer cylinder [8] (TASK 07-10-03-580-801).

SUBTASK 32-11-21-020-028

(8) Remove the gland nut [19]:

(a) Remove the bolts [26], washers [25], and nuts [24] for the lockplate [23] to release the gland nut [19].

(b) Wrap a cloth around the chrome surface of the inner cylinder [12].

NOTE: This will prevent damage to the inner cylinder [12] when the gland nut [19] moves down.

(c) Use the gland nut wrench adapter, SPL-10997, to remove the gland nut [19] from the outer cylinder [8].

1) Loosen the gland nut [19] until it is disconnected from the outer cylinder [8].

a) Do not move the gland nut [19] down the inner cylinder [12].

(d) Lower the axle jack until the gland nut [19] is approximately 3 in. (7.6 cm) below the outer cylinder [8].

SUBTASK 32-11-21-020-029

(9) Remove the damaged scraper ring [20], but do not damage the inner cylinder [12] or the lower bearing carrier [11] in the removal process.

SUBTASK 32-11-21-640-004

(10) Lubricate the replacement wave cut split scraper ring [20] with fluid, D00467, or AFS-682 lubricant, D00589.

SUBTASK 32-11-21-420-049

(11) Install the wave cut split scraper ring [20] with a scraper ring plunger (lower bearing seal replacement equipment, SPL-9507, C32064-9), into the gland nut [19].

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SUBTASK 32-11-21-400-001

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(12) Apply a thin layer of corrosion inhibiting material, G50136, to these items:

(a) The threads and thread reliefs of the bolts [26].

(b) The threads of the nuts [24].

(c) The splines and faces of the washers [25].

SUBTASK 32-11-21-620-008

WARNING

DO NOT GET THE CORROSION-INHIBITING COMPOUND ON YOUR SKIN, IN YOUR EYES, OR ON YOUR CLOTHES. THE COMPOUND IS POISONOUS AND FLAMMABLE. USE APPROVED GLOVES, AND EYE PROTECTION. MAKE SURE THAT THERE IS GOOD AIRFLOW IN THE WORK AREA. KEEP THE COMPOUND AWAY FROM SPARKS AND FLAMES. THE COMPOUND CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(13) Apply a thick layer of corrosion preventive Cor-Ban 27L Compound, G50237 to these components:

(a) The gland nut [19] thread major and minor diameters.

(b) The mating outer cylinder [8] thread major and minor diameters.

(c) The outer cylinder [8] thread relief.

SUBTASK 32-11-21-160-005

WARNING

DO NOT GET THE CORROSION-INHIBITING COMPOUND ON YOUR SKIN, IN YOUR EYES, OR ON YOUR CLOTHES. THE COMPOUND IS POISONOUS AND FLAMMABLE. USE APPROVED GLOVES, AND EYE PROTECTION. MAKE SURE THAT THERE IS GOOD AIRFLOW IN THE WORK AREA. KEEP THE COMPOUND AWAY FROM SPARKS AND FLAMES. THE COMPOUND CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(14) Remove all unwanted corrosion inhibiting material, G50136, or corrosion preventive Cor-Ban 27L Compound, G50237.

SUBTASK 32-11-21-400-002

(15) Install the gland nut [19].

(a) Move the gland nut [19] up to the outer cylinder [8] and engage the gland nut [19] threads.

(b) Use the gland nut wrench adapter, SPL-10997, to torque the gland nut [19] to 125 ft-lb (169 N·m) - 150 ft-lb (203 N·m).

SUBTASK 32-11-21-400-003

(16) Install the lockplate [23] for the gland nut [19].

(a) Install the lockplate [23] in the holes that will put the tab of the lockplate [23] as close to a slot in the gland nut [19] as possible.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) If it is necessary, loosen the gland nut [19] a small amount to help with the alignment.

(b) Install the bolts [26], washers [25] and nuts [24].

I. Put the Airplane Back to Its Usual Condition

SUBTASK 32-11-21-020-051

(1) Remove the rope that holds the disconnected end of the upper torsion link to the outer cylinder [8].

SUBTASK 32-11-21-400-005

(2) Do this task: Main Landing Gear Torsion Link Connection, TASK 32-11-51-400-803.

SUBTASK 32-11-21-400-004

(3) Do this task: Main Landing Gear Brake Installation, TASK 32-41-41-400-801.

NOTE: This procedure will also install the wheel and tire assemblies.

SUBTASK 32-11-21-580-011

(4) Remove the axle jack (TASK 07-10-03-580-801).

SUBTASK 32-11-21-580-012

CAUTION

MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION

MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(5) Lower the airplane and remove the jacks (TASK 07-10-01-580-802).

SUBTASK 32-11-21-400-006

(6) Install the dust cap [5] loosely.

SUBTASK 32-11-21-610-003

(7) Do this task: Main Landing Gear Shock Strut Servicing, Airplane on the Ground, TASK 12-15-31-610-802.

SUBTASK 32-11-21-710-001

(8) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1
B	16	C01346	LANDING GEAR PARKING BRAKE
B	17	C00129	LANDING GEAR LATCH & PRESS WARN
C	15	C01355	LANDING GEAR AIR/GND SYS 2
C	16	C01356	LANDING GEAR AIR/GND SYS 1
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN
D	15	C01401	LANDING GEAR AIR/GND RELAY
D	16	C01432	LANDING GEAR ALTN EXTEND SOL

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(Continued)

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	16	C00196	LANDING GEAR ANTISKID INBD
E	18	C00195	LANDING GEAR ANTISKID OUTBD

——— END OF TASK ———

TASK 32-11-21-200-801

5. Main Landing Gear Shock Strut Seal Leakage Check

A. General

(1) This task gives the instructions to make sure that the shock strut or upper shock strut is serviceable when you find fluid leakage on the inner cylinder. This task requires a repeated wipe and observation period in order to determine if there is leakage.

NOTE: The presence of a fluid film on the inner cylinder after landings is normal. For new landing gear, overhauled landing gear or when a seal change occurred, it is usual to have hydraulic fluid streaks on the shock strut during the initial landing cycles. These streaks are caused by the hydraulic fluid applied to the shock strut seals during seal installation and will stop after more landing cycles occur. The fluid film is different from the actual continuous drips that are the indication of shock strut seal leakage.

B. References

Reference	Title
05-51-01-210-804	Phase IA Inspection (P/B 201)
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
07-10-03-580-802	Lift the Main Landing Gear Axles on Axle Jacks-Handling (P/B 201)
09-10-00-580-801	Nose Gear Maintenance Towing (P/B 201)
12-15-31-610-801	Main Landing Gear Shock Strut Fluid Check (P/B 301)
12-15-31-610-802	Main Landing Gear Shock Strut Servicing, Airplane on the Ground (P/B 301)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
143	Area Below Aft Cargo Compartment - Left
144	Area Below Aft Cargo Compartment - Right
731	Left Main Landing Gear - Outboard Door
741	Right Main Landing Gear - Outboard Door

D. Main Landing Gear Shock Strut Seal Leakage Check

NOTE: If the inspection finds a fault on the scraper ring, you will have to replace the damaged scraper ring.

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SUBTASK 32-11-21-480-041

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-11-21-210-001

(2) Examine the shock struts for leakage.

NOTE: A small quantity of hydraulic fluid that is isolated to the movable surface of the inner cylinder is not leakage.

SUBTASK 32-11-21-210-002

(3) If you find leakage of the shock strut fluid on the inner cylinder, do the steps that follow to make sure that the shock strut is serviceable:

(a) Do a check for leaks for 5 minutes.

1) Wipe the surface of the inner cylinder with a clean cloth to remove all of the fluid from the leak.

2) Monitor the inner cylinder where it meets the outer cylinder for 5 minutes.

3) Make sure that the extension of the inner cylinder does not change during this time.

4) Make a record of the location and the number of drops that come from the seals during the 5 minutes.

(b) Jack the inner gear cylinder by the gear jack pad (TASK 07-10-01-580-801 or TASK 07-10-03-580-802) or tow the airplane (TASK 09-10-00-580-801) to cause movement between the inner cylinder and outer cylinder.

NOTE: If the airplane is towed, the minimum distance the airplane must be moved is two airplane lengths.

1) In cold weather, locally heat the inner cylinder for at least 15 minutes.

(c) Do a check for leaks for 5 minutes:

1) Wipe the surface of the inner cylinder with a clean cloth to remove all of the fluid from the leak.

2) Monitor the inner cylinder where it meets the outer cylinder for 5 minutes.

NOTE: The extension of the inner cylinder must not be changed during this time.

3) Make a record of the location and the number of drops that come from the seals during the 5 minutes.

(d) Calculate the average of the number of drops that came from the seals during both of the 5 minute measurements.

SUBTASK 32-11-21-210-003

(4) If the leakage stopped, do the steps that follow:

(a) Continue to monitor the strut for leakage on subsequent landings.

1) If you see more signs of leakage, do the entire leak check procedure again.

NOTE: If you find a continuous leakage flow on subsequent landings, the shock strut is not serviceable.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION: IF THE AIRPLANE MADE A LANDING WITH A DEFLATED OR FLAT STRUT, DO THE HARD LANDING INSPECTION. MAKE SURE THAT NO DAMAGE OCCURRED. A DEFLATED OR FLAT STRUT CAN CAUSE DAMAGE DURING A LANDING.

(b) If you do not find leakage on subsequent landings, examine the fluid level of the shock strut within 5 days or 15 flight cycles (TASK 12-15-31-610-801).

NOTE: If the leakage has stopped and the fluid level is correct, there is no additional action necessary.

1) If the shock strut is deflated or flat, do this task: Phase IA Inspection, TASK 05-51-01-210-804.

SUBTASK 32-11-21-210-004

(5) If there was leakage but the average was less than 50 drops during the two 5 minute periods, do the steps that follow:

NOTE: 20 drops is equal to 1.0 cc.

(a) Continue to monitor the strut for leakage on subsequent landings.

1) If the leakage continues or increases do the entire seal check procedure again.

2) Continue to monitor the fluid level of the shock strut at no more than 5 day or 15 flight cycle intervals, until you replace the seals (TASK 12-15-31-610-801).

CAUTION: IF THE AIRPLANE MADE A LANDING WITH A DEFLATED OR FLAT STRUT, DO THE HARD LANDING INSPECTION. MAKE SURE THAT NO DAMAGE OCCURRED. A DEFLATED OR FLAT STRUT CAN CAUSE DAMAGE DURING A LANDING.

(b) If you do not find leakage on subsequent landings, examine the fluid level of the shock strut within 3 days or 10 flight cycles (TASK 12-15-31-610-801).

NOTE: If the leakage has stopped and the fluid level is correct, there is no additional action necessary.

1) If the shock strut is deflated or flat, do this task: Phase IA Inspection, TASK 05-51-01-210-804.

SUBTASK 32-11-21-210-005

(6) If the average leakage was more than 50 drops during the two 5 minute periods, but was not continuous, do these steps as follows:

NOTE: 20 drops is equal to 1.0 cc.

CAUTION: IF THE AIRPLANE MADE A LANDING WITH A DEFLATED OR FLAT STRUT, DO THE HARD LANDING INSPECTION. MAKE SURE THAT NO DAMAGE OCCURRED. A DEFLATED OR FLAT STRUT CAN CAUSE DAMAGE DURING A LANDING.

(a) Do this task: Main Landing Gear Shock Strut Servicing, Airplane on the Ground, TASK 12-15-31-610-802.

1) If the shock strut is deflated or flat, do this task: Phase IA Inspection, TASK 05-51-01-210-804.

(b) Continue to monitor the fluid level of the shock strut after each flight until you replace the seals.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-11-21-210-006

(7) If the fluid leaked continuously during the 5 minute period, the shock strut is not serviceable and you must replace the shock strut seals.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 32-11-21-580-022

(1) If it is necessary, lower the inner gear cylinder from the gear jack pad (TASK 07-10-01-580-802 or TASK 07-10-03-580-802).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN LANDING GEAR CORROSION PREVENTION - MAINTENANCE PRACTICES

1. General

A. The main landing gear (MLG) fittings, especially at the attachment lugs, can be susceptible to corrosion. Damaged paint finishes and plating are attributed to exposure to the weather elements and runway debris. Service wear and hard landings contribute to galling of lugs and lug faces.

B. Specific corrosion problems may be found with the aft center door hinge connecting the inner and center door. Corrosion can be attributed to the accumulation of debris in the recessed area of the attachments to the center door.

C. Damaged finishes on the outboard aft lug of the main landing gear upper strut can contribute to stress corrosion cracking of the steel strut.

D. Cracks and damaged finishes can be on the upper torsion link pins. Cracking can be attributed to stress corrosion.

E. Corrosion and flaking or missing chrome can be on the torsion link pins.

F. Stress corrosion cracking can be on the MLG brake assembly shear studs. Cracked or broken attaching shear studs can reduce the structural integrity of the brake attaching assembly.

TASK 32-11-37-600-801

2. Main Landing Gear Corrosion Prevention

A. Consumable Materials

Reference	Description	Specification
G00009	Compound - Organic Corrosion Inhibiting	BMS3-23
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

B. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

C. Main Landing Gear Corrosion Prevention Practices

SUBTASK 32-11-37-610-001

(1) Do these steps to prevent corrosion in the main landing gear:

(a) Make regular inspections to prevent or find the start of corrosion.

NOTE: Fasteners that are gone, white powdery or other deposits are signs of corrosion.

(b) After you clean the areas, make sure that protective finishes stay serviceable.

(c) If you find corrosion (bulges of the skin or white deposits of corrosion materials at fastener heads or joint edges), refer to SRM SUBJECT 51-10-02 for details of corrosion removal.

(d) For small quantities of corrosion, to decrease the downtime of the airplane, do these steps:

1) Clean off the corrosion materials.

2) Apply a corrosion inhibiting compound into the affected area to stop the corrosion process.

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3) Repair the finish system when the maintenance schedule permits.

D. Main Landing Gear Corrosion Prevention Treatment

SUBTASK 32-11-37-610-002

(1) Do these steps for prevention treatment:

(a) At the next maintenance opportunity, corrosion prevention treatment should be accomplished to the main landing gear.

CAUTION

DO NOT APPLY CORROSION INHIBITING COMPOUND ON GREASED JOINTS OR SEALED BEARINGS. THESE COMPOUNDS DISSOLVE GREASE AND OTHER LUBRICANTS. THEY ARE PENETRATING COMPOUNDS AND CAN GET AROUND THE SEALS AND INTO THE BEARINGS.

(b) After application of corrosion inhibiting compound, all grease fittings in the treated areas should be greased.

(c) If you cleaned with high pressure water, apply the corrosion inhibitor again.

NOTE: Make sure that the pressure of the water is not more than 50 psi.

(d) Shock Struts (Oleo):

1) Apply the water displacing corrosion inhibiting compound to exterior areas of the inner and outer cylinder with broken finish systems.

2) Spray all lugs, lug faces, connecting pin fasteners, and strut door attachments with corrosion inhibitor.

(e) Axles:

1) Apply the water displacing corrosion inhibiting compound to the outside surfaces of the axles, except the journal and bearing surfaces.

2) Make a suitable nozzle extension and spray the inside surfaces of the axles with corrosion inhibitor after protecting the electrical wire and connectors.

(f) Side Struts:

1) Apply the water displacing corrosion inhibiting compound to the surface areas with broken finish systems on the upper and lower side struts.

2) Spray all lugs, lug faces, connecting pins and fasteners with corrosion inhibitor.

(g) Walking Beam:

1) Apply the water displacing corrosion inhibiting compound to the surface areas with broken finish systems.

(h) Door Hinges:

1) Remove debris from the recessed areas of the door and apply the water displacing corrosion inhibiting compound to all hinge fittings.

2) Lubricate the hinge pins, if it is necessary, after application of the corrosion inhibitor.

(i) Miscellaneous:

1) Spray the lugs, lug faces and connecting pins of links, actuators and lock mechanisms connecting above mentioned components with the corrosion inhibitor.

(j) Install mylar tape under the corrosion resistant steel clamps at overhaul to minimize risk of galvanic corrosion should the finish be damaged.

NOTE: This step is recommended.

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E. Main Landing Gear Corrosion Prevention Frequency of Application

SUBTASK 32-11-37-610-003

(1) Frequency of corrosion prevention application is as follows:

(a) Periodic inspection is necessary for areas identified as susceptible to corrosion and should be consistent to the schedules identified in the Maintenance Planning Document.

NOTE: Operators must be aware of reported problems and areas of occurrence.

WARNING

DO NOT GET THE CORROSION-INHIBITING COMPOUND ON YOUR SKIN, IN YOUR EYES, OR ON YOUR CLOTHES. THE COMPOUND IS POISONOUS AND FLAMMABLE. USE APPROVED GLOVES, AND EYE PROTECTION. MAKE SURE THAT THERE IS GOOD AIRFLOW IN THE WORK AREA. KEEP THE COMPOUND AWAY FROM SPARKS AND FLAMES. THE COMPOUND CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(b) Annual application of corrosion inhibiting compound, G00009, is necessary to areas identified or as required by washing cycles.

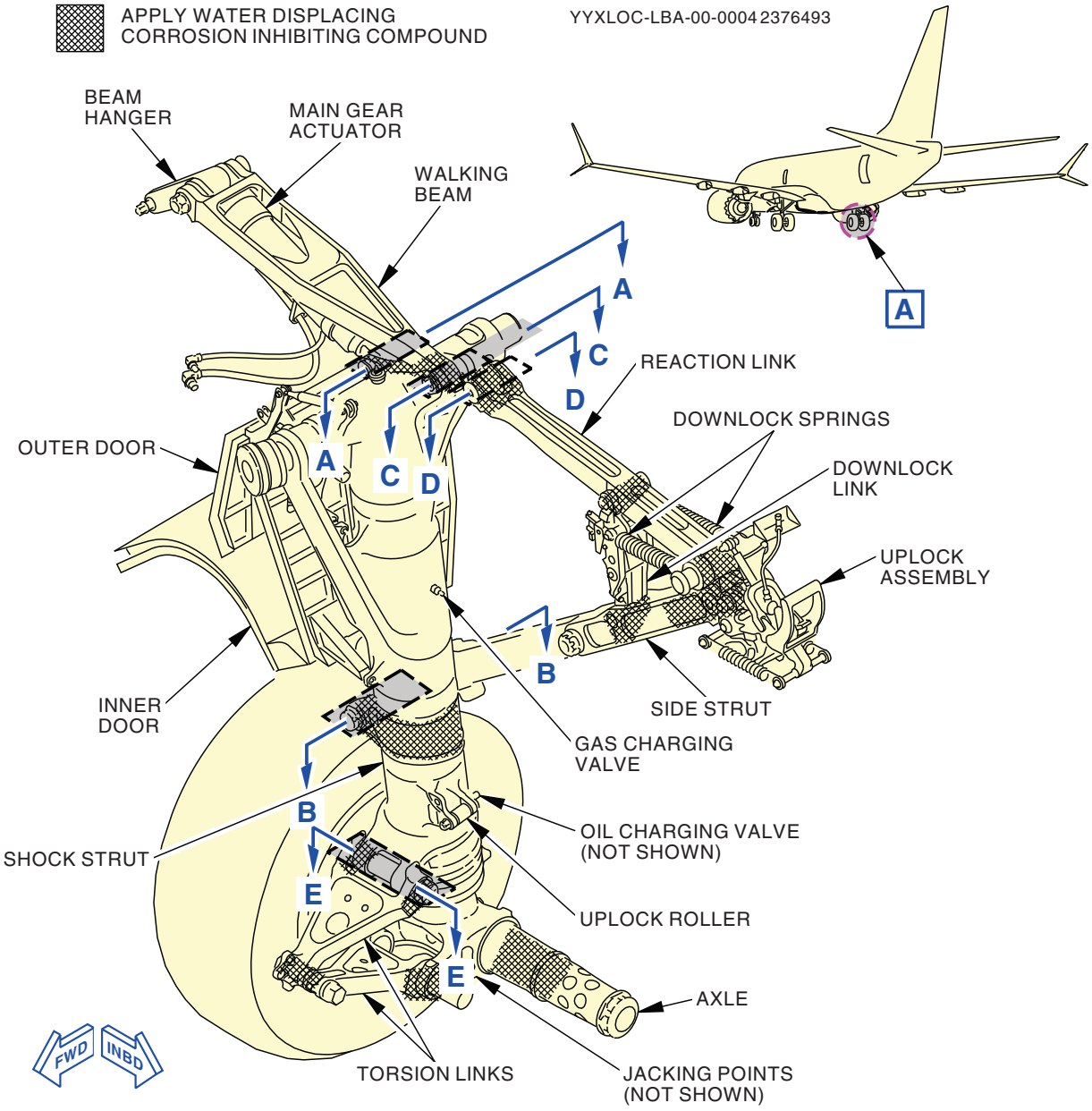
WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(c) Apply corrosion inhibiting material, G50136, to the areas identified as required.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



PREVENTIVE MAINTENANCE
LANDING GEAR

A

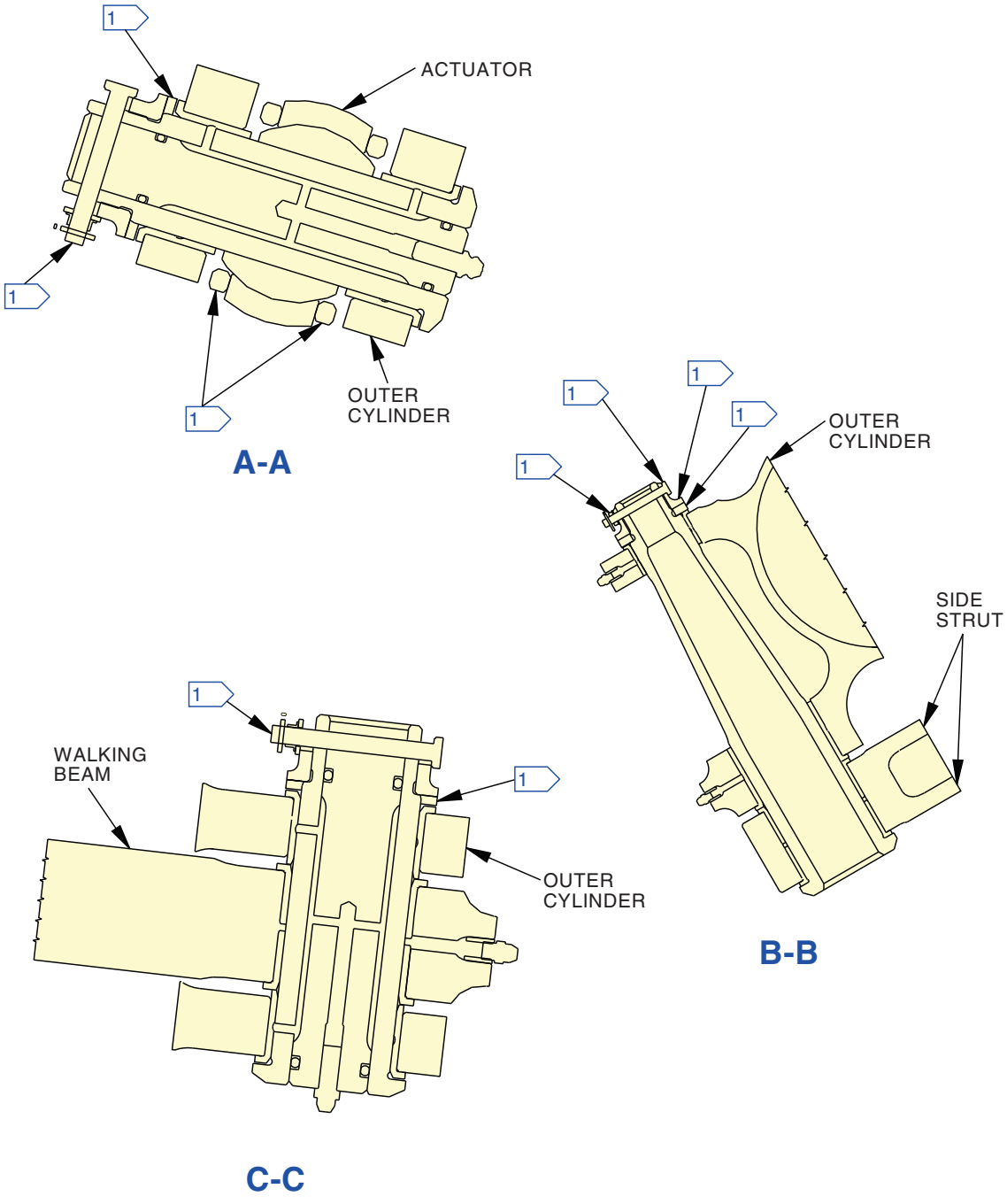
NOTE:

APPLY CORROSION INHIBITING COMPOUND (CIC) BMS3-23, TO EXPOSED OUTER SURFACES ONLY. DO NOT DISASSEMBLE PARTS TO APPLY CIC. DO NOT SPRAY INTO JOINTS, LINKAGES, PULLEYS, CABLES OR BEARINGS.

2414073 S00061534337_V2

Main Landing Gear Corrosion Prevention
Figure 201/32-11-37-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

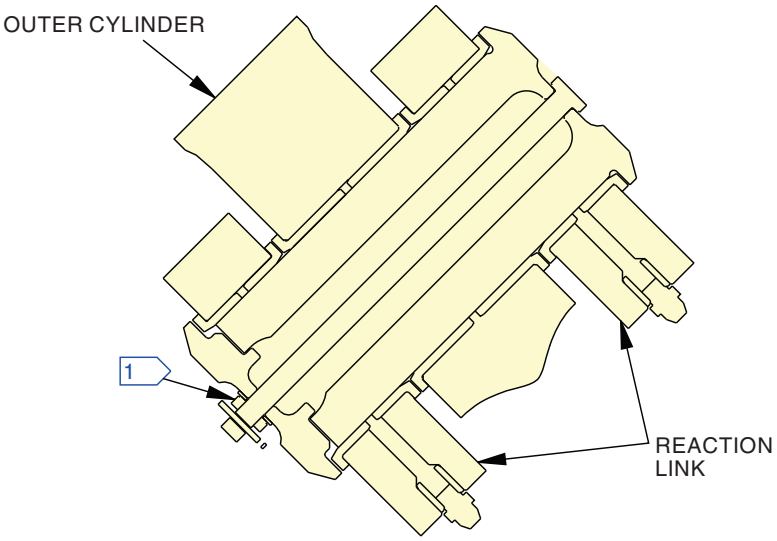


1 APPLY BMS 3-38 TO THESE AREAS.

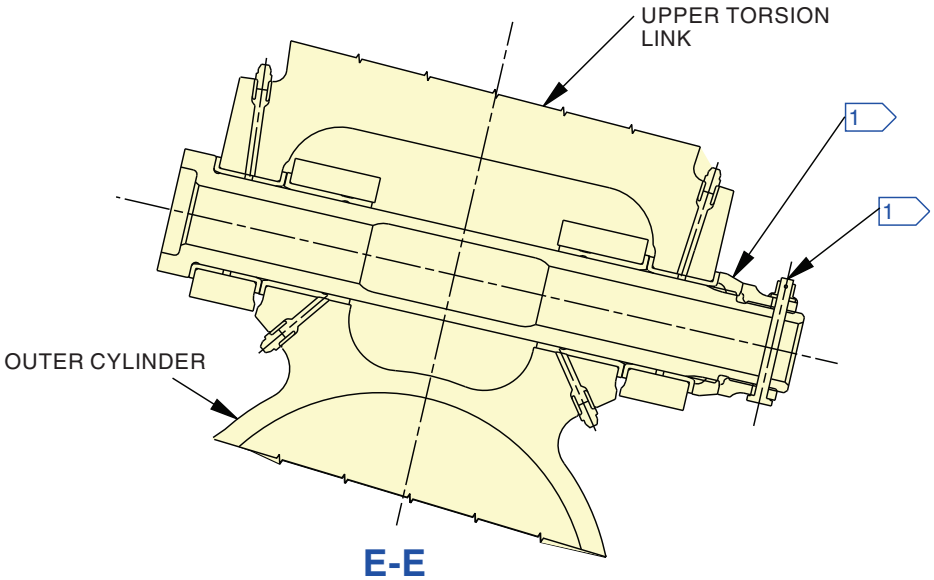
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Main Landing Gear Corrosion Prevention
Figure 201/32-11-37-990-801 (Sheet 2 of 3)

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D-D



E-E

1 APPLY BMS 3-38 TO THESE AREAS.

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Main Landing Gear Corrosion Prevention
Figure 201/32-11-37-990-801 (Sheet 3 of 3)

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MAIN LANDING GEAR TORSION LINKS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the upper torsion link from the main landing gear

(2) A removal of the lower torsion link from the main landing gear

(3) The disconnection of the torsion links for the main landing gear

(4) An installation of the upper torsion link for the main landing gear

(5) An installation of the lower torsion link for the main landing gear

(6) The connection of the torsion links for the main landing gear

TASK 32-11-51-000-801

2. Main Landing Gear Upper Torsion Link Removal

(Figure 401)

A. General

(1) This task provides instructions to remove the upper torsion link.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-11-81-000-804	Main Landing Gear Shimmy Damper Removal (P/B 401)
32-45-11-000-801	Main Landing Gear Wheel and Tire Assembly Removal (P/B 401)

C. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Prepare for the Removal

SUBTASK 32-11-51-480-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED IN ALL OF THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801

SUBTASK 32-11-51-020-002

(2) Remove the left hand inboard or the right hand outboard tire of the main landing gear. To do this, do this task: Main Landing Gear Wheel and Tire Assembly Removal, TASK 32-45-11-000-801.

E. Main Landing Gear Upper Torsion Link Removal

SUBTASK 32-11-51-020-003

(1) Remove the support bracket [4] and support bracket [6] from the upper torsion link [5]:

NOTE: Remove the support brackets with the sensor targets attached.

(a) Remove the nuts [2] and the washers [3] from the bolts [1].

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(b) Remove the bolts [1] that hold the support bracket [4] and support bracket [6] to the upper torsion link [5].

(c) Remove the support bracket [4] and support bracket [6] from the work area.

SUBTASK 32-11-51-020-004

(2) Remove the bracket [19] from the upper torsion link [5]:

NOTE: Remove the bracket [19] with the electrical conduit and the clamps attached.

(a) Remove the nuts [10] and the washers [9] from the bolts [7].

(b) Remove the bolts [7] to disconnect the aft side of the bracket [19] from the upper torsion link [5].

(c) Remove the washers [8] from the bolts [7].

(d) Remove the nut [14], and the washer [13] from the bolt [21].

(e) Remove the bolt [21] to disconnect the forward side of the bracket [19] from the upper torsion link [5].

NOTE: Removal of the bolt [21] will also disconnect one side of the hose guide bracket [12] from the upper torsion link [5].

(f) Remove the washer [20] from the bolt [21].

(g) Isolate the bracket [19] from the upper torsion link [5].

SUBTASK 32-11-51-020-005

(3) Remove the hose guide bracket [12] from the upper torsion link [5]:

NOTE: Remove the hose guide bracket [12] with the hose guides and the brake hoses attached.

(a) Remove the nut [15] and the washer [16] from the bolt [18].

(b) Remove the bolt [18] to disconnect the hose guide bracket [12] from the upper torsion link [5].

(c) Remove the washer [17] from the bolt [18].

(d) Isolate the hose guide bracket [12] from the upper torsion link [5].

SUBTASK 32-11-51-480-002

(4) Use a rope to hold the forward side of the lower torsion link [22] to the outer cylinder [33].

NOTE: This step will hold the lower torsion link [22] in its position when you do the subsequent steps.

SUBTASK 32-11-51-420-001

(5) Disconnect the upper torsion link [5] and the lower torsion link [22]:

(a) Remove the cotter pin [26], nut [27], and the washer [25] from the bolt [23].

(b) Remove the bolt [23] and washer [69] from the apex pin [31].

(c) Remove the apex nut [24] from the apex pin [31].

(d) Remove the bracket [28] and the washer [29] and the tapered sleeve [54] from the apex pin [31].

(e) Isolate the bracket [28] from the apex joint.

NOTE: Isolate the bracket [28] with the electrical conduit and the clamps attached.

(f) Remove the apex pin [31] and straight sleeve [71] to disconnect the upper torsion link [5] from the lower torsion link [22].

(g) Remove the spacer [30] between the upper torsion link [5] and the lower torsion link [22].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-11-51-020-001

(6) If necessary, remove the shimmy damper (TASK 32-11-81-000-804).

SUBTASK 32-11-51-020-006

(7) Disconnect the upper torsion link [5] from the outer cylinder [33]:

(a) Remove the cotter pin [38], nut [39], and the washer [37] from the bolt [35].

(b) Remove the bolt [35] from the upper torsion pin [32].

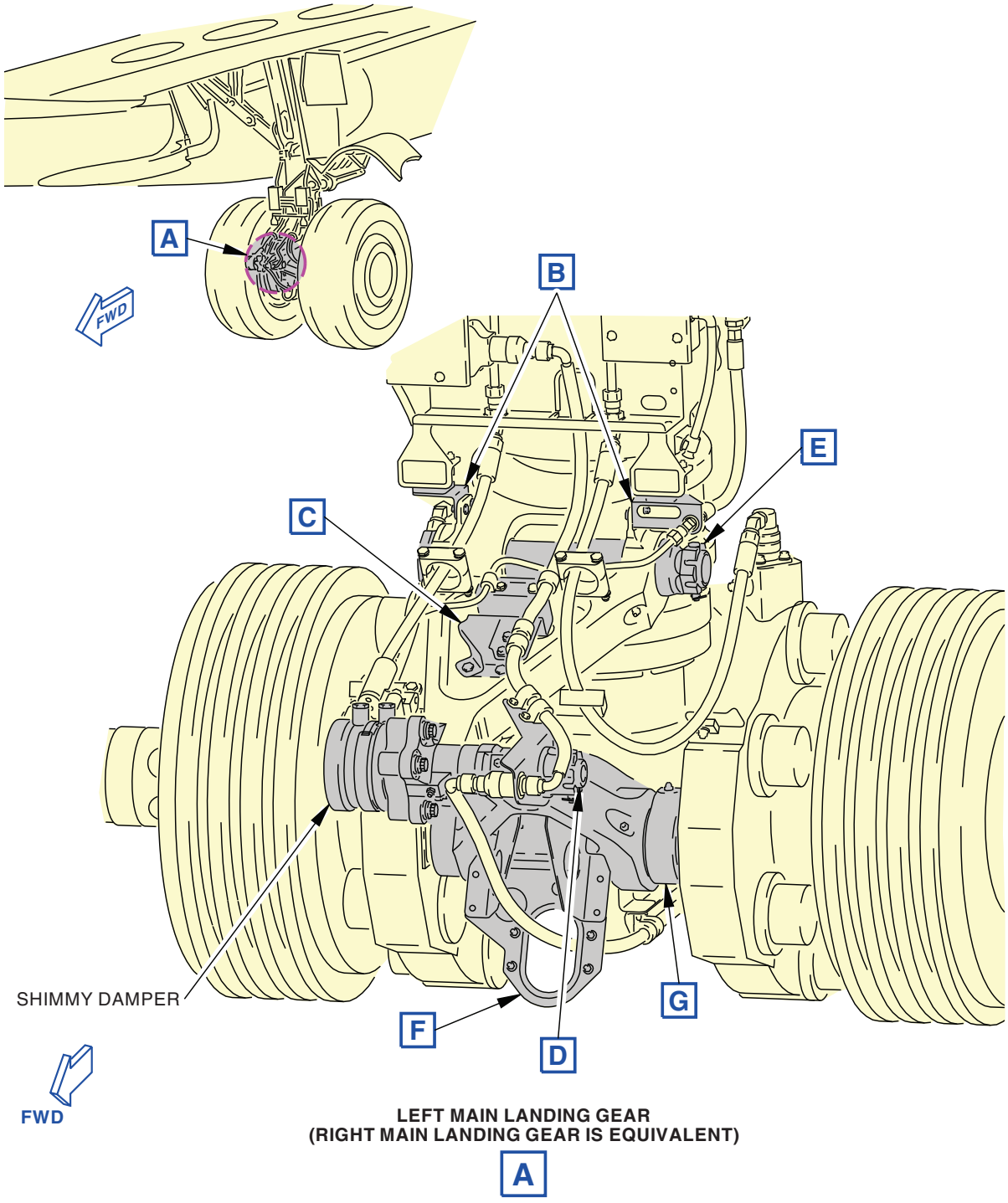
(c) Remove the nut [36] and the washer [34] from the upper torsion pin [32].

(d) Hold the upper torsion link [5] and remove the upper torsion pin [32] to disconnect the upper torsion link [5] from the outer cylinder [33].

(e) Remove the upper torsion link [5] from the airplane.

——— END OF TASK ———

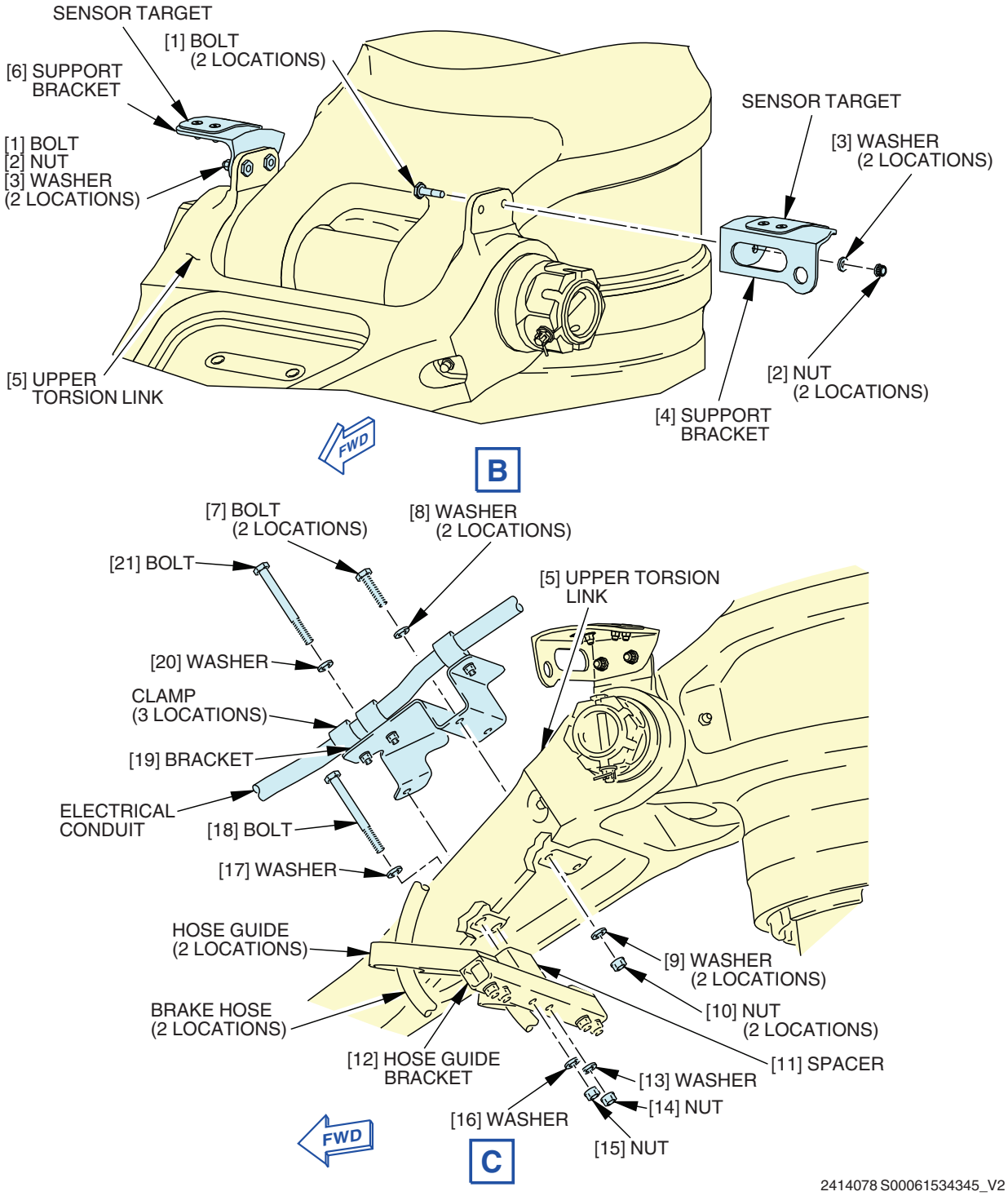
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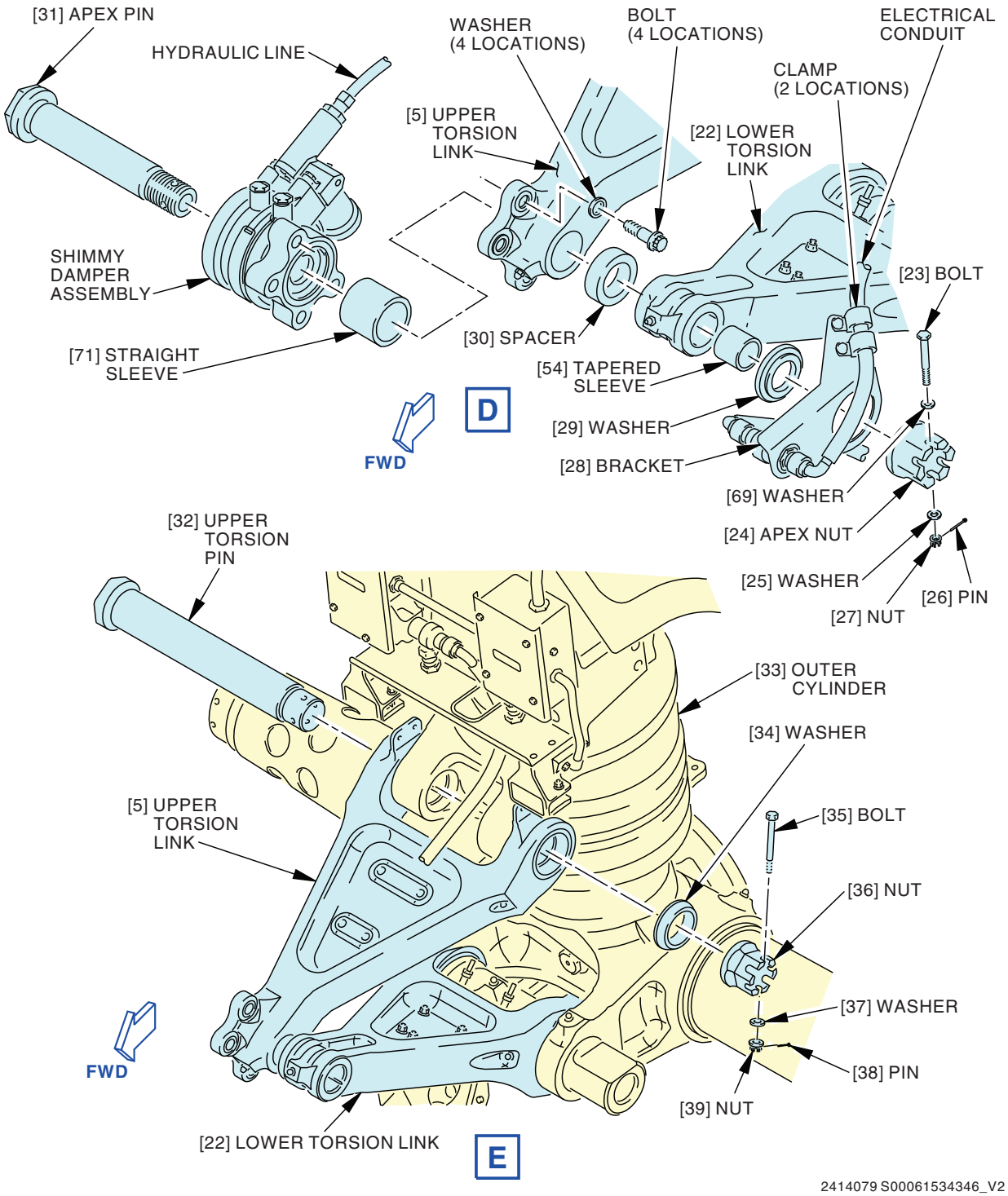
Main Landing Gear Torsion Link Installation
Figure 401/32-11-51-990-801 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



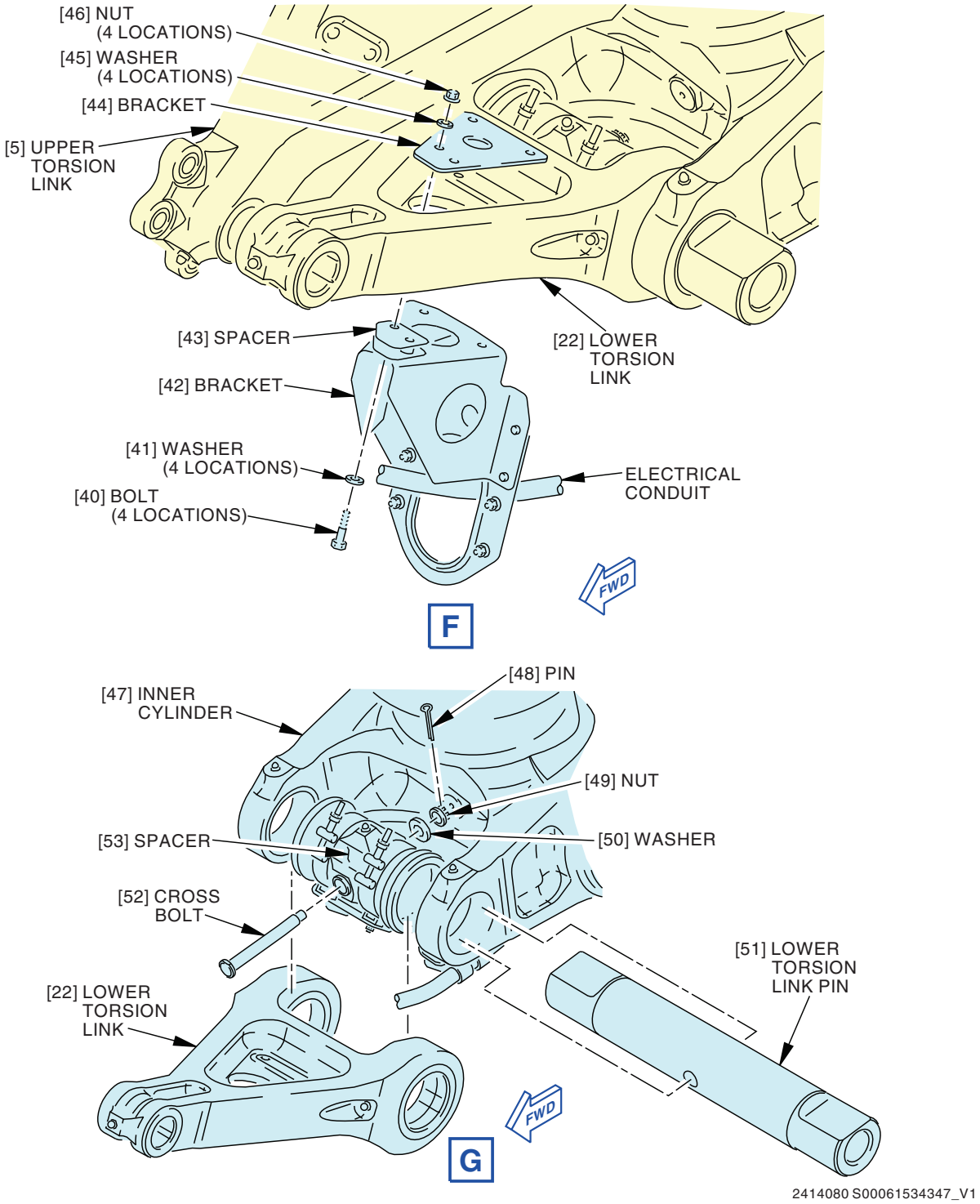
Main Landing Gear Torsion Link Installation
Figure 401/32-11-51-990-801 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Torsion Link Installation
Figure 401/32-11-51-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Torsion Link Installation
Figure 401/32-11-51-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-11-51-000-802

3. Main Landing Gear Lower Torsion Link Removal

(Figure 401)

A. General

(1) This task provides instructions to remove the lower torsion link.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-41-41-000-801	Main Landing Gear Brake Removal (P/B 401)

C. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Prepare for the Removal

SUBTASK 32-11-51-480-008

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-11-51-020-008

(2) Do this task: Main Landing Gear Brake Removal, TASK 32-41-41-000-801.

NOTE: This procedure will also remove the wheel and tire assemblies.

E. Main Landing Gear Lower Torsion Link Removal

SUBTASK 32-11-51-020-009

(1) Remove the bracket [42] and bracket [44] from the lower torsion link [22]:

NOTE: Remove the bracket [42] with the electrical conduit attached.

(a) Remove the nuts [46], and the washers [45] from the bolts [40].

(b) Remove the bolts [40] to disconnect the bracket [42], and bracket [44] from the lower torsion link [22].

(c) Remove the washers [41] from the bolts [40].

(d) Remove the bracket [44] and the spacer [43] from the lower torsion link [22].

(e) Isolate the bracket [42] from the lower torsion link [22].

SUBTASK 32-11-51-480-004

(2) Use a rope to hold the forward side of the upper torsion link [5] to the outer cylinder [33].

NOTE: This step will hold the upper torsion link [5] in its position when you do the subsequent steps.

SUBTASK 32-11-51-420-002

(3) Disconnect the upper torsion link [5] and the lower torsion link [22]:

(a) Remove the cotter pin [26], nut [27], and the washer [25] from the bolt [23].

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(b) Remove the bolt [23] and washer [69] from the apex pin [31].

(c) Remove the apex nut [24] from the apex pin [31].

(d) Remove the bracket [28] and the washer [29] and the tapered sleeve [54] from the apex pin [31].

(e) Isolate the bracket [28] from the apex joint.

NOTE: Isolate the bracket [28] with the electrical conduit and the clamps attached.

(f) Remove the apex pin [31] and straight sleeve [71] to disconnect the upper torsion link [5] from the lower torsion link [22].

(g) Remove the spacer [30] between the upper torsion link [5] and the lower torsion link [22].

SUBTASK 32-11-51-020-010

(4) Disconnect the lower torsion link [22] from the inner cylinder [47]:

(a) Remove the cotter pin [48], nut [49], and washer [50] from the cross bolt [52].

(b) Remove the cross bolt [52] that holds the lower torsion pin [51] to the spacer [53].

(c) Hold the lower torsion link [22] and remove the lower torsion pin [51] to disconnect the lower torsion link [22] from the inner cylinder [47].

SUBTASK 32-11-51-480-005

(5) Use a rope to hold the spacer [53] to the inner cylinder [47].

NOTE: This step will hold the spacer [53] in its position when you remove and install the lower torsion link [22] in the subsequent steps.

——— END OF TASK ———

TASK 32-11-51-020-801

4. Main Landing Gear Torsion Link Disconnection

(Figure 401)

A. General

(1) This task provides instructions to disconnect the torsion links.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-45-11-000-801	Main Landing Gear Wheel and Tire Assembly Removal (P/B 401)

C. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

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D. Prepare for Disconnection

SUBTASK 32-11-51-480-009

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-11-51-020-031

(2) Remove the left hand inboard or the right hand outboard tire of the main landing gear. To do this, do this task: Main Landing Gear Wheel and Tire Assembly Removal, TASK 32-45-11-000-801.

E. Main Landing Gear Torsion Link Disconnection

SUBTASK 32-11-51-420-003

(1) Disconnect the upper torsion link [5] and the lower torsion link [22]:

(a) Remove the cotter pin [26], nut [27], and the washer [25] from the bolt [23].

(b) Remove the bolt [23] and washer [69] from the apex pin [31].

(c) Remove the apex nut [24] from the apex pin [31].

(d) Remove the bracket [28] and the washer [29] and the tapered sleeve [54] from the apex pin [31].

(e) Isolate the bracket [28] from the apex joint.

NOTE: Isolate the bracket [28] with the electrical conduit and the clamps attached.

(f) Remove the apex pin [31] and straight sleeve [71] to disconnect the upper torsion link [5] from the lower torsion link [22].

(g) Remove the spacer [30] between the upper torsion link [5] and the lower torsion link [22].

——— END OF TASK ———

TASK 32-11-51-400-801

5. Main Landing Gear Upper Torsion Link Installation

(Figure 401)

A. General

(1) This task provides instructions to install the upper torsion link.

B. References

Reference	Title
12-21-11-640-802	Main Landing Gear Lower End Components Servicing (P/B 301)
32-11-81-400-804	Main Landing Gear Shimmy Damper Installation (P/B 401)
32-45-11-400-801	Main Landing Gear Wheel and Tire Assembly - Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
D00633	Grease - Aircraft General Purpose	BMS3-33

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(Continued)

Reference	Description	Specification
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
5	Upper torsion link	32-11-51-01-155	PTF ALL
26	Cotter pin	32-11-81-01-020	PTF ALL
38	Cotter pin	32-11-51-01-010	PTF ALL

E. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

F. Main Landing Gear Upper Torsion Link Installation

SUBTASK 32-11-51-620-001

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(1) Apply a thin layer of corrosion inhibiting material, G50136, to these items:

(a) The threads and thread reliefs of the bolt [23], nut [27], apex nut [24], bolt [35], nut [39], and nut [36].

(b) The faces of the washer [25], washer [69], washer [29], and washer [37].

(c) The pin shank and threads of the apex pin [31].

(d) The inside diameter of spacer [30], tapered sleeve [54], and straight sleeve [71].

(e) The splines and faces of the washer [34].

(f) The new cotter pin [26] and new cotter pin [38].

SUBTASK 32-11-51-160-001

(2) Remove any excess corrosion inhibiting material, G50136.

SUBTASK 32-11-51-640-001

(3) Apply grease, D00633, to the following parts and areas:

(a) Apply to the outside diameter of the tapered sleeve [54] and straight sleeve [71].

(b) The chrome plated surfaces of the upper torsion pin [32].

SUBTASK 32-11-51-160-002

(4) Remove all unwanted grease, D00633.

SUBTASK 32-11-51-420-004

(5) Connect the upper torsion link [5] to the outer cylinder [33]:

(a) Put the upper torsion link [5] in its position on the clevis of the outer cylinder [33].

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(b) Put the upper torsion pin [32] through the upper torsion link [5] and clevis of the outer cylinder [33].

(c) Install the washer [34] and nut [36] on the upper torsion pin [32].

(d) Tighten the nut [36] to 95 ft-lb (128.8 N·m) - 115 ft-lb (155.9 N·m).

(e) If it is necessary, loosen the nut [36] to the nearest castellation to align the holes for the bolt [35].

(f) Install the bolt [35] in the upper torsion pin [32].

(g) Install the washer [37] and nut [39] on the bolt [35].

(h) Install the new cotter pin [38] in the bolt [35].

1) If it is necessary, loosen the nut [39] to the nearest castellation to align the hole for the new cotter pin [38].

SUBTASK 32-11-51-080-001

(6) Remove the rope that holds the lower torsion link [22] in its position to the outer cylinder [33].

SUBTASK 32-11-51-420-009

(7) If necessary, install the shimmy damper (TASK 32-11-81-400-804).

SUBTASK 32-11-51-420-005

(8) Connect the upper torsion link [5] and lower torsion link [22]:

(a) Align the apex holes in the upper torsion link [5] and lower torsion link [22].

(b) Put the spacer [30] in its position between the upper torsion link [5] and lower torsion link [22].

(c) Put the apex pin [31] through the upper torsion link [5], straight sleeve [71], spacer [30], and lower torsion link [22].

CAUTION

MAKE SURE THAT YOU INSTALL THE SLEEVES IN THE CORRECT LOCATIONS. IF YOU INSTALL THE SLEEVES IN THE INCORRECT LOCATIONS, VIBRATION CAN CAUSE DAMAGE TO COMPONENTS OF LANDING GEAR.

(d) Install the tapered sleeve [54] in the lower torsion link [22], washer [29], and bracket [28] on the apex pin [31].

NOTE: There may be a gap between the torsion link and the apex joint. This condition is part of a normal shimmy damper installation.

NOTE: The shimmy damper installation has a straight sleeve and a tapered sleeve. The straight sleeve is longer than the tapered sleeve.

(e) Install the apex nut [24] on the apex pin [31].

(f) Tighten the apex nut [24] to 50 ft-lb (67.8 N·m) - 58 ft-lb (78.6 N·m).

1) Advance the apex nut [24] to the nearest castellation that aligns with the hole in the apex pin [31].

2) Make sure that the final torque is not more than 150 ft-lb (203.4 N·m).

(g) Install the retention bolt [23] and washer [69] through the apex nut [24] and apex pin [31].

(h) Make sure that the electrical conduit and apex nut [24] do not contact one another.

1) Make sure that the electrical conduit is at an angle of 50 +0/-5 degrees from the bottom of the bracket [28].

(i) Install the washer [25] and nut [27] on the bolt [23].

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(j) Install the new cotter pin [26] in the bolt [23].

1) If it is necessary, loosen the nut [27] to the nearest castellation to align the hole for the new cotter pin [26].

SUBTASK 32-11-51-420-006

(9) Connect the hose guide bracket [12] to the upper torsion link [5]:

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin layer of corrosion inhibiting material, G50136, to the bolt [18], washer [16], washer [17], and nut [15].

(b) Install the washer [17] on the bolt [18].

(c) Put the hose guide bracket [12] in its position below the upper torsion link [5].

(d) Put the spacer [11] in its position between the hose guide bracket [12] and upper torsion link [5].

(e) Put the bolt [18] through the upper torsion link [5], spacer [11], and hose guide bracket [12].

NOTE: This step will connect one side of the hose guide bracket to the upper torsion link.

(f) Install the washer [16] and nut [15] on the bolt [18].

SUBTASK 32-11-51-420-007

(10) Connect the bracket [19] to the upper torsion link [5]:

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin layer ofcorrosion inhibiting material, G50136, to the bolt [7], bolt [21], washers [8], washers [9], washer [13], washer [20], nuts [10], and nut [14].

(b) Install the washers [8] on the bolts [7].

(c) Put the bracket [19] in its position on the upper torsion link [5].

(d) Put the bolt [7] through the bracket [19] and upper torsion link [5].

NOTE: This step will connect the aft side of the bracket to the upper torsion link.

(e) Install the washers [9] and nuts [10] on the bolts [7].

(f) Install the washer [20] on the bolt [21].

(g) Put the bolt [21] through the bracket [19], upper torsion link [5], spacer [11], and hose guide bracket [12].

(h) Install the washer [13] and nut [14] on the bolt [21].

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SUBTASK 32-11-51-390-001

(11) Fay seal the spacer [11] with sealant, A00247.

SUBTASK 32-11-51-420-008

(12) Connect the support bracket [4] and support bracket [6] to the upper torsion link [5]:

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin layer of corrosion inhibiting material, G50136, to the bolts [1], nuts [2], and washers [3].

(b) Put the bolts [1] through the upper torsion link [5], support bracket [4], and support bracket [6].

(c) Install the washers [3] and nuts [2] on the bolts [1].

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-11-51-640-002

(1) For the grease fittings on the upper torsion link [5] and lower torsion link [22], do this task: Main Landing Gear Lower End Components Servicing, TASK 12-21-11-640-802.

SUBTASK 32-11-51-420-010

(2) Install the left hand inboard or the right hand outboard tire of the main landing gear, do this task: Main Landing Gear Wheel and Tire Assembly - Installation, TASK 32-45-11-400-801.

——— END OF TASK ———

TASK 32-11-51-400-802

6. Main Landing Gear Lower Torsion Link Installation

(Figure 401)

A. General

(1) This task provides instructions to install the lower torsion link.

B. References

Reference	Title
12-21-11-640-802	Main Landing Gear Lower End Components Servicing (P/B 301)
32-11-81-400-804	Main Landing Gear Shimmy Damper Installation (P/B 401)
32-41-41-400-801	Main Landing Gear Brake Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
22	Lower torsion link	32-11-51-01-195	PTF ALL

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(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
26	Cotter pin	32-11-81-01-020	PTF ALL
48	Cotter pin	32-11-51-01-015	PTF ALL

E. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

F. Main Landing Gear Lower Torsion Link Installation

SUBTASK 32-11-51-620-002



WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(1) Apply a thin layer of the corrosion inhibiting material, G50136, to these items:
   (a) The threads and thread reliefs of the bolt [23], nut [27], apex nut [24], and nut [49].
   (b) The faces of the washer [25], washer [69], washer [29], and washer [50].
   (c) The pin shank and threads of the apex pin [31].
   (d) The inside diameter of spacer [30], tapered sleeve [54], and straight sleeve [71].
   (e) The new cotter pin [26].

SUBTASK 32-11-51-160-003

(2) Remove any excess corrosion inhibiting material, G50136.

SUBTASK 32-11-51-640-003

(3) Apply grease, D00633, to the following parts and areas:
   (a) Apply to the outside diameter of the tapered sleeve [54] and straight sleeve [71].
   (b) The chrome plated surfaces of the lower torsion pin [51] and cross bolt [52].

SUBTASK 32-11-51-160-004

(4) Remove all unwanted grease, D00633.

SUBTASK 32-11-51-080-002

(5) Remove the rope that holds the spacer [53] to the inner cylinder [47].

SUBTASK 32-11-51-420-011

(6) Connect the lower torsion link [22] to the inner cylinder [47]:
   (a) Put the lower torsion link [22] in its position on the inner cylinder [47].
   (b) Put the lower torsion pin [51] through the clevis on the inner cylinder [47] and spacer [53].
   (c) Align the hole in the lower torsion pin [51] with hole in the spacer [53] for the installation of the cross bolt [52].
   (d) Put the cross bolt [52] through the spacer [53] and lower torsion pin [51].
   (e) Install the washer [50] and nut [49] on the cross bolt [52].
   (f) Tighten the nut [49] to 160 in-lb (18.1 N·m) - 190 in-lb (21.5 N·m).

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(g) Install the new cotter pin [48] in the cross bolt [52].

1) If it is necessary, loosen the nut [49] to the nearest castellation to align the hole for the new cotter pin [48].

SUBTASK 32-11-51-080-003

(7) Remove the rope that holds the upper torsion link [5] to the outer cylinder [33].

SUBTASK 32-11-51-420-013

(8) If necessary, install the shimmy damper (TASK 32-11-81-400-804).

SUBTASK 32-11-51-420-012

(9) Connect the upper torsion link [5] and lower torsion link [22]:

(a) Align the apex holes in the upper torsion link [5] and lower torsion link [22].

(b) Put the spacer [30] in its position between the upper torsion link [5] and lower torsion link [22].

(c) Put the apex pin [31] through the upper torsion link [5], straight sleeve [71], spacer [30], and lower torsion link [22].

CAUTION

MAKE SURE THAT YOU INSTALL THE SLEEVES IN THE CORRECT LOCATIONS. IF YOU INSTALL THE SLEEVES IN THE INCORRECT LOCATIONS, VIBRATION CAN CAUSE DAMAGE TO COMPONENTS OF LANDING GEAR.

(d) Install the tapered sleeve [54] in the lower torsion link [22], washer [29], bracket [28] and apex nut [24] on the apex pin [31].

NOTE: There may be a gap between the torsion link and the apex joint. This condition is part of a normal shimmy damper installation.

NOTE: The shimmy damper installation has a straight sleeve and a tapered sleeve. The straight sleeve is longer than the tapered sleeve.

(e) Tighten the apex nut [24] to 50 ft-lb (67.8 N·m) - 58 ft-lb (78.6 N·m).

1) Advance the apex nut [24] to the nearest castellation that aligns with the hole in the apex pin [31].

2) Make sure that the final torque is not more than 150 ft-lb (203.4 N·m).

(f) Install the retention bolt [23] and washer [69] through the apex nut [24] and apex pin [31].

(g) Make sure that the electrical conduit and apex nut [24] do not contact one another.

1) Make sure that the electrical conduit is at an angle of 50 +0/-5 degrees from the bottom of the bracket [28].

(h) Install the washer [25] and nut [27] on the bolt [23].

(i) Install the new cotter pin [26] in the bolt [23].

1) If it is necessary, loosen the nut [27] to the nearest castellation to align the hole for the new cotter pin [26].

SUBTASK 32-11-51-420-014

(10) Connect the bracket [42] and bracket [44] to the lower torsion link [22]:

(a) Put bracket [42], bracket [44], and spacer [43] in their position on the lower torsion link [22].

(b) Install the washers [41] on the bolts [40].

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(c) Put the bolts [40] through the bracket [42], spacer [43], lower torsion link [22], and bracket [44].

(d) Install the washers [45] and nuts [46] on the bolts [40].

SUBTASK 32-11-51-420-015

(11) Do this task: Main Landing Gear Brake Installation, TASK 32-41-41-400-801

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-11-51-640-004

(1) For the grease fittings on the upper torsion link [5] and lower torsion link [22], do this task: Main Landing Gear Lower End Components Servicing, TASK 12-21-11-640-802.

——— END OF TASK ———

TASK 32-11-51-400-803

7. Main Landing Gear Torsion Link Connection

(Figure 401)

A. General

(1) This task provides instructions to connect the torsion links.

B. References

Reference	Title
12-21-11-640-802	Main Landing Gear Lower End Components Servicing (P/B 301)
32-11-81-400-804	Main Landing Gear Shimmy Damper Installation (P/B 401)
32-45-11-400-801	Main Landing Gear Wheel and Tire Assembly - Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
5	Upper torsion link	32-11-51-01-155	PTF ALL
22	Lower torsion link	32-11-51-01-195	PTF ALL
26	Cotter pin	32-11-81-01-020	PTF ALL

E. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

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F. Main Landing Gear Torsion Link Connection

SUBTASK 32-11-51-620-003

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(1) Apply a thin layer of the corrosion inhibiting material, G50136, to these items:

(a) The threads and thread reliefs of the bolt [23]

(b) The threads and thread reliefs of the apex nut [24] and nut [27]

(c) The faces of the washer [25], washer [69], and washer [29]

(d) The pin shank and threads of the apex pin [31]

(e) The inside diameter of the spacer [30], tapered sleeve [54], and straight sleeve [71]

(f) The new cotter pin [26].

SUBTASK 32-11-51-160-005

(2) Remove any excess corrosion inhibiting material, G50136.

SUBTASK 32-11-51-640-005

(3) Apply grease, D00633, to the outer diameter of the tapered sleeve [54] and straight sleeve [71].

SUBTASK 32-11-51-160-006

(4) Remove all unwanted grease, D00633.

SUBTASK 32-11-51-420-016

(5) If necessary, install the shimmy damper (TASK 32-11-81-400-804).

SUBTASK 32-11-51-420-017

(6) Connect the upper torsion link [5] and lower torsion link [22]:

(a) Align the apex holes in the upper torsion link [5] and lower torsion link [22].

(b) Put the spacer [30] in its position between the upper torsion link [5] and lower torsion link [22].

(c) Put the apex pin [31] through the upper torsion link [5], straight sleeve [71], spacer [30], and lower torsion link [22].

CAUTION

MAKE SURE THAT YOU INSTALL THE SLEEVES IN THE CORRECT LOCATIONS. IF YOU INSTALL THE SLEEVES IN THE INCORRECT LOCATIONS, VIBRATION CAN CAUSE DAMAGE TO COMPONENTS OF LANDING GEAR.

(d) Install the tapered sleeve [54], washer [29], bracket [28], and apex nut [24] on the apex pin [31].

NOTE: There may be a gap between the torsion link and the apex joint. This condition is part of the normal shimmy damper installation.

NOTE: The shimmy damper installation has a straight sleeve and a tapered sleeve. The straight sleeve is longer than the tapered sleeve.

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(e) Tighten the apex nut [24] to 50 ft-lb (67.8 N·m) - 58 ft-lb (78.6 N·m).

1) Advance the apex nut [24] to the nearest castellation that aligns with the hole in the apex pin [31].

2) Make sure that the final torque is not more than 150 ft-lb (203.4 N·m).

(f) Install the retention bolt [23] and washer [69] through the apex nut [24] and apex pin [31].

(g) Make sure that the electrical conduit and apex nut [24] do not contact one another.

1) Make sure that the electrical conduit is at an angle of 50 +0/-5 degrees from the bottom of the bracket [28].

(h) Install the washer [25] and nut [27] on the bolt [23].

(i) Install the new cotter pin [26] in the bolt [23].

1) If it is necessary, loosen the nut [27] to the nearest castellation to align the hole for the new cotter pin [26].

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-11-51-420-037

(1) Install the wheel and tire assembly (TASK 32-45-11-400-801).

SUBTASK 32-11-51-640-006

(2) For the grease fittings on the upper torsion link [5] and lower torsion link [22], do this task: Main Landing Gear Lower End Components Servicing, TASK 12-21-11-640-802.

——— END OF TASK ———

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MAIN LANDING GEAR TORSION LINKS - INSPECTION/CHECK

1. General

A. This task supplies the data for a dimensional inspection of the torsion links for the main landing gear.

TASK 32-11-51-200-801

2. Main Landing Gear Torsion Links Inspection

(Figure 601)

A. General

(1) This procedure gives wear limits and repair data for the torsion links on the main landing gear.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-11-51-000-801	Main Landing Gear Upper Torsion Link Removal (P/B 401)
32-11-51-000-802	Main Landing Gear Lower Torsion Link Removal (P/B 401)
32-11-51-400-801	Main Landing Gear Upper Torsion Link Installation (P/B 401)
32-11-51-400-802	Main Landing Gear Lower Torsion Link Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-1096	Micrometer - Depth, 0-1 inch, Readable to 1/1000 inch
STD-1097	Caliper - Vernier, 0-6 inch, Readable to 1/1000 inch

D. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Procedure

SUBTASK 32-11-51-480-007

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-11-51-020-012

(2) To remove or to get access to the torsion links of the main landing gear do the tasks that follows Figure 601:

(a) Do this task: Main Landing Gear Upper Torsion Link Removal, TASK 32-11-51-000-801.

(b) Do this task: Main Landing Gear Lower Torsion Link Removal, TASK 32-11-51-000-802.

SUBTASK 32-11-51-220-001

(3) Examine the parts of the main landing gear torsion links for worn areas:

(a) Remove the parts to be checked.

(b) Measure the parts as necessary with a micrometer (0-1 inch, readable to 1/1000 inch), STD-1096 or a vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

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(c) Compare the dimensions you measured with the permitted wear dimensions in Table 601.

(d) If the parts are not in the tolerance, replace or repair the part as shown in Table 601.

Table 601/32-11-51-993-801 Wear Limits

NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE DIAMETRIC CLEARANCE
#1 BUSHING	I.D.	2.0000 in. (50.8000 mm)	2.0015 in. (50.8381 mm)	2.0053 in. (50.9346 mm)	0.0063 in. (0.1600 mm)
#1 PIN	O.D.	1.9980 in. (50.7492 mm)	1.9990 in. (50.7746 mm)	1.9952 in. (50.6781 mm)	
#2 BUSHING	I.D.	2.0000 in. (50.8000 mm)	2.0015 in. (50.8381 mm)	2.0053 in. (50.9346 mm)	0.0063 in. (0.1600 mm)
#2 PIN	O.D.	1.9980 in. (50.7492 mm)	1.9990 in. (50.7746 mm)	1.9952 in. (50.6781 mm)	
#3 BUSHING	I.D.	1.4390 in. (36.5506 mm)	1.4402 in. (36.5811 mm)	1.4446 in. (36.6928 mm)	0.0066 in. (0.1676 mm)
#3 SLEEVE	O.D.	1.4360 in. (36.4744 mm)	1.4380 in. (36.5252 mm)	1.4336 in. (36.4134 mm)	
#4 BUSHING	I.D	1.4980 in. (38.0492 mm)	1.4992 in. (38.0797 mm)	1.5036 in. (38.1914 mm)	0.0076 in. (0.1930 mm)
#4 SLEEVE	O.D.	1.4940 in. (37.9476 mm)	1.4960 in. (37.9984 mm)	1.4916 in. (37.8866 mm)	
#5 BUSHING	I.D.	2.5000 in. (63.5000 mm)	2.5015 in. (63.5381 mm)	2.5062 in. (63.6575 mm)	0.0072 in. (0.1829 mm)
#5 PIN	O.D.	2.4975 in. (63.4365 mm)	2.4990 in. (63.4746 mm)	2.4943 in. (63.3552 mm)	
#6 BUSHING	I.D.	2.5000 in. (63.5000 mm)	2.5015 in. (63.5381 mm)	2.5062 in. (63.6575 mm)	0.0072 in. (0.1829 mm)
#6 PIN	O.D.	2.4975 in. (63.4365 mm)	2.4990 in. (63.4746 mm)	2.4943 in. (63.3552 mm)	

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Table 601/32-11-51-993-801 Wear Limits (Continued)

NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE DIAMETRIC CLEARANCE
#7 BUSHING	I.D.	2.5000 in. (63.5000 mm)	2.5015 in. (63.5381 mm)	2.5062 in. (63.6575 mm)	0.0072 in. (0.1829 mm)
#7 PIN	O.D.	2.4975 in. (63.4365 mm)	2.4990 in. (63.4746 mm)	2.4943 in. (63.3552 mm)	

SUBTASK 32-11-51-420-018

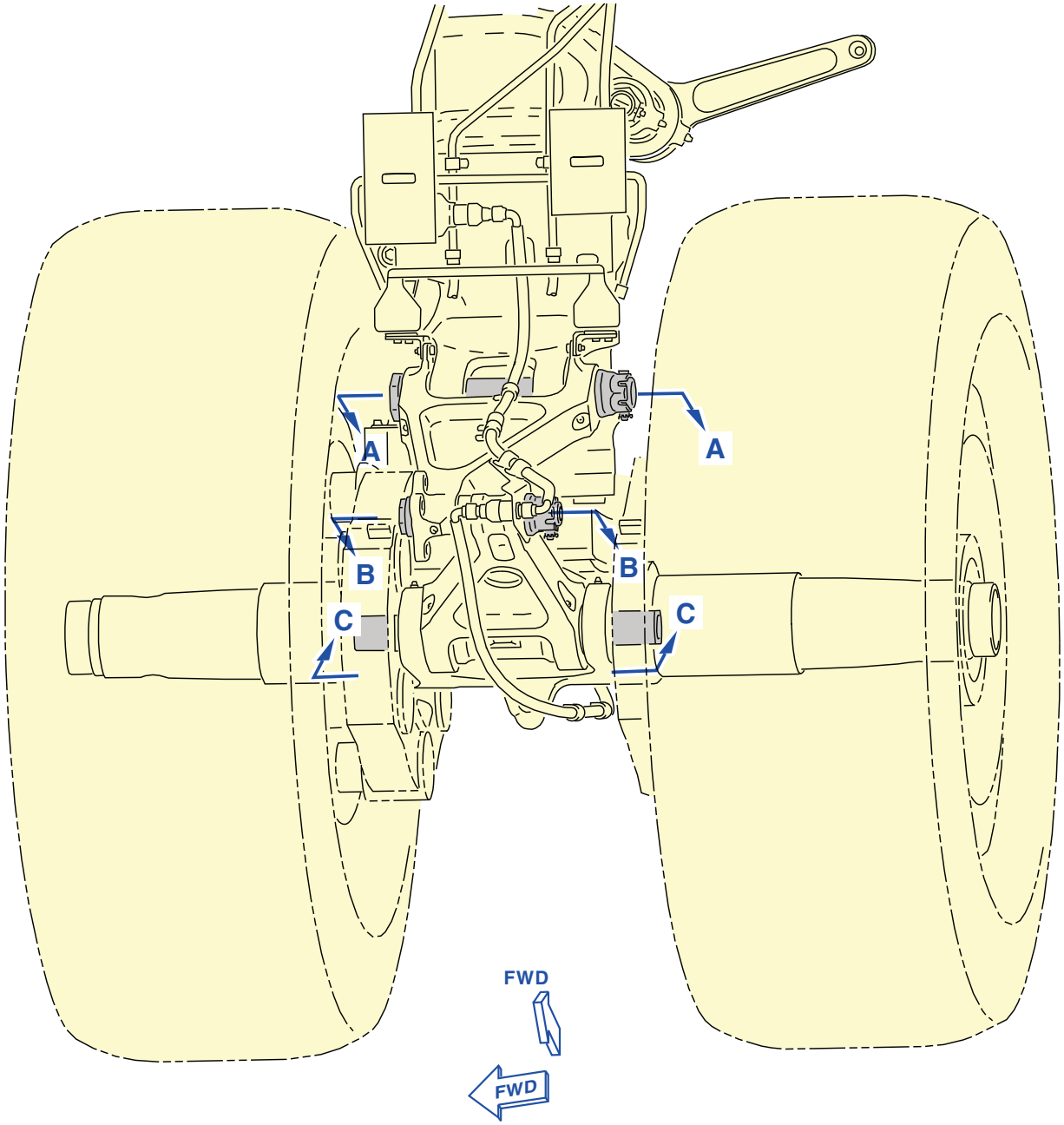
(4) To install the torsion links or parts for the main landing gear do the task that follows:

(a) Do this task: Main Landing Gear Upper Torsion Link Installation, TASK 32-11-51-400-801.

(b) Do this task: Main Landing Gear Lower Torsion Link Installation, TASK 32-11-51-400-802.

——— END OF TASK ———

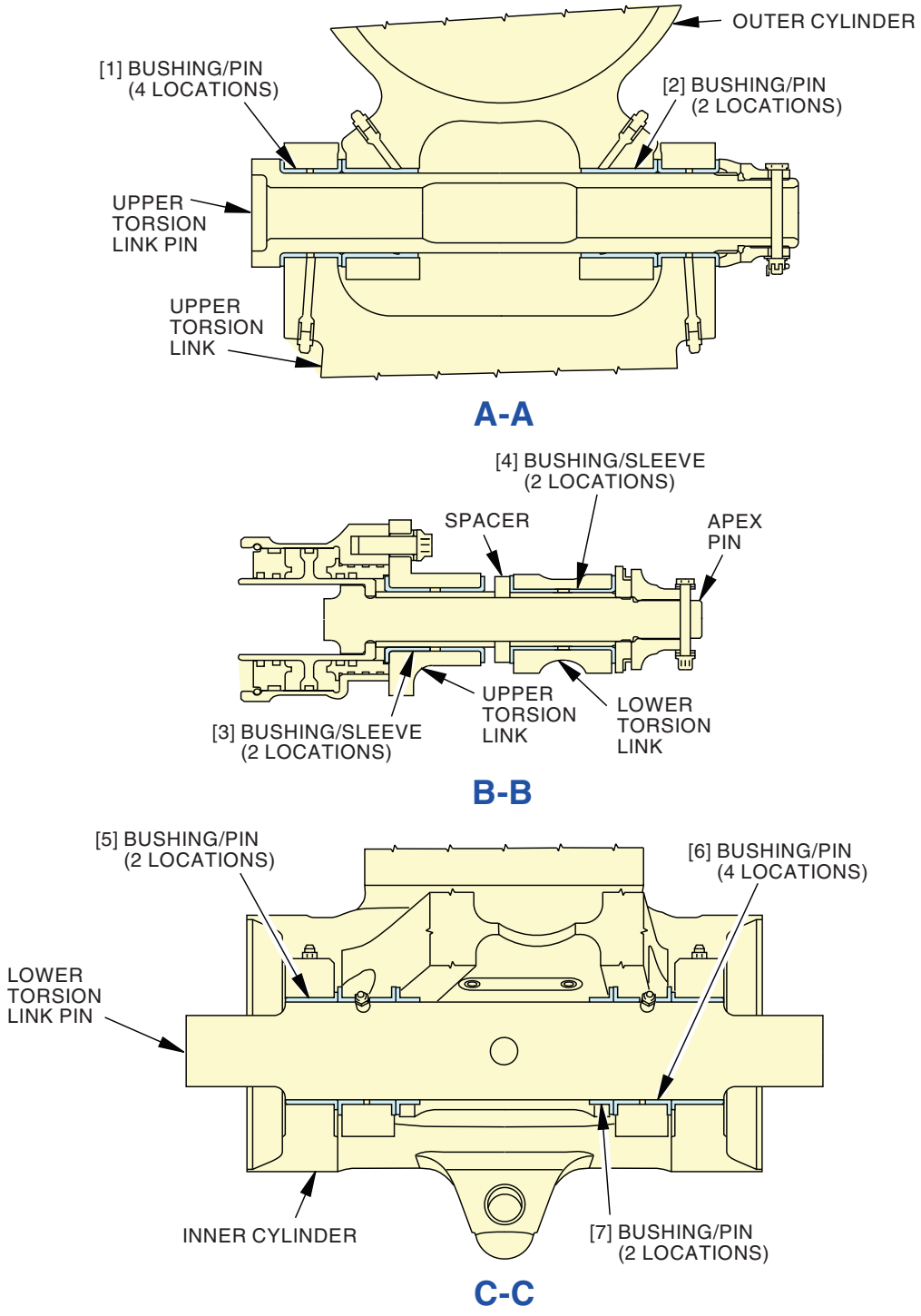
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Main Landing Gear Torsion Wear Limits
Figure 601/32-11-51-990-802 (Sheet 1 of 2)

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Main Landing Gear Torsion Wear Limits
Figure 601/32-11-51-990-802 (Sheet 2 of 2)

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MAIN LANDING GEAR SIDE STRUT ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Main Landing Gear Upper Side Strut Removal

(2) Main Landing Gear Upper Side Strut Installation

(3) Main Landing Gear Lower Side Strut Removal

(4) Main Landing Gear Lower Side Strut Installation

(5) Main Landing Gear Side Strut Removal

(6) Main Landing Gear Side Strut Installation.

B. The lower side strut and the upper side strut are referred to as the "side strut" in this procedure.

TASK 32-11-61-000-801

2. Main Landing Gear Upper Side Strut Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the upper side strut.

B. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-32-91-000-801	Main Gear Downlock Springs Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1359	Wrench Adapter, C32029-3 or C32029-93 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1367	Slug Driver, C32029-11 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1368	Thread Protector, C32029-12 or C32029-101 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1369	Thread Protector , C32029-13 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left

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(Continued)

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Prepare for the Removal

SUBTASK 32-11-61-480-010

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-11-61-580-001

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(2) Do this task: Lift the Airplane on Jacks, TASK 07-10-01-580-801.

SUBTASK 32-11-61-860-001

(3) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-11-61-020-001

(4) Do this task: Main Gear Downlock Springs Removal, TASK 32-32-91-000-801.

F. Main Landing Gear Upper Side Strut Removal

SUBTASK 32-11-61-020-002

(1) Disconnect the upper side strut [3] from the downlock strut [1]:

(a) Remove the cotter pin [6] from the downlock link pin [2].

(b) Remove the nut [5] and washer [4] from the downlock link pin [2].

(c) Remove the downlock link pin [2] to disconnect the upper side strut [3] from the downlock strut [1].

NOTE: Use the thread protector, C32029-13, SPL-1369, for the downlock link pin [2] when you do this step.

SUBTASK 32-11-61-020-003

(2) Disconnect the upper side strut [3] from the lower side strut [8]:

(a) Remove the cotter pin [13], nut [14], and washer [12] from the bolt [11].

(b) Remove the bolt [11] from the side strut pin [7].

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(c) Use a rope to support the upper end of the lower side strut [8] and attach the rope to the outboard side of the reaction link assembly [15].

NOTE: The lower side strut weighs approximately 48.3 lb (21.91 kg).

(d) Remove the end cap [9] from the side strut pin [7].

(e) Remove the side strut pin [7] to disconnect the upper side strut [3] from the lower side strut [8].

NOTE: Use the slug driver, C32029-11, SPL-1367, for the side strut pin [7] when you do this step.

SUBTASK 32-11-61-020-004

(3) Disconnect the upper side strut [3] from the reaction link assembly [15]:

(a) Remove the cotter pin [19], nut [18], and washer [17] from cross bolt [16].

(b) Remove the cross bolt [16] from the side strut pin [22].

(c) Remove the side strut nut [20] and side strut washer [21] from the side strut pin [22].

NOTE: Use wrench adapter, C32029-3 or C32029-93, SPL-1359, to hold the head of the side strut pin [22] when you loosen the side strut nut [20].

(d) Remove the side strut pin [22] that holds the upper side strut [3] to the reaction link assembly [15].

NOTE: Use the thread protector, C32029-12 or C32029-101, SPL-1368, for the side strut pin [22] when you do this step.

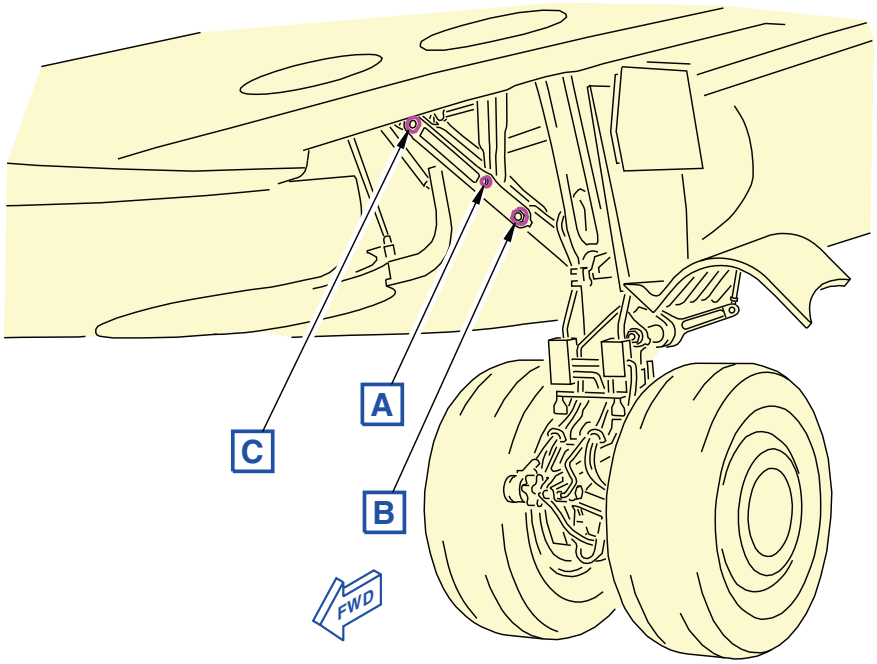
SUBTASK 32-11-61-020-005

(4) Lower the upper side strut [3] carefully and remove it from the airplane.

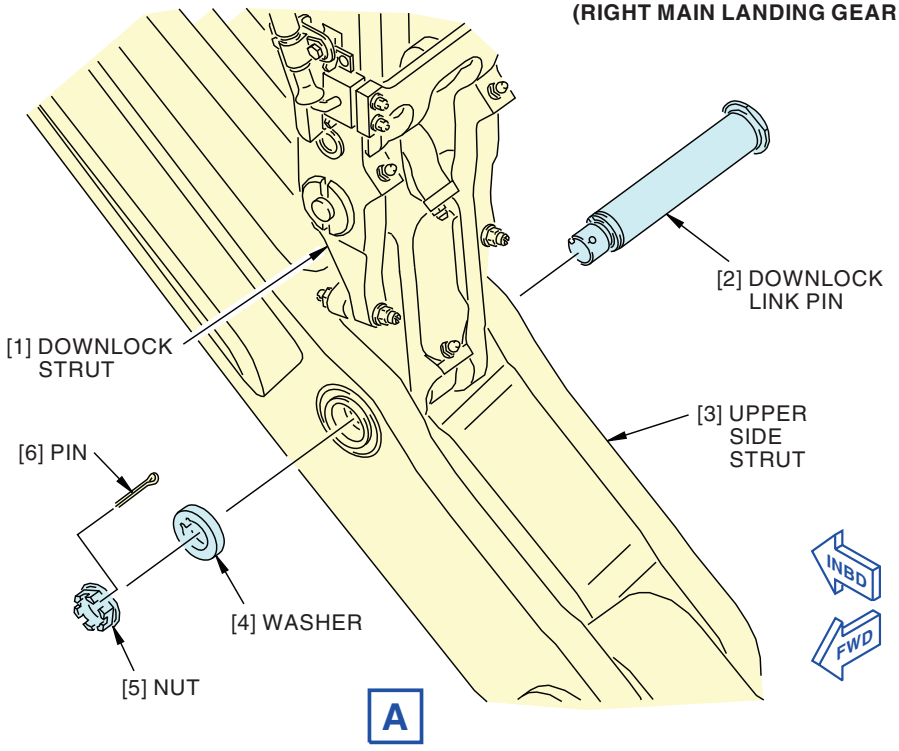
NOTE: Make sure that the upper side strut is supported carefully when you lower it. The upper side strut weighs approximately 42.8 lb (19.41 kg).

——— END OF TASK ———

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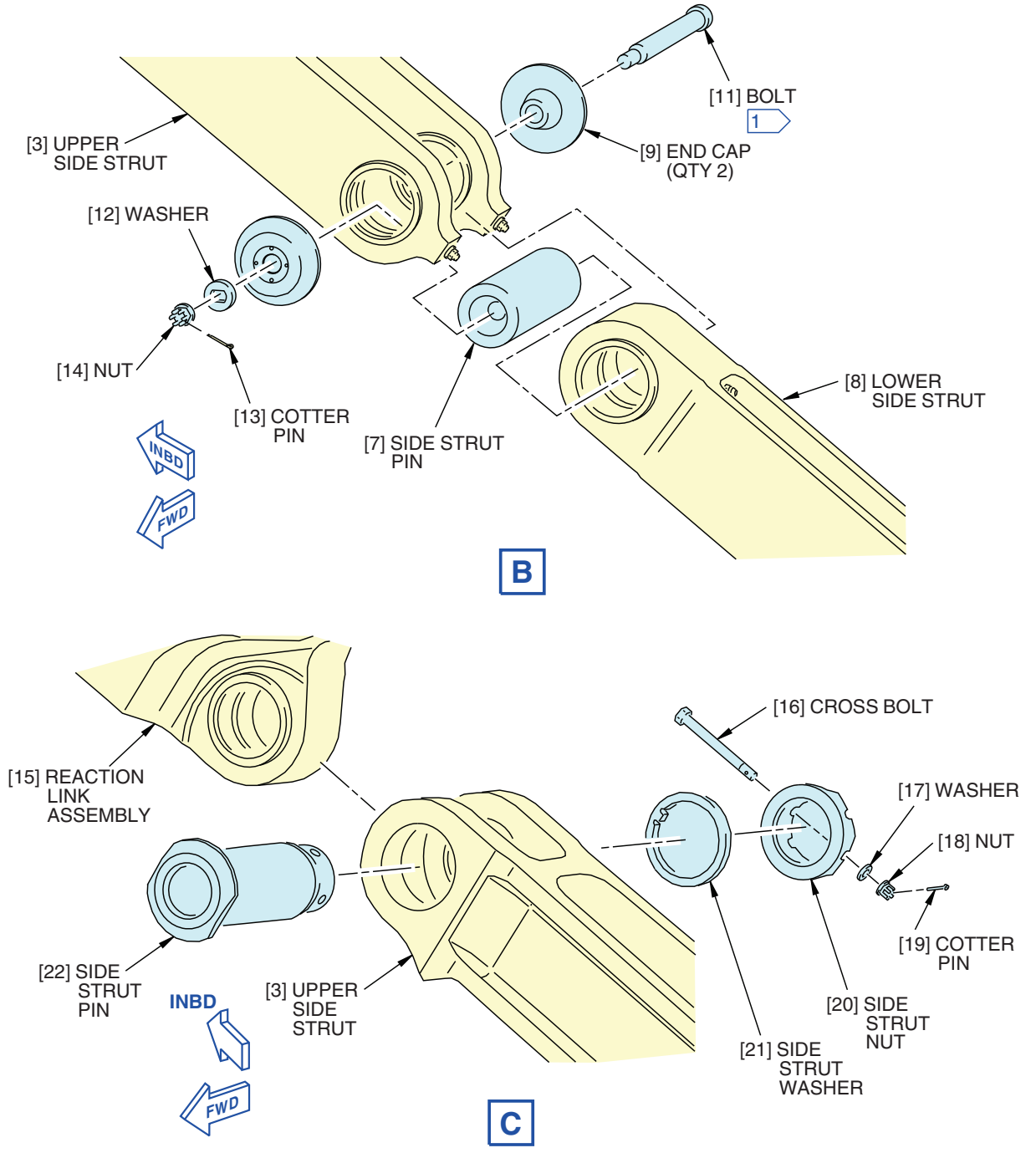
LEFT MAIN LANDING GEAR
(RIGHT MAIN LANDING GEAR IS EQUIVALENT)



2414083 S00061534360_V1

Main Landing Gear Upper Side Strut Installation
Figure 401/32-11-61-990-801 (Sheet 1 of 2)

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1 MAKE SURE THAT THE BOLT HEAD POINTS IN THE AFT DIRECTION.

2414084 S00061534361_V2

Main Landing Gear Upper Side Strut Installation
Figure 401/32-11-61-990-801 (Sheet 2 of 2)

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TASK 32-11-61-400-801

3. Main Landing Gear Upper Side Strut Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the upper side strut.

B. References

Reference	Title
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
12-21-11-640-801	Main Landing Gear Upper End Components Servicing (P/B 301)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-32-00-710-802	Main Landing Gear - Operational Test (P/B 501)
32-32-91-400-801	Main Gear Downlock Spring Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1359	Wrench Adapter, C32029-3 or C32029-93 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1368	Thread Protector, C32029-12 or C32029-101 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1369	Thread Protector , C32029-13 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Upper side strut	32-11-61-01-310	PTF ALL
6	Cotter pin	32-11-61-01-040	PTF ALL
13	Cotter pin	32-11-61-01-295	PTF ALL
19	Cotter pin	32-11-71-01-085	PTF ALL

F. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
713	Nose Landing Gear

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(Continued)

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

G. Prepare for the Installation

SUBTASK 32-11-61-480-011

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

H. Main Landing Gear Upper Side Strut Installation

SUBTASK 32-11-61-420-001

(1) Connect the upper side strut [3] to the reaction link assembly [15]:

(a) Lubricate the chrome plated surfaces of the side strut pin [22] with grease, D00633.

(b) Remove all unwanted grease, D00633.

(c) Put the upper side strut [3] in its position for connection to the reaction link assembly [15].

(d) Put the side strut pin [22] through the upper side strut [3] and the reaction link assembly [15].

NOTE: Use the thread protector, C32029-12 or C32029-101, SPL-1368, for the side strut pin [22] when you do this step.

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(e) Apply a thin layer of the corrosion inhibiting material, G50136 to these items:

NOTE: Do not apply corrosion inhibiting material to the chrome plated surfaces.

1) The threads and thread reliefs of the side strut pin [22] and cross bolt [16].

2) The threads of the side strut nut [20] and nut [18].

3) The faces of the side strut washer [21] and washer [17].

(f) Remove all unwanted corrosion inhibiting material, G50136.

(g) Install the side strut washer [21] and side strut nut [20] on the side strut pin [22].

(h) Tighten the side strut nut [20] to 600 in-lb (67.79 N·m) to 900 in-lb (101.69 N·m).

NOTE: Use the wrench adapter, C32029-3 or C32029-93, SPL-1359, to hold the head of the side strut pin [22] when you tighten the side strut nut [20].

(i) If it is necessary, loosen the side strut nut [20] to the nearest castellation to align the hole for the cross bolt [16].

(j) Install the cross bolt [16] in the side strut pin [22].

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(k) Install the washer [17] and nut [18] on the cross bolt [16].

(l) Tighten the nut [18] to 30 in-lb (3.39 N·m) to 50 in-lb (5.65 N·m).

(m) If it is necessary, loosen the nut [18] to the nearest castellation to align the holes for the new cotter pin [19].

(n) Install the new cotter pin [19] in the cross bolt [16].

SUBTASK 32-11-61-620-001

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(2) Apply a thin layer of the corrosion inhibiting material, G50136 to these items:

NOTE: Do not apply corrosion inhibiting material to the chrome plated surfaces.

(a) The inside diameter of the side strut pin [7], the inside unplated surface of the end cap [9]

(b) The threads of the nut [14]

(c) The face of the washer [12].

SUBTASK 32-11-61-160-001

(3) Remove all unwanted corrosion inhibiting material, G50136.

(a) Make sure that the drainage holes in the end cap [9] are clear of the corrosion inhibiting material, G50136 after the application.

SUBTASK 32-11-61-620-002

(4) Lubricate the chrome plated surfaces of the side strut pin [7] and the surfaces of the end cap [9] with grease, D00633.

SUBTASK 32-11-61-160-005

(5) Remove all unwanted grease, D00633.

SUBTASK 32-11-61-420-002

(6) Connect the upper side strut [3] to the lower side strut [8]:

(a) Remove the rope that supports the lower side strut [8] to the reaction link assembly [15].

(b) Put the upper side strut [3] in its position for connection to the lower side strut [8].

(c) Put the side strut pin [7] through the upper side strut [3] and the lower side strut [8].

(d) Install the end cap [9] on the side strut pin [7].

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WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(e) Apply a thin layer of the corrosion inhibiting material, G50136, to the threads, thread reliefs of the bolt [11], and its shank.

NOTE: Do not apply corrosion inhibiting material to the chrome plated surfaces.

CAUTION

MAKE SURE THAT THE HEAD OF THE BOLT POINTS IN THE AFT DIRECTION. IF THE HEAD OF THE BOLT POINTS FORWARD, DAMAGE TO COMPONENTS AND THE AIRPLANE STRUCTURE WILL OCCUR.

(f) Install the bolt [11] in the side strut pin [7].

(g) Install the washer [12] and nut [14] on the bolt [11].

(h) Tighten the nut [14] to 30 in-lb (3.39 N·m) to 50 in-lb (5.65 N·m) more than the run on torque.

(i) If it is necessary, loosen the nut [14] to the nearest castellation to align the hole for the new cotter pin [13].

(j) Install the new cotter pin [13] in the bolt [11].

SUBTASK 32-11-61-420-003

(7) Connect the upper side strut [3] to the downlock strut [1]:

(a) Lubricate the chrome plated surfaces of the downlock link pin [2] with grease, D00633.

(b) Remove all unwanted grease, D00633.

(c) Put the downlock strut [1] in its position for connection to the upper side strut [3].

(d) Put the downlock link pin [2] through the upper side strut [3] and the downlock strut [1].

NOTE: Use the thread protector, C32029-13, SPL-1369, for the downlock link pin [2] when you do this step.

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(e) Apply a thin layer of the corrosion inhibiting material, G50136 to these items:

NOTE: Do not apply corrosion inhibiting material to the chrome plated surfaces.

1) The threads and thread reliefs of the downlock link pin [2]

2) The threads of the nut [5]

3) The faces of the washer [4].

(f) Remove all unwanted corrosion inhibiting material, G50136.

(g) Install the washer [4] and nut [5] on the downlock link pin [2].

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(h) Tighten the nut [5] to 360 in-lb (40.67 N·m) to 600 in-lb (67.79 N·m).

(i) If it is necessary, loosen the nut [5] to the nearest castellation to align the hole for the new cotter pin [6].

(j) Install the new cotter pin [6] in the downlock link pin [2].

SUBTASK 32-11-61-420-004

(8) Do this task: Main Gear Downlock Spring Installation, TASK 32-32-91-400-801.

SUBTASK 32-11-61-640-001

(9) Lubricate the lube fittings on the upper side strut. To lubricate them, refer to Main Landing Gear Upper End Components Servicing, TASK 12-21-11-640-801.

SUBTASK 32-11-61-780-001

(10) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-11-61-710-001

(11) Do this task: Main Landing Gear - Operational Test, TASK 32-32-00-710-802.

SUBTASK 32-11-61-580-002

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(12) Do this task: Lower the Airplane off Jacks, TASK 07-10-01-580-802.

——— END OF TASK ———

TASK 32-11-61-000-802

4. Main Landing Gear Lower Side Strut Removal

(Figure 402)

A. General

(1) This task gives the instructions to remove the lower side strut.

B. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-32-91-000-801	Main Gear Downlock Springs Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

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Reference	Description
SPL-1367	Slug Driver, C32029-11 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1370	Thread Protector, C32029-14 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Prepare for the Removal

SUBTASK 32-11-61-480-012

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-11-61-580-003

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(2) Do this task: Lift the Airplane on Jacks, TASK 07-10-01-580-801.

SUBTASK 32-11-61-860-002

(3) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-11-61-020-006

(4) Do this task: Main Gear Downlock Springs Removal, TASK 32-32-91-000-801.

F. Main Landing Gear Lower Side Strut Removal

SUBTASK 32-11-61-020-007

(1) Disconnect the lower side strut [8] from the pushrod [41]:

(a) Remove the cotter pin [56], nut [42], and washer [43] from the bolt [46].

1) Discard the cotter pin [56].

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(b) Remove the bolt [46], washer [44], and bushing [45] to disconnect the lower side strut [8] from the pushrod [41].

(c) Isolate the pushrod [41] from the lower side strut [8].

SUBTASK 32-11-61-480-004

(2) Use a rope to hold the pushrod [41] and the inner door in its position.

SUBTASK 32-11-61-020-008

(3) Disconnect the lower side strut [8] from the shock strut [51]:

(a) Remove the cotter pin [49], nut [50], and washer [48] from the cross bolt [47].

1) Discard the cotter pin [49].

(b) Remove the cross bolt [47] from the side strut pin [52].

(c) Remove the side strut nut [54] and side strut washer [53] from the side strut pin [52].

(d) Remove the side strut pin [52] to disconnect the lower side strut [8] from the shock strut [51].

1) Use the thread protector, C32029-14, SPL-1370, for the side strut pin [52] removal.

SUBTASK 32-11-61-020-009

(4) Disconnect the lower side strut [8] from the upper side strut [3]:

(a) Remove the cotter pin [13], nut [14], and washer [12] from the bolt [11].

1) Discard the cotter pin [13].

(b) Remove the bolt [11] from the side strut pin [7].

(c) Remove the end cap [9] from the side strut pin [7].

(d) Use a rope to support the upper end of the lower side strut [8] and attach the rope to the outboard side of the reaction link assembly [15].

NOTE: The lower side strut weighs approximately 48.3 lb (21.91 kg).

(e) Remove the side strut pin [7] to disconnect the upper side strut [3] from the lower side strut [8].

1) Use the slug driver, C32029-11, SPL-1367, for the side strut pin [7] removal.

SUBTASK 32-11-61-020-010

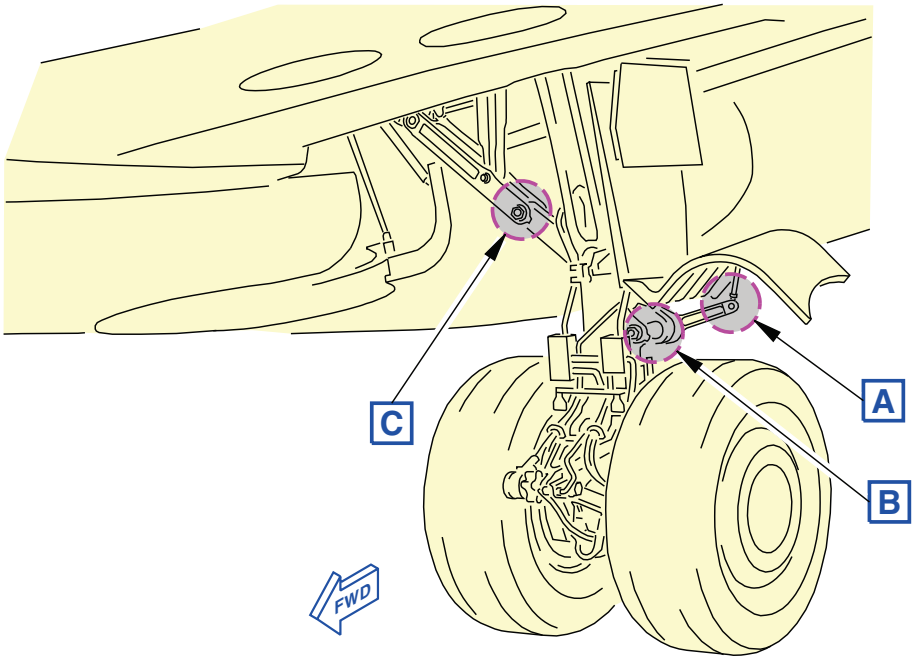
(5) Carefully lower and remove the lower side strut [8] from the airplane.

NOTE: The lower side strut weighs approximately 48.3 lb (21.91 kg).

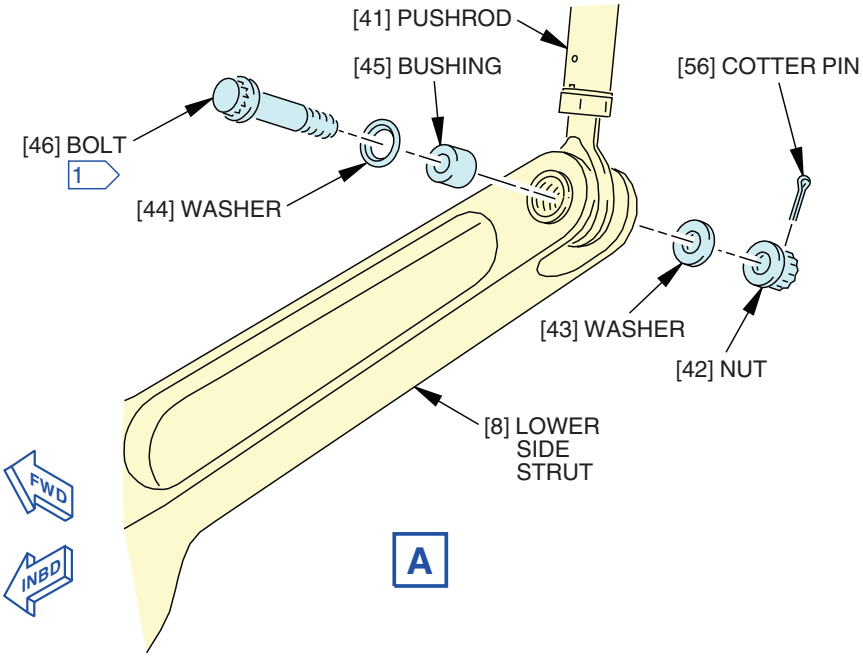
(a) Make sure that the lower side strut [8] has sufficient support.

——— END OF TASK ———

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LEFT MAIN LANDING GEAR
(RIGHT MAIN LANDING GEAR IS OPPOSITE)

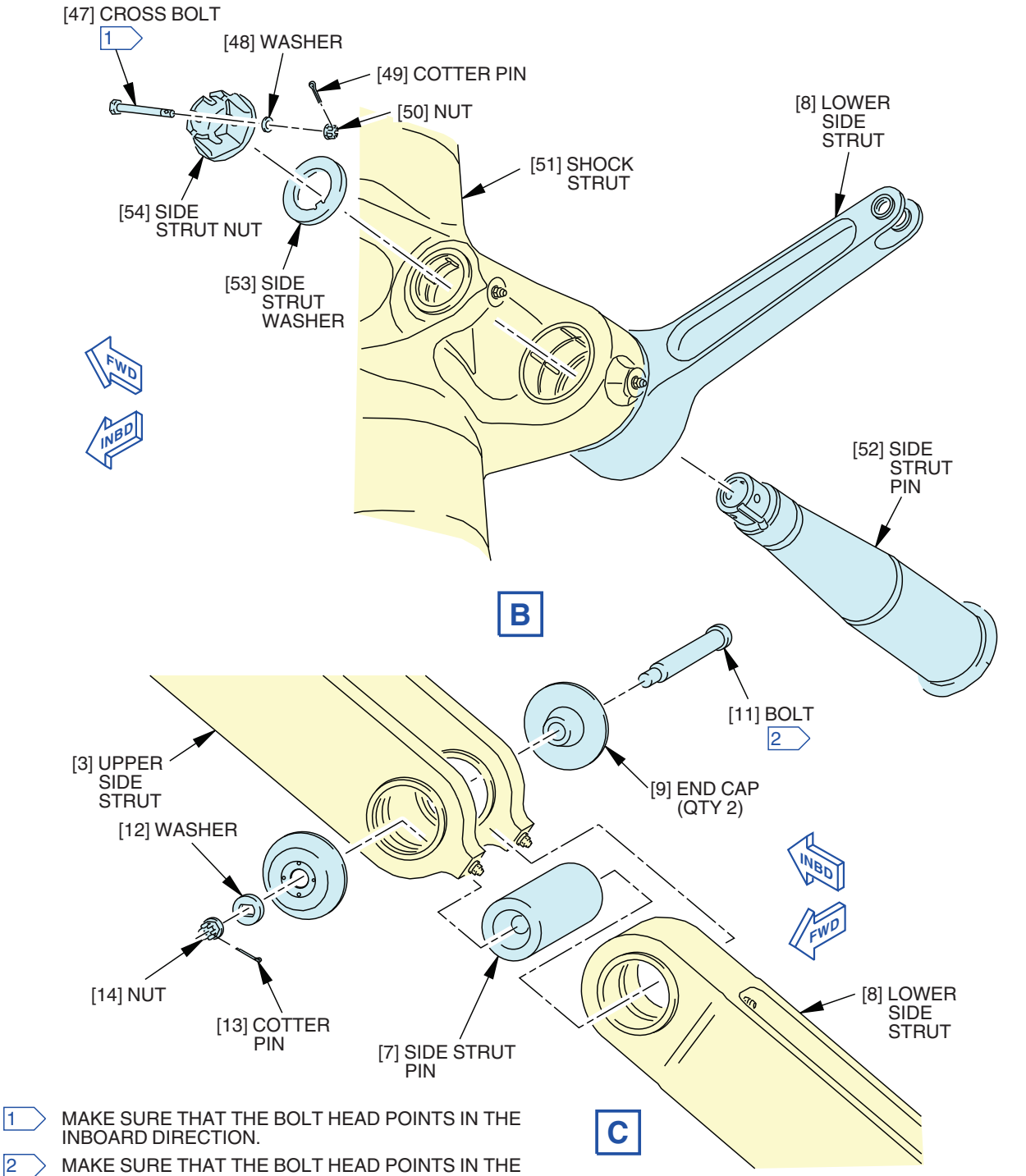


1 FOR THE RIGHT MAIN LANDING GEAR, MAKE SURE THAT THE BOLT HEAD POINTS IN THE AFT DIRECTION.

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Main Landing Gear Lower Side Strut Installation
Figure 402/32-11-61-990-802 (Sheet 1 of 2)

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2414086 S00061534365_V3

Main Landing Gear Lower Side Strut Installation
Figure 402/32-11-61-990-802 (Sheet 2 of 2)

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TASK 32-11-61-400-802

5. Main Landing Gear Lower Side Strut Installation

(Figure 402)

A. General

(1) This task gives the instructions to install the lower side strut.

B. References

Reference	Title
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
12-21-11-640-801	Main Landing Gear Upper End Components Servicing (P/B 301)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-13-11-820-801	Main Landing Gear Shock Strut Door Adjustment (P/B 501)
32-32-00-710-802	Main Landing Gear - Operational Test (P/B 501)
32-32-91-400-801	Main Gear Downlock Spring Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1361	Wrench Adapter, C32029-39 or C32029-95 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1370	Thread Protector, C32029-14 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
8	Lower side strut	32-11-61-01-350	PTF ALL
13	Cotter pin	32-11-61-01-295	PTF ALL
49	Cotter pin	32-11-61-05-040	PTF ALL
56	Cotter pin	32-13-21-01-020	PTF ALL

F. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
713	Nose Landing Gear
734	Left Main Landing Gear

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(Continued)

Zone	Area
744	Right Main Landing Gear

G. Prepare for the Installation

SUBTASK 32-11-61-480-013

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

H. Main Landing Gear Lower Side Strut Installation

SUBTASK 32-11-61-620-003

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(1) Apply a thin layer of corrosion inhibiting material, G50136, to these items:

NOTE: Do not apply corrosion inhibiting material to the chrome plated surfaces.

(a) The inside diameter of the side strut pin [7], the inside unplated surface of the end cap [9]

(b) The threads of the nut [14]

(c) The face of the washer [12].

SUBTASK 32-11-61-160-003

(2) Remove all unwanted corrosion inhibiting material, G50136.

(a) Make sure that the drainage holes in the end cap [9] are clear of corrosion inhibiting material, G50136, after the application.

SUBTASK 32-11-61-620-004

(3) Lubricate the chrome plated surfaces of the side strut pin [7] and surfaces of the end cap [9] with grease, D00633.

SUBTASK 32-11-61-160-004

(4) Remove all unwanted grease, D00633.

SUBTASK 32-11-61-420-005

(5) Connect the upper side strut [3] to the lower side strut [8]:

(a) Remove the rope that supports the lower side strut [8] to the reaction link assembly [15].

(b) Put the upper side strut [3] in its position for connection to the lower side strut [8].

(c) Put the side strut pin [7] through the upper side strut [3] and lower side strut [8].

(d) Install the end cap [9] on the side strut pin [7].

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WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(e) Apply a thin layer of corrosion inhibiting material, G50136, to the threads, thread reliefs of the bolt [11], and its shank.

NOTE: Do not apply corrosion preventive material to the chrome plated surfaces.

CAUTION

MAKE SURE THAT THE HEAD OF THE BOLT POINTS IN THE AFT DIRECTION. IF THE HEAD OF THE BOLT POINTS FORWARD, DAMAGE TO COMPONENTS AND THE AIRPLANE STRUCTURE WILL OCCUR.

(f) Install the bolt [11] in the side strut pin [7].

(g) Install the washer [12] and nut [14] on the bolt [11].

(h) Tighten the nut [14] to 30 in-lb (3.39 N·m) - 50 in-lb (5.65 N·m) more than the run on torque.

(i) If it is necessary, loosen the nut [14] to the nearest castellation to align the hole for the new cotter pin [13].

(j) Install the new cotter pin [13] in the bolt [11].

SUBTASK 32-11-61-420-006

(6) Connect the lower side strut [8] to the shock strut [51]:

(a) Lubricate the chrome plated surfaces of the side strut pin [52] with grease, D00633.

(b) Remove all unwanted grease, D00633.

(c) Put the lower side strut [8] in its position for attachment to the shock strut [51].

(d) Put the side strut pin [52] through the lower side strut [8] and clevis on the shock strut [51].

1) Use the thread protector, C32029-14, SPL-1370, for the side strut pin [52] installation.

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(e) Apply a thin layer of corrosion inhibiting material, G50136, to these items:

NOTE: Do not apply corrosion inhibiting material to the chrome plated surfaces.

1) The threads and thread reliefs of the side strut pin [52], nut [50], and cross bolt [47]

2) The threads of the side strut nut [54] and nut [50]

3) The faces of the side strut washer [53] and washer [48].

(f) Remove all unwanted corrosion inhibiting material, G50136.

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(g) Install the side strut washer [53] and side strut nut [54] on the side strut pin [52].

(h) Tighten the side strut nut [54] to 95 ft-lb (128.8 N·m) - 115 ft-lb (155.9 N·m).

1) Use the wrench adapter, C32029-39 or C32029-95, SPL-1361, to hold the head of the side strut pin [52] when you tighten the side strut nut [54].

(i) If it is necessary, loosen the side strut nut [54] to the nearest castellation to align the hole for the cross bolt [47].

(j) Install the cross bolt [47] with the bolt head inboard in the side strut pin [52].

1) Make sure that the head of the cross bolt [47] points inboard.

(k) Install the washer [48] and nut [50] on the cross bolt [47].

(l) Tighten the nut [50] to 20 in-lb (2.26 N·m) - 24 in-lb (2.71 N·m).

(m) If it is necessary, loosen the nut [50] to the nearest castellation to align the hole for the cotter pin [49].

CAUTION

MAKE SURE THAT THE COTTER PIN IS INSTALLED CORRECT. A CHAFF CONDITION CAN OCCUR TO THE SHEATHING ON THE CABLE OF THE GROUND SPOILER INTERLOCK. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(n) Install the new cotter pin [49] in the cross bolt [47].

1) Make sure that the cotter pin [49] is installed in accordance with the type B preferred installation method (TASK 20-10-44-400-801).

(o) Check the interior of the side strut pin [52] and repair the application of corrosion inhibiting material, G50136, as required.

SUBTASK 32-11-61-080-001

(7) Remove the rope that holds the pushrod [41] in its position.

SUBTASK 32-11-61-420-015

(8) Connect the lower side strut [8] to the pushrod [41]:

(a) Align the hole in the pushrod [41] with the hole in the lower side strut [8].

(b) Put the bushing [45] through the lower side strut [8] and pushrod [41].

(c) Install the washer [44] on the bolt [46].

(d) Put the bolt [46] through the lower side strut [8] and pushrod [41].

(e) Install the washer [43] and nut [42] on the bolt [46].

(f) Hand tighten the nut [42].

NOTE: Final torque to the nut will be done after final adjustments to the doors are made in the subsequent step.

SUBTASK 32-11-61-820-001

(9) If the length of pushrod [41] is changed, do the step that follows:

NOTE: No adjustment to the doors is necessary if the pushrod was not changed.

(a) Do this task: Main Landing Gear Shock Strut Door Adjustment, TASK 32-13-11-820-801.

SUBTASK 32-11-61-420-036

(10) Tighten the nut [42].

(a) Install the new cotter pin [56] in the hole of the bolt [46].

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(b) If it is necessary, loosen the nut [42] to the nearest castellation to align the holes for the cotter pin [56].

I. Put the Airplane Back to Its Usual Condition

SUBTASK 32-11-61-420-008

(1) Do this task: Main Gear Downlock Spring Installation, TASK 32-32-91-400-801.

SUBTASK 32-11-61-640-002

(2) Lubricate the lube fittings on the lower side strut, do this task: Main Landing Gear Upper End Components Servicing, TASK 12-21-11-640-801.

SUBTASK 32-11-61-780-002

(3) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-11-61-710-002

(4) Do this task: Main Landing Gear - Operational Test, TASK 32-32-00-710-802.

SUBTASK 32-11-61-580-004

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(5) Do this task: Lower the Airplane off Jacks, TASK 07-10-01-580-802.

——— END OF TASK ———

TASK 32-11-61-000-803

6. Main Landing Gear Side Strut Removal

(Figure 403)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions to remove the side strut.

B. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-32-91-000-801	Main Gear Downlock Springs Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

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Reference	Description
SPL-1359	Wrench Adapter, C32029-3 or C32029-93 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1368	Thread Protector, C32029-12 or C32029-101 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1369	Thread Protector , C32029-13 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1370	Thread Protector, C32029-14 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Prepare for the Removal

SUBTASK 32-11-61-480-014

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-11-61-580-005

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(2) Do this task: Lift the Airplane on Jacks, TASK 07-10-01-580-801.

SUBTASK 32-11-61-860-003

(3) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-11-61-020-011

(4) Do this task: Main Gear Downlock Springs Removal, TASK 32-32-91-000-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Main Landing Gear Side Strut Removal

SUBTASK 32-11-61-020-012

(1) Disconnect the upper side strut [3] from the downlock strut [1]:

(a) Remove the cotter pin [6] from the downlock link pin [2].

1) Discard the cotter pin [6].

(b) Remove the nut [5] and washer [4] from the downlock link pin [2].

(c) Remove the downlock link pin [2] to disconnect the upper side strut [3] from the downlock strut [1].

1) Use the thread protector, C32029-13, SPL-1369, for the downlock link pin [2] removal.

SUBTASK 32-11-61-020-013

(2) Disconnect the lower side strut [8] from the pushrod [41]:

(a) Remove the cotter pin [56], nut [42], and washer [43] from the bolt [46].

1) Discard the cotter pin [56].

(b) Remove the bolt [46], washer [44], and bushing [45] to disconnect the lower side strut [8] from the pushrod [41].

(c) Isolate the pushrod [41] from the lower side strut [8].

SUBTASK 32-11-61-480-007

(3) Use a rope to hold the pushrod [41] and the inner door in its position.

SUBTASK 32-11-61-020-014

(4) Disconnect the lower side strut [8] from the shock strut [51]:

(a) Remove the cotter pin [49], nut [50], and washer [48] from the cross bolt [47].

1) Discard the cotter pin [49].

(b) Remove the cross bolt [47] from the side strut pin [52].

(c) Remove the side strut nut [54] and side strut washer [53] from the side strut pin [52].

(d) Remove the side strut pin [52] to disconnect the lower side strut [8] from the shock strut [51].

1) Use the thread protector, C32029-14, SPL-1370, for the side strut pin [52] removal.

SUBTASK 32-11-61-020-015

(5) Disconnect the upper side strut [3] from the reaction link assembly [15]:

(a) Remove the cotter pin [19], nut [18], and washer [17] from cross bolt [16].

1) Discard the cotter pin [19].

(b) Remove the cross bolt [16] from the side strut pin [22].

(c) Remove the side strut nut [20] and side strut washer [21] from the side strut pin [22].

1) Use the wrench adapter, C32029-3 or C32029-93, SPL-1359, to hold the head of the side strut pin [22] when you loosen the side strut nut [20].

(d) Remove the side strut pin [22] that holds the upper side strut [3] to the reaction link assembly [15].

1) Use the thread protector, C32029-12 or C32029-101, SPL-1368, for the side strut pin [22] removal.

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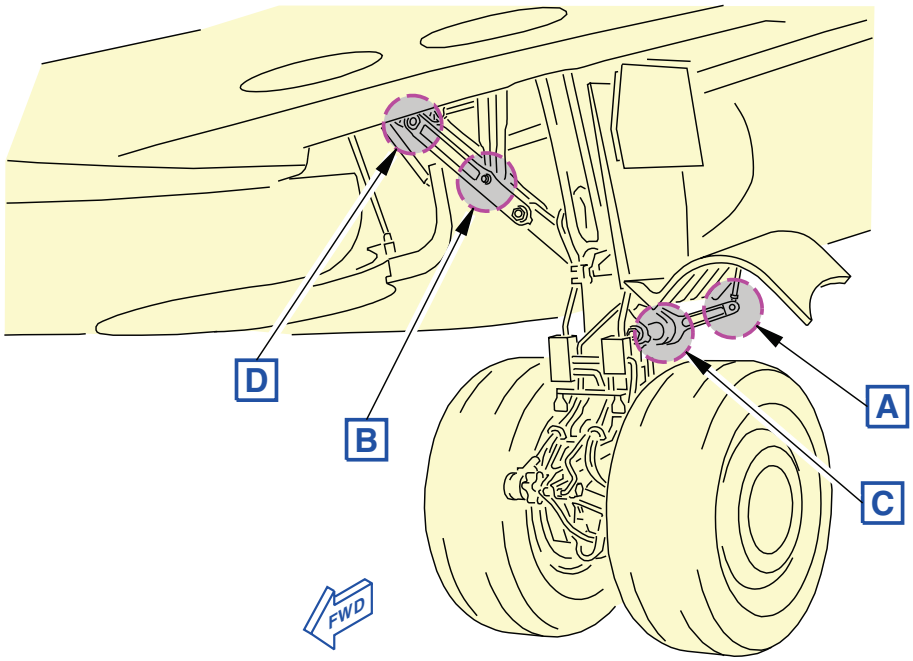
SUBTASK 32-11-61-020-016

(6) Remove the assembly of the upper side strut [3] and the lower side strut [8] from the airplane.

NOTE: The side strut assembly weighs approximately 90.11 lb (40.87 kg). Make sure a minimum of two person will hold and lower the side strut when you remove it from the airplane.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

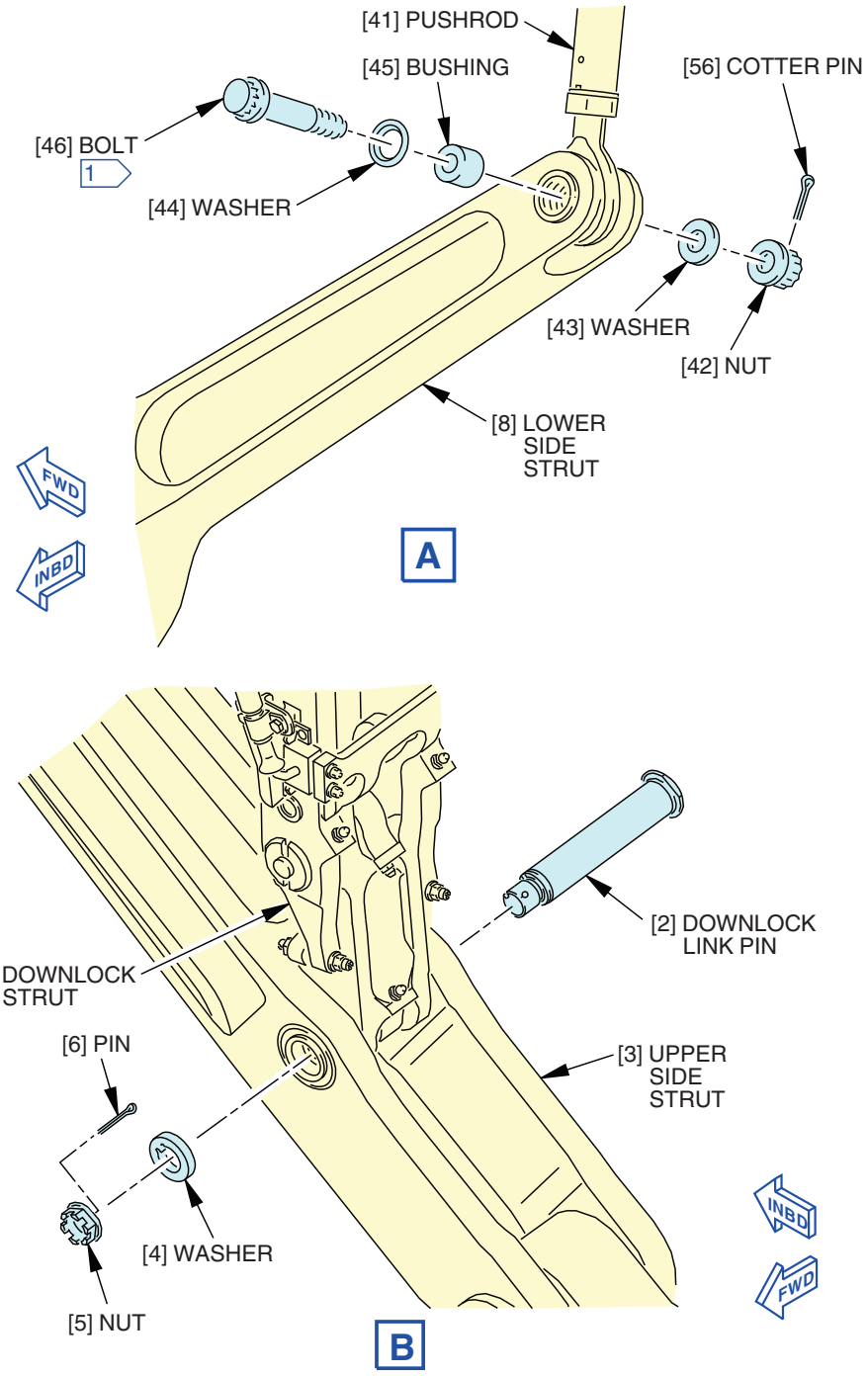


LEFT MAIN LANDING GEAR
(RIGHT MAIN LANDING GEAR IS OPPOSITE)

2414087 S00061534368_V2

Main Landing Gear Side Strut Installation
Figure 403/32-11-61-990-803 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

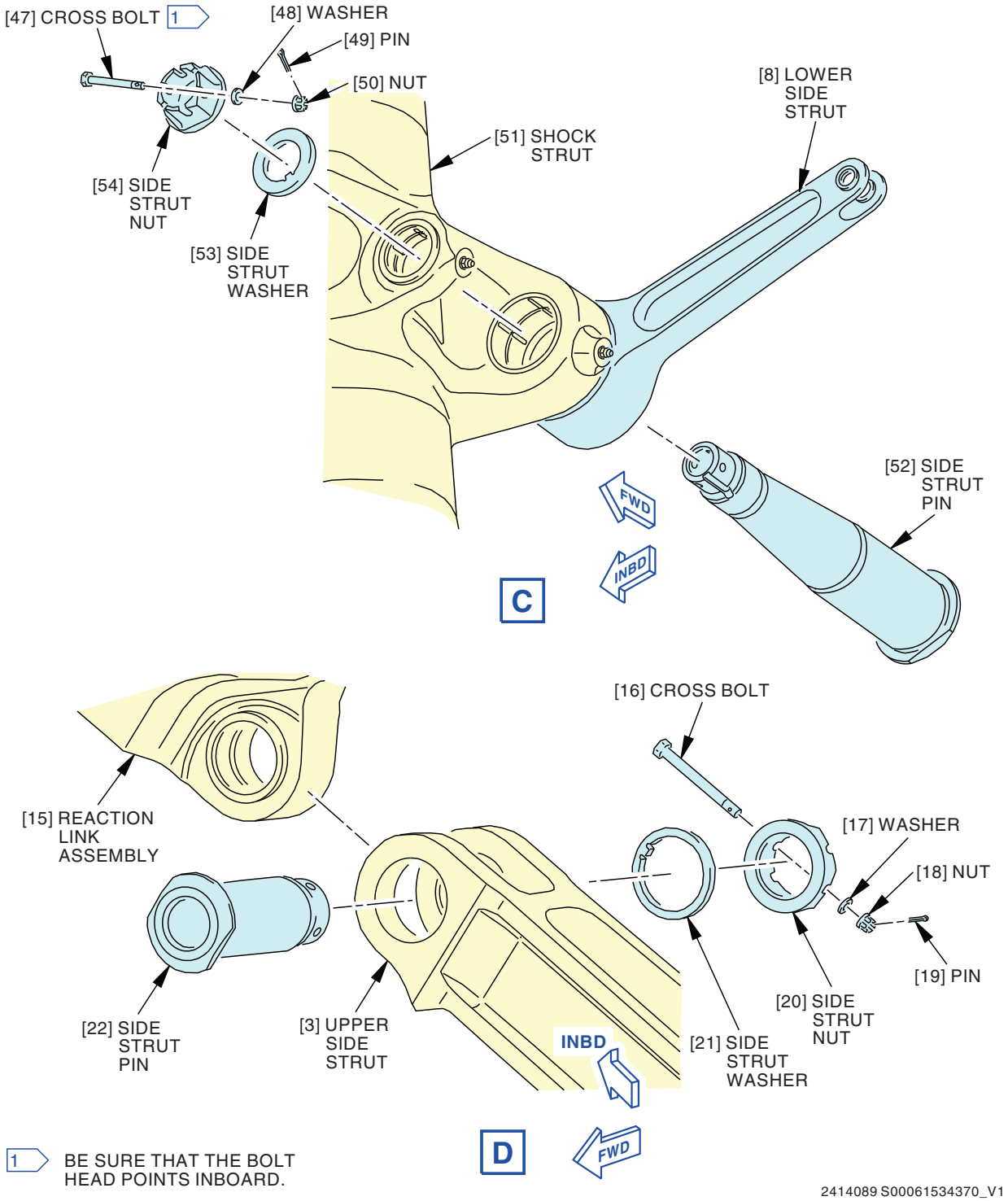


1 FOR THE RIGHT MAIN LANDING GEAR, MAKE SURE THAT THE BOLT HEAD POINTS IN THE AFT DIRECTION.

2414088 S00061534369_V3

Main Landing Gear Side Strut Installation
Figure 403/32-11-61-990-803 (Sheet 2 of 3)

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Main Landing Gear Side Strut Installation
Figure 403/32-11-61-990-803 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-11-61-400-803

7. Main Landing Gear Side Strut Installation

(Figure 403)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions to install the side strut.

B. References

Reference	Title
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
12-21-11-640-801	Main Landing Gear Upper End Components Servicing (P/B 301)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-13-11-820-801	Main Landing Gear Shock Strut Door Adjustment (P/B 501)
32-32-00-710-802	Main Landing Gear - Operational Test (P/B 501)
32-32-91-400-801	Main Gear Downlock Spring Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1359	Wrench Adapter, C32029-3 or C32029-93 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1361	Wrench Adapter, C32029-39 or C32029-95 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1368	Thread Protector, C32029-12 or C32029-101 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1369	Thread Protector , C32029-13 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1370	Thread Protector, C32029-14 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

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E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Upper side strut	32-11-61-01-310	PTF ALL
6	Cotter pin	32-11-61-01-040	PTF ALL
8	Lower side strut	32-11-61-01-350	PTF ALL
19	Cotter pin	32-11-71-01-085	PTF ALL
49	Cotter pin	32-11-61-05-040	PTF ALL
56	Cotter pin	32-13-21-01-020	PTF ALL

F. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

G. Prepare for the Installation

SUBTASK 32-11-61-480-015

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

H. Main Landing Gear Side Strut Installation

SUBTASK 32-11-61-420-009

(1) Connect the upper side strut [3] to the reaction link assembly [15]:

(a) Lubricate the chrome plated surfaces of the side strut pin [22] with grease, D00633.

(b) Remove all unwanted grease, D00633.

(c) Put the upper side strut [3] in its position for connection to the reaction link assembly [15].

(d) Put the side strut pin [22] through the upper side strut [3] and reaction link assembly [15].

1) Use the thread protector, C32029-12 or C32029-101, SPL-1368, for the side strut pin [22] installation.

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(e) Apply a thin layer of corrosion inhibiting material, G50136, to these items:

NOTE: Do not apply corrosion inhibiting materal to the chrome plated surfaces.

1) The threads and thread reliefs of the side strut pin [22] and cross bolt [16]

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) The threads of the side strut nut [20] and nut [18]

3) The faces of the side strut washer [21] and washer [17].

(f) Remove all unwanted corrosion inhibiting material, G50136.

(g) Install the side strut washer [21] and side strut nut [20] on the side strut pin [22].

(h) Tighten the side strut nut [20] to 600 in-lb (67.8 N·m) - 900 in-lb (101.7 N·m).

1) Use the wrench adapter, C32029-3 or C32029-93, SPL-1359, to hold the head of the side strut pin [22] when you tighten the side strut nut [20].

(i) If it is necessary, loosen the side strut nut [20] to the nearest castellation to align the hole for the cross bolt [16].

(j) Install the cross bolt [16] in the side strut pin [22].

(k) Install the washer [17] and nut [18] on the cross bolt [16].

(l) Tighten the nut [18] to 30 in-lb (3.39 N·m) - 50 in-lb (5.65 N·m).

(m) If it is necessary, loosen the nut [18] to the nearest castellation to align the holes for the new cotter pin [19].

(n) Install the new cotter pin [19] in the cross bolt [16].

SUBTASK 32-11-61-420-010

(2) Connect the lower side strut [8] to the shock strut [51]:

(a) Lubricate the chrome plated surfaces of the side strut pin [52] with grease, D00633.

(b) Remove all unwanted grease, D00633.

(c) Put the lower side strut [8] in its position for attachment to the shock strut [51].

(d) Put the side strut pin [52] through the lower side strut [8] and clevis on the shock strut [51].

1) Use the thread protector, C32029-14, SPL-1370, for the side strut pin [52] installation.

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(e) Apply a thin layer of corrosion inhibiting material, G50136, to these items:

NOTE: Do not apply corrosion inhibiting material to the chrome plated surfaces.

1) The threads and thread reliefs of the side strut pin [52] and cross bolt [47].

2) The threads of the side strut nut [54] and nut [50].

3) The faces of the side strut washer [53] and washer [48].

(f) Remove all unwanted corrosion inhibiting material, G50136.

(g) Install the side strut washer [53] and side strut nut [54] on the side strut pin [52].

(h) Tighten the side strut nut [54] to 95 ft-lb (128.8 N·m) - 115 ft-lb (155.9 N·m).

1) Use the wrench adapter, C32029-39 or C32029-95, SPL-1361, to hold the head of the side strut pin [52] when you tighten the side strut nut [54].

(i) Install the cross bolt [47] with the bolt head inboard in the side strut pin [52].

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1) Make sure that the head of the cross bolt [47] points inboard.

2) If it is necessary, loosen the side strut nut [54] to the nearest castellation to align the hole for the cross bolt [47].

(j) Install the washer [48] and nut [50] on the cross bolt [47].

(k) Tighten the nut [50] to 20 in-lb (2.26 N·m) - 24 in-lb (2.71 N·m).

(l) If it is necessary, loosen the nut [50] to the nearest castellation to align the hole for the new cotter pin [49].

CAUTION MAKE SURE THAT THE COTTER PIN IS INSTALLED CORRECT. A CHAFF CONDITION CAN OCCUR TO THE SHEATHING ON THE CABLE OF THE GROUND SPOILER INTERLOCK. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(m) Install the new cotter pin [49] in the cross bolt [47].

1) Make sure that the cotter pin [49] is installed in accordance with the type B preferred installation method (TASK 20-10-44-400-801).

(n) Check the interior of side strut pin [52] and reapply corrosion inhibiting material, G50136, as required.

SUBTASK 32-11-61-080-002

(3) Remove the rope that holds the pushrod [41] in its position.

SUBTASK 32-11-61-420-011

(4) Connect the lower side strut [8] to the pushrod [41]:

(a) Align the hole in the pushrod [41] with the hole in the lower side strut [8].

(b) Put the bushing [45] through the lower side strut [8] and pushrod [41].

(c) Install the washer [44] on the bolt [46].

(d) Put the bolt [46] through the lower side strut [8] and pushrod [41].

(e) Install the washer [43] and nut [42] on the bolt [46].

(f) Hand tighten the nut [42].

NOTE: Final torque to the nut will be done after final adjustments to the doors are made in the subsequent step.

SUBTASK 32-11-61-420-012

(5) Connect the upper side strut [3] to the downlock strut [1]:

(a) Lubricate the chrome plated surfaces of the downlock link pin [2] with grease, D00633.

(b) Remove all unwanted grease, D00633.

(c) Put the downlock strut [1] in its position for connection to the upper side strut [3].

(d) Put the downlock link pin [2] through the upper side strut [3] and downlock strut [1].

1) Use the thread protector, C32029-13, SPL-1369, for the downlock link pin [2] installation.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(e) Apply a thin layer of corrosion inhibiting material, G50136, to these items:

NOTE: Do not apply corrosion inhibiting material to the chrome plated surfaces.

1) The threads and thread reliefs of the downlock link pin [2]
2) The threads of the nut [5]
3) The faces of the washer [4].

(f) Remove all unwanted corrosion inhibiting material, G50136.

(g) Install the washer [4] and nut [5] on the downlock link pin [2].

(h) Tighten the nut [5] to 360 in-lb (40.67 N·m) - 600 in-lb (67.79 N·m).

(i) If it is necessary, loosen the nut [5] to the nearest castellation to align the hole for the new cotter pin [6].

(j) Install the new cotter pin [6] in the downlock link pin [2].

SUBTASK 32-11-61-820-002

(6) If the length of pushrod [41] is changed, do the step that follows:

NOTE: No adjustment to the doors is necessary if the pushrod was not changed.

(a) Do this task: Main Landing Gear Shock Strut Door Adjustment, TASK 32-13-11-820-801.

SUBTASK 32-11-61-420-037

(7) Tighten the nut [42].

(a) Install the new cotter pin [56] in the hole of the bolt [46].

(b) If it is necessary, loosen the nut [42] to the nearest castellation to align the holes for the cotter pin [56].

I. Put the Airplane Back to Its Usual Condition

SUBTASK 32-11-61-420-013

(1) Do this task: Main Gear Downlock Spring Installation, TASK 32-32-91-400-801.

SUBTASK 32-11-61-640-003

(2) Lubricate the lube fittings on the side strut, do this task: Main Landing Gear Upper End Components Servicing, TASK 12-21-11-640-801.

SUBTASK 32-11-61-780-003

(3) For hydraulic systems A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-11-61-710-003

(4) Do this task: Main Landing Gear - Operational Test, TASK 32-32-00-710-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-11-61-580-006

CAUTION

MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION

MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(5) Do this task: Lower the Airplane off Jacks, TASK 07-10-01-580-802.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN LANDING GEAR SIDE STRUT ASSEMBLY - INSPECTION/CHECK

1. General

A. This task supplies the data for a dimensional inspection of the side strut assembly for the main landing gear.

B. This task does not remove or install the side strut assembly. Refer to the MAIN LANDING GEAR SIDE STRUT ASSEMBLY - REMOVAL/INSTALLATION, PAGEBLOCK 32-11-61/401 for tasks to remove and install the side strut assembly for the main landing gear.

TASK 32-11-61-200-801

2. Main Landing Gear Side Strut Assembly Inspection

A. General

(1) This task gives the maximum design and wear clearances for the side strut assembly for the main landing gear.

NOTE: A certain quantity of free play is permitted as a result of design tolerances for all joints on the side strut assembly.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Main Landing Gear Side Strut Assembly Inspection

SUBTASK 32-11-61-480-009

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-11-61-200-001

(2) Examine the joints on the side strut assembly for the main landing gear.

(a) Make sure the all joints on the side strut assembly for the main landing gear are within the maximum design or wear clearances.

(b) If the joint exceeds the maximum wear clearance, then replace or repair the applicable part.

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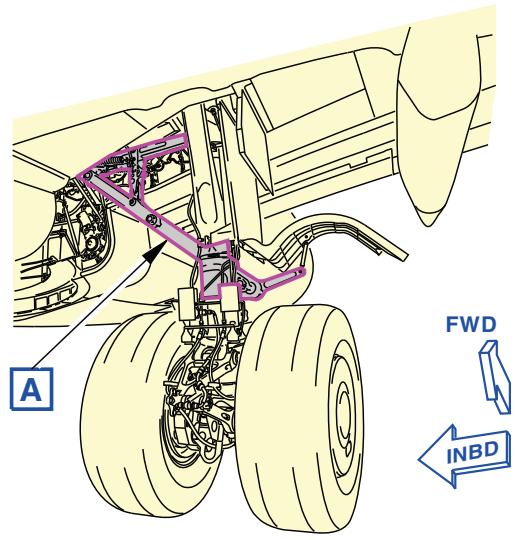
Main Landing Gear Side Strut Design and Wear Clearances

Joint (Main Landing Gear Side Strut Design Clearance/Figure 601)	Maximum Design Clearances inches (mm)	Maximum Wear Clearance inches (mm)
A *[1]	0.041 in. (1.041 mm)	0.060 in. (1.524 mm)
B1**[2]	0.105 in. (2.667 mm)	0.150 in. (3.810 mm)
B2 *[3]	0.201 in. (5.105 mm)	0.240 in. (6.096 mm)
C *[4]	0.105 in. (2.667 mm)	0.150 in. (3.810 mm)
D *[5]	0.050 in. (1.270 mm)	0.070 in. (1.778 mm)
E *[6]	0.041 in. (1.041 mm)	0.061 in. (1.549 mm)
F *[7]	0.021 in. (0.533 mm)	0.043 in. (1.092 mm)
G *[8]	0.068 in. (1.727 mm)	0.098 in. (2.489 mm)

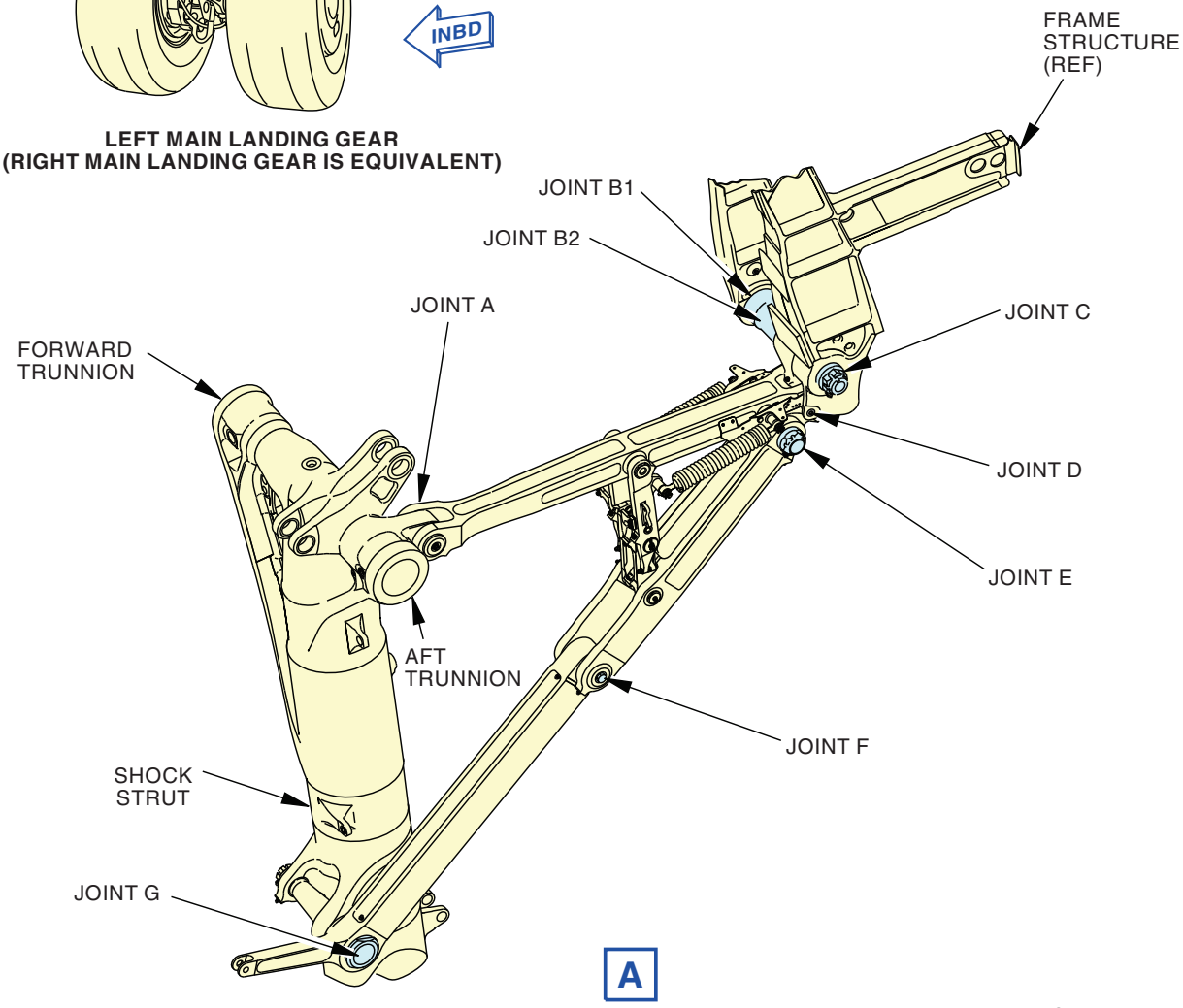
*[1] Distance between the outboard end of the reaction link and outer cylinder.
*[2] Distance between flange face of hanger link to end of spacer.
*[3] Distance between the hanger link and the fuselage attach fitting.
*[4] Maximum axial free play for the side of body joint pin.
*[5] Distance between the inboard end of the reaction link and the lower end of the hanger link.
*[6] Distance between the upper end of the upper side strut and the reaction link.
*[7] Distance between the upper and lower side strut.
*[8] Distance between the lower end of the lower side strut and the outer cylinder.

END OF TASK

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LEFT MAIN LANDING GEAR
(RIGHT MAIN LANDING GEAR IS EQUIVALENT)



Main Landing Gear Side Strut Design Clearance
Figure 601/32-11-61-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN LANDING GEAR SIDE BRACE ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Main Landing Gear Side Brace Removal

(2) Main Landing Gear Side Brace Installation.

TASK 32-11-62-000-801

2. Main Landing Gear Side Brace Assembly Removal

A. General

(1) This task gives the instructions to remove the Main Landing Gear (MLG) side strut assembly along with the MLG reaction link assembly.

(2) The MLG side strut assembly along with the MLG reaction link assembly is referred to as the MLG side brace assembly.

B. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1364	Slug Driver Assy, C32029-8 (include in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1370	Thread Protector, C32029-14 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-8264	Transportation Equipment - Tower Jack or Holding Fixture, MLG Part #: J32062-26 Supplier: 81205 Opt Part #: J32062-1 Supplier: 81205
SPL-11946	Wrench Adapter, C32029-31 or C32029-99 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-19644	Dolly - Side Strut, MLG Part #: C32078-1 Supplier: 81205
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
713	Nose Landing Gear

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(Continued)

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Prepare for the Removal

SUBTASK 32-11-62-480-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-11-62-580-001

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(2) Do this task: Lift the Airplane on Jacks, TASK 07-10-01-580-801.

SUBTASK 32-11-62-860-001

(3) Remove the power from the hydraulic systems A and B (TASK 29-11-00-860-805).

SUBTASK 32-11-62-860-002

(4) Remove the pressure from the hydraulic reservoirs (TASK 29-09-00-860-802).

F. Main Landing Gear Side Brace Assembly Removal

SUBTASK 32-11-62-020-001

(1) Do these steps to disconnect the lower side strut [2] from the pushrod [3].

(a) Remove the nut [4] and washer [5] from the bolt [8].

(b) Remove the bolt [8], washer [6], and bushing [7] to disconnect the lower side strut [2] from the pushrod [3].

(c) Isolate the pushrod [3] from the lower side strut [2].

(d) Use a rope to hold the pushrod [3] and the inner door in its position.

SUBTASK 32-11-62-020-002

(2) Do these steps to disconnect the lower side strut [2] from the shock strut [13].

(a) Remove the cotter pin [11], nut [12], and washer [10] from the cross bolt [9].

(b) Remove the cross bolt [9] from the side strut pin [14].

(c) Remove the side strut nut [16] and side strut washer [15] from the side strut pin [14].

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(d) Remove the side strut pin [14] to disconnect the lower side strut [2] from the shock strut [13].

NOTE: Use the thread protector, C32029-14, SPL-1370, for the side strut pin [14] when you do this step.

(e) Use the strap assembly to temporarily hold lower side strut [2] in its position.

SUBTASK 32-11-62-480-002

(3) Do these steps to install the strut support assembly to the side brace assembly [1].

NOTE: The strut support assembly is part of the side strut dolly, SPL-19644. For installation instructions, refer to the tool placard.

(a) Install the protective pad on the lower side strut [2].

(b) Use the transportation equipment, SPL-8264 to lift the strut support assembly into position under the side brace assembly [1].

1) Install the ball lock pins to lock the strut support assembly.

NOTE: The two side ball lock pins lock the strut support assembly to the transportation equipment, SPL-8264. The top ball lock pin locks the rotation of the strut support assembly. You can remove the top ball lock pin to rotate the strut support assembly.

2) If the strut support assembly is installed on the cart assembly, remove the thumb screws before you lift the strut support assembly away from the cart assembly.

NOTE: Do not lift the cart assembly with the strut support assembly.

(c) Secure the side brace assembly [1] to the strut support assembly.

1) Install the strap assembly that holds the side brace assembly [1] to the strut support assembly.

2) Tighten the swivel screws that secure the side brace assembly [1] to the strut support assembly.

SUBTASK 32-11-62-020-003

(4) Do these steps to disconnect the reaction link assembly [17] from the trunnion [18]:

(a) Remove the cotter pin [23], nut [22], and washer [21] from the cross bolt [19].

(b) Remove the cross bolt [19] from the trunnion pin [20].

(c) Remove the end cap [24] from the trunnion pin [20].

(d) Use the slug driver assy, C32029-8, SPL-1364, to remove the trunnion pin [20] that attaches the reaction link assembly [17] to the trunnion [18].

SUBTASK 32-11-62-020-020

(5) Remove and discard the lockwire from the electrical harnesses [26].

SUBTASK 32-11-62-020-004

(6) Disconnect the electrical harnesses [26].

SUBTASK 32-11-62-020-005

(7) Do these steps to disconnect the hydraulic line [28] and hydraulic line [28A] from the hydraulic tubes [27]:

(a) Disconnect the hydraulic lines [28] and hydraulic line [28A] from the hydraulic tubes [27].

(b) Install caps/plugs, STD-14686 on the hydraulic line [28] and hydraulic line [28A].

(c) Put tags on the hydraulic line [28] and hydraulic line [28A] to prevent the crossing of the hydraulic lines during the installation.

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(d) Install caps/plugs, STD-14686 on the hydraulic tubes [27].

SUBTASK 32-11-62-020-006

(8) Do these steps to remove the hanger link assembly [29] from the supporting structure:

(a) Remove the cotter pin [30], nut [31], and washer [32] from the cross bolt [33].

(b) Remove the cross bolt [33] from the hanger link assembly pin [36].

(c) Remove the nut [34] and washer [35] from the hanger link assembly pin [36] with wrench adapter, C32029-31 or C32029-99, SPL-11946.

(d) Remove the hanger link assembly pin [36] and bushing [37] that attaches the hanger link assembly [29] to the supporting structure.

(e) Use the strap assembly to hold lower side strut [2] to the hanger link assembly [29].

SUBTASK 32-11-62-020-007

(9) Use the transportation equipment, SPL-8264 to lower the strut support assembly along with the side brace assembly [1] into position on the cart assembly.

(a) If it is necessary, disconnect the lower end of the support rod [50] and support rod [51] from the wheel well structure.

1) Remove the bolt [52], washer [53], bushing [54], washer [55], and nut [56].

(b) Install the top ball lock pin to lock the rotation of the strut support assembly.

(c) Install the thumb screws to secure the strut support assembly to the cart assembly.

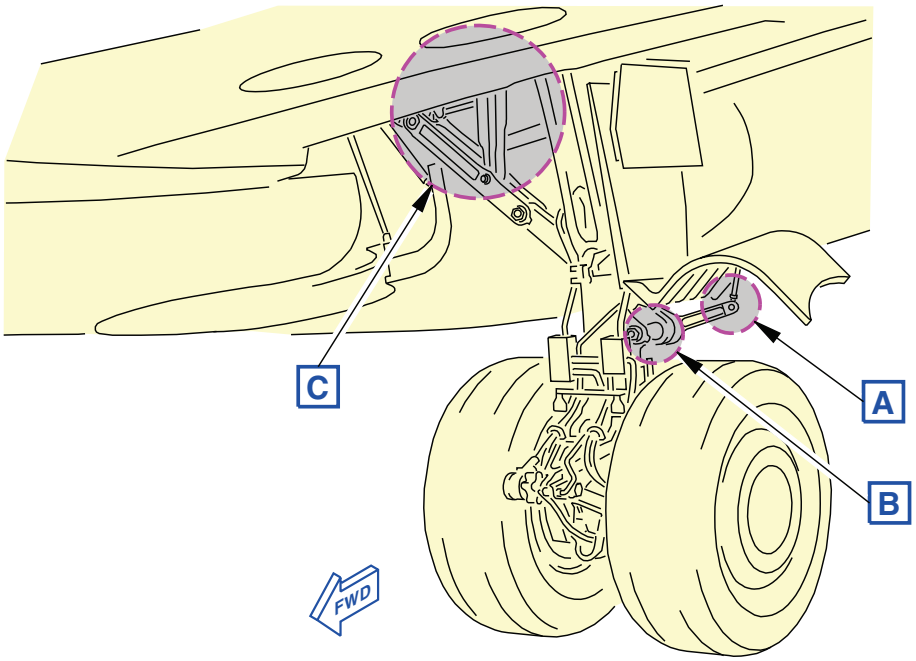
(d) Remove the two side ball lock pins.

NOTE: The two side ball lock pins lock the strut support assembly to the transportation equipment, SPL-8264.

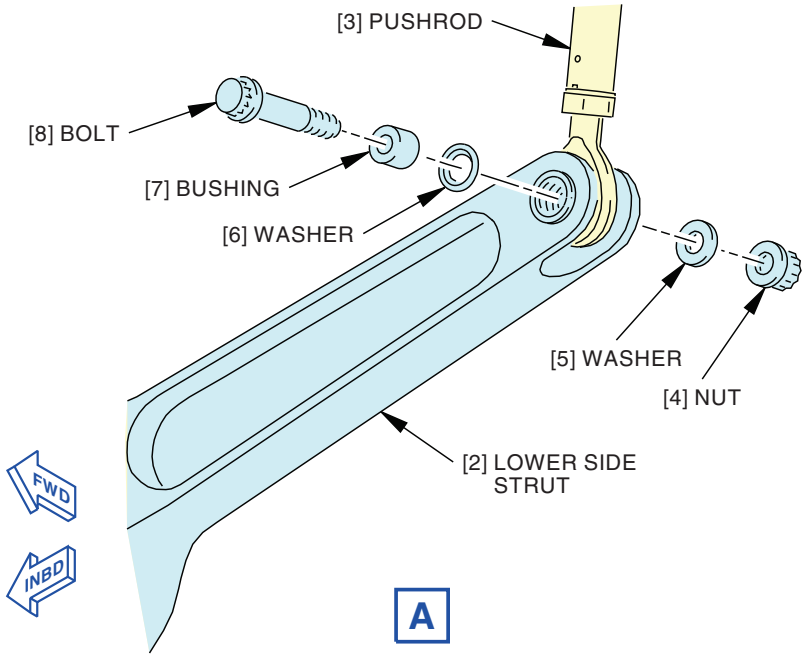
(e) Remove the transportation equipment, SPL-8264.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



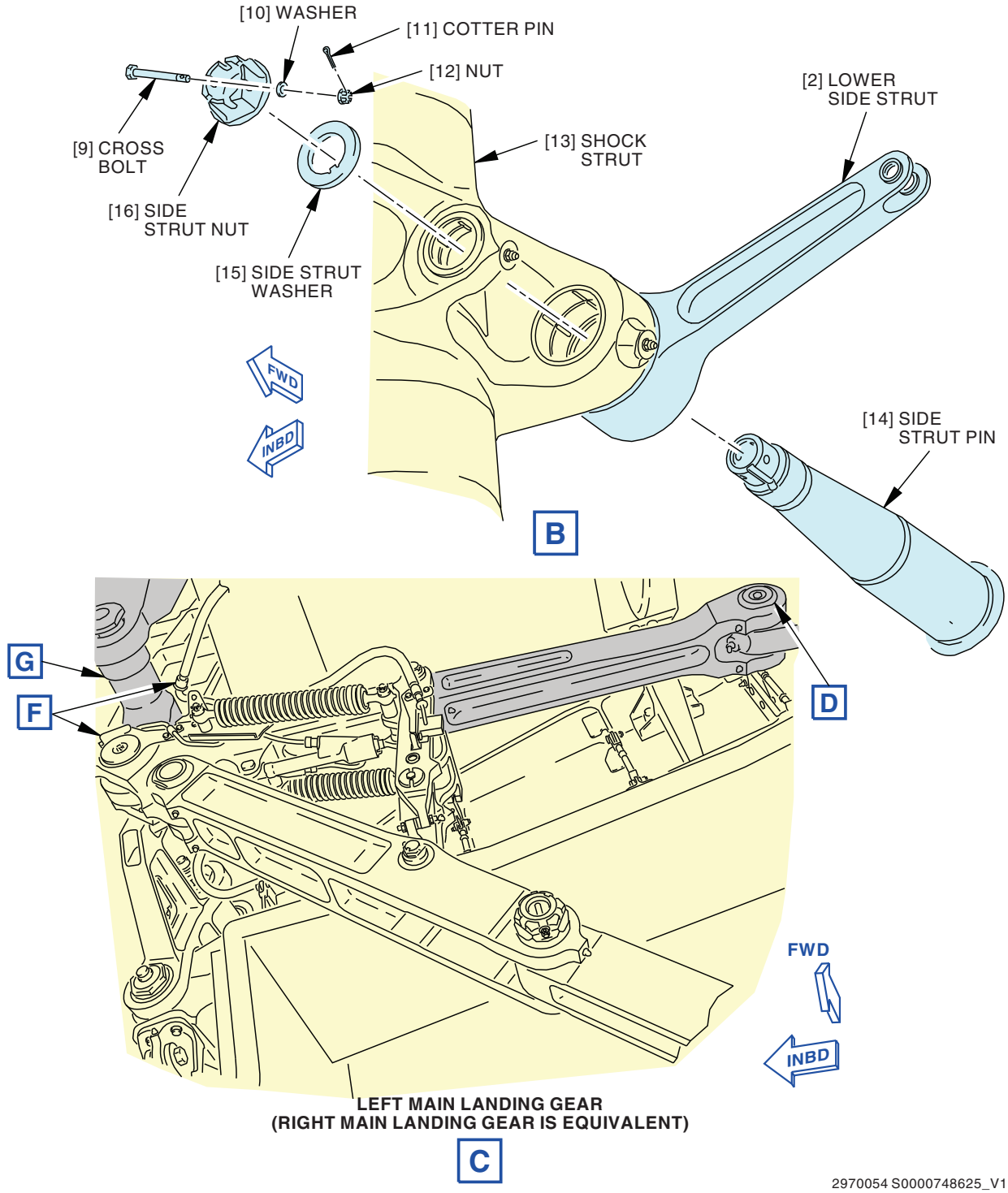
LEFT MAIN LANDING GEAR
(RIGHT MAIN LANDING GEAR IS EQUIVALENT)



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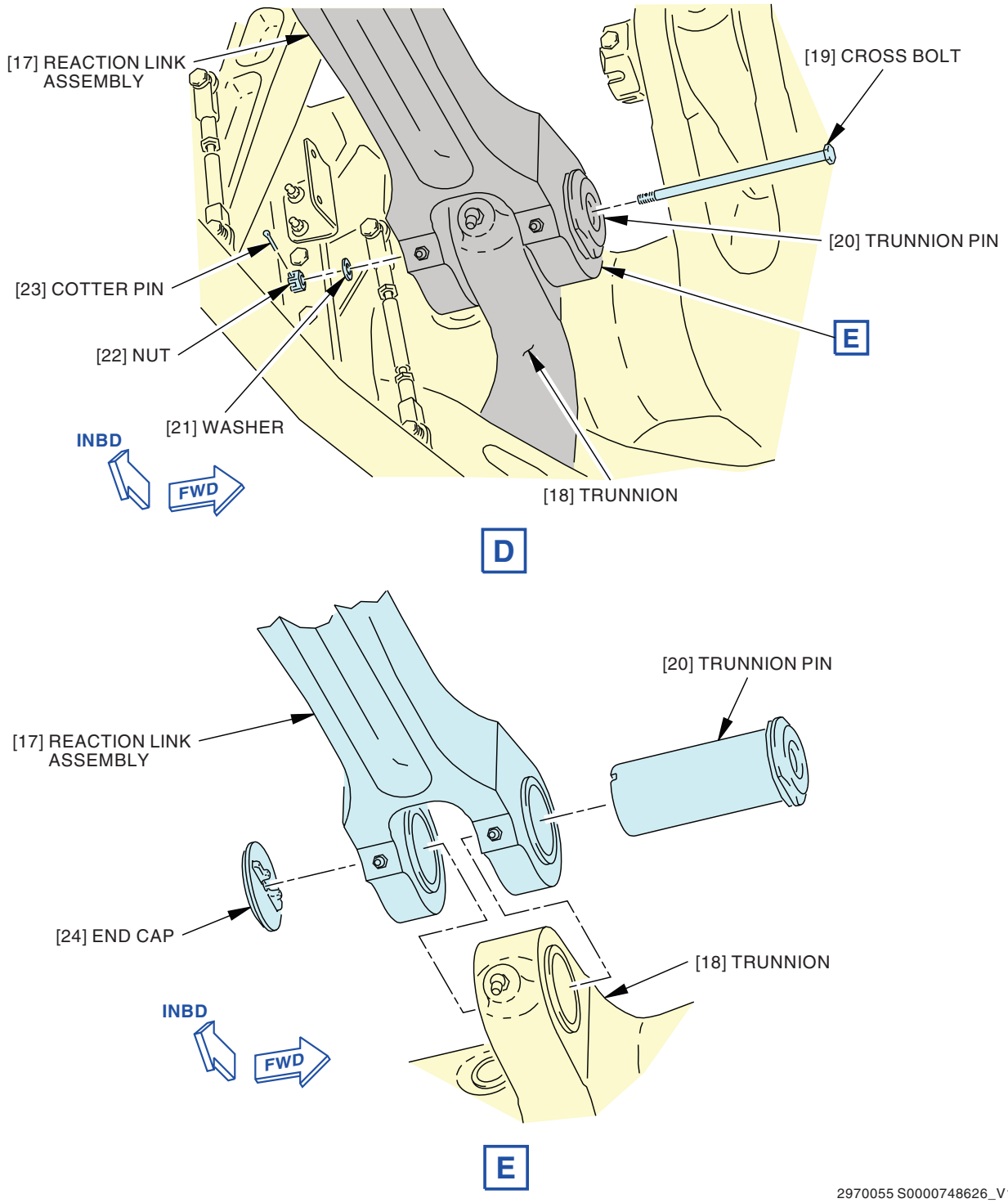
Main Landing Gear Side Brace Assembly Installation
Figure 401/32-11-62-990-801 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



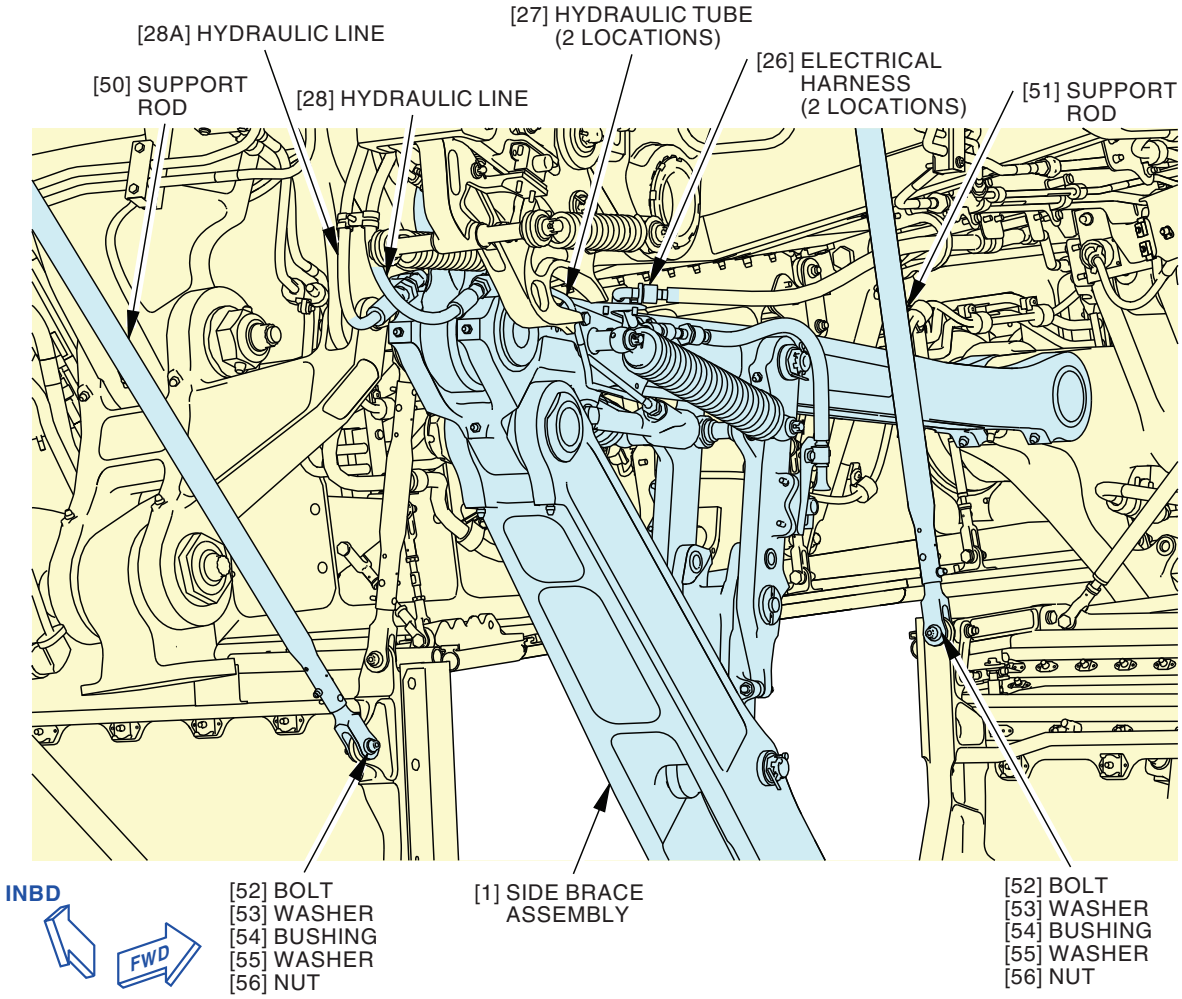
Main Landing Gear Side Brace Assembly Installation
Figure 401/32-11-62-990-801 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Side Brace Assembly Installation
Figure 401/32-11-62-990-801 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

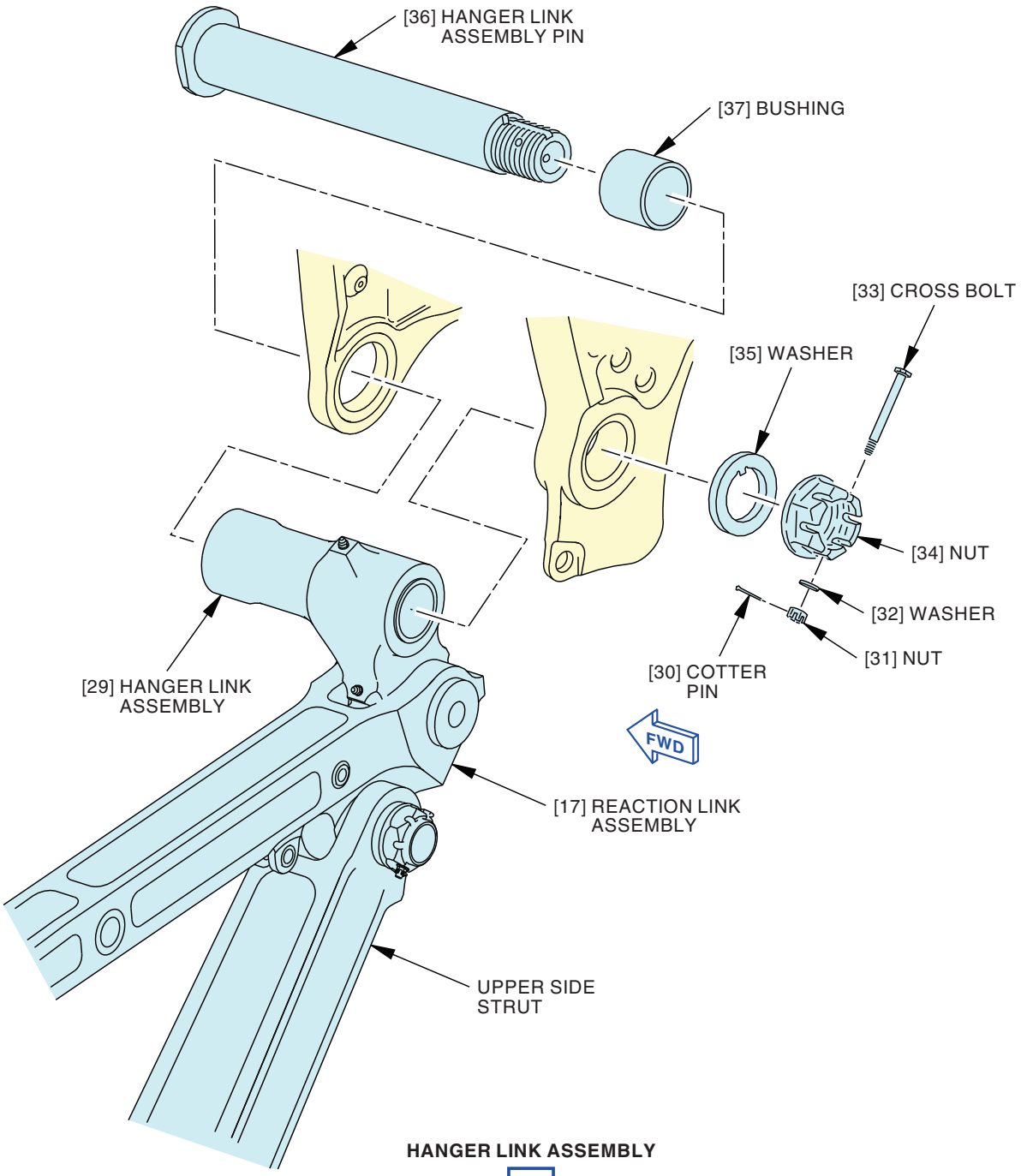


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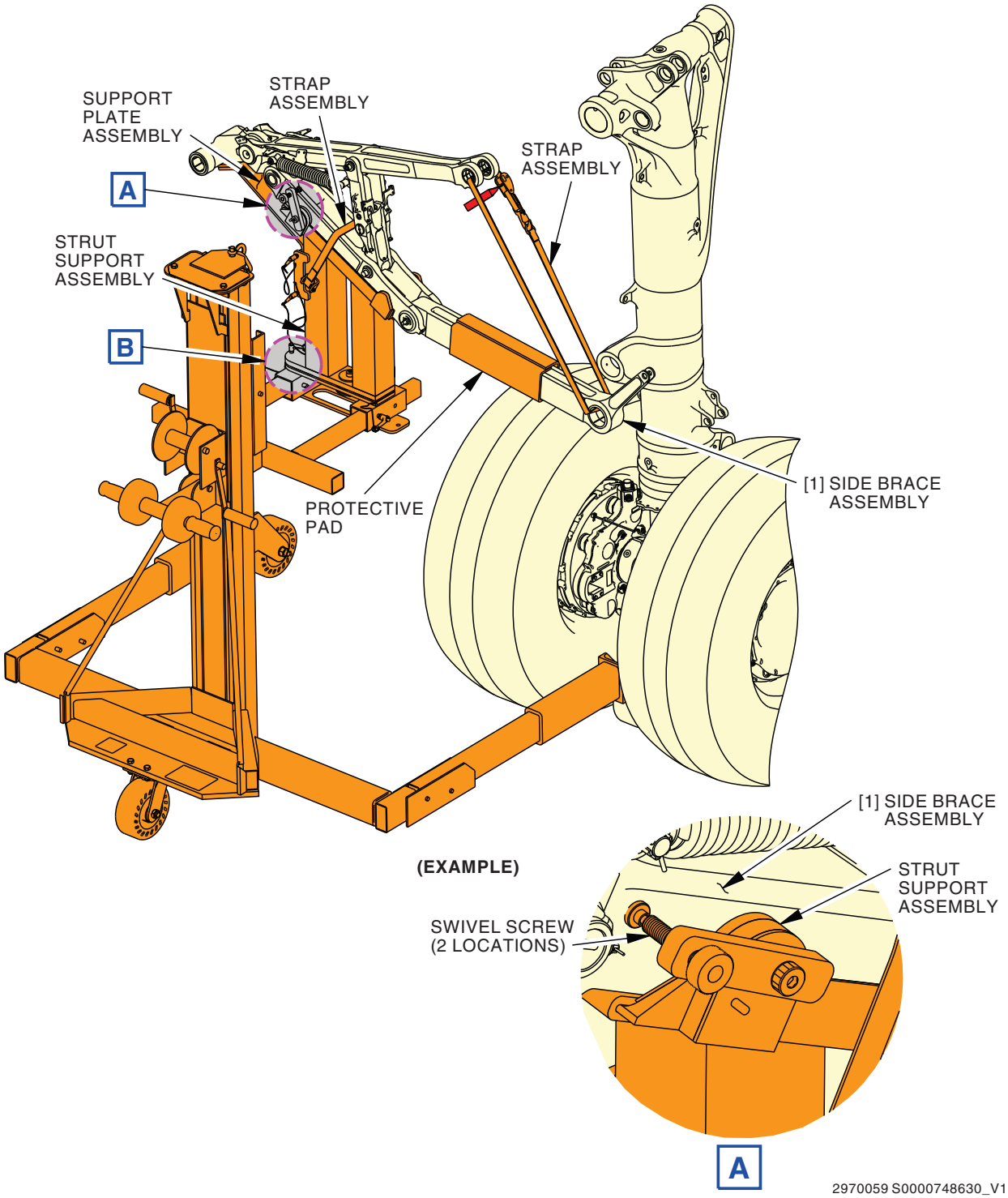
Main Landing Gear Side Brace Assembly Installation
Figure 401/32-11-62-990-801 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



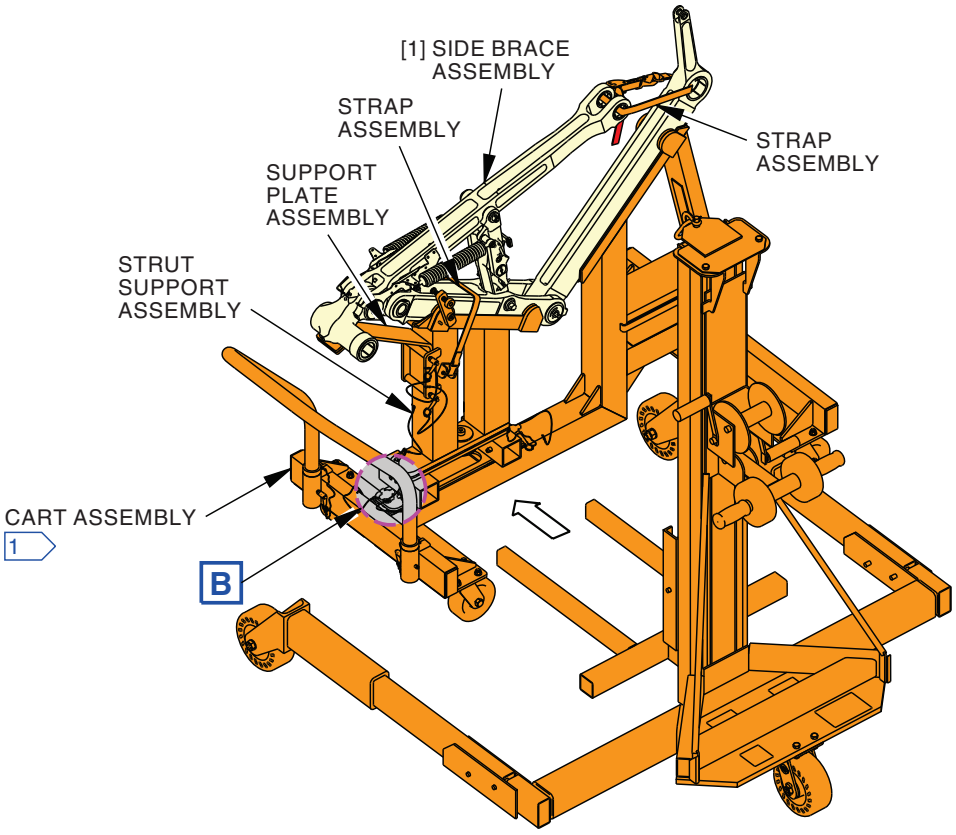
Main Landing Gear Side Brace Assembly Installation
Figure 401/32-11-62-990-801 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

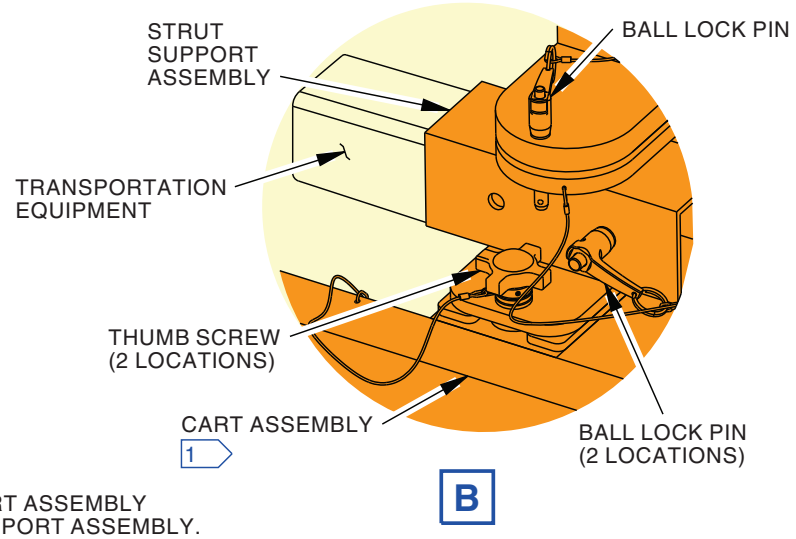


Main Landing Gear Side Brace Assembly Rigging Equipment
Figure 402/32-11-62-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)



1 DO NOT LIFT THE CART ASSEMBLY WITH THE STRUT SUPPORT ASSEMBLY.

2970060 S0000748631_V1

Main Landing Gear Side Brace Assembly Rigging Equipment
Figure 402/32-11-62-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-11-62-400-801

3. Main Landing Gear Side Brace Assembly Installation

A. General

(1) This task gives the instructions to install the Main Landing Gear (MLG) side strut assembly along with the MLG reaction link assembly.

(2) The MLG side strut assembly along with the MLG reaction link assembly is referred to as the MLG side brace assembly.

B. References

Reference	Title
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
12-21-11-640-801	Main Landing Gear Upper End Components Servicing (P/B 301)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-13-11-820-801	Main Landing Gear Shock Strut Door Adjustment (P/B 501)
32-32-00-710-802	Main Landing Gear - Operational Test (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1361	Wrench Adapter, C32029-39 or C32029-95 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1370	Thread Protector, C32029-14 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-8264	Transportation Equipment - Tower Jack or Holding Fixture, MLG Part #: J32062-26 Supplier: 81205 Opt Part #: J32062-1 Supplier: 81205
SPL-11946	Wrench Adapter, C32029-31 or C32029-99 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-19644	Dolly - Side Strut, MLG Part #: C32078-1 Supplier: 81205
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

D. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G02166	Lockwire - MS20995NC20, Monel - 0.020 Inch (0.508 mm) Diameter	NASM20995
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

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E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

F. Prepare for the Installation

SUBTASK 32-11-62-480-003

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

G. Main Landing Gear Side Brace Assembly Installation

SUBTASK 32-11-62-480-004

CAUTION BE CAREFUL WHEN YOU MOVE THE SIDE BRACE ASSEMBLY INTO ITS POSITION. IF YOU DO NOT OBEY, DAMAGE TO THE COMPONENTS CAN OCCUR.

(1) Use the transportation equipment, SPL-8264, to lift the strut support assembly along with the side brace assembly [1] into position on the MLG.

NOTE: The strut support assembly is part of the side strut dolly, SPL-19644. For installation instructions, refer to the tool placard.

(a) If not already done, install the protective pad on the lower side strut [2].

(b) If not already done, secure the side brace assembly [1] to the strut support assembly.

1) Install the strap assembly to hold the side brace assembly [1] to the strut support assembly.

2) Tighten the swivel screws to secure the side brace assembly [1] to the strut support assembly.

(c) Install the ball lock pins to lock the strut support assembly.

NOTE: The two side ball lock pins lock the strut support assembly to the transportation equipment, SPL-8264. The top ball lock pin locks the rotation of the strut support assembly. You can remove the top ball lock pin to rotate the strut support assembly.

(d) If the strut support assembly is installed on the cart assembly, remove the thumb screws before you lift the strut support assembly away from the cart assembly.

NOTE: Do not lift the cart assembly with the strut support assembly.

SUBTASK 32-11-62-420-001

(2) Do these steps to connect the reaction link assembly [17] to the trunnion [18]:

(a) Lubricate the chrome plated surfaces of the trunnion pin [20] and end cap [24] with grease, D00633.

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(b) Position the outboard end of the reaction link assembly [17] for attachment to the trunnion [18].

(c) Put the trunnion pin [20] through the reaction link assembly [17] and trunnion [18].

(d) Install the end cap [24] on the trunnion pin [20].

(e) Put the cross bolt [19] through the trunnion pin [20] and end cap [24].

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(f) Apply a thin layer of corrosion inhibiting material, G50136, to these items:

1) Do not apply corrosion inhibiting material to the chrome plated surfaces.
2) The threads and thread reliefs of the cross bolt [19].
3) The threads of the nut [22].
4) The faces of the washer [21].

(g) Remove all unwanted corrosion inhibiting material, G50136.

(h) Install the washer [21] and nut [22] on the cross bolt [19].

(i) If it is necessary, loosen the nut [22] to the nearest castellation that aligns with the hole for the cotter pin [23].

(j) Install the cotter pin [23] in the cross bolt [19].

SUBTASK 32-11-62-420-002

(3) Do these steps to install the hanger link assembly [29] into the supporting structure:

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin layer of corrosion inhibiting material, G50136, to these items:

1) Do not apply corrosion inhibiting material to the chrome plated surfaces.
2) The threads and thread reliefs of the hanger link assembly pin [36] and cross bolt [33].
3) The threads of the nut [34] and nut [31].
4) The faces of the washer [35] and washer [32].
5) Remove all unwanted corrosion inhibiting material, G50136.

(b) Prior to installation of the hanger link assembly pin [36], lubricate chrome plated surfaces with grease, D00633.

(c) Install the hanger link assembly pin [36] and bushing [37] that supports the hanger link assembly [29] to the supporting structure, through the hanger link assembly [29].

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(d) Install the washer [35] and nut [34] on the hanger link assembly pin [36].

NOTE: Do not tighten the nut at this time.

SUBTASK 32-11-62-420-003

(4) Do these steps to connect the hydraulic line [28] and hydraulic line [28A] to the hydraulic tubes [27]:

(a) Remove the caps/plugs, STD-14686, from the hydraulic line [28], hydraulic line [28A], and hydraulic tubes [27].

(b) Connect the hydraulic line [28] and hydraulic line [28A] to the hydraulic tubes [27].

(c) Tighten the hydraulic line [28] to 140 ±7 in-lb (15.8 ±0.8 N·m).

(d) Tighten the hydraulic line [28A] to 270 ±14 in-lb (30.5 ±1.6 N·m).

(e) Remove the tags from the hydraulic line [28] and hydraulic line [28A].

SUBTASK 32-11-62-420-020

(5) Connect the electrical harnesses [26].

SUBTASK 32-11-62-420-022

(6) Install MS20995NC20 lockwire, G02166, or MS20995NC32 lockwire, G01912, on each electrical harness [26] (TASK 20-10-44-400-801).

SUBTASK 32-11-62-420-004

(7) Do these steps to connect the lower side strut [2] to the shock strut [13]:

(a) Lubricate the chrome plated surfaces of the side strut pin [14] with grease, D00633.

(b) Remove all unwanted grease, D00633.

(c) Put the lower side strut [2] in its position for attachment to the shock strut [13].

(d) Put the side strut pin [14] through the lower side strut [2] and clevis on the shock strut [13].

NOTE: Use the thread protector, C32029-14, SPL-1370, for the side strut pin [14] when you do this task.

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(e) Apply a thin layer of the corrosion inhibiting material, G50136, to these items:

1) Do not apply corrosion inhibiting material to the chrome plated surfaces.

2) The threads and thread reliefs of the side strut pin [14] and cross bolt [9].

3) The threads of the side strut nut [16] and nut [12].

4) The faces of the side strut washer [15] and washer [10].

(f) Remove all unwanted corrosion inhibiting material, G50136.

(g) Install the side strut washer [15] and side strut nut [16] on the side strut pin [14].

NOTE: Do not tighten the side strut nut at this time.

(h) Check the interior of the side strut pin [14] and repair the application of corrosion inhibiting material, G50136, as required.

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SUBTASK 32-11-62-420-005

(8) Do these steps to tighten the fasteners at the hanger link assembly pin [36]:

(a) Use the wrench adapter, C32029-31 or C32029-99, SPL-11946, to hold the hanger link assembly pin [36] to tighten nut [34].

(b) Tighten the nut [34] to 95 ft-lb (128.8 N·m) - 115 ft-lb (155.9 N·m).

(c) If it is necessary, loosen the nut [34] to the nearest castellation to align the hole for the cross bolt [33].

(d) Install the cross bolt [33] in the hanger link assembly pin [36].

(e) Install the nut [31] and washer [32] to the cross bolt [33].

(f) Tighten the nut [31] to 20 in-lb (2.3 N·m) - 24 in-lb (2.7 N·m) above run-on torque.

(g) If it is necessary, loosen the nut [31] to the nearest castellation that aligns with the hole for the cotter pin [30].

(h) Install the cotter pin [30] in the cross bolt [33].

SUBTASK 32-11-62-420-006

(9) Do these steps to tighten the fasteners at the side strut pin [14]:

(a) Tighten the side strut nut [16] to 95 ft-lb (128.8 N·m) - 115 ft-lb (155.9 N·m).

NOTE: Use the wrench adapter, C32029-39 or C32029-95, SPL-1361, to hold the head of the side strut pin when you tighten the side strut nut.

(b) If it is necessary, loosen the side strut nut [16] to the nearest castellation to align the hole for the cross bolt [9].

(c) Install the cross bolt [9] with the bolt head inboard in the side strut pin [14].

1) Make sure that the bolt head points inboard.

(d) Install the washer [10] and nut [12] on the cross bolt [9].

(e) Tighten the nut [12] to 20 in-lb (2.3 N·m) - 24 in-lb (2.7 N·m).

(f) If it is necessary, loosen the nut [12] to the nearest castellation to align the hole for the cotter pin [11].

(g) Install the cotter pin [11] in the cross bolt [9].

1) Make sure that the cotter pin [11] is installed in accordance with TASK 20-10-44-400-801.

(h) Check the interior of the side strut pin [14] and repair the application of corrosion inhibiting material, G50136, as required.

SUBTASK 32-11-62-080-001

(10) Do these steps to remove the strut support assembly from the side brace assembly [1]:

(a) Remove the strap assembly that holds the side brace assembly [1] to the strut support assembly.

(b) Loosen the swivel screws that secure the side brace assembly [1] to the strut support assembly.

SUBTASK 32-11-62-080-002

(11) Use the transportation equipment, SPL-8264, to lower the strut support assembly into position on the cart assembly.

(a) Install the top ball lock pin to lock the rotation of the strut support assembly.

(b) Install the thumb screws to secure the strut support assembly to the cart assembly.

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(c) Remove the two side ball lock pins.

NOTE: The two side ball lock pins lock the strut support assembly to the transportation equipment, SPL-8264.

(d) Remove the transportation equipment, SPL-8264.

SUBTASK 32-11-62-420-007

(12) If removed, connect the lower end of the support rod [50] and support rod [51] from the wheel well structure.

(a) Install the bolt [52], washer [53], bushing [54], washer [55], and nut [56].

(b) Make sure that you install the countersunk washer under the bolt head and the plain washer under the nut.

SUBTASK 32-11-62-420-008

(13) Do these steps to connect the lower side strut [2] to the pushrod [3]:

(a) Remove the rope that holds the pushrod [3] in its position.

(b) Lubricate the nut [4], washer [5], and bolt [8] with grease, D00633.

(c) Remove all unwanted grease, D00633.

(d) Align the hole in the pushrod [3] with the hole in the lower side strut [2].

(e) Put the bushing [7] through the lower side strut [2] and pushrod [3].

(f) Install the washer [6] on the bolt [8].

(g) Put the bolt [8] through the lower side strut [2] and pushrod [3].

(h) Install the washer [5] and nut [4] on the bolt [8].

(i) Hand tighten the nut [4] until after final adjustments to the doors are made in the subsequent steps.

SUBTASK 32-11-62-820-001

(14) If you changed the length of pushrod [3], do the step that follows:

NOTE: No adjustment to the doors is necessary if the pushrod was not changed.

(a) Do this task: Main Landing Gear Shock Strut Door Adjustment, TASK 32-13-11-820-801.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-11-62-640-001

(1) Lubricate the lube fittings on the MLG side brace assembly, do this task: Main Landing Gear Upper End Components Servicing, TASK 12-21-11-640-801.

SUBTASK 32-11-62-860-004

(2) For the hydraulic systems A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-11-62-710-001

(3) Do this task: Main Landing Gear - Operational Test, TASK 32-32-00-710-802.

SUBTASK 32-11-62-790-002

(4) Do a check of the hydraulic connections for leakage.

SUBTASK 32-11-62-580-002



CAUTION

MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

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(CAUTION PRECEDES)

CAUTION	MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(5) Do this task: Lower the Airplane off Jacks, TASK 07-10-01-580-802.

——— END OF TASK ———

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MAIN LANDING GEAR REACTION LINK ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the main landing gear reaction link assembly

(2) An installation of the main landing gear reaction link assembly.

TASK 32-11-71-000-801

2. Main Landing Gear Reaction Link Assembly Removal

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-32-91-000-801	Main Gear Downlock Springs Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1359	Wrench Adapter, C32029-3 or C32029-93 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1360	Wrench Adapter, C32029-4 or C32029-94 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1364	Slug Driver Assy, C32029-8 (include in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1368	Thread Protector, C32029-12 or C32029-101 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1369	Thread Protector , C32029-13 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
713	Nose Landing Gear

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(Continued)

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Prepare for the Removal

SUBTASK 32-11-71-480-003

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-11-71-580-004

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(2) Do this task: Lift the Airplane on Jacks, TASK 07-10-01-580-801.

SUBTASK 32-11-71-860-001

(3) Remove the power from the hydraulic systems A and B (TASK 29-11-00-860-805).

SUBTASK 32-11-71-860-008

(4) Remove the pressure from the hydraulic reservoirs (TASK 29-09-00-860-802).

E. Main Landing Gear Reaction Link Assembly Removal

SUBTASK 32-11-71-020-001

(1) Do this task: Main Gear Downlock Springs Removal, TASK 32-32-91-000-801.

SUBTASK 32-11-71-020-002

(2) Do these steps to disconnect the hydraulic line [37] and hydraulic line [37A] from the hydraulic tubes [2]:

(a) Disconnect the hydraulic line [37] and hydraulic line [37A] from the hydraulic tubes [2] attached to the bracket assembly [38].

(b) Install caps/plugs, STD-14686, on the hydraulic line [37] and hydraulic line [37A].

(c) Put tags on the hydraulic line [37] and hydraulic line [37A] to prevent the crossing of the hydraulic line [37] and hydraulic line [37A] during the installation.

(d) Put tags on the hydraulic tubes [2] to prevent the crossing of the hydraulic tubes [2] during the installation.

(e) Disconnect the hydraulic tubes [2] from the downlock actuator [4].

(f) Install caps/plugs, STD-14686, on the hydraulic tubes [2].

(g) Remove the unions [68] from the downlock actuator [4].

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(h) Remove the packings [3] from the ports of the downlock actuator [4] and discard them.

(i) Install caps/plugs, STD-14686, in the ports of the downlock actuator [4].

SUBTASK 32-11-71-020-003

(3) Do these steps to disconnect the reaction link assembly [1] from the downlock actuator [4]:

(a) Remove the pin [11], nut [10], and washer [9] from the rod end pin [6].

(b) Remove the rod end pin [6] that attaches the head end of the downlock actuator [4] to the reaction link assembly [1].

(c) Remove the washers [7] and washer [8].

(d) Slowly lower the head end of the downlock actuator [4] to the upper end of the upper side strut [21].

(e) Use a tape to attach the head end of the downlock actuator [4] to the upper end of the upper side strut [21].

NOTE: This task will prevent the downlock actuator from moving when you do the subsequent steps.

SUBTASK 32-11-71-020-004

(4) Do these steps to disconnect the reaction link assembly [1] from the upper lock link [5]:

(a) Remove the pin [16] from the downlock link pin [13].

(b) Remove the nut [15] and washer [14] from the downlock link pin [13].

1) Use the wrench adapter, C32029-4 or C32029-94, SPL-1360, to hold the head of the downlock link pin [13] when you loosen the nut [15].

(c) Remove the downlock link pin [13] that attaches the reaction link assembly [1] to the upper lock link [5].

(d) Use the thread protector, C32029-13, SPL-1369, when you do this step.

SUBTASK 32-11-71-020-005

(5) Do these steps to disconnect the reaction link assembly [1] from the upper side strut [21]:

(a) Remove the pin [23], nut [22], and washer [24] from crossbolt [17].

(b) Remove the crossbolt [17] from the side strut pin [20].

(c) Remove the side strut nut [18] and the side strut washer [19] from the side strut pin [20].

1) Use the wrench adapter, C32029-3 or C32029-93, SPL-1359, to hold the head of the side strut pin [20] when you loosen the side strut nut [18].

(d) Attach a strap to the upper end of the upper side strut [21].

NOTE: The assembly of the upper side strut, lower side strut, and the lower lock link weigh approximately 130 lb (59 kg).

1) Lower the upper side strut [21] sufficiently for the removal of the reaction link assembly [1] in the subsequent steps.

2) Attach the upper end of the upper side strut [21] to the hanger link assembly [35].

(e) Use the thread protector, C32029-12 or C32029-101, SPL-1368, to remove the side strut pin [20] that attaches the reaction link assembly [1] to the upper side strut [21].

SUBTASK 32-11-71-020-006

(6) Do these steps to disconnect the reaction link assembly [1] from the hanger link assembly [35]:

(a) Remove the pin assembly [31] from the reaction link assembly [1].

(b) Remove the pin [28], nut [29], and washer [30] from the crossbolt [25].

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(c) Remove the crossbolt [25] from the end cap [27].

(d) Remove the washer [26] from the crossbolt [25].

(e) Isolate the end cap [27], bracket assembly [38], hydraulic tube [2], electrical harness [12], spacers [32], and spacer [33] as an assembly from the reaction link assembly [1].

(f) If it is necessary to disconnect the electrical harness [12], do this step:

1) Remove the bolts for the bracket which holds the sensor to the upper lock link [5] to disconnect the electrical harness [12].

(g) Use the slug driver assy, C32029-8, SPL-1364, to remove the hanger link pin [34] that attaches the reaction link assembly [1] to the hanger link assembly [35].

SUBTASK 32-11-71-020-007

(7) If it is necessary, do these steps to remove the hanger link assembly [35] from the supporting structure:

(a) Disconnect the lower end of the support rod [69] and support rod [70] from the wheel well structure (View J, Figure 401).

1) Remove the bolt [71], washer [72], bushing [73], washer [74], and nut [75].

(b) Remove the pin [53], nut [52], and washer [51] from the bolt [50].

(c) Remove the bolt [50] from the hanger link assembly pin [46].

(d) Remove the nut [49] and washer [48] from the hanger link assembly pin [46].

NOTE: It is not necessary to remove the nut [76], washer [77], and washer [78] for the hanger link assembly pin [46] removal.

(e) Remove the hanger link assembly pin [46] and spacer [47] that attach the hanger link assembly [35] to the supporting structure.

(f) Remove the hanger link assembly [35].

SUBTASK 32-11-71-020-008

(8) Do these steps to disconnect the reaction link assembly [1] from the trunnion [41]:

(a) Remove the pin [44], nut [43], and washer [42] from the crossbolt [39].

(b) Remove the crossbolt [39] from the trunnion pin [40].

(c) Remove the end cap [45] from the trunnion pin [40].

(d) Use the slug driver assy, C32029-8, SPL-1364, to remove the trunnion pin [40] that attaches the reaction link assembly [1] to the trunnion [41].

SUBTASK 32-11-71-020-009

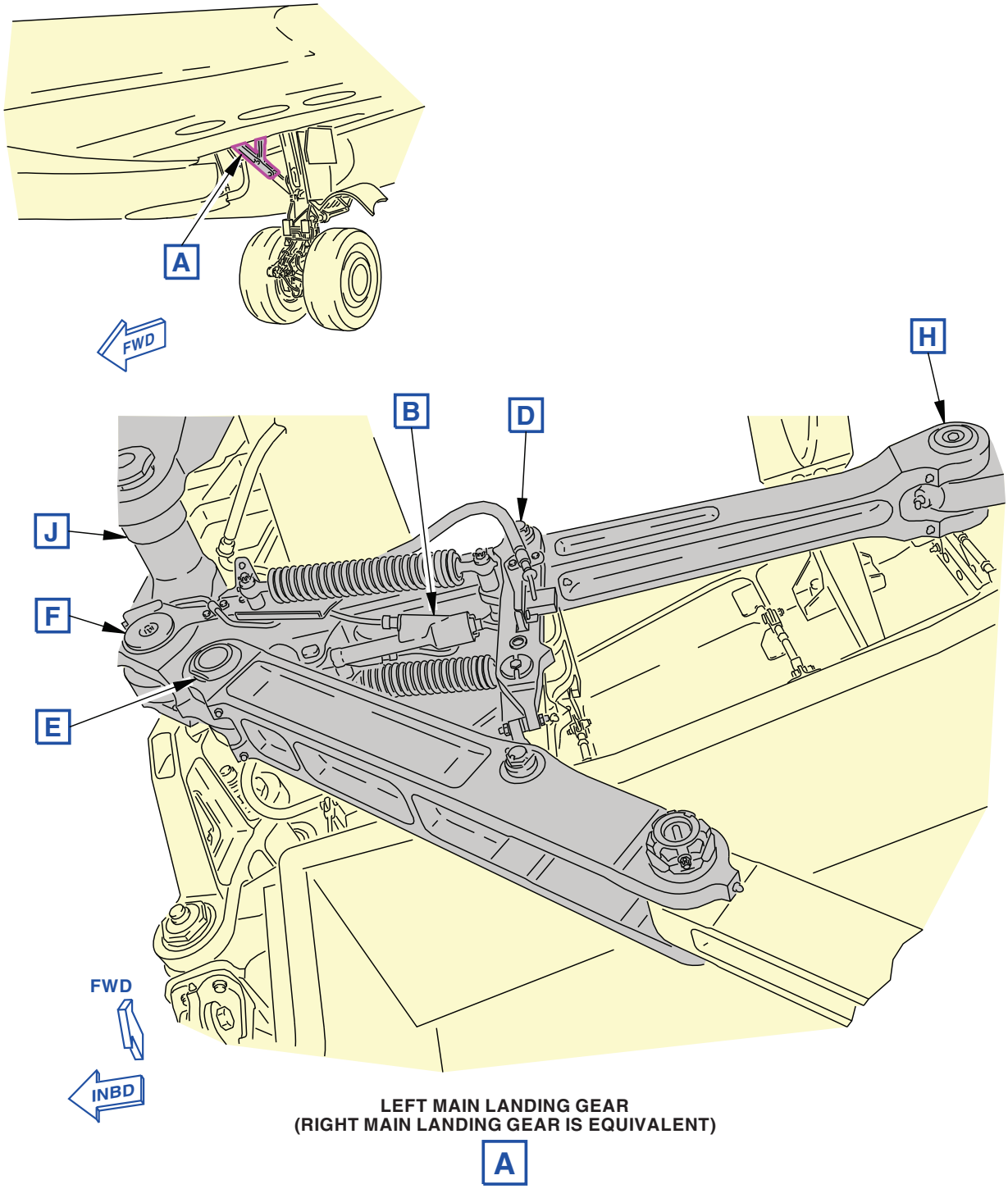
(9) Remove the reaction link assembly [1] from the airplane.

(a) Make sure that a minimum of two persons will hold the reaction link assembly [1] when you remove it from the airplane.

NOTE: The reaction link assembly weighs approximately 63.2 lb (28.7 kg).

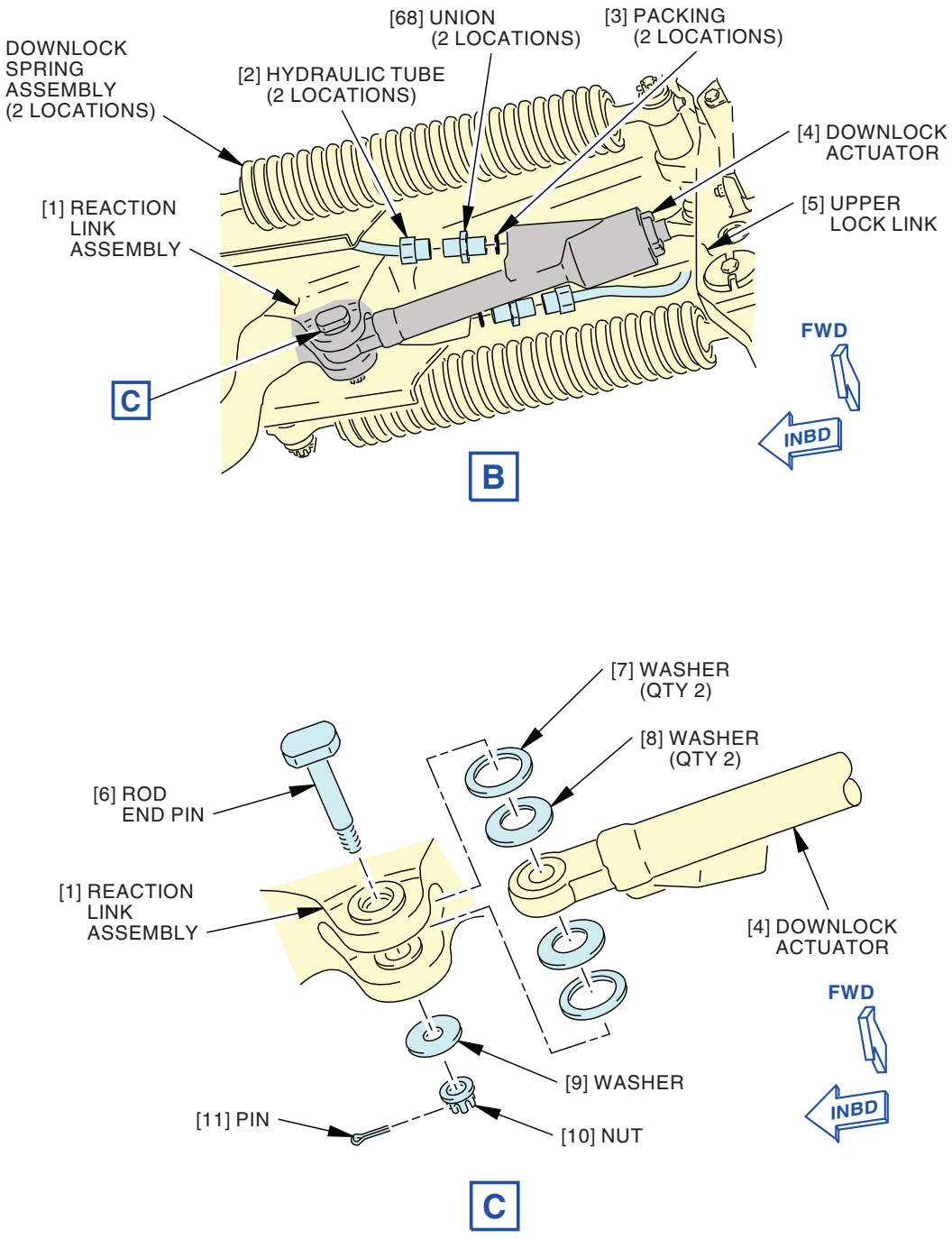
——— END OF TASK ———

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Main Landing Gear Reaction Link Assembly Installation
Figure 401/32-11-71-990-801 (Sheet 1 of 7)

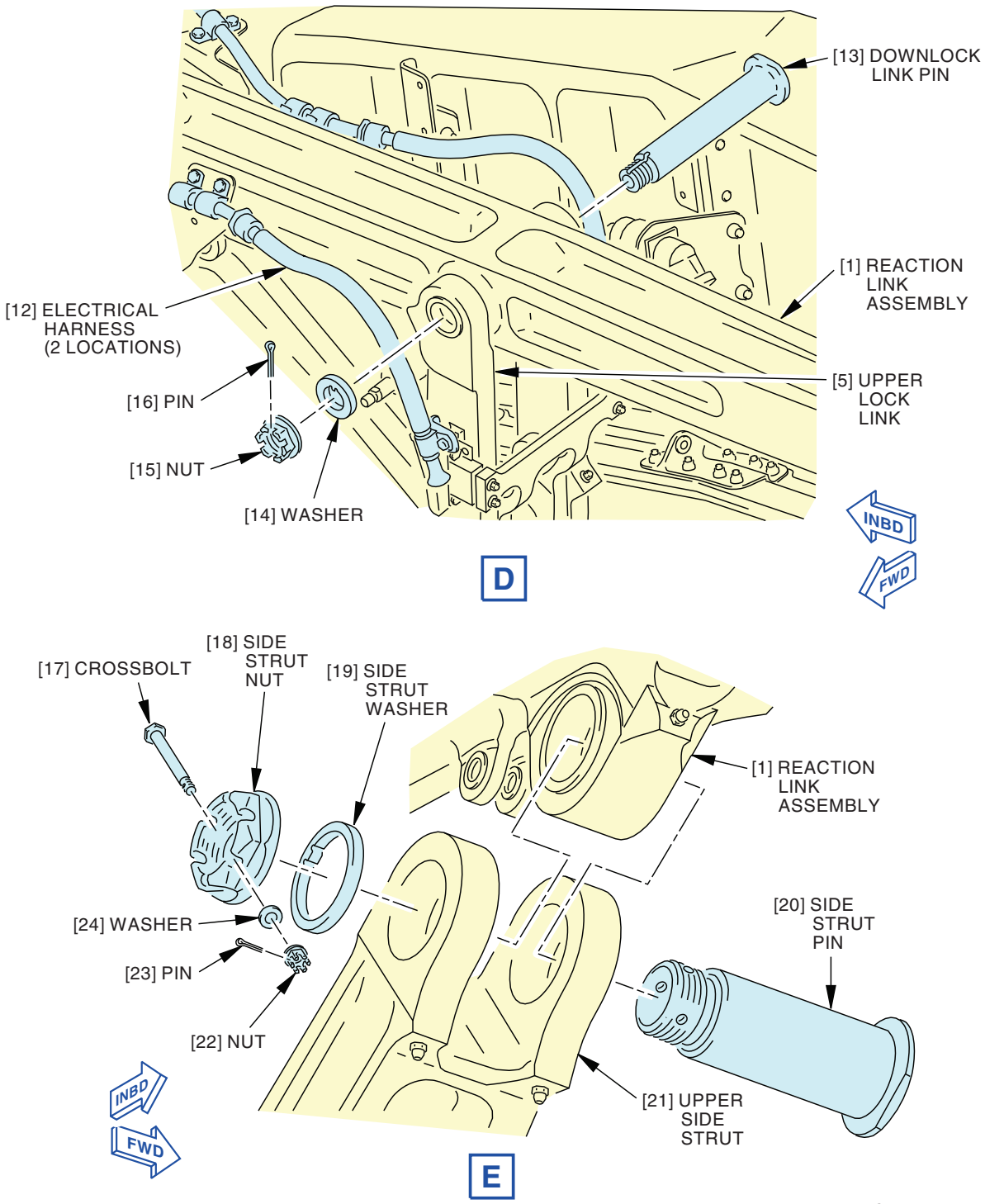
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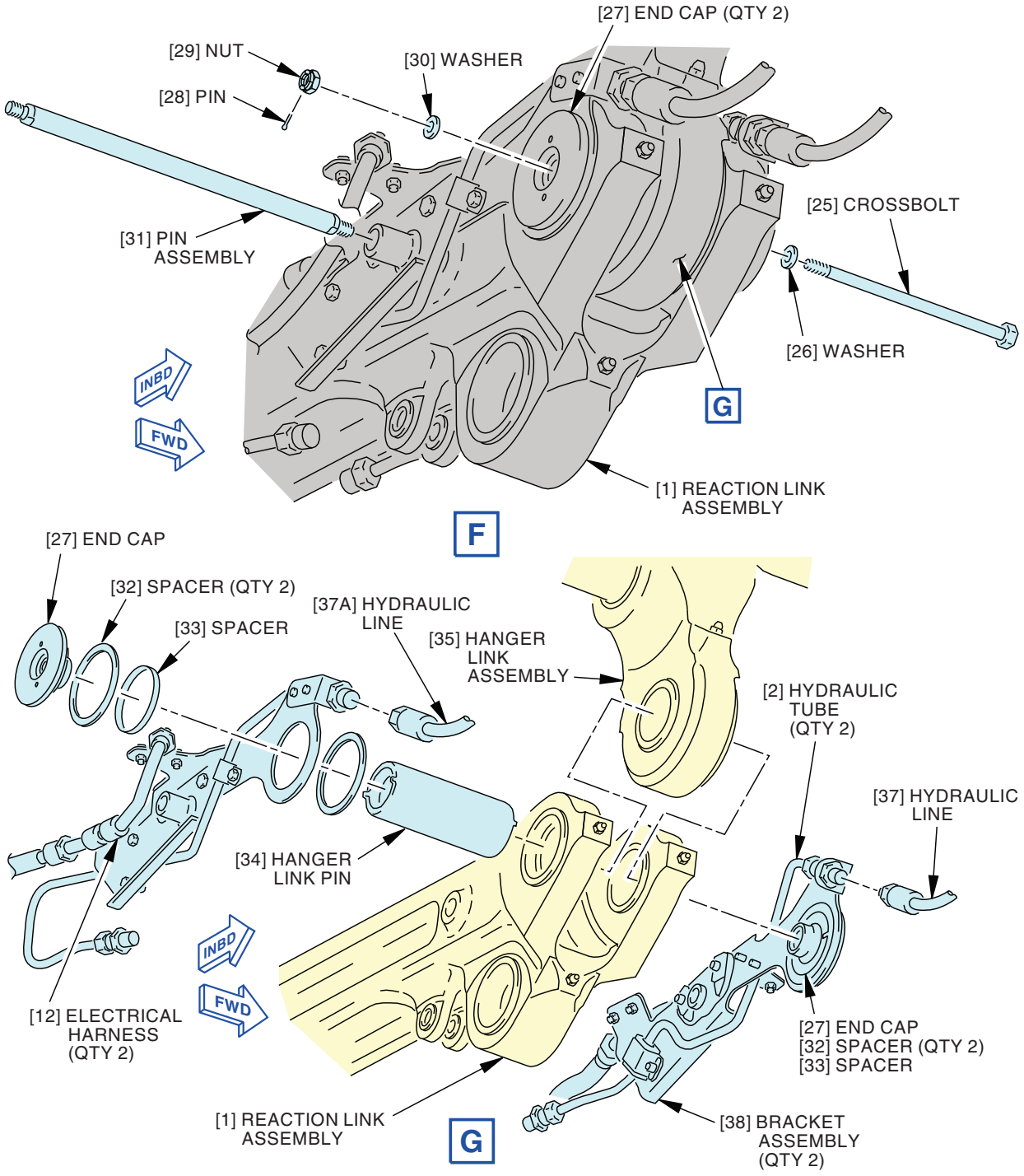
Main Landing Gear Reaction Link Assembly Installation
Figure 401/32-11-71-990-801 (Sheet 2 of 7)

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Main Landing Gear Reaction Link Assembly Installation
Figure 401/32-11-71-990-801 (Sheet 3 of 7)

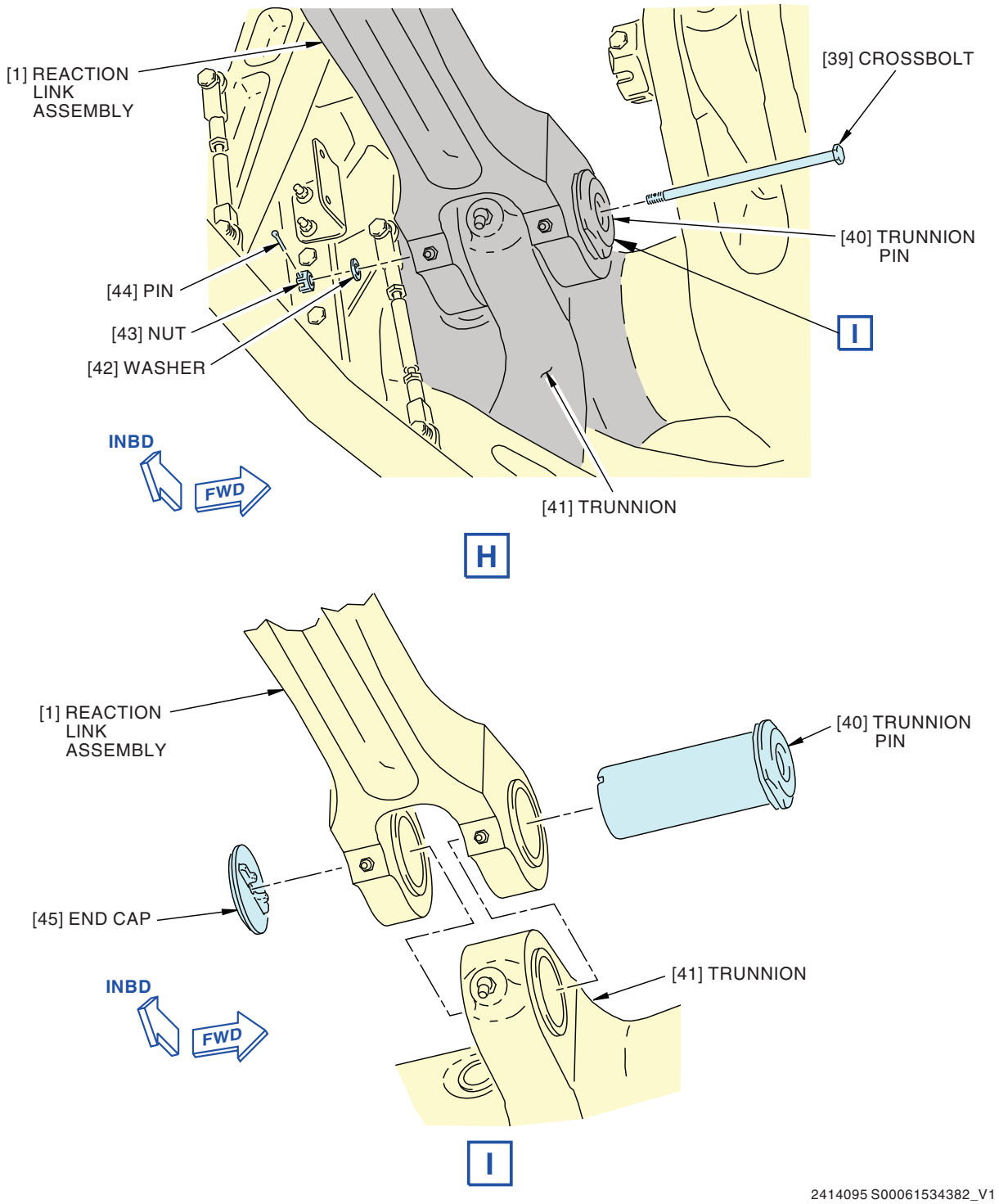
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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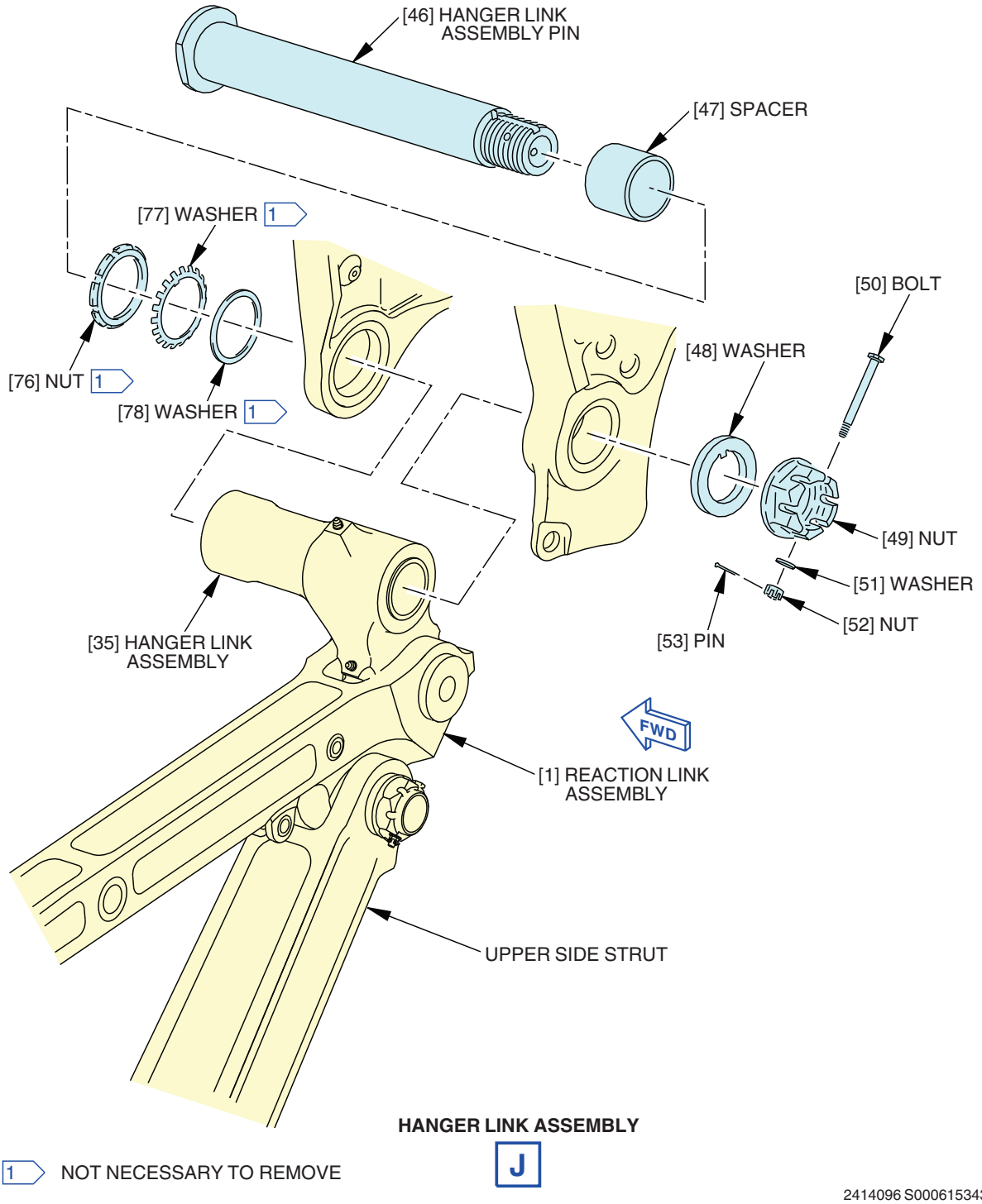
Main Landing Gear Reaction Link Assembly Installation
Figure 401/32-11-71-990-801 (Sheet 4 of 7)

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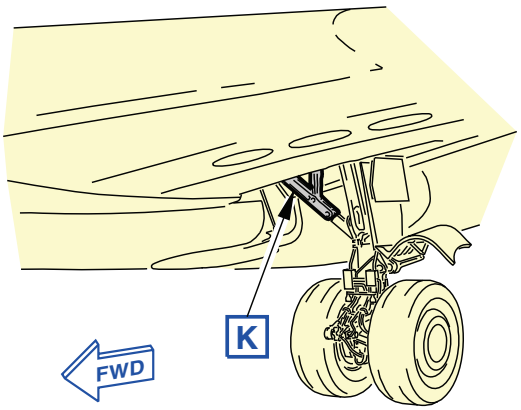
Main Landing Gear Reaction Link Assembly Installation
Figure 401/32-11-71-990-801 (Sheet 5 of 7)

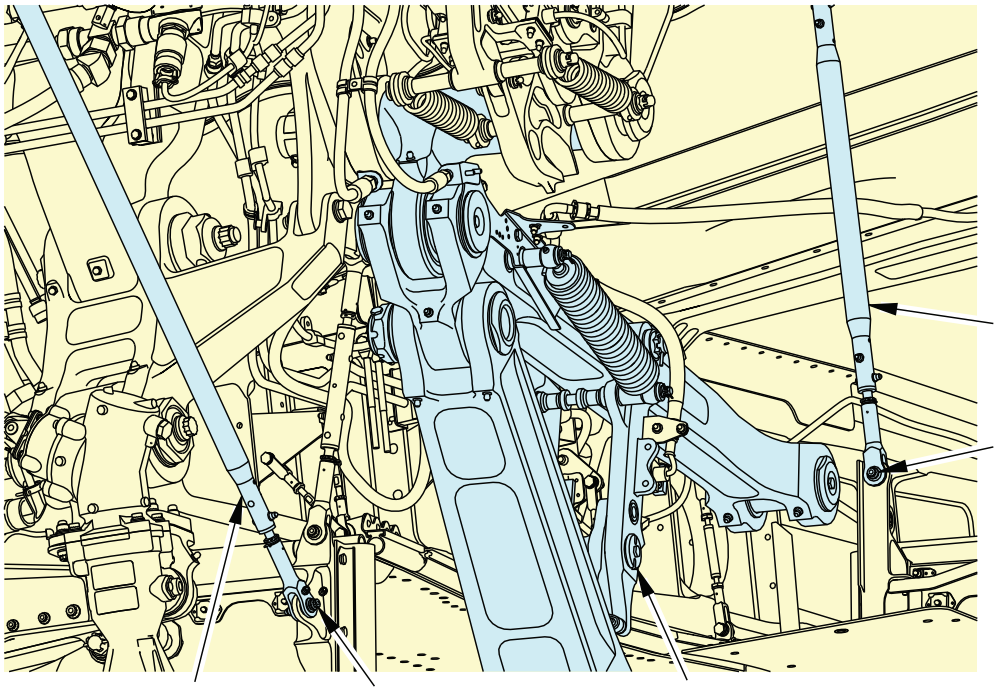
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Reaction Link Assembly Installation
Figure 401/32-11-71-990-801 (Sheet 6 of 7)

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LEFT MAIN LANDING GEAR
(RIGHT MAIN LANDING GEAR IS EQUIVALENT)

K

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Main Landing Gear Reaction Link Assembly Installation
Figure 401/32-11-71-990-801 (Sheet 7 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-11-71-420-801

3. Main Landing Gear Reaction Link Assembly Installation

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
29-09-00-860-801	Hydraulic Reservoirs Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-32-00-710-801	Main Landing Gear Test - Component Replacement (P/B 501)
32-32-00-710-802	Main Landing Gear - Operational Test (P/B 501)
32-32-51-400-801	Main Gear Downlock Actuator Installation (P/B 401)
32-32-91-400-801	Main Gear Downlock Spring Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1359	Wrench Adapter, C32029-3 or C32029-93 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1360	Wrench Adapter, C32029-4 or C32029-94 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1368	Thread Protector, C32029-12 or C32029-101 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1369	Thread Protector , C32029-13 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
D00633	Grease - Aircraft General Purpose	BMS3-33
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Reaction link assembly	32-11-71-01-105	PTF ALL
		32-11-71-01-160	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Packing	32-32-51-01A-055	PTF ALL
11	Pin	32-32-51-01-003	PTF ALL
		32-32-51-01A-005	PTF ALL
16	Pin	32-11-71-01-010	PTF ALL
23	Pin	32-11-71-01-085	PTF ALL
28	Pin	32-11-71-01-055	PTF ALL
44	Pin	32-11-61-05-010	PTF ALL
53	Pin	32-11-61-05-075	PTF ALL

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

F. Prepare for the Installation

SUBTASK 32-11-71-480-004

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

G. Main Landing Gear Reaction Link Assembly Installation

SUBTASK 32-11-71-420-001

(1) Do these steps to connect the reaction link assembly [1] to the trunnion [41]:

(a) Lubricate the chrome plated surfaces of the trunnion pin [40] and end cap [45] with grease, D00633.

(b) Put the outboard end of the reaction link assembly [1] in position for attachment to the trunnion [41].

1) Make sure that a minimum of two persons will hold the reaction link assembly [1] when you install it to the airplane.

NOTE: The reaction link assembly weighs approximately 63.2 lb (28.7 kg).

(c) Put the trunnion pin [40] through the reaction link assembly [1] and trunnion [41].

(d) Install the end cap [45] on the trunnion pin [40].

(e) Put the crossbolt [39] through the trunnion pin [40] and end cap [45].

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WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(f) Apply a thin layer of corrosion inhibiting material, G50136, to these items:

1) Do not apply corrosion inhibiting material to the chrome plated surfaces.
2) The threads and thread reliefs of the crossbolt [39]
3) The threads of the nut [43]
4) The faces of the washer [42].

(g) Remove all unwanted corrosion inhibiting material, G50136.

(h) Install the washer [42] and nut [43] on the crossbolt [39].

(i) If it is necessary, loosen the nut [43] to the nearest castellation that aligns with the hole for the new pin [44].

(j) Install the new pin [44] in the crossbolt [39].

SUBTASK 32-11-71-020-010

(2) If previously removed, do these steps to install the hanger link assembly [35] into the supporting structure:

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin layer of corrosion inhibiting material, G50136, to these items:

1) Do not apply corrosion inhibiting material to the chrome plated surfaces.
2) The threads and thread reliefs of the hanger link assembly pin [46] and bolt [50]
3) The threads of the nut [49] and nut [52]
4) The faces of the washer [48] and washer [51].
5) Remove all unwanted corrosion inhibiting material, G50136.

(b) Prior to installation of the hanger link assembly pin [46], lubricate chrome plated surfaces with grease, D00633.

(c) Install the hanger link assembly pin [46] and spacer [47] that supports the hanger link assembly [35] to the supporting structure, through the hanger link assembly [35].

(d) Install the washer [48] and nut [49] on the hanger link assembly pin [46].

(e) Tighten the nut [49] to 95 ft-lb (128.8 N·m) - 115 ft-lb (155.9 N·m).

(f) If it is necessary, loosen the nut [49] to align the nearest castellation.

(g) Install the bolt [50] in the hanger link assembly pin [46].

(h) Install the nut [52] and washer [51] to the bolt [50].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(i) Tighten the nut [52] until the nut [52] and head of the bolt [50] are fully seated.

(j) If it is necessary, loosen the nut [52] to the nearest castellation that aligns with the hole for the new pin [53].

(k) Install the new pin [53] in the bolt [50].

(l) Remove all unwanted corrosion inhibiting material, G50136.

(m) Connect the lower end of the support rod [69] and support rod [70] to the wheel well structure.

1) Install the bolt [71], washer [72], bushing [73], washer [74], and nut [75].

2) Make sure that you install the countersunk washer under the bolt head and the plain washer under the nut.

SUBTASK 32-11-71-420-002

(3) Do these steps to connect the reaction link assembly [1] to the hanger link assembly [35]:

(a) Lubricate the chrome plated surfaces of the hanger link pin [34] and end cap [27] with grease, D00633.

(b) Put the inboard end of the reaction link assembly [1] in its position for attachment to the hanger link assembly [35].

(c) Put the hanger link pin [34] through the reaction link assembly [1] and hanger link assembly [35].

(d) Attach the assembly of the end cap [27], bracket assembly [38], hydraulic tube [2], electrical harness [12], spacers [32], and spacer [33] on each end of the hanger link pin [34].

(e) If it is necessary to connect the electrical harness [12], do this step:

1) Install the bolts for the bracket that holds the sensor to the upper lock link [5] to connect the electrical harness [12].

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(f) Apply a thin layer of corrosion inhibiting material, G50136, to the faces of the washer [26].

(g) Put the washer [26] on the crossbolt [25].

(h) Put the crossbolt [25] through the end caps [27].

(i) Apply a thin layer of corrosion inhibiting material, G50136, to these items:

1) Do not apply corrosion inhibiting material to the chrome plated surfaces.

2) The threads and thread reliefs of the crossbolt [25]

3) The threads of the nut [29]

4) The faces of the washer [30].

(j) Remove all unwanted corrosion inhibiting material, G50136.

(k) Install the washer [30] and nut [29] on the crossbolt [25].

(l) Tighten the nut [29] to 40 ±10 in-lb (4.5 ±1.1 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(m) If it is necessary, loosen the nut [29] to the nearest castellation that aligns with the hole for the new pin [28].

(n) Install the new pin [28] in the crossbolt [25].

(o) Put the pin assembly [31] through the bracket assembly [38] and reaction link assembly [1].

SUBTASK 32-11-71-420-003

(4) Do these steps to connect the reaction link assembly [1] to the upper side strut [21]:

(a) Lubricate the chrome plated surfaces of the side strut pin [20] with grease, D00633.

(b) Remove the strap that supports the upper side strut [21] to the hanger link assembly [35].

(c) Put the upper side strut [21] in position for attachment to the reaction link assembly [1].

(d) Use the thread protector, C32029-12 or C32029-101, SPL-1368, to put the side strut pin [20] through the upper side strut [21] and reaction link assembly [1].

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(e) Apply a thin layer of corrosion inhibiting material, G50136, to these items:

1) Do not apply corrosion inhibiting material to the chrome plated surfaces.

2) The threads and thread reliefs of the side strut pin [20] and crossbolt [17]

3) The threads of the side strut nut [18] and nut [22]

4) The faces of the side strut washer [19] and washer [24].

(f) Remove all unwanted corrosion inhibiting material, G50136.

(g) Install the side strut washer [19] and side strut nut [18] on the side strut pin [20].

(h) Tighten the side strut nut [18] to 750 ±150 in-lb (84.7 ±16.9 N·m).

1) Use the wrench adapter, C32029-3 or C32029-93, SPL-1359, to hold the head of the side strut pin [20] when you tighten the side strut nut [18].

(i) If it is necessary, loosen the side strut nut [18] to the nearest castellation that aligns with the hole for the crossbolt [17].

(j) Install the crossbolt [17] in the side strut pin [20].

(k) Install the washer [24] and nut [22] on the crossbolt [17].

(l) Tighten the nut [22] to 40 ±10 in-lb (4.5 ±1.1 N·m).

(m) If it is necessary, loosen the nut [22] to the nearest castellation that aligns with the hole for the new pin [23].

(n) Install the new pin [23] in the crossbolt [17].

SUBTASK 32-11-71-420-004

(5) Do these steps to connect the reaction link assembly [1] to the upper lock link [5]:

(a) Lubricate the chrome plated surfaces of the downlock link pin [13] with grease, D00633.

(b) Put the upper lock link [5] in position for attachment to the reaction link assembly [1].

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(c) Use the thread protector, C32029-13, SPL-1369, to put the downlock link pin [13] through the upper lock link [5] and reaction link assembly [1].

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(d) Apply a thin layer of corrosion inhibiting material, G50136, to these items:

1) Do not apply corrosion inhibiting material to the chrome plated surfaces.
2) The threads and thread reliefs of the downlock link pin [13]
3) The threads of the nut [15]
4) The faces of the washer [14].

(e) Remove all unwanted corrosion inhibiting material, G50136.

(f) Install the washer [14] and nut [15] on the downlock link pin [13].

(g) Tighten the nut [15] to 480 ±120 in-lb (54.2 ±13.6 N·m).

1) Use the wrench adapter, C32029-4 or C32029-94, SPL-1360, to hold the head of the downlock link pin [13] when you tighten the nut [15].

(h) If it is necessary, loosen the nut [15] to the nearest castellation that aligns with the hole for the new pin [16].

(i) Install the new pin [16] in the downlock link pin [13].

SUBTASK 32-11-71-640-001

(6) Do these steps to connect the downlock actuator [4] to the reaction link assembly [1]:

(a) Lubricate the inside surface of the lug, head end of the downlock actuator [4], and rod end pin [6] with grease, D00633.

(b) Put the head end of the downlock actuator [4] in position for attachment to the reaction link assembly [1].

1) If it is necessary, retract or extend the downlock actuator [4] to align the head end to the reaction link assembly [1].

(c) Put the rod end pin [6] through the reaction link assembly [1], washers [7], washers [8], and head end of the downlock actuator [4].

(d) Install the washer [9] and nut [10] on the rod end pin [6].

(e) Tighten the nut [10] to 20 ±5 in-lb (2.3 ±0.6 N·m).

(f) If it is necessary, loosen the nut [10] to the nearest castellation that aligns with the hole for the new pin [11].

(g) Install the new pin [11] in the rod end pin [6].

SUBTASK 32-11-71-420-005

(7) Do these steps to connect the hydraulic tubes [2] to the downlock actuator [4]:

(a) Lubricate these items with hydraulic fluid, D00153:

1) The new packings [3]
2) The unions [68].

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(b) Remove the caps/plugs, STD-14686, from the ports on the downlock actuator [4].

(c) Install the packings [3] and unions [68] in the downlock actuator [4].

(d) Remove the caps/plugs, STD-14686, from the hydraulic tubes [2].

(e) Connect the hydraulic tubes [2] to the downlock actuator [4].

(f) Tighten the B-nuts on the hydraulic tubes [2] to 140 in-lb (15.8 N·m).

(g) Loosen the B-nuts on the hydraulic tubes [2] and re-tighten them to 140 ±7 in-lb (15.8 ±0.8 N·m).

(h) Remove the tags from the hydraulic tubes [2].

SUBTASK 32-11-71-420-006

(8) Do these steps to connect the hydraulic line [37] and hydraulic line [37A] to the hydraulic tubes [2] that are attached to the bracket assembly [38]:

(a) Remove the caps/plugs, STD-14686, from the hydraulic line [37] and hydraulic line [37A].

(b) Connect the hydraulic line [37] and hydraulic line [37A] to the hydraulic tubes [2].

(c) Tighten the hydraulic line [37] to 140 ±7 in-lb (15.8 ±0.8 N·m).

(d) Tighten the hydraulic line [37A] to 270 ±14 in-lb (30.5 ±1.6 N·m).

(e) Remove the tags from the hydraulic line [37] and hydraulic line [37A].

H. Main Landing Gear Reaction Link Assembly Installation Test

SUBTASK 32-11-71-420-007

(1) Do this task: Main Gear Downlock Spring Installation, TASK 32-32-91-400-801.

SUBTASK 32-11-71-710-001

(2) Do this task: Main Gear Downlock Actuator Installation, TASK 32-32-51-400-801.

SUBTASK 32-11-71-860-009

(3) For hydraulic system A, do this task: Hydraulic Reservoirs Pressurization, TASK 29-09-00-860-801.

SUBTASK 32-11-71-860-003

(4) Do this task: Main Landing Gear - Operational Test, TASK 32-32-00-710-802.

NOTE: If a hydraulic cart is not available, it is optional to perform the Main Landing Gear Test - Component Replacement, TASK 32-32-00-710-801 instead of the operational test.

SUBTASK 32-11-71-210-001

(5) Do a check of the hydraulic connections for leakage.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 32-11-71-860-004

(1) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-11-71-580-006

CAUTION

MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

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(CAUTION PRECEDES)

CAUTION	MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(2) Do this task: Lower the Airplane off Jacks, TASK 07-10-01-580-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN LANDING GEAR HYDRAULIC SHIMMY DAMPER COMPONENTS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the compensator assembly for the hydraulic shimmy damper on the main landing gear

(2) Installation of the compensator assembly for the hydraulic shimmy damper on the main landing gear

(3) Removal of the manifold assembly for the hydraulic shimmy damper on the main landing gear

(4) Installation of the manifold assembly for the hydraulic shimmy damper on the main landing gear

(5) Removal of the end gland assembly for the hydraulic shimmy damper on the main landing gear

(6) Installation of the end gland assembly for the hydraulic shimmy damper on the main landing gear

TASK 32-11-81-000-801

2. Main Landing Gear Shimmy Damper Compensator Removal

(Figure 201)

A. General

(1) This procedure has instructions to remove the compensator assembly from the hydraulic shimmy damper on the main landing gear torsion links.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Prepare for the Removal

SUBTASK 32-11-81-480-028

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-11-81-480-002

(2) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-11-81-860-001

(3) For the A and B hydraulic systems, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

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SUBTASK 32-11-81-860-002

(4) For the A and B hydraulic systems, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

E. Main Landing Gear Shimmy Damper Compensator Removal

SUBTASK 32-11-81-020-001

(1) Remove the lockwire from the compensator assembly [16].

SUBTASK 32-11-81-020-002

(2) Remove the compensator assembly [16] from the manifold assembly [4] on the shimmy damper [1].

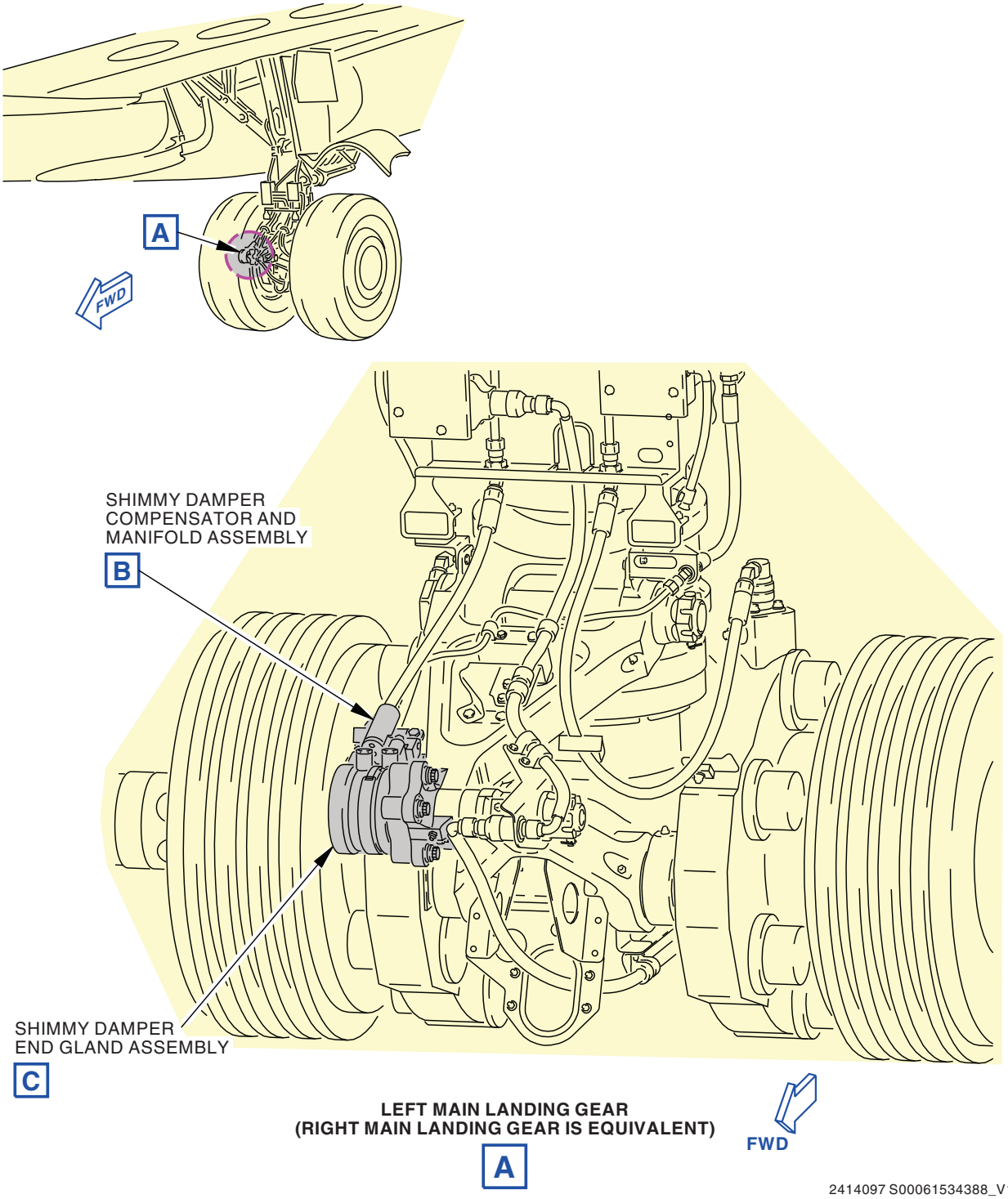
(a) Remove the packing [14], backup ring [13], and backup ring [15] from the compensator assembly [16] and discard.

SUBTASK 32-11-81-480-003

(3) Install a plug in the port of the manifold assembly [4] for the compensator assembly [16].

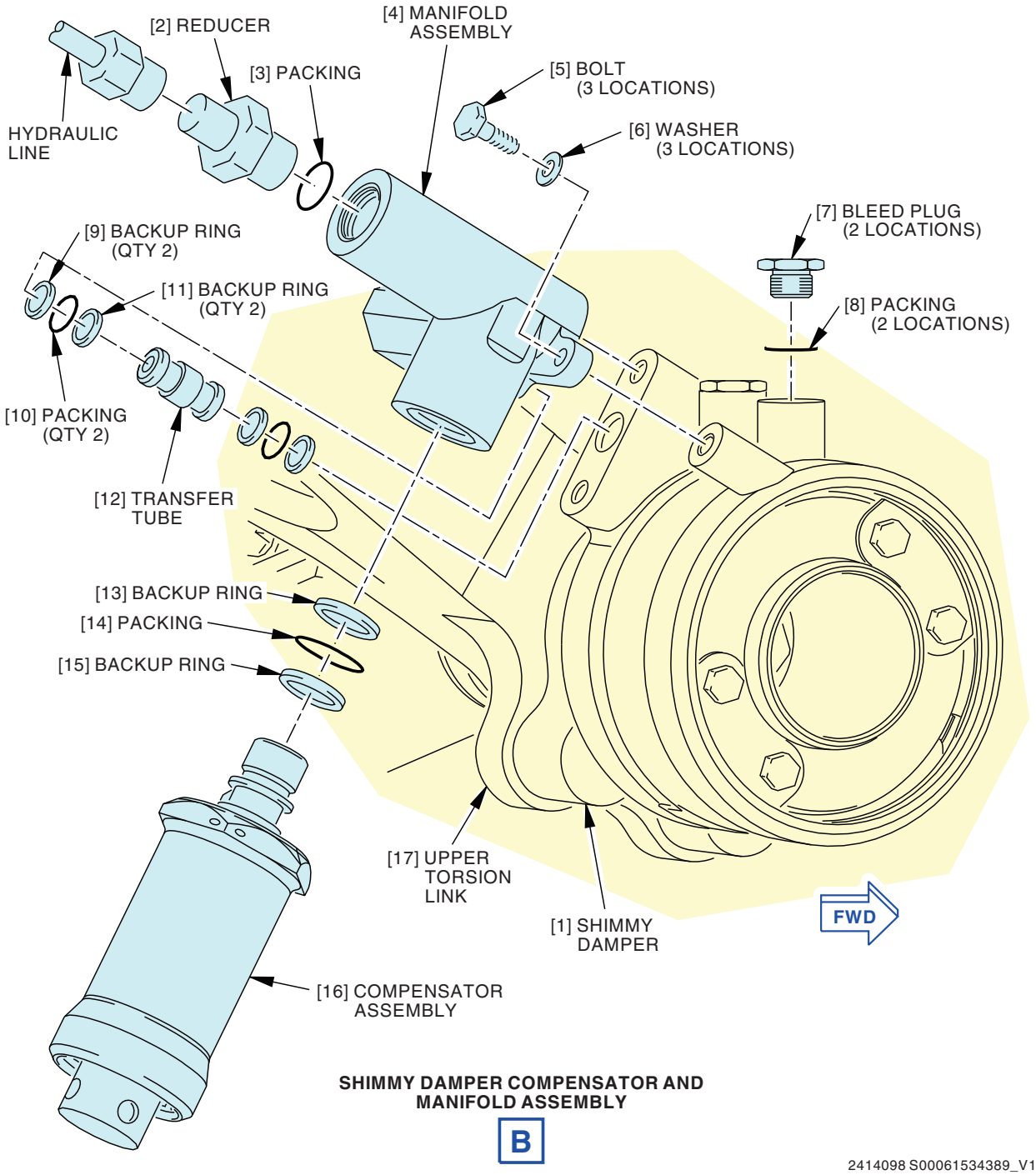
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



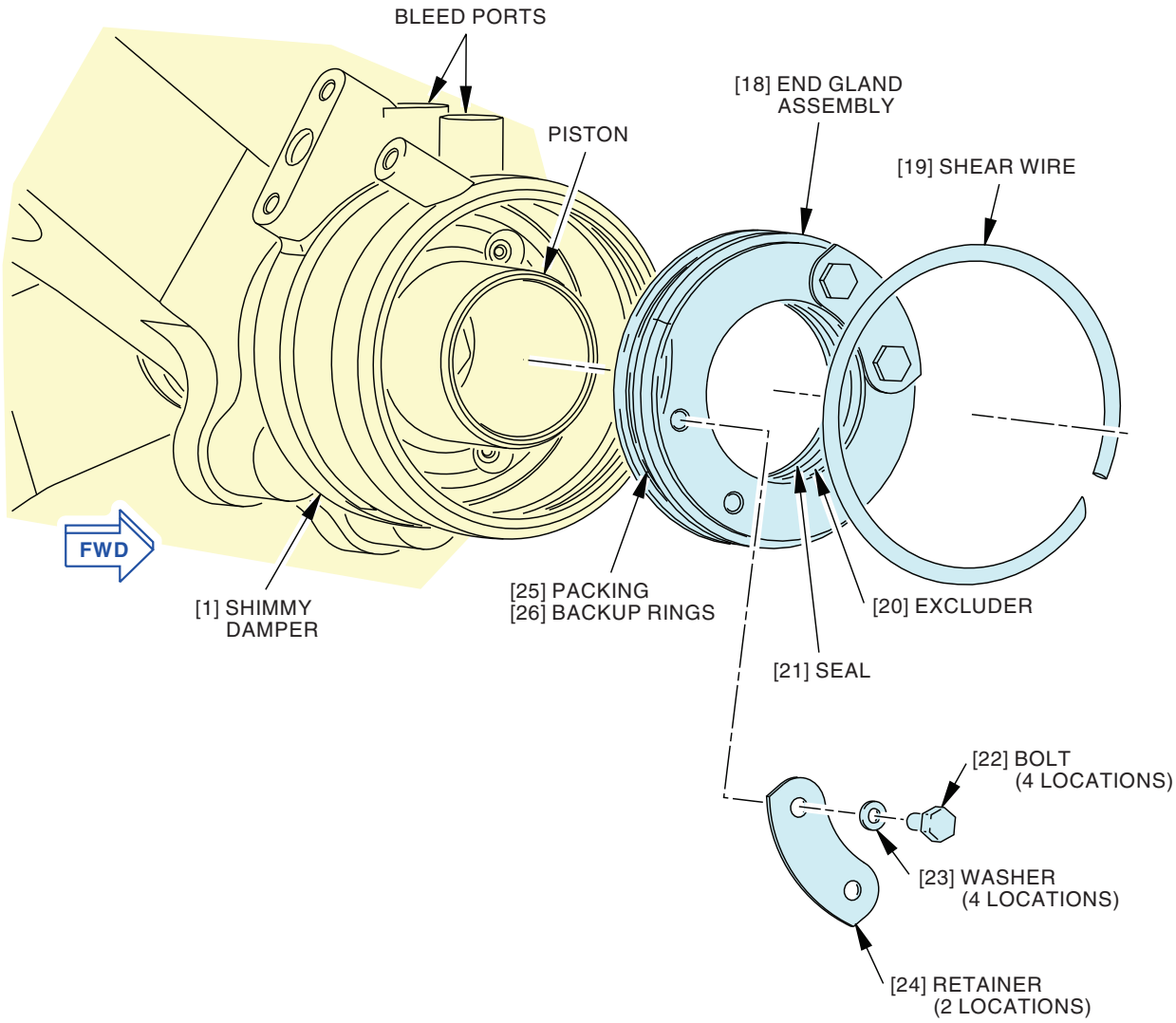
Main Landing Gear Shimmy Damper Component Installation
Figure 201/32-11-81-990-801 (Sheet 1 of 3)

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Main Landing Gear Shimmy Damper Component Installation
Figure 201/32-11-81-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



SHIMMY DAMPER END GLAND ASSEMBLY

C

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Main Landing Gear Shimmy Damper Component Installation
Figure 201/32-11-81-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-11-81-400-801

3. Main Landing Gear Shimmy Damper Compensator Installation

(Figure 201)

A. General

(1) This task has instructions to install the compensator assembly for the hydraulic shimmy damper on the main landing gear torsion links.

B. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-11-81-870-801	Hydraulic Shimmy Damper Bleeding (P/B 401)

C. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Shimmy damper	32-11-81-01-105	PTF ALL
		32-11-81-01-370	PTF ALL
13	Backup ring	32-11-81-01-197	PTF ALL
14	Packing	32-11-81-01-195	PTF ALL
15	Backup ring	32-11-81-01-197	PTF ALL
16	Compensator assembly	32-11-81-01-235	PTF ALL

E. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

F. Main Landing Gear Shimmy Damper Compensator Installation

SUBTASK 32-11-81-480-004

(1) Remove the plug from the manifold assembly [4] on the shimmy damper [1], if one is installed.

SUBTASK 32-11-81-420-001

(2) Make sure that the new packing [14], new backup ring [13], and new backup ring [15] are installed on the compensator assembly [16].

SUBTASK 32-11-81-420-002

(3) Lubricate the compensator assembly [16] thread area, new packing [14], new backup ring [13] and new backup ring [15] with hydraulic fluid, D00153.

SUBTASK 32-11-81-420-003

(4) Install the compensator assembly [16] in its location in the manifold assembly [4].

SUBTASK 32-11-81-420-004

(5) Tighten thecompensator assembly [16] to 385 ±19 in-lb (43.5 ±2.1 N·m).

SUBTASK 32-11-81-420-005

(6) Install lockwire on the compensator assembly [16] compensator to the closest bolt [5] for the manifold assembly [4].

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G. Main Landing Gear Shimmy Damper Compensator Installation Test

SUBTASK 32-11-81-420-006

(1) Do this task: Hydraulic Shimmy Damper Bleeding, TASK 32-11-81-870-801.

(a) Make sure that there are no leaks at the compensator connection when you bleed the shimmy damper [1].

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-11-81-480-005

(1) Remove the chocks from around the landing gear tires.

SUBTASK 32-11-81-860-003

(2) For the A and B hydraulic systems, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

END OF TASK

TASK 32-11-81-000-802

4. Main Landing Gear Shimmy Damper Manifold Assembly Removal

(Figure 201)

A. General

(1) This task has instructions to remove the manifold assembly from the hydraulic shimmy damper on the main landing gear torsion links.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Prepare for the Removal

SUBTASK 32-11-81-480-029

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-11-81-480-007

(2) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-11-81-860-004

(3) For the A and B hydraulic systems, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

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SUBTASK 32-11-81-860-005

(4) For the A and B hydraulic systems, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

E. Main Landing Gear Shimmy Damper Manifold Assembly Removal

SUBTASK 32-11-81-020-003

(1) Disconnect the hydraulic line from the manifold assembly [4] on the shimmy damper [1].

SUBTASK 32-11-81-020-004

(2) If the replacement manifold assembly [4] does not have a reducer [2] installed, do these steps:

(a) Remove the reducer [2] and packing [3].

(b) Discard the packing [3].

SUBTASK 32-11-81-480-008

(3) Do these steps to remove the manifold assembly [4]:

(a) Remove the bolts [5] and washers [6] that connect the manifold assembly [4] to the shimmy damper [1].

(b) Remove the manifold assembly [4] from the shimmy damper [1].

(c) Remove the transfer tube [12] from its location.

NOTE: The transfer tube [12] can be in the manifold assembly [4] or the port for the manifold assembly [4].

(d) Remove the packings [10], backup rings [9], and backup rings [11] from the transfer tube [12] and discard.

——— END OF TASK ———

TASK 32-11-81-400-802

5. Main Landing Gear Shimmy Damper Manifold Assembly Installation

(Figure 201)

A. General

(1) This task has instructions to install the manifold assembly on the hydraulic shimmy damper on the main landing gear.

B. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-11-81-870-801	Hydraulic Shimmy Damper Bleeding (P/B 401)

C. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Shimmy damper	32-11-81-01-105	PTF ALL
		32-11-81-01-370	PTF ALL
3	Packing	32-11-81-01-100	PTF ALL
4	Manifold assembly	32-11-81-01-200	PTF ALL

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(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
9	Backup ring	32-11-81-01-290	PTF ALL
10	Packing	32-11-81-01-285	PTF ALL
11	Backup ring	32-11-81-01-290	PTF ALL

E. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

F. Main Landing Gear Shimmy Damper Manifold Assembly Installation

SUBTASK 32-11-81-420-007

(1) Make sure that the mating surfaces on the manifold assembly [4] and on the shimmy damper [1] are clean.

SUBTASK 32-11-81-480-009

(2) Remove the plugs from the new manifold assembly [4] and from the shimmy damper [1], if they are installed.

SUBTASK 32-11-81-480-010

(3) Do these steps to install the transfer tube [12]:

(a) Install the new packing [10], new backup ring [9], and new backup ring [11] on the transfer tube [12] in their two locations.

(b) Lubricate the transfer tube [12], new packings [10], new backup rings [9], and new backup rings [11] with hydraulic fluid, D00153.

(c) Install the transfer tube [12] in its location in the shimmy damper [1].

SUBTASK 32-11-81-480-011

(4) Do these steps to install the manifold assembly [4]:

(a) Put the manifold assembly [4] in its location on the shimmy damper [1] (and the transfer tube [12]).

(b) Install the bolts [5] and washers [6] to connect the manifold assembly [4] to the shimmy damper [1].

NOTE: The shimmy damper [1] housing contains inserts at the manifold bolt [5] bolt locations.

(c) Install lockwire on the compensator assembly [16] to the closest bolt [5] for the manifold assembly [4].

SUBTASK 32-11-81-020-005

(5) If the new manifold assembly [4] does not have a reducer [2] installed, do these steps:

(a) Remove the plug from the port in the manifold assembly [4], if one is installed.

(b) Lubricate the reducer [2] and the new packing [3] with hydraulic fluid, D00153.

(c) Install the new packing [3] on the reducer [2].

(d) Install the reducer [2] in the manifold assembly [4].

SUBTASK 32-11-81-480-012

(6) Remove the cap from the reducer [2], if one is installed.

SUBTASK 32-11-81-420-008

(7) Connect the hydraulic line to the manifold assembly [4] on the shimmy damper [1].

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(a) Tighten the B-nut on the hydraulic line to 140 ±7 in-lb (16 ±1 N·m).

G. Main Landing Gear Shimmy Damper Manifold Assembly Installation Test

SUBTASK 32-11-81-420-009

(1) Do this task: Hydraulic Shimmy Damper Bleeding, TASK 32-11-81-870-801.

(a) Make sure that there are no leaks at the manifold connection and the hydraulic line connection when you bleed the shimmy damper [1].

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-11-81-480-013

(1) Remove the chocks from around the landing gear tires.

SUBTASK 32-11-81-860-006

(2) For the A and B hydraulic systems, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

TASK 32-11-81-000-803

6. Main Landing Gear Shimmy Damper End Gland Assembly Removal

(Figure 201)

A. General

(1) This task has instructions to remove the end gland assembly from the hydraulic shimmy damper on the main landing gear torsion links.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Prepare for the Removal

SUBTASK 32-11-81-480-030

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-11-81-480-015

(2) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-11-81-860-007

(3) For the A and B hydraulic systems, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-11-81-860-008

(4) For the A and B hydraulic systems, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

E. Main Landing Gear Shimmy Damper End Gland Assembly Removal

SUBTASK 32-11-81-020-020

(1) Carefully loosen and remove the bleed plugs [7] to release the pressure in the shimmy damper [1].

(a) Remove and discard the packings [8].

SUBTASK 32-11-81-020-006

(2) Remove the bolts [22] and washers [23] that hold the retainers [24] to the end gland assembly [18].

SUBTASK 32-11-81-020-007

(3) Remove the retainers [24] from the end gland assembly [18].

SUBTASK 32-11-81-020-008

(4) Remove the shear wire [19] from the inner diameter of the shimmy damper [1].

(a) Discard the shear wire [19].

SUBTASK 32-11-81-020-009

(5) Remove the end gland assembly [18] from the shimmy damper [1].

SUBTASK 32-11-81-480-016

(6) Put a covering on the end of the shimmy damper [1] for protection.

——— END OF TASK ———

TASK 32-11-81-400-803

7. Main Landing Gear Shimmy Damper End Gland Assembly Installation

(Figure 201)

A. General

(1) This task has instructions to install the end gland assembly for the hydraulic shimmy damper on the main landing gear torsion links.

B. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-11-81-870-801	Hydraulic Shimmy Damper Bleeding (P/B 401)

C. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Shimmy damper	32-11-81-01-105	PTF ALL
		32-11-81-01-370	PTF ALL
8	Packing	32-11-81-01-305	PTF ALL
18	End gland assembly	32-11-81-01-355	PTF ALL

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(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
18 (cont.)		32-11-81-01-365	PTF ALL
19	Shear wire	32-11-81-01-310	PTF ALL
20	Excluder	32-11-81-01-330	PTF ALL
21	Seal	32-11-81-01-335	PTF ALL
25	Packing	32-11-81-01-315	PTF ALL
26	Backup ring	32-11-81-01-320	PTF ALL

E. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

F. Main Landing Gear Shimmy Damper End Gland Assembly Installation

SUBTASK 32-11-81-480-017

(1) If the seal [21] or packing [25] for the end gland assembly [18] are not installed, do these steps:

(a) Install the new excluder [20] in the inside groove closest to the outside face of the end gland assembly [18].

(b) Install the new seal [21] in the inside groove closest to the inside face of the end gland assembly [18].

(c) Install the new packing [25] and the new backup rings [26] in the outside groove on the end gland assembly [18].

SUBTASK 32-11-81-480-018

(2) Remove the covering from the end of the shimmy damper [1], if it is installed.

SUBTASK 32-11-81-480-019

(3) Lubricate all of the new seal [21] and new packing [25] for the end gland assembly [18] with hydraulic fluid, D00153.

SUBTASK 32-11-81-020-010

(4) Install the end gland assembly [18] in the shimmy damper [1].

(a) Make sure that the new backup rings [26] stay in their positions.

SUBTASK 32-11-81-020-011

(5) Install the new shear wire [19] on the inner diameter of the shimmy damper [1].

NOTE: The shear wire must be in the groove in the end gland assembly, and the groove in the inner diameter of the shimmy damper.

SUBTASK 32-11-81-020-012

(6) Put the retainers [24] in their position on the end gland assembly [18].

SUBTASK 32-11-81-020-013

(7) Install the bolts [22] and washers [23] to hold the retainers [24] to the end gland assembly [18].

SUBTASK 32-11-81-390-001

(8) Lightly lubricate the new packings [8] with hydraulic fluid, D00153.

SUBTASK 32-11-81-420-021

(9) Install the packings [8] on the bleed plugs [7].

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SUBTASK 32-11-81-420-022

(10) Install the bleed plugs [7] on the shimmy damper [1].

G. Main Landing Gear Shimmy Damper End Gland Assembly Installation Test

SUBTASK 32-11-81-420-010

(1) Do this task: Hydraulic Shimmy Damper Bleeding, TASK 32-11-81-870-801.

(a) Make sure that there are no leaks at the end gland assembly [18] interface when you bleed the shimmy damper [1].

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-11-81-480-020

(1) Remove the chocks from around the landing gear tires.

SUBTASK 32-11-81-860-009

(2) For the A and B hydraulic systems, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN LANDING GEAR HYDRAULIC SHIMMY DAMPER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the hydraulic shimmy damper from the main landing gear

(2) Installation of the hydraulic shimmy damper on the main landing gear

(3) Bleed the hydraulic shimmy damper on the main landing gear.

TASK 32-11-81-000-804

2. Main Landing Gear Shimmy Damper Removal

(Figure 401)

A. General

(1) This task provides instructions to remove the hydraulic shimmy damper from the torsion links.

B. References

Reference	Title
07-10-03-580-801	Lift the Airplane on Axle Jacks-Handling (P/B 201)
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-11-51-020-801	Main Landing Gear Torsion Link Disconnection (P/B 401)
32-45-11-000-801	Main Landing Gear Wheel and Tire Assembly Removal (P/B 401)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Packing	32-11-81-01-100	PTF ALL

D. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Prepare for the Removal

SUBTASK 32-11-81-480-031

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-11-81-480-022

(2) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-11-81-860-010

(3) For the A and B hydraulic systems, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

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SUBTASK 32-11-81-860-011

(4) For the A and B hydraulic system, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

F. Main Landing Gear Shimmy Damper Removal

SUBTASK 32-11-81-020-014

(1) Disconnect the hydraulic line from the manifold assembly [2] on the shimmy damper assembly [1].

SUBTASK 32-11-81-020-015

(2) If the replacement shimmy damper assembly [1] does not have a reducer [3] installed in the manifold assembly [2], do these steps:

(a) Remove the reducer [3] and packing [4] from the shimmy damper assembly [1] you removed.

(b) Discard the packing [4].

(c) Install the reducer [3] with a cap and a new packing [4] in the port of the manifold assembly [2] of the replacement shimmy damper assembly [1].

(d) As an alternative, install a plug in the port of the manifold assembly [2].

SUBTASK 32-11-81-580-001

(3) Lift the wheel and tire assemblies with jacks until they are clear of the ground (TASK 07-10-03-580-801).

SUBTASK 32-11-81-020-019

(4) Remove the left hand inboard or the right hand outboard tire of the main landing gear (Main Landing Gear Wheel and Tire Assembly Removal, TASK 32-45-11-000-801).

SUBTASK 32-11-81-020-016

(5) Do this task: Main Landing Gear Torsion Link Disconnection, TASK 32-11-51-020-801.

SUBTASK 32-11-81-020-017

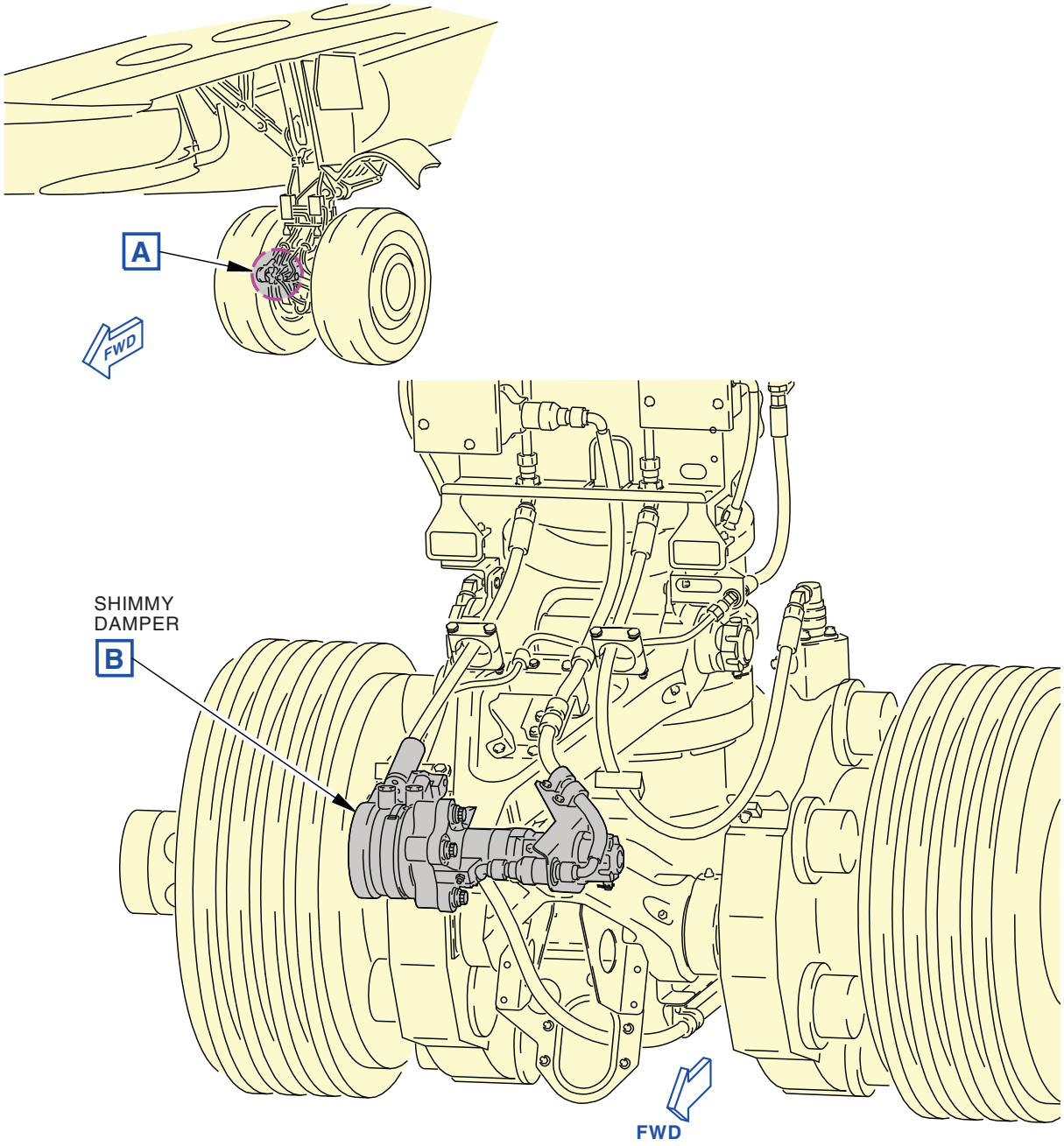
(6) Remove the shimmy damper assembly [1] from the upper torsion link [18].

(a) Remove lockwire from bolts [16].

(b) Remove the bolts [16], washers [17], and straight sleeve [21] that holds the shimmy damper assembly [1] to the upper torsion link [18].

——— END OF TASK ———

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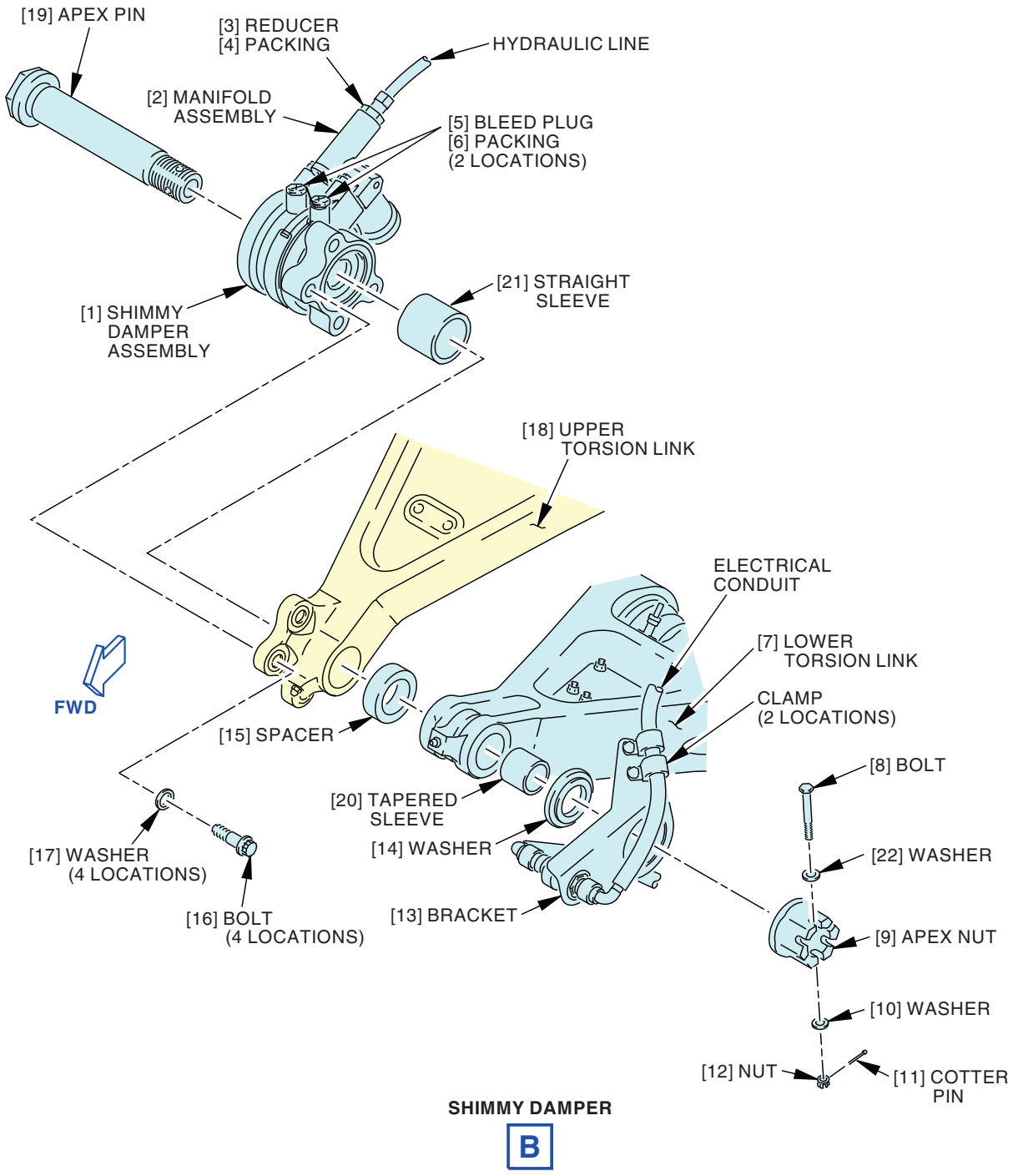
LEFT MAIN LANDING GEAR
(RIGHT MAIN LANDING GEAR IS EQUIVALENT)

A

2414100 S00061534398_V1

Main Landing Gear Shimmy Damper Installation
Figure 401/32-11-81-990-802 (Sheet 1 of 2)

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2414101 S00061534399_V3

Main Landing Gear Shimmy Damper Installation
Figure 401/32-11-81-990-802 (Sheet 2 of 2)

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TASK 32-11-81-400-804

3. Main Landing Gear Shimmy Damper Installation

(Figure 401)

A. General

(1) This task has instructions to install the hydraulic shimmy damper for the main landing gear on the torsion links.

B. References

Reference	Title
07-10-03-580-801	Lift the Airplane on Axle Jacks-Handling (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-11-51-400-803	Main Landing Gear Torsion Link Connection (P/B 401)
32-45-11-400-801	Main Landing Gear Wheel and Tire Assembly - Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
D00633	Grease - Aircraft General Purpose	BMS3-33
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50347	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Shimmy damper assembly	32-11-81-01-370	PTF ALL
4	Packing	32-11-81-01-100	PTF ALL
11	Cotter pin	32-11-81-01-020	PTF ALL

E. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

F. Main Landing Gear Shimmy Damper Installation

SUBTASK 32-11-81-420-011

(1) Prepare these parts for installation:

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WARNING USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin layer of corrosion inhibiting material, G50136, to the thread reliefs, threads of the bolt [16], faces of the washer [17], inside diameter of the straight sleeve [21] and threads inside the shimmy damper assembly [1] attach holes.

(b) Apply a thin layer of grease, D00633, to the outside diameter of the straight sleeve [21].

1) Make sure that you install the bolts [16], washers [17], and straight sleeve [21] immediately while the grease, D00633, is wet.

(c) Apply a thin layer of corrosion inhibiting material, G50136, to the shimmy damper assembly [1] face common to the upper torsion link [18].

1) Make sure that no grease, D00633, is in or on the piston shaft of the damper.

(d) Remove excess grease, D00633.

SUBTASK 32-11-81-420-012

(2) Hold the shimmy damper assembly [1] in its position on the upper torsion link [18].

SUBTASK 32-11-81-420-013

CAUTION MAKE SURE THAT YOU INSTALL THE SLEEVES IN THE CORRECT LOCATIONS. IF YOU INSTALL THE SLEEVES IN THE INCORRECT LOCATIONS, VIBRATION CAN CAUSE DAMAGE TO COMPONENTS OF LANDING GEAR.

(3) Loosely install the bolts [16], washers [17], and straight sleeve [21] to hold the shimmy damper assembly [1] to the upper torsion link [18].

NOTE: The upper torsion link has a straight sleeve and the lower torsion link has a tapered sleeve. The straight sleeve is longer than the tapered sleeve.

SUBTASK 32-11-81-420-014

CAUTION MAKE SURE THAT YOU INSTALL THE SLEEVES IN THE CORRECT LOCATIONS. IF YOU INSTALL THE SLEEVES IN THE INCORRECT LOCATIONS, VIBRATION CAN CAUSE DAMAGE TO COMPONENTS OF LANDING GEAR.

(4) Do this task: Main Landing Gear Torsion Link Connection, TASK 32-11-51-400-803.

(a) When you connect the lower torsion link [7] and upper torsion link [18], do not tighten the apex nut [9] completely or install the bolt [8] for the apex nut [9].

SUBTASK 32-11-81-420-015

(5) Tighten the bolts [16] for the shimmy damper assembly [1], in a cross pattern, to 650 ±65 in-lb (73.4 ±7.4 N·m).

(a) Install MS20995NC32 lockwire, G50347, on the bolts [16] with the double-twist method.

SUBTASK 32-11-81-420-016

(6) Tighten the apex nut [9] to 54 ±4 ft-lb (73.2 ±5.4 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Advance the apex nut [9] to the nearest castellation that aligns with the hole in the apex pin [19].

(b) Tighten the apex nut [9] to a final torque of 150 ft-lb (203.4 N·m) maximum.

(c) Install the bolt [8], washer [10], washer [22], nut [12], and new cotter pin [11].

SUBTASK 32-11-81-020-018

(7) If the new shimmy damper assembly [1] does not have a reducer [3] and new packing [4] installed in the manifold assembly [2], do these steps:

(a) Remove the plug from the port in the manifold assembly [2], if one is installed.

(b) Lubricate the reducer [3] and new packing [4] with hydraulic fluid, D00153.

(c) Install the new packing [4] on the reducer [3].

(d) Install the reducer [3] in the manifold assembly [2].

SUBTASK 32-11-81-480-023

(8) Remove the cap on the reducer [3] , if one is installed.

SUBTASK 32-11-81-420-017

(9) Connect the hydraulic line to the reducer [3] on the shimmy damper assembly [1].

(a) Tighten the B-nut on the hydraulic line to 140 ±7 in-lb (16 ±1 N·m).

SUBTASK 32-11-81-420-020

(10) Install the wheel and tire assembly (TASK 32-45-11-400-801).

SUBTASK 32-11-81-080-001

(11) Remove the axle jacks from the main landing gear (TASK 07-10-03-580-801).

G. Main Landing Gear Shimmy Damper Installation Test

SUBTASK 32-11-81-420-018

(1) Do this task: Hydraulic Shimmy Damper Bleeding, TASK 32-11-81-870-801.

(a) Make sure that there are no leaks at the hydraulic connection when you bleed the shimmy damper assembly [1].

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-11-81-860-012

(1) For the A and B hydraulic systems, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

TASK 32-11-81-870-801

4. Hydraulic Shimmy Damper Bleeding

(Figure 401)

A. General

(1) This task has instructions to bleed the hydraulic shimmy damper.

(2) The hydraulic shimmy damper has an internal piston with areas for hydraulic fluid on the two sides of the piston. To bleed the shimmy damper it is necessary to remove the air from the two fluid areas.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Consumable Materials

Reference	Description	Specification
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
5	Bleed plug	32-11-81-01-300	PTF ALL
6	Packing	32-11-81-01-305	PTF ALL

E. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

F. Hydraulic Shimmy Damper Bleeding

SUBTASK 32-11-81-480-032

WARNING

MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-11-81-860-013

(2) If the bleed plugs [5] need new packings [6], do these steps:

(a) For the A hydraulic system, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

(b) Remove lockwire from the bleed plugs [5].

(c) Remove the bleed plug [5] from the shimmy damper assembly [1].

NOTE: Hydraulic return pressure that remains after you have depressurized system A will force fluid out the bleed port when you remove the bleed plug [5].

(d) Install the new packing [6] on the bleed plug [5].

(e) Install the bleed plug [5] in its location on the shimmy damper assembly [1].

SUBTASK 32-11-81-790-001

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) For the hydraulic system (system A), do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-11-81-480-027

(4) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-11-81-840-001

(5) Do the steps that follow to bleed the air from the shimmy damper assembly [1] at the two bleed plugs [5]:

(a) Bleed one side of the shimmy damper assembly [1] piston:

NOTE: The shimmy damper assembly [1] hydraulic pressure is supplied from the return lines of hydraulic system (system A), to bleed the shimmy damper assembly [1] it is necessary to cycle the landing gear control lever.

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

1) Move the landing gear control handle to the UP position.
2) Open one of the two bleed plugs [5].

NOTE: You can open either the left or the right bleed plug [5].

3) Move the landing gear control handle to the DN position.
4) Close the bleed plug [5].

(b) Continue to move the landing gear control handle and open and close the bleed plug [5] as stated above until the air is gone from the fluid flow.

(c) Bleed the other side of the shimmy damper assembly [1] piston through the bleed plug [5] that remains:

1) Do the steps that follow to bleed the shimmy damper assembly [1] from the other bleed plug [5]:
   a) Move the landing gear control handle to the UP position.
   b) Open the other bleed plug [5].
   c) Move the landing gear control handle to the DN position.
   d) Close the bleed plug [5].

(d) Continue to move the landing gear control handle and open and close the bleed plug [5] as stated above until the air is gone from the fluid flow.

(e) Install MS20995C32 lockwire, G01048 on the bleed plugs [5] with the double-twist method.

SUBTASK 32-11-81-860-014

(6) For the hydraulic system (system A), do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-11-81-860-015

(7) Do a check of the PSEU for faults generated by the bleeding procedure and clear them.

NOTE: When you move the landing gear control lever to bleed the shimmy damper assembly [1], the gear lock sensors will not transition their state. This will cause faults to be stored in the PSEU and cause the MAINT light to come on.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN LANDING GEAR SHIMMY DAMPER AND TORSION LINK APEX ASSEMBLY – INSPECTION/CHECK

1. General

A. This procedure gives the task to do a dimensional inspection of the wear limits for the hydraulic shimmy damper and torsion link apex assembly of the main landing gear.

TASK 32-11-81-200-801

2. Main Landing Gear Hydraulic Shimmy Damper and Torsion Link Assembly Inspection

(Figure 601)

A. General

(1) This procedure gives the wear limits and repair data for the shimmy damper and the torsion link apex assembly of the main landing gear.

(2) This task does not remove or install the hydraulic shimmy damper. Refer to the Main Landing Gear Hydraulic Shimmy Damper - Removal/Installation for procedures to remove and install the hydraulic shimmy damper of the main landing gear.

(3) This task does not remove or install the torsion link apex assembly. Refer to the Main Landing Gear Torsion Link Assembly - Removal/Installation for procedures to remove and install the torsion link apex assembly.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-11-51-020-801	Main Landing Gear Torsion Link Disconnection (P/B 401)
32-11-51-400-803	Main Landing Gear Torsion Link Connection (P/B 401)

C. Tools/Equipment

Reference	Description
STD-1096	Micrometer - Depth, 0-1 inch, Readable to 1/1000 inch
STD-1097	Caliper - Vernier, 0-6 inch, Readable to 1/1000 inch

D. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Shimmy Damper and Torsion Link Assembly Inspection

SUBTASK 32-11-81-480-033

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, then do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-11-81-020-021

(2) To get access to the inspection areas of the torsion link assembly, do this task: Main Landing Gear Torsion Link Disconnection, TASK 32-11-51-020-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-11-81-020-022

(3) Do these steps to examine the parts of the main landing gear shimmy damper and the torsion link apex assembly for worn areas:

NOTE: The index numbers in Table 601 can be identified in Figure 601, view A-A.

(a) Remove the parts to be examined.

(b) Measure the parts as necessary with a micrometer (0-1 inch, readable to 1/1000 inch), STD-1096 or a vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

(c) Compare the dimensions that you measured with the permitted wear limits in Table 601.

(d) If the parts do not agree with the limits, then repair or replace the part as shown in Table 601.

Table 601/32-11-81-993-801 Main Landing Gear Shimmy Damper Inspection/Check

INDEX NO.	PART NAME	DIM.	DESIGN LIMITS DIAMETER		WEAR LIMITS		REPLACE WORN PART	REPAIR WORN PART	REPAIR INSTR
			MINIMUM INCHES (mm)	MAXIMUM INCHES (mm)	PERMITTED WEAR DIMENSIONS INCHES (mm)	MAXIMUM DIAMETER CLEARANCE INCHES (mm)			
[1]	BUSHINGS	WIDTH	2.4955 in. (63.386 mm)	2.5000 in. (63.500 mm)	2.4900 in. (63.246 mm)	N/A	X		*[1]
[2]	SPACER	WIDTH	0.3500 in. (8.890 mm)	0.3600 in. (9.144 mm)	0.3500 in. (8.890 mm)	N/A	X		
[3]	STRAIGHT SLEEVE	LENGTH	2.9900 in. (75.946 mm)	3.0000 in. (76.200 mm)	2.9900 in. (75.946 mm)	N/A	X		
[4]	BUSHING	WIDTH	2.6455 in. (67.196 mm)	2.6490 in. (67.285 mm)	2.6400 in. (67.056 mm)	N/A	X		*[2]
	TAPERED SLEEVE	LENGTH	2.6550 in. (67.437 mm)	2.6650 in. (67.691 mm)	2.6550 in. (67.437 mm)	N/A	X		
[5]	BOLT	OD	1.1850 in. (30.099 mm)	1.1865 in. (30.137 mm)	1.1828 in. (30.043 mm)	0.0077 in. (0.196 mm)		X	*[3]
	STRAIGHT SLEEVE	ID	1.1885 in. (30.188 mm)	1.1905 in. (30.239 mm)	1.1942 in. (30.333 mm)				
[6]	STRAIGHT SLEEVE	OD	1.4360 in. (36.474 mm)	1.4380 in. (36.525 mm)	1.4336 in. (36.413 mm)	0.0076 in. (0.193 mm)	X		
	BUSHING	ID	1.4390 in. (36.551 mm)	1.4402 in. (36.581 mm)	1.4436 in. (36.667 mm)		X		
[7]	BOLT	OD	1.1850 in. (30.099 mm)	1.1865 in. (30.137 mm)	1.1828 in. (30.043 mm)	0.0077 in. (0.196 mm)		X	*[3]
	TAPERED SLEEVE	ID	1.1885 in. (30.188 mm)	1.1905 in. (30.239 mm)	1.1942 in. (30.333 mm)		X		

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 601/32-11-81-993-801 Main Landing Gear Shimmy Damper Inspection/Check (Continued)

[8]	TAPERED SLEEVE	OD	1.4940 in. (37.948 mm)	1.4960 in. (37.998 mm)	1.4916 in. (37.887 mm)	0.0066 in. (0.168 mm)	X		
	BUSHING	ID	1.4980 in. (38.049 mm)	1.4992 in. (38.080 mm)	1.5036 in. (38.191 mm)		X		
[9]	DAMPER STROKE	GAP	0.0000 in. (0.000 mm)	0.5045 in. (12.814 mm)	N/A	1.0090 in. (25.629 mm)	X		*[4]
[10]	LOWER TORSION LINK GAP	GAP	0.0060 in. (0.152 mm)	0.0195 in. (0.495 mm)	N/A	0.0250 in. (0.635 mm)	X		*[5]

*[1] Width is the distance between the bushing flange faces of the upper torsion link.

*[2] Width is the distance between the bushing flange faces of the lower torsion link.

*[3] The worn part is repairable. Refer to the SOPM.

*[4] The gap can have a range within the limits shown and is dependent on the position of the shimmy damper piston along its stroke.

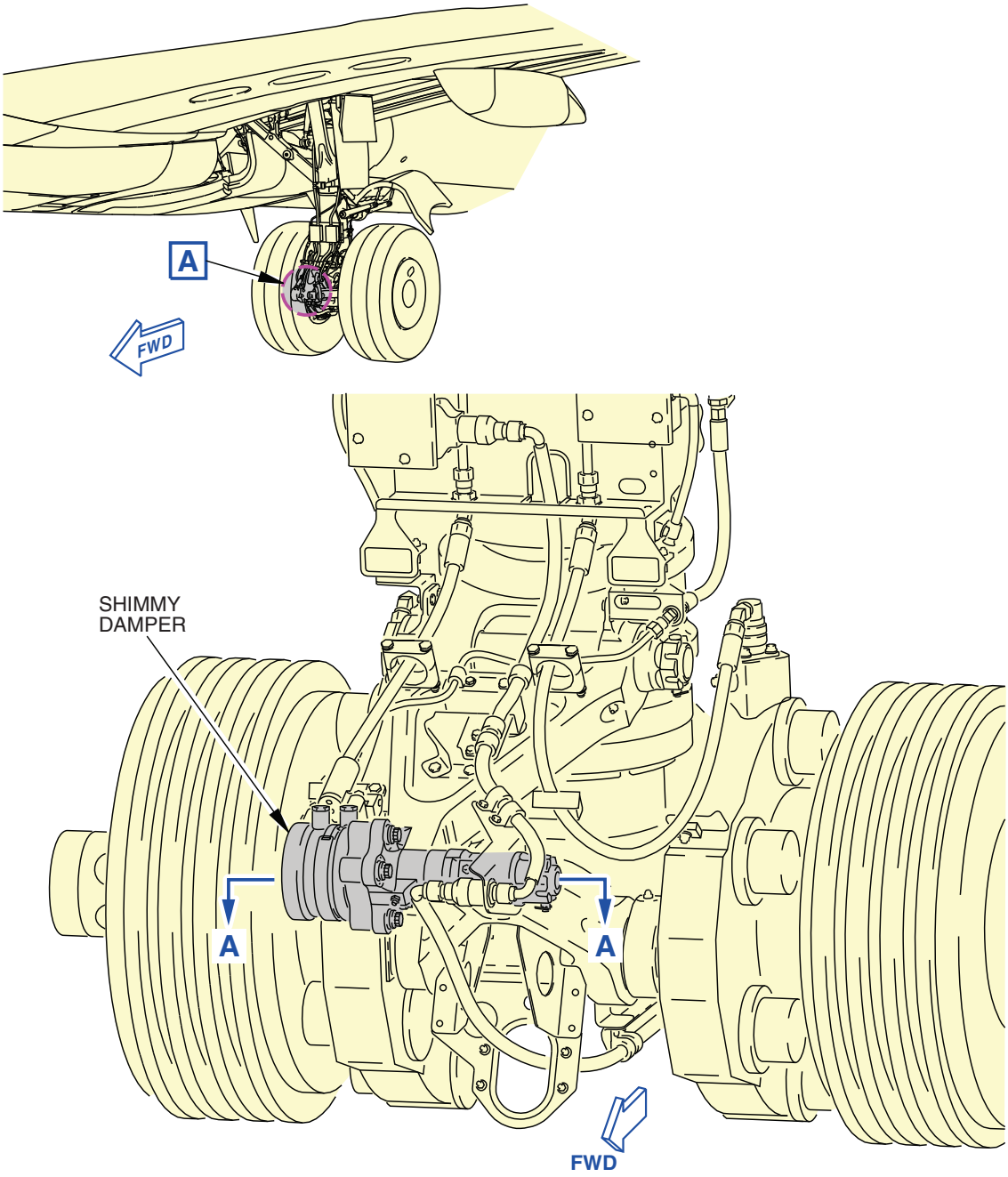
*[5] The maximum gap is measured with a feeler gage.

SUBTASK 32-11-81-420-023

(4) Do this task: Main Landing Gear Torsion Link Connection, TASK 32-11-51-400-803.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



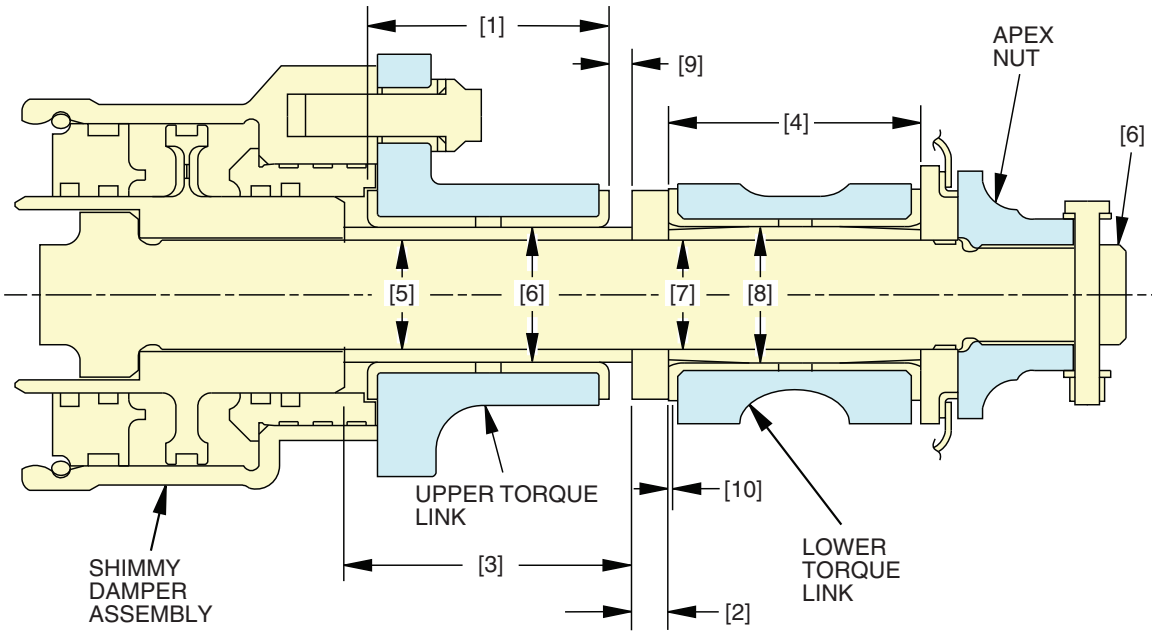
LEFT MAIN LANDING GEAR
(RIGHT MAIN LANDING GEAR IS EQUIVALENT)

A

3077131 S0000834705_V1

Main Landing Gear Shimmy Damper Inspection/Check
Figure 601/32-11-81-990-805 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



A-A

3077140 S0000834707_V1

Main Landing Gear Shimmy Damper Inspection/Check
Figure 601/32-11-81-990-805 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN LANDING GEAR FORWARD TRUNNION BEARING ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Main Landing Gear Forward Trunnion Bearing Assembly Removal

(2) Main Landing Gear Forward Trunnion Bearing Assembly Installation.

TASK 32-11-83-000-801

2. Main Landing Gear Forward Trunnion Bearing Assembly Removal

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
57-16-01-000-801	MLG Forward Trunnion Housing Assembly Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-4360	Wrench - Trunnion Spherical Bearing, MLG Part #: C32023-12 Supplier: 81205 Opt Part #: C32023-1 Supplier: 81205

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Prepare for the Removal

SUBTASK 32-11-83-020-008

(1) Do this task: MLG Forward Trunnion Housing Assembly Removal, TASK 57-16-01-000-801.

E. Main Landing Gear Forward Trunnion Bearing Assembly Removal

SUBTASK 32-11-83-020-009

(1) Do these steps to remove the bearing assembly [1] from the housing assembly [7]:

(a) Remove the lockwires from the nut retainer [6].

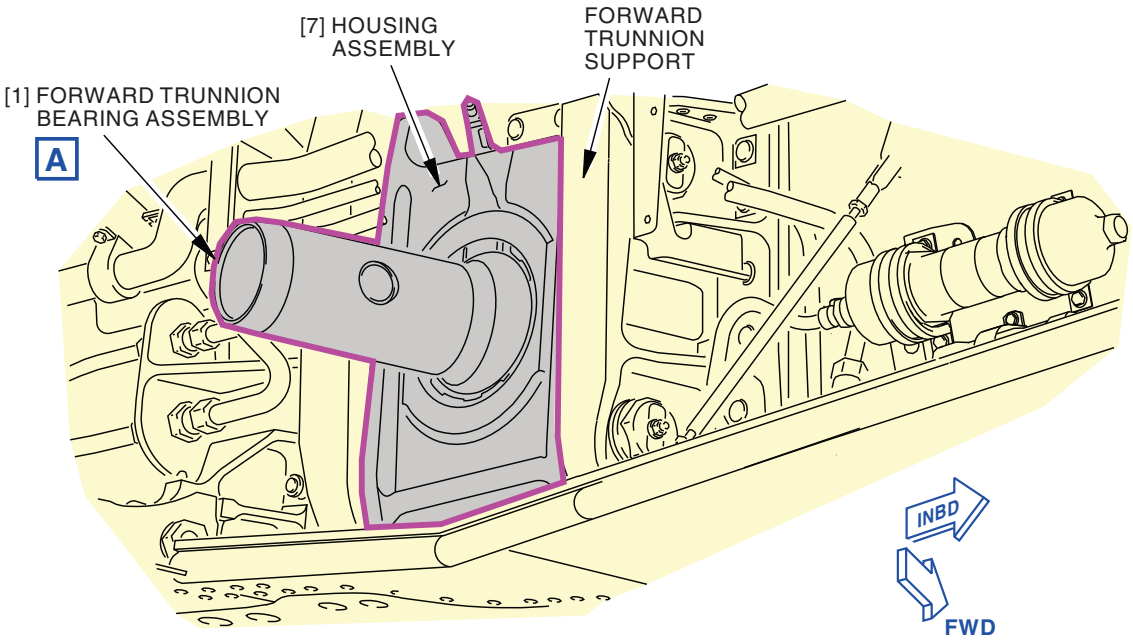
(b) Remove the nut retainer [6], pin assembly [4], and race assembly [5].

1) Use wrench, SPL-4360, or equivalent, to remove the nut retainer [6].

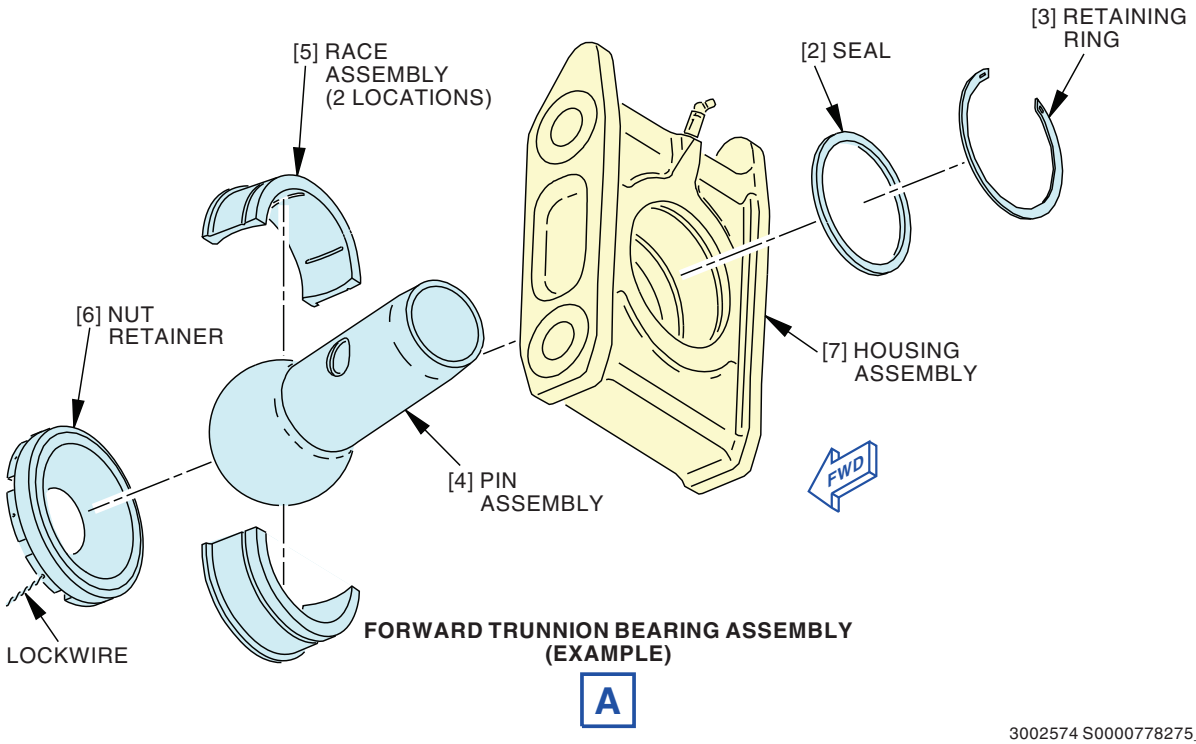
(c) Remove the retaining ring [3] and seal [2].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FORWARD TRUNNION BEARING ASSEMBLY (MAIN LANDING GEAR REMOVED)



FORWARD TRUNNION BEARING ASSEMBLY (EXAMPLE)

3002574 S0000778275_V1

Main Landing Gear Forward Trunnion Bearing Assembly Installation
Figure 401/32-11-83-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-11-83-400-801

3. Main Landing Gear Forward Trunnion Bearing Assembly Installation

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
57-16-01-400-801	MLG Forward Trunnion Housing Assembly Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-4360	Wrench - Trunnion Spherical Bearing, MLG Part #: C32023-12 Supplier: 81205 Opt Part #: C32023-1 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Bearing assembly	32-11-83-01-055	PTF ALL

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
734	Left Main Landing Gear
744	Right Main Landing Gear

F. Main Landing Gear Forward Trunnion Bearing Assembly Installation

SUBTASK 32-11-83-420-009

(1) Install the forward trunnion bearing assembly [1], do these steps:

(a) Install the pin assembly [4] and race assembly [5] into the housing assembly [7].

1) Apply a coating of grease, D00633, to these parts before assembly.

2) Make sure that the splines of the race assembly [5] are in full contact with the housing assembly [7] splines before you install the nut retainer [6].

(b) Install the nut retainer [6].

1) Apply grease, D00633, to the nut retainer [6] threads.

2) Tighten the nut retainer [6] to 50 ft-lb (68 N·m) - 75 ft-lb (102 N·m).

a) Use the wrench, SPL-4360, or equivalent, to tighten the nut retainer [6].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION

MAKE SURE THAT THE CLEARANCE BETWEEN THE RACE ASSEMBLY AND THE HOUSING ASSEMBLY IS LESS THAN 0.0050 INCH (0.1270 MM). IF THE CLEARANCE IS MORE THAN 0.0050 INCH (0.1270 MM) DURING THE INSTALLATION OF THE FORWARD TRUNNION BEARINGS, DAMAGE TO THE AIRPLANE CAN OCCUR.

3) Make sure that the clearance between the aft face of the race assembly [5] and the housing assembly [7] is 0.0050 in. (0.1270 mm) or less.

NOTE: It is not required to continue with the installation until a correct clearance is measured after tightening the nut retainer.

a) If a clearance of more than 0.0050 in. (0.1270 mm) is found, disassemble the nut retainer [6], pin assembly [4], and race assembly [5] to look for unwanted material between the mated parts.

(c) Install MS20995NC32 lockwire, G01912, onto the nut retainer [6] and housing assembly [7].

1) Install MS20995NC32 lockwire, G01912, with the double twist method.

(d) Apply a coating of grease, D00633, to the seal [2] and retaining ring [3].

(e) Install the seal [2] and retaining ring [3].

(f) Put grease, D00633, into the grease fitting on the housing assembly [7] until the grease can be seen at the aft edge of the bearing race.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-11-83-420-010

(1) Do this task: MLG Forward Trunnion Housing Assembly Installation, TASK 57-16-01-400-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN LANDING GEAR AXLE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the axle (referred to as the axle assembly [1]) for the main landing gear

(2) An installation of the axle for the main landing gear.

(3) A removal of the axle sleeve for the main landing gear

(4) An installation of the axle sleeve for the main landing gear.

TASK 32-11-85-000-801

2. Main Landing Gear Axle Removal

(Figure 401)

A. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-41-41-000-801	Main Landing Gear Brake Removal (P/B 401)
32-42-11-000-801	Antiskid Wheel Speed Transducer Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1864	Puller Equipment - Axle Sleeve Assembly, MLG Part #: C32032-125 Supplier: 81205 Opt Part #: C32032-98 Supplier: 81205

C. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Prepare for the Removal

SUBTASK 32-11-85-860-003

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-11-85-500-001

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(2) Do this task: Lift the Airplane on Jacks, TASK 07-10-01-580-801.

E. Main Landing Gear Axle Removal

SUBTASK 32-11-85-020-001

(1) Do this task: Main Landing Gear Brake Removal, TASK 32-41-41-000-801.

NOTE: This procedure will also remove the wheel and tire assemblies.

SUBTASK 32-11-85-020-019

(2) Do this taskAntiskid Wheel Speed Transducer Removal, TASK 32-42-11-000-801

NOTE: Make sure that you attach a label to the wires to indicate the left and right connectors.

SUBTASK 32-11-85-020-003

(3) Do these steps to remove the wiring assembly from the axle assembly [1]:

(a) Remove the lockwire from the nut [5].

(b) Loosen the elbow [4] for the electrical connector.

(c) Loosen the nut [5] to disconnect the electrical connector from the inner cylinder assembly [3].

(d) Pull the wiring assembly completely out of the axle assembly [1].

1) Make sure that you attach a label to the wires to indicate the left and the right connectors.

SUBTASK 32-11-85-020-004

CAUTION

DO NOT APPLY TOO MUCH FORCE WHEN YOU REMOVE THE BRAKE SLEEVE. IF YOU USE TOO MUCH FORCE, DAMAGE TO THE AIRPLANE AND THE SUPPORT EQUIPMENT CAN OCCUR.

(4) Use the puller equipment, SPL-1864, to remove the brake sleeve [2] from the axle assembly [1].

NOTE: Applied heat can decrease the force necessary for the removal.

SUBTASK 32-11-85-020-005

(5) Do these steps to remove the bracket [10]:

(a) Remove the cotter pin [9] from the pin [6].

(b) Remove the nut [8] and washer [7] from the pin [6].

(c) Remove the nuts [11] and washers [12] that hold the bracket [10] to the mounting studs.

(d) Slowly lower the bracket [10] to the ground.

SUBTASK 32-11-85-020-006

(6) Do these steps to remove the lower torsion link pin [16].

NOTE: It is necessary to remove the lower torsion link pin to get access to the pin that holds the axle to the inner cylinder assembly.

(a) Disconnect the lower torsion link [18] from the inner cylinder assembly [3] as follows:

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1) Remove pin [13], nut [14] and washer [15] from the cross bolt [17].
2) Remove the cross bolt [17] that holds the lower torsion link pin [16] to the spacer [19].
3) Hold the lower torsion link [18] and remove the lower torsion link pin [16] to disconnect the lower torsion link [18] from the inner cylinder assembly [3].
4) Remove the spacer [19] from the lower torsion link [18].

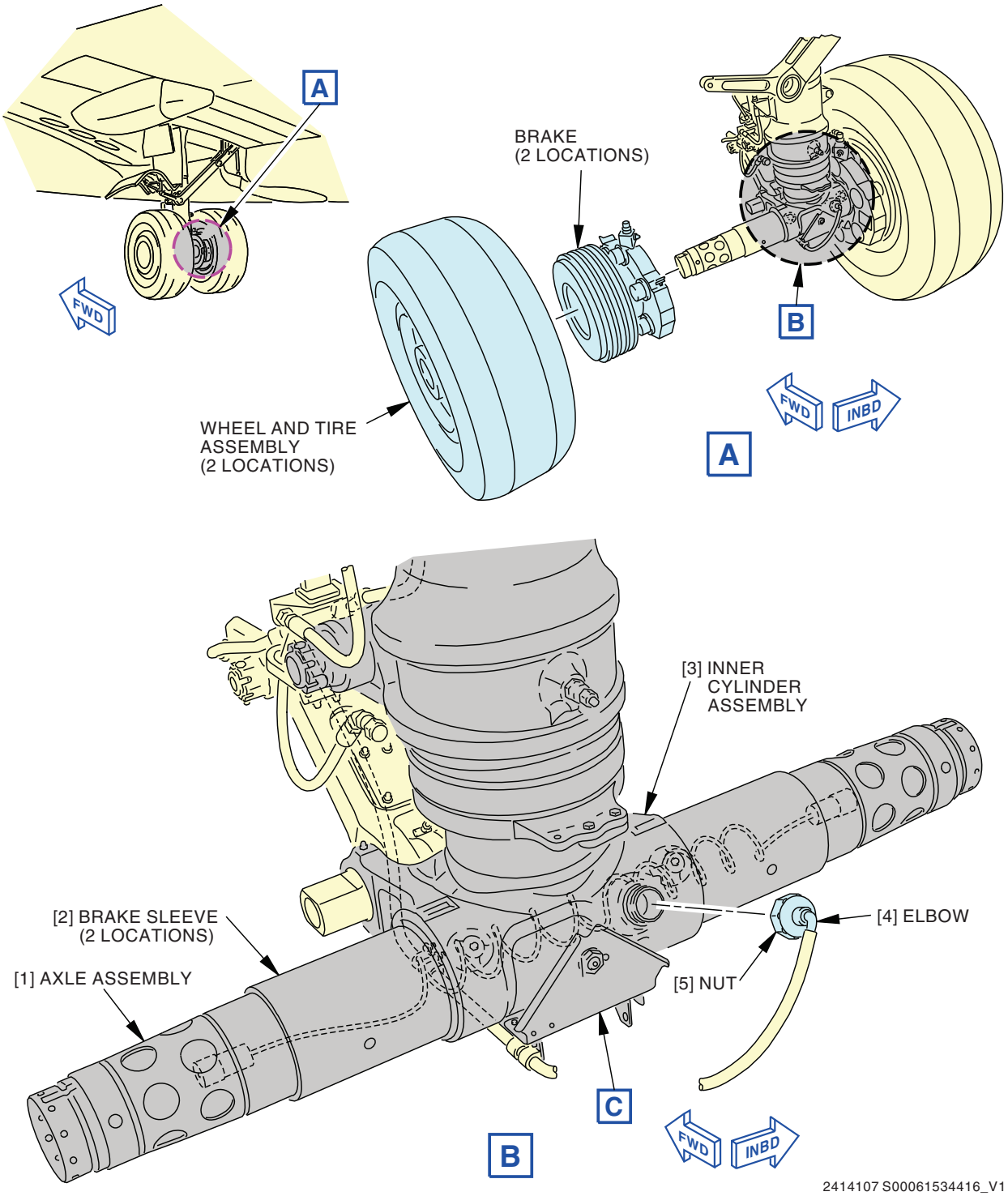
SUBTASK 32-11-85-020-007

(7) Do these steps to remove the axle assembly [1]:
   (a) Remove the pin [6] that holds the axle assembly [1] to the inner cylinder assembly [3].
   (b) Remove the axle assembly [1] from the airplane.

   NOTE: If the removal of the axle assembly [1] is difficult, applied heat may be necessary. Use puller assembly, puller equipment, SPL-1864.

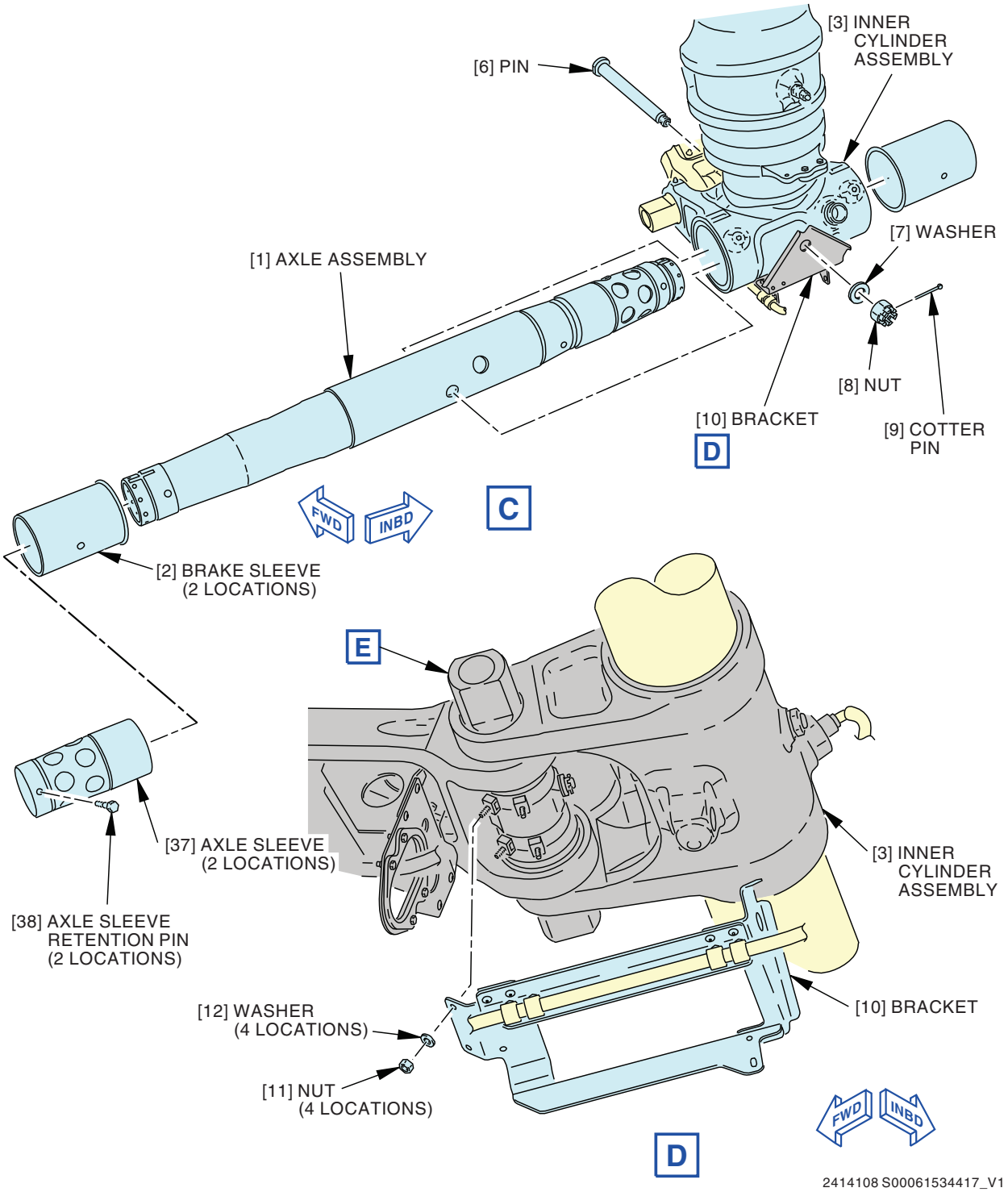
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



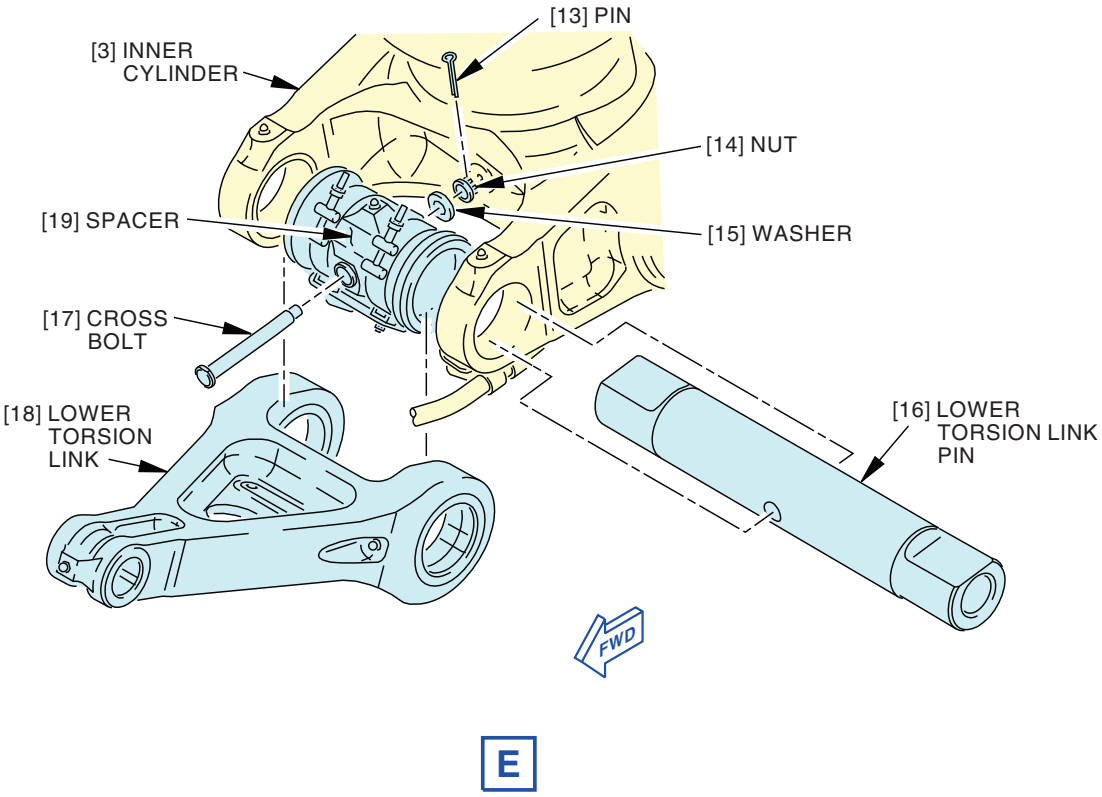
Main Landing Gear Axle Installation
Figure 401/32-11-85-990-801 (Sheet 1 of 3)

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Main Landing Gear Axle Installation
Figure 401/32-11-85-990-801 (Sheet 2 of 3)

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Main Landing Gear Axle Installation
Figure 401/32-11-85-990-801 (Sheet 3 of 3)

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TASK 32-11-85-420-801

3. Main Landing Gear Axle Installation

(Figure 401)

A. References

Reference	Title
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
12-21-11-640-802	Main Landing Gear Lower End Components Servicing (P/B 301)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-41-41-400-801	Main Landing Gear Brake Installation (P/B 401)
32-42-00-710-803	Transducer Operational Test (P/B 501)
32-42-00-720-803	Antiskid Valve Operational Test (P/B 501)
32-42-11-400-801	Antiskid Wheel Speed Transducer Installation (P/B 401)
32-46-00-730-801	Tire and Brake Monitoring System - System Test (P/B 501)

B. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33
G02166	Lockwire - MS20995NC20, Monel - 0.020 Inch (0.508 mm) Diameter	NASM20995
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Axle assembly	32-11-85-01-045	PTF ALL
9	Cotter pin	32-11-85-01-005	PTF ALL
13	Pin	32-11-51-01-015	PTF ALL

D. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Prepare for the Installation

SUBTASK 32-11-85-860-002

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Main Landing Gear Axle Installation

SUBTASK 32-11-85-420-001

(1) Lightly lubricate the bushing bores of the inner cylinder assembly [3] with grease, D00633.

SUBTASK 32-11-85-420-002

(2) Put the axle assembly [1] through the inner cylinder assembly [3].

SUBTASK 32-11-85-420-003

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(3) Apply a thin layer of corrosion inhibiting material, G50136, or Cor-Ban 27L Compound, G50237, to these items:

NOTE: Do not apply corrosion inhibiting material, G50136, or Cor-Ban 27L Compound, G50237, to the chrome plated surfaces of the pin [6].

(a) The head face of the pin [6]

(b) The threads and thread reliefs of the pin [6]

(c) The threads of the nut [8]

(d) The faces of the washer [7].

SUBTASK 32-11-85-020-008

(4) Remove all unwanted corrosion inhibiting material, G50136.

SUBTASK 32-11-85-420-004

(5) Lubricate the chrome plated surfaces of pin [6] with grease, D00633.

SUBTASK 32-11-85-420-005

(6) Align the hole on the axle assembly [1] with the holes on the inner cylinder assembly [3] for the installation of the pin [6].

SUBTASK 32-11-85-420-006

(7) Put the pin [6] through the inner cylinder assembly [3] and axle assembly [1].

SUBTASK 32-11-85-420-007

(8) Do these steps to install the lower torsion link pin [16].

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin layer of corrosion inhibiting material, G50136, or Cor-Ban 27L Compound, G50237, to these items:

1) The threads of nut [14].

2) The face of washer [15].

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(b) Remove all excess corrosion inhibiting material, G50136.

(c) Lubricate the chrome plated surfaces of the lower torsion link pin [16] and cross bolt [17] with grease, D00633.

(d) Remove all unwanted grease, D00633.

(e) Connect the lower torsion link [18] to the inner cylinder assembly [3]:

1) Put the lower torsion link [18] in its position on the inner cylinder assembly [3].

2) Put the lower torsion link pin [16] through the clevis on the inner cylinder assembly [3] and spacer [19].

3) Align the hole in the lower torsion link pin [16] with hole in the spacer [19] for the installation of the cross bolt [17].

4) Put the cross bolt [17] through the spacer [19] and lower torsion link pin [16].

5) Install the washer [15] and nut [14] on the cross bolt [17].

6) Tighten the nut [14] to 160 in-lb (18.1 N·m) - 190 in-lb (21.5 N·m).

7) Install the new pin [13] in the cross bolt [17].

a) If it is necessary, loosen the nut [14] to the nearest castellation to align the hole for the new pin [13].

SUBTASK 32-11-85-420-008

(9) Put the bracket [10] in its position on the pin [6] for attachment to the mounting studs.

SUBTASK 32-11-85-420-009

(10) Install the washers [12] and nuts [11] that hold the bracket [10] to the mounting studs.

(a) Tighten the nuts [11] with your hand.

SUBTASK 32-11-85-420-010

(11) Install the washer [7] and nut [8] on the pin [6].

SUBTASK 32-11-85-420-011

(12) Tighten the nut [8] to 160 in-lb (18.1 N·m) - 190 in-lb (21.5 N·m).

(a) If it is necessary, loosen the nut [8] to the nearest castellation to align the holes for the new cotter pin [9].

SUBTASK 32-11-85-420-012

(13) Install the new cotter pin [9] in the pin [6].

SUBTASK 32-11-85-640-001

(14) Do these steps to install the brake sleeve [2]:

(a) Apply a light coating of grease, D00633, to the lands of the axle assembly [1] and axle assembly axle sleeve, where the brake sleeve [2] inside diameters set.

NOTE: Avoid use of too much grease, D00633, on the brake sleeve [2].

(b) Heat the brake sleeve [2] to 375 ±25°F (191 ±14°C).

(c) Install the brake sleeve [2] on the axle assembly [1].

(d) Remove all unwanted grease, D00633.

SUBTASK 32-11-85-420-013

(15) Do these steps to install the wiring assembly in the axle assembly [1]:

(a) Make sure that you connect the correct wires to the correct in-axle assembly.

(b) Before you connect the electrical connector to the inner cylinder assembly [3], examine the electrical connector for corrosion.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING

DO THE NEXT TWO STEPS BELOW IF THE AIRPLANE OPERATES WHERE DEICING FLUID THAT CONTAINS POTASSIUM FORMATE OR POTASSIUM ACETATE. ALSO, DO THE STEPS FOR ALL AIRPLANES THAT YOU FOUND CORROSION IN THE ELECTRICAL CONNECTORS IN THE MAIN WHEEL WELL. THE ELECTRICAL CONNECTORS ARE IN A SYSTEM THAT IS NECESSARY FOR SAFE FLIGHT. IF YOU DO NOT OBEY, AN IN-FLIGHT EMERGENCY CAN OCCUR.

(c) If there was corrosion, refer to SWPM 20-60-03 to correct the problem.

(d) Apply the D5026NS or ZC-026 compound, G50171, to the electrical connector (SWPM 20-60-03).

(e) Push the wiring through the hole on the inner cylinder assembly [3] and into the axle assembly [1].

(f) Tighten the nut [5] to 70 in-lb (7.9 N·m) - 90 in-lb (10.2 N·m) to connect the electrical connector to the inner cylinder assembly [3].

(g) Tighten the elbow [4] to 45 in-lb (5.1 N·m) - 55 in-lb (6.2 N·m).

(h) Install MS20995NC20 lockwire, G02166, on the nut [5] (TASK 20-10-44-400-801).

SUBTASK 32-11-85-420-014

(16) Install the in-axle assemblies (TASK 32-42-11-400-801).

(a) Make sure that the correct wires connected to the correct in-axle assembly.

SUBTASK 32-11-85-420-015

(17) Do this task: Main Landing Gear Brake Installation, TASK 32-41-41-400-801.

NOTE: This procedure will also install the wheel and tire assemblies.

SUBTASK 32-11-85-600-001

(18) For the grease fittings on the upper torsion link and lower torsion link [18], do this task: Main Landing Gear Lower End Components Servicing, TASK 12-21-11-640-802.

SUBTASK 32-11-85-720-004

(19) Do this task: Antiskid Valve Operational Test, TASK 32-42-00-720-803.

SUBTASK 32-11-85-710-001

(20) Do a test of the transducer: Transducer Operational Test, TASK 32-42-00-710-803.

SUBTASK 32-11-85-730-001

(21) Do this task: Tire and Brake Monitoring System - System Test, TASK 32-46-00-730-801.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-11-85-580-001

CAUTION

MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(1) Do this task: Lower the Airplane off Jacks, TASK 07-10-01-580-802.

——— END OF TASK ———

TASK 32-11-85-000-802

4. Main Landing Gear Axle Sleeve Removal

(Figure 401)

A. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-41-41-000-801	Main Landing Gear Brake Removal (P/B 401)
32-42-11-000-801	Antiskid Wheel Speed Transducer Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1864	Puller Equipment - Axle Sleeve Assembly, MLG Part #: C32032-125 Supplier: 81205 Opt Part #: C32032-98 Supplier: 81205

C. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Prepare for the Removal

SUBTASK 32-11-85-480-002

WARNING

MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-11-85-580-002

CAUTION

MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(2) Do this task: Lift the Airplane on Jacks, TASK 07-10-01-580-801.

E. Main Landing Gear Axle Sleeve Removal

SUBTASK 32-11-85-020-009

(1) Do this task: Main Landing Gear Brake Removal, TASK 32-41-41-000-801.

NOTE: This procedure will also remove the wheel and tire assemblies.

SUBTASK 32-11-85-020-010

(2) Remove the in-axle assemblies (Antiskid Wheel Speed Transducer Removal, TASK 32-42-11-000-801).

(a) Make sure you attach a label to the wires to indicate the left and right connectors.

SUBTASK 32-11-85-860-006

(3) Make sure that the wiring assembly is stowed inside the axle assembly [1].

SUBTASK 32-11-85-020-012

CAUTION

DO NOT APPLY TOO MUCH FORCE WHEN YOU REMOVE THE BRAKE SLEEVE. IF YOU USE TOO MUCH FORCE, DAMAGE TO THE AIRPLANE AND THE SUPPORT EQUIPMENT CAN OCCUR.

(4) Remove the brake sleeve [2] from the axle assembly [1] with the puller equipment, SPL-1864.

NOTE: Applied heat can decrease the force necessary for the removal.

SUBTASK 32-11-85-020-013

(5) Do these steps to remove the axle sleeve [37] from the axle assembly [1]:

(a) Remove the axle sleeve retention pin [38].

1) Use the puller equipment, SPL-1864, RC-50 assembly to push the axle sleeve retention pin [38] out of the axle assembly [1].

(b) Remove the axle sleeve [37] with the puller equipment, SPL-1864.

NOTE: Applied heat may be required for removal.

——— END OF TASK ———

TASK 32-11-85-420-802

5. Main Landing Gear Axle Sleeve Installation

(Figure 401)

A. References

Reference	Title
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
12-21-11-640-802	Main Landing Gear Lower End Components Servicing (P/B 301)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-41-41-400-801	Main Landing Gear Brake Installation (P/B 401)
32-42-11-400-801	Antiskid Wheel Speed Transducer Installation (P/B 401)

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B. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
37	Axle sleeve	32-11-85-01-035	PTF ALL

D. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Prepare for the Installation

SUBTASK 32-11-85-860-004

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

F. Main Landing Gear Axle Sleeve Installation

SUBTASK 32-11-85-420-016

(1) Do the steps that follow steps to install the axle sleeve [37]:

(a) Apply a generous coating of grease, D00633, to the lands of the axle assembly [1], where the axle sleeve [37] inside diameters sit.

(b) Heat the axle sleeve [37] to 375 ±25°F (191 ±14°C).

(c) Align the retention pin holes in the axle sleeve [37] and the axle assembly [1] before installing the axle sleeve [37] on the axle assembly [1].

NOTE: The axle sleeve [37] will rapidly shrink to normal size and seize the axle assembly [1] , preventing further pin alignment efforts.

SUBTASK 32-11-85-420-017

(2) Do the steps that follow to install the axle sleeve retention pin [38]:

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply corrosion inhibiting material, G50136 or Cor-Ban 27L Compound, G50237, to the retention pin hole in the axle assembly [1], and the hole in the axle sleeve [37].

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(b) Lightly press or tap the axle sleeve retention pin [38] to install the axle sleeve retention pin [38] into position.

NOTE: The head of the axle sleeve retention pin [38] must be flush or below the outside diameter of the axle sleeve [37].

SUBTASK 32-11-85-020-014

(3) Remove all unwanted corrosion inhibiting material, G50136 or Cor-Ban 27L Compound, G50237.

SUBTASK 32-11-85-420-018

(4) Do these steps to install the brake sleeve [2]:

(a) Apply a light coating of grease, D00633, to the lands of the axle assembly [1] and the axle assembly sleeve, where the brake sleeve [2] inside diameters sit.

NOTE: Avoid use of too much grease, D00633, on the brake sleeve [2].

(b) Heat the brake sleeve [2] to 375 ±25°F (191 ±14°C).

(c) Install the brake sleeve [2] on the axle assembly [1].

(d) Remove all unwanted grease, D00633.

SUBTASK 32-11-85-420-020

(5) Install the in-axle assemblies (Antiskid Wheel Speed Transducer Installation, TASK 32-42-11-400-801).

SUBTASK 32-11-85-420-021

(6) Do this task: Main Landing Gear Brake Installation, TASK 32-41-41-400-801.

NOTE: This procedure will also install the wheel and tire assemblies.

(a) Make sure that you connect the correct wires to the correct in-axle assembly.

SUBTASK 32-11-85-600-002

(7) For the grease fittings on the upper torsion link and the lower torsion link [18], do this task: Main Landing Gear Lower End Components Servicing, TASK 12-21-11-640-802.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-11-85-580-003

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(1) Do this task: Lower the Airplane off Jacks, TASK 07-10-01-580-802.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN LANDING GEAR AXLE - REPAIRS

1. General

A. This procedure contains these tasks:

(1) Main landing Gear Axle Repair

(2) Repair of the Axle Sleeve of the Main Landing Gear

(3) Repair of the Brake Sleeve of the Main Landing Gear

B. You can remove the cadium plating to inspect the base metal for heat damage on airplanes on which heavy braking gives sufficient heat to melt the cadium plating on the axle.

C. The repairs to the axle sleeves will be the blending out of scratches, scoring, and galling in the limits given.

TASK 32-11-85-300-801

2. Main Landing Gear Axle Repair

(Figure 801, Figure 802)

A. References

Reference	Title
05-51-07 P/B 201	HIGH ENERGY STOP/HEAT DAMAGE CONDITION - MAINTENANCE PRACTICES
32-11-85-000-802	Main Landing Gear Axle Sleeve Removal (P/B 401)
32-41-42-000-801	Main Landing Gear Brake Sleeve Removal (P/B 401)

B. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

C. Main Landing Gear Axle Repair

SUBTASK 32-11-85-200-001

(1) Refer to HIGH ENERGY STOP/HEAT DAMAGE CONDITION - MAINTENANCE PRACTICES, PAGEBLOCK 05-51-07/201 for axle heat damage inspection. If necessary, contact Boeing Engineering for an analysis of repair.

(2) Refer to Main Landing Gear Axle Sleeve Removal, TASK 32-11-85-000-802 and Main Landing Gear Brake Sleeve Removal, TASK 32-41-42-000-801.

SUBTASK 32-11-85-300-008

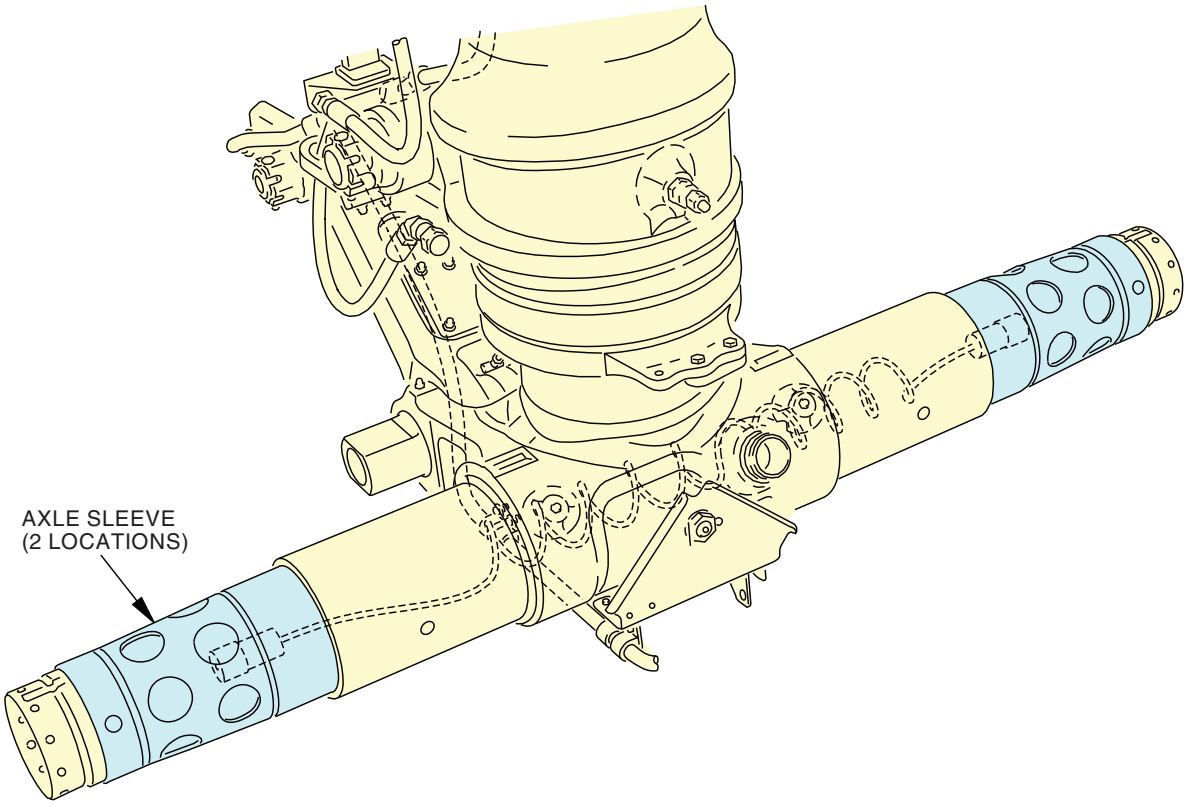
(3) Do a check of the diameter of the bearing supports after the axle sleeve installation.

(a) Make sure the outer diameter of the small bearing support is 4.3266 in. (109.8956 mm) to 4.3293 in. (109.9642 mm).

(b) Make sure the outer diameter of large bearing support is 4.7466 in. (120.5636 mm) to 4.7491 in. (120.6271 mm).

——— END OF TASK ———

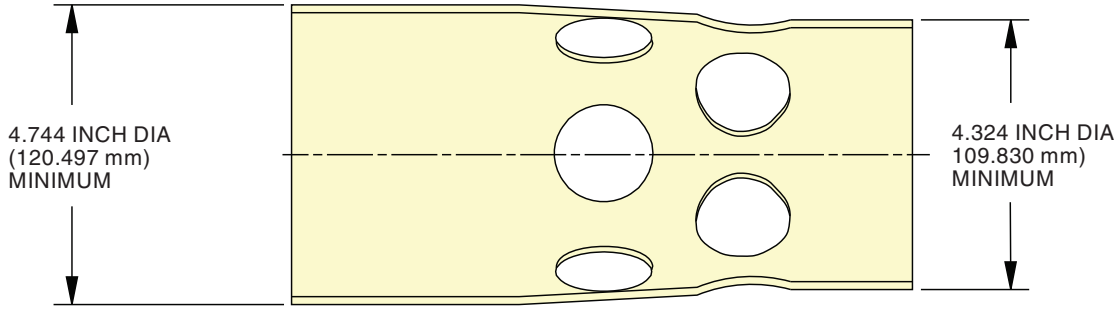
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Axle Sleeve Installation
Figure 801/32-11-85-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE: THE DIMENSIONS OF THE AXLE SLEEVES YOU WILL INSTALL WILL BE DIFFERENT THAN THE AXLE SLEEVES YOU WILL REMOVE BECAUSE OF THE INTERFERENCE FIT.

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Permitted Wear Limits of Installed Axle Sleeves
Figure 802/32-11-85-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-11-85-300-802

3. Repair of the Axle Sleeve of the Main Landing Gear

(Figure 801, Figure 802)

A. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

B. Procedure to Repair the Axle Sleeve of the Main Landing Gear.

SUBTASK 32-11-85-300-009

(1) Do a check on the wear limits of the axle sleeve bearing boss.

(a) Examine the diameters of the inboard and the outboard bearing boss for worn areas as shown on Figure 802.

1) If worn areas on the bearing boss is more than the specified wear limits and the clearance between the wheel bearing and the boss is more than 0.006 in. (0.152 mm), you must remove the axle sleeve and replace or rework.

2) If worn areas on the bearing boss is more than the specified wear limits and the clearance between the wheel bearing and the boss is 0.006 in. (0.152 mm) or less, the axle sleeve can continue to be in service for 65 hours flying time before removal for replacement or repairs.

SUBTASK 32-11-85-300-010

(2) Repair the local damage on the bearing surfaces.

(a) Remove the sharp edges and make a smooth finish on the damage to a 0.50 in. (12.70 mm) radius.

NOTE: Do not reduce the wall thickness by more than 0.025 in. (0.635 mm) or more than 30 % of the total bearing area of any 90 degree segment when you rework.

(b) If damage is on the anti-rotation bolt hole, replace the sleeve or contact BOEING to make an analysis of a possible repair.

——— END OF TASK ———

TASK 32-11-85-300-803

4. Repair of the Brake Sleeve of the Main Landing Gear

(Figure 803, Figure 804)

A. References

Reference	Title
32-41-42-000-801	Main Landing Gear Brake Sleeve Removal (P/B 401)
32-41-42-420-801	Main Landing Gear Brake Sleeve Installation (P/B 401)

B. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

C. Procedure to Repair the Brake Sleeve of the Main Landing Gear.

SUBTASK 32-11-85-220-001

(1) Do a check of the wear limits of the sleeve.

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(a) Measure the sleeve diameter in the worn and unworn areas of the sleeve.

1) Make sure that the difference in the sleeve diameter is 0.003 in. (0.076 mm) or less.

2) Make sure that the base metal is not exposed.

SUBTASK 32-11-85-020-018

(2) Replace the Brake Sleeve if the worn areas are more than specified or the base metal is exposed. Contact Boeing Engineering to perform an analysis of repair.

(a) Do the replacement of the brake sleeve in less than 65 hours of flight time, if the worn areas are more than the specified limits.

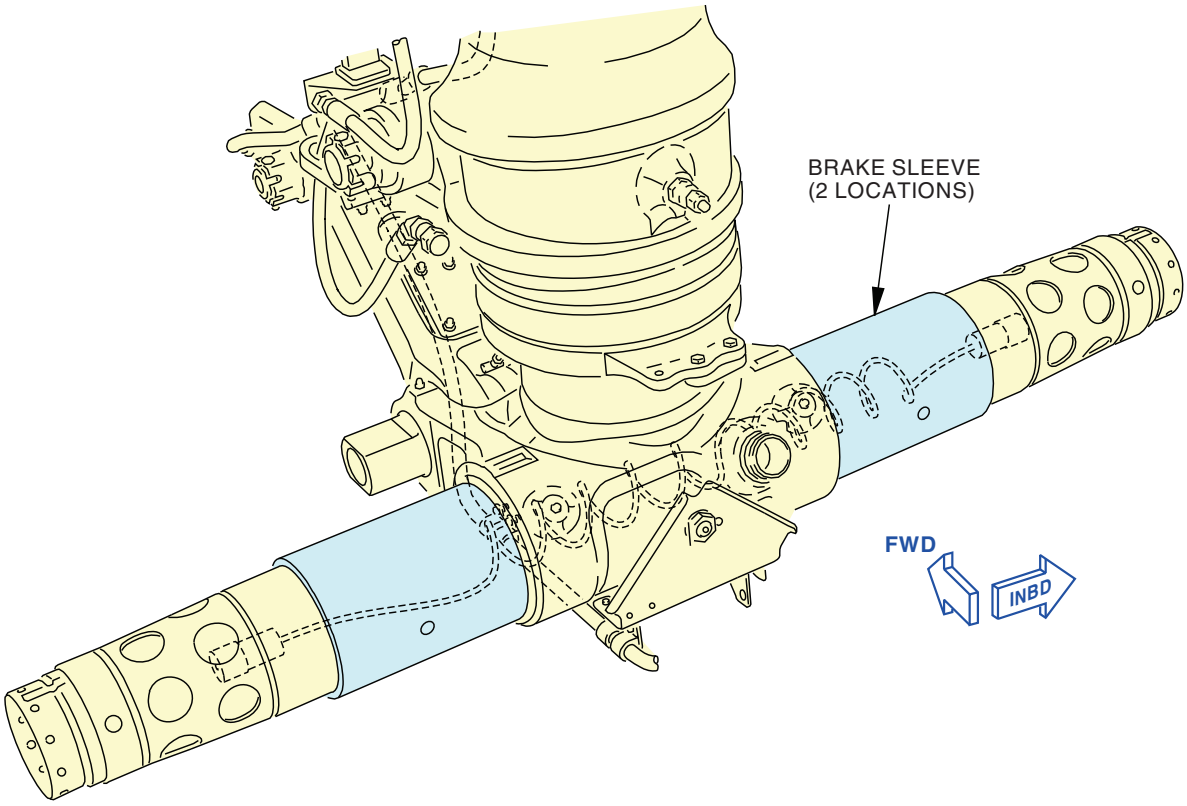
These are the tasks:

Main Landing Gear Brake Sleeve Removal, TASK 32-41-42-000-801.

Main Landing Gear Brake Sleeve Installation, TASK 32-41-42-420-801.

——— END OF TASK ———

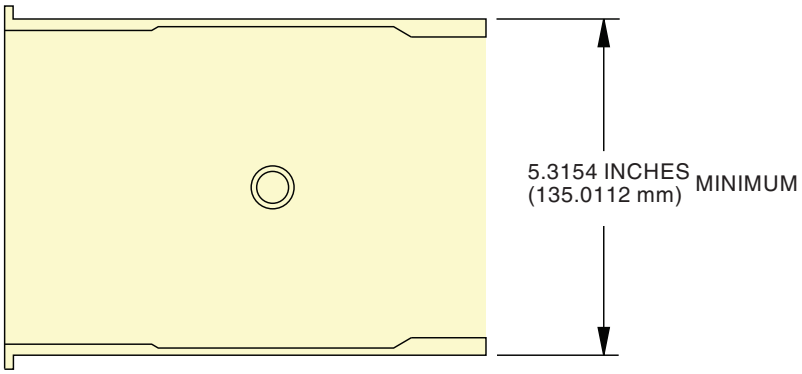
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414112 S00061534430_V1

Brake Sleeve Installation
Figure 803/32-11-85-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE: THE DIMENSIONS OF THE BRAKE SLEEVE YOU WILL INSTALL WILL BE DIFFERENT THAN THE BRAKE SLEEVE YOU WILL REMOVE BECAUSE OF THE INTERFERENCE FIT.

2414113 S00061534431_V1

Permitted Wear LImits of Installed Brake Sleeve
Figure 804/32-11-85-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN LANDING GEAR DOWNLOCK STRUT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the reaction link assembly [6] and the link arm assembly for the main landing gear

(2) An installation of the reaction link assembly [6] and the link arm assembly for the main landing gear.

B. The assembly of the reaction link assembly [6] and the link arm assembly are referred to as the downlock strut [5] in this procedure.

TASK 32-11-89-000-801

2. Main Landing Gear Downlock Strut Removal

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
07-10-03-580-801	Lift the Airplane on Axle Jacks-Handling (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-32-91-000-801	Main Gear Downlock Springs Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1360	Wrench Adapter, C32029-4 or C32029-94 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1369	Thread Protector , C32029-13 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-12498	Release Lever - Downlock, MLG downlock Part #: C32052-1 Supplier: 81205
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

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D. Prepare for the Removal

SUBTASK 32-11-89-480-003

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-11-89-860-001

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(2) Lift the airplane on jacks until the main landing gear wheels are off of the ground (TASK 07-10-01-580-801).

SUBTASK 32-11-89-860-002

(3) Remove the power from hydraulic system A and B (TASK 29-11-00-860-805).

SUBTASK 32-11-89-020-001

(4) Do this task: Main Gear Downlock Springs Removal, TASK 32-32-91-000-801.

E. Main Landing Gear Downlock Strut Removal

SUBTASK 32-11-89-020-002

(1) Do these steps to disconnect the hydraulic tubes [2] from the downlock actuator [4]:

(a) Put tags on the hydraulic tubes [2] to prevent the crossing of the hydraulic tubes [2] during the installation.

(b) Disconnect the hydraulic tubes [2] from the downlock actuator [4].

(c) Install the caps/plugs, STD-14686, on the hydraulic tubes [2].

(d) Remove the unions [28] from the downlock actuator [4].

(e) Remove the packings [3] from the ports of the downlock actuator [4] and discard them.

(f) Install the caps/plugs, STD-14686, in the ports of the downlock actuator [4].

SUBTASK 32-11-89-020-003

(2) Use the downlock release lever, SPL-12498, to manually release the Main Landing Gear (MLG) downlock mechanism.

SUBTASK 32-11-89-020-004

(3) Do these steps to disconnect the rod end of the downlock actuator [4] from the downlock strut [5]:

(a) Remove the spacer [7] from the rod end pin [8].

(b) From the forward side, remove the rod end pin [8] that holds the rod end of the downlock actuator [4] to the downlock strut [5].

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(c) Remove the spacers [9] and washers [10].

(d) Slowly lower the rod end of the downlock actuator [4] to the upper end of the upper side strut.

(e) Use tape to attach the rod end of the downlock actuator [4] to the upper end of the upper side strut.

NOTE: This will prevent the downlock actuator from moving when the subsequent steps are done.

SUBTASK 32-11-89-020-005

(4) Do these steps to disconnect the electrical harnesses [11] from the downlock strut [5]:

(a) Remove the nuts [20], washers [19], and bolts [18] to disconnect the brackets [17] from the downlock strut [5].

1) Remove the brackets [17] with the sensors attached.

(b) Isolate the brackets [17] from the downlock strut [5].

SUBTASK 32-11-89-080-001

(5) Remove the downlock pin from the MLG in order to remove the downlock strut [5] (TASK 32-00-01-080-801).

SUBTASK 32-11-89-020-006

(6) Do these steps to disconnect the downlock strut [5] from the reaction link assembly [6]:

(a) Remove the pin [22] from the downlock link pin [12].

(b) Remove the nut [21] and washer [23] from the downlock link pin [12].

1) Use a wrench adapter, C32029-4 or C32029-94, SPL-1360, to hold the head of the downlock link pin [12] when the nut [21] is loosen.

(c) Remove the downlock link pin [12] that holds the downlock strut [5] to the reaction link assembly [6].

1) Use a thread protector, C32029-13, SPL-1369.

SUBTASK 32-11-89-020-007

(7) Do these steps to disconnect the downlock strut [5] from the upper side strut:

(a) Remove the pin [16] from the downlock link pin [13].

(b) Remove the nut [15] and washer [14] from the downlock link pin [13].

(c) Remove the downlock link pin [13].

1) Use a thread protector, C32029-13, SPL-1369.

SUBTASK 32-11-89-020-008

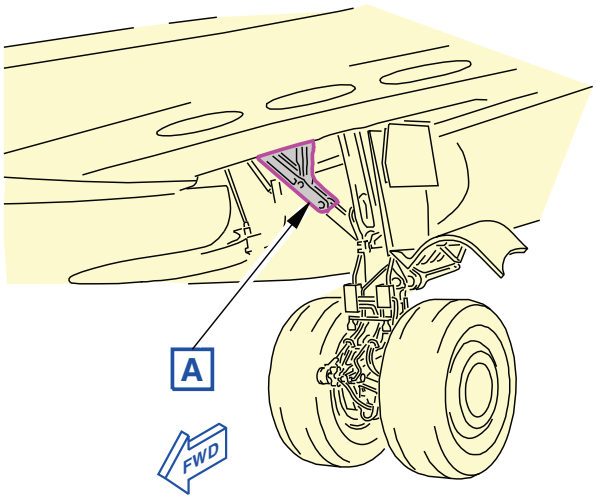
(8) Remove the downlock strut [5] from the airplane.

SUBTASK 32-11-89-580-001

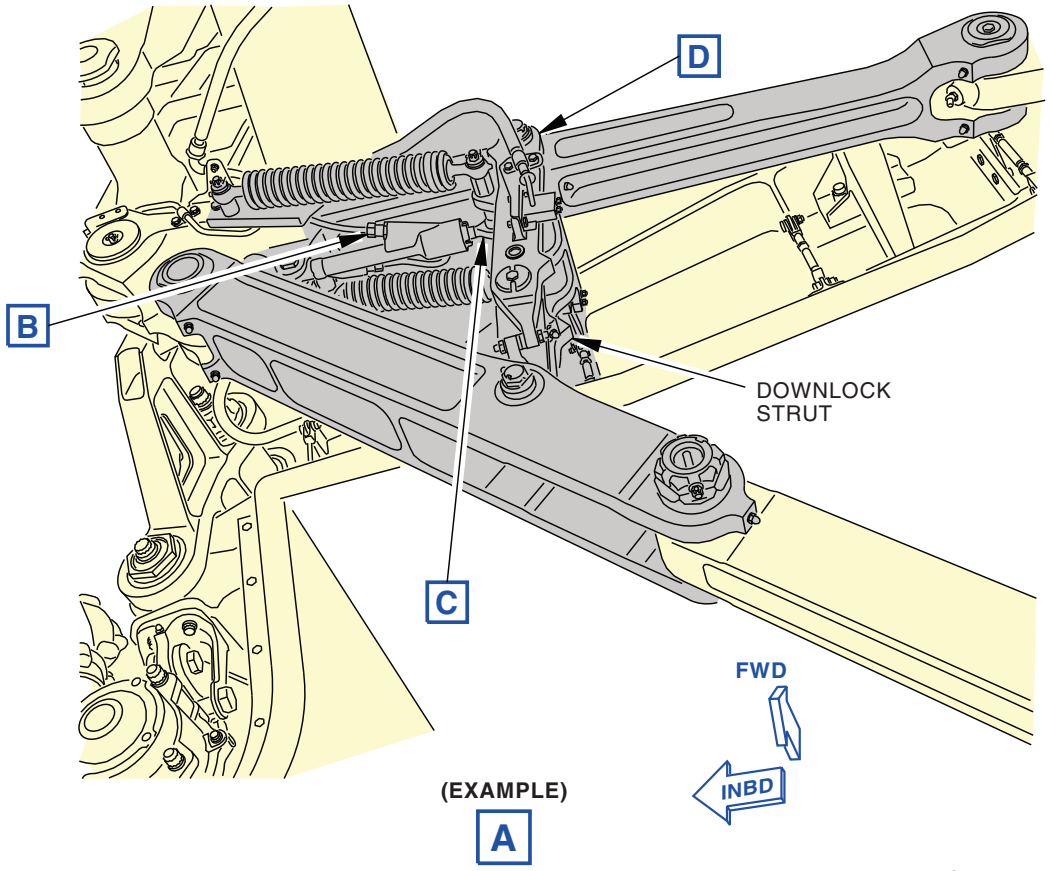
(9) If the removal of the downlock strut [5] is difficult, do this task to decrease the weight on the gear: Lift the Airplane on Axle Jacks-Handling, TASK 07-10-03-580-801.

——— END OF TASK ———

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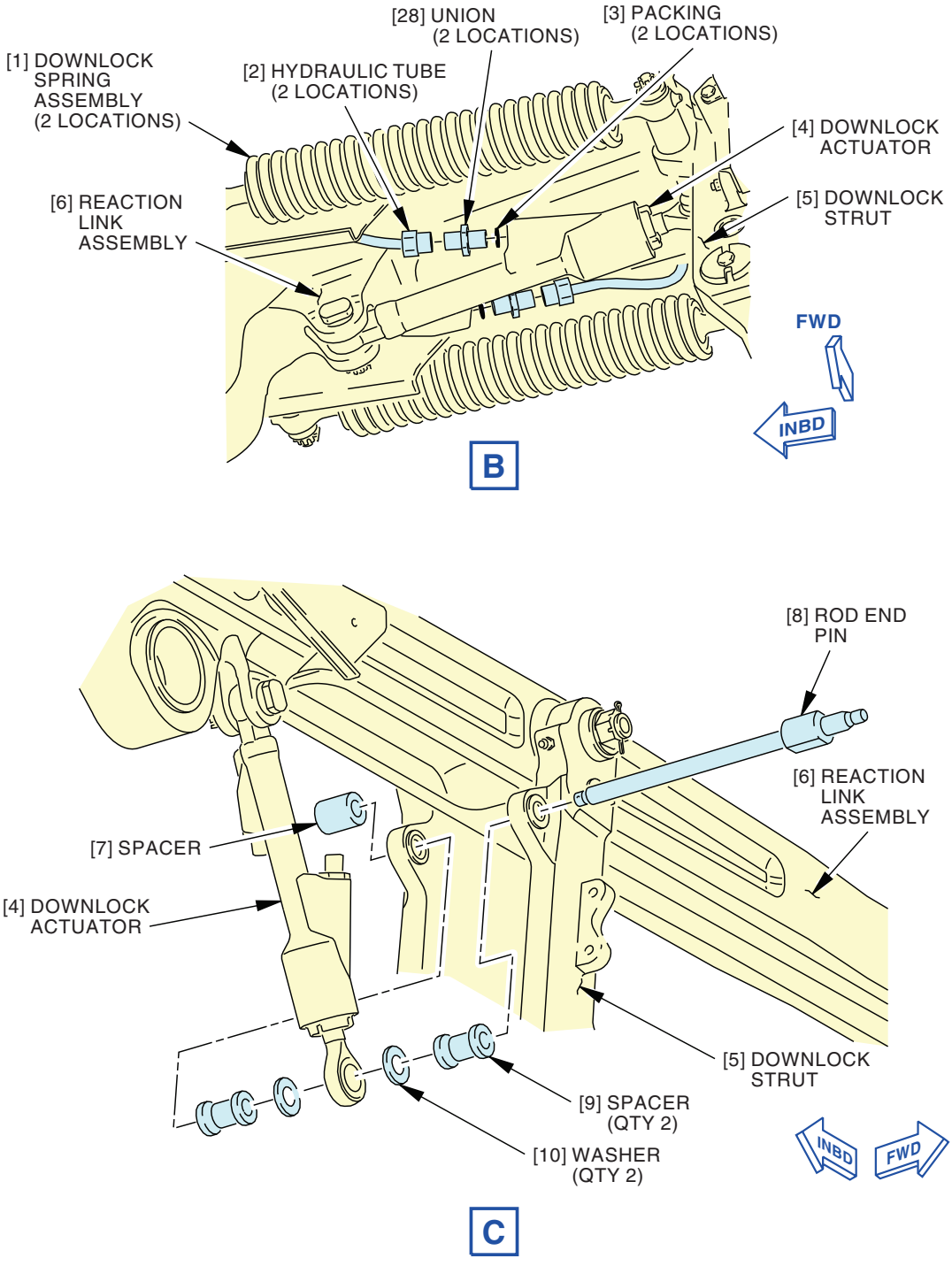
LEFT MAIN LANDING GEAR
(RIGHT MAIN LANDING GEAR IS OPPOSITE)



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Main Landing Downlock Strut Installation
Figure 401/32-11-89-990-801 (Sheet 1 of 3)

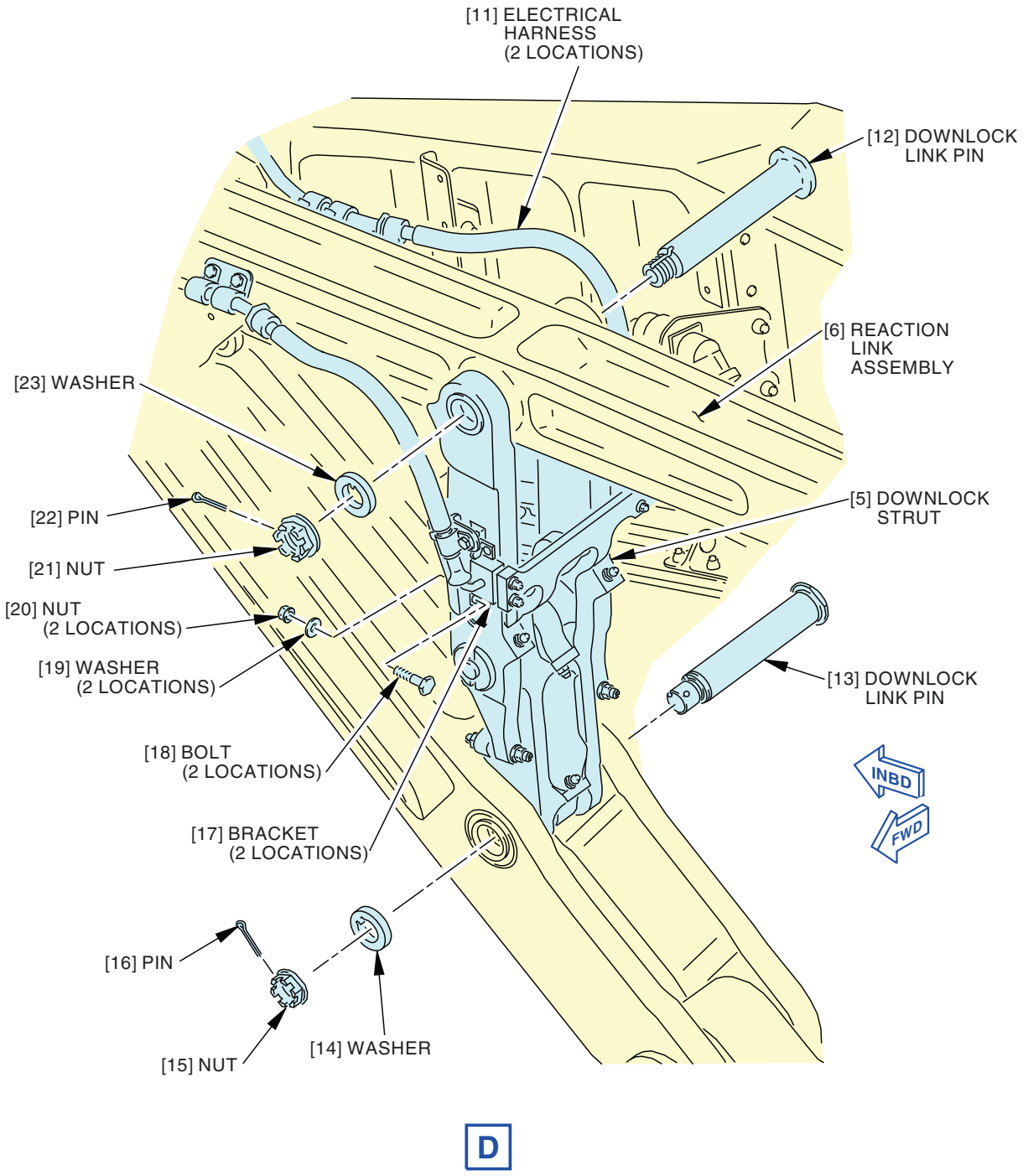
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2414115 S00061534436_V3

Main Landing Downlock Strut Installation
Figure 401/32-11-89-990-801 (Sheet 2 of 3)

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Main Landing Downlock Strut Installation
Figure 401/32-11-89-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-11-89-420-801

3. Main Landing Gear Downlock Strut Installation

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
12-21-11-640-801	Main Landing Gear Upper End Components Servicing (P/B 301)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-32-51-400-801	Main Gear Downlock Actuator Installation (P/B 401)
32-32-91-400-801	Main Gear Downlock Spring Installation (P/B 401)
32-61-31-220-801	Main Landing Gear Down-and-Locked Sensor Clearance Measurement (P/B 501)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1360	Wrench Adapter, C32029-4 or C32029-94 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-1369	Thread Protector , C32029-13 (included in C32029 kit) Part #: C32029-87 Supplier: 81205 Opt Part #: C32029-45 Supplier: 81205
SPL-20083	Wrench Adapter, C32029-55 or C32029-90 (included in C32029 kit) Part #: C32029-87 Supplier: 81205
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
D00633	Grease - Aircraft General Purpose	BMS3-33
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Main Landing Gear Downlock Strut Installation

SUBTASK 32-11-89-420-001

(1) Do these steps to connect the downlock strut [5] to the upper side strut:

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin layer of the corrosion inhibiting material, G50136, to these items:

1) The threads and thread reliefs of the downlock link pin [13]
2) The threads of the nut [15]
3) The faces of the washer [14].

(b) Do not apply corrosion inhibiting material, G50136, to the chrome plated surfaces.

(c) Remove any excess corrosion inhibiting material, G50136.

(d) Lubricate the chrome plated surfaces of the downlock link pin [13] with grease, D00633.

(e) Put the downlock strut [5] in its position for attachment to the upper side strut.

(f) Put the downlock link pin [13] through the upper side strut and the downlock strut [5].

1) Use a thread protector, C32029-13, SPL-1369 when you do this task.

(g) Install the washer [14] and nut [15] on the downlock link pin [13].

(h) Tighten the nut [15] to 480 ±120 in-lb (54 ±14 N·m).

1) Use a wrench adapter, C32029-55 or C32029-90, SPL-20083, to hold the head of the downlock link pin [13] when you tighten the nut [15].

(i) Install the new pin [16] in the downlock link pin [13].

1) If it is necessary, loosen the nut [15] to the nearest castellation that aligns with the holes for the new pin [16].

SUBTASK 32-11-89-420-002

(2) Do these steps to connect the downlock strut [5] to the reaction link assembly [6]:

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin layer of the corrosion inhibiting material, G50136, to these items:

1) The threads and thread reliefs of the downlock link pin [12]
2) The threads of the nut [21]
3) The faces of the washer [23].

(b) Do not apply corrosion inhibiting material, G50136, to the chrome plated surfaces.

(c) Remove all excess corrosion inhibiting material, G50136.

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(d) Lubricate the chrome plated surfaces of the downlock link pin [12] with grease, D00633.

(e) Put the downlock strut [5] in its position for attachment to the reaction link assembly [6].

(f) Put the downlock link pin [12] through the downlock strut [5] and the reaction link assembly [6].

1) Use a thread protector, C32029-13, SPL-1369, when you do this task.

(g) Install the washer [23] and nut [21] on the downlock link pin [12].

(h) Tighten the nut [21] to 480 ±120 in-lb (54 ±14 N·m).

1) Use a wrench adapter, C32029-4 or C32029-94, SPL-1360, to hold the head of the downlock link pin [12] when you tighten the nut [21].

(i) Install the new pin [22] in the downlock link pin [12].

1) If it is necessary, loosen the nut [21] to the nearest castellation that aligns with the holes for the new pin [22].

SUBTASK 32-11-89-640-001

(3) Do these steps to connect the rod end of the downlock actuator [4] to the downlock strut [5]:

(a) Lubricate the inside surface of the rod end pin [8] with grease, D00633.

1) Do not lubricate the downlock actuator [4] rod and head end bearing inside diameter.

(b) Put the rod end of the downlock actuator [4] in its position for attachment to the downlock strut [5].

1) If it is necessary, retract or extend the downlock actuator [4] to align the rod end of the downlock actuator [4] to the lug on the downlock strut [5].

(c) From the forward side, put the rod end pin [8] through the downlock strut [5], spacers [9], washers [10], and the rod end of the downlock actuator [4].

(d) Install the spacer [7] on the rod end pin [8].

(e) Remove all unwanted grease, D00633.

SUBTASK 32-11-89-420-003

(4) Do these steps to connect the hydraulic tubes [2] to the downlock actuator [4]:

(a) Lubricate these items with hydraulic fluid, D00153:

1) new packings [3]

2) unions [28].

(b) Remove the caps/plugs, STD-14686, from the ports on the downlock actuator [4].

(c) Install the new packings [3] and unions [28] in the downlock actuator [4].

(d) Remove the caps/plugs, STD-14686, from the hydraulic tubes [2].

(e) Connect the hydraulic tubes [2] to the downlock actuator [4].

(f) Tighten the B-nuts on hydraulic tubes [2] to 140 in-lb (16 N·m).

(g) Loosen the B-nuts on hydraulic tubes [2] and re-tighten them to 140 ±7 in-lb (16 ±1 N·m).

(h) Remove the tags from the hydraulic tubes [2].

SUBTASK 32-11-89-420-004

(5) Do this step to connect the electrical harnesses [11] to the downlock strut [5]:

(a) Install the nuts [20], washers [19], and bolts [18] to connect the brackets [17] to the downlock strut [5].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-11-89-820-001

(6) Do this task: Main Landing Gear Down-and-Locked Sensor Clearance Measurement, TASK 32-61-31-220-801

F. Main Landing Gear Downlock Strut Installation Test

SUBTASK 32-11-89-420-005

(1) Do this task: Main Gear Downlock Spring Installation, TASK 32-32-91-400-801.

SUBTASK 32-11-89-610-001

(2) Lubricate the downlock strut [5] (TASK 12-21-11-640-801).

SUBTASK 32-11-89-860-003

(3) For hydraulic systems A and B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-11-89-710-001

(4) Do a test of the downlock actuator [4] (TASK 32-32-51-400-801).

SUBTASK 32-11-89-790-001

(5) Examine the connections of the hydraulic tubes [2] for leakage.

G. Put the Airplane Back to Its Usual Condition.

SUBTASK 32-11-89-080-002

CAUTION

MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION

MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(1) Do this task: Lower the Airplane off Jacks, TASK 07-10-01-580-802, if it is necessary.

SUBTASK 32-11-89-800-001

(2) Remove the power from hydraulic systems A and B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN LANDING GEAR WHEEL WELL BLADE SEAL - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the blade seal from the wheel well of the main landing gear

(2) An installation of the blade seal on the wheel well of the main landing gear.

B. The blade seal can be replaced in segments or all at one time.

TASK 32-12-12-000-801

2. Main Landing Gear Wheel Well Blade Seal - Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

C. Prepare for the Removal

SUBTASK 32-12-12-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801

SUBTASK 32-12-12-480-002

(2) Put chocks around the tires of all the landing gear (PAGEBLOCK 10-11-05/201).

D. Main Landing Gear Wheel Well Blade Seal Removal

SUBTASK 32-12-12-020-001

(1) Do these steps to remove all the blade seal [4] from the wheel well:

(a) Remove the screws [1] and washers [2] that hold the blade seal [4] between the upper seal retainer [3] and the lower seal retainer [5].

(b) Isolate the upper seal retainer [3] from the blade seal [4] and the lower seal retainer [5].

(c) Remove the blade seal [4] from the wheel well.

(d) Hold the lower seal retainer [5] in place with tape.

SUBTASK 32-12-12-020-002

(2) Do these steps to remove a segment or segments of the blade seal [4] from the wheel well:

(a) Remove the screws [1] and washers [2] that hold the segment or segments of the blade seal [4] between the upper seal retainer [3] and the lower seal retainer [5].

(b) Loosen the adjacent screws to a segment or segments of the blade seal [4] you will remove to help in the removal of the blade seal [4].

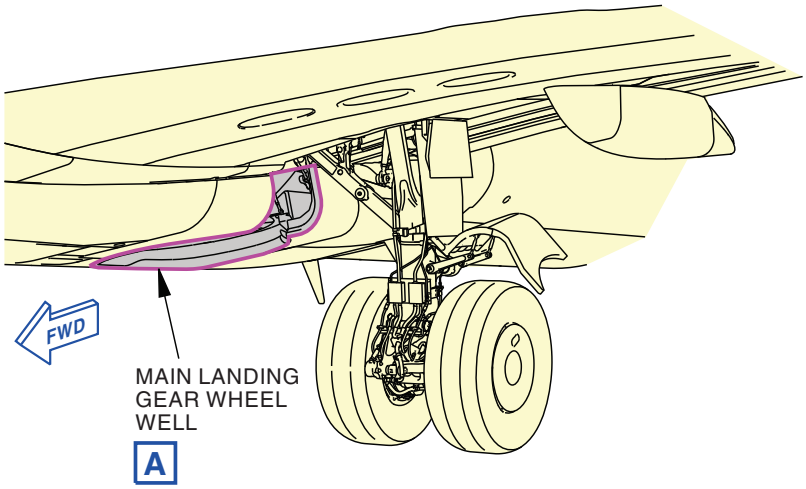
ECCN 9E991 BOEING PROPRIETARY - See title page for details

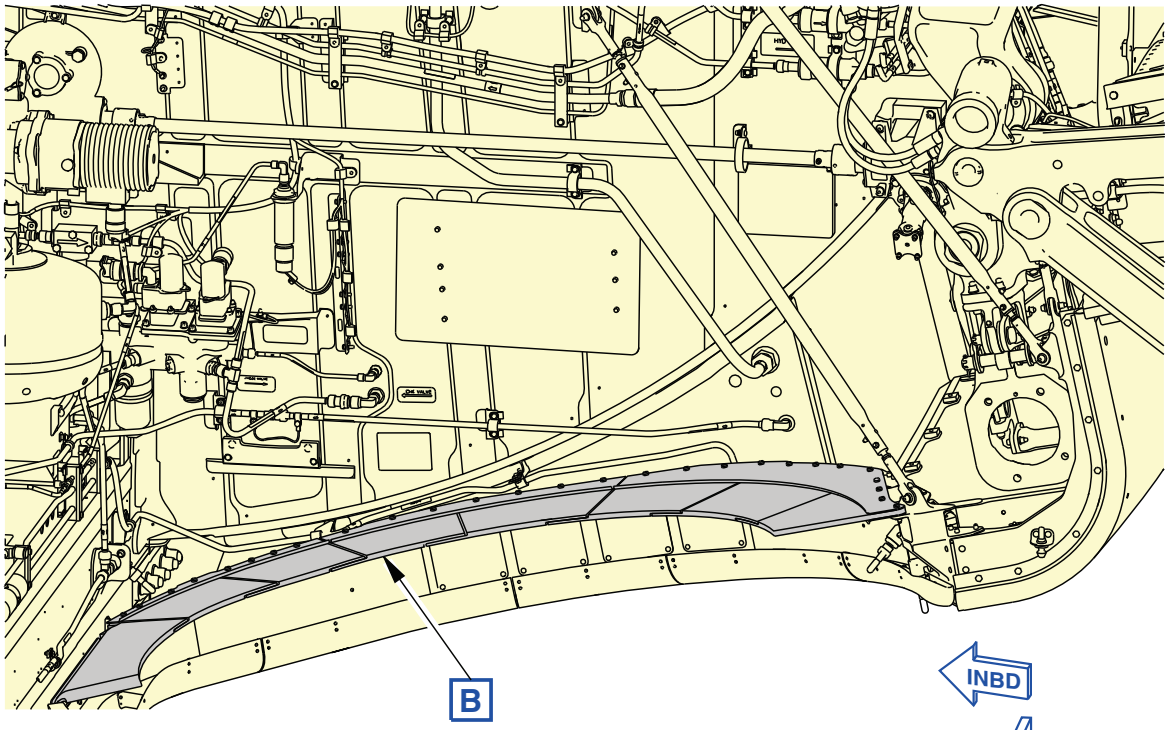
(c) Lift the upper seal retainer [3] lightly until you can remove a segment or segments of the blade seal [4].

(d) Remove a segment or segments of the blade seal [4] from the wheel well.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





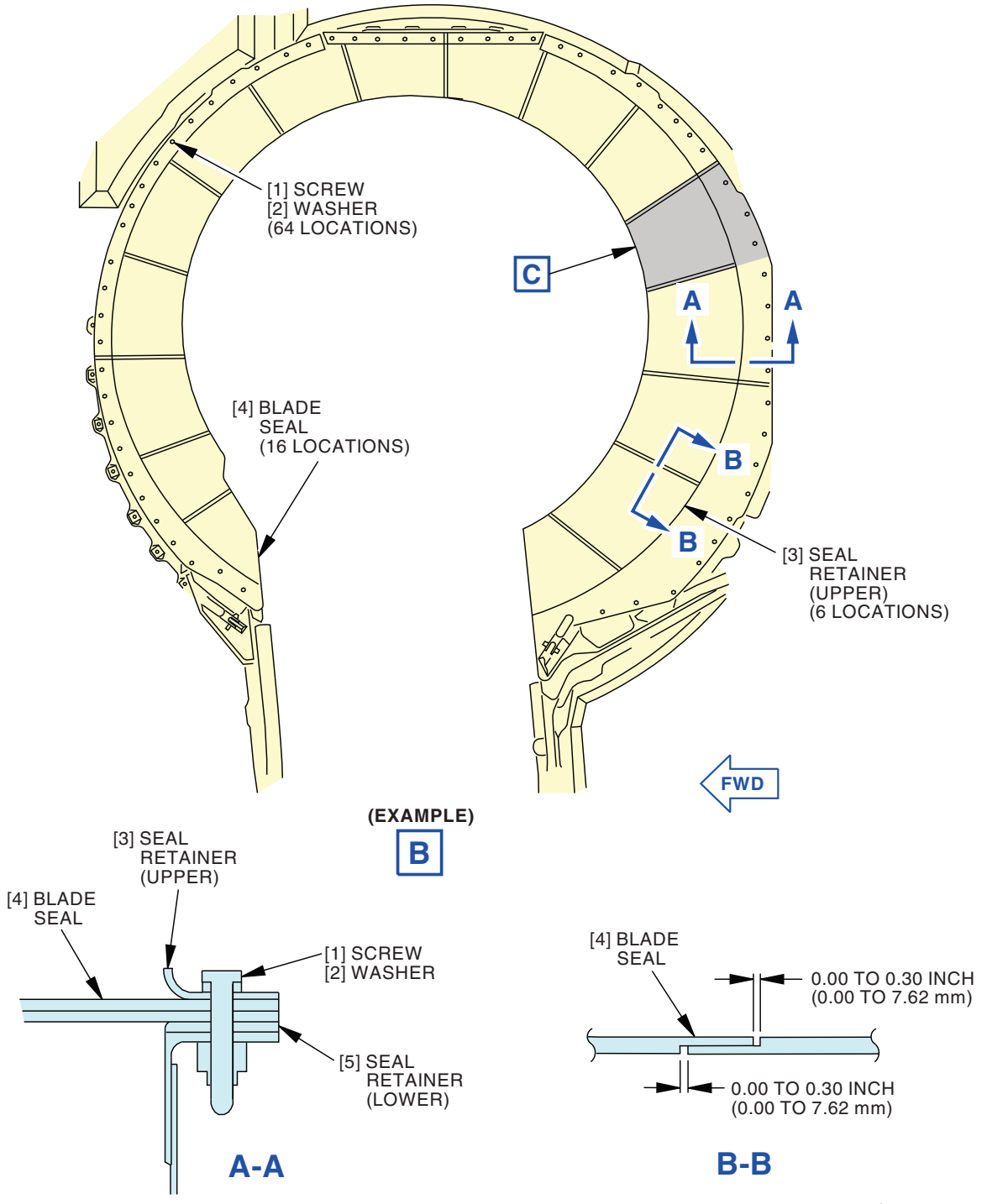
MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE IS SHOWN, RIGHT SIDE IS EQUIVALENT)

A

2884755 S0000684972_V1

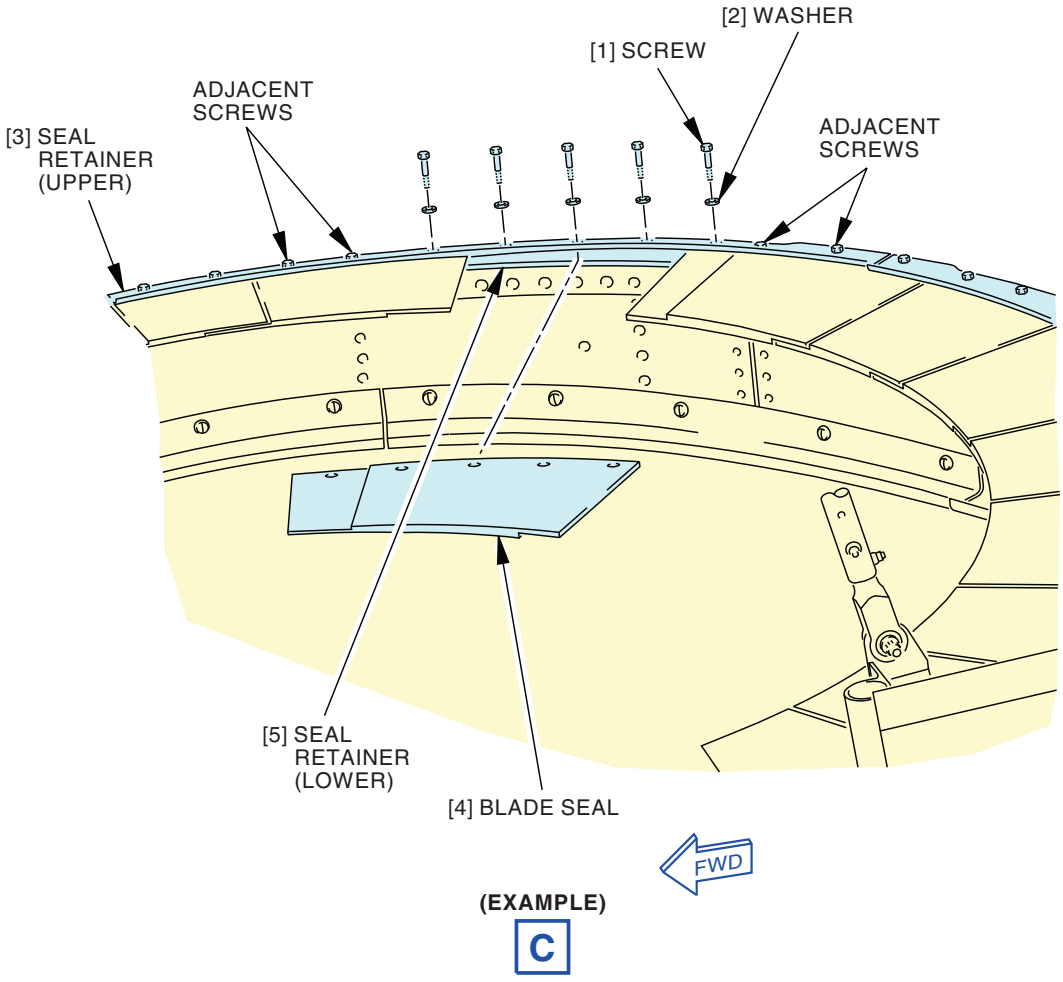
Main Landing Gear Wheel Well Blade Seal Installation
Figure 401/32-12-12-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Wheel Well Blade Seal Installation
Figure 401/32-12-12-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414119 S00061534445_V2

Main Landing Gear Wheel Well Blade Seal Installation
Figure 401/32-12-12-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-12-12-400-801

3. Main Landing Gear Wheel Well Blade Seal Installation

(Figure 401)

A. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Blade seal	53-14-51-10A-025	PTF ALL
		53-14-51-10A-030	PTF ALL
		53-14-51-10A-035	PTF ALL
		53-14-51-10A-050	PTF ALL
		53-14-51-10A-055	PTF ALL
		53-14-51-10A-060	PTF ALL
		53-14-51-10A-065	PTF ALL
		53-14-51-10A-080	PTF ALL
		53-14-51-10A-085	PTF ALL
		53-14-51-10A-095	PTF ALL
		53-14-51-10A-100	PTF ALL
		53-14-51-10A-115	PTF ALL
		53-14-51-10A-120	PTF ALL
		53-14-51-10A-135	PTF ALL
		53-14-51-10A-140	PTF ALL
		53-14-51-10A-145	PTF ALL

B. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

C. Main Landing Gear Wheel Well Blade Seal Installation

SUBTASK 32-12-12-420-001

(1) Do these steps to install all the blade seal [4] on the wheel well:

(a) Remove the tape that holds the lower seal retainer [5] in position.

(b) Put the blade seal [4] and the upper seal retainer [3] in position on the lower seal retainer [5].

NOTE: Each blade seal [4] and seal retainer was designed to fit a specified area. Make sure you install the blade seal [4] and the seal retainer on the same position from which they were removed.

(c) Align the fastener holes on the blade seal [4] with the fastener holes on the upper seal retainer [3] and lower seal retainer [5].

(d) Install the screws [1] and washers [2] to hold the blade seal [4] between the upper seal retainer [3] and the lower seal retainer [5].

(e) Tighten the screws [1] to 15 in-lb (2 N·m) maximum.

1) Make sure there is a two thread minimum protrusion of the screw [1] through the nut plate element.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-12-12-420-002

(2) Do these steps to install a segment or segments of the blade seal [4] on the wheel well:

(a) Lift the upper seal retainer [3] lightly until you can install a segment or segments of a blade seal [4] between the upper seal retainer [3] and the lower seal retainer [5].

(b) Align the fastener holes on the blade seal [4] with the fastener holes on the upper seal retainer [3] and the lower seal retainer [5].

(c) Install the screws [1] and washers [2] to hold the blade seal [4] between the upper seal retainer [3] and the lower seal retainer [5].

(d) Tighten the screws [1] to 15 in-lb (2 N·m) maximum.

1) Make sure there is a two thread minimum protrusion of the screw [1] through the nut plate element.

D. Put the Airplane Back to Its Usual Condition

SUBTASK 32-12-12-080-001

(1) Remove chocks around the tires of all the landing gear.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN LANDING GEAR WHEEL WELL BLADE SEAL - INSPECTION/CHECK

1. General

A. This procedure contains the task to examine the blade seal on the wheel well of the main landing gear.

TASK 32-12-12-200-801

2. Main Landing Gear Wheel Well Blade Seal Inspection

(Figure 601)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-12-12-000-801	Main Landing Gear Wheel Well Blade Seal - Removal (P/B 401)
32-12-12-400-801	Main Landing Gear Wheel Well Blade Seal Installation (P/B 401)
32-41-93-710-802	Gear Retract Braking Operational Test (P/B 501)

B. Tools/Equipment

Reference	Description
STD-552	Knife - Utility
STD-1074	Scissors

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

D. Prepare for the Inspection

SUBTASK 32-12-12-210-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-12-12-480-003

(2) Put chocks around the tires of all the landing gear (PAGEBLOCK 10-11-05/201).

E. Main Landing Gear Wheel Well Blade Seal Inspection

SUBTASK 32-12-12-020-003

(1) Examine the blade seal segments for tears in the half-thickness (overlap) areas.

(a) If one segment of the overlap area has a tear, do the steps that follow:

1) Cut a sufficient area from the overlap to fully remove the tear with a scissors, STD-1074, or an utility knife, STD-552.

NOTE: The area of damage removal must be over or under an undamaged area of the adjacent blade seal segment.

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(b) If both segments of the overlap area have a tear, do the steps that follow:

1) Replace one or both blade seal segments.

NOTE: If one blade seal segment is replaced, refer to the above step to repair the remaining damaged blade seal segment.

SUBTASK 32-12-12-020-004

(2) Examine the blade seal segments for tears in the full-thickness (non-overlap) areas.

(a) If you find a tear in a blade seal segment, replace the segment (TASK 32-12-12-000-801, TASK 32-12-12-400-801).

SUBTASK 32-12-12-020-005

(3) Examine the seal segments for damage possibly caused by friction wear when a tire enters the wheel well.

NOTE: Worn edges from friction with tires is usual, and local fraying of fabric along the worn edges is permitted. If the worn edge has no tears or separated sections, then there is no limit to the depth of wear and operators can replace blade seals at their discretion. Be advised, however, that blade seals which have worn beyond the point of contact with the tire could contribute to drag in the wheel well area.

(a) If you find damage to the blade seals which appears to have been caused by spinning tires, do this task: Gear Retract Braking Operational Test, TASK 32-41-93-710-802.

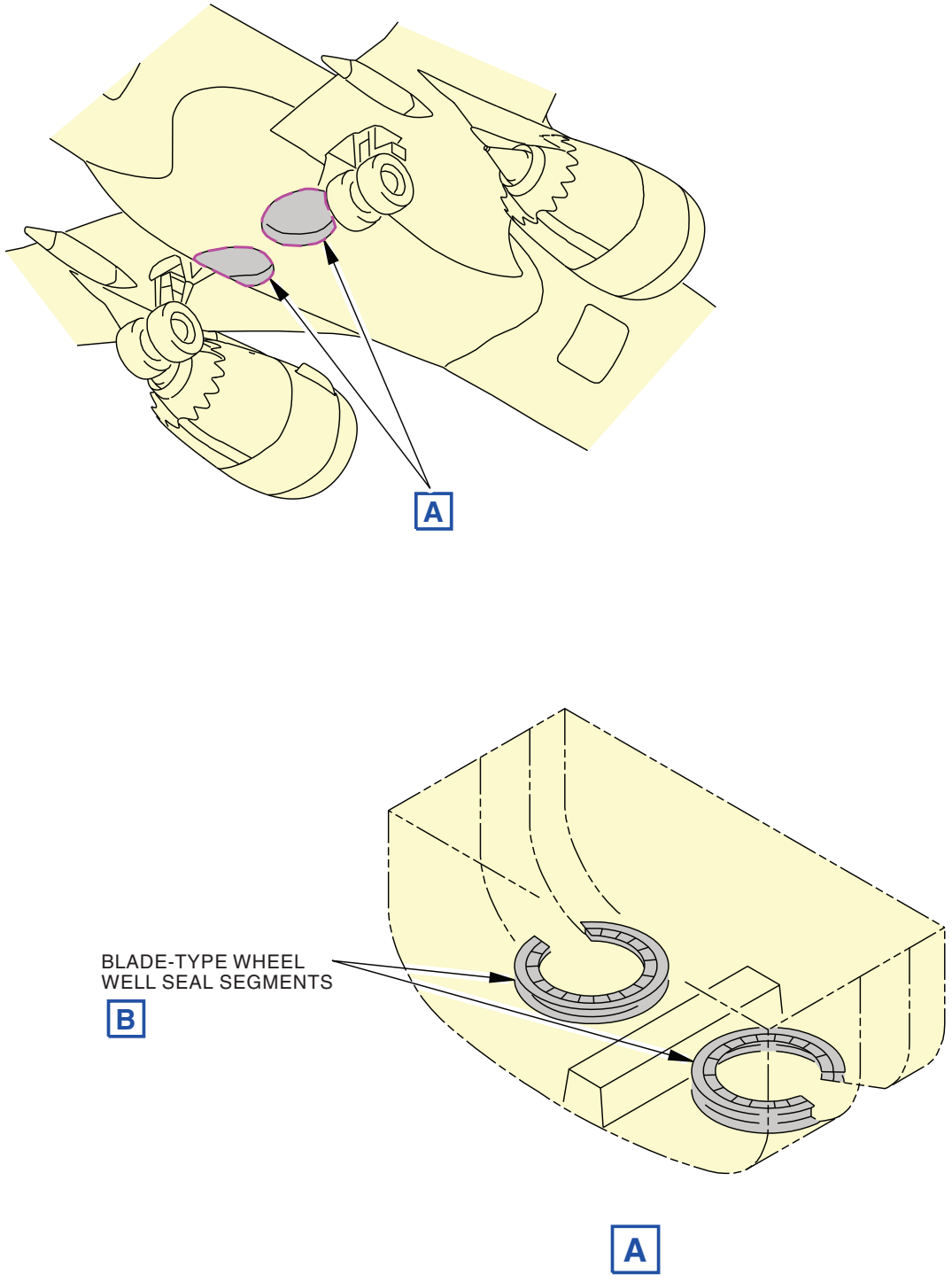
F. Put the Airplane Back to Its Usual Condition

SUBTASK 32-12-12-080-002

(1) Remove the chocks around the tires of all the landing gear.

——— END OF TASK ———

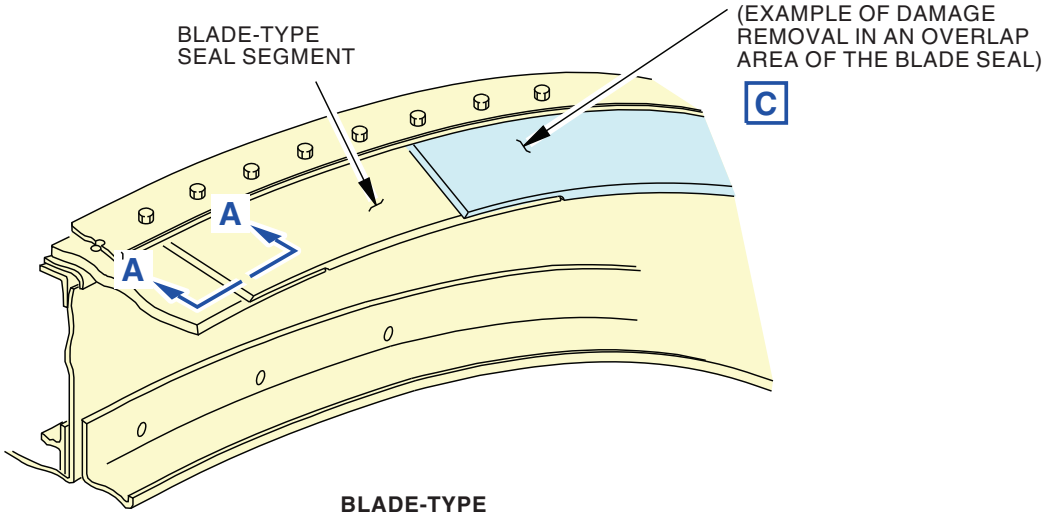
ECCN 9E991 BOEING PROPRIETARY - See title page for details

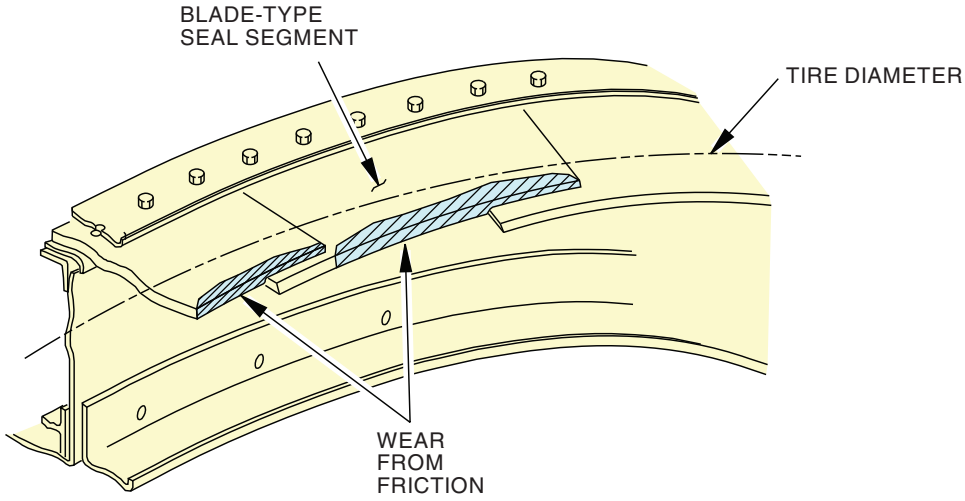


2414120 S00061534449_V1

Main Landing Gear Wheel Well Blade Seal Inspection/Check
Figure 601/32-12-12-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

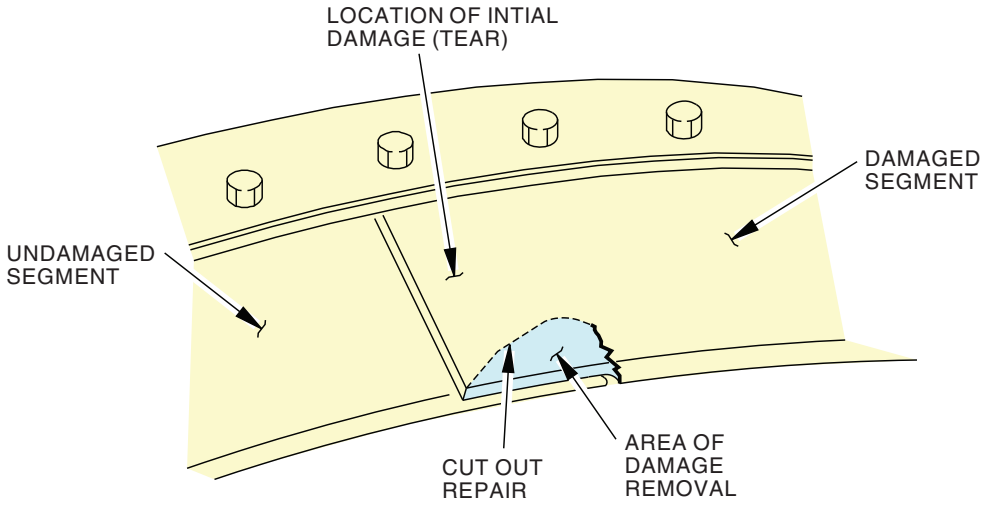




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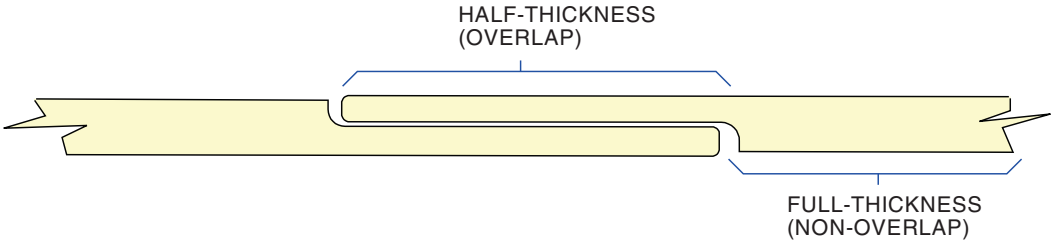
Main Landing Gear Wheel Well Blade Seal Inspection/Check
Figure 601/32-12-12-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE OF DAMAGE REMOVAL IN AN OVERLAP AREA OF A WHEEL WELL BLADE SEAL)

A



(EXAMPLE)

A-A

2414122 S00061534451_V1

Main Landing Gear Wheel Well Blade Seal Inspection/Check
Figure 601/32-12-12-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN LANDING GEAR SHOCK STRUT DOORS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the center door from the shock strut of the main landing gear

(2) Installation of the center door on the shock strut of the main landing gear

(3) Removal of the inner door from the shock strut of the main landing gear

(4) Installation of the inner door on the shock strut of the main landing gear

(5) Removal of the door assembly from the shock strut of the main landing gear

(6) Installation of the door assembly on the shock strut of the main landing gear

B. The assembly of the inner door and the center door is referred to as the "door assembly" in this procedure.

TASK 32-13-11-000-801

2. Main Landing Gear Center Door Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the center door [1].

NOTE: The inner door and center door are components of the door assembly. The hinge fittings installed on the inner door and center door can change the part number of the door assembly. Do a check of the IPC to make sure that the inner door and center door components are correct for the door assembly installed on the airplane.

(2) It is not necessary to do the Main Landing Gear Shock Strut Door Adjustment, TASK 32-13-11-820-801, if you can do these steps:

(a) Make sure that the inner door [2] and center door [1] are being reinstalled on the same Main Landing Gear (MLG).

(b) Make sure that the inner door [2], center door [1], and pushrod [45] that are being installed are the same ones that were removed.

(c) Make sure that no adjustment was made on the existing pushrod [45].

(d) Make sure that no adjustments have been made to the hinges [20], no shims added or removed.

(e) Make sure that there is no damage to the lockwires on the pushrod [45].

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-13-11-820-801	Main Landing Gear Shock Strut Door Adjustment (P/B 501)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
732	Left Main Landing Gear - Center Door
733	Left Main Landing Gear - Inboard Door

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(Continued)

Zone	Area
734	Left Main Landing Gear
742	Right Main Landing Gear - Center Door
743	Right Main Landing Gear - Inboard Door
744	Right Main Landing Gear

D. Prepare for the Removal

SUBTASK 32-13-11-480-001

WARNING OBEY THE PROCEDURE FOR THE INSTALLATION OF THE DOWNLOCK PINS. IF YOU MOVE THE CONTROL LEVER FOR THE LANDING GEAR TO THE UP POSITION, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-13-11-480-002

(2) Put chocks around the tires of all the landing gear (PAGEBLOCK 10-11-05/201).

E. Main Landing Gear Center Door Removal

SUBTASK 32-13-11-020-001

(1) Do these steps to remove the center door [1]:

(a) Remove the screw [111], nut [113], and washers [112] to disconnect the jumper cable [106] and jumper cable [28] from the center door [1] (View J).

(b) Remove the pins [24], nuts [23], washers [22], bolts [21], and bushing [27] from the forward and aft hinges [20] to disconnect the center door [1] from the inner door [2] (View D and E).

1) If it is necessary, remove the bolts [26] and pull the seal [25] to the down position for easy removal of the bolt [21].

(c) For the top aft hinge, remove the pin [10], nut [9], washer [77], spacers [7], washer [5], and bolt [4] to disconnect the tie rod [3] from the tray assembly [6] on the shock strut [8] (View B).

(d) For the top forward hinge, remove the pin [12], nut [13], washer [78], spacers [14], washer [15], and bolt [16] to disconnect from the tie rod [11] from the tray assembly [6] on the shock strut [8] (View C).

(e) Do these steps to remove the anti-rotation plate [68] and upper plate [88] (View D, E, A-A, and B-B):

1) Remove the bolt [70], washer [71], washer [58], and nut [59].

2) Remove the plate [68].

3) Remove the bolts [84], washers [85], washers [57], and nuts [87].

4) Remove the upper plate [88], shim [90], and spacer [91].

(f) For the bottom aft hinge, remove the bolt [17], clip [69], washer [18], barrel nut [47], and retainer [48] to disconnect the center door [1] from the shock strut [8] (View A-A).

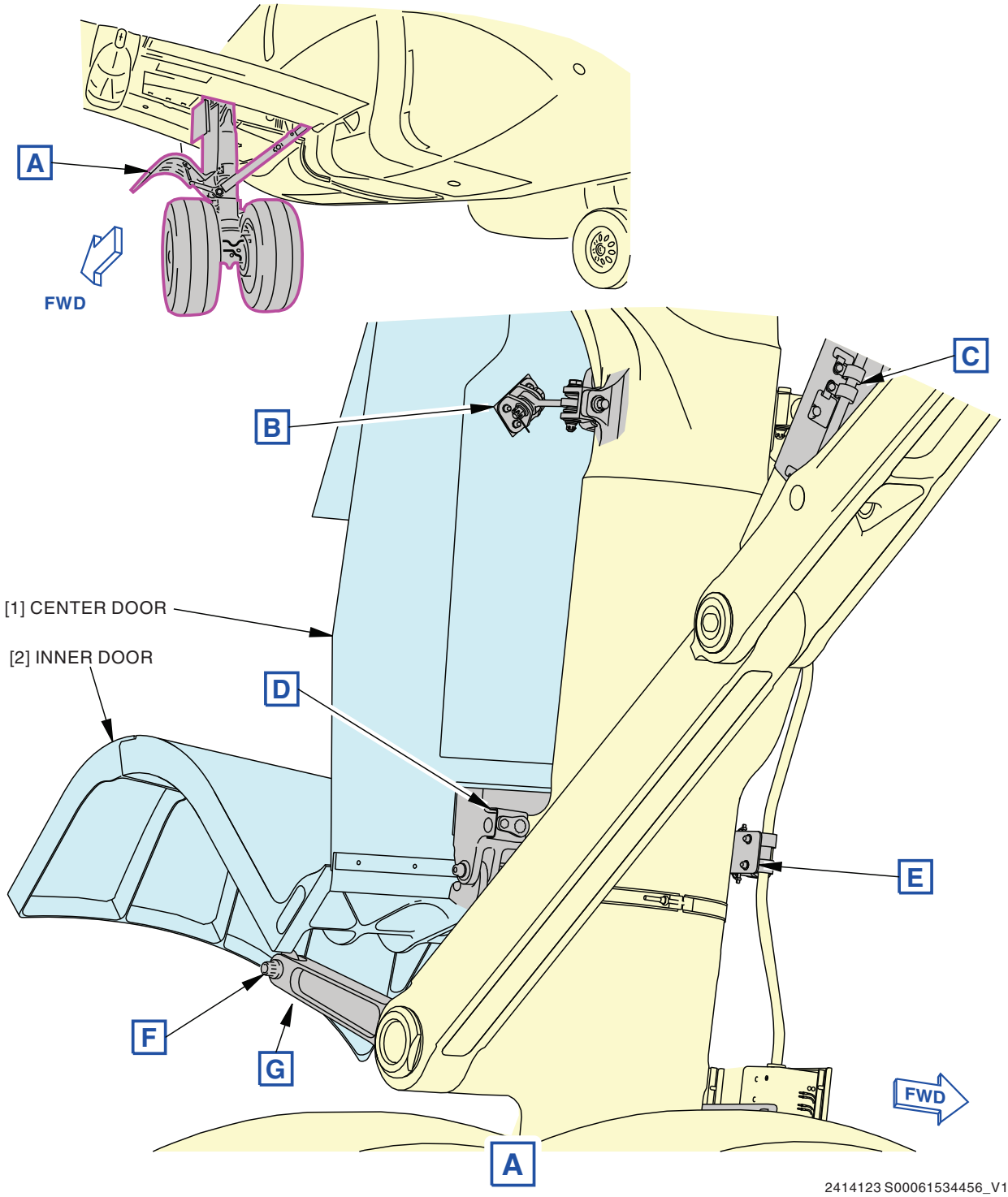
(g) For the bottom forward hinge, remove the bolt [29], washer [30], washer [67], barrel nut [49], and retainer [50] to disconnect the center door [1] from the shock strut [8] (View B-B).

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(h) Remove the center door [1] from the airplane.

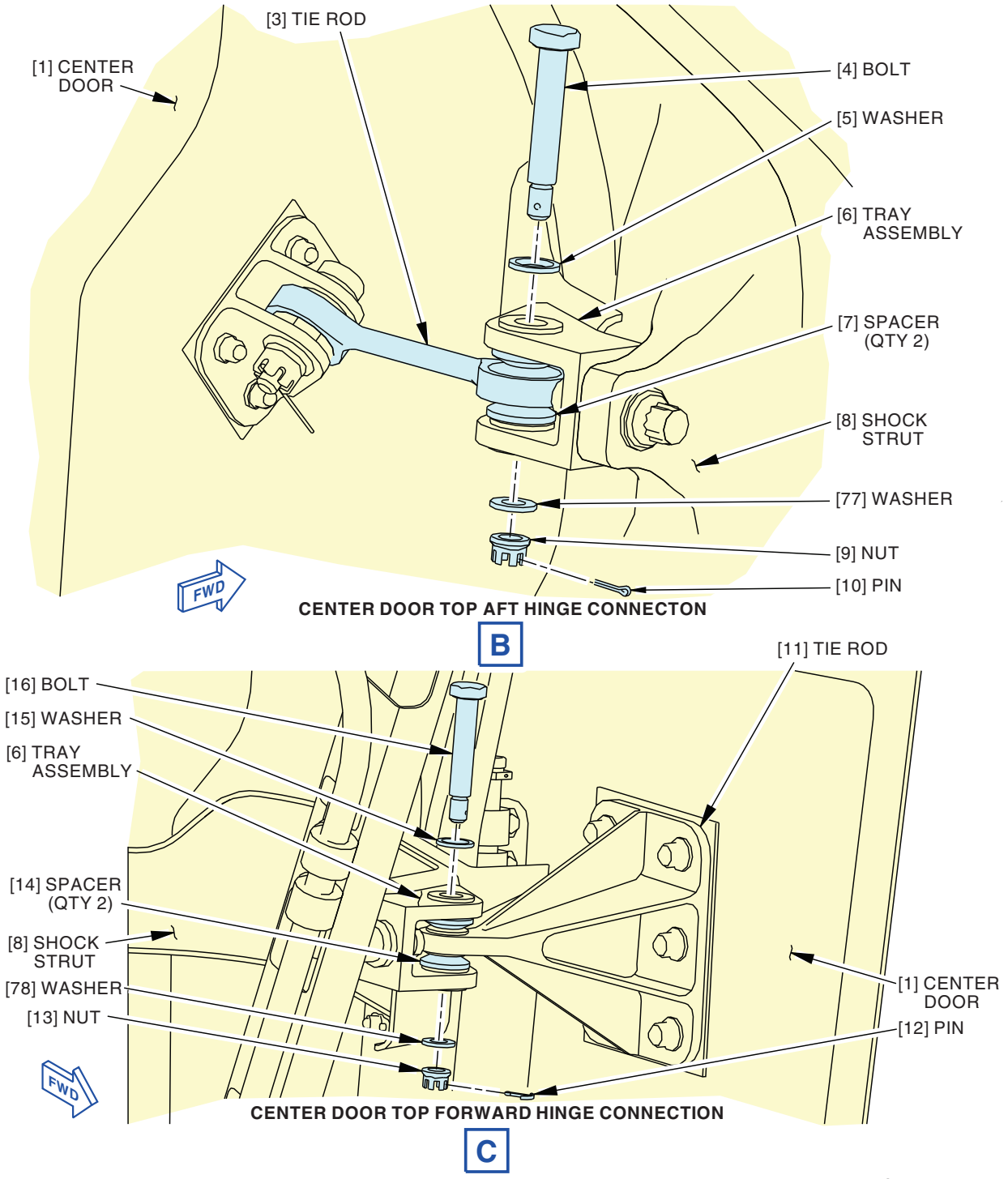
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



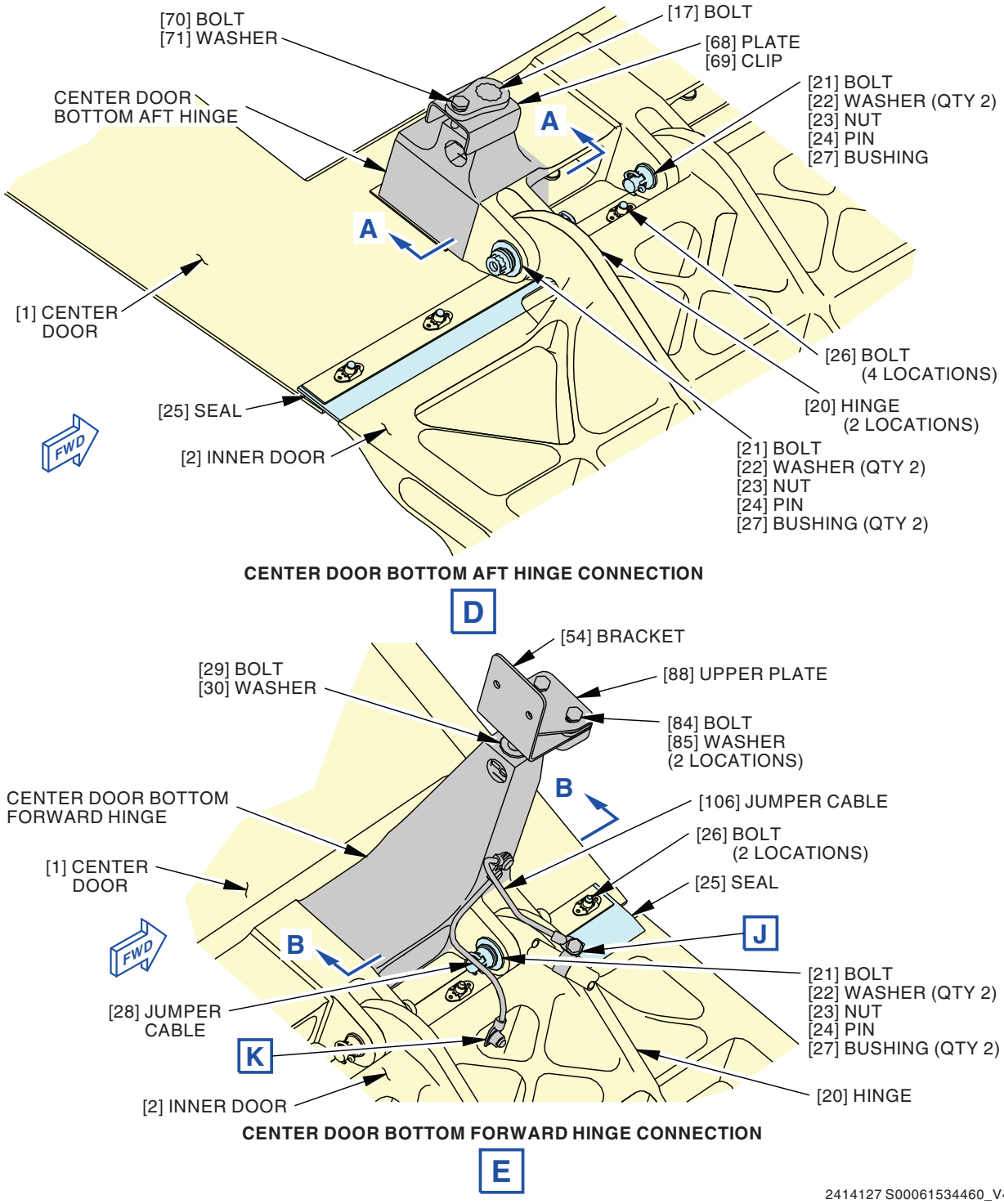
Main Landing Gear Shock Strut Doors Installation
Figure 401/32-13-11-990-801 (Sheet 1 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



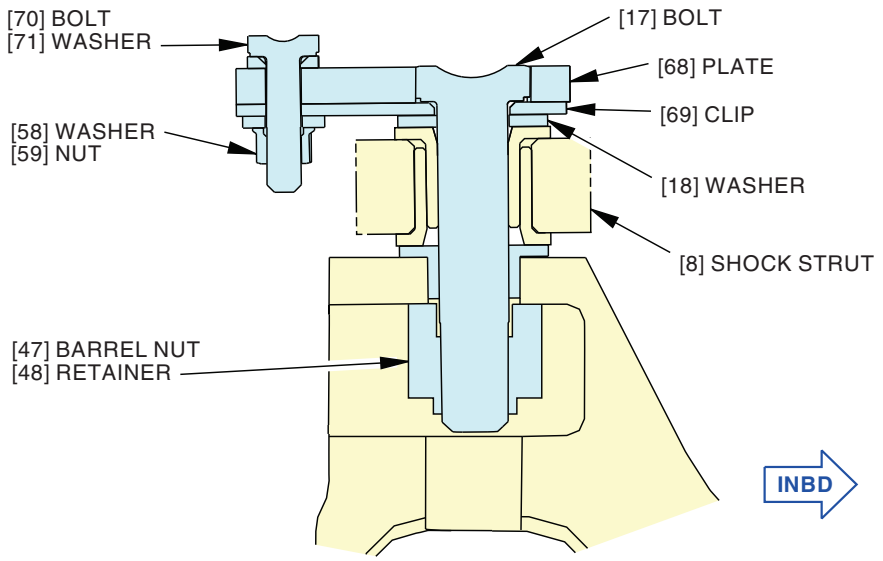
Main Landing Gear Shock Strut Doors Installation
Figure 401/32-13-11-990-801 (Sheet 2 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



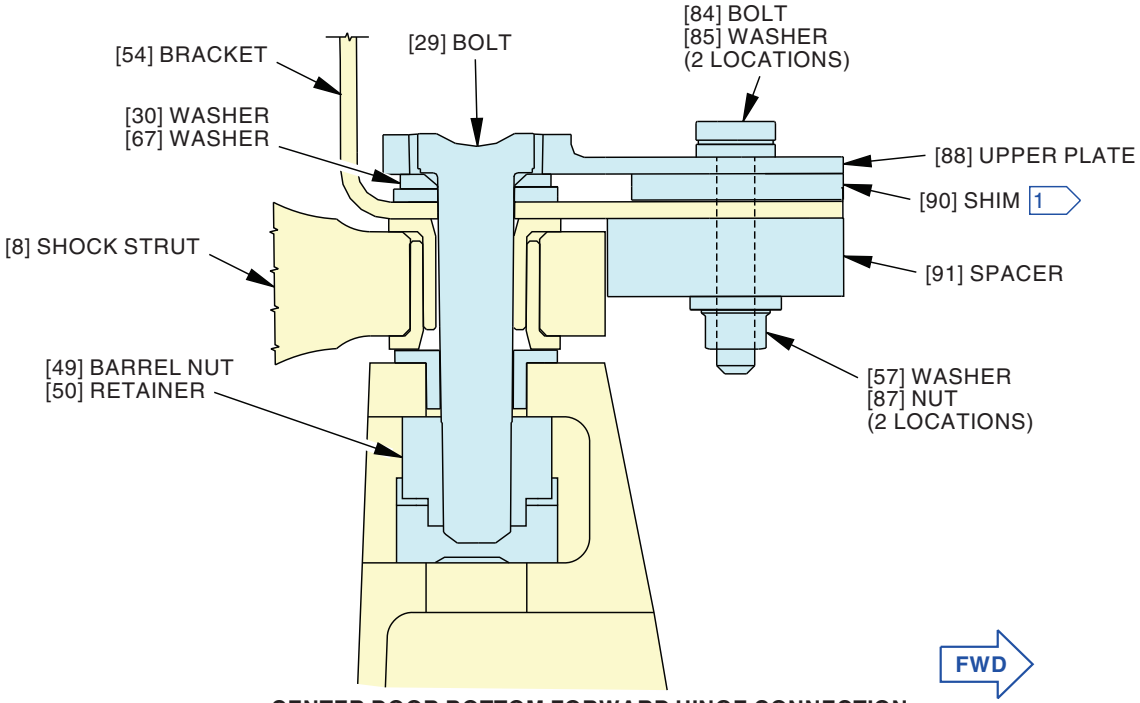
Main Landing Gear Shock Strut Doors Installation
Figure 401/32-13-11-990-801 (Sheet 3 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CENTER DOOR BOTTOM AFT HINGE CONNECTION

A-A



CENTER DOOR BOTTOM FORWARD HINGE CONNECTION

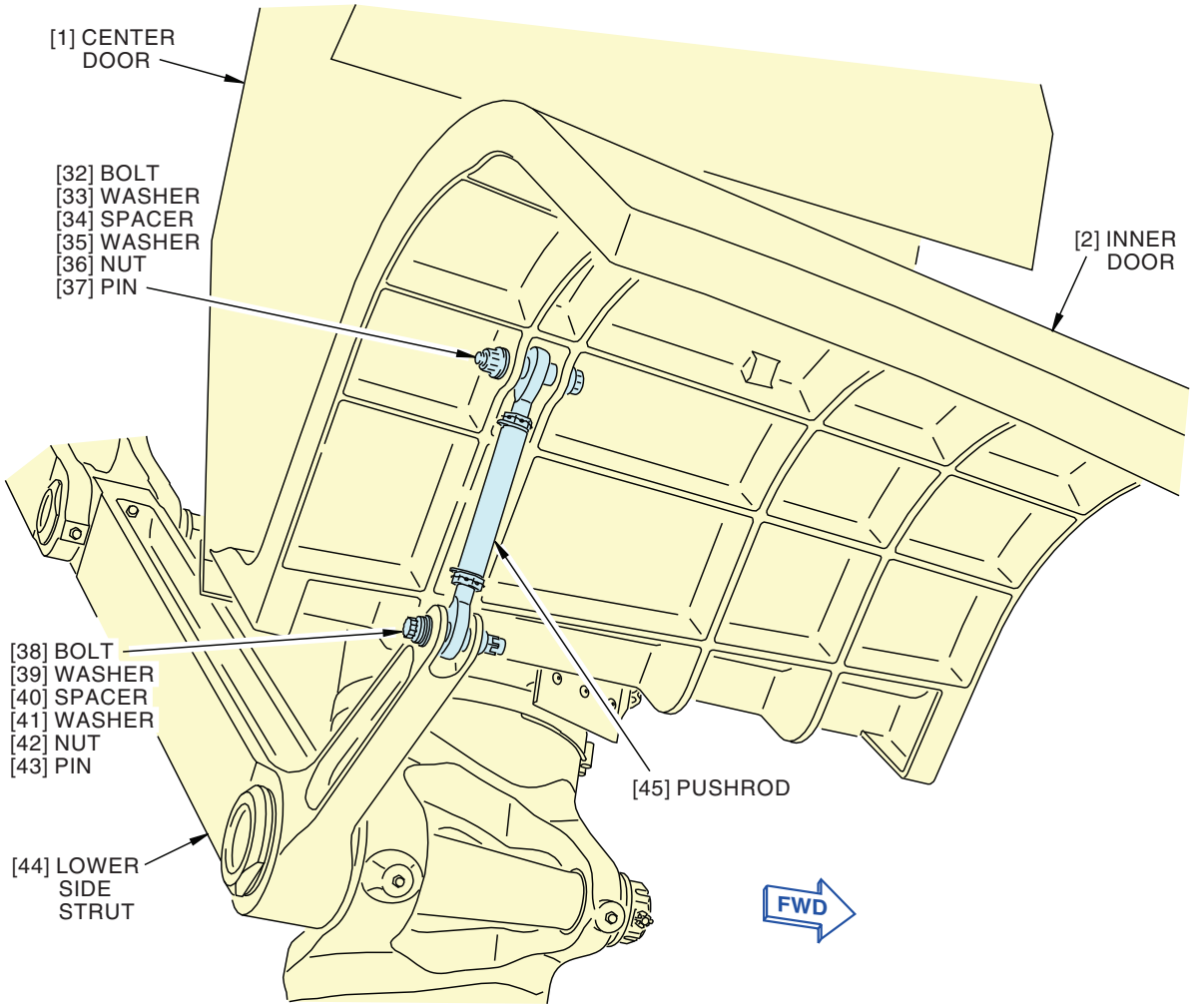
B-B

1 ADJUST SHIM THICKNESS TO MATCH WASHERS.

2414128 S00061534461_V3

Main Landing Gear Shock Strut Doors Installation
Figure 401/32-13-11-990-801 (Sheet 4 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



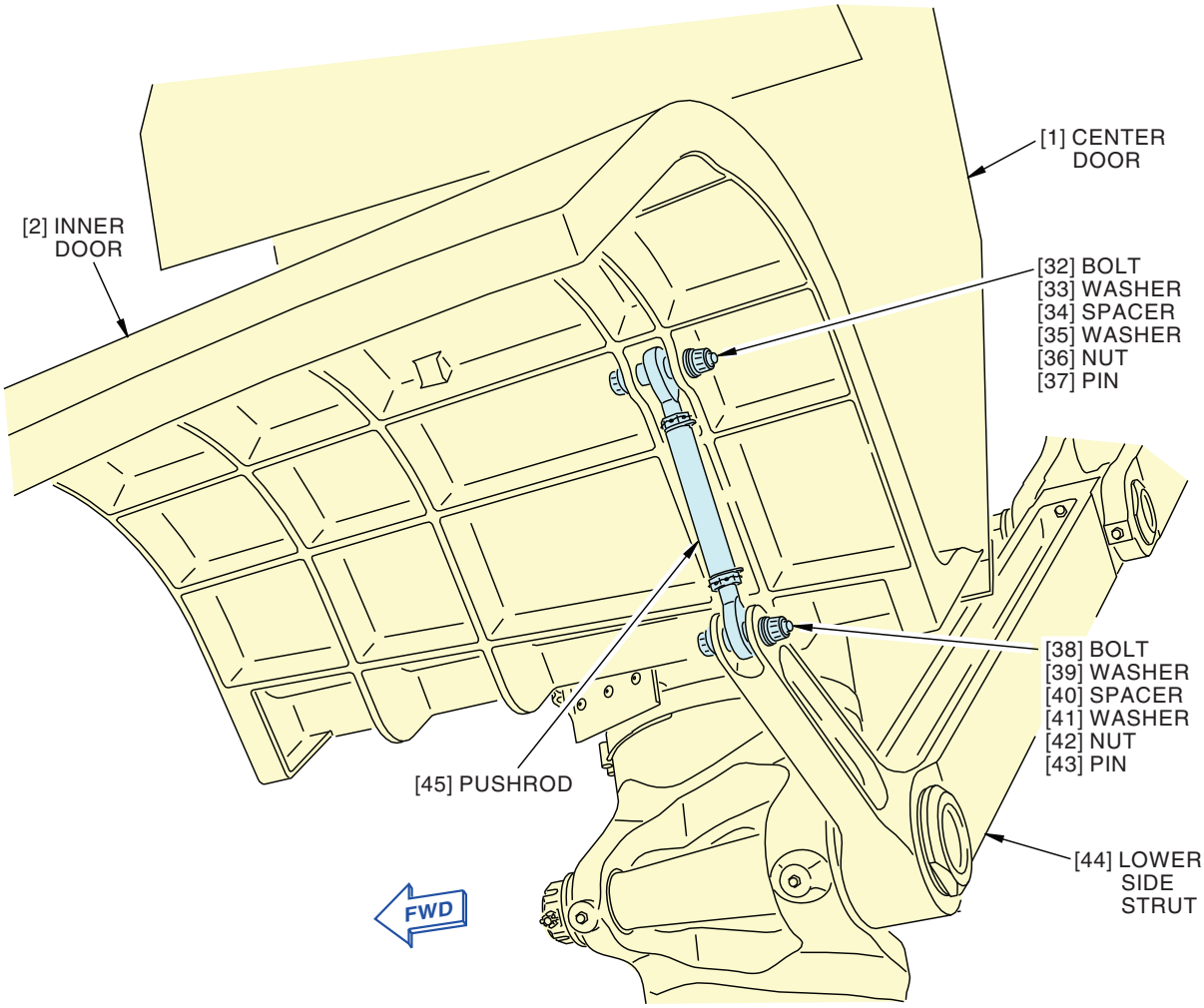
INNER DOOR PUSHROD CONNECTION
(RIGHT MAIN LANDING GEAR)

F

2414129 S00061534462_V1

Main Landing Gear Shock Strut Doors Installation
Figure 401/32-13-11-990-801 (Sheet 5 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



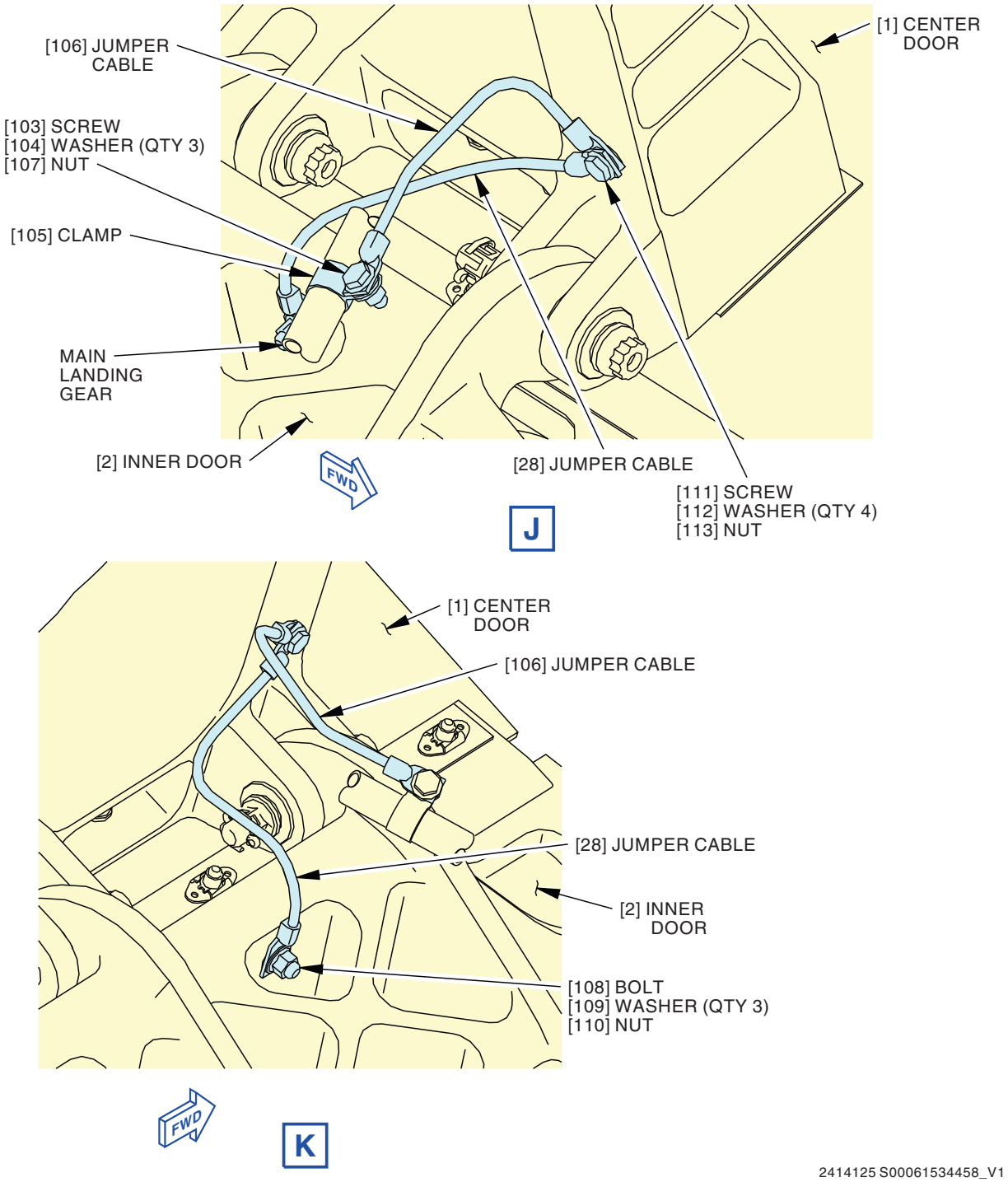
INNER DOOR PUSHROD CONNECTION
(LEFT MAIN LANDING GEAR)

G

2414130 S00061534463_V1

Main Landing Gear Shock Strut Doors Installation
Figure 401/32-13-11-990-801 (Sheet 6 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Shock Strut Doors Installation
Figure 401/32-13-11-990-801 (Sheet 7 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-13-11-420-801

3. Main Landing Gear Center Door Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the center door [1].

NOTE: The inner door and center door are components of the door assembly. The hinge fittings installed on the inner door and center door can change the part number of the door assembly. Do a check of the IPC to make sure that the inner door and center door components are correct for the door assembly installed on the airplane.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-13-11-820-801	Main Landing Gear Shock Strut Door Adjustment (P/B 501)
51-21-72-370-801	BMS10-79 Primer - Application (P/B 701)
51-21-73-370-802	BMS10-60 Topcoat Application (P/B 701)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
C00175	Primer - Urethane Compatible, Corrosion Resistant	BMS10-79 Type III
C00700	Coating - Protective Enamel (BAC 707 Gray Color, Gloss)	BMS10-60 Type I
G50398	Pad - Adbrasive, Scotch-Brite Type S, Abrasive Pad	
G50492	Pad - 3M Scotch Brite 7448 Ultra Fine Pad	MIL-A-9962A Type III Grade AAA

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E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Center door	32-13-21-01-385	PTF ALL
		32-13-21-01-388	PTF ALL
		32-13-21-01-390	PTF ALL
		32-13-21-01-395	PTF ALL
		32-13-21-20-070	PTF ALL
		32-13-21-20-075	PTF ALL
10	Pin	32-13-21-01-280	PTF ALL
12	Pin	32-13-21-01-280	PTF ALL
24	Pin	32-13-21-20-020	PTF ALL

F. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
732	Left Main Landing Gear - Center Door
733	Left Main Landing Gear - Inboard Door
734	Left Main Landing Gear
742	Right Main Landing Gear - Center Door
743	Right Main Landing Gear - Inboard Door
744	Right Main Landing Gear

G. Prepare for the Installation

SUBTASK 32-13-11-480-003

WARNING OBEY THE PROCEDURE FOR THE INSTALLATION OF THE DOWNLOCK PINS. IF YOU MOVE THE CONTROL LEVER FOR THE LANDING GEAR TO THE UP POSITION, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-13-11-480-004

(2) Put chocks around the tires of all the landing gear (PAGEBLOCK 10-11-05/201).

H. Main Landing Gear Center Door Installation

SUBTASK 32-13-11-400-002

(1) Do these steps to install the center door [1] (Views D, E, A-A and B-B):

(a) Put the center door [1] in its position for connection to the shock strut [8] and the inner door [2].

(b) For the bottom aft hinge, install the bolt [17], plate [68], clip [69], washer [18], barrel nut [47], and retainer [48].

(c) For the bottom forward hinge, install the bolt [29], washer [30], washer [67], barrel nut [49], and retainer [50].

SUBTASK 32-13-11-400-003

(2) Continue the center door [1] installation (Views B, C, D, and E):

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(a) For the top aft hinge, install the bolt [4], washer [5], spacers [7], washer [77], and nut [9] to connect the tie rod [3] to the tray assembly [6] on the shock strut [8].

1) Do not tighten the nut [9].

NOTE: Final torque to the nut and installation of the pin will be done after final adjustments to the doors are made in the subsequent step.

(b) For the top forward hinge, install the bolt [16], washer [15], spacers [14], washer [78], and nut [13] to connect the tie rod [11] to the tray assembly [6] on the shock strut [8].

1) Do not tighten the nut [13].

NOTE: Final torque to the nut and installation of the pin will be done after final adjustments to the doors are made in the subsequent step.

(c) Install the bolts [21] to attach the center door [1] to the inner door [2]:

1) If it is necessary, remove the bolts [26] and pull the seal [25] slightly to the down position for easy installation of the bolt [21].

2) If removed, install the bolts [26].

(d) Install the bushings [27], washers [22], and nuts [23] on the bolts [21].

1) Do not tighten the nuts [23].

NOTE: Final torque to the nuts and installation of the pins will be done after final adjustments to the doors are made in the subsequent step.

SUBTASK 32-13-11-820-001

(3) Do this task: Main Landing Gear Shock Strut Door Adjustment, TASK 32-13-11-820-801.

(a) It is not necessary to do this task if you can do these steps:

1) Make sure that the inner door [2] and center door [1] are being reinstalled on the same Main Landing Gear (MLG).

2) Make sure that the inner door [2], center door [1], and pushrod [45] that are being installed are the same ones that were removed.

3) Make sure that no adjustment was made on the existing pushrod [45].

4) Make sure that no adjustments have been made to the hinges [20], no shims added or removed.

5) Make sure that there is no damage to the lockwires on the pushrod [45].

SUBTASK 32-13-11-420-001

(4) Do these steps to safety the fasteners on the center door [1] (Views D, E, A-A and B-B):

(a) Tighten the bolt [17] and bolt [29] to between 398 in-lb (45.0 N·m) to 422 in-lb (47.7 N·m).

1) Tighten the bolt [17] and bolt [29] to the low torque value.

2) Tighten the bolts to align the head with the anti-rotation plate [68] and upper plate [88], if it is necessary.

(b) Adjust the thickness of shim [90] to match washer thickness.

(c) Install the bolt [70], washer [71], washer [58], and nut [59].

(d) Install the upper plate [88], shim [90], spacer [91], bolt [84], washers [85], washers [57], and nuts [87].

SUBTASK 32-13-11-420-003

(5) Tighten the nut [9] and install the pin [10] in the hole on the bolt [4] (View B).

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(a) If it is necessary, loosen the nut [9] to the nearest castellation to align the holes for the pin [10].

SUBTASK 32-13-11-420-004

(6) Tighten the nut [13] and install the pin [12] in the hole on the bolt [16] (View C).

(a) If it is necessary, loosen the nut [13] to the nearest castellation to align the holes for the pin [12].

SUBTASK 32-13-11-420-005

(7) Tighten the nuts [23] and install the pins [24] in the hole on the bolts [21] (View D and E).

(a) If it is necessary, loosen the nuts [23] to the nearest castellation to align the holes for the pins [24].

(b) Make sure that there is a minimum gap of 0.030 in. (0.762 mm) between the washer [22] and center door [1] hinge bushing.

NOTE: Only required where the bushing is installed.

SUBTASK 32-13-11-400-004

(8) Do these steps to prepare the jumper cable [106], jumper cable [28], and center door [1] for installation.

(a) Use Cleaning Procedure 1 to prepare the attachment surface for the jumper cable [106] and jumper cable [28], on the center door [1] (SWPM 20-20-00).

1) Use a very fine Scotch-Brite Type S pad, G50398, or an ultrafine Scotch-Brite 7448 pad, G50492.

2) Twist the pad with a circular movement while you push with your thumb.

(b) Manually apply coating, C00064 to the abraded surface on the center door [1].

SUBTASK 32-13-11-420-006

(9) Install the jumper cable [106], jumper cable [28], screw [111], washers [112], and nut [113] on the center door [1] (View J).

SUBTASK 32-13-11-280-001

(10) Use an intrinsically safe approved bonding meter, COM-1550, to measure the resistance between the jumper cable [106], jumper cable [28] and the center door [1](SWPM 20-20-10).

(a) Make sure that the electrical bonding resistance is 0.001 ohm (1.0 milliohms) or less.

SUBTASK 32-13-11-110-002

(11) Refinish the abraded surfaces that you can see.

(a) Apply one coat of primer, C00175 (TASK 51-21-72-370-801).

(b) Apply coating, C00700 (TASK 51-21-73-370-802).

SUBTASK 32-13-11-390-001

(12) Apply a fillet seal with sealant, A00247, to the jumper cable [106], jumper cable [28], and fasteners (SWPM 20-20-10).

SUBTASK 32-13-11-420-042

(13) If the screw [103] was removed, do these steps to install the jumper cable [106] to the MLG.

(a) Do these steps to prepare the jumper cable [106] and the MLG for installation.

1) Use solvent cleaning procedure to prepare the attachment surface for the jumper cable [106] on the MLG (SWPM 20-20-00).

(b) Install the jumper cable [106], screw [103], washers [104], clamp [105], and nut [107] on the MLG (View J).

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(c) Use an intrinsically safe approved bonding meter, COM-1550, to measure the resistance between the jumper cable [106] and the MLG (SWPM 20-20-10).

1) Make sure that the electrical bonding resistance is 0.002 ohm (2.0 milliohms) or less.

(d) Apply a fillet seal with sealant, A00247, to the jumper cable [106] and fasteners (SWPM 20-20-10).

SUBTASK 32-13-11-420-043

(14) If the screw [108] was removed, do these steps to install the jumper cable [28] to the inner door [2]:

(a) Do these steps to prepare the jumper cable [28] and the inner door [2] for installation.

1) Use Cleaning Procedure 1 to prepare the attachment surface for the jumper cable [28] on the inner door [2] (SWPM 20-20-00).

a) Use a very fine Scotch-Brite Type S pad, G50398, or an ultrafine Scotch-Brite 7448 pad, G50492.

b) Twist the pad with a circular movement while you push with your thumb.

2) Manually apply coating, C00064 to the abraded surface on the inner door [2].

(b) Install the jumper cable [28], screw [108], washers [109], and nut [110] on the inner door [2] (View K).

(c) Use an intrinsically safe approved bonding meter, COM-1550, to measure the resistance between the jumper cable [28] and the inner door [2] (SWPM 20-20-10).

1) Make sure that the electrical bonding resistance is 0.001 ohm (1.0 milliohms) or less.

(d) Refinish the abraded surfaces that you can see.

1) Apply one coat of primer, C00175 (TASK 51-21-72-370-801).

2) Apply coating, C00700 (TASK 51-21-73-370-802).

(e) Apply a fillet seal with sealant, A00247, to the jumper cable [28] and fasteners (SWPM 20-20-10).

——— END OF TASK ———

TASK 32-13-11-000-802

4. Main Landing Gear Inner Door Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the inner door [2].

NOTE: The inner door and center door are components of the door assembly. The hinge fittings installed on the inner door and center door can change the part number of the door assembly. Do a check of the illustrated parts catalog to make sure that the inner door and center door components are correct for the door assembly installed on the airplane.

(2) It is not necessary to do the Main Landing Gear Shock Strut Door Adjustment, TASK 32-13-11-820-801, if you can do these steps:

(a) Make sure that the inner door [2] and center door [1] are being reinstalled on the same Main Landing Gear (MLG).

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(b) Make sure that the inner door [2], center door [1], and pushrod [45] that are being installed are the same ones that were removed.

(c) Make sure that no adjustment was made on the existing pushrod [45].

(d) Make sure that no adjustments have been made to the hinges [20], no shims added or removed.

(e) Make sure that there is no damage to the lockwires on the pushrod [45].

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-13-11-820-801	Main Landing Gear Shock Strut Door Adjustment (P/B 501)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
732	Left Main Landing Gear - Center Door
733	Left Main Landing Gear - Inboard Door
734	Left Main Landing Gear
742	Right Main Landing Gear - Center Door
743	Right Main Landing Gear - Inboard Door
744	Right Main Landing Gear

D. Prepare for the Removal

SUBTASK 32-13-11-480-005

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-13-11-480-006

(2) Put chocks around the tires of all the landing gear (PAGEBLOCK 10-11-05/201).

E. Main Landing Gear Inner Door Removal

SUBTASK 32-13-11-020-002

(1) Do these steps to remove the inner door [2]:

(a) Remove the screw [108], washers [109], and nut [110] to disconnect the jumper cable [28] from the inner door [2] (View K).

(b) If you need to replace the pushrod [45] due to excessive wear, do these steps (View F or G):

1) Remove the pin [43], nut [42], washer [41], spacer [40], washer [39], and bolt [38] to disconnect the pushrod [45] from the lower side strut [44].

2) Remove the pin [37], nut [36], washer [35], spacer [34], washer [33], and bolt [32] to disconnect the inner door [2] from the pushrod [45].

(c) If you do not need to replace the pushrod [45], do this step (View F or G):

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1) Remove the pin [37], nut [36], washer [35], spacer [34], washer [33], and bolt [32] to disconnect the inner door [2] from the pushrod [45].

(d) Remove the pins [24], nuts [23], washers [22], bolts [21], and bushing [27] from the forward and aft hinges [20] to disconnect the center door [1] from the inner door [2] (View D and E).

1) If it is necessary, remove the bolts [26] and pull the seal [25] to the down position for easy removal of the bolt [21].

(e) Remove the inner door [2] from the airplane.

——— END OF TASK ———

TASK 32-13-11-420-802

5. Main Landing Gear Inner Door Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the inner door [2].

NOTE: The inner door and center door are components of the door assembly. The hinge fittings installed on the inner door and center door can change the part number of the door assembly. Do a check of the illustrated parts catalog to make sure that the inner door and center door components are correct for the door assembly installed on the airplane.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-13-11-820-801	Main Landing Gear Shock Strut Door Adjustment (P/B 501)
51-21-72-370-801	BMS10-79 Primer - Application (P/B 701)
51-21-73-370-802	BMS10-60 Topcoat Application (P/B 701)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95

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(Continued)

Reference	Description	Specification
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
C00175	Primer - Urethane Compatible, Corrosion Resistant	BMS10-79 Type III
C00700	Coating - Protective Enamel (BAC 707 Gray Color, Gloss)	BMS10-60 Type I
G50398	Pad - Adbrasive, Scotch-Brite Type S, Abrasive Pad	
G50492	Pad - 3M Scotch Brite 7448 Ultra Fine Pad	MIL-A-9962A Type III Grade AAA

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Inner door	32-13-21-01-375	PTF ALL
		32-13-21-01-380	PTF ALL
		32-13-21-20-410	PTF ALL
		32-13-21-20-415	PTF ALL
24	Pin	32-13-21-20-020	PTF ALL
37	Pin	32-13-21-01-020	PTF ALL
43	Pin	32-13-21-01-020	PTF ALL

F. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
732	Left Main Landing Gear - Center Door
733	Left Main Landing Gear - Inboard Door
734	Left Main Landing Gear
742	Right Main Landing Gear - Center Door
743	Right Main Landing Gear - Inboard Door
744	Right Main Landing Gear

G. Prepare for the Installation

SUBTASK 32-13-11-480-007

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-13-11-480-008

(2) Put chocks around the tires of all the landing gear (PAGEBLOCK 10-11-05/201).

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H. Main Landing Gear Inner Door Installation

SUBTASK 32-13-11-420-007

(1) Do these steps to install the inner door [2]:

(a) Put the inner door [2] in its position for connection to the center door [1].

(b) Install the nuts [23], washers [22], bolts [21], and bushings [27] to the forward and aft hinges [20] to connect the inner door [2] to the center door [1] (Views D and E).

1) If the bolts [26] are installed, remove them and pull the seal [25] slightly to the down position for easy installation of the bolts [21].

2) Do not tighten the nuts [23].

NOTE: Final torque to the nuts and installation of the pins will be done after final adjustments to the doors are made in the subsequent step.

(c) To connect the inner door [2] to the pushrod [45], do these steps (View F or G):

1) Install the bolt [32], washer [33], spacer [34], washer [35], and nut [36] to hold the inner door [2] to the pushrod [45].

a) Hand-tighten the nut [36].

NOTE: Final torque to the nut and installation of the pin will be done after final adjustments to the doors are made in the subsequent step.

2) If you need to connect the pushrod [45] to the lower side strut [44], do these steps:

a) Install the bolt [38], washer [39], spacer [40], washer [41], and nut [42] to connect the pushrod [45] to the lower side strut [44].

<1> Hand-tighten the nut [42].

NOTE: Final torque to the nut and installation of the pin will be done after final adjustments to the doors are made in the subsequent step.

(d) Do these steps to prepare the jumper cable [28] and the inner door [2] for installation:

1) Use Cleaning Procedure 1 to prepare the attachment surface for the jumper cable [28] on the inner door [2] (SWPM 20-20-00).

a) Use a very fine Scotch-Brite Type S pad, G50398, or an ultrafine Scotch-Brite 7448 pad, G50492.

b) Twist the pad with a circular movement while you push with your thumb.

2) Manually apply coating, C00064 to the abraded surface on the inner door [2].

(e) Install the jumper cable [28], screw [108], washers [109], and nut [110] on the inner door [2] (View K).

(f) Use an intrinsically safe approved bonding meter, COM-1550, to measure the resistance between the jumper cable [28] and the inner door [2] (SWPM 20-20-10).

1) Make sure that the electrical bonding resistance is 0.001 ohm (1.0 milliohms) or less.

(g) Refinish the abraded surfaces that you can see.

1) Apply one coat of primer, C00175 (TASK 51-21-72-370-801).

2) Apply coating, C00700 (TASK 51-21-73-370-802).

(h) Apply a fillet seal with sealant, A00247, to the jumper cable [28] and fasteners (SWPM 20-20-10).

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SUBTASK 32-13-11-820-002

(2) Do this task: Main Landing Gear Shock Strut Door Adjustment, TASK 32-13-11-820-801.

(a) It is not necessary to do this task if you can do these steps:

1) Make sure that the inner door [2] and center door [1] are being reinstalled on the same Main Landing Gear (MLG).

2) Make sure that the inner door [2], center door [1] and pushrod [45] that are being installed are the same ones that were removed.

3) Make sure that no adjustment was made on the existing pushrod [45].

4) Make sure that no adjustments have been made to the hinges [20], no shims added or removed.

5) Make sure that there is no damage to the lockwires on the pushrod [45].

6) Make sure that the MLG center door [1] has not been adjusted since the inner door [2] was removed.

SUBTASK 32-13-11-420-008

(3) Tighten the nut [23] and install the pin [24] in the bolt [21] (Views D and E).

(a) If it is necessary, loosen the nut [23] to the nearest castellation to align the holes for the pin [24].

(b) Make sure that there is a minimum gap of 0.030 in. (0.762 mm) between the washer [22] and center door [1] hinge bushing.

NOTE: Only required where the bushing is installed.

SUBTASK 32-13-11-420-009

(4) Tighten the nut [36] and install the pin [37] in the hole in the bolt [32] (Views F and G).

(a) If it is necessary, loosen the nut [36] to the nearest castellation to align the holes for the pin [37].

SUBTASK 32-13-11-420-010

(5) Tighten the nut [42] and install the pin [43] in the hole in the bolt [38] (Views F and G).

(a) If it is necessary, loosen the nut [42] to the nearest castellation to align the holes for the pin [43].

——— END OF TASK ———

TASK 32-13-11-000-803

6. Main Landing Gear Door Assembly Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the door assembly.

(a) It will give you the instruction to remove the inner door [2].

(b) It will then give you the instruction to remove the center door [1].

(c) The inner door [2] and center door [1] will stay attached.

(2) It is not necessary to do the Main Landing Gear Shock Strut Door Adjustment, TASK 32-13-11-820-801, if you can do these steps:

(a) Make sure that the inner door [2] and center door [1] are being reinstalled on the same Main Landing Gear (MLG).

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(b) Make sure that the inner door [2], center door [1], and pushrod [45] that are being installed are the same ones that were removed.

(c) Make sure that no adjustment was made on the existing pushrod [45].

(d) Make sure that no adjustments have been made to the hinges [20], no shims added or removed.

(e) Make sure that there is no damage to the lockwires on the pushrod [45].

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-13-11-820-801	Main Landing Gear Shock Strut Door Adjustment (P/B 501)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
732	Left Main Landing Gear - Center Door
733	Left Main Landing Gear - Inboard Door
734	Left Main Landing Gear
742	Right Main Landing Gear - Center Door
743	Right Main Landing Gear - Inboard Door
744	Right Main Landing Gear

D. Prepare for the Removal

SUBTASK 32-13-11-480-009

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-13-11-480-010

(2) Put chocks around the tires of all the landing gear (PAGEBLOCK 10-11-05/201).

E. Main Landing Gear Door Assembly Removal

SUBTASK 32-13-11-020-003

(1) Do these steps to remove the door assembly at the inner door [2] location:

(a) If you need to replace the pushrod [45] due to excessive wear, do these steps (View F or G):

1) Remove the pin [43], nut [42], washer [41], spacer [40], washer [39], and bolt [38] to disconnect the pushrod [45] from the lower side strut [44].

2) Remove the pin [37], nut [36], washer [35], spacer [34], washer [33], and bolt [32] to disconnect the inner door [2] from the pushrod [45].

(b) If you do not need to replace the pushrod [45], do this step (View F or G):

1) Remove the pin [37], nut [36], washer [35], spacer [34], washer [33], and bolt [32] to disconnect the inner door [2] from the pushrod [45].

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SUBTASK 32-13-11-020-004

(2) Do these steps to remove the door assembly at the center door [1] location:

(a) Remove the screw [103], washers [104], clamp [105], and nut [107] to disconnect the jumper cable [106] from the MLG (View J).

(b) For the top aft hinge, remove the pin [10], nut [9], washer [77], spacers [7], washer [5], and bolt [4] to disconnect the tie rod [3] from the tray assembly [6] on the shock strut [8] (View B).

(c) For the top forward hinge, remove the pin [12], nut [13], washer [78], spacers [14], washer [15], and bolt [16] to disconnect the tie rod [11] from the tray assembly [6] on the shock strut [8] (View C).

(d) Do these steps to remove the anti-rotation plate [68] and upper plate [88] (Views D, E, A-A, and B-B):

1) Remove the bolt [70], washer [71], washer [58], and nut [59].

2) Remove the plate [68].

3) Remove the bolts [84], washers [85], washers [57], and nuts [87].

4) Remove the upper plate [88], shim [90], and spacer [91].

(e) For the bottom aft hinge, remove the bolt [17], washer [18], clip [69], barrel nut [47], and retainer [48] to disconnect the center door [1] from the shock strut [8] (View A-A).

(f) For the bottom forward hinge, remove the bolt [29], washer [30], washer [67], barrel nut [49], and retainer [50] to disconnect the center door [1] from the shock strut [8] (View B-B).

(g) Remove the door assembly (center door [1] and inner door [2]) from the airplane.

——— END OF TASK ———

TASK 32-13-11-420-803

7. Main Landing Gear Door Assembly Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the door assembly.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-13-11-820-801	Main Landing Gear Shock Strut Door Adjustment (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

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Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Center door	32-13-21-01-385	PTF ALL
		32-13-21-01-388	PTF ALL
		32-13-21-01-390	PTF ALL
		32-13-21-01-395	PTF ALL
		32-13-21-20-070	PTF ALL
		32-13-21-20-075	PTF ALL
2	Inner door	32-13-21-01-375	PTF ALL
		32-13-21-01-380	PTF ALL
		32-13-21-20-410	PTF ALL
		32-13-21-20-415	PTF ALL
10	Pin	32-13-21-01-280	PTF ALL
12	Pin	32-13-21-01-280	PTF ALL
37	Pin	32-13-21-01-020	PTF ALL
43	Pin	32-13-21-01-020	PTF ALL

F. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
732	Left Main Landing Gear - Center Door
733	Left Main Landing Gear - Inboard Door
734	Left Main Landing Gear
742	Right Main Landing Gear - Center Door
743	Right Main Landing Gear - Inboard Door
744	Right Main Landing Gear

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G. Prepare for the Installation

SUBTASK 32-13-11-480-011

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-13-11-480-012

(2) Put chocks around the tires of all the landing gear (PAGEBLOCK 10-11-05/201).

H. Main Landing Gear Door Assembly Installation

SUBTASK 32-13-11-400-005

(1) Do these steps to install the door assembly (center door [1] and inner door [2]) (View A-A and B-B):

(a) Put the door assembly (center door [1] and inner door [2]) in their position for connection to the shock strut [8] and the lower side strut [44].

(b) For the center door [1], bottom aft hinge, install the bolt [17], plate [68], clip [69], washer [18], barrel nut [47], and retainer [48].

(c) For the center door [1], bottom forward hinge, install the bolt [29], washer [30], washer [67], barrel nut [49], and retainer [50].

SUBTASK 32-13-11-400-006

(2) Continue the door assembly (center door [1] and inner door [2]) installation:

(a) For the center door [1], top aft hinge, install the bolt [4], washer [5], spacers [7], washer [77], and nut [9] to connect the tie rod [3] to the tray assembly [6] on the shock strut [8] (View B).

1) Do not tighten the nut [9].

NOTE: Final torque to the nut and installation of the pin will be done after final adjustments to the doors are made in the subsequent step.

(b) For the center door [1], top forward hinge, install the bolt [16], washer [15], spacers [14], washer [78], and nut [13] to connect the tie rod [11] to the tray assembly [6] on the shock strut [8] (View C).

1) Do not tighten the nut [13].

NOTE: Final torque to the nut and installation of the pin will be done after final adjustments to the doors are made in the subsequent step.

(c) To connect the inner door [2] to the pushrod [45], do these steps (View F or G):

1) Install the bolt [32], washer [33], spacer [34], washer [35], and nut [36] to hold the inner door [2] to the pushrod [45].

a) Hand-tighten the nut [36].

NOTE: Final torque to the nut and installation of the pin will be done after final adjustments to the doors are made in the subsequent step.

2) If you need to connect the pushrod [45] to the lower side strut [44], do these steps:

a) Install the bolt [38], washer [39], spacer [40], washer [41], and nut [42] to connect the pushrod [45] to the lower side strut [44].

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<1> Hand-tighten the nut [42].

NOTE: Final torque to the nut and installation of the pin will be done after final adjustments to the doors are made in the subsequent step.

SUBTASK 32-13-11-820-003

(3) Do this task: Main Landing Gear Shock Strut Door Adjustment, TASK 32-13-11-820-801.

(a) It is not necessary to do this task if you can do these steps:

1) Make sure that the inner door [2] and center door [1] are being reinstalled on the same Main Landing Gear (MLG).

2) Make sure that the inner door [2], center door [1], and pushrod [45] that are being installed are the same ones that were removed.

3) Make sure that no adjustment was made on the existing pushrod [45].

4) Make sure that no adjustments have been made to the hinges [20], no shims added or removed.

5) Make sure that there is no damage to the lockwires on the pushrod [45].

SUBTASK 32-13-11-420-011

(4) Do these steps to safety the fasteners on the center door [1] (Views D, E, A-A, and B-B):

(a) Tighten the bolt [17] and bolt [29] to 398 in-lb (45.0 N·m) - 422 in-lb (47.7 N·m).

1) Tighten the bolt [17] and bolt [29] to the low torque value.

2) Tighten the bolts to align the head with the anti-rotation plate [68] and upper plate [88], if it is necessary.

(b) Adjust the thickness of shim [90] to match washer thickness.

(c) Install the bolt [70], washer [71], washer [58], and nut [59].

(d) Install the upper plate [88], shim [90], spacer [91], bolt [84], washer [85], washer [57], and nut [87].

SUBTASK 32-13-11-420-013

(5) Tighten the nut [9] and install the pin [10] in the hole in the bolt [4] (View B).

(a) If it is necessary, loosen the nut [9] to the nearest castellation to align the holes for the pin [10].

SUBTASK 32-13-11-420-014

(6) Tighten the nut [13] and install the pin [12] in the hole in the bolt [16]. (View C).

(a) If it is necessary, loosen the nut [13] to the nearest castellation to align the holes for the pin [12].

SUBTASK 32-13-11-420-015

(7) Tighten the nut [36] and install the pin [37] in the hole in the bolt [32] (View F or G).

(a) If it is necessary, loosen the nut [36] to the nearest castellation to align the holes for the pin [37].

SUBTASK 32-13-11-420-016

(8) Tighten the nut [42] and install the pin [43] in the hole in the bolt [38] (View F or G).

(a) If it is necessary, loosen the nut [42] to the nearest castellation to align the holes for the pin [43].

SUBTASK 32-13-11-420-017

(9) Do these steps to prepare the jumper cable [106] and MLG for installation.

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(a) Use solvent cleaning procedure to prepare the attachment surface for the jumper cable [106] on the MLG (SWPM 20-20-00).

SUBTASK 32-13-11-420-018

(10) Install the jumper cable [106], screw [103], washers [104], clamp [105], and nut [107] on the MLG (View J).

SUBTASK 32-13-11-280-002

(11) Use an intrinsically safe approved bonding meter, COM-1550, to measure the resistance between the jumper cable [106] and the MLG (SWPM 20-20-10).

(a) Make sure that the electrical bonding resistance is 0.002 ohm (2.0 milliohms) or less.

SUBTASK 32-13-11-390-002

(12) Apply a fillet seal with sealant, A00247, to the jumper cable [106] and fasteners (SWPM 20-20-10).

——— END OF TASK ———

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MAIN LANDING GEAR SHOCK STRUT DOOR - ADJUSTMENT/TEST

1. General

A. This procedure has this task:

(1) An adjustment of the main landing gear shock strut door.

TASK 32-13-11-820-801

2. Main Landing Gear Shock Strut Door Adjustment

(Figure 501, Figure 502, Figure 503, and Figure 504)

A. General

(1) This task gives the instructions to adjust the Main Landing Gear (MLG) shock strut door.

(2) This task is not necessary if you can do these steps:

(a) Make sure that the inner door [2] and center door [1] are being reinstalled on the same MLG.

(b) Make sure that the inner door [2], center door [1], and pushrod that are being installed are the same ones that were removed.

(c) Make sure that no adjustment was made on the existing pushrod.

(d) Make sure that no adjustments have been made to the hinges, no shims added or removed.

(e) Make sure that there is no damage to the lockwires on the pushrod.

B. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
29-11-00-860-802	Hydraulic System A or B Pressurization with a Portable Hydraulic Cart (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-13-21-000-802	Main Landing Gear Wing Door Adjustment (P/B 501)
32-32-00-710-801	Main Landing Gear Test - Component Replacement (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

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Reference	Description
COM-163	Cart - Hydraulic, Portable 3000 PSI, 30 GPM Part #: 5833 Supplier: 59603 Part #: 673A5FCBBCLMNY4 Supplier: 59603 Part #: 9780-0327 Supplier: 05172 Part #: HGPU50-30-1SX Supplier: 0022N Part #: HPU2 Supplier: 0T652 Part #: HPU3.3 Supplier: 0T652 Part #: HPU4 Supplier: 0T652 Part #: HT2000-1-E/1-S Supplier: H6394 Part #: HT2000-1-E/1-S2 Supplier: H6394 Part #: HT2000-1-F-S Supplier: H6394 Part #: HT2000-1-F-S2 Supplier: H6394 Part #: PH50D Supplier: 10000 Part #: PH50E Supplier: 10000 Part #: TMH3000 Supplier: L1238 Opt Part #: 57301FC3BBCILMNY Supplier: 59603 Opt Part #: 57305KV3BBCI-A1 Supplier: 59603
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
G00440	Lockwire - MS20995C41, Corrosion Resistant Steel - 0.041 Inch (1.0414 mm) Diameter	NASM20995

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
731	Left Main Landing Gear - Outboard Door
734	Left Main Landing Gear
741	Right Main Landing Gear - Outboard Door
744	Right Main Landing Gear

F. Prepare for the Adjustment

SUBTASK 32-13-11-480-015

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-13-11-860-004

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(2) Lift the airplane with jacks until the MLG wheels are above the ground (TASK 07-10-01-580-801).

NOTE: Lift the airplane to a height that provides the clearance to retract the MLG.

SUBTASK 32-13-11-840-005

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Connect the portable hydraulic cart, COM-163, to the ground service disconnect module for hydraulic system A (TASK 29-11-00-860-802).

NOTE: Alternatively, the MLG can also be extended/retracted with the Electric Motor-Driven Pump (EMDP) (Main Landing Gear Test - Component Replacement, TASK 32-32-00-710-801).

G. Main Landing Gear Shock Strut Door Adjustment

SUBTASK 32-13-11-860-005

(1) Do the steps that follow to retract the MLG:

(a) Remove the DO NOT OPERATE tag, STD-858 from the control lever for the landing gear.

(b) For the applicable adjusted MLG door, remove the downlock pin (TASK 32-00-01-080-801).

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE RETRACTION PATH FOR THE APPLICABLE LANDING GEAR. THE LANDING GEAR AND DOORS MOVE QUICKLY. THE MOVEMENT OF THE LANDING GEAR AND DOORS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(c) Move the control lever for the landing gear to the UP position.

(d) Attach a DO NOT OPERATE tag, STD-858, to the control lever for the landing gear.

(e) Close the MLG door as follows:

1) Slowly increase the hydraulic pressure from the portable hydraulic cart, COM-163, until the MLG is in the up and locked position (TASK 29-11-00-860-802).

2) Make sure that the hydraulic pressure supplied by the portable hydraulic cart, COM-163, does not go above 3200 psi (22,063 kPa).

3) With the MLG in the up and locked position, decrease the hydraulic pressure from the portable hydraulic cart, COM-163, to zero.

SUBTASK 32-13-11-820-009

(2) Calculate the Net Effect Value (NEV) of the center door [1].

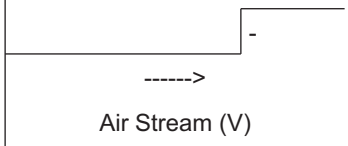
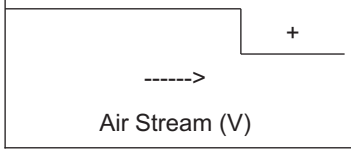
(a) Measure and record the fit or clearance values at each of the edge zones in Table 501.

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(b) Record the measured fit or clearance values.

(c) Compare the measured fit values to the values in Table 501.

Table 501/32-13-11-993-803 MAIN LANDING GEAR CENTER DOOR CLEARANCE AND FLUSHNESS VALUES

EDGE ZONE	CLEARANCE	FLUSHNESS
A	0.40 ±0.23 in. (10.160 ±5.842 mm)	+0.03 in. (0.76 mm)/-0.05 in. (-1.27 mm)
B	0.33 ±0.10 in. (8.382 ±2.540 mm)	NO ADJUSTMENT NECESSARY
C	0.40 ±0.23 in. (10.160 ±5.842 mm)	+0.03 in. (0.76 mm)/-0.05 in. (-1.27 mm)
DOOR FLUSHNESS DIAGRAM		
	(NEGATIVE MISFAIR)	A negative tolerance (-) indicates the face of the mismatch faces aft.
	(POSITIVE MISFAIR)	A positive tolerance (+) indicates the face of the mismatch is into the air stream.

SUBTASK 32-13-11-970-002

(3) If one or more of the measured values of the edge zone are more than the tolerances shown, find and convert all of the measured step (misfair) or clearance values into a NEV (Figure 502).

(a) Calculate the NEV for the interface.

1) Add all of the calculated NEV values, both those in tolerance and those that are not in the tolerance and divide the result by the total number of measurements.

NOTE: The result is the total NEV for the interface.

(b) Do a check of the NEV result against Net Effect Limit (NEL) for the interface given in the chart.

NOTE: This value is usually 1.0. If the NEV for the interface is less than or equal to the NEL, the interface is aerodynamically acceptable.

(c) Make sure that the calculated NEV is within the NEL value (the interface meets the aerosmoothness requirements).

SUBTASK 32-13-11-840-006

(4) If the fit is not in tolerance, do the steps that follow:

(a) Remove the DO NOT OPERATE tag, STD-858 from the landing gear control lever.

(b) Operate the portable hydraulic cart, COM-163, to supply hydraulic pressure to the MLG.

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WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE RETRACTION PATH FOR THE APPLICABLE LANDING GEAR. THE LANDING GEAR AND DOORS MOVE QUICKLY. THE MOVEMENT OF THE LANDING GEAR AND DOORS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(c) Move the landing gear control lever to the DN position.

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(d) Install the downlock pins on all the landing gears (TASK 32-00-01-480-801).

(e) Attach a DO NOT OPERATE tag, STD-858, to the control lever for the landing gear.

(f) Decrease the pressure from the portable hydraulic cart, COM-163, to zero.

(g) Adjust the center door [1] as follows:

NOTE: A change in the adjustment of the center door can have an effect on the adjustment of the wing door.

1) Do a check of the wing door, do this task: Main Landing Gear Wing Door Adjustment, TASK 32-13-21-000-802.

2) To adjust the door at the bottom hinge line, adjust the washer stackup at the two mounting points.

NOTE: The maximum total combined washer thickness shall not exceed the thickness of three BACW10P533CA washers.

3) If it is necessary, adjust the grip length of the bolt.

4) Tighten the bolts, after the washers have been adjusted for fair.

5) Do a check again to make sure that the center door [1] is still in tolerance (Table 501).

SUBTASK 32-13-11-220-003

(5) Measure and record the clearance A between the inner door [2] and fairing panel [9] and the clearance B between the inner door [2] and fairing panel [4] (Figure 503).

(a) Make sure that the clearances are in tolerance (Table 502).

Table 502/32-13-11-993-805 MAIN LANDING GEAR SHOCK STRUT DOORS AND BONDING PARTS FIT CLEARANCES

Permitted Clearances Between the Inner Door and Fairing Panels
A: 0.4000 ±0.1200 in. (10.1600 ±3.0480 mm)
B: 0.4000 ±0.1200 in. (10.1600 ±3.0480 mm)

SUBTASK 32-13-11-820-010

(6) Measure the fit or clearance values for the inner door [2] at the edge zones in Table 503.

(a) Record the measured fit or clearance values.

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(b) For all airplane MLG inner door [2] clearance and flushness checks, compare the measured fit values to the values in Table 503.

NOTE: After the inboard door rigging has been completed and with the door in the stowed position. The leading and trailing edges of the exterior surface of the inboard MLG door zones D and E inboard, should lie inside the exterior surface of the wing-to-body fairing.

Table 503/32-13-11-993-804 MAIN LANDING GEAR INNER DOOR CLEARANCE AND FLUSHNESS VALUES

EDGE ZONE	CLEARANCE	FLUSHNESS
D(1)	0.40 ±0.12 in. (10.16 ±3.05 mm)	-0.0694 ±0.0500 in. (-1.76 ±1.27 mm)
D(2)	0.425 ±0.120 in. (10.79 ±3.05 mm)	-0.0985 ±0.0500 in. (-2.50 ±1.27 mm)
E(2)	0.40 ±0.12 in. (10.16 ±3.05 mm)	-0.3526 ±0.0500 in. (-8.96 ±1.27 mm)
E(1)	0.40 ±0.12 in. (10.16 ±3.05 mm)	-0.3021 ±0.0500 in. (-7.67 ±1.27 mm)
F	0.340 ±0.100 in. (8.64 ±2.54 mm)	-0.3526 ±0.0500 in. (-8.96 ±1.27 mm)
DOOR FLUSHNESS DIAGRAM		
------> Air Stream (V)	(NEGATIVE MISFAIR)	A negative tolerance (-) indicates the face of the mismatch faces aft.
------> Air Stream (V)	(POSITIVE MISFAIR)	A positive tolerance (+) indicates the face of the mismatch is into the air stream.

(c) Make sure that the inner door clearance and flushness measured values are within the limits in Table 503.

SUBTASK 32-13-11-840-007

(7) If the fit is not in tolerance, do the steps that follow:

(a) Remove the DO NOT OPERATE tag, STD-858 from the landing gear control lever.

(b) Operate the portable hydraulic cart, COM-163, to supply hydraulic pressure to the MLG.

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE RETRACTION PATH FOR THE APPLICABLE LANDING GEAR. THE LANDING GEAR AND DOORS MOVE QUICKLY. THE MOVEMENT OF THE LANDING GEAR AND DOORS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

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(WARNING PRECEDES)

CAUTION ONLY EXTEND THE MAIN GEAR IF THE CENTER DOOR IS ADJUSTED ZERO TO POSITIVE. NEGATIVE ADJUSTMENT CAN CAUSE THE OUTER DOOR TO HIT THE CENTER DOOR. DAMAGE CAN OCCUR TO THE TWO DOORS.

(c) Move the landing gear control lever to the DN position.

(d) Attach a DO NOT OPERATE tag, STD-858, to the landing gear control lever.

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(e) Install the downlock pins on all the landing gears (TASK 32-00-01-480-801).

(f) Decrease the pressure from the portable hydraulic cart, COM-163, to zero.

(g) Remove the lockwires from the jamnuts [5] on pushrod [6].

(h) Loosen the jamnuts [5].

(i) Turn the pushrod [6] on either direction to adjust the pushrod [6].

(j) After the final adjustment, insert a 0.050 in. (1.270 mm) pin or wire into the inspection hole.

(k) Make sure that the wire is stopped by the rod end for adjustment to be acceptable.

(l) Tighten the jamnut [5] at each end.

(m) Install MS20995C41 lockwire, G00440, on the jamnuts [5].

(n) Do a check again to make sure that the inner door [2] is still in tolerance (Table 503).

SUBTASK 32-13-11-220-004

(8) Make sure that the clearance A between the inner door [2] and fairing panel [9] and the clearance B between the inner door [2] and fairing panel [4] are in tolerance (Table 502).

SUBTASK 32-13-11-820-011

(9) If it is necessary, adjust the pushrods [6] until the fit tolerances meet the tolerances (Table 501 or Table 503).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-13-11-840-008

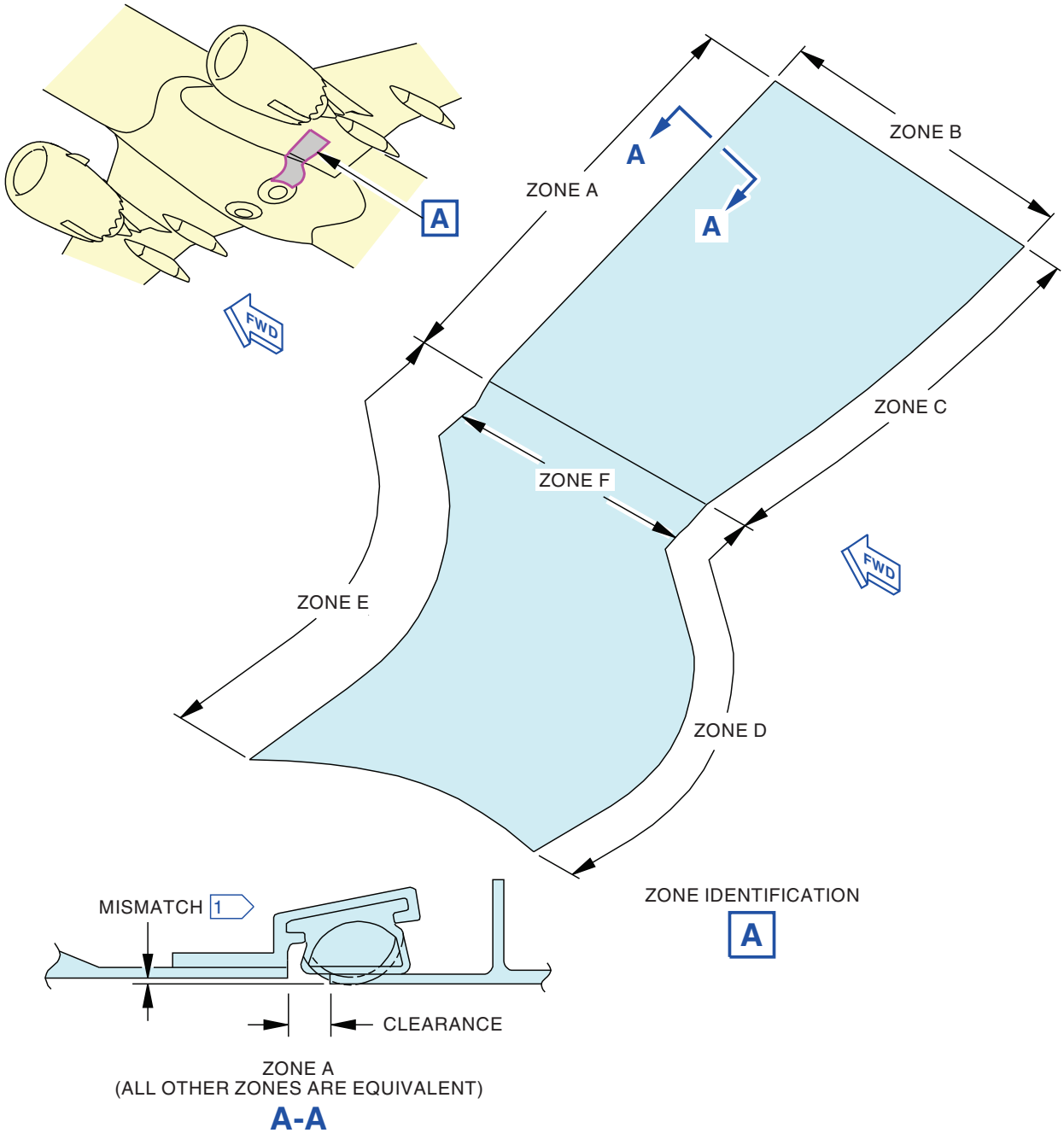
(1) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-13-11-580-002

(2) Lower the airplane and remove the jacks (TASK 07-10-01-580-802).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

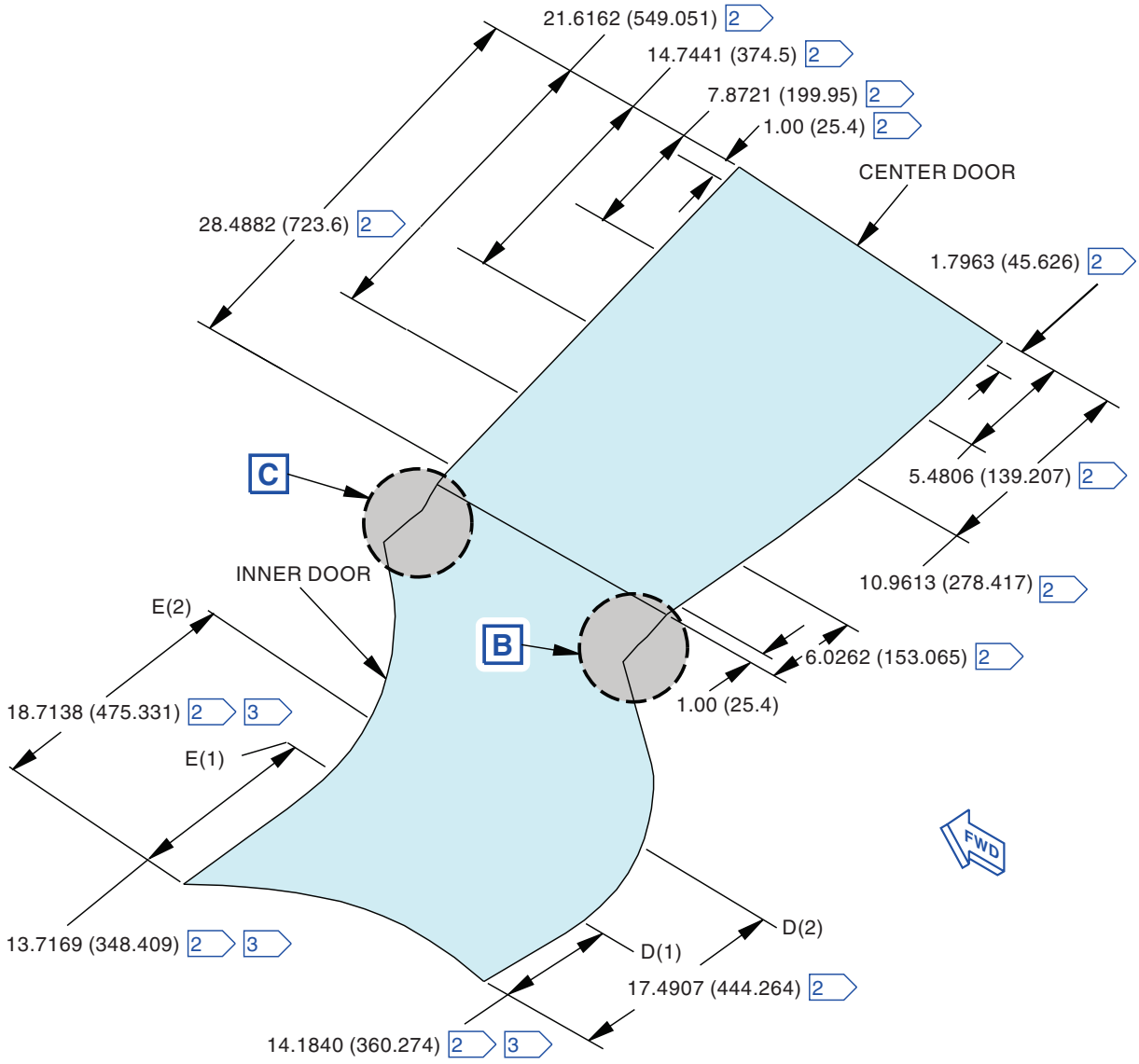


1 A POSITIVE TOLERANCE (+) INDICATES THE FACE OF THE MISMATCH IS FORWARD (INTO THE AIRSTREAM). A NEGATIVE TOLERANCE (-) INDICATES THE FACE OF THE MISMATCH FACES AFT.

2414131 S00061534471_V2

Main Landing Gear Shock Strut Doors Fit Tolerances
Figure 501/32-13-11-990-806 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



DIMENSIONS

A

NOTE:

ALL DIMENSIONS ARE IN INCHES (MILLIMETERS ARE IN PARENTHESES). THESE MEASUREMENTS LOCATIONS ARE FOR STEP/MISFAIR BETWEEN THE DOOR AND THE PANEL.

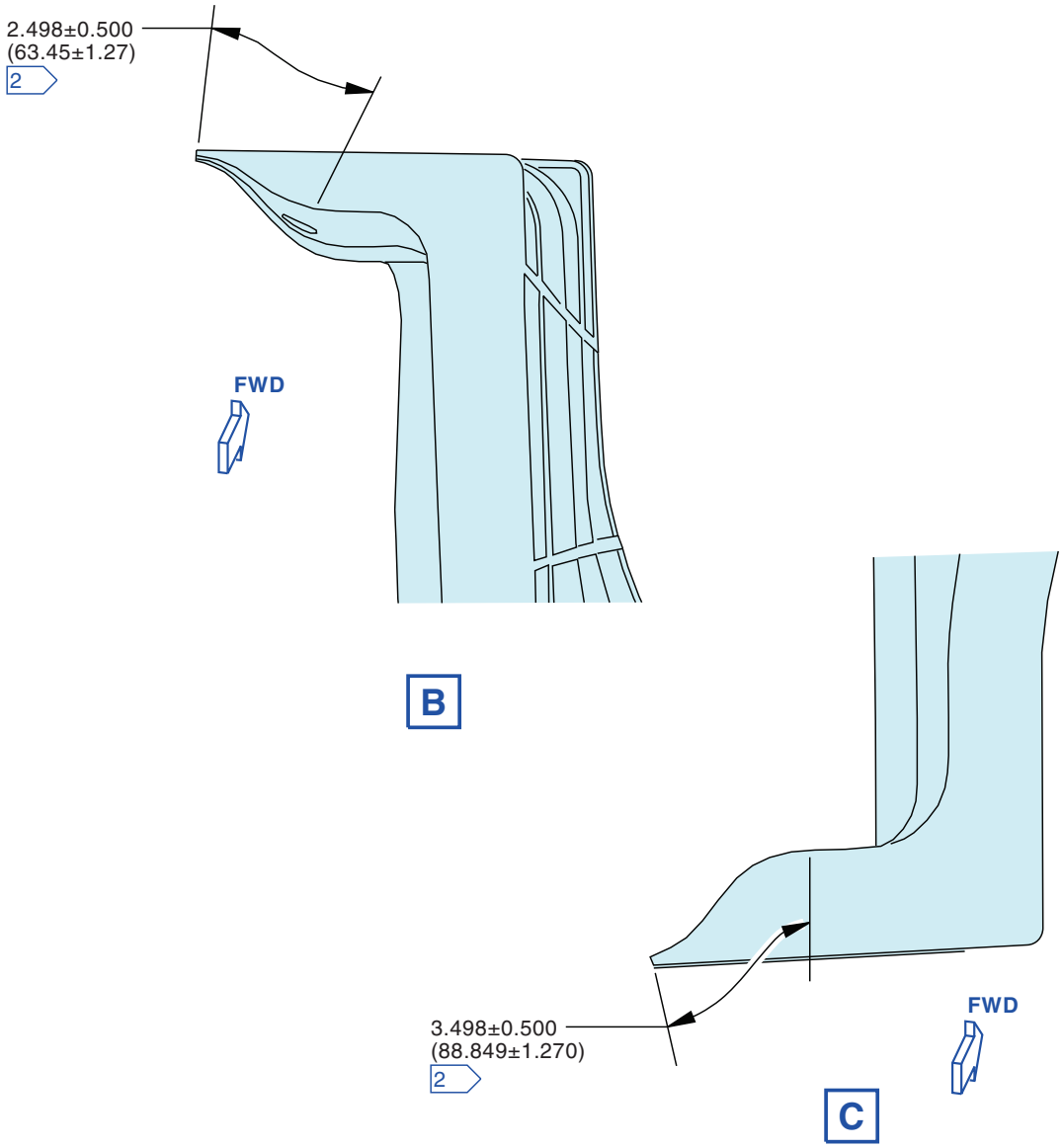
2 THE DIMENSIONAL TOLERANCE IS ±0.5000 INCH (12.7 mm).

3 NEW CALCULATIONS ARE NOT REQUIRED.

2414132 S00061534472_V2

Main Landing Gear Shock Strut Doors Fit Tolerances
Figure 501/32-13-11-990-806 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



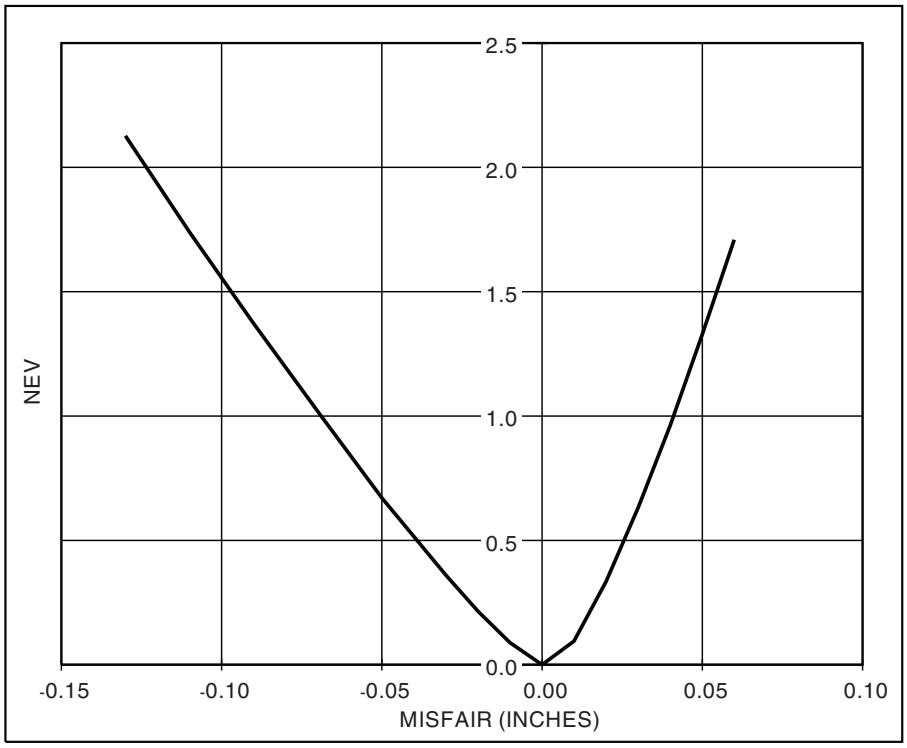
NOTE:

ALL DIMENSIONS ARE IN INCHES (MILLIMETERS ARE IN PARENTHESES). THESE MEASUREMENT LOCATIONS ARE FOR STEP/MISFAIR BETWEEN THE DOOR AND THE PANEL.

2414133 S00061534473_V2

Main Landing Gear Shock Strut Doors Fit Tolerances
Figure 501/32-13-11-990-806 (Sheet 3 of 3)

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MISFAIR	NEV
-0.130	2.127
-0.110	1.741
-0.090	1.373
-0.070	1.019
-0.050	0.671
-0.030	0.361
-0.020	0.215
-0.010	0.089
0.000	0.000
0.010	0.095
0.020	0.335
0.030	0.633
0.040	0.962
0.050	1.329
0.060	1.709

1

MAIN LANDING GEAR WING DOOR NET EFFECT VALUE CHART

NOTE:

THE NET EFFECT LIMIT (NEL) FOR THIS ITEM IS 1.0.

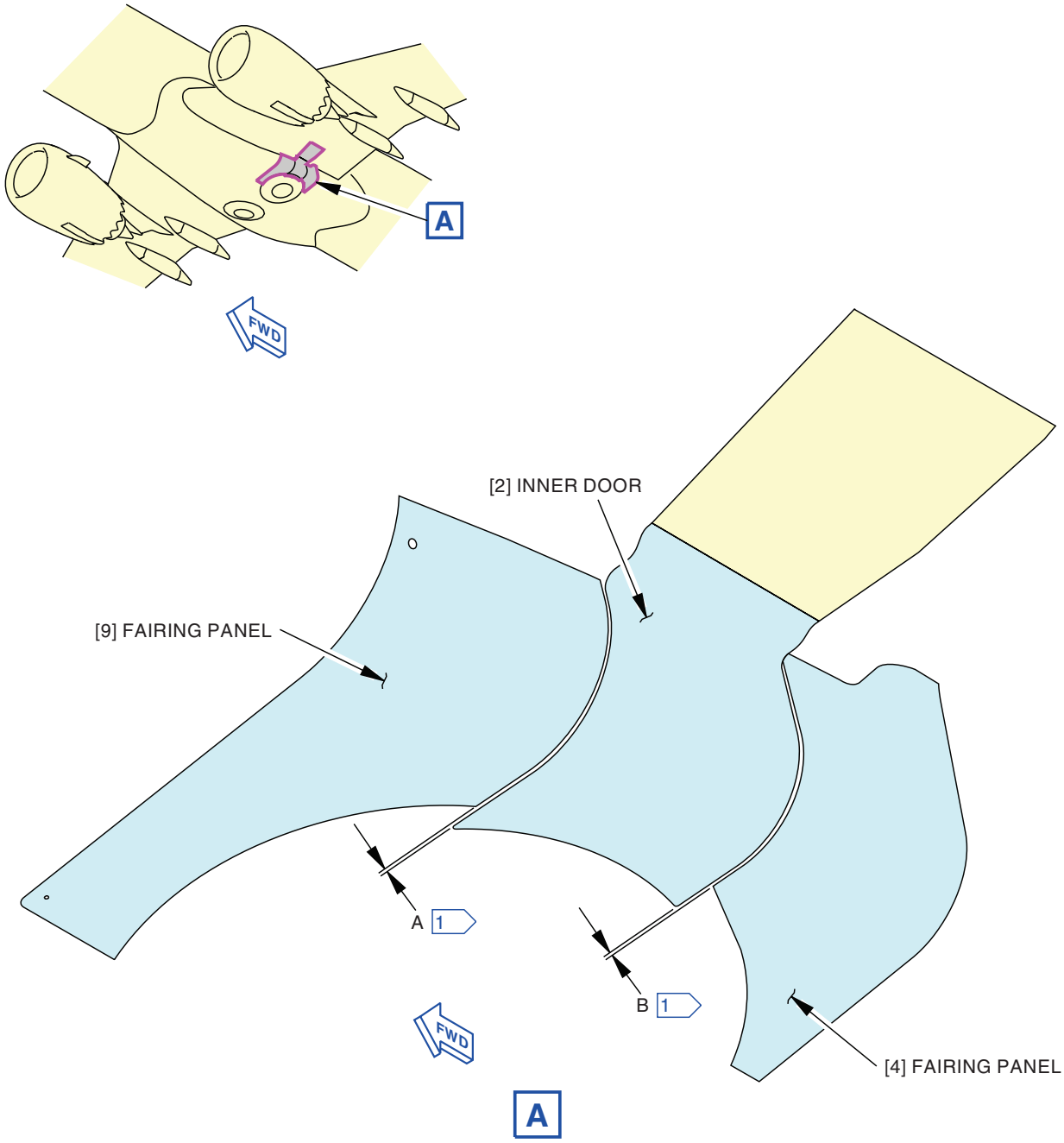
IF ALL MEASURED STEP VALUES FOR THIS ITEM ARE BETWEEN-0.041 AND -0.069 THEN PASSAGE OF NEL IS ASSURED AND AN ACTUAL NEL CALCULATION IS NOT REQUIRED.

1 NEV OF 1.0 CORRESPONDS TO MISFAIR OF 0.041 AND -0.069.

2414134 S00061534474_V1

Main Landing Gear Wing Door Net Effect Value Chart
Figure 502/32-13-11-990-807

ECCN 9E991 BOEING PROPRIETARY - See title page for details

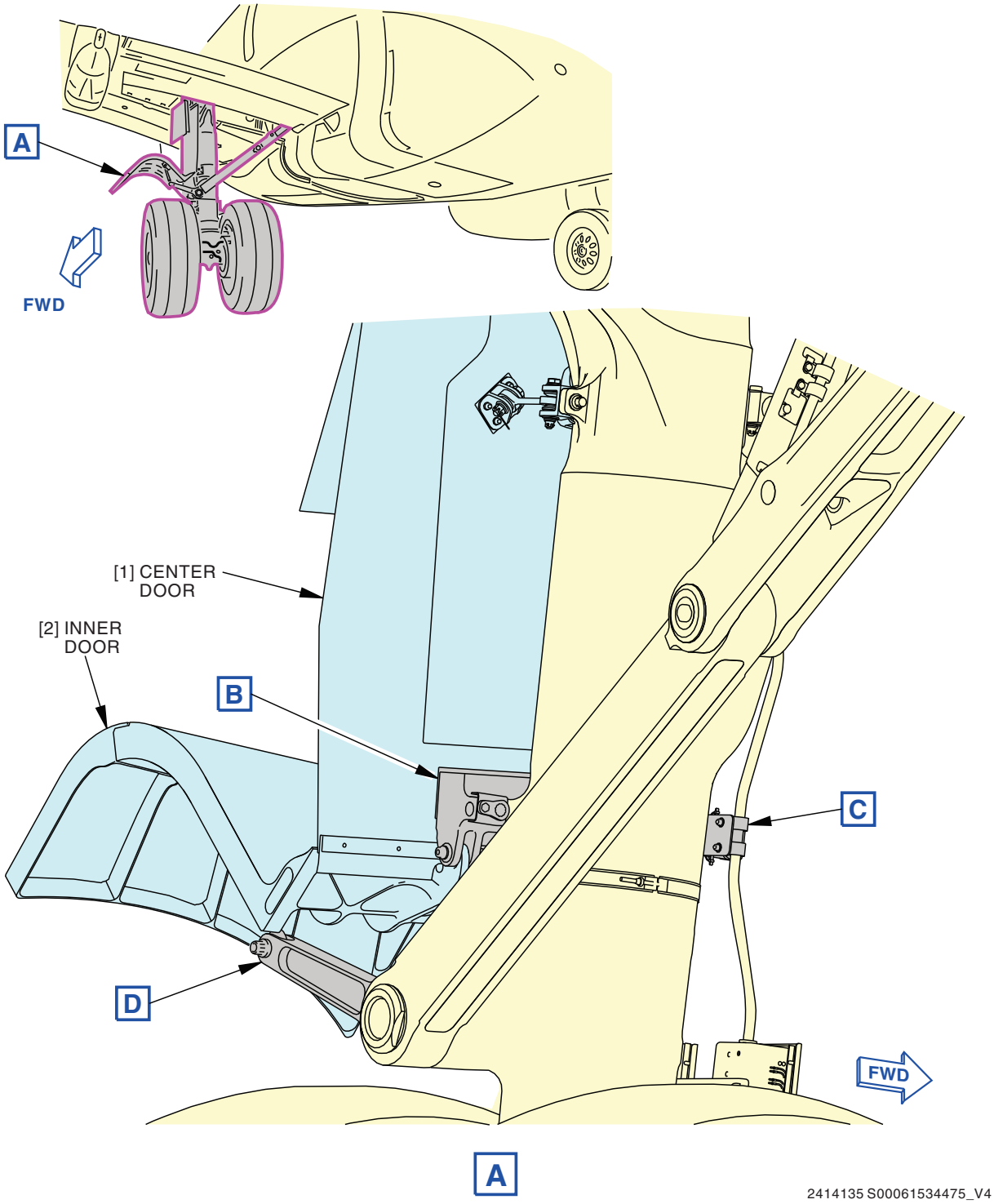


1 CLEARANCE BETWEEN THE INNER DOOR AND THE FAIRING PANEL

2806891 S0000644950_V5

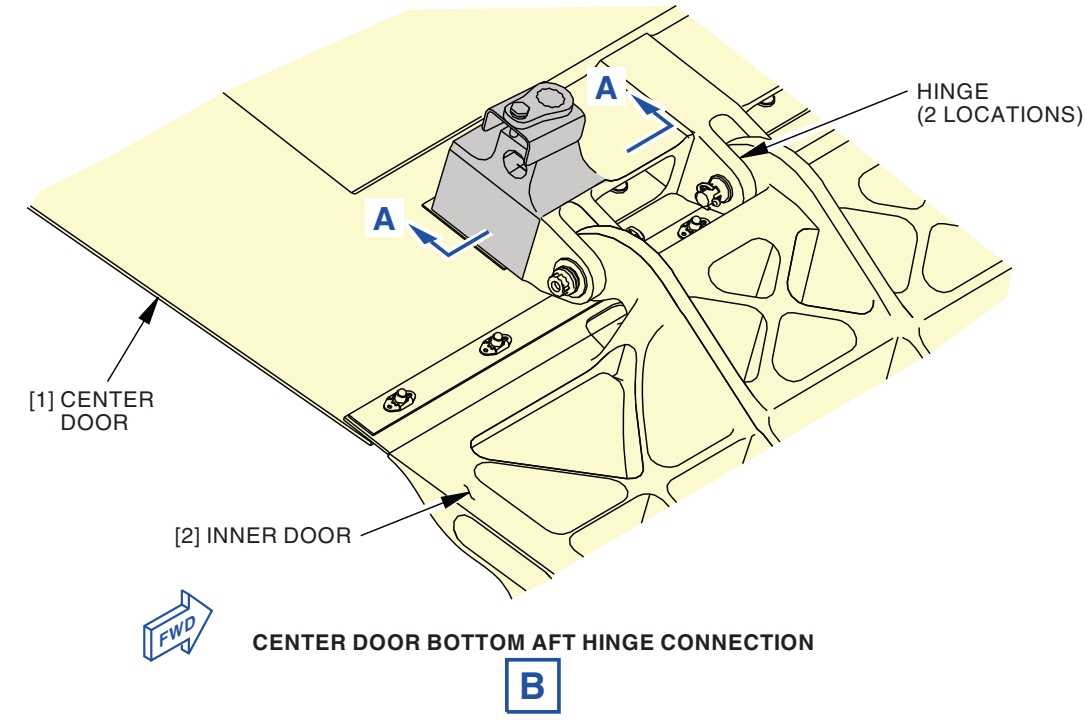
Main Landing Gear Shock Strut Doors and Bonding Parts Fit Clearances
Figure 503/32-13-11-990-808

ECCN 9E991 BOEING PROPRIETARY - See title page for details



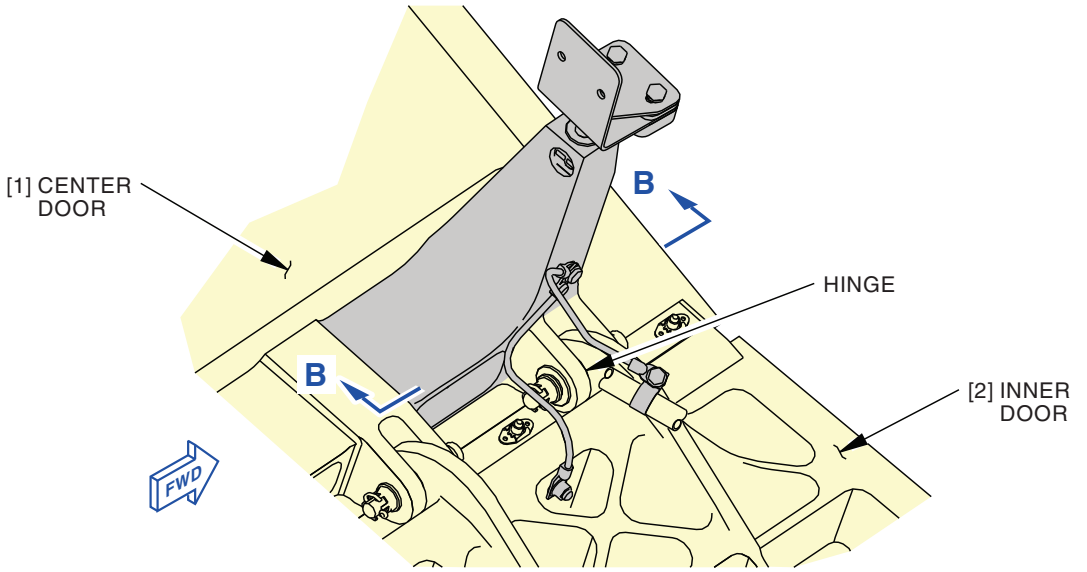
Main Landing Gear Shock Strut Doors Adjustment
Figure 504/32-13-11-990-809 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CENTER DOOR BOTTOM AFT HINGE CONNECTION

B



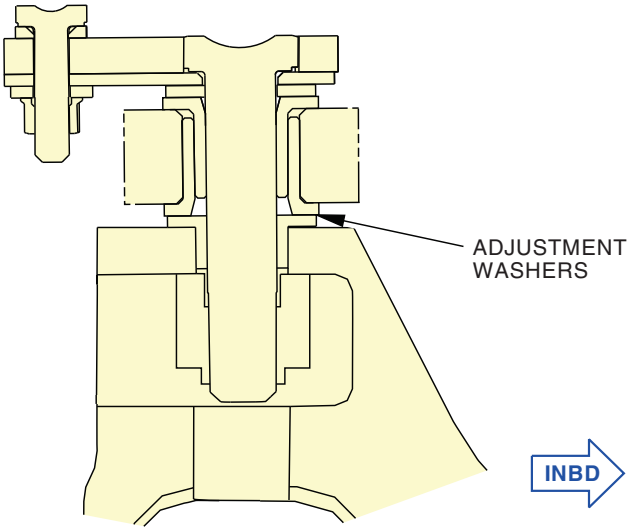
CENTER DOOR BOTTOM FORWARD HINGE CONNECTION

C

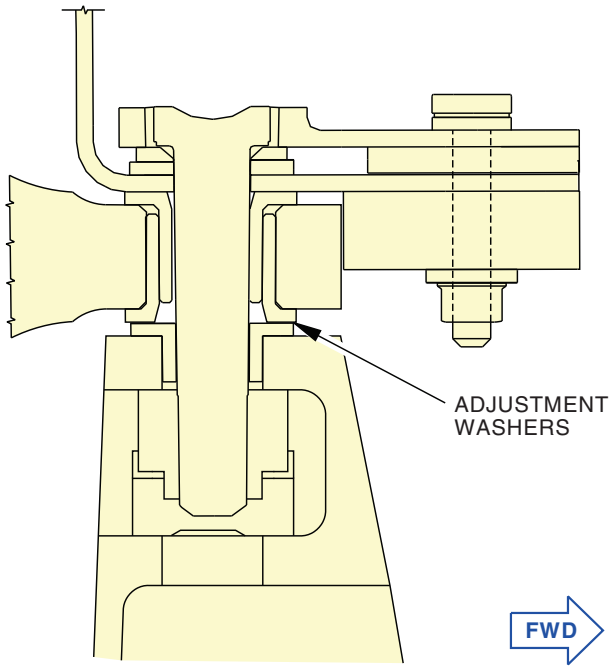
2937787 S0000713136_V1

Main Landing Gear Shock Strut Doors Adjustment
Figure 504/32-13-11-990-809 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CENTER DOOR BOTTOM AFT HINGE CONNECTION
A-A

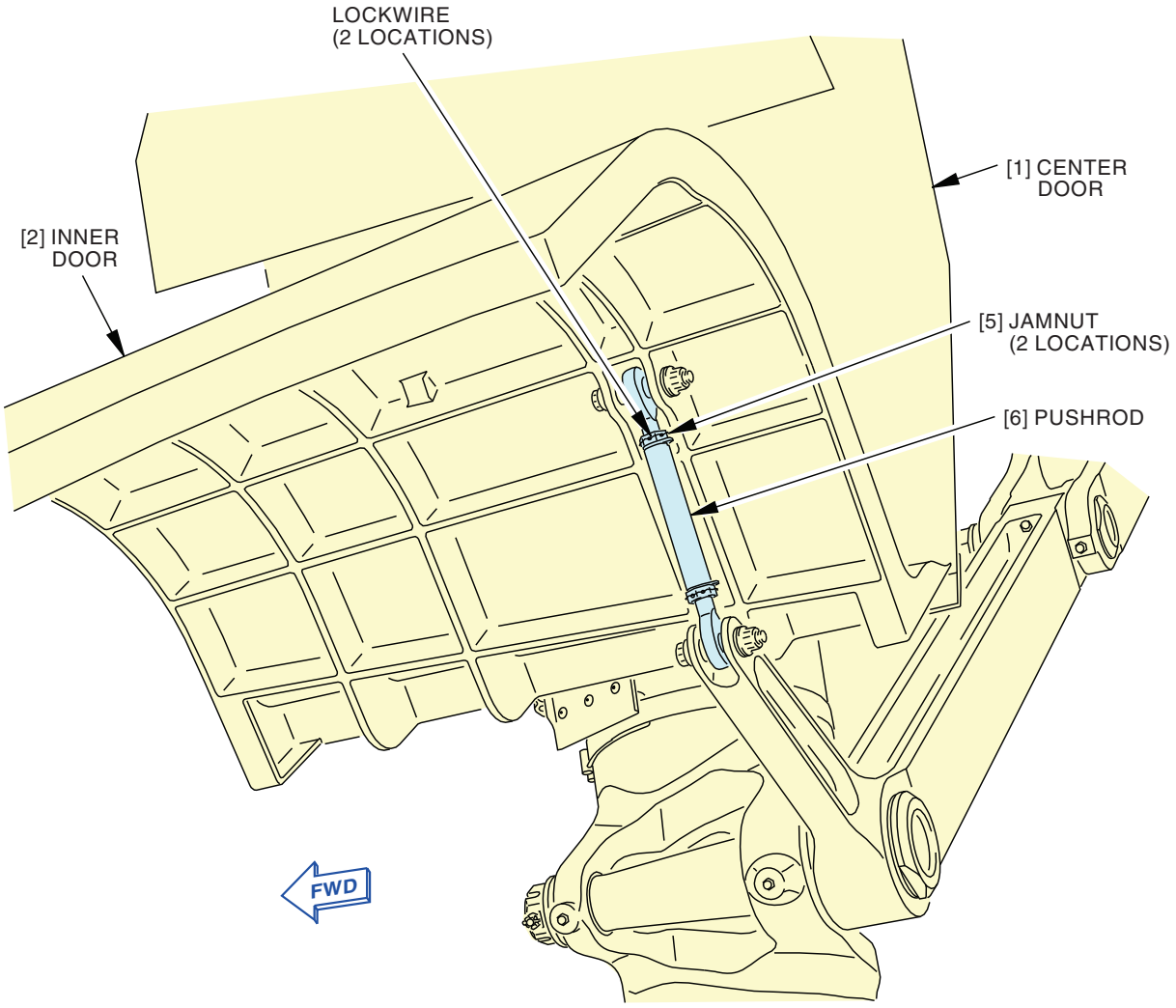


CENTER DOOR BOTTOM FORWARD HINGE CONNECTION
B-B

2937788 S0000713137_V1

Main Landing Gear Shock Strut Doors Adjustment
Figure 504/32-13-11-990-809 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



INNER DOOR PUSHROD

D

2414140 S00061534480_V3

Main Landing Gear Shock Strut Doors Adjustment
Figure 504/32-13-11-990-809 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN LANDING GEAR WING DOOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the main landing gear wing door

(2) An installation of the main landing gear wing door.

TASK 32-13-21-000-801

2. Main Landing Gear Wing Door Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Consumable Materials

Reference	Description	Specification
G50079	Scraper - Wood or Plastic	

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
731	Left Main Landing Gear - Outboard Door
734	Left Main Landing Gear
741	Right Main Landing Gear - Outboard Door
744	Right Main Landing Gear

D. Access Panels

Number	Name/Location
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

E. Prepare for the Removal

SUBTASK 32-13-21-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-13-21-480-002

(2) Put chocks around the tires of all the landing gear (PAGEBLOCK 10-11-05/201).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-13-21-010-001

(3) For the left main landing gear, open this access panel:

Number	Name/Location
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

SUBTASK 32-13-21-010-002

(4) For the right main landing gear, open this access panel:

Number	Name/Location
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

F. Main Landing Gear Wing Door Removal

SUBTASK 32-13-21-020-001

(1) Disconnect the outer door [16].

(a) Examine the condition of the pushrod [3] for wear and do the subsequent steps to disconnect the outer door [16].

(b) If you need to replace the pushrod [3], do the steps that follow:

1) Remove the cotter pin [5], nut [4], and washer [6] from the bolt [9].

2) Remove the bolt [9], washer [8], and bushing [7] to disconnect the pushrod [3] from the door actuator bracket [10].

3) Remove the washer [8] from the bolt [9].

(c) If you do not need to replace the pushrod [3], do the steps that follow:

1) Remove the cotter pin [13], nut [12], and washer [11] from the bolt [2].

2) Remove the bolt [2] and washer [1] to disconnect the forward hinge [14] from the pushrod [3].

3) Remove the washer [1] from the bolt [2].

SUBTASK 32-13-21-020-002

(2) Disconnect the outer door [16] from the hinge fitting [19] and hinge fitting [23]:

(a) Remove the bolt [30], washers [31], and nut [33] to disconnect the jumper cable [32] from the stabilizer link assembly.

1) Use scraper, G50079, to remove sealant from the end of the jumper cable [32].

(b) Remove the cotter pin [15], nut [17], and washer [18] from the bolt [22].

(c) Remove the bolt [22], washer [21], and bushing [20] to disconnect the forward hinge [14] from the hinge fitting [19].

(d) Remove the washer [21] from the bolt [22].

(e) Remove the cotter pin [29], nut [28], and washer [27] from the bolt [25].

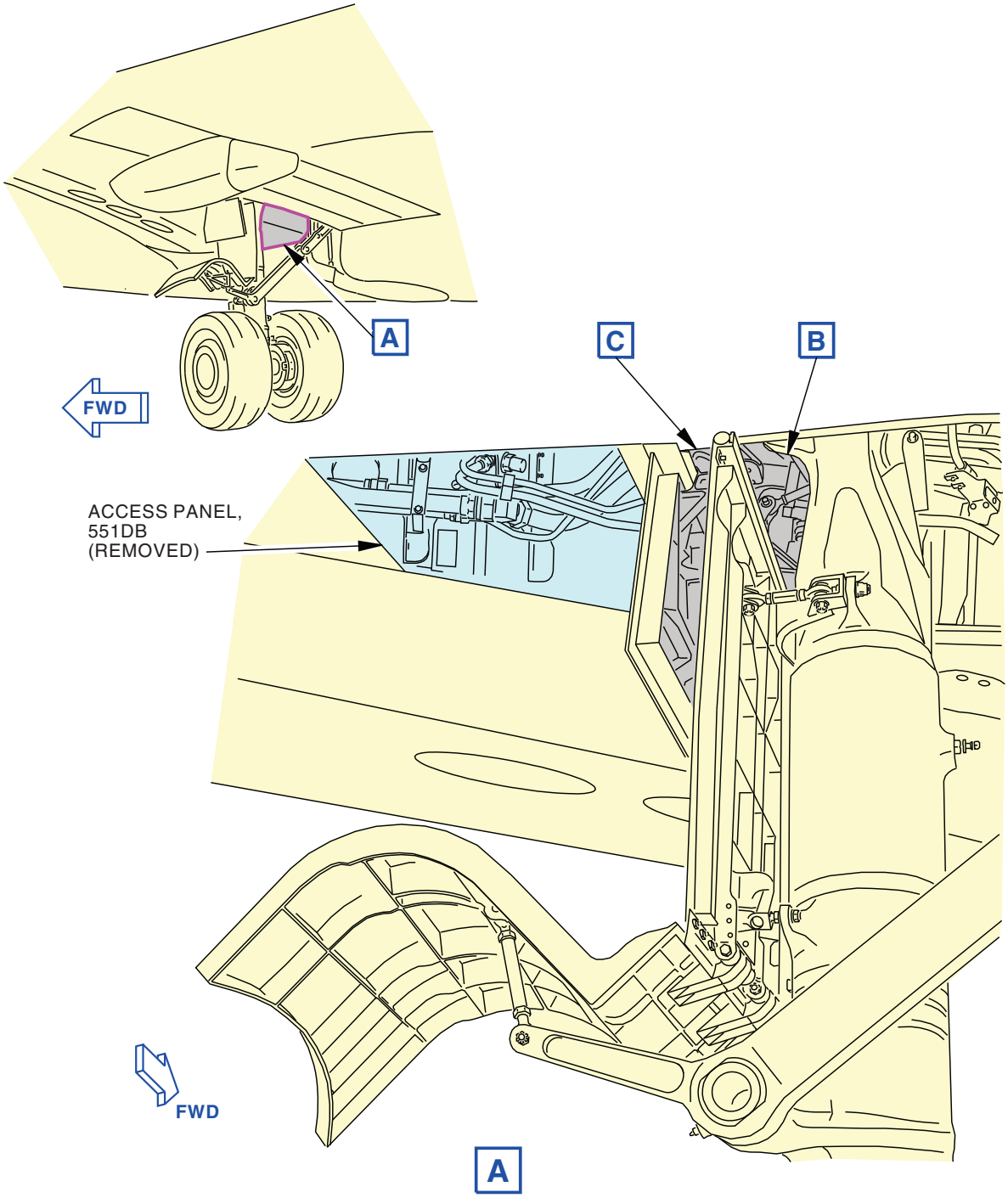
(f) Remove the bolt [25] and washer [24] to disconnect the aft hinge [26] from the hinge fitting [23].

(g) Remove the washer [24] from the bolt [25].

(h) Remove the outer door [16] from the airplane.

END OF TASK

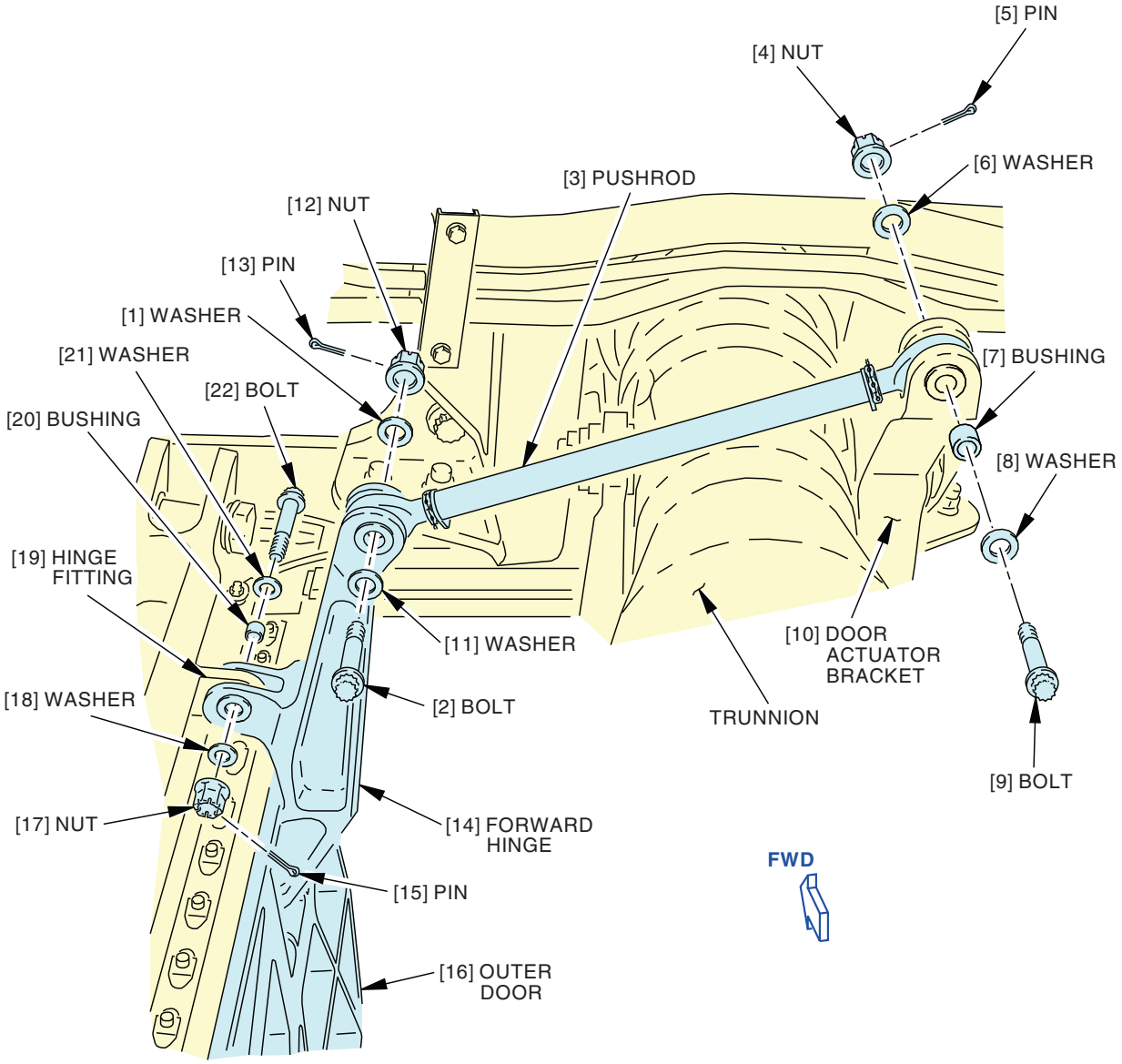
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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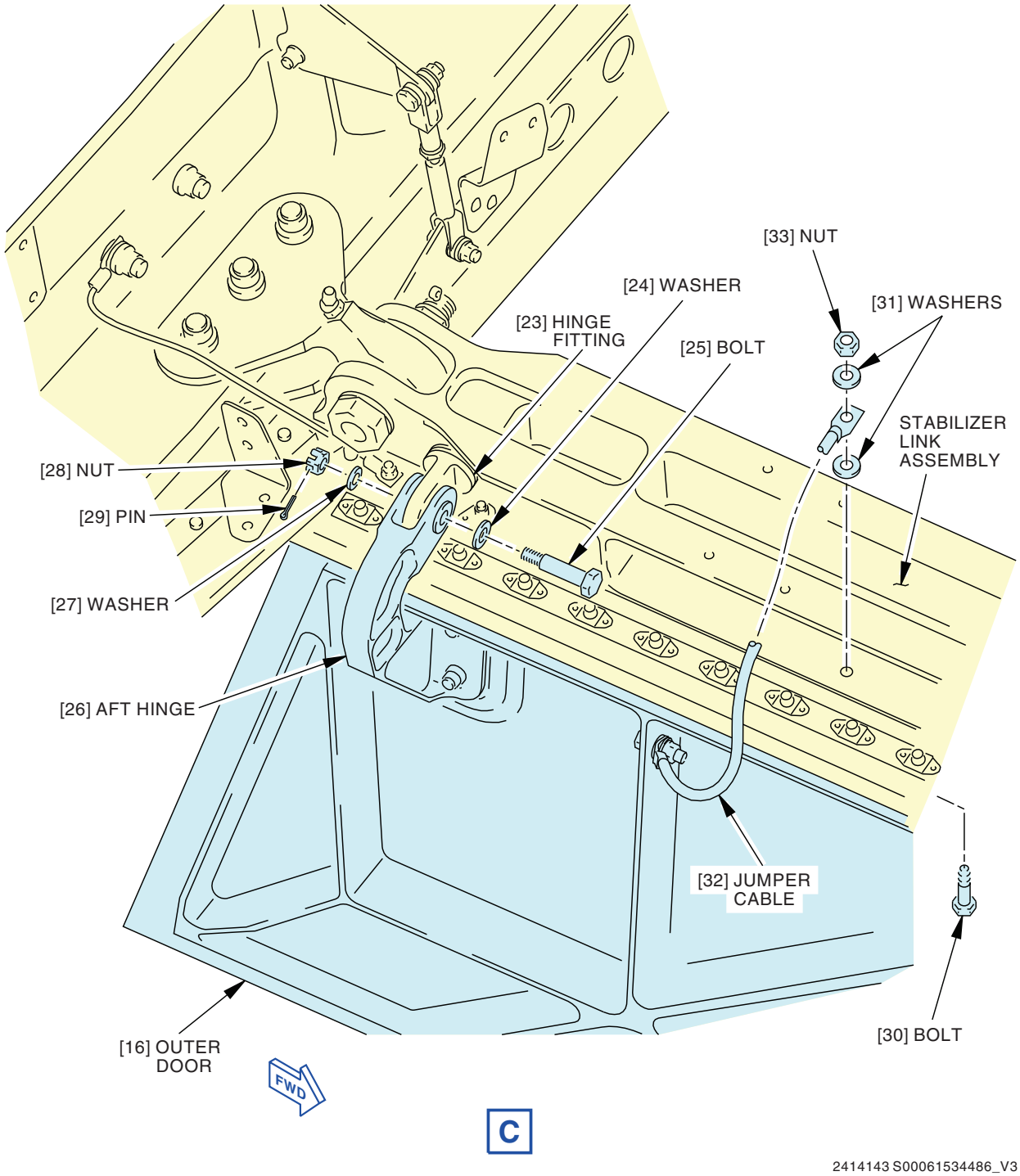
Main Landing Gear Wing Door Installation
Figure 401/32-13-21-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Wing Door Installation
Figure 401/32-13-21-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Wing Door Installation
Figure 401/32-13-21-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-13-21-420-801

3. Main Landing Gear Wing Door Installation

(Figure 401)

A. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-13-21-000-802	Main Landing Gear Wing Door Adjustment (P/B 501)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50337	Sealant - Fuel Tank	BMS5-45 Class B
A50359	Sealant - Low Density, Non-Chromate Type	BMS5-142 Type II Class B
B00130	Alcohol - Isopropyl	TT-I-735
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
D00633	Grease - Aircraft General Purpose	BMS3-33
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50262	Wiper - Cleaning	BMS15-5
G51464	Abrasive - Aluminum Oxide paper, 60 grit or finer	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
5	Pin	32-13-21-01-020	PTF ALL

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(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
13	Pin	32-13-21-01-025	PTF ALL
15	Pin	32-13-21-01-280	PTF ALL
16	Outer door	32-13-21-01-375	PTF ALL
		32-13-21-01-380	PTF ALL
29	Pin	32-13-21-01-285	PTF ALL

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
731	Left Main Landing Gear - Outboard Door
734	Left Main Landing Gear
741	Right Main Landing Gear - Outboard Door
744	Right Main Landing Gear

F. Access Panels

Number	Name/Location
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

G. Prepare for the Installation

SUBTASK 32-13-21-480-003

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

H. Main Landing Gear Wing Door Installation

SUBTASK 32-13-21-420-001

(1) Connect the outer door [16] to the hinge fitting [23] and hinge fitting [19]:

(a) Put the outer door [16] in its position in the hinge fitting [23] and hinge fitting [19].

(b) Install the washer [24] on the bolt [25].

(c) Align the hole in the aft hinge [26] to the hole in the hinge fitting [23].

(d) Put the bolt [25] through the aft hinge [26] and hinge fitting [23].

(e) Install the washer [27] and nut [28] on the bolt [25].

(f) Install the washer [21] on the bolt [22].

(g) Align the hole in the forward hinge [14] to the hole in the hinge fitting [19].

(h) Put the bolt [22] through the forward hinge [14] and hinge fitting [19].

(i) Install the washer [18] and nut [17] on the bolt [22].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(j) Hand tighten the nut [28] and nut [17] until final adjustment to the outer door [16] is made in the subsequent step.

1) After final adjustment to the outer door [16] and final torques to the nut [28] and nut [17] are applied, install the pin [29] and pin [15] in the bolt [25] and bolt [22].

SUBTASK 32-13-21-420-002

(2) Connect the outer door [16]:

(a) If you need to connect the outer door [16] to the pushrod [3], do the steps that follow:

1) Put the outer door [16] in its position for connection to the pushrod [3].
2) Install the washer [1] on the bolt [2].
3) Align the hole in the forward hinge [14] to the hole in the pushrod [3].
4) Apply grease, D00633, to the shank of the bolt [2].
5) Put the bolt [2] through the forward hinge [14] and pushrod [3].
6) Install the washer [11] and nut [12] on the bolt [2].
7) Hand tighten the nut [12] until final adjustment to the outer door [16] is made in the subsequent step.

a) After final adjustment to the outer door [16] and final torque to the nut [12] is applied, install the pin [13] in the bolt [2].

(b) If you need to connect the pushrod [3] to the door actuator bracket [10], do the steps that follow:

1) Put the pushrod [3] in its position for connection to the door actuator bracket [10].
2) Install the washer [8] on the bolt [9].
3) Align the hole in the pushrod [3] to the hole in the door actuator bracket [10].
4) Apply grease, D00633, to the shank of the bolt [9].
5) Put the bushing [7] through the door actuator bracket [10] and pushrod [3].
6) Put the bolt [9] through the door actuator bracket [10] and pushrod [3].
7) Install the washer [6] and nut [4] on the bolt [9].
8) Hand tighten the nut [4] until final adjustment to the outer door [16] is made in the subsequent step.

a) After final adjustment to the outer door [16] and final torque to the nut [4] is applied, install the pin [5] in the bolt [9].

(c) Install the jumper cable [32] as follows:

1) Clean the mating surface of the stabilizer link assembly with 60 grit or finer wet/dry aluminum oxide sandpaper, G51464.
2) Use cotton wiper, G00034, moist in alcohol, B00130, to clean the mating surfaces of the bolt [30], washers [31], nut [33], and jumper cable [32].

a) If it is necessary, use wiper, G50262, to remove remaining solvent.

3) Apply sealant, A00247, to the under-head surface of the bolt [30].
4) Install the bolt [30], washers [31], nut [33] that attach the jumper cable [32] to the stabilizer link assembly.

a) Tighten the nut [33] to 28 in-lb (3 N·m) - 35 in-lb (4 N·m).
b) Make sure that the squeeze-out of sealant, A00247, between the bolt [30] and structure is continuous.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

c) Make sure that the squeeze-out of sealant, A00247, is not more than 0.2 in. (5.1 mm) in all directions.

<1> If it is necessary, use cotton wiper, G00034, moist in solvent, B00184, to remove excess of sealant, A00247.

5) Measure the electrical bonding resistance between the jumper cable [32] and stabilizer link assembly (SWPM 20-20-00).

a) Use the non-intrinsically safe bonding meter, COM-614.

b) Make sure that the electrical bonding resistance is 0.001 ohm (1 milliohm) or less.

6) Apply a fillet seal with sealant, A50337, or sealant, A50231, or sealant, A50359, to cover the bolt [30], washers [31], nut [33], and jumper cable [32].

a) Make sure that there is no excess sealant on the bolt [30] head.

SUBTASK 32-13-21-820-001

(3) Do this task: Main Landing Gear Wing Door Adjustment, TASK 32-13-21-000-802.

SUBTASK 32-13-21-420-003

(4) Tighten the nut [4] and install the pin [5] in the hole in the bolt [9].

(a) If it is necessary, loosen the nut [4] to the nearest castellation to align the hole for the pin [5].

SUBTASK 32-13-21-420-004

(5) Tighten the nut [12] and install the pin [13] in the hole in the bolt [2].

(a) If it is necessary, loosen the nut [12] to the nearest castellation to align the hole for the pin [13].

SUBTASK 32-13-21-420-005

(6) Tighten the nut [17] and install the pin [15] in the hole in the bolt [22].

(a) If it is necessary, loosen the nut [17] to the nearest castellation to align the hole for the pin [15].

SUBTASK 32-13-21-420-006

(7) Tighten the nut [28] and install the pin [29] in the hole in the bolt [25].

(a) If it is necessary, loosen the nut [28] to the nearest castellation to align the hole for the pin [29].

I. Put the Airplane Back to Its Usual Condition

SUBTASK 32-13-21-410-001

(1) For the left main landing gear, do this step:

(a) Close this access panel:

Number	Name/Location
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

SUBTASK 32-13-21-410-002

(2) For the right main landing gear, do this step:

(a) Close this access panel:

Number	Name/Location
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN LANDING GEAR WING DOOR - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) An adjustment of the wing door for the main landing gear.

B. This task is not necessary if:

(1) The same wing door that is removed is re-installed without modifications.

(2) The door attachment rod-end or hinge fittings were not disturbed during the wing door removal/installation procedures.

TASK 32-13-21-000-802

2. Main Landing Gear Wing Door Adjustment

(Figure 501, Figure 502)

A. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
29-11-00-860-802	Hydraulic System A or B Pressurization with a Portable Hydraulic Cart (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-13-11-820-801	Main Landing Gear Shock Strut Door Adjustment (P/B 501)
32-32-00-710-801	Main Landing Gear Test - Component Replacement (P/B 501)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-163	Cart - Hydraulic, Portable 3000 PSI, 30 GPM Part #: 5833 Supplier: 59603 Part #: 673A5FCBBCLMNY4 Supplier: 59603 Part #: 9780-0327 Supplier: 05172 Part #: HGPU50-30-1SX Supplier: 0022N Part #: HPU2 Supplier: 0T652 Part #: HPU3.3 Supplier: 0T652 Part #: HPU4 Supplier: 0T652 Part #: HT2000-1-E/1-S Supplier: H6394 Part #: HT2000-1-E/1-S2 Supplier: H6394 Part #: HT2000-1-F-S Supplier: H6394 Part #: HT2000-1-F-S2 Supplier: H6394 Part #: PH50D Supplier: 10000 Part #: PH50E Supplier: 10000 Part #: TMH3000 Supplier: L1238 Opt Part #: 57301FC3BBCILMNY Supplier: 59603 Opt Part #: 57305KV3BBCI-A1 Supplier: 59603
STD-858	Tag - DO NOT OPERATE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
731	Left Main Landing Gear - Outboard Door
734	Left Main Landing Gear
741	Right Main Landing Gear - Outboard Door
744	Right Main Landing Gear

D. Prepare for the Adjustment

SUBTASK 32-13-21-480-004

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-13-21-860-001

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(2) Lift the airplane with jacks until the Main Landing Gear (MLG) wheels are above the ground and there is enough clearance to retract the MLG (TASK 07-10-01-580-801).

SUBTASK 32-13-21-840-001

WARNING KEEP PERSONS AND EQUIPMENT AWAY FROM ALL CONTROL SURFACES AND THE NOSE GEAR WHEN HYDRAULIC POWER IS SUPPLIED. THE AILERONS, ELEVATORS, RUDDER, FLAPS, SLATS, SPOILERS, AND THE NOSE GEAR ARE SUPPLIED WITH POWER BY THE HYDRAULIC SYSTEMS. INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR WHEN HYDRAULIC POWER IS SUPPLIED.

(3) Pressurize the hydraulic system A with a portable hydraulic cart, COM-163 (TASK 29-11-00-860-802).

NOTE: Alternatively, the MLG can also be extended/retracted with the Electric Motor-Driven Pump (EMDP) (Main Landing Gear Test - Component Replacement, TASK 32-32-00-710-801).

E. Main Landing Gear Wing Door Adjustment

SUBTASK 32-13-21-860-003

(1) Do the steps that follow to retract the MLG:

(a) Remove the DO NOT OPERATE tag, from the control lever for the landing gear.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) For the applicable MLG for the door you are adjusting, do this task: Landing Gear Downlock Pins Removal, TASK 32-00-01-080-801.

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE RETRACTION PATH FOR THE APPLICABLE LANDING GEAR. THE LANDING GEAR AND DOORS MOVE QUICKLY. THE MOVEMENT OF THE LANDING GEAR AND DOORS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(c) Move the control lever for the landing gear to the UP position to retract the MLG.

(d) Attach a DO NOT OPERATE tag, STD-858, to the control lever for the landing gear.

(e) Do the steps that follow to close the MLG wing door:

1) To retract the MLG, slowly increase the hydraulic pressure that is supplied by the portable hydraulic cart, COM-163, until the MLG is in the up and locked position.

2) Make sure that the hydraulic pressure supplied by the portable hydraulic cart, COM-163, does not go above 3200 psi (22,063 kPa).

3) With the gear in the up and locked position, adjust the hydraulic pressure supplied by the portable hydraulic cart, COM-163, to zero.

SUBTASK 32-13-21-820-002

(2) Make sure that the fit of the wing door [11] is in tolerance (Table 501).

(a) Make a written record of the fit values that are not in tolerance.

(b) Use the "ALL" tolerances, in (Table 501), for all airplane MLG wing door clearance and flushness checks.

Table 501/32-13-21-993-802 MAIN LANDING GEAR WING DOOR CLEARANCE AND FLUSHNESS VALUES

AIRPLANE EFFECTIVITY	EDGE ZONE	CLEARANCE	FLUSHNESS
ALL	A	0.340 ±0.100 in. (8.636 ±2.540 mm)	
ALL	B	0.400 ±0.230 in. (10.160 ±5.842 mm)	+0.041 in. (1.04 mm)/-0.069 in. (1.75 mm) (NEL=1.0)
ALL	C	0.330 ±0.100 in. (8.382 ±2.540 mm)	
ALL	D	0.400 ±0.230 in. (10.160 ±5.842 mm)	+0.041 in. (1.04 mm)/-0.069 in. (1.75 mm) (NEL=1.0)

DOOR FLUSHNESS DIAGRAM



(c) Compare the fit values that are not in tolerance with the Net Effect Value (NEV) chart (Figure 501).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-13-21-840-002

(3) If the fit is not in tolerance, do the steps that follow:

(a) Extend the MLG:

1) Remove the DO NOT OPERATE tag, from the control lever for the landing gear.

2) Operate the portable hydraulic cart, COM-163, to supply hydraulic pressure to the MLG.

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE RETRACTION PATH FOR THE APPLICABLE LANDING GEAR. THE LANDING GEAR AND DOORS MOVE QUICKLY. THE MOVEMENT OF THE LANDING GEAR AND DOORS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

3) Move the control lever for the landing gear to the DN position to extend the MLG.

4) Attach a DO NOT OPERATE tag, STD-858, to the control lever for the landing gear.

5) For the applicable MLG, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

6) Adjust the hydraulic pressure supplied by the portable hydraulic cart, COM-163, to zero.

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

7) Install the downlock pins on the MLG (TASK 32-00-01-480-801).

SUBTASK 32-13-21-820-003

(4) Adjust the eyebolts of the hinge fittings [14] (Figure 502):

NOTE: A change in the adjustment of the outer door can have an effect on the adjustment of the shock strut doors. If it is necessary, do a check on the adjustment of the shock strut doors (TASK 32-13-11-820-801).

(a) Remove the pins [10], nuts [9], washers [8], washers [7], bushing [6] (on forward hinge [4] only), and bolts [5] to disconnect the wing door [11] from the lower stabilizer link [15].

(b) Remove the pins [19], washers [18], washers [17], and nuts [16] from the hinge fittings [14].

(c) Remove the washer [12] and washer [13] located on the shank of the eyebolt from the inboard side, under the lug, of the lower stabilizing beam to the outboard side, under the nut [16].

(d) Install the washers [17], washers [18], and nut [16] to the hinge fittings [14].

1) Tighten the nuts [16] to 700 in-lb (79.1 N·m).

2) Increase torque until first slot in the nut [16] aligns with the hinge bolt [5].

3) Make sure that the torque is 920 in-lb (103.9 N·m) or less.

4) Install the pin [19].

(e) Install the bushing [6] (for the forward hinge [4] only), bolts [5], washers [7], washers [8], and nuts [9] to connect the wing door [11] to the lower stabilizer link [15].

1) Install the pins [10].

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SUBTASK 32-13-21-220-001

(5) Make sure that the wing door [11] is still in tolerance (Table 501).

SUBTASK 32-13-21-970-001

(6) If one or more of the measured values are more than the tolerances shown in the chart (Figure 501), convert the measured step (misfair) into a NEV shown in the chart (Figure 501).

(a) Find the NEV for the interface:

1) Add all of the calculated NEV values, both those in tolerance and those that are not in the tolerance and divide the result by the total number of measurements.

NOTE: The result is the NEV for the interface.

2) Example of the NEV calculation:

<Flushness> (Point A: -0.03 in, Point B: 0.04 in, Point C: -0.04 in, Point D: 0.01 in) (NEV'A' + NEV'B' + NEV'C' + NEV'D')/4 =(0.55 + 1.5 + 0.8 + 0.3)/4=0.788 < 1.0 (Within Limit).

(b) Do a check of the NEV result against Net Effect Limit (NEL) for the interface given in the chart (Figure 501).

NOTE: This value is usually 1.0. If the NEV for the interface is less than or equal to the NEL, the interface is aerodynamically acceptable.

(c) If the calculated NEV is more than the NEL (the interface does not meet the aerosmoothness requirements), do the steps that follow:

1) Adjust the length of the pushrods [1] (Figure 502):

a) Loosen the jamnut [2] on the pushrod [1] at either end and turn the pushrod [1] in either direction.

b) After the final adjustment of the pushrod [1], insert a 0.050 in. (1.270 mm) diameter pin or wire into the inspection hole.

c) Make sure the wire that you insert into the inspection hole touches and is stopped by the pushrod [1] end.

NOTE: The wire must be stopped by the pushrod [1] end for the adjustment to be acceptable.

d) Tighten the jamnuts [2] and washers [3].

e) Install the lockwire.

(d) Make sure that the wing door [11] is still in tolerance (Table 501).

1) If it is necessary, re-adjust the pushrods [1] or eyebolts of the hinge fittings [14].

F. Put the Airplane Back to Its Usual Condition

SUBTASK 32-13-21-840-003

(1) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-13-21-580-001

CAUTION

MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

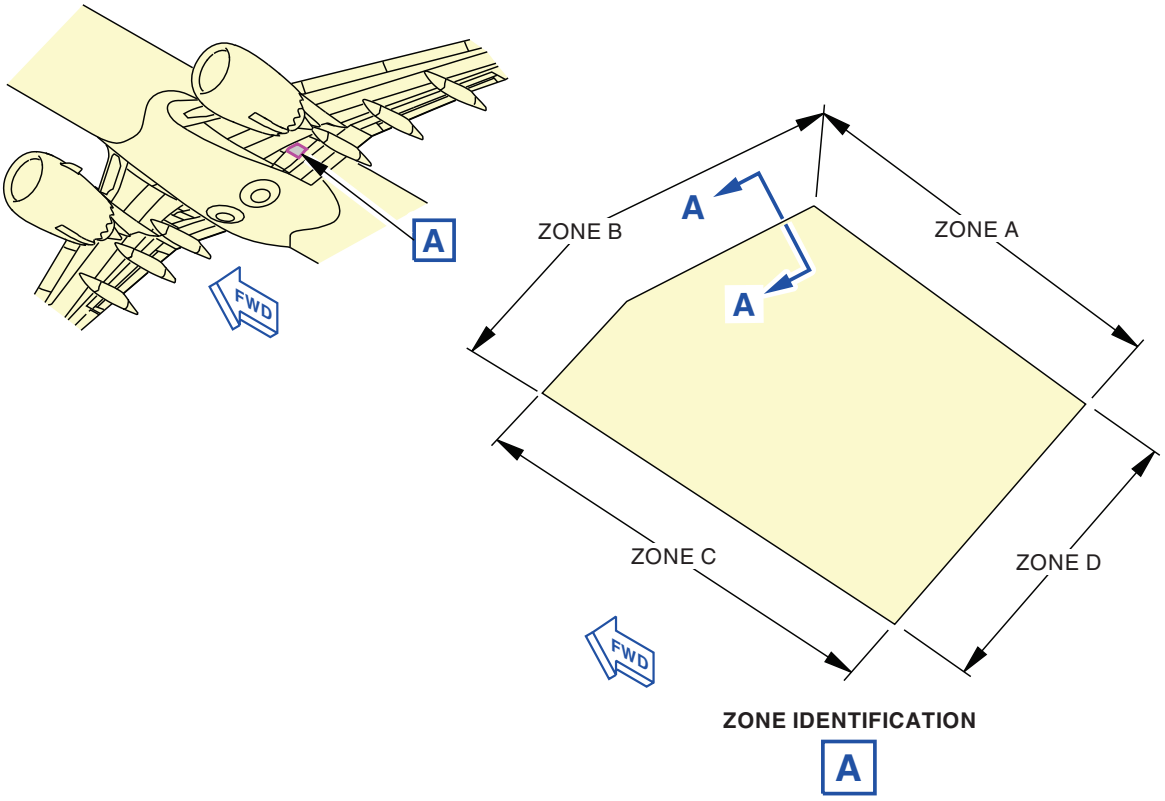
(CAUTION PRECEDES)

CAUTION	MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(2) To lower the airplane and remove the jacks, do this task: Lower the Airplane off Jacks, TASK 07-10-01-580-802.

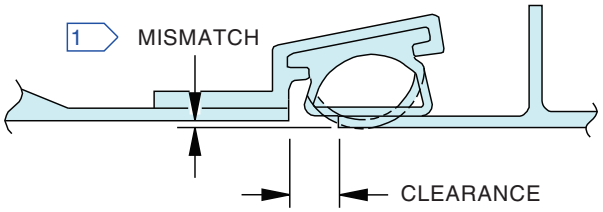
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ZONE IDENTIFICATION

A



ZONE B
(ALL OTHER ZONES ARE EQUIVALENT)

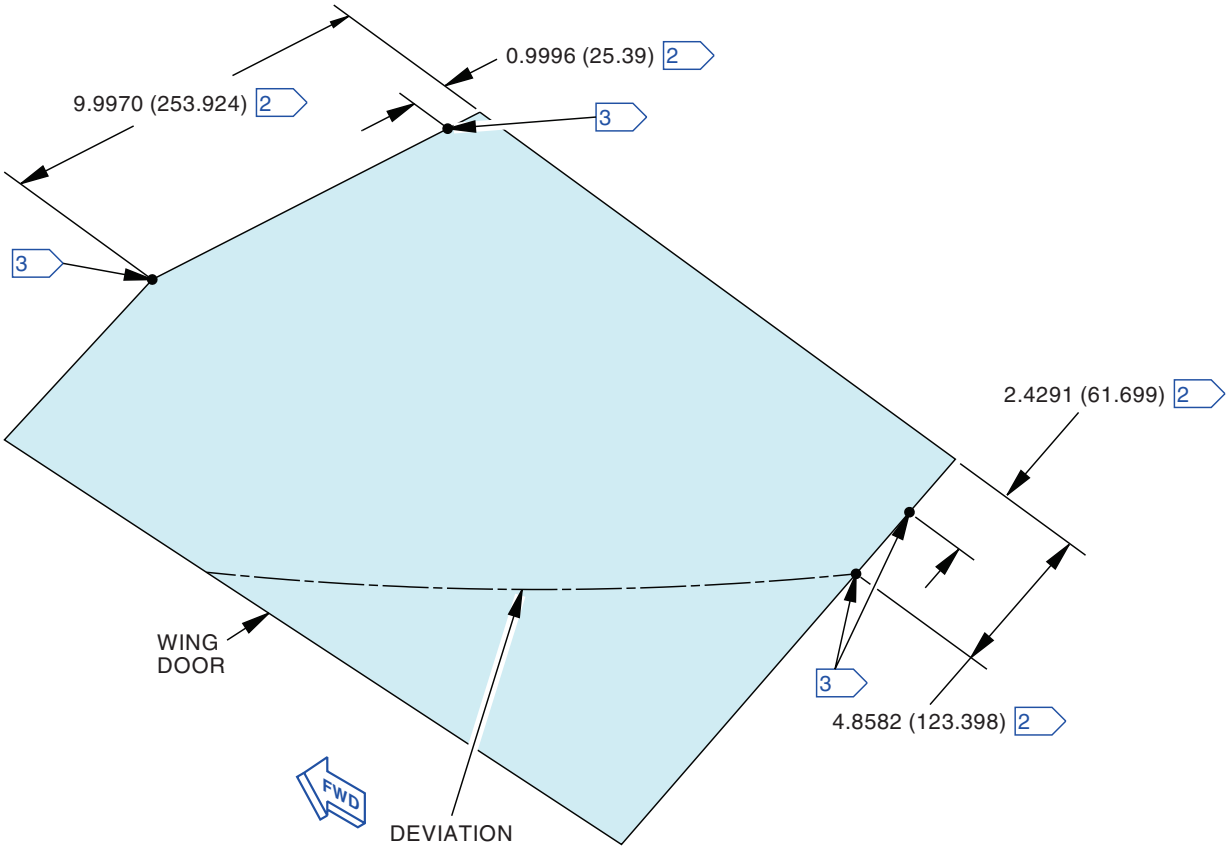
A-A

1 A POSITIVE TOLERANCE (+) INDICATES THE FACE OF THE MISMATCH IS FORWARD (INTO THE AIRSTREAM). A NEGATIVE TOLERANCE (-) INDICATES THE FACE OF THE MISMATCH FACES AFT.

2414144 S00061534490_V1

Main Landing Gear Wing Door Fit Tolerances
Figure 501/32-13-21-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



MAIN LANDING GEAR WING DOOR DIMENSIONAL TOLERANCES

A

NOTE:

ALL DIMENSIONS ARE IN INCHES (MILLIMETERS ARE IN PARENTHESES).

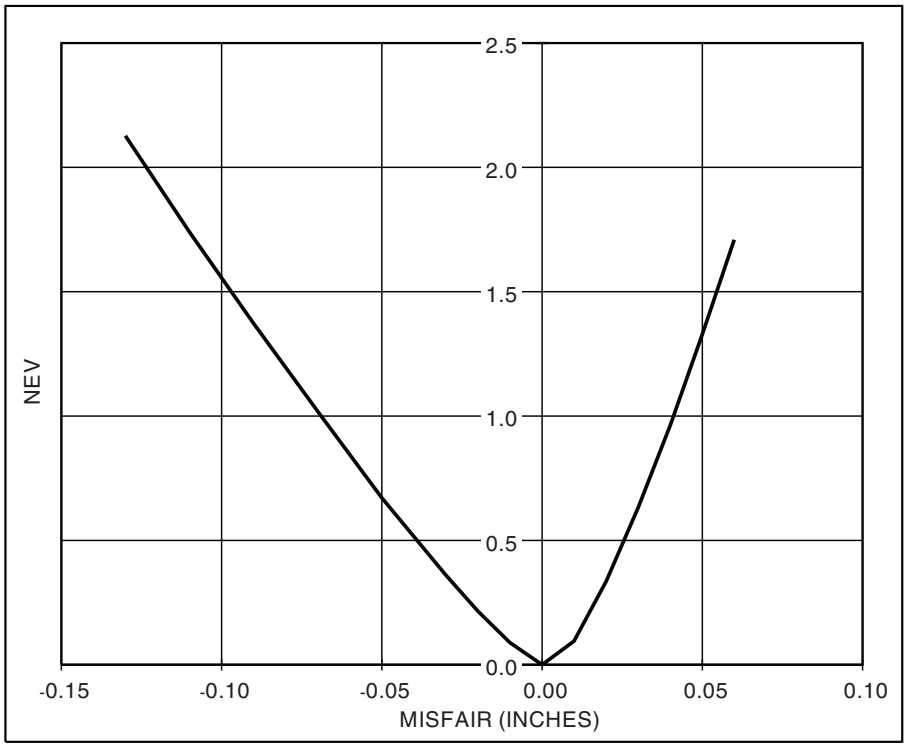
2 THE DIMENSIONAL TOLERANCE IS ±0.5000 INCH (12.7 mm).

3 KEY MISFAIR INSPECTION POINTS.

2784082 S0000631550_V2

Main Landing Gear Wing Door Fit Tolerances
Figure 501/32-13-21-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



MISFAIR	NEV
-0.130	2.127
-0.110	1.741
-0.090	1.373
-0.070	1.019
-0.050	0.671
-0.030	0.361
-0.020	0.215
-0.010	0.089
0.000	0.000
0.010	0.095
0.020	0.335
0.030	0.633
0.040	0.962
0.050	1.329
0.060	1.709

4

MAIN LANDING GEAR WING DOOR NET EFFECT VALUE CHART

NOTE:

THE NET EFFECT LIMIT (NEL) FOR THIS ITEM IS 1.0.

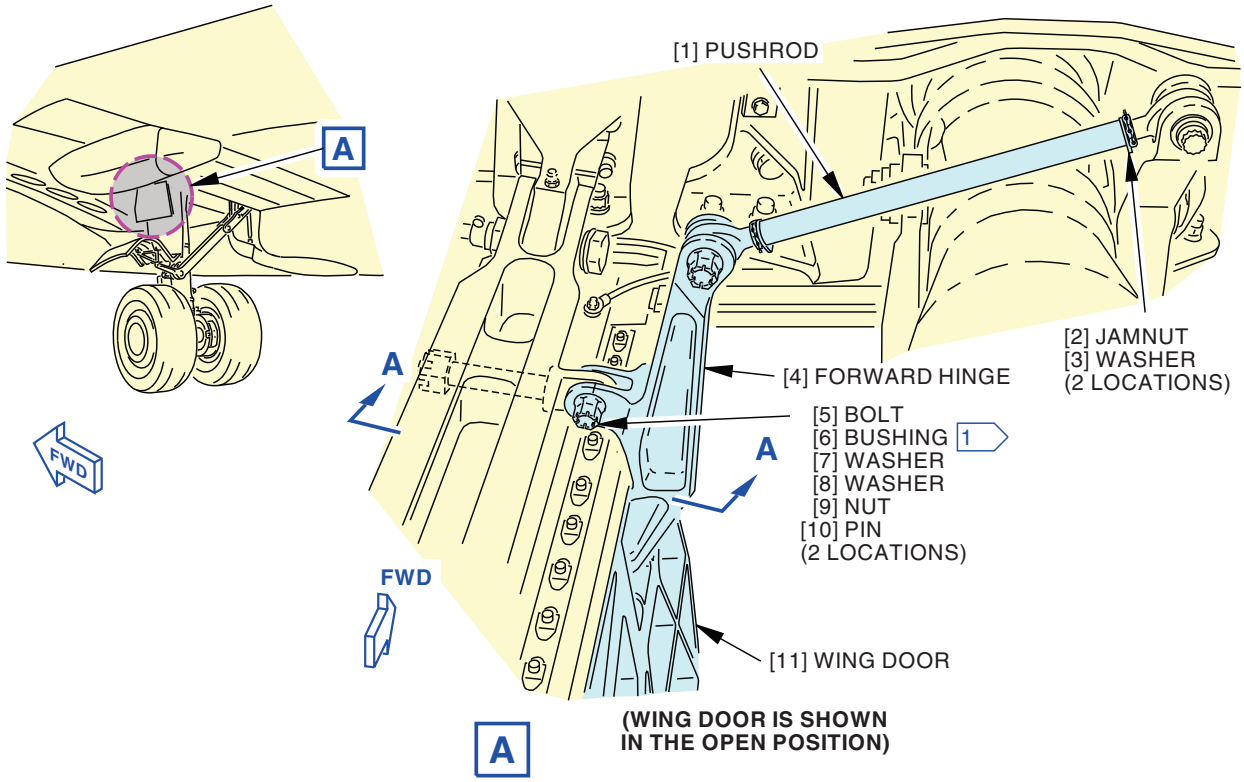
IF ALL MEASURED STEP VALUES FOR THIS ITEM ARE BETWEEN-0.041 AND -0.069 THEN PASSAGE OF NEL IS ASSURED AND AN ACTUAL NEL CALCULATION IS NOT REQUIRED.

4 NEV OF 1.0 CORRESPONDS TO MISFAIR OF 0.041 AND -0.069.

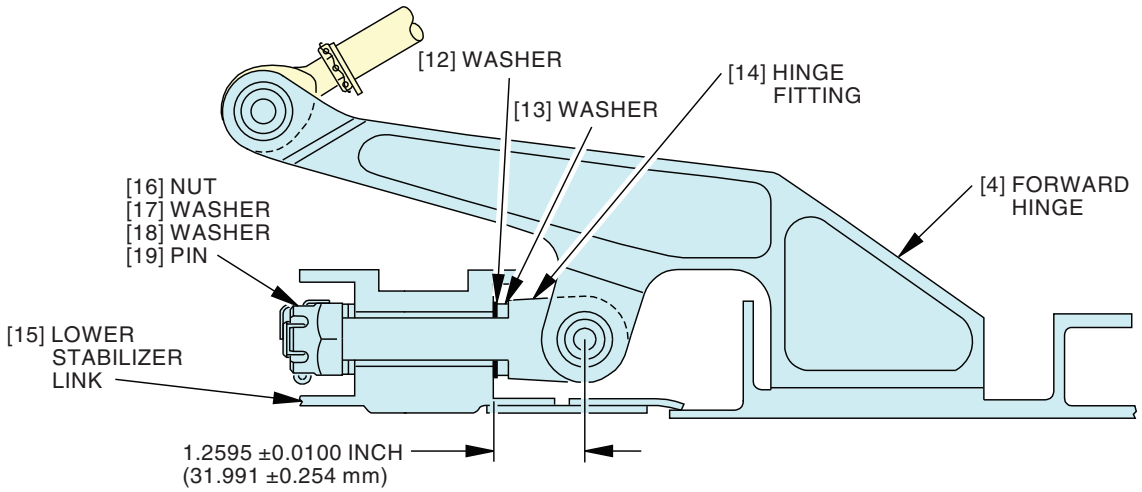
2840917 S0000662597_V1

Main Landing Gear Wing Door Fit Tolerances
Figure 501/32-13-21-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(WING DOOR IS SHOWN IN THE OPEN POSITION)



FORWARD CONNECTION IS SHOWN (AFT CONNECTION IS EQUIVALENT)

A-A

1 ON FORWARD HINGE ONLY.

2414147 S00061534493_V3

Main Landing Gear Wing Door Adjustment
Figure 502/32-13-21-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE LANDING GEAR - MAINTENANCE PRACTICES

1. General

A. This procedure contains these tasks:

(1) The removal of the gas charging valve on the shock strut of the nose landing gear.

(2) The installation of the gas charging valve on the shock strut of the nose landing gear.

(3) The removal of the oil charging valve on the shock strut of the nose landing gear.

(4) The installation of the oil charging valve on the shock strut of the nose landing gear.

B. The gas charging valve is on the top of the shock strut of the nose landing gear. The gas charging valve is on the center of the top of the shock strut.

TASK 32-21-00-000-801

2. Nose Landing Gear Gas Charging Valve Removal

(Figure 201)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
713	Nose Landing Gear

C. Prepare for the Removal

SUBTASK 32-21-00-860-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-21-00-860-002

(2) Make sure the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-21-00-860-003

(3) Make sure that there is no pressure in the hydraulic systems A and B (TASK 29-11-00-860-805).

D. Nose Landing Gear Gas Charging Valve Removal

SUBTASK 32-21-00-630-001

WARNING DO NOT REMOVE THE VALVE BODY UNTIL YOU DEFLATE THE SHOCK STRUT FULLY. THE AIR PRESSURE CAN BLOW THE VALVE BODY OUT AND CAUSE INJURIES TO PERSONNEL.

(1) Deflate the shock strut [1] for the nose landing gear fully:

(a) Remove the lockwire [6] from the gas charging valve [4] and the shock strut [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Discard the lockwire [6].

(b) Remove the dust cap [2] for the gas charging valve [4].

(c) Loosen the swivel nut [3] of the gas charging valve [4] for a maximum of two turns.

NOTE: Fluid in the shock strut [1] will have bubbles when you release the pressure.

1) Deflate the shock strut [1] slowly to prevent the leakage of the fluid through the gas charging valve [4].

(d) Loosen the swivel nut [3] of the gas charging valve [4] fully when all of the pressure in the shock strut [1] is released.

SUBTASK 32-21-00-020-001

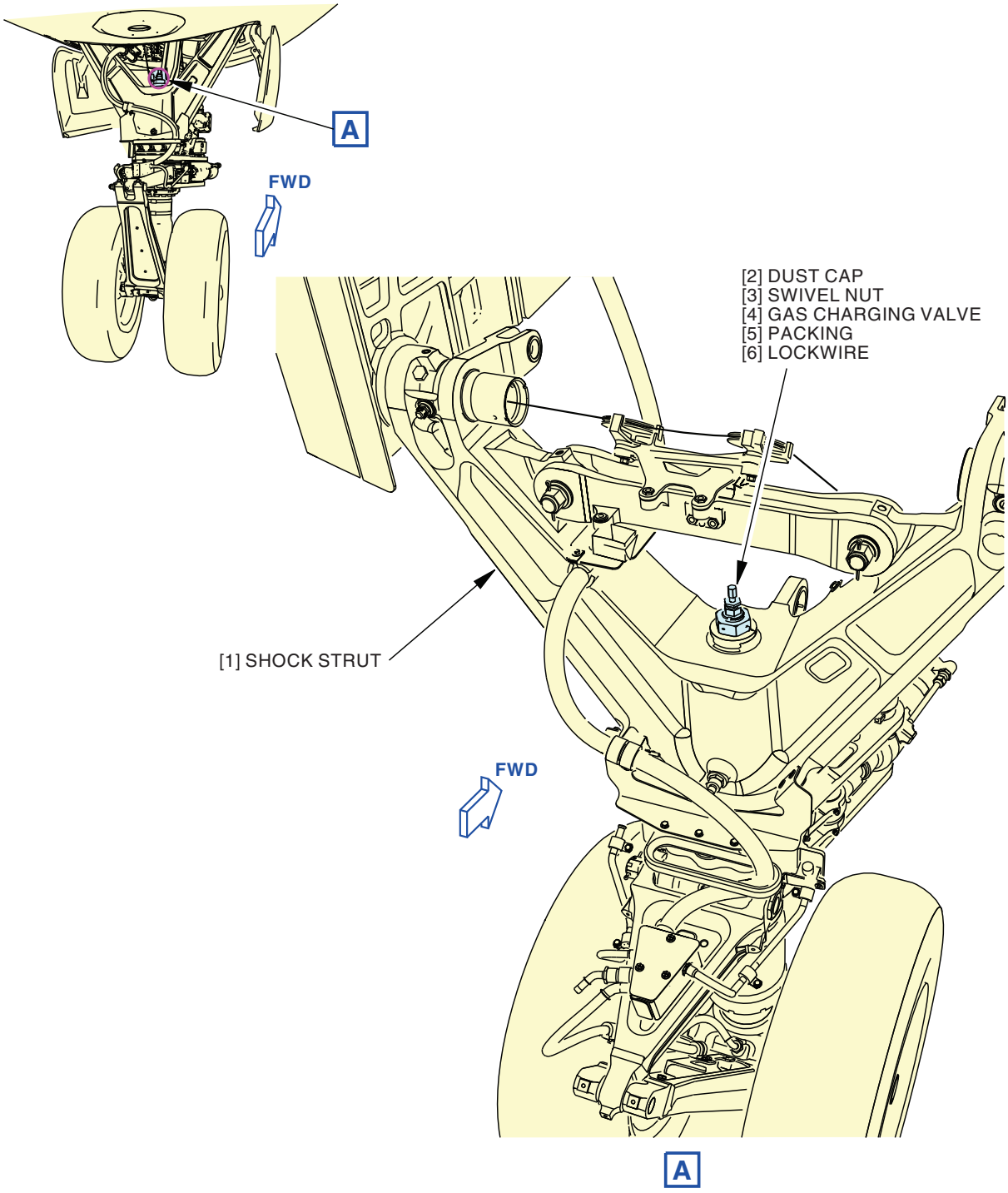
(2) Remove the gas charging valve [4].

(a) Remove the packing [5] from the gas charging valve [4].

(b) Discard the packing [5].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Landing Gear Gas Charging Valve
Figure 201/32-21-00-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-21-00-400-801

3. Nose Landing Gear Gas Charging Valve Installation

(Figure 201)

A. References

Reference	Title
12-15-41-610-801	Nose Landing Gear Shock Strut Fluid Check (P/B 301)
24-22-00-860-804	Remove External Power (P/B 201)

B. Consumable Materials

Reference	Description	Specification
A00226	Compound - Tamper-Proof Putty	BMS8-45
D00467	Fluid - Landing Gear Shock Strut	BMS3-32 Type II
D50022	Fluid - Landing Gear Shock Strut (Specifically For Preservation)	BMS3-32 Type I
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Gas charging valve	32-21-00-01-360	PTF ALL
5	Packing	32-21-00-01-363	PTF ALL

D. Location Zones

Zone	Area
713	Nose Landing Gear

E. Nose Landing Gear Gas Charging Valve Installation

SUBTASK 32-21-00-600-001

(1) To install the new packing [5] on the gas charging valve [4], do these steps.

(a) Lubricate the new packing [5] with fluid, D00467, or fluid, D50022.

(b) Lubricate the surface of the gas charging valve [4] which the packing slides against with fluid, D00467, or fluid, D50022.

(c) Install the new packing [5] on the gas charging valve [4].

SUBTASK 32-21-00-600-002

(2) Lubricate the threads of the gas charging valve [4] with the fluid for the shock strut [1].

(a) Refer to Nose Landing Gear Shock Strut Fluid Check, TASK 12-15-41-610-801 for the fluid.

SUBTASK 32-21-00-420-001

(3) Install the gas charging valve [4] on the shock strut [1].

NOTE: The dust cap [2] and the swivel nut [3] of the gas charging valve [4] are installed when the shock strut [1] is serviced.

(a) Tighten the body of the gas charging valve [4] to 12 ±1 ft-lb (16 ±2 N·m).

(b) Install MS20995C32 lockwire, G01048, on the gas charging valve [4] and the shock strut [1] with the double twist method.

(c) Apply compound, A00226, on the gas charging valve [4] and the shock strut [1].

NOTE: This will make sure that the seal made by the compound, A00226, will break if the body of the gas charging valve [4] is loosened.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-21-00-610-001

(4) Service the shock strut [1] (TASK 12-15-41-610-801).

F. Put the Airplane Back to Its Usual Condition

SUBTASK 32-21-00-860-007

(1) If electrical power is not necessary, do this task: Remove External Power, TASK 24-22-00-860-804.

——— END OF TASK ———

TASK 32-21-00-000-803

4. Nose Landing Gear Oil Charging Valve Removal

(Figure 202)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1829	Oil Drain Valve - Shock Strut, LG Part #: J32108-16 Supplier: 81205 Opt Part #: A32066-1 Supplier: 81205
STD-1110	Container - Hydraulic Fluid Resistant, 5 Gallon (19 Liter)
STD-1140	Hose - Flexible, 1/2 Inch ID, BMS 3-11 Resistant, 10 Foot

C. Location Zones

Zone	Area
713	Nose Landing Gear

D. Prepare for the Removal

SUBTASK 32-21-00-800-001

CAUTION DO NOT DEFLATE OR SERVICE THE SHOCK STRUTS WHILE THE TAIL STAND IS INSTALLED. IF YOU DEFLATE OR SERVICE THE SHOCK STRUTS, THE LOAD ON THE TAIL STAND CAN BE TOO LARGE. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Do not deflate or service the shock struts while the tail stand is installed.

SUBTASK 32-21-00-480-016

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(2) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-21-00-860-028

(3) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-21-00-860-020

(4) Make sure that there is no pressure in the hydraulic systems A and B (TASK 29-11-00-860-805).

E. Nose Landing Gear Oil Charging Valve Removal

SUBTASK 32-21-00-020-018

WARNING CLEAR THE AREA BELOW THE NOSE BEFORE YOU DEFLATE THE SHOCK STRUT. IF YOU DEFLATE SHOCK STRUT, THE NOSE CAN MOVE DOWN AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

WARNING DO NOT REMOVE THE VALVE BODY UNTIL YOU DEFLATE THE SHOCK STRUT FULLY. THE AIR PRESSURE CAN BLOW THE VALVE BODY OUT AND CAUSE INJURIES TO PERSONNEL.

(1) Deflate the shock strut [1] for the nose landing gear fully:

(a) Remove the dust cap [2] for the gas charging valve [4].

(b) Loosen the swivel nut [3] a maximum of two turns.

NOTE: Fluid in the shock strut [1] can have bubbles when you release the pressure. Deflate the shock strut slowly to prevent the leakage of the fluid through the gas charging valve [4].

(c) Loosen the swivel nut [3] fully when all of the pressure in the shock strut [1] is released.

NOTE: The shock strut [1] is fully deflated when the dimension from the lower surface of the lower steering plate to the upper surface of the tow fitting lug is 13.8 in. (351 mm).

SUBTASK 32-21-00-680-001

(2) Drain the oil from the shock strut [1]:

(a) Remove the cap assembly [8] from the oil charging valve [9].

(b) Put a 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110 in a position to catch the oil when the oil charging valve [9] is opened.

(c) Install the drain equipment on the oil charging valve [9]:

NOTE: The tool, landing gear shock strut drain valve, SPL-1829, could also be used, but it will take much longer to drain the shock strut.

1) Cut a length of ten foot hose (1/2 inch ID), STD-1140, long enough to reach the container on the floor.

2) Insert a small allen wrench in the end of the cut length of tubing so that the long end of the allen wrench is flush with the end of the tube and the short end goes through the wall of the tube.

3) Install the cut length of tubing onto the oil charging valve [9] so that the allen wrench goes into the check valve and holds it open to drain the oil.

(d) Remove the drain equipment when you have removed all of the oil.

SUBTASK 32-21-00-020-019

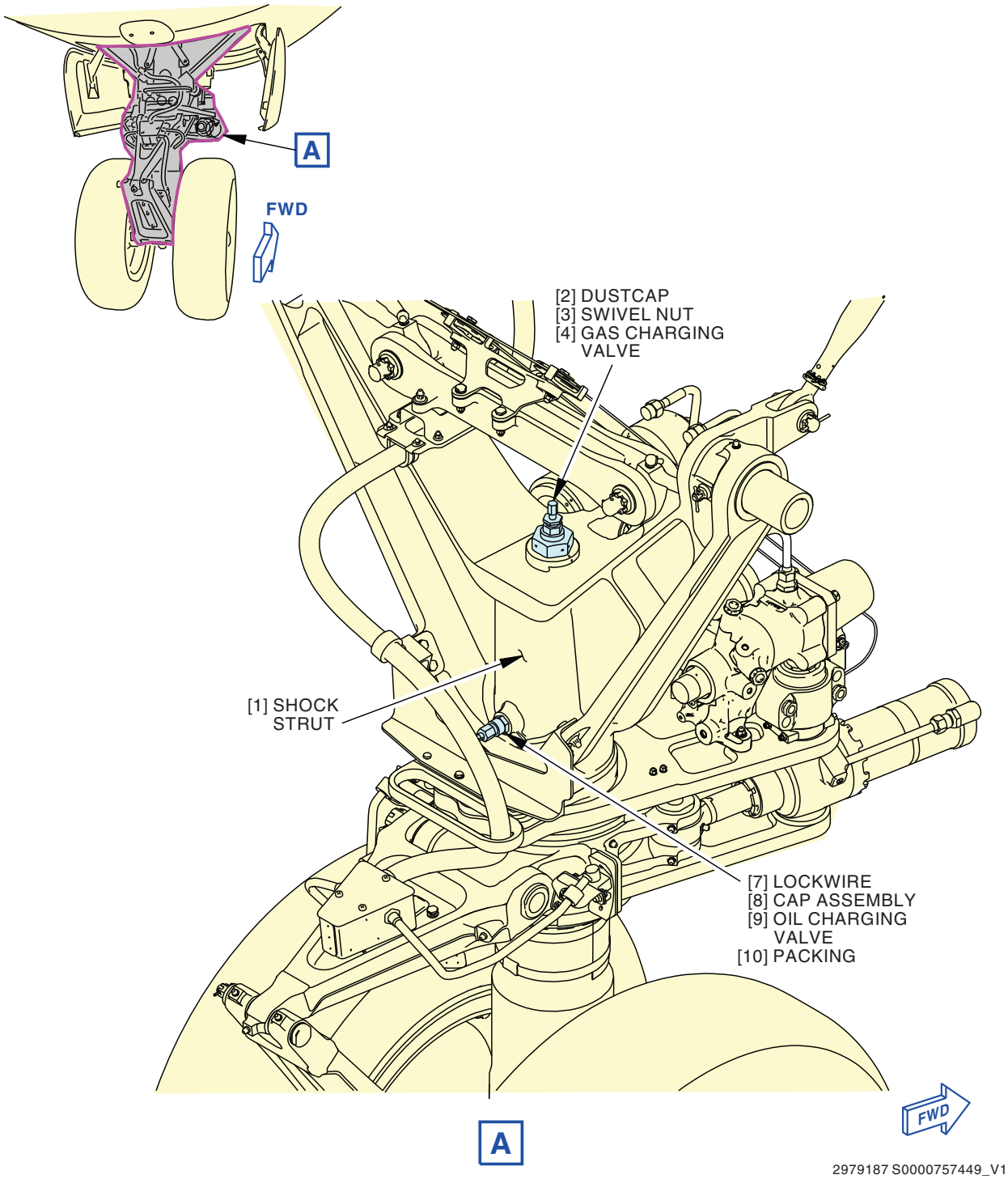
(3) Remove the oil charging valve [9] as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the lockwire [7] from the oil charging valve [9].
    1) Discard the lockwire [7].
(b) Remove the oil charging valve [9].
(c) Remove the packing [10] from the oil charging valve [9].
    1) Discard the packing [10].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Landing Gear Oil Charging Valve Installation
Figure 202/32-21-00-990-809

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-21-00-400-803

5. Nose Landing Gear Oil Charging Valve Installation

(Figure 202)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
12-15-41-610-802	Nose Landing Gear Shock Strut Servicing, Airplane on the Ground (P/B 301)
24-22-00-860-802	Remove Electrical Power (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Consumable Materials

Reference	Description	Specification
A00226	Compound - Tamper-Proof Putty	BMS8-45
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
9	Oil charging valve	32-21-00-01-035	PTF ALL
10	Packing	32-21-00-01-030	PTF ALL

D. Location Zones

Zone	Area
713	Nose Landing Gear

E. Prepare for the Installation

SUBTASK 32-21-00-480-017

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-21-00-860-023

(2) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

F. Nose Landing Gear Oil Charging Valve Installation

SUBTASK 32-21-00-420-019

(1) To install the new packing [10] on the oil charging valve [9], do these steps:

(a) Lubricate the new packing [10] with the fluid for the shock strut [1].

1) Refer to Nose Landing Gear Shock Strut Servicing, Airplane on the Ground, TASK 12-15-41-610-802 for the fluid.

(b) Lubricate the surface of the oil charging valve [9] which the packing slides against with the fluid for the shock strut [1].

1) Refer to Nose Landing Gear Shock Strut Servicing, Airplane on the Ground, TASK 12-15-41-610-802 for the fluid.

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(c) Install the new packing [10] on the oil charging valve [9].

SUBTASK 32-21-00-640-005

(2) Lubricate the threads of the oil charging valve [9] with the fluid for the shock strut [1].

(a) Refer to Nose Landing Gear Shock Strut Servicing, Airplane on the Ground, TASK 12-15-41-610-802 for the fluid.

SUBTASK 32-21-00-420-021

(3) Install the oil charging valve [9] on the shock strut [1].

NOTE: The cap assembly [8] is installed when the shock strut [1] is serviced.

(a) Torque the body of the oil charging valve [9] to 22 ft-lb (30 N·m) – 25 ft-lb (34 N·m).

(b) Install the lockwire [7] (MS20995C32 lockwire, G01048) on the oil charging valve [9] and the shock strut [1] using the double twist method.

(c) Apply compound, A00226 to oil charging valve [9] and the shock strut [1].

NOTE: This will make sure that the seal made by the compound, A00226, will break if the body of the oil charging valve [9] is loosened.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-21-00-600-003

(1) Service the shock strut [1], do this task: Nose Landing Gear Shock Strut Servicing, Airplane on the Ground, TASK 12-15-41-610-802.

SUBTASK 32-21-00-860-026

(2) If electrical power is not necessary, do this task: Remove Electrical Power, TASK 24-22-00-860-802.

SUBTASK 32-21-00-860-027

(3) Remove the downlock pins, do this task: Landing Gear Downlock Pins Removal, TASK 32-00-01-080-801.

——— END OF TASK ———

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NOSE LANDING GEAR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the nose landing gear

(2) An installation of the nose landing gear.

TASK 32-21-00-000-802

2. Nose Landing Gear Removal

(Figure 401, Figure 402)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
07-10-03 P/B 201	JACK AIRPLANE AXLES - MAINTENANCE PRACTICES
07-10-03-580-803	Lift the Nose Landing Gear Axle With Axle Jacks-Handling (P/B 201)
07-10-21-580-801	Lift the Airplane Nose with the Nose Jack at Jack Point D (P/B 201)
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-45-21-000-801	Nose Landing Gear Wheel and Tire Assembly Removal (P/B 401)
32-51-00-820-801	General Check of the Nose Wheel Steering System Adjustment (P/B 501)
32-51-31-000-802	Nose Gear Steering System Cables Removal (P/B 401)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-4938	Insertion/Removal Tool - Contact (size 16, backshell side) Part #: M81969/1-03 Supplier: 11851 Part #: M81969/14-03 Supplier: 11139
SPL-1559	Adjustable Spanner Wrench (2.00 to 4.00 Inch Dia. Retainer) Part #: F72959-34 Supplier: 81205 Part #: F72959-35 Supplier: 81205 Part #: F72959-36 Supplier: 81205 Part #: F72959-41 Supplier: 81205 Opt Part #: F72959-5 Supplier: 81205 Opt Part #: F72959-6 Supplier: 81205 Opt Part #: F72959-7 Supplier: 81205 Opt Part #: F72959-8 Supplier: 81205

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(Continued)

Reference	Description
SPL-1821	Adjustable Spanner Wrench (0.75 to 2.00 Inch Dia. Retainer, 0.12 x 0.12 Key Arm) Part #: F72959-40 Supplier: 81205 Opt Part #: F72959-4 Supplier: 81205
SPL-1871	Retention Straps - Shock Strut, NLG/MLG Part #: C32030 -31 Supplier: 81205 Opt Part #: C32030-10 Supplier: 81205
SPL-10305	Removal/Installation Equipment, Outrigger - NLG Part #: C32049-1 Supplier: 81205
SPL-14481	Removal/Installation Equipment - Trunnion Pin, NLG Part #: C32054-1 Supplier: 81205
STD-837	String
STD-6545	Rope - Nylon
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)
STD-15079	R/I Equipment - Trunnion Pin, NLG (Note: C32054-3 Slide Hammer Assembly is included in SPL-14481 kit)
STD-15082	R/I Equipment - Trunnion Pin, NLG (Note: C32054-7 Puller Pin Assembly is included in SPL-14481 kit)
STD-15083	R/I Equipment - Trunnion Pin, NLG (Note: C32054-8 Cross Pin is included in SPL-14481 kit)

C. Consumable Materials

Reference	Description	Specification
G50980	Tape - 3M Scotch 811 Removable Tape	

D. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
713	Nose Landing Gear

E. Access Panels

Number	Name/Location
113BW	Forward Nose Wheel Well Panel

F. Prepare for the Removal

SUBTASK 32-21-00-480-012

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-21-00-860-008

(2) For hydraulic system A, make sure that the hydraulic power is removed (TASK 29-11-00-860-805).

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SUBTASK 32-21-00-860-009

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	17	C00129	LANDING GEAR LATCH & PRESS WARN
C	15	C01355	LANDING GEAR AIR/GND SYS 2
C	16	C01356	LANDING GEAR AIR/GND SYS 1
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN
D	15	C01401	LANDING GEAR AIR/GND RELAY
D	16	C01432	LANDING GEAR ALTN EXTEND SOL
D	17	C01027	LANDING GEAR NOSE GEAR STEER
D	18	C00451	LANDING GEAR AURAL WARN

SUBTASK 32-21-00-020-002

(4) Disconnect the nwsa/nwsb cables [49] (TASK 32-51-31-000-802).

(a) Install the rig pin NS2 in the drum for the Captain's control wheel (TASK 32-51-00-820-801).

(b) To get access to the cable turnbuckle [1], do this step:

1) Open this access panel:

(Nose Wheel Well Access Panels Removal, TASK 53-14-01-020-801)

Number	Name/Location
113BW	Forward Nose Wheel Well Panel

(c) Remove the locking clip [2] for the cable turnbuckle [1] (View B, Figure 401).

(d) Loosen the tension at the cable turnbuckle [1].

(e) Disconnect the nwsa/nwsb cables [49] at the cable turnbuckle [1] (View K, Figure 401).

(f) Remove the bolts [88], washers [89], and pulleys [87] from the bracket that is outboard of the left trunnion pin [63].

1) Remove the nwsa/nwsb cables [49] from the guard [90] and grooves of the pulleys [87].

SUBTASK 32-21-00-020-003

(5) Disconnect the conduit [4] from the junction box [5].

(a) Remove the lockwire on the conduit [4] and to the junction box [5].

(b) Disconnect the conduit [4] from the junction box [5].

(c) Remove the screws [10] and washers [3] for the junction box cover [9] (View C, Figure 401).

(d) Disconnect the plug [6] from the connector on the wheel well overhead (View D, Figure 401).

(e) Identify the wires in the plug [6] that go to the landing gear and label them or make a drawing to be used during installation.

(f) Use a contact insertion / removal tool, COM-4938, tool to remove the wires from the plug [6].

(g) Attach a string, STD-837, long enough to feed through the rigid conduit [7] to the junction box [5], to each wire removed from the plug [6].

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(h) Pull the wires through to the junction box [5].

1) Do not pull the string, STD-837, all the way through the conduit.

NOTE: It will be more easier to re-insert the wires later.

(i) Untie the string, STD-837, from the wires and leave hanging.

SUBTASK 32-21-00-020-020

(6) To disconnect the down lock sensor plug [80] and rigid conduit [81] from the wheel well, do these steps:

(a) Disconnect the plug [80] from the connector on the wheel well wall (View M, Figure 401).

(b) Remove the screws [82] and the clamps [83].

(c) Remove the screws [84], washers [85] and clamp [86].

(d) Remove the rigid conduit [81] from the clamps [83] and clamp [86].

(e) Install a protective cap on the plug [80].

G. Nose Landing Gear Removal

SUBTASK 32-21-00-020-004

(1) Disconnect all of the hydraulic lines [25] (View F, Figure 401).

(a) Loosen the swivel nut to disconnect each hydraulic line [25] from the swivel [24].

(b) Put caps/plugs, STD-14686, on the hydraulic lines [25].

(c) Install caps/plugs, STD-14686, on each of the hydraulic ports of the swivel [24].

(d) Attach tags to identify the swivel port and hydraulic lines [25] to prevent the crossing of the lines during the installation.

SUBTASK 32-21-00-020-005

(2) Disconnect the right upper rod assembly [26] (View G, Figure 401).

(a) Remove the pin [31], nut [32], and washer [30].

(b) Remove the bolt [27], washer [28], and bushing [29] from the clevis of the right trunnion [23].

(c) Use a nylon rope, STD-6545, to hold the upper rod assembly [26] away from the work area.

SUBTASK 32-21-00-020-006

(3) Disconnect the left upper rod assembly [41] (View I, Figure 401).

(a) Remove the pin [43], nut [42], and washer [44].

(b) Remove the bolt [47], washer [46], and bushing [45] from the clevis of the left trunnion [40].

(c) Use a nylon rope, STD-6545, to hold the upper rod assembly [41] away from the work area.

SUBTASK 32-21-00-020-007

(4) Remove each pulley [50] (View J, Figure 401).

(a) Remove the nut [52] and washer [53] from the bolt [55].

(b) Remove the bolt [55] and washer [48] that hold the pulley [50] from the bracket [51].

(c) Remove the washer [48] from the bolt [55].

(d) Remove the pulley [50] from the bracket [51].

(e) Remove the nwsa/nwsb cables [49] from the groove of the pulley [50].

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SUBTASK 32-21-00-020-016

(5) Remove the bracket [51] (View J, Figure 401).

(a) Remove the bolts [75], washers [76], and nuts [77].

SUBTASK 32-21-00-020-008

(6) Remove the seal [65] (View K, Figure 401).

(a) Remove the lockwire from the screw [58] and lock [57].

(b) Remove the screw [58] that holds the lock [57] to the left trunnion pin [63].

(c) Remove the lock [57] from the left trunnion pin [63].

(d) Use the spanner wrench, SPL-1821 (face lug adapter), or spanner wrench set, SPL-1559, to remove the trunnion pin nut [68] from the left trunnion pin [63].

(e) Pull the seal ring [69] and seal [65] from the left trunnion pin [63].

1) If it is difficult to pull the seal [65], hold the eyebolt [67] and pull the seal [65] out of the left trunnion pin [63].

(f) Remove the nut [64], washer [56], and inboard retainer [66] from the eyebolt [67].

(g) Remove the eyebolt [67] from the left trunnion pin [63].

(h) Remove the outboard retainer [66] from the eyebolt [67].

(i) Isolate the segments of the seal [65] from the nwsa/nwsb cables [49].

(j) Pull the nwsa/nwsb cables [49] from the trunnion pin nut [68], seal ring [69], and left trunnion pin [63].

(k) Attach identification tags to the nwsa/nwsb cables [49].

(l) Wind the nwsa/nwsb cables [49] into a coil and attach them to the trunnion of the gear with Scotch 811 removable tape, G50980.

SUBTASK 32-21-00-020-009

WARNING DO NOT REMOVE THE VALVE BODY UNTIL YOU DEFLATE THE SHOCK STRUT FULLY. THE AIR PRESSURE CAN BLOW THE VALVE BODY OUT AND CAUSE INJURIES TO PERSONNEL.

(7) Deflate the shock strut [17] (View E, Figure 401).

(a) Remove the dust cap [14] for the gas charging valve [16].

(b) Loosen the swivel nut [15] a maximum of two turns.

(c) Deflate the shock strut slowly to prevent the leakage of the fluid through the gas valve.

NOTE: Fluid in the shock strut will have bubbles when the pressure is released.

(d) Loosen the swivel nut [15] fully when all of the pressure in the shock strut [17] is released.

SUBTASK 32-21-00-480-002

(8) Install the retention strap, SPL-1871, with the marker facing up to hold the shock strut [17] in its compressed position.

SUBTASK 32-21-00-580-001

(9) Lift the airplane nose (TASK 07-10-21-580-801).

(a) Make sure that the chocks are installed at the main landing gear after the nose landing gear is lifted.

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SUBTASK 32-21-00-020-010

(10) Remove the nose gear wheel and tire assembly (TASK 32-45-21-000-801).

NOTE: If the Outrigger R/I Equipment, SPL-10305, will be used, the wheel and tire assembly must not be removed from the nose gear.

SUBTASK 32-21-00-020-011

(11) Disconnect the lower drag strut [19] (View E, Figure 401).

(a) Remove the pin [11], nut [22], washer [21], and bolt [13] from the pin [18].

(b) Remove the nut [12] and washer [20] from the pin [18].

(c) Hold the lower drag strut [19] in position and remove the pin [18].

(d) Use a nylon rope, STD-6545, to hold the lower drag strut [19] away from the work area.

NOTE: When the pin is removed the lower drag strut will move forward.

SUBTASK 32-21-00-580-002

(12) If it is necessary, lift the gear with an axle jack to reduce the weight of the gear when the left trunnion pin [63] and right trunnion pin [37] are removed (TASK 07-10-03-580-803).

SUBTASK 32-21-00-020-012

(13) Remove the right trunnion pin [37] (View H, Figure 401).

(a) Remove the pin [35], nut [36], washer [34], and bolt [33].

(b) Remove the swivel [24] from the right trunnion pin [37].

(c) Use the nose landing gear trunnion pin alignment pin assembly, SPL-14481, puller pin assembly, STD-15082, slide hammer assembly, STD-15079, and cross pin, STD-15083, to remove the right trunnion pin [37] from the right trunnion [23].

1) Make sure that the spacer [39] is caught that is between the sidewall of the nose wheel well and right trunnion [23].

(d) Discard the O-ring [74].

SUBTASK 32-21-00-020-013

(14) Remove the left trunnion pin [63] (Figure 402).

(a) Remove the pin [59], nut [60], washer [78], and bolt [62] (View K, Figure 401).

(b) Use the nose landing gear trunnion pin alignment pin assembly, SPL-14481, puller pin assembly, STD-15082, slide hammer assembly, STD-15079, and cross pin, STD-15083, to remove the left trunnion pin [63] from the left trunnion [40].

1) Make sure that the spacer [71] is caught that is between the sidewall of the nose wheel well and left trunnion [40] (View L, Figure 401).

(c) Remove the cable guard [70] from the left trunnion pin [63].

(d) Discard the O-ring [74].

SUBTASK 32-21-00-860-010

(15) Measure the thickness of the spacer [39] and spacer [71] that were between the sidewalls of the wheel well and nose landing gear trunnion.

NOTE: The thickness of the washer stacks are special for each airplane.

(a) Make a record of the measurement of the washer stack for use when the washer stack is installed.

SUBTASK 32-21-00-020-014

(16) Remove the nose landing gear from the airplane.

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(a) Move the nose landing gear forward in the wheel well.

(b) Turn the gear approximately 90 degrees such that it is clear of the wheel well sidewalls and nose wheel well doors.

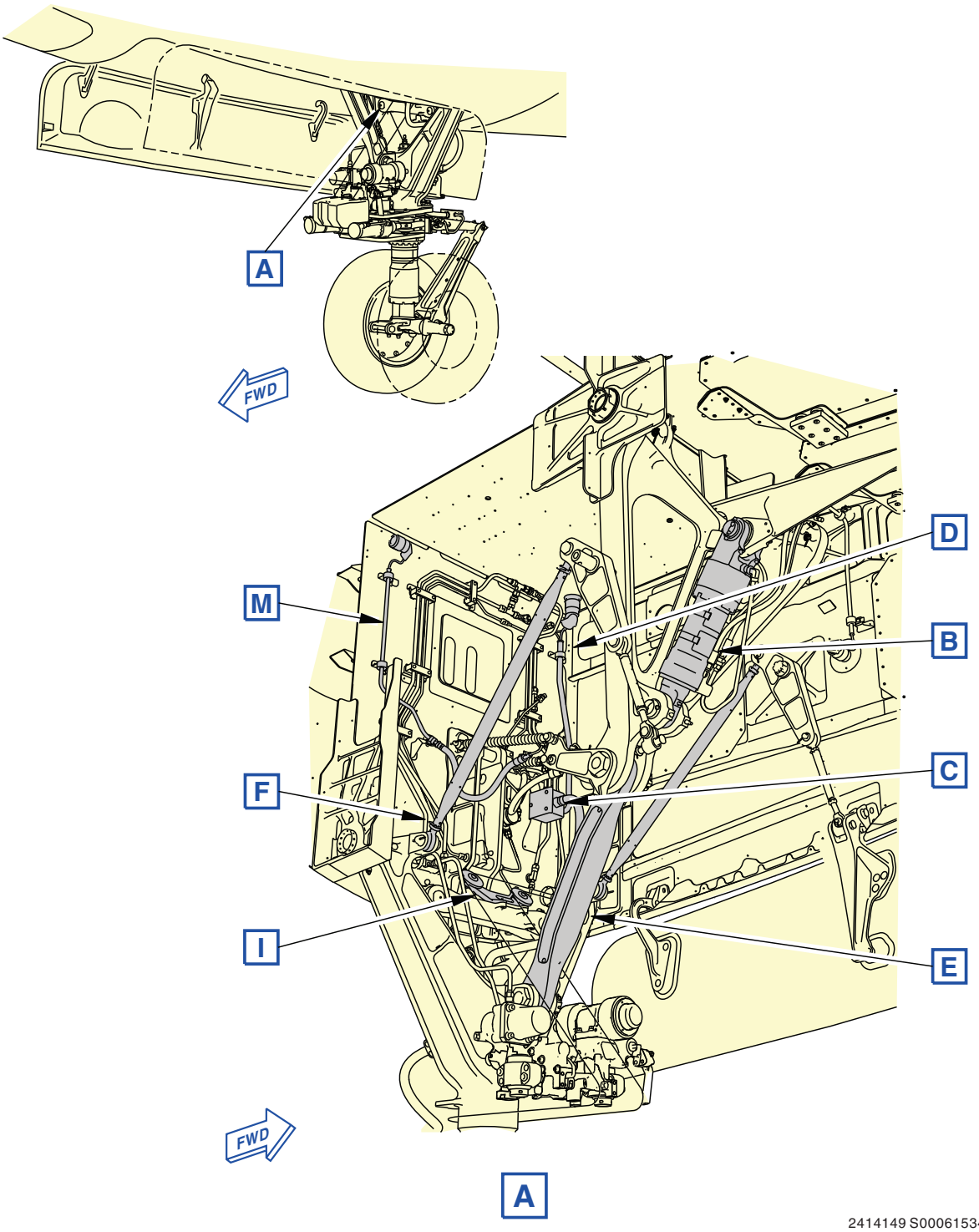
(c) Lower the nose landing gear at point E until it clears the nose wheel well (PAGEBLOCK 07-10-03/201).

(d) If it is necessary, use a dolly or the Outrigger R/I Equipment, SPL-10305, to transport the nose landing gear.

1) When the Outrigger R/I Equipment, SPL-10305, is installed and will be left unattended, chock the nose landing gear tires and lock the caster brake (PAGEBLOCK 10-11-05/201).

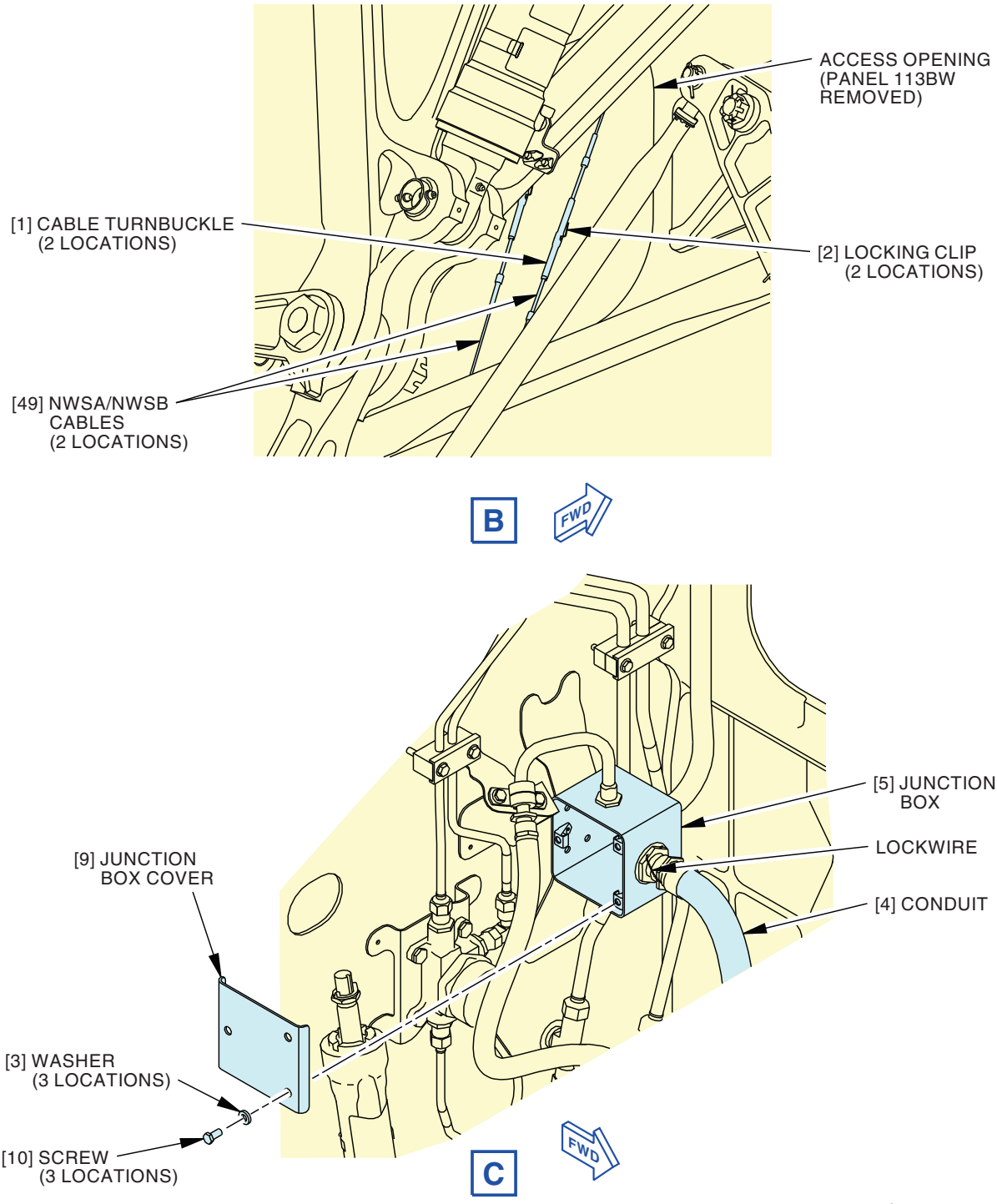
——— END OF TASK ———

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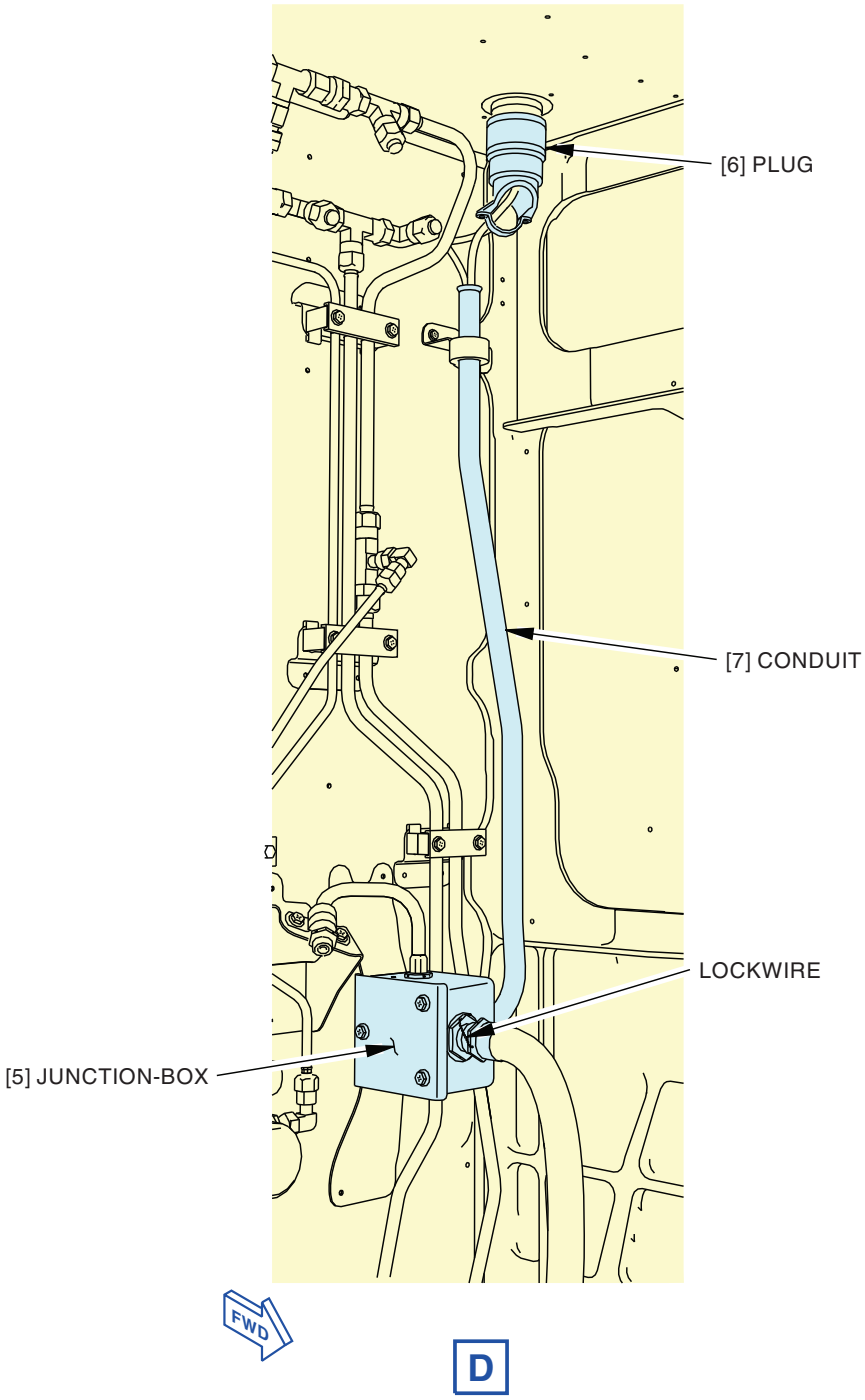
Nose Landing Gear Installation
Figure 401/32-21-00-990-802 (Sheet 1 of 9)

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Nose Landing Gear Installation
Figure 401/32-21-00-990-802 (Sheet 2 of 9)

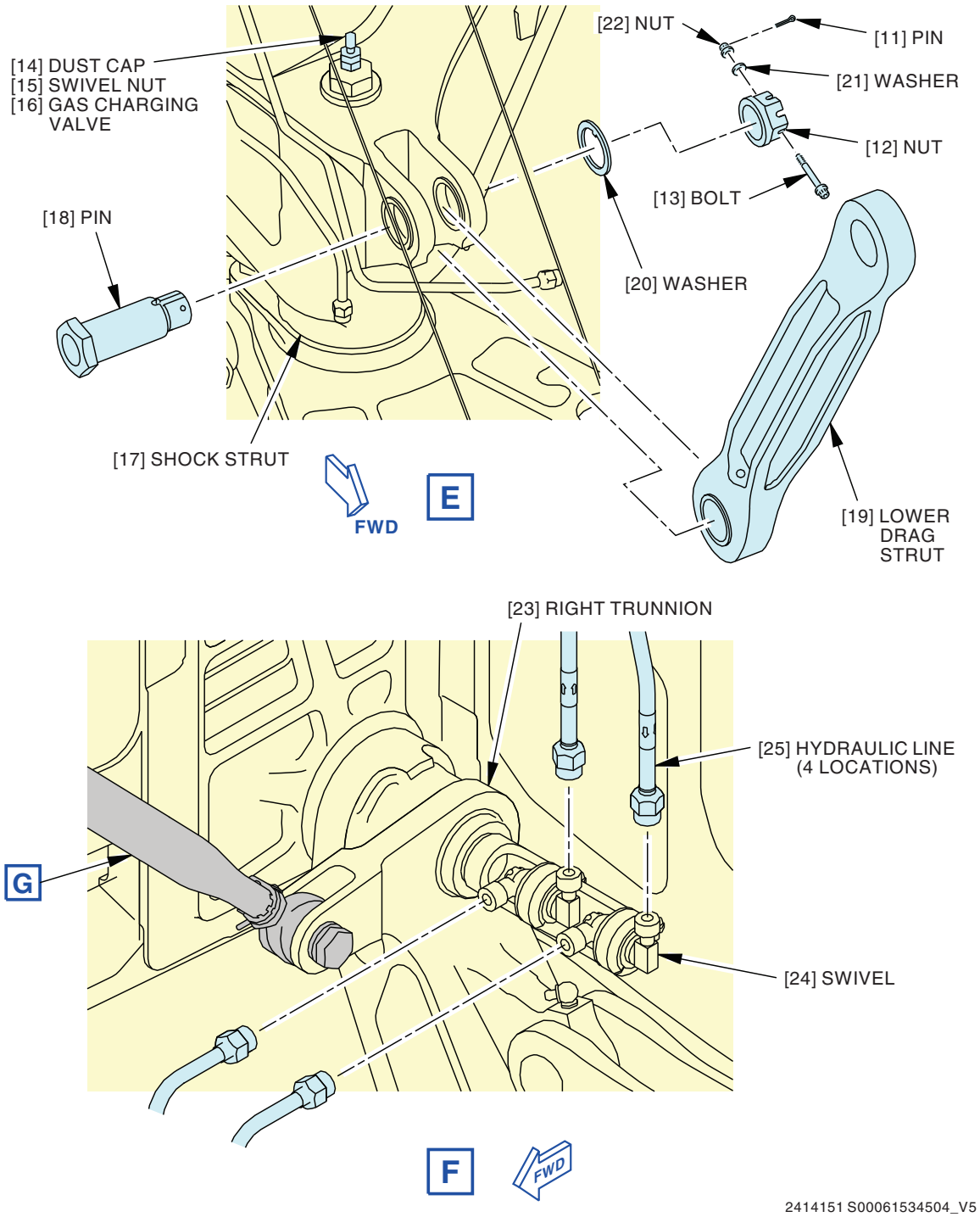
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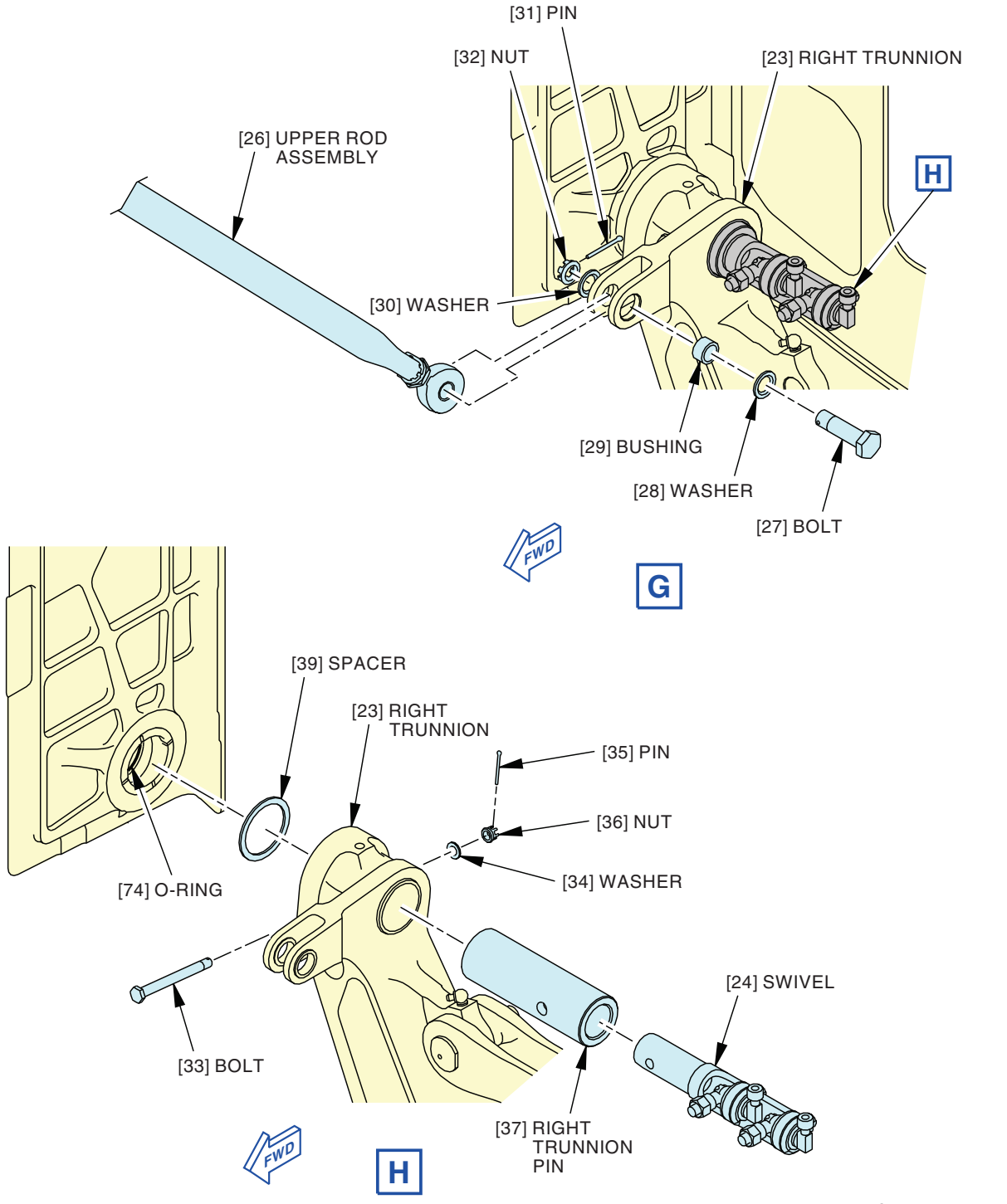
Nose Landing Gear Installation
Figure 401/32-21-00-990-802 (Sheet 3 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



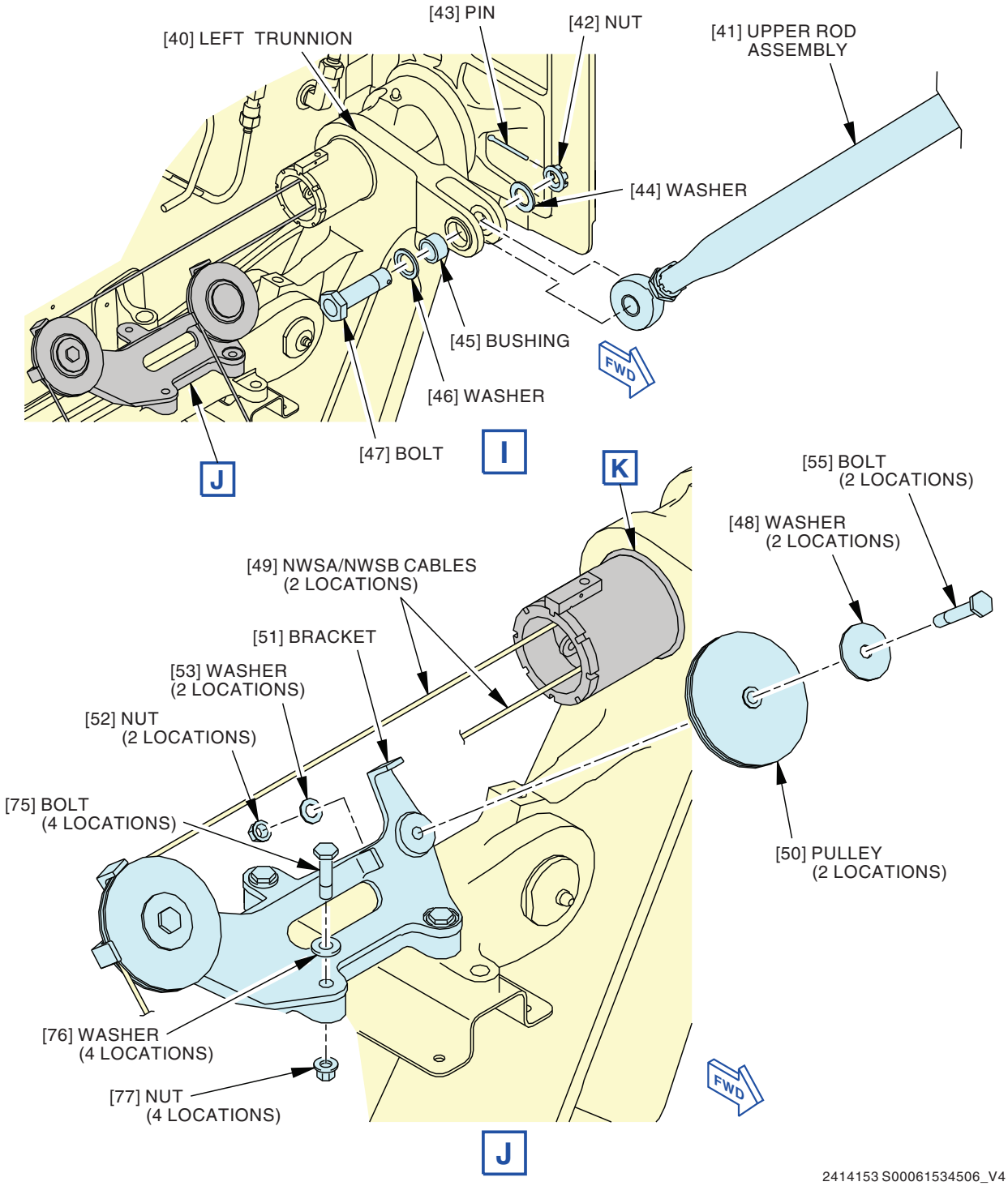
Nose Landing Gear Installation
Figure 401/32-21-00-990-802 (Sheet 4 of 9)

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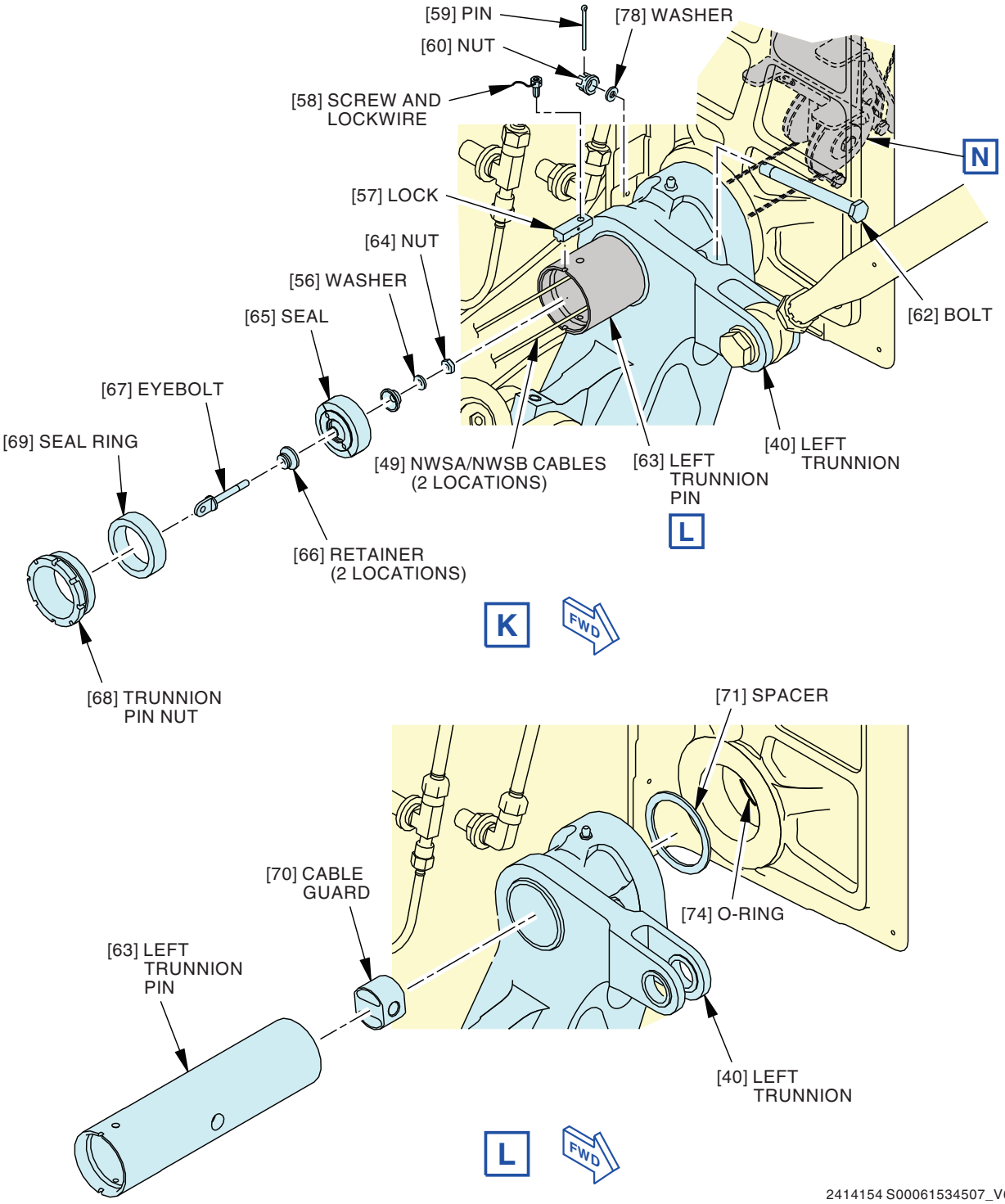
Nose Landing Gear Installation
Figure 401/32-21-00-990-802 (Sheet 5 of 9)

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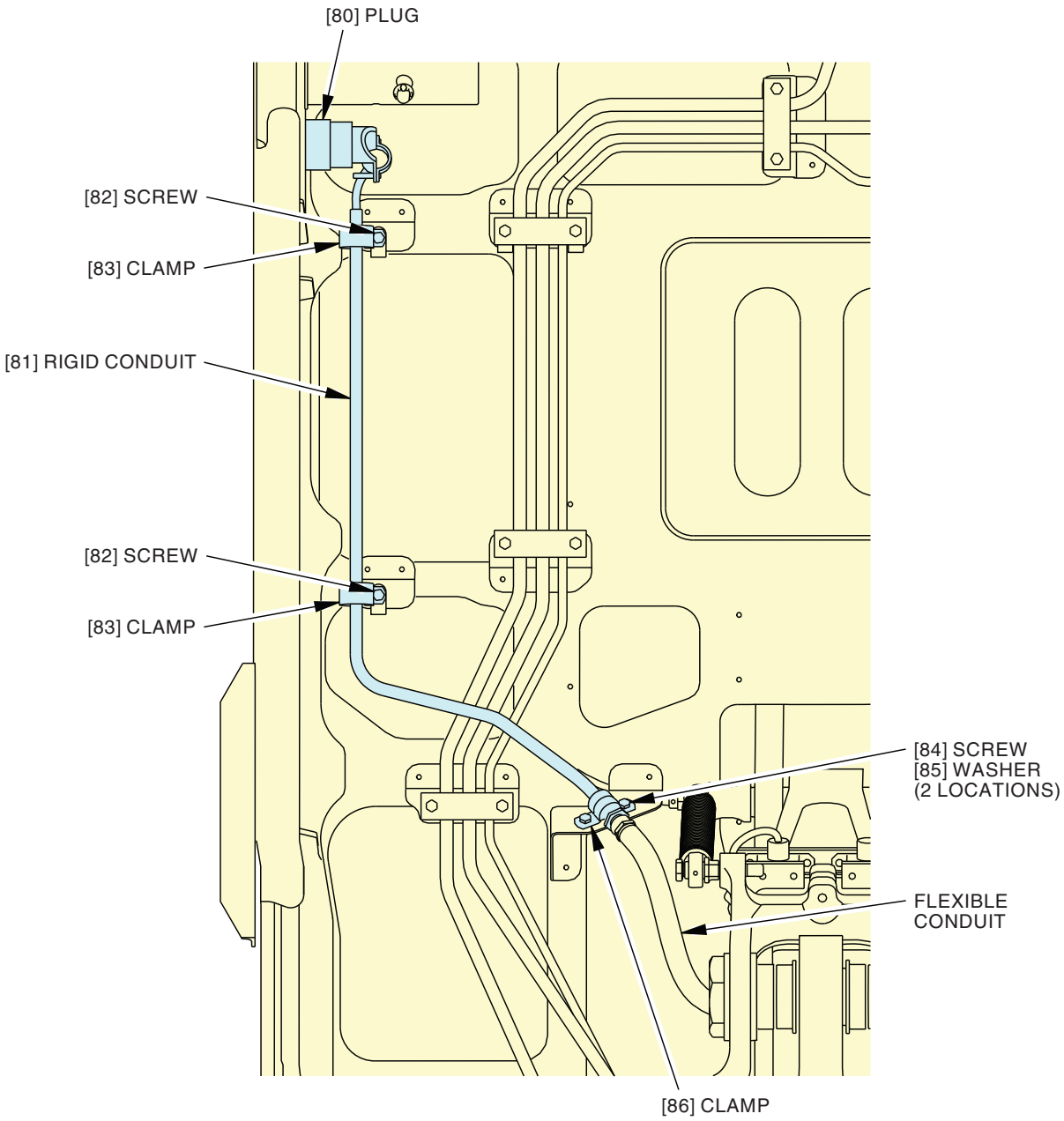
Nose Landing Gear Installation
Figure 401/32-21-00-990-802 (Sheet 6 of 9)

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Nose Landing Gear Installation
Figure 401/32-21-00-990-802 (Sheet 7 of 9)

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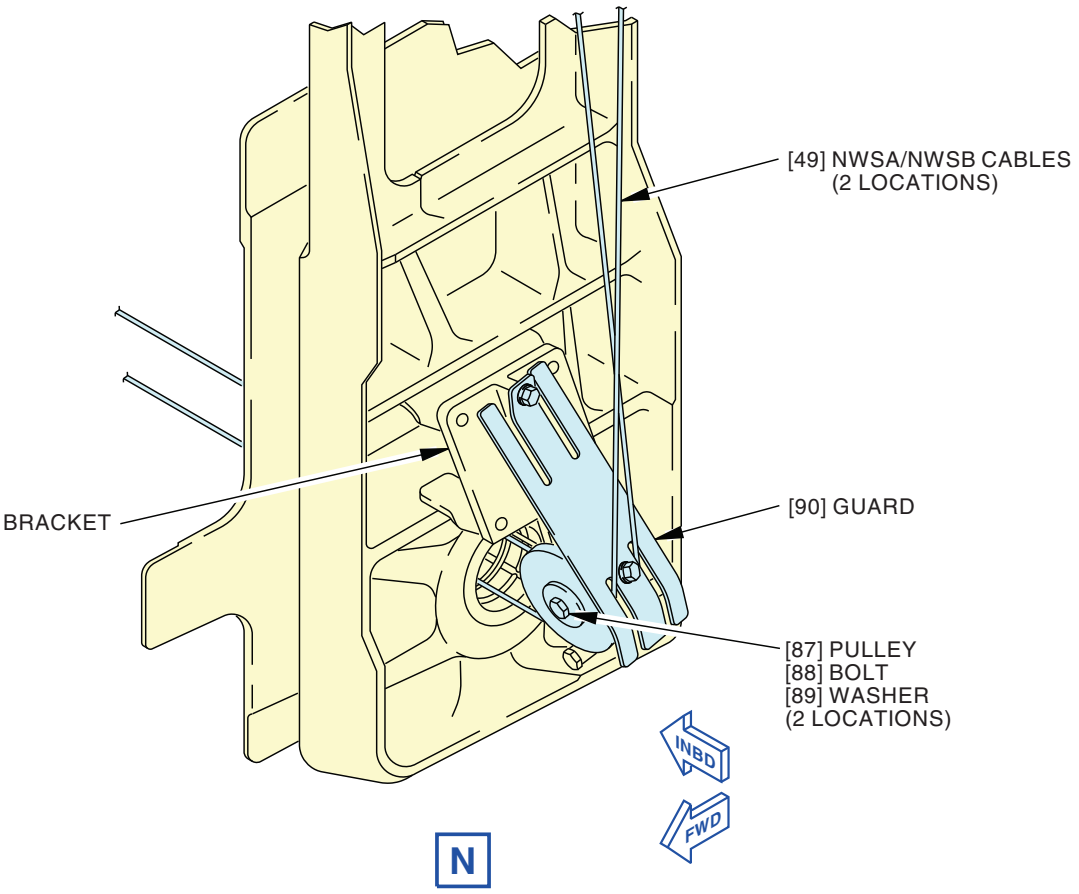
(VIEW IN THE AFT DIRECTION)

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Nose Landing Gear Installation
Figure 401/32-21-00-990-802 (Sheet 8 of 9)

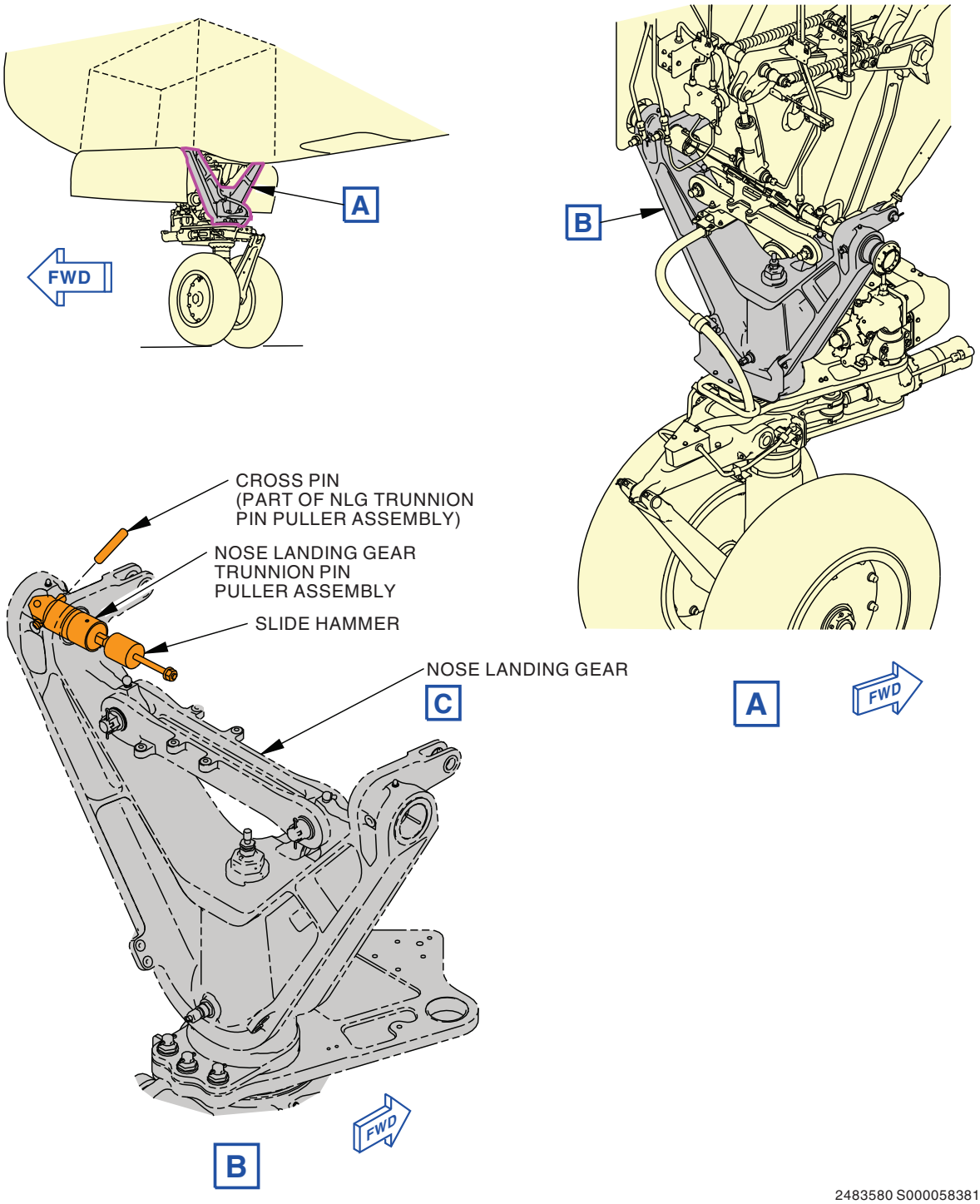
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Nose Landing Gear Installation
Figure 401/32-21-00-990-802 (Sheet 9 of 9)

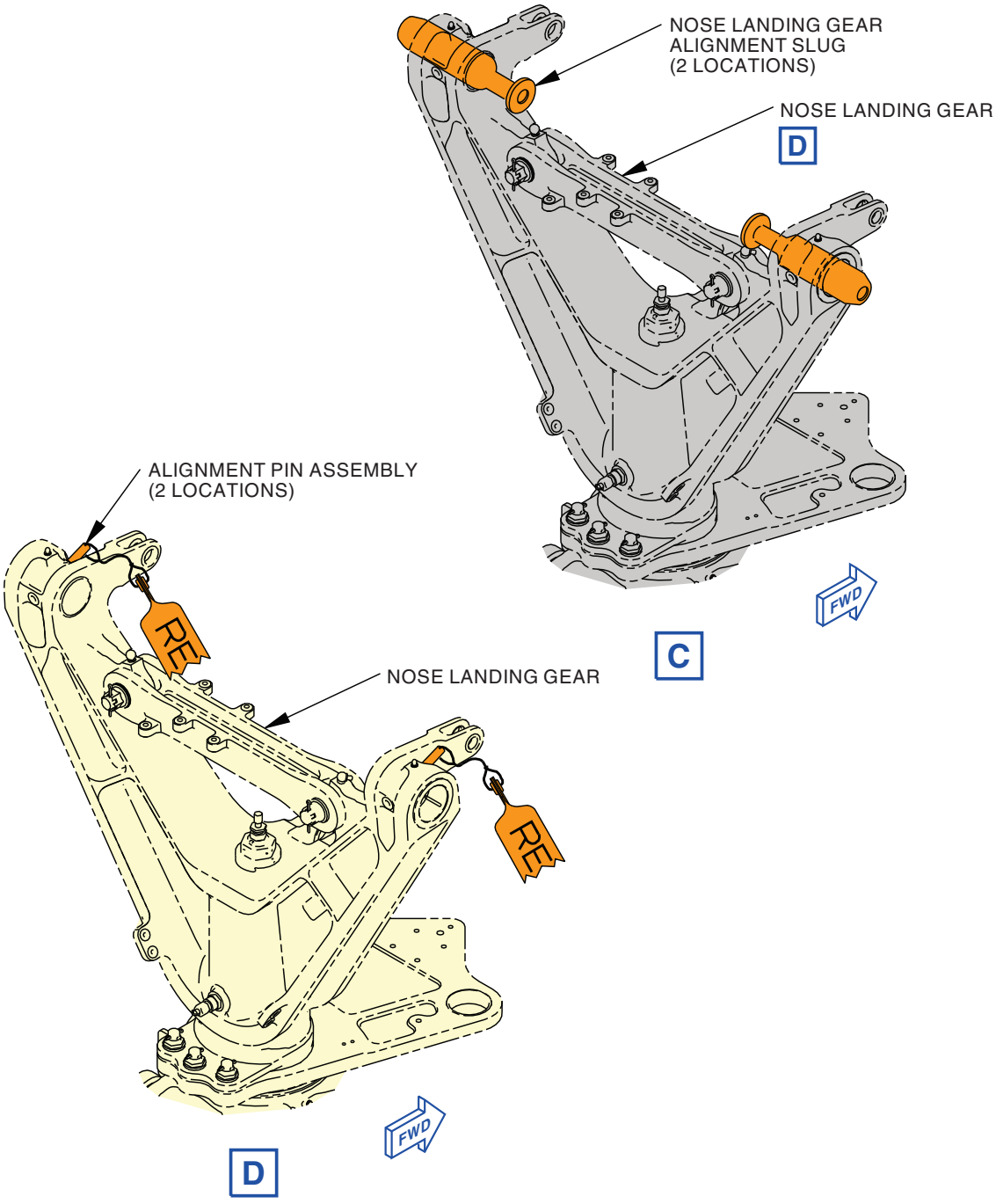
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Trunnion Pin, Nose Landing Gear Removal/Installation Equipment
Figure 402/32-21-00-990-808 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2483630 S0000583819_V1

Trunnion Pin, Nose Landing Gear Removal/Installation Equipment
Figure 402/32-21-00-990-808 (Sheet 2 of 2)

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TASK 32-21-00-400-802

3. Nose Landing Gear Installation

(Figure 401, Figure 402)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
05-51-91-790-801	Cabin Pressure Leak Test (P/B 201)
07-10-03-580-803	Lift the Nose Landing Gear Axle With Axle Jacks-Handling (P/B 201)
07-10-21-580-802	Lower the Airplane Nose, Jack Point D (P/B 201)
10-11-05-480-801	Chock Installation During Engine Operation (P/B 201)
10-11-05-500-801	Chock Installation Winds/Gusts Maximum 40 mph (35 Knots) (P/B 201)
10-11-05-500-802	Chock Installation in Winds More Than 35 Knots - Handling (P/B 201)
12-15-41-610-801	Nose Landing Gear Shock Strut Fluid Check (P/B 301)
12-15-41-610-802	Nose Landing Gear Shock Strut Servicing, Airplane on the Ground (P/B 301)
12-15-41-610-803	Nose Landing Gear Shock Strut Servicing, Airplane on Jacks (P/B 301)
12-21-21-640-801	Nose Landing Gear Upper End Components Servicing (P/B 301)
12-21-21-640-802	Nose Landing Gear Lower End Components Servicing (P/B 301)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-21-00-710-801	Nose Landing Gear Steering Broken Cable Compensator Operational Check (P/B 601)
32-33-00-710-802	Nose Landing Gear Operational Test (P/B 501)
32-35-00-730-801	Nose Gear Manual Extension System Test - Airplane on Jacks (P/B 501)
32-45-21-400-801	Nose Landing Gear Wheel and Tire Assembly Installation (P/B 401)
32-51-00-700-801	Nose Wheel Steering System Test (Airplane on Ground) (P/B 501)
32-51-00-820-801	General Check of the Nose Wheel Steering System Adjustment (P/B 501)
32-51-00-820-805	Nose Wheel Steering System Adjustment (P/B 501)
32-51-31-400-802	Nose Gear Steering System Cables Installation (P/B 401)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

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Reference	Description
COM-4938	Insertion/Removal Tool - Contact (size 16, backshell side) Part #: M81969/1-03 Supplier: 11851 Part #: M81969/14-03 Supplier: 11139
SPL-1559	Adjustable Spanner Wrench (2.00 to 4.00 Inch Dia. Retainer) Part #: F72959-34 Supplier: 81205 Part #: F72959-35 Supplier: 81205 Part #: F72959-36 Supplier: 81205 Part #: F72959-41 Supplier: 81205 Opt Part #: F72959-5 Supplier: 81205 Opt Part #: F72959-6 Supplier: 81205 Opt Part #: F72959-7 Supplier: 81205 Opt Part #: F72959-8 Supplier: 81205
SPL-1821	Adjustable Spanner Wrench (0.75 to 2.00 Inch Dia. Retainer, 0.12 x 0.12 Key Arm) Part #: F72959-40 Supplier: 81205 Opt Part #: F72959-4 Supplier: 81205
SPL-1871	Retention Straps - Shock Strut, NLG/MLG Part #: C32030 -31 Supplier: 81205 Opt Part #: C32030-10 Supplier: 81205
SPL-10305	Removal/Installation Equipment, Outrigger - NLG Part #: C32049-1 Supplier: 81205
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)
STD-15080	R/I Equipment - Trunnion Pin, NLG (Note: C32054-4 Alignment Pin Assembly is included in SPL-14481 kit)
STD-15081	R/I Equipment - Trunnion Pin, NLG (Note: C32054-6 Trunnion Dummy Pin is included in SPL-14481 kit)

C. Consumable Materials

Reference	Description	Specification
C00308	Compound - Corrosion Preventive, Petrolatum Hot Application	MIL-C-11796
C00528	Compound - Corrosion Preventive, Petroleum Hot Application (Soft Film)	MIL-C-11796 Class III
D00633	Grease - Aircraft General Purpose	BMS3-33
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50225	Lockwire - MS20995C20, Corrosion Resistant Steel - 0.020 Inch (0.508 mm) Diameter	NASM20995
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
11	Pin	32-21-00-02-010	PTF ALL
31	Pin	32-22-11-03-005	PTF ALL
35	Pin	32-21-00-02-045	PTF ALL

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(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
43	Pin	32-22-11-03-005	PTF ALL
59	Pin	32-21-00-02-045	PTF ALL
65	Seal	32-21-21-01-040	PTF ALL
74	O-ring	32-21-00-02-070	PTF ALL

E. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
713	Nose Landing Gear

F. Access Panels

Number	Name/Location
113BW	Forward Nose Wheel Well Panel

G. Prepare for the Installation

SUBTASK 32-21-00-480-003

(1) If it is necessary, use a dolly or the Outrigger R/I Equipment, SPL-10305, to move the nose landing gear.

(a) When the Outrigger R/I Equipment, SPL-10305, is installed and will be left unattended, do these steps:

1) Chock the nose landing gear tires, do one of these tasks:

- Chock Installation During Engine Operation, TASK 10-11-05-480-801
- Chock Installation Winds/Gusts Maximum 40 mph (35 Knots), TASK 10-11-05-500-801
- Chock Installation in Winds More Than 35 Knots - Handling, TASK 10-11-05-500-802.

2) Lock the caster brake.

SUBTASK 32-21-00-480-004

(2) Install the retention strap, SPL-1871, with the marker facing up to hold the shock strut [17] in its compressed position.

SUBTASK 32-21-00-620-001

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(3) Apply a thin layer of the corrosion inhibiting material, G50136, to these items:

(a) The threads and thread reliefs of the bolt [62] and bolt [33]

(b) The threads of the nut [60] and nut [36]

(c) The faces of the washer [78] and washer [34].

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SUBTASK 32-21-00-160-001

(4) Remove all unwanted corrosion inhibiting material, G50136.

SUBTASK 32-21-00-640-001

(5) Lubricate the following items with grease, D00633:

(a) The shank of the bolt [62] and bolt [33]

(b) The shank of left trunnion pin [63] and right trunnion pin [37]

(c) The spacer [39] and spacer [71]

(d) The cable guard [70].

SUBTASK 32-21-00-160-002

(6) Remove all unwanted grease, D00633.

H. Nose Landing Gear Installation

SUBTASK 32-21-00-560-002

(1) Put the nose landing gear in its position for installation on the sidewall of the nose wheel well.

SUBTASK 32-21-00-580-003

(2) Lift the nose landing gear with an axle jack until the right trunnion [23] and left trunnion [40] align with the mating hole in the sidewall of the nose wheel well (TASK 07-10-03-580-803).

SUBTASK 32-21-00-420-002

(3) Install the right trunnion pin [37] (View H, Figure 401).

(a) Put the spacer [39] in its position between the right trunnion [23] and sidewall of the nose wheel well.

(b) Apply grease, D00633, and install the new O-ring [74] inside nose wheel well trunnion bushing.

(c) Use the trunnion dummy pin, STD-15081, to put the right trunnion pin [37] through the right trunnion [23], spacer [39], and sidewall of the nose wheel well (Figure 402).

(d) Use the alignment pin assembly, STD-15080, to make sure that the hole in the right trunnion pin [37] for the bolt [33] aligns with the mating hole in the right trunnion [23] (Figure 402).

(e) Put the right trunnion pin [37] through the right trunnion [23], spacer [39], and sidewall of the nose wheel well.

1) Make sure that the hole in the right trunnion pin [37] for the bolt [33] aligns with the mating hole in the right trunnion [23].

(f) Put the swivel [24] through the right trunnion pin [37].

1) Make sure that the hole in the swivel [24] for the bolt [33] aligns with the mating hole in the right trunnion pin [37] and right trunnion [23].

(g) Put the bolt [33] through the right trunnion [23], right trunnion pin [37], and swivel [24].

(h) Install the washer [34] and nut [36] on the bolt [33].

(i) Tighten the nut [36] to 20 in-lb (2.3 N·m) - 24 in-lb (2.7 N·m) plus the run-on torque.

(j) Install the pin [35] in the bolt [33].

1) If it is necessary, loosen the nut [36] to the nearest castellation to align the holes for the pin [35].

SUBTASK 32-21-00-420-003

(4) Install the left trunnion pin [63] (View K, Figure 401).

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(a) Put the spacer [71] in its position between the left trunnion [40] and sidewall of the nose wheel well (View L, Figure 401).

(b) Apply grease, D00633, and install the new O-ring [74] inside nose wheel well trunnion bushing.

(c) Use the trunnion dummy pin, STD-15081, to put the left trunnion pin [63] through the left trunnion [40], spacer [71], and in the left sidewall of the nose wheel well (Figure 402).

(d) Use the alignment pin assembly, STD-15080, to make sure that the hole in the left trunnion pin [63] for the bolt [62] aligns with the mating hole in the left trunnion [40] (Figure 402).

(e) Put the left trunnion pin [63] through the left trunnion [40], spacer [71], and in the left sidewall of the nose wheel well.

(f) Put the cable guard [70] into the left trunnion pin [63].

1) Make sure that the hole in the cable guard [70] for the bolt [62] that aligns with the mating hole in the left trunnion pin [63] and left trunnion [40].

(g) Put the bolt [62] through the left trunnion [40], left trunnion pin [63], and cable guard [70].

(h) Install the washer [78] and nut [60] on the bolt [62].

(i) Tighten the nut [60] to 20 in-lb (2.3 N·m) - 24 in-lb (2.7 N·m) plus the run-on torque.

(j) Install the pin [59] in the bolt [62].

1) If it is necessary, loosen the nut [60] to the nearest castellation to align the holes for the pin [59].

SUBTASK 32-21-00-220-001

(5) Measure the total axial free play in the trunnion.

NOTE: If the total axial free play is not correct, this will not cause any damage to the landing gear structure, but there may be a loud noise during landing gear extension/retraction.

(a) Use a thickness gage to measure the clearance between the spacer [39], spacer [71], and bushing faces of the sidewall or the trunnion on the two sides of the nose landing gear.

(b) Make sure that the combined clearance of the two trunnions is 0.008 in. (0.203 mm) - 0.015 in. (0.381 mm).

(c) If it is necessary to get the correct clearance, adjust the spacer [39] and spacer [71] stack thickness.

SUBTASK 32-21-00-080-001

(6) Remove the axle jack after the correct clearance has been achieved.

SUBTASK 32-21-00-420-014

(7) Install the bracket [51] (View J, Figure 401).

(a) Install the bolt [75], washer [76], and nut [77].

SUBTASK 32-21-00-420-004

(8) Install the pulleys [50] (View J, Figure 401).

(a) Put the nwsa/nwsb cable [49] in their position on the groove of the pulley [50].

(b) Put each pulley [50] in its position on the bracket [51].

(c) Install the washer [48] on the bolt [55].

(d) Put the bolt [55] through the pulley [50] and bracket [51].

(e) Install the washer [53] and nut [52] on the bolt [55].

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SUBTASK 32-21-00-710-005

(9) Do the operational check of the broken cable compensator, do this task: Nose Landing Gear Steering Broken Cable Compensator Operational Check, TASK 32-21-00-710-801.

SUBTASK 32-21-00-420-005

CAUTION MAKE SURE THAT THE STEERING CABLES DO NOT CROSS. CROSSED CABLES CAN INCREASE CABLE FRICTION WHICH CAN CAUSE DAMAGE TO THE CABLES.

(10) Connect the nwsa/nwsb cables [49] (View K, Figure 401) (TASK 32-51-31-400-802).

(a) Make sure that the rig pin NS2 is in the drum for the Captain's control wheel (TASK 32-51-00-820-801).

(b) Put the nwsa/nwsb cable [49] through the inboard side of the trunnion pin nut [68], seal ring [69], and left trunnion pin [63].

1) Make sure that one nwsa/nwsb cable [49] goes over the trunnion pin bolt and one nwsa/nwsb cable [49] goes under the trunnion pin bolt where the nwsa/nwsb cables [49] go through the trunnion.

(c) For the pulleys [87] on the bracket that is outboard of the left trunnion pin [63], do these steps:

1) Put the nwsa/nwsb cables [49] in the grooves of the pulleys [87] and through the guard [90].

2) Install the pulleys [87], bolts [88], and washers [89].

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

a) Install the bolts [88] with Cor-Ban 27L Compound, G50237 (preferred), or corrosion inhibiting material, G50136 (optional).

b) Tighten the bolts [88] to 73 in-lb (8 N·m) - 77 in-lb (9 N·m).

(d) Apply a thin layer of corrosion preventive compound, C00308, or compound, C00528, to the threads on the internal and external surfaces of the mating adjustable parts on the cable turnbuckles [1].

(e) Connect each nwsa/nwsb cable [49] to the cable turnbuckle [1] (View B, Figure 401).

(f) Remove the identification tags attached to the nwsa/nwsb cables [49].

(g) Install the locking clips [2] on the cable turnbuckles [1].

SUBTASK 32-21-00-410-001

(11) Install the seal [65] (View K, Figure 401).

(a) Assemble the seal [65] around the nwsa/nwsb cables [49] at the inboard end of the left trunnion pin [63].

(b) Install one retainer [66] on the eyebolt [67].

(c) Put the eyebolt [67] through the seal [65].

(d) Install the other retainer [66], washer [56], and nut [64] on the eyebolt [67].

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(e) Tighten the nut [64] sufficiently to clamp the seal [65].

(f) Loosen the nut [64] to approximately 1/4 turn until you can turn the eyebolt [67] with your fingers.

(g) Put the seal [65] into the left trunnion pin [63] until it touches the stop.

(h) Install the seal ring [69] into the left trunnion pin [63].

SUBTASK 32-21-00-620-002

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(12) Apply a thin layer of the corrosion inhibiting material, G50136, to the trunnion pin nut [68].

SUBTASK 32-21-00-420-006

(13) Install the trunnion pin nut [68] (View K, Figure 401).

(a) Install the trunnion pin nut [68] on the inboard side of the left trunnion pin [63].

(b) Use the spanner wrench, SPL-1821, or spanner wrench set, SPL-1559, to tighten the trunnion pin nut [68] to 100 in-lb (11 N·m) - 150 in-lb (17 N·m).

(c) If it is necessary, loosen the trunnion pin nut [68] to align with the vernier of the lock holes.

(d) Install the lock [57] on the inboard side of the left trunnion pin [63].

(e) Install the screw [58] that holds the lock [57] to the left trunnion pin [63].

(f) Install the MS20995NC32 lockwire, G01912, on the screw [58].

SUBTASK 32-21-00-710-001

(14) Make sure that the seal [65] will turn freely from the extended cable position of the gear to a simulated gear retracted cable position.

NOTE: This step will make sure that the cables will not become twisted when the gear is retracted.

SUBTASK 32-21-00-620-003

WARNING

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(15) Apply a thin layer of the corrosion inhibiting material, G50136, to these items:

(a) The threads and thread reliefs of the bolt [27] and bolt [47]

(b) The threads of the nut [32] and nut [42]

(c) The faces of the washer [28], washer [30], washer [44], and washer [46].

SUBTASK 32-21-00-160-003

(16) Remove all unwanted corrosion inhibiting material, G50136.

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SUBTASK 32-21-00-640-002

(17) Lubricate the chrome plated surfaces of the bolt [27], bolt [47], bushing [29], and bushing [45] with grease, D00633.

SUBTASK 32-21-00-160-004

(18) Remove all unwanted grease, D00633.

SUBTASK 32-21-00-420-007

(19) Connect the left upper rod assembly [41] (View I, Figure 401).

(a) Put the upper rod assembly [41] in its position on the clevis of the left trunnion [40].

(b) Put the bushing [45] through the clevis of the left trunnion [40] and rod end of the upper rod assembly [41].

(c) Install the washer [46] on the bolt [47].

(d) Put the bolt [47] through the clevis of the left trunnion [40] and rod end of the upper rod assembly [41].

(e) Install the washer [44] and nut [42] on the bolt [47].

(f) Tighten the nut [42] to 325 in-lb (36.7 N·m) - 360 in-lb (40.7 N·m).

(g) Install the pin [43] in the bolt [47].

1) If it is necessary, loosen the nut [42] to the nearest castellation to align the holes for the pin [43].

SUBTASK 32-21-00-420-008

(20) Connect the right upper rod assembly [26] (View G, Figure 401).

(a) Put the upper rod assembly [26] to the right trunnion [23].

(b) Put the bushing [29] through the clevis of the right trunnion [23] and rod end of the upper rod assembly [26].

(c) Install the washer [28] on the bolt [27].

(d) Put the bolt [27] through the clevis of the right trunnion [23] and the rod end of the upper rod assembly [26].

(e) Install the washer [30] and nut [32] on the bolt [27].

(f) Tighten the nut [32] to 325 in-lb (36.7 N·m) - 360 in-lb (40.7 N·m).

(g) Install the pin [31] in the bolt [27].

1) If it is necessary, loosen the nut [32] to the nearest castellation to align the holes for the pin [31].

SUBTASK 32-21-00-420-009

(21) Connect the hydraulic lines [25] (View F, Figure 401).

(a) Remove the plug from the hydraulic ports of the swivel [24].

(b) Remove the caps/plugs, STD-14686, from the hydraulic lines [25].

(c) Connect the hydraulic lines [25] to the swivel [24].

(d) Remove the tags from the swivel ports and the hydraulic lines [25].

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SUBTASK 32-21-00-620-004

WARNING

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(22) Apply a thin layer of the corrosion inhibiting material, G50136, to these items:

(a) The threads and thread reliefs of the pin [18]

(b) The threads of the nut [12]

(c) The faces of the washer [20] and washer [21]

(d) The shank of bolt [13].

SUBTASK 32-21-00-160-005

(23) Remove all unwanted corrosion inhibiting material, G50136.

SUBTASK 32-21-00-640-003

(24) Lubricate the shank of the pin [18] with grease, D00633.

SUBTASK 32-21-00-160-006

(25) Remove all unwanted grease, D00633.

SUBTASK 32-21-00-420-010

(26) Connect the lower drag strut [19] (View E, Figure 401).

(a) Make sure that the drag brace lubrication fittings are on the right side when looking forward.

(b) Put the lower drag strut [19] in its position on the clevis of the shock strut [17].

(c) Put the pin [18] through the clevis of the shock strut [17] and lower drag strut [19].

(d) Install the washer [20] and nut [12] on the pin [18].

(e) Tighten the nut [12] to 600 in-lb (67.8 N·m) - 700 in-lb (79.1 N·m).

1) Loosen the nut [12] to the nearest castellation to align the holes for the bolt [13].

(f) Install the bolt [13] in the pin [18].

(g) Install the washer [21] on the bolt [13].

(h) Install the nut [22] on the bolt [13].

(i) Install the cotter pin [11] in the bolt [13].

1) If it is necessary, loosen the nut [22] to the nearest castellation to align the holes for the cotter pin [11].

SUBTASK 32-21-00-420-011

(27) Do this task: Nose Landing Gear Wheel and Tire Assembly Installation, TASK 32-45-21-400-801.

SUBTASK 32-21-00-420-012

(28) Connect the wires to the plug [6] (View D, Figure 401).

(a) Pull the wires through the conduit.

1) Connect the conduit [4] to the junction box [5] (View C, Figure 401).

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2) Install MS20995C20 lockwire, G50225, on the conduit [4] and to the junction box [5].

3) Attach the strings hanging out of the end of the rigid conduit [7] that lead to the junction box [5] to the wires coming out of the conduit [4].

4) Pull the wires all the way through the rigid conduit [7].

(b) Connect the wires to the plug [6] with a contact insertion / removal tool, COM-4938, tool using the drawing you made during the removal.

(c) Install the screws [10] and washers [3] to attach the junction box cover [9] on the junction box [5].

SUBTASK 32-21-00-420-022

(29) To connect the down lock sensor plug [80] and rigid conduit [81] to the wheel well, do these steps:

(a) Remove the protective cap from the plug [80].

(b) Put the rigid conduit [81] into position (View M, Figure 401).

(c) Install the clamp [86] with the screw [84] and washer [85].

(d) Install the clamps [83] with the screws [82].

(e) Connect the plug [80] to the connector on the wheel well wall.

I. Nose Landing Gear Installation Test

SUBTASK 32-21-00-410-002

(1) Close this access panel:

(Nose Wheel Well Access Panels Installation, TASK 53-14-01-420-801)

Number	Name/Location
113BW	Forward Nose Wheel Well Panel

(a) Do this task: Cabin Pressure Leak Test, TASK 05-51-91-790-801.

SUBTASK 32-21-00-080-002

(2) Remove the retention strap, SPL-1871, from the nose landing gear.

SUBTASK 32-21-00-610-002

(3) Lubricate the nose landing gear (TASK 12-21-21-640-801, TASK 12-21-21-640-802).

SUBTASK 32-21-00-610-005

(4) If it is necessary, do this task: Nose Landing Gear Shock Strut Servicing, Airplane on Jacks, TASK 12-15-41-610-803.

SUBTASK 32-21-00-860-011

(5) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-21-00-790-001

(6) Do a check of the swivel [24] for leakage.

SUBTASK 32-21-00-860-012

(7) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	17	C00129	LANDING GEAR LATCH & PRESS WARN
C	15	C01355	LANDING GEAR AIR/GND SYS 2

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(Continued)

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	16	C01356	LANDING GEAR AIR/GND SYS 1
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN
D	15	C01401	LANDING GEAR AIR/GND RELAY
D	16	C01432	LANDING GEAR ALTN EXTEND SOL
D	17	C01027	LANDING GEAR NOSE GEAR STEER
D	18	C00451	LANDING GEAR AURAL WARN

SUBTASK 32-21-00-710-002

(8) Do this task: Nose Landing Gear Operational Test, TASK 32-33-00-710-802.

SUBTASK 32-21-00-710-003

(9) Do this task: Nose Gear Manual Extension System Test - Airplane on Jacks, TASK 32-35-00-730-801.

SUBTASK 32-21-00-820-001

(10) Adjust the nwsa/nwsb cables [49] (TASK 32-51-00-820-805).

SUBTASK 32-21-00-710-004

(11) Do this task: Nose Wheel Steering System Test (Airplane on Ground), TASK 32-51-00-700-801.

SUBTASK 32-21-00-860-013

(12) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-21-00-860-014

(13) Do this task: Lower the Airplane Nose, Jack Point D, TASK 07-10-21-580-802.

SUBTASK 32-21-00-200-001

(14) Do a check of the pressure and extension of the shock strut.

(a) Measure the pressure and extension of the shock strut.

1) Make sure that the pressure you measure is still correct for the extension of the shock strut (TASK 12-15-41-610-801).

2) If the pressure is not correct for the extension of the shock strut, fully service the shock strut (TASK 12-15-41-610-802).

NOTE: If the pressure is not correct for the extension of the shock strut, the shock strut does not have the proper amount of fluid.

END OF TASK

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NOSE LANDING GEAR - INSPECTION/CHECK

1. General

A. This procedure contains these tasks:

(1) An inspection of the nose landing gear

(2) An inspection of the nose landing gear torsional freeplay

(3) An operational check of the nose landing gear steering broken cable compensator.

B. This procedure does not remove or install the nose landing gear. Refer to (Nose Landing Gear Gas Charging Valve Removal, TASK 32-21-00-000-801) and (Nose Landing Gear Gas Charging Valve Installation, TASK 32-21-00-400-801) to remove and install the nose landing gear.

TASK 32-21-00-200-801

2. Nose Landing Gear - Inspection

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task provides the regular inspection of the nose landing gear.

(2) Refer to these tasks: Phase I Inspection, TASK 05-51-37-200-804, High Energy Stop, TASK 05-51-07-210-801, Brake Seizure Conditional Inspection, TASK 05-51-15-210-801 and Overweight Landing Inspection, TASK 05-51-35-210-803, for special inspection procedures on these conditions:

(a) After a hard or overweight landing.

(b) After a brake seizure when the airplane is towing or is moving at the high speed on the ground.

(c) After the gear has collided with an obstacle.

(d) After a high energy stop.

B. References

Reference	Title
05-51-07-210-801	High Energy Stop (P/B 201)
05-51-15-210-801	Brake Seizure Conditional Inspection (P/B 201)
05-51-35-210-803	Overweight Landing Inspection (P/B 201)
05-51-37-200-804	Phase I Inspection (P/B 201)
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-00-10-211-801	Landing Gear Inner Cylinder Chrome Inspection (P/B 601)
32-00-11-100-802	Nose Landing Gear Steering Actuators Chrome Surfaces Cleaning (P/B 701)
32-00-11-100-803	Nose Landing Gear Retract Actuator Chrome Surfaces Cleaning (P/B 701)
32-21-11-200-801	Nose Landing Gear Shock Strut Seal Leakage Check (P/B 801)
32-45-00-700-801	Wheels Fast Check (Wheel Installed on the Airplane) (P/B 601)
32-45-00-700-803	Landing Gear Tire Inspection (P/B 601)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Consumable Materials

Reference	Description	Specification
A00226	Compound - Tamper-Proof Putty	BMS8-45
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95

D. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
713	Nose Landing Gear

E. Nose Landing Gear Inspection

SUBTASK 32-21-00-480-018

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-21-00-480-006

(2) Put chocks around the tires of all the landing gear (PAGEBLOCK 10-11-05/201).

SUBTASK 32-21-00-210-001

(3) Examine the upper components of the nose landing gear:

(a) Examine the drag strut for cracks, damage, and worn areas on the attach points.

(b) Examine the trunnion pins of the drag strut for worn areas and looseness.

(c) For the retract actuator, do these steps:

1) Examine the retract actuator for these items:

a) Damage, looseness, and security of actuator components and attachment bolts.

b) Dirt, scratches, and debris on the exposed piston chrome and actuator body rod end nut.

c) Leaks and loose hydraulic connections.

2) Do this task: Nose Landing Gear Retract Actuator Chrome Surfaces Cleaning, TASK 32-00-11-100-803.

(d) Examine the lock links for cracks, damage, and worn areas on the attach points.

(e) Examine the lock link springs for worn areas, damage, and security of attachments.

(f) Examine the lock actuator and hydraulic transfer cylinder for loose mounting bolts and leaks.

(g) Do these steps to examine the shock strut:

1) Examine the shock strut for leaks and specified extension, do this task: Nose Landing Gear Shock Strut Seal Leakage Check, TASK 32-21-11-200-801.

2) Examine the shock strut for cracks on the outer cylinder lugs.

3) Examine the external surface of the inner cylinder for dirt, scratches, or galling, do this task: Landing Gear Inner Cylinder Chrome Inspection, TASK 32-00-10-211-801.

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(h) Examine the manual extension release mechanism.

(i) Make sure that the wheel well doors are correctly attached to the push rods and door hinges.

(j) Examine the door linkage for worn areas, damage, and security of attachments.

SUBTASK 32-21-00-210-002

(4) Examine the lower end components of the nose landing gear:

(a) Remove the summing mechanism cover.

(b) Examine the summing mechanism, do these steps:

1) Examine the summing mechanism cover for cracks, damage, and worn areas on the attach points.

2) Examine the summing mechanism for damage and worn areas.

3) Examine the steering metering valve for damage, leaks, and security of attachments.

4) For the steering actuators, do these steps:

a) Examine the steering actuators for these items:

<1> Damage, looseness, and security of actuator components and attachment bolts.

<2> Dirt, scratches, and debris on exposed piston chrome and actuator body rod end nut.

<3> Leaks at the rod end seal and trunnion swivels.

b) Do this task: Nose Landing Gear Steering Actuators Chrome Surfaces Cleaning, TASK 32-00-11-100-802.

5) Examine the hydraulic connections for leaks.

6) Examine the steering cables to cable pulleys attachment.

(c) Examine the upper and lower steering sleeve:

1) Make sure that the top steering sleeve is securely attached to the outer cylinder.

a) Examine the mounting bolts that attach the top steering sleeve to the outer cylinder for looseness or wear below the bolt heads.

b) Look for cracks in compound, A00226, at the mounting nuts that attach the top steering sleeve to the outer cylinder.

c) Look for cracks in sealant, A00247, at the interface between the top steering sleeve and outer cylinder.

NOTE: Cracks in the sealant at the interface between the top steering sleeve and outer cylinder could let moisture enter and cause corrosion.

2) Make sure that the lower steering plate and retainer nut on the steering collar are tightly attached with no unusual or excessive play in the assembled joint.

a) Examine the lower steering plate for cracks, dirt, or worn areas.

(d) Install the summing mechanism cover.

(e) Examine the tires and wheels of the nose landing gear, do these tasks: Landing Gear Tire Inspection, TASK 32-45-00-700-803 and Wheels Fast Check (Wheel Installed on the Airplane), TASK 32-45-00-700-801.

(f) Examine the safety sensors and all electrical wiring for security and chafing.

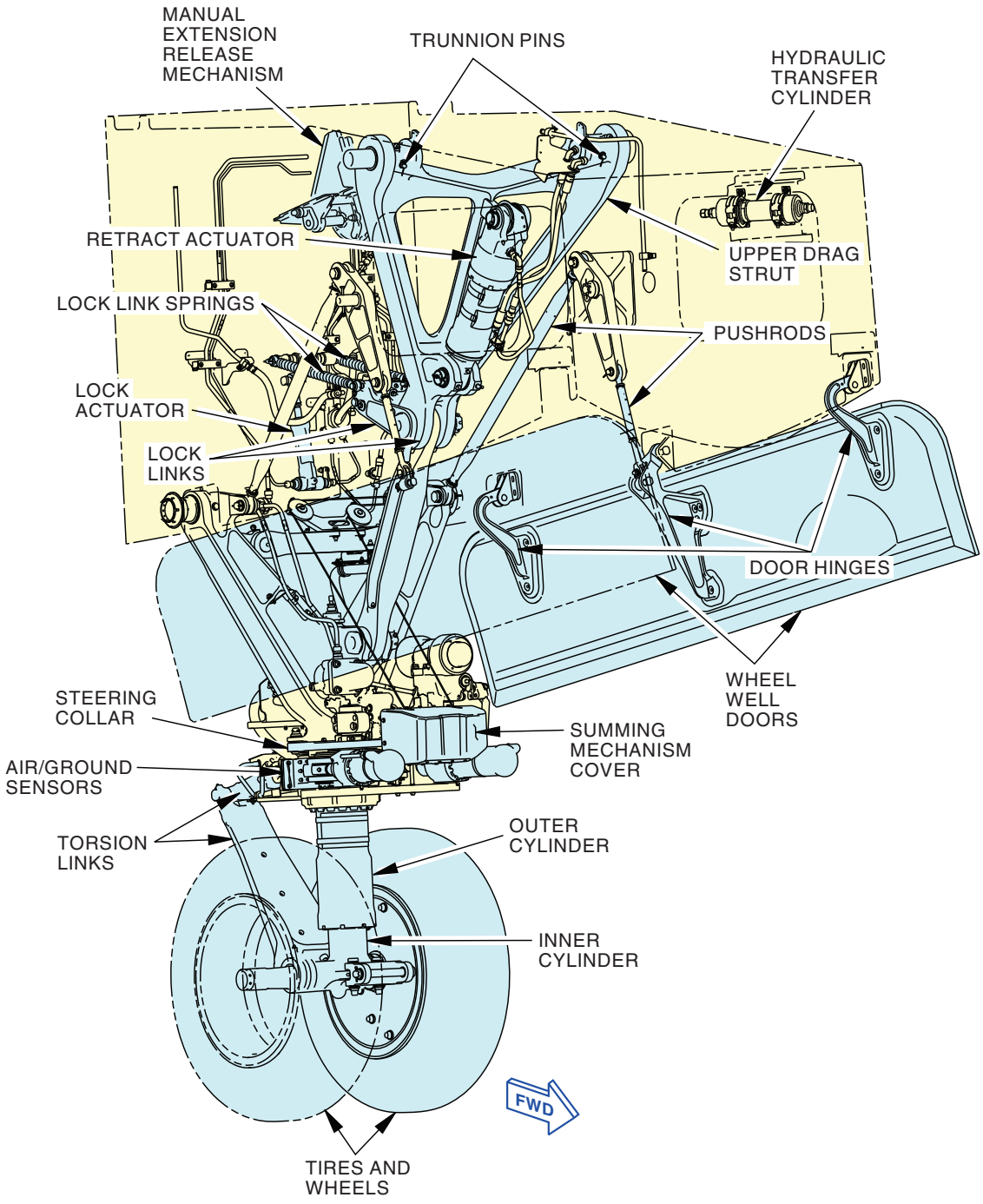
ECCN 9E991 BOEING PROPRIETARY - See title page for details

(g) Examine the torsion links:

1) Make sure that the attachments to the shock strut are correct.
2) Examine the attachment lugs of the torsion links for cracks.
3) Examine the torsion links, for cracks and loose mounting bolts.

——— END OF TASK ———

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Nose Landing Gear Inspection
Figure 601/32-21-00-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-21-00-700-801

3. Nose Landing Gear Torsional Freeplay Inspection

(Figure 601, Figure 602)

A. References

Reference	Title
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
07-10-03-580-803	Lift the Nose Landing Gear Axle With Axle Jacks-Handling (P/B 201)
07-10-21-580-801	Lift the Airplane Nose with the Nose Jack at Jack Point D (P/B 201)
12-15-41-610-803	Nose Landing Gear Shock Strut Servicing, Airplane on Jacks (P/B 301)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1557	Gauge - Force Part #: DG-200 Supplier: 92456 Part #: FDIX 100 Supplier: 0BFD9 Part #: FDIX 50 Supplier: 0BFD9 Part #: FG-60KBM Supplier: 45KA8 Part #: LG-050 Supplier: 92456 Part #: LG-100 Supplier: 92456 Opt Part #: DPP-500G Supplier: 92456 Opt Part #: DPPH-150 Supplier: 92456 Opt Part #: DPPH-200 Supplier: 92456 Opt Part #: DPPH-50 Supplier: 92456 Opt Part #: FDI 100 Supplier: 0BFD9 Opt Part #: FDI 50 Supplier: 0BFD9 Opt Part #: FDV 100 Supplier: 0BFD9 Opt Part #: FDV 50 Supplier: 0BFD9
SPL-1891	Freeplay Equipment - NLG Part #: C32044-1 Supplier: 81205

C. Prepare for the Inspection

SUBTASK 32-21-00-860-015

(1) Remove the pressure from the hydraulic systems A and B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-21-00-860-016

(2) Lift the nose of the airplane with jacks until the shock strut is fully extended (TASK 07-10-21-580-801).

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SUBTASK 32-21-00-020-015

WARNING CLEAR THE AREA BELOW THE NOSE BEFORE YOU DEFLATE THE SHOCK STRUT. IF YOU DEFLATE SHOCK STRUT, THE NOSE CAN MOVE DOWN AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(3) Deflate the shock strut for the nose landing gear:

(a) Remove the cap for the gas valve which is on the top of the shock strut.

WARNING DO NOT REMOVE THE VALVE BODY UNTIL YOU DEFLATE THE SHOCK STRUT FULLY. THE AIR PRESSURE CAN BLOW THE VALVE BODY OUT AND CAUSE INJURIES TO PERSONNEL.

(b) Loosen the gas valve swivel nut a maximum of two turns.

NOTE: Fluid in the shock strut will have bubbles when you release the pressure. Deflate the shock strut slowly to prevent the leakage of the fluid through the gas valve.

(c) Loosen the gas valve swivel nut fully when all of the pressure in the shock strut is released.

NOTE: The shock strut is fully deflated when the dimension "X" is 13.8 in. (350.5 mm).

SUBTASK 32-21-00-580-004

(4) Lift the nose landing gear with the axle jack until you see approximately 4 in. (101.6 mm) of the chrome surface on the inner cylinder, do this task: Lift the Nose Landing Gear Axle With Axle Jacks-Handling, TASK 07-10-03-580-803.

SUBTASK 32-21-00-480-007

(5) Install the freeplay equipment, SPL-1891, on the nose landing gear.

(a) Install the collar assemblies, C32044-4 and C32044-5, on the two actuator rods of the steering cylinders.

(b) Set the collar assemblies, C32044-4 and C32044-5, tightly against the actuator end plates to prevent any movement of the actuator rods.

SUBTASK 32-21-00-480-008

(6) Set the dial indicator gage on the platform on the forward side of the wheel on the axle centerline to read the front and the aft motion of the right wheel rim.

NOTE: The gage must be installed on the forward side of the wheel on the axle centerline as near to the outboard edge of the rim as possible.

SUBTASK 32-21-00-480-009

(7) Install the lever arm assembly, C32044-2, on the tow fitting between the wheels on the nose landing gear.

D. Nose Landing Gear Torsional Freeplay Inspection

SUBTASK 32-21-00-720-001

(1) Do these steps to examine the torsional free play of the nose landing gear:

(a) Measure 60 in. (1524 mm) on the lever arm assembly, C32044-2, from the shock strut centerline.

(b) From the 60 in. (1524 mm) you measured on the lever arm assembly, C32044-2, use the force gauge, COM-1557, to apply a clockwise pull of 30 lbf (133 N).

(c) Set the dial indicator gage to "0" and loosen the load.

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(d) Apply a counterclockwise pull of 30 lbf (133 N) on the lever arm assembly, C32044-2, as in step (b).

(e) Write the value that you get on the dial indicator gage and loosen the load.

(f) Do the above five steps for a total of five cycles.

(g) Get the average value of the torsional free play.

1) Make sure that the average value is less than 0.20 in. (5.1 mm).

NOTE: The free play agrees with 1°arc (0.017 rad).

SUBTASK 32-21-00-700-001

(2) If the torsional free play is satisfactory, put the airplane in its usual condition.

SUBTASK 32-21-00-700-002

(3) If the torsional free play stays more than the approved maximum, contact Boeing Service Engineering.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 32-21-00-080-003

(1) Remove the dial indicator gage, dial indicator platform, and the lever arm assembly, C32044-2.

SUBTASK 32-21-00-080-004

(2) Remove the collar assemblies, C32044-4 and C32044-5, from the two actuator rods of the steering cylinders.

SUBTASK 32-21-00-860-017

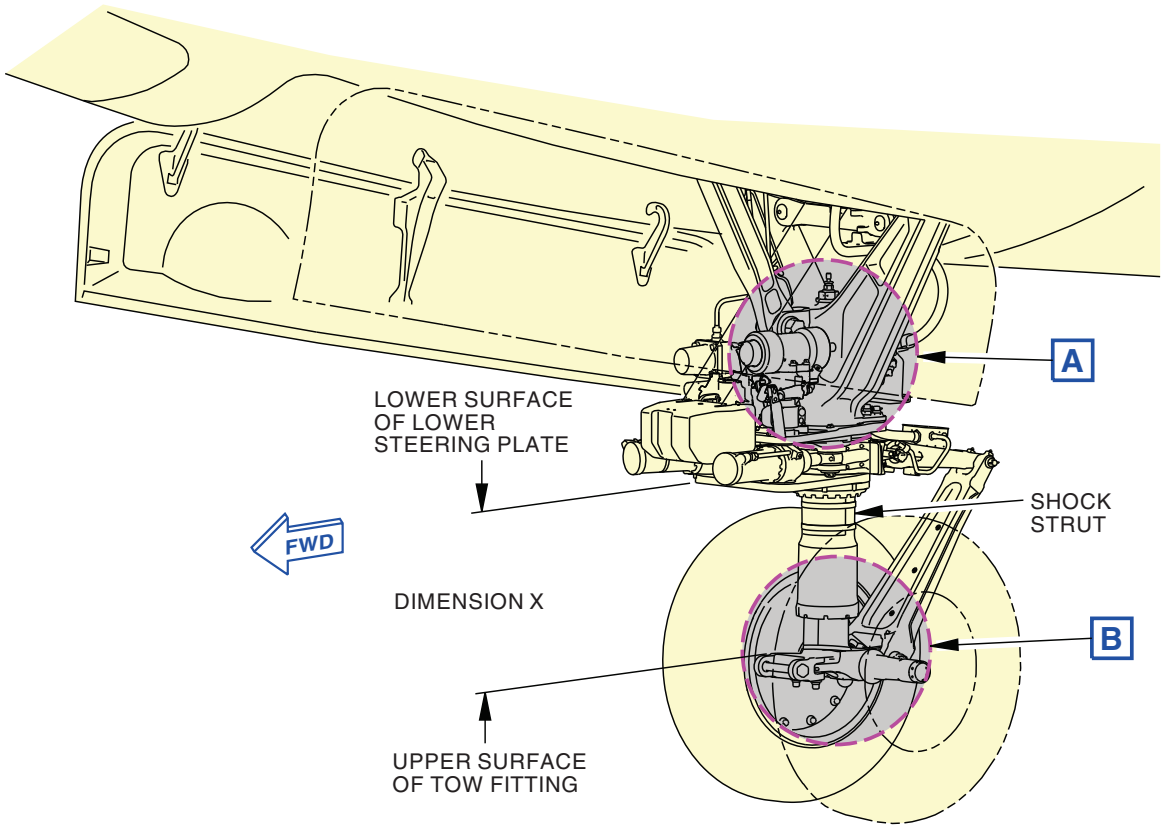
(3) Inflate the shock strut for the nose landing gear (TASK 12-15-41-610-803).

SUBTASK 32-21-00-860-018

(4) Do this task: Lower the Airplane off Jacks, TASK 07-10-01-580-802.

——— END OF TASK ———

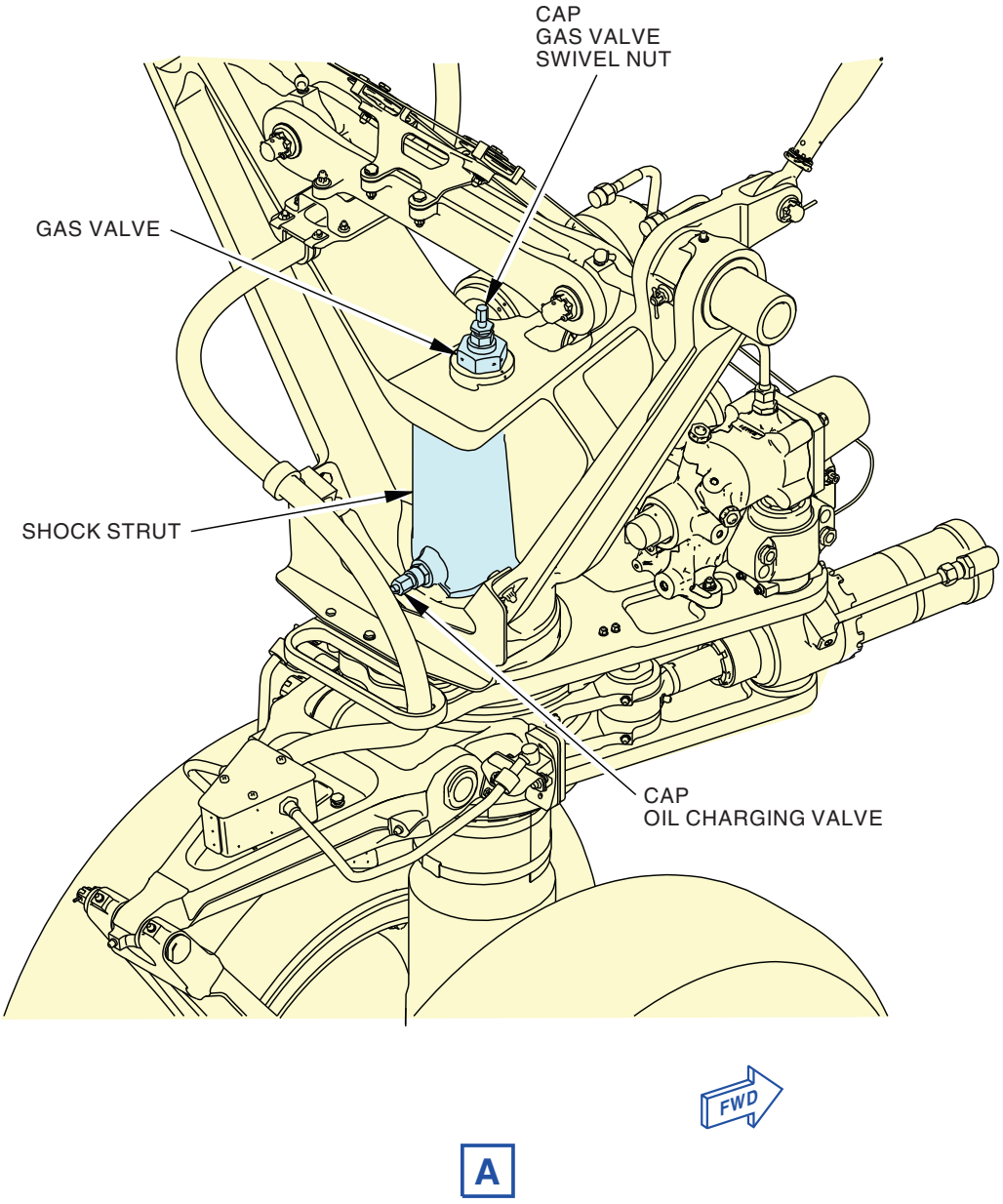
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Nose Landing Gear Torsional Free Play Inspection
Figure 602/32-21-00-990-804 (Sheet 1 of 4)

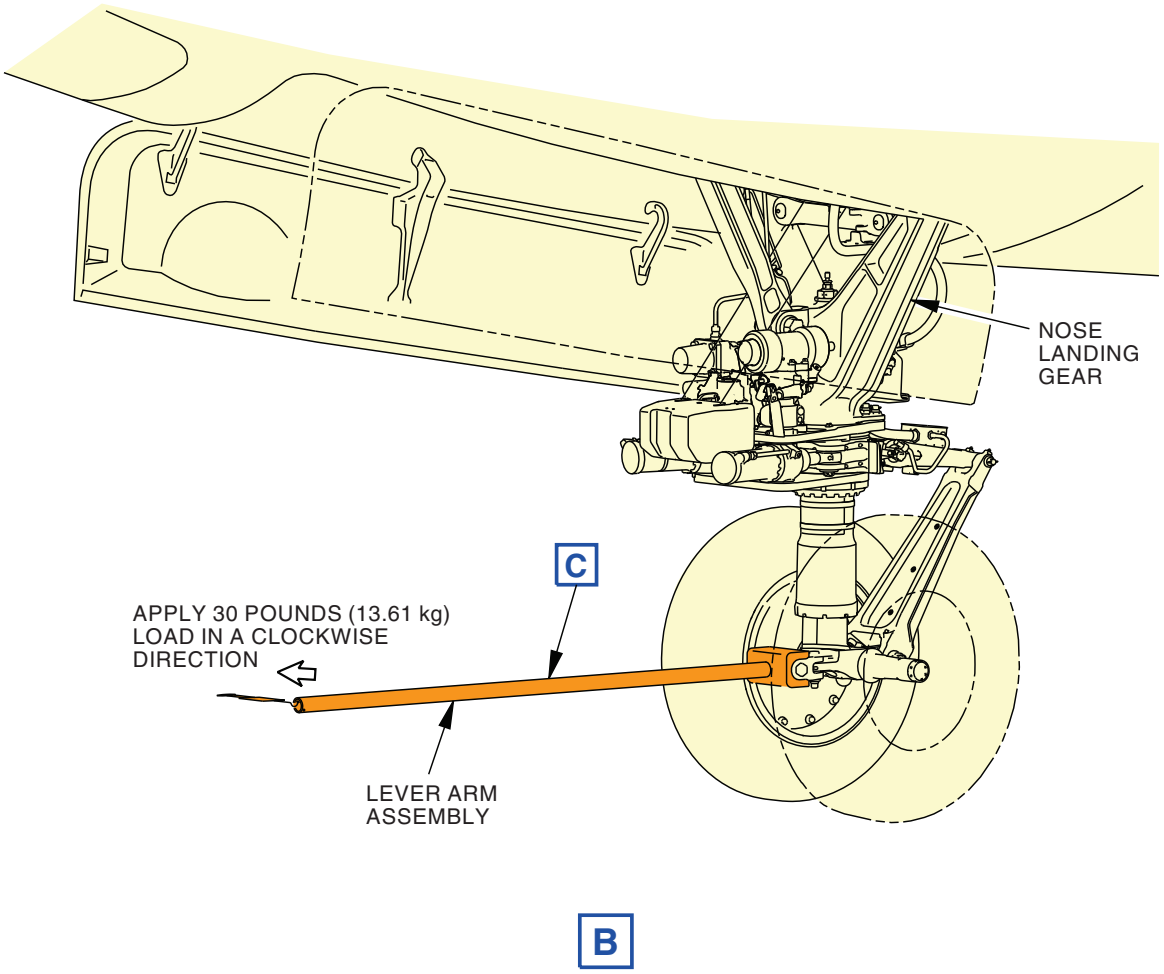
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Nose Landing Gear Torsional Free Play Inspection
Figure 602/32-21-00-990-804 (Sheet 2 of 4)

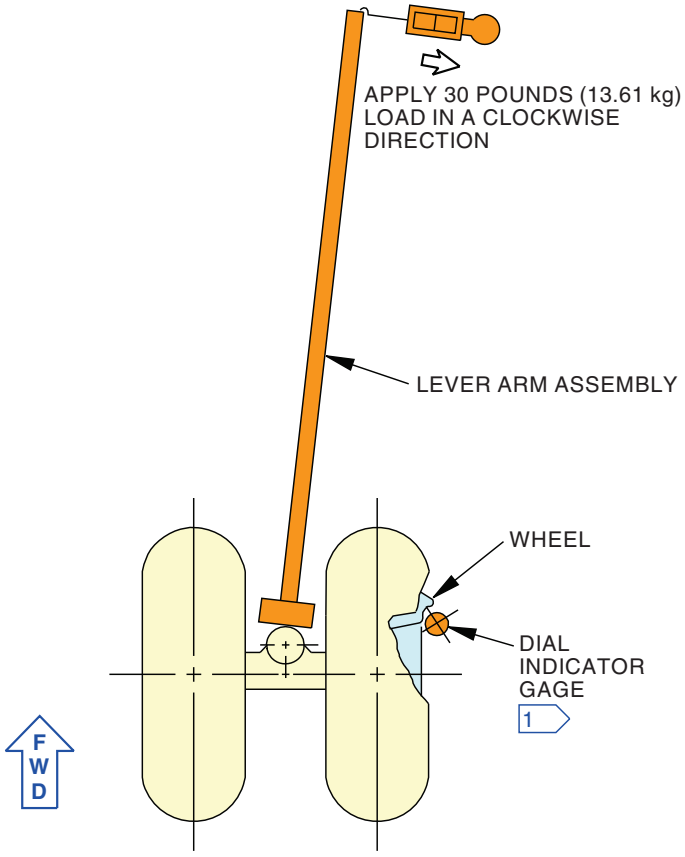
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Nose Landing Gear Torsional Free Play Inspection
Figure 602/32-21-00-990-804 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 POSITION GAGE IN FORE AND AFT DIRECTION

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Nose Landing Gear Torsional Free Play Inspection
Figure 602/32-21-00-990-804 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-21-00-710-801

4. Nose Landing Gear Steering Broken Cable Compensator Operational Check

(Figure 603)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives instructions on how to do an operational check for the Nose Landing Gear Steering Broken Cable Compensator.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-51-00-700-801	Nose Wheel Steering System Test (Airplane on Ground) (P/B 501)
32-51-00-820-801	General Check of the Nose Wheel Steering System Adjustment (P/B 501)
32-51-31-400-802	Nose Gear Steering System Cables Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

D. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
713	Nose Landing Gear

E. Prepare for the Operational Check

SUBTASK 32-21-00-420-018

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-21-00-420-016

(2) Install chocks around all landing gear tires (PAGEBLOCK 10-11-05/201).

F. Nose Landing Gear Steering Broken Cable Compensator Operational Check

SUBTASK 32-21-00-010-001

(1) Remove the summing mechanism cover to get access to the summing levers.

SUBTASK 32-21-00-480-013

(2) Install these rig pins from the rigging pin kit, SPL-1585 (TASK 32-51-00-820-801).

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(a) The rig pin in the captain's steering tiller.

(b) The rig pin in the rudder pedal interconnect quadrant.

SUBTASK 32-21-00-020-017

(3) Disconnect the turnbuckles for the lower section of the steering cables.

(a) Make sure there is enough slack in the steering cables that the summing levers can move freely.

NOTE: Removal of the entire steering cable is not necessary.

SUBTASK 32-21-00-710-006

(4) Hold the left summing lever and while you move the right summing lever.

(a) Make sure that the compensator mechanism moves freely with no binding or sticking.

SUBTASK 32-21-00-710-007

(5) Hold the right summing lever while you move the left summing lever.

(a) Make sure that the compensator mechanism moves freely with no binding or sticking.

SUBTASK 32-21-00-710-008

(6) Move both compensator summing levers so that the compensator cam roller is free of the cam.

(a) Make sure that the cam roller turns freely with no sticking or binding.

SUBTASK 32-21-00-420-017

(7) Connect the turnbuckles for the lower section of the steering cables.

SUBTASK 32-21-00-820-002

(8) Rig the nose gear steering system cables to their original configuration (TASK 32-51-31-400-802).

SUBTASK 32-21-00-080-007

(9) Remove these rig pins:

(a) The rig pin in the captain's steering tiller.

(b) The rig pin in the rudder pedal interconnect quadrant.

SUBTASK 32-21-00-710-009

(10) Do a check of the nose wheel steering system (TASK 32-51-00-700-801).

SUBTASK 32-21-00-410-003

(11) Install the summing mechanism cover.

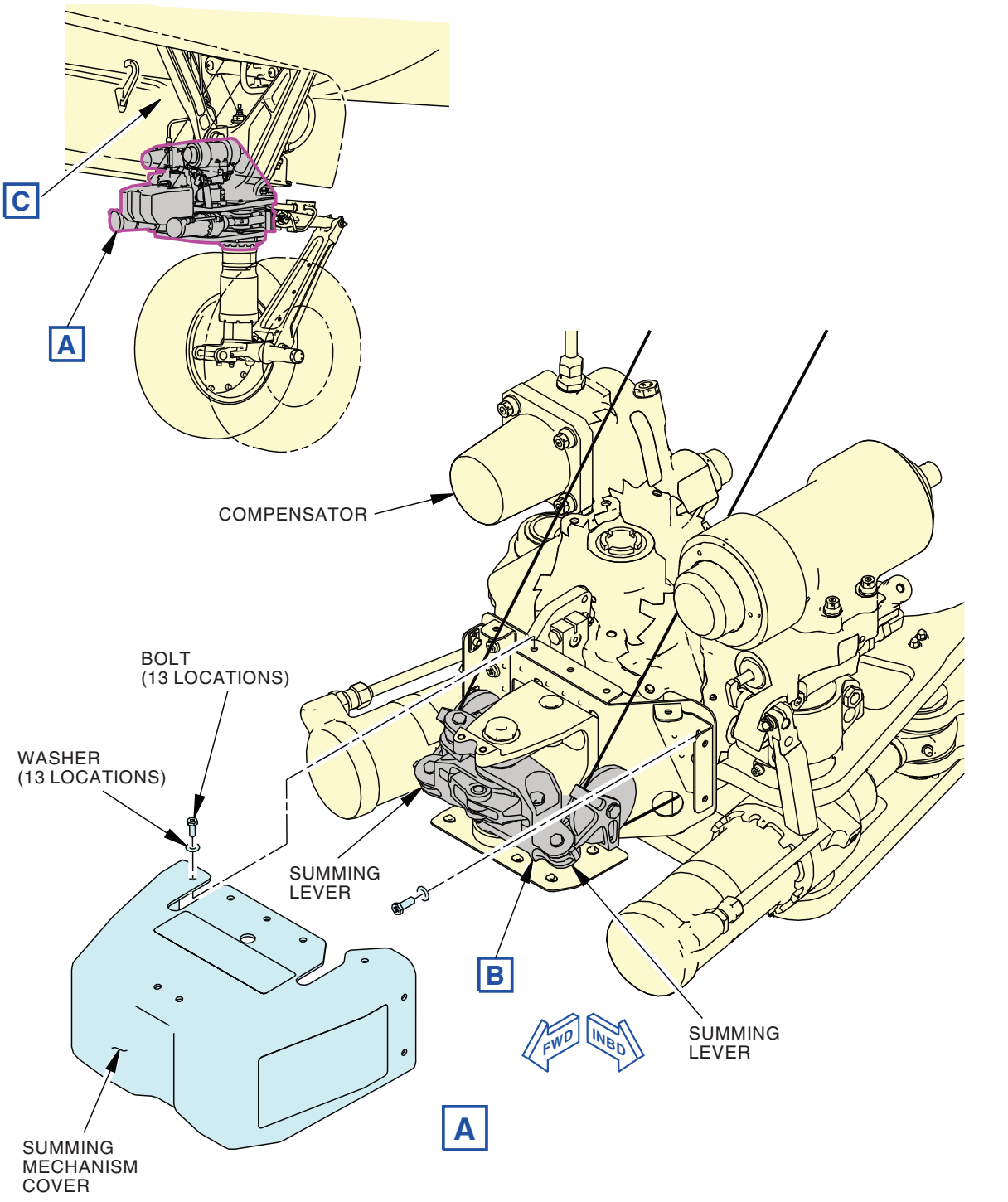
G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-21-00-080-005

(1) Remove chocks from all landing gear tires, if it is necessary.

——— END OF TASK ———

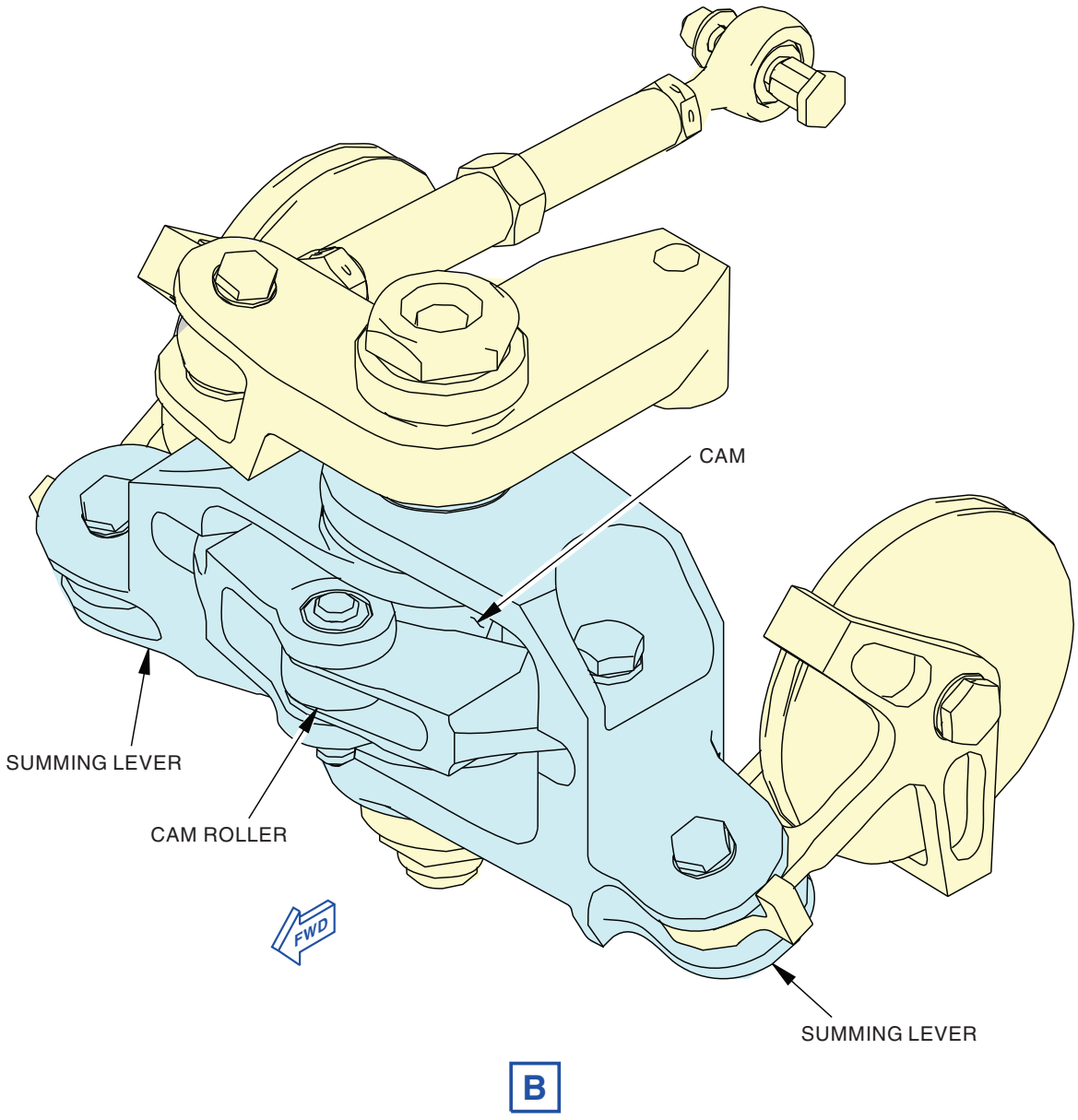
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Nose Landing Gear Steering Broken Cable Compensator Operational Check
Figure 603/32-21-00-990-807 (Sheet 1 of 3)

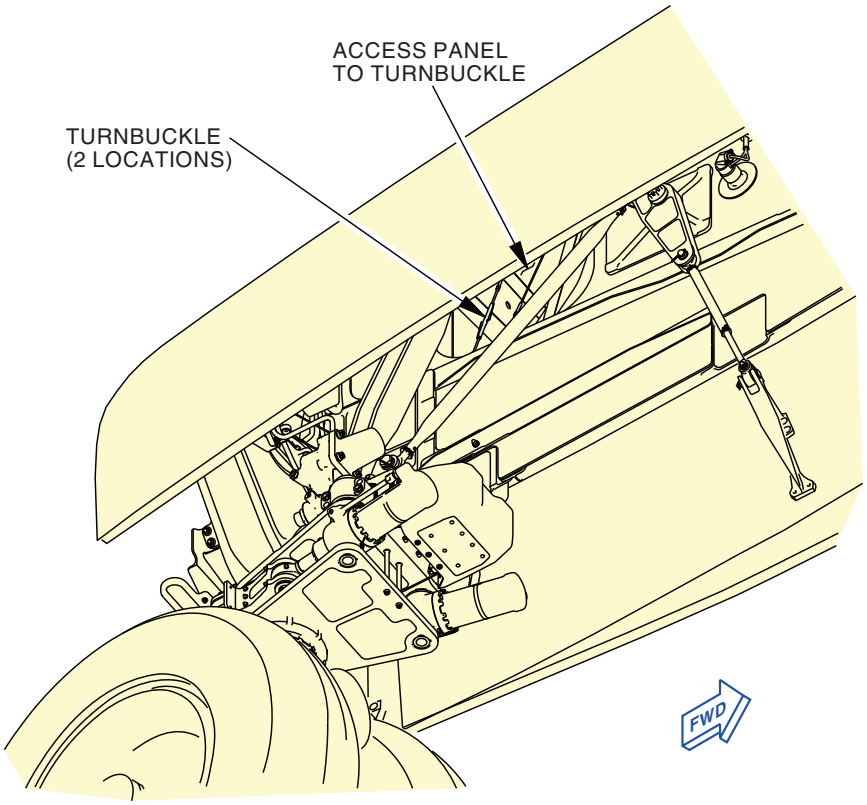
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Nose Landing Gear Steering Broken Cable Compensator Operational Check
Figure 603/32-21-00-990-807 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



C

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Nose Landing Gear Steering Broken Cable Compensator Operational Check
Figure 603/32-21-00-990-807 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE LANDING GEAR - REPAIRS

1. General

A. This procedure contains these tasks:

(1) Repair of the flexible conduit on the nose landing gear.

TASK 32-21-00-300-801

2. Flexible Conduit on the Nose Landing Gear Repair

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
SWPM 20-10-91	Repair of Flexible Electrical conduit

B. Consumable Materials

Reference	Description	Specification
B00067	Alcohol - Denatured, Ethyl (Ethanol)	AMS 3002 (Supersedes O-E-760)
B50073	Alcohol - Isopropyl	ASTM D 770
G00150	Tape - Nitto P-421 NAT (Formerly Permacel) PTFE Film Tape	
G50392	Sleeving - Expandable, Braided (Teflon, PFA)	BMS13-52 Type IV

C. Prepare for the Flexible Conduit on the Nose Landing Gear Repair

SUBTASK 32-21-00-480-010

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-21-00-480-011

(2) Put chocks around the tires of all the landing gear (PAGEBLOCK 10-11-05/201).

D. Flexible Conduit on the Nose Landing Gear Repair

SUBTASK 32-21-00-960-001

(1) To replace the Teflon tape [4], do these steps:

(a) Remove the Teflon tape [4] which extends approximately 12 in. (305 mm) on the flexible conduit [1].

(b) Clean the conduit with alcohol, B00067, or alcohol, B50073.

(c) Wind the Teflon tape [4] (Nitto P-421 tape, G00150) on the area where the flexible conduit [1] touches the bracket [2].

1) Make sure the Teflon tape [4] does not extend more than 3 in. (76 mm) from the bracket [2].

(d) Apply a second layer of Teflon tape [4] which winds at the opposite start/end.

(e) Attach a string at the two ends of the Teflon tape [4].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-21-00-300-001

(2) Use these steps to find the necessary repair if the flexible conduit [1] is damaged:

(a) Clean the flexible conduit [1] with alcohol, B00067, or alcohol, B50073.

(b) Examine the worn area.

1) Replace the flexible conduit [1] if damage extends through the outer silicone rubber, through the stainless steel and into the Teflon tubing at the next maintenance opportunity.

2) Replace or repair the flexible conduit [1] if there are worn or broken strands to the stainless steel braid.

3) If there are tears or cuts which do not extend through the outer silicone rubber layer, no repair is necessary.

SUBTASK 32-21-00-300-002

(3) Follow these steps if a repair of the flexible conduit [1] is necessary (SWPM 20-10-91):

NOTE: A protective outer sleeve, G50392, with a 0.75 in. (19.05 mm) diameter and natural color can be installed on the flexible conduit [1] at this time if desired.

(a) Cut frayed material from the damaged area of the flexible conduit [1] to prevent damage to the inner conduit layers.

(b) Wind the area with Teflon tape [4] (Nitto P-421 tape, G00150) with a minimum of three wrappings.

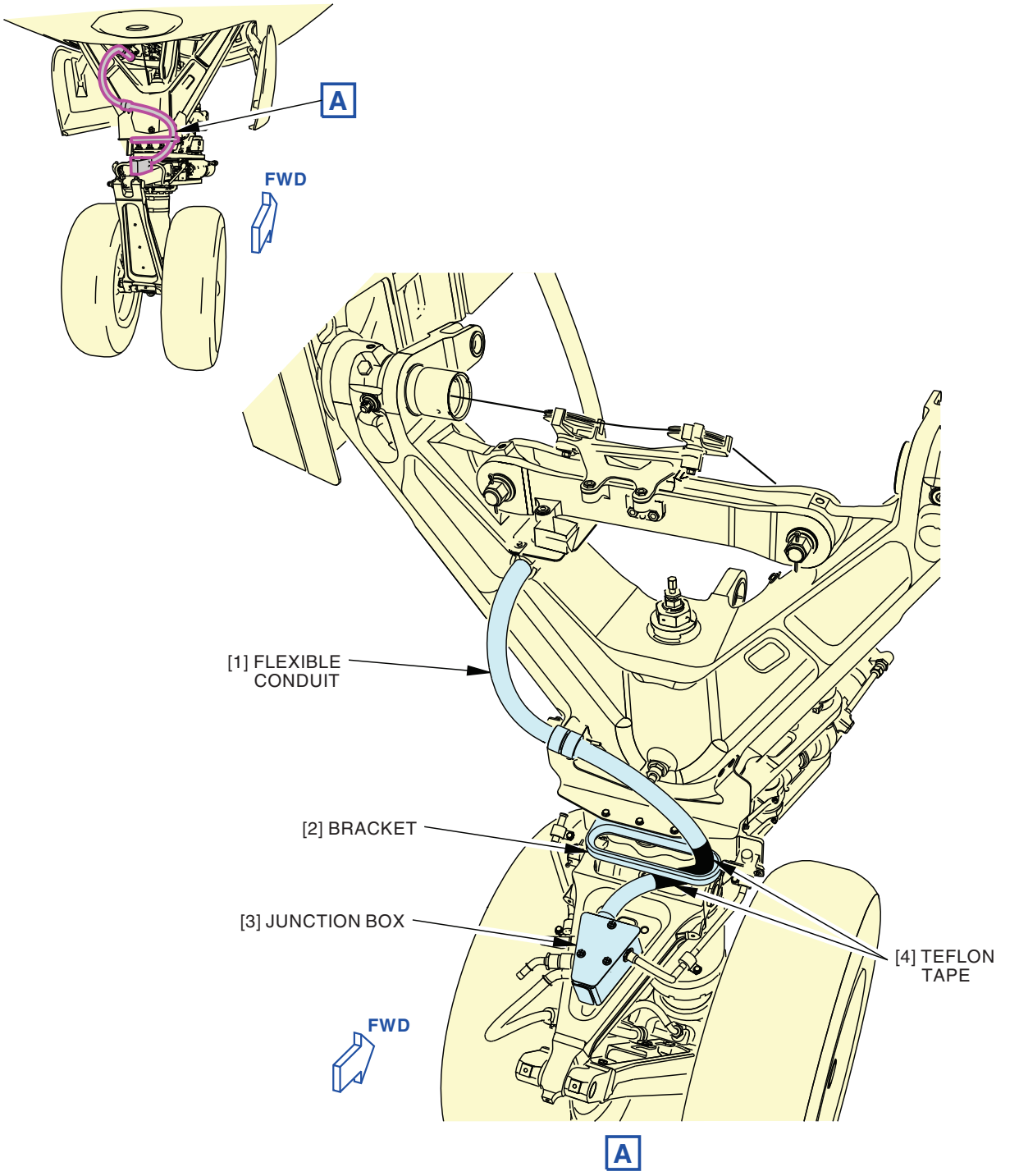
(c) Wind each layer in the opposite direction of the previous layer.

(d) Extend the Teflon tape [4] wrapping 1 in. (25 mm) or more from each end of the worn or damaged area.

(e) Attach a string around the two ends of the Teflon tape [4].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Gear Flexible Conduit
Figure 801/32-21-00-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE LANDING GEAR SHOCK STRUT SEALS - REPAIRS

1. General

A. This procedure has these tasks:

(1) A replacement of the active seals with the spare seals on the lower bearing carrier. The inner cylinder is partly removed from the outer cylinder, the leaky, active seals are removed, and spare seals are installed.

(2) An installation of new dynamic seals, new static seals, and new spare seals on the lower bearing carrier. The inner cylinder is fully removed from the outer cylinder, the lower bearing carrier is removed, and all of the seals are removed and replaced.

(a) This task contains steps to replace the metering pin seal on the inner cylinder.

(3) A replacement of the scraper ring. This task is used when the scraper ring is damaged and inner cylinder removal is not possible.

(4) A check for hydraulic leaks on the shock strut of the nose landing gear.

(5) A check for hydraulic leaks on the shock strut metering pin of the nose landing gear.

B. Make sure that all working areas, tools, and your hands are clean when you handle the elastomer seals. Some greases can cause the elastomer seals to swell too much. This can cause the seals to leak.

C. It is acceptable to install the spare dynamic seal in place of the active dynamic seal (cap seal assembly) to reduce leakage past the shock strut dynamic seal in cold weather. Re-install the cap seal assembly when warm weather returns.

NOTE: If the airplane will be operated for an extended period of time with the spare dynamic seal installed in the active dynamic seal position, it is recommended to keep a spare dynamic seal in the spare dynamic seal position in case the active seal develops a leak and needs to be replaced.

A daily cleaning of the inner cylinder chrome is recommended, when operating the airplane with the spare dynamic seal in the active position (PAGEBLOCK 32-00-10/701). The cold weather dynamic seal is installed on all current production airplanes and has helped reduce this issue.

TASK 32-21-11-960-801

2. Active Seals with the Spare Seals Replacement

(Figure 801, Figure 802, Figure 803, Figure 804)

A. General

(1) This task replaces the active static seals and the active dynamic seals with the spare seals on the lower bearing carrier of the nose landing gear.

(2) For this task you will partly disassemble the nose gear strut inner cylinder from the outer cylinder, pull the lower bearing carrier out of the outer cylinder and replace the active seals with the spare seals that are already installed.

(3) It is recommended that the spare seals only be used to replace the active seals if there are no replacement seals available or if the inner cylinder cannot be removed from the outer cylinder.

B. References

Reference	Title
07-10-03-580-803	Lift the Nose Landing Gear Axle With Axle Jacks-Handling (P/B 201)

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(Continued)

Reference	Title
07-10-06-580-803	Lower the Airplane Nose (P/B 201)
07-10-21-580-801	Lift the Airplane Nose with the Nose Jack at Jack Point D (P/B 201)
07-10-21-580-802	Lower the Airplane Nose, Jack Point D (P/B 201)
12-15-41-610-802	Nose Landing Gear Shock Strut Servicing, Airplane on the Ground (P/B 301)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-21-31-000-803	Nose Landing Gear Torsion Link Disconnection (P/B 401)
32-21-31-400-803	Nose Landing Gear Torsion Link Connection (P/B 401)
32-45-21-000-801	Nose Landing Gear Wheel and Tire Assembly Removal (P/B 401)
32-45-21-400-801	Nose Landing Gear Wheel and Tire Assembly Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1376	Scraper Plunger, C32016-2 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-1377	Scraper Reducer, C32016-3 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-1378	Scraper Retainer, C32016-4 or -67 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-1379	Scraper Lock Assy, C32016-5 or -68 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-1381	Dynamic Seal Stuffer, C32016-7 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-1829	Oil Drain Valve - Shock Strut, LG Part #: J32108-16 Supplier: 81205 Opt Part #: A32066-1 Supplier: 81205
SPL-1867	Wrench - Gland Nut, Shock Strut NLG Part #: C32025-7 Supplier: 81205 Opt Part #: C32025-1 Supplier: 81205

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(Continued)

Reference	Description
SPL-4047	Seal Puller Assy, C32016-6 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-12809	Strap Assembly, C32016-59 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
STD-1110	Container - Hydraulic Fluid Resistant, 5 Gallon (19 Liter)

D. Consumable Materials

Reference	Description	Specification
A00226	Compound - Tamper-Proof Putty	BMS8-45
D00467	Fluid - Landing Gear Shock Strut	BMS3-32 Type II
D00589	Lubricant - Hydraulic Assembly - AFS-682	
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

E. Location Zones

Zone	Area
711	Nose Landing Gear Door - Left

F. Prepare for the Replacement

SUBTASK 32-21-11-420-040

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-21-11-860-001

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

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SUBTASK 32-21-11-020-001

WARNING CLEAR THE AREA BELOW THE NOSE BEFORE YOU DEFLATE THE SHOCK STRUT. IF YOU DEFLATE SHOCK STRUT, THE NOSE CAN MOVE DOWN AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(3) Deflate the shock strut [4] for the nose landing gear:

(a) Remove the cap [1] for the gas charging valve [2], that is on the top of the shock strut [4].

WARNING DO NOT REMOVE THE VALVE BODY UNTIL YOU DEFLATE THE SHOCK STRUT FULLY. THE AIR PRESSURE CAN BLOW THE VALVE BODY OUT AND CAUSE INJURIES TO PERSONNEL.

(b) Loosen the swivel nut [3] a maximum of two turns.

NOTE: Fluid in the shock strut can have bubbles when you release the pressure.

1) Deflate the shock strut [4] slowly to prevent the leakage of the fluid through the gas charging valve [2].

(c) Loosen the swivel nut [3] fully when all of the pressure in the shock strut [4] is released.

NOTE: The shock strut is fully deflated when the dimension "X" is 13.8 in. (35.1 cm).

SUBTASK 32-21-11-680-001

(4) Drain the hydraulic fluid from the shock strut [4]:

(a) Remove the cap [5] from the oil charging valve [6].

(b) Put a 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110, in a position to catch the hydraulic fluid when the oil charging valve [6] is opened.

(c) Install the drain equipment on the oil charging valve [6]:

NOTE: The landing gear shock strut drain valve, SPL-1829, could also be used, but it will take much longer to drain the shock strut.

1) Cut a length of plastic tubing, sufficiently long to reach the container on the floor.

2) Insert a small allen wrench in the end of the length of tubing so that the long end of the allen wrench is flush with the end of the tube and the short end goes through the wall of the tube.

3) Install the tubing on the oil charging valve [6] so that the allen wrench goes into the check valve and holds it open to drain the hydraulic fluid.

(d) Remove the drain equipment when you have removed all of the hydraulic fluid.

SUBTASK 32-21-11-020-002

(5) Do this task: Nose Landing Gear Torsion Link Disconnection, TASK 32-21-31-000-803.

SUBTASK 32-21-11-480-002

(6) Lift the nose of the airplane until there is more than enough room to install the seal puller assembly [73] (seal puller assembly, C32016-6, SPL-4047) on the inner cylinder [12] after the gland nut [17] has been removed (TASK 07-10-21-580-801).

SUBTASK 32-21-11-580-001

(7) Use axle jack to hold the inner cylinder [12] in its position in the outer cylinder [9] (TASK 07-10-03-580-803).

SUBTASK 32-21-11-020-003

(8) Do this task: Nose Landing Gear Wheel and Tire Assembly Removal, TASK 32-45-21-000-801.

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SUBTASK 32-21-11-020-004

(9) Disconnect the gland nut [17] from the outer cylinder [9]:

(a) Remove the bolt [14], washer [15], and lockplate [16] to release the gland nut [17].

(b) Use the gland nut wrench, SPL-1867, to disconnect the gland nut [17] from the outer cylinder [9].

1) Keep the gland nut [17] in its position against the outer cylinder [9] after you have disconnected it from the outer cylinder [9].

SUBTASK 32-21-11-580-002

(10) Lift the nose of the airplane until there is enough room to install the scraper retainer [79] (scraper retainer, C32016-4 or -67, SPL-1378) and the scraper lock assembly [74] (scraper lock assembly, C32016-5 or -68, SPL-1379) on the gland nut [17] (TASK 07-10-21-580-801).

SUBTASK 32-21-11-480-003

(11) Install the scraper retainer [79] (scraper retainer, C32016-4 or -67, SPL-1378) and the scraper lock assembly [74] (scraper lock assembly, C32016-5 or -68, SPL-1379) on the gland nut [17].

NOTE: This will make sure that the scraper ring for the gland nut does not come out of its groove.

(a) Install the scraper retainer [79] (scraper retainer, C32016-4 or -67, SPL-1378) on the gland nut [17].

(b) Install the scraper lock assembly [74] (scraper lock assembly, C32016-5 or -68, SPL-1379) to hold the scraper retainer [79] (scraper retainer, C32016-4 or -67, SPL-1378) in its position on the gland nut [17].

SUBTASK 32-21-11-020-005

(12) Move the gland nut [17] down at the inner cylinder [12].

(a) If the scraper ring [24] comes out of its groove, use the scraper plunger [80] (scraper plunger, C32016-2, SPL-1376) and the scraper reducer [81] (scraper reducer, C32016-3, SPL-1377) to install the scraper ring [24] back in its groove.

SUBTASK 32-21-11-020-006

(13) Move the lower bearing carrier [23]:

(a) Loosely install the seal puller assembly [73] (seal puller assembly, C32016-6, SPL-4047) around the inner cylinder [12].

(b) Push the seal puller assembly [73] (seal puller assembly, C32016-6, SPL-4047) up, so that you can install it correctly at the bottom of the outer cylinder [9].

(c) Align the holes in the seal puller assembly [73] (seal puller assembly, C32016-6, SPL-4047) with the holes in the lower bearing carrier [23] for the puller screws [78].

(d) Install the puller screws [78] to hold the seal puller assembly [73] (seal puller assembly, C32016-6, SPL-4047) to the lower bearing carrier [23].

(e) Tighten the retainer screws on each side of the seal puller assembly [73] (seal puller assembly, C32016-6, SPL-4047).

(f) Install the strap assembly [75] (strap assembly, C32016-59, SPL-12809) around the axle and connect the ends to the shoulder bolts on each side of the seal puller assembly [73] (seal puller assembly, C32016-6, SPL-4047).

SUBTASK 32-21-11-580-003

(14) Lift the airplane until the lower bearing carrier [23] is completely out of the outer cylinder [9] (TASK 07-10-21-580-801).

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G. Active Static and Active Dynamic Seal with the Spare Seals Replacement

SUBTASK 32-21-11-900-001

(1) Replace the active static seal [28] with a spare static seal [27] (Figure 801, Figure 802, Figure 803):

NOTE: Some greases can cause the elastomer seals to swell too much. This can cause the seals to leak.

(a) Make sure that your hands are clean when you handle the elastomer seals.

(b) Remove the two backup rings [29] from the active static seal [28].

(c) Make sure that the two backup rings [29] are serviceable.

NOTE: The recommended installation is to replace all the old backup rings with new backup rings. Use the old backup rings only after a full inspection, and new rings are not available.

1) Replace the two backup rings [29] if they have nicks, scratches or other damage.

(d) Use a plastic cutting tool with a backup strip to cut the active static seal [28] from its outer groove in the lower bearing carrier [23].

(e) Move the spare static seal [27] out of the spare seal groove in the lower bearing carrier [23].

CAUTION

DO NOT APPLY VASELINE OR PETROLATUM TO THE SEALS. THIS CAN CAUSE THE SEAL MATERIAL TO BECOME LARGER. DAMAGE TO THE SEALS CAN OCCUR.

(f) Wipe the spare static seal [27] and the backup rings [29] to remove unwanted hydraulic fluid and assembly lubricant.

(g) Put the spare static seal [27] in the static seal groove in the lower bearing carrier [23].

(h) Put the white backup rings [29] on the spare static seal [27] with the rounded edge of the backup rings [29] against the seal.

SUBTASK 32-21-11-020-007

(2) Remove the seal retainer [18]:

(a) Remove the bearing carrier pins [30] for the seal retainer [18].

(b) Move the seal retainer [18] away from the lower bearing carrier [23] to get access to the elastomer ring [19].

(c) Use tape to hold the seal retainer [18] in position on the inner cylinder [12], and to prevent it from moving down on the lower bearing carrier [23].

SUBTASK 32-21-11-900-002

(3) Replace the elastomer ring [19] with the spare dynamic seal [26]:

(a) Make sure that your hands are clean when you handle the elastomer seals.

NOTE: Some greases can cause the elastomer seals to swell too much. This can cause the seals to leak.

(b) Move the lower bearing carrier [23] up and down the inner cylinder [12] to move the elastomer ring [19], backup ring [21], and cap seal [20] from the inner groove of the lower bearing carrier [23].

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(c) If the elastomer ring [19] does not come out of the inner groove of the lower bearing carrier [23], remove it with a tool that is not sharp.

NOTE: Hardened steel cotter pin removal tools work well for removing the dynamic seal.

1) Remove the backup ring [21], and cap seal [20] from the elastomer ring [19] and discard.

NOTE: Use a plastic cutting tool to cut the cap seal.

2) Use a plastic cutting tool with a back-up strip to cut the elastomer ring [19] from the inner groove of the lower bearing carrier [23] and discard.

(d) Remove the four backup rings [22] from the spare dynamic seal groove and retain for installation.

(e) Remove the spare dynamic seal [26] from its groove in the lower bearing carrier [23] and move it down on the inner cylinder [12].

NOTE: The round sealing surface of the seal must be against the chrome on the inner cylinder.

CAUTION

DO NOT APPLY VASELINE OR PETROLATUM TO THE SEALS. THIS CAN CAUSE THE SEAL MATERIAL TO BECOME LARGER. DAMAGE TO THE SEALS CAN OCCUR.

(f) Apply a thin layer of fluid, D00467, or AFS-682 lubricant, D00589, to the spare dynamic seal [26] and the backup rings [22].

NOTE: AFS-682 lubricant, D00589, will help keep the backup rings in their positions.

(g) Put the white backup ring [22] on the spare dynamic seal [26] with the rounded edge of the backup ring [22] against the spare dynamic seal [26].

(h) Put one brown backup ring [22] on the outside of each white backup ring [22].

1) Put the split in the brown backup rings [22] in the opposite positions on the diameter from the split in the adjacent white backup rings [22].

CAUTION

MAKE SURE THAT THE BACKUP RINGS AND THE SEAL STAY TOGETHER WHEN YOU INSTALL THE SEAL. IF THE BACKUP RINGS AND THE SEAL MOVE APART, THEY WILL NOT INSTALL CORRECTLY. THIS CAN CAUSE DAMAGE TO THE SEAL AND THE BACKUP RINGS.

(i) Use the dynamic seal stuffer [82] (dynamic seal stuffer, C32016-7, SPL-1381) to install the spare dynamic seal [26], and the backup rings [22] in the active seal location in the lower bearing carrier [23] (View F-F, Figure 802).

(j) Put the seal retainer [18] in its position on the lower bearing carrier [23].

(k) Make sure that the seal retainer [18] is flush to the lower bearing carrier [23] all around the circumference before installing the bearing carrier pins [30].

NOTE: If the seal retainer [18] is not flush, the spare dynamic seal [26] and backup rings [22] may not be installed correctly.

(l) If it is necessary, apply a thin layer of AFS-682 lubricant, D00589, to the bearing carrier pins [30] to hold them in position while the lower bearing carrier [23] is re-installed into the outer cylinder [9].

NOTE: AFS-682 lubricant, D00589, will help keep the bearing carrier pin in position. The material is a very thick, sticky material and heavy quantities can cause the lower bearing carrier to be difficult to re-install.

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1) Use the minimum quantity of AFS-682 lubricant, D00589, necessary to retain the bearing carrier pins [30].

2) Remove excess AFS-682 lubricant, D00589.

CAUTION

DO NOT USE FORCE TO INSTALL THE RETENTION PINS INTO THEIR POSITIONS. THE PINS WILL INSTALL FREELY. TOO MUCH FORCE CAN CAUSE DAMAGE TO THE SEAL RETAINER.

(m) Install the bearing carrier pins [30] to retain the seal retainer [18] to the lower bearing carrier [23].

1) If the bearing carrier pins [30] are not free during installation, then make sure that the seal retainer [18] is flush to the lower bearing carrier [23], then reinsert the bearing carrier pins [30].

NOTE: If the seal retainer is not flush, the spare dynamic seal and backup rings may not have been correctly installed.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-21-11-420-001

(1) Move the lower bearing carrier [23] up the inner cylinder [12] into the outer cylinder [9].

SUBTASK 32-21-11-640-001

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(2) Apply a thin layer of corrosion inhibiting material, G50136, to the threads of the gland nut [17] and to the threads of the outer cylinder [9].

SUBTASK 32-21-11-420-002

(3) Install the gland nut [17]:

(a) Move the gland nut [17] up to the outer cylinder [9] and engage the threads.

(b) Use the gland nut wrench, SPL-1867, to tighten the gland nut [17] to 100 ft-lb (136 N·m) - 125 ft-lb (169 N·m).

(c) Remove excess corrosion inhibiting compound.

SUBTASK 32-21-11-420-003

(4) Install the lockplate [16] for the gland nut [17] (Figure 802):

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin layer of corrosion inhibiting material, G50136, to the shank and threads of bolts [14] and washers [15].

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(b) Install the lockplate [16] in the holes that will put the tab of the lockplate [16] as close to a outer cylinder [9] slot as possible.

1) If it is necessary, loosen the gland nut [17] slightly to align slot on the end of outer cylinder [9] with the lockplate [16] tab.

(c) Install the bolts [14] and washers [15].

(d) Tighten the bolts [14].

(e) Install MS20995C32 lockwire, G01048, on the bolts [14].

SUBTASK 32-21-11-420-004

(5) Install the oil charging valve [6]:

(a) Remove and discard the old packing [7].

(b) Apply a thin layer of fluid, D00467, to a new packing [7].

(c) Install the new packing [7] on the oil charging valve [6].

(d) Install the oil charging valve [6] and tighten it to 22 ft-lb (30 N·m) - 25 ft-lb (34 N·m).

(e) Install the cap [5] loosely.

NOTE: You will tighten it in the shock strut servicing task.

(f) Apply compound, A00226, to the oil charging valve [6].

NOTE: This will make sure that the seal made by compound, A00226, will break if the oil charging valve becomes loose.

SUBTASK 32-21-11-410-007

(6) Do this task: Nose Landing Gear Wheel and Tire Assembly Installation, TASK 32-45-21-400-801.

SUBTASK 32-21-11-080-001

(7) Lower and remove the axle jack (TASK 07-10-06-580-803).

SUBTASK 32-21-11-080-002

(8) Lower the airplane nose and remove the jacks (TASK 07-10-21-580-802).

SUBTASK 32-21-11-420-005

(9) Do this task: Nose Landing Gear Torsion Link Connection, TASK 32-21-31-400-803.

SUBTASK 32-21-11-600-001

(10) Do this task: Nose Landing Gear Shock Strut Servicing, Airplane on the Ground, TASK 12-15-41-610-802.

SUBTASK 32-21-11-420-006

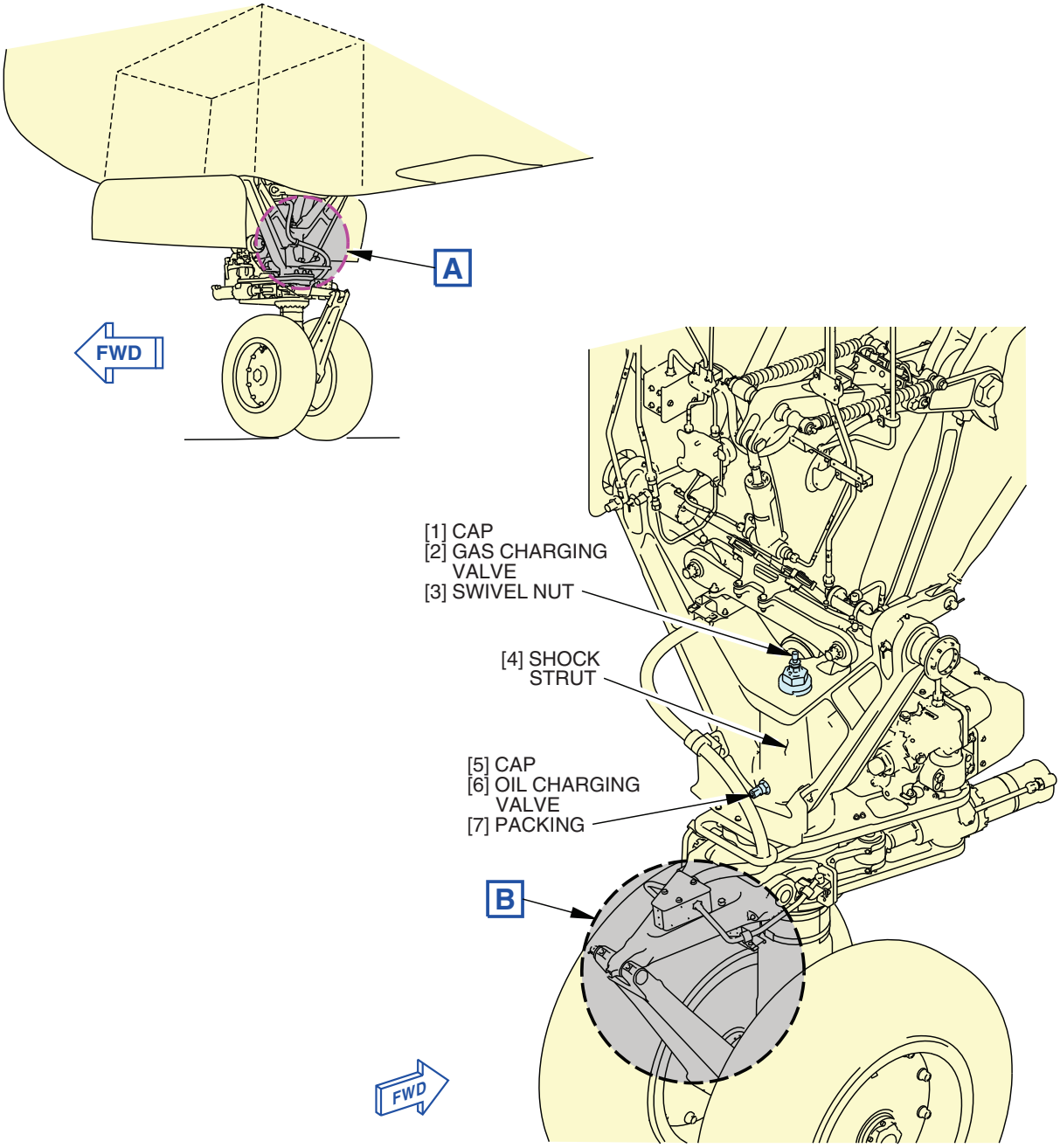
(11) Make sure that the swivel nut [3] is tightened to 6 ±1 ft-lb (8 ±1 N·m).

SUBTASK 32-21-11-420-026

(12) Make sure that the cap [1] is installed for the gas charging valve [2].

——— END OF TASK ———

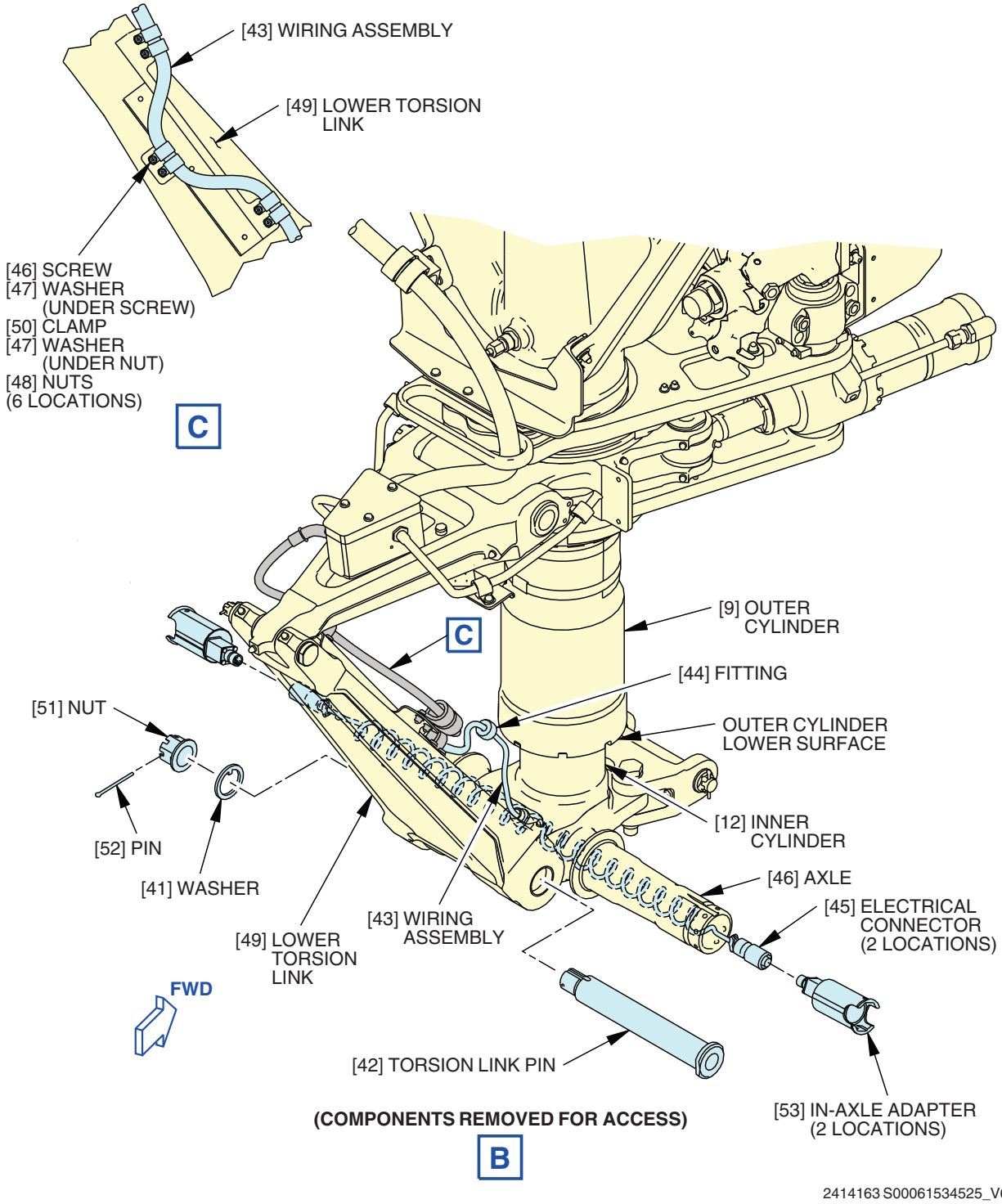
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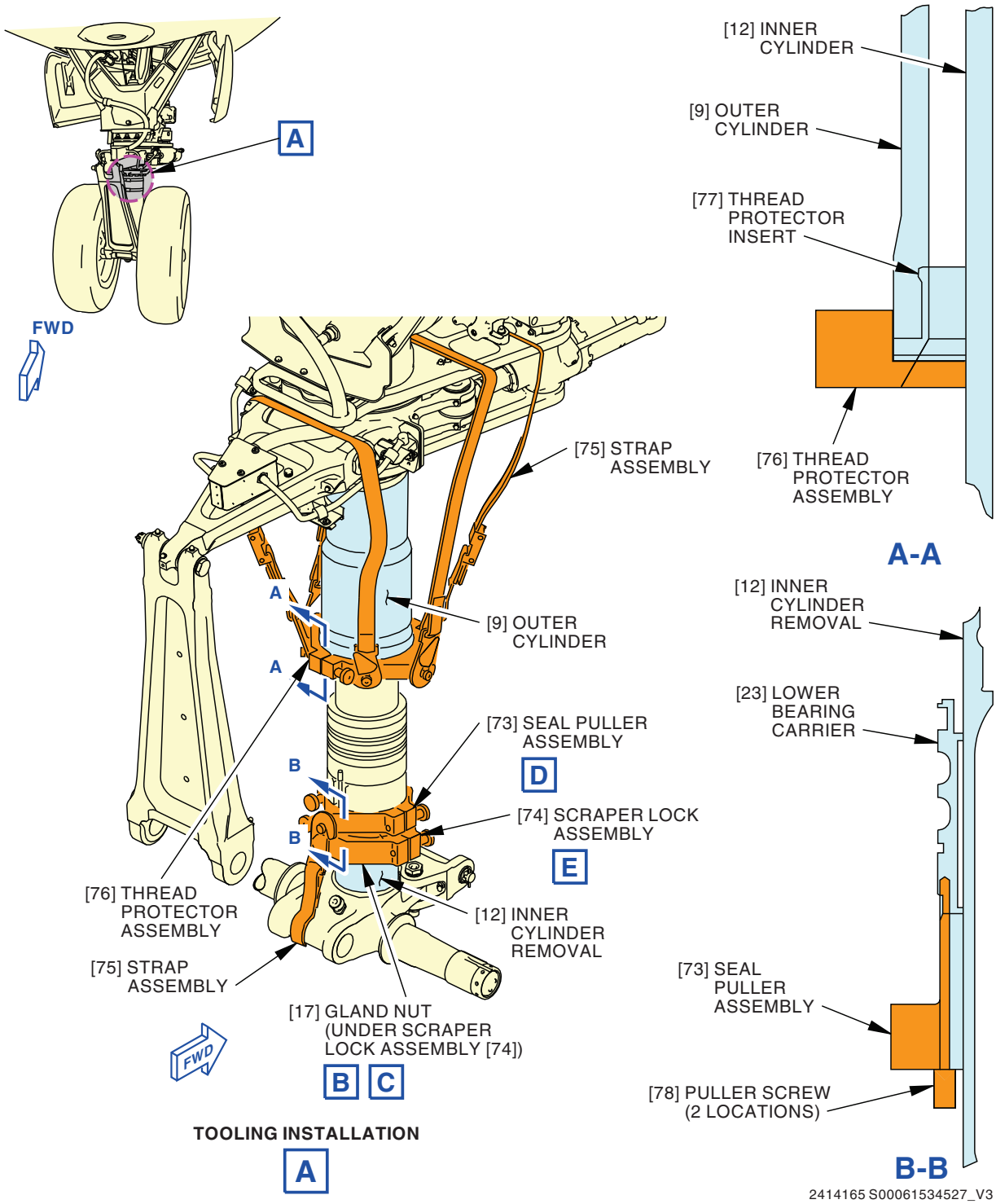
Nose Landing Gear Shock Strut Seal Replacement
Figure 801/32-21-11-990-801 (Sheet 1 of 2)

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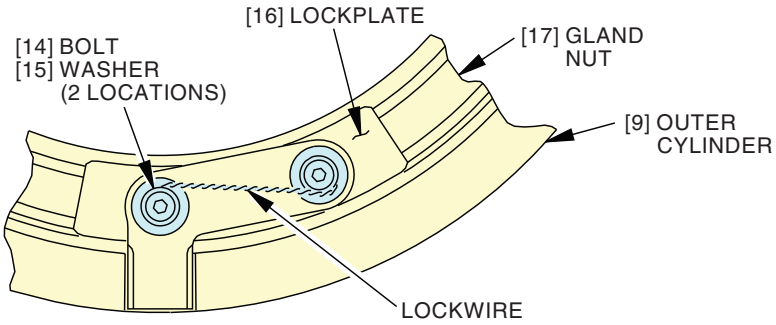
Nose Landing Gear Shock Strut Seal Replacement
Figure 801/32-21-11-990-801 (Sheet 2 of 2)

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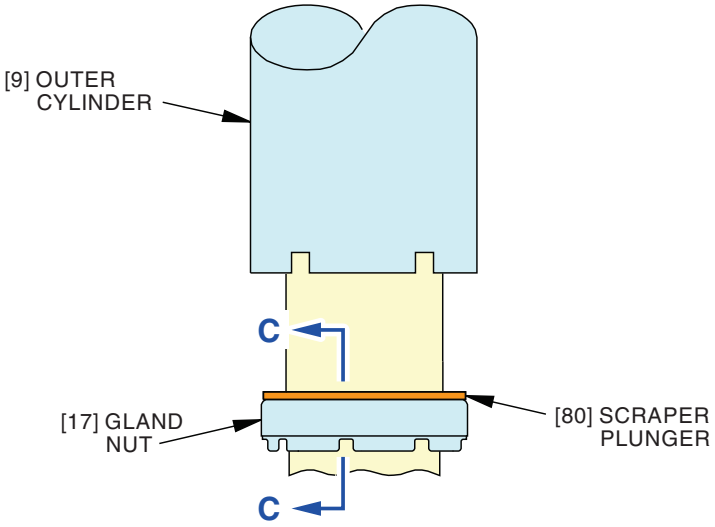


Removal of the Inner Cylinder from the Outer Cylinder
Figure 802/32-21-11-990-802 (Sheet 1 of 7)

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B



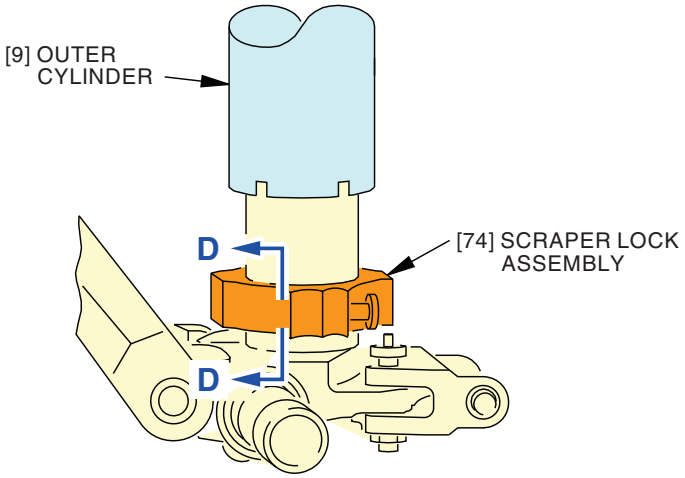
OPTIONAL - USE SCRAPER PLUNGER TO REINSTALL SCRAPER RING [24]

C

2414166 S00061534528_V2

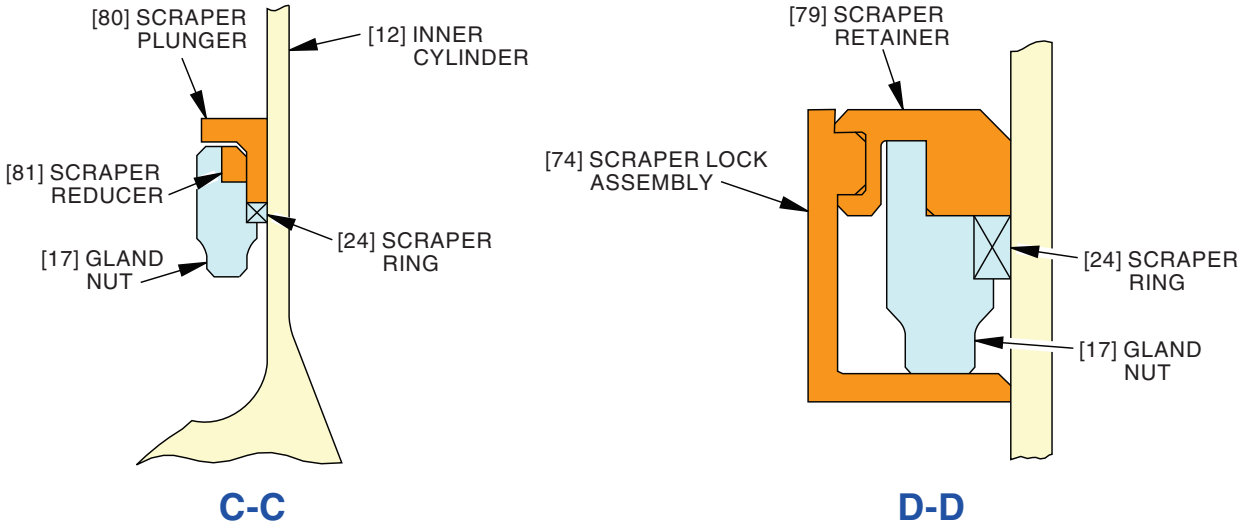
Removal of the Inner Cylinder from the Outer Cylinder
Figure 802/32-21-11-990-802 (Sheet 2 of 7)

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INSTALL THE GLAND NUT SCRAPER RING RETAINER

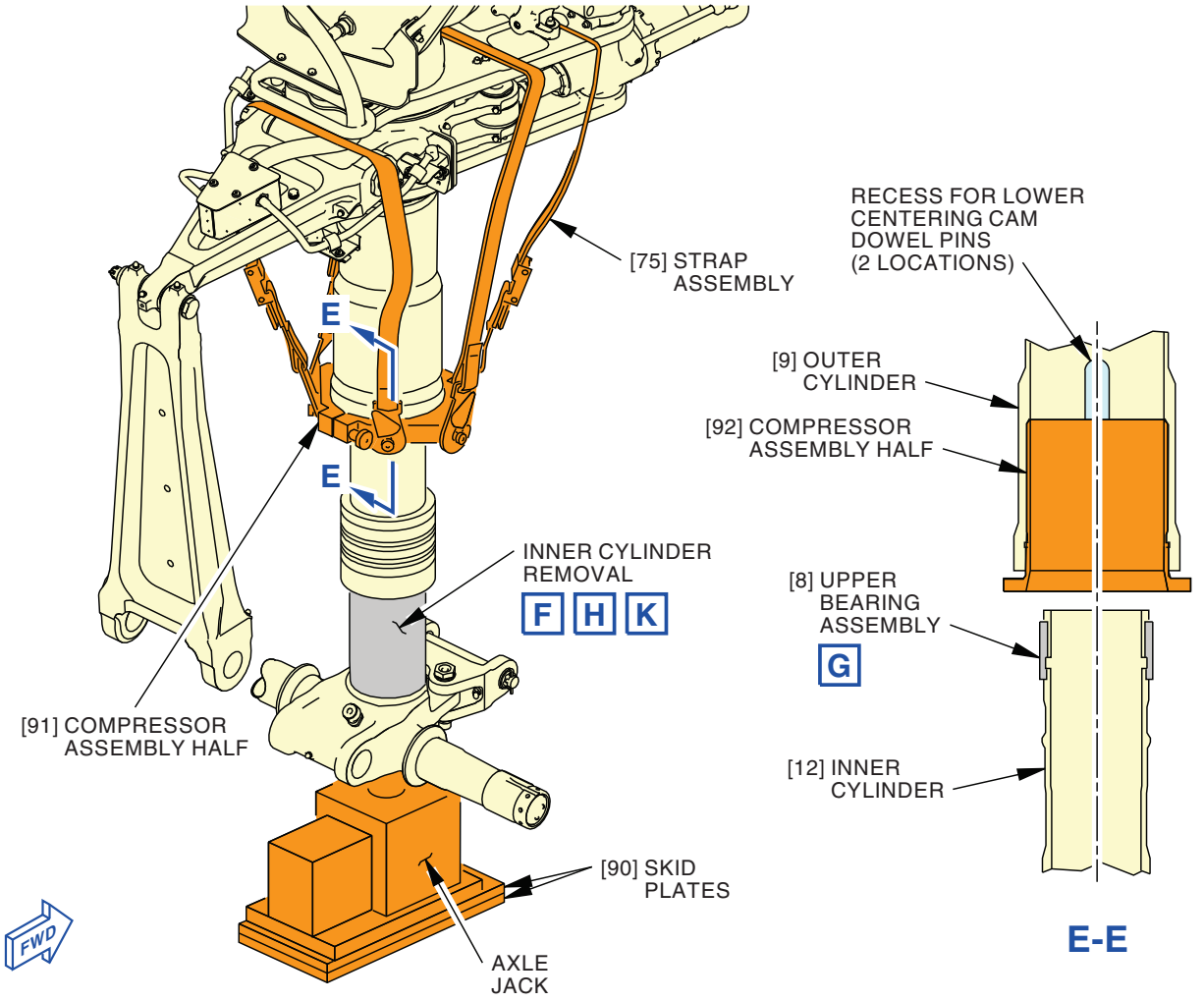
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Removal of the Inner Cylinder from the Outer Cylinder
Figure 802/32-21-11-990-802 (Sheet 3 of 7)

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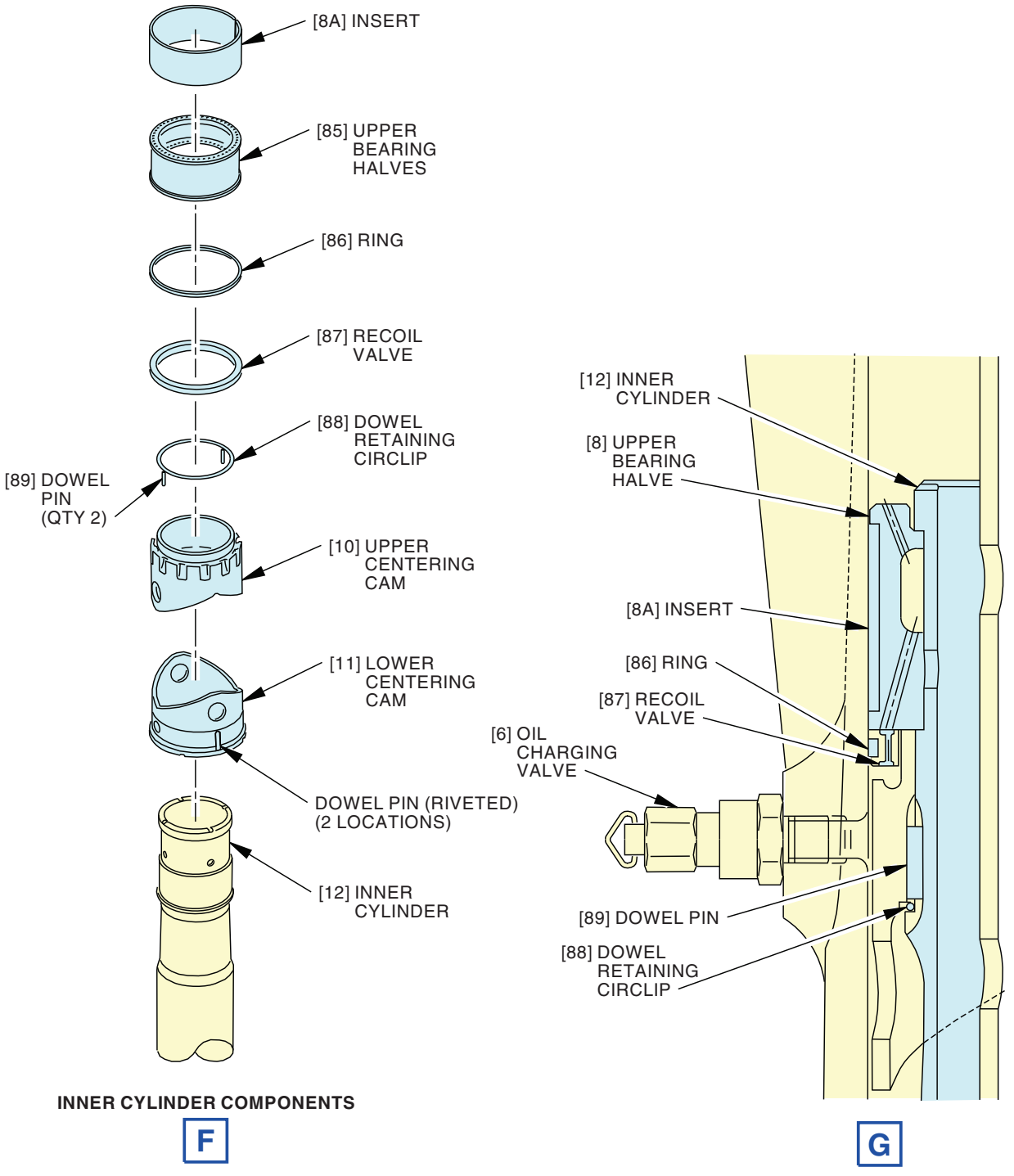
INNER CYLINDER REMOVAL FROM THE OUTER CYLINDER

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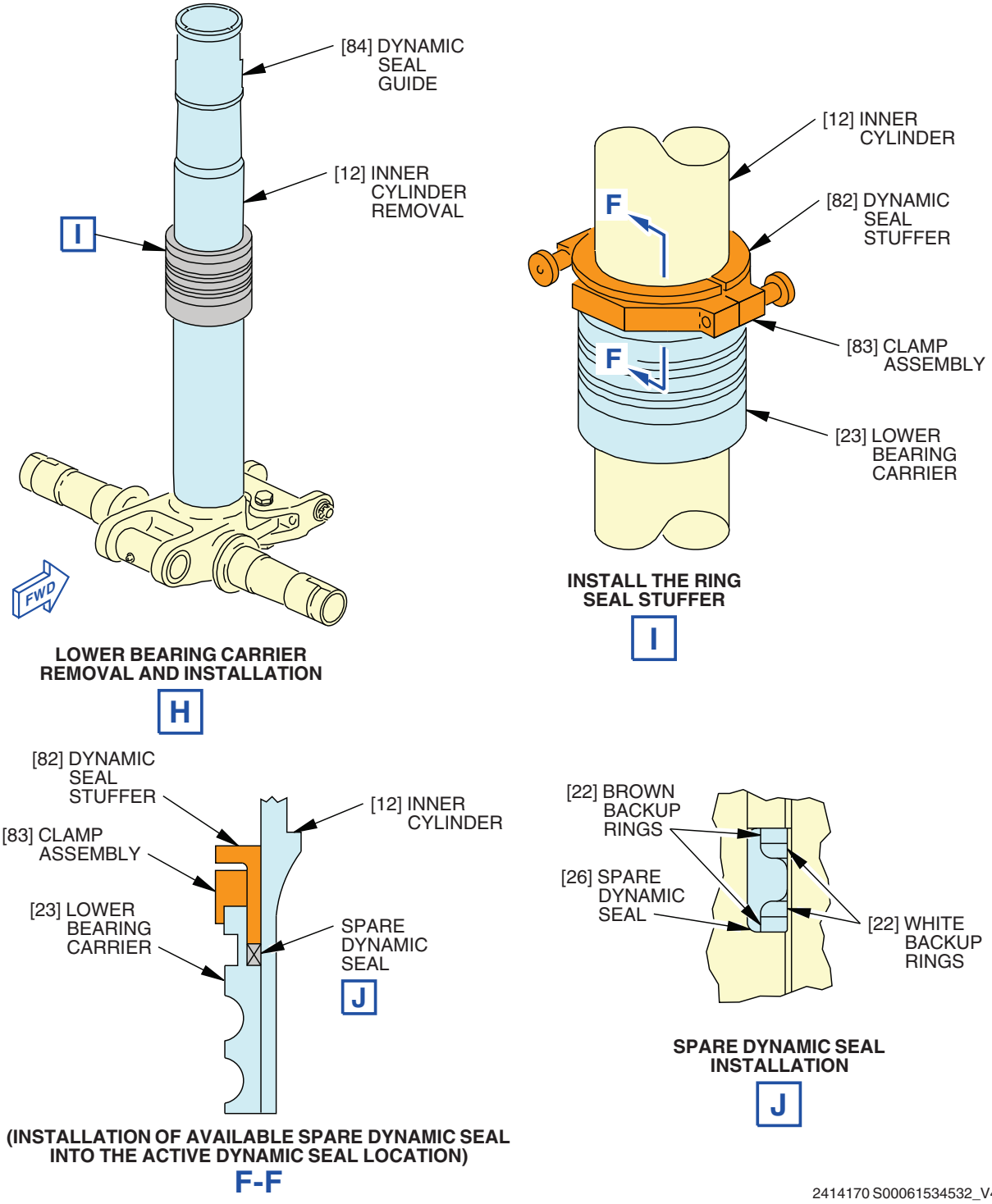
Removal of the Inner Cylinder from the Outer Cylinder
Figure 802/32-21-11-990-802 (Sheet 4 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



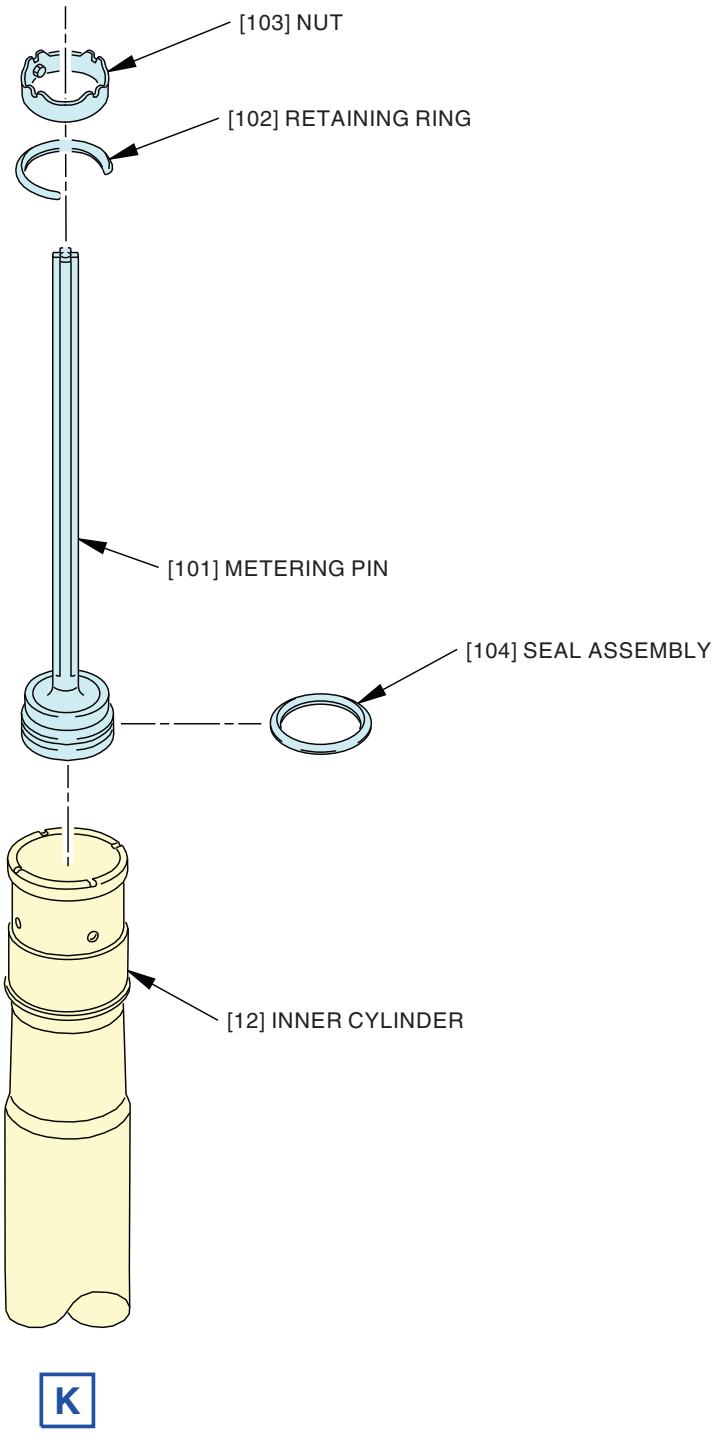
Removal of the Inner Cylinder from the Outer Cylinder
Figure 802/32-21-11-990-802 (Sheet 5 of 7)

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Removal of the Inner Cylinder from the Outer Cylinder
Figure 802/32-21-11-990-802 (Sheet 6 of 7)

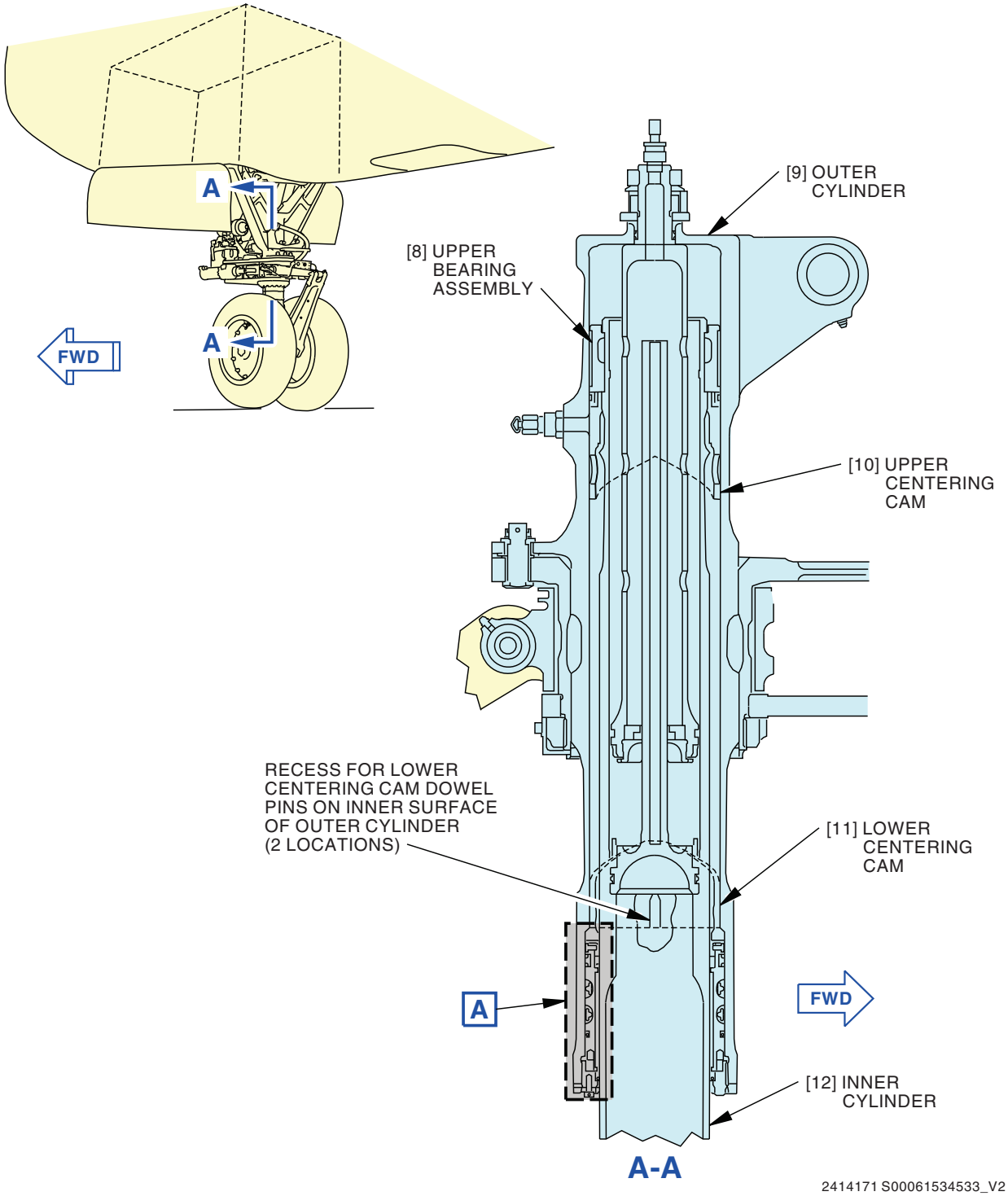
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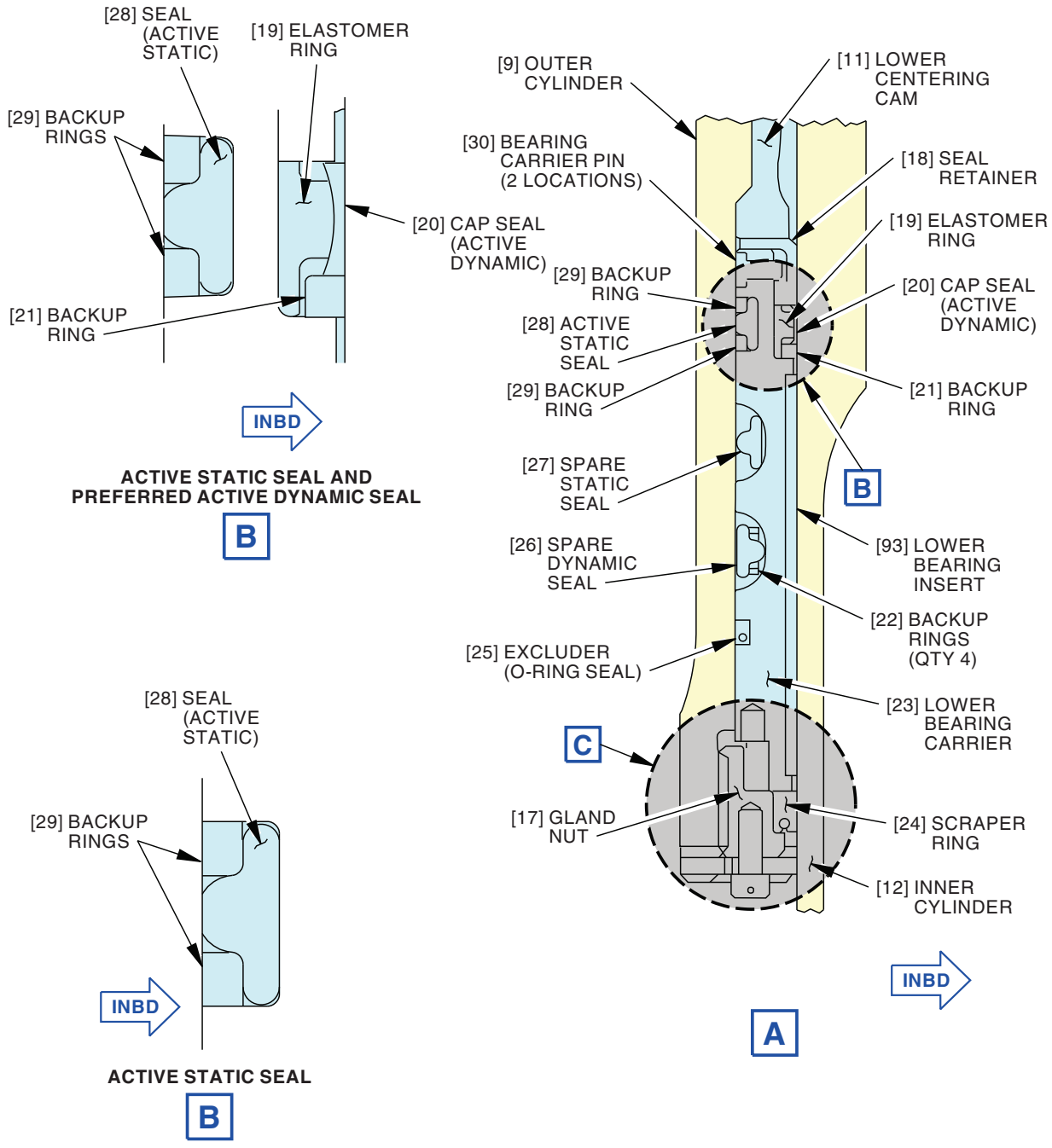
Removal of the Inner Cylinder from the Outer Cylinder
Figure 802/32-21-11-990-802 (Sheet 7 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Lower Bearing Carrier Seal Replacement
Figure 803/32-21-11-990-803 (Sheet 1 of 3)

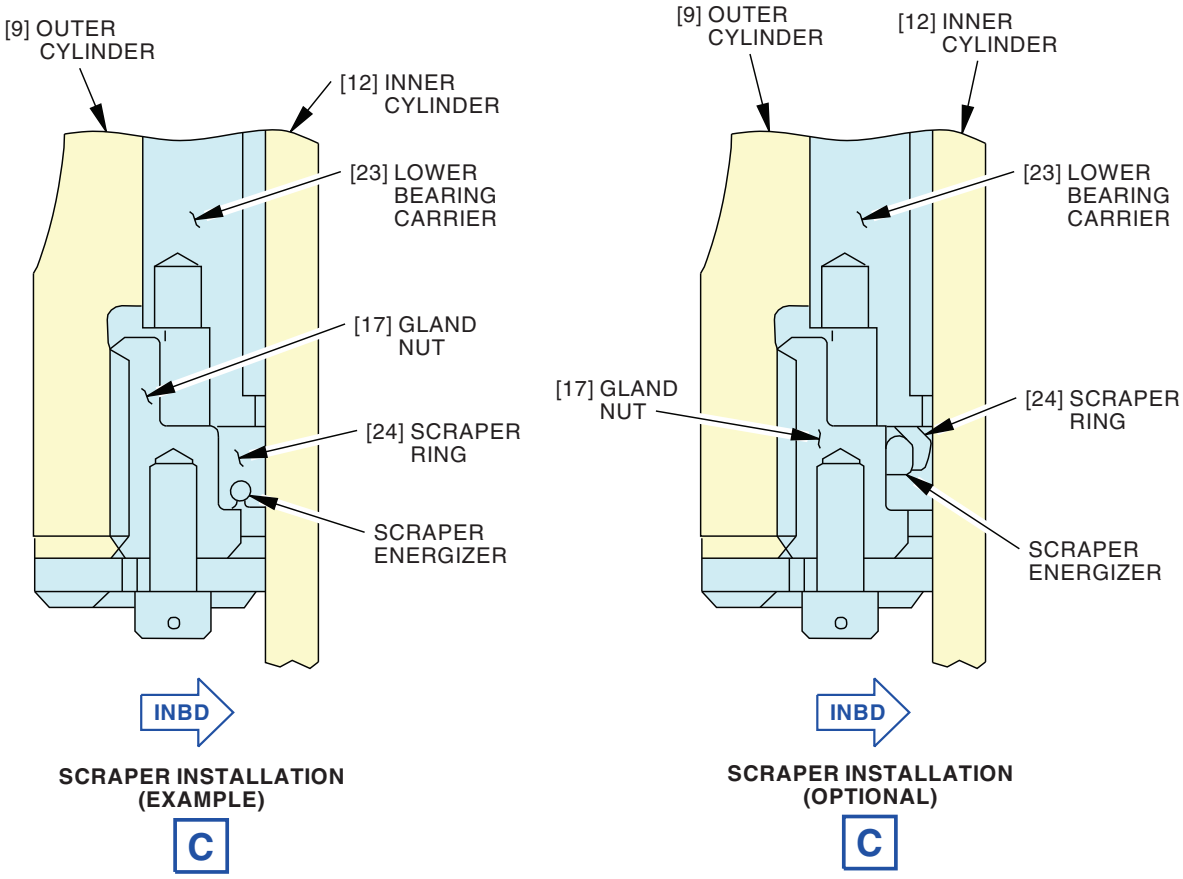
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414172 S00061534534_V3

Lower Bearing Carrier Seal Replacement
Figure 803/32-21-11-990-803 (Sheet 2 of 3)

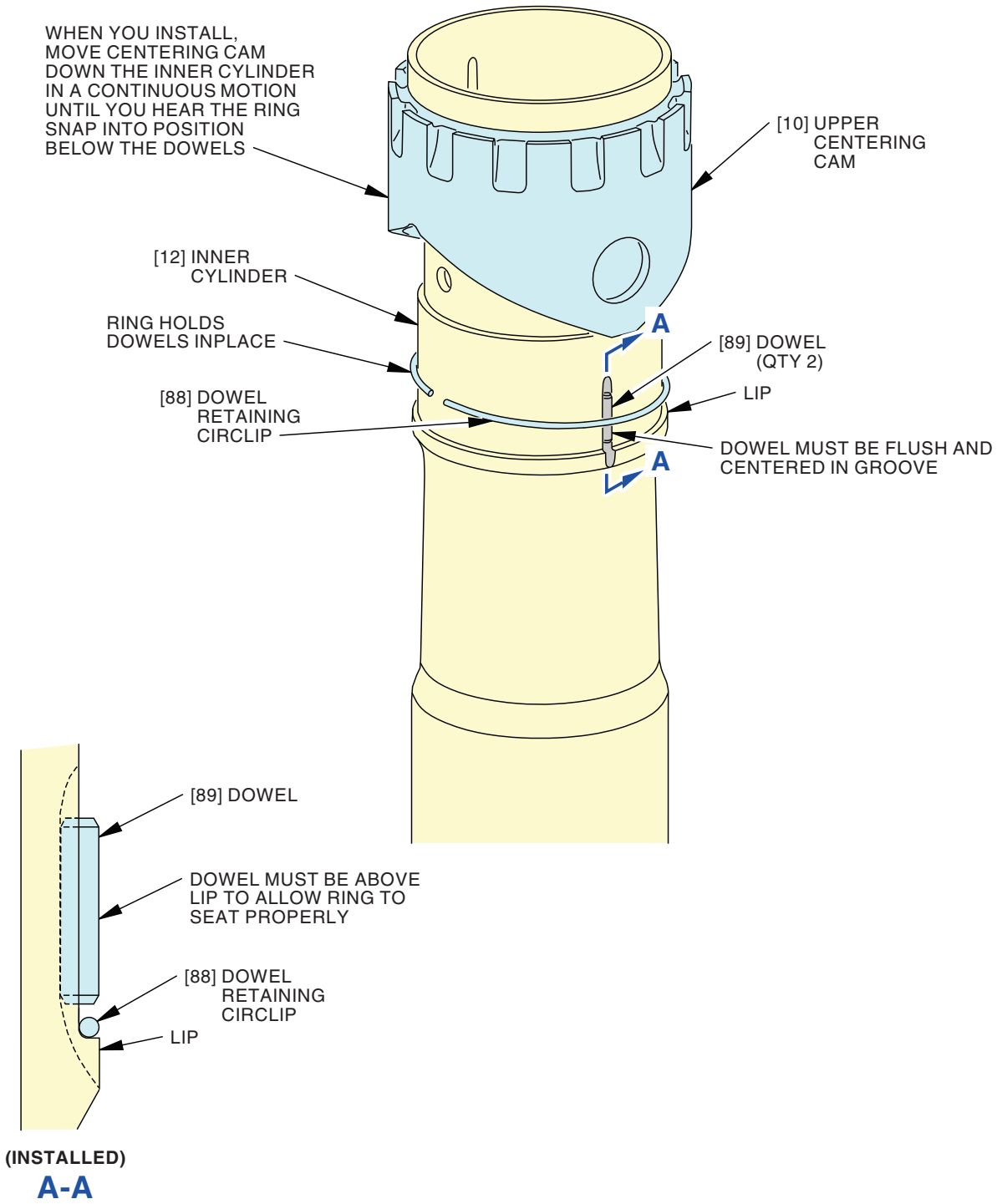
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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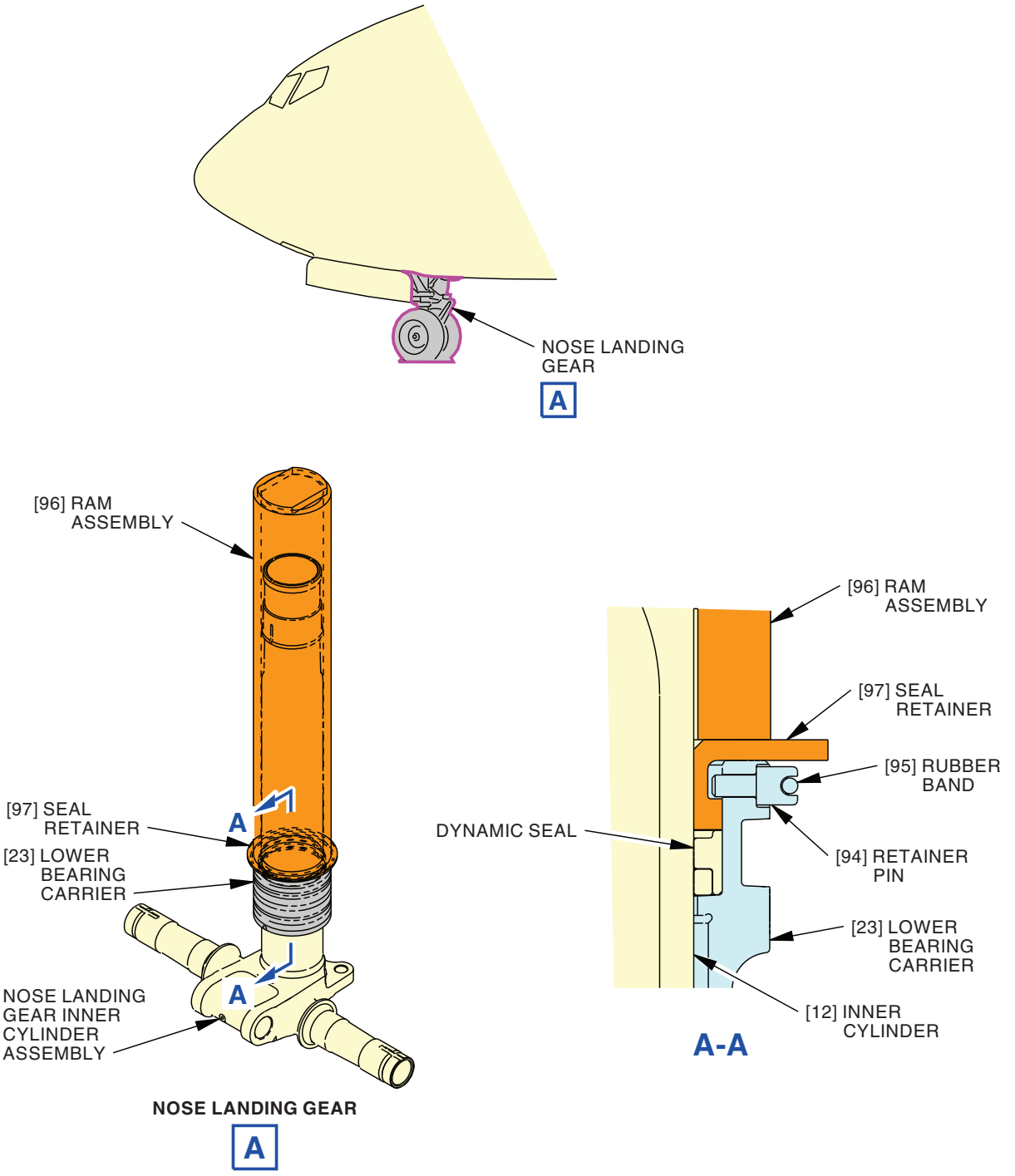
Lower Bearing Carrier Seal Replacement
Figure 803/32-21-11-990-803 (Sheet 3 of 3)

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Upper Centering Cam Replacement
Figure 804/32-21-11-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414175 S00061534537_V4

Dynamic Seal Installation using Optional Tooling
Figure 805/32-21-11-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-21-11-960-802

3. Seals and Spare Seals Replacement

(Figure 801, Figure 802, Figure 803, Figure 804, Figure 805)

A. General

(1) This task provides instructions to replace the active seals and spare seals of the nose landing gear.

NOTE: To complete this task it is necessary to disassemble the inner cylinder from the outer cylinder.

(2) Use this task when there are no spare seals available on the lower bearing carrier to replace the active seals, and you will install a new 3-piece active dynamic seal.

(3) It is recommended that the spare seals only be used to replace the active seals if there are no replacement seals available or if the inner cylinder cannot be removed from the outer cylinder.

B. References

Reference	Title
07-10-03-580-803	Lift the Nose Landing Gear Axle With Axle Jacks-Handling (P/B 201)
07-10-06-580-803	Lower the Airplane Nose (P/B 201)
07-10-21-580-801	Lift the Airplane Nose with the Nose Jack at Jack Point D (P/B 201)
07-10-21-580-802	Lower the Airplane Nose, Jack Point D (P/B 201)
12-15-41-610-802	Nose Landing Gear Shock Strut Servicing, Airplane on the Ground (P/B 301)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-45-21-000-801	Nose Landing Gear Wheel and Tire Assembly Removal (P/B 401)
32-45-21-400-801	Nose Landing Gear Wheel and Tire Assembly Installation (P/B 401)
32-49-03-000-803	Nose Landing Gear In-Axle Adapter Removal (P/B 401)
32-49-03-400-803	Nose Landing Gear In-Axle Adapter Installation (P/B 401)
32-51-00-700-801	Nose Wheel Steering System Test (Airplane on Ground) (P/B 501)
SWPM 20-60-03	Special Protection of Electrical Connectors
SWPM 20-85-12	Assembly of Landing Gear Cables

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1376	Scraper Plunger, C32016-2 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205

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(Continued)

Reference	Description
SPL-1377	Scraper Reducer, C32016-3 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-1378	Scraper Retainer, C32016-4 or -67 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-1379	Scraper Lock Assy, C32016-5 or -68 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-1382	Dynamic Seal Guide, C32016-45 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-1383	Compressor Assy Half, C32016-10 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-1384	Thread Protector Assy, C32016-11 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-1385	Thread Protector Half, C32016-21 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-1829	Oil Drain Valve - Shock Strut, LG Part #: J32108-16 Supplier: 81205 Opt Part #: A32066-1 Supplier: 81205
SPL-1867	Wrench - Gland Nut, Shock Strut NLG Part #: C32025-7 Supplier: 81205 Opt Part #: C32025-1 Supplier: 81205
SPL-1895	Removal/Installation Equipment - Metering Pin and Retainer Ring, NLG Part #: C32035-1 Supplier: 81205
SPL-3927	Strap Assy, C32017-60 (included in C32017 Eqpt) Part #: C32017-67 Supplier: 81205 Opt Part #: C32017-52 Supplier: 81205 Opt Part #: C32017-53 Supplier: 81205
SPL-4047	Seal Puller Assy, C32016-6 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205

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(Continued)

Reference	Description
SPL-12809	Strap Assembly, C32016-59 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-13124	Removal/Installation Equipment - Lower Seals, NLG Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-13125	Retainer Pin, C32016-56 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-13126	Ram Assembly, C32016-47 (included in C32016 kit) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-13129	Rubber Band, C32016-58 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-23426	Seal Retainer, C32016-55 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
STD-1110	Container - Hydraulic Fluid Resistant, 5 Gallon (19 Liter)

D. Consumable Materials

Reference	Description	Specification
A00226	Compound - Tamper-Proof Putty	BMS8-45
D00467	Fluid - Landing Gear Shock Strut	BMS3-32 Type II
D00589	Lubricant - Hydraulic Assembly - AFS-682	
D00633	Grease - Aircraft General Purpose	BMS3-33
D50022	Fluid - Landing Gear Shock Strut (Specifically For Preservation)	BMS3-32 Type I
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	
G50225	Lockwire - MS20995C20, Corrosion Resistant Steel - 0.020 Inch (0.508 mm) Diameter	NASM20995

E. Location Zones

Zone	Area
711	Nose Landing Gear Door - Left

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F. Prepare for the Replacement

SUBTASK 32-21-11-420-041

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-21-11-860-002

(2) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-21-11-020-008

WARNING CLEAR THE AREA BELOW THE NOSE BEFORE YOU DEFLATE THE SHOCK STRUT. IF YOU DEFLATE SHOCK STRUT, THE NOSE CAN MOVE DOWN AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(3) Deflate the shock strut [4] for the nose landing gear, do these steps:

(a) Remove the cap [1] for the gas charging valve [2], that is on the top of the shock strut [4].

WARNING DO NOT REMOVE THE VALVE BODY UNTIL YOU DEFLATE THE SHOCK STRUT FULLY. THE AIR PRESSURE CAN BLOW THE VALVE BODY OUT AND CAUSE INJURIES TO PERSONNEL.

(b) Loosen the swivel nut [3] a maximum of two turns.

NOTE: Fluid in the shock strut can have bubbles when you release the pressure.

1) Deflate the shock strut [4] slowly to prevent the leakage of the fluid through the gas charging valve [2].

(c) Loosen the swivel nut [3] fully when all of the pressure in the shock strut [4] is released.

SUBTASK 32-21-11-680-002

(4) Drain the hydraulic fluid from the shock strut [4], do these steps:

(a) Remove the cap [5] from the oil charging valve [6].

(b) Put a 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110, in a position to catch the hydraulic fluid when the oil charging valve [6] is opened.

(c) Install the drain equipment on the oil charging valve [6], do these steps:

NOTE: The landing gear shock strut drain valve, SPL-1829, could also be used, but it will take much longer to drain the shock strut.

1) Cut a length of plastic tubing, sufficiently long to reach the container on the floor.

2) Insert a small Allen wrench in the end of the length of tubing, so that the long end of the Allen wrench is flush with the end of the tube and short end goes through the wall of the tube.

3) Install the tubing on the oil charging valve [6] so that the Allen wrench goes into the check valve and holds it open to drain the hydraulic fluid.

(d) Remove the drain equipment when you have removed all of the hydraulic fluid.

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SUBTASK 32-21-11-580-004

WARNING DO NOT LIFT THE NOSE OF THE AIRPLANE MORE THAN 84 INCHES (2.13 METERS). THE AIRPLANE WILL NOT BE STABLE IF YOU LIFT THE NOSE MORE THAN 84 INCHES (2.13 METERS). DAMAGE TO EQUIPMENT AND PERSONNEL CAN OCCUR.

(5) Lift the nose of the airplane until the lower surface of the shock strut outer cylinder [9] is approximately 36 in. (914 mm) above the ground (TASK 07-10-21-580-801).

NOTE: If lifting the nose outside in windy conditions, or if other maintenance is being performed at the same time as lifting the airplane nose, you may want to use a tail support to stabilize the airplane.

SUBTASK 32-21-11-010-001

(6) Remove the wheels (TASK 32-45-21-000-801).

SUBTASK 32-21-11-020-032

(7) Disconnect the electrical connectors [45] and wiring assembly [43], do these steps:

(a) Remove the in-axle adapters [53] (TASK 32-49-03-000-803).

(b) Remove the electrical connectors [45] from the wiring assembly [43] (SWPM 20-85-12).

(c) Remove the screws [46], washers [47], nuts [48], and clamps [50] from the lower torsion link [49].

(d) Remove lockwire from the fitting [44].

(e) Loosen the fitting [44] from the inner cylinder [12].

(f) Pull the wiring assembly [43] completely out of the axle [46] and inner cylinder [12].

1) Make sure that you attach a label to the wires to indicate the left and right connectors.

G. Strut Torsion Links Disassembly

SUBTASK 32-21-11-020-010

(1) Disconnect the lower torsion link [49] from the inner cylinder [12], do these steps:

(a) Use a jack to hold the inner cylinder [12] in its position in the outer cylinder [9], or when the inner cylinder [12] is moved out of or into the outer cylinder [9] unless otherwise directed (TASK 07-10-03-580-803).

(b) Remove the pin [52], nut [51], and washer [41] from the torsion link pin [42].

(c) Remove the torsion link pin [42] to disconnect the lower torsion link [49] from the inner cylinder [12].

(d) Supply a rope or other suitable tieback for the torsion links to hold them out of the work area.

SUBTASK 32-21-11-020-011

WARNING WHEN THE GLAND NUT IS LOOSENED AND MOVED DOWN THE INNER CYLINDER, IT CAN CAUSE A SHOCK-STRUT FLUID LEAK. USE LINT-FREE CLOTH PADS TO ABSORB LEAKAGE. SHOCK STRUT FLUID CAN CAUSE INJURY TO PERSONNEL.

(2) Disconnect the gland nut [17] from the outer cylinder [9], do these steps:

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(a) Measure and write down, the dimension from the gland nut [17] lower surface to the outer cylinder [9] lower surface.

NOTE: This measurement will be used to make sure that the stackup of the lower bearing carrier components upon reinstallation in the outer cylinder is correct.

(b) Remove the bolt [14], washer [15], and lockplate [16] to release the gland nut [17].

(c) Wrap towels around the inner cylinder [12], one on top of the other, to absorb fluid when you slowly loosen the gland nut [17].

1) Let the fluid drip from the towels into a bucket.

(d) Use the gland nut wrench, SPL-1867, to slowly loosen the gland nut [17] until the fluid begins to flow from the outer cylinder [9].

1) When the fluid flow stops, disconnect the gland nut [17] from the outer cylinder [9].

(e) Hold the gland nut [17] in its position against the outer cylinder [9] after you have disconnected it from the outer cylinder [9].

SUBTASK 32-21-11-480-005

(3) Install the scraper retainer [79] (scraper retainer, C32016-4 or -67, SPL-1378) and scraper lock assembly [74] (scraper lock assembly, C32016-5 or -68, SPL-1379) on the gland nut [17], do these steps:

NOTE: This will make sure that the scraper ring for the gland nut does not come out of its groove.

(a) Install the scraper retainer [79] (scraper retainer, C32016-4 or -67, SPL-1378) on the gland nut [17].

(b) Install the scraper lock assembly [74] (scraper lock assembly, C32016-5 or -68, SPL-1379) to hold the scraper retainer [79] (scraper retainer, C32016-4 or -67, SPL-1378) in its position on the gland nut [17].

SUBTASK 32-21-11-020-012

(4) Move the gland nut [17] down to the bottom of the inner cylinder [12].

(a) If the scraper ring [24] comes out of its groove, use the scraper plunger [80] (scraper plunger, C32016-2, SPL-1376) and scraper reducer [81] (scraper reducer, C32016-3, SPL-1377) to install the scraper ring [24] back in its groove.

SUBTASK 32-21-11-020-013

(5) Move and lower the lower bearing carrier [23], do these steps:

(a) Loosely install the seal puller assembly [73] (seal puller assembly, C32016-6, SPL-4047) around the inner cylinder [12].

(b) Push the seal puller assembly [73] (seal puller assembly, C32016-6, SPL-4047) up, such that it will install correctly at the bottom of the outer cylinder [9].

(c) Align the holes in the seal puller assembly [73] (seal puller assembly, C32016-6, SPL-4047) with the holes in the lower bearing carrier [23] for the puller screws [78].

(d) Install the puller screws [78] to hold the seal puller assembly [73] (seal puller assembly, C32016-6, SPL-4047) to the lower bearing carrier [23].

(e) Tighten the retainer puller screw [78] on each side of the seal puller assembly [73] (seal puller assembly, C32016-6, SPL-4047).

(f) Use the jack to lift the inner cylinder [12] until the scraper lock assembly [74] is approximately 6 in. (152 mm) below the seal puller assembly [73].

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(g) Install the strap assembly [75] (strap assembly, C32016-59, SPL-12809) around the axle and connect the ends to the shoulder bolts on each side of the seal puller assembly [73] (seal puller assembly, C32016-6, SPL-4047.

(h) Use the jack to lower the inner cylinder [12] until the lower bearing carrier [23] is completely out of the outer cylinder [9].

1) Do not lower the inner cylinder [12] more than necessary.

(i) Move the lower bearing carrier [23] by hand down on top of the gland nut [17].

NOTE: The bearing carrier pins may fall to ground during the movement of the lower bearing carrier down the inner cylinder.

(j) Remove the bearing carrier pins [30].

1) Keep the bearing carrier pin [30] for the re-installation.

(k) Remove the inner cylinder [12] from the outer cylinder [9], do these steps:

WARNING MAKE SURE THAT A MINIMUM OF TWO PERSONS WILL HOLD THE INNER CYLINDER WHEN YOU DO THE SUBSEQUENT STEPS. THE INNER CYLINDER IS HEAVY. INJURY TO PERSONS OR DAMAGE TO THE EQUIPMENT CAN OCCUR.

1) Use two people to lift the inner cylinder [12] off of the jack.

2) Move the jack away from the area.

WARNING MAKE SURE TO KEEP THE CYLINDER VERTICAL. THE INNER CYLINDER CONTAINS SHOCK STRUT FLUID WHICH CAN FLOW OUT. USE A CONTAINER TO CONTAIN THE REMAINING FLUID. THE FLUID CAN CAUSE INJURY TO PERSONNEL OR EQUIPMENT.

3) Lower the inner cylinder [12] out of the outer cylinder [9].

(l) Pour the remaining shock strut fluid into a suitable 5 gal (19 l) container.

NOTE: Total strut fluid is approximately 0.88 gal (3.3 l).

1) If it is necessary, flush the inner cylinder [12] may flushed with fluid, D00467, to make sure that there is no debris upon reassembly.

NOTE: Debris from normal operation sometimes accumulates in the shock strut fluid.

(m) Move the inner cylinder [12], lower bearing carrier [23], and gland nut [17] to a suitable location for further disassembly.

NOTE: Two persons can transport the inner cylinder by holding the assembly by the axles and holding the inner cylinder upright.

SUBTASK 32-21-11-840-001

(6) Use a plastic bag of suitable proportions to cover the open inner cylinder [12] and outer cylinder [9] when you are not directly working on them.

NOTE: This will keep dirt or debris from getting inside the inner cylinder and outer cylinder.

H. Metering Pin Seal Replacement

SUBTASK 32-21-11-960-003

(1) Do these steps if it is required to replace the metering pin seal assembly [104] (Figure 802).

NOTE: The metering pin seal assembly should be replaced if the leakage exceed the rates in Table 801.

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(a) Remove the metering pin [101] from the inner cylinder [12], do these steps:

1) Use the metering pin and retainer ring equipment, SPL-1895, to remove the nut [103] and retaining ring [102] from the inner cylinder [12].
2) Remove the metering pin [101] from the inner cylinder [12].
3) Remove the seal assembly [104] from the metering pin [101].

(b) Install the metering pin [101] in the inner cylinder [12], do these steps:

1) Cover the seal assembly [104] with fluid, D00467.
2) Install the seal assembly [104] on the metering pin [101].
3) Apply a layer of hydraulic fluid, D00467, to the threads of nut [103].
4) Put the flat lip of the retaining ring [102] against the mating face of the metering pin [101].
5) Use the metering pin and retainer ring equipment, SPL-1895, to compress the retaining ring [102] on the metering pin [101].
6) Install the nut [103] to hold the retaining ring [102] compressed between the metering pin [101] and nut [103].
7) Install the metering pin [101], retaining ring [102] and nut [103] in the inner cylinder [12].
8) Use the metering pin and retainer ring equipment, SPL-1895, to loosen the nut [103] and let the retaining ring [102] expand into the groove in the inner cylinder [12].
9) Use the metering pin and retainer ring equipment, SPL-1895, to torque the nut [103] to 75 ft-lb (101.7 N·m) - 100 ft-lb (135.6 N·m).

I. Inner Cylinder Component Disassembly

SUBTASK 32-21-11-020-014

(1) Remove the upper bearing assembly [8] and recoil valve [87] from the top of the inner cylinder [12], do these steps:

(a) Remove the insert [8A] from the upper bearing halfs [85].

NOTE: Pull the insert apart slightly and slide it up and off the upper bearing halfs.

1) Inspect the insert [8A] for damage and excessive wear.

a) Make sure that the insert [8A] thickness is greater than or equal to 0.0865 in. (2.1971 mm).

b) If it is necessary, replace the insert [8A].

(b) Remove the upper bearing halfs [85] from the inner cylinder [12].

(c) Remove the recoil valve [87] and ring [86].

SUBTASK 32-21-11-020-015

(2) Remove the upper centering cam [10] and lower centering cam [11] from the inner cylinder [12].

(a) Remove the dowel retaining circlip [88] and dowels [89] from the inner cylinder [12].

SUBTASK 32-21-11-020-016

(3) Remove the lower bearing carrier [23] from the inner cylinder [12], do these steps:

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(a) Use rubber mallet and lightly hit the bottom of the seal puller assembly [73] (seal puller assembly, C32016-6, SPL-4047) until the lower bearing carrier [23] can be pulled off at the top of the inner cylinder [12].

NOTE: The lower bearing carrier is coated with a dry lubricant film. The film may be worn and appear uneven, this is not cause for the lower bearing carrier replacement.

(b) Remove the puller screw [78] to disconnect the seal puller assembly [73] (seal puller assembly, C32016-6, SPL-4047) from the lower bearing carrier [23].

(c) Remove the seal puller assembly [73] (seal puller assembly, C32016-6, SPL-4047) from the lower bearing carrier [23].

(d) Remove the lower bearing insert [93] from the lower bearing carrier [23].

1) Inspect the lower bearing insert [93] for damage and excessive wear.

a) Make sure that the lower bearing insert [93] thickness is greater than or equal to 0.0865 in. (2.1971 mm).

b) If it is necessary, replace the lower bearing insert [93].

(e) Install the lower bearing insert [93] on the lower bearing carrier [23].

SUBTASK 32-21-11-020-017

(4) Remove the seal retainer [18]:

(a) Move the seal retainer [18] away from the lower bearing carrier [23] to get access to the elastomer ring [19].

J. Active Dynamic Seal, Active Static Seal, and Spare Seal Replacement

NOTE: Do this task in a sheltered and clean environment. Make sure that all fay surfaces of components have been cleaned and are free of debris before re-assembly.

NOTE: You may disassemble the lower bearing carrier components all at once to aid in cleaning of the lower bearing carrier before reassembly.

SUBTASK 32-21-11-900-003

(1) If there is any sign of damage or wear on the old scraper ring [24], install a new scraper ring [24] on the gland nut [17], do these steps:

NOTE: The scraper ring prevents dirt and debris from entering the shock strut when the inner cylinder and outer cylinder compress.

(a) Remove the gland nut [17] from the inner cylinder [12].

(b) Remove the scraper retainer [79] (scraper retainer, C32016-4 or -67, SPL-1378) and scraper lock assembly [74] (scraper lock assembly, C32016-5 or -68, SPL-1379) from the gland nut [17].

(c) Remove the old scraper ring [24] from its groove on the gland nut [17].

(d) Install the new scraper ring [24] in its groove on the gland nut [17].

1) Make sure to use the correct scraper configuration to be installed in its groove on the gland nut [17].

NOTE: There are two possible scraper configurations used for installation (Figure 803).

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SUBTASK 32-21-11-480-006

(2) Install the scraper retainer [79] (scraper retainer, C32016-4 or -67, SPL-1378) and scraper lock assembly [74] (scraper lock assembly, C32016-5 or -68, SPL-1379) on the gland nut [17], do these steps:

NOTE: This will make sure that the new scraper ring for the gland nut does not come out of its groove.

(a) Install the scraper retainer [79] (scraper retainer, C32016-4 or -67, SPL-1378) on the gland nut [17].

(b) Install the scraper lock assembly [74] (scraper lock assembly, C32016-5 or -68, SPL-1379) to hold the scraper retainer [79] (scraper retainer, C32016-4 or -67, SPL-1378) in its position on the gland nut [17].

SUBTASK 32-21-11-900-004

(3) If there is any sign of damage or wear on the old excluder [25], replace the excluder [25] on the lower bearing carrier [23], do these steps:

NOTE: The excluder prevents dirt and debris from damaging the active static seal.

(a) Remove the excluder [25] from its groove in the lower bearing carrier [23].

(b) Install the new excluder [25] in its groove in the lower bearing carrier [23], do these steps:

1) Put the excluder [25] in a sealed plastic bag and submerge the bag in hot water (not boiling) for 3-5 minutes.

2) Apply a thin layer of fluid, D50022, (preferred) or fluid, D00467, (alternative) onto the excluder [25].

3) Put fluid, D50022, (preferred) or fluid, D00467, (alternative) on the surface on which the excluder [25] will move.

4) Put the excluder [25] in position evenly on the carrier with the spring or elastomer facing down (toward the gland nut [17]).

NOTE: Zip ties can be used to help with this step.

5) Carefully slide the excluder [25] into position in its carrier groove.

6) Make sure that you move the seal evenly down the carrier and do not let it roll over.

7) Make sure that the excluder [25] stays in its correct position until it reaches room temperature and has conformed to its original shape.

NOTE: A compression ring can be used to lightly hold the excluder in its position until it reaches room temperature.

SUBTASK 32-21-11-900-005

(4) Replace the active static seal [28], do these steps (Figure 803):

(a) Make sure that your hands are clean when you handle the elastomer seals.

NOTE: Some greases can cause the elastomer seals to swell too much. This can cause the seals to leak.

(b) Replace the backup rings [29] from the active static seal [28].

(c) Remove the active static seal [28] from its groove on the lower bearing carrier [23].

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CAUTION DO NOT APPLY VASELINE OR PETROLATUM TO THE SEALS. THIS CAN CAUSE THE SEAL MATERIAL TO BECOME LARGER. DAMAGE TO THE SEALS CAN OCCUR.

(d) You may apply a very thin layer of fluid, D00467, or AFS-682 lubricant, D00589, to the new active static seal [28] and backup rings [29] to help keep the rings in their position during installation.

NOTE: The material is a very thick and sticky and large quantities can cause the lower bearing carrier to be difficult to re-install.

1) Use the minimum quantity of AFS-682 lubricant, D00589, necessary to keep the parts in position.

2) Remove excess AFS-682 lubricant, D00589.

NOTE: Use of excess AFS-682 lubricant, D00589, can cause problems with assembly of the components.

(e) Install the new active static seal [28] in its groove on the lower bearing carrier [23].

CAUTION MAKE SURE THAT THE RADIUS OF THE BACKUP RING IS AGAINST THE SEAL RADIUS. IF THE BACKUP RINGS ARE NOT INSTALLED CORRECTLY, DAMAGE CAN OCCUR.

(f) Install the first backup ring [29] on the active static seal [28] with the rounded edge of the backup ring [29] against the seal.

(g) Install the second backup ring [29] with the split in the opposite position on the diameter of the lower bearing carrier [23] as the split of the first backup ring [29] installed.

SUBTASK 32-21-11-900-006

(5) Replace the spare static seals [27], do these steps:

(a) Make sure that your hands are clean when you handle the elastomer seals.

NOTE: Some greases can cause the elastomer seals to swell too much. This can cause the seals to leak.

(b) Install the new spare static seals [27] in its groove on the lower bearing carrier [23].

NOTE: The spare static seals should be installed as shown in (Figure 803).

SUBTASK 32-21-11-900-007

CAUTION MAKE SURE THAT THE BACKUP RING, THE CAP, AND THE SEAL STAY TOGETHER WHILE YOU INSTALL THE SEAL. IF THE BACKUP RING, THE CAP, AND THE SEAL MOVE APART, THEY WILL NOT INSTALL CORRECTLY. THIS CAN CAUSE DAMAGE TO THE SEAL, THE CAP, AND THE BACKUP RING.

(6) Replace the active dynamic seal, do these steps:

(a) Remove the cap seal [20], backup ring [21], and elastomer ring [19] from the inner groove on the lower bearing carrier [23].

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CAUTION DO NOT APPLY VASELINE OR PETROLATUM TO THE SEALS. THIS CAN CAUSE THE SEAL MATERIAL TO BECOME LARGER. DAMAGE TO THE SEALS CAN OCCUR.

(b) You can apply a thin layer of fluid, D00467, or AFS-682 lubricant, D00589, to the elastomer ring [19], cap seal [20], and backup ring [21] to help keep the backup rings [21] in position during installation.

NOTE: The material is very thick and sticky and large quantities can cause the lower bearing carrier to be difficult to re-install.

1) Use the minimum quantity of AFS-682 lubricant, D00589, necessary to keep the parts in position.

2) Remove excess AFS-682 lubricant, D00589.

NOTE: Use of excess AFS-682 lubricant, D00589, can cause problems with assembly of the components.

(c) The three parts of the active dynamic seal can be assembled on the inner cylinder [12] and then put in the dynamic seal groove of the lower bearing carrier [23] or the parts can be installed in the groove one at a time.

(d) Install the backup ring [21] of the new dynamic seal in the inner groove for the active dynamic seal.

NOTE: The edge radius should be facing up and towards the elastomer ring.

CAUTION DO NOT PUT THE CAP SEAL INTO THE SEAL RETAINER WHILE IT MOVES DOWN THE INNER CYLINDER. IF YOU DO NOT OBEY, DAMAGE TO THE SEAL CAN OCCUR.

(e) Install the cap seal [20] in the inner groove for the active dynamic seal.

(f) Install the elastomer ring [19] in its groove on the lower bearing carrier [23] (View B, Figure 803).

NOTE: The wide seal edge is pointed toward the bottom of the lower bearing carrier.

SUBTASK 32-21-11-900-008

(7) Replace the spare dynamic seal [26], do these steps:

(a) Install a spare dynamic seal [26] if there is no spare dynamic seal in the spare seal groove on the lower bearing carrier [23].

(b) Install the spare dynamic seal backup rings [22] in the spare seal groove on the lower bearing carrier [23].

SUBTASK 32-21-11-420-008

(8) Install the seal retainer [18] or optional seal retainer [97], do these steps:

NOTE: You may use optional seal retainer (seal retainer, C32016-55, SPL-23426) to help install the dynamic seal correctly.

(a) Put the seal retainer [18] in its position on the lower bearing carrier [23].

(b) Make sure that the seal retainer [18] is flush to the lower bearing carrier [23] all around the circumference before you install the bearing carrier pins [30].

NOTE: If the seal retainer is not flush, the active dynamic seal and the backup rings may not be installed correctly.

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(c) You may apply a thin layer of AFS-682 lubricant, D00589, to the bearing carrier pins [30] to hold them in position while the lower bearing carrier [23] is re-installed into the outer cylinder [9].

NOTE: AFS-682 lubricant, D00589, will help keep the bearing carrier pins in position.

1) Use the minimum quantity of AFS-682 lubricant, D00589, necessary to retain the bearing carrier pins [30].

NOTE: The material is very thick and sticky and large quantities can cause the lower bearing carrier to be difficult to re-install.

2) Remove excess AFS-682 lubricant, D00589.

CAUTION

DO NOT FORCE THE PINS INTO THEIR POSITION. THE PINS MUST INSTALL FREELY. IF YOU FORCE THE PINS INTO THEIR POSITION, IT CAN CAUSE DAMAGE TO THE SEAL RETAINER.

(d) Install the bearing carrier pins [30] to keep the seal retainer [18] in the lower bearing carrier [23].

1) If the bearing carrier pins [30] are not free during installation, make sure that the seal retainer [18] is flush to the lower bearing carrier [23], then reinsert the bearing carrier pins [30].

NOTE: If the seal retainer is not flush, the active dynamic seal may have not been installed correctly.

(e) Install the optional seal retainer [97], do these steps:

1) Put the optional seal retainer [97] in its position on the lower bearing carrier [23].

2) Install the retainer pins [94] to connect the seal retainer to the lower bearing carrier [23].

3) Install the rubber band [95] to hold the retainer pins [94] in position.

SUBTASK 32-21-11-840-002

WARNING

MAKE SURE THAT A MINIMUM OF TWO PERSONS WILL HOLD THE INNER CYLINDER WHEN YOU DO THE SUBSEQUENT STEPS. THE INNER CYLINDER IS HEAVY. INJURY TO PERSONS OR DAMAGE TO THE EQUIPMENT CAN OCCUR.

(9) Hold the inner cylinder [12] in a vertical position.

SUBTASK 32-21-11-480-007

(10) Coat the dynamic seal guide [84] (dynamic seal guide, C32016-45, SPL-1382) and lower bearing carrier [23] with fluid, D00467.

SUBTASK 32-21-11-480-008

(11) Install the dynamic seal guide [84] (dynamic seal guide, C32016-45, SPL-1382) at the top of the inner cylinder [12].

SUBTASK 32-21-11-420-009

(12) Install the lower bearing carrier [23], do these steps:

NOTE: You can install the lower bearing carrier into the dynamic seal guide (dynamic seal guide, C32016-45, SPL-1382), at the workbench and then install the dynamic seal guide (dynamic seal guide, C32016-45, SPL-1382) with the lower bearing carrier onto the top of the inner cylinder.

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CAUTION

MAKE SURE THAT THE BACKUP RINGS AND THE SEAL STAY TOGETHER WHEN YOU MOVE THE LOWER BEARING CARRIER. IF THE BACKUP RINGS AND THE SEAL MOVE APART, THEY WILL NOT INSTALL CORRECTLY. THIS CAN CAUSE DAMAGE TO THE SEAL AND THE BACKUP RINGS.

(a) Put the lower bearing carrier [23] in its position for installation on the inner cylinder [12], do these steps:

(b) Use a rubber mallet and lightly hit the top of the lower bearing carrier [23] until it passes the dynamic seal guide [84] (dynamic seal guide, C32016-45, SPL-1382).

(c) Move the lower bearing carrier [23] down to the bottom of the inner cylinder [12].

1) Use ram assembly, C32016-47, SPL-13126, as necessary to help install the lower bearing carrier [23] onto the inner cylinder [12].

2) If you used optional seal retainer [97] (lower seal R/I equipment, SPL-13124) for the installation, remove the rubber band [95] (rubber band, C32016-58, SPL-13129), retainer pin [94] (retainer pin, C32016-56, SPL-13125) and seal retainer [97] (lower seal R/I equipment, SPL-13124).

3) After you remove the optional seal retainer [97] (lower seal R/I equipment, SPL-13124), look down into the lower bearing carrier [23] and examine the dynamic seal.

a) Make sure that the dynamic seal appears fully seated.

4) Install the production seal retainer [18] and bearing carrier pins [30] onto the lower bearing carrier [23].

SUBTASK 32-21-11-020-018

(13) Remove the dynamic seal guide [84] (dynamic seal guide, C32016-45, SPL-1382) from the inner cylinder [12].

K. Inner Cylinder Components Assembly

SUBTASK 32-21-11-420-010

(1) Install these items:

(a) Install the lower centering cam [11] on the inner cylinder [12].

NOTE: There are two dowel pins loosely attached to the lower centering cam. These dowel pins must be aligned with the recesses inside the outer cylinder when you assemble the inner cylinder and outer cylinder assemblies in the subsequent steps. Align the centering cam dowel pins with the strut axle, or gland nut lockable recesses on the outer cylinder to align the pins to the pin recesses. The dowel pins are designed to be loosely held in place with 0.005 in. (0.13 mm) – 0.015 in. (0.38 mm) dowel movement after the rivet is in position.

(b) Put the upper centering cam dowels [89] in position on the inner cylinder [12] (Figure 804).

1) Make sure that each dowel [89] is centered and flush in it's centering cam groove.

2) Make sure that the dowels [89] is in position above the lip, on the inner cylinder [12].

NOTE: This will let the dowel retaining circlip to later move into its installed position.

(c) Put the dowel retaining circlip [88] approximately mid-point on the dowels [89] to hold the dowels [89] in position on the inner cylinder [12].

(d) Install the upper centering cam [10] on the inner cylinder [12].

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1) Carefully push the dowel retaining circlip [88] down the dowels [89], with the upper centering cam [10], until the dowels [89] are in position in the upper centering cam [10].

NOTE: The dowel retaining circlip will snap over the bottom of the dowels into the circlip groove, that retains the dowels up inside the upper centering cam.

(e) Install the upper bearing assembly [8] and recoil valve [87] on the inner cylinder [12], do these steps:

1) Install the ring [86] on the recoil valve [87], if it was removed.

2) Install the recoil valve [87] and ring [86] on top of the upper centering cam [10].

3) Install the upper bearing half [85] on the inner cylinder [12].

(f) Install the insert [8A] on the upper bearing halfs [85].

1) Use hand pressure to make an overlap of ¼ in. (6.4 mm) to the insert [8A] to keep its round shape.

NOTE: This will keep the insert snug against the upper bearing halfs.

2) Use the compressor assembly half [92] (compressor assy half, C32016-10, SPL-1383) to hold the bearing closed during inner cylinder [12] installation.

SUBTASK 32-21-11-420-011

CAUTION

MAKE SURE THAT YOU INSTALL THE COMPRESSOR ASSEMBLY BEFORE YOU INSTALL THE INNER CYLINDER. IF YOU DO NOT, THE TOOL CAN CAUSE DAMAGE TO THE THREADS OF THE OUTER CYLINDER AND UPPER BEARING ASSEMBLY.

(2) Install the compressor assembly half [92] (compressor assy half, C32016-10, SPL-1383), do these steps:

(a) Install the compressor assembly half [92] (compressor assy half, C32016-10, SPL-1383), such that it fits at the bottom of the outer cylinder [9].

(b) Install the strap assembly [75] (strap assembly, C32016-59, SPL-12809) around the outer cylinder [9] and connect the ends to the shoulder bolts on the side of the compressor assembly half [92] (compressor assy half, C32016-10, SPL-1383).

1) Tighten the strap assembly [75] (strap assy, C32017-60, SPL-3927).

SUBTASK 32-21-11-420-012

CAUTION

DO NOT LIFT THE AIRPLANE WITH THE AXLE JACK. IF YOU LIFT THE AIRPLANE WITH THE AXLE JACK, IT CAN CAUSE DAMAGE TO EQUIPMENT.

(3) Install the inner cylinder [12] in the outer cylinder [9], do these steps:

WARNING

MAKE SURE THAT A MINIMUM OF TWO PERSONS WILL HOLD THE INNER CYLINDER WHEN YOU DO THE SUBSEQUENT STEPS. THE INNER CYLINDER IS HEAVY. INJURY TO PERSONS OR DAMAGE TO THE EQUIPMENT CAN OCCUR.

CAUTION

BE CAREFUL NOT TO CAUSE DAMAGE TO THE INNER CYLINDER OR THE METERING PIN. DAMAGE TO THESE PARTS CAN CAUSE DAMAGE TO THE SEALS.

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(CAUTION PRECEDES)

CAUTION

MAKE SURE THAT YOU HAVE THE DOWEL PINS ON THE LOWER CENTERING CAM ALIGNED WITH THE RECESSES IN THE OUTER CYLINDER OR THE AXLE CENTERLINE. IF NOT ALIGNED, IT CAN CAUSE DAMAGE TO THE DOWEL PINS AND THE INNER CYLINDER.

(a) Put the inner cylinder [12] in position to install in the outer cylinder [9].

1) You can use the Centering Assembly C32016-35, or manually move the lower centering cam [11], to help align the lower centering cam dowel pins with the dowel pin recesses inside the outer cylinder [9].

2) Align the dowel pins with the axle centerline or the gland nut tablock recesses on the outer cylinder [9].

(b) Manually lift the inner cylinder [12], into the outer cylinder [9], until the top edge of the chrome surface of the inner cylinder [12] is level with the lower surface of the compressor assembly half [92] (compressor assy half, C32016-10, SPL-1383).

NOTE: Use great care in installing the inner cylinder into the outer cylinder to keep from damaging the lower bearing carrier dowel pins that are loosely riveted to the carrier, and are easily caused to be detached from the carrier if acted upon with much force.

(c) Use a floor jack to hold the inner cylinder [12] in position.

SUBTASK 32-21-11-080-003

(4) Remove the compressor assembly half [92] (compressor assy half, C32016-10, SPL-1383), do these steps:

(a) Remove the strap assembly [75] (strap assembly, C32016-59, SPL-12809) from the outer cylinder [9] and compressor assembly half [92] (compressor assy half, C32016-10, SPL-1383).

(b) Remove the compressor assembly half [92] (compressor assy half, C32016-10, SPL-1383) from the outer cylinder [9].

SUBTASK 32-21-11-420-013

CAUTION

MAKE SURE THAT YOU HAVE THE DOWEL PINS ON THE LOWER CENTERING CAM ALIGNED WITH THE RECESSES IN THE OUTER CYLINDER OR THE AXLE CENTERLINE. IF NOT ALIGNED, IT CAN CAUSE DAMAGE TO THE DOWEL PINS AND THE INNER CYLINDER.

(5) Carefully put the thread protector insert [77] (thread protector half, C32016-21, SPL-1385) between the outer cylinder [9] and inner cylinder [12].

NOTE: The thread protector will expand and push against the outer cylinder, this will protect the static seals from the gland nut threads on the outer cylinder.

SUBTASK 32-21-11-480-009

(6) Install the thread protector assembly [76] (thread protector assembly, C32016-11, SPL-1384), do these steps:

(a) Install the thread protector assembly [76] (thread protector assembly, C32016-11, SPL-1384), so that it fits at the bottom of the outer cylinder [9].

(b) Install the strap assembly [75] (strap assembly, C32016-59, SPL-12809) around the outer cylinder [9] and connect the ends to the shoulder bolts on the side of the thread protector assembly [76] (thread protector assembly, C32016-11, SPL-1384).

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1) Tighten the strap assembly [75] (strap assembly, C32016-59, SPL-12809).

SUBTASK 32-21-11-480-010

(7) Install the seal puller assembly [73] (seal puller assembly, C32016-6, SPL-4047) around the inner cylinder [12] to support the lower bearing carrier [23].

NOTE: It is not necessary to install the two puller screws, as the inner cylinder will push on the seal puller assembly and lower bearing carrier as the inner cylinder is pushed into position. The two plastic skid plates can be put upright, between the lower bearing carrier and scraper lock assembly, to give extra height to move the lower bearing carrier up into the outer cylinder.

SUBTASK 32-21-11-580-006

CAUTION

MAKE SURE THAT YOU HAVE THE DOWEL PINS ON THE LOWER CENTERING CAM ALIGNED WITH THE RECESSES IN THE OUTER CYLINDER OR THE AXLE CENTERLINE. IF NOT ALIGNED, IT CAN CAUSE DAMAGE TO THE DOWEL PINS AND THE INNER CYLINDER.

(8) Use two people to lift the inner cylinder [12] until you feel the dowels [89] of the lower centering cam [11] engage the groves in the outer cylinder [9].

(a) If it is necessary, with the lower centering cam dowel pins engaged in their recesses, use a jack to push on and fully seat the lower bearing carrier [23] in the outer cylinder [9].

(b) Make sure that the static seals and backup rings stay in their positions as the lower bearing carrier [23] is installed.

1) You may apply a thin layer of fluid, D00467, or AFS-682 lubricant, D00589, to the elastomer ring [19], cap seal [20], and backup ring [21] to help keep the backup ring [21] in position during the installation.

NOTE: AFS-682 lubricant, D00589, will help keep the backup ring in its position.

NOTE: The material is a very thick, tacky material and a large quantity can make the bearing carrier hard to install.

2) Use the minimum quantity of AFS-682 lubricant, D00589, necessary to keep the parts in their positions.

3) Remove unwanted AFS-682 lubricant, D00589.

NOTE: If you apply too much AFS-682 lubricant, D00589, this can cause problems with assembly of the components.

CAUTION

DO NOT LET THE INNER CYLINDER LOWER BACK DOWN WITHOUT THE THREAD PROTECTOR INSTALLED. DAMAGE TO THE SEALS CAN OCCUR.

(c) Remove the thread protector assembly [76] when the lower surface of the lower bearing carrier [23], is aligned with the lower surface of the thread protector assembly [76].

NOTE: As you lift the inner cylinder, adjust the supporting jack to stay within ½ in. (13 mm) of the inner cylinder.

SUBTASK 32-21-11-020-019

(9) Remove the seal puller assembly [73] and skid plates.

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SUBTASK 32-21-11-020-020

(10) Remove the scraper retainer [79] (scraper retainer, C32016-4 or -67, SPL-1378) and scraper lock assembly [74] (scraper lock assembly, C32016-5 or -68, SPL-1379) from the gland nut [17].

SUBTASK 32-21-11-640-002

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(11) Apply a liberal quantity of corrosion inhibiting material, G50136, to the threads of the gland nut [17] and threads of the outer cylinder [9].

SUBTASK 32-21-11-420-014

(12) Install the gland nut [17], do these steps:

(a) Move the gland nut [17] up on the inner cylinder [12] and engage the threads to the outer cylinder [9].

(b) Use the gland nut wrench, SPL-1867, to torque the gland nut [17] to 100 ft-lb (135.6 N·m) - 125 ft-lb (169.5 N·m).

(c) Remove all excess corrosion inhibiting material, G50136.

SUBTASK 32-21-11-420-015

(13) Install the lockplate [16] for the gland nut [17] (Figure 802):

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin coat of corrosion inhibiting material, G50136, to the shank of the unplated bolts, threads, thread reliefs, and washer faces prior to installation.

(b) Install the lockplate [16] in the holes that will put the tab of the lockplate [16] as close as possible to a slot in the outer cylinder [9].

1) If it is necessary, loosen the gland nut [17] slightly to align lockplate [16] with the slot in the outer cylinder [9].

(c) Install the bolt [14] and washer [15].

(d) Tighten the bolt [14].

(e) Install MS20995C32 lockwire, G01048, on the bolt [14].

SUBTASK 32-21-11-420-038

(14) Before connecting the electrical connector [45] and wiring assembly [43], examine the electrical connector [45] for corrosion.

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WARNING

DO THE STEPS BELOW IF THE AIRPLANE OPERATES WHERE DEICING FLUID THAT CONTAINS POTASSIUM FORMATE OR POTASSIUM ACETATE IS USED. ALSO, DO THE STEPS FOR ALL AIRPLANES THAT YOU FOUND CORROSION IN THE ELECTRICAL CONNECTORS IN THE MAIN WHEEL WELL. THE ELECTRICAL CONNECTORS ARE IN A SYSTEM THAT IS NECESSARY FOR SAFE FLIGHT.

(a) If there was corrosion, correct the problem (SWPM 20-60-03).

(b) Apply D5026NS or ZC-026 compound, G50171, to the electrical connector [45] (SWPM 20-60-03).

SUBTASK 32-21-11-420-039

(15) Connect the wiring assembly [43] and electrical connectors [45], do these steps:

(a) Put the wiring assembly [43] into the inner cylinder [12] and axle [46].

(b) Tighten the fitting [44] to 45 in-lb (5.1 N·m) - 55 in-lb (6.2 N·m), to connect the wiring assembly [43] to the inner cylinder [12].

(c) Install MS20995C20 lockwire, G50225, on the fitting [44].

(d) Put the wiring assembly [43] in its position for the attachment to the clamps [50].

(e) Install the washers [47], screws [46], and nuts [48], to hold the electrical harness to the clamps [50].

(f) Connect the electrical connectors [45] to the wiring assembly [43] (SWPM 20-85-12).

(g) Install the in-axle adapters [53] (TASK 32-49-03-400-803).

SUBTASK 32-21-11-420-016

(16) Install components removed for access.

SUBTASK 32-21-11-420-017

(17) Connect the lower torsion link [49] to the inner cylinder [12], do these steps:

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin layer of corrosion inhibiting material, G50136, to these items:

1) The threads and thread reliefs of the torsion link pin [42]

2) The threads of the nut [51]

3) The faces of the washer [41]

4) The mating surfaces of the torsion link pin [42].

5) Apply a thin layer of grease, D00633, to the chromed surfaces of the torsion link pin [42].

(b) Remove all unwanted corrosion inhibiting material, G50136.

(c) Put the lower torsion link [49] in its position on the inner cylinder [12].

(d) Put the torsion link pin [42] through the lower torsion link [49] and inner cylinder [12].

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(e) Install the washer [41] and nut [51] on the torsion link pin [42].

(f) Torque the nut [51] to 250 in-lb (28.2 N·m) - 300 in-lb (33.9 N·m) more than the run-on torque.

(g) If it is necessary, loosen the nut [51] to the nearest castellation to align the holes for the pin [52].

1) If there is no clearance, loosen the nut [51] to one more castellation and do a check again.

(h) Install the pin [52] in the torsion link pin [42].

SUBTASK 32-21-11-980-001

CAUTION

DO NOT LIFT THE AIRPLANE NOSE OFF THE AIRPLANE NOSE JACK WHEN YOU MOVE THE INNER CYLINDER TO THE FULLY COMPRESSED POSITION. STOP MOVEMENT OF THE INNER CYLINDER IF THERE IS RESISTANCE. DAMAGE TO EQUIPMENT CAN OCCUR.

(18) If it is not fully compressed, use a jack to move the inner cylinder [12] from the fully extended to the fully compressed position.

NOTE: There is a total travel is 15.50 in. (39.4 cm) for a fully compressed strut.

SUBTASK 32-21-11-080-004

(19) Lower and remove the axle jack, do this task: Lower the Airplane Nose, TASK 07-10-06-580-803.

SUBTASK 32-21-11-410-001

(20) Install the wheels (TASK 32-45-21-400-801).

SUBTASK 32-21-11-080-005

(21) Lower the airplane nose and remove the jacks (TASK 07-10-21-580-802).

SUBTASK 32-21-11-420-018

(22) If it is removed, install the oil charging valve [6], do these steps:

(a) Remove and discard the old packing [7].

(b) Apply a thin layer of fluid, D00467, on the new packing [7].

(c) Install the new packing [7] on the oil charging valve [6].

(d) Install the oil charging valve [6] and torque it to 22 ft-lb (30 N·m) - 25 ft-lb (34 N·m).

(e) Install the cap [5] loosely.

NOTE: You will tighten it in the shock strut servicing task.

(f) Apply compound, A00226, to the oil charging valve [6].

NOTE: This will make sure that the seal made by the compound, A00226, will break if the oil changing valve becomes loose.

SUBTASK 32-21-11-600-002

(23) Do this task: Nose Landing Gear Shock Strut Servicing, Airplane on the Ground, TASK 12-15-41-610-802.

SUBTASK 32-21-11-420-019

(24) Make sure that the swivel nut [3] is torqued to 5 ft-lb (6.8 N·m) - 7 ft-lb (9.5 N·m) and the cap [1] is installed for the gas charging valve [2].

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SUBTASK 32-21-11-710-001

(25) Do the subsequent sections in this task: Nose Wheel Steering System Test (Airplane on Ground), TASK 32-51-00-700-801:

(a) General

(b) Prepare to Test the Steering System of the Nose Wheel

(c) Do the Steering Centering Test

(d) Put the Airplane Back to Its Usual Condition.

——— END OF TASK ———

TASK 32-21-11-960-803

4. Scraper Ring Replacement

(Figure 801)

A. General

(1) This task is used when the scraper ring [24] is damaged and replacement is necessary but removal of the inner cylinder [12] is not possible.

(2) It is acceptable to operate the aircraft with a missing section of the energizer ring or other damage to the scraper ring. The scraper ring should be replaced with a new solid scraper ring or split scraper ring at the earliest maintenance opportunity to avoid excessive damage to the dynamic seal. If you do not have a split scraper ring available, you can cut a solid scraper ring or the damaged scraper ring to re-install it.

NOTE: The installation of a split scraper ring is also a temporary repair and should be replaced when the nose landing gear is disassembled for other maintenance or overhaul.

B. References

Reference	Title
07-10-03-580-803	Lift the Nose Landing Gear Axle With Axle Jacks-Handling (P/B 201)
07-10-21-580-801	Lift the Airplane Nose with the Nose Jack at Jack Point D (P/B 201)
07-10-21-580-802	Lower the Airplane Nose, Jack Point D (P/B 201)
12-15-41-610-802	Nose Landing Gear Shock Strut Servicing, Airplane on the Ground (P/B 301)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-21-31-000-803	Nose Landing Gear Torsion Link Disconnection (P/B 401)
32-21-31-400-803	Nose Landing Gear Torsion Link Connection (P/B 401)
32-45-21-000-801	Nose Landing Gear Wheel and Tire Assembly Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

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Reference	Description
SPL-1378	Scraper Retainer, C32016-4 or -67 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-1379	Scraper Lock Assy, C32016-5 or -68 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-1829	Oil Drain Valve - Shock Strut, LG Part #: J32108-16 Supplier: 81205 Opt Part #: A32066-1 Supplier: 81205
SPL-1867	Wrench - Gland Nut, Shock Strut NLG Part #: C32025-7 Supplier: 81205 Opt Part #: C32025-1 Supplier: 81205
STD-1110	Container - Hydraulic Fluid Resistant, 5 Gallon (19 Liter)

D. Consumable Materials

Reference	Description	Specification
A00226	Compound - Tamper-Proof Putty	BMS8-45
D00467	Fluid - Landing Gear Shock Strut	BMS3-32 Type II
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

E. Location Zones

Zone	Area
711	Nose Landing Gear Door - Left

F. Prepare for the Repair

SUBTASK 32-21-11-420-042

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-21-11-710-002

(2) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-21-11-710-003

(3) Deflate the shock strut [4] for the nose landing gear:

(a) Remove the cap [1] for the gas charging valve [2] that is on the top of the shock strut [4].

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WARNING DO NOT REMOVE THE VALVE BODY UNTIL YOU DEFLATE THE SHOCK STRUT FULLY. THE AIR PRESSURE CAN BLOW THE VALVE BODY OUT AND CAUSE INJURIES TO PERSONNEL.

(b) Loosen the swivel nut [3] a maximum of two turns.

NOTE: Fluid in the shock strut can have bubbles when you release the pressure.

1) Deflate the shock strut [4] slowly to prevent the leakage of the fluid through the gas charging valve [2].

(c) Loosen the swivel nut [3] fully when all of the pressure in the shock strut [4] is released.

NOTE: The shock strut is fully deflated when the dimension "X" is 13.8 in. (350.5 mm).

SUBTASK 32-21-11-170-001

(4) Drain the hydraulic fluid from the shock strut [4]:

(a) Remove the cap [5] from the oil charging valve [6].

(b) Put a 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110, in a position to catch the hydraulic fluid when the oil charging valve [6] is opened.

(c) Install the drain equipment on the oil charging valve [6]:

NOTE: The landing gear shock strut drain valve, SPL-1829, could also be used, but it will take much longer to drain the shock strut.

1) Cut a length of plastic tubing, sufficiently long to reach the container on the floor.

2) Insert a small allen wrench in the end of the length of tubing so that the long end of the allen wrench is flush with the end of the tube and the short end goes through the wall of the tube.

3) Install the tubing on the oil charging valve [6] so that the allen wrench goes into the check valve and holds it open to drain the hydraulic fluid.

(d) Remove the drain equipment when you have removed all of the hydraulic fluid.

SUBTASK 32-21-11-020-022

(5) Do this task: Nose Landing Gear Torsion Link Disconnection, TASK 32-21-31-000-803.

SUBTASK 32-21-11-580-008

(6) Lift the nose of the airplane until there is about five inches of chrome exposed (TASK 07-10-21-580-801).

SUBTASK 32-21-11-020-023

(7) Use an axle jack to hold the inner cylinder [12] in its position in the outer cylinder [9] (TASK 07-10-03-580-803).

SUBTASK 32-21-11-020-024

(8) Do this task: Nose Landing Gear Wheel and Tire Assembly Removal, TASK 32-45-21-000-801.

G. Scraper Ring Replacement

SUBTASK 32-21-11-020-025

(1) Disconnect the gland nut [17] from the outer cylinder [9]:

(a) Remove the bolt [14], washer [15], and lockplate [16] to release the gland nut [17].

(b) Use the gland nut wrench, SPL-1867, to disconnect the gland nut [17] from the outer cylinder [9].

(c) Remove the scraper ring [24] from the gland nut [17].

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(d) Remove the damaged scraper ring [24], but do not damage the inner cylinder [12] in the removal process.

(e) Install the wave cut scraper ring [24] on the inner cylinder [12] and slide it into the gland nut [17].

SUBTASK 32-21-11-420-020

(2) Install the scraper retainer [79] (scraper retainer, C32016-4 or -67, SPL-1378), and the scraper lock assembly [74] (scraper lock assembly, C32016-5 or -68, SPL-1379), on the gland nut [17].

NOTE: This will make sure that the scraper ring for the gland nut does not come out of its groove.

(a) Install the scraper retainer [79] (scraper retainer, C32016-4 or -67, SPL-1378), on the gland nut [17].

(b) Install the scraper lock assembly [74] (scraper lock assembly, C32016-5 or -68, SPL-1379), to hold the scraper retainer [79] (scraper retainer, C32016-4 or -67, SPL-1378), in its position on the gland nut [17].

SUBTASK 32-21-11-600-003

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(3) Apply a thin layer of corrosion inhibiting material, G50136, to the threads of the gland nut [17] and to the threads of the outer cylinder [9].

SUBTASK 32-21-11-420-021

(4) Install the gland nut [17]:

(a) Move the gland nut [17] up to the outer cylinder [9] and engage the threads.

(b) Use the gland nut wrench, SPL-1867, to tighten the gland nut [17] to 100 ft-lb (136 N·m) - 125 ft-lb (169 N·m).

(c) Remove all excess corrosion inhibiting compound.

SUBTASK 32-21-11-420-022

(5) Install the lockplate [16] for the gland nut [17] (Figure 802):

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin layer of corrosion inhibiting material, G50136, to the shank and threads of bolts [14] and washers [15].

(b) Install the lockplate [16] in the holes that will put the tab of the lockplate [16] as close to an outer cylinder [9] slot as possible.

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1) If it is necessary, loosen the gland nut [17] slightly to align slot in the outer cylinder [9] with the lockplate [16].

(c) Install the bolts [14] and washers [15].

(d) Tighten the bolts [14].

(e) Install MS20995C32 lockwire, G01048, on the bolts [14].

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-21-11-420-023

(1) Install the oil charging valve [6]:

(a) Remove and discard the old packing [7].

(b) Apply a thin layer of fluid, D00467, on a new packing [7].

(c) Install the new packing [7] on the oil charging valve [6].

(d) Install the oil charging valve [6] and tighten it to 22 ft-lb (30 N·m) - 25 ft-lb (34 N·m).

(e) Install the cap [5] loosely.

NOTE: You will tighten it in the shock strut servicing task.

(f) Apply compound, A00226, to the oil charging valve [6].

NOTE: This will make sure that the seal made by compound, A00226, will break if the oil charging valve becomes loose.

SUBTASK 32-21-11-580-010

(2) Lower the airplane nose and remove the jacks (TASK 07-10-21-580-802).

SUBTASK 32-21-11-580-011

(3) Do this task: Nose Landing Gear Torsion Link Connection, TASK 32-21-31-400-803.

SUBTASK 32-21-11-420-024

(4) Do this task: Nose Landing Gear Shock Strut Servicing, Airplane on the Ground, TASK 12-15-41-610-802.

SUBTASK 32-21-11-420-025

(5) Make sure that the swivel nut [3] is tightened to 5 ft-lb (6.8 N·m) - 7 ft-lb (9.5 N·m).

SUBTASK 32-21-11-420-027

(6) Make sure that the cap [1] is installed for the gas charging valve [2].

——— END OF TASK ———

TASK 32-21-11-200-801

5. Nose Landing Gear Shock Strut Seal Leakage Check

(Figure 801, Figure 802, Figure 803)

A. General

(1) This task gives the instructions to make sure a shock strut [4] is serviceable when you find fluid leakage on the inner cylinder [12].

NOTE: During the initial landing cycles of a new landing gear it is usual to have hydraulic fluid streaks on the shock strut [4]. These streaks are caused by hydraulic fluid applied to the shock strut seals during seal installation and will stop after more landing cycles occur. These streaks are different from the actual continuous drips that are the indication of shock strut seal leakage.

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B. References

Reference	Title
05-51-01-210-804	Phase IA Inspection (P/B 201)
09-10-00-580-801	Nose Gear Maintenance Towing (P/B 201)
12-15-41-610-801	Nose Landing Gear Shock Strut Fluid Check (P/B 301)
12-21-21-640-802	Nose Landing Gear Lower End Components Servicing (P/B 301)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
711	Nose Landing Gear Door - Left

D. Shock Strut Seal Leak Check

NOTE: If the inspection finds a fault on the scraper ring [24], you must replace the damaged scraper ring [24].

NOTE: The inner cylinder [12] must be removed from the outer cylinder [9] to replace the production scraper ring [24]. If the inner cylinder [12] cannot be removed from the outer cylinder [9] at this time, then refer to task Scraper Ring Replacement, TASK 32-21-11-960-803

SUBTASK 32-21-11-480-011

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-21-11-210-001

(2) Inspect the shock struts [4] for leakage.

SUBTASK 32-21-11-210-002

(3) If you find leakage of the shock strut fluid on the inner cylinder [12], do the steps that follow to make sure the shock strut [4] is serviceable:

(a) Do a check for leaks for 5 minutes:

1) Wipe the surface of the inner cylinder [12] with a clean cloth to remove all of the fluid from the leak.

2) Monitor the inner cylinder [12] where it touches the outer cylinder [9] for 5 minutes.

a) Make sure the extension of the inner cylinder [12] does not change during this time.

3) Make a record of the location and the number of drops which come from the seals during the 5 minutes.

(b) Tow the airplane to cause movement between the inner cylinder [12] and the outer cylinder [9] (TASK 09-10-00-580-801).

NOTE: The minimum distance the airplane must be moved is two airplane lengths.

1) In cold weather, locally heat the inner cylinder for at least 15 minutes.

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(c) Do a check for leaks for 5 minutes:

1) Wipe the surface of the inner cylinder [12] with a clean cloth to remove all of the fluid from the leak.

2) Monitor the inner cylinder [12] where it touches the outer cylinder [9] for 5 minutes.

a) Make sure the extension of the inner cylinder [12] does not change during this time.

3) Make a record of the location and the number of drops which come from the seals during the 5 minutes.

(d) Calculate the average of the number of drops which came from the seals during both of the 5 minute measurements.

SUBTASK 32-21-11-210-003

(4) If the leakage stopped, do the steps that follow:

(a) Continue to monitor the shock strut [4] for leakage on subsequent landings.

1) If you see more signs of leakage, do the entire leak check procedure again.

a) If you find a continuous leakage flow on subsequent landings, the shock strut [4] is not serviceable.

CAUTION

IF THE AIRPLANE MADE A LANDING WITH A DEFLATED OR FLAT STRUT, DO THE HARD LANDING INSPECTION. MAKE SURE THAT NO DAMAGE OCCURRED. A DEFLATED OR FLAT STRUT CAN CAUSE DAMAGE DURING A LANDING.

2) If you do not find leakage on subsequent landings, continue to examine the fluid level of the shock strut [4].

a) You must examine the fluid level in less than 3 days or 10 flight cycles, (TASK 12-15-41-610-801).

b) If the shock strut [4] has low pressure or is deflated, do this task: Phase IA Inspection, TASK 05-51-01-210-804.

SUBTASK 32-21-11-210-004

(5) If there was leakage, but the average was less than 20 drops during the two 5 minute periods (Table 801), do the steps that follow:

NOTE: 20 drops is equal to 1.0 cc.

(a) Continue to monitor the shock strut [4] for leakage on subsequent landings until you replace the seals.

1) If the leakage continues or increases, do the entire seal leak check procedure again.

CAUTION

IF THE AIRPLANE MADE A LANDING WITH A DEFLATED OR FLAT STRUT, DO THE HARD LANDING INSPECTION. MAKE SURE THAT NO DAMAGE OCCURRED. A DEFLATED OR FLAT STRUT CAN CAUSE DAMAGE DURING A LANDING.

(b) If you do not find leakage on subsequent landings, examine the fluid level of the shock strut [4] within 3 days or 10 flight cycles (TASK 12-15-41-610-801).

1) If the shock strut [4] has low pressure or is deflated, do this task: Phase IA Inspection, TASK 05-51-01-210-804.

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(c) Continue to monitor the fluid level of the shock strut [4] at no more than 3 day or 10 flight cycle intervals, until you replace the seals (TASK 12-15-41-610-801).

SUBTASK 32-21-11-210-005

(6) If the average leakage was more than 20 drops during the two 5 minute periods, but was not continuous (Table 801), do the steps that follow:

NOTE: 20 drops is equal to 1.0 cc.

CAUTION

IF THE AIRPLANE MADE A LANDING WITH A DEFLATED OR FLAT STRUT, DO THE HARD LANDING INSPECTION. MAKE SURE THAT NO DAMAGE OCCURRED. A DEFLATED OR FLAT STRUT CAN CAUSE DAMAGE DURING A LANDING.

(a) Do this task: Nose Landing Gear Shock Strut Fluid Check, TASK 12-15-41-610-801.

1) If the shock strut [4] has low pressure or is deflated, do this task: Phase IA Inspection, TASK 05-51-01-210-804.

(b) Continue to monitor the fluid level of the shock strut [4] after each flight until you replace the seals.

SUBTASK 32-21-11-210-006

(7) If the fluid leaked continuously during the 5 minute period (Table 801), the shock strut [4] is not serviceable and you must replace the shock strut seals.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 32-21-11-640-003

(1) Do this task: Nose Landing Gear Lower End Components Servicing, TASK 12-21-21-640-802.

——— END OF TASK ———

TASK 32-21-11-200-802

6. Nose Landing Gear Shock Strut Metering Pin Seal Leakage Check

(Figure 801, Figure 802, Figure 803)

A. General

(1) This task gives the instructions to a shock strut [4] is serviceable when you find fluid leakage on the lower portion of the inner cylinder [12] coming from the drain hole and collecting on the jack ball or adjacent surfaces.

NOTE: It is common, especially with elevated ambient temperatures, to have the Corrosion Inhibiting Compound (CIC) applied to the inner wall of the Inner Cylinder become less viscous and form indications on in the Inner cylinder drain hole and areas adjacent to the drain hole including the jack ball. If this occurs, indications of CIC will be thicker and have a wax like consistency. Indications fitting this description should not be considered a leak.

NOTE: This task has associated leak rates for determining action by the operator. This Task assumes that no other leaks have been noted elsewhere on the strut including static and dynamic seals, fill and charging valves, and orifice support tube static seal located at the top of the strut adjacent to the charging valve. If multiple areas of leakage are found, contact Service Engineering.

B. References

Reference	Title
05-51-01-210-804	Phase IA Inspection (P/B 201)

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(Continued)

Reference	Title
12-15-41-610-801	Nose Landing Gear Shock Strut Fluid Check (P/B 301)
12-21-21-640-802	Nose Landing Gear Lower End Components Servicing (P/B 301)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
711	Nose Landing Gear Door - Left

D. Shock Strut Metering Pin Seal Leak Check

SUBTASK 32-21-11-480-018

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-21-11-210-007

(2) Inspect the shock strut [4] for leakage.

SUBTASK 32-21-11-211-001

(3) If you find leakage of the shock strut fluid on the inner cylinder [12] drain hole and jack ball, do the steps that follow to make sure the shock strut [4] is serviceable:

(a) Do a check for leaks for 5 minutes:

1) Wipe the surface of the inner cylinder [12] drain hole and surfaces adjacent to the drain hole and jack ball with a clean cloth to remove all of the fluid from the leak.

2) Monitor the inner cylinder [12] drain hole and surfaces adjacent to the drain hole and jack ball for 5 minutes.

a) Make sure the extension of the inner cylinder [12] does not change during this time.

3) Make a record of the location and the number of drops which come from the drain hole during the 5 minutes.

(b) Do this task: Nose Landing Gear Lower End Components Servicing, TASK 12-21-21-640-802

NOTE: This step is intended to depressurize the strut and repressurize to reposition the seal contact interface with the sealing surface.

(c) Do a check for leaks for 5 minutes:

1) Wipe the surface of the inner cylinder [12] drain hole and surfaces adjacent to the drain hole and jack ball with a clean cloth to remove all of the fluid from the leak.

2) Monitor the inner cylinder [12] drain hole and surfaces adjacent to the drain hole and jack ball for 5 minutes.

a) Make sure the extension of the inner cylinder [12] does not change during this time.

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3) Make a record of the location and the number of drops which come from the drain hole during the 5 minutes.

(d) Calculate the average of the number of drops which came from the drain hole during both of the 5 minute measurements.

SUBTASK 32-21-11-211-002

(4) If the leakage stopped, do the steps that follow:

(a) Continue to monitor the shock strut [4] for leakage on subsequent landings.

1) If you see more signs of leakage, do the entire leak check procedure again.

a) If you find a continuous leakage flow on subsequent landings, the shock strut [4] is not serviceable.

CAUTION

IF THE AIRPLANE MADE A LANDING WITH A DEFLATED OR FLAT STRUT, DO THE HARD LANDING INSPECTION. MAKE SURE THAT NO DAMAGE OCCURRED. A DEFLATED OR FLAT STRUT CAN CAUSE DAMAGE DURING A LANDING.

(b) If you do not find leakage on subsequent landings, examine the fluid level of the shock strut [4] within 3 days or 10 flight cycles (TASK 12-15-41-610-801).

NOTE: If the leakage has stopped, there is no addition action necessary.

1) If the shock strut [4] has low pressure or is deflated, do this task: Phase IA Inspection, TASK 05-51-01-210-804.

SUBTASK 32-21-11-211-003

(5) If there was leakage, but the average was less than 20 drops during the two 5 minute periods (Table 801), do the steps that follow:

Table 801/32-21-11-993-801 Nose Landing Gear Strut Leakage Rates

COMPONENTS	NORMAL OPERATION LIMITS *[1]	DISPATCH LIMITS TO AVOID DELAY *[1]
Shock Strut Seals	Less than 20 drops	More than 20 drops but not continuous
Metering Pin Seal	Less than 20 drops	More than 20 drops but not continuous

*[1] There are approximately 20 drops in a cubic centimeter or 75600 drops in a gallon.

NOTE: 20 drops is equal to 1.0 cc.

(a) Continue to monitor the shock strut [4] for leakage on subsequent landings until you replace the seal.

1) If the leakage continues or increases, do the entire seal leak check procedure again.

CAUTION

IF THE AIRPLANE MADE A LANDING WITH A DEFLATED OR FLAT STRUT, DO THE HARD LANDING INSPECTION. MAKE SURE THAT NO DAMAGE OCCURRED. A DEFLATED OR FLAT STRUT CAN CAUSE DAMAGE DURING A LANDING.

(b) If you do not find leakage on subsequent landings, examine the fluid level of the shock strut [4] within 3 days or 10 flight cycles (TASK 12-15-41-610-801).

1) If the shock strut [4] has low pressure or is deflated, do this task: Phase IA Inspection, TASK 05-51-01-210-804.

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(c) Continue to monitor the fluid level of the shock strut [4] at no more than 3 day or 10 flight cycle intervals, until you replace the seal (TASK 12-15-41-610-801).

SUBTASK 32-21-11-210-008

(6) If the average leakage was more than 20 drops during the two 5 minute periods, but was not continuous (Table 801), do the steps that follow:

NOTE: 20 drops is equal to 1.0 cc.

CAUTION

IF THE AIRPLANE MADE A LANDING WITH A DEFLATED OR FLAT STRUT, DO THE HARD LANDING INSPECTION. MAKE SURE THAT NO DAMAGE OCCURRED. A DEFLATED OR FLAT STRUT CAN CAUSE DAMAGE DURING A LANDING.

(a) Do this task: Nose Landing Gear Shock Strut Fluid Check, TASK 12-15-41-610-801.

1) If the shock strut [4] has low pressure or is deflated, do this task: Phase IA Inspection, TASK 05-51-01-210-804.

(b) Continue to monitor the fluid level of the shock strut [4] after each flight until you replace the seal.

SUBTASK 32-21-11-020-031

(7) If the fluid leaked continuously during the 5 minute period (Table 801), the shock strut [4] is not serviceable and you must replace the seal assembly [104].

(a) Do this task to replace the active static seal [28], cap seal [20] and metering pin seal assembly [104]: Seals and Spare Seals Replacement, TASK 32-21-11-960-802.

(b) Do a visual check on the spare static seal [27], spare dynamic seal [26] and the scraper ring [24].

1) Replace if necessary.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 32-21-11-610-003

(1) Do this task: Nose Landing Gear Lower End Components Servicing, TASK 12-21-21-640-802.

——— END OF TASK ———

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NOSE LANDING GEAR TOW FITTING - REMOVAL/INSTALLATION

1. General

A. This procedure contains two tasks. The first task removes the tow fitting from the nose landing gear. The second task installs the tow fitting on the nose landing gear.

TASK 32-21-14-000-801

2. Nose Landing Gear Tow Fitting Removal

A. General

(1) This procedure supplies instructions to remove the nose landing gear tow fitting.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
713	Nose Landing Gear

D. Prepare for the Removal

SUBTASK 32-21-14-480-001

WARNING MAKE CERTAIN LANDING GEAR DOWNLOCK PINS ARE INSTALLED TO PREVENT INADVERTENT RETRACTION OF LANDING GEAR, WITH POSSIBLE INJURY TO PERSONNEL.

(1) Make sure the downlocks are installed on the nose and main landing gear (TASK 32-00-01-480-801).

E. Nose Landing Gear Tow Fitting Removal

SUBTASK 32-21-14-020-002

(1) Remove these parts:

(a) The nuts [9].

(b) The washers [10].

(c) The bolts [3].

(d) The washers [4].

SUBTASK 32-21-14-020-003

(2) If removal of the tow fitting bolt [1] is necessary, remove these parts:

(a) The pin [7].

(b) The nut [6].

(c) The washer [5].

(d) The bolt [1].

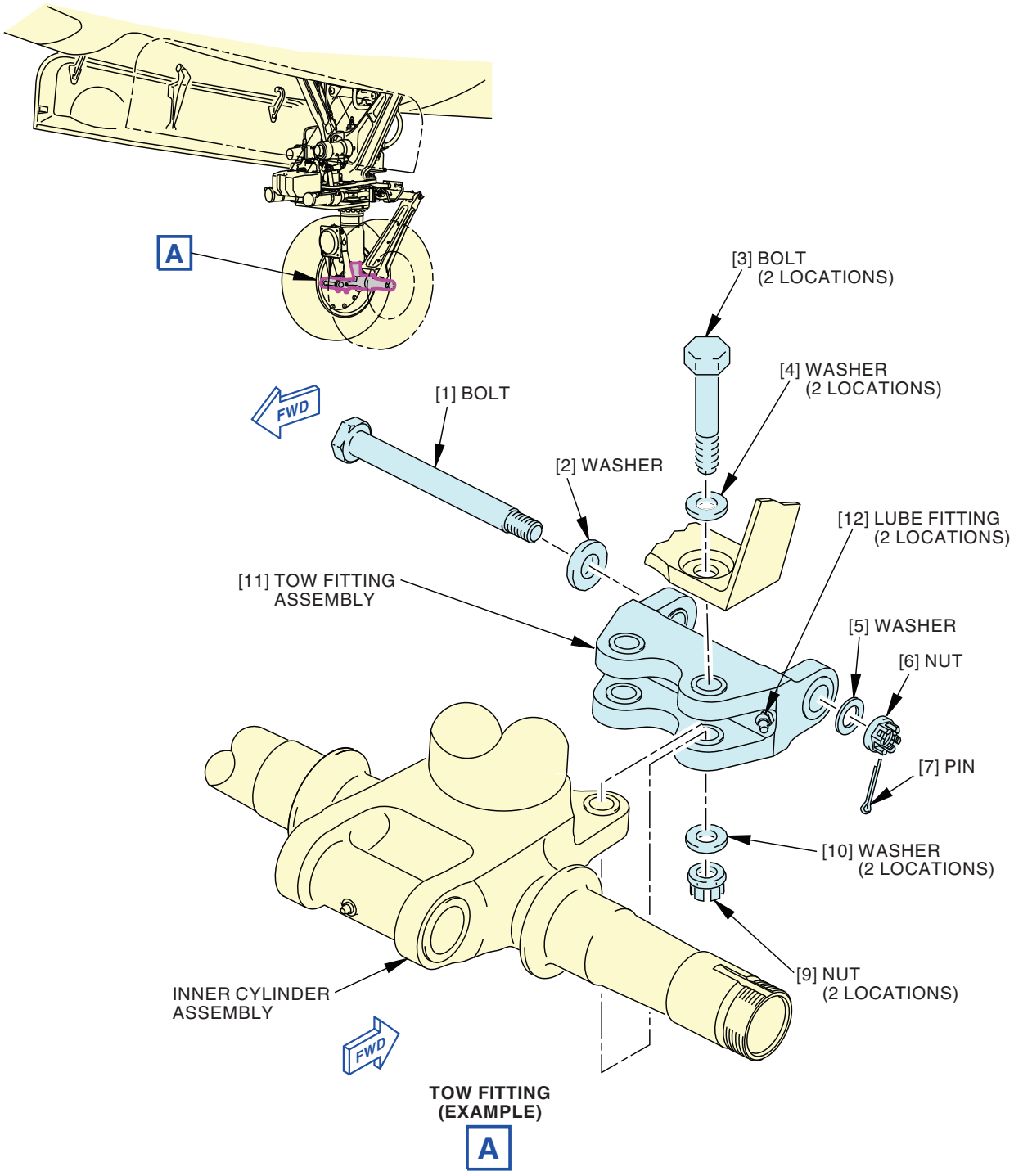
(e) The washer [2].

SUBTASK 32-21-14-020-004

(3) Remove the tow fitting assembly [11] from the inner cylinder assembly.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Landing Gear Tow Fitting Installation
Figure 401/32-21-14-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-21-14-400-801

3. Nose Landing Gear Tow Fitting Installation

A. General

(1) This procedure supplies instructions to install the tow fitting and tow fitting bolt.

B. Consumable Materials

Reference	Description	Specification
D00013	Grease - Aircraft And Instrument Grease	MIL-PRF-23827 (NATO G-354) (Supersedes MIL-G-23827)
D00633	Grease - Aircraft General Purpose	BMS3-33
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
7	Pin	32-21-00-01-040	PTF ALL
11	Tow fitting assembly	32-21-00-01-070	PTF ALL

D. Location Zones

Zone	Area
713	Nose Landing Gear

E. Nose Gear Tow Fitting and Tow Fitting Bolt Installation

SUBTASK 32-21-14-916-001

(1) If the tow fitting assembly [11] installation is necessary, do these steps:

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin coat of corrosion inhibiting material, G50136 to the shank of the unplated bolt [1], the bolt threads and thread reliefs, and to the faces of washer [2] and washer [5] prior to installation. Wipe off the excess.

(b) Insert the tow fitting bolt [1] into the washer [2] and then into the tow fitting assembly [11].

(c) Install the washer [5] and the nut [6] to connect the tow fitting bolt [1] to the tow fitting assembly [11].

(d) Tighten the nut [6] to 160 in-lb (18 N·m) – 190 in-lb (21 N·m) above run-on torque and back off to align the nearest castellation.

(e) Install the pin [7].

(f) Lubricate the two lube fittings [12] with grease, D00633, or alternate grease, D00013.

SUBTASK 32-21-14-916-002

(2) Do the steps that follow to install the tow fitting assembly [11] on the inner cylinder assembly:

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WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin coat of corrosion inhibiting material, G50136 to the shank of the unplated bolts [3], the bolt threads and bolt reliefs, and to the faces of the washers [4] and washers [10] prior to installation. Wipe off the excess corrosion inhibiting material, G50136.

(b) Hold the tow fitting assembly [11] in position.

(c) Install the bolts [3] into the washers [4] and then into the tow fitting assembly [11].

(d) Install the washers [10] and nuts [9] to the bolts [3].

(e) Install self-locking nuts [9] and tighten the nuts [9] to 250 in-lb (28 N·m) – 300 in-lb (34 N·m) above the run-on torque.

NOTE: If a BACN10JC self-locking nut is used on a bolt that has a hole for a cotter pin, it is recommended that the nut not be used again after removal. The edges of the hole will make the self-locking feature unserviceable.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE LANDING GEAR DRAG STRUT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the drag strut for the nose landing gear

(2) Installation of the drag strut for the nose landing gear.

TASK 32-21-21-000-801

2. Nose Landing Gear Drag Strut Removal

(Figure 401, Figure 402)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
07-10-03-580-803	Lift the Nose Landing Gear Axle With Axle Jacks-Handling (P/B 201)
07-10-21-580-801	Lift the Airplane Nose with the Nose Jack at Jack Point D (P/B 201)
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-33-52-000-801	Nose Gear Lock Spring Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1871	Retention Straps - Shock Strut, NLG/MLG Part #: C32030 -31 Supplier: 81205 Opt Part #: C32030-10 Supplier: 81205
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)
STD-15078	R/I Equipment - Trunnion Pin, NLG (Note: C32054-2 Puller Assembly is included in SPL-14481 kit)

C. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
713	Nose Landing Gear

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D. Prepare for the Removal

SUBTASK 32-21-21-420-010

WARNING

MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-21-21-580-001

(2) Do this task: Lift the Airplane Nose with the Nose Jack at Jack Point D, TASK 07-10-21-580-801.

NOTE: Lower the jack that supports the tail of the airplane while you lift the nose landing gear.

SUBTASK 32-21-21-580-002

(3) If it is necessary to lift the gear with an axle jack to reduce the weight of the gear when you remove the right trunnion pin [1] and left trunnion pin [2], do this task: Lift the Nose Landing Gear Axle With Axle Jacks-Handling, TASK 07-10-03-580-803.

SUBTASK 32-21-21-860-001

(4) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-21-21-860-002

(5) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	17	C00129	LANDING GEAR LATCH & PRESS WARN
C	15	C01355	LANDING GEAR AIR/GND SYS 2
C	16	C01356	LANDING GEAR AIR/GND SYS 1
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN
D	15	C01401	LANDING GEAR AIR/GND RELAY
D	16	C01432	LANDING GEAR ALTN EXTEND SOL
D	17	C01027	LANDING GEAR NOSE GEAR STEER
D	18	C00451	LANDING GEAR AURAL WARN

SUBTASK 32-21-21-480-002

(6) Install chocks around the tires of all the landing gear (PAGEBLOCK 10-11-05/201).

SUBTASK 32-21-21-020-001

(7) Do this task: Nose Gear Lock Spring Removal, TASK 32-33-52-000-801.

SUBTASK 32-21-21-080-001

(8) Remove the downlock pin for the nose landing gear (TASK 32-00-01-080-801).

E. Nose Landing Gear Drag Strut Removal

SUBTASK 32-21-21-020-002

(1) If it is necessary, disconnect the hydraulic hose connection from the rod end of the retract actuator [3]:

NOTE: These steps are only necessary if the retract actuator will be removed.

(a) Disconnect the hydraulic UP-hose at the fitting on the retract actuator [3].

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(b) Disconnect the hydraulic DOWN-hose at the reducer [4] and tube [8].

1) If it is necessary, remove the clamp [7], screws [5], and washers [6] from the bracket.

(c) Install the caps/plugs, STD-14686, on the hydraulic hoses and the fittings on the retract actuator [3].

SUBTASK 32-21-21-020-003

(2) Disconnect the rod end of the retract actuator [3]:

(a) Remove the nut [12] and washer [14] from each of the bolts [16].

(b) Remove the bolts [16] from the nut [15].

(c) Remove the nut [15] from the pin [9].

(d) Remove the pin [9] that holds the rod end of the retract actuator [3] and the washer [18] or washers [18] from the upper drag strut [17].

SUBTASK 32-21-21-020-004

(3) Disconnect the forward lock link [20]:

NOTE: This step will also disconnect the lower drag strut from the upper drag strut.

(a) Remove the pin [13] from the bolt [22].

(b) Remove the nut [25] and washer [24] from the bolt [22].

(c) Remove the bolt [22] from the pin [19].

(d) Remove the nut [23] and washer [21] from the pin [19].

(e) Remove the pin [19] to disconnect the forward lock link [20] from the upper drag strut [17] and lower drag strut [10].

SUBTASK 32-21-21-020-005

WARNING FULLY EXTEND THE SHOCK STRUT BEFORE YOU DISCONNECT THE DRAG BRACE. IF IT IS NOT FULLY EXTENDED, THE COMPRESSED SHOCK STRUT CAN CAUSE INJURIES.

(4) Disconnect the lower drag strut [10] from the shock strut [29]:

(a) Remove the pin [26] from the bolt [28].

(b) Remove the nut [33] and washer [32] from the bolt [28].

(c) Remove the bolt [28] from the pin [30].

(d) Remove the nut [27] and washer [31] from the pin [30].

(e) Remove the pin [30] that holds the lower drag strut [10] from the shock strut [29].

SUBTASK 32-21-21-480-003

(5) Use the retention strap, SPL-1871, to hold the shock strut [29] in the compressed position.

SUBTASK 32-21-21-020-006

(6) Disconnect the upper drag strut [17] from the bracket [43]:

NOTE: This step will also remove the bracket, that contains the down position sensor target. Keep the bracket for the installation procedure.

(a) Remove the nut [34] and washer [35] from the bolt [36].

(b) Remove the bolt [36] and washer [35] to disconnect the upper drag strut [17] from the bracket [43].

(c) Remove the washer [35] from the bolt [36].

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SUBTASK 32-21-21-020-007

(7) Remove the right trunnion pin [1] and left trunnion pin [2]:

(a) Remove the pin [38] from each bolt [40].

(b) Remove the nut [37] and washer [39] from each bolt [40].

(c) Remove the bolts [40] that hold the right trunnion pin [1] and left trunnion pin [2] to the sidewall of the nose wheel well.

(d) Remove the right trunnion pin [1] and left trunnion pin [2] to disconnect the upper drag strut [17] from the sidewall of the nose wheel well.

1) Use the puller assembly, STD-15078, to remove the right trunnion pin [1] and left trunnion pin [2] from the inboard side of the upper drag strut [17].

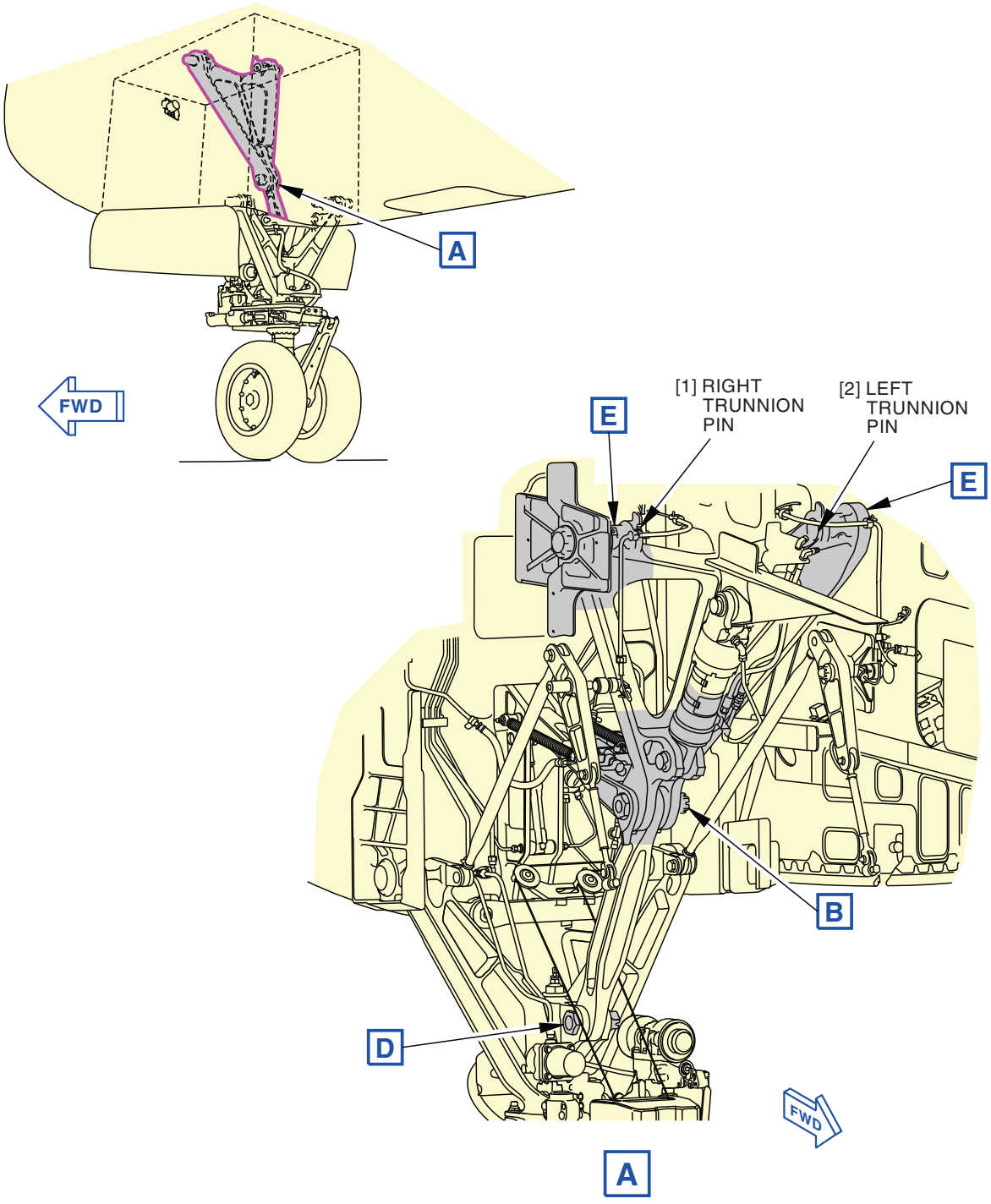
(e) Remove and discard the O-rings [42].

SUBTASK 32-21-21-020-008

(8) Remove the upper drag strut [17] from the airplane.

——— END OF TASK ———

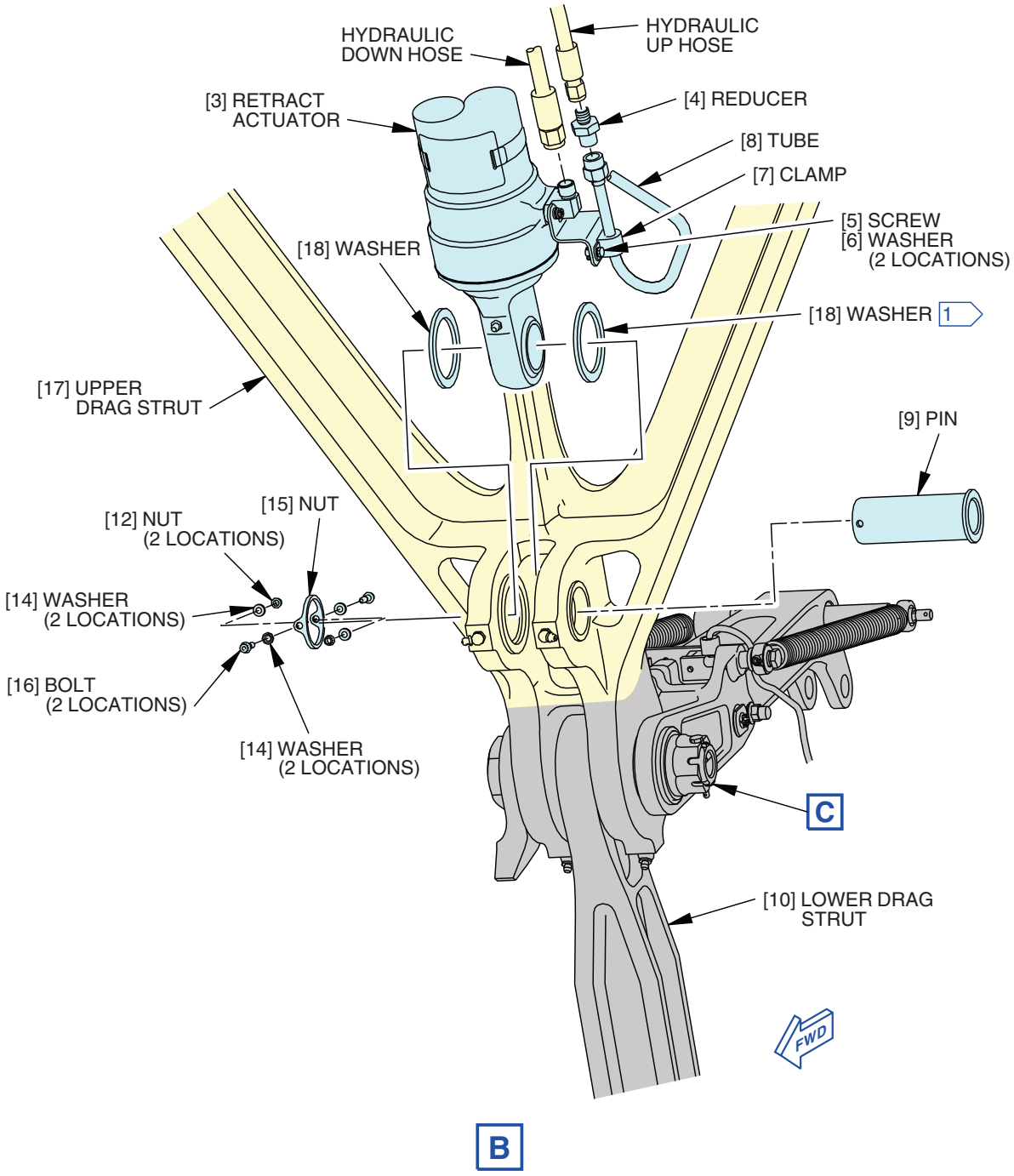
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414177 S00061534549_V1

Nose Landing Gear Drag Strut Installation
Figure 401/32-21-21-990-801 (Sheet 1 of 5)

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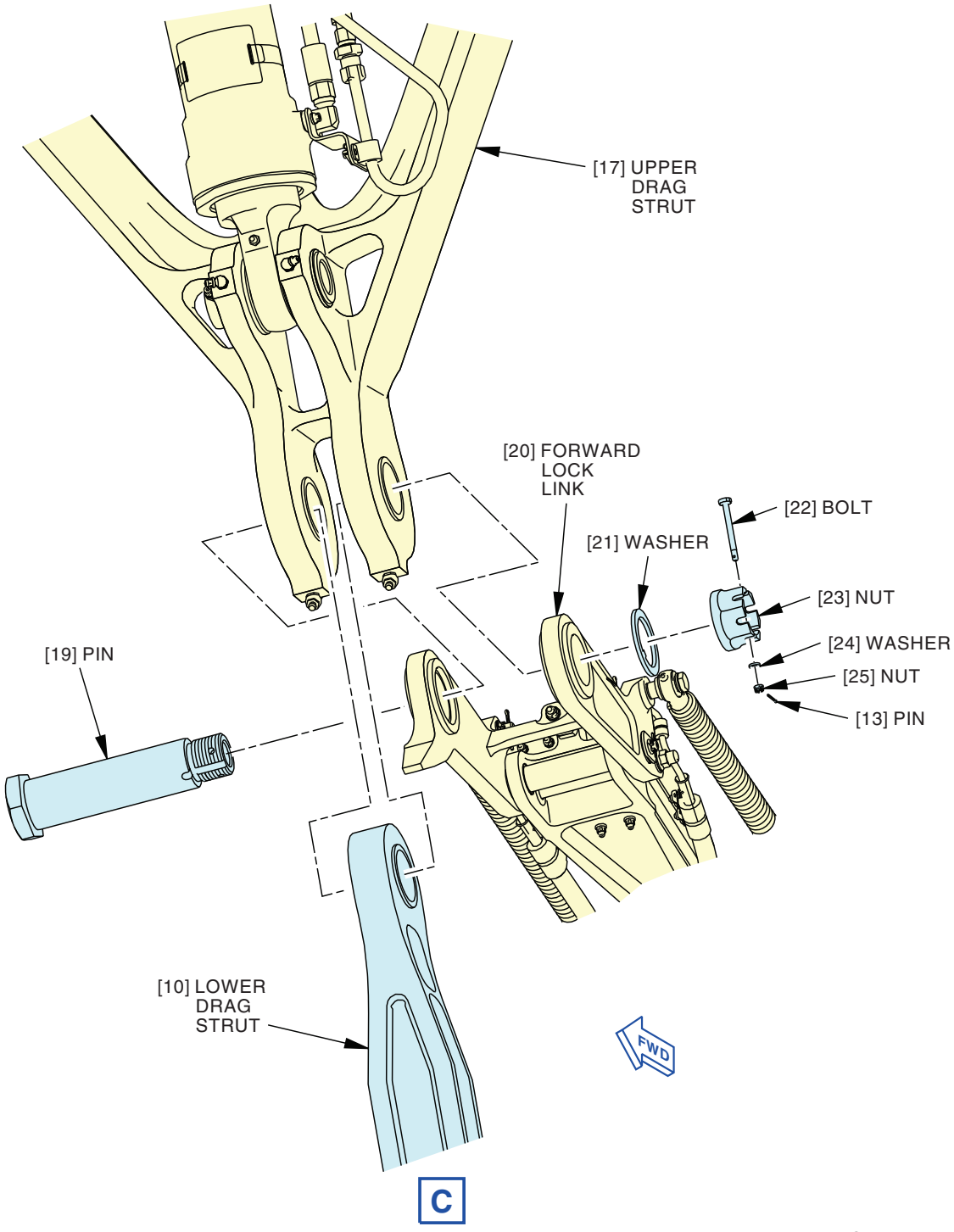


1 NOT INSTALLED ON ALL AIRCRAFT

2414178 S00061534550_V2

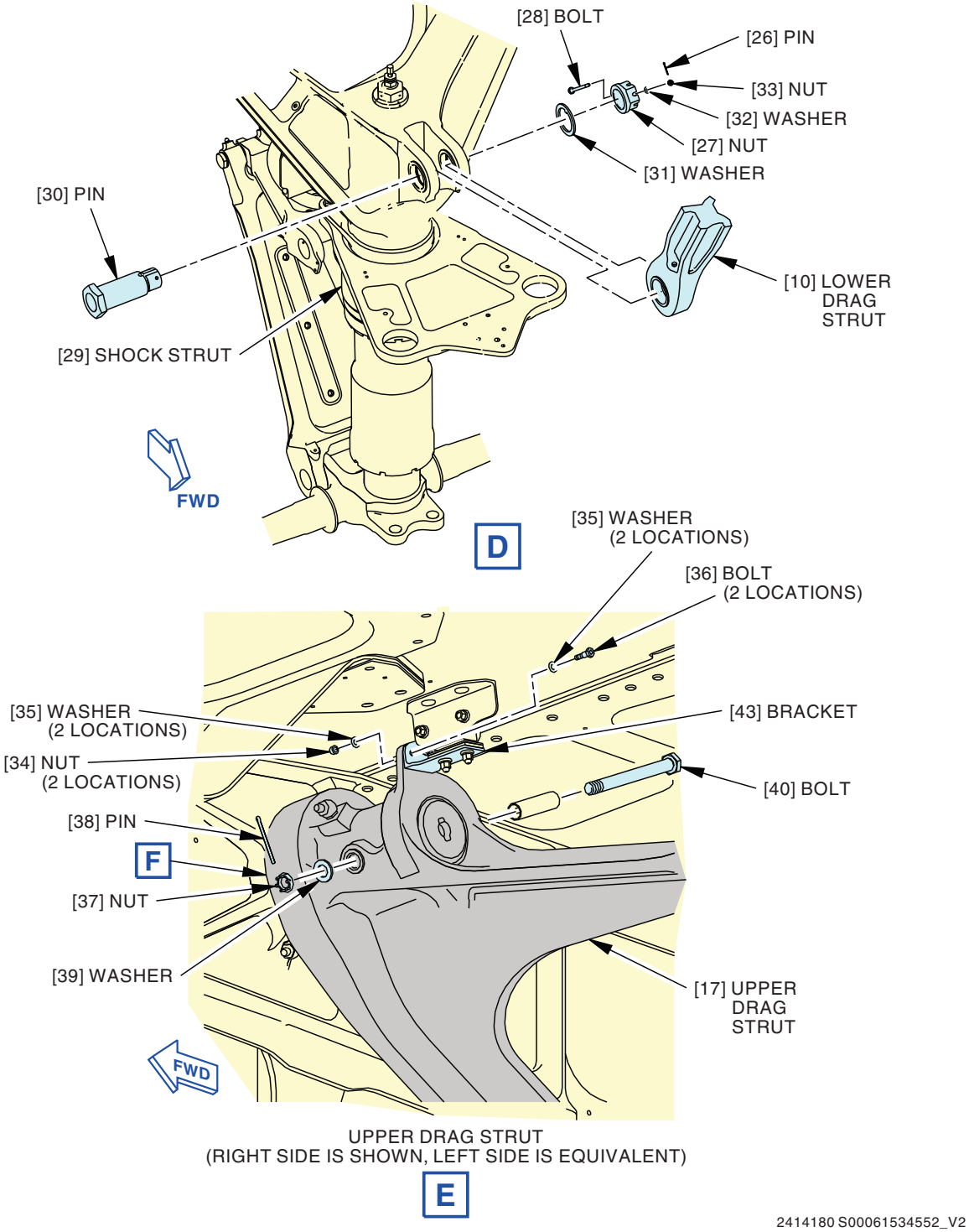
Nose Landing Gear Drag Strut Installation
Figure 401/32-21-21-990-801 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



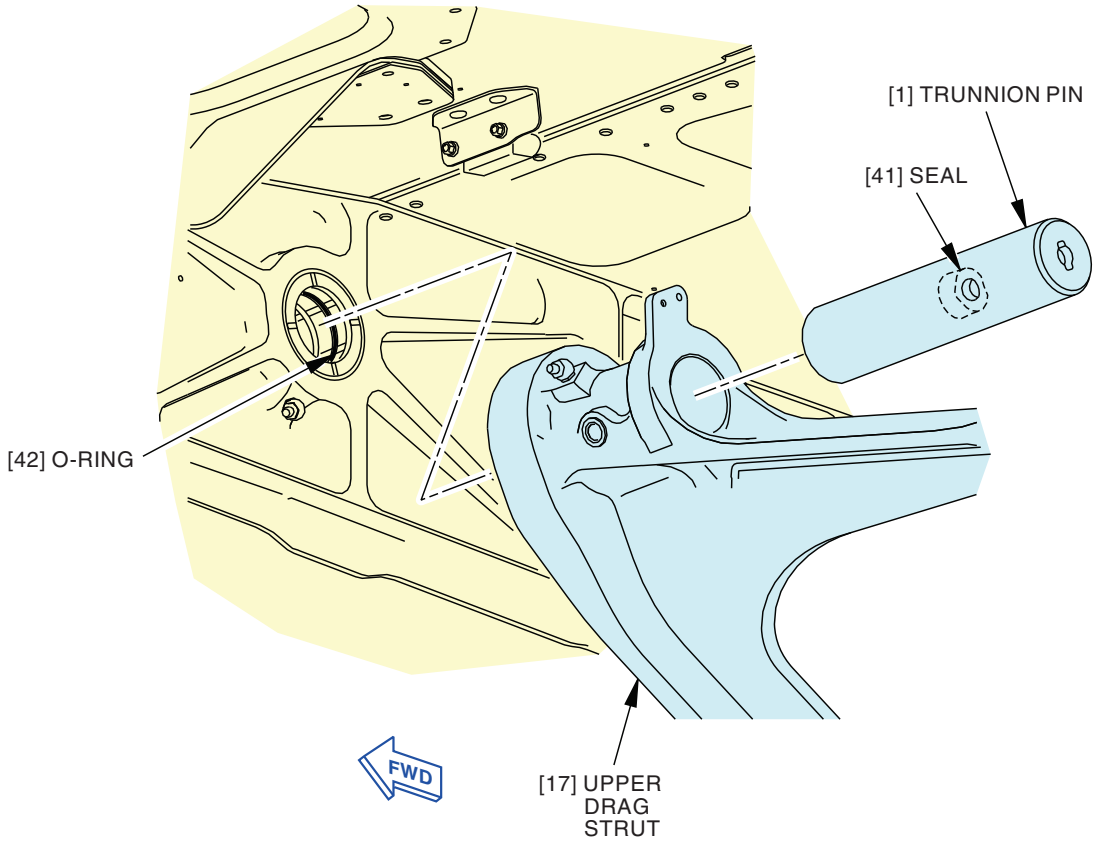
Nose Landing Gear Drag Strut Installation
Figure 401/32-21-21-990-801 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Landing Gear Drag Strut Installation
Figure 401/32-21-21-990-801 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



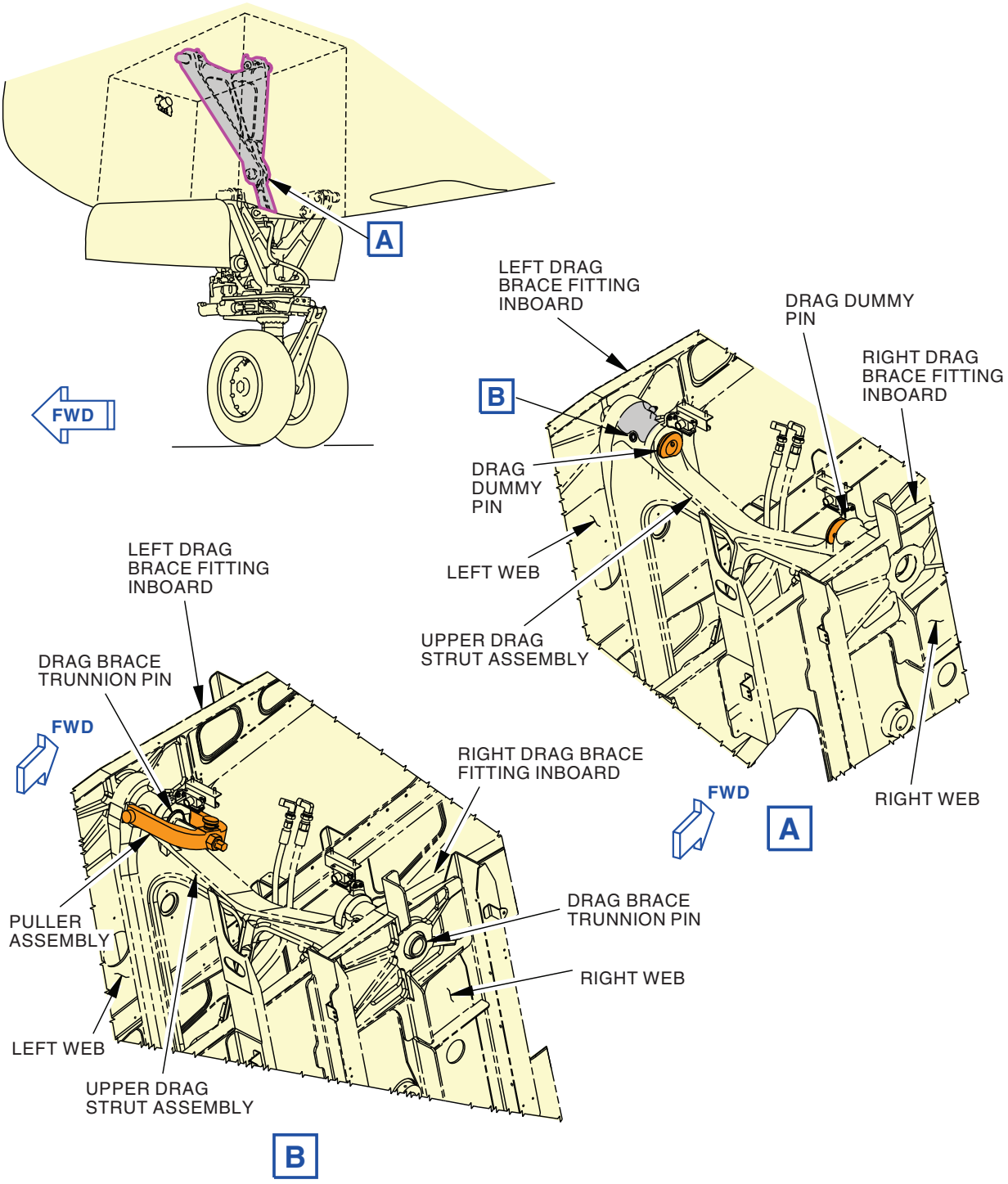
UPPER DRAG STRUT
(RIGHT SIDE IS SHOWN, LEFT SIDE IS EQUIVALENT)

F

2414181 S00061534553_V2

Nose Landing Gear Drag Strut Installation
Figure 401/32-21-21-990-801 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414182 S00061534554_V3

Drag Strut Trunnion Pin Removal and Installation Equipment
Figure 402/32-21-21-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-21-21-400-801

3. Nose Landing Gear Drag Strut Installation

(Figure 401, Figure 402)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
07-10-21-580-802	Lower the Airplane Nose, Jack Point D (P/B 201)
12-21-21-640-801	Nose Landing Gear Upper End Components Servicing (P/B 301)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
32-33-00-710-802	Nose Landing Gear Operational Test (P/B 501)
32-33-52-400-801	Nose Gear Lock Spring Installation (P/B 401)
32-35-00-730-801	Nose Gear Manual Extension System Test - Airplane on Jacks (P/B 501)
32-51-00-700-801	Nose Wheel Steering System Test (Airplane on Ground) (P/B 501)
32-61-51-400-801	Nose Landing Gear Down Position Sensor Installation (P/B 401)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)
STD-15084	R/I Equipment - Trunnion Pin, NLG (Note: C32054-9 Drag Dummy Pin is included in SPL-14481 kit)

C. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50346	Compound - Corrosion Preventive	BMS3-26 Type II

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
10	Lower drag strut	32-21-21-01-175	PTF ALL
17	Upper drag strut	32-21-21-01-150	PTF ALL
42	O-ring	32-21-00-02-070	PTF ALL

E. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
713	Nose Landing Gear

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Installation

SUBTASK 32-21-21-620-001

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(1) Apply a thin layer of corrosion inhibiting material, G50136, to these items:

(a) The shank and threads of the bolts [40]

1) Apply a thick layer of corrosion inhibiting material, G50136, to the bolts [40].

(b) The threads and thread reliefs of the nuts [37]

(c) The faces of the washers [39]

(d) The pins [38].

SUBTASK 32-21-21-160-001

(2) Remove all unwanted corrosion inhibiting material, G50136.

SUBTASK 32-21-21-640-001

(3) Lubricate the chrome plated surfaces of the right trunnion pin [1] and left trunnion pin [2] with grease, D00633.

SUBTASK 32-21-21-160-002

(4) Remove all unwanted grease, D00633.

G. Nose Landing Gear Drag Strut Installation

SUBTASK 32-21-21-560-001

(1) Put the upper drag strut [17] in its position for installation on the sidewall of the nose wheel well.

SUBTASK 32-21-21-580-003

(2) Align the upper drag strut [17] with the mating hole in the sidewall of the nose wheel well.

SUBTASK 32-21-21-420-001

(3) Install the right trunnion pin [1] and left trunnion pin [2]:

(a) Make sure that the seal [41] is installed in the right trunnion pin [1] and left trunnion pin [2].

(b) Apply grease, D00633, to the new O-rings [42].

(c) Install the O-rings [42] inside nose wheel well trunnion bushing.

(d) Use the drag dummy pins, STD-15084, to align the upper drag strut [17] and sidewall of the nose wheel well (Figure 402).

(e) Put the right trunnion pin [1] and left trunnion pin [2] through the upper drag strut [17] and sidewall of the nose wheel well.

(f) Put the bolts [40] through the right trunnion pin [1] and left trunnion pin [2].

(g) Install the washer [39] and nut [37] on each bolt [40].

(h) Tighten each nut [37] to 20 in-lb (2.3 N·m) - 24 in-lb (2.7 N·m) more than the run-on torque.

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(i) If it is necessary, loosen the nut [37] to the nearest castellation to align the holes for each pin [38].

(j) Install the pin [38] in each bolt [40].

(k) Turn the seal [41] as necessary to fully seat the seal [41] against the crossbolt.

CAUTION

WEAR SAFETY GLASSES DURING APPLICATION, AND IF APPLICATION OF SOLVENT RESULTS IN IRRITATION, USE AN AIR PURIFYING ORGANIC VAPOR CARTRIDGE RESPIRATOR. USE NITRILE GLOVES AND WASH YOUR HANDS WITH MOISTURE CREAM AFTER APPLICATION. STORE DPM 6380-4 CLEANER/SOLVENT IN FLAMMABLE RESISTANT CONTAINERS IN QUANTITIES NO LARGER THAN 1 GALLON.

(l) Reapply compound, G50346, if it is necessary, to coat the interior surface of the right trunnion pin [1] and left trunnion pin [2].

SUBTASK 32-21-21-420-002

(4) Connect the upper drag strut [17] to the bracket [43]:

(a) Put the washer [35] on each bolt [36].

(b) Align the hole in the upper drag strut [17] to the hole in the bracket [43].

(c) Put the bolts [36] through the bracket [43] and upper drag strut [17].

(d) Install the washer [35] and nut [34] on each bolt [36].

SUBTASK 32-21-21-620-002

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(5) Apply a thin layer of corrosion inhibiting material, G50136, to these items:

(a) The threads and thread reliefs of the pin [30] and bolt [28]

(b) The threads of the nut [27] and nut [33]

(c) The faces of the washer [31] and washer [32]

(d) The pin [26].

SUBTASK 32-21-21-160-003

(6) Remove all unwanted corrosion inhibiting material, G50136.

SUBTASK 32-21-21-640-002

(7) Lubricate the shank of the pin [30] and bolt [28] with grease, D00633.

SUBTASK 32-21-21-160-004

(8) Remove all unwanted grease, D00633.

SUBTASK 32-21-21-420-003

(9) Connect the lower drag strut [10]:

(a) Make sure that the drag brace lubrication fittings are on the right side when looking forward.

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(b) Put the lower drag strut [10] in its position on the shock strut [29].

(c) Put the pin [30] through the shock strut [29] and lower drag strut [10].

(d) Install the washer [31] and nut [27] on the pin [30].

(e) Tighten the nut [27] to 650 ±50 in-lb (73.4 ±5.6 N·m).

(f) Loosen the nut [27] to the nearest castellation to align the holes for the bolt [28].

(g) Install the bolt [28] in the pin [30].

(h) Install the washer [32] on the bolt [28].

(i) Install the nut [33] on the bolt [28].

(j) If it is necessary, loosen the nut [33] to the nearest castellation to align the holes for the pin [26].

(k) Install the pin [26] in the bolt [28].

SUBTASK 32-21-21-620-003

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(10) Apply a thin layer of corrosion inhibiting material, G50136, to these items:

(a) The threads and thread reliefs of the pin [19] and bolt [22]

(b) The threads of the nut [23] and nut [25]

(c) The faces of the washer [21] and washer [24]

(d) The pin [13].

SUBTASK 32-21-21-160-005

(11) Remove all unwanted corrosion inhibiting material, G50136.

SUBTASK 32-21-21-420-009

(12) If a nut and bolt is installed in the downlock pin hole, remove it and install a downlock pin in that location.

SUBTASK 32-21-21-420-004

(13) Connect the forward lock link [20] and lower drag strut [10]:

NOTE: This step will also connect the lower drag strut to the upper drag strut.

(a) Put the lower drag strut [10] and forward lock link [20] in their positions on the upper drag strut [17].

(b) Put the pin [19] through the forward lock link [20], upper drag strut [17], and lower drag strut [10].

(c) Install the washer [21] and nut [23] on the pin [19].

I (d) Tighten the nut [23] to 95 in-lb (10.7 N·m) - 115 in-lb (13.0 N·m).

(e) Loosen the nut [23] to the nearest castellation to align the hole for the bolt [22].

(f) Install the bolt [22] in the pin [19].

(g) Install the washer [24] and nut [25] on the bolt [22].

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(h) If it is necessary, loosen the nut [25] to the nearest castellation to align the holes for the pin [13].

(i) Install the pin [13] in the bolt [22].

SUBTASK 32-21-21-420-005

(14) If the retract actuator [3] was removed, hold the retract actuator [3] and do these steps to connect the retract actuator [3] to the upper drag strut [17]:

(a) Put the rod end of the retract actuator [3] and washer [18] or washers [18] in their position on the upper drag strut [17].

(b) Put the pin [9] through the upper drag strut [17], rod end of the retract actuator [3] and washer [18] or washers [18].

(c) Install the nut [15] on the pin [9].

(d) Install the bolts [16] , washers [14], and nuts [15] in the pin [9].

SUBTASK 32-21-21-080-002

(15) If they were installed, remove the caps/plugs, STD-14686, from the hydraulic hoses.

SUBTASK 32-21-21-420-006

(16) If the retract actuator [3] was removed, connect the hydraulic hoses to the retract actuator [3].

(a) Tighten the B-nut on the hydraulic DOWN-hose to 270 ±14 in-lb (30.5 ±1.5 N·m).

(b) Tighten the B-nut on the hydraulic UP-hose to 140 ±7 in-lb (15.8 ±0.8 N·m).

(c) If it is necessary for the hydraulic UP-hose, install the clamp [7], screws [5], and washers [6] at the bracket to attach the tube [8] and reducer [4].

SUBTASK 32-21-21-720-001

(17) Do a post installation check of the nose landing gear down position sensor (TASK 32-61-51-400-801).

NOTE: This step is necessary because the bracket, that contains the down position sensor target, was removed.

H. Nose Landing Gear Drag Strut Installation Test

SUBTASK 32-21-21-420-007

(1) Do this task: Nose Gear Lock Spring Installation, TASK 32-33-52-400-801.

SUBTASK 32-21-21-610-001

(2) Lubricate the upper components of the nose landing gear (TASK 12-21-21-640-801).

SUBTASK 32-21-21-860-003

(3) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	17	C00129	LANDING GEAR LATCH & PRESS WARN
C	15	C01355	LANDING GEAR AIR/GND SYS 2
C	16	C01356	LANDING GEAR AIR/GND SYS 1
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN
D	15	C01401	LANDING GEAR AIR/GND RELAY
D	16	C01432	LANDING GEAR ALTN EXTEND SOL
D	17	C01027	LANDING GEAR NOSE GEAR STEER
D	18	C00451	LANDING GEAR AURAL WARN

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SUBTASK 32-21-21-860-004

(4) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-21-21-710-001

(5) Do this task: Nose Landing Gear Operational Test, TASK 32-33-00-710-802.

SUBTASK 32-21-21-710-002

(6) Do this task: Nose Gear Manual Extension System Test - Airplane on Jacks, TASK 32-35-00-730-801.

SUBTASK 32-21-21-710-003

(7) Do this task: Nose Wheel Steering System Test (Airplane on Ground), TASK 32-51-00-700-801.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 32-21-21-860-005

(1) Do this task: Lower the Airplane Nose, Jack Point D, TASK 07-10-21-580-802.

——— END OF TASK ———

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NOSE LANDING GEAR DRAG STRUT - INSPECTION/CHECK

1. General

A. This task is a visual inspection of the upper drag strut of the nose landing gear.

B. This task does not remove or install the drag strut. Refer to NOSE LANDING GEAR DRAG STRUT - REMOVAL/INSTALLATION, PAGEBLOCK 32-21-21/401, for procedures to remove and install the drag strut of the nose landing gear.

TASK 32-21-21-200-801

2. Nose Landing Gear Drag Strut Inspection

A. General

(1) This procedure give wear limits and repair data for the drag strut of the nose landing gear.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-21-21-000-801	Nose Landing Gear Drag Strut Removal (P/B 401)

C. Tools/Equipment

Reference	Description
STD-1107	Gauge - Feeler, 0.0 - 0.5 Inch, Readable to 1/1000th

D. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
713	Nose Landing Gear

E. Nose Landing Gear Drag Strut Inspection

SUBTASK 32-21-21-480-004

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-21-21-020-009

(2) Examine the upper drag brace bushings for these forms of damage:

NOTE: It is not necessary to remove the nose landing gear drag brace strut assembly for a visual inspection for these conditions:

(a) corrosion

(b) cracks in bushing or fillet seal

(c) scratches along the rim of the bushing

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SUBTASK 32-21-21-410-001

(3) Use a 0.0 - 0.5 Inch feeler gauge, STD-1107, to do a check for 0.156 in. (3.962 mm) to 0.200 in. (5.080 mm) clearance between the structure support bushings and the nose landing gear drag strut upper trunnion bushings assemblies.

NOTE: To have satisfactory results the 0.0 - 0.5 Inch feeler gauge, STD-1107, must show smooth transition while being installed or removed.

(a) If the 0.0 - 0.5 Inch feeler gauge, STD-1107, can be installed and removed without force, the check is satisfactory.

(b) If the check is satisfactory, put the airplane back to its usual condition.

(c) If the bushing shows damage or if smooth transition of the 0.0 - 0.5 Inch feeler gauge, STD-1107, can not be satisfied, do this task: Nose Landing Gear Drag Strut Removal, TASK 32-21-21-000-801.

SUBTASK 32-21-21-200-001

(4) Refer to SRM SUBJECT 53-10-90 for the inspection and removal of damage to the nose landing gear drag strut fittings.

——— END OF TASK ———

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NLG SHOCK STRUT INNER CYLINDER - REMOVAL/INSTALLATION

1. General

A. This procedure has these two tasks:

(1) Removal of the inner cylinder for the nose landing gear

(2) Installation of the inner cylinder for the nose landing gear

TASK 32-21-24-000-801

2. NLG Inner Cylinder Removal

(Figure 401)

A. References

Reference	Title
07-10-21-580-801	Lift the Airplane Nose with the Nose Jack at Jack Point D (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-21-11-960-802	Seals and Spare Seals Replacement (P/B 801)
32-21-14-000-801	Nose Landing Gear Tow Fitting Removal (P/B 401)
32-21-31-000-802	Nose Landing Gear Lower Torsion Link Removal (P/B 401)
32-45-21-000-801	Nose Landing Gear Wheel and Tire Assembly Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1376	Scraper Plunger, C32016-2 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-1377	Scraper Reducer, C32016-3 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-1378	Scraper Retainer, C32016-4 or -67 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-1379	Scraper Lock Assy, C32016-5 or -68 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-1829	Oil Drain Valve - Shock Strut, LG Part #: J32108-16 Supplier: 81205 Opt Part #: A32066-1 Supplier: 81205

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(Continued)

Reference	Description
SPL-1867	Wrench - Gland Nut, Shock Strut NLG Part #: C32025-7 Supplier: 81205 Opt Part #: C32025-1 Supplier: 81205
SPL-1895	Removal/Installation Equipment - Metering Pin and Retainer Ring, NLG Part #: C32035-1 Supplier: 81205
SPL-3927	Strap Assy, C32017-60 (included in C32017 Eqpt) Part #: C32017-67 Supplier: 81205 Opt Part #: C32017-52 Supplier: 81205 Opt Part #: C32017-53 Supplier: 81205
SPL-4047	Seal Puller Assy, C32016-6 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
STD-1110	Container - Hydraulic Fluid Resistant, 5 Gallon (19 Liter)
STD-3906	Mallet - Rubber
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

D. Prepare for the Removal

SUBTASK 32-21-24-480-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-21-24-860-001

(2) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-21-24-020-001

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT ARE AWAY FROM THE AREA BELOW THE NOSE BEFORE YOU DEFLATE THE SHOCK STRUT. WHEN YOU DEFLATE THE SHOCK STRUT, THE NOSE CAN MOVE DOWN. THIS CAN CAUSE INJURIES TO PERSONNEL OR DAMAGE TO EQUIPMENT.

(3) Deflate the shock strut [4] for the nose landing gear:

(a) Remove the cap [1] for the gas charging valve [2] which is on the top of the shock strut [4].

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WARNING DO NOT REMOVE THE VALVE BODY UNTIL YOU DEFLATE THE SHOCK STRUT FULLY. THE AIR PRESSURE CAN BLOW THE VALVE BODY OUT AND CAUSE INJURIES TO PERSONNEL.

(b) Loosen the swivel nut [3] a maximum of two turns.

1) Deflate the shock strut [4] slowly to prevent the leakage of the fluid through the gas charging valve [2].

NOTE: Fluid in the shock strut [4] can have bubbles when you release the pressure.

(c) Loosen the swivel nut [3] fully when all of the pressure in the shock strut [4] is released.

NOTE: The shock strut [4] is fully deflated when the dimension "X" is13.8 in. (350.5 mm).

SUBTASK 32-21-24-680-001

(4) Drain the hydraulic fluid from the shock strut [4]:

NOTE: Very little fluid will come out.

(a) Remove the cap [5] from the oil charging valve [6].

(b) Put a 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110, in a position to catch the hydraulic fluid when the oil charging valve [6] is opened.

(c) Install the drain equipment on the oil charging valve [6]:

NOTE: The tool, landing gear shock strut drain valve, SPL-1829, could also be used, but it will take much longer to drain the shock strut [4].

1) Cut a length of plastic tubing, long enough to reach the container on the floor.

2) Insert a small Allen wrench in the end of the length of tubing, such that the long end of the Allen wrench is flush with the end of the tube and the short end penetrates the wall of the tube.

3) Install the tubing on the oil charging valve [6] such that the Allen wrench goes into the check valve and holds it open to drain the hydraulic fluid.

(d) Remove the drain equipment when you have removed all of the hydraulic fluid.

SUBTASK 32-21-24-580-001

(5) Lift the nose of the airplane until the shock strut [4] extends approximately 10 in. (254 mm) (TASK 07-10-21-580-801).

SUBTASK 32-21-24-020-002

(6) Remove the tow fitting (TASK 32-21-14-000-801).

SUBTASK 32-21-24-020-003

(7) Remove the NLG wheel and tire (TASK 32-45-21-000-801).

SUBTASK 32-21-24-020-004

(8) Do this task: Nose Landing Gear Lower Torsion Link Removal, TASK 32-21-31-000-802.

E. NLG Inner Cylinder Removal

(Figure 401)

SUBTASK 32-21-24-020-005

(1) Disconnect the gland nut [18] from the outer cylinder [9] (Figure 401 (Sheet 3)):

(a) Remove the bolt [15], washer [16], and lockplate [17] to release the gland nut [18].

(b) Use the gland nut wrench, SPL-1867, to remove the gland nut [18] from the outer cylinder [9].

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1) Keep the gland nut [18] in its position against the outer cylinder [9] after you have removed it from the outer cylinder [9].

SUBTASK 32-21-24-580-002

(2) Lift the nose of the airplane until there is enough room to install the scraper retainer, C32016-4 or -67, SPL-1378 (scraper retainer [61]), and the scraper lock assembly, C32016-5 or -68, SPL-1379 (scraper lock assembly [56]), on the gland nut [18] (TASK 07-10-21-580-801) (Figure 401 (Sheet 7)).

SUBTASK 32-21-24-480-002

(3) Install the scraper retainer, C32016-4 or -67, SPL-1378 (scraper retainer [61]), and the scraper lock assembly, C32016-5 or -68, SPL-1379 (scraper lock assembly [56]), on the gland nut [18] (Figure 401 (Sheet 7)).

NOTE: This will make sure the scraper ring [21] for the gland nut [18] does not come out of its groove.

(a) Install the scraper retainer, C32016-4 or -67, SPL-1378 (scraper retainer [61]), on the gland nut [18].

(b) Install the scraper lock assembly, C32016-5 or -68, SPL-1379 (scraper lock assembly [56]), to hold the scraper retainer, C32016-4 or -67, SPL-1378 (scraper retainer [61]), in its position on the gland nut [18].

SUBTASK 32-21-24-020-006

(4) Move the gland nut [18] down at the inner cylinder [14] (Figure 401 (Sheet 7)).

(a) If the scraper ring [21] comes out of its groove, use the scraper plunger, C32016-2, SPL-1376 (scraper plunger [62]), and the scraper reducer, C32016-3, SPL-1377 (scraper reducer [63]), to install the scraper ring [21] back in its groove.

SUBTASK 32-21-24-020-007

(5) Move the lower bearing carrier [20] (Figure 401 (Sheet 6)):

(a) Loosely install the seal puller assembly, C32016-6, SPL-4047 (seal puller assembly [55]) around the inner cylinder [14].

(b) Push the seal puller assembly, C32016-6, SPL-4047 (seal puller assembly [55]), up, such that it will install correctly at the bottom of the outer cylinder [9].

(c) Align the holes in the seal puller assembly, C32016-6, SPL-4047 (seal puller assembly [55]), with the holes in the lower bearing carrier [20] for the puller screws [60].

(d) Install the puller screws [60] to hold the seal puller assembly, C32016-6, SPL-4047 (seal puller assembly [55]), to the lower bearing carrier [20].

(e) Tighten the retainer screws on each side of the seal puller assembly, C32016-6, SPL-4047 (seal puller assembly [55]).

(f) Install the strap assy, C32017-60, SPL-3927 (strap assembly [57]), around the axle and connect the ends to the shoulder bolts on each side of the seal puller assembly, C32016-6, SPL-4047 (seal puller assembly [55]).

SUBTASK 32-21-24-580-003

(6) Lift the airplane until the lower bearing carrier [20] is completely out of the outer cylinder [9] (TASK 07-10-21-580-801).

SUBTASK 32-21-24-020-008

(7) Remove the inner cylinder [14] from the outer cylinder [9] (Figure 401 (Sheet 12)):

(a) Remove the downlock pin from the nose landing gear (TASK 32-00-01-080-801).

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WARNING DO NOT PUT A PLUG IN THE OPEN PORTS OF THE RETRACT ACTUATOR. THE HYDRAULIC LINES CAN BREAK AND CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(b) Disconnect the hydraulic UP-hose at the fitting on the retract actuator.

(c) Disconnect the hydraulic DOWN-hose at the bracket on the retract actuator.

(d) Put caps/plugs, STD-14686, on the hydraulic lines.

(e) Push up on the lock link to move it from the locked overcenter position.

(f) Optional: Connect a hydraulic pump to the retract actuator.

NOTE: The pressure line from the pump must be connected to the Up port of the actuator. The return line to the pump must be connected to the Down port of the actuator.

WARNING HOLD THE INNER CYLINDER WITH A ROPE BEFORE YOU CONNECT OR DISCONNECT THE CYLINDERS. IF YOU DO NOT HOLD THE INNER CYLINDER IN ITS POSITION, IT WILL FALL. THE INNER CYLINDER IS HEAVY AND CAN CAUSE INJURIES TO PERSONNEL OR DAMAGE TO EQUIPMENT.

CAUTION BE CAREFUL NOT TO CAUSE DAMAGE TO THE INNER CYLINDER OR THE METERING PIN. DAMAGE TO THESE PARTS CAN CAUSE DAMAGE TO THE SEALS.

(g) Slowly apply pressure to the retract actuator to remove the inner cylinder [14] from the outer cylinder [9].

NOTE: This will move the nose landing gear forward.

(h) Remove the inner cylinder [14].

SUBTASK 32-21-24-840-001

WARNING MAKE SURE THAT A MINIMUM OF TWO PERSONS WILL HOLD THE INNER CYLINDER WHEN YOU DO THE SUBSEQUENT STEPS. THE INNER CYLINDER IS HEAVY. INJURY TO PERSONS OR DAMAGE TO THE EQUIPMENT CAN OCCUR.

(8) Hold the inner cylinder [14] in a vertical position.

F. Inner Cylinder Component Removal

NOTE: Do these steps if the inner cylinder [14] you will install does not have these components installed.

SUBTASK 32-21-24-020-009

(1) Remove the metering pin [11] from the inner cylinder [14] (Figure 401 (Sheet 9)):

(a) Use the metering pin and retainer ring equipment, SPL-1895, to remove the nut [12] and retaining ring [67] from the inner cylinder [14].

(b) Remove the metering pin [11] from the inner cylinder [14].

(c) Remove the seal assembly [75] from the metering pin [11].

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SUBTASK 32-21-24-020-010

(2) Remove the upper bearing [8] and the recoil valve [70] from the top of the inner cylinder [14] (Figure 401 (Sheet 10)):

(a) Remove the upper bearing [8] from the upper bearing carrier [68].

1) Pull the upper bearing [8] apart slightly and slide it up and off the upper bearing carrier [68].

(b) Remove the upper bearing carrier [68] halves from the inner cylinder [14].

(c) Remove the recoil valve ring [69].

(d) Remove the recoil valve [70] from the upper centering cam [10].

SUBTASK 32-21-24-020-011

(3) Remove the upper centering cam [10] and lower centering cam [13] from the inner cylinder [14] (Figure 401 (Sheet 10)).

(a) Remove the upper centering cam [10] from the inner cylinder [14].

(b) Remove the dowel retaining circlip [71] and dowel [72] from the inner cylinder [14].

(c) Remove the lower centering cam [13] from the inner cylinder [14].

SUBTASK 32-21-24-020-012

(4) Remove the lower bearing carrier [20] from the inner cylinder [14] (Figure 401 (Sheet 6)):

(a) Use a rubber mallet, STD-3906, and lightly hit the bottom of the seal puller assembly, C32016-6, SPL-4047 (seal puller assembly [55]), until the lower bearing carrier [20] can be pulled out off the top of the inner cylinder [14].

NOTE: The lower bearing carrier [20] is coated with a dry lube film. The film may be worn and appear uneven, this is not cause for the lower bearing carrier [20] replacement.

(b) Remove the puller screw [60] to disconnect the seal puller assembly, C32016-6, SPL-4047 (seal puller assembly [55]), from the lower bearing carrier [20].

(c) Remove the seal puller assembly, C32016-6, SPL-4047 (seal puller assembly [55]) from the inner cylinder [14] .

(d) Put the lower bearing carrier [20] on a table or bench.

SUBTASK 32-21-24-020-013

(5) Remove the seal retainer [19] (Figure 401 (Sheet 3)):

(a) Remove the bearing carrier pin [22] for the seal retainer [19].

(b) Move the seal retainer [19] away from the lower bearing carrier [20] to get access to the active dynamic seal [23].

NOTE: It is recommended, if replacement seals are available, that all active and spare seals be replaced while the inner cylinder is removed. See task: Seals and Spare Seals Replacement, TASK 32-21-11-960-802

SUBTASK 32-21-24-900-001

(6) Install a new scraper ring [21] on the gland nut [18] if there is any sign of damage or wear on the old scraper ring [21] (Figure 401 (Sheet 3)):

NOTE: The scraper ring [21] prevents dirt and debris from entering the shock strut [4] when the inner cylinder [14] and outer cylinder [9] compress.

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(a) Make sure to use the correct scraper configuration to be installed in its groove on the gland nut [18].

NOTE: There are two possible scraper configurations used for installation (Figure 401 (Sheet 4)).

(b) Remove the gland nut [18] from the inner cylinder [14] (Figure 401 (Sheet 7)).

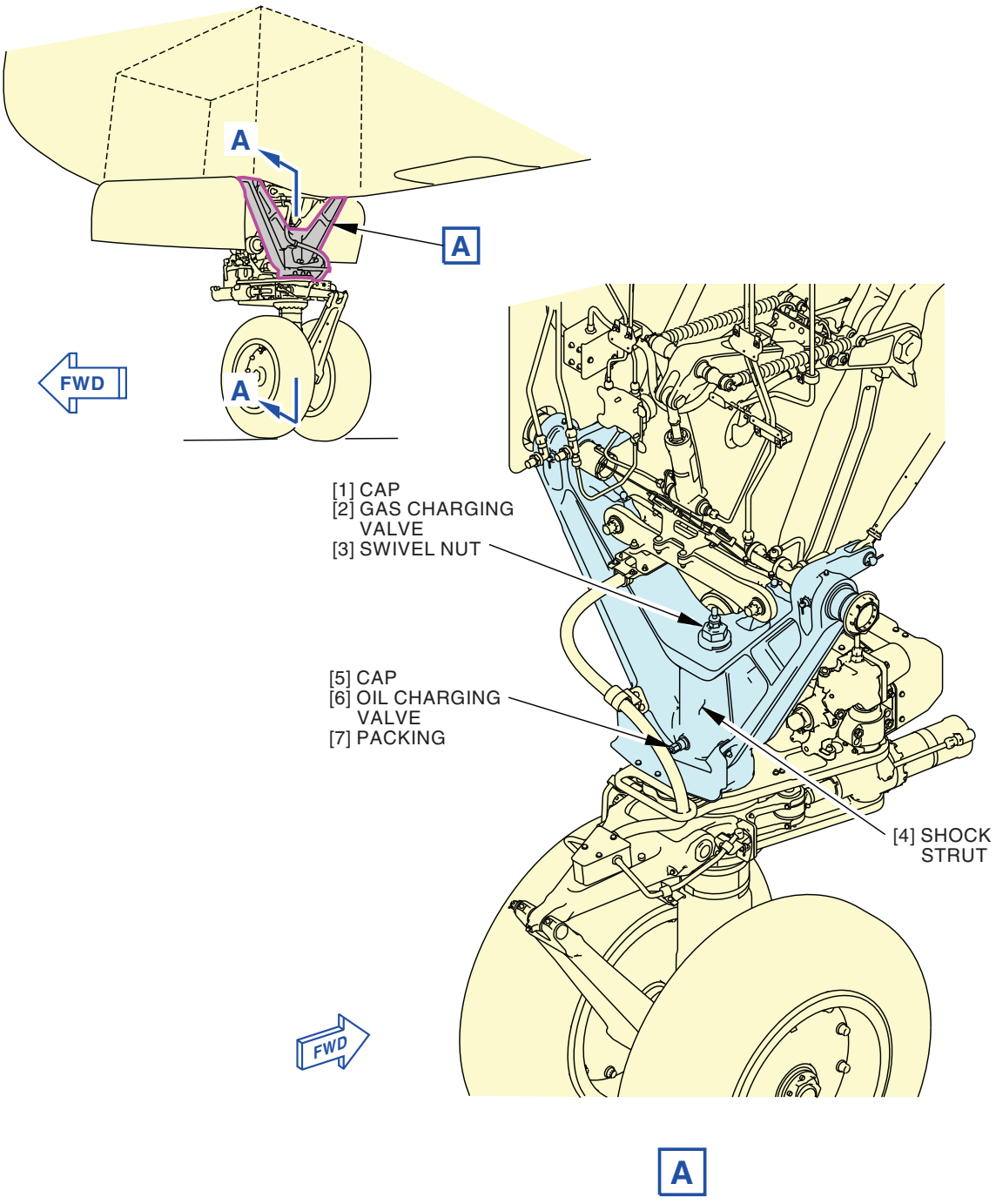
(c) Remove the scraper retainer, C32016-4 or -67, SPL-1378 (scraper retainer [61]), and the scraper lock assembly, C32016-5 or -68, SPL-1379 (scraper lock assembly [56]), from the gland nut [18].

(d) Remove the old scraper ring [21] from its groove on the gland nut [18].

(e) Install the new scraper ring [21] in its groove on the gland nut [18].

——— END OF TASK ———

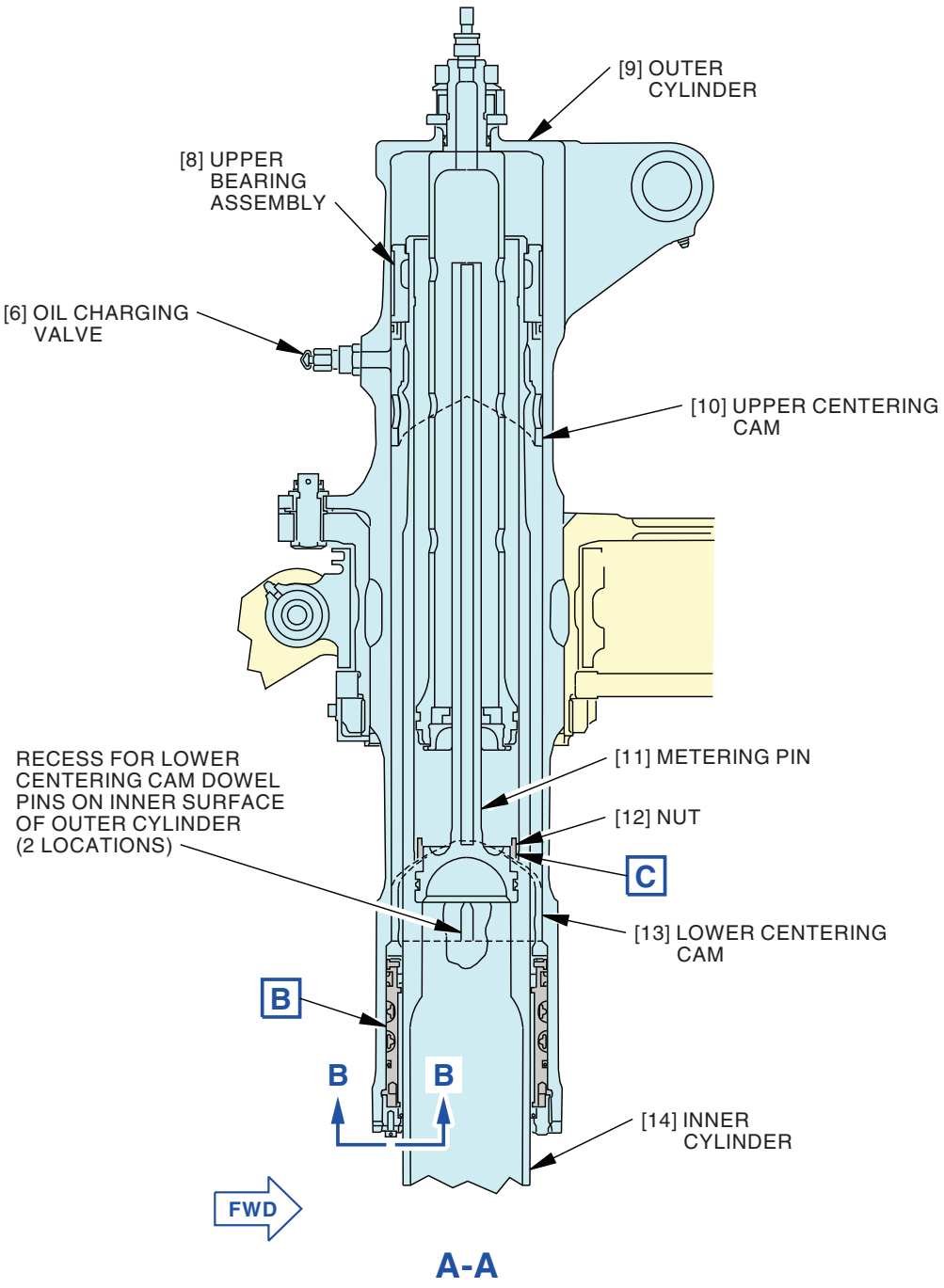
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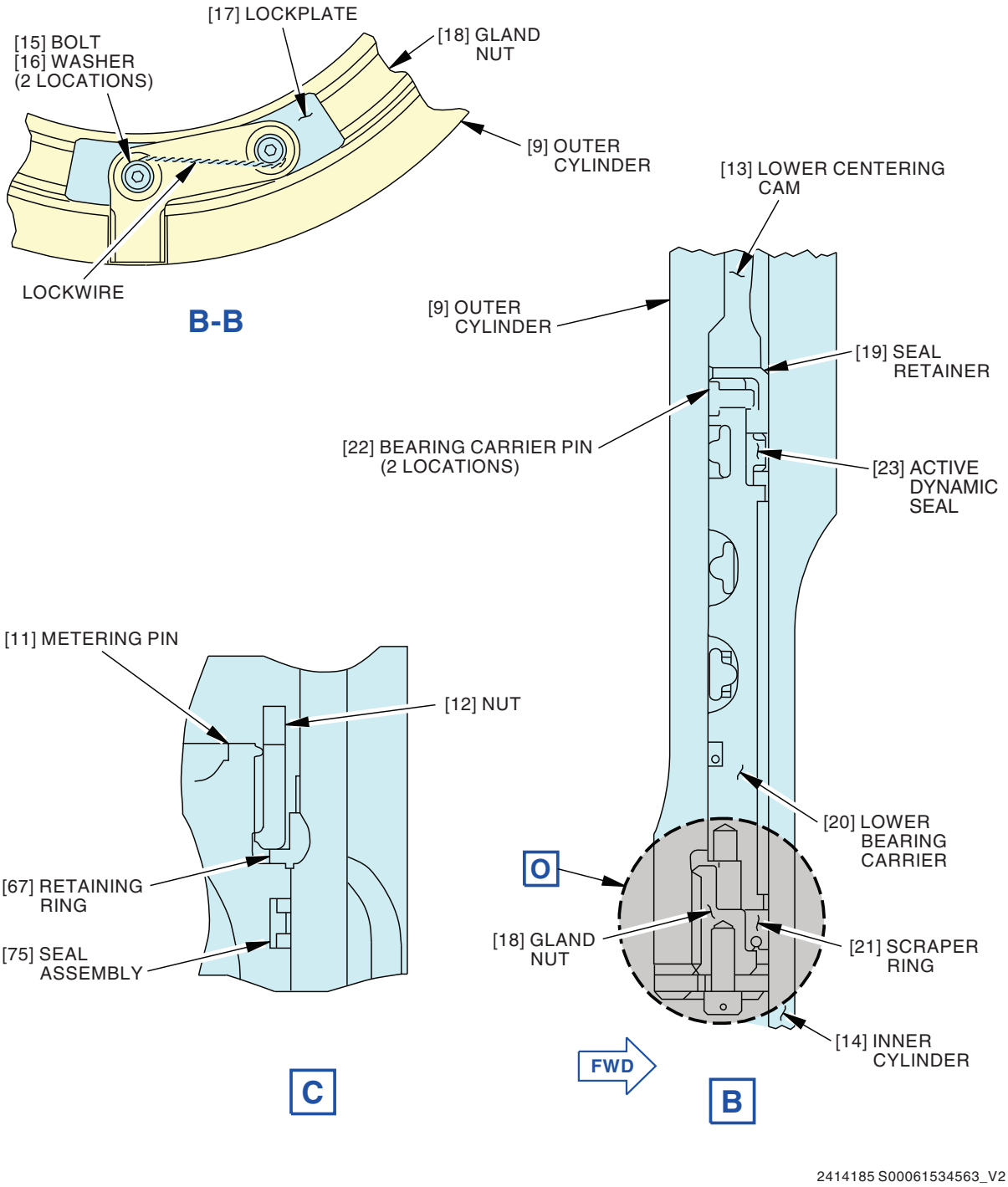
Nose Landing Gear Shock Strut Inner Cylinder Installation
Figure 401/32-21-24-990-801 (Sheet 1 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



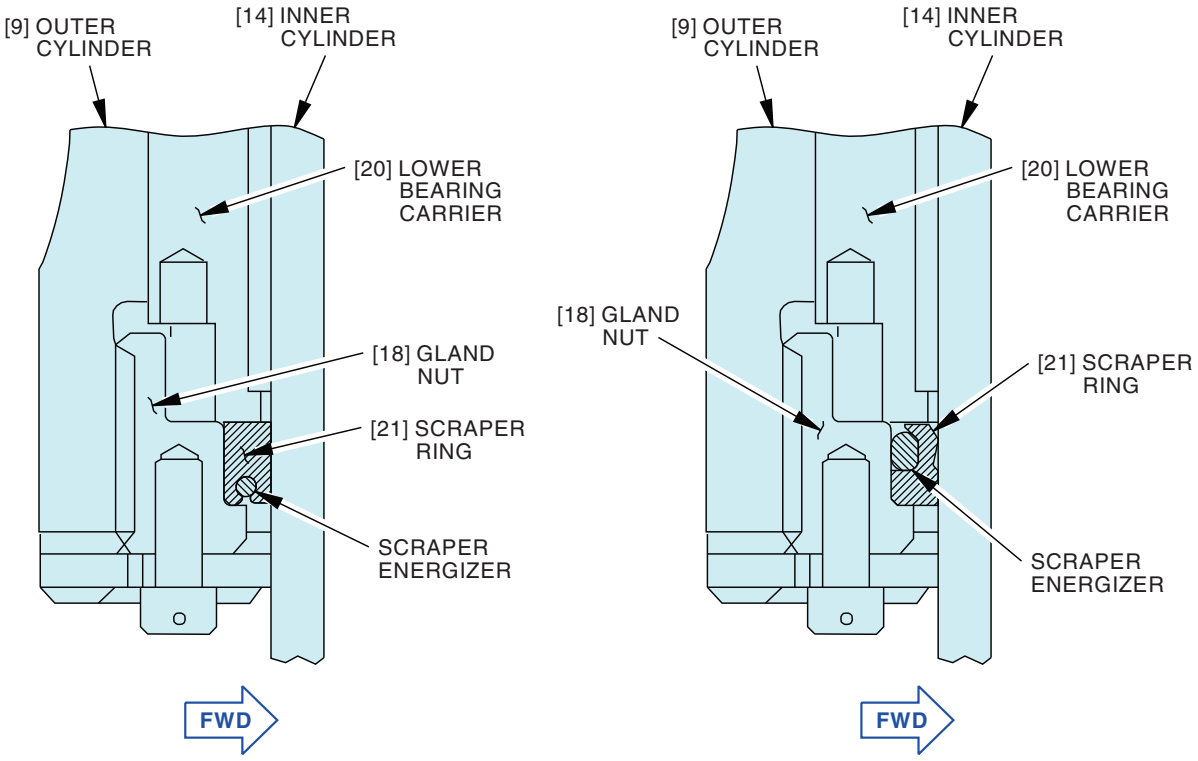
Nose Landing Gear Shock Strut Inner Cylinder Installation
Figure 401/32-21-24-990-801 (Sheet 2 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Landing Gear Shock Strut Inner Cylinder Installation
Figure 401/32-21-24-990-801 (Sheet 3 of 12)

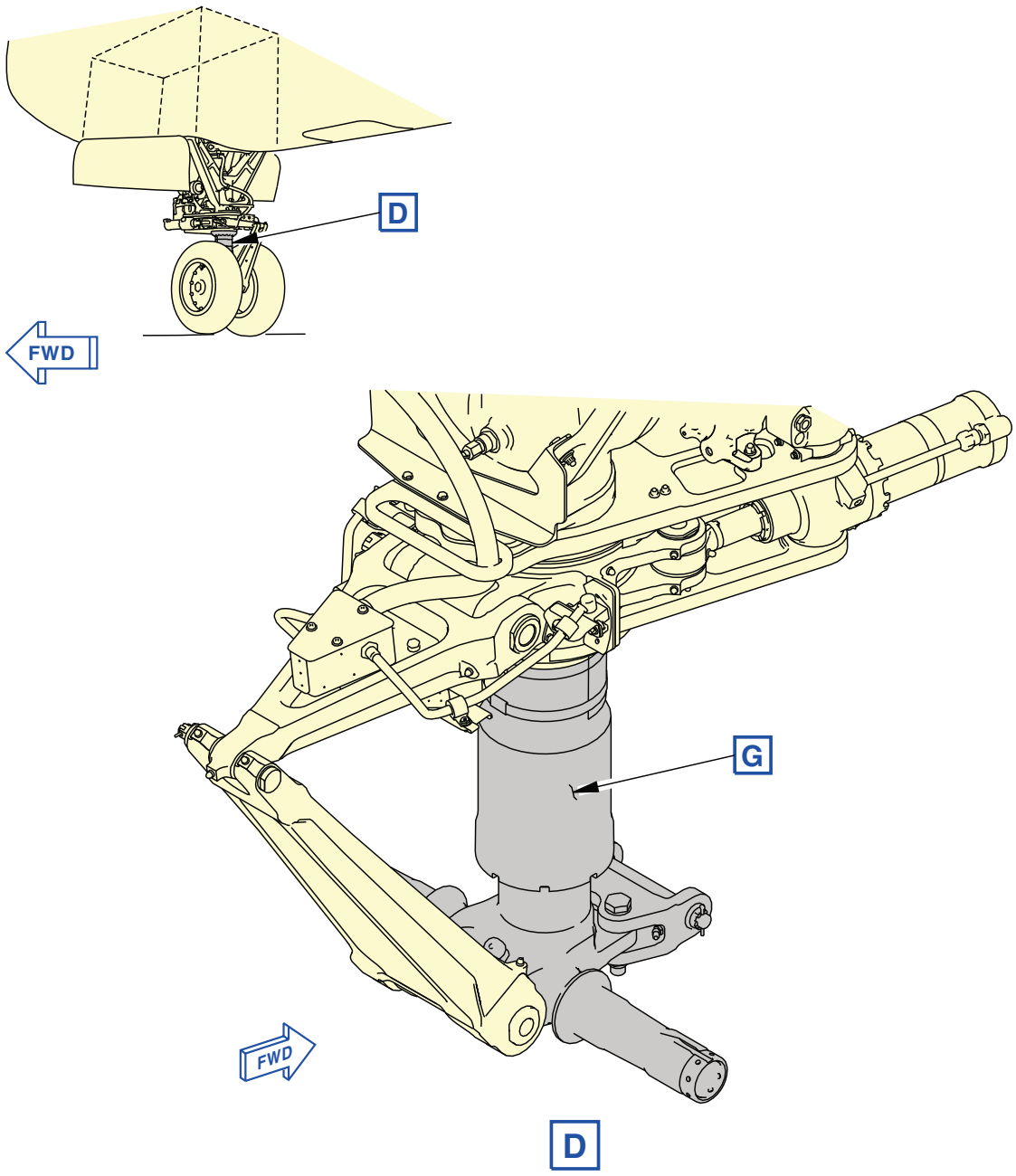
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Nose Landing Gear Shock Strut Inner Cylinder Installation
Figure 401/32-21-24-990-801 (Sheet 4 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

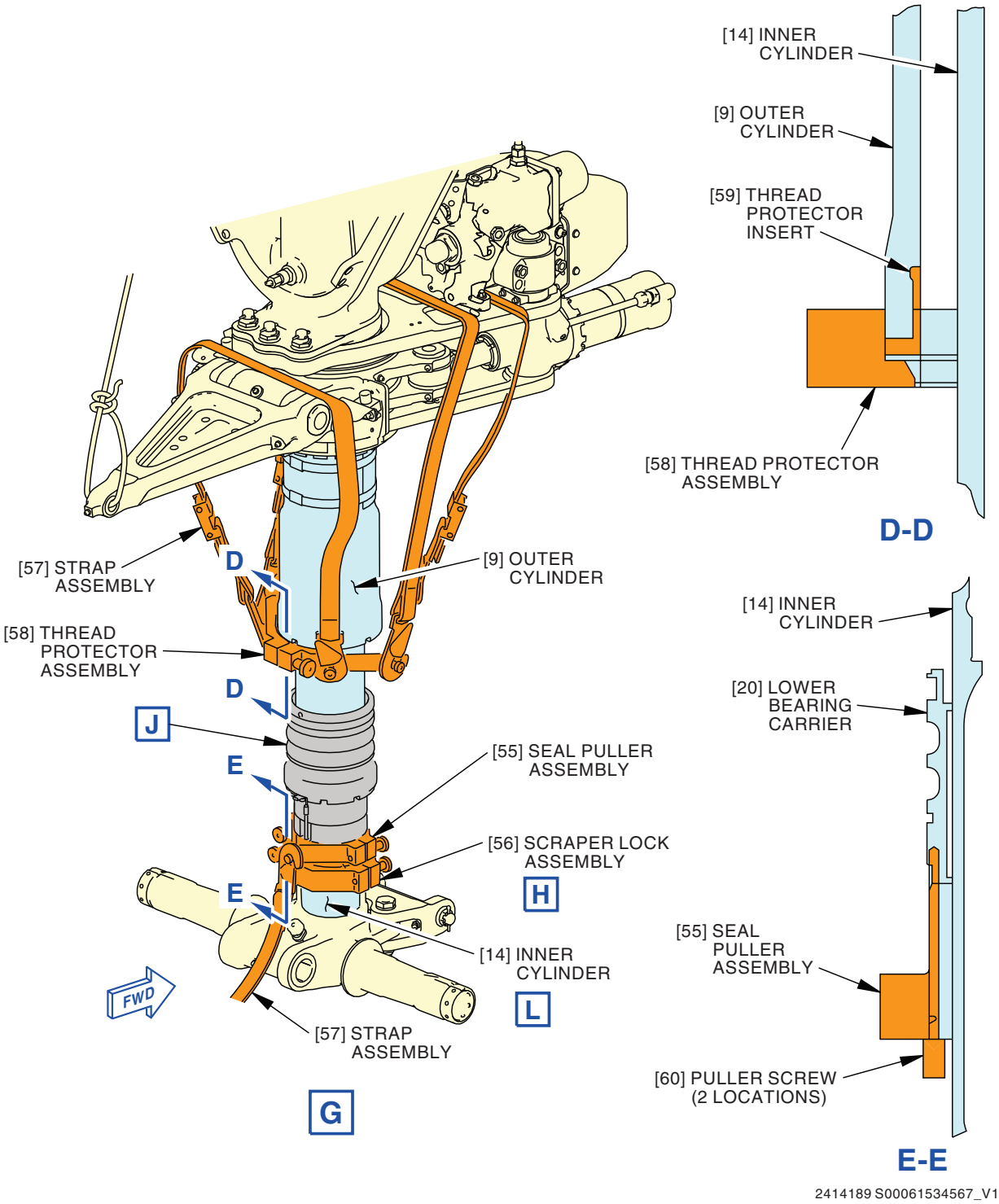


NOTE: WHEELS AND TIRES REMOVED
FOR CLARITY.

2414187 S00061534565_V1

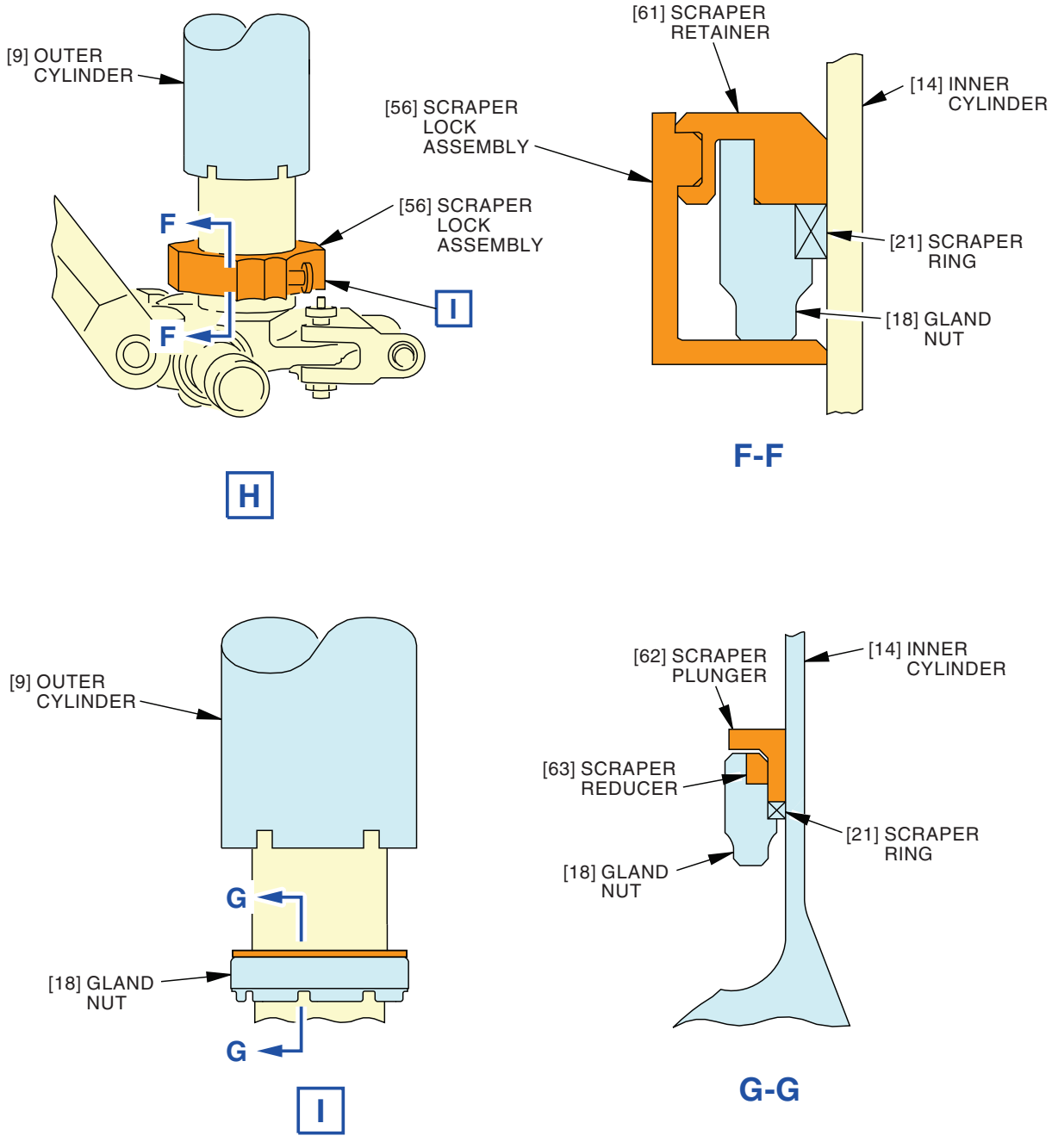
Nose Landing Gear Shock Strut Inner Cylinder Installation
Figure 401/32-21-24-990-801 (Sheet 5 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



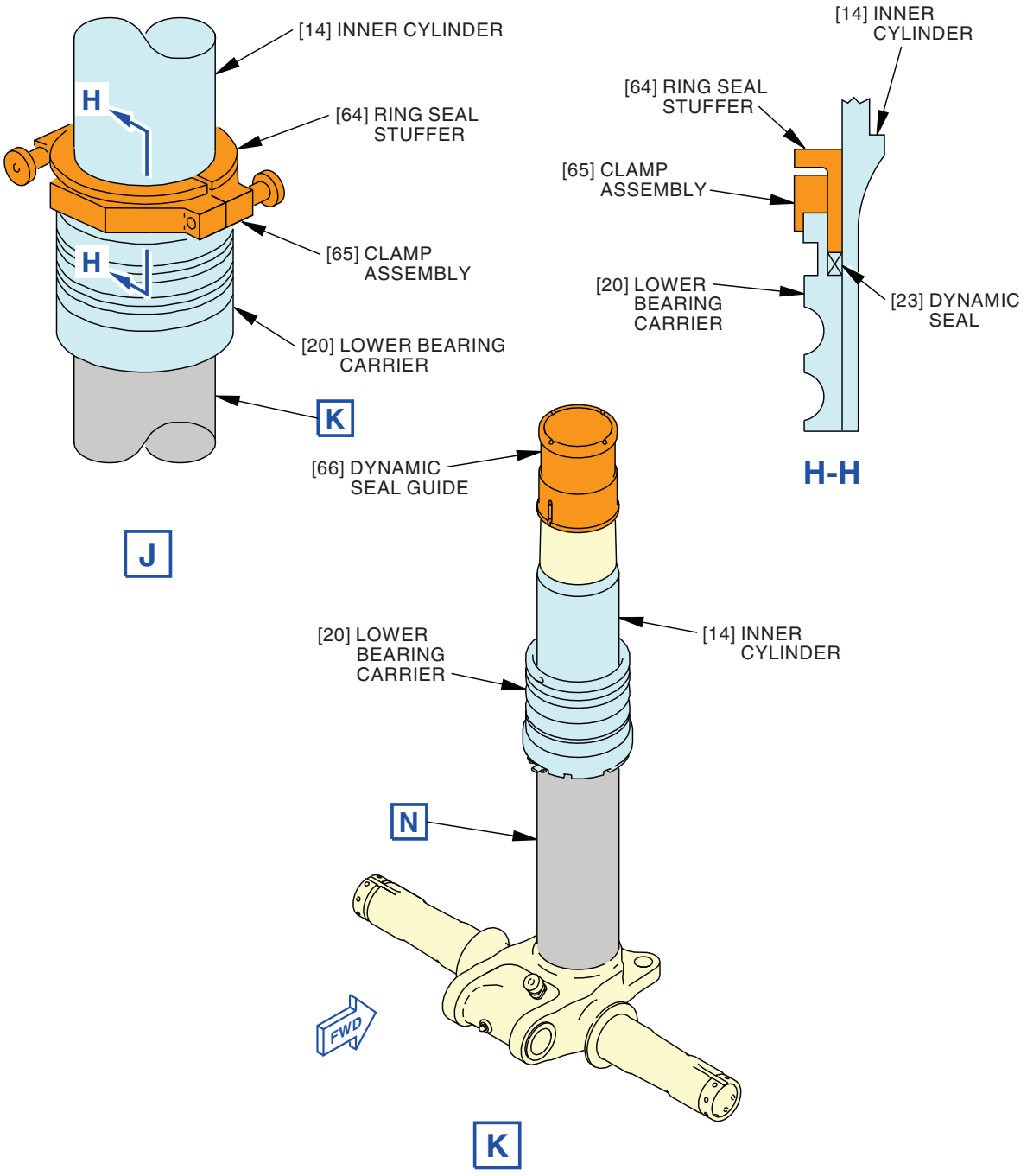
Nose Landing Gear Shock Strut Inner Cylinder Installation
Figure 401/32-21-24-990-801 (Sheet 6 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



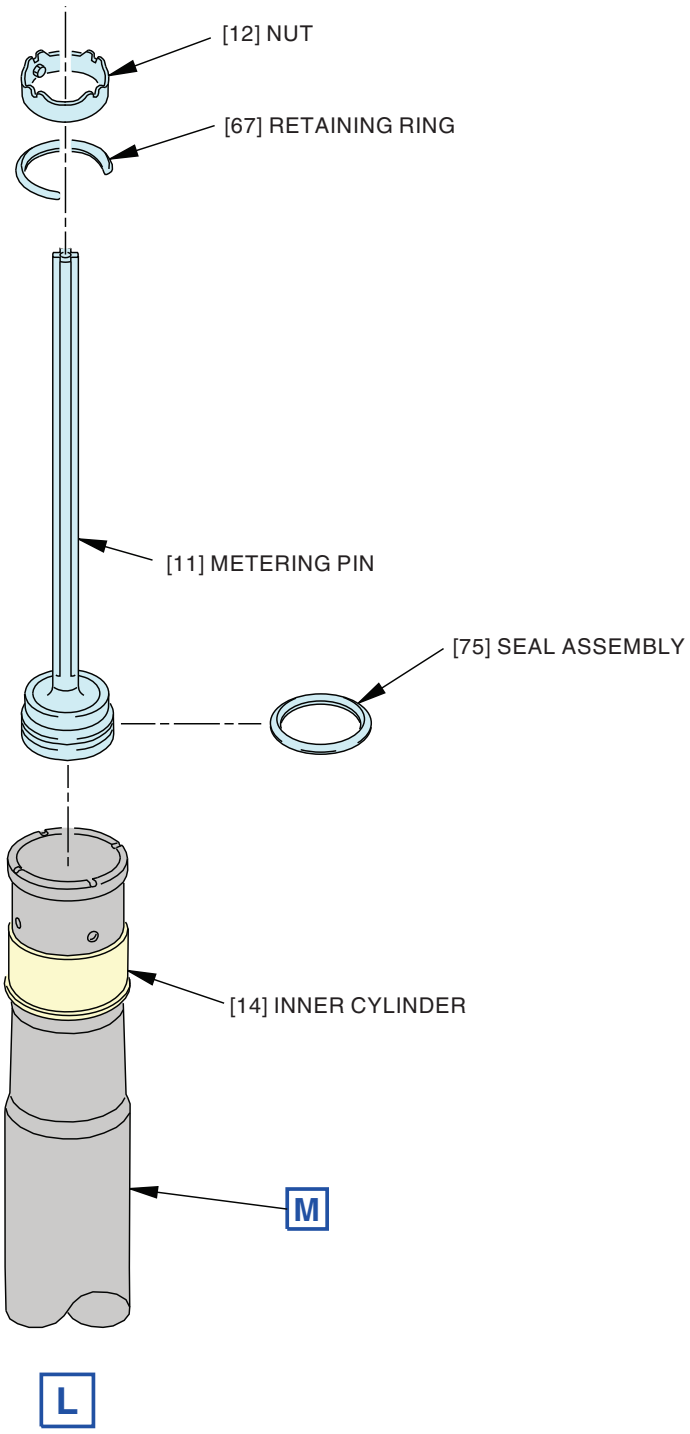
Nose Landing Gear Shock Strut Inner Cylinder Installation
Figure 401/32-21-24-990-801 (Sheet 7 of 12)

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Nose Landing Gear Shock Strut Inner Cylinder Installation
Figure 401/32-21-24-990-801 (Sheet 8 of 12)

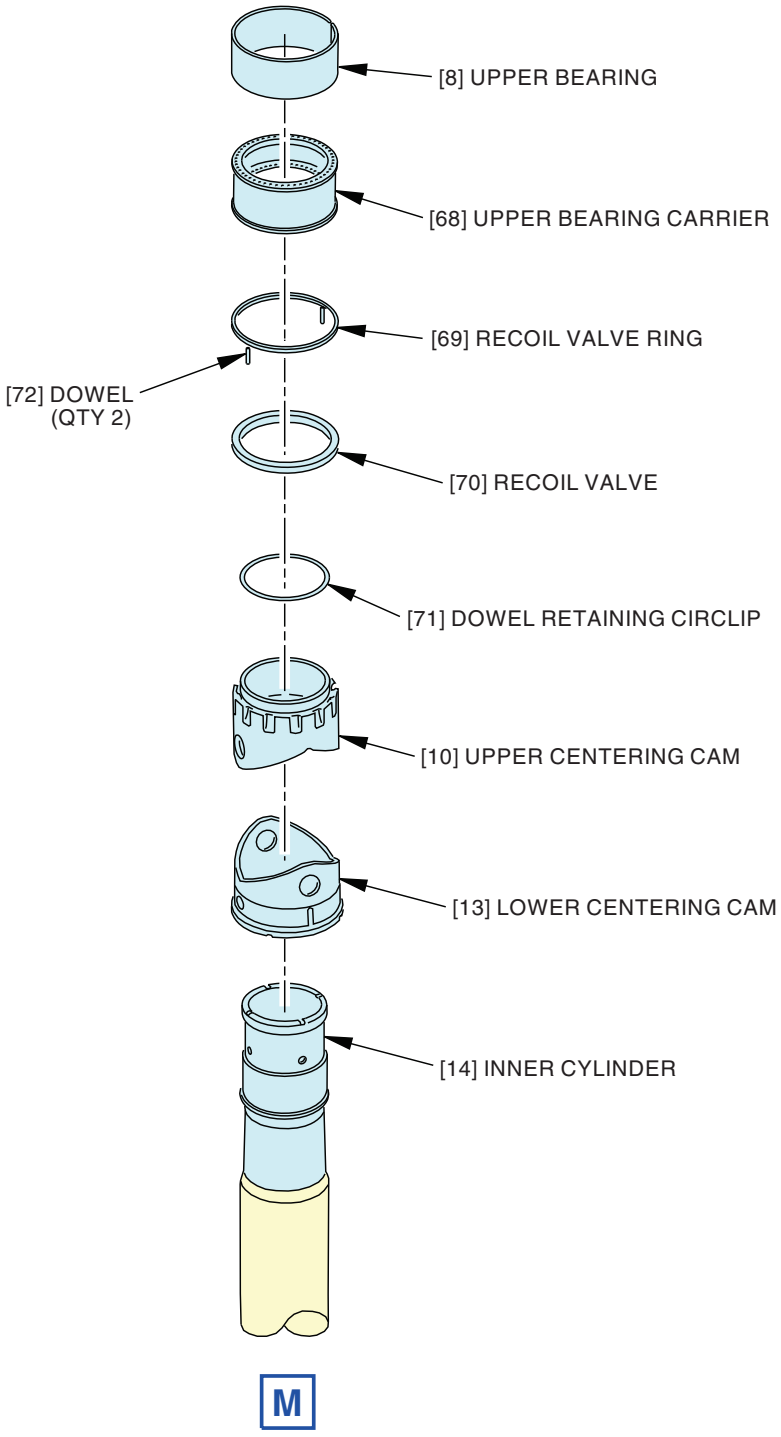
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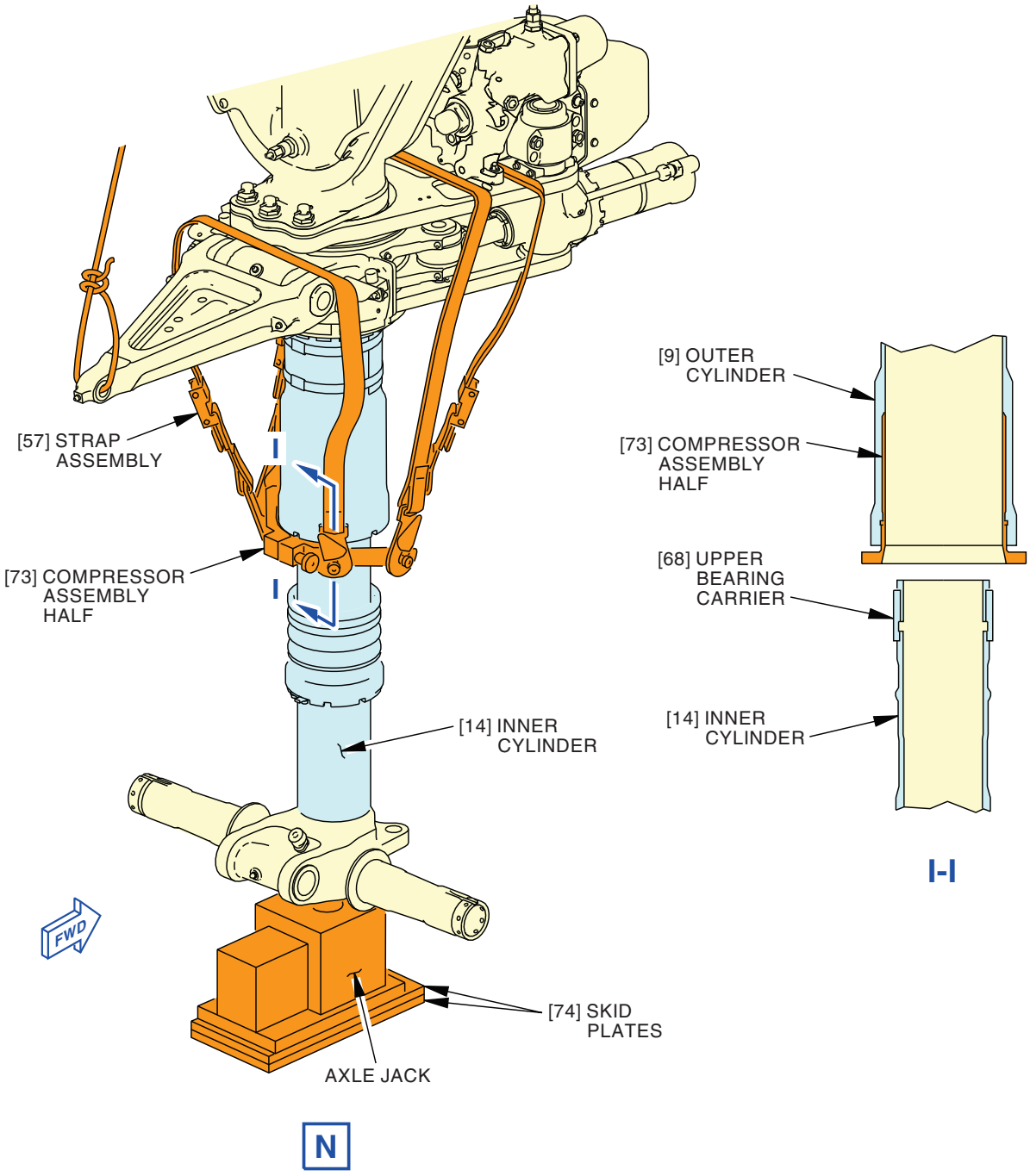
Nose Landing Gear Shock Strut Inner Cylinder Installation
Figure 401/32-21-24-990-801 (Sheet 9 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Landing Gear Shock Strut Inner Cylinder Installation
Figure 401/32-21-24-990-801 (Sheet 10 of 12)

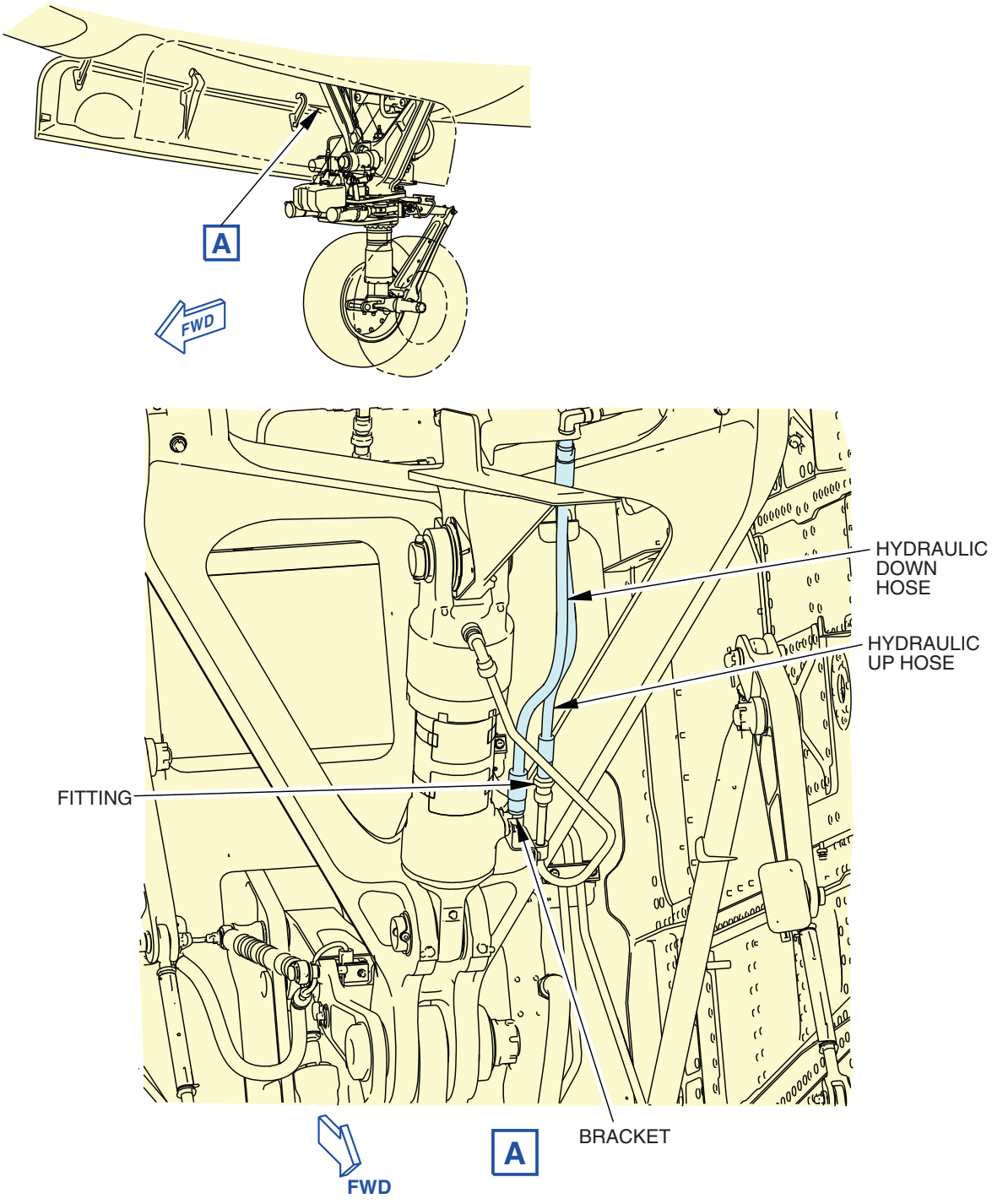
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Nose Landing Gear Shock Strut Inner Cylinder Installation
Figure 401/32-21-24-990-801 (Sheet 11 of 12)

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Nose Landing Gear Shock Strut Inner Cylinder Installation
Figure 401/32-21-24-990-801 (Sheet 12 of 12)

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TASK 32-21-24-400-801

3. NLG Inner Cylinder Installation

(Figure 401)

A. References

Reference	Title
07-10-03-580-803	Lift the Nose Landing Gear Axle With Axle Jacks-Handling (P/B 201)
07-10-21-580-802	Lower the Airplane Nose, Jack Point D (P/B 201)
12-15-41-610-802	Nose Landing Gear Shock Strut Servicing, Airplane on the Ground (P/B 301)
12-21-21-640-802	Nose Landing Gear Lower End Components Servicing (P/B 301)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-21-14-400-801	Nose Landing Gear Tow Fitting Installation (P/B 401)
32-21-31-400-802	Nose Landing Gear Lower Torsion Link Installation (P/B 401)
32-45-21-400-801	Nose Landing Gear Wheel and Tire Assembly Installation (P/B 401)
32-51-00-700-801	Nose Wheel Steering System Test (Airplane on Ground) (P/B 501)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1376	Scraper Plunger, C32016-2 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-1377	Scraper Reducer, C32016-3 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-1378	Scraper Retainer, C32016-4 or -67 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-1379	Scraper Lock Assy, C32016-5 or -68 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-1382	Dynamic Seal Guide, C32016-45 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205

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(Continued)

Reference	Description
SPL-1383	Compressor Assy Half, C32016-10 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-1384	Thread Protector Assy, C32016-11 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-1385	Thread Protector Half, C32016-21 (included in C32016 eqpt) Part #: C32016-66 Supplier: 81205 Opt Part #: C32016-33 Supplier: 81205 Opt Part #: C32016-46 Supplier: 81205
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205
SPL-1867	Wrench - Gland Nut, Shock Strut NLG Part #: C32025-7 Supplier: 81205 Opt Part #: C32025-1 Supplier: 81205
SPL-1895	Removal/Installation Equipment - Metering Pin and Retainer Ring, NLG Part #: C32035-1 Supplier: 81205
SPL-3927	Strap Assy, C32017-60 (included in C32017 Eqpt) Part #: C32017-67 Supplier: 81205 Opt Part #: C32017-52 Supplier: 81205 Opt Part #: C32017-53 Supplier: 81205
STD-3906	Mallet - Rubber
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Consumable Materials

Reference	Description	Specification
A00226	Compound - Tamper-Proof Putty	BMS8-45
D00467	Fluid - Landing Gear Shock Strut	BMS3-32 Type II
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Gas charging valve	32-21-00-01-360	PTF ALL
6	Oil charging valve	32-21-00-01-035	PTF ALL
7	Packing	32-21-00-01-030	PTF ALL

E. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

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F. Components on the Inner Cylinder Installation

NOTE: Do these steps if the inner cylinder [14] you will install does not have these components installed.

SUBTASK 32-21-24-420-001

(1) Install the gland nut [18] on the inner cylinder [14] (Figure 401 (Sheet 7)).

(a) If the scraper ring [21] comes out of its groove, use the scraper plunger, C32016-2, SPL-1376 (scraper plunger [62]) and the scraper reducer, C32016-3, SPL-1377 (scraper reducer [63]), to install the scraper ring [21] back in its groove.

SUBTASK 32-21-24-420-002

(2) Install the seal retainer [19] (Figure 401 (Sheet 3)):

(a) Put the seal retainer [19] in its position on the lower bearing carrier [20].

(b) Install the bearing carrier pins [22] to connect the seal retainer [19] to the lower bearing carrier [20].

SUBTASK 32-21-24-840-002

WARNING MAKE SURE THAT A MINIMUM OF TWO PERSONS WILL HOLD THE INNER CYLINDER WHEN YOU DO THE SUBSEQUENT STEPS. THE INNER CYLINDER IS HEAVY. INJURY TO PERSONS OR DAMAGE TO THE EQUIPMENT CAN OCCUR.

(3) Hold the inner cylinder [14] in a vertical position.

SUBTASK 32-21-24-480-003

(4) Install the dynamic seal guide, C32016-45, SPL-1382 (dynamic seal guide [66]), at the top of the inner cylinder [14] (Figure 401 (Sheet 8)).

SUBTASK 32-21-24-420-003

(5) Install the lower bearing carrier [20]:

CAUTION MAKE SURE THAT THE BACKUP RINGS AND THE SEAL STAY TOGETHER WHEN YOU MOVE THE LOWER BEARING CARRIER. IF THE BACKUP RINGS AND THE SEAL MOVE APART, THEY WILL NOT INSTALL CORRECTLY. THIS CAN CAUSE DAMAGE TO THE SEAL AND THE BACKUP RINGS.

(a) Put the lower bearing carrier [20] in its position for installation on the inner cylinder [14].

(b) Use rubber mallet, STD-3906, and lightly hit the top of the lower bearing carrier [20] until it passes the dynamic seal guide, C32016-45, SPL-1382 (dynamic seal guide [66]).

(c) Move the lower bearing carrier [20] down to the bottom of the inner cylinder [14] .

SUBTASK 32-21-24-020-014

(6) Remove the dynamic seal guide, C32016-45, SPL-1382 (dynamic seal guide [66]), from the inner cylinder [14].

SUBTASK 32-21-24-420-004

(7) Install these items (Figure 401 (Sheet 10)):

(a) Install the lower centering cam [13] on the inner cylinder [14].

(b) Install the dowels [72] in their position on the inner cylinder [14].

(c) Install the dowel retaining circlip [71] on the inner cylinder [14] and on the dowels [72] to hold the dowels [72] in position.

(d) Install the upper centering cam [10] on the inner cylinder [14].

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(e) Verify the dowels [72] line up with the grooves on the upper centering cam [10] as it is installed on the inner cylinder [14].

(f) Verify dowel retaining circlip [71] snaps into the groove just below the dowels [72] on the inner cylinder [14].

(g) Install the upper bearing carrier [68] and the recoil valve [70] on the inner cylinder [14]:

1) Install the recoil valve [70] on the top of the upper centering cam [10].

2) Install the recoil valve ring [69] on the recoil valve [70] if it was removed.

3) Install the upper bearing carrier [68] on the inner cylinder [14].

(h) Install the upper bearing [8] on the upper bearing carrier [68] halves (Figure 401 (Sheet 11)).

1) If the upper bearing [8] does not fit tightly around the upper bearing carrier [68], use the compressor assy half, C32016-10, SPL-1383 (compressor assembly half [73]), to hold the upper bearing [8] closed during inner cylinder [14] installation.

2) Make an overlap of 0.25 in. (6.35 mm) to the upper bearing [8] to keep its round shape.

SUBTASK 32-21-24-420-005

(8) Install the metering pin [11] in the inner cylinder [14] (Figure 401 (Sheet 9)):

(a) Cover the seal assembly [75] fully with fluid, D00467.

(b) Install the seal assembly [75] on the metering pin [11].

(c) Install the metering pin [11], retaining ring [67], and nut [12] in the inner cylinder [14].

1) Install the retaining ring [67] on the metering pin [11] with the flange side first (Figure 401).

2) Install the nut [12] to hold the compressed retainer ring on the metering pin [11].

(d) Use the metering pin and retainer ring equipment, SPL-1895, to loosen the nut [12] and let the retaining ring [67] expand into the groove in the inner cylinder [14].

(e) Use the metering pin and retainer ring equipment, SPL-1895, to tighten the nut [12] to 75 ft-lb (101.7 N·m) - 100 ft-lb (135.6 N·m).

G. Inner Cylinder in the Outer Cylinder Installation

SUBTASK 32-21-24-420-006

(1) Install the compressor assy half, C32016-10, SPL-1383 (compressor assembly half [73]) (Figure 401 (Sheet 11)):

(a) Install the compressor assy half, C32016-10, SPL-1383 (compressor assembly half [73]), such that it fits at the bottom of the outer cylinder [9].

(b) Install the strap assy, C32017-60, SPL-3927 (strap assembly [57]), around the outer cylinder [9] and connect the ends to the shoulder bolts on the side of the compressor assy half, C32016-10, SPL-1383 (compressor assembly half [73]).

1) Tighten the strap assy, C32017-60, SPL-3927 (strap assembly [57]).

SUBTASK 32-21-24-420-007

(2) Install the inner cylinder [14] in the outer cylinder [9] (Figure 401 (Sheet 12)):

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CAUTION

BE CAREFUL NOT TO CAUSE DAMAGE TO THE INNER CYLINDER OR THE METERING PIN. DAMAGE TO THESE PARTS CAN CAUSE DAMAGE TO THE SEALS.

(a) Put the inner cylinder [14] in position to install in the outer cylinder [9].

(b) Disconnect the hydraulic pump from the retract actuator.

(c) Remove the caps/plugs, STD-14686, from the hydraulic lines.

(d) Connect the hydraulic UP-hose at the fitting on the retract actuator.

1) Tighten the B-nut on the hydraulic UP-hose to 140 ±7 in-lb (15.8 ±0.8 N·m).

(e) Connect the hydraulic DOWN-hose at the bracket on the retract actuator.

1) Tighten the B-nut on the hydraulic DOWN-hose to 270 ±14 in-lb (30.5 ±1.5 N·m).

(f) Move the nose landing gear aft while you put the inner cylinder [14] into the outer cylinder [9].

(g) Push down on the lock link to move it to the locked overcenter position.

(h) Install the downlock pin in the nose landing gear (TASK 32-00-01-480-801).

SUBTASK 32-21-24-080-001

(3) Remove the compressor assy half, C32016-10, SPL-1383 (compressor assembly half [73]):

(a) Remove the strap assy, C32017-60, SPL-3927 (strap assembly [57]), around the outer cylinder [9] and to the shoulder bolts on the side of the compressor assy half, C32016-10, SPL-1383 (compressor assembly half [73]).

(b) Remove the compressor assy half, C32016-10, SPL-1383 (compressor assembly half [73]), from the outer cylinder [9].

SUBTASK 32-21-24-420-008

(4) Carefully put the thread protector half, C32016-21, SPL-1385 (thread protector insert [59]), between the outer cylinder [9] and the inner cylinder [14] (Figure 401 (Sheet 6)).

NOTE: The thread protector insert [59] will expand and push against the outer cylinder [9]. This will protect the static seals from the gland nut [18] threads on the outer cylinder [9].

SUBTASK 32-21-24-480-004

(5) Install the thread protector assembly, C32016-11, SPL-1384 (thread protector assembly [58]) (Figure 401 (Sheet 6)):

(a) Install the thread protector assembly, C32016-11, SPL-1384 (thread protector assembly [58]), such that it fits at the bottom of the outer cylinder [9].

(b) Install the strap assy, C32017-60, SPL-3927 (strap assembly [57]), around the outer cylinder [9] and connect the ends to the shoulder bolts on the side of the thread protector assembly, C32016-11, SPL-1384 (thread protector assembly [58]).

1) Tighten the strap assy, C32017-60, SPL-3927 (strap assembly [57]).

SUBTASK 32-21-24-580-004

(6) Lift the nose landing gear with an axle jack until the lower bearing carrier [20] moves into the recess between the inner cylinder [14] and outer cylinder [9] (TASK 07-10-03-580-803) .

(a) Make sure the static seals stay in their positions as the lower bearing carrier [20] is installed.

(b) Continue to lift the nose landing gear with an axle jack until the shock strut extension is reduced to 3.5 ±0.5 in. (88.9 ±12.7 mm) (TASK 07-10-03-580-803) .

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SUBTASK 32-21-24-020-015

(7) Remove the thread protector assembly, C32016-11, SPL-1384 (thread protector assembly [58]), and the thread protector half, C32016-21, SPL-1385 (thread protector insert [59]) (Figure 401 (Sheet 6)).

SUBTASK 32-21-24-020-016

(8) Remove the scraper retainer, C32016-4 or -67, SPL-1378 (scraper retainer [61]), and the scraper lock assembly, C32016-5 or -68, SPL-1379 (scraper lock assembly [56]), from the gland nut [18] (Figure 401 (Sheet 7)).

SUBTASK 32-21-24-640-001

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(9) Apply a liberal quantity of corrosion inhibiting material, G50136, to the threads of the gland nut [18] and the threads of the outer cylinder [9].

SUBTASK 32-21-24-420-009

(10) Install the gland nut [18]:

(a) Move the gland nut [18] up on the inner cylinder [14] and engage the threads to the outer cylinder [9].

(b) Use the gland nut wrench, SPL-1867, for the gland nut [18] to tighten the gland nut [18] to 100 ft-lb (135.6 N·m) - 125 ft-lb (169.5 N·m).

SUBTASK 32-21-24-420-010

(11) Install the lockplate [17] for the gland nut [18] (Figure 401 (Sheet 3)):

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin layer of corrosion inhibiting material, G50136, to the shank and threads of bolts [15] and washers [16].

(b) Install the lockplate [17] in the holes such that it requires the least amount of back off of the gland nut [18] to align the tab of the lockplate [17] with the slot on the outer cylinder [9].

1) If it is necessary, loosen the gland nut [18] slightly to align slot in the outer cylinder [9] with the lockplate [17] tab.

(c) Install the bolts [15] and washers [16].

(d) Tighten the bolts [15].

(e) Install lockwire on the bolts [15].

(f) Remove all excess corrosion inhibiting material, G50136.

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SUBTASK 32-21-24-420-011

(12) Do this task: Nose Landing Gear Lower Torsion Link Installation, TASK 32-21-31-400-802.

SUBTASK 32-21-24-420-015

(13) Do this task: Nose Landing Gear Tow Fitting Installation, TASK 32-21-14-400-801.

SUBTASK 32-21-24-420-019

(14) Do this task:Nose Landing Gear Wheel and Tire Assembly Installation, TASK 32-45-21-400-801.

SUBTASK 32-21-24-980-001

(15) Move the inner cylinder [14] from the fully extended to the fully compressed position.

NOTE: Make sure the total travel is 15.50 in. (39.4 cm).

SUBTASK 32-21-24-860-002

(16) Remove the NLG towing lever pin, SPL-1499 from the towing lever.

SUBTASK 32-21-24-420-017

(17) Install the oil charging valve [6], if removed:

(a) Remove and discard the old packing [7].

(b) Apply a thin layer of fluid, D00467, on the new packing [7].

(c) Install the new packing [7] on the oil charging valve [6].

(d) Install the oil charging valve [6] and tighten it to 22 ft-lb (29.8 N·m) - 25 ft-lb (33.9 N·m).

(e) Install the cap [5] loosely.

NOTE: You will tighten it in the shock strut servicing task.

(f) Apply compound, A00226, to the oil charging valve [6].

NOTE: Apply it to make sure the seal made by the compound, A00226, will break if the oil charging valve [6] becomes loose.

SUBTASK 32-21-24-600-001

(18) Do this task: Nose Landing Gear Shock Strut Servicing, Airplane on the Ground, TASK 12-15-41-610-802.

SUBTASK 32-21-24-420-018

(19) Make sure the swivel nut [3] is tightened to 5 ft-lb (6.8 N·m) - 7 ft-lb (9.5 N·m) and the cap [1] is installed for the gas charging valve [2].

SUBTASK 32-21-24-710-002

(20) Do a test on the steering system of the nose wheel (TASK 32-51-00-700-801).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-21-24-580-005

(1) Do this task: Lower the Airplane Nose, Jack Point D, TASK 07-10-21-580-802.

SUBTASK 32-21-24-640-002

(2) Lubricate the lube fittings on the torsion links (TASK 12-21-21-640-802).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE LANDING GEAR TORSION LINKS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the upper torsion link (referred to as the upper torsion link [15]) from the nose landing gear

(2) An installation of the upper torsion link on the nose landing gear.

(3) A removal of the lower torsion link (referred to as the lower torsion link [11]) from the nose landing gear

(4) An installation of the lower torsion link on the nose landing gear.

(5) The disconnection of the torsion links of the nose landing gear

(6) The connection of the torsion links of the nose landing gear.

TASK 32-21-31-000-801

2. Nose Landing Gear Upper Torsion Link Removal

(Figure 401)

A. General

(1) This procedure supplies instructions to remove the upper torsion link.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205
STD-837	String
STD-6545	Rope - Nylon

D. Location Zones

Zone	Area
713	Nose Landing Gear

E. Prepare for the Removal

SUBTASK 32-21-31-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

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SUBTASK 32-21-31-480-008

WARNING ONLY USE THE CORRECT PIN FOR THE AIRPLANE MODEL. IF YOU USE AN INCORRECT PIN, THE HYDRAULIC STEERING CAN OPERATE. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) Install the NLG towing lever pin, SPL-1499, for the towing lever in the metering valve for the nose wheel steering.

F. Nose Landing Gear Upper Torsion Link Removal

SUBTASK 32-21-31-020-001

(1) Disconnect the box assembly [38] from the upper torsion link [15]:

NOTE: Disconnect the box assembly [38] with the electrical conduits [22]attached.

(a) Remove the screws [44] and washers [45] to remove the cover [46] from the box assembly [38].

(b) Remove the nuts [33] and washers [34] from the bolt [36].

(c) Remove the bolts [36] and washers [35] from the upper torsion link [15].

(d) Remove the washers [35] from the bolts [36].

SUBTASK 32-21-31-020-002

(2) Remove the clamps [25] that hold the electrical conduits [22] to the bracket [26]:

(a) Remove the screws [23] that hold the clamps [25] to the bracket [26].

(b) Remove the clamps [25] and the nut clips [24] from the bracket [26].

SUBTASK 32-21-31-020-003

(3) Remove the bracket [26] from the upper torsion link [15]:

(a) Remove the nut [43] and washers [42] from the bolts [39].

(b) Remove the bracket [26] and washers [41] from the bolts [39].

(c) Remove the bolts [39] and washers [40] from the upper torsion link [15].

(d) Remove the washers [40] from the bolts [39].

SUBTASK 32-21-31-020-004

(4) Disconnect the sensor brackets [17] from the upper torsion link [15]:

NOTE: Disconnect the sensor brackets [17] with the sensors [16] and the electrical conduits [22] attached.

(a) Remove the nuts [21] and washers [20] from the bolts [18].

(b) Remove the bolts [18] that hold the sensor brackets [17] to the upper torsion link [15].

(c) Remove the washers [19] from the bolts [18].

(d) Isolate the sensor brackets [17] from the upper torsion link [15].

1) If it is necessary, use a string, STD-837, to hold the sensor brackets [17] and attach the string, STD-837, over the top of the outer cylinder.

SUBTASK 32-21-31-480-002

(5) Use a nylon rope, STD-6545, to hold the upper end of the lower torsion link [11] to the outer cylinder.

NOTE: This task will hold the lower torsion link [11] in its position when you remove and install the upper torsion link [15] in the subsequent steps.

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SUBTASK 32-21-31-020-005

(6) Remove the upper torsion link [15] from the nose landing gear:

(a) Remove the pin [14], nut [13] and washer [12] from the apex pin [6].

(b) Remove the apex pin [6] to disconnect the upper torsion link [15] from the lower torsion link [11].

(c) Remove the pin [2], nut [1] and washer [3] from the torsion link pin [5].

(d) Remove the torsion link pin [5] to disconnect the upper torsion link [15] from the steering collar [4].

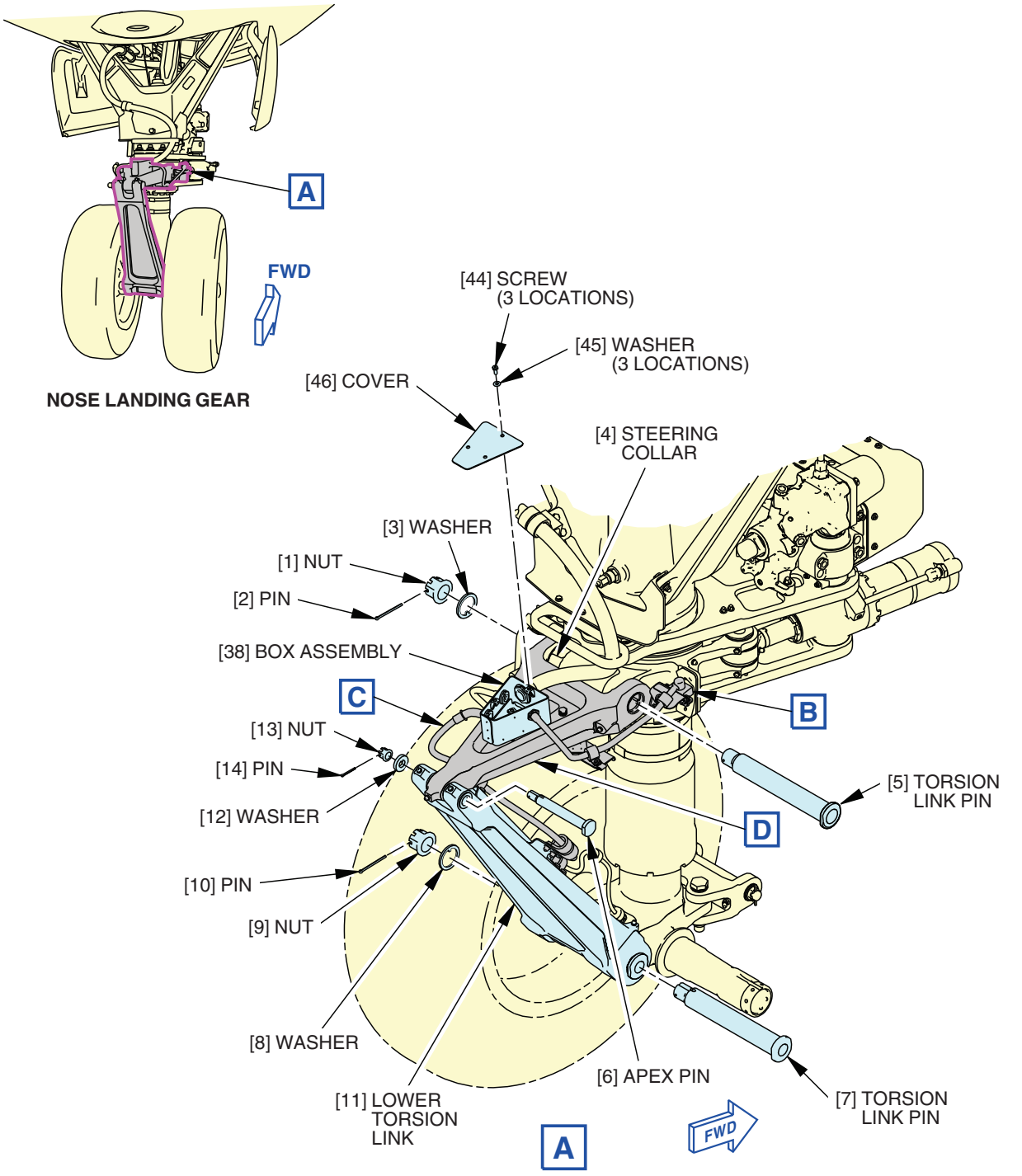
(e) Isolate the box assembly [38] from the upper torsion link [15].

1) If it is necessary, use a string, STD-837, to hold the box assembly [38] and attach the string, STD-837, to the outer cylinder.

(f) Remove the upper torsion link [15] from the airplane.

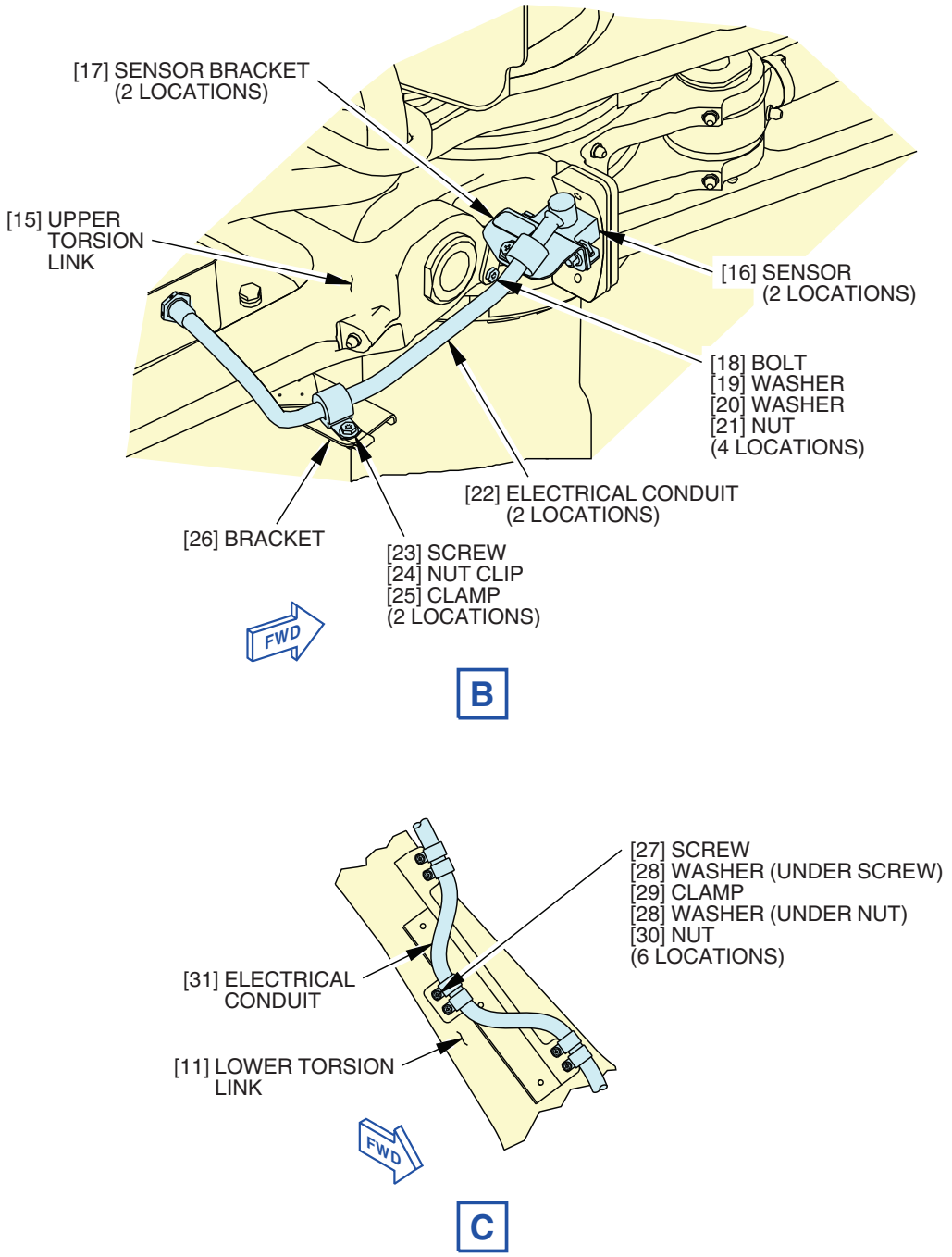
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Landing Gear Torsion Links Installation
Figure 401/32-21-31-990-804 (Sheet 1 of 3)

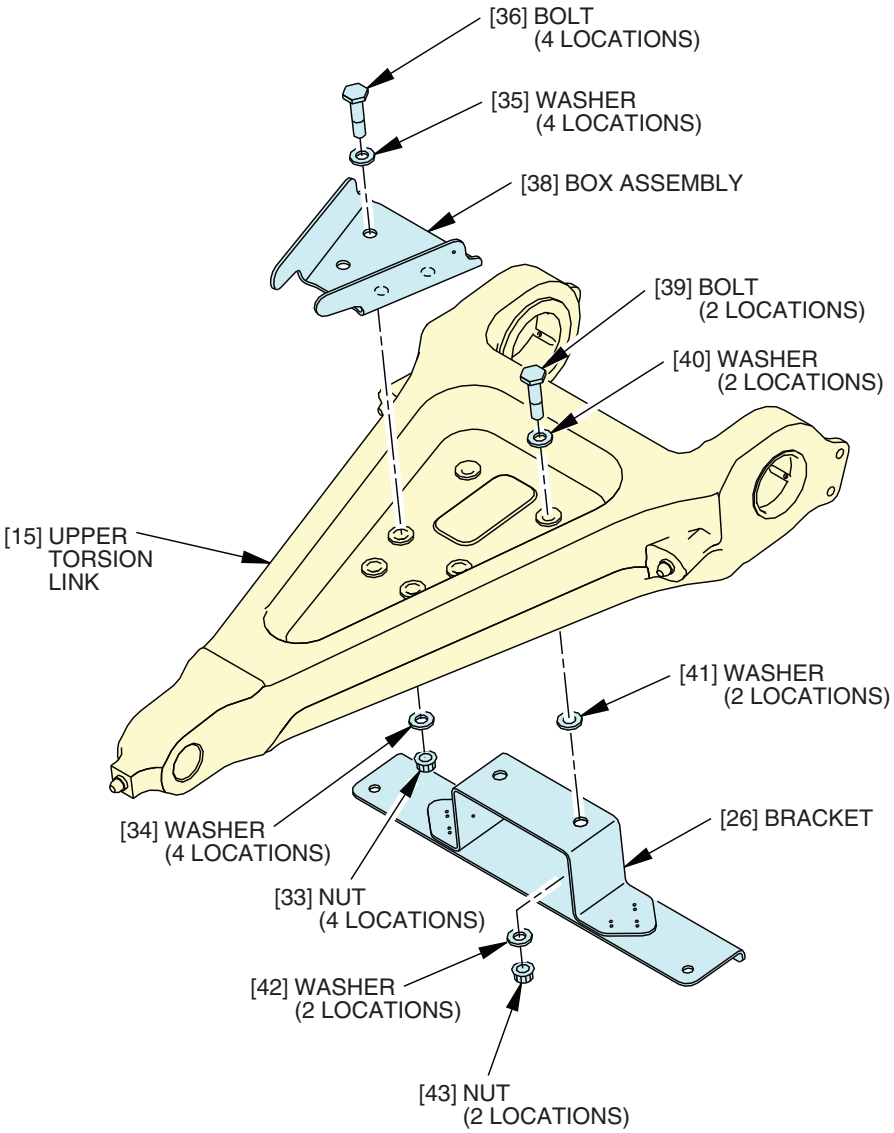
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Nose Landing Gear Torsion Links Installation
Figure 401/32-21-31-990-804 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Nose Landing Gear Torsion Links Installation
Figure 401/32-21-31-990-804 (Sheet 3 of 3)

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TASK 32-21-31-400-801

3. Nose Landing Gear Upper Torsion Link Installation

(Figure 401)

A. General

(1) This procedure supplies instructions to install the upper torsion link [15].

B. References

Reference	Title
12-21-21-640-802	Nose Landing Gear Lower End Components Servicing (P/B 301)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205
STD-837	String
STD-6545	Rope - Nylon

D. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
11	Lower torsion link	32-21-31-01-100	PTF ALL
14	Pin	32-21-31-01-010	PTF ALL
15	Upper torsion link	32-21-31-01-070	PTF ALL
38	Box assembly	32-21-51-15-125	PTF ALL
46	Cover	32-21-51-11-080	PTF ALL

F. Location Zones

Zone	Area
713	Nose Landing Gear

G. Nose Landing Gear Upper Torsion Link Installation

SUBTASK 32-21-31-420-001

(1) Connect the upper torsion link [15] to the steering collar [4]:

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WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin layer of the corrosion inhibiting material, G50136, to these items:

1) The threads and thread reliefs of the torsion link pin [5]
2) The threads of the nut [1]
3) The faces of the washer [3]
4) The mating surfaces of the upper torsion link [15] and the steering collar [4].

(b) Remove excess corrosion inhibiting material, G50136.

(c) Put the upper torsion link [15] in its position on the steering collar [4].

(d) Put the torsion link pin [5] through the upper torsion link [15] and the steering collar [4].

(e) Install the washer [3] and the nut [1] on the torsion link pin [5]].

(f) Tighten the nut [1] to 21 ft-lb (28.5 N·m) - 25 ft-lb (33.9 N·m) more than the run-on torque.

(g) If it is necessary, loosen the nut [1] to the nearest castellation to align the holes for the pin [2].

(h) Install the pin [2] in the torsion link pin [5].

SUBTASK 32-21-31-420-002

(2) Connect the upper torsion link [15] to the lower torsion link [11]:

(a) Lubricate the chrome plated of the apex pin [6] with grease, D00633.

(b) Remove any excess grease, D00633.

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(c) Apply a thin layer of the corrosion inhibiting material, G50136, to these items:

1) The threads and thread reliefs of the apex pin [6]
2) The threads of the nut [13]
3) The faces of the washer [12]
4) The mating surfaces of the upper torsion link [15] and the lower torsion link [11].

(d) Remove excess corrosion inhibiting material, G50136.

(e) Align the apex holes in the upper torsion link [15], and the lower torsion link [11].

(f) Put the apex pin [6] through the lower torsion link [11] and the upper torsion link [15].

(g) Install the washer [12] and the nut [13] on the apex pin [6].

(h) Tighten the nut [13] to 160 in-lb (18.1 N·m) - 190 in-lb (21.5 N·m) more than the run-on torque.

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(i) Install the pin [14] in the apex pin [6].

1) If it is necessary, loosen the nut [13] to the nearest castellation to align the holes for the pin [14].

SUBTASK 32-21-31-080-001

(3) Remove the nylon rope, STD-6545,that holds the upper end of the lower torsion link [11] to the outer cylinder.

SUBTASK 32-21-31-020-006

(4) Connect the sensor brackets [17] to the upper torsion link [15]:

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin layer of the corrosion inhibiting material, G50136, to these items:

1) The threads and thread reliefs of the bolts [18]

2) The threads of the nuts [21]

3) The faces of the washers [19] and washers [20]

(b) Remove any excess corrosion inhibiting material, G50136.

(c) If a string, STD-837, is attached to the sensor brackets [17] and the outer cylinder, remove the string, STD-837.

(d) Put the sensor brackets [17] in their position on the upper torsion link [15].

(e) Install the washers [19] on the bolts [18].

(f) Put the bolts [18] through the upper torsion link [15], and the sensor brackets [17].

(g) Install the washers [20] and the nuts [21] on the bolts [18].

(h) Tighten the nuts [21].

SUBTASK 32-21-31-020-007

(5) Connect the box assembly [38] to the upper torsion link [15]:

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin layer of the corrosion inhibiting material, G50136, to these items:

1) The threads and shank of the bolts [36]

2) The threads of the nut [33]

3) The faces of the washers [34] and washers [35].

(b) Remove excess corrosion inhibiting material, G50136.

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(c) If a string, STD-837, is attached to the box assembly [38] and the outer cylinder, remove the string, STD-837.

(d) Put the box assembly [38] in its position on the upper torsion link [15].

(e) Install the washers [35] on the bolts [36].

(f) Put the bolts [36] through the box assembly [38] and the upper torsion link [15].

(g) Install the washers [34] and the nuts [33] on the bolts [36].

(h) Install the cover [46] on the box assembly [38].

(i) Install the screws [44] and washers [45].

SUBTASK 32-21-31-020-008

(6) Install the bracket [26] on the upper torsion link [15]:

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin layer of the corrosion inhibiting material, G50136, to these items:

1) The threads and shank of the bolts [39]

2) The threads of the nuts [43]

3) The faces of the washers [40], washers [41], washers [42].

(b) Remove excess corrosion inhibiting material, G50136.

(c) Install the washers [40] on the bolts [39].

(d) Put the bolts [39] through the upper torsion link [15], washers [41] and the bracket [26].

(e) Install the washers [42] and nuts [43] on the bolts [39].

SUBTASK 32-21-31-420-003

(7) Install the clamps [25] that hold the electrical conduits [22] to the bracket [26]:

(a) Put the electrical conduits [22] in their position on the bracket [26].

(b) Install the clamps [25] and the nut clips [24] on the bracket [26].

(c) Put the screws [23] through the clamps [25], nut clips [24] and bracket [26].

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-21-31-640-001

(1) Lubricate the lube fittings on the upper torsion link [15] (TASK 12-21-21-640-802).

SUBTASK 32-21-31-080-004

(2) Remove the NLG towing lever pin, SPL-1499, for the towing lever from the metering valve for nose wheel steering.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-21-31-000-802

4. Nose Landing Gear Lower Torsion Link Removal

(Figure 401)

A. General

(1) This procedure supplies instructions to remove the lower torsion link.

B. References

Reference	Title
07-10-03-580-803	Lift the Nose Landing Gear Axle With Axle Jacks-Handling (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-45-21-000-801	Nose Landing Gear Wheel and Tire Assembly Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205
STD-6545	Rope - Nylon

D. Location Zones

Zone	Area
713	Nose Landing Gear

E. Prepare for the Removal

SUBTASK 32-21-31-480-007

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-21-31-480-010

WARNING ONLY USE THE CORRECT PIN FOR THE AIRPLANE MODEL. IF YOU USE AN INCORRECT PIN, THE HYDRAULIC STEERING CAN OPERATE. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) Install the NLG towing lever pin, SPL-1499, for the towing lever in the metering valve for the nose wheel steering.

SUBTASK 32-21-31-580-001

(3) Lift the nose landing gear with an axle jack (TASK 07-10-03-580-803).

SUBTASK 32-21-31-010-001

(4) Remove the left or right wheel and tire of the nose landing gear (TASK 32-45-21-000-801).

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F. Nose Landing Gear Lower Torsion Link Removal

SUBTASK 32-21-31-020-015

(1) Remove the clamps [29] that hold the electrical conduit [31] to the lower torsion link [11].

(a) Remove the screws [27], washers [28], and nuts [30].

SUBTASK 32-21-31-480-004

(2) Use a nylon rope, STD-6545, to hold the aft end of the upper torsion link [15] to the outer cylinder.

NOTE: This step will hold the upper torsion link in its position when you remove and install the lower torsion link in the subsequent steps.

SUBTASK 32-21-31-020-009

(3) Remove the lower torsion link [11] from the nose landing gear:

(a) Remove the pin [14], nut [13], and washer [12] from the apex pin [6].

(b) Remove the apex pin [6] to disconnect the lower torsion link [11] from the upper torsion link [15].

(c) Remove the pin [10], nut [9], and washer [8] from the torsion link pin [7].

(d) Remove the torsion link pin [7] to disconnect the lower torsion link [11] from the inner cylinder.

(e) Remove the lower torsion link [11] from the airplane.

——— END OF TASK ———

TASK 32-21-31-400-802

5. Nose Landing Gear Lower Torsion Link Installation

(Figure 401)

A. General

(1) This procedure supplies instructions to install the lower torsion link.

B. References

Reference	Title
07-10-21-580-802	Lower the Airplane Nose, Jack Point D (P/B 201)
12-21-21-640-802	Nose Landing Gear Lower End Components Servicing (P/B 301)
32-45-21-400-801	Nose Landing Gear Wheel and Tire Assembly Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33

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(Continued)

Reference	Description	Specification
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
11	Lower torsion link	32-21-31-01-100	PTF ALL
14	Pin	32-21-31-01-010	PTF ALL
15	Upper torsion link	32-21-31-01-070	PTF ALL

F. Location Zones

Zone	Area
713	Nose Landing Gear

G. Nose Landing Gear Lower Torsion Link Installation

SUBTASK 32-21-31-420-004

(1) Connect the lower torsion link [11] to the inner cylinder:

(a) Lubricate the chrome plated surfaces of the torsion link pin [7], lower torsion link [11], and mating surfaces of the inner cylinder with grease, D00633.

(b) Remove all unwanted grease, D00633.

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(c) Apply a thin layer of corrosion inhibiting material, G50136, to the threads of the nut [9] and faces of the washer [8].

(d) Remove all unwanted corrosion inhibiting material, G50136.

(e) Put the lower torsion link [11] in its position on the inner cylinder.

(f) Put the torsion link pin [7] through the lower torsion link [11] and inner cylinder.

(g) Install the washer [8] and nut [9] on the torsion link pin [7].

(h) Tighten the nut [9] to 250 in-lb (28.2 N·m) - 300 in-lb (33.9 N·m) more than the run-on torque.

(i) If it is necessary, loosen the nut [9] to the nearest castellation to align the holes for the pin [10].

(j) Install the pin [10] in the torsion link pin [7].

SUBTASK 32-21-31-420-005

(2) Connect the lower torsion link [11] to the upper torsion link [15]:

(a) Lubricate the chrome plated of the apex pin [6] with grease, D00633.

(b) Remove any excess grease, D00633.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(c) Apply a thin layer of corrosion inhibiting material, G50136, to these items:

1) The threads and thread reliefs of the apex pin [6]
2) The threads of the nut [13]
3) The faces of the washer [12]
4) The mating surfaces of the lower torsion link [11] and upper torsion link [15].

(d) Remove all unwanted corrosion inhibiting material, G50136.

(e) Align the apex holes in the lower torsion link [11] and upper torsion link [15].

(f) Put the apex pin [6] through the lower torsion link [11] and upper torsion link [15].

(g) Install the washer [12] and nut [13] on the apex pin [6].

(h) Tighten the nut [13] to 160 in-lb (18.1 N·m) - 190 in-lb (21.5 N·m) more than the run-on torque.

(i) Install the cotter pin [14] in the apex pin [6].

1) If it is necessary, loosen the nut [13] to the nearest castellation to align the holes for the cotter pin [14].

SUBTASK 32-21-31-420-009

(3) Install the electrical conduit [31] on the lower torsion link [11].

(a) Put the electrical conduit [31] in its position for the attachment to the clamps [29].

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(b) Apply a thin layer of corrosion inhibiting material, G50136, to the threads of the screws [27] and nuts [30].

(c) Remove all unwanted corrosion inhibiting material, G50136.

(d) Install the screws [27], washers [28], and nuts [30], to hold the electrical conduit [31] to the clamps [29].

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-21-31-420-006

(1) Do this task: Nose Landing Gear Wheel and Tire Assembly Installation, TASK 32-45-21-400-801.

SUBTASK 32-21-31-080-002

(2) Lower the axle jack for the nose landing gear and remove the jack (TASK 07-10-21-580-802).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-21-31-640-002

(3) Lubricate the lube fittings on the lower torsion link [11] (TASK 12-21-21-640-802).

SUBTASK 32-21-31-080-005

(4) Remove the NLG towing lever pin, SPL-1499, for the towing lever from the metering valve for nose wheel steering.

——— END OF TASK ———

TASK 32-21-31-000-803

6. Nose Landing Gear Torsion Link Disconnection

(Figure 401)

A. General

(1) This procedure supplies instructions to disconnect the torsion links.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205

D. Location Zones

Zone	Area
713	Nose Landing Gear

E. Prepare for the Disconnection

SUBTASK 32-21-31-480-005

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-21-31-480-009

WARNING ONLY USE THE CORRECT PIN FOR THE AIRPLANE MODEL. IF YOU USE AN INCORRECT PIN, THE HYDRAULIC STEERING CAN OPERATE. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) Install the NLG towing lever pin, SPL-1499, for the towing lever in the metering valve for the nose wheel steering.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Nose Landing Gear Torsion Link Disconnection

SUBTASK 32-21-31-020-010

(1) Disconnect the apex joint of the upper torsion link [15] and the lower torsion link [11]:

(a) Remove the pin [14], nut [13] and washer [12] from the apex pin [6].

1) Discard the pin [14].

(b) Remove the apex pin [6] to disconnect the lower torsion link [11] from the upper torsion link [15].

——— END OF TASK ———

TASK 32-21-31-400-803

7. Nose Landing Gear Torsion Link Connection

(Figure 401)

A. General

(1) This task gives the instructions to connect the torsion links.

B. References

Reference	Title
12-21-21-640-802	Nose Landing Gear Lower End Components Servicing (P/B 301)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
14	Pin	32-21-31-01-010	PTF ALL

F. Location Zones

Zone	Area
713	Nose Landing Gear

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Prepare for Nose Landing Gear Torsion Link Connection

SUBTASK 32-21-31-400-001

WARNING MAKE SURE THAT YOU INSTALL THE NLG TOWING-LEVER PIN CORRECTLY. IF NOT, SUDDEN MOVEMENT OF THE STEERING COLLAR CAN CAUSE INJURY TO PERSONNEL.

(1) If the NLG towing lever pin, SPL-1499, is not installed, complete the following steps:

(a) Move the towing lever on the metering valve module to the TOW position.

(b) Install the NLG towing lever pin, SPL-1499.

NOTE: The NLG towing lever pin, SPL-1499, holds the towing lever in the TOW position.

H. Nose Landing Gear Torsion Link Connection

SUBTASK 32-21-31-420-007

(1) Connect the apex joint of the upper torsion link [15], and the lower torsion link [11]:

(a) Lubricate the chrome plated of the apex pin [6] with grease, D00633.

(b) Remove any excess grease, D00633.

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(c) Apply a thin layer of the corrosion inhibiting material, G50136, to these items:

1) The threads and thread reliefs of the apex pin [6]

2) The threads of the nut [13]

3) The faces of the washer [12]

4) The mating surfaces of the upper torsion link [15] and the lower torsion link [11].

(d) Remove all unwanted corrosion inhibiting material, G50136.

(e) Align the apex holes in the upper torsion link [15] and the lower torsion link [11].

(f) Put the apex pin [6] through the lower torsion link [11] and the upper torsion link [15].

(g) Install the washer [12] and the nut [13] on the apex pin [6].

(h) Tighten the nut [13] to 160 in-lb (18.1 N·m) - 190 in-lb (21.5 N·m) more than the run-on torque.

(i) Install the new pin [14] in the apex pin [6].

1) If it is necessary, loosen the nut [13] to the nearest castellation to align the holes for the pin [14].

I. Put the Airplane Back to Its Usual Condition

SUBTASK 32-21-31-640-003

(1) Lubricate the lube fittings on the apex joint of the lower torsion link [11] and the upper torsion link [15] (TASK 12-21-21-640-802).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-21-31-080-003

(2) Remove the NLG towing lever pin, SPL-1499, for the towing lever from the metering valve for nose wheel steering.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE LANDING GEAR TORSION LINK - REPAIRS

1. General

A. This procedure has this task:

(1) Nose Landing Gear Lower Torsion Link Repair

TASK 32-21-31-350-801

2. Nose Landing Gear Lower Torsion Link Repair

(Figure 801)

A. General

(1) This repair permits blends on the outer surfaces only. One blend per flange is permitted.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
51-00-58	STANDARD TREATMENT METHODS

C. Consumable Materials

Reference	Description	Specification
C00033	Coating - Protective Enamel, Flexibility Use	BMS10-60 Type II
C00175	Primer - Urethane Compatible, Corrosion Resistant	BMS10-79 Type III
C50005	Coating - Chemical Conversion - Bonderite M-CR 1200S Aero (Formerly Alodine 1200S)	BAC5719 Class A, C or D, MIL-DTL-81706 Type I Class 1A or 3

D. Location Zones

Zone	Area
713	Nose Landing Gear

E. Prepare for the Repair

SUBTASK 32-21-31-420-010

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

F. Nose Landing Gear Lower Torsion Link Repair

SUBTASK 32-21-31-350-001

(1) Blend out the defects within the limits and areas shown.

(a) Blends can be on the left flange or the right flange, or both, but only one blend is permitted on each flange.

(b) Blend out the defects 20:1 with the adjacent surfaces.

(c) If blends are necessary outside the limits or areas shown, ask Boeing for advice.

(d) See the figure for permitted blends (Figure 801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-21-31-350-002

(2) Do a penetrant inspection (NDI) of the blended areas.

SUBTASK 32-21-31-350-004

(3) Shot peen or flapper/roto-peen the blended areas.

(a) Intensity 0.008-0.013A2.

(b) Coverage 2.0.

SUBTASK 32-21-31-350-003

(4) Apply Bonderite M-CR 1200S Aero coating, C50005, to all blended areas (SUBJECT 51-00-58).

SUBTASK 32-21-31-350-005

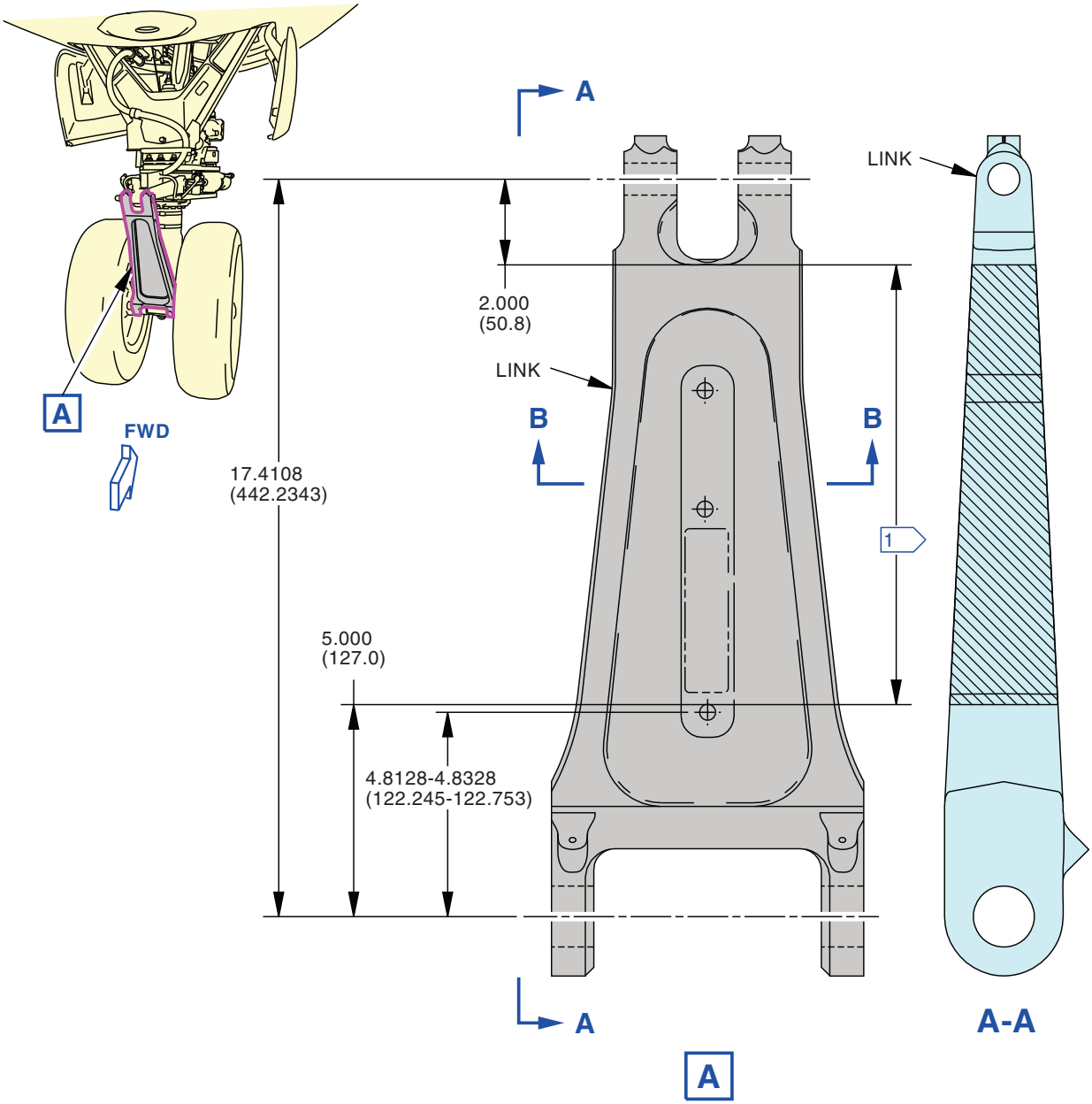
(5) Apply one coat of BMS 10-79, Type III primer, C00175, to the reworked areas.

SUBTASK 32-21-31-350-006

(6) Apply BMS 10-60 Type II gloss enamel coating, C00033, with a dry film thickness of 0.254 mm (0.010 in.) - 0.305 mm (0.012 in.) to the reworked areas.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

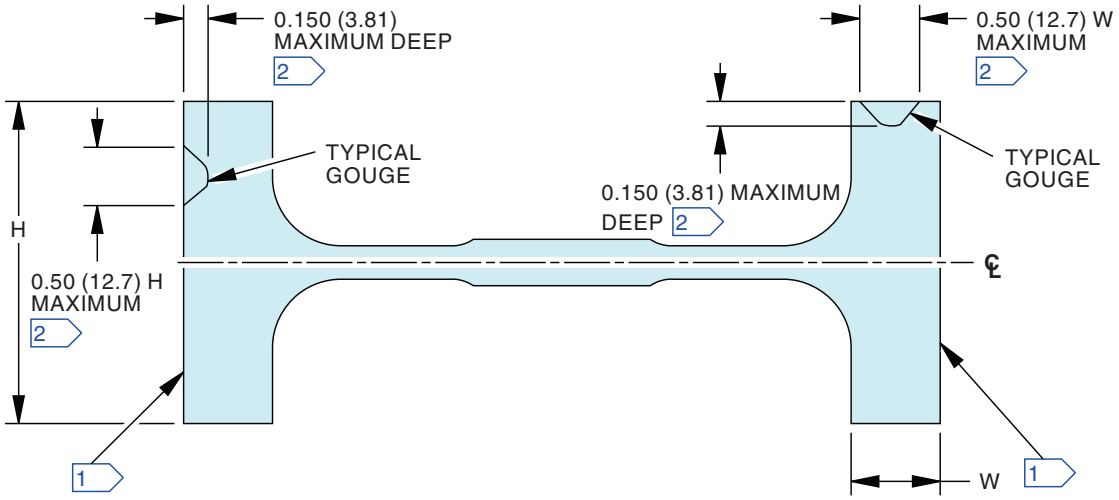
ALL DIMENSIONS ARE IN INCHES (mm).

1 BLENDS PERMITTED ONLY ON THESE OUTER SURFACES. ONLY ONE BLEND PERMITTED ON EACH FLANGE, ON ONLY ONE OF THE TOP, SIDE, OR BOTTOM SURFACES

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Nose Landing Gear Lower Torsion Link Repair
Figure 801/32-21-31-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



B-B

NOTE:

ALL DIMENSIONS ARE IN INCHES (mm).

1 BLENDS PERMITTED ONLY ON THESE OUTER SURFACES. ONLY ONE BLEND PERMITTED ON EACH FLANGE, ON ONLY ONE OF THE TOP, SIDE, OR BOTTOM SURFACES

2 LIMIT FOR DEFECT REMOVAL

2414201 S00061534592_V1

Nose Landing Gear Lower Torsion Link Repair
Figure 801/32-21-31-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE LANDING GEAR CORROSION PREVENTION - MAINTENANCE PRACTICES

1. General

A. This pageblock contains this task:

(1) Corrosion prevention of the nose landing gear.

TASK 32-21-37-600-801

2. Nose Landing Gear Corrosion Prevention

A. General

(1) The basic corrosion prevention philosophy is to make periodic inspections to preclude or detect the early stages of corrosion. Missing fasteners, white powdery or discolored deposits are evidence of the existence of corrosion which should alert operators that some corrective action is necessary. A corrosion prevention program should be initiated to prevent the accumulation of corrosive products in order to minimize the occurrence of corrosion.

(2) After the suspect areas are cleaned, a full inspection helps to make sure that protective finishes applied during manufacture remain intact.

(3) Where corrosion exists (noticeable bulges of the skin or white deposits of corrosion products at fastener heads of joint edges), refer to Structural Repair Manual for details of corrosion removal.

(4) For minor corrosion, to minimize the downtime of the airplane, the corrosion products should be cleaned off, followed by the application of a corrosion inhibiting compound into the affected area to slow the corrosion process. The finish system should be restored at the next maintenance opportunity.

B. Consumable Materials

Reference	Description	Specification
G00009	Compound - Organic Corrosion Inhibiting	BMS3-23
G01314	Tape - 3M Polyester Film Tape 8412	

C. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right

D. Corrosion Prevention Treatment

SUBTASK 32-21-37-610-001

(1) At the next maintenance opportunity, do this corrosion prevention treatment to the nose landing gear.

SUBTASK 32-21-37-610-002

CAUTION

DO NOT APPLY CORROSION INHIBITING COMPOUND ON GREASED JOINTS OR SEALED BEARINGS. THESE COMPOUNDS DISSOLVE GREASE AND OTHER LUBRICANTS. THEY ARE PENETRATING COMPOUNDS AND CAN GET AROUND THE SEALS AND INTO THE BEARINGS.

(2) Do these corrosion preventive measures to these components (Figure 201).

(a) Shock struts (Oleo).

1) Apply corrosion inhibiting compound, G00009, to the exterior areas of the inner and outer cylinder with broken finish systems.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Spray all lugs, lug faces, connecting pins and fasteners should be sprayed with corrosion inhibiting compound, G00009.

(b) Upper Drag Brace Only.

1) Apply corrosion inhibiting compound, G00009, to the exterior surface areas with broken finish systems.

2) Spray all lugs, lug faces, connecting pins, and fasteners with corrosion inhibiting compound, G00009.

(c) Axle.

1) Apply corrosion inhibiting compound, G00009, to outside surface areas of the axle.

2) Make suitable nozzle extension and spray the inside surfaces of the axle.

3) At the wheel removal, spray the exterior surfaces of axle cover by the wheel except at the bearing or journal surfaces.

(d) Torsion Links.

1) Apply corrosion inhibiting compound, G00009, to the surface areas with broken finish systems.

2) Spray the lugs, lug faces, and connecting pins with corrosion inhibiting compound.

(e) Trunnion.

1) Apply corrosion inhibiting compound, G00009, to the exterior surface areas with broken finish systems.

2) Spray the lugs, lug faces, connecting pins, fasteners, and trunnion bearing caps with corrosion inhibiting compound, G00009.

(f) Lock Brace Assembly.

1) Apply corrosion inhibiting compound, G00009, to the surface areas of all components of the locking linkages with broken finish systems.

2) Also spray the lugs, lug faces, and connecting pins.

SUBTASK 32-21-37-610-003

(3) Install 3M 8412 tape, G01314, under the corrosion resistant steel clamps at overhaul (recommended) to minimize the risk of galvanic corrosion if the finish is damaged .

E. Frequency of Application

SUBTASK 32-21-37-610-004

(1) Periodically inspect the areas identified as susceptible to corrosion.

NOTE: Operators must be aware of reported problems and areas of occurrences.

(a) Do this consistent to the schedules specified in the Maintenance Planning Document.

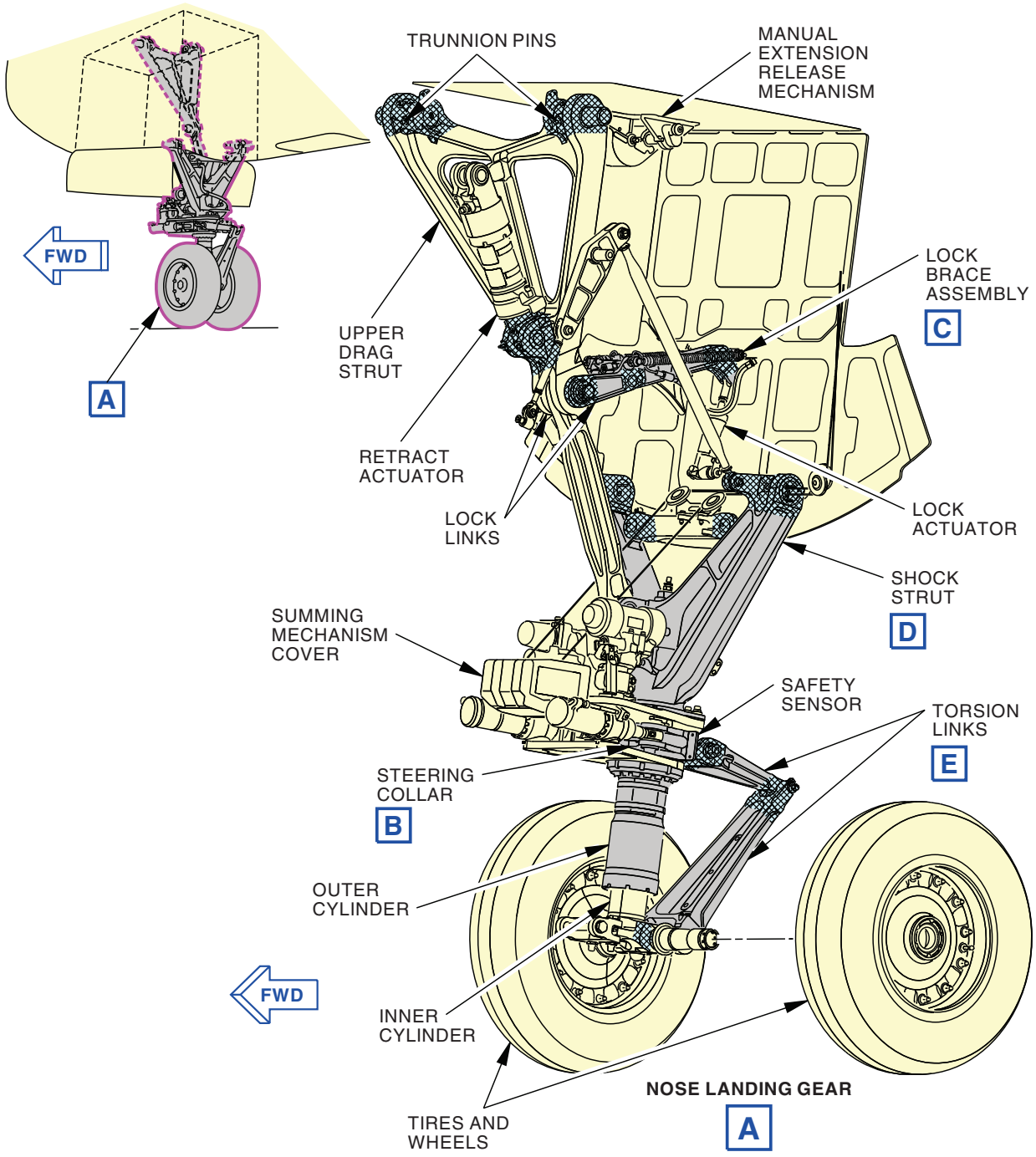
SUBTASK 32-21-37-610-005

(2) Periodically apply the corrosion inhibiting compound, G00009, to the areas identified.

(a) Do this consistent to the schedules specified in the Maintenance Planning Document.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

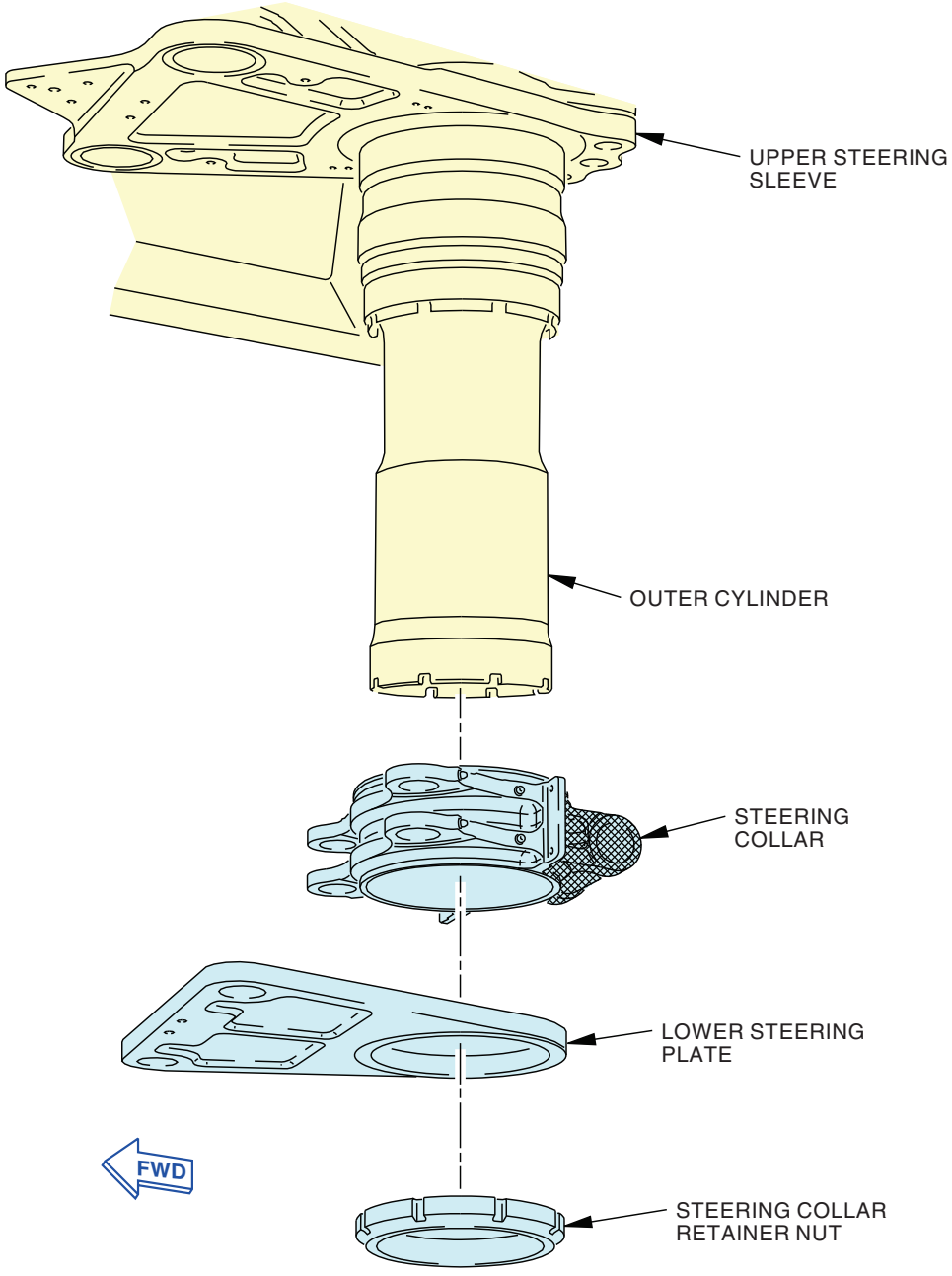


APPLY CORROSION INHIBITING COMPOUND (CIC) BMS3-23, TO EXPOSED OUTER SURFACES ONLY. DO NOT DISASSEMBLE PARTS TO APPLY CIC. DO NOT SPRAY INTO JOINTS, LINKAGES, PULLEYS, CABLES OR BEARINGS.

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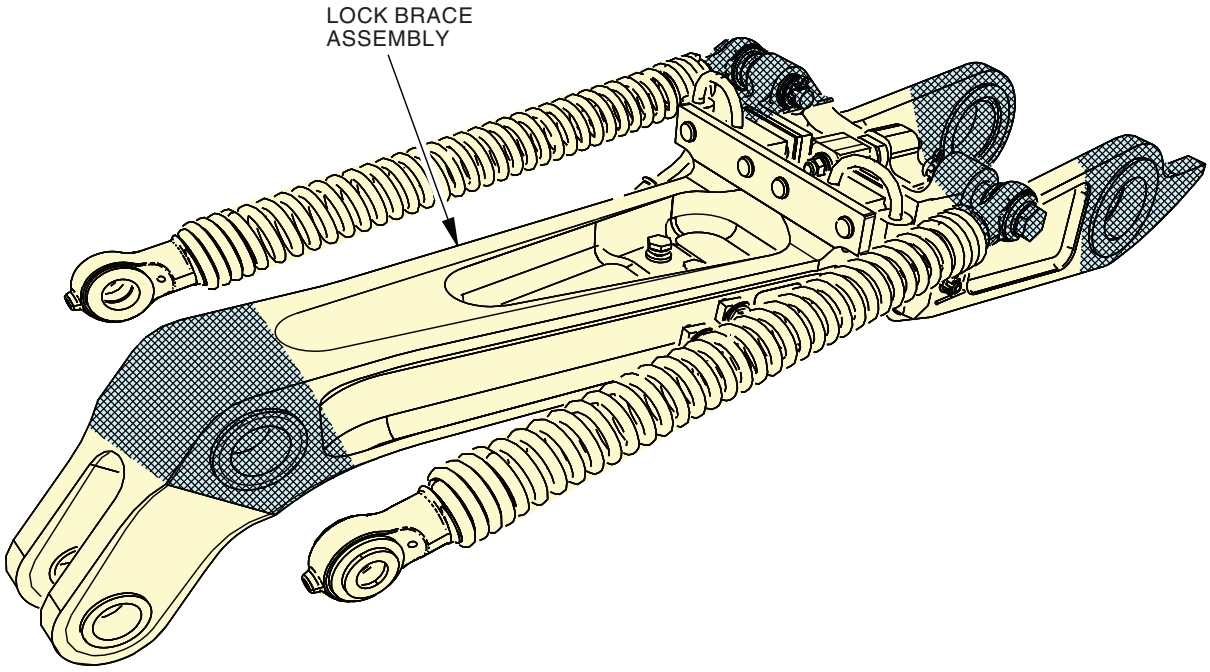
Nose Landing Gear Corrosion Prevention
Figure 201/32-21-37-990-801 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Landing Gear Corrosion Prevention
Figure 201/32-21-37-990-801 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



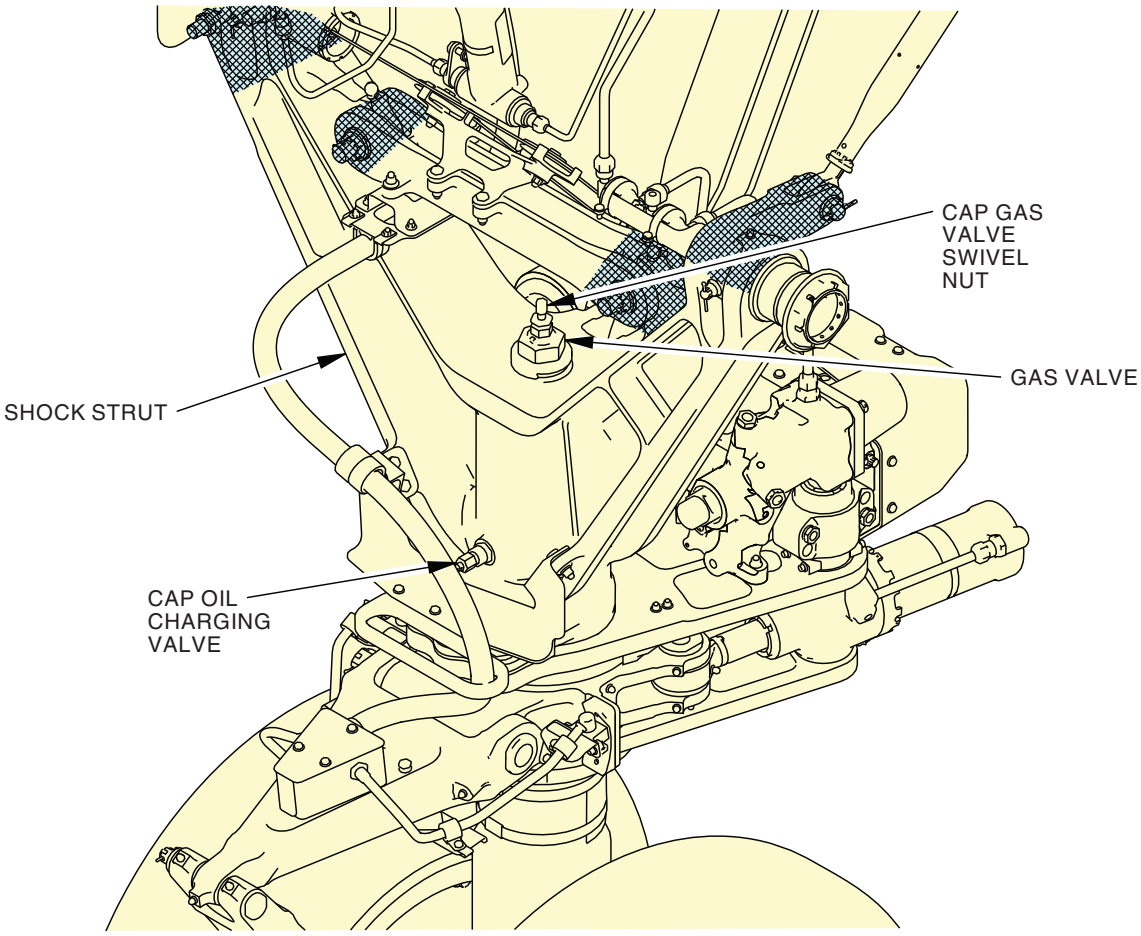
LOCK BRACE ASSEMBLY

C

2414204 S00061534598_V2

Nose Landing Gear Corrosion Prevention
Figure 201/32-21-37-990-801 (Sheet 3 of 5)

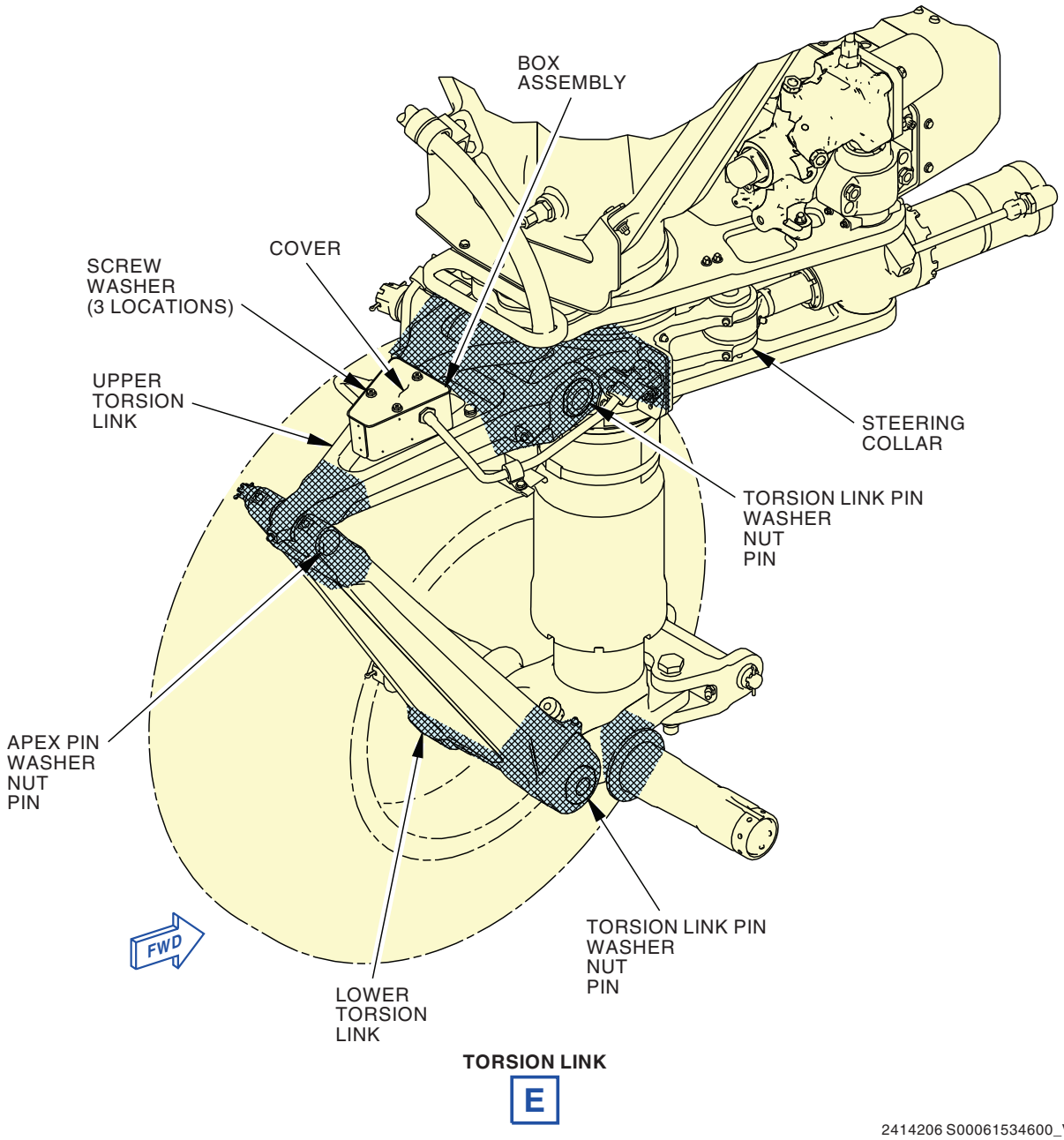
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2414205 S00061534599_V2

Nose Landing Gear Corrosion Prevention
Figure 201/32-21-37-990-801 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Landing Gear Corrosion Prevention
Figure 201/32-21-37-990-801 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE LANDING GEAR AXLE - INSPECTION/CHECK

1. General

A. This task supplies the inspection of the nose landing gear axles for corrosion.

TASK 32-21-71-200-801

2. Nose Gear Axle Detailed Inspection for Corrosion

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions to do the detailed inspection of the nose gear axles for corrosion.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-45-21-000-801	Nose Landing Gear Wheel and Tire Assembly Removal (P/B 401)
32-45-21-400-801	Nose Landing Gear Wheel and Tire Assembly Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
G00440	Lockwire - MS20995C41, Corrosion Resistant Steel - 0.041 Inch (1.0414 mm) Diameter	NASM20995
G01043	Cloth - Lint-free	
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

D. Location Zones

Zone	Area
713	Nose Landing Gear

E. Nose Gear Axle Detailed Inspection

SUBTASK 32-21-71-490-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-21-71-000-001

(2) To remove the nose landing gear wheel and tire assembly, do this task: Nose Landing Gear Wheel and Tire Assembly Removal, TASK 32-45-21-000-801.

SUBTASK 32-21-71-000-002

(3) For airplanes that use cotter pins, remove the cotter pins in the axle spacer and remove the spacer from the axle.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-21-71-010-001

(4) For airplanes that use lockwire, remove the MS20995C41 lockwire, G00440, in the axle spacer and remove the spacer from the axle.

SUBTASK 32-21-71-200-001

(5) Examine the nose gear axles.

NOTE: Corrosion can cause the nose gear axle to fracture and the wheel to depart the airplane.

(a) Make sure that the spacer is removed when you inspect for corrosion on the nose landing gear axle.

(b) Use a clean lint-free cloth, G01043, to remove grease and debris from the nose gear axle.

1) Do not use solvents or detergents to clean the nose gear axle.

(c) Do a Detailed Visual Inspection aided with a magnifying glass in the Nose Landing Gear (NLG) axles for corrosion or cracks caused by corrosion.

1) Pay special attention to the area that was under the axle spacer, and most importantly, at the 6 o'clock position on the axle.

2) If corrosion is detected on a nose landing gear axle, contact Boeing for disposition.

NOTE: No corrosion is permitted on the axle. Corrosion can cause the axle to fracture and the wheel to depart the airplane.

SUBTASK 32-21-71-600-001

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(6) Apply a light layer of corrosion inhibiting material, G50136, to the area underneath the axle spacer.

SUBTASK 32-21-71-420-001

(7) For airplanes that use cotter pins, make sure that the three cotter pins are installed to hold the spacer on the nose gear axle.

(a) Make sure that the three cotter pins installed on the spacer do not touch the lower torsion link lug during wheel installation.

SUBTASK 32-21-71-420-002

(8) For airplanes that use lockwire, make sure that MS20995C41 lockwire, G00440, is installed.

(a) Make sure to bend the lockwire end pigtails away from the inboard edge of the spacer (Figure 601).

(b) Examine the completed installation to make sure that the lockwire ends do not touch the nose gear axle or the torsion link lugs.

(c) Adjust the installed lockwire to make sure that the lockwire does not hit the landing gear parts during landing gear operations.

NOTE: Spacers have six lockwire holes.

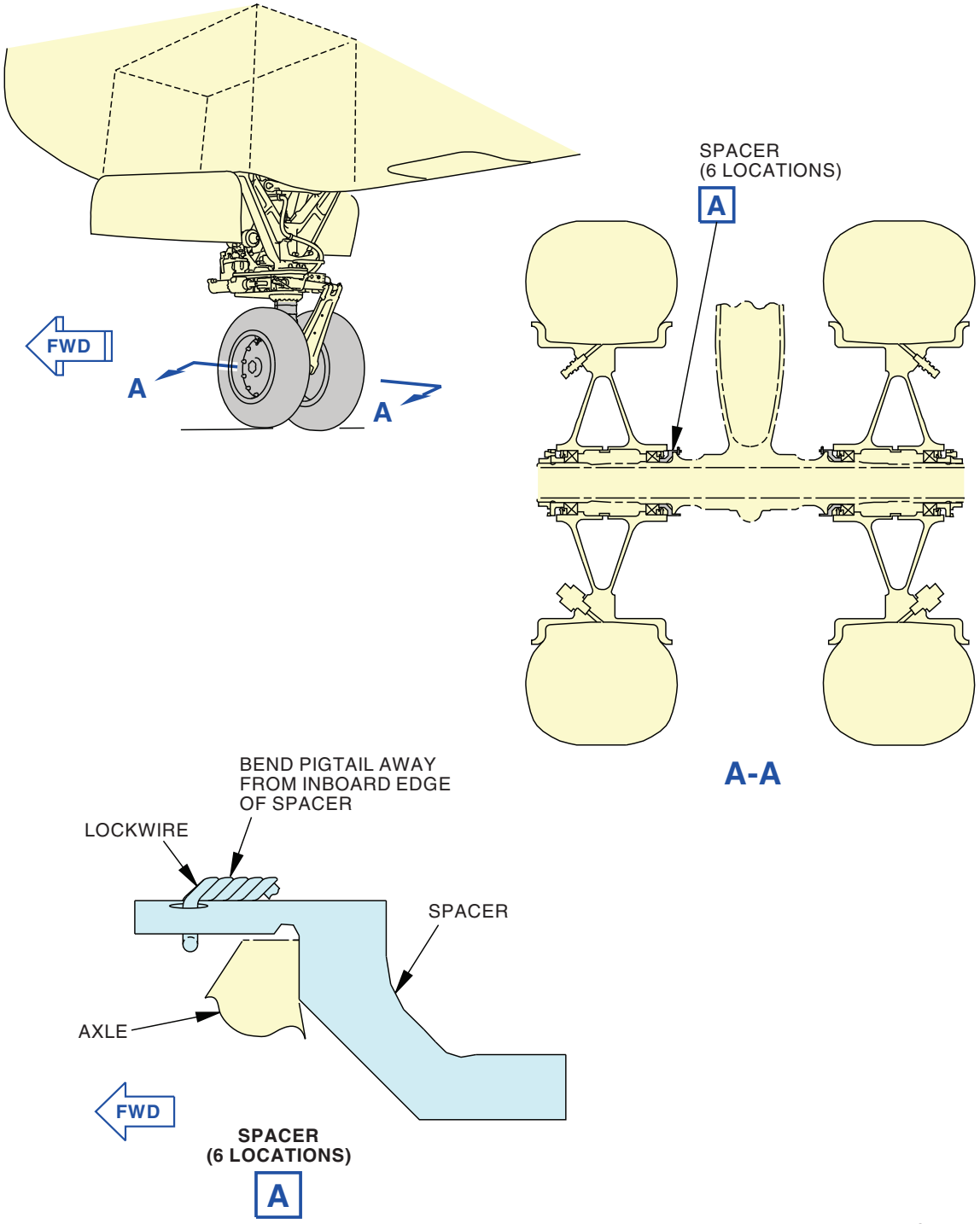
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-21-71-400-001

(9) To install the nose landing gear wheel and tire assembly, do this task: Nose Landing Gear Wheel and Tire Assembly Installation, TASK 32-45-21-400-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414207 S00061534604_V1

Nose Gear Axle and Spacer Lockwire Installation
Figure 601/32-21-71-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE LANDING GEAR WHEEL WELL DOORS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the wheel well doors, referred to as door assembly [3], for the nose landing gear

(2) An installation of the wheel well doors for the nose landing gear.

TASK 32-22-11-000-801

2. Nose Landing Gear Wheel Well Doors Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right

C. Prepare for the Removal

SUBTASK 32-22-11-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-22-11-480-002

(2) Install chocks around the tires of the nose landing gear (PAGEBLOCK 10-11-05/201).

D. Nose Landing Gear Wheel Well Doors Removal

SUBTASK 32-22-11-020-001

(1) Disconnect the bonding jumper [14] from the door hinge [1]:

(a) Remove the nut [13] from the bolt [10].

(b) Remove the washers [12] and bonding jumper [14] from the bolt [10].

(c) Remove the bolt [10], and washer [11] from the door hinge [1].

(d) Remove the washer [11] from the bolt [10].

SUBTASK 32-22-11-020-002

(2) Disconnect the center hinge [2] from the lower push rod [18]:

(a) Remove the pin [19], nut [20], and the washer [21] from the bolt [22].

(b) Remove the bolt [22] to disconnect the center hinge [2] from the lower push rod [18].

SUBTASK 32-22-11-480-003

(3) Use a rope and attach the lower push rod [18] to the upper rod assembly.

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SUBTASK 32-22-11-020-003

(4) Disconnect the door assembly [3] from the hinge bracket [26]:

(a) Remove the pin [17], nut [16], and washer [15] from the bolt [23].

(b) Remove the bolt [23], washer [24], and bushing [25] to disconnect the center hinge [2] from the hinge bracket [26].

SUBTASK 32-22-11-020-004

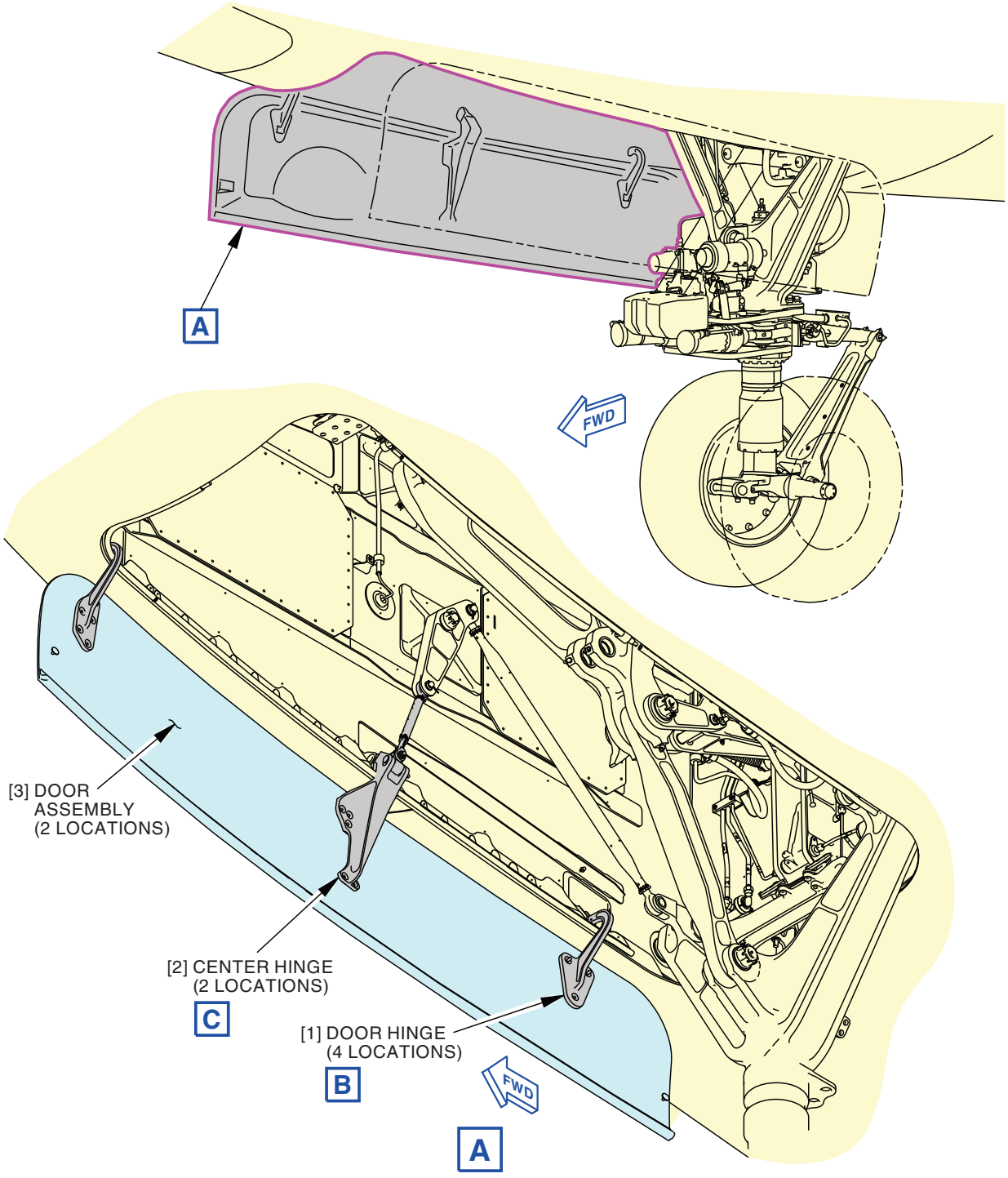
(5) Disconnect the door assembly [3] from the forward and aft hinge brackets [9]:

(a) Remove the pin [7], nut [8], and washer [6] from the bolt [4].

(b) Remove the bolt [4] and washer [5] to disconnect the door hinge [1] from the hinge bracket [9].

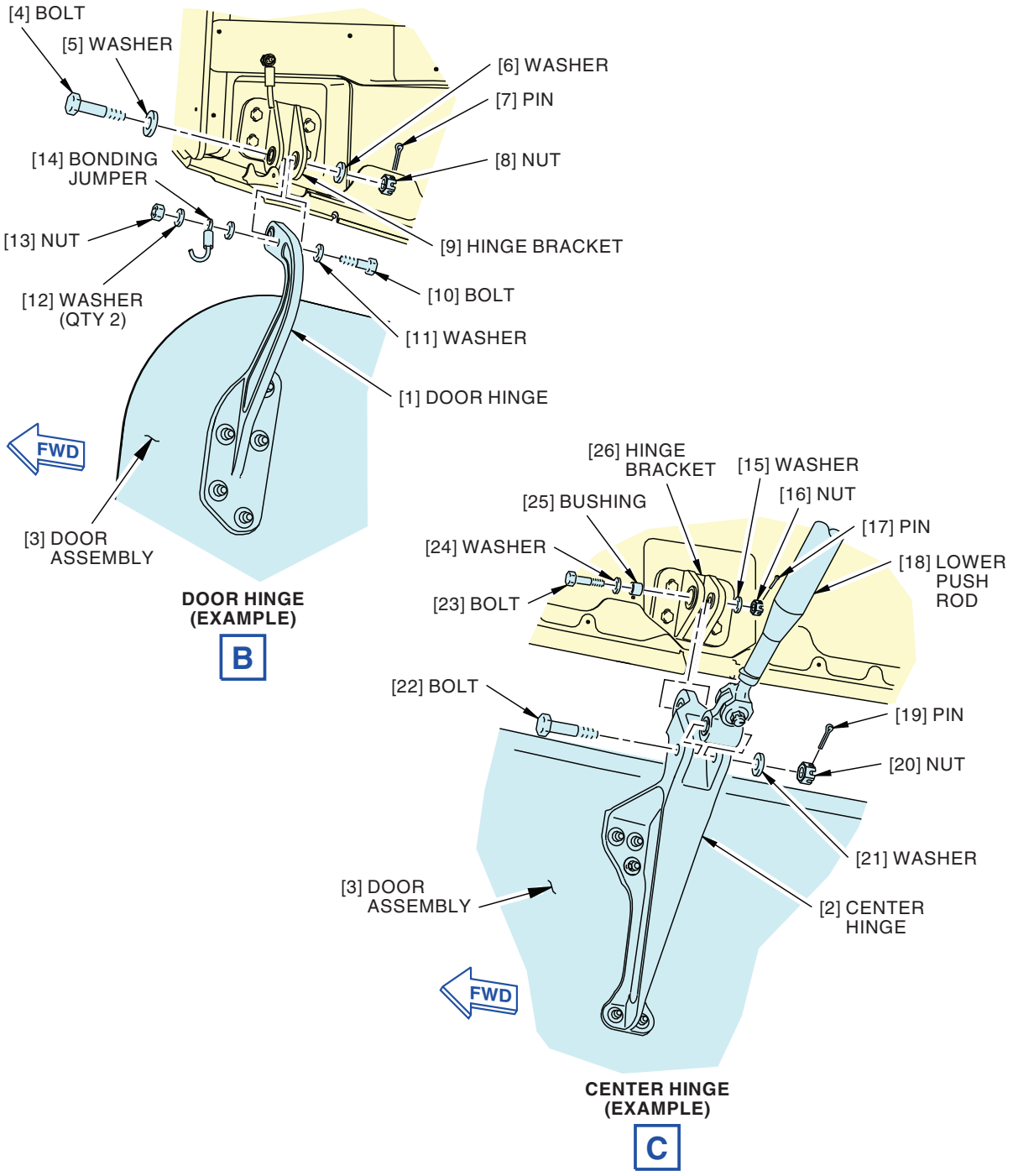
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Landing Gear Wheel Well Doors Installation
Figure 401/32-22-11-990-801 (Sheet 1 of 2)

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Nose Landing Gear Wheel Well Doors Installation
Figure 401/32-22-11-990-801 (Sheet 2 of 2)

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TASK 32-22-11-400-801

3. Nose Landing Gear Wheel Well Doors Installation

(Figure 401)

A. References

Reference	Title
20-10-34-120-801	Hand Clean Metal Surfaces with Abrasives (P/B 701)
32-22-11-000-802	Nose Landing Gear Wheel Well Doors Adjustment (P/B 501)
32-33-61-820-801	Nose Gear Wheel Well Door Linkage Adjustment and Check (P/B 501)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

C. Consumable Materials

Reference	Description	Specification
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50337	Sealant - Fuel Tank	BMS5-45 Class B
A50359	Sealant - Low Density, Non-Chromate Type	BMS5-142 Type II Class B
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
C00175	Primer - Urethane Compatible, Corrosion Resistant	BMS10-79 Type III
C00700	Coating - Protective Enamel (BAC 707 Gray Color, Gloss)	BMS10-60 Type I
C50475	Coating - Chemical Conversion - Bonderite M-CR 600 Aero	BAC5719 Type II Class D
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

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D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Door assembly	32-22-11-01-370	PTF ALL
7	Pin	32-22-11-01-045	PTF ALL
17	Pin	32-22-11-01-050	PTF ALL
19	Pin	32-22-11-03-065	PTF ALL

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

F. Nose Wheel Well Doors Installation

NOTE: A NLG door rigging check is not necessary if reinstalling the same NLG door that was removed and no changes to the adjustments to the door stops, door seals, door hinges, or door pushrods were made.

SUBTASK 32-22-11-420-001

(1) Connect the door assembly [3] to the forward and aft hinge brackets [9]:

(a) Hold the door assembly [3] in its position in the hinge bracket [9].

(b) Align the hole in the door hinge [1] to the hole in the hinge bracket [9].

(c) Put the bolt [4], with the washer [5] installed, through the hinge bracket [9] and the door hinge [1].

(d) Install the washer [6] and nut [8] on the bolt [4].

NOTE: The new pin [7] is installed after final adjustments to the door assembly [3] are made in the subsequent step.

(e) Hand tighten the nut [8] until after final adjustments to the door assembly [3] are made in the subsequent step.

SUBTASK 32-22-11-420-002

(2) Connect the door assembly [3] to the center hinge bracket [26]:

(a) Put the center hinge [2] in its position in the center hinge bracket [26].

(b) Align the hole in the center hinge [2] to the hole in the center hinge bracket [26].

(c) Put the bolt [23] through the center hinge bracket [26] and the center hinge [2].

(d) Install the washer [24], bushing [25], washer [15], and nut [16] on the bolt [23].

NOTE: The new pin [17] is installed after final adjustments to the door assembly [3] are made in the subsequent step.

(e) Hand tighten the nut [16] until after final adjustments to the door assembly [3] are made in the subsequent step.

SUBTASK 32-22-11-420-003

(3) Connect the center hinge [2] to the lower push rod [18]:

(a) Put the end of the lower push rod [18] in its position on the center hinge [2].

(b) Align the end of the lower new pin [17] to the hole in the center hinge [2].

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WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(c) Apply a thin layer of corrosion inhibiting material, G50136, to these items:

1) The threads and thread reliefs of the bolt [22] and the nut [20] and the faces of the washer [21].

(d) Put the bolt [22] through the center hinge [2] and the lower push rod [18].

(e) Install the washer [21] and nut [20] on the bolt [22].

NOTE: The new pin [19] is installed after final adjustments to the door assembly [3] are made in the subsequent step.

(f) Hand tighten the nut [20] until after final adjustments to the door assembly [3] are made in the subsequent step.

SUBTASK 32-22-11-120-001

(4) Prepare the mating surfaces of the door hinge [1], bolt [10], washer [11], washers [12], nut [13] and bonding jumper [14]:

(a) To fully clean the mating surface of the door hinge [1], do this task: Hand Clean Metal Surfaces with Abrasives, TASK 20-10-34-120-801.

(b) Use the cotton wiper, G00034, moist in solvent, B00184, to clean the mating surfaces of the bolt [10], washer [11], washers [12], nut [13] and bonding jumper [14].

SUBTASK 32-22-11-020-005

(5) Connect the bonding jumper [14] to the door hinge [1]:

(a) Put the bonding jumper [14] in its position on the door hinge [1].

(b) Install the washers [12] on the bolt [10].

(c) Put the bolt [10] through the door hinge [1], washers [12], bonding jumper [14], and washer [11].

(d) Install the nut [13] on the bolt [10].

(e) Tighten the nut [13] to 28 in-lb (3 N·m) - 35 in-lb (4 N·m).

SUBTASK 32-22-11-280-001

(6) Measure the electrical bonding resistance between the door hinge [1] and the bonding jumper [14] (SWPM 20-20-00).

(a) Use the intrinsically safe approved bonding meter, COM-1550.

(b) Make sure that the electrical bonding resistance is 0.001 ohm (1.0 milliohms) or less.

SUBTASK 32-22-11-390-001

(7) Apply a fillet seal with sealant, A50337, or sealant, A50231, or sealant, A50359, to cover the bolt [10], washers [12], washer [11], and nut [13].

SUBTASK 32-22-11-110-001

(8) Refinish the abraded surfaces as follows:

(a) Manually apply coating, C00064, or Bonderite M-CR 600 Aero coating, C50475.

(b) Apply one coat of primer, C00175.

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(c) Apply coating, C00700.

SUBTASK 32-22-11-420-004

(9) Adjust the wheel well doors for the nose landing gear (TASK 32-22-11-000-802).

SUBTASK 32-22-11-420-005

(10) Adjust the lower push rod [18] (TASK 32-33-61-820-801).

SUBTASK 32-22-11-420-006

(11) Tighten the nut [8] and install the new pin [7] in the hole in the bolt [4].

(a) If it is necessary, loosen the nut [8] to the nearest castellation to align the hole in the bolt [4].

SUBTASK 32-22-11-420-007

(12) Tighten the nut [16] and install the new pin [17] in the hole in the bolt [23].

(a) If it is necessary, loosen the nut [16] to the nearest castellation to align the hole in the bolt [23].

SUBTASK 32-22-11-420-008

(13) Tighten the nut [20] to 75 ±15 in-lb (8.5 ±1.7 N·m).

(a) Install the new pin [19] in the hole in the bolt [22].

(b) If it is necessary, loosen the nut [20] to the nearest castellation to align the hole in the bolt [22].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE LANDING GEAR WHEEL WELL DOORS - ADJUSTMENT/TEST

1. General

A. This procedure has this task:

(1) An adjustment of the wheel well doors for the nose landing gear.

TASK 32-22-11-000-802

2. Nose Landing Gear Wheel Well Doors Adjustment

(Figure 501)

A. General

(1) No adjustments will be made to the hinge fittings mounted on the doors during the procedure.

NOTE: The door seal installed incorrectly could prevent the wheel well retract to the normal close position.

B. References

Reference	Title
07-10-21-580-801	Lift the Airplane Nose with the Nose Jack at Jack Point D (P/B 201)
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-33-00-710-802	Nose Landing Gear Operational Test (P/B 501)
32-33-61-820-801	Nose Gear Wheel Well Door Linkage Adjustment and Check (P/B 501)

C. Consumable Materials

Reference	Description	Specification
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

D. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right

E. Prepare for the Adjustment

SUBTASK 32-22-11-480-004

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-22-11-580-001

(2) Do this task: Lift the Airplane Nose with the Nose Jack at Jack Point D, TASK 07-10-21-580-801.

NOTE: Lower the jack that supports the tail of the airplane while you lift the nose landing gear.

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SUBTASK 32-22-11-480-005

(3) Install chocks around the tires of the main landing gear (PAGEBLOCK 10-11-05/201).

SUBTASK 32-22-11-020-006

(4) Disconnect the lower pushrod [12] from the center door hinge [24].

(a) Remove the pin [19], nut [18], and washer [17] from the bolt [16].

(b) Remove the bolt [16] to disconnect the lower pushrod [12] from the center door hinge [24].

SUBTASK 32-22-11-080-002

(5) Remove the downlock pin for the nose landing gear (TASK 32-00-01-080-801).

SUBTASK 32-22-11-860-001

(6) Retract the nose landing gear (TASK 32-33-00-710-802).

F. Nose Landing Gear Wheel Well Doors Adjustment

SUBTASK 32-22-11-210-001

(1) Make sure that you can turn the bolt [20] by hand.

(a) If not, adjust the center hinge bracket [25] as follows:

1) Loosen the bolts [10] in the center hinge bracket [25].

2) Move the center hinge bracket [25] up or down until you can move the bolt [20] by hand.

3) Tighten the bolts [10] in the center hinge bracket [25].

NOTE: The bolt [20] must be free to move with hand pressure after you tighten the bolts [10].

SUBTASK 32-22-11-210-002

(2) Make sure that you can turn the bolt [20] by hand.

(a) If not, adjust the center hinge bracket [25] as follows:

1) Remove the pin [22], nut [21], bushing [37], washer [38], and washer [23] from the bolt [20].

2) Remove the bolt [20] that connects the center door hinge [24] to the center hinge bracket [25].

3) Remove the bolts [10] and washers [11] to disconnect the center hinge bracket [25] from the bulkhead.

NOTE: The nutplates [7] for the bolts [10] are on the opposite side of the bulkhead.

4) Remove the plate [9] and shim [8] from the bulkhead.

a) If the shim [8] was not installed, install a shim [8] to get the correct adjustment.

b) If the shim [8] was installed, remove the shim [8] to get the correct adjustment.

5) Install the plate [9] and the center hinge bracket [25] on the bulkhead.

6) Install the bolts [10] and washers [11].

a) Make sure that the bolts [10] are only hand tight.

NOTE: The bolts [10] will be tightened after the center hinge bracket [25] is adjusted.

7) Install the bolt [20], washer [23],bushing [37], washer [38], and nut [21] that connect the center door hinge [24] to the center hinge bracket [25].

8) Tighten the nut [21] and install the pin [22].

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a) If it is necessary, loosen the nut [21] to the nearest castellation to align the hole in the bolt [20].

9) If it is necessary, move the center hinge bracket [25] up or down until you can turn the bolt [20] by hand.

10) Tighten the bolts [10] in the center hinge bracket [25].

a) Make sure that the bolt [20] is free to move with hand pressure after you tighten the bolts [10].

SUBTASK 32-22-11-820-001

(3) Manually close the door assembly [4] that you will adjust.

SUBTASK 32-22-11-820-002

(4) Make sure that the fit on the forward edge and aft edge of the door assembly [4], and the surrounding structure is in the tolerance as follows:

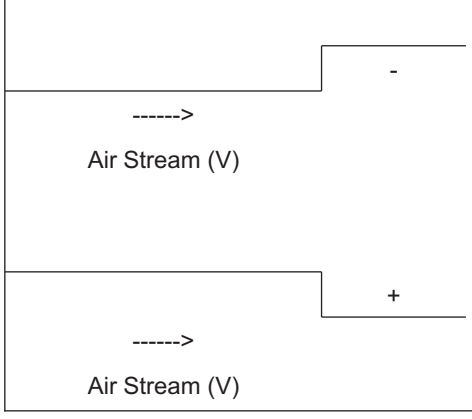
(a) Record the fit values that are not in tolerance.

(b) Use the "ALL" tolerances for all airplane forward door flushness checks (Table 501).

Table 501/32-22-11-993-801 Nose Landing Gear Wheel Well Door Flushness Values

AIRPLANE EFFECTIVITY	EDGE ZONE	FLUSHNESS (MISMATCH) IN INCHES (mm)
ALL	A-A	0.00 ±0.06 in. (0.00 ±1.52 mm) (In Streamwise Direction)
ALL	B	0.0019 ±0.0600 in. (0.04826 ±1.52400 mm) (In Streamwise Direction)
ALL	C	0.0031 in. (0.079 mm) + 0.03 in. (0.76 mm) (Door outside wheel well) - -0.06 in. (1.524 mm) (Door inside wheel well)
ALL	D	0.0015 in. (0.0381 mm) +0.03 in. (0.76 mm) (Door inside wheel well) - -0.06 in. (1.52 mm) (Door outside wheel well)

WHEEL WELL DOOR FLUSHNESS DIAGRAM



A negative tolerance (-) indicates the face of the mismatch faces aft.

A positive tolerance (+) indicates the face of the mismatch is into the air stream.

SUBTASK 32-22-11-020-007

(5) If the mismatch along the forward and aft edge is not in tolerance, do the steps that follow:

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(a) Remove the pin retainer spring [2] from the stop pin [1].

NOTE: The stop pin [1] is on the wheel well bulkhead.

(b) Retract the stop pin [1] for the door assembly [4].

(c) Adjust the stop pin [1] on the forward wheel well bulkhead until it just touches the forward/inboard door stop pad [3] on the forward edge of the door assembly [4].

SUBTASK 32-22-11-420-009

(6) Install the pin retainer spring [2] on the stop pin [1].

SUBTASK 32-22-11-210-003

(7) Make sure that the fit along the outboard edges of the door assembly [4] and surrounding structure is in the tolerance as follows:

(a) Record the fit values that are not in tolerance.

SUBTASK 32-22-11-820-003

(8) Make sure that the fit around the door assembly [4] is in the tolerance (Figure 501).

NOTE: This should only be necessary for a new door assembly [4].

(a) Use the "ALL" tolerances for the airplane door clearance checks (Table 502).

Table 502/32-22-11-993-802 Nose Landing Gear Wheel Well Door Clearance Values

AIRPLANE EFFECTIVITY	EDGE ZONE	CLEARANCE
ALL	A-A	0.10 +0.10 / -0.05 in. (2.54 +2.54 / -1.27 mm)
ALL	B	0.1995 +0.0500 / -0.0200 in. (5.067 +1.270 / -0.508 mm)
ALL	C	0.1 +0.1 / -0.0 in. (2.540 +1.524 / -1.016 mm)
ALL	D	0.15 +0.06 / -0.09 in. (3.81 +1.52 / -2.29 mm)
ALL	E	0.31 +0.04 / -0.25 in. (7.87 +1.02 / -6.35 mm)

SUBTASK 32-22-11-820-004

(9) If the mismatch along the outboard edge is not in tolerance, do the steps that follow:

(a) Remove the bolts [29] and washers [30] for the forward hinge bracket [36].

(b) Remove the shims [27] or add the correct thickness of the shims [27] below the plate [28] of the forward hinge bracket [36].

(c) Put the bolt [31] through the forward hinge bracket [36] and the door hinge [35].

(d) Install the washer [32], washer [39], and nut [33] on the bolt [31].

(e) Tighten the nut [33] and install the pin [34].

(f) Close the door assembly [4] with your hand.

SUBTASK 32-22-11-820-005

(10) Do the same steps for the aft hinge bracket.

SUBTASK 32-22-11-820-006

(11) Align the external outboard surface of the door with the fuselage skin surface.

SUBTASK 32-22-11-020-008

(12) Loosen the four bolts [29] in each of the aft and forward hinge brackets [36] of the aft door.

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(a) Move each hinge bracket until the serrations of the hinge bracket align with the serrations in the plate [28].

NOTE: The outboard external edge of the door must align with the adjacent fuselage skin.

SUBTASK 32-22-11-420-010

(13) Tighten the four bolts [29] in each of the aft and forward hinge brackets [36].

SUBTASK 32-22-11-420-011

(14) If it is not connected, connect the center door hinge [24] to the lower pushrod [12] as follows:

(a) Put the end of the lower pushrod [12] in its position on the center door hinge [24].

(b) Align the end of the lower pushrod [12] to the hole in the center door hinge [24].

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(c) Apply a thin layer of corrosion inhibiting material, G50136, to these items:

1) The threads and thread reliefs of the bolt [16], nut [18], and the faces of the washer [17].

(d) Put the bolt [16] through the center door hinge [24] and lower pushrod [12].

(e) Install the washer [17] and nut [18] on the bolt [16].

(f) Hand tighten the nut [18] until after final adjustments to the lower pushrod [12] are made in the subsequent step.

(g) Make sure that the upper pushrod dimension of 30.77 ±0.01 in. (781.56 ±0.25 mm) is maintained and threads are visible through the witness holes to reach adjustment values.

NOTE: Threads of the rods ends do not need to cover the witness holes.

SUBTASK 32-22-11-420-012

(15) Do this task: Nose Gear Wheel Well Door Linkage Adjustment and Check, TASK 32-33-61-820-801.

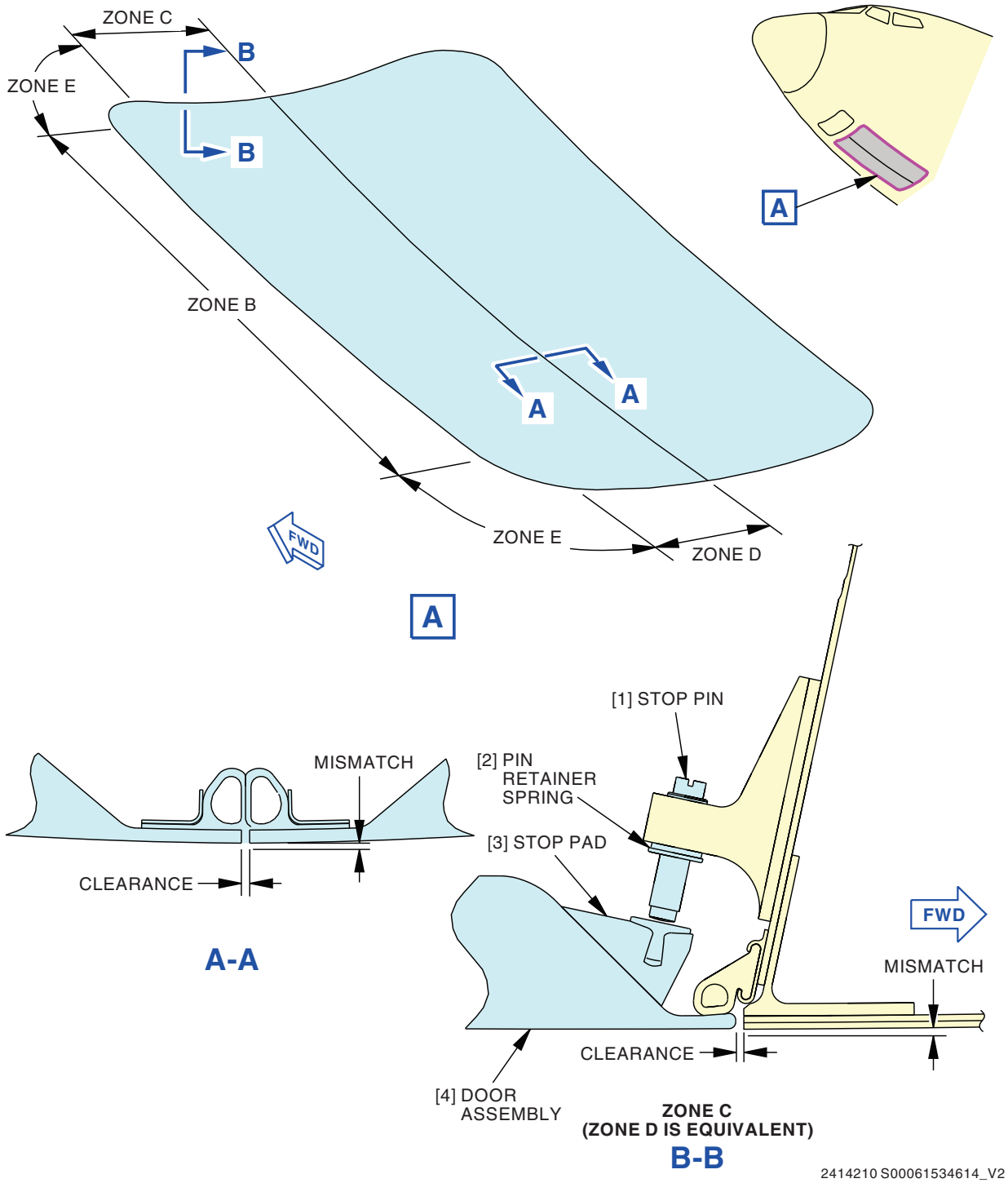
SUBTASK 32-22-11-420-013

(16) Tighten the nut [18] and install the pin [19] in the hole in the bolt [16].

(a) If it is necessary, loosen the nut [18] to the nearest castellation to align the hole in the bolt [16].

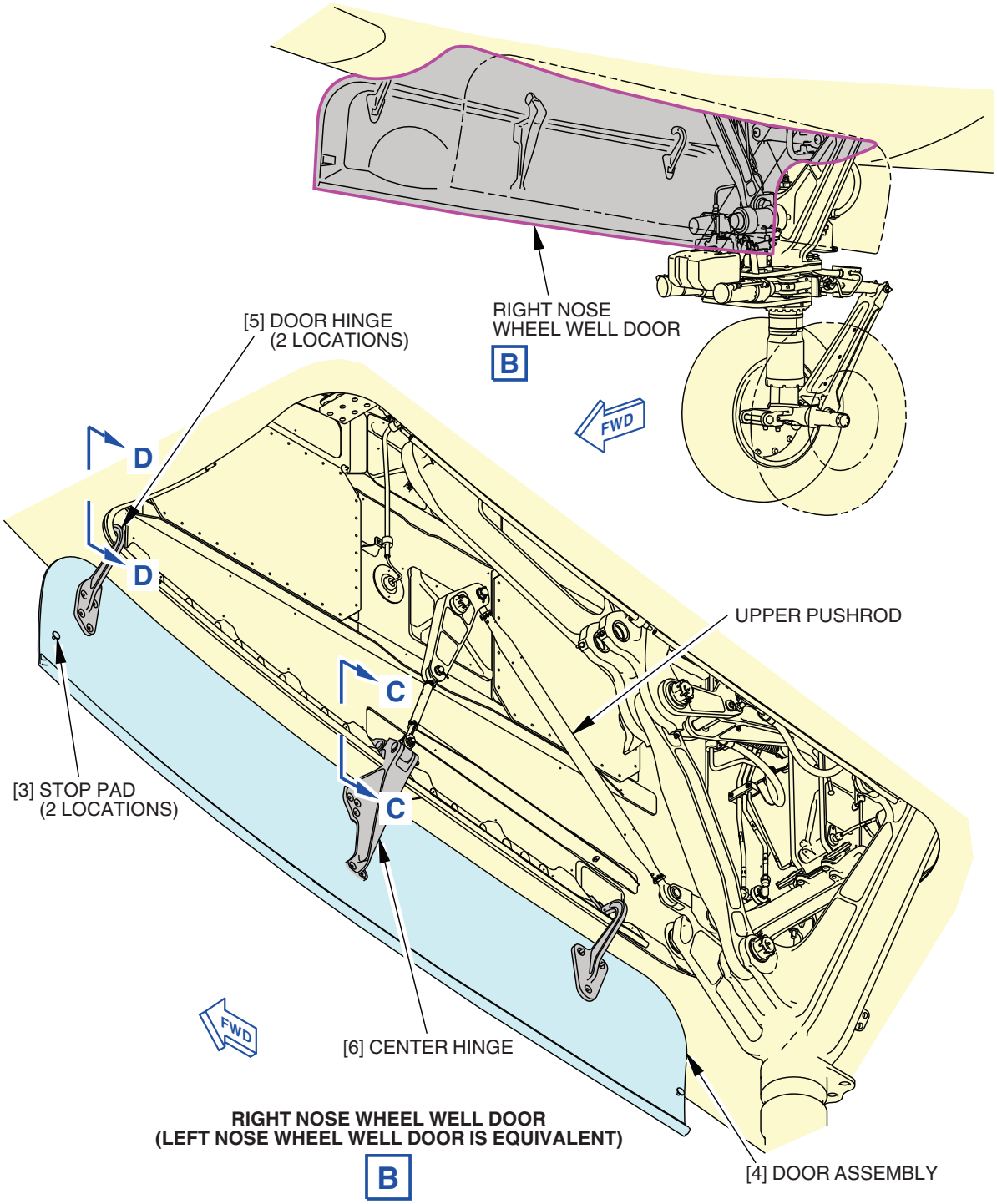
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



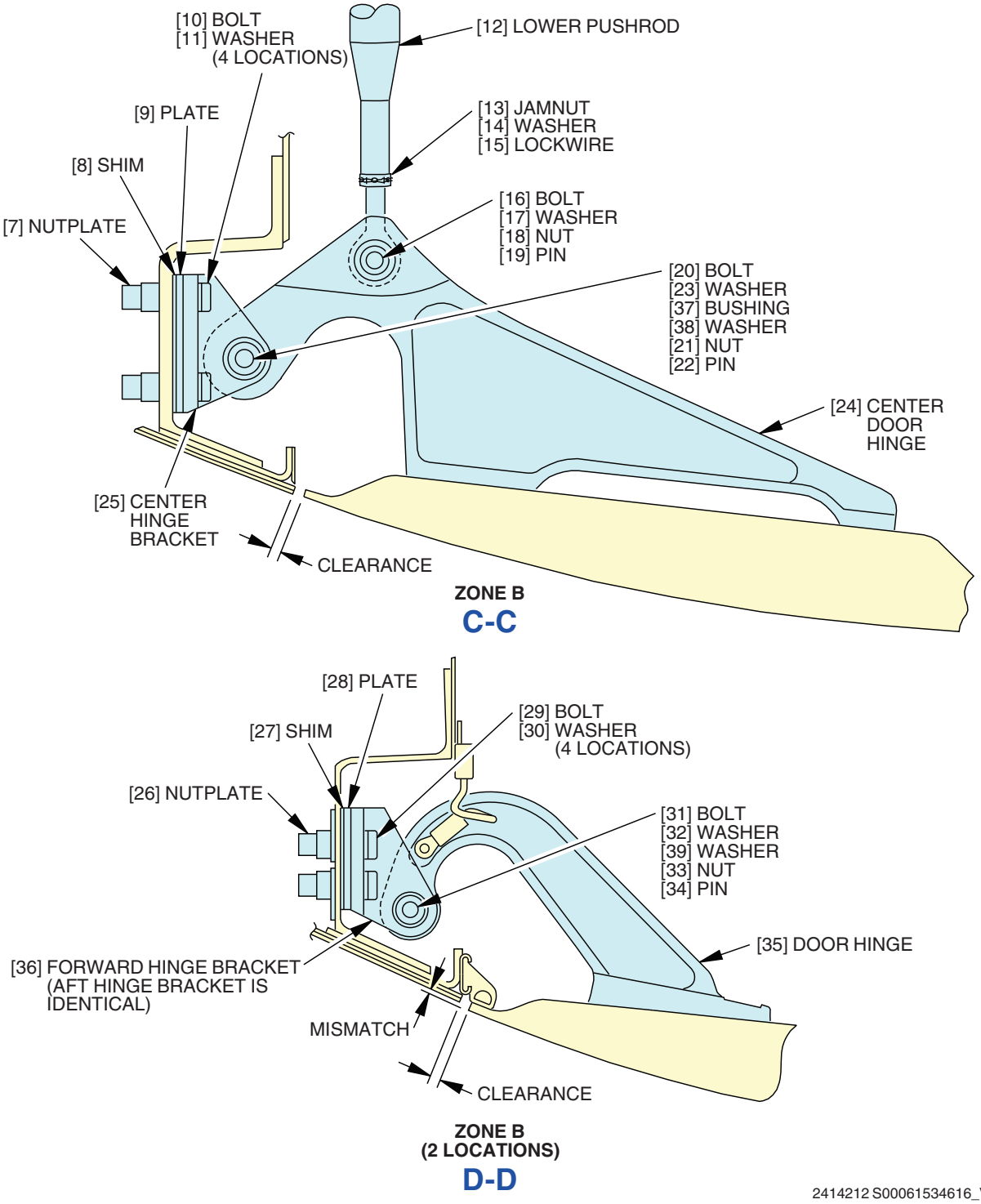
Nose Landing Gear Wheel Well Doors Adjustment
Figure 501/32-22-11-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Landing Gear Wheel Well Doors Adjustment
Figure 501/32-22-11-990-802 (Sheet 2 of 3)

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Nose Landing Gear Wheel Well Doors Adjustment
Figure 501/32-22-11-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LANDING GEAR CONTROL SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure contains this task:

(1) A test of the Landing Gear Control System. The airplane is not jacked for this test. This test checks the electrical connection of the landing control module and verifies the landing gear control lever operation.

TASK 32-31-00-730-801

2. System Test - Landing Gear Control System

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1690	Actuators/Deactuators Set - Proximity Sensor Part #: 8-758-01 Supplier: 08748 Part #: A27092-139 Supplier: 81205 Part #: A27092-146 Supplier: 81205 Opt Part #: A27092-106 Supplier: 81205

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the System Test

SUBTASK 32-31-00-480-001



WARNING OBEY THE PROCEDURE FOR THE INSTALLATION OF THE DOWNLOCK PINS. IF YOU MOVE THE CONTROL LEVER FOR THE LANDING GEAR TO THE UP POSITION, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear.

(a) If it is necessary, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-31-00-860-002

(2) Supply pressure to hydraulic system A (TASK 29-11-00-860-801).

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SUBTASK 32-31-00-860-004

(3) Open this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN

SUBTASK 32-31-00-700-001

(4) Make sure that these busses are powered:

- 28 V DC Bus 1
- 28 V DC Bat Bus
- 28 V AC Transfer Bus 1

E. Landing Gear Control Lever Module Test

SUBTASK 32-31-00-860-006

(1) Move the landing gear lever to the UP position.

(a) Do not operate the LOCK OVRD button for the landing gear lever.

SUBTASK 32-31-00-700-008

(2) Make sure that the landing gear lever will not go to the UP position.

SUBTASK 32-31-00-480-002

(3) Push the LOCK OVRD button for the landing gear control lever and move the control lever to the UP position.

SUBTASK 32-31-00-700-009

(4) Make sure that the landing gear control lever moves freely to the UP position.

SUBTASK 32-31-00-860-062

(5) Move the landing gear lever to the DN position.

SUBTASK 32-31-00-720-001

(6) Install deactuators on the Air/Ground proximity sensors on the landing gear that follow (TASK 32-09-00-860-801):

NOTE: Use the deactuator plate that is provided with the proximity sensor test set, SPL-1690, to put the airplane in the air mode.

NOTE: As an option, deactuator plate for rectangular sensor can be made from 5052, 6061, 7075 or 2024 aluminum with a dimension of 1.5 in. (38 mm) x 0.75 in. (19 mm) x 0.05 in. (1.27 mm). Deactuator plate for round sensor can be made from copper sheet with a dimension of 1.0 in. (25 mm) x 1.0 in. (25 mm) x 0.025 in. (0.64 mm).

- Proximity switch S1010, RIGHT GEAR ON GND
- Proximity switch S1012, LEFT GEAR ON GND
- Proximity switch S1011, RIGHT GEAR ON GND
- Proximity switch S1013, LEFT GEAR ON GND.

SUBTASK 32-31-00-860-063

(7) Push the SYS button on the Multi Function Panel (MFP).

SUBTASK 32-31-00-860-064

(8) To show the PSEU Maintenance page on the Multi Function Display (MFD), do these steps:

(a) Select MAINT.

(b) Select MAINT DATA PGS.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Select 32 PSEU - DISPLAY.

SUBTASK 32-31-00-720-002

(9) Move the control lever for the landing gear to the UP position.

(a) Do not operate the LOCK OVRD button for the landing gear control lever.

SUBTASK 32-31-00-700-010

(10) Make sure that the landing gear control lever goes to the UP position.

SUBTASK 32-31-00-700-028

(11) Make sure that the LG LEVER UP status shows UP for PSEU SYS 1 and PSEU SYS 2.

SUBTASK 32-31-00-700-029

(12) Make sure that the LG LEVER DN PWR status shows NOT DOWN for PSEU SYS 1 and PSEU SYS 2.

SUBTASK 32-31-00-700-030

(13) Make sure that the LEFT GEAR, RIGHT GEAR and NOSE GEAR indications come on in RED.

SUBTASK 32-31-00-730-001

(14) Move the landing gear control lever to the DN position.

SUBTASK 32-31-00-700-031

(15) Make sure that the LG LEVER UP status shows NOT UP for PSEU SYS 1 and PSEU SYS 2.

SUBTASK 32-31-00-700-032

(16) Make sure that the LG LEVER DN PWR status shows DOWN for PSEU SYS 1 and PSEU SYS 2.

SUBTASK 32-31-00-700-033

(17) Make sure that the LEFT GEAR, RIGHT GEAR and NOSE GEAR indications go off.

SUBTASK 32-31-00-730-002

(18) Remove the deactuators from the Air/Ground proximity sensors on the landing gear (TASK 32-09-00-860-802).

SUBTASK 32-31-00-860-008

(19) Supply pressure to hydraulic system B (TASK 29-11-00-860-805).

F. Electrical Connection Test

SUBTASK 32-31-00-700-003

(1) Select NEXT PAGE on the MFD.

(a) Make sure that the IO MATRIX is shown.

SUBTASK 32-31-00-860-065

(2) Set the NOSE WHEEL STEER switch to the ALT position.

SUBTASK 32-31-00-700-013

(3) Make sure that D10982 Pin 39 shows 1.

SUBTASK 32-31-00-700-014

(4) Make sure that LG HYDRAULIC TRANSFER status shows ALT NOSE STEER.

SUBTASK 32-31-00-860-066

(5) Set the NOSE WHEEL STEER switch to the NORM position.

SUBTASK 32-31-00-700-004

(6) Make sure that D10982 Pin 39 shows 0.

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SUBTASK 32-31-00-700-005

(7) Make sure that LG HYDRAULIC TRANSFER status shows ---.

SUBTASK 32-31-00-700-006

(8) Make sure that the rotary switch on the Antiskid/Autobrake Control Unit (AACU) is in the NORM position.

SUBTASK 32-31-00-860-067

(9) Set the autobrake selector switch, on the P9 panel, to the 1 position.

SUBTASK 32-31-00-700-018

(10) Push and release the RESET button on the AACU.

SUBTASK 32-31-00-700-019

(11) Make sure that the AACU display shows MEM CLR and then clears.

SUBTASK 32-31-00-700-020

(12) Push and hold the ENABLE/VERIFY button on the AACU.

SUBTASK 32-31-00-700-021

(13) Push and hold the VERIFY button on the AACU.

SUBTASK 32-31-00-700-022

(14) Release both ENABLE/VERIFY and VERIFY buttons.

NOTE: The AACU display will flash WAIT during the test.

SUBTASK 32-31-00-210-001

(15) Make sure that the AACU display does not show GEARSW1 or GEARSW2.

(a) Push the VERIFY button to toggle through all the status messages until the display shows TEST END.

SUBTASK 32-31-00-860-058

(16) Move the landing gear control lever to the UP position.

SUBTASK 32-31-00-860-059

(17) Push and hold the ENABLE/VERIFY button on the AACU.

SUBTASK 32-31-00-860-060

(18) Push and hold the VERIFY button on the AACU.

SUBTASK 32-31-00-860-061

(19) Release both ENABLE/VERIFY and VERIFY buttons.

NOTE: The AACU display will flash WAIT during the test.

SUBTASK 32-31-00-700-023

(20) Make sure that the AACU display shows GEARSW1 or GEARSW2.

SUBTASK 32-31-00-700-024

(21) Push the VERIFY button until the AACU display shows TEST END.

SUBTASK 32-31-00-700-007

(22) Set the FASTEN BELTS switch, on the P5 panel, to the AUTO position.

SUBTASK 32-31-00-700-025

(23) Make sure that the fasten seat belt signs are off.

SUBTASK 32-31-00-700-026

(24) Move the landing gear control lever to the DN position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-31-00-700-027

(25) Make sure that the fasten seat belt signs come on.

G. Landing Gear Control Lever Discrete Test

SUBTASK 32-31-00-860-068

(1) Push the MENU button on the Control Display Unit (CDU).

SUBTASK 32-31-00-700-034

(2) Do a check of the Digital Flight Data Acquisition Unit (DFDAU) discretes on the CDU as follows:

(a) Select ACMS.

(b) Select MAINTENANCE.

(c) Select INTERFACE TESTS.

(d) Select RECORDER.

(e) Select CONFIGURE DAR OUTPUT.

1) Make sure that DAR OUTPUT shows ON.

a) If DAR OUTPUT shows OFF, select ON.

2) Make sure that the SUBFRAME value increases

(f) Select RETURN until the DFDAU MAIN MENU shows.

(g) Select DISPLAY DATA.

(h) Select ANALOG DISPLAY.

(i) Enter B111/B into the CDU scratchpad.

(j) Push the L1 Line Select Key (LSK).

1) Make sure that the DATA value adjacent to B111 is 1.

(k) Enter B112/B into the CDU scratchpad.

(l) Push the L2 LSK.

1) Make sure that the DATA value adjacent to B112 is 0.

(m) Move the landing gear control lever to the UP position.

1) Make sure that the DATA value adjacent to B111 is 0.

2) Make sure that the DATA value adjacent to B112 is 1.

(n) Move the landing gear control lever to the DN position.

(o) Select RETURN until the DFDAU MAIN MENU shows.

(p) Select MAINTENANCE.

(q) Select INTERFACE TESTS.

(r) Select RECORDER.

(s) Select CONFIGURE DAR OUTPUT.

(t) Select OFF.

1) Make sure that DAR OUTPUT shows OFF.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-31-00-860-070

(1) Set the autobrake selector switch, on the P9 panel, to the OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-31-00-860-071

(2) Remove pressure from hydraulic systems A and B (TASK 29-11-00-860-805).

SUBTASK 32-31-00-860-009

(3) Close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN

——— END OF TASK ———

TASK 32-31-00-710-801

3. Landing Gear Control Electrical Circuits - Operational Test

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) There are two procedures to do this operational test. Each procedure is optional to the other. Do only one of the procedures that follow:

(a) Operational test with the airplane on jacks.

(b) Operational test with the airplane on the ground.

B. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1690	Actuators/Deactuators Set - Proximity Sensor Part #: 8-758-01 Supplier: 08748 Part #: A27092-139 Supplier: 81205 Part #: A27092-146 Supplier: 81205 Opt Part #: A27092-106 Supplier: 81205

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Operational Test with the Airplane on Jacks

SUBTASK 32-31-00-860-018

CAUTION	MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION	MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

WARNING	MAKE SURE THAT THE AREA AROUND THE MAIN LANDING GEAR IS CLEAR OF PERSONNEL, AND EQUIPMENT. THE DOORS CLOSE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(1) Do this task: Lift the Airplane on Jacks, TASK 07-10-01-580-801.

NOTE: This test is for the landing gear control circuits. It is not specific to the NLG or MLG.

SUBTASK 32-31-00-860-015

(2) Pressurize the hydraulic system A (TASK 29-11-00-860-801).

SUBTASK 32-31-00-860-010

(3) Open this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	18	C04020	LANDING GEAR CONTROL 1

SUBTASK 32-31-00-860-016

(4) Put the landing gear lever in the UP position.

SUBTASK 32-31-00-860-017

(5) Make sure that the landing gear retract.

SUBTASK 32-31-00-860-011

(6) Open this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	19	C04021	LANDING GEAR CONTROL 2

SUBTASK 32-31-00-860-019

(7) Put the landing gear lever in the DN position.

SUBTASK 32-31-00-860-020

(8) Make sure that the Main Landing Gear (MLG) do not extend.

SUBTASK 32-31-00-860-021

(9) Put the landing gear lever in the UP position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-31-00-860-012

(10) Close this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	19	C04021	LANDING GEAR CONTROL 2

SUBTASK 32-31-00-860-022

(11) Put the landing gear lever in the DN position.

SUBTASK 32-31-00-860-023

(12) Make sure that the landing gear extend.

SUBTASK 32-31-00-860-013

(13) Open this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	19	C04021	LANDING GEAR CONTROL 2

SUBTASK 32-31-00-860-014

(14) Close this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	18	C04020	LANDING GEAR CONTROL 1

SUBTASK 32-31-00-860-024

(15) Put the landing gear lever in the UP position.

SUBTASK 32-31-00-860-025

(16) Make sure that the landing gear retract.

SUBTASK 32-31-00-860-026

(17) Put the landing gear lever in the DN position.

SUBTASK 32-31-00-860-027

(18) Make sure that the landing gear extend.

SUBTASK 32-31-00-860-030

(19) For hydraulic systems A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-31-00-860-054

CAUTION

MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION

MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(20) Lower the airplane off the jacks (TASK 07-10-01-580-802).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-31-00-860-055

(21) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	18	C04020	LANDING GEAR CONTROL 1
D	19	C04021	LANDING GEAR CONTROL 2

F. Operational Test with the Airplane on Ground

SUBTASK 32-31-00-860-031

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-31-00-860-032

NOTE: This test is for the landing gear control circuits. It is not specific to the Nose Landing Gear (NLG) or MLG.

(2) Pressurize the hydraulic system A (TASK 29-11-00-860-801).

SUBTASK 32-31-00-860-056

(3) Do this task: Put the Airplane in the Air Mode, TASK 32-09-00-860-801.

(a) Install deactuators on the Air/Ground proximity sensors on the landing gear that follow:

NOTE: Use the deactuator plate that is provided with the proximity sensor test set, SPL-1690, to put the airplane in the air mode.

NOTE: As an option, deactuator plate for rectangular sensor can be made from 5052, 6061, 7075 or 2024 aluminum with a dimension of 1.5 in. (38 mm) x 0.75 in. (19 mm) x 0.05 in. (1.27 mm). Deactuator plate for round sensor can be made from copper sheet with a dimension of 1.0 in. (25 mm) x 1.0 in. (25 mm) x 0.025 in. (0.64 mm).

1) SYS 1 has the following proximity sensors:

- proximity switch S1010, RIGHT GEAR ON GND.
- proximity switch S1012, LEFT GEAR ON GND.

2) SYS 2 has the following proximity sensors:

- proximity switch S1011, RIGHT GEAR ON GND.
- proximity switch S1013, LEFT GEAR ON GND.

NOTE: AIR/GND override cannot be used to simulate air mode.

SUBTASK 32-31-00-860-033

(4) Open this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	18	C04020	LANDING GEAR CONTROL 1

SUBTASK 32-31-00-860-034

(5) Put the landing gear lever in the UP position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-31-00-860-035

(6) Make sure that the rod end of one of the two MLG retract actuator tries to extend.

SUBTASK 32-31-00-860-036

(7) Open this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	19	C04021	LANDING GEAR CONTROL 2

SUBTASK 32-31-00-860-037

(8) Make sure that the MLG retract actuator rod end retracts a small quantity.

SUBTASK 32-31-00-860-038

(9) Put the landing gear lever in the DN position.

SUBTASK 32-31-00-860-039

(10) Make sure that the MLG retract actuator rod end does not move.

SUBTASK 32-31-00-860-040

(11) Close this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	19	C04021	LANDING GEAR CONTROL 2

SUBTASK 32-31-00-860-041

(12) Make sure that the MLG retract actuator rod end tries to retract.

SUBTASK 32-31-00-860-042

(13) Open this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	19	C04021	LANDING GEAR CONTROL 2

SUBTASK 32-31-00-860-043

(14) Close this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	18	C04020	LANDING GEAR CONTROL 1

SUBTASK 32-31-00-860-044

(15) Put the landing gear lever in the UP position.

SUBTASK 32-31-00-860-045

(16) Make sure that the rod end of a MLG retract actuator tries to extend.

SUBTASK 32-31-00-860-046

(17) Open this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	18	C04020	LANDING GEAR CONTROL 1

SUBTASK 32-31-00-860-047

(18) Make sure that the MLG retract actuator rod end retracts a small quantity.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-31-00-860-048

(19) Put the landing gear lever in the DN position.

SUBTASK 32-31-00-860-049

(20) Make sure that the MLG retract actuator rod end does not move.

SUBTASK 32-31-00-860-050

(21) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	18	C04020	LANDING GEAR CONTROL 1

SUBTASK 32-31-00-860-051

(22) Make sure that the MLG retract actuator rod end tries to retract.

SUBTASK 32-31-00-860-052

(23) Do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

SUBTASK 32-31-00-860-053

(24) For hydraulic systems A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-31-00-860-057

(25) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	18	C04020	LANDING GEAR CONTROL 1
D	19	C04021	LANDING GEAR CONTROL 2

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LANDING GEAR CONTROL LEVER MODULE - REMOVAL/INSTALLATION

1. General

A. This procedure has these two tasks:

(1) Removal of the landing gear control lever module (referred to as landing gear module [1]).

(2) Installation of the landing gear module [1].

TASK 32-31-11-020-801

2. Landing Gear Control Lever Module Removal

(Figure 401)

A. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-840-801	Prepare to Put the Airplane in the Air Mode (P/B 201)
34-24-01-000-802	Integrated Standby Flight Display (ISFD) Removal (P/B 401)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Removal

SUBTASK 32-31-11-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-31-11-040-002

(2) Do this task: Integrated Standby Flight Display (ISFD) Removal, TASK 34-24-01-000-802.

SUBTASK 32-31-11-860-015

(3) Do this task: Prepare to Put the Airplane in the Air Mode, TASK 32-09-00-840-801.

SUBTASK 32-31-11-860-001

(4) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	8	C01551	ISFD

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	17	C00129	LANDING GEAR LATCH & PRESS WARN
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

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F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	18	C04020	LANDING GEAR CONTROL 1
D	19	C04021	LANDING GEAR CONTROL 2

SUBTASK 32-31-11-860-002

(5) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-31-11-020-005

(6) Disconnect the control lever module [10] from the landing gear module [1] before you remove the landing gear module [1].

NOTE: There is an interference between the landing gear module [1] and the floodlight on the glare shield [2] if you do not disconnect the control lever module [10] from the landing gear module [1] before you remove it.

(a) Loosen the captive screws [13] on the lightplate [12] on the lower front of the landing gear module [1] panel to remove the lightplate [12].

(b) Remove the screws [9] that connect the control lever module [10] to the landing gear module [1].

D. Landing Gear Control Lever Module Removal

SUBTASK 32-31-11-010-001

(1) Loosen the captive fasteners [8] which hold the landing gear module [1] to the forward panel structure [7].

SUBTASK 32-31-11-020-001

(2) Move the landing gear module [1] out of the forward panel structure [7] to get access to the electrical connectors [14].

SUBTASK 32-31-11-020-002

(3) Disconnect the electrical connectors [14] from the landing gear module [1].

SUBTASK 32-31-11-020-003

(4) Remove the landing gear module [1] from the airplane.

SUBTASK 32-31-11-020-007

(5) Move the control lever module [10] out of the forward panel structure [7] to get access to the electrical connectors [11].

SUBTASK 32-31-11-020-008

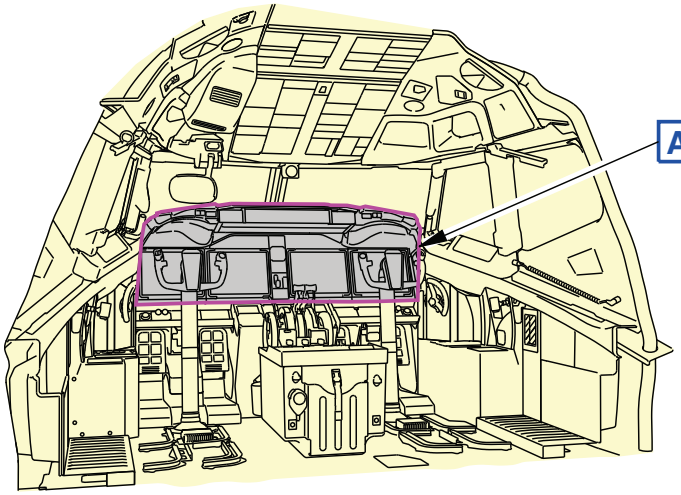
(6) Disconnect the electrical connectors [11] from the control lever module [10].

SUBTASK 32-31-11-020-009

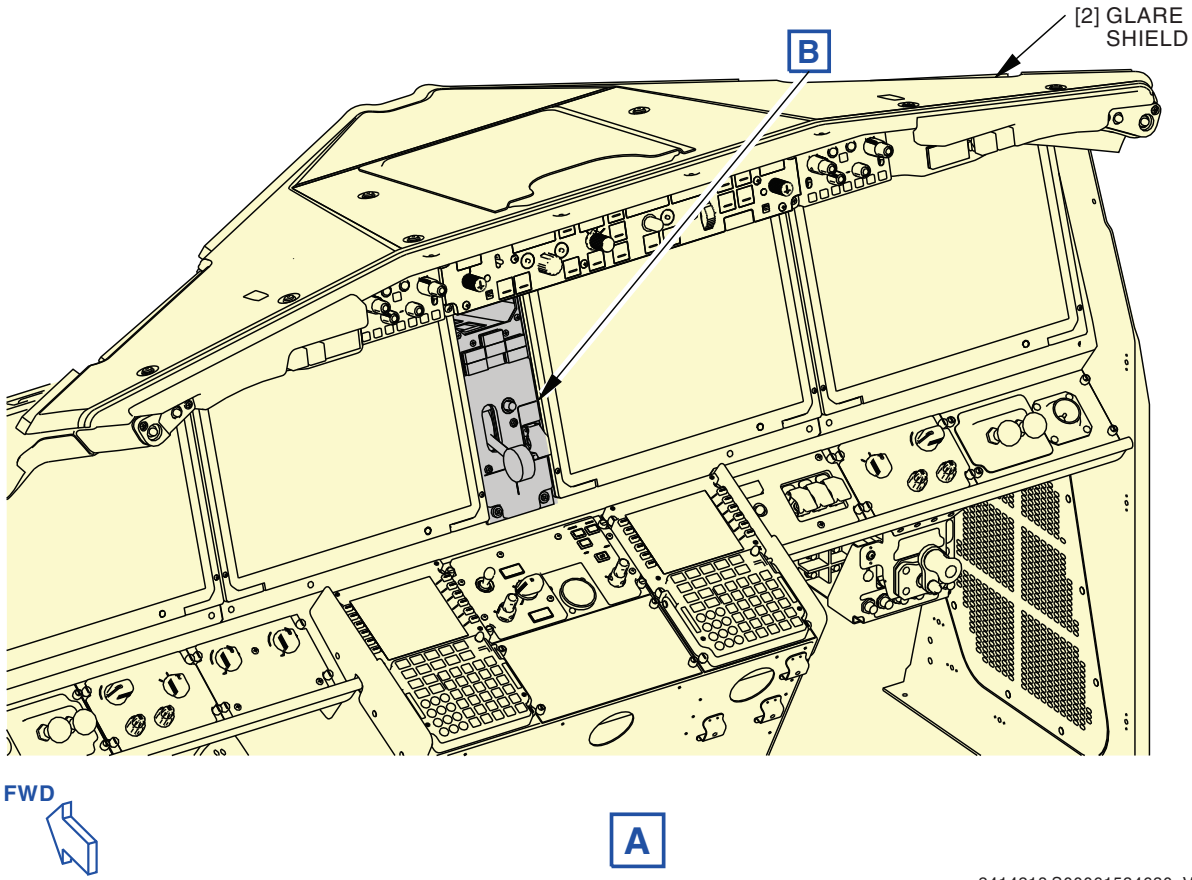
(7) Remove the control lever module [10] from the airplane.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



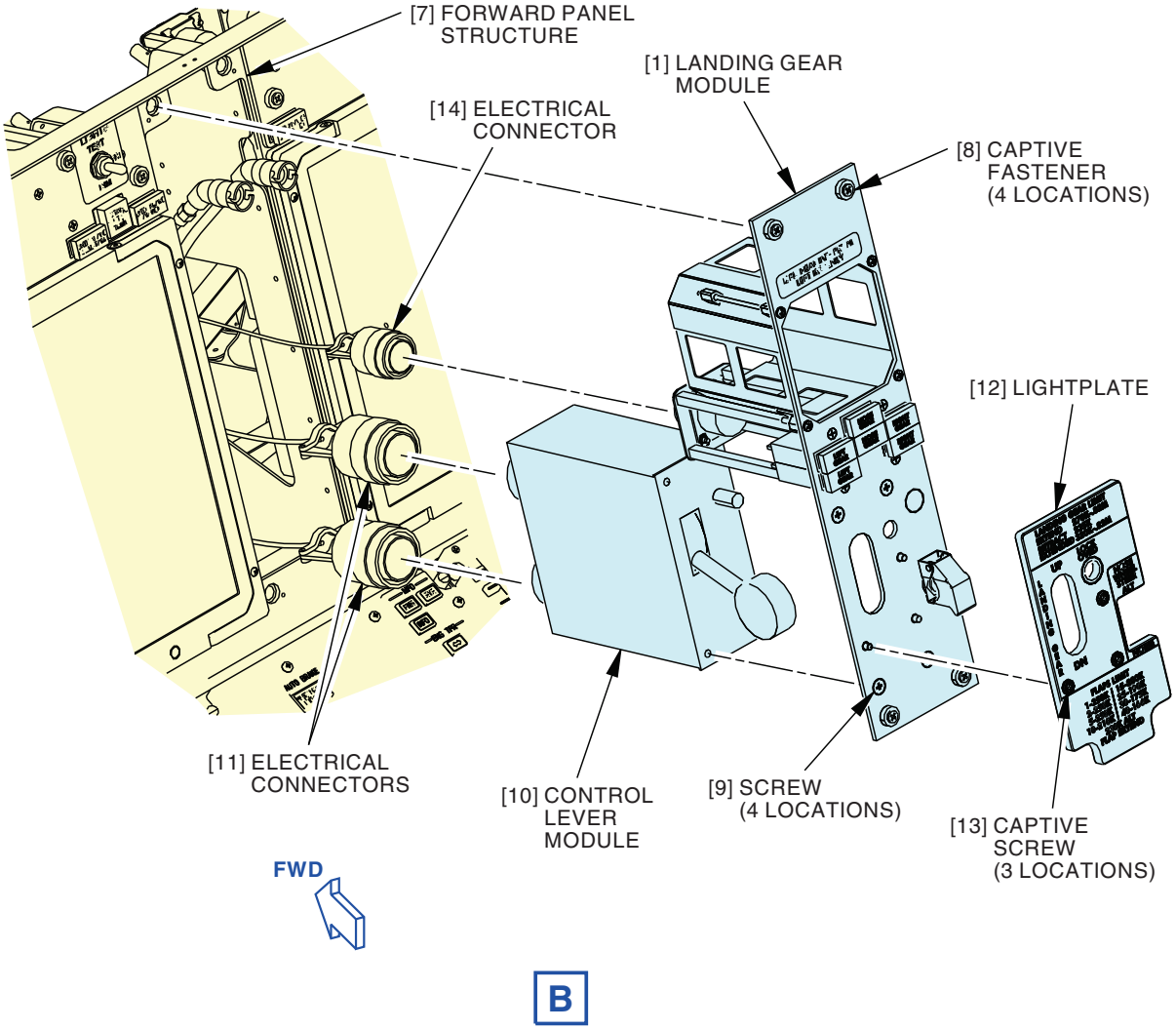
FLIGHT COMPARTMENT



2414218 S00061534630_V3

Landing Gear Control Lever Module Installation
Figure 401/32-31-11-990-801 (Sheet 1 of 2)

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2414220 S00061534632_V4

Landing Gear Control Lever Module Installation
Figure 401/32-31-11-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-31-11-400-801

3. Landing Gear Control Lever Module Installation

(Figure 401)

A. References

Reference	Title
20-60-07-910-802	Inspection, Cleaning and Sealing of Electrical Connectors (P/B 201)
32-09-00-840-802	Return the Airplane Systems Back to Their Normal on Ground Condition (P/B 201)
32-31-11-710-801	Landing Gear Control Lever Module - Operational Test (P/B 501)
34-24-01-400-802	Integrated Standby Flight Display (ISFD) Installation (P/B 401)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Landing Gear Control Lever Module Installation

SUBTASK 32-31-11-420-001

(1) Before you connect the electrical connectors [11] to the landing gear module [1], examine the electrical connectors [11] for corrosion (TASK 20-60-07-910-802).

SUBTASK 32-31-11-020-006

(2) If the control lever module [10] connected to the landing gear module [1], do these steps:

NOTE: There is an interference between the landing gear module [1] and the floodlight on the glare shield [2] if you do not disconnect the control lever module [10] from the landing gear module [1] before you install it.

(a) Loosen the captive screws [13] on the lightplate [12] on the lower front of the landing gear module [1] panel to remove the lightplate [12].

(b) Remove the screws [9] that connect the control lever module [10] to the landing gear module [1].

(c) Disconnect the control lever module [10] from the landing gear module [1].

SUBTASK 32-31-11-430-001

(3) Connect the electrical connectors [11] to the control lever module [10].

SUBTASK 32-31-11-420-005

(4) Put the control lever module [10] into position in the forward panel structure [7].

SUBTASK 32-31-11-420-007

(5) Connect the electrical connectors [14] to the landing gear module [1].

SUBTASK 32-31-11-420-002

(6) Move the landing gear module [1] into position in the forward panel structure [7].

SUBTASK 32-31-11-420-008

(7) Install the screws [9] that connect the control lever module [10] to the landing gear module [1].

SUBTASK 32-31-11-420-010

(8) Tighten the captive screws [13] to install the lightplate [12] on the lower front of the landing gear module [1].

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SUBTASK 32-31-11-420-009

(9) Tighten the captive fasteners [8] that attach the landing gear module [1] to the forward panel structure [7].

SUBTASK 32-31-11-400-001

(10) Do this task: Integrated Standby Flight Display (ISFD) Installation, TASK 34-24-01-400-802.

D. Landing Gear Control Lever Module Installation Test

SUBTASK 32-31-11-860-004

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	8	C01551	ISFD

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	17	C00129	LANDING GEAR LATCH & PRESS WARN
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	18	C04020	LANDING GEAR CONTROL 1
D	19	C04021	LANDING GEAR CONTROL 2

SUBTASK 32-31-11-860-016

(2) Do this task: Return the Airplane Systems Back to Their Normal on Ground Condition, TASK 32-09-00-840-802.

SUBTASK 32-31-11-710-001

(3) Do this task: Landing Gear Control Lever Module - Operational Test, TASK 32-31-11-710-801.

——— END OF TASK ———

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LANDING GEAR CONTROL LEVER MODULE - ADJUSTMENT/TEST

1. General

A. This procedure tests the landing gear control lever module.

TASK 32-31-11-710-801

2. Landing Gear Control Lever Module - Operational Test

A. General

(1) This task tests the landing gear control lever module.

(2) The landing gear control lever module is installed in the P2 panel.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)
32-42-00-740-801	Antiskid/Autobrake Control Unit BITE Procedure (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1690	Actuators/Deactuators Set - Proximity Sensor Part #: 8-758-01 Supplier: 08748 Part #: A27092-139 Supplier: 81205 Part #: A27092-146 Supplier: 81205 Opt Part #: A27092-106 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Prepare for the Operational Test

SUBTASK 32-31-11-860-014

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

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SUBTASK 32-31-11-840-001

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN

SUBTASK 32-31-11-860-006

(3) Set the MAX Display System (MDS) to the MAINT -> MAINT DATA PGS -> 32 PSEU, DISPLAY.

F. Landing Gear Control Lever Module Test

SUBTASK 32-31-11-860-007

(1) On the PSEU maint page on the MDS, make sure that the SYS 1 and SYS 2 are in the GND mode for all three gear (TASK 32-09-00-860-802).

(a) Make sure that the three green landing gear indication lights are illuminated.

SUBTASK 32-31-11-860-008

(2) Do not operate the lock override button for the landing gear lever and move the landing gear lever to the UP position.

(a) Verify that the landing gear lever will not go to the UP position.

SUBTASK 32-31-11-860-009

(3) Press and hold the lock override button for the landing gear control lever and move the control lever to the UP position.

(a) Make sure that the landing gear control lever moves freely into the UP position.

(b) On the PSEU maint page on the MDS, make sure that the lever position has changed.

(c) Make sure that the red landing gear indication lights are illuminated.

SUBTASK 32-31-11-860-010

(4) Move the lever to the DOWN position.

(a) On the PSEU maint page on the MDS, make sure that the lever position has changed.

(b) Make sure that the green landing gear indication lights are illuminated.

SUBTASK 32-31-11-840-002

(5) Prepare to put the airplane in the AIR mode (TASK 32-09-00-860-801).

(a) Install deactuators on all six Air/Ground proximity sensors on the landing gear that follow:

NOTE: Use the deactuator plate that is provided with proximity sensor test set, SPL-1690, to put the airplane in the AIR mode.

NOTE: As an option, deactuator plates for rectangular sensors can be made from 5052, 6061, or 2024 aluminum with a dimension of 1.5 in. (38 mm) x 0.75 in. (19 mm) x 0.05 in. (1.27 mm). Deactuator plates for round sensors can be made from copper sheet with a dimension of 1.0 in. (25 mm) x 1.0 in. (25 mm) x 0.025 in. (0.64 mm).

1) Sys 1 has the following proximity sensors:

- proximity switch S1010, RIGHT GEAR ON GND

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- proximity switch S1012, LEFT GEAR ON GND
- proximity switch S1014, NOSE GEAR ON GND.

2) SYS 2 has the following proximity sensors:
- proximity switch S1011, RIGHT GEAR ON GND
- proximity switch S1013, LEFT GEAR ON GND
- proximity switch S1015, NOSE GEAR ON GND.

NOTE: The recommended method of AIR/GND override cannot be used to simulate air mode for this test.

SUBTASK 32-31-11-860-011

(6) Do not operate the lock override for the landing gear control lever and move the control lever for the landing gear to the UP position.
(a) Make sure that the landing gear control lever moves freely into the UP position.
(b) On the PSEU maint page on the MDS, make sure that the lever position has changed.
(c) Make sure that the red landing gear indication lights are illuminated.

SUBTASK 32-31-11-860-012

(7) Move the landing gear control lever to the DN position.
(a) On the PSEU maint page on the MDS, make sure that the lever position has changed.
(b) Make sure that the green landing gear indication lights are illuminated.

SUBTASK 32-31-11-840-005

(8) Return the AIR/GND SYSTEM to the ground mode (TASK 32-09-00-860-802).

SUBTASK 32-31-11-840-003

(9) Pressurize the hydraulic system B (TASK 29-11-00-860-801).

SUBTASK 32-31-11-860-013

(10) Put the NOSE WHEEL STEER switch to the ALT position.
(a) From the Multi Function Display (MFD), select MAINT DATA PGS -> SELECT PSEU.
(b) SELECT NEXT PAGE until D10982 is displayed.
(c) Make sure that the display for CONNECTOR D10982 PIN 39 reads 1.

SUBTASK 32-31-11-840-004

(11) Put the NOSE WHEEL STEER switch to the Norm position.
(a) Make sure that the display for connector D10982 PIN 39 shows 0.

SUBTASK 32-31-11-840-007

(12) Remove hydraulic power from the hydraulic system B (TASK 29-11-00-860-805).

SUBTASK 32-31-11-740-002

(13) Do the BITE test of the Antiskid/Autobrake Control Unit (AACU) (TASK 32-42-00-740-801).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-31-11-840-006

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

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F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN

——— END OF TASK ———

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LANDING GEAR SELECTOR VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the selector valve for the landing gear (referred to as the selector valve [8])

(2) An installation of the selector valve for the landing gear.

(3) Operational test of the Selector/Bypass Valve. (airplane jacked).

TASK 32-31-51-020-801

2. Landing Gear Selector Valve Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

D. Prepare for the Removal

SUBTASK 32-31-51-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-31-51-860-001

(2) For hydraulic system A, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 32-31-51-580-001

(3) Install chocks around the tires of the main landing gear (PAGEBLOCK 10-11-05/201).

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SUBTASK 32-31-51-040-001

(4) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	17	C00129	LANDING GEAR LATCH & PRESS WARN

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	18	C04020	LANDING GEAR CONTROL 1
D	19	C04021	LANDING GEAR CONTROL 2

E. Landing Gear Selector Valve Removal

SUBTASK 32-31-51-480-002

(1) Disconnect the hydraulic lines from the selector valve [8].

SUBTASK 32-31-51-480-003

(2) Install caps/plugs, STD-14686, on the hydraulic lines and the ports on the selector valve [8].

SUBTASK 32-31-51-480-004

(3) Disconnect the electrical connectors from the selector valve [8].

SUBTASK 32-31-51-020-001

(4) Remove the washers [6] and bolts [7] which attach the selector valve [8] to the structure.

SUBTASK 32-31-51-020-002

(5) Remove the selector valve [8] from the airplane.

SUBTASK 32-31-51-080-001

(6) Remove the caps/plugs, STD-14686, from the ports on the selector valve [8] and drain the hydraulic fluid.

SUBTASK 32-31-51-020-003

(7) Do these steps if the replacement selector valve [8] does not have the unions [1] installed:

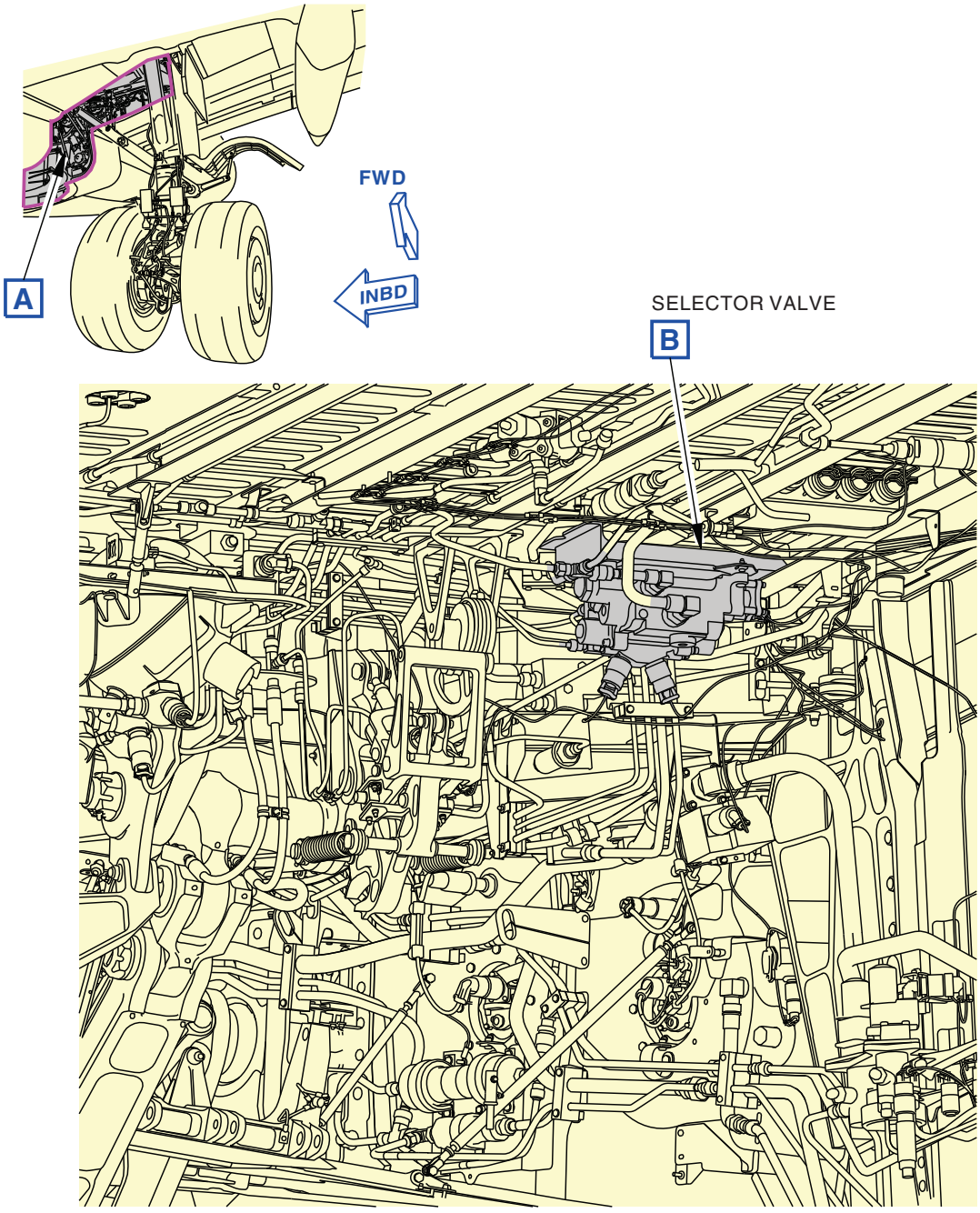
(a) Remove the unions [1] from the selector valve [8].

(b) Remove and discard the packings [2].

(c) Install plugs in the ports on the selector valve [8].

——— END OF TASK ———

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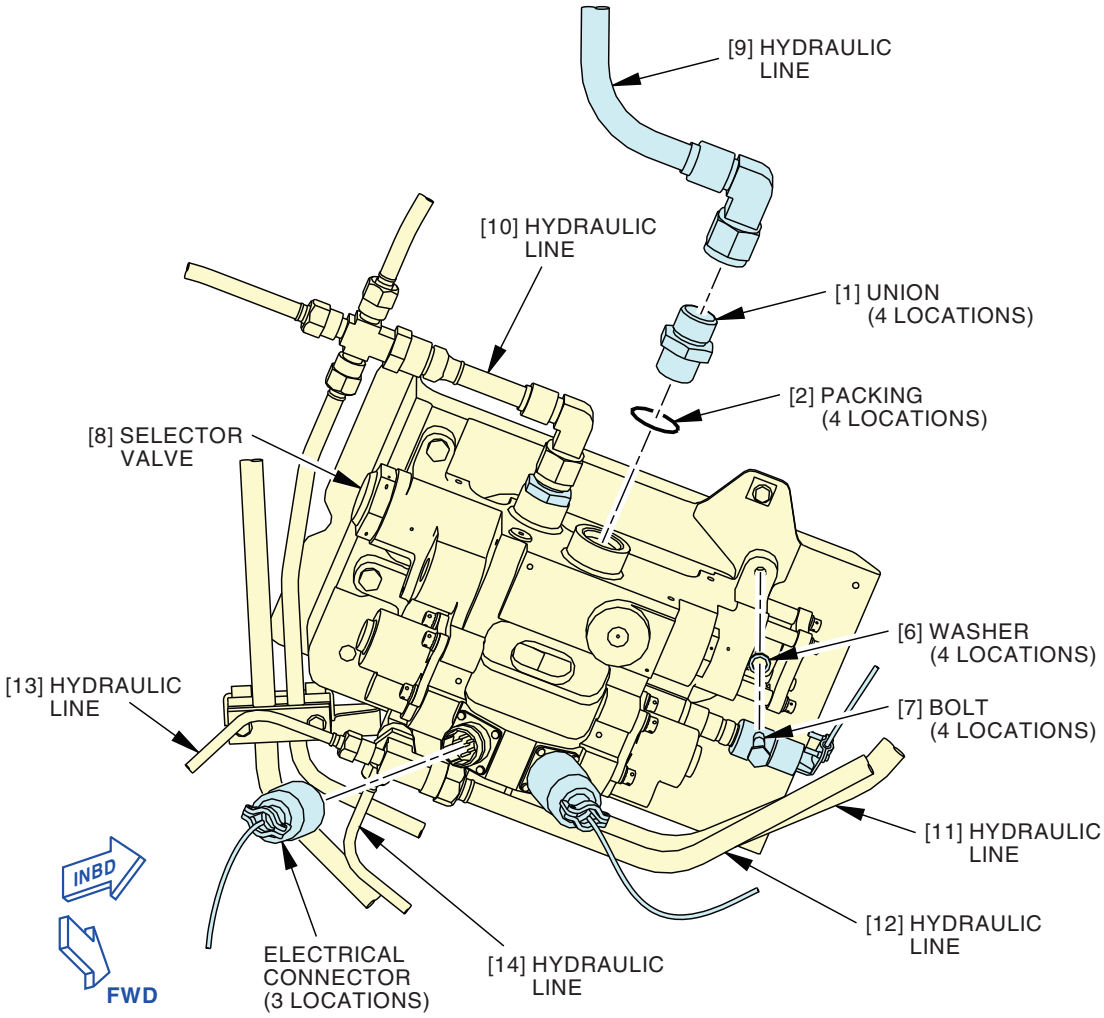
MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE IS SHOWN, RIGHT SIDE IS EQUIVALENT)

A

2414232 S00061534669_V1

Landing Gear Selector Valve Installation
Figure 401/32-31-51-990-801 (Sheet 1 of 2)

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SELECTOR VALVE

B

2414233 S00061534670_V2

Landing Gear Selector Valve Installation
Figure 401/32-31-51-990-801 (Sheet 2 of 2)

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TASK 32-31-51-400-801

3. Landing Gear Selector Valve Installation

(Figure 401)

A. References

Reference	Title
20-10-34-120-801	Hand Clean Metal Surfaces with Abrasives (P/B 701)
32-31-00-730-801	System Test - Landing Gear Control System (P/B 501)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
SWPM 20-60-03	Special Protection of Electrical Connectors

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
8	Selector valve	32-31-51-01-040	PTF ALL

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

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F. Landing Gear Selector Valve Installation

SUBTASK 32-31-51-420-001

(1) Do these steps if the replacement selector valve [8] does not have the unions [1] installed:

(a) Lubricate these items with hydraulic fluid, D00153:

1) The packings [2].

2) The unions [1].

(b) Remove the plugs from the ports on the selector valve [8].

(c) Install the packings [2] and unions [1] in the selector valve [8].

SUBTASK 32-31-51-610-001

(2) Fill the selector valve [8] with hydraulic fluid, D00153.

(a) Put caps/plugs, STD-14686, on the unions [1].

SUBTASK 32-31-51-100-001

(3) Clean the surface of the bracket where the selector valve [8] will attach (TASK 20-10-34-120-801).

SUBTASK 32-31-51-100-002

(4) Clean the surface of the selector valve [8] that will attach to the bracket (TASK 20-10-34-120-801).

SUBTASK 32-31-51-110-001

(5) Apply coating, C00064, to all of the areas that you cleaned on the bracket and the selector valve [8].

SUBTASK 32-31-51-420-002

(6) Put the selector valve [8] in its position on the bracket.

SUBTASK 32-31-51-420-003

(7) Install the bolts [7] and washers [6].

SUBTASK 32-31-51-760-001

(8) Use an intrinsically safe approved bonding meter, COM-1550, to measure the electrical resistance between the selector valve [8] and the bracket (SWPM 20-20-00).

(a) Make sure the electrical bonding resistance is less than 0.0025 ohm (2.5 milliohms).

1) If the electrical bonding resistance is 0.0025 ohm (2.5 milliohms) or more, clean the mating surfaces of the bracket and the selector valve [8], and apply the coating, C00064, again.

SUBTASK 32-31-51-110-002

(9) If necessary, repair the damaged finish on the selector valve [8] with a manual application of coating, C00064.

SUBTASK 32-31-51-390-001

(10) Apply a fillet seal of sealant, A02315, to the joints between the selector valve [8] and bracket, the heads and exposed threads of the bolts [7].

SUBTASK 32-31-51-080-002

(11) Remove the caps/plugs, STD-14686, from the unions [1] and connect the hydraulic lines to the selector valve [8].

SUBTASK 32-31-51-420-004

(12) Torque the B-nuts of the hydraulic line [9] and hydraulic line [11] to 700 ±35 in-lb (79 ±4 N·m).

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SUBTASK 32-31-51-420-008

(13) Torque the B-nuts of the hydraulic line [10] and hydraulic line [12] to 500 ±25 in-lb (56 ±3 N·m).

SUBTASK 32-31-51-420-009

(14) Torque the B-nuts of the hydraulic line [13] and hydraulic line [14] to 140 ±7 in-lb (16 ±1 N·m).

SUBTASK 32-31-51-420-005

(15) Before you connect the electrical connectors to the selector valve [8], examine the electrical connectors for corrosion.

WARNING

DO THE STEPS BELOW IF THE AIRPLANE OPERATES WHERE DEICING FLUID THAT CONTAINS POTASSIUM FORMATE OR POTASSIUM ACETATE IS USED. ALSO, DO THE STEPS FOR ALL AIRPLANES THAT YOU FOUND CORROSION IN THE ELECTRICAL CONNECTORS IN THE MAIN WHEEL WELL. THE ELECTRICAL CONNECTORS ARE IN A SYSTEM THAT IS NECESSARY FOR SAFE FLIGHT.

(a) If there is corrosion, refer to SWPM 20-60-03 to correct the problem.

(b) Apply the D5026NS or ZC-026 compound, G50171, to the electrical connectors (SWPM 20-60-03).

(16) Connect the electrical connectors to the selector valve [8].

SUBTASK 32-31-51-010-001

(17) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	17	C00129	LANDING GEAR LATCH & PRESS WARN

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	18	C04020	LANDING GEAR CONTROL 1
D	19	C04021	LANDING GEAR CONTROL 2

G. Landing Gear Selector Valve Installation Test

SUBTASK 32-31-51-700-001

(1) Do the test of the control system for the landing gear (TASK 32-31-00-730-801).

——— END OF TASK ———

TASK 32-31-51-700-801

4. Selector/Bypass Valve Operational Test

A. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

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B. Prepare for Test

SUBTASK 32-31-51-820-001

(1) Move the LANDING GEAR control lever to the DOWN position.

SUBTASK 32-31-51-480-007

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(2) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-31-51-910-001

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(3) Jack the aircraft Lift the Airplane on Jacks, TASK 07-10-01-580-801

SUBTASK 32-31-51-000-001

(4) Close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	3	C01312	ENGINE 1 RUN/PWR
D	1	C04030	DSP CAPT EFIS CONT PNL
D	2	C04003	MULTI FUNCTION PANEL
D	3	C04026	DISPLAY LEFT INBOARD
D	4	C04027	DISPLAY LEFT OUTBOARD
D	5	C04023	DISPLAY DPC 1 PRI

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	9	C04024	DISPLAY DPC 2 HOLDUP
D	10	C04022	DISPLAY DPC 1 HOLDUP
D	11	C04025	DISPLAY DPC 2 PRI
E	10	C04029	DISPLAY RIGHT OUTBOARD
E	11	C04028	DISPLAY RIGHT INBOARD
E	12	C04002	MAINT/BITE PANEL
E	13	C04031	DSP F/O EFIS CONT PNL

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	17	C00129	LANDING GEAR LATCH & PRESS WARN

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(Continued)

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	15	C01355	LANDING GEAR AIR/GND SYS 2
C	16	C01356	LANDING GEAR AIR/GND SYS 1
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN
D	16	C01432	LANDING GEAR ALTN EXTEND SOL
E	12	C00314	INDICATOR MASTER DIM SECT 2
F	11	C00317	INDICATOR MASTER DIM SECT 5
F	13	C01179	INDICATOR MASTER DIM SECT 7

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	18	C04020	LANDING GEAR CONTROL 1

SUBTASK 32-31-51-040-002

(5) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	19	C04021	LANDING GEAR CONTROL 2

C. Test procedure

SUBTASK 32-31-51-910-002

(1) Remove the downlock pins from the landing gear (Landing Gear Downlock Pins Removal, TASK 32-00-01-080-801).

SUBTASK 32-31-51-710-001

(2) Move the engine No. 1 and No. 2 throttles completely forward.

NOTE: Throttles in the advanced position prevent the landing gear warning (red) lights from remaining on after gear is up and locked.

SUBTASK 32-31-51-710-002

(3) Put the Master Dim and Test switch in the bright position.

SUBTASK 32-31-51-710-003

(4) Ensure the manual extension door is fully closed.

SUBTASK 32-31-51-710-004

(5) Move the landing gear control lever UP and DOWN 2 times, allowing the gear to fully extend and retract each cycle.

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(a) Make sure that there is no rubbing or binding of the main gear retract actuator hoses during gear swing.

(6) Make sure that all cycles complete and that lights are OFF when gear is UP, GREEN lights are illuminated when gear is DOWN.

(7) Make sure that the nose gear completes cycle before main gear, and that left and right gear complete less than one second apart.

(8) Make sure that there is no rubbing of hoses or doors, no unusual grinding, popping, or scraping sounds.

SUBTASK 32-31-51-710-005

(9) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	18	C04020	LANDING GEAR CONTROL 1

SUBTASK 32-31-51-710-006

(10) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	19	C04021	LANDING GEAR CONTROL 2

SUBTASK 32-31-51-710-007

(11) Move the landing gear control lever UP and DOWN (2) times, allowing the gear to fully extend and retract each time.

(12) Make sure that all cycles complete and that all lights are OFF when gear is UP, and GREEN lights are illuminated when gear is DOWN.

(13) Make sure that the nose gear completes cycle before main gear and that left and right gear complete less than 1 second apart.

SUBTASK 32-31-51-710-008

(14) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	18	C04020	LANDING GEAR CONTROL 1

SUBTASK 32-31-51-710-009

(15) Move the landing gear control lever UP and Down 2 times, allowing the gear to fully extend and retract each time.

SUBTASK 32-31-51-710-010

(16) Make sure that the landing gear retraction and extension is smooth.

(17) Make sure that all cycles complete and that all lights are OFF when gear is UP, and GREEN lights are illuminated when gear is DOWN.

(18) Make sure that the nose gear completes cycle before main gear, and that left and right gear complete less than 1 second apart.

SUBTASK 32-31-51-710-011

(19) Move the landing gear control lever to the UP position, then:

(a) When the landing gear is 75% up, Move the landing gear control lever to the DOWN position.

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(b) When the landing gear is 75% down, Move the landing gear control lever to the UP position.

(c) When the landing gear is 75% up, Move the landing gear control lever to the DOWN position.

SUBTASK 32-31-51-710-012

(20) Move the landing gear lever to the UP position.

(a) Wait for the gear to fully retract and all lights to extinguish.

SUBTASK 32-31-51-710-013

(21) Verify that the landing gear changed direction with no hydraulic leaks or mechanical damage. Repair any leaks.

SUBTASK 32-31-51-710-014

(22) Open the door for the landing gear manual extension handles in the flight deck floor.

SUBTASK 32-31-51-710-015

(23) Pull the NOSE gear manual extension handle at an angle as if sitting in the first officer's seat.

NOTE: Do not pull handle straight up.

SUBTASK 32-31-51-710-016

(24) Verify the NOSE GEAR light on the P2-3 panel is illuminated when the gear is fully extended.

SUBTASK 32-31-51-710-017

(25) Pull the LEFT main gear manual Extension handle at an angle as if sitting in the first officer's seat.

NOTE: Do not pull handle straight up.

SUBTASK 32-31-51-710-018

(26) Verify the LEFT GEAR green light on the P2-3 panel is illuminated when the gear is fully extended.

SUBTASK 32-31-51-710-019

(27) Pull the RIGHT GEAR manual extension handle at an angle as if sitting in the first officer's seat.

NOTE: Do not pull the handle straight up.

SUBTASK 32-31-51-710-020

(28) Verify the RIGHT GEAR green light on the P2-3 panel is illuminated when the gear is fully extended.

SUBTASK 32-31-51-710-021

(29) Close the door for the LANDING GEAR manual extension handles in the flight deck floor.

SUBTASK 32-31-51-710-022

(30) Move the landing gear control lever to the DOWN position.

SUBTASK 32-31-51-420-007

(31) Install the downlock pins in the landing gear (Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801).

SUBTASK 32-31-51-040-003

(32) Remove pressure to the 'A' hydraulic system (Hydraulic System A or B Power Removal, TASK 29-11-00-860-805).

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SUBTASK 32-31-51-040-004

(33) Return airplane to the pretest configuration.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LANDING GEAR SELECTOR VALVE - INSPECTION/CHECK

1. General

A. This procedure has these tasks:

(1) An inspection of the landing gear selector valve.

(2) An inspection of the landing gear selector valve secondary end cap.

TASK 32-31-51-200-801

2. Landing Gear Selector Valve - Inspection

(Figure 601)

A. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

C. Landing Gear Selector Valve Inspection

SUBTASK 32-31-51-480-005

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-31-51-860-002

(2) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

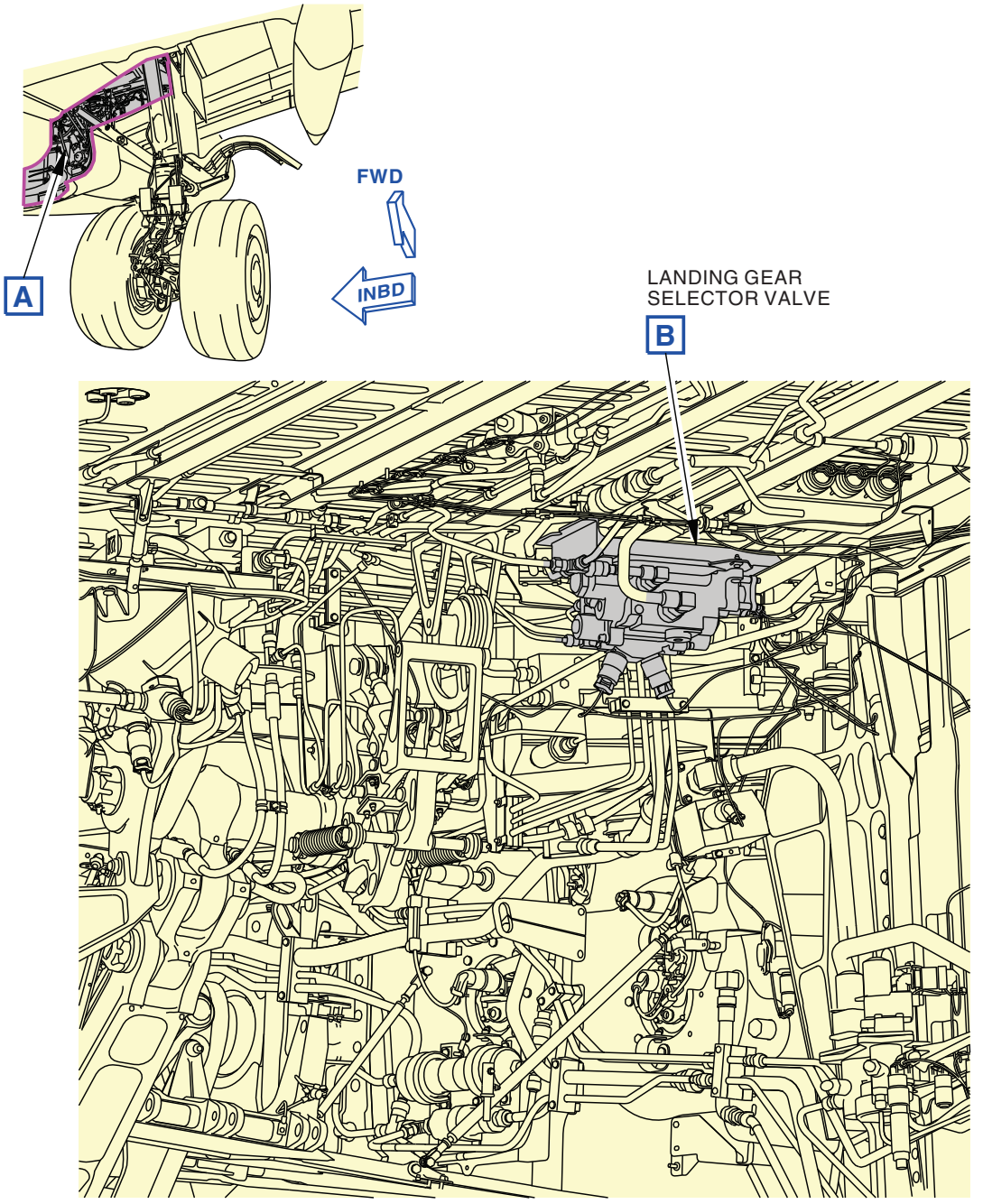
SUBTASK 32-31-51-200-001

(3) Do a detailed visual examination of the landing gear selector valve:

(a) Look at the visible part of the valve piston to make sure there are no leaks or damage.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



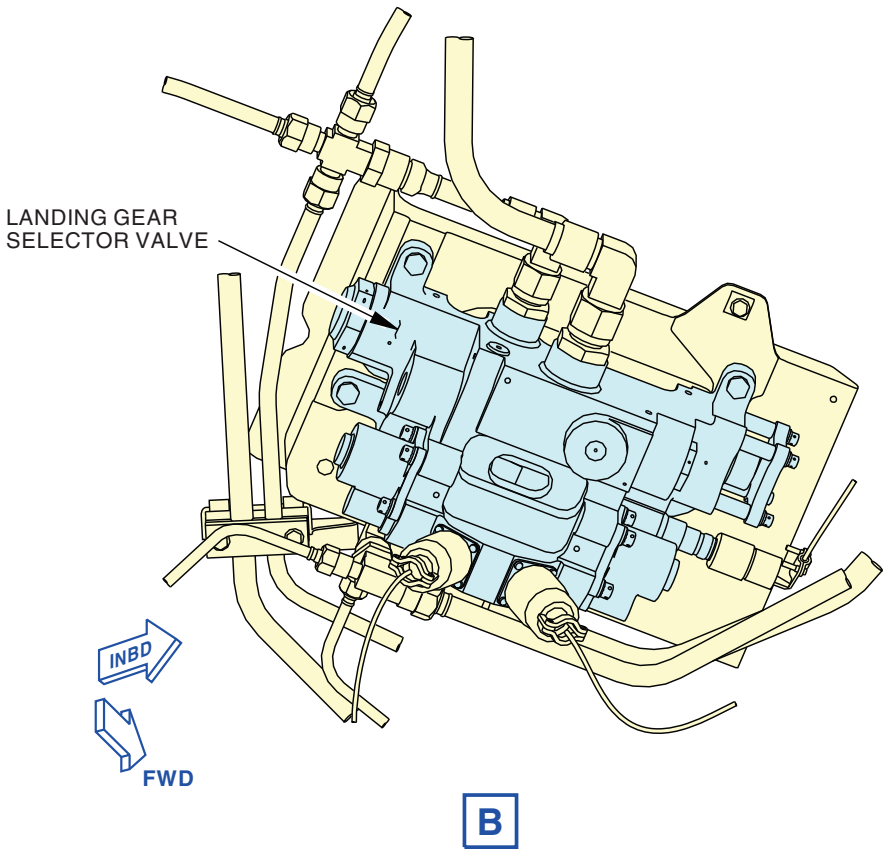
MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE IS SHOWN, RIGHT SIDE IS EQUIVALENT)

A

2536511 S0000602082_V1

Landing Gear Selector Valve Inspection
Figure 601/32-31-51-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414234 S00061534674_V1

Landing Gear Selector Valve Inspection
Figure 601/32-31-51-990-802 (Sheet 2 of 2)

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TASK 32-31-51-200-802

3. Landing Gear Selector Valve Secondary End Cap - Inspection

(Figure 602)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

C. Landing Gear Selector Valve Secondary End Cap Inspection

SUBTASK 32-31-51-480-006

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-31-51-860-003

(2) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

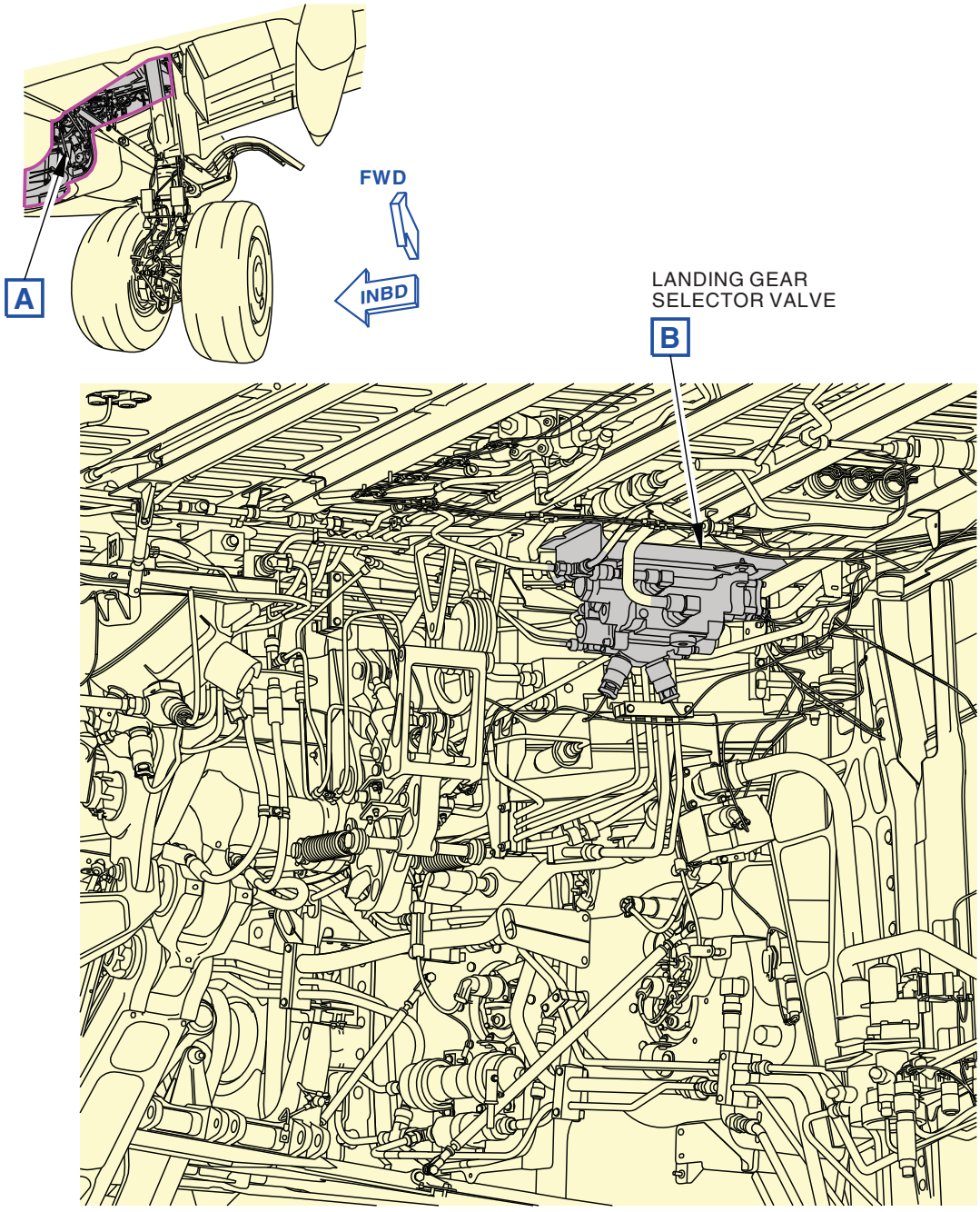
SUBTASK 32-31-51-200-002

(3) Do a detailed visual examination of the landing gear selector valve secondary end cap.

(a) Look at the visible part of the end cap to make sure there are no leaks or damage.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



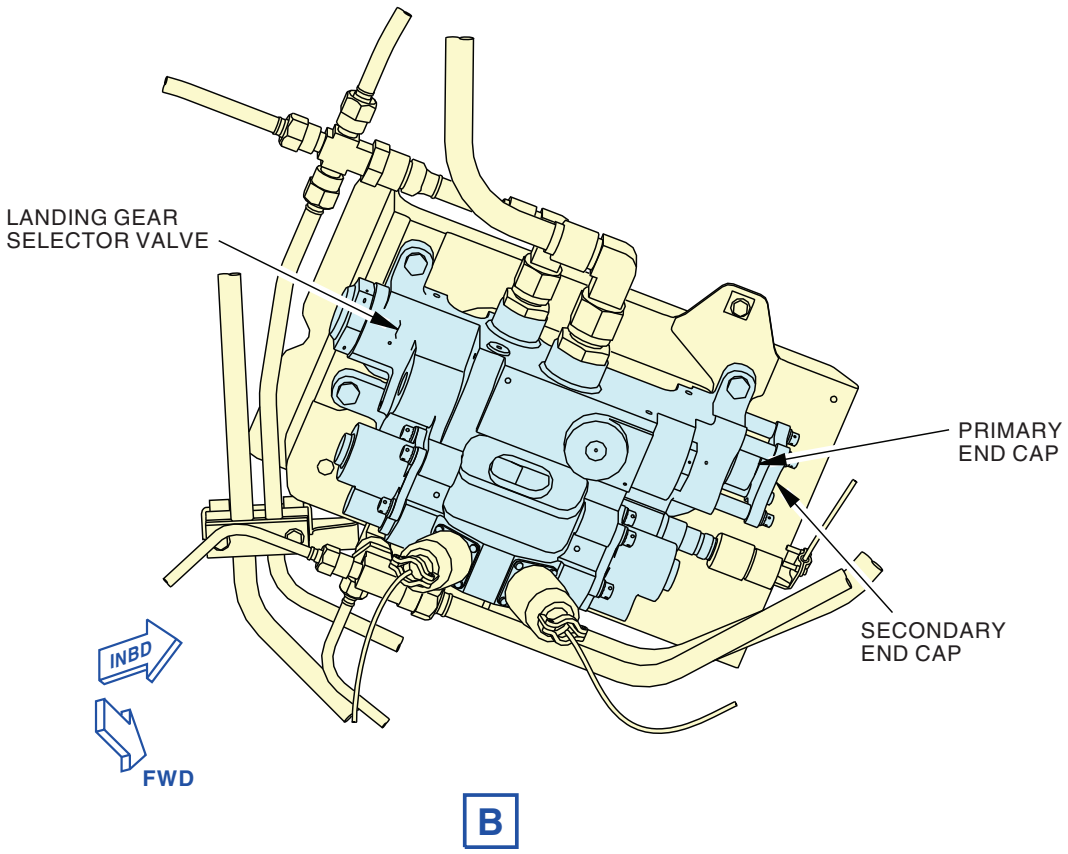
MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE IS SHOWN, RIGHT SIDE IS EQUIVALENT)

A

2536511 S0000602082_V1

Landing Gear Selector Valve Secondary End Cap Inspection
Figure 602/32-31-51-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2546003 S0000605161_V1

Landing Gear Selector Valve Secondary End Cap Inspection
Figure 602/32-31-51-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LANDING GEAR TRANSFER VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the transfer valve for the landing gear

(2) An installation of the transfer valve for the landing gear

(3) A system test of the transfer valve for the landing gear

(4) An operational test of the transfer valve for the landing gear.

TASK 32-31-71-020-801

2. Landing Gear Transfer Valve Removal

(Figure 401)

A. General

(1) The landing gear transfer valve is referred to as the valve assembly in this task.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Prepare for the Removal

SUBTASK 32-31-71-480-005

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-31-71-860-001

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

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SUBTASK 32-31-71-860-002

(3) Remove the pressure from hydraulic system A and B (TASK 29-09-00-860-802).

SUBTASK 32-31-71-580-001

(4) Install chocks around the tires of the main landing gear (PAGEBLOCK 10-11-05/201).

F. Landing Gear Transfer Valve Removal

SUBTASK 32-31-71-020-001

(1) Disconnect the electrical connector [11] from the valve assembly [1].

SUBTASK 32-31-71-930-001

(2) Put tags on the hydraulic lines [2] to identify the correct locations of the lines.

SUBTASK 32-31-71-020-002

(3) Disconnect the hydraulic lines [2] from the valve assembly [1].

SUBTASK 32-31-71-480-002

(4) Install caps/plugs, STD-14686, on the hydraulic lines [2] and ports on the valve assembly [1].

SUBTASK 32-31-71-020-003

(5) Remove the bolt [5], washer [6], and washer [8] to disconnect the jumper assembly [7] from the valve assembly [1].

SUBTASK 32-31-71-020-004

(6) Remove the washers [12] and bolts [13] that attach the valve assembly [1] to the support bracket [9].

SUBTASK 32-31-71-020-005

(7) Remove the valve assembly [1] from the airplane.

SUBTASK 32-31-71-080-001

(8) Remove the caps/plugs, STD-14686, from the ports on the valve assembly [1] and drain the hydraulic fluid.

SUBTASK 32-31-71-020-006

(9) Do these steps if the replacement valve assembly [1] does not have unions [10] or reducer [3] installed:

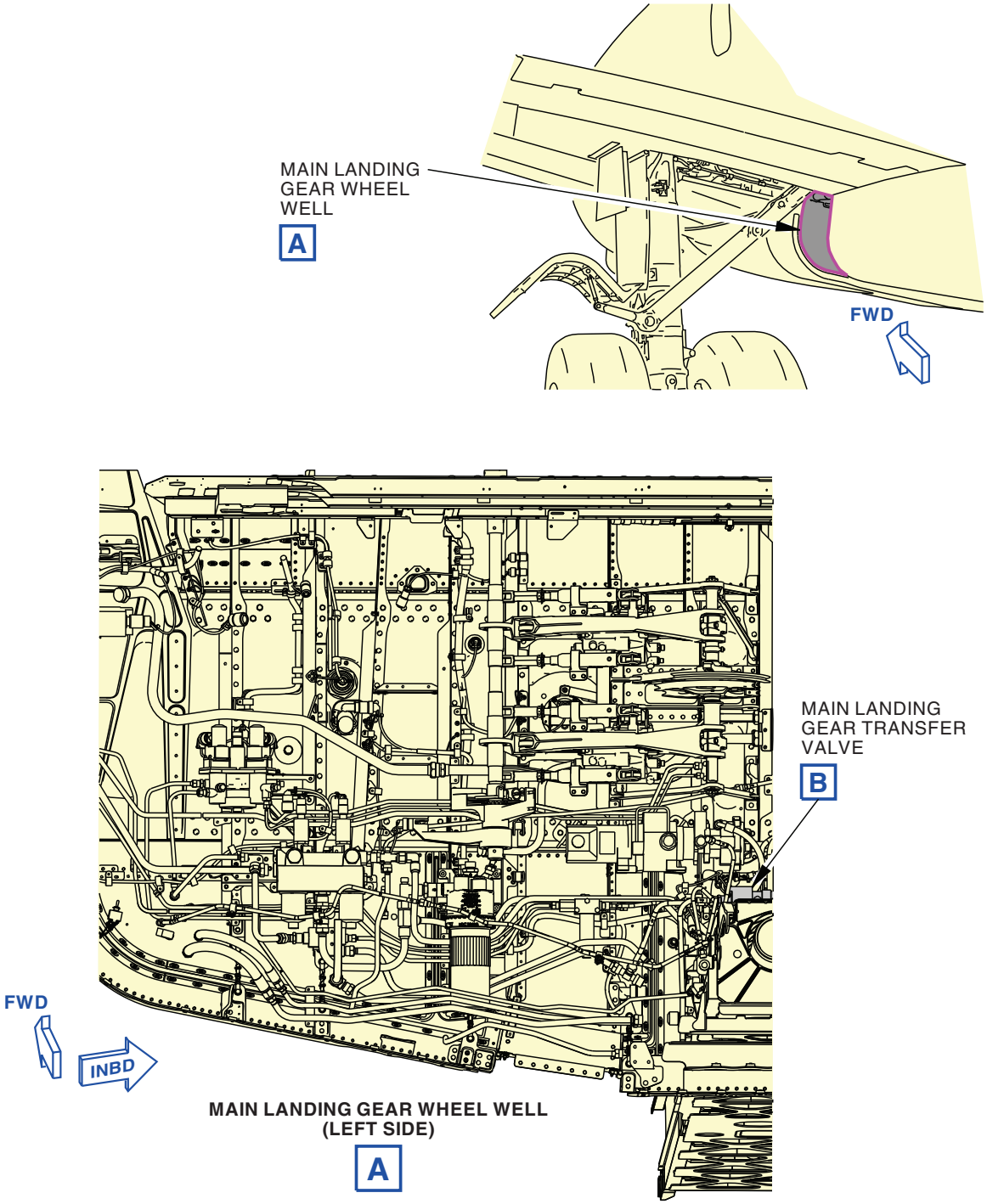
(a) Remove the unions [10] and reducer [3].

(b) Remove and discard the packings [4].

(c) Install caps/plugs, STD-14686, in the ports on the valve assembly [1].

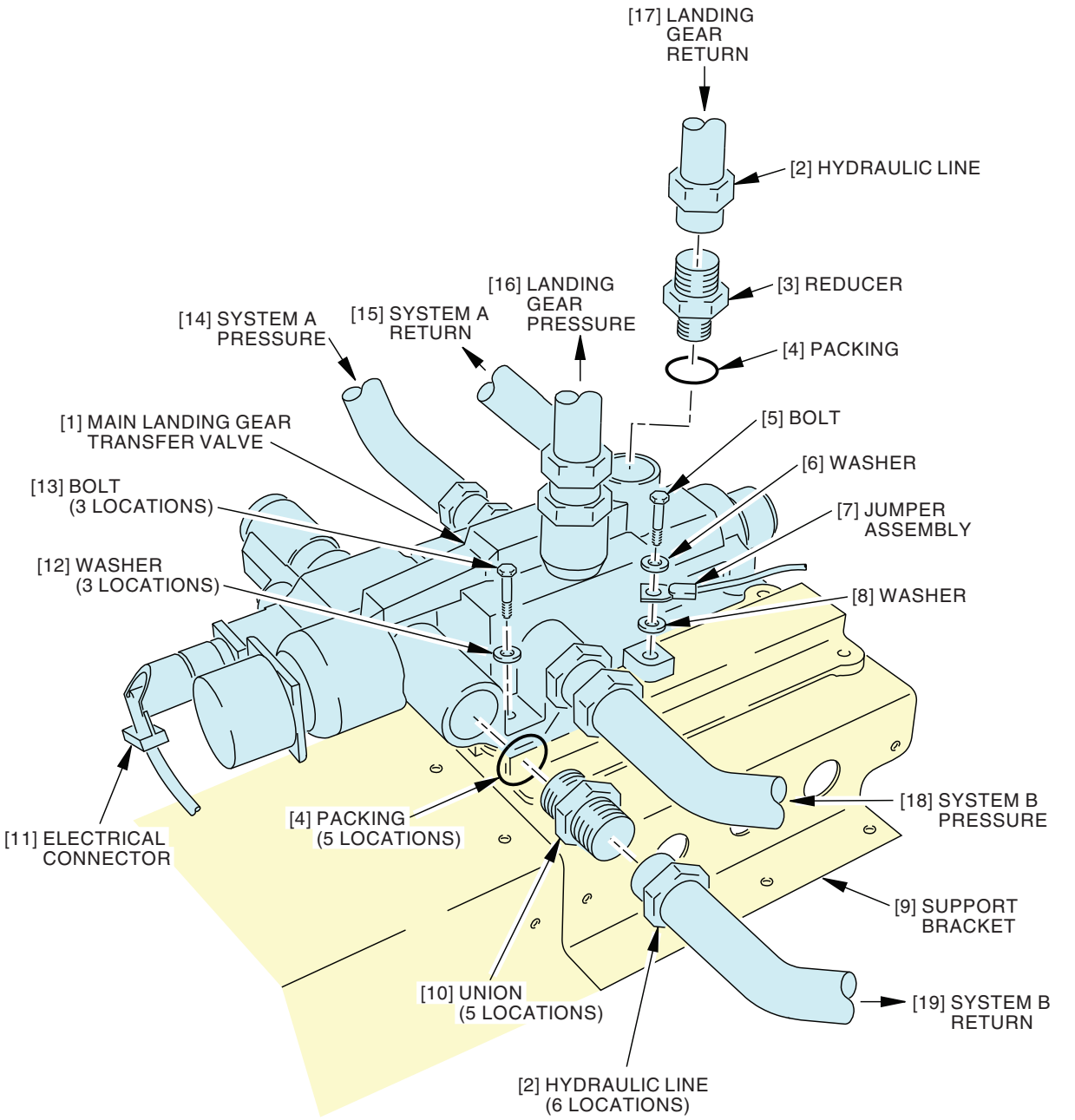
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Transfer Valve Installation
Figure 401/32-31-71-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



MAIN LANDING GEAR TRANSFER VALVE

B

2414243 S00061534693_V3

Main Landing Gear Transfer Valve Installation
Figure 401/32-31-71-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-31-71-400-801

3. Landing Gear Transfer Valve Installation

(Figure 401)

A. General

(1) After the installation task for the landing gear transfer valve assembly, perform the operational test.

B. References

Reference	Title
20-10-34-120-801	Hand Clean Metal Surfaces with Abrasives (P/B 701)
20-10-37-120-801	Electrical Bonding Inspection (P/B 601)
29-00-00-870-801	Bleed the Hydraulic Systems (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
51-22-00-390-806	Fillet Seal Application (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
SWPM 20-60-03	Special Protection of Electrical Connectors

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

D. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	

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E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Valve assembly	32-31-71-01-045	PTF ALL
4	Packing	32-31-71-01-060	PTF ALL

F. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

G. Landing Gear Transfer Valve Installation

SUBTASK 32-31-71-420-001

(1) Do these steps if the replacement valve assembly [1] does not have the unions [10] or reducer [3] installed:

(a) Lubricate these items with hydraulic fluid, D00153:

1) The new packings [4]

2) The reducer [3]

3) The unions [10].

(b) Remove the caps/plugs, STD-14686, from the ports on the valve assembly [1].

(c) Install the packings [4], reducer [3], and unions [10] on the valve assembly [1].

(d) Tighten the reducer [3] and unions [10] to 360 ±18 in-lb (40.7 ±2.0 N·m).

SUBTASK 32-31-71-100-001

(2) Clean the surface of the support bracket [9], where the valve assembly [1] will be attached (TASK 20-10-34-120-801).

SUBTASK 32-31-71-100-002

(3) Clean the surface of the valve assembly [1] that will attach to the support bracket [9] (TASK 20-10-34-120-801).

SUBTASK 32-31-71-110-001

(4) Apply coating, C00064, to all of the areas that you cleaned on the support bracket [9] and valve assembly [1].

SUBTASK 32-31-71-600-001

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(5) Apply corrosion inhibiting material, G50136, to the shank and threads of the bolt [5] and bolts [13].

SUBTASK 32-31-71-420-002

(6) Install the bolts [13] and washers [12] that attach the valve assembly [1] to the support bracket [9].

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SUBTASK 32-31-71-100-003

(7) Clean the surface of the valve assembly [1] where the jumper assembly [7] connects (TASK 20-10-34-120-801).

SUBTASK 32-31-71-100-004

(8) Clean the surface of the jumper assembly [7], washer [6], and washer [8] that will attach to the valve assembly [1] (SWPM 20-20-00):

(a) Use Cleaning Procedure 5 to clean the jumper assembly [7], washer [6], and washer [8].

SUBTASK 32-31-71-110-002

(9) Apply coating, C00064, to all of the cleaned areas on the jumper assembly [7] and valve assembly [1].

SUBTASK 32-31-71-420-003

(10) Install the washer [6], washer [8], and bolt [5] to attach the jumper assembly [7] to the valve assembly [1].

SUBTASK 32-31-71-760-001

(11) Measure the electrical bonding resistance between the valve assembly [1] and support bracket [9] (TASK 20-10-37-120-801).

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure that the electrical bonding resistance is 0.001 ohm (1 milliohm) or less.

SUBTASK 32-31-71-420-004

(12) Apply a fillet seal along the edges of the joint between the valve assembly [1] and support bracket [9] (TASK 51-22-00-390-806).

SUBTASK 32-31-71-390-001

(13) Apply a fillet seal of sealant, A02315, to the terminal connections of the jumper assembly [7] (TASK 51-22-00-390-806).

SUBTASK 32-31-71-080-002

(14) Remove the caps/plugs, STD-14686, from the unions [10] and reducer [3] on the valve assembly [1].

SUBTASK 32-31-71-420-005

(15) Connect the hydraulic lines to the valve assembly [1].

(a) Tighten the B-nut on System A Pressure hydraulic line [14] to 700 ±35 in-lb (79 ±4 N·m).

(b) Tighten the B-nut on System A Return hydraulic line [15] to 700 ±35 in-lb (79 ±4 N·m).

(c) Tighten the B-nut on Landing Gear Pressure hydraulic line [16] to 700 ±35 in-lb (79 ±4 N·m).

(d) Tighten the B-nut on Landing Gear Return hydraulic line [17] to 500 ±25 in-lb (56 ±3 N·m).

(e) Tighten the B-nut on System B Pressure hydraulic line [18]to 700 ±35 in-lb (79 ±4 N·m).

(f) Tighten the B-nut on System B Return hydraulic line [19]to 700 ±35 in-lb (79 ±4 N·m).

SUBTASK 32-31-71-840-001

(16) Remove the tags on the hydraulic lines [2] to identify the correct location.

SUBTASK 32-31-71-420-007

(17) Before the electrical connector [11] connections to the valve assembly [1], examine the electrical connector [11] for corrosion.

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WARNING

DO THE STEPS BELOW IF THE AIRPLANE OPERATES WHERE DEICING FLUID THAT CONTAINS POTASSIUM FORMATE OR POTASSIUM ACETATE IS USED. ALSO, DO THE STEPS FOR ALL AIRPLANES THAT YOU FOUND CORROSION IN THE ELECTRICAL CONNECTORS IN THE MAIN WHEEL WELL. THE ELECTRICAL CONNECTORS ARE IN A SYSTEM THAT IS NECESSARY FOR SAFE FLIGHT.

(a) If there is corrosion, correct the problem (SWPM 20-60-03).

(b) Apply D5026NS or ZC-026 compound, G50171, to the connector (SWPM 20-60-03).

(c) Connect the electrical connector [11] to the valve assembly [1].

SUBTASK 32-31-71-860-003

(18) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

SUBTASK 32-31-71-860-023

(19) Do this task: Bleed the Hydraulic Systems, TASK 29-00-00-870-801.

SUBTASK 32-31-71-860-004

(20) For hydraulic systems A and B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-31-71-210-001

(21) Make sure that there are no leaks at the hydraulic line connections.

SUBTASK 32-31-71-710-002

(22) Do the operational test of the valve assembly [1] (TASK 32-31-71-700-801).

——— END OF TASK ———

TASK 32-31-71-730-801

4. Landing Gear Transfer Valve System Test

A. General

(1) This task does an operational test of the transfer valve solenoid, logic check, and transfer valve gear retraction test.

B. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
29-11-00-860-802	Hydraulic System A or B Pressurization with a Portable Hydraulic Cart (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

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C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

D. Prepare for the Operational Test

SUBTASK 32-31-71-480-003

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-31-71-860-005

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(2) Lift the airplane on jacks (TASK 07-10-01-580-801).

SUBTASK 32-31-71-860-006

(3) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	1	C01316	ENGINE 1 START LEVER CHAN A
B	3	C01312	ENGINE 1 RUN/PWR
D	1	C04030	DSP CAPT EFIS CONT PNL
D	2	C04003	MULTI FUNCTION PANEL
D	3	C04026	DISPLAY LEFT INBOARD
D	4	C04027	DISPLAY LEFT OUTBOARD
D	5	C04023	DISPLAY DPC 1 PRI

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	9	C04024	DISPLAY DPC 2 HOLDUP
D	10	C04022	DISPLAY DPC 1 HOLDUP
D	11	C04025	DISPLAY DPC 2 PRI
E	10	C04029	DISPLAY RIGHT OUTBOARD

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(Continued)

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	11	C04028	DISPLAY RIGHT INBOARD
E	12	C04002	MAINT/BITE PANEL
E	13	C04031	DSP F/O EFIS CONT PNL

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	6	C01318	ENGINE 2 START LEVER CHAN A
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	17	C00129	LANDING GEAR LATCH & PRESS WARN
C	15	C01355	LANDING GEAR AIR/GND SYS 2
C	16	C01356	LANDING GEAR AIR/GND SYS 1
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN
E	12	C00314	INDICATOR MASTER DIM SECT 2
F	11	C00317	INDICATOR MASTER DIM SECT 5

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	18	C04020	LANDING GEAR CONTROL 1
D	19	C04021	LANDING GEAR CONTROL 2

E. Transfer Valve Operational Test

SUBTASK 32-31-71-860-010

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN

SUBTASK 32-31-71-860-007

(2) Make sure that the control lever for the landing gear is in the DN position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-31-71-860-008

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Remove the landing gear downlock pins (TASK 32-00-01-080-801).

SUBTASK 32-31-71-860-009

(4) Move both Engine No. 1 and No. 2 throttles completely forward.

NOTE: Throttles in the advanced position prevent the landing gear warning (red) lights from remaining on after gear is up and locked.

SUBTASK 32-31-71-710-003

(5) Put the Master Dim and Test switch in the bright position.

SUBTASK 32-31-71-710-004

(6) Set the GND TESTS switch to ENABLE.

NOTE: The switch is located on the P61-4 panel.

SUBTASK 32-31-71-710-005

(7) Supply 3000 ±150 psi (20,684 ±1034 kPa) pressure to the hydraulic systems A and B (TASK 29-11-00-860-802).

SUBTASK 32-31-71-710-006

(8) Start the PSEU LG Transfer Valve test.

(a) Select SYS on the Multi-Function Display Panel.

(b) Select MAINT -> ONBD MAINT -> LINE MAINT -> GROUND TESTS -> ATA SYSTEM 32: PSEU -> OPERATIONAL TEST.

(c) Select LGTV TEST and then select CONTINUE.

(d) Select START TEST from the PSEU LGTV test screen.

(e) Make sure that the display shows TEST CONDITION: PASS.

SUBTASK 32-31-71-710-007

(9) Select STOP FUNCTION from the on-screen display.

SUBTASK 32-31-71-710-042

(10) Move Fuel switches to the ON position.

SUBTASK 32-31-71-710-009

(11) Wait five (5) minutes.

NOTE: This simulates engine running and energizes relay R564 to the PSEU.

SUBTASK 32-31-71-710-043

(12) Select the MAINT DATA PGS tab from the Multi-Function Display.

SUBTASK 32-31-71-710-011

(13) Select PSEU from the Multi-Function Display.

SUBTASK 32-31-71-710-012

(14) Select NEXT PAGE until IO MATRIX is visible.

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SUBTASK 32-31-71-710-013

(15) Make sure that the display shows 1 for D10986 pin 39 (left engine running).

SUBTASK 32-31-71-710-015

(16) Make sure that the display shows 0 for D10984 pin 57 (VALVE OFF).

SUBTASK 32-31-71-710-044

(17) Select MAINT -> ONBD MAINT -> OTHER FUNCTIONS -> SPECIAL FUNCTIONS -> ATA 32: PSEU from the on-screen display.

SUBTASK 32-31-71-710-017

(18) Select AIR/GND OVERRIDE and then select CONTINUE from the on-screen menu.

SUBTASK 32-31-71-710-047

(19) Select CONTINUE after reading through the precautions screen.

SUBTASK 32-31-71-710-018

(20) Select START FUNCTION on the display.

SUBTASK 32-31-71-710-019

(21) Select Set SYS 1 and 2 ON GND from the on-screen menu and then select CONTINUE.

SUBTASK 32-31-71-710-020

(22) Move the NOSE WHEEL STEER switch to the ALT position.

SUBTASK 32-31-71-710-021

(23) Make sure that the display shows 1 for D10984 for pin 57 (VALVE ON).

SUBTASK 32-31-71-710-022

(24) Make sure that the landing gear control lever is in the DN position and 'A' and 'B' hydraulic system are pressurized.

SUBTASK 32-31-71-710-023

(25) Remove pressure from the hydraulic system A through the ground service disconnect.

(a) Keep the hydraulic system B pressurized at 3000 ±150 psi (20,684 ±1034 kPa).

SUBTASK 32-31-71-710-024

(26) Push the LOCK OVRD button.

SUBTASK 32-31-71-710-025

(27) Prepare to measure the landing gear retract/extension times.

NOTE: The landing gear retraction time is measured from the time the landing gear control lever is in the 'UP' position until the gear red lights go off.

SUBTASK 32-31-71-710-027

(28) Move the landing gear control lever to the UP position.

(a) Make sure that the elapsed time from the landing gear control lever UP until all red lights go off is less than 9.0 seconds.

NOTE: The nose landing gear light should go off at least 1 second before the main landing gear lights. The main landing gear lights should go off within 1 second of each other. If these conditions are not true, the timing test should be considered to have FAILED.

SUBTASK 32-31-71-710-028

(29) Make sure that the GEAR does not extend.

SUBTASK 32-31-71-710-030

(30) Make sure that all landing gear green and red lights are off.

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SUBTASK 32-31-71-710-031

(31) Wait 15 seconds.

NOTE: This lets the auto-off trigger and end the gear up command.

SUBTASK 32-31-71-710-032

(32) Supply 3000 ±150 psi (20,684 ±1034 kPa) pressure to the hydraulic system A through the ground service disconnect (TASK 29-11-00-860-802).

(a) Keep the hydraulic system B pressurized at 3000 ±150 psi (20,684 ±1034 kPa).

SUBTASK 32-31-71-710-033

(33) Set the NOSE WHEEL STEER switch to NORM.

SUBTASK 32-31-71-710-034

(34) Select MAINT ->ONBD MAINT -> OTHER FUNCTIONS -> SPECIAL FUNCTIONS -> ATA System 32: PSEU from the on-screen display.

SUBTASK 32-31-71-710-035

(35) Select AIR/GND OVERRIDE and then select CONTINUE from the on-screen menu.

SUBTASK 32-31-71-710-048

(36) Select CONTINUE after reading through the precautions screen.

SUBTASK 32-31-71-710-036

(37) Select START FUNCTION on the display.

SUBTASK 32-31-71-710-045

(38) Select RESET OVERRIDE on the display.

SUBTASK 32-31-71-710-037

(39) Select STOP FUNCTION from the MFD Test Screen.

SUBTASK 32-31-71-710-038

(40) Make sure that the display shows 0 for D10984 pin 57 (VALVE OFF).

F. Put the Airplane to Its Usual Condition

SUBTASK 32-31-71-710-040

(1) Move the landing gear control lever to the DN position.

(a) Make sure that the elapsed time from the landing gear control lever DN until all green lights come on is less than 13.0 seconds.

SUBTASK 32-31-71-710-041

(2) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-31-71-940-001

WARNING MAKE SURE THAT THE GROUND LOCK PINS ARE INSTALLED IN ALL OF THE LANDING GEAR. WITHOUT THE GROUND LOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(3) Install the landing gear ground locks (TASK 32-00-01-480-801).

SUBTASK 32-31-71-940-002

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(4) Lower the airplane off the jacks (TASK 07-10-01-580-802).

SUBTASK 32-31-71-860-015

(5) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN

——— END OF TASK ———

TASK 32-31-71-700-801

5. Landing Gear Transfer Valve Operational Test

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task has the steps to use the Proximity Switch Electronic Unit (PSEU) to do the operational test of the landing gear transfer valve.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-09-10-740-801	Proximity Switch Electronics Unit - Operational Test (P/B 501)

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Operational Test

SUBTASK 32-31-71-860-011

(1) Set the GND TEST switch, on the P61 maintenance bite panel, to the ENABLE position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-31-71-860-013

(2) Supply Hydraulic System A and System B power (TASK 29-11-00-860-801).

SUBTASK 32-31-71-710-046

(3) On the MAX Display System (MDS) Multi Function Display (MFD), use the PSEU ground test function to do the LGTV Test (TASK 32-09-10-740-801).

(a) Make these selections on the MFD main function menu:

1) ONBOARD MAINTENANCE
2) LINE MAINTENANCE
3) GROUND TESTS
4) 32 PROXIMITY SWITCH ELECTRONIC UNIT
5) OPERATIONAL TEST
6) LGTV TEST
7) CONTINUE.

(b) Do the instructions that show on the MFD PRECONDITIONS and then make these selections:

NOTE: Make sure that you verify and follow the inhibited conditions instructions closely. If one of the inhibited conditions is not met, the test will be failed.

1) CONTINUE
2) START TEST.

(c) When the test is completed, make sure that the screen displays TEST CONDITION: PASS.

(d) If FAILED shows, refer to the IFIM and do the applicable procedure(s).

SUBTASK 32-31-71-860-014

WARNING

MAKE SURE THAT EQUIPMENT AND PERSONNEL ARE CLEAR OF THE FLIGHT CONTROLS SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(4) For Hydraulic System A and System B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-31-71-860-012

(5) Set the GND TEST switch, on the P61 maintenance bite panel, to the NORM position.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LANDING GEAR TRANSFER VALVE - ADJUSTMENT/TEST

1. General

A. This procedure has this task:

(1) Alternate landing gear transfer valve operational test.

TASK 32-31-71-700-802

2. Alternate Landing Gear Transfer Valve Operational Test

A. General

(1) This procedure provides the alternate method to do the landing gear transfer valve operational test.

(2) The landing gear control lever module is installed in the P2 panel.

B. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Operational Test

SUBTASK 32-31-71-480-004

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gears (TASK 32-00-01-480-801).

SUBTASK 32-31-71-840-002

WARNING OBEY THE PROCEDURE THAT PUTS THE AIRPLANE IN THE AIR MODE. IF YOU DO THE PROCEDURE INCORRECTLY, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Put the airplane in the Air Mode using the alternate method (TASK 32-09-00-860-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Alternate Landing Gear Transfer Valve Operational Test

SUBTASK 32-31-71-860-016

(1) Pressurize Hydraulic System B (TASK 29-11-00-860-801).

SUBTASK 32-31-71-410-001

(2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC

SUBTASK 32-31-71-860-019

(3) Move the landing gear control lever to the UP position.

(a) Make sure that the landing gear tries to retract.

NOTE: You will hear the “whoosh” sound and the actuators will strain. This verifies that one of the coil windings is getting power.

SUBTASK 32-31-71-860-020

(4) Move the landing gear control lever to the DN position.

SUBTASK 32-31-71-410-002

(5) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC

SUBTASK 32-31-71-410-003

(6) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

SUBTASK 32-31-71-860-021

(7) Move the landing gear control lever to the UP position.

(a) Make sure that the landing gear tries to retract.

NOTE: You will hear the “whoosh” sound and the actuators will strain. This verifies that one of the coil windings is getting power.

SUBTASK 32-31-71-860-022

(8) Move the landing gear control lever to the DN position.

SUBTASK 32-31-71-410-004

(9) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

F. Put the Airplane Back to Its Usual Condition

SUBTASK 32-31-71-860-018

(1) Remove pressure from the Hydraulic System B (TASK 29-11-00-860-805).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-31-71-860-017

(2) Return the airplane to the Ground Mode (TASK 32-09-00-860-802).

SUBTASK 32-31-71-840-003

(3) Examine the hydraulic reservoirs for the correct servicing (TASK 12-12-00-610-801).

(a) Do the servicing if it is necessary.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN GEAR EXTENSION AND RETRACTION - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) The first task does an extension/retraction test of the main landing gear after you replaced or adjusted a landing gear component.

(2) The second task does an operational test of the main landing gear.

(3) The third task does an operational test of the transfer valve.

B. The times for gear retraction in the operational test depend on the hydraulic flow rate. The retraction times given in this procedure are based on a minimum flow rate of 19 gpm (72 l/min) at a hydraulic system pressure of 3000 ±150 psi (20,684 ±1034 kPa). Hydraulic pressure can be supplied by a hydraulic test bench to give this minimum flow rate.

NOTE: System losses due to pressure drops and overall hydraulic system internal leakage may require a higher flow rate to achieve optimum retraction times. A hydraulic cart with a higher flow rate capacity is necessary to achieve the stated retraction times.

C. The Electric Motor-Driven Pump (EMDP) can be used for the component replacement test. When you use the EMDP operate only one landing gear at a time. The EMDP gives a hydraulic flow rate of approximately 6 gpm (23 l/min). This lower rate causes slower landing gear operation, especially if you operate more than one gear at the same time. Do not use the EMDP to measure extension/retraction times.

D. The component replacement test is satisfactory in most cases after replacing a landing gear structural or system component. If you are troubleshooting slow landing gear extension, it is recommended to do the operational test.

TASK 32-32-00-710-801

2. Main Landing Gear Test - Component Replacement

(Figure 501)

A. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
29-11-00-860-802	Hydraulic System A or B Pressurization with a Portable Hydraulic Cart (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-10-740-801	Proximity Switch Electronics Unit - Operational Test (P/B 501)

B. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Prepare for the Test

SUBTASK 32-32-00-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-32-00-580-001

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(2) Do this task: Lift the Airplane on Jacks, TASK 07-10-01-580-801.

SUBTASK 32-32-00-860-003

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN

SUBTASK 32-32-00-860-004

(4) Make sure that the control lever for the landing gear is in the DN position.

SUBTASK 32-32-00-080-001

(5) Remove the downlock pin from the main landing gear that you will test (TASK 32-00-01-080-801).

SUBTASK 32-32-00-860-005

(6) Make sure that all the gear red lights on the P2-3 panel are off.

SUBTASK 32-32-00-860-006

(7) Make sure that all the gear green lights on the P2-3 panel are on.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-32-00-860-007

(8) For hydraulic system A, do this task: Hydraulic System A or B Pressurization with a Portable Hydraulic Cart, TASK 29-11-00-860-802.

SUBTASK 32-32-00-860-008

WARNING MAKE SURE THAT YOU OPEN THE CIRCUIT BREAKER FOR THE WEATHER RADAR SYSTEM. THE FORWARD MOVEMENT OF A THRUST LEVER CAN CAUSE THE AUTOMATIC OPERATION OF THE SYSTEM. THE OPERATION OF THIS SYSTEM CAN CAUSE SERIOUS INJURY TO PERSONS AND DAMAGE TO EQUIPMENT IN THE AREA OF THE NOSE RADOME.

(9) Move the two thrust levers to the full forward position.

NOTE: Throttles in the advanced position prevent the landing gear warning (red) lights from remaining on after gear is up and locked.

D. Main Landing Gear Test - Component Replacement (Normal Extension and Retraction)

SUBTASK 32-32-00-860-024

(1) If component replacements for the landing gear selector valve, the gear lever or PSEU are necessary, then do these steps:

Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	19	C04021	LANDING GEAR CONTROL 2

(a) Cycle the gear on Landing Gear Control 1:

WARNING MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE MAIN LANDING GEAR. FAST MOVEMENT OF THE MAIN LANDING GEAR CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

1) Move the control lever for the landing gear to UP and make sure that the landing gear retracts.

NOTE: Do not use the lever lock override trigger.

2) Make sure that the green and red lights for the landing gear you retracted are off.

3) Move the control lever to the DN position and make sure that the landing gear extends.

4) Make sure that the landing gear is down and locked.

5) Make sure that the three green lights are on and the three red lights are off for the landing gear.

(b) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	19	C04021	LANDING GEAR CONTROL 2

(c) Cycle the gear on Landing Gear Control 2:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that this circuit breaker is open and has safety tag:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	18	C04020	LANDING GEAR CONTROL 1

WARNING MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE MAIN LANDING GEAR. FAST MOVEMENT OF THE MAIN LANDING GEAR CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

2) Move the control lever for the landing gear to UP and make sure that the landing gear retracts.

NOTE: Do not use the lever lock override trigger.

3) Make sure that the green and red lights for the landing gear you retracted are off.

4) Move the control lever to the DN position and make sure that the landing gear extends.

5) Make sure that the landing gear is down and locked.

6) Make sure that the three green lights are on and the three red lights are off for the landing gear.

7) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	18	C04020	LANDING GEAR CONTROL 1

SUBTASK 32-32-00-860-025

(2) If component replacements for an actuator or hydraulic component are necessary, then do these steps:

(a) Cycle the gear a minimum of 6 times to try to get the air out. To cycle the gear, do these steps:

WARNING MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE MAIN LANDING GEAR. FAST MOVEMENT OF THE MAIN LANDING GEAR CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

1) Move the control lever for the landing gear to UP and make sure that the landing gear retracts.

NOTE: Do not use the lever lock override trigger.

2) Make sure that the green and red lights for the landing gear you retracted are off.

3) Move the control lever to the DN position and make sure that the landing gear extends.

4) Make sure that the landing gear is down and locked.

5) Make sure that the three green lights are on and the three red lights are off for the landing gear.

SUBTASK 32-32-00-480-002

(3) Install the down lock pin in the main landing gear that you tested (TASK 32-00-01-480-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-32-00-580-002

CAUTION

MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION

MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(4) Do this task: Lower the Airplane off Jacks, TASK 07-10-01-580-802.

SUBTASK 32-32-00-860-010

(5) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN

SUBTASK 32-32-00-860-011

(6) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-32-00-070-001

(7) To do a check of the PSEU for stored faults with the SELF-TEST, do this task: Proximity Switch Electronics Unit - Operational Test, TASK 32-09-10-740-801.

——— END OF TASK ———

TASK 32-32-00-710-802

3. Main Landing Gear - Operational Test

(Figure 501)

A. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
12-15-31-610-802	Main Landing Gear Shock Strut Servicing, Airplane on the Ground (P/B 301)
29-11-00-860-802	Hydraulic System A or B Pressurization with a Portable Hydraulic Cart (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

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B. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Test

SUBTASK 32-32-00-860-026

WARNING MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE MAIN LANDING GEAR. FAST MOVEMENT OF THE MAIN LANDING GEAR CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Move the control lever to the DN position.

SUBTASK 32-32-00-480-003

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-32-00-710-034

(3) Make sure that the shock struts' pressure of the main landing gear is correct (TASK 12-15-31-610-802).

SUBTASK 32-32-00-580-003

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(4) Do this task: Lift the Airplane on Jacks, TASK 07-10-01-580-801.

SUBTASK 32-32-00-860-013

(5) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	3	C01312	ENGINE 1 RUN/PWR
D	1	C04030	DSP CAPT EFIS CONT PNL
D	2	C04003	MULTI FUNCTION PANEL
D	3	C04026	DISPLAY LEFT INBOARD
D	4	C04027	DISPLAY LEFT OUTBOARD

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	5	C04023	DISPLAY DPC 1 PRI

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	9	C04024	DISPLAY DPC 2 HOLDUP
D	10	C04022	DISPLAY DPC 1 HOLDUP
D	11	C04025	DISPLAY DPC 2 PRI
E	10	C04029	DISPLAY RIGHT OUTBOARD
E	11	C04028	DISPLAY RIGHT INBOARD
E	12	C04002	MAINT/BITE PANEL
E	13	C04031	DSP F/O EFIS CONT PNL

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	17	C00129	LANDING GEAR LATCH & PRESS WARN
C	15	C01355	LANDING GEAR AIR/GND SYS 2
C	16	C01356	LANDING GEAR AIR/GND SYS 1
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN
D	15	C01401	LANDING GEAR AIR/GND RELAY
D	16	C01432	LANDING GEAR ALTN EXTEND SOL
E	12	C00314	INDICATOR MASTER DIM SECT 2
F	11	C00317	INDICATOR MASTER DIM SECT 5
F	13	C01179	INDICATOR MASTER DIM SECT 7

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	1	C01958	NETWORK FILE SERVER
D	18	C04020	LANDING GEAR CONTROL 1
D	19	C04021	LANDING GEAR CONTROL 2

SUBTASK 32-32-00-860-014

(6) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN

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SUBTASK 32-32-00-860-017

(7) Make sure that the control lever for the landing gear is in the DN position.

SUBTASK 32-32-00-860-015

(8) Move the two thrust levers to the full forward position.

SUBTASK 32-32-00-860-016

(9) Put the master dim and test switch in the bright position.

SUBTASK 32-32-00-080-002

(10) Remove the downlock pins from the nose and main landing gear (TASK 32-00-01-080-801).

SUBTASK 32-32-00-860-018

(11) Make sure that all the gear red lights on the P2-3 panel are off.

SUBTASK 32-32-00-860-019

(12) Make sure that all the gear green lights on the P2-3 and P5 aft overhead panel are on.

SUBTASK 32-32-00-860-020

(13) For hydraulic system A, do this task: Hydraulic System A or B Pressurization with a Portable Hydraulic Cart, TASK 29-11-00-860-802.

D. Main Landing Gear (Normal Extension and Retraction) Operational Test

SUBTASK 32-32-00-710-003

WARNING MAKE SURE THAT THE AREA AROUND THE MAIN LANDING GEAR IS CLEAR OF PERSONNEL, AND EQUIPMENT. THE DOORS CLOSE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(1) Move the control lever for the landing gear to UP and measure the time for the retraction.

NOTE: Measure the landing gear retraction time from the time the control lever is put in the UP position until all the landing gear red "disagree" lights go off.

NOTE: Do not use the lever lock override trigger during the operational test.

NOTE: Do not use the Electric Motor-Driven Pump (EMDP) to measure extension / retraction times.

(a) Make sure that the landing gear is up-and-locked in 9.0 seconds or less.

SUBTASK 32-32-00-860-021

(2) Make sure that the three green lights and three red lights for the landing gear are off.

SUBTASK 32-32-00-710-004

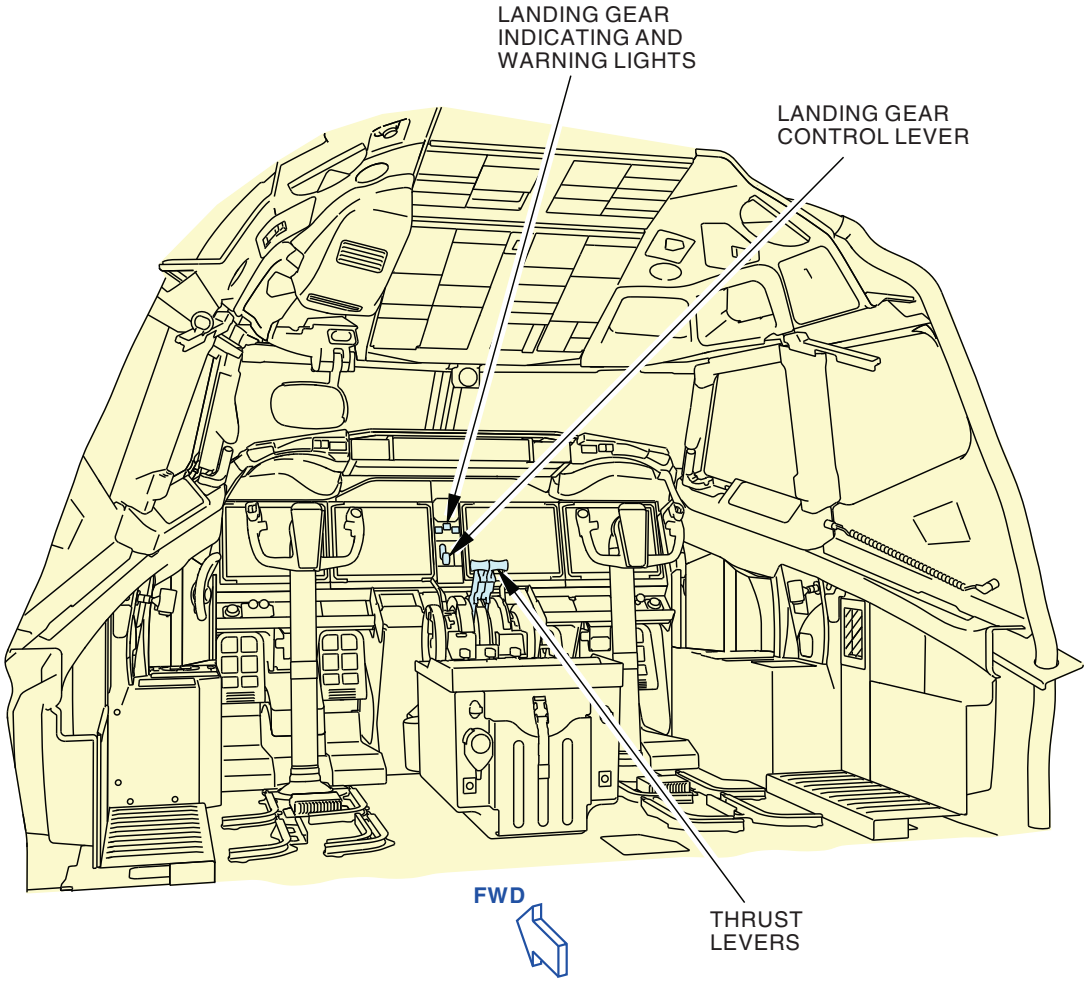
(3) Move the control lever to the DN position and measure the time for extension.

NOTE: Measure the landing gear extension time from the time the control lever is put in the DN position until all the landing gear green lights come on.

(a) Make sure that the landing gear is down and locked in 13.0 seconds or less.

END OF TASK

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FLIGHT COMPARTMENT

2414020 S00061534175_V1

Main Gear Extension and Retraction Test
Figure 501/32-32-00-990-801

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TASK 32-32-00-710-803

4. Transfer Valve - Operational Test

A. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
29-11-00-860-802	Hydraulic System A or B Pressurization with a Portable Hydraulic Cart (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Test

SUBTASK 32-32-00-710-005

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED IN ALL OF THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-32-00-710-006

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(2) Do this task: Lift the Airplane on Jacks, TASK 07-10-01-580-801.

SUBTASK 32-32-00-710-007

(3) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	1	C01316	ENGINE 1 START LEVER CHAN A
B	3	C01312	ENGINE 1 RUN/PWR
D	1	C04030	DSP CAPT EFIS CONT PNL
D	2	C04003	MULTI FUNCTION PANEL
D	3	C04026	DISPLAY LEFT INBOARD
D	4	C04027	DISPLAY LEFT OUTBOARD
D	5	C04023	DISPLAY DPC 1 PRI

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F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	9	C04024	DISPLAY DPC 2 HOLDUP
D	10	C04022	DISPLAY DPC 1 HOLDUP
D	11	C04025	DISPLAY DPC 2 PRI
E	10	C04029	DISPLAY RIGHT OUTBOARD
E	11	C04028	DISPLAY RIGHT INBOARD
E	12	C04002	MAINT/BITE PANEL
E	13	C04031	DSP F/O EFIS CONT PNL

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	6	C01318	ENGINE 2 START LEVER CHAN A
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	17	C00129	LANDING GEAR LATCH & PRESS WARN
C	15	C01355	LANDING GEAR AIR/GND SYS 2
C	16	C01356	LANDING GEAR AIR/GND SYS 1
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN
D	15	C01401	LANDING GEAR AIR/GND RELAY
E	12	C00314	INDICATOR MASTER DIM SECT 2
F	11	C00317	INDICATOR MASTER DIM SECT 5

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	1	C01958	NETWORK FILE SERVER
D	18	C04020	LANDING GEAR CONTROL 1
D	19	C04021	LANDING GEAR CONTROL 2

SUBTASK 32-32-00-710-008

(4) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

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F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN

D. Transfer Valve - Operational Test

SUBTASK 32-32-00-710-009

(1) Do this task: Landing Gear Downlock Pins Removal, TASK 32-00-01-080-801.

SUBTASK 32-32-00-710-010

(2) Move the two engine throttles fully forward.

NOTE: Throttles in the advanced position prevent the landing gear warning (red) lights from remaining on after gear is up and locked.

SUBTASK 32-32-00-710-011

(3) Put the Master Dim and Test switch in the bright position.

SUBTASK 32-32-00-710-012

(4) Supply hydraulic power to the hydraulic systems A and B (TASK 29-11-00-860-802).

SUBTASK 32-32-00-710-013

(5) Set the GND TESTS switch to ENABLE.

NOTE: The switch is on the P61-4 panel.

SUBTASK 32-32-00-710-014

(6) Start the PSEU LG Transfer Valve Test:

(a) Select SYS on the MFD (Multi Function Display) Panel.

(b) Select MAINT-> ONBD MAINT -> LINE MAINT ->GROUND TESTS -> ATA SYSTEM 32: PSEU -> OPERATIONAL TEST.

(c) Select LGTV TEST -> CONTINUE.

(d) Select START TEST from the PSEU LGTV test screen.

(e) Make sure that the screen displays 'TEST CONDITION: PASS'.

(f) Select RETURN AIRPLANE TO NORMAL from the on-screen display.

SUBTASK 32-32-00-710-015

(7) Set the Fuel Switches to the ON position.

NOTE: This applies if there is a replacement of an engine, the PSEU or the DPC. This does not apply to the selector valve, hydraulics or the landing gear lever.

(a) Wait 5 minutes.

NOTE: This simulates Engine Running and energizes relay R564 to the PSEU.

SUBTASK 32-32-00-710-016

(8) Select the MAINT DATA PGS tab from the MFD.

(a) Select PSEU.

(b) Select NEXT PAGE until I/O MATRIX shows.

(c) Make sure the display shows 1 for D10986 pin 39 (left engine running).

(d) Make sure the display shows 0 for D10984 pin 57 (valve off).

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SUBTASK 32-32-00-710-018

(9) Set the Fuel Switches to the OFF position.

NOTE: With the airplane IN AIR and no engine running signal, the Landing Gear Transfer Valve will set and the system will retract on B pressure alone.

SUBTASK 32-32-00-710-019

(10) Make sure the display shows 1 for D10984 pin 57 (valve on).

SUBTASK 32-32-00-710-020

(11) Make sure that the landing control lever is in the DN position.

(a) Make sure that the hydraulic systems A and B are pressurized.

SUBTASK 32-32-00-710-021

(12) Remove the pressure from the hydraulic system A.

(a) Keep the hydraulic system B pressure (TASK 29-11-00-860-805).

SUBTASK 32-32-00-710-023

(13) Prepare to measure landing gear retraction/extension times.

NOTE: Landing gear retraction time is measured from the time that the gear control lever is in the 'UP' position until all red lights are off.

SUBTASK 32-32-00-710-024

(14) Move the landing gear control lever to the UP position.

(a) Make sure that the elapsed time from gear lever UP until all red lights are off is less than 9.0 seconds.

SUBTASK 32-32-00-710-025

(15) Move the landing gear control lever to the DN position.

(a) Make sure that the gear does not extend.

SUBTASK 32-32-00-710-026

(16) Move the landing gear control lever to the UP position.

(a) Make sure that all landing gear green and red lights are off.

SUBTASK 32-32-00-710-028

(17) Supply pressure to the hydraulic system A (TASK 29-11-00-860-802).

(a) Keep the hydraulic system B pressurized at 3000 psi (20,684 kPa).

SUBTASK 32-32-00-710-030

(18) Make sure the display shows 0 for D10984 pin 57 (valve off).

SUBTASK 32-32-00-710-031

(19) Move the landing gear control lever to the DN position.

(a) Make sure that the elapsed time from gear lever DN until all green lights come on is less than 13.0 seconds.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 32-32-00-710-032

(1) Remove pressure from the hydraulic systems A and B (TASK 29-11-00-860-805).

(a) Install the downlock pins in all landing gear (TASK 32-00-01-480-801).

——— END OF TASK ———

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MAIN GEAR ACTUATOR ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure contains scheduled maintenance task data.

B. This procedure has these tasks:

(1) A removal of the actuator assembly for the main landing gear (referred to as the retract actuator [2])

(2) An installation of the actuator assembly for the main landing gear.

TASK 32-32-11-000-801

2. Main Gear Actuator Assembly Removal

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1870	Hoist - Retract Actuator, Main Landing Gear Part #: C32028-109 Supplier: 81205 Part #: C32028-110 Supplier: 81205 Part #: C32028-111 Supplier: 81205 Opt Part #: C32028-1 Supplier: 81205 Opt Part #: C32028-3 Supplier: 81205 Opt Part #: C32028-56 Supplier: 81205 Opt Part #: C32028-58 Supplier: 81205 Opt Part #: C32028-59 Supplier: 81205
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
551	Left Wing - Rear Spar To Landing Gear Support Beam
651	Right Wing - Rear Spar to Landing Gear Support Beam
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

D. Access Panels

Number	Name/Location
551BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel

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(Continued)

Number	Name/Location
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel
651BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

E. Prepare for the Removal

SUBTASK 32-32-11-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-32-11-860-001

(2) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-32-11-860-021

(3) Do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 32-32-11-580-001

(4) Install chocks around the tires of the main landing gear (PAGEBLOCK 10-11-05/201).

SUBTASK 32-32-11-010-001

(5) Open the applicable access panels:

Number	Name/Location
551BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel
651BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

F. Main Gear Actuator Assembly Removal

SUBTASK 32-32-11-020-001

(1) Disconnect the hydraulic UP and DN hoses from the retract actuator [2].

SUBTASK 32-32-11-480-002

(2) Install caps/plugs, STD-14686, in the hydraulic hoses.

SUBTASK 32-32-11-480-003

(3) Install the hoist, SPL-1870.

SUBTASK 32-32-11-980-001

(4) Hold the head end of the retract actuator [2] in position with the hoist, SPL-1870.

SUBTASK 32-32-11-020-002

(5) Remove these parts to disconnect the beam hanger [3] from the structure:

(a) pin [5]

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(b) nut [6]

(c) washer [7]

(d) pin [4].

SUBTASK 32-32-11-020-003

(6) Hold the walking beam [1] and retract actuator [2] and remove these parts to disconnect the walking beam [1] from the main gear trunnion:

(a) cotter pin [12], nut [11], washer [10], and bolt [17]

(b) nut [18] and washer [19]

(c) walking beam attach pin [16].

SUBTASK 32-32-11-020-004

(7) Hold the walking beam [1] and retract actuator [2], and remove these parts to disconnect the retract actuator [2] from the main gear trunnion:

(a) cotter pin [12], nut [11], washer [10], and bolt [8]

(b) nut [9] and washer [13]

(c) spacers [14]

(d) actuator attach pin [15].

SUBTASK 32-32-11-020-005

(8) Remove the walking beam [1] and retract actuator [2] with the hoist, SPL-1870.

SUBTASK 32-32-11-080-001

(9) Remove the hoist, SPL-1870, from the walking beam [1] and retract actuator [2].

SUBTASK 32-32-11-020-006

(10) Do these steps to remove the retract actuator [2] from the walking beam [1] and the beam hanger [3]:

(a) Remove the cotter pin [12], nut [11], washer [10], and bolt [17].

(b) Remove the nut [21] and washer [22].

(c) Remove the pin [20].

(d) Remove the retract actuator [2].

SUBTASK 32-32-11-020-007

(11) Do these steps if the replacement retract actuator [2] does not have a restrictor [23] and union [26] installed:

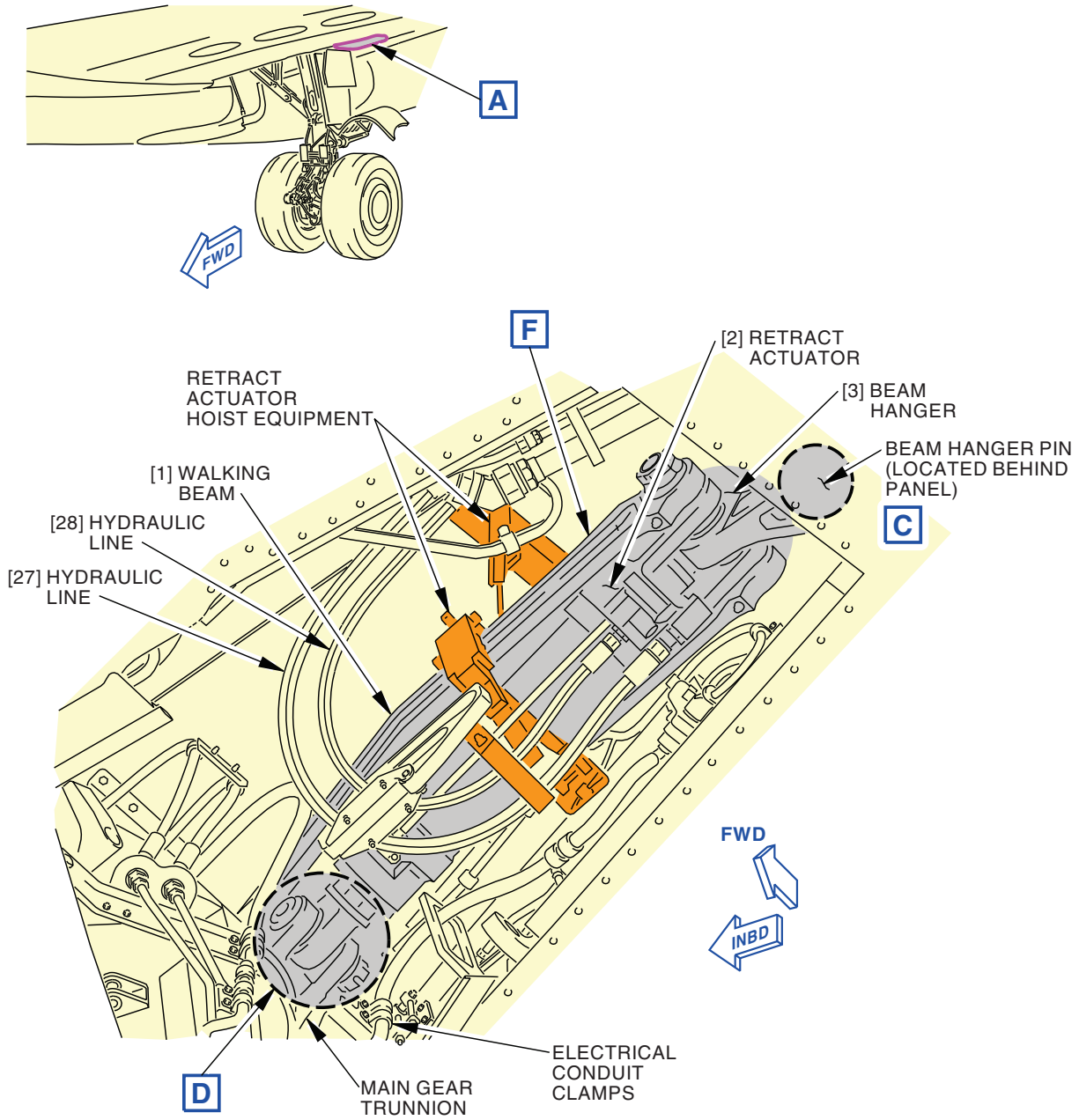
(a) Remove the restrictor [23] and the union [26] from the retract actuator [2].

(b) Remove and discard the packing [24] and packing [25].

(c) Install caps/plugs, STD-14686, in the ports on the retract actuator [2].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



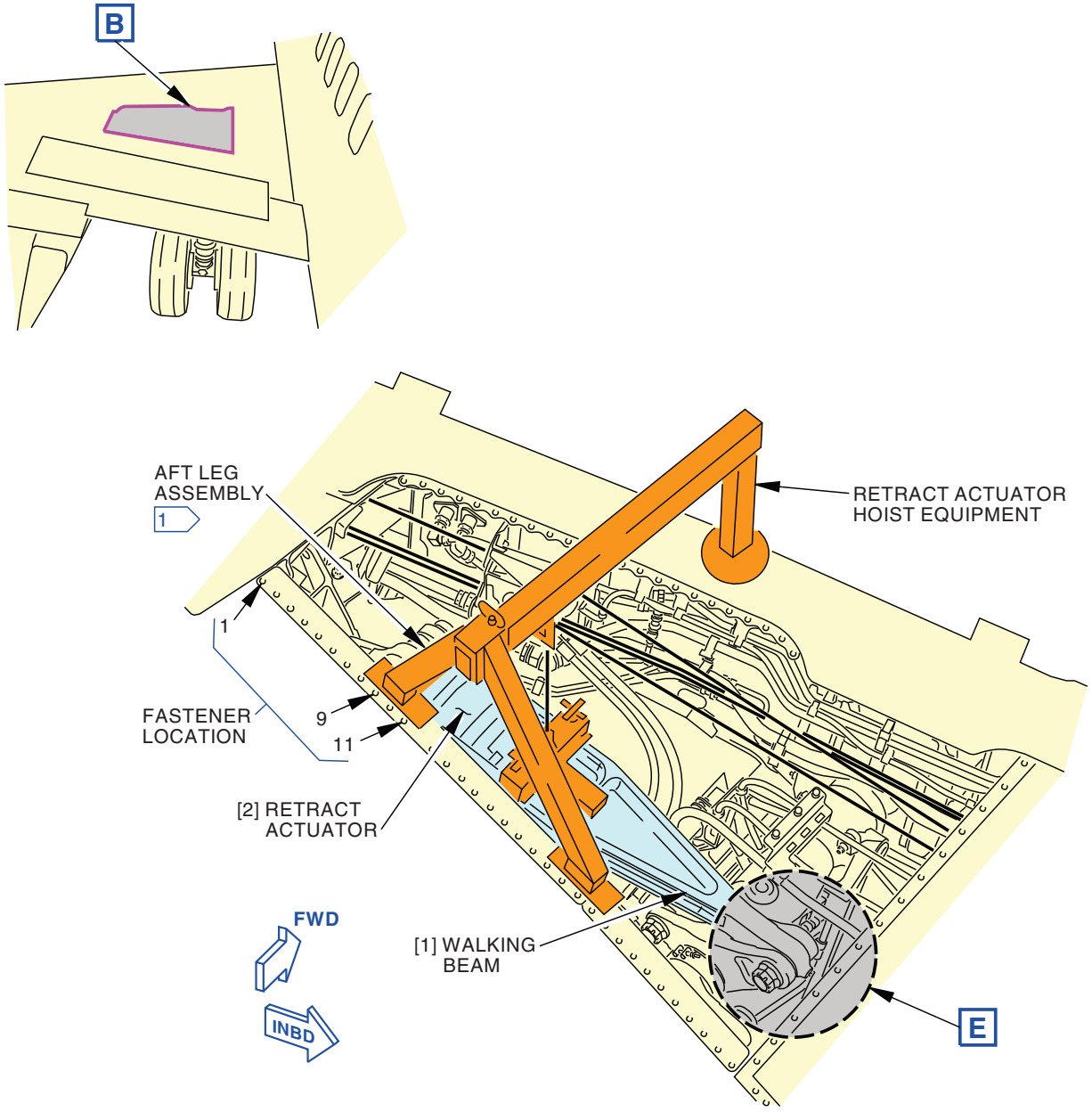
MAIN LANDING GEAR ACTUATOR (ACCESS PANEL, OUTER WING DOOR, AND SHOCK STRUT DOOR REMOVED)

A

2414245 S00061534705_V2

Main Gear Actuator Assembly Installation
Figure 401/32-32-11-990-801 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



MAIN LANDING GEAR ACTUATOR (ACCESS PANEL REMOVED AND RETRACT ACTUATOR HOIST TOOL INSTALLED)

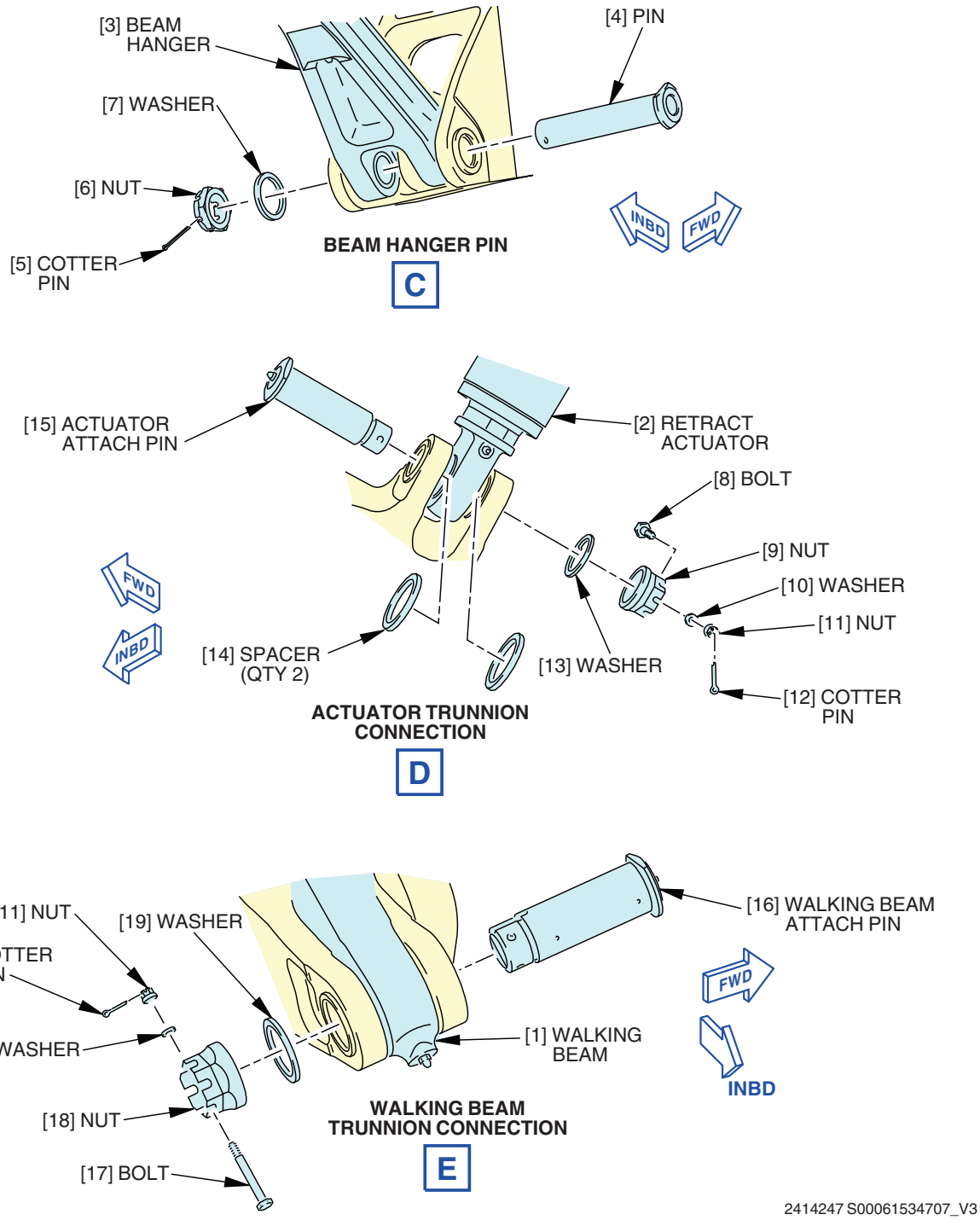
1 USE FASTENER LOCATIONS NUMBERS 9 AND 11 FOR THE AFT OUTBOARD LEG OF THE HOIST EQUIPMENT.

B

2414246 S00061534706_V1

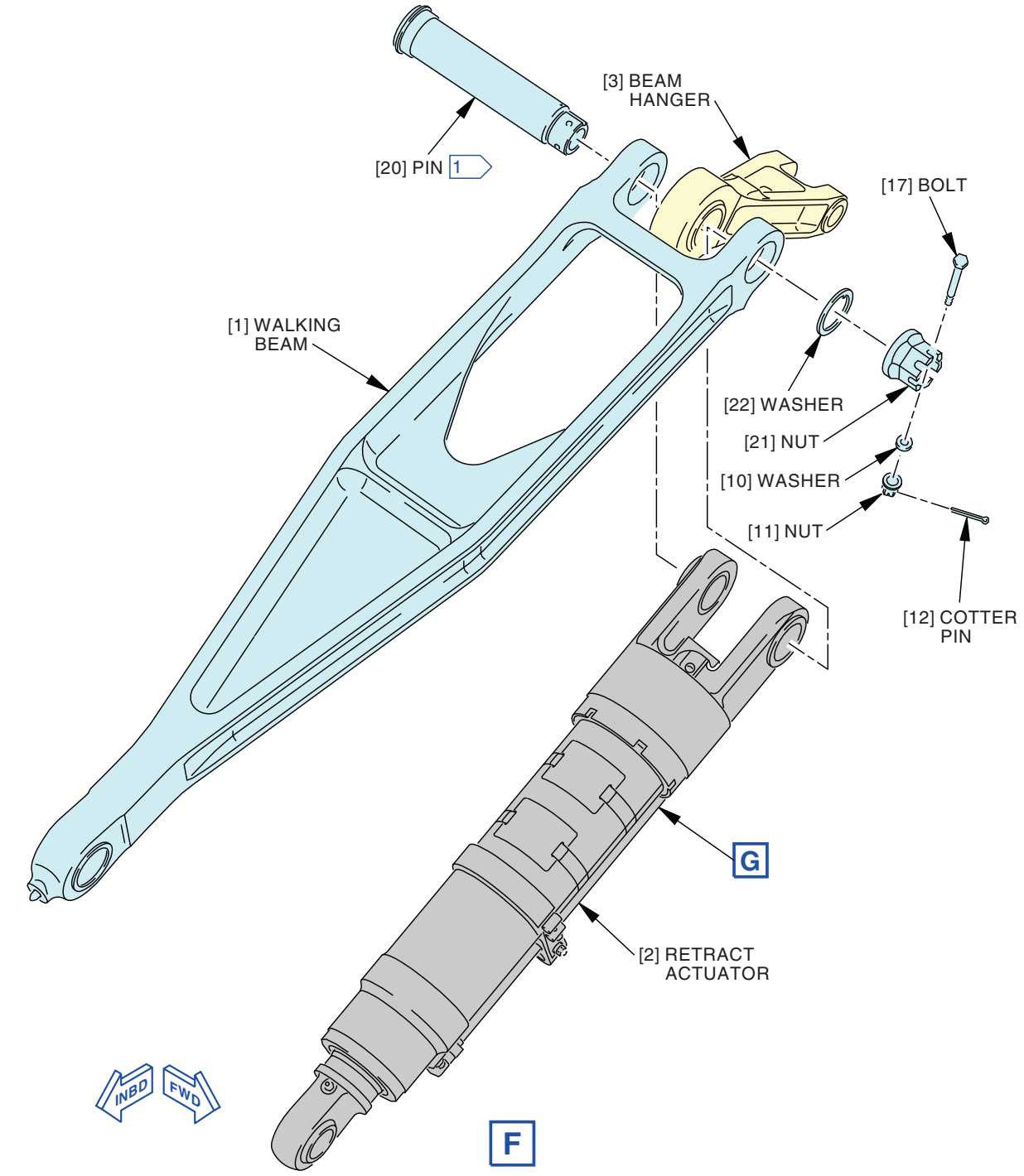
Main Gear Actuator Assembly Installation
Figure 401/32-32-11-990-801 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Gear Actuator Assembly Installation
Figure 401/32-32-11-990-801 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

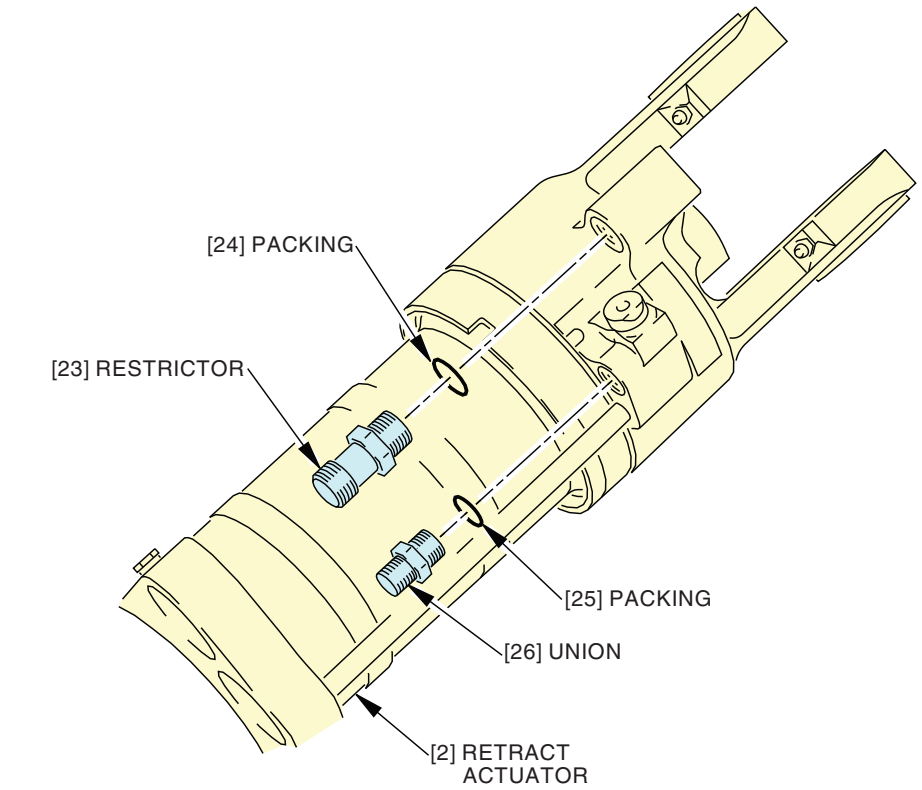


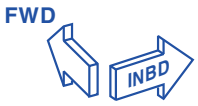
1 MAKE SURE THE HEAD OF THE PIN IS POINTED AFT.

2876729 S0000680290_V2

Main Gear Actuator Assembly Installation
Figure 401/32-32-11-990-801 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





RETRACT ACTUATOR

G

2414249 S00061534709_V1

Main Gear Actuator Assembly Installation
Figure 401/32-32-11-990-801 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-32-11-400-801

3. Main Gear Actuator Assembly Installation

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
12-21-11-620-801	Main Landing Gear Bushings Servicing (P/B 301)
12-21-11-640-801	Main Landing Gear Upper End Components Servicing (P/B 301)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-32-00-710-801	Main Landing Gear Test - Component Replacement (P/B 501)
32-32-00-710-802	Main Landing Gear - Operational Test (P/B 501)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1870	Hoist - Retract Actuator, Main Landing Gear Part #: C32028-109 Supplier: 81205 Part #: C32028-110 Supplier: 81205 Part #: C32028-111 Supplier: 81205 Opt Part #: C32028-1 Supplier: 81205 Opt Part #: C32028-3 Supplier: 81205 Opt Part #: C32028-56 Supplier: 81205 Opt Part #: C32028-58 Supplier: 81205 Opt Part #: C32028-59 Supplier: 81205
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
D00633	Grease - Aircraft General Purpose	BMS3-33
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Retract actuator	32-32-11-01-135	PTF ALL
5	Cotter pin	32-32-93-01-010	PTF ALL
12	Cotter pin	32-32-11-01-010	PTF ALL
		32-32-11-01-095	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
12 (cont.)		32-32-93-01-030	PTF ALL
23	Restrictor	32-32-11-01-195	PTF ALL
24	Packing	32-32-11-01-205	PTF ALL
25	Packing	32-32-11-01-200	PTF ALL
26	Union	32-32-11-01-190	PTF ALL

E. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
551	Left Wing - Rear Spar To Landing Gear Support Beam
651	Right Wing - Rear Spar to Landing Gear Support Beam
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

F. Access Panels

Number	Name/Location
551BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel
651BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

G. Prepare for the Installation

SUBTASK 32-32-11-100-001

(1) Clean the surfaces of these items that will attach the beam hanger [3] to the structure:
   (a) attachment fitting on the beam hanger [3]
   (b) cotter pin [5], pin [4], washer [7], and nut [6].

SUBTASK 32-32-11-100-002

(2) Clean these items that will attach the retract actuator [2] to the main gear trunnion:
   (a) attachment fitting on the main gear trunnion
   (b) actuator attach pin [15]
   (c) spacers [14], washer [13], and nut [9]
   (d) cotter pin [12], nut [11], washer [10], and bolt [8].

SUBTASK 32-32-11-100-003

(3) Clean the surfaces of these items that will attach the walking beam [1] and retract actuator [2] to the main gear trunnion:
   (a) attachment fitting on the main gear trunnion
   (b) walking beam attach pin [16]
   (c) washer [19] and nut [18]
   (d) cotter pin [12], nut [11], washer [10], and bolt [17].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-32-11-600-001

(4) Apply grease, D00633, to the chrome-plated surfaces of the pin [4], pin [20], actuator attach pin [15], and walking beam attach pin [16].

SUBTASK 32-32-11-600-002

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(5) Apply a thin layer of corrosion inhibiting material, G50136, to the thread reliefs and threads of the pin [4] and nut [6], faces of the washer [7].

SUBTASK 32-32-11-600-003

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(6) Apply a thin layer of corrosion inhibiting material, G50136, to these parts:

(a) the faces of the washer [13] and washer [19]

(b) the threads and thread reliefs on the actuator attach pin [15] and walking beam attach pin [16]

(c) the threads and thread reliefs of the bolt [8] and bolt [17], faces of the washers [10], threads and thread reliefs of the nuts [11] and cotter pin [12].

SUBTASK 32-32-11-420-001

(7) Do these steps if the replacement retract actuator [2] does not have the restrictor [23] and union [26] installed:

(a) Lubricate these items with hydraulic fluid, D00153:

1) packing [24]

2) packing [25]

3) restrictor [23]

4) union [26].

(b) Remove the caps/plugs, STD-14686, from the ports on the retract actuator [2].

(c) Install the packing [24], packing [25], restrictor [23], and union [26] in the retract actuator [2].

H. Main Gear Actuator Assembly Installation

SUBTASK 32-32-11-420-002

(1) Do these steps to connect the retract actuator [2] to the walking beam [1]:

(a) Install the retract actuator [2] between the beam hanger [3] and walking beam [1] with the hydraulic ports down.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION

MAKE SURE THAT THE HEAD OF THE PIN IS POINTED AFT. THE INCORRECT ORIENTATION OF THE PIN CAN CAUSE DAMAGE TO THE MAIN-LANDING-GEAR (MLG) SUPPORT BEAM.

(b) Install the pin [20].

(c) Install the washer [22] and nut [21].

(d) Tighten the nut [21] to 42.5 ±7.5 ft-lb (57.6 ±10.2 N·m).

(e) If it is necessary, loosen the nut [21] to align the nearest castellation.

(f) Install the bolt [17], washer [10], and nut [11].

(g) Tighten the nut [11] to 62.5 ±12.5 in-lb (7.1 ±1.5 N·m).

(h) If it is necessary, loosen the nut [11] to align the nearest castellation.

(i) Install the cotter pin [12].

SUBTASK 32-32-11-480-004

(2) Connect the walking beam [1] and retract actuator [2] to the hoist, SPL-1870.

SUBTASK 32-32-11-420-003

(3) Put the beam hanger [3] in its position in the bracket on the structure.

SUBTASK 32-32-11-420-004

(4) Install these parts to connect the beam hanger [3] to the bracket on the wing structure:

(a) pin [4], washer [7], and nut [6]

1) Tighten the nut [6] to 600 ±50 in-lb (68 ±6 N·m).

a) If it is necessary, you may tighten by the pin head if access to the nut [6] is restricted.

2) If it is necessary, loosen the nut [6] to align the nearest castellation.

(b) cotter pin [5].

SUBTASK 32-32-11-420-005

(5) Install these parts to connect the retract actuator [2] to the main gear trunnion:

(a) actuator attach pin [15], spacers [14], washer [13], and nut [9]

1) Remove the clamps for the electrical conduit on the inboard side of the nut [9] and move the conduit to get clearance to tighten the nut [9].

2) Tighten the nut [9] to 62.5 ±12.5 ft-lb (85 ±17 N·m).

3) If it is necessary, loosen the nut [9] to align the nearest castellation.

(b) bolt [8], washer [10], and nut [11]

1) Tighten the nut [11] to 50 ±5 in-lb (5.6 ±0.6 N·m).

2) If it is necessary, loosen the nut [11] to align the nearest castellation.

(c) cotter pin [12].

SUBTASK 32-32-11-410-001

(6) Put the electrical conduit back in its position and install the clamps.

SUBTASK 32-32-11-420-006

(7) Hold the walking beam [1], retract actuator [2], and install these parts to connect the walking beam [1] to the main gear trunnion:

(a) walking beam attach pin [16] with the pin head forward

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) washer [19], nut [18]

1) Tighten the nut [18] to 62.5 ±12.5 ft-lb (84.7 ±16.9 N·m).

2) If it is necessary, loosen the nut [18] to align the nearest castellation.

(c) bolt [17], washer [10], and nut [11]

1) Tighten the nut [11] to 50 ±5 in-lb (5.6 ±0.6 N·m).

2) If it is necessary, loosen the nut [11] to align the nearest castellation.

(d) cotter pin [12].

SUBTASK 32-32-11-080-002

(8) Remove the hoist, SPL-1870.

SUBTASK 32-32-11-080-003

(9) Remove the caps/plugs, STD-14686, from the hydraulic hoses.

SUBTASK 32-32-11-420-007

(10) Connect the hydraulic UP and DN hoses to the retract actuator [2].

(a) Tighten the B-nut on hydraulic line [27] to 500 ±25 in-lb (56 ±3 N·m).

(b) Tighten the B-nut on hydraulic line [28] to 270 ±14 in-lb (31 ±2 N·m).

I. Main Gear Actuator Assembly Bleeding (Airplane on Jacks)

SUBTASK 32-32-11-580-002

CAUTION

MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION

MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

WARNING

MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Lift the airplane for landing gear retraction (TASK 07-10-01-580-801).

SUBTASK 32-32-11-640-005

(2) Lubricate the attach points (TASK 12-21-11-620-801 or TASK 12-21-11-640-801).

SUBTASK 32-32-11-080-004

(3) Remove the downlock pins from the main landing gear (TASK 32-00-01-080-801).

SUBTASK 32-32-11-860-002

(4) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-32-11-700-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED IN THE NOSE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE NOSE LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(5) Do a retraction test of the main landing gear, do this task: Main Landing Gear - Operational Test, TASK 32-32-00-710-802 or Main Landing Gear Test - Component Replacement, TASK 32-32-00-710-801.

SUBTASK 32-32-11-480-005

(6) Extend the main landing gear and install the downlock pins (TASK 32-00-01-480-801).

SUBTASK 32-32-11-210-001

(7) Examine the hydraulic connections for leaks.

SUBTASK 32-32-11-210-002

(8) Make sure that there is no interference of the hoses and retract actuator [2] during gear operation.

SUBTASK 32-32-11-860-003

(9) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-32-11-580-003

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(10) Do this task: Lower the Airplane off Jacks, TASK 07-10-01-580-802.

J. Main Gear Actuator Assembly Bleeding (Airplane on Ground Without Jacks) Alternate Method

SUBTASK 32-32-11-210-003

(1) Make sure that the hydraulic lines are installed correctly and they are not twisted.

SUBTASK 32-32-11-480-007

WARNING OBEY THE PROCEDURE FOR THE INSTALLATION OF THE DOWNLOCK PINS. IF YOU MOVE THE CONTROL LEVER FOR THE LANDING GEAR TO THE UP POSITION, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(2) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801

SUBTASK 32-32-11-600-004

(3) Lubricate the attach points (TASK 12-21-11-620-801 or TASK 12-21-11-640-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-32-11-860-004

(4) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-32-11-860-005

(5) Push the override button, and move the control lever for the landing gear from the DN position to the UP position and to the DN position three times.

(a) Hold the control lever in each position for three seconds before you move the control lever to the subsequent position.

SUBTASK 32-32-11-860-006

(6) Move the control lever to the DN position.

SUBTASK 32-32-11-200-001

(7) Make sure that there are no leaks at the hydraulic connections.

SUBTASK 32-32-11-860-009

(8) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-32-11-410-002

(9) Install the hydraulic access panel.

K. Put the Airplane Back to Its Usual Condition

SUBTASK 32-32-11-410-003

(1) Close the applicable access panel:

Number	Name/Location
551BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel
651BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN GEAR RETRACT PRESSURE FUSES - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the retract pressure fuses for the main landing gear

(2) An installation of the retract pressure fuses for the main landing gear

TASK 32-32-21-000-801

2. Main Gear Retract Pressure Fuses Removal

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

D. Prepare for the Removal

SUBTASK 32-32-21-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-32-21-860-001

(2) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-32-21-580-001

(3) Install chocks around the tires of the main landing gear (PAGEBLOCK 10-11-05/201).

E. Main Gear Retract Pressure Fuse Removal

SUBTASK 32-32-21-020-001

(1) Remove the screws [1], washers [2], spacers [5], and the clipnuts [3] that hold the hydraulic lines to the structure on the two ends of the retract pressure fuse [4].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-32-21-020-002

(2) Disconnect the hydraulic lines from the retract pressure fuse [4].

SUBTASK 32-32-21-480-002

(3) Install caps/plugs, STD-14686, on the hydraulic lines.

SUBTASK 32-32-21-020-003

(4) Remove the retract pressure fuse [4] from the airplane:

(a) If necessary, remove the reducer fitting [7] from one end of the retract pressure fuse [4].

F. Main Gear Retract Pressure Hydraulic Fuse Functional Test

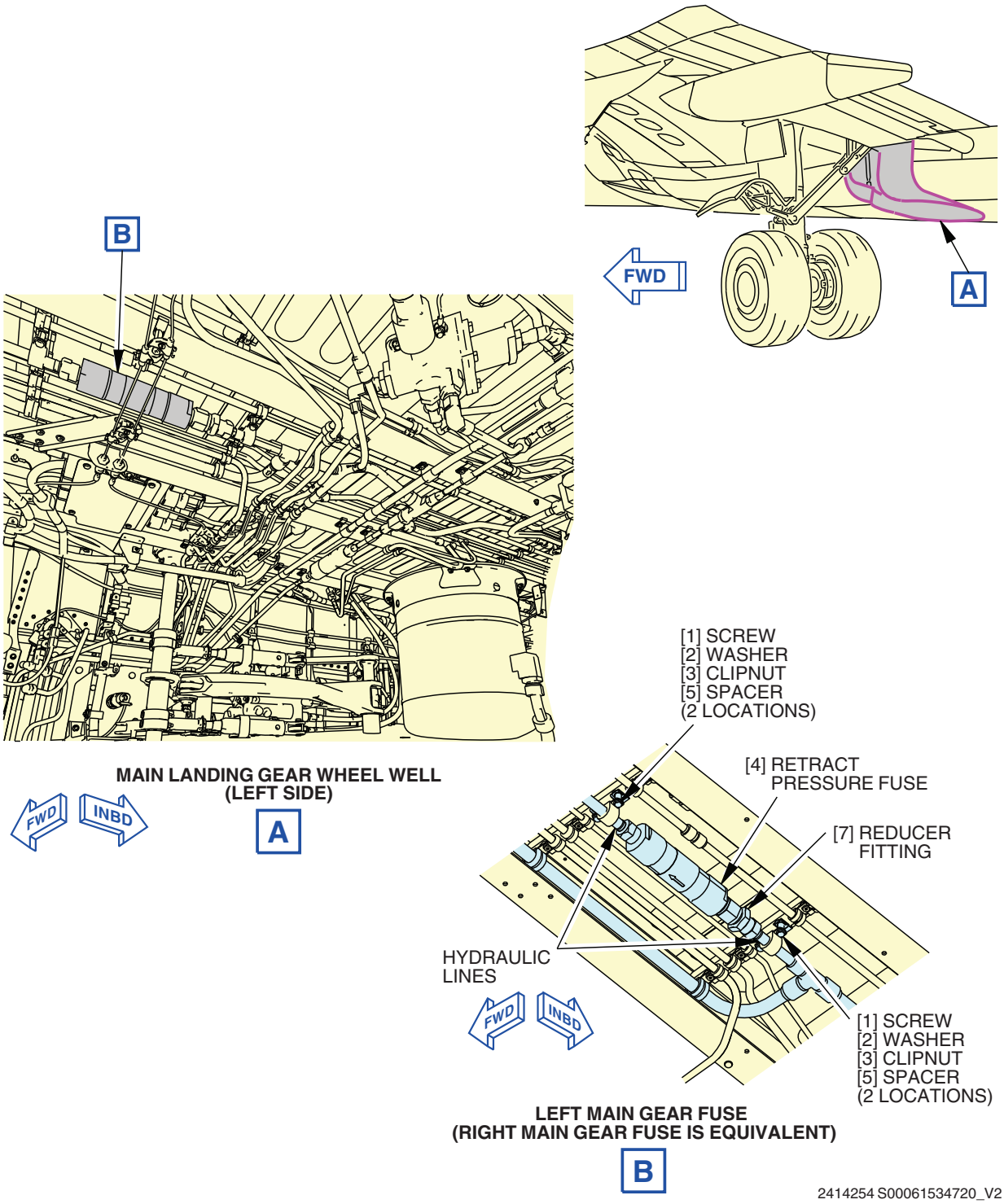
SUBTASK 32-32-21-720-001

(1) Do the functional test of the retract pressure fuse [4] with the supplier's recommended component maintenance test instructions and test equipment.

NOTE: The supplier's instructions give a reverse flow test, internal leakage test, pressure drop test, volumetric capacity test and reset test for the fuse. This is an off-airplane bench test.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Gear Retract Pressure Fuse Installation
Figure 401/32-32-21-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-32-21-400-801

3. Main Gear Retract Pressure Fuses - Installation

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Consumable Materials

Reference	Description	Specification
D00054	Fluid - Hydraulic, Assembly Lubricant, MCS 352B	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Retract pressure fuse	32-32-21-01-015	PTF ALL

E. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

F. Main Gear Retract Pressure Fuse Installation

SUBTASK 32-32-21-080-001

(1) Remove the caps/plugs, STD-14686, from the unions and the hydraulic lines that will connect to the retract pressure fuse [4].

SUBTASK 32-32-21-420-001

(2) Put the retract pressure fuse [4] in position to align it with the hydraulic lines:

(a) If necessary, install the reducer fitting [7] to the retract pressure fuse [4].

1) Tighten the reducer fitting [7] to 700 ±35 in-lb (79 ±4 N·m)

SUBTASK 32-32-21-420-002

(3) Connect the hydraulic lines to the retract pressure fuse [4]:

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(a) For the hydraulic line connecting to the outlet fitting end of the retract pressure fuse [4], apply MCS 352B fluid, D00054, to the entire circumference of the external fitting threads, the shoulder of the sleeve in contact with the coupling nut, and to the sealing surface of the sleeve.

(b) Tighten the B-nuts on the hydraulic lines to 500 ±25 in-lb (56 ±3 N·m).

SUBTASK 32-32-21-420-004

(4) Install the screws [1], washers [2], spacers [5], and the clipnuts [3] for the clamps that hold the hydraulic lines to the structure on the two ends of the retract pressure fuse [4].

G. Main Gear Retract Pressure Fuse Installation Test

SUBTASK 32-32-21-480-003

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-32-21-860-002

(2) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-32-21-700-001

(3) Use the override trigger to move the control lever for the landing gear from the DN position to the UP position.

SUBTASK 32-32-21-710-001

(4) Make sure the UP hose of the main gear retract actuator moves to show that hydraulic pressure is applied when the control lever is moved to the UP position.

SUBTASK 32-32-21-700-002

(5) Move the control lever to the DN position.

SUBTASK 32-32-21-700-003

(6) Make sure the DOWN hose on the main gear retract actuator moves when you move the control lever from the UP position to the DN position.

SUBTASK 32-32-21-210-001

(7) Do a check for leakage with the hydraulic pressure applied.

SUBTASK 32-32-21-210-002

(8) Make sure there are no leaks in the hydraulic connections.

SUBTASK 32-32-21-860-003

(9) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-32-21-840-001

(10) Examine the hydraulic reservoirs for the correct servicing (TASK 12-12-00-610-801).

(a) Do the servicing if it is necessary.

END OF TASK

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MAIN GEAR UPLOCK ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure has these two tasks:

(1) A removal of the uplock assembly for the main landing gear

(2) An installation of the uplock assembly for the main landing gear.

TASK 32-32-31-000-801

2. Main Gear Uplock Assembly Removal

(Figure 401, Figure 402)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
10-11-05-500-801	Chock Installation Winds/Gusts Maximum 40 mph (35 Knots) (P/B 201)
10-11-05-500-802	Chock Installation in Winds More Than 35 Knots - Handling (P/B 201)
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-34-21-000-801	Main Gear Manual Extension Mechanism Removal (P/B 401)
32-61-21-020-801	Main Landing Gear Uplock Sensor Removal (P/B 401)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

D. Prepare for the Removal

SUBTASK 32-32-31-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed in the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-32-31-860-001

(2) Release the pressure from the hydraulic reservoirs, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

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SUBTASK 32-32-31-580-001

(3) Install chocks around the tires of the main landing gear (TASK 10-11-05-500-801 or TASK 10-11-05-500-802).

SUBTASK 32-32-31-020-001

(4) Remove the uplock sensors from the uplock assembly [9] (TASK 32-61-21-020-801).

E. Main Gear Uplock Assembly Removal

SUBTASK 32-32-31-020-002

(1) Put tags on the two hydraulic lines to show their connection locations.

SUBTASK 32-32-31-020-003

(2) Disconnect the two hydraulic lines from the uplock actuator.

SUBTASK 32-32-31-480-002

(3) Install caps/plugs, STD-14686, in the two hydraulic lines.

SUBTASK 32-32-31-020-021

WARNING MAKE SURE THAT YOU INSTALL THE RIG PIN IN THE MANUAL EXTENSION MECHANISM. IF YOU DO NOT OBEY, THE SPRING IN THE MECHANISM COULD UNWIND AND CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(4) Disconnect the control rod of the manual extension for the uplock assembly [9] (TASK 32-34-21-000-801).

SUBTASK 32-32-31-020-004

(5) If it is necessary to remove the uplock hanger assembly [19], remove the cotter pins [13], nuts [12], uplock washers [11], and uplock hanger pins [10] to disconnect the uplock hanger assembly [19].

SUBTASK 32-32-31-020-007

(6) Remove the cotter pin [5], nut [4], washer [3], bushing [2], and bolt [1] to disconnect the uplock hanger assembly [19] from the structure.

SUBTASK 32-32-31-020-008

(7) Remove the uplock hanger assembly [19].

SUBTASK 32-32-31-020-005

(8) Remove the cotter pin [8], nut [7], uplock washer [6], and uplock attach pin [14] to disconnect the uplock assembly [9] from the structure.

SUBTASK 32-32-31-020-006

(9) Remove the uplock assembly [9].

SUBTASK 32-32-31-020-009

(10) If the uplock actuator on the replacement uplock assembly does not have hydraulic fittings installed, do these steps:

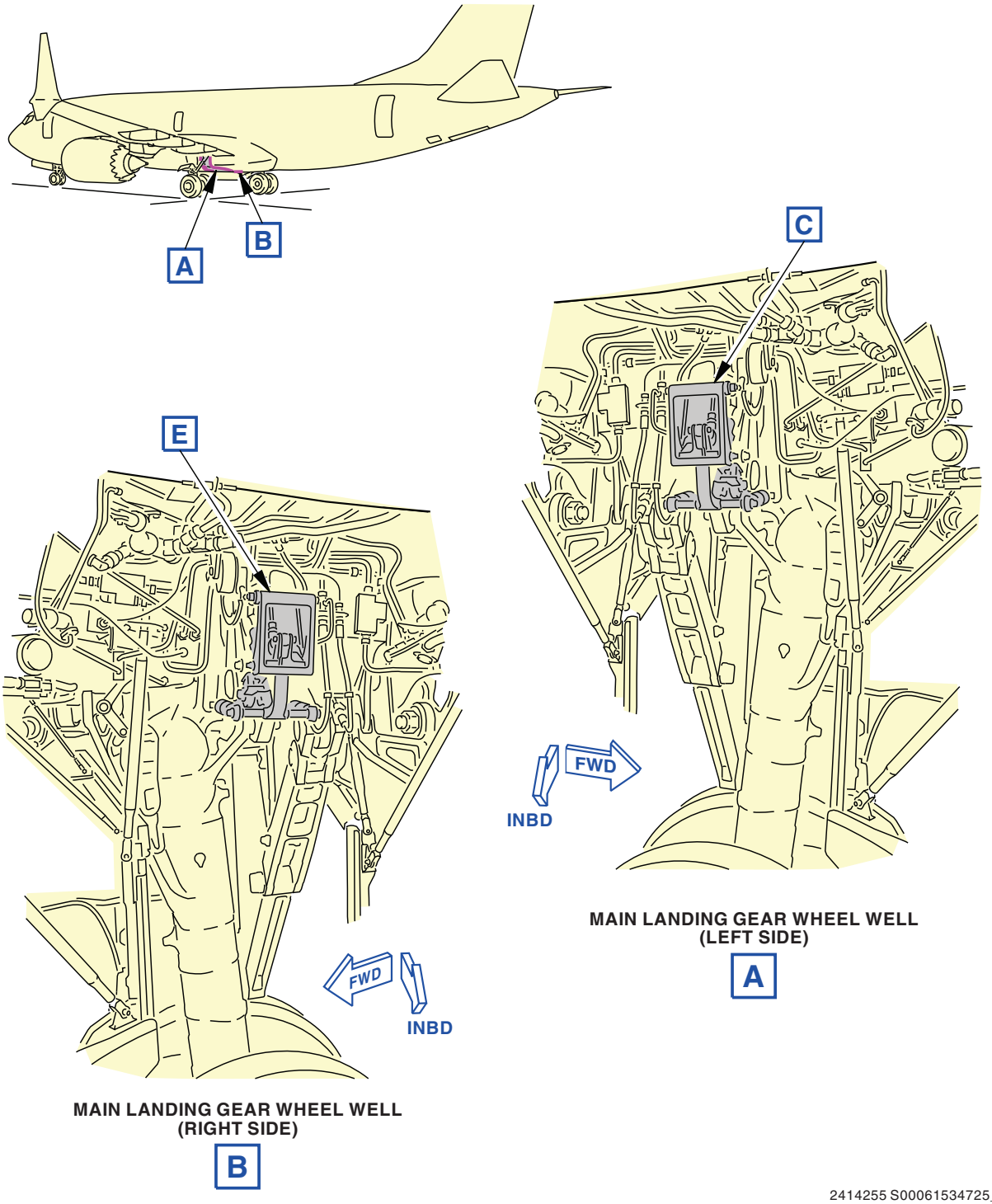
(a) Remove the union [18] and the restrictor [16] from the uplock actuator.

(b) Remove and discard the packing [15], and packing [17].

(c) Install caps/plugs, STD-14686, in the ports on the uplock actuator.

END OF TASK

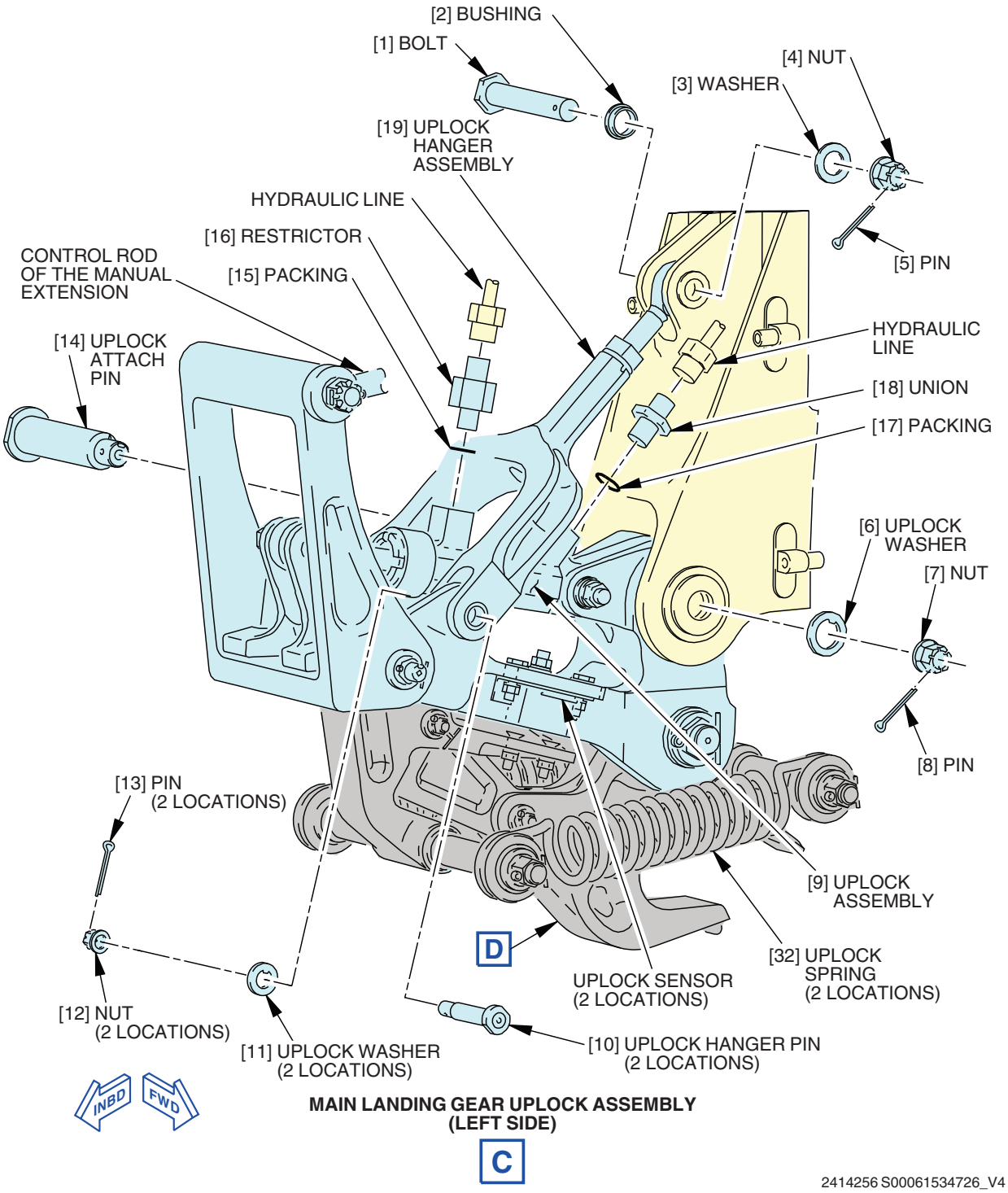
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414255 S00061534725_V2

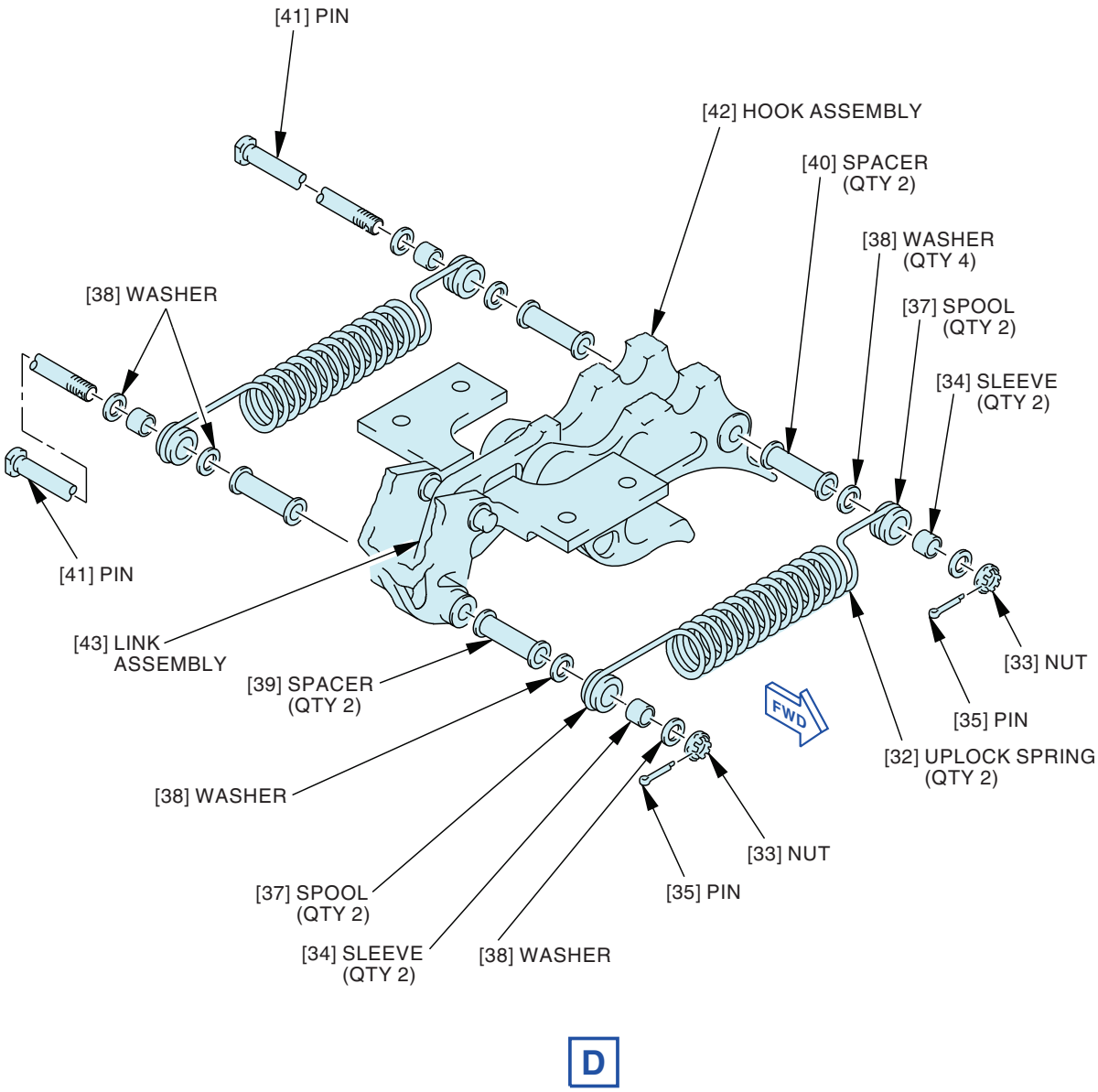
Main Landing Gear Uplock Assembly Installation
Figure 401/32-32-31-990-801 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Uplock Assembly Installation
Figure 401/32-32-31-990-801 (Sheet 2 of 5)

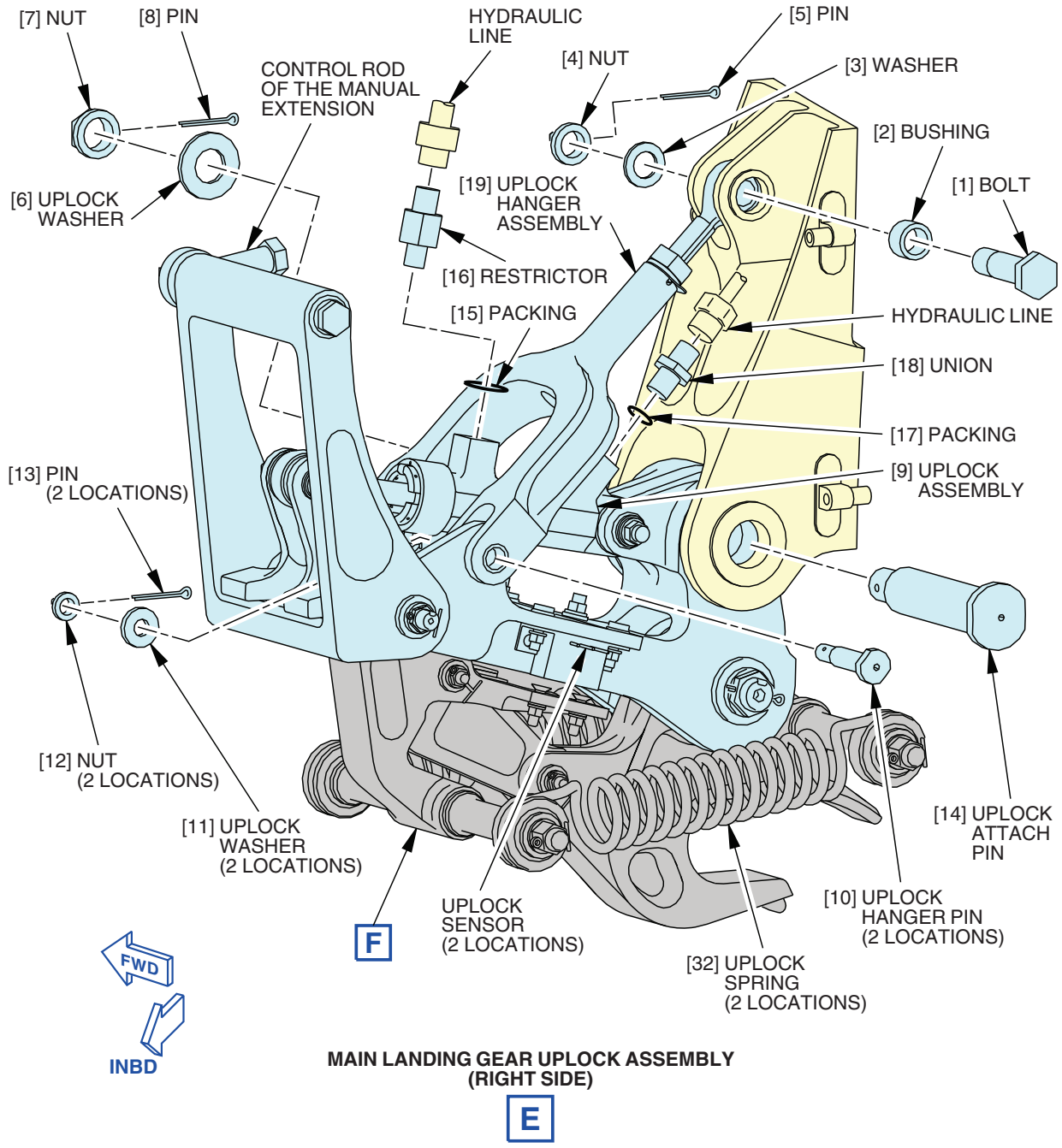
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414257 S00061534727_V3

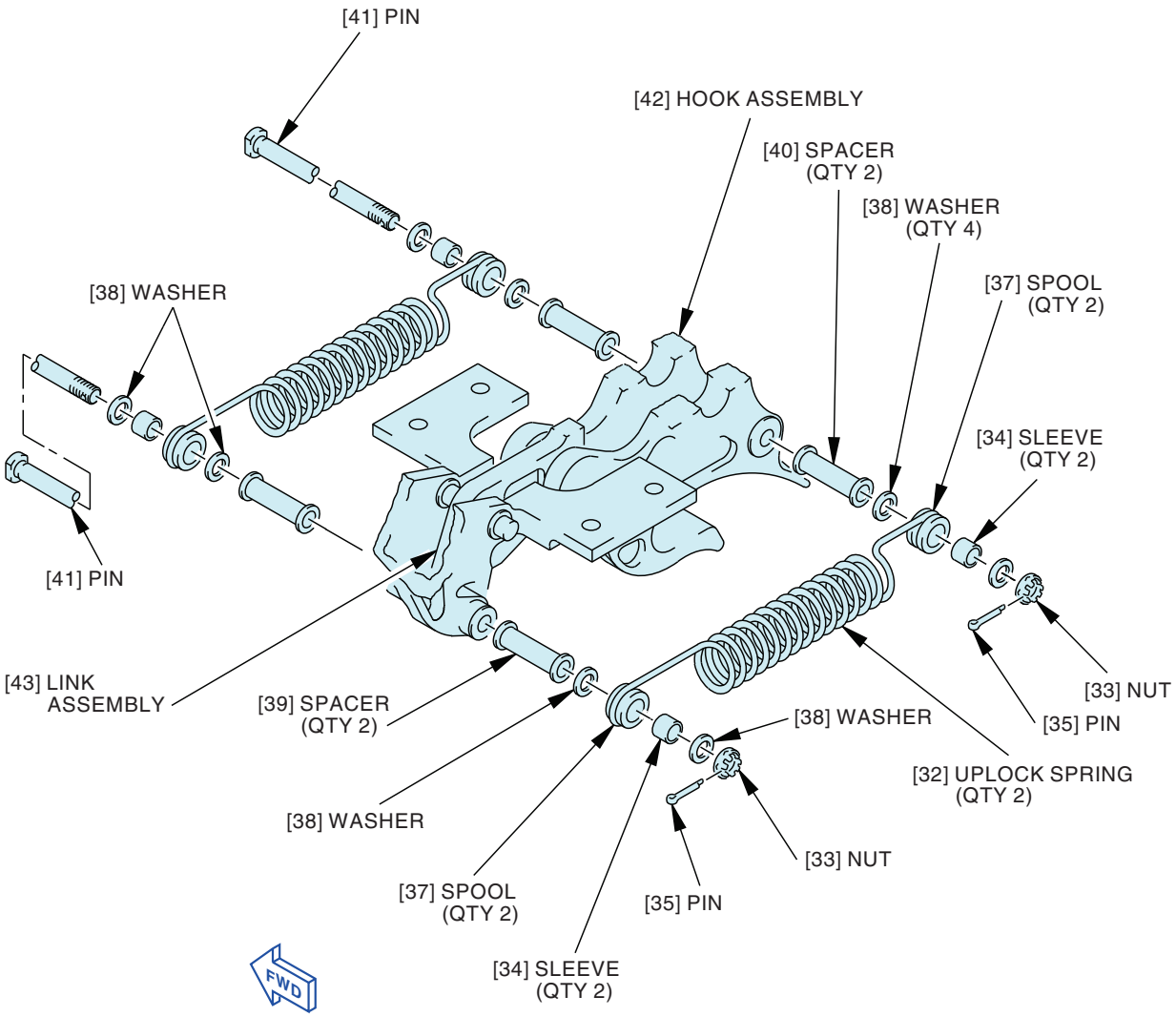
Main Landing Gear Uplock Assembly Installation
Figure 401/32-32-31-990-801 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Uplock Assembly Installation
Figure 401/32-32-31-990-801 (Sheet 4 of 5)

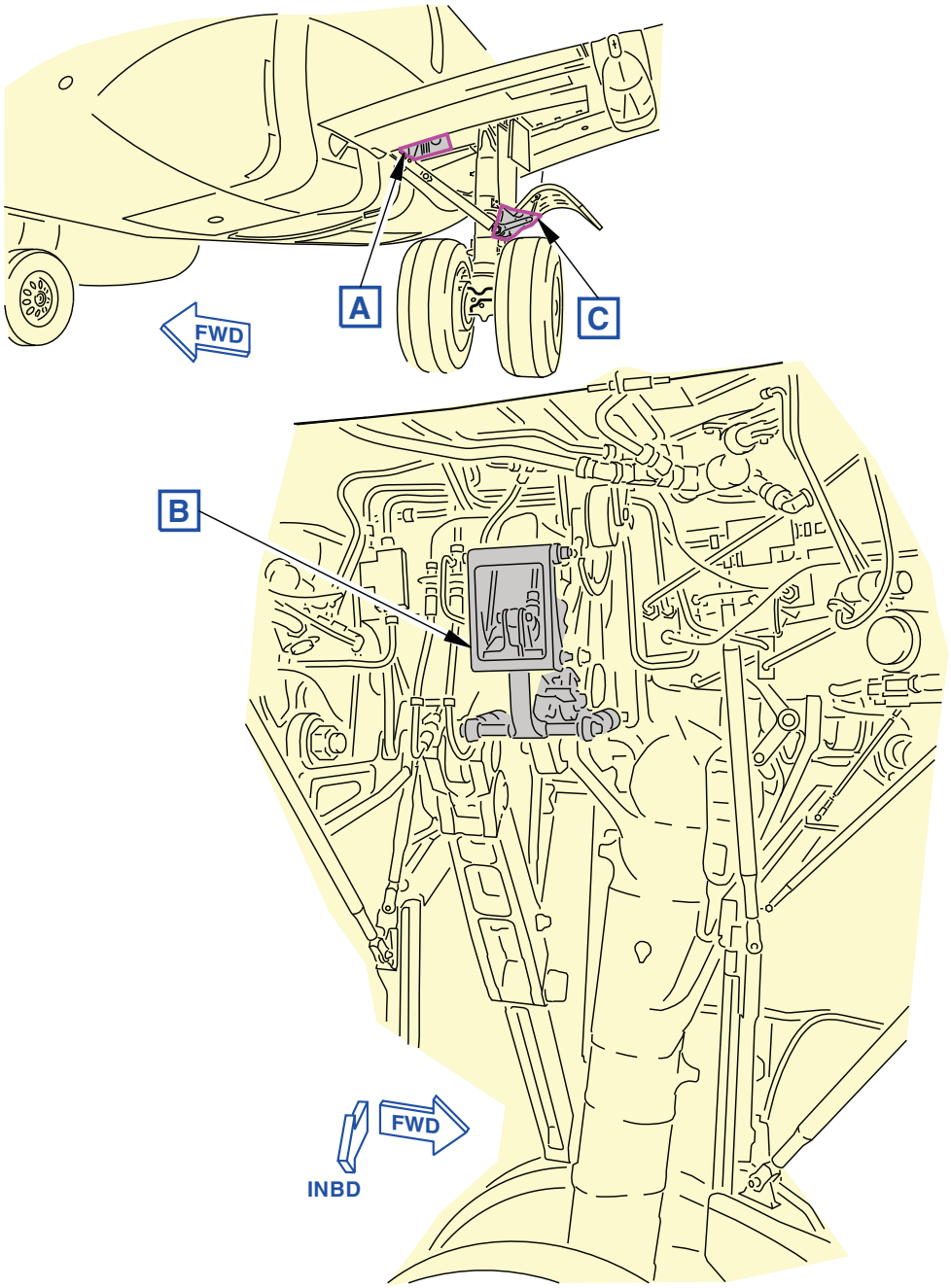
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2973714 S0000752856_V2

Main Landing Gear Uplock Assembly Installation
Figure 401/32-32-31-990-801 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

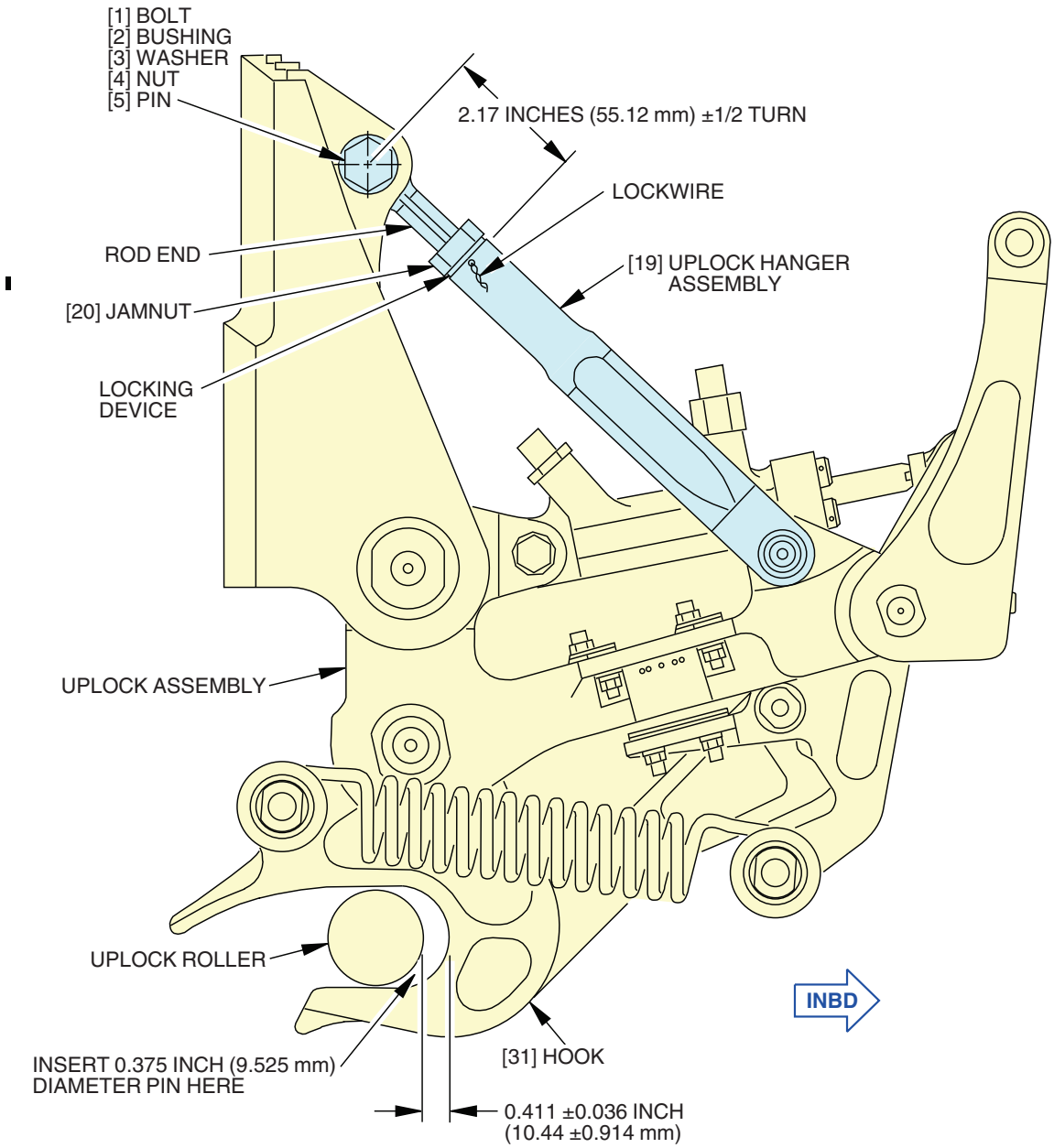


MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE IS SHOWN, RIGHT SIDE IS EQUIVALENT)

2414258 S00061534728_V2

Main Landing Gear Uplock Assembly Adjustment
Figure 402/32-32-31-990-802 (Sheet 1 of 3)

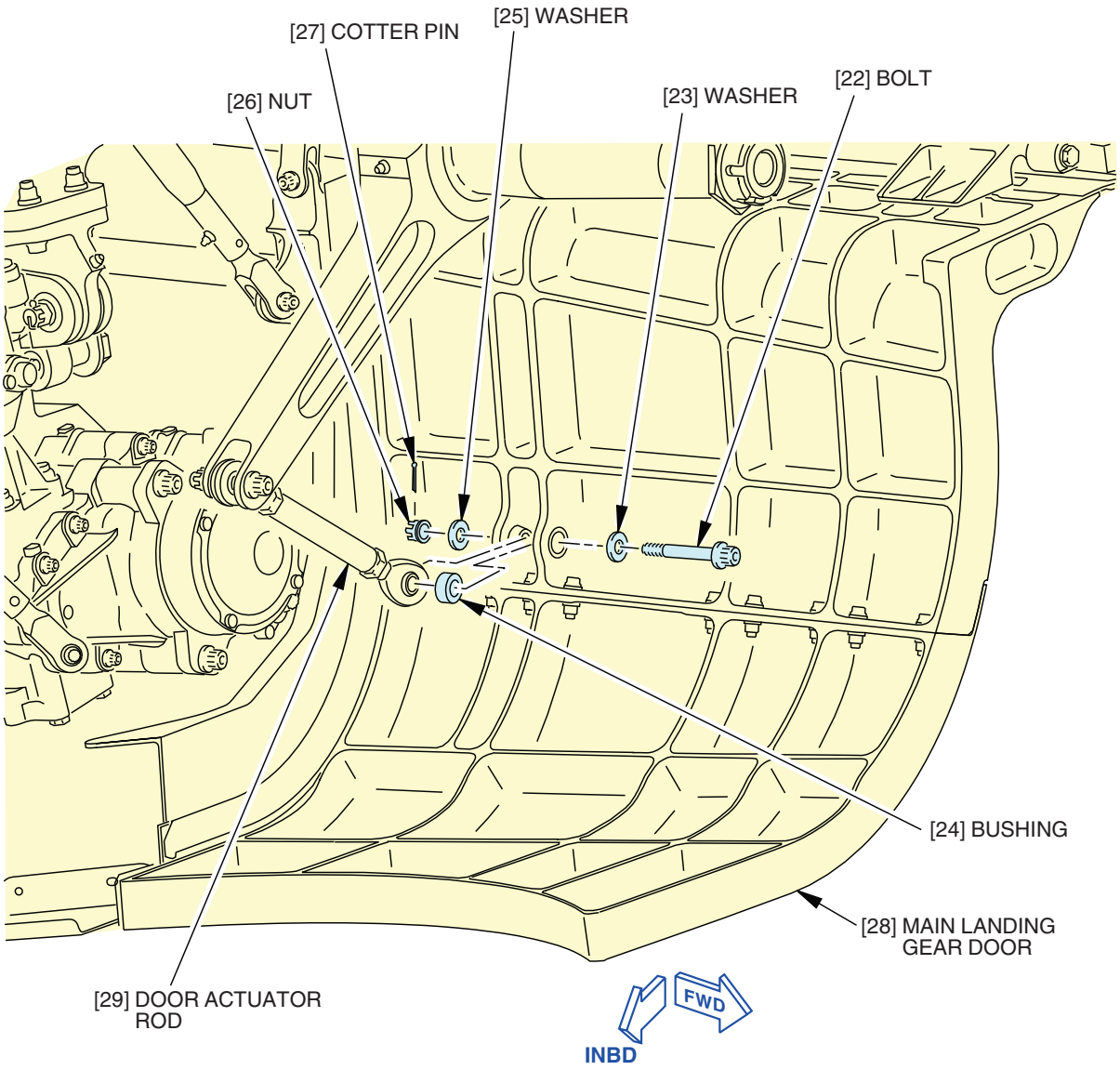
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414259 S00061534729_V3

Main Landing Gear Uplock Assembly Adjustment
Figure 402/32-32-31-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414260 S00061534730_V2

Main Landing Gear Uplock Assembly Adjustment
Figure 402/32-32-31-990-802 (Sheet 3 of 3)

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TASK 32-32-31-400-801

3. Main Gear Uplock Assembly Installation

(Figure 401, Figure 402)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-13-21-000-802	Main Landing Gear Wing Door Adjustment (P/B 501)
32-34-21-400-801	Main Gear Manual Extension Mechanism Installation (P/B 401)
32-61-21-400-801	Main Landing Gear Uplock Sensor Installation (P/B 401)
32-61-21-820-801	Main Landing Gear Uplock Sensor Clearance Adjustment (P/B 501)

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Consumable Materials

Reference	Description	Specification
D00054	Fluid - Hydraulic, Assembly Lubricant, MCS 352B	
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
D00633	Grease - Aircraft General Purpose	BMS3-33
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G51180	3M EC1252 or Equivalent, Torque Seal, Pink or Yellow.	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
5	Cotter pin	32-32-31-01-030	PTF ALL
8	Cotter pin	32-32-31-01-010	PTF ALL
9	Uplock assembly	32-32-31-01-060	PTF ALL
13	Cotter pin	32-32-31-01-125	PTF ALL
15	Packing	32-32-41-01-035	PTF ALL
17	Packing	32-32-41-01-030	PTF ALL
19	Uplock hanger assembly	32-32-31-01-130	PTF ALL
		32-32-31-01-155	PTF ALL

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E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

F. Main Gear Uplock Assembly Installation

SUBTASK 32-32-31-420-001

(1) Do these steps if the replacement uplock actuator on the uplock assembly [9] does not have the union [18] or restrictor [16] installed:

(a) Lubricate the new packing [15], new packing [17], union [18], and restrictor [16] with hydraulic fluid, D00153 or MCS 352B fluid, D00054.

(b) Remove the caps/plugs, STD-14686, from the ports on the uplock actuator.

(c) Install the packing [15], packing [17], union [18], and restrictor [16] in the uplock actuator.

SUBTASK 32-32-31-600-001

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(2) Apply a thin layer of corrosion inhibiting material, G50136, on the parts that follow:

(a) The threads and thread reliefs of the uplock attach pin [14] and nut [7].

(b) The faces of the uplock washer [6].

(c) The new cotter pin [8].

SUBTASK 32-32-31-640-001

(3) Lubricate the shank of the uplock attach pin [14] with grease, D00633.

SUBTASK 32-32-31-420-002

(4) Put the uplock assembly [9] in position on the structure.

SUBTASK 32-32-31-420-003

(5) Install the uplock attach pin [14], uplock washer [6], and nut [7] to attach the uplock assembly [9] to the structure.

SUBTASK 32-32-31-420-004

(6) Tighten the nut [7] within the torque range 350 in-lb (40 N·m) to 400 in-lb (45 N·m).

(a) Loosen to the nearest castellation, if it is necessary, and install the new cotter pin [8].

SUBTASK 32-32-31-420-022

(7) If the uplock hanger assembly [19] was removed, do these steps:

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WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin layer of corrosion inhibiting material, G50136, to the parts that follow:

1) The threads and thread reliefs of the uplock hanger pins [10] and nuts [12].
2) The faces of the uplock washers [11].
3) The new cotter pins [13].

(b) Lubricate the shank of the uplock hanger pin [10] with grease, D00633.

(c) Install the uplock hanger pin [10], uplock washers [11], and nuts [12] to connect the uplock hanger assembly [19] to the uplock assembly [9].

(d) Tighten the nuts [12] to 22.0 ±2.0 in-lb (2.5 ±0.2 N·m) above the run-on torque.

1) Loosen to the nearest castellation, if it is necessary, and install the new cotter pins [13].

SUBTASK 32-32-31-820-001

(8) If necessary, loosen the jamnut [20] on the rod end of the uplock hanger assembly [19].

(a) Adjust the rod end until the distance from the center of the bolt [1] to the top of the threaded end of the uplock hanger assembly [19] is 2.17 in. (55.1 mm).

SUBTASK 32-32-31-600-003

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(9) Apply a thin layer of or corrosion inhibiting material, G50136, on the parts that follow:

(a) The threads and thread reliefs of the bolt [1] and nut [4].

(b) The faces of the washer [3].

(c) The new cotter pin [5].

SUBTASK 32-32-31-640-003

(10) Lubricate the shanks of the bolt [1] with a light film of grease, D00633.

(a) Wipe off excess grease, D00633.

SUBTASK 32-32-31-420-007

(11) Install the bolt [1], bushing [2], washer [3], nut [4], and new cotter pin [5] to attach the uplock hanger assembly [19] to the structure.

NOTE: The installation of the hanger assembly rod end does not require the pin to engage to the self locking nut castellations.

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SUBTASK 32-32-31-420-008

(12) Make sure that the 2.17 in. (55.12 mm) dimension from the center of bolt [1], to the top of the threaded end of the hanger under the jamnut [20] is 2.17 in. (55.12 mm).

SUBTASK 32-32-31-420-009

(13) Tighten the jamnut [20] on the rod end of the uplock hanger assembly [19] to 225 ±25 in-lb (25 ±3 N·m).

SUBTASK 32-32-31-420-010

(14) Install lockwire on the jamnut [20].

SUBTASK 32-32-31-420-021

(15) Connect the control rod of the manual extension mechanism to the uplock assembly [9] (TASK 32-34-21-400-801).

SUBTASK 32-32-31-420-011

(16) Do these steps to connect the two hydraulic lines to the uplock actuator:

(a) Remove the caps/plugs, STD-14686, from the two hydraulic lines.

(b) Connect the two hydraulic lines to the uplock actuator.

(c) Tighten the B-nuts on the hydraulic lines to 140 in-lb (15.8 N·m), back off fitting and then tighten the fitting again to 140 ±7 in-lb (15.8 ±0.8 N·m).

(d) Apply a strip of torque seal, G51180, to the B-nuts on the hydraulic lines.

(e) Remove the tags from the two hydraulic lines.

SUBTASK 32-32-31-420-012

(17) Install uplock sensors on the uplock assembly [9], do this task: Main Landing Gear Uplock Sensor Installation, TASK 32-61-21-400-801.

SUBTASK 32-32-31-420-013

(18) Do the adjustment for the uplock sensors on the uplock assembly [9], do this task: Main Landing Gear Uplock Sensor Clearance Adjustment, TASK 32-61-21-820-801.

G. Main Gear Uplock Assembly Installation Test

SUBTASK 32-32-31-580-002

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE APPLICABLE LANDING GEAR. THE LANDING GEAR EXTENDS AND RETRACTS QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(1) Do this task: Lift the Airplane on Jacks, TASK 07-10-01-580-801.

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SUBTASK 32-32-31-020-015

(2) Remove the pin [27], nut [26], washer [25], bushing [24], washer [23], and bolt [22] to disconnect the main gear door rod actuator [29] from the main gear door [28].

NOTE: The main gear doors will remain hanging during the installation test.

SUBTASK 32-32-31-080-001

(3) Remove the downlock pin only from the main gear you are working on (TASK 32-00-01-080-801) and leave the pin in the other main gear and the nose gear.

SUBTASK 32-32-31-860-002

(4) Pressurize the hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-32-31-870-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED IN ALL OF THE LANDING GEAR BEFORE YOU MOVE THE CONTROL LEVER. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(5) Extend and retract the main landing gear you are working on several times.

NOTE: It will bleed the air from the uplock actuator and the lines.

(a) Examine for the correct gear operation.

SUBTASK 32-32-31-040-001

(6) Retract the main landing gear you are working on:

NOTE: The main gear doors will remain hanging during the installation test.

(a) Install a DO NOT OPERATE tag, STD-858, on the landing gear handle.

SUBTASK 32-32-31-200-001

WARNING MAKE SURE THAT THE MAIN LANDING GEAR IS RETRACTED AND THAT YOU INSTALL A “DO NOT OPERATE” TAG ON THE LANDING GEAR LEVER. ACCIDENTAL MOVEMENT OF THE LANDING GEAR CAN CAUSE INJURIES TO PERSONNEL OR DAMAGE TO EQUIPMENT.

(7) Make sure that the gap between the uplock roller and the uplock assembly hook is 0.411 ±0.036 in. (10.44 ±0.91 mm) (Figure 402).

(a) Use a 0.375 in. (9.525 mm) diameter pin, or equivalent, inserted between the uplock roller and the back of the uplock assembly hook.

(b) If an adjustment is necessary, extend the landing gear.

WARNING MAKE SURE THAT YOU INSTALL THE DOWNLOCK PINS ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

1) Make sure that the downlock pins are installed in the main landing gear (TASK 32-00-01-480-801).

2) Adjust the length of the uplock hanger assembly [19], plus or minus one half turn, until the gap requirement is met.

a) Loosen the jamnut [20].

b) Tighten the jamnut [20] to 200 in-lb (23 N·m) - 250 in-lb (28 N·m).

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c) Remove the downlock pin from the gear being worked on only and repeat this step until the gap requirement is met.

d) Install lockwire between the jamnut [20] and uplock hanger assembly [19].

SUBTASK 32-32-31-800-001

(8) Extend the landing gear:

(a) Remove the DO NOT OPERATE tag, STD-858, from the landing gear handle.

(b) Move the landing gear lever to the DN position.

(c) Install the downlock pins in the main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-32-31-420-018

(9) Install the bolt [22], washer [23], bushing [24], washer [25], nut [26], and pin [27] to reconnect the main gear door rod actuator [29] to the main gear door [28].

SUBTASK 32-32-31-420-019

(10) Do a check of the main gear door adjustment.

(a) To adjust the door, do this task: Main Landing Gear Wing Door Adjustment, TASK 32-13-21-000-802.

SUBTASK 32-32-31-210-001

(11) Examine the hydraulic connections for leaks.

SUBTASK 32-32-31-860-003

(12) Remove power to hydraulic system A (TASK 29-11-00-860-805).

SUBTASK 32-32-31-580-003

CAUTION

MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION

MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(13) Do this task: Lower the Airplane off Jacks, TASK 07-10-01-580-802.

SUBTASK 32-32-31-840-001

(14) Examine the hydraulic reservoirs for the correct servicing (TASK 12-12-00-610-801).

(a) Do a servicing if it is necessary.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN GEAR UPLOCK ACTUATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the uplock actuator for the main landing gear

(2) An installation of the uplock actuator for the main landing gear.

TASK 32-32-41-000-801

2. Main Gear Uplock Actuator Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

D. Prepare for the Removal

SUBTASK 32-32-41-480-007

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-32-41-860-001

(2) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-32-41-580-001

(3) Install chocks around the tires of the main landing gear (PAGEBLOCK 10-11-05/201).

E. Main Gear Uplock Actuator Removal

SUBTASK 32-32-41-480-002

(1) Put tags on the hydraulic line [14] and hydraulic line [15] to show their connection locations.

SUBTASK 32-32-41-020-001

(2) Disconnect the hydraulic line [14] and hydraulic line [15] from the uplock actuator [1].

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SUBTASK 32-32-41-480-003

(3) Install caps/plugs, STD-14686, in the hydraulic line [14] and hydraulic line [15].

SUBTASK 32-32-41-020-005

(4) If it is necessary, remove and keep the union [7] and restrictor [12].

(a) Remove and discard the packing [5] and packing [11].

SUBTASK 32-32-41-480-008

(5) Install caps/plugs, STD-14686, in the ports on the uplock actuator [1].

SUBTASK 32-32-41-020-002

(6) Remove the nut [10], washer [6], washers [9], sleeve [3], washer [2], and bolt [8] from the rod end of the uplock actuator [1].

SUBTASK 32-32-41-020-003

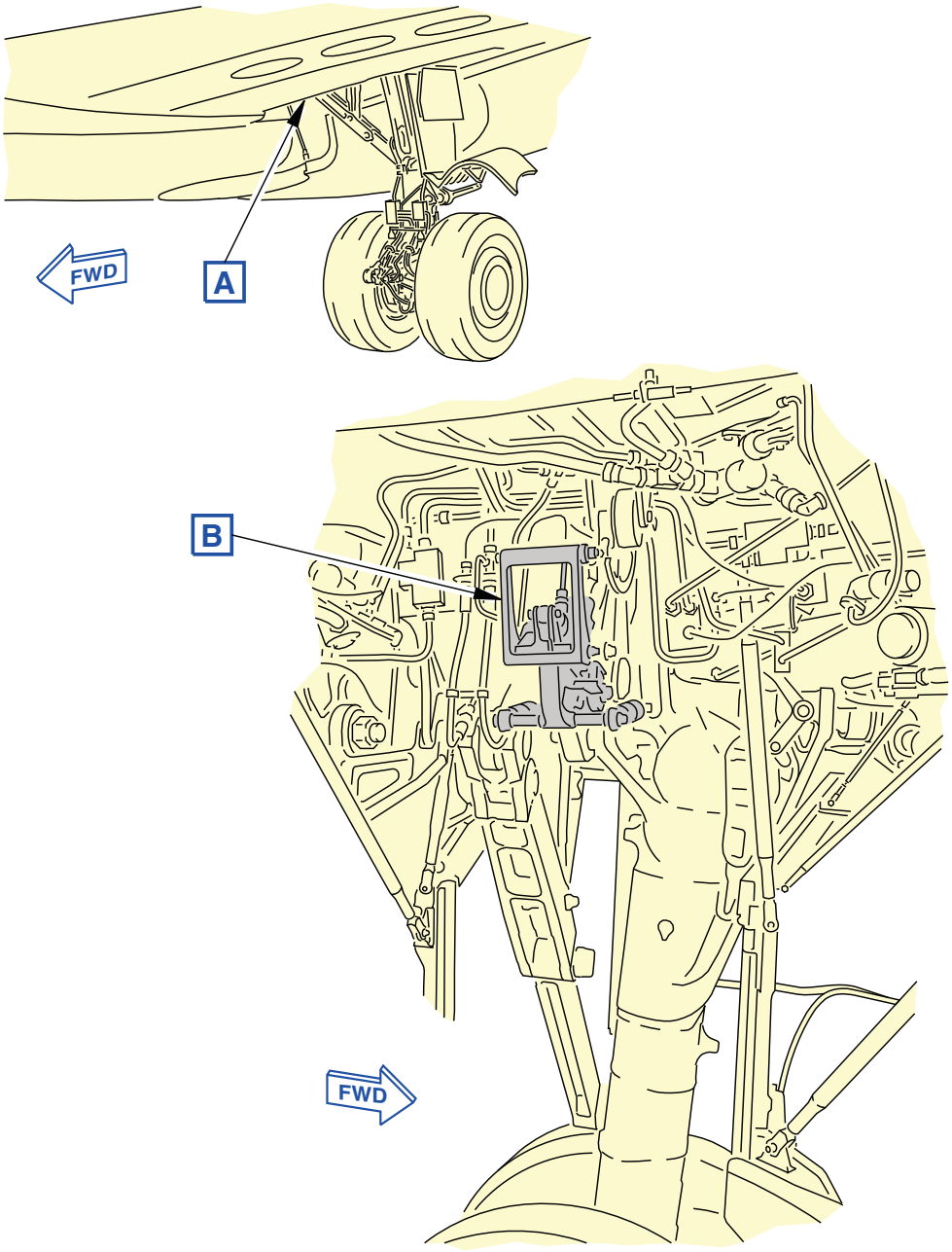
(7) Remove the nut [10], washer [6], washers [4], sleeve [3], washer [2], and bolt [13] from the head end of the uplock actuator [1].

SUBTASK 32-32-41-020-004

(8) Remove the uplock actuator [1] from the airplane.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

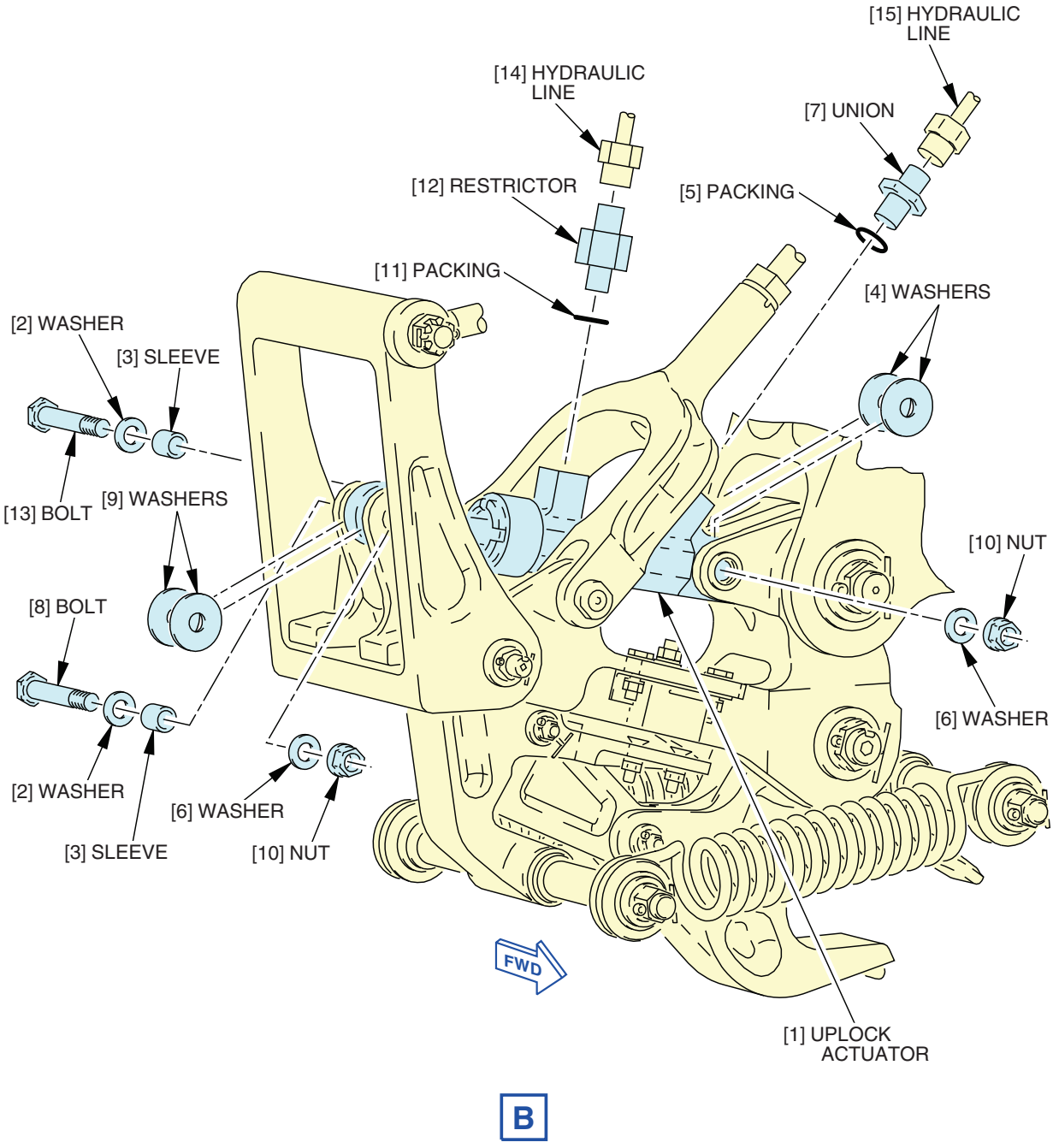


MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE IS SHOWN, RIGHT SIDE IS EQUIVALENT)

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Main Landing Gear Uplock Actuator Installation
Figure 401/32-32-41-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Main Landing Gear Uplock Actuator Installation
Figure 401/32-32-41-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-32-41-400-801

3. Main Gear Uplock Actuator Installation

(Figure 401)

A. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-4896	Adapter Assembly - Uplock, MLG Part #: C32048-1 Supplier: 81205
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
D00633	Grease - Aircraft General Purpose	BMS3-33
G51180	3M EC1252 or Equivalent, Torque Seal, Pink or Yellow.	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
5	Packing	32-32-41-01-030	PTF ALL
11	Packing	32-32-41-01-035	PTF ALL

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

F. Main Gear Uplock Actuator Installation

SUBTASK 32-32-41-420-001

(1) Do these steps if the replacement uplock actuator [1] does not have the union [7] or restrictor [12] installed:

(a) Lubricate these items with hydraulic fluid, D00153:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) packing [5],
2) union [7], restrictor [12]

(b) Remove the caps/plugs, STD-14686, from the ports on the uplock actuator [1].

(c) Install the packing [5], packing [11], union [7], and restrictor [12] in the uplock actuator [1].

SUBTASK 32-32-41-610-001

(2) Extend the uplock actuator [1].

(a) Fill the uplock actuator [1] from the two ends with hydraulic fluid, D00153.

(b) Put cap/plug, STD-14686, on the union [7] and restrictor [12].

SUBTASK 32-32-41-420-002

(3) Put the uplock actuator [1] in its position with the rod end inboard.

SUBTASK 32-32-41-640-001

(4) Lubricate the chrome plated surfaces of the sleeves [3] with grease, D00633.

SUBTASK 32-32-41-420-003

(5) Install these items to connect the head end of the uplock actuator [1] to the uplock actuator [1]:

(a) sleeve [3]

(b) bolt [13], washer [2] and washer [4]

(c) washer [4], washer [6] and nut [10]

(d) Tighten the nut [10].

SUBTASK 32-32-41-420-004

(6) Install these items to connect the rod end of the uplock actuator [1]to the uplock actuator [1]:

(a) sleeve [3]

(b) bolt [8], washer [2] and washer [9]

(c) washer [9], washer [6] and nut [10]

(d) Tighten the nut [10].

SUBTASK 32-32-41-420-005

(7) Remove the caps/plugs, STD-14686, from the union [7] and the restrictor [12] and connect the hydraulic lines to the uplock actuator [1].

(a) Connect the two hydraulic lines to the uplock actuator [1].

(b) Tighten the B-nuts on the hydraulic lines to 140 ±7 in-lb (16 ±1 N·m).

(c) Apply a strip of torque seal, G51180, to the fitting.

(d) Remove the tags from the two hydraulic lines.

G. Main Gear Uplock Actuator Installation Test (Airplane on Jacks)

SUBTASK 32-32-41-580-002

CAUTION

MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

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(CAUTION PRECEDES)

CAUTION

MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

WARNING

MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF CONTROL SURFACES, CABLES, AND LANDING GEAR AREAS. PRESSURIZATION OF A HYDRAULIC SYSTEM CAN MOVE CONTROL SURFACES, CABLES, LANDING GEAR, AND LANDING GEAR DOORS. MOVEMENT OF CONTROL SURFACES, CABLES, LANDING GEAR, AND LANDING GEAR DOORS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Lift the Airplane on Jacks, TASK 07-10-01-580-801.

SUBTASK 32-32-41-860-002

(2) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-32-41-860-003

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

SUBTASK 32-32-41-080-001

(4) For the main landing gear, do this task: Landing Gear Downlock Pins Removal, TASK 32-00-01-080-801

SUBTASK 32-32-41-870-001

WARNING

MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED IN THE NOSE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE NOSE LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(5) Extend and retract the main landing gear several times.

NOTE: It will bleed the air from the uplock actuator and the lines. Examine for the correct gear operation.

SUBTASK 32-32-41-210-001

(6) Examine the hydraulic connections for leaks.

SUBTASK 32-32-41-480-004

(7) Extend the main landing gear and install the down lock pins (TASK 32-00-01-480-801).

SUBTASK 32-32-41-840-001

(8) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

SUBTASK 32-32-41-860-004

(9) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-32-41-580-003

CAUTION

MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION

MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(10) Do this task: Lower the Airplane off Jacks, TASK 07-10-01-580-802.

H. Main Gear Uplock Actuator Installation Test (Airplane not on Jacks)

NOTE: This test gives does a check of the uplock actuator [1] with the airplane on the ground, if jacks are not available.

SUBTASK 32-32-41-210-002

(1) Make sure that you installed the hydraulic hoses correctly.

SUBTASK 32-32-41-480-006

WARNING

MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(2) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-32-41-860-005

(3) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-32-41-700-001

(4) Use the override trigger to move the landing gear lever from the DN position to the UP position for three times.

(a) Hold the landing gear lever in each position for three seconds before you move the landing gear lever to the subsequent position.

SUBTASK 32-32-41-700-002

(5) Move the landing gear lever to the UP position.

SUBTASK 32-32-41-860-006

(6) Turn the uplock hook to the closed position with the MLG uplock adapter assembly, SPL-4896.

SUBTASK 32-32-41-700-003

(7) Move the landing gear lever to the DN position.

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(a) Make sure that the uplock hook turns to the open position.

SUBTASK 32-32-41-210-003

(8) Do a check for leakage with the hydraulic pressure applied.

SUBTASK 32-32-41-860-007

(9) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-32-41-210-004

(10) Make sure that there are no leaks in the hydraulic connections.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 32-32-41-200-001

(1) Service the reservoirs if it is necessary (TASK 12-12-00-610-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN GEAR DOWNLOCK ACTUATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the downlock actuator for the main landing gear

(2) An installation of the downlock actuator for the main landing gear.

TASK 32-32-51-000-801

2. Main Gear Downlock Actuator Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-32-91-000-801	Main Gear Downlock Springs Removal (P/B 401)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

D. Prepare for the Removal

SUBTASK 32-32-51-480-007

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-32-51-860-001

(2) For hydraulic system A, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 32-32-51-580-001

(3) Install chocks around the tires of the main landing gear (PAGEBLOCK 10-11-05/201).

E. Main Gear Downlock Actuator Removal

SUBTASK 32-32-51-480-002

(1) Put tags on the lines to show their connection locations.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-32-51-020-001

(2) Disconnect the hydraulic line [8] and hydraulic line [9] from the downlock actuator [1].

SUBTASK 32-32-51-480-003

(3) Install the caps/plugs, STD-14686, in the hydraulic line [8] and hydraulic line [9].

SUBTASK 32-32-51-020-002

(4) Remove the springs [6] (TASK 32-32-91-000-801).

SUBTASK 32-32-51-020-003

(5) Remove the rod end pin [5], spacer [4], spacers [10], and washers [11] that attach the rod end of the downlock actuator [1] to the side strut assembly.

SUBTASK 32-32-51-020-004

(6) Remove the pin [7], nut [13], special washer [14], washers [15], washers [16], and head end pin [12] that attach the head end of the downlock actuator [1] to the side strut assembly.

SUBTASK 32-32-51-020-005

(7) Remove the downlock actuator [1] from the airplane.

SUBTASK 32-32-51-020-006

(8) If the replacement downlock actuator [1] does not have unions [2] installed, do these steps:

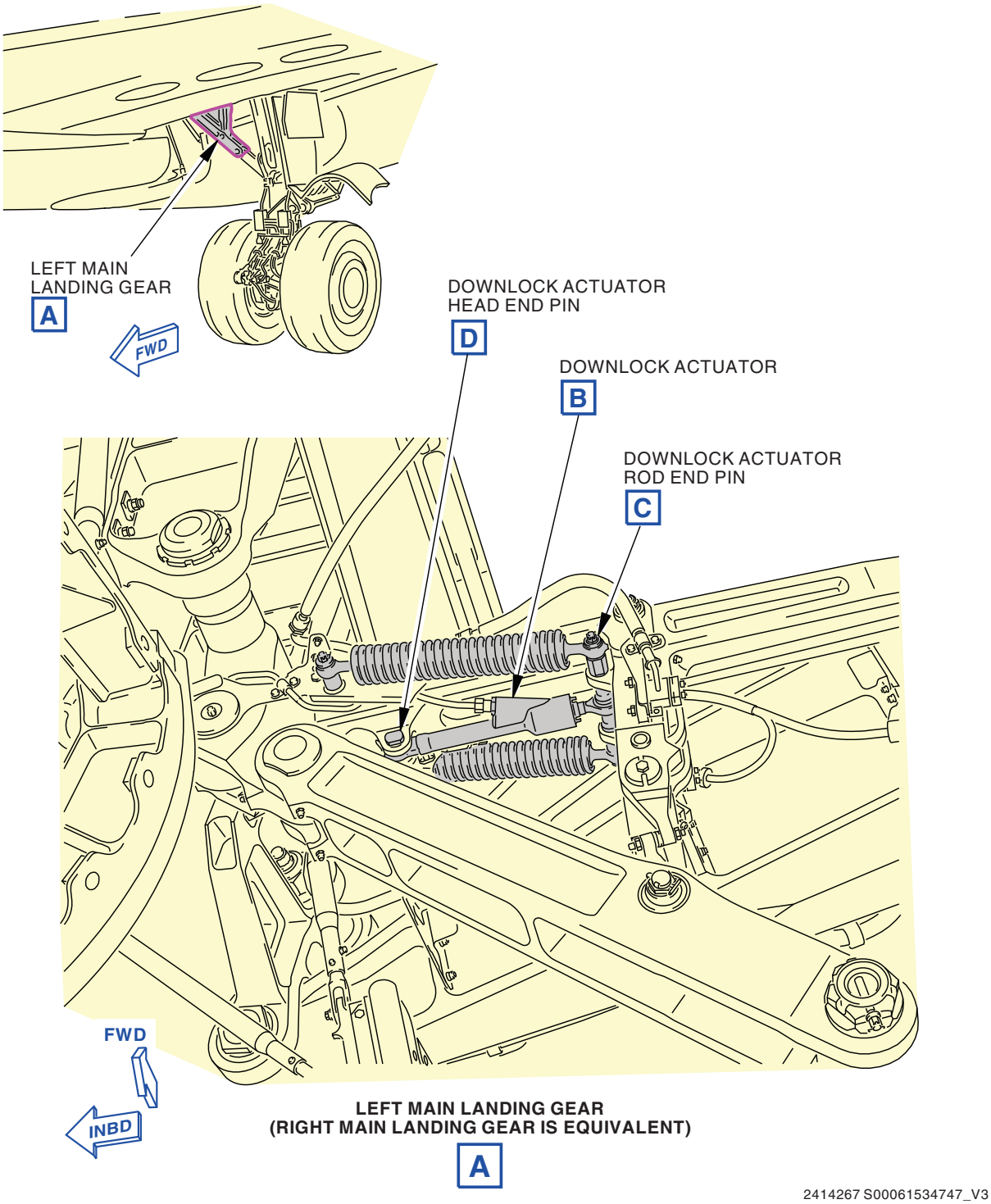
(a) Remove the unions [2] from the downlock actuator [1].

(b) Remove and discard the packings [3].

(c) Install the caps/plugs, STD-14686, in the ports on the downlock actuator [1].

——— END OF TASK ———

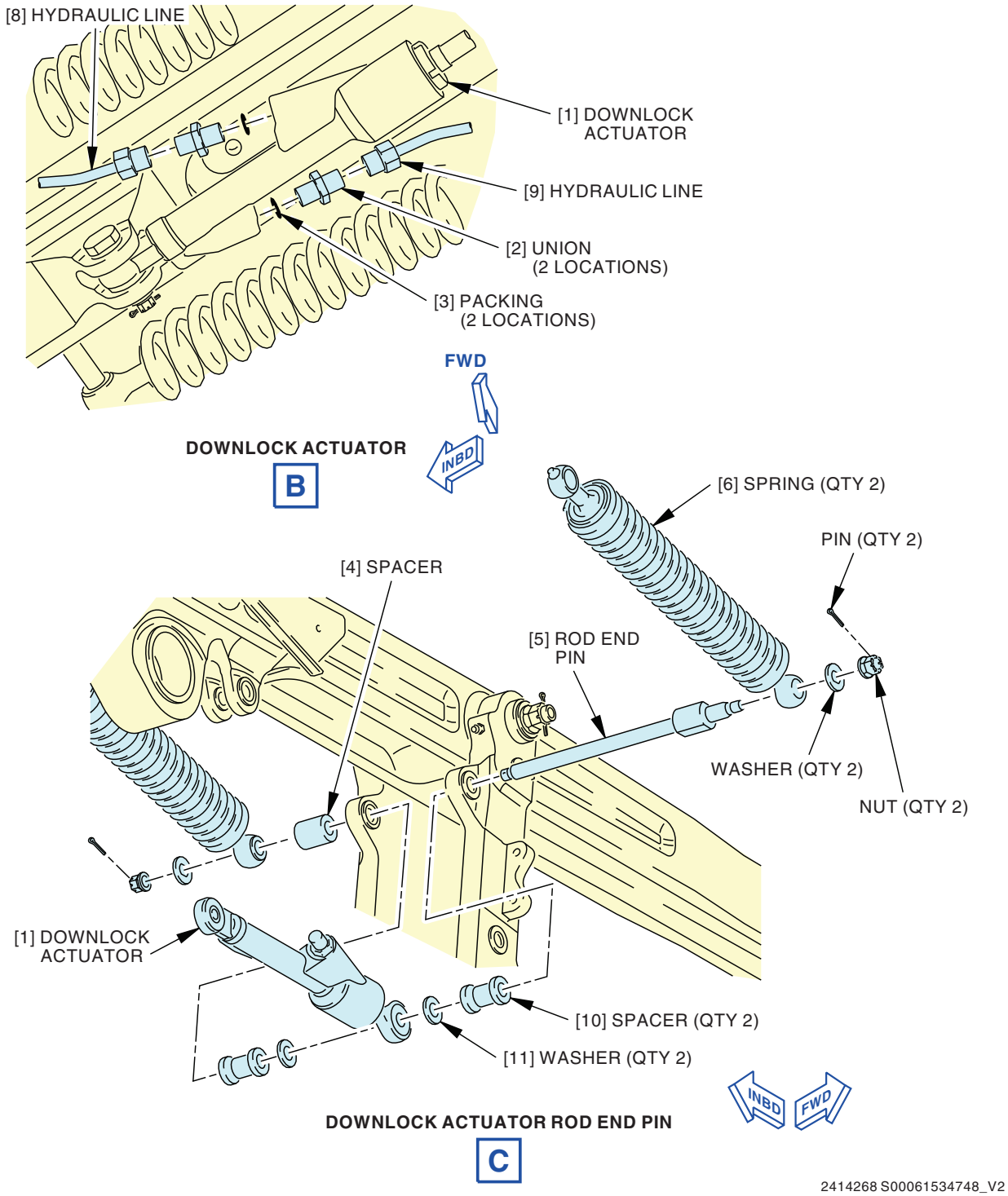
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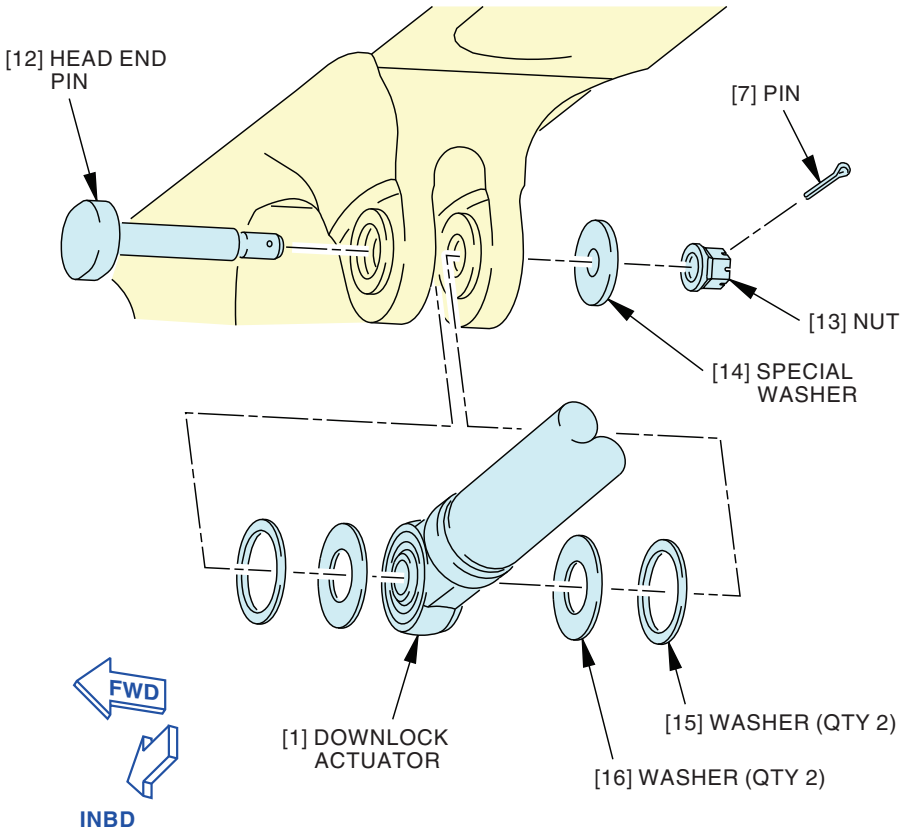
Main Landing Gear Downlock Actuator Installation
Figure 401/32-32-51-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Downlock Actuator Installation
Figure 401/32-32-51-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



DOWNLOCK ACTUATOR HEAD END PIN

D

2414269 S00061534749_V1

Main Landing Gear Downlock Actuator Installation
Figure 401/32-32-51-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-32-51-400-801

3. Main Gear Downlock Actuator Installation

(Figure 401)

A. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-32-91-400-801	Main Gear Downlock Spring Installation (P/B 401)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
D00633	Grease - Aircraft General Purpose	BMS3-33
D50396	Grease - Aircraft and Instrument Grease	MIL-PRF-23827
G51180	3M EC1252 or Equivalent, Torque Seal, Pink or Yellow.	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Downlock actuator	32-32-51-01-018	PTF ALL
		32-32-51-01A-040	PTF ALL
2	Union	32-32-51-01-022	PTF ALL
		32-32-51-01A-050	PTF ALL
3	Packing	32-32-51-01-024	PTF ALL
		32-32-51-01A-055	PTF ALL
7	Pin	32-32-51-01-003	PTF ALL
		32-32-51-01A-005	PTF ALL

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Main Gear Downlock Actuator Installation

SUBTASK 32-32-51-420-001

(1) If the replacement downlock actuator [1] does not have the unions [2] installed, do these steps:

(a) Lubricate these items with hydraulic fluid, D00153:

1) new packings [3]

2) unions [2].

(b) Remove the caps/plugs, STD-14686, from the ports on the downlock actuator [1].

(c) Install the packings [3] and unions [2] in the downlock actuator [1].

SUBTASK 32-32-51-610-001

(2) Retract the downlock actuator [1] and fill it with hydraulic fluid, D00153.

(a) Put the caps/plugs, STD-14686, on the unions [2].

SUBTASK 32-32-51-420-010

(3) Do these steps to connect the head end of the downlock actuator [1] to the side strut assembly:

(a) Apply grease, D00633, or grease, D50396, to the head end pin [12] diameter, and to the inside surface of the lug on the side strut assembly.

1) Do not lubricate the downlock actuator [1] head end bearing inside diameter.

(b) Put the downlock actuator [1] in position with the head end inboard and the ports on the top part of the downlock actuator [1].

(c) Install the head end pin [12], washers [15], washers [16], special washer [14], and nut [13].

(d) Tighten the nut [13] to 20 ±5 in-lb (2.3 ±0.6 N·m) above run-on torque.

1) If it is necessary, loosen the nut [13] to the nearest castellation.

2) Install the new pin [7] on the head end pin [12].

SUBTASK 32-32-51-420-011

(4) Do these steps to connect the rod end of the downlock actuator [1] to the side strut assembly:

(a) Lubricate the chrome plated or thermal spay coated surfaces of the rod end pin [5] with grease, D00633.

(b) Put the rod end of the downlock actuator [1] in its position for attachment to the side strut assembly.

1) If it is necessary, retract or extend the downlock actuator [1] to align the rod end of the downlock actuator [1] to the lug on the side strut assembly.

(c) Install the rod end pin [5], spacer [4], spacers [10], and washers [11].

SUBTASK 32-32-51-420-009

(5) Install the springs [6] (TASK 32-32-91-400-801).

SUBTASK 32-32-51-080-002

(6) Remove the caps/plugs, STD-14686, from the unions [2], hydraulic line [8], and hydraulic line [9].

SUBTASK 32-32-51-420-012

(7) Connect the hydraulic line [8] and hydraulic line [9] to the downlock actuator [1].

(a) Tighten the B-nuts on the hydraulic line [8] and hydraulic line [9] to 140 in-lb (15.8 N·m).

(b) Back off the B-nuts on the hydraulic line [8] and hydraulic line [9].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Tighten the B-nuts on the hydraulic line [8] and hydraulic line [9] to 140 ±7 in-lb (15.8 ±0.8 N·m).

1) Make sure that the applied torque value is correct.

(d) Apply a yellow stripe of torque seal, G51180, to the B-nuts on the hydraulic line [8] and hydraulic line [9].

SUBTASK 32-32-51-860-006

(8) Remove the tags installed on the hydraulic lines.

G. Main Gear Downlock Actuator Installation Test (Airplane on Jacks)

SUBTASK 32-32-51-580-002

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

WARNING MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF CONTROL SURFACES, CABLES, AND LANDING GEAR AREAS. PRESSURIZATION OF A HYDRAULIC SYSTEM CAN MOVE CONTROL SURFACES, CABLES, LANDING GEAR, AND LANDING GEAR DOORS. MOVEMENT OF CONTROL SURFACES, CABLES, LANDING GEAR, AND LANDING GEAR DOORS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Lift the Airplane on Jacks, TASK 07-10-01-580-801.

SUBTASK 32-32-51-080-003

(2) Remove the downlock pins for the main landing gear (TASK 32-00-01-080-801).

SUBTASK 32-32-51-860-002

(3) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-32-51-870-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED IN THE NOSE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE NOSE LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(4) Extend and retract the main landing gear several times.

NOTE: It will bleed the air from the downlock actuator and lines.

(a) Make sure that the main landing gear operates correctly.

SUBTASK 32-32-51-210-001

(5) Examine the hydraulic connections for leaks.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-32-51-480-004

(6) Extend the main landing gear and install the downlock pins, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-32-51-860-003

(7) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-32-51-580-003

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(8) Do this task: Lower the Airplane off Jacks, TASK 07-10-01-580-802.

H. Installation Test for the Main Gear Downlock Actuator (Airplane not on Jacks)

SUBTASK 32-32-51-210-002

(1) Make sure that you installed the hydraulic hoses correctly.

SUBTASK 32-32-51-480-006

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(2) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-32-51-860-004

(3) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-32-51-700-001

(4) Use the override trigger to move the landing gear lever from the DN position to the UP position for three times.

(a) Hold the landing gear lever in each position for three seconds before you move the landing gear lever to the subsequent position.

SUBTASK 32-32-51-700-002

(5) Make sure that the rod end of the downlock actuator [1] tries to extend when you move the landing gear lever from the DOWN position to the UP position.

SUBTASK 32-32-51-700-003

(6) Move the landing gear lever to the DN position.

SUBTASK 32-32-51-210-003

(7) Do a check for leakage with the hydraulic pressure applied.

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SUBTASK 32-32-51-860-005

(8) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-32-51-210-004

(9) Make sure that there are no leaks in the hydraulic connections.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 32-32-51-200-001

(1) If it is necessary, service the reservoirs (TASK 12-12-00-610-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN GEAR FRANGIBLE FITTING ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

NOTE: If the entire assembly is damaged, it is necessary to replace the entire frangible fitting assembly. If only the frangible fitting is damaged, it is necessary to replace only that component.

(1) A removal of the main gear frangible fitting assembly

(2) An installation of the main gear frangible fitting assembly

(3) A removal of the main gear frangible fitting

(4) An installation of the main gear frangible fitting.

TASK 32-32-52-000-801

2. Main Gear Frangible Fitting Assembly Removal

(Figure 401)

A. General

(1) This task is for the removal of main gear frangible fitting assembly.

B. References

Reference	Title
10-11-05-500-801	Chock Installation Winds/Gusts Maximum 40 mph (35 Knots) (P/B 201)
10-11-05-500-802	Chock Installation in Winds More Than 35 Knots - Handling (P/B 201)
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

E. Prepare for the Removal

SUBTASK 32-32-52-480-017

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-32-52-860-001

(2) Depressurize hydraulic system A (TASK 29-09-00-860-802).

SUBTASK 32-32-52-580-001

(3) Install wheel chocks for the main landing gear (TASK 10-11-05-500-801 or TASK 10-11-05-500-802).

F. Removal of the Frangible Fitting Assembly

SUBTASK 32-32-52-020-001

(1) Disconnect the hydraulic line [6] from the frangible fitting assembly [1].

SUBTASK 32-32-52-480-002

(2) Install a cap/plug, STD-14686, on the hydraulic line [6].

SUBTASK 32-32-52-020-002

(3) Remove the union [5] and packing [4] from the frangible fitting assembly [1].

(a) Discard the packing [4].

SUBTASK 32-32-52-480-003

(4) Install a cap/plug, STD-14686, in the port on the frangible fitting assembly [1].

SUBTASK 32-32-52-020-003

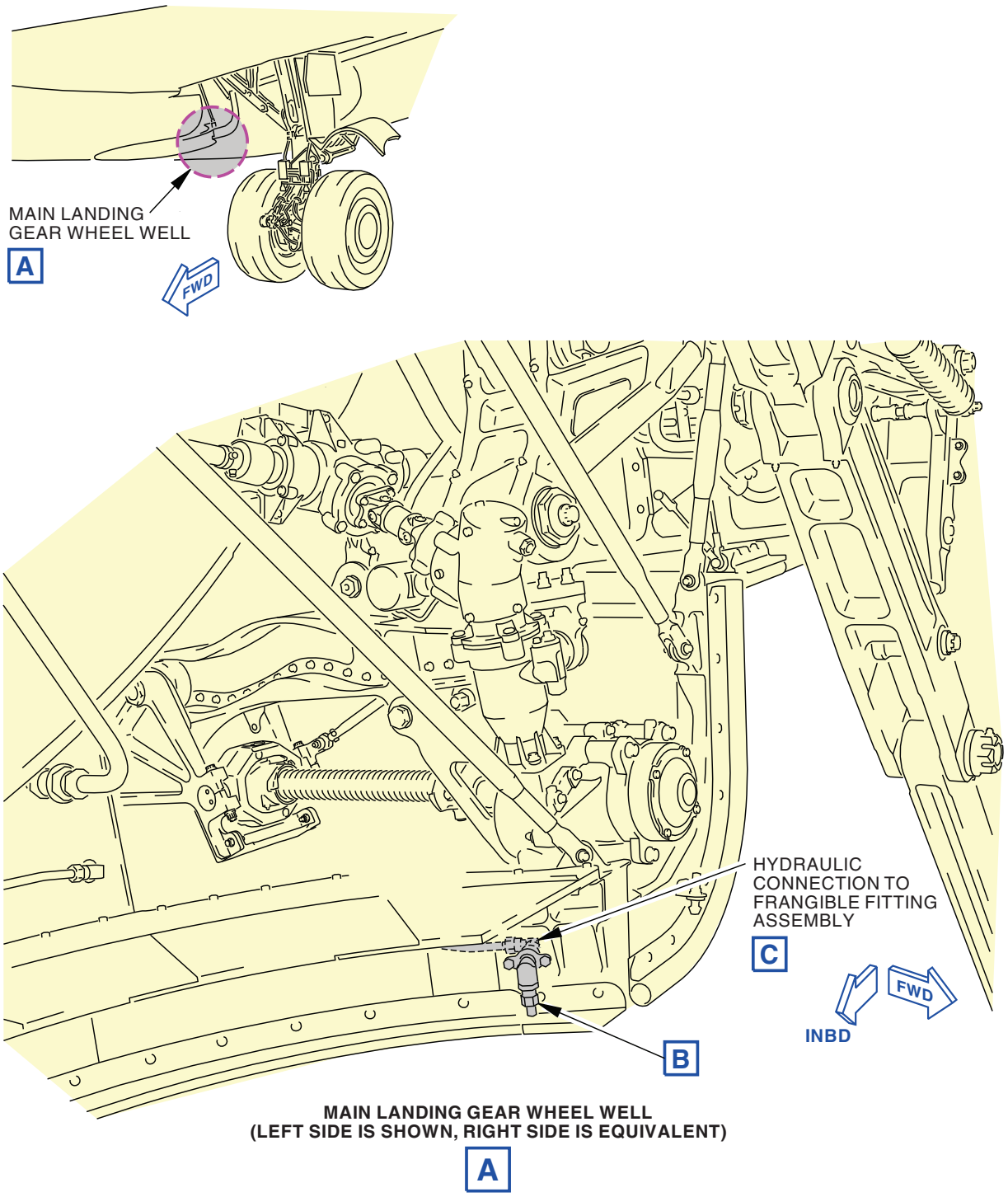
(5) Remove the bolts [3] and washers [2] that attach the frangible fitting assembly [1] to the structure.

SUBTASK 32-32-52-020-004

(6) Remove the frangible fitting assembly [1] from the airplane.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



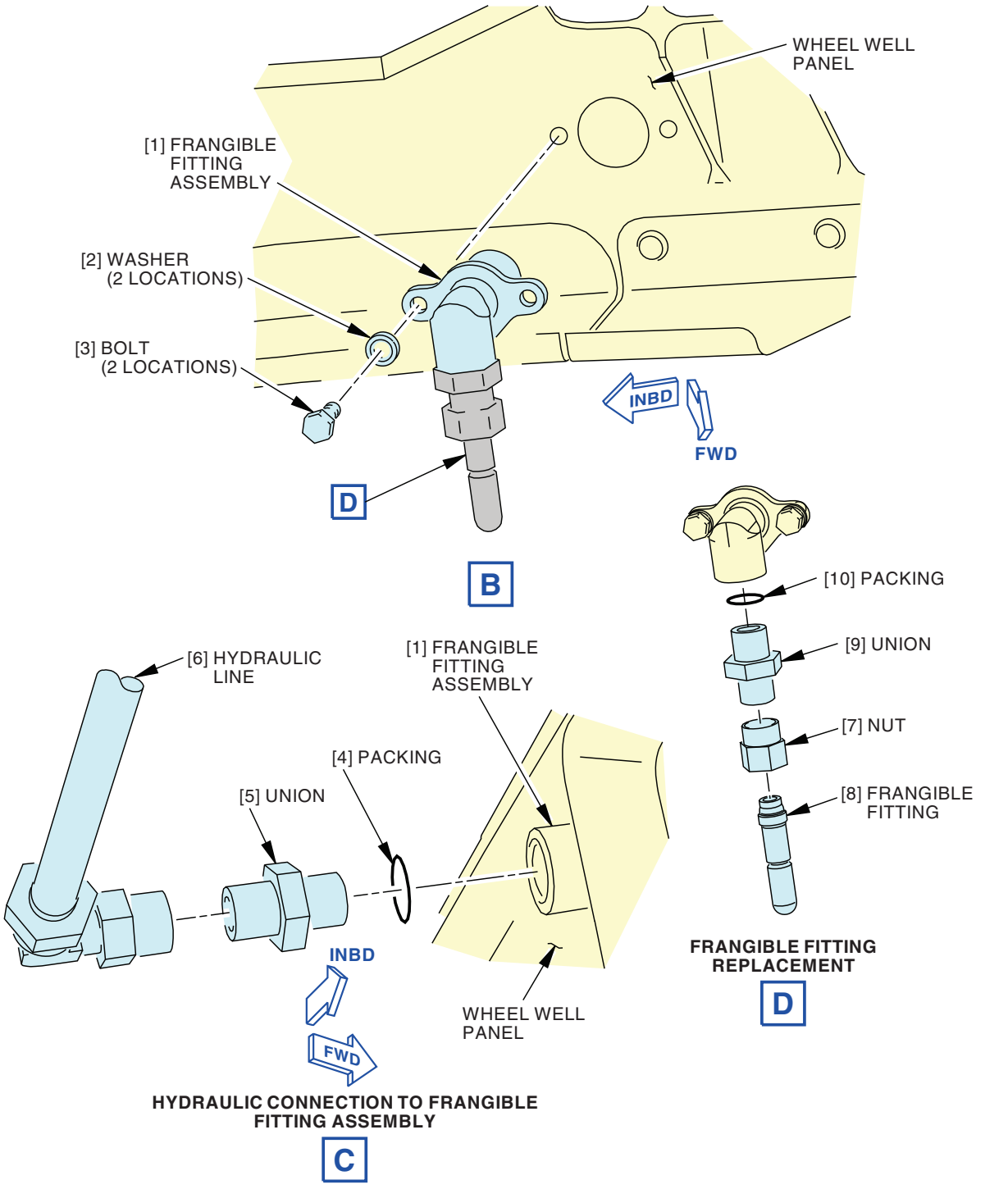
MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE IS SHOWN, RIGHT SIDE IS EQUIVALENT)

A

2414270 S00061534754_V1

Main Gear Frangible Fitting Assembly
Figure 401/32-32-52-990-801 (Sheet 1 of 2)

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2414271 S00061534755_V2

Main Gear Frangible Fitting Assembly
Figure 401/32-32-52-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-32-52-400-801

3. Main Gear Frangible Fitting Assembly Installation

(Figure 401)

A. General

(1) This task is for the installation of the main gear frangible fitting assembly.

B. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

C. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

D. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Packing	32-32-52-02-085	PTF ALL

F. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

G. Frangible Fitting Assembly Installation

SUBTASK 32-32-52-420-001

(1) Lubricate the new packing [4] and union [5] with hydraulic fluid, D00153.

SUBTASK 32-32-52-420-002

(2) Put the frangible fitting assembly [1] in its position in the structure.

SUBTASK 32-32-52-420-003

(3) Install the bolts [3] and washers [2].

SUBTASK 32-32-52-390-001

(4) Fay surface seal the frangible fitting assembly [1] with sealant, A00247.

SUBTASK 32-32-52-080-001

(5) Remove the cap/plug, STD-14686, from the ports of the frangible fitting assembly [1].

SUBTASK 32-32-52-420-004

(6) Install the packing [4] and union [5] on the frangible fitting assembly [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-32-52-420-009

(7) Tighten the union [5] to 215 ±11 in-lb (24.3 ±1.2 N·m).

SUBTASK 32-32-52-080-002

(8) Remove the cap/plug, STD-14686, from the hydraulic line [6].

SUBTASK 32-32-52-420-010

(9) Connect the hydraulic line [6] to the union [5].

SUBTASK 32-32-52-420-011

(10) Tighten the B-nut on the hydraulic line [6] to 270 ±14 in-lb (31 ±2 N·m).

H. Frangible Fitting Assembly Installation Test

SUBTASK 32-32-52-860-002

(1) Pressurize hydraulic system A (TASK 29-11-00-860-801).

SUBTASK 32-32-52-700-001

(2) Use the override trigger to move the landing gear lever from the DN position to the UP position three times as follows:

(a) Hold the landing gear lever in each position for three seconds before you move the landing gear lever to the subsequent position.

SUBTASK 32-32-52-700-002

(3) Move the landing gear lever to the DN position.

SUBTASK 32-32-52-210-001

(4) Do a check for leakage with the hydraulic pressure applied.

SUBTASK 32-32-52-860-003

(5) Depressurize hydraulic system A (TASK 29-11-00-860-805).

SUBTASK 32-32-52-210-002

(6) Make sure that there are no leaks in the hydraulic connections.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 32-32-52-200-001

(1) If it is necessary, service the reservoirs (TASK 12-12-00-610-801).

——— END OF TASK ———

TASK 32-32-52-000-802

4. Main Gear Frangible Fitting Removal

(Figure 401)

A. General

(1) This task is for the removal of the main gear frangible fitting.

B. References

Reference	Title
10-11-05-500-801	Chock Installation Winds/Gusts Maximum 40 mph (35 Knots) (P/B 201)
10-11-05-500-802	Chock Installation in Winds More Than 35 Knots - Handling (P/B 201)
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

E. Prepare for the Removal

SUBTASK 32-32-52-480-018

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-32-52-860-004

(2) Depressurize hydraulic system A (TASK 29-09-00-860-802).

SUBTASK 32-32-52-580-002

(3) Install wheel chocks for the main landing gear (TASK 10-11-05-500-801 or TASK 10-11-05-500-802).

F. Main Gear Frangible Fitting Removal

SUBTASK 32-32-52-020-005

(1) Disconnect the nut [7] with the frangible fitting [8] from the union [9].

SUBTASK 32-32-52-020-006

(2) Remove the frangible fitting [8].

SUBTASK 32-32-52-211-001

(3) Examine the frangible fitting assembly [1] for damage.

SUBTASK 32-32-52-960-001

(4) If there is damage to the frangible fitting assembly [1], replace it.

NOTE: You must replace the entire frangible fitting assembly, not just the frangible fitting.

SUBTASK 32-32-52-020-008

(5) If it is necessary, remove the union [9] and packing [10] from the frangible fitting assembly [1].

SUBTASK 32-32-52-910-002

(6) Discard the packing [10].

SUBTASK 32-32-52-480-006

(7) Install the cap/plug, STD-14686, in the ports on the frangible fitting assembly [1].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-32-52-400-802

5. Main Gear Frangible Fitting Installation

(Figure 401)

A. General

(1) This task is for the installation of the main gear frangible fitting.

B. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

C. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

D. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
8	Frangible fitting	32-32-52-02-065	PTF ALL

F. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

G. Main Gear Frangible Fitting Installation

SUBTASK 32-32-52-080-003

(1) Remove the cap/plug, STD-14686, from the port(s) on the frangible fitting assembly [1].

SUBTASK 32-32-52-420-008

(2) If it is necessary to install the union [9], do these steps:

(a) Lubricate the packing [10] and union [9] with hydraulic fluid, D00153.

(b) Install the union [9] with the packing [10] on the frangible fitting assembly [1]

(c) Tighten the union [9] to 215.0 ±10.8 in-lb (24.3 ±1.2 N·m).

SUBTASK 32-32-52-420-005

(3) Put the frangible fitting [8] to the nut [7].

SUBTASK 32-32-52-420-006

(4) Install the nut [7] with the frangible fitting [8] to the union [9].

SUBTASK 32-32-52-420-007

(5) Tighten the nut [7] to 270.0 ±14.0 in-lb (30.5 ±1.6 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Main Gear Frangible Fitting Installation Test

SUBTASK 32-32-52-860-005

(1) Pressurize hydraulic system A (TASK 29-11-00-860-801).

SUBTASK 32-32-52-700-003

(2) Use the override trigger to move the landing gear lever from the DN position to the UP position three times as follows:

(a) Hold the landing gear lever in each position for three seconds before you move the landing gear lever to the subsequent position.

SUBTASK 32-32-52-700-004

(3) Move the landing gear lever to the DN position.

SUBTASK 32-32-52-210-003

(4) Do a check for leakage with the hydraulic pressure applied.

SUBTASK 32-32-52-860-006

(5) Remove the pressure from hydraulic system A (TASK 29-11-00-860-805).

SUBTASK 32-32-52-210-004

(6) Make sure that there are no leaks in the hydraulic connections.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 32-32-52-200-002

(1) Service the reservoirs if it is necessary (TASK 12-12-00-610-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN GEAR TRANSFER CYLINDER - REMOVAL/INSTALLATION

1. General

A. This procedure has these two tasks:

(1) A removal of the transfer cylinder assembly [1] for the main landing gear

(2) An installation of the transfer cylinder assembly [1] for the main landing gear.

TASK 32-32-71-000-801

2. Main Gear Transfer Cylinder Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
551	Left Wing - Rear Spar To Landing Gear Support Beam
651	Right Wing - Rear Spar to Landing Gear Support Beam

D. Prepare for the Removal

SUBTASK 32-32-71-480-007

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-32-71-860-001

(2) For hydraulic system A, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 32-32-71-580-001

(3) Install chocks around the tires of the main landing gear (PAGEBLOCK 10-11-05/201).

E. Main Gear Transfer Cylinder Removal

SUBTASK 32-32-71-020-001

(1) Disconnect the hydraulic lines from the transfer cylinder assembly [1]:

(a) Disconnect the tee [8] from the union [6].

(b) Disconnect the hydraulic line [9] from the union [6].

SUBTASK 32-32-71-480-002

(2) Install caps/plugs, STD-14686, on the hydraulic line [9] and the tee [8].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-32-71-020-002

(3) Remove the screws [3], washers [4], and spacers [7] from the clamps [2].

SUBTASK 32-32-71-020-003

(4) Remove the clamps [2].

SUBTASK 32-32-71-020-004

(5) Remove the transfer cylinder assembly [1].

SUBTASK 32-32-71-020-005

(6) Do these steps if the replacement transfer cylinder assembly [1] does not have the unions [6] installed:

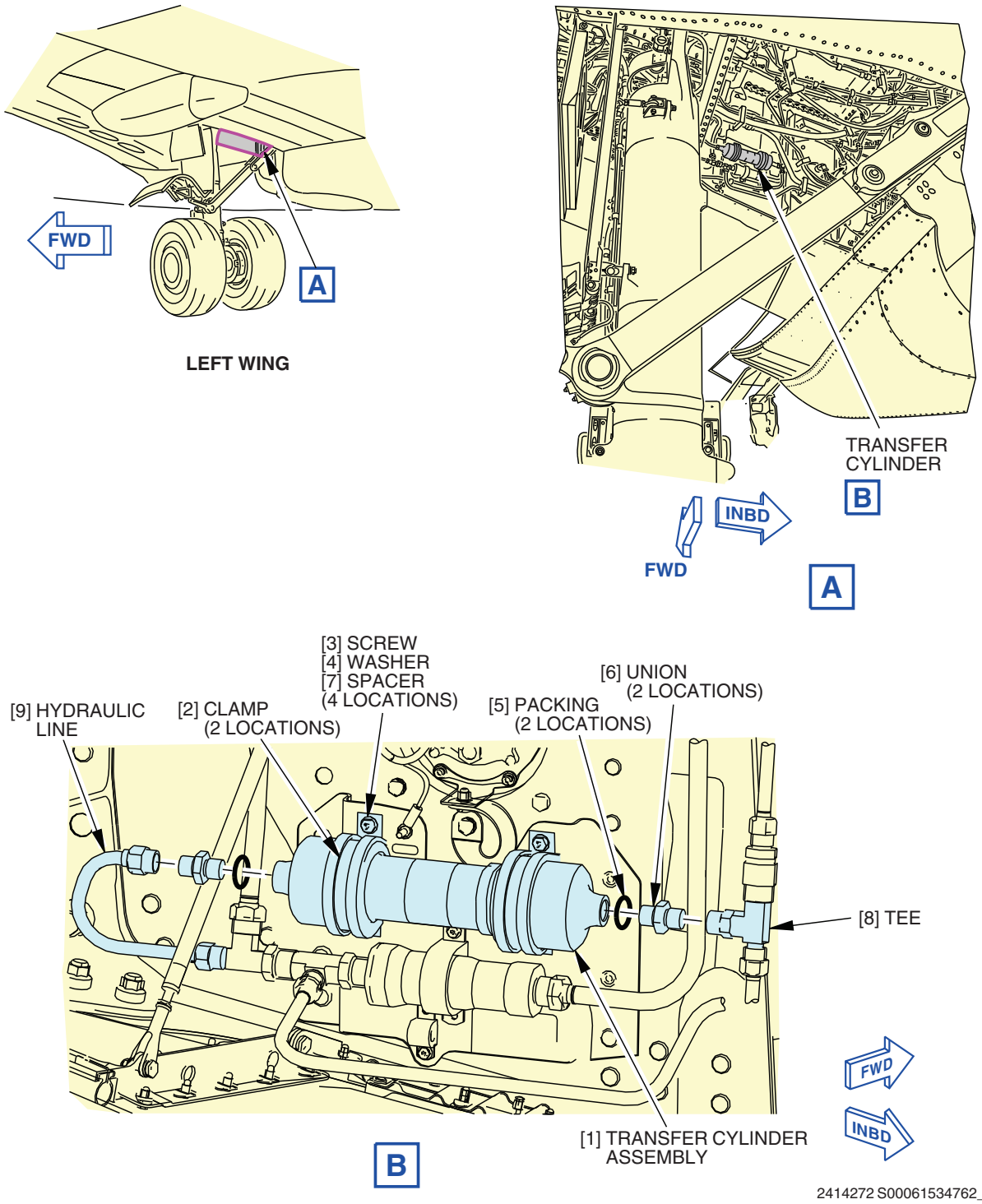
(a) Remove the unions [6] from the transfer cylinder assembly [1].

(b) Remove and discard the packings [5].

(c) Install caps/plugs, STD-14686, in the ports on the transfer cylinder assembly [1].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Transfer Cylinder Installation
Figure 401/32-32-71-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-32-71-400-801

3. Main Gear Transfer Cylinder Installation

(Figure 401)

A. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Transfer cylinder assembly	32-32-71-02-040	PTF ALL
5	Packing	32-32-71-02-075	PTF ALL

E. Location Zones

Zone	Area
551	Left Wing - Rear Spar To Landing Gear Support Beam
651	Right Wing - Rear Spar to Landing Gear Support Beam

F. Main Gear Transfer Cylinder Installation

SUBTASK 32-32-71-420-001

(1) Do these steps if the replacement transfer cylinder assembly [1] does not have the unions [6] installed:

(a) Apply a thin layer of hydraulic fluid, D00153, to the new packings [5].

(b) Put the new packings [5] in the grooves of the unions [6].

(c) Remove the caps/plugs, STD-14686, from the ports on the transfer cylinder assembly [1].

(d) Install the new packings [5] and unions [6] in the transfer cylinder assembly [1].

(e) Fill the transfer cylinder assembly [1] from the two ends with hydraulic fluid, D00153.

1) Put caps/plugs, STD-14686, on the unions [6]

SUBTASK 32-32-71-420-002

(2) Put the transfer cylinder assembly [1] in the clamps [2].

SUBTASK 32-32-71-420-003

(3) Adjust the position of the transfer cylinder assembly [1] to align it with the hydraulic lines.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-32-71-080-001

(4) Remove the caps/plugs, STD-14686 from the hydraulic line [9] and the tee [8]:

SUBTASK 32-32-71-420-006

(5) Connect the tee [8] to the union [6].

(a) Tighten the B-nut on the tee [8] to 270 ±14 in-lb (31 ±2 N·m).

SUBTASK 32-32-71-420-007

(6) Connect the hydraulic line [9] to the union [6].

(a) Tighten the B-nut on the hydraulic line [9] to 270 ±14 in-lb (31 ±2 N·m).

SUBTASK 32-32-71-420-004

(7) Install the screws [3], washers [4], and spacers [7] on the clamps [2].

(a) Tighten the screws [3] to 50 in-lb (5.6 N·m) - 65 in-lb (7.3 N·m).

G. Main Gear Transfer Cylinder Installation Test (Airplane on Jacks)

NOTE: This test does a check of the transfer cylinder assembly [1] with the airplane lifted on jacks.

SUBTASK 32-32-71-580-002

CAUTION

MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION

MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

WARNING

MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF CONTROL SURFACES, CABLES, AND LANDING GEAR AREAS. PRESSURIZATION OF A HYDRAULIC SYSTEM CAN MOVE CONTROL SURFACES, CABLES, LANDING GEAR, AND LANDING GEAR DOORS. MOVEMENT OF CONTROL SURFACES, CABLES, LANDING GEAR, AND LANDING GEAR DOORS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Lift the Airplane on Jacks, TASK 07-10-01-580-801.

SUBTASK 32-32-71-860-002

(2) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-32-71-480-003

WARNING

MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED IN THE NOSE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE NOSE LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Keep the downlock pin installed in the nose landing gear if maintenance work is necessary in the nose wheel well area or on the nose landing gear (TASK 32-00-01-480-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-32-71-080-002

(4) Remove the downlock pin for the nose landing gear if no maintenance work is necessary in the nose wheel well area or on the nose landing gear (TASK 32-00-01-080-801).

SUBTASK 32-32-71-080-003

(5) Remove the downlock pins for the main landing gear (TASK 32-00-01-080-801).

SUBTASK 32-32-71-870-001

(6) Extend and retract the main landing gear several times.

NOTE: It will bleed the air from the transfer cylinder and the lines.

SUBTASK 32-32-71-210-001

(7) Make sure that there are no leaks in the hydraulic connections.

SUBTASK 32-32-71-480-004

(8) Extend the main landing gear and install the downlock pins (TASK 32-00-01-480-801).

SUBTASK 32-32-71-860-003

(9) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-32-71-580-003

CAUTION

MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION

MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(10) Do this task: Lower the Airplane off Jacks, TASK 07-10-01-580-802.

H. Main Gear Transfer Cylinder Installation Test (Airplane not on Jacks)

NOTE: This test does a check of the transfer cylinder assembly [1] with the airplane on the ground, if jacks are not available.

SUBTASK 32-32-71-480-006

WARNING

MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-32-71-860-004

(2) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-32-71-700-001

(3) Use the override trigger to move the landing gear lever from the DN position to the UP position for three times.

(a) Hold the landing gear lever in each position for three seconds before you move the landing gear lever to the subsequent position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-32-71-700-002

(4) Move the landing gear lever to the DN position.

SUBTASK 32-32-71-210-002

(5) Make sure that there are no leaks in the hydraulic connections.

SUBTASK 32-32-71-860-005

(6) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 32-32-71-200-001

(1) Service the reservoirs if it is necessary (TASK 12-12-00-610-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN LANDING GEAR HYDRAULIC FLOW REGULATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the hydraulic flow regulator for the main landing gear

(2) An installation of the hydraulic flow regulator for the main landing gear

(a) The installation of the hydraulic flow regulator for the main landing gear includes an installation test of the hydraulic flow regulator.

TASK 32-32-72-000-801

2. Main Gear Hydraulic Flow Regulator Removal

(Figure 401)

A. References

Reference	Title
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
551	Left Wing - Rear Spar To Landing Gear Support Beam
651	Right Wing - Rear Spar to Landing Gear Support Beam

D. Prepare for the Removal

SUBTASK 32-32-72-480-005

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-32-72-860-001

(2) For hydraulic system A, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

E. Main Gear Hydraulic Flow Regulator Removal

SUBTASK 32-32-72-020-001

(1) Loosen the hydraulic lines to the hydraulic flow regulator [3].

SUBTASK 32-32-72-020-002

(2) Remove the screws [1], washers [2], and the clamp [8] that hold the hydraulic flow regulator [3] to the structure.

SUBTASK 32-32-72-020-003

(3) Disconnect the hydraulic line [4] from the hydraulic flow regulator [3].

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SUBTASK 32-32-72-020-006

(4) Disconnect the tee [9] from of the hydraulic flow regulator [3].

SUBTASK 32-32-72-480-004

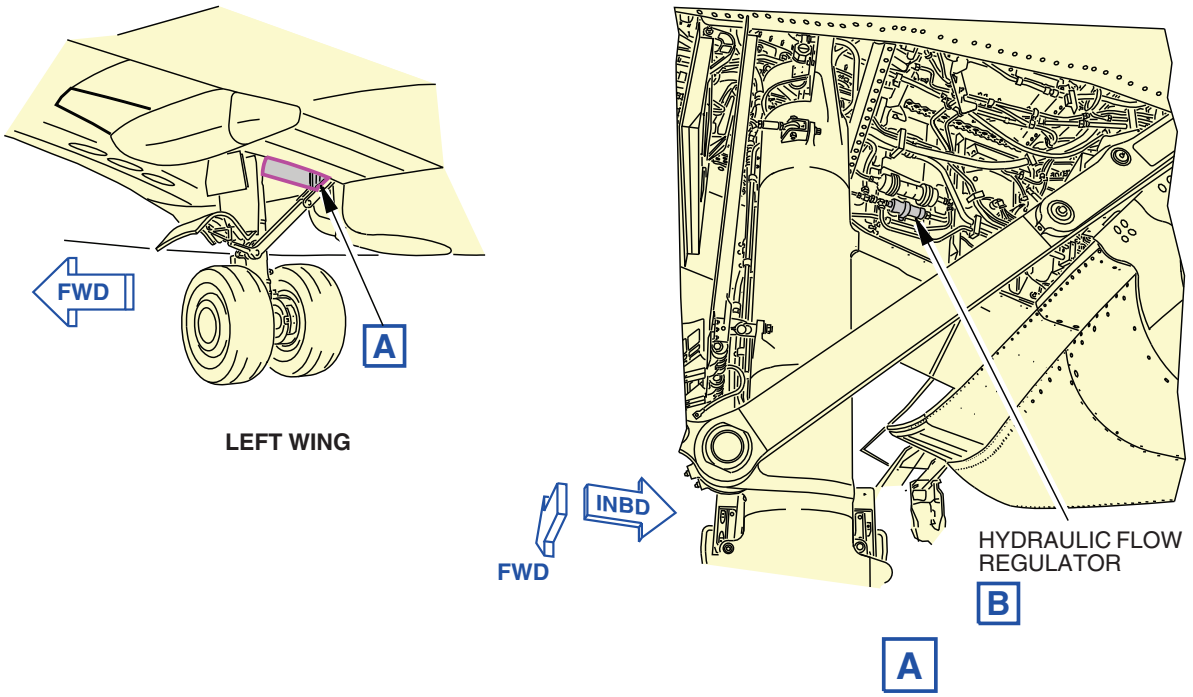
(5) Install the caps/plugs, STD-14686, on the hydraulic line [4] and the tee [9].

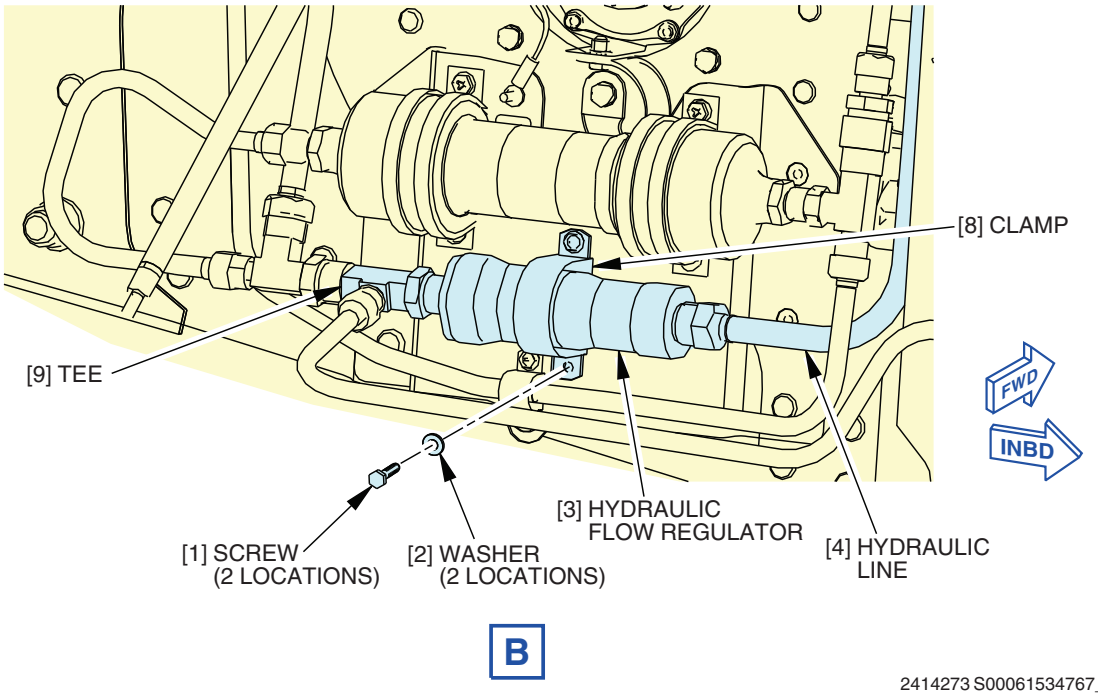
SUBTASK 32-32-72-020-005

(6) Remove the hydraulic flow regulator [3].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





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Main Gear Flow Regulator Installation
Figure 401/32-32-72-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-32-72-400-801

3. Main Gear Hydraulic Flow Regulator Installation

(Figure 401)

A. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Hydraulic flow regulator	32-32-72-01-040	PTF ALL
		32-32-72-02-010	PTF ALL

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

E. Main Gear Hydraulic Flow Regulator Installation

SUBTASK 32-32-72-420-001

(1) Put the hydraulic flow regulator [3] in position to align it with the hydraulic lines.

SUBTASK 32-32-72-080-001

(2) Remove the caps/plugs, STD-14686, from the hydraulic lines and connect the hydraulic lines to the hydraulic flow regulator [3].

(a) Tighten the B-nut on hydraulic line [4] to 500 ±25 in-lb (56 ±3 N·m).

(b) Tighten the B-nut on hydraulic line [5] to 500 ±25 in-lb (56 ±3 N·m).

(c) Tighten the B-nut on hydraulic line [6] to 270 ±14 in-lb (31 ±2 N·m).

(d) Tighten the B-nut on hydraulic line [7] to 270 ±14 in-lb (31 ±2 N·m).

SUBTASK 32-32-72-420-002

(3) Install the screws [1] and washers [2] in the clamps for the hydraulic lines.

SUBTASK 32-32-72-420-003

(4) Tighten the hydraulic lines.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Main Gear Hydraulic Flow Regulator Installation Test

SUBTASK 32-32-72-480-003

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-32-72-860-002

(2) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-32-72-700-001

(3) Move the landing gear lever to the UP position, then to the DN position three times to cycle the hydraulic pressure.

NOTE: This will bleed the air from the regulator and the lines.

NOTE: Use the override for the landing gear lever lock to move the control lever for the landing gear from the DN position to the UP position.

(a) Hold the landing gear lever in each position for three seconds before the landing gear lever is moved to the next position.

(b) Make sure that the UP hose of the retract actuator moves to show that hydraulic pressure is applied when the control lever is moved to the UP position.

(c) Move the control lever to the DN position.

(d) Make sure that the DOWN hose on the retract actuator moves when the control lever is moved from the UP position to the DN position.

SUBTASK 32-32-72-210-001

(4) Do a check for leakage with the hydraulic pressure applied.

SUBTASK 32-32-72-860-003

(5) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-32-72-210-002

(6) Make sure that there are no leaks in the hydraulic connections.

SUBTASK 32-32-72-840-001

(7) Examine the hydraulic reservoirs for the correct servicing (TASK 12-12-00-610-801).

(a) Do the servicing if it is necessary.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN GEAR DOWNLOCK SPRINGS - REMOVAL/INSTALLATION

1. General

A. This procedure has these two tasks:

(1) A removal of the main gear downlock springs

(2) An installation of the main gear downlock springs.

TASK 32-32-91-000-801

2. Main Gear Downlock Springs Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
57-50-02-000-801	Inboard Fixed Trailing Edge Lower Panel Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1866	Expander Set - Spring, Main Landing Gear and Nose Landing Gear Part #: C32014-29 Supplier: 81205 Opt Part #: C32014-1 Supplier: 81205 Opt Part #: C32014-20 Supplier: 81205

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

D. Access Panels

Number	Name/Location
193CL	Wheel Well Panel - Aft Outboard
193CR	Wheel Well Panel - Aft Outboard
551FB	Lower Inboard Fixed Trailing Edge, Landing Gear Access Panel
651FB	Lower Inboard Fixed Trailing Edge, Landing Gear Access Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare for the Removal

SUBTASK 32-32-91-480-005

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-32-91-860-001

(2) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-32-91-580-001

(3) Install chocks around the tires of the main landing gear (PAGEBLOCK 10-11-05/201).

F. Main Gear Downlock Springs Removal

SUBTASK 32-32-91-020-001

(1) Do these steps to remove the forward downlock spring assembly [1]:

(a) Remove the pin [3], nut [4], and washer [2] from the outboard attachment end of the forward downlock spring assembly [1].

(b) Remove the pin [3], nut [4], and washer [2] from the inboard attachment end of the forward downlock spring assembly [1].

(c) Install the expander set, SPL-1866, on the forward downlock spring assembly [1] to release the tension.

(d) Install a plastic cap or tape on the threads on the inboard and outboard pins.

NOTE: This will protect the threads when you remove the forward downlock spring assembly.

(e) Disconnect the inboard and outboard ends of the forward downlock spring assembly [1].

SUBTASK 32-32-91-020-010

(2) Do these steps to remove the aft downlock spring assembly [1]:

(a) Remove the applicable access panel:

1) For the left wing, remove this access panel:

(TASK 57-50-02-000-801)

Number	Name/Location
551FB	Lower Inboard Fixed Trailing Edge, Landing Gear Access Panel

2) For the right wing, remove this access panel:

(TASK 57-50-02-000-801)

Number	Name/Location
651FB	Lower Inboard Fixed Trailing Edge, Landing Gear Access Panel

(b) If it is necessary, remove the applicable access panel to get better access to the aft outboard downlock spring assembly [36]:

1) For the left wing, open this access panel:

Number	Name/Location
193CL	Wheel Well Panel - Aft Outboard

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) For the right wing, open this access panel:

Number	Name/Location
193CR	Wheel Well Panel - Aft Outboard

(c) Remove the pin [3], nut [4], and washer [2] from the outboard attachment end of the aft downlock spring assembly [1].

(d) Remove the pin [3], nut [4], and washer [2] from the inboard attachment end of the aft downlock spring assembly [1].

(e) Install the expander set, SPL-1866, on the aft downlock spring assembly [1] to release the tension.

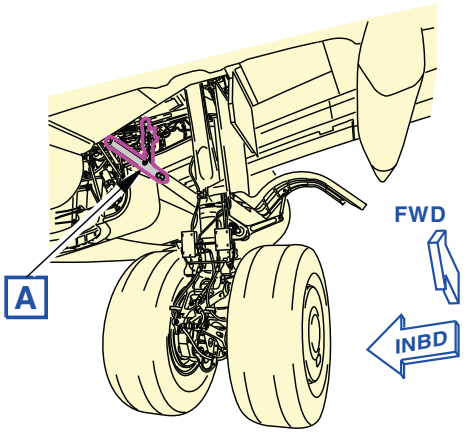
(f) Install a plastic cap or tape on the threads on the inboard and outboard pins.

NOTE: This will protect the threads when you remove the aft downlock spring assembly.

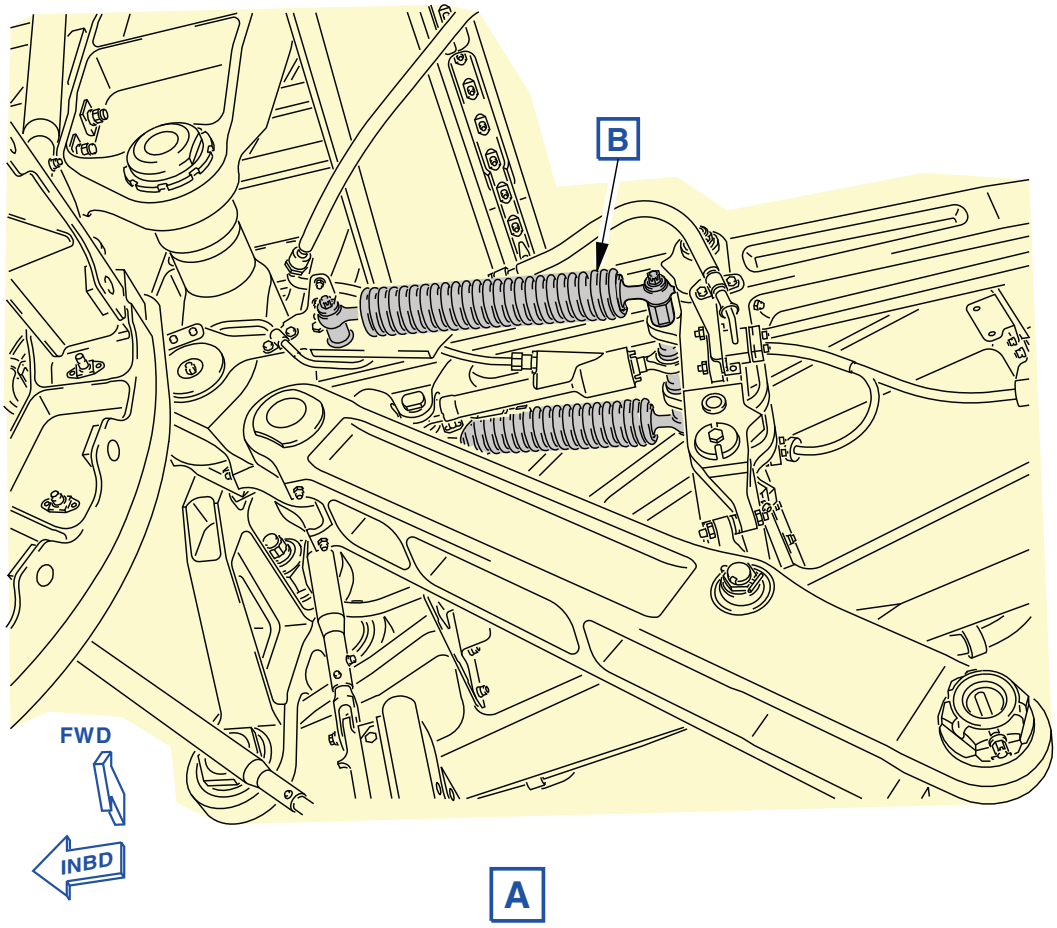
(g) Disconnect the inboard and outboard ends of the aft downlock spring assembly [1].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



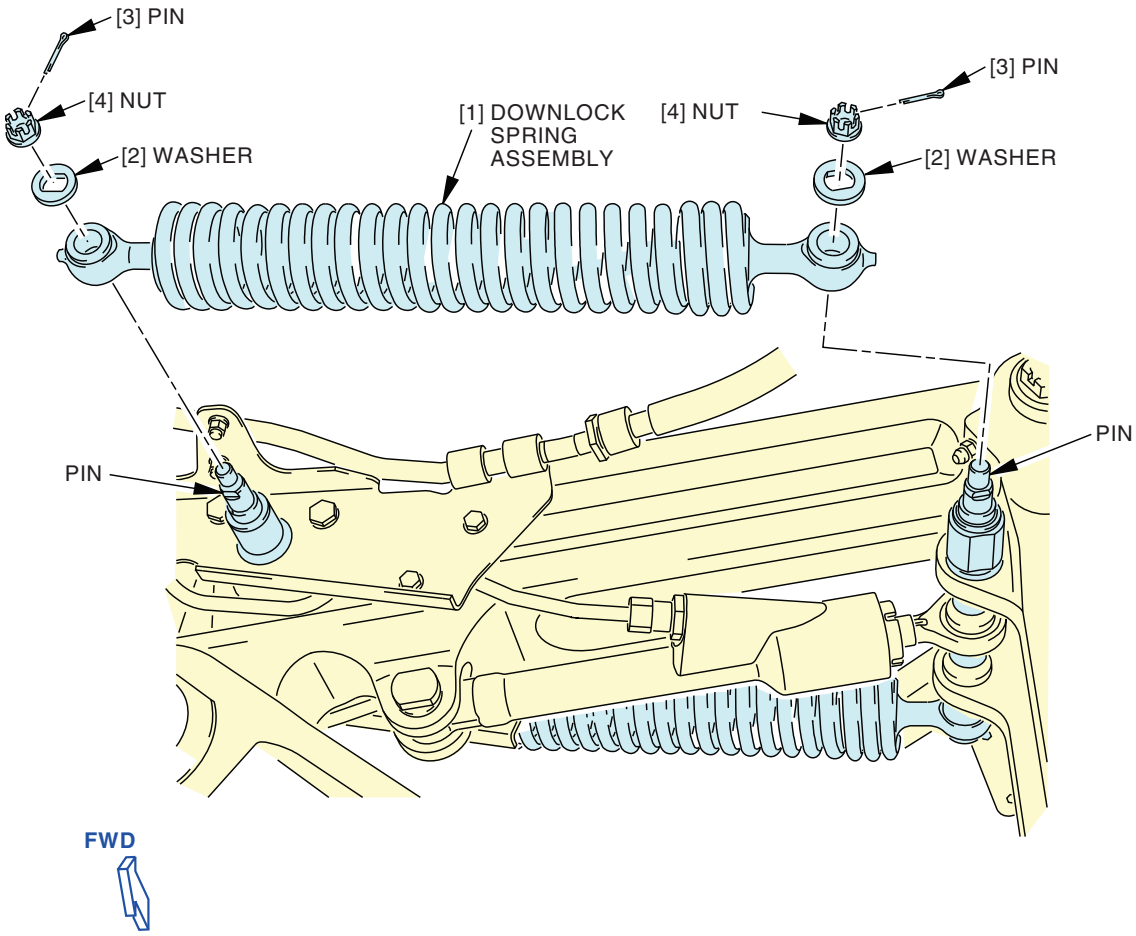
LEFT MAIN LANDING GEAR
(RIGHT MAIN LANDING GEAR IS EQUIVALENT)



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Main Landing Gear Downlock Springs Installation
Figure 401/32-32-91-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Main Landing Gear Downlock Springs Installation
Figure 401/32-32-91-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-32-91-400-801

3. Main Gear Downlock Spring Installation

(Figure 401)

A. References

Reference	Title
57-50-02-400-801	Inboard Fixed Trailing Edge Lower Panel Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1866	Expander Set - Spring, Main Landing Gear and Nose Landing Gear Part #: C32014-29 Supplier: 81205 Opt Part #: C32014-1 Supplier: 81205 Opt Part #: C32014-20 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Downlock spring assembly	32-32-91-01-045	PTF ALL
3	Pin	32-32-91-01-010	PTF ALL

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

F. Access Panels

Number	Name/Location
193CL	Wheel Well Panel - Aft Outboard
193CR	Wheel Well Panel - Aft Outboard
551FB	Lower Inboard Fixed Trailing Edge, Landing Gear Access Panel
651FB	Lower Inboard Fixed Trailing Edge, Landing Gear Access Panel

G. Main Gear Downlock Spring Installation

SUBTASK 32-32-91-020-008

(1) Remove the plastic cap or tape on the threads on the inboard and outboard pins.

SUBTASK 32-32-91-420-001

(2) Do these steps to install the downlock spring assembly [1]:

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WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin layer of corrosion inhibiting material, G50136, on the parts for the downlock spring assembly [1] that follow:

1) the threads and thread reliefs of the inboard and outboard attach points
2) washers [2]
3) nuts [4].

(b) Lubricate the chrome plated or thermal spray coated surfaces of attach pins with grease, D00633.

(c) Install the expander set, SPL-1866, on the forward downlock spring assembly [1].

(d) Put the forward downlock spring assembly [1] in position at the inboard and outboard attach points.

(e) Install the washers [2] and nuts [4] on the inboard and outboard attach points for the forward downlock spring assembly [1].

(f) Tighten the nuts [4] to 20 ±5 in-lb (2.3 ±0.6 N·m).

(g) Loosen to the nearest castellation, if it is necessary, and install the pins [3].

(h) Install the expander set, SPL-1866, on the aft downlock spring assembly [1] and extend it to attach on the outboard end.

(i) Put the aft downlock spring assembly [1] in position at the inboard and outboard attach points.

(j) Install the washers [2] and nuts [4] on the inboard and outboard attach points for the aft downlock spring assembly [1].

(k) Tighten the nuts [4] to 20 ±5 in-lb (2.3 ±0.6 N·m).

(l) Loosen to the nearest castellation, if it is necessary, and install the pins [3].

(m) Wipe off any excess corrosion inhibiting material, G50136.

SUBTASK 32-32-91-410-001

(3) Install the applicable access panel:

(a) For the left wing, install this access panel:

(TASK 57-50-02-400-801)

Number	Name/Location
551FB	Lower Inboard Fixed Trailing Edge, Landing Gear Access Panel

(b) For the right wing, install this access panel:

(TASK 57-50-02-400-801)

Number	Name/Location
651FB	Lower Inboard Fixed Trailing Edge, Landing Gear Access Panel

SUBTASK 32-32-91-410-002

(4) If removed, install the applicable access panel:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) For the left wing, close this access panel:

Number	Name/Location
193CL	Wheel Well Panel - Aft Outboard

(b) For the right wing, close this access panel:

Number	Name/Location
193CR	Wheel Well Panel - Aft Outboard

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN GEAR BEAM HANGER PIN - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the beam hanger pin for the main landing gear

(2) An installation of the beam hanger pin for the main landing gear.

B. The beam hanger pin is referred to as the pin in this procedure.

TASK 32-32-93-000-801

2. Main Gear Beam Hanger Pin Removal

(Figure 401)

A. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1870	Hoist - Retract Actuator, Main Landing Gear Part #: C32028-109 Supplier: 81205 Part #: C32028-110 Supplier: 81205 Part #: C32028-111 Supplier: 81205 Opt Part #: C32028-1 Supplier: 81205 Opt Part #: C32028-3 Supplier: 81205 Opt Part #: C32028-56 Supplier: 81205 Opt Part #: C32028-58 Supplier: 81205 Opt Part #: C32028-59 Supplier: 81205

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
551	Left Wing - Rear Spar To Landing Gear Support Beam
651	Right Wing - Rear Spar to Landing Gear Support Beam
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

D. Access Panels

Number	Name/Location
551BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel
651BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare for the Removal

SUBTASK 32-32-93-480-004

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-32-93-860-001

(2) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-32-93-010-001

(3) Open the applicable access panel:

Number	Name/Location
551BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel
651BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

F. Main Gear Beam Hanger Pin Removal

SUBTASK 32-32-93-980-001

(1) Use a hoist, SPL-1870, to hold the head end of the retract actuator in position.

SUBTASK 32-32-93-020-001

(2) Remove these parts to disconnect the pin [1] from the structure:

(a) cotter pin [4]

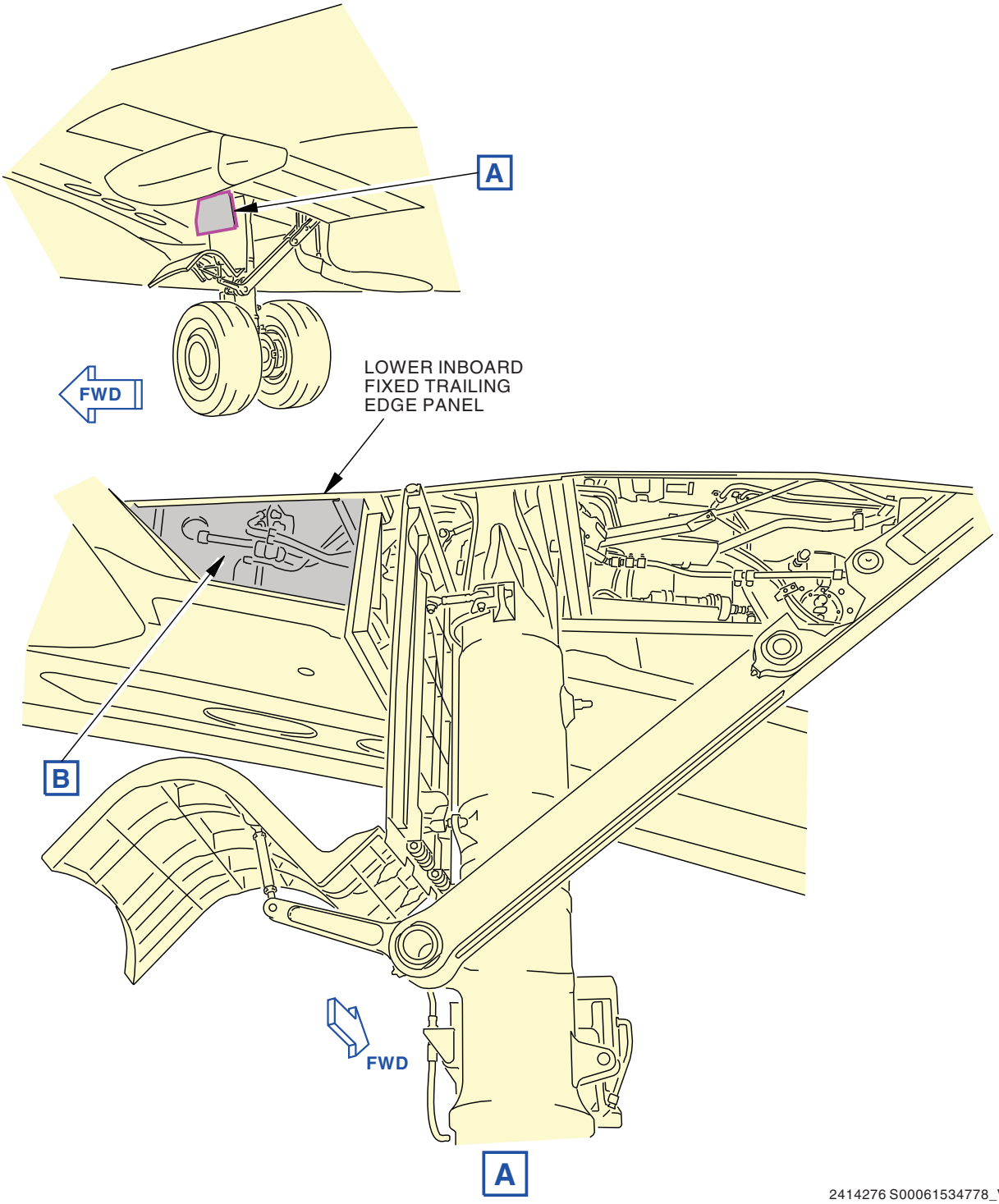
(b) nut [3]

(c) washer [2]

(d) pin [1].

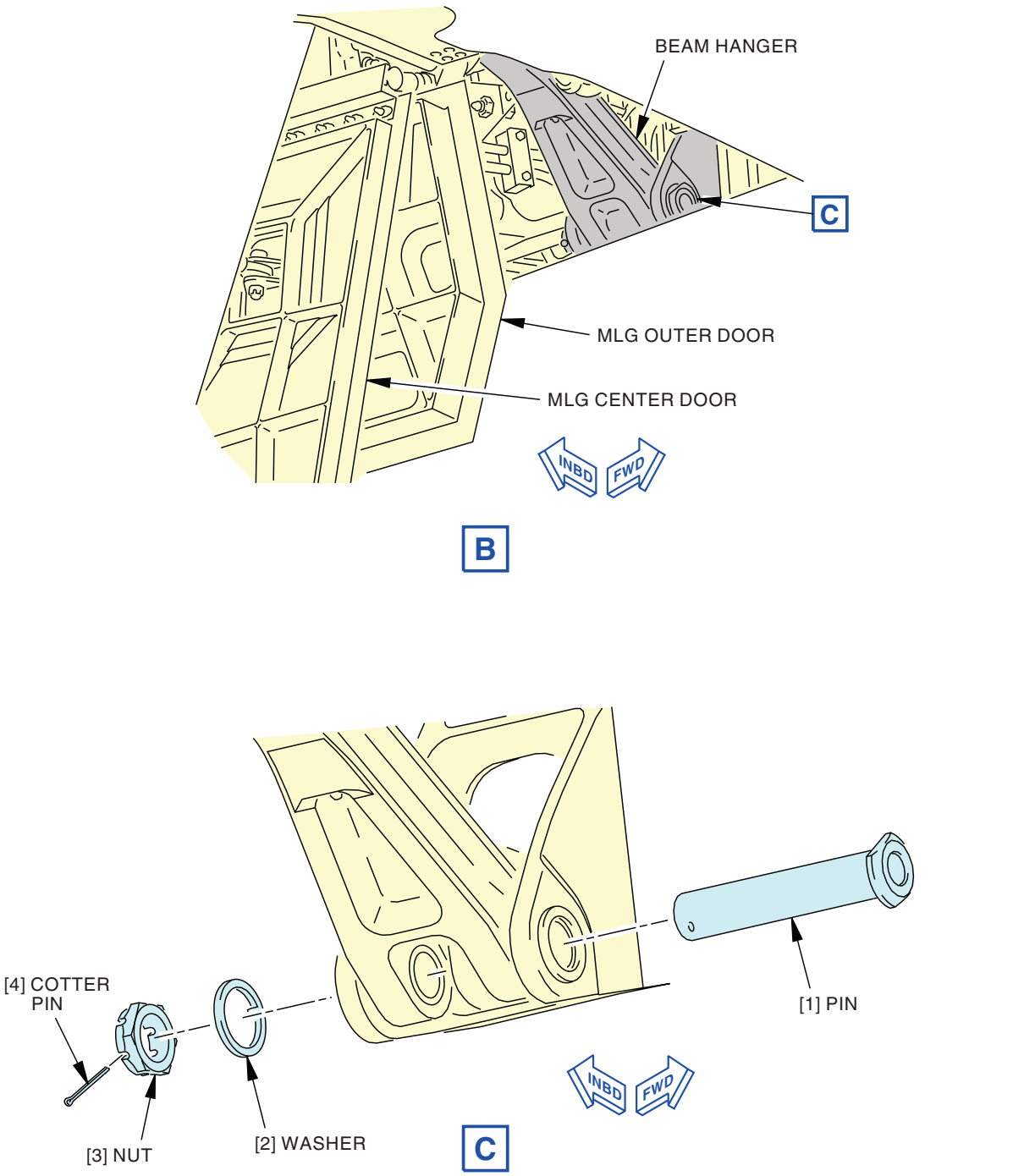
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Beam Hanger Pin Installation
Figure 401/32-32-93-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Beam Hanger Pin Installation
Figure 401/32-32-93-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-32-93-400-801

3. Main Gear Beam Hanger Pin Installation

(Figure 401)

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-32-00-710-802	Main Landing Gear - Operational Test (P/B 501)

B. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Pin	32-32-93-01-015	PTF ALL
4	Cotter pin	32-32-93-01-010	PTF ALL

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
551	Left Wing - Rear Spar To Landing Gear Support Beam
651	Right Wing - Rear Spar to Landing Gear Support Beam
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

E. Access Panels

Number	Name/Location
551BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel
651BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

F. Main Gear Beam Hanger Pin Installation

SUBTASK 32-32-93-100-001

(1) Clean the surfaces of these items (Refer to BAC5750):

(a) Attachment fitting on the beam hanger

(b) pin [1]

(c) washer [2]

(d) nut [3].

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SUBTASK 32-32-93-600-001

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(2) Apply a thin layer of corrosion inhibiting material, G50136, to the thread reliefs and threads on the pin [1], nut [3], and face of the washer [2].

SUBTASK 32-32-93-640-001

(3) Apply grease, D00633, to the chrome surface on the pin [1].

SUBTASK 32-32-93-420-001

(4) Put the beam hanger in its position in the bracket on the structure.

SUBTASK 32-32-93-420-002

(5) Install these parts that connect the beam hanger to the bracket on the wing structure:

(a) pin [1]

(b) washer [2]

(c) nut [3].

1) Tighten the nut [3] to 600 ±50 in-lb (68 ±6 N·m).

NOTE: If you do not have access to the nut, you can apply the torque on the pin head.

2) If it is necessary, loosen the nut [3] to align the nearest castellation.

(d) Install the new cotter pin [4].

SUBTASK 32-32-93-100-002

(6) Wipe off the excess of corrosion inhibiting material, G50136.

SUBTASK 32-32-93-080-001

(7) Remove the hoist.

SUBTASK 32-32-93-410-001

(8) Close the applicable access panel:

Number	Name/Location
551BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel
651BT	Upper Inboard Fixed Trailing Edge, MLG Actuator Access Panel

G. Main Gear Beam Hanger Pin Installation Test

SUBTASK 32-32-93-860-002

(1) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-32-93-080-002

(2) For the main landing gear, do this task: Landing Gear Downlock Pins Removal, TASK 32-00-01-080-801.

SUBTASK 32-32-93-870-001

(3) Do this task: Main Landing Gear - Operational Test, TASK 32-32-00-710-802.

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H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-32-93-480-002

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) For the main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-32-93-860-003

(2) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-32-93-410-002

(3) Close the applicable access panel:

Number	Name/Location
551DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel
651DB	Lower Inboard Fixed Trailing Edge, Lube Actuator & MLG Beam Outboard Attach Pin Access Panel

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE GEAR EXTENSION AND RETRACTION - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) Extension/retraction test of the nose landing gear after you replaced a component in the extension/retraction system.

(2) Operational test of the nose landing gear.

B. The times for gear retraction in the operational test depend on the hydraulic flow rate. The retraction times given in this procedure are based on a minimum flow rate of 3 gpm (11 l/min) at a hydraulic system pressure of 3000 ±150 psi (20,684 ±1034 kPa). Hydraulic power can be supplied by a hydraulic test bench or the engine driven pumps to give this minimum flow rate.

NOTE: System losses due to pressure drops and overall hydraulic system internal leakage may require a higher flow rate to achieve optimum retraction times. A hydraulic cart with a higher flow rate capacity is necessary to achieve the stated retraction times.

C. For the nose gear, the electric motor driven pump (EMDP) can be used for the component replacement or operational tests since the EMDP provides a hydraulic flow rate of approximately 6 gpm (23 l/min). When you use the EMDP, operate only one landing gear at a time because the lower flow rate of the EMDP causes slower landing gear operation if you operate more than one gear at the same time. Do not use the EMDP to measure extension/retraction times for the nose gear and main gears operated together or the main gears, operated individually or together.

D. The component replacement test is satisfactory in most cases after replacing a landing gear structural or system component. If you are troubleshooting slow landing gear extension, it is recommended to do the operational test.

TASK 32-33-00-710-801

2. Nose Landing Gear Test - Component Replacement

(Figure 501)

A. References

Reference	Title
07-10-21-580-801	Lift the Airplane Nose with the Nose Jack at Jack Point D (P/B 201)
07-10-21-580-802	Lower the Airplane Nose, Jack Point D (P/B 201)
20-40-11-910-801	Static Grounding (P/B 201)
29-00-00-993-801	Table: Hydraulic Fluid Leakage Limits (P/B 601)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

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C. Prepare for the Test

SUBTASK 32-33-00-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-33-00-760-001

(2) Do this task: Static Grounding, TASK 20-40-11-910-801.

SUBTASK 32-33-00-580-001

(3) Do this task: Lift the Airplane Nose with the Nose Jack at Jack Point D, TASK 07-10-21-580-801.

SUBTASK 32-33-00-860-003

(4) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN

SUBTASK 32-33-00-080-001

(5) For the nose landing gear, do this task: Landing Gear Downlock Pins Removal, TASK 32-00-01-080-801.

SUBTASK 32-33-00-860-004

(6) Put the master dim & test switch in the bright position.

SUBTASK 32-33-00-860-006

(7) Make sure the control lever for the landing gear is in the DN position.

SUBTASK 32-33-00-860-007

(8) Make sure that all the gear red lights on the P2-3 panel are off.

SUBTASK 32-33-00-860-008

(9) Make sure that all the gear green lights on the P2-3 panel are on.

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SUBTASK 32-33-00-710-001

WARNING

MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF CONTROL SURFACES, CABLES, AND LANDING GEAR AREAS. PRESSURIZATION OF A HYDRAULIC SYSTEM CAN MOVE CONTROL SURFACES, CABLES, LANDING GEAR, AND LANDING GEAR DOORS. MOVEMENT OF CONTROL SURFACES, CABLES, LANDING GEAR, AND LANDING GEAR DOORS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(10) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-33-00-860-009

WARNING

MAKE SURE THAT YOU OPEN THE CIRCUIT BREAKER FOR THE WEATHER RADAR SYSTEM. THE FORWARD MOVEMENT OF A THRUST LEVER CAN CAUSE THE AUTOMATIC OPERATION OF THE SYSTEM. THE OPERATION OF THIS SYSTEM CAN CAUSE SERIOUS INJURY TO PERSONS AND DAMAGE TO EQUIPMENT IN THE AREA OF THE NOSE RADOME.

(11) Move the two thrust levers to the full forward position.

D. Nose Landing Gear Test (Normal Extension and Retraction)

SUBTASK 32-33-00-710-002

WARNING

MAKE SURE ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE NOSE LANDING GEAR. FAST MOVEMENT OF THE NOSE LANDING GEAR CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Move the control lever for the landing gear to UP.

(a) Make sure the nose landing gear retracts.

SUBTASK 32-33-00-860-010

(2) Make sure the green light and the red light for the nose landing gear are off.

SUBTASK 32-33-00-710-003

(3) Move the control lever to the DN position.

(a) Make sure the nose landing gear extends.

SUBTASK 32-33-00-710-004

(4) Move the landing gear control handle UP and DOWN 5 times, to let the nose landing gear fully extend and retract.

(a) Make sure the nose landing gear retraction and extension is smooth.

(b) Make sure there are no hydraulic leaks.

NOTE: The presence of a fluid film on the actuator rod is normal. This film is different from the actual continuous drips that are the indication of actuator seal leakage. (Table 29-00-00-993-801)

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E. Put the Airplane Back to Its Usual Condition

SUBTASK 32-33-00-840-001

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN

SUBTASK 32-33-00-860-011

(2) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-33-00-480-002

(3) For the nose landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-33-00-580-002

(4) Do this task: Lower the Airplane Nose, Jack Point D, TASK 07-10-21-580-802.

——— END OF TASK ———

TASK 32-33-00-710-802

3. Nose Landing Gear Operational Test

(Figure 501)

A. References

Reference	Title
07-10-21-580-801	Lift the Airplane Nose with the Nose Jack at Jack Point D (P/B 201)
07-10-21-580-802	Lower the Airplane Nose, Jack Point D (P/B 201)
12-15-41-610-802	Nose Landing Gear Shock Strut Servicing, Airplane on the Ground (P/B 301)
20-40-11-910-801	Static Grounding (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
117	Electrical and Electronics Compartment - Left

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(Continued)

Zone	Area
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Test

SUBTASK 32-33-00-480-003

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-33-00-760-002

(2) Do this task: Static Grounding, TASK 20-40-11-910-801.

SUBTASK 32-33-00-710-008

(3) Make sure that the shock strut's pressure of the nose landing gear is correct (TASK 12-15-41-610-802).

SUBTASK 32-33-00-580-003

(4) Do this task: Lift the Airplane Nose with the Nose Jack at Jack Point D, TASK 07-10-21-580-801.

SUBTASK 32-33-00-860-015

(5) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN

SUBTASK 32-33-00-080-002

(6) For the nose landing gear, do this task: Landing Gear Downlock Pins Removal, TASK 32-00-01-080-801.

SUBTASK 32-33-00-860-016

(7) Make sure that the control lever for the landing gear is in the DN position.

SUBTASK 32-33-00-860-017

(8) Make sure that all the gear red lights on the P2-3 panel are off.

SUBTASK 32-33-00-860-018

(9) Make sure that all the gear green lights on the P2-3 panel are on.

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SUBTASK 32-33-00-710-005

(10) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-33-00-860-019

WARNING MAKE SURE THAT YOU OPEN THE CIRCUIT BREAKER FOR THE WEATHER RADAR SYSTEM. THE FORWARD MOVEMENT OF A THRUST LEVER CAN CAUSE THE AUTOMATIC OPERATION OF THE SYSTEM. THE OPERATION OF THIS SYSTEM CAN CAUSE SERIOUS INJURY TO PERSONS AND DAMAGE TO EQUIPMENT IN THE AREA OF THE NOSE RADOME.

(11) Move the two thrust levers to the full forward position.

SUBTASK 32-33-00-860-020

(12) Put the master dim and test switch in the bright position.

D. Nose Landing Gear Operational Test

SUBTASK 32-33-00-710-006

WARNING MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE NOSE LANDING GEAR. FAST MOVEMENT OF THE NOSE LANDING GEAR CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Operate the override trigger and move the control lever for the landing gear to UP.

(a) Measure the time for retraction of the nose gear.

NOTE: Measure the nose gear retraction time from the time the control lever is put in the UP position until the red "disagree" light for the nose gear goes off. Use the nose gear red "disagree" light to measure the retraction time.

(b) Make sure that the nose landing gear is up and locked in 9.0 seconds or less.

SUBTASK 32-33-00-860-022

(2) Make sure that the green light and the red light for the nose landing gear are off.

SUBTASK 32-33-00-710-007

(3) Move the control lever to the DN position and measure the time for extension of the nose gear.

NOTE: Measure the nose gear extension time from the time the control lever is put in the DN position until the landing gear green light for the nose gear comes on. Use the nose gear green light to measure the extension time.

(a) Make sure that the nose landing gear is down and locked in 13.0 seconds or less.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 32-33-00-860-023

(1) For hydraulic systems A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-33-00-480-004

(2) For the nose landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-33-00-580-004

(3) Do this task: Lower the Airplane Nose, Jack Point D, TASK 07-10-21-580-802.

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SUBTASK 32-33-00-860-024

(4) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

F/O Electrical System Panel, P6-2

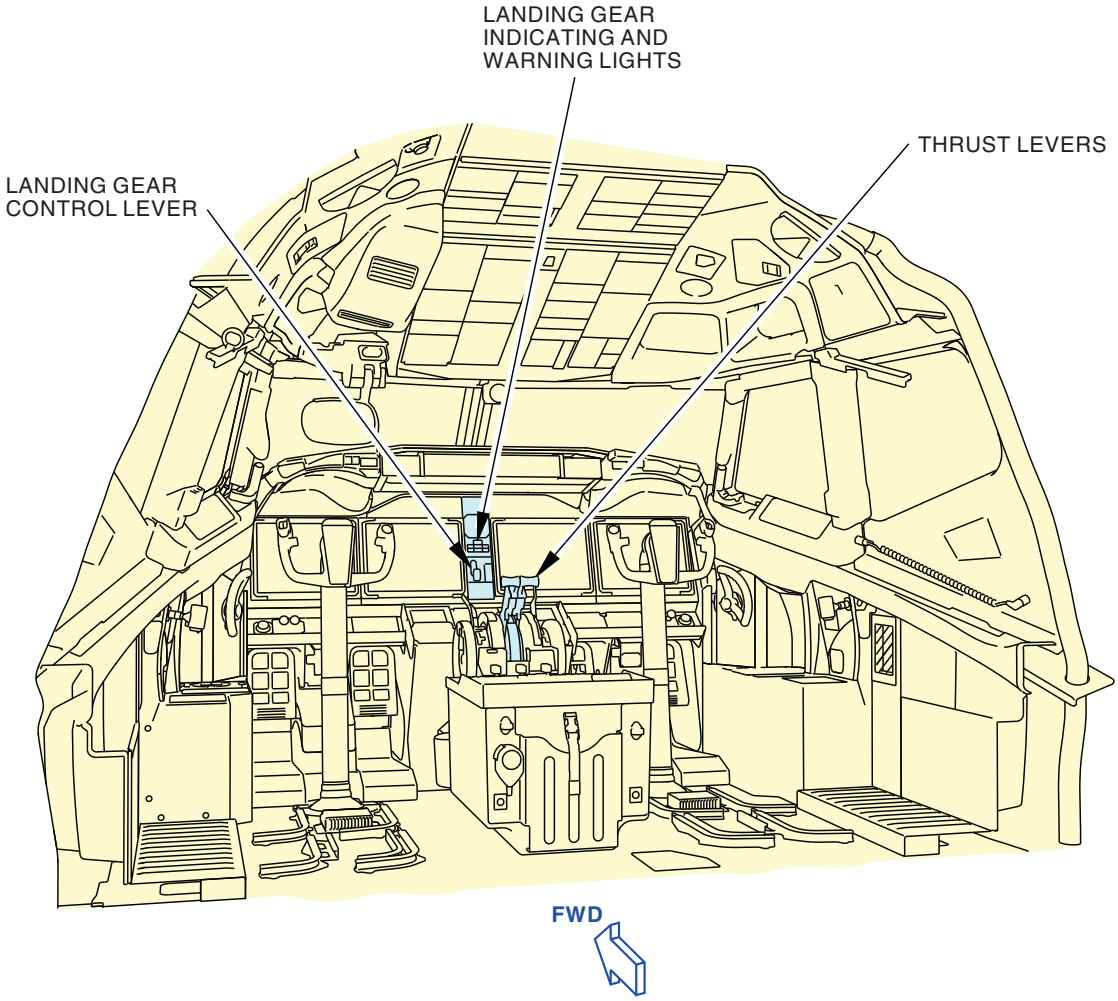
Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN

END OF TASK

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FLIGHT COMPARTMENT

2414278 S00061534786_V1

Nose Gear Extension and Retraction Test
Figure 501/32-33-00-990-801

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NOSE GEAR RETRACT ACTUATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Nose Gear Retract Actuator Removal

(2) Nose Gear Retract Actuator Installation.

(a) The installation of the retract actuator for the nose gear includes two installation tests. An installation test with the airplane on jacks or an installation test with the airplane off jacks. You can choose to use either of the installation test methods.

TASK 32-33-11-000-801

2. Nose Gear Retract Actuator Removal

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

D. Prepare for the Removal

SUBTASK 32-33-11-480-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-33-11-860-001

(2) For hydraulic system A, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 32-33-11-580-001

(3) Install chocks around the tires of the nose landing gear (PAGEBLOCK 10-11-05/201).

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E. Nose Gear Retract Actuator Removal

SUBTASK 32-33-11-020-001

(1) Disconnect the hydraulic UP-hose at the bracket on the retract actuator [1].

SUBTASK 32-33-11-020-002

(2) Disconnect the hydraulic DOWN-hose at the fitting on the retract actuator [1].

SUBTASK 32-33-11-480-002

(3) Install caps/plugs, STD-14686, on the hydraulic hoses and the fittings on the retract actuator [1].

SUBTASK 32-33-11-020-003

(4) Hold the retract actuator [1] in its position and disconnect the rod end of the retract actuator [1] from the drag brace (View B):

(a) Remove the bolts [18], washers [19], washers [20], nuts [21], and retainer [22] from the pin [23].

(b) Remove the washers [15] and pin [23] from the airplane.

SUBTASK 32-33-11-020-004

(5) Hold the retract actuator [1] in its position and disconnect the head end of the retract actuator [1] from the top fitting (View C):

(a) Remove the bolts [18], washers [19], washers [20], nuts [21], and retainer [22] from the pin [23].

(b) Remove the washers [17] and pin [24] from the airplane.

SUBTASK 32-33-11-020-005

(6) Remove the retract actuator [1] from the airplane.

SUBTASK 32-33-11-020-006

(7) If the replacement retract actuator [1] does not have the tube assembly installed, remove these parts from the retract actuator [1] you removed:

(a) The reducer [2].

(b) The screws [4], washers [5], and clamp [3].

(c) The tube assembly [6].

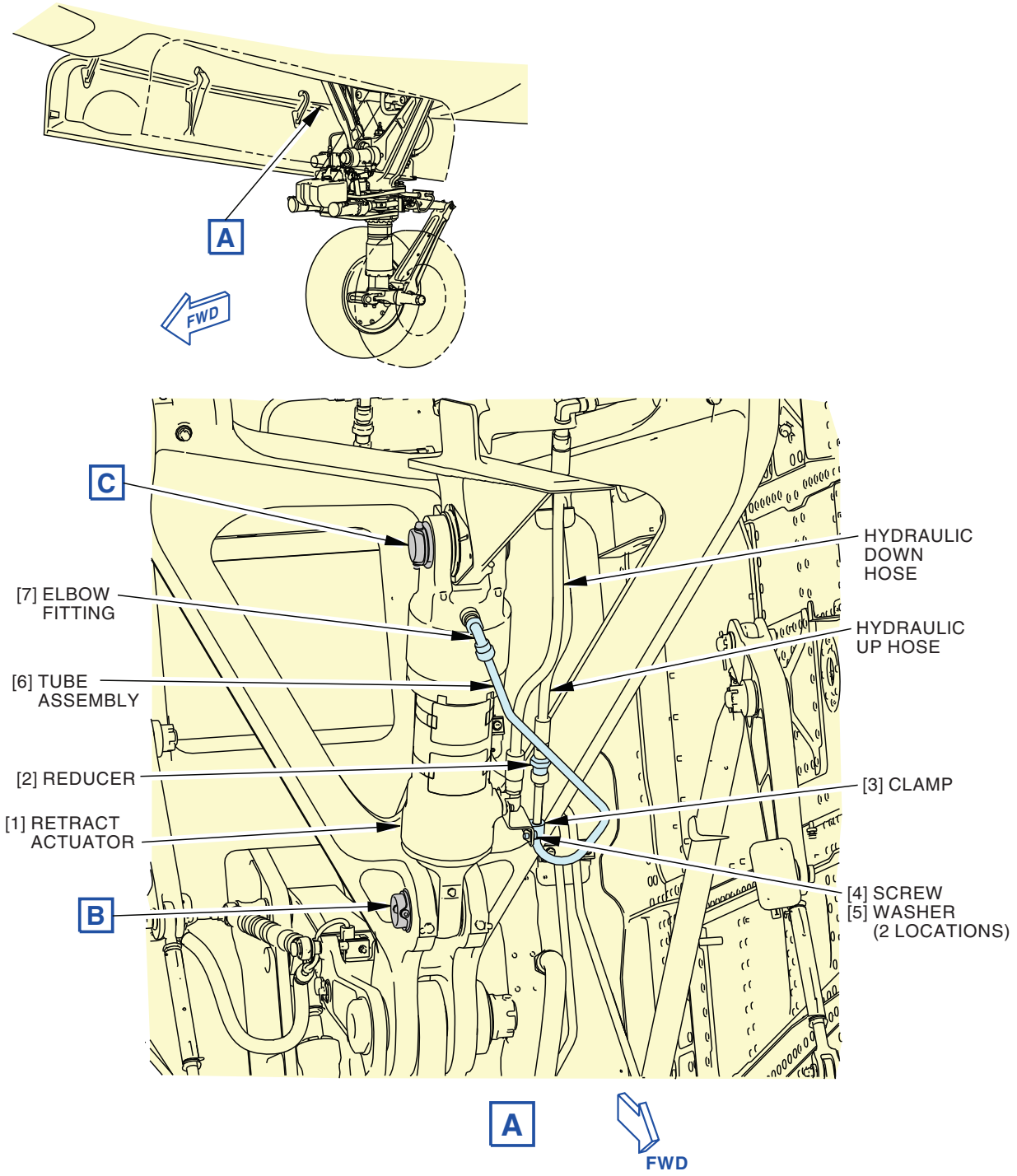
(d) The elbow fitting [7].

SUBTASK 32-33-11-480-003

(8) Install a cap/plug, STD-14686, in the extend port of the retract actuator [1].

——— END OF TASK ———

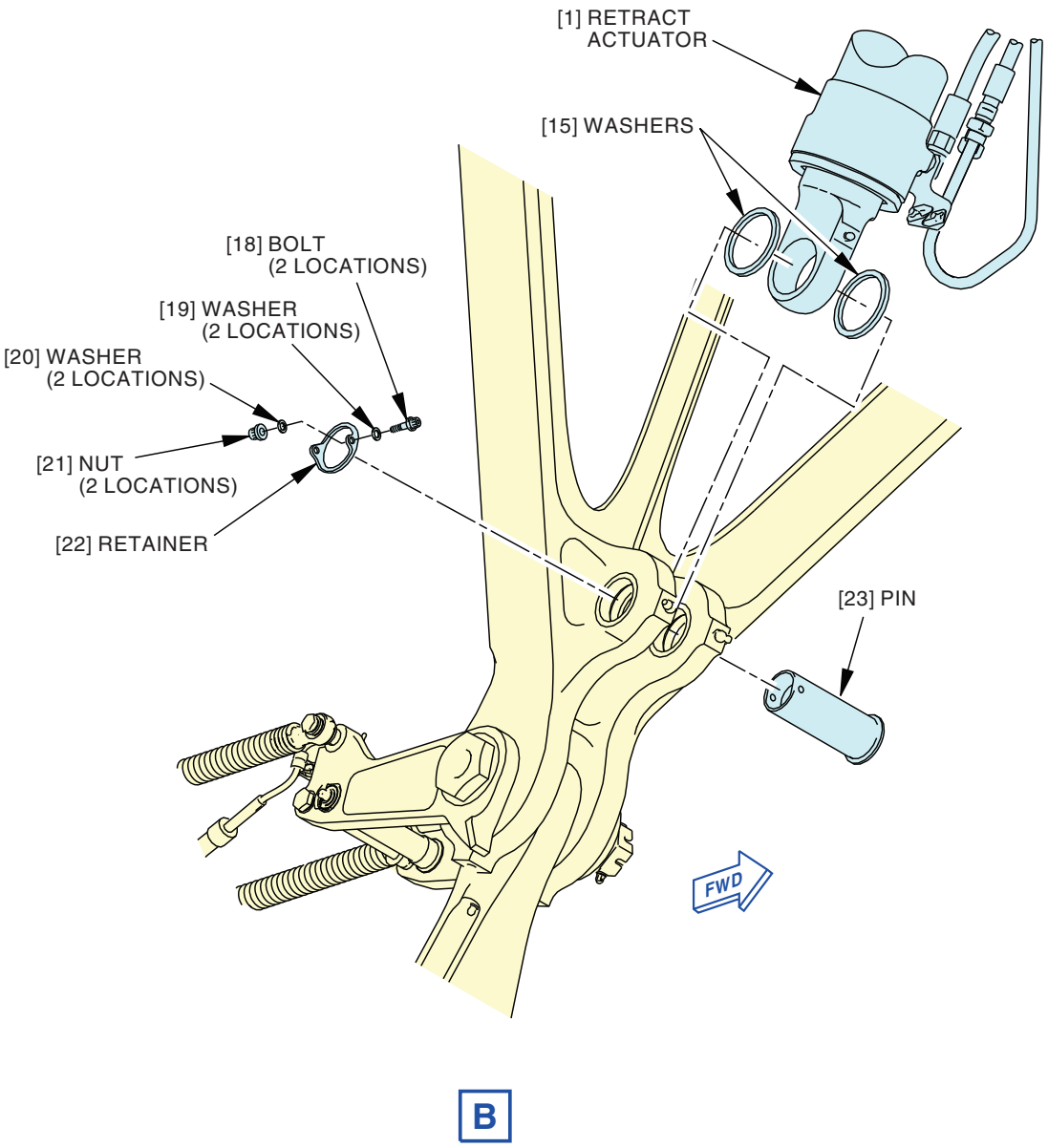
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Nose Gear Retract Actuator Installation
Figure 401/32-33-11-990-801 (Sheet 1 of 3)

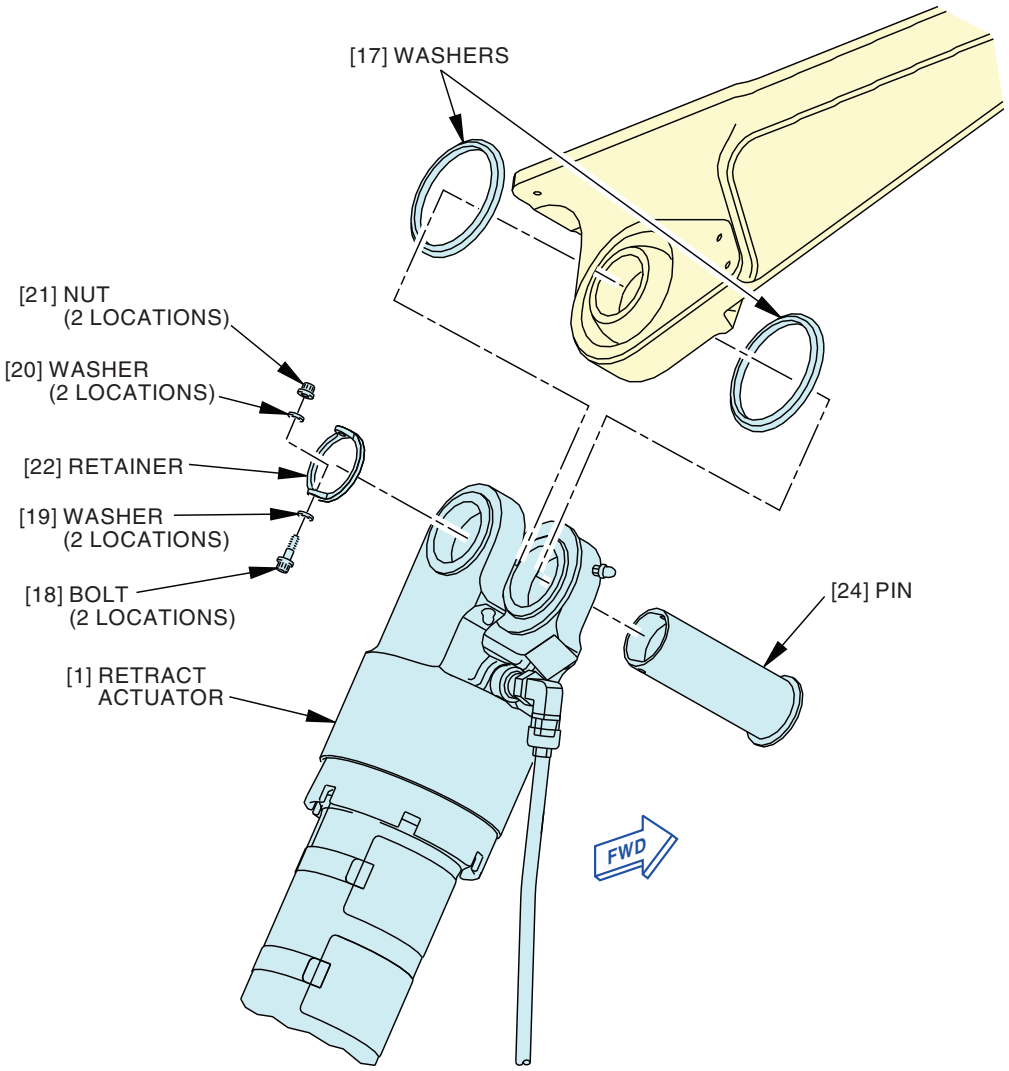
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Nose Gear Retract Actuator Installation
Figure 401/32-33-11-990-801 (Sheet 2 of 3)

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Nose Gear Retract Actuator Installation
Figure 401/32-33-11-990-801 (Sheet 3 of 3)

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TASK 32-33-11-400-801

3. Nose Gear Retract Actuator Installation

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
07-10-21-580-801	Lift the Airplane Nose with the Nose Jack at Jack Point D (P/B 201)
07-10-21-580-802	Lower the Airplane Nose, Jack Point D (P/B 201)
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
29-00-00-993-801	Table: Hydraulic Fluid Leakage Limits (P/B 601)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
D00633	Grease - Aircraft General Purpose	BMS3-33

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Retract actuator	32-33-11-01-036	PTF ALL

E. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

F. Nose Gear Retract Actuator Installation

SUBTASK 32-33-11-080-001

(1) Remove the caps/plugs, STD-14686, from the port and the fitting on the retract actuator [1].

SUBTASK 32-33-11-020-007

(2) If the replacement retract actuator [1] does not have the tube assembly installed, install these parts from the retract actuator [1] you removed:

(a) The elbow fitting [7].

(b) The tube assembly [6].

(c) The screws [4], washers [5], and clamp [3].

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(d) The reducer [2].

SUBTASK 32-33-11-980-001

(3) Make sure that the retract actuator [1] is in the retracted position.

SUBTASK 32-33-11-610-001

(4) Make sure that the retract actuator [1] is full of hydraulic fluid, D00153.

(a) Put a cap/plug, STD-14686, on the hydraulic fitting on the retract actuator [1] where the DOWN-hose will attach.

(b) Put a cap/plug, STD-14686, on the fitting on the retract actuator [1] where the UP-hose will attach.

SUBTASK 32-33-11-600-001

(5) Prepare the retract actuator [1] for installation on the top fitting:

(a) Apply grease, D00633, to the bushings of the retract actuator [1] and the shaft of the pin [24].

SUBTASK 32-33-11-420-001

(6) Hold the retract actuator [1] in its position and do these steps to connect the head end of the retract actuator [1] to the top fitting (View C, Figure 401):

(a) Install the pin [24] and the washers [17].

NOTE: The head end of pin must be on the same side as the hydraulic hoses to prevent physical interference.

(b) Install the retainer [22] with the flat surface touching the retract actuator [1] on the pin [24].

(c) Install the bolts [18], washers [19], washers [20], and nuts [21] on the retainer [22] and the pin [24].

SUBTASK 32-33-11-420-002

(7) Prepare the retract actuator [1] for installation on the drag brace:

(a) Apply grease, D00633, to the bushings of the retract actuator [1] and the shaft of the pin [23].

SUBTASK 32-33-11-420-003

(8) Hold the retract actuator [1] in its position and do these steps to connect the rod end of the retract actuator [1] to the drag brace (View B, Figure 401):

(a) Install the pin [23] and washers [15].

(b) Install the retainer [22] with the flat surface touching the retract actuator [1] on the pin [23].

(c) Install the bolts [18], washers [19], washers [20], and nuts [21] on the retainer [22] and the pin [23].

SUBTASK 32-33-11-080-002

(9) Remove the caps/plugs, STD-14686, from the hydraulic hoses.

SUBTASK 32-33-11-420-004

(10) Connect the hydraulic UP-hose at the bracket on the retract actuator [1].

(a) Tighten the B-nut on the hydraulic UP-hose to 133 in-lb (15.0 N·m) - 147 in-lb (16.6 N·m).

SUBTASK 32-33-11-420-005

(11) Connect the hydraulic DOWN-hose at the fitting on the retract actuator [1].

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(a) Tighten the B-nut on the hydraulic DOWN-hose to 256.5 in-lb (29.0 N·m) - 283.5 in-lb (32.0 N·m).

SUBTASK 32-33-11-600-002

(12) Apply grease, D00633, to the lubrication fittings to lubricate these items:

(a) The head end bushings of the retract actuator [1].

(b) The bearing of the attach fitting for the head end.

(c) The rod end bearing of the retract actuator [1].

(d) The bushings of the attach fitting for the rod end.

G. Nose Gear Retract Actuator Installation Test

NOTE: This test gives two methods to do a check of the nose gear retract actuator. Method one is with the airplane lifted on jacks and method two is with the airplane on the ground if jacks are not available.

SUBTASK 32-33-11-700-001

(1) Installation test with the airplane on jacks (method one).

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE APPLICABLE LANDING GEAR. THE LANDING GEAR EXTENDS AND RETRACTS QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(a) Lift the airplane for nose gear retraction (TASK 07-10-21-580-801).

(b) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

(c) Remove the downlock pin for the nose landing gear (TASK 32-00-01-080-801).

WARNING MAKE SURE THAT YOU INSTALL THE DOWNLOCK PINS IN THE MAIN LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE MAIN LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(d) Use the override trigger and move the control lever for the landing gear from the DN position to the UP position and back to DN seven times.

NOTE: It will bleed the air from the retract actuator and the lines. Examine for the correct gear operation.

(e) Examine the hydraulic connections for leaks.

NOTE: The presence of a fluid film on the actuator rod is normal. This film is different from the actual continuous drips that are the indication of actuator seal leakage. (Table 29-00-00-993-801)

(f) Extend the nose landing gear and install the downlock pin (TASK 32-00-01-480-801).

(g) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

(h) Do this task: Lower the Airplane Nose, Jack Point D, TASK 07-10-21-580-802.

(i) Examine the hydraulic reservoirs for the correct servicing (TASK 12-12-00-610-801).

1) Do a servicing if it is necessary.

SUBTASK 32-33-11-700-002

(2) Installation test with the airplane off jacks (method two).

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(a) Make sure that you installed the hydraulic hoses correctly.

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(b) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

(c) Use the override trigger to move the control lever for the landing gear from the DN position to the UP position for three times.

1) Hold the lever in each position for three seconds before you move the landing gear lever to the subsequent position.

(d) Make sure that the retract actuator [1] tries to extend when you select the UP position.

(e) Move the control lever for the landing gear to the DN position.

(f) Do a check for leakage with the hydraulic pressure applied.

NOTE: The presence of a fluid film on the actuator rod is normal. This film is different from the actual continuous drips that are the indication of actuator seal leakage. (Table 29-00-00-993-801)

(g) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

(h) Make sure that there are no leaks in the hydraulic connections.

(i) Examine the hydraulic reservoirs for the correct servicing (TASK 12-12-00-610-801).

1) Do a servicing if it is necessary.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE GEAR LOCK ACTUATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Nose Landing Gear Lock Actuator Removal

(2) Nose Landing Gear Lock Actuator Installation.

(a) The installation of the lock actuator for the nose landing gear includes an installation test of the lock actuator for the nose landing gear with the airplane off the jacks.

TASK 32-33-21-000-801

2. Nose Landing Gear Lock Actuator Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

D. Prepare for the Removal

SUBTASK 32-33-21-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-33-21-860-001

(2) For hydraulic system A, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 32-33-21-580-001

(3) Install chocks around the tires of the nose landing gear (PAGEBLOCK 10-11-05/201).

E. Nose Landing Gear Lock Actuator Removal

SUBTASK 32-33-21-020-001

(1) Disconnect hydraulic line [12] and hydraulic line [13] from the lock actuator [2].

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SUBTASK 32-33-21-480-002

(2) Install caps/plugs, STD-14686, on the hydraulic lines.

SUBTASK 32-33-21-020-002

(3) Remove the bolt [8], washer [9], and nut [10] from the bottom pin [7].

SUBTASK 32-33-21-020-003

(4) Remove the bottom pin [7].

SUBTASK 32-33-21-020-004

(5) Remove the pin [4], nut [3], washer [5], and washer [6] for the top pin [1].

SUBTASK 32-33-21-020-005

(6) Remove the top pin [1].

SUBTASK 32-33-21-020-006

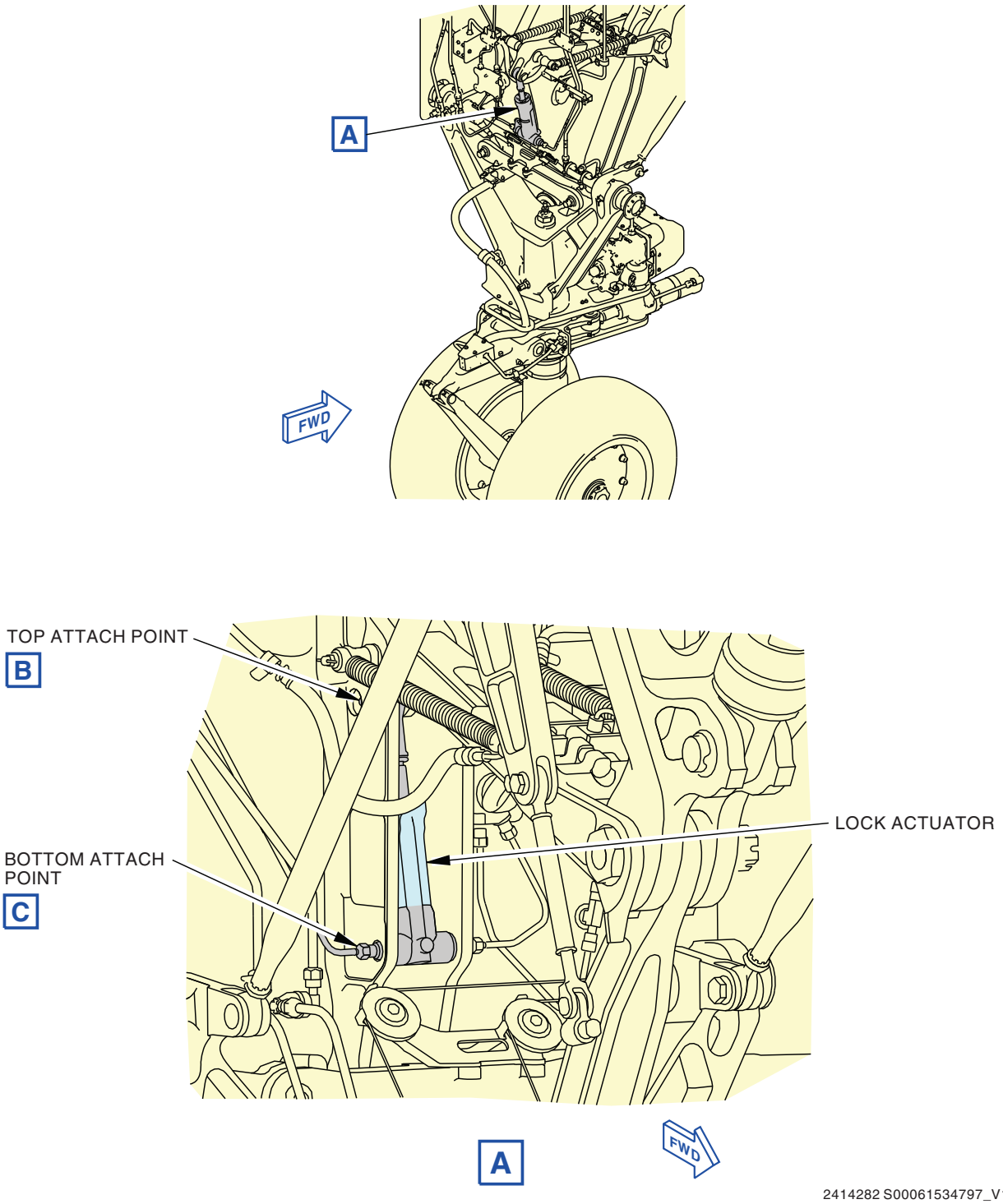
(7) Remove the two washers [11].

SUBTASK 32-33-21-020-007

(8) Remove the lock actuator [2].

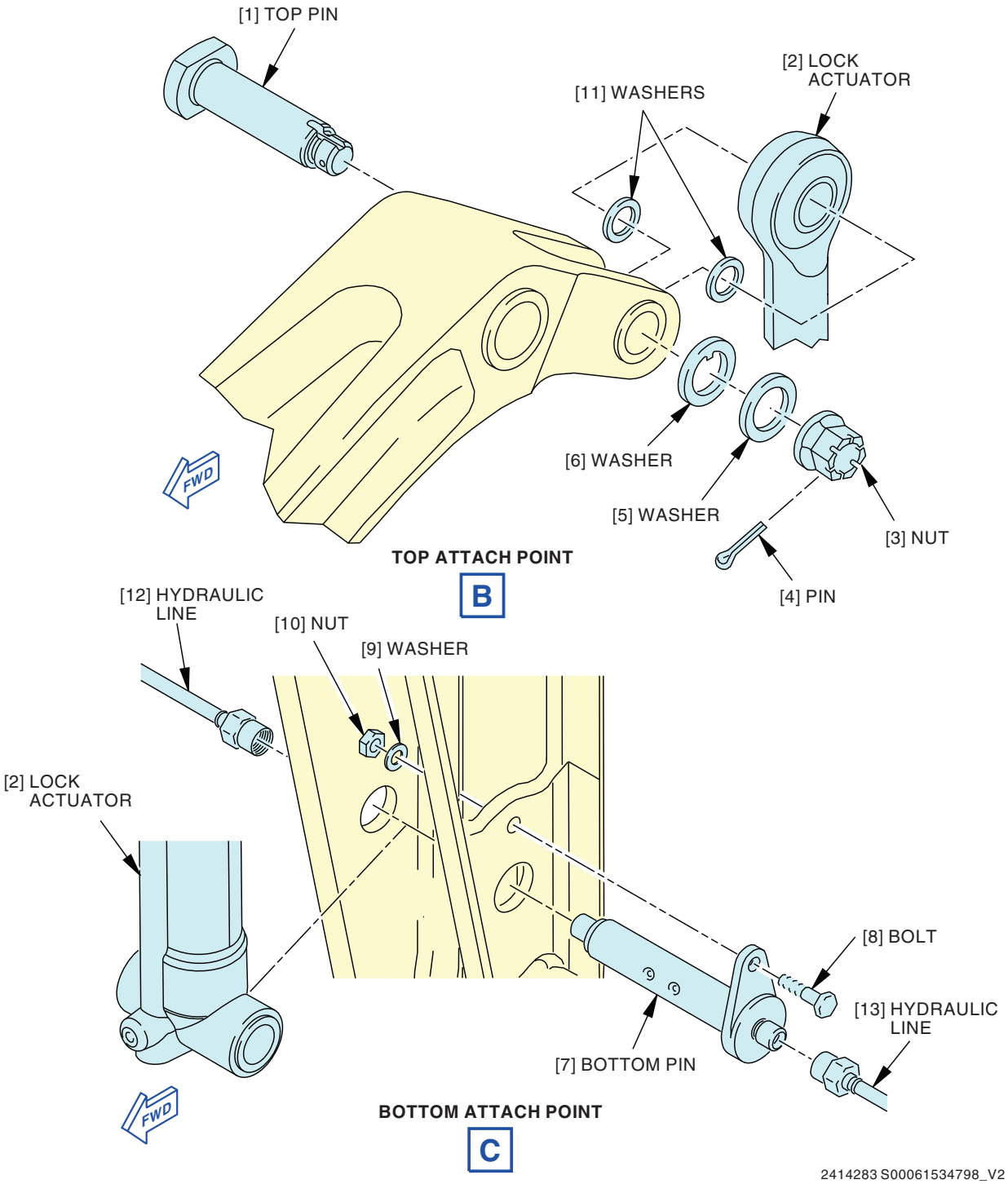
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Landing Gear Lock Actuator Installation
Figure 401/32-33-21-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Landing Gear Lock Actuator Installation
Figure 401/32-33-21-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-33-21-400-801

3. Nose Landing Gear Lock Actuator Installation

(Figure 401)

A. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
D00054	Fluid - Hydraulic, Assembly Lubricant, MCS 352B	
D00633	Grease - Aircraft General Purpose	BMS3-33
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Lock actuator	32-33-21-01-055	PTF ALL
4	Pin	32-33-21-01-050	PTF ALL

E. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

F. Nose Landing Gear Lock Actuator Installation

SUBTASK 32-33-21-620-001

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(1) Put a thin layer of corrosion inhibiting material, G50136, on these items:

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(a) The thread relief and threads of the top pin [1].

(b) The face of the washer [5].

(c) The new pin [4].

SUBTASK 32-33-21-640-001

(2) Apply a layer of grease, D00633, to the top pin [1] and mating surfaces.

(a) Do not apply it to the threads or thread relief.

SUBTASK 32-33-21-420-001

(3) Put the two washers [11] in their positions.

SUBTASK 32-33-21-420-002

(4) Put the lock actuator [2] in its position with the hydraulic boss in the forward direction (Figure 401).

SUBTASK 32-33-21-420-003

(5) Install the top pin [1].

SUBTASK 32-33-21-420-004

(6) Install the nut [3], washer [5], and washer [6].

SUBTASK 32-33-21-210-001

(7) Make sure that the bottom pin [7] and holes in the bottom pin [7] are clean.

SUBTASK 32-33-21-640-002

(8) Apply a thin layer of assembly lubricant MCS 352B fluid, D00054, to these parts:

(a) The bottom pin [7].

(b) Inside the bore of the trunnion on the lock actuator [2].

1) Make sure that the seals in the trunnion of the lock actuator [2] are lubricated and in the correct position.

SUBTASK 32-33-21-420-005

(9) Install the bottom pin [7].

SUBTASK 32-33-21-420-006

(10) Install the bolt [8], washer [9], and nut [10].

SUBTASK 32-33-21-640-003

(11) Use sealant, A02315, to apply a fillet seal around the area where the bottom pin [7] touches the aft bulkhead nose wheel well.

SUBTASK 32-33-21-080-001

(12) Remove the caps/plugs, STD-14686, from the hydraulic lines.

SUBTASK 32-33-21-420-007

(13) Connect hydraulic line [12] and hydraulic line [13] to the lock actuator [2].

(a) Tighten the B-nuts on hydraulic line [12] an hydraulic line [13] to 140 ±7 in-lb (16 ±1 N·m).

SUBTASK 32-33-21-420-008

(14) Tighten the nut [3] to 21 ±3 ft-lb (28 ±5 N·m).

SUBTASK 32-33-21-420-009

(15) Install the pin [4].

(a) Loosen the nut [3] if it is necessary to align the nut [3] with the hole for the pin [4].

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G. Nose Landing Gear Lock Actuator Installation Test

SUBTASK 32-33-21-700-001

(1) Do these steps to do an installation test of the lock actuator [2] with the airplane off jacks:

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) If the downlock pins are not installed, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

(b) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

(c) Use the override trigger to move the landing gear lever from the DN position to the UP position for three times.

1) Hold the landing gear lever in each position for three seconds before you move the landing gear lever to the subsequent position.

2) Watch for leaks each time the landing gear lever is moved.

(d) Move the landing gear lever to the DN position.

(e) Make sure that the nose lock actuator tries to extend.

NOTE: The hydraulic line to the actuator should pulse.

(f) Move the landing gear lever to the UP position.

(g) Make sure that the nose lock actuator tries to retract.

NOTE: The hydraulic line to the actuator should pulse.

(h) Move the landing gear control lever to the DOWN position.

(i) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

(j) Examine the hydraulic reservoirs for the correct servicing (TASK 12-12-00-610-801).

1) Do the servicing if it is necessary.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE GEAR LOCK VALVE MANIFOLD - REMOVAL/INSTALLATION

1. General

A. This procedure has these two tasks:

(1) Nose Gear Lock Valve Manifold Removal

(2) Nose Gear Lock Valve Manifold Installation.

TASK 32-33-31-000-801

2. Nose Gear Lock Valve Manifold Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

D. Prepare for the Removal

SUBTASK 32-33-31-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-33-31-860-001

(2) Remove the pressure from hydraulic system A (TASK 29-09-00-860-802).

SUBTASK 32-33-31-580-001

(3) Install chocks around the tires of the nose landing gear PAGEBLOCK 10-11-05/201).

E. Nose Gear Lock Valve Manifold Removal

SUBTASK 32-33-31-020-001

(1) Disconnect the hydraulic lines from the manifold assembly [1].

SUBTASK 32-33-31-480-002

(2) Install caps/plugs, STD-14686, on the hydraulic lines.

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SUBTASK 32-33-31-020-002

(3) Remove the bolts [2] and washers [3] that hold the manifold assembly [1] on the structure.

SUBTASK 32-33-31-020-003

(4) Remove the manifold assembly [1] from the airplane.

SUBTASK 32-33-31-020-004

(5) Do these steps if the replacement manifold assembly [1] does not have hydraulic unions [5] installed:

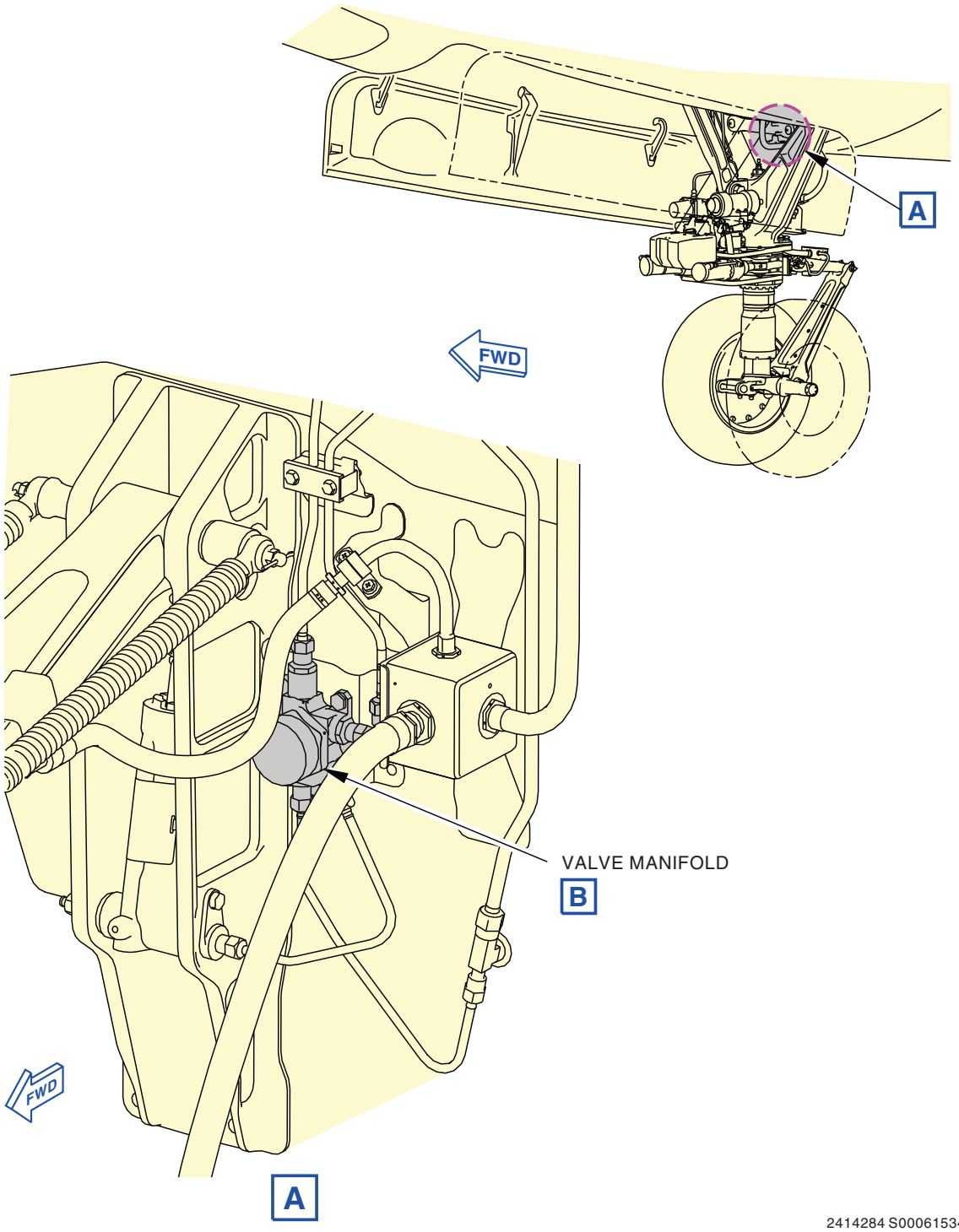
(a) Remove the hydraulic unions [5] from the manifold assembly [1].

(b) Remove and discard the packings [4].

(c) Install plugs in the ports of the manifold assembly [1].

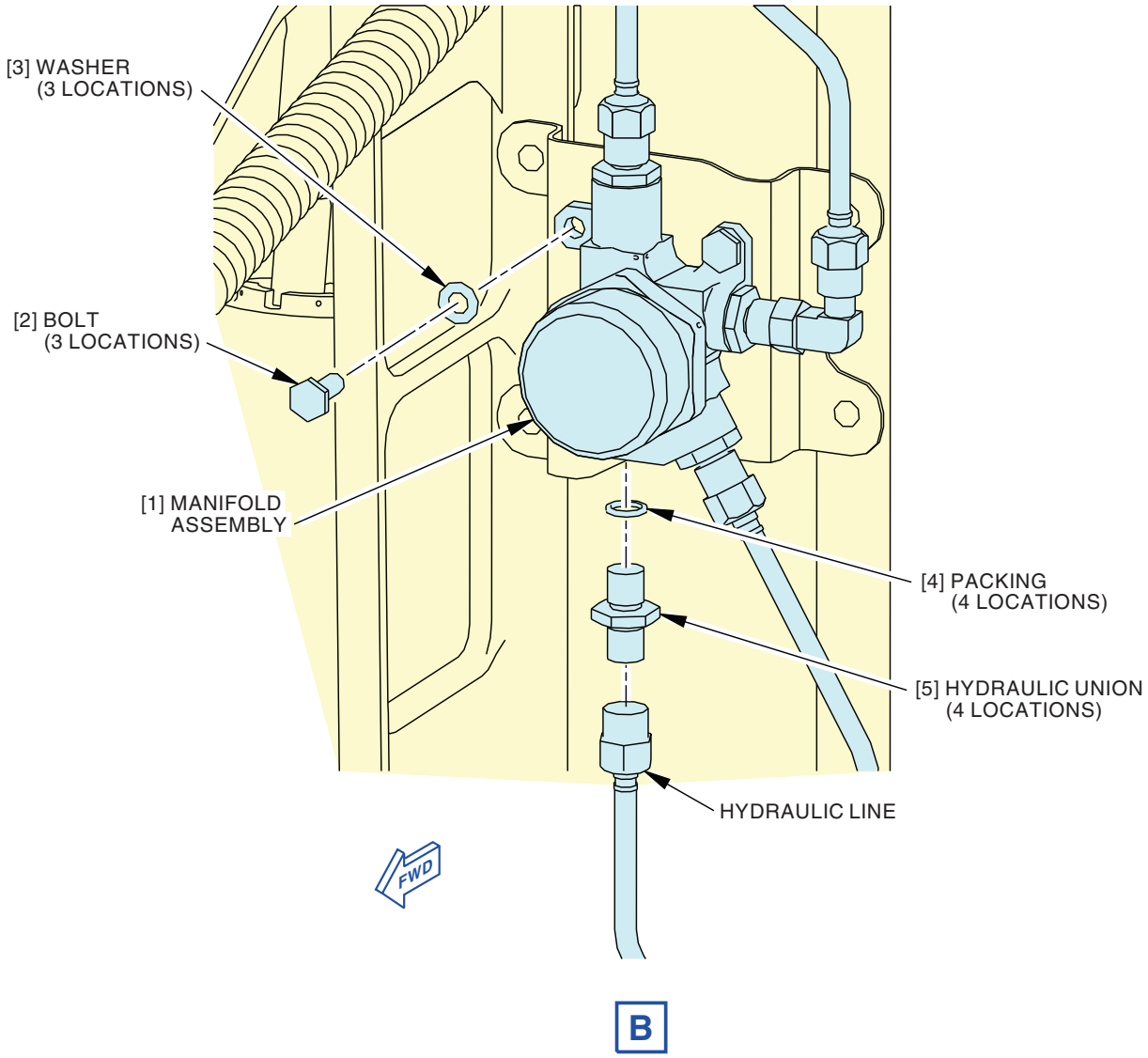
——— END OF TASK ———

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Nose Landing Gear Valve Manifold Installation
Figure 401/32-33-31-990-801 (Sheet 1 of 2)

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Nose Landing Gear Valve Manifold Installation
Figure 401/32-33-31-990-801 (Sheet 2 of 2)

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TASK 32-33-31-400-801

3. Nose Gear Lock Valve Manifold Installation

(Figure 401)

A. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Manifold assembly	32-33-31-01-005	PTF ALL
4	Packing	32-33-31-01-040	PTF ALL

E. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

F. Nose Gear Lock Valve Manifold Installation

SUBTASK 32-33-31-420-001

(1) Do these steps if the manifold assembly [1] does not have the hydraulic unions [5] installed:

(a) Lubricate these items with hydraulic fluid, D00153:

1) The packings [4].

2) The hydraulic unions [5].

(b) Remove the plugs from the ports on the manifold assembly [1].

(c) Install the packings [4] and the hydraulic unions [5] in the manifold assembly [1].

SUBTASK 32-33-31-610-001

(2) Fill the manifold assembly [1] with hydraulic fluid, D00153.

NOTE: Put a cap on the hydraulic unions [5].

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SUBTASK 32-33-31-420-002

(3) Put the manifold assembly [1] in its position.

SUBTASK 32-33-31-420-003

(4) Adjust the position of the manifold assembly [1] to align it with the hydraulic lines.

SUBTASK 32-33-31-080-001

(5) Remove the caps/plugs, STD-14686, from the hydraulic unions [5] and connect the hydraulic lines to the manifold assembly [1].

(a) Tighten the B-nuts on the hydraulic lines to 140 ±7 in-lb (16 ±1 N·m).

SUBTASK 32-33-31-420-004

(6) Install a washer [3] on each bolt [2].

SUBTASK 32-33-31-420-005

(7) Install the bolts [2] and washers [3] that attach the manifold assembly [1] to the structure.

G. Nose Gear Lock Valve Manifold Installation Test (Airplane on Jacks)

NOTE: This test does a check of the manifold assembly [1] with the airplane lifted on jacks.

SUBTASK 32-33-31-580-002

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

WARNING MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE NOSE LANDING GEAR. FAST MOVEMENT OF THE NOSE LANDING GEAR CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Lift the airplane for nose gear retraction (TASK 07-10-01-580-801).

SUBTASK 32-33-31-860-002

(2) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-33-31-080-002

(3) Remove the downlock pin for the nose landing gear (TASK 32-00-01-080-801).

SUBTASK 32-33-31-870-001

WARNING MAKE SURE THAT YOU INSTALL THE DOWNLOCK PINS IN THE MAIN LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE MAIN LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(4) Extend and retract the nose landing gear several times.

NOTE: This will bleed the air from the manifold assembly [1] and the lines. Look for the correct gear operation.

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